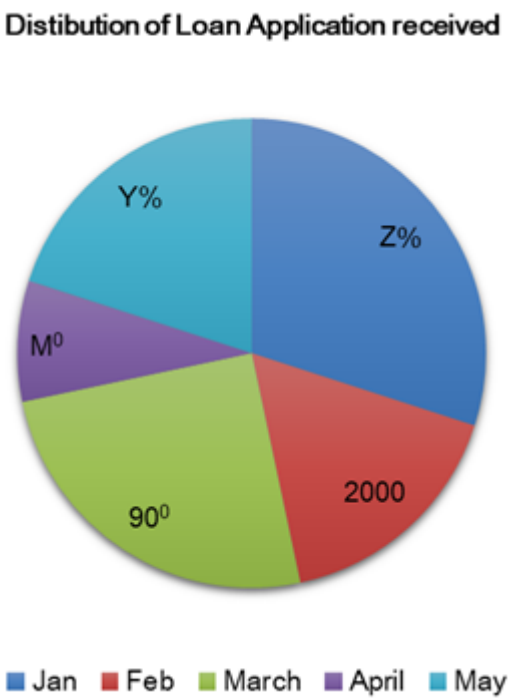
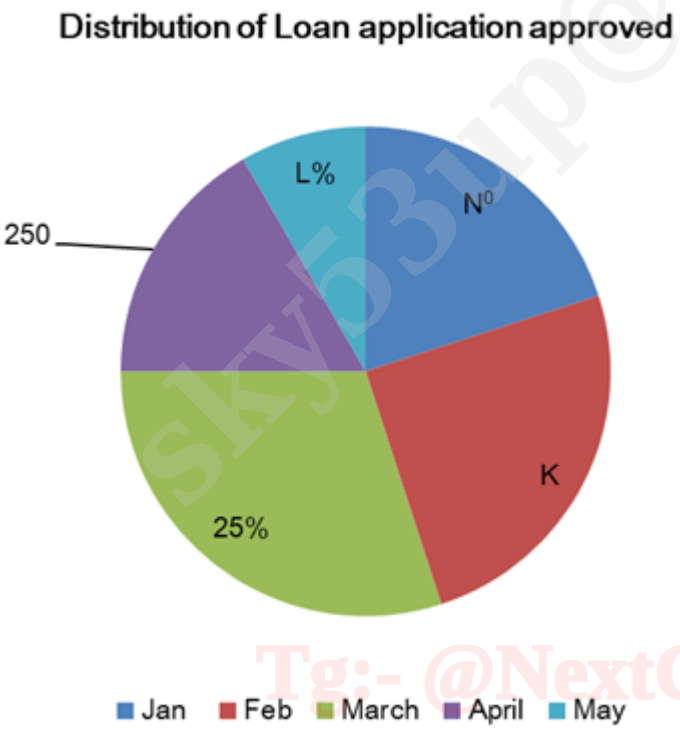
1. Questions

Read the following information carefully and answer the questions based on it.

The charts given below show the distribution of the number of loan applications received and approved in five different months (January to May) of 2022, represented as percentages, degrees or numbers.





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Note:

a). Number of loan applications received in January is 50% more than that in May.

b). Z = M, and in each month, the ratio of total applications received to total applications approved is 4:1.

For the month of January, out of the total number of loan applications approved, (M+10)% are

home loan applications, while the rest are personal loan applications. Find the number of personal loan applications rejected, if the total number of home loan applications (rejected + Approved) is the same as the total number of loan applications in the month of February?

a. 1020

b. 1640

c. 1240

d. 1080

e. None of these

2. Questions

Out of the total number of loan applications received in April, Y% of applications are from females. The maximum number of rejected male loan applications is A, such that the difference between A and 2K is _____.

a. 250

b. 200

c. 450

d. 400

e. None of these

3. Questions

Quantity I:

The difference between the number of loan applications approved in January and March month.

Quantity II:

Out of the total number of applications in the month of May, B% are approved, and out of the total number of approved applications, 3B% are male applications. Find the number of female approved applications in the month of May.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II

d. Quantity I ≥ Quantity II

e. Quantity I ≤ Quantity II

4. Questions

The number of loan applications received from January to May is twice that of June, out of which (3Y + 10) % are rejected. Out of the total number of approved applications in June, (3.5L + 5) %

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are home loan applications, while the rest are personal loan applications. Find the difference between the number of personal loan applications approved in June and March, if the ratio of personal to home loan applications approved in March is N:162.

a. 300

b. 150

c. 0

d. 90

e. None of these

5. Questions

Which of the following statement(s) is/are must be true.

I). The total number of loan applications rejected = 9000

II). The difference between the number of loan applications approved in the months of January and May > 300

III). The difference between the value of L in degrees and the value of N is Y% more than the value of M

a. I and II only

b. II and III only

c. I, II, and III

d. I and III only

e. None of these

6. Questions

Z years ago, the age of Q was 50% more than that of P , while Y years from now, the age of Q will be 12.5% more than that of P. The average present age of P and Q together is half of Q's age after Y years.

Find which of the following is/ are true.

I). Y = 3Z

II). Age of Q is Y/4, when P was born

III). Present age of P is 20% less than that of Q.

a. II and III only

b. I and III only

c. I and II only

d. I, II, and III

e. None of these

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7. Questions

A certain sum Q is invested at (3S – 10) % per annum compound interest for 2 years, amounting to Rs. (1.5Q + R). When Rs. 4Z is invested at (S + 5) % per annum, then compound interest earned after 2 years is Rs. A, while the simple interest earned on Rs. 2Z at S% per annum after three years is Rs. B. Find A is how much % more than that of B, if R is 75% of Q.

a. 75%

b. 93.33%

c. 87.5%

d. 62.5%

e. None of these

8. Questions

P and Q started a business with initial investments of Rs. Z and Rs. (Z + 4M/3), respectively. After four months, R joined with an initial investment of Rs. Y. After six months from the start of the business, P increased his investment by Rs. M. At the end of the year, it was noticed that the profit share of P is 37.5% less than that of R, and the difference between the profit share of Q and R is Rs. 24000. Find the profit share of Q, if after the increment in investment of P, it becomes half of that of R.

a. Rs. 80000

b. Rs. 120000

c. Rs. 96000

d. Rs. 144000

e. None of these

9. Questions

In a company, there are 12Z employees, and each employee likes either coffee or tea. The number of male employees who like coffee is 25% less than the number of male employees who like tea. The number of female employees who like tea is one- third of the number of male employees who like tea. The ratio of male to female employees who like coffee is 3:5. Find the number of male employees in the company.

a. 6.3Z

b. 4.2Z

c. 5.6Z

d. 3.6Z

e. None of these

10. Questions

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The time taken by boat P to travel 90 km downstream is T hours, while the time taken by boat Q to travel 180 km upstream is (T + 10) hours. The speed of boat P in still water is Z km/h, and the speed of the stream for each boat is 6 km/h, while the time taken by boat Q to travel 360 downstream is_____. The downstream speed of boat P is the same as the speed of boat Q in still water.

a. (2T + 10) hours

b. 2T hours

c. (4T – 10) hours

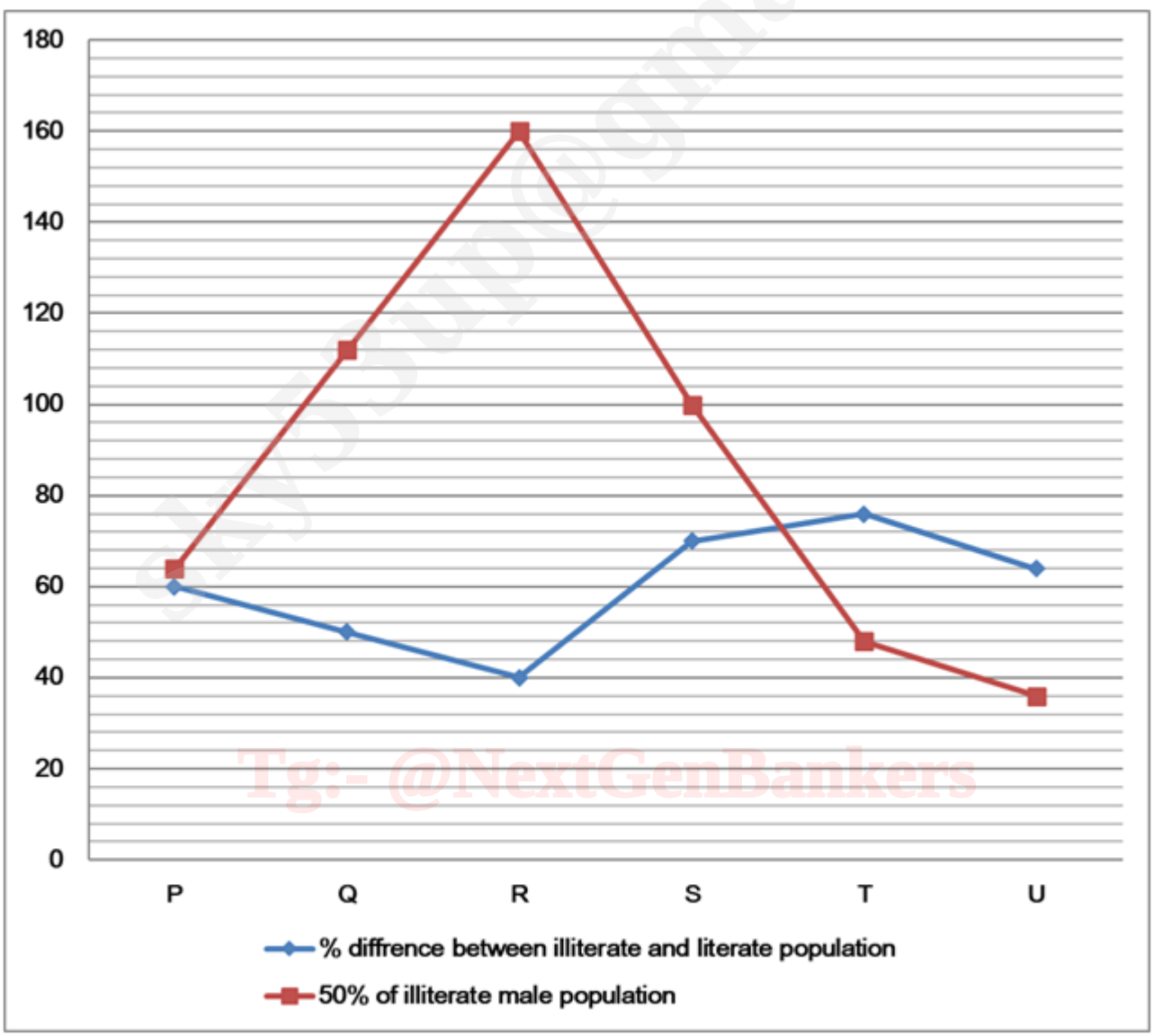
d. (T + 10) hours

e. None of these

11. Questions

Read the following information carefully and answer the below questions based on it.

The chart given below shows the difference between the number of illiterate population and the number of literate population of a village in percentage, as well as half the number of illiterate males in that village.



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Note: In each village number of literates > number of illiterates

Total population = Number of literates + Number of illiterates = Number of males + Number of females.

In village R, the number of illiterate females is 400, which is 640 less than the male population in the village. The half of the literate female population is how much % more or less than the illiterate female population.

a. 20%

b. 16.66%

c. 25%

d. 32.5%

e. None of these

12. Questions

The number of literates in village P is A, while the number of illiterates in the village is B, if the difference between A and B is 1152, then find the number of illiterate female population of village P?

a. 256

b. 252

c. 248

d. 264

e. None of these

13. Questions

If the ratio of illiterate females to illiterate males in village U is 9:4 respectively, then find the literate population of the village

a. 1148

b. 738

c. 984

d. 1066

e. None of these

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14. Questions

In village S, the literate population is 2448, and the number of female illiterates is (10Z + 182). Find the value of Z.

a. 8

b. 12

c. 5

d. 6

e. None of these

15. Questions

In village Q, the number of female population is 50% more than the number of male population. Out of the total literate population, 40% are males. Find the literate population of the village?

a. 1600

b. 1680

c. 1620

d. 1520

e. None of these

16. Questions

Read the following information carefully and answer the below questions based on it.

The table given below shows the number of employees in four different companies – P, Q, R, and S, and the ratio of males to females in each company.

Name of the company	% of employees	Male: Female
P	30%	3: A
Q	Z%	(2A +1):3
R	Y%	B:5
S	L%	(C – 3):1

Note:

a). The total number of employees in all companies together is 800, and in each company, the number of male employees > the number of female employees

b). The difference between the number of employees in Q and R (Q > R) is 60% of the difference between the number of employees in P and S.

c). A, B and C are natural numbers < 10, and the number of employees in P is thrice that of S. C is not greater than B

The difference between the number of male employees in company P and Q is what % of the difference between the number of male and female employees in company S?

a. 87.5%

b. 80%

c. 93.75%

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d. Either 75% or 90%

e. 82%

17. Questions

The average number of employees in companies P, R and T together is 264, and the number of male employees in company T is (5Y + 72), Find the number of female employees in company T?

a. 192

b. 180

c. 168

d. 208

e. None of these

18. Questions

Quantity I:

The number of male managers in company P is 5L% more than that of female managers, while the number of male non - managers and female non – managers are D and E respectively. Find D is how much % more or less than that of E.

Quantity II:

The difference between the number of male employees in company S and R is (D + 2). Find the value of D.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity IIOr No relation

d. Quantity I ≥ Quantity II

e. Quantity I ≤ Quantity II

19. Questions

Find the maximum possible difference between the total number of male and female employees in all companies together?

a. 200

b. 192

c. 208

d. 216

e. None of these

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20. Questions

Find the value of (3A + 5B + 8C)?

a. 89

b. 87

c. 97

d. Both (a) and (C)

e. None of these

Explanations:

1. Questions

In each month, the number of loan applications received= 4 x loan applications approved.

Number of loan applications received in Feb = 2000

So, the number of loan applications approved in Feb = 2000 x 1/4 = 500 = K

Number of loan applications approved in April = 250

So, the number of loan applications received in April = 250 x 4 = 1000 = M^0 x total loan applications

Z = M, that means the ratio of the number of loan applications in April to Jan = 5:18

So, loan applications in Jan = 1000 x 18/5 = 3600

So, loan applications in the month of May = 3600 x 2/3 = 2400

So, 75% x total number of loan applications = 2000 + 3600 + 1000 + 2400 = 9000

So, the total number of loan applications = 9000/75 x 100 = 12000

So, value of Z = 3600/12000 x 100 = 30 = M

Value of Y = 2400/12000 x 100 = 20 = L

Value of N = 30/100 x 360 = 108

Now we can find all the related data.

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Month	Total applications	Applications approved
Jan	3600	900
Feb	2000	500
March	3000	750
April	1000	250
May	2400	600
Total	12000	3000

Answer: E

According to the question,

For the month of January,

Number of personal loan application = 3600 – 2000 = 1600

Number of personal loan applications approved = 900 x 60% = 540

So, number of personal loan applications rejected = 1600 – 540 = 1060

Hence answer is option E

2. Questions

In each month, the number of loan applications received= 4 x loan applications approved.

Number of loan applications received in Feb = 2000

So, the number of loan applications approved in Feb = 2000 x 1/4 = 500 = K

Number of loan applications approved in April = 250

So, the number of loan applications received in April = 250 x 4 = 1000 = M^0 x total loan applications

Z = M, that means the ratio of the number of loan applications in April to Jan = 5:18

So, loan applications in Jan = 1000 x 18/5 = 3600

So, loan applications in the month of May = 3600 x 2/3 = 2400

So, 75% x total number of loan applications = 2000 + 3600 + 1000 + 2400 = 9000

So, the total number of loan applications = 9000/75 x 100 = 12000

So, value of Z = 36000/12000 x 100 = 30 = M

Value of Y = 2400/12000 x 100 = 20 = L

Value of N = 30/100 x 360 = 108

Now we can find all the related data.

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Month	Total applications	Applications approved
Jan	3600	900
Feb	2000	500
March	3000	750
April	1000	250
May	2400	600
Total	12000	3000

Answer: A

Number of loan applications in the month of April = 1000

Number of female loan applications = 20% x 1000 = 200

Number of male loan applications = 1000 – 200 = 800

Total number of male applications – number of male applications approved = number of male applications rejected.

So, maximum number of female loan applications approved = 200

So, minimum number of male loan applications approved = 250 – 200 = 50

So, maximum number of male loan applications rejected = 800 – 50 = 750 = A

Required difference = 2K – A = 2 x 500 – 750 = 250

Hence answer is option A

3. Questions

In each month, the number of loan applications received= 4 x loan applications approved.

Number of loan applications received in Feb = 2000

So, the number of loan applications approved in Feb = 2000 x 1/4 = 500 = K

Number of loan applications approved in April = 250

So, the number of loan applications received in April = 250 x 4 = 1000 = M^0 x total loan applications

Z = M, that means the ratio of the number of loan applications in April to Jan = 5:18

So, loan applications in Jan = 1000 x 18/5 = 3600

So, loan applications in the month of May = 3600 x 2/3 = 2400

So, 75% x total number of loan applications = 2000 + 3600 + 1000 + 2400 = 9000

So, the total number of loan applications = 9000/75 x 100 = 12000

So, value of Z = 36000/12000 x 100 = 30 = M

Value of Y = 2400/12000 x 100 = 20 = L

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Value of N = 30/100 x 360 = 108

Now we can find all the related data.

Month	Total applications	Applications approved
Jan	3600	900
Feb	2000	500
March	3000	750
April	1000	250
May	2400	600
Total	12000	3000

Answer: C

Quantity I.

Required difference = 900 – 750 = 150

Quantity II.

Value of B = 25%

So, the number of female loan applications approved = 600 x 25% = 150

So, Quantity I = Quantity II

Hence answer is option C

4. Questions

In each month, the number of loan applications received= 4 x loan applications approved.

Number of loan applications received in Feb = 2000

So, the number of loan applications approved in Feb = 2000 x 1/4 = 500 = K

Number of loan applications approved in April = 250

So, the number of loan applications received in April = 250 x 4 = 1000 = M^0 x total loan applications

Z = M, that means the ratio of the number of loan applications in April to Jan = 5:18

So, loan applications in Jan = 1000 x 18/5 = 3600

So, loan applications in the month of May = 3600 x 2/3 = 2400

So, 75% x total number of loan applications = 2000 + 3600 + 1000 + 2400 = 9000

So, the total number of loan applications = 9000/75 x 100 = 12000

So, value of Z = 36000/12000 x 100 = 30 = M

Value of Y = 2400/12000 x 100 = 20 = L

Value of N = 30/100 x 360 = 108

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Now we can find all the related data.

Month	Total applications	Applications approved
Jan	3600	900
Feb	2000	500
March	3000	750
April	1000	250
May	2400	600
Total	12000	3000

Answer: B

Loan applications in June = 12000 x 50% = 6000

Loan applications approved in June = 30% x 6000 = 1800

Personal loan applications approved in June = 1800 x 25% = 450

Ratio of Personal to home loan applications approved in March = 108:162 = 2:3

So, personal loan applications approved in March = 2/5 x 750 = 300

Required difference = 450 – 300 = 150

Hence answer is option B

5. Questions

In each month, the number of loan applications received= 4 x loan applications approved.

Number of loan applications received in Feb = 2000

So, the number of loan applications approved in Feb = 2000 x 1/4 = 500 = K

Number of loan applications approved in April = 250

So, the number of loan applications received in April = 250 x 4 = 1000 = M^0 x total loan applications

Z = M, that means the ratio of the number of loan applications in April to Jan = 5:18

So, loan applications in Jan = 1000 x 18/5 = 3600

So, loan applications in the month of May = 3600 x 2/3 = 2400

So, 75% x total number of loan applications = 2000 + 3600 + 1000 + 2400 = 9000

So, the total number of loan applications = 9000/75 x 100 = 12000

So, value of Z = 36000/12000 x 100 = 30 = M

Value of Y = 2400/12000 x 100 = 20 = L

Value of N = 30/100 x 360 = 108

Now we can find all the related data.

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Month	Total applications	Applications approved
Jan	3600	900
Feb	2000	500
March	3000	750
April	1000	250
May	2400	600
Total	12000	3000

Answer: D

I). Total number of loan applications rejected = 9000

Required value = 12000 – 3000 = 9000

This statement is true.

II). Difference between the number of loan applications approved in the months of January and May > 300

Required difference = 900 – 600 = 300

This statement is false.

III). Difference between the value of L in degrees and the value of N is Y% more than the value of M.

Value of L in degrees = 20/100 x 360 = 72

Difference between L and N = 108 – 72 = 36

Required % change = (36 – 30)/30 x 100 = 20% = Y%

This statement is true.

So, only I and III are true

Hence answer is option D

6. Questions

Answer: A

Let, age of P and Q before Z years be 2a, and 3a, respectively.

Now,

[2a + (Z + Y)] / [3a + (Z + Y)] = 8/9

6a = (Z + Y) …… (1)

Also,

2a + Z + 3a + Z = 3a + Z + Y

2a = (Y – Z) …….. (2)

From (1) and (2), we get

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$3Y - 3Z = Z + Y$

$Y = 2Z$

So, $2a = 2Z - Z = Z$

So, present age of $P = Z + Z = 2Z$ years = Y years

Present age of $Q = 3Z/2 + Z = 2.5Z$ years = $5Y/4$ years

I). $Y = 3Z$

$Y = 2Z$

This statement is false.

II). Age of Q is Y/4, when P was born.

Present age of $P = 2Z$ years

So, age of Q when P was born = $2.5Z - 2Z = 0.5Z = Y/4$

This statement is true.

III). Present age of P is 20% less than that of Q.

Required % = $0.5Z/2.5Z \times 100 = 20\%$

Hence answer is option A

7. Questions

Answer: C

$R = 0.75Q$

Amount after 2 years = $(1.5Q + 0.75Q) = 2.25Q$

So,

$2.25Q = Q \times [1 + (3S - 10)/100]^2$

$3/2 - 1 = (3S - 10)/100$

Value of S = 20

Value of $A = 4Z \times (5/4 \times 5/4 - 1) = 9Z/4$

Value of $B = 2Z \times 20\% \times 3 = 1.2Z$

Ratio of A to B = 15:8

Required % change = $(15 - 8)/8 \times 100 = 87.5\%$

Hence answer is option C

8. Questions

Answer: B

$Z + M = Y/2$ …… (1)

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Ratio of profit share of P, Q, and R = (Z x 3 + Y/2 x 3): (Z + 4M/3) x 6: Y x 4

Ratio of profit share of P to R = 5:8

Now,

(3Z + 3Y/2) / 4Y = 5/8

6Z + 3Y = 5Y

2Y = 6Z

Y = 3Z

So, value of M = 3Z/2 – Z = 0.5Z

So, ratio of profit share of P, Q, and R = (3Z + 9Z/2): (Z + 2Z/3)*6: 12Z

= 15Z/2: 10Z: 12Z

= 15:20:24

So, profit share of R = [24000/ (24 – 20)] x 20 = Rs. 120000

Hence Answer is option B

9. Questions

Answer: A

Let, number of male employees like tea = 4a

So, number of male employees like coffee = 75% x 4a = 3a

So, number of female employees like tea = 4a/3

So, number of female employees like coffee = 5/3 x 3a = 5a

Now,

(4a + 3a) + (4a/3) + 5a = 12Z

40a/3 = 12Z

Value of a = 12Z x 3/40 = 9Z/10

So, number of male employees in the company = 7 x 9Z/10 = 6.3Z

Hence answer is option A

10. Questions

Answer: D

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Downstream speed of boat P = Z + 6

Now,

180/ (Z + 6 – 6) – 90/ (Z + 6) = 10

$Z^2 – 3Z – 108 = 0$

$(Z - 12)(Z + 9) = 0$

So, value of Z = 12

Value of T = 90/18 = 5

Downstream speed of boat Q = 12 + 6 + 6 = 24

Required time = 360/24 = 15 hours

Hence answer is option D

11. Questions

Answer: A

In village R

Literates + illiterates = 100%

Literates – Illiterates = 40%

So, literate population = (100 + 40)/2 = 70% x total population

Illiterate population = 30% x total population

Illiterate males = 160 x 2 = 320

So, 30% x total population = 320 + 400 = 720

So, total population of village = 720/30 x 100 = 2400

Literate population of village = 2400 – 720 = 1680

Total males = 400 + 640 = 1040

Literate males = 1040 – 320 = 720

Literate females = 1680 – 720 = 960

Required % change = (480 – 400)/400 x 100 = 20%

Hence answer is option A

12. Questions

Answer: A

In village P

literate -illiterate = 60%

literate + illiterate = 100%

so, literate population = 80%

illiterate population = 20%

Now,

80% - 20% = 1152

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60% x total population = 1152

So, 20% x total population = illiterate population = 1152/60 x 20 = 384

Illiterate male population = 64 x 2 = 128

So, illiterate female population = 384 – 128 = 256

Hence answer is option A

13. Questions

Answer: D

In village U,

Number of illiterate males = 72

So, number of illiterate females = 72/4 x 9 = 162

So, total illiterate population = 162 + 72 = 234

illiterate population = (100 - 64)/2 = 18%

So, total population = 234/18 x 100 = 1300

So, literate population = 1300 x 82% = 1066

Hence answer is option D

14. Questions

Answer: C

In village S

Literate population = (100 + 70)/2 = 85%

Illiterate population = 15%

So, illiterate population of village = 2448/85 x 15 = 432

so, number of illiterate females = 432 – 100 x 2 = 232

(10Z + 182) = 232

Value of Z = 5

Hence answer is option C

15. Questions

Answer: B

In village Q

Literate population = (100 + 50)/2 = 75%

Illiterate population = (100 – 50)/2 = 25%

Ratio of female to male = 3:2

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So, female population = 60%

Male population = 40%

Male literates = 40% x 75% = 30%

Male illiterates = 40% - 30% = 10%

So, 10% x total population = 2 x 112

So, literate population = 224/10 x 75 = 1680

Hence answer is option B

16. Questions

Number of employees in P = 30%

So, number of employees in S = 30%/3 = 10%

Now,

Q + R = 100 – 40 = 60%..... (1)

Also,

(Q – R) = 60% x (30 – 10) % = 12%

So, number of employees in Q = (60 + 12)/2 = 36%

So, number of employees in R = (60 – 12)/2 = 24%

For company P,

Total number of employees = 30% x 800 = 240

Ratio of male to female employees = 3: A

So, possible value of A = 1 or 2

If A = 1

So, ratio of male to female employees in Q = 3:3 (not possible, because males > females)

So, value of A = 2

Number of male employees = 3/5 x 240 = 144

Number of female employees = 2/5 x 240 = 96

For company Q

Total employees = 36% x 800 = 288

Ratio of male to female employees = 5:3

So, male employees = 5/8 x 288 = 180

So, female employees = 3/8 x 288 = 108

For company R

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Total employees = 24% x 800 = 192

Ratio of male to female employees = B:5

So, B should be more than 5

If B = 6, then the total number of employees is a multiple of 11 (not possible)

If B = 7, then the total number of employees is a multiple of 12 (possible)

So, number of male employees = 7/12 x 192 = 112

So, number of female employees = 192 – 112 = 80

For company S,

Total employees = 10% x 800 = 80

Ratio of male to female employees = (C – 3): 1

C – 3 > 1

C > 4

C = 5, 6, 7

On putting C =5, so total number of employees is a multiple of 3 (not possible)

If C = 6

Ratio of male to female employees = 3:1

Number of male employees = ¾ x 80 = 60

Number of female employees = 80 – 60 = 20

If C = 7

Ratio of male to female employees = 4:1

Number of male employees= 4/5 x 80 = 64

Number of female employees = 80 – 64 = 16

Name of the company	Total employees	Male employees	Female employees
P	240	144	96
Q	288	180	108
R	192	112	80
S	80	60/64	20/16

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Answer: D

According to the question,

Difference between male employees in companies P and Q = 180 – 144 = 36

Difference between male and female employees in company S = (60 – 20) OR (64 – 16) = 40 OR 48

Required % = 36/40 x 100 OR 36/48 x 100 = 90% OR 75%

Hence answer is option D.

17. Questions

Number of employees in P = 30%

So, number of employees in S = 30%/3 = 10%

Now,

Q + R = 100 – 40 = 60%..... (1)

Also,

(Q – R) = 60% x (30 – 10) % = 12%

So, number of employees in Q = (60 + 12)/2 = 36%

So, number of employees in R = (60 – 12)/2 = 24%

For company P,

Total number of employees = 30% x 800 = 240

Ratio of male to female employees = 3: A

So, possible value of A = 1 or 2

If A = 1

So, ratio of male to female employees in Q = 3:3 (not possible, because males > females)

So, value of A = 2

Number of male employees = 3/5 x 240 = 144

Number of female employees = 2/5 x 240 = 96

For company Q

Total employees = 36% x 800 = 288

Ratio of male to female employees = 5:3

So, male employees = 5/8 x 288 = 180

So, female employees = 3/8 x 288 = 108

For company R

Total employees = 24% x 800 = 192

Ratio of male to female employees = B:5

So, B should be more than 5

If B = 6, then the total number of employees is a multiple of 11 (not possible)

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If B = 7, then the total number of employees is a multiple of 12 (possible)

So, number of male employees = 7/12 x 192 = 112

So, number of female employees = 192 – 112 = 80

For company S,

Total employees = 10% x 800 = 80

Ratio of male to female employees = (C – 3): 1

C – 3 > 1

C > 4

C = 5, 6, 7

On putting C =5, so total number of employees is a multiple of 3 (not possible)

If C = 6

Ratio of male to female employees = 3:1

Number of male employees = ¾ x 80 = 60

Number of female employees = 80 – 60 = 20

If C = 7

Ratio of male to female employees = 4:1

Number of male employees= 4/5 x 80 = 64

Number of female employees = 80 – 64 = 16

Name of the company	Total employees	Male employees	Female employees
P	240	144	96
Q	288	180	108
R	192	112	80
S	80	60/64	20/16

Answer: C

Number of employees in company T = 264 x 3 – 240 – 192 = 360

Number of male employees in company T = 24 x 5 + 72 = 192

So, number of female employees in company T = 360 – 192 = 168

Hence answer is option C

18. Questions

Number of employees in P = 30%

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So, number of employees in S = 30%/3 = 10%

Now,

Q + R = 100 – 40 = 60%..... (1)

Also,

(Q – R) = 60% x (30 – 10) % = 12%

So, number of employees in Q = (60 + 12)/2 = 36%

So, number of employees in R = (60 – 12)/2 = 24%

For company P,

Total number of employees = 30% x 800 = 240

Ratio of male to female employees = 3: A

So, possible value of A = 1 or 2

If A = 1

So, ratio of male to female employees in Q = 3:3 (not possible, because males > females)

So, value of A = 2

Number of male employees = 3/5 x 240 = 144

Number of female employees = 2/5 x 240 = 96

For company Q

Total employees = 36% x 800 = 288

Ratio of male to female employees = 5:3

So, male employees = 5/8 x 288 = 180

So, female employees = 3/8 x 288 = 108

For company R

Total employees = 24% x 800 = 192

Ratio of male to female employees = B:5

So, B should be more than 5

If B = 6, then the total number of employees is a multiple of 11 (not possible)

If B = 7, then the total number of employees is a multiple of 12 (possible)

So, number of male employees = 7/12 x 192 = 112

So, number of female employees = 192 – 112 = 80

For company S,

Total employees = 10% x 800 = 80

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Ratio of male to female employees = (C – 3): 1

C – 3 > 1

C > 4

C = 5, 6, 7

On putting C =5, so total number of employees is a multiple of 3 (not possible)

If C = 6

Ratio of male to female employees = 3:1

Number of male employees = ¾ x 80 = 60

Number of female employees = 80 – 60 = 20

If C = 7

Ratio of male to female employees = 4:1

Number of male employees= 4/5 x 80 = 64

Number of female employees = 80 – 64 = 16

Name of the company	Total employees	Male employees	Female employees
P	240	144	96
Q	288	180	108
R	192	112	80
S	80	60/64	20/16

Answer: D

Quantity I.

Ratio of Male to female = 3:2

Ratio of male manager to female manager = 3:2

So, ratio of male to female non – managers must be = 3:2

So, required % change = (3 – 2)/2 x 100 = 50%

Quantity II.

Required difference = 112 – 60 OR 112 – 64 = 52 OR 48

So, value of D = 50 or 46

So, Quantity I ≥ Quantity II

Hence answer is option D

19. Questions

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Number of employees in P = 30%

So, number of employees in S = 30%/3 = 10%

Now,

Q + R = 100 – 40 = 60%..... (1)

Also,

(Q – R) = 60% x (30 – 10) % = 12%

So, number of employees in Q = (60 + 12)/2 = 36%

So, number of employees in R = (60 – 12)/2 = 24%

For company P,

Total number of employees = 30% x 800 = 240

Ratio of male to female employees = 3: A

So, possible value of A = 1 or 2

If A = 1

So, ratio of male to female employees in Q = 3:3 (not possible, because males > females)

So, value of A = 2

Number of male employees = 3/5 x 240 = 144

Number of female employees = 2/5 x 240 = 96

For company Q

Total employees = 36% x 800 = 288

Ratio of male to female employees = 5:3

So, male employees = 5/8 x 288 = 180

So, female employees = 3/8 x 288 = 108

For company R

Total employees = 24% x 800 = 192

Ratio of male to female employees = B:5

So, B should be more than 5

If B = 6, then the total number of employees is a multiple of 11 (not possible)

If B = 7, then the total number of employees is a multiple of 12 (possible)

So, number of male employees = 7/12 x 192 = 112

So, number of female employees = 192 – 112 = 80

For company S,

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Total employees = 10% x 800 = 80

Ratio of male to female employees = (C – 3): 1

C – 3 > 1

C > 4

C = 5, 6, 7

On putting C =5, so total number of employees is a multiple of 3 (not possible)

If C = 6

Ratio of male to female employees = 3:1

Number of male employees = ¾ x 80 = 60

Number of female employees = 80 – 60 = 20

If C = 7

Ratio of male to female employees = 4:1

Number of male employees= 4/5 x 80 = 64

Number of female employees = 80 – 64 = 16

Name of the company	Total employees	Male employees	Female employees
P	240	144	96
Q	288	180	108
R	192	112	80
S	80	60/64	20/16

Answer: A

Total employees = 800

Number of male employees = 144 + 180 + 112 + 60/64 = 496/500

Number of female employees = 304/300

Possible difference = (496 – 304)/ (500 – 300) = 192/200

So, Maximum possible difference = 200

Hence answer is option A

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20. Questions

Number of employees in P = 30%

So, number of employees in S = 30%/3 = 10%

Now,

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Q + R = 100 – 40 = 60%..... (1)

Also,

(Q – R) = 60% x (30 – 10) % = 12%

So, number of employees in Q = (60 + 12)/2 = 36%

So, number of employees in R = (60 – 12)/2 = 24%

For company P,

Total number of employees = 30% x 800 = 240

Ratio of male to female employees = 3: A

So, possible value of A = 1 or 2

If A = 1

So, ratio of male to female employees in Q = 3:3 (not possible, because males > females)

So, value of A = 2

Number of male employees = 3/5 x 240 = 144

Number of female employees = 2/5 x 240 = 96

For company Q

Total employees = 36% x 800 = 288

Ratio of male to female employees = 5:3

So, male employees = 5/8 x 288 = 180

So, female employees = 3/8 x 288 = 108

For company R

Total employees = 24% x 800 = 192

Ratio of male to female employees = B:5

So, B should be more than 5

If B = 6, then the total number of employees is a multiple of 11 (not possible)

If B = 7, then the total number of employees is a multiple of 12 (possible)

So, number of male employees = 7/12 x 192 = 112

So, number of female employees = 192 – 112 = 80

For company S,

Total employees = 10% x 800 = 80

Ratio of male to female employees = (C – 3): 1

C – 3 > 1

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C > 4

C = 5, 6, 7

On putting C =5, so total number of employees is a multiple of 3 (not possible)

If C = 6

Ratio of male to female employees = 3:1

Number of male employees = ¾ x 80 = 60

Number of female employees = 80 – 60 = 20

If C = 7

Ratio of male to female employees = 4:1

Number of male employees= 4/5 x 80 = 64

Number of female employees = 80 – 64 = 16

Name of the company	Total employees	Male employees	Female employees
P	240	144	96
Q	288	180	108
R	192	112	80
S	80	60/64	20/16

Answer: D

Required value = (3 x 2 + 5 x 7 + 8 x 6) Or (3 x 2 + 5 x 7 + 8 x 7)

= 89 Or 97

Hence answer is option D

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1. Questions

The cost price of a radio is Rs.800. A shopkeeper marked it ____ % above the cost price and offers a discount of ____ % on the marked price to get a profit of Rs.160.

i). 50%, 20%

ii). 40%, 14.28%

iii). 30%, 5%

a. Only ii and iii are correct

b. Only i and iii are correct

c. Only i and ii are correct

d. All are correct

e. None of these

2. Questions

A can complete a task in 'P' days. B and C together can complete the same task in 28 days, and A and C together can complete the task in 32 days. The efficiency of A is 20% less than the efficiency of B. Find the value of P.

a. 56

b. 64

c. 52

d. 55

e. None of these

3. Questions

The total surface area of a cylinder of radius 'k' meters and height 42 meters is 4928 m^2. Find the volume of the cylinder.

a. 25374 m^3

b. 25864 m^3

c. 25872 m^3

d. 25776 m^3

e. None of these

4. Questions

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A boat can cover ___ m downstream in 25seconds and ___ m upstream in 35 seconds. The time taken by the boat to go 540 m downstream is 20% more than that taken by the boat to go 300 m upstream. If the speed of the boat in still water were 20% more, then its downstream speed was 21 m/s.

Quantity I: Find the value which will be filled in the 1st blank?

Quantity II: Find the value which will be filled in the 2nd blank?

a. Quantity I > Quantity II
b. Quantity I = Quantity II or the relation can't be established
c. Quantity I ≤ Quantity II
d. Quantity I < Quantity II
e. Quantity I ≥ Quantity II

5. Questions

Average of present ages of A, B and C is 23 years and A's present age is ___% of D's present age. After 6 years, the ratio of A's age to B's age will be 2: 3 and C's present age is 70% of B's present age. The difference between the present ages of B and D is 10 years.

Quantity I: Find the value which will be filled in the blank?

Quantity II: A's present age is what per cent of B's present age?

a. Quantity I > Quantity II
b. Quantity I = Quantity II or the relation can't be established
c. Quantity I ≤ Quantity II
d. Quantity I < Quantity II
e. Quantity I ≥ Quantity II

6. Questions

Time taken by a car, a bike and a bus to cross a 240 m long bridge is 't' seconds, (t + 5) seconds and (t + 2) seconds respectively. The speed of the car is 50% more than that of the bike and also 4 m/s more than that of the bus.

Quantity I: If the speed of the bus were ___% of its original speed, it would have crossed the bridge in 16 seconds.

Quantity II: If the speed of the bike were ___% of its original speed, it would have crossed the bridge in 20 seconds.

a. Quantity I < Quantity II
b. Quantity I ≤ Quantity II

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c. Quantity I > Quantity II

d. Quantity I = Quantity II or the relation can't be established

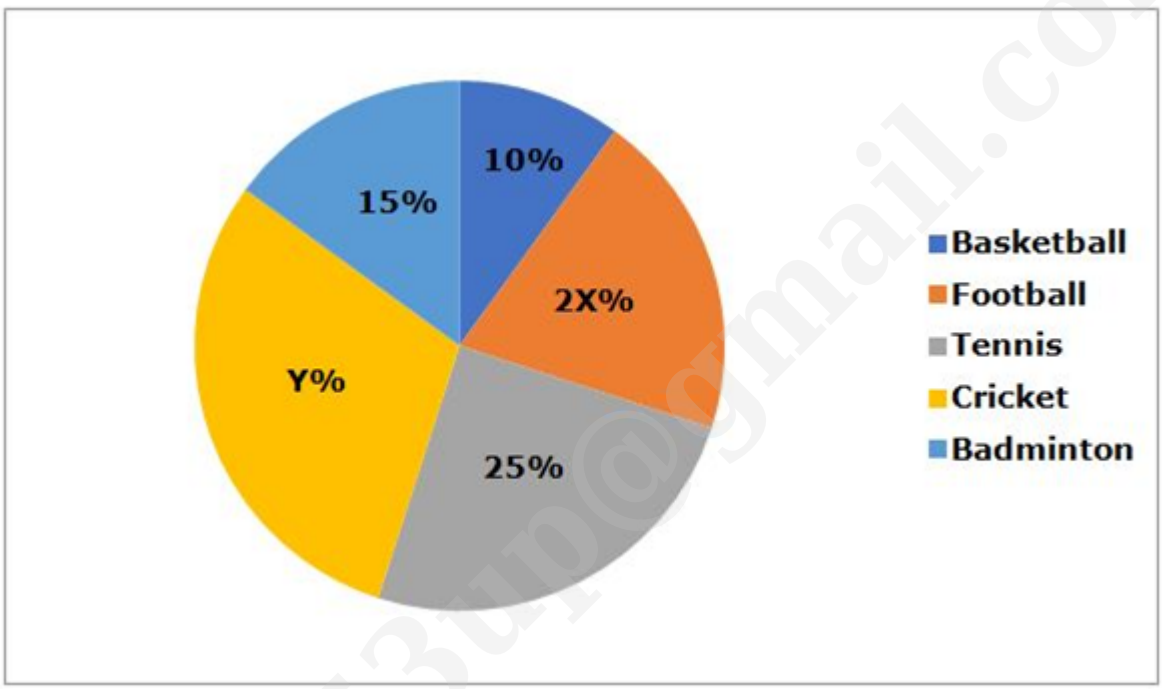
e. Quantity I ≥ Quantity II

7. Questions

Study the following data carefully and answer the questions:

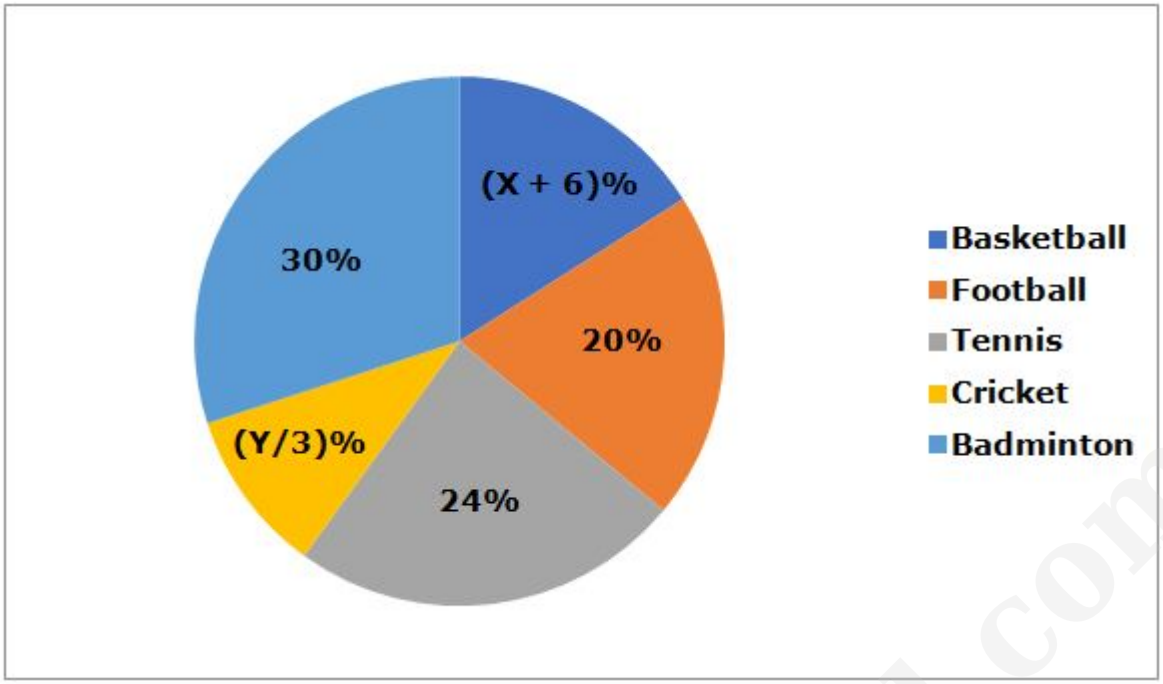
Data given below is related to number of boys and girls in a school, who play 5 different sports Basketball, Football, Tennis, Cricket and Badminton.

The pie chart given below shows the percentage distribution of the number of boys, who play the given 5 sports.



Pie chart given below shows the percentage distribution of the number of girls, who play the given 5 sports.

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Note:

1). Difference between the number of boys, who play basketball and those, who play football, is 80.

2). 36% of the total number of students, who play football, are girls.

Total number of students, who play basketball, is approximately what per cent of the total number of students, who play badminton?

a. 55.6%

b. 65.6%

c. 59.6%

d. 63.6%

e. 57.6%

8. Questions

Find the ratio of total number of boys, who play football, tennis and cricket to the total number of girls, who play basketball, football and tennis?

a. 25:18

b. 20:9

c. 60:29

d. 3:1

e. 15:8

9. Questions

Which of the following is/are true?

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A). Difference between number of boys and girls, who play cricket, is 185.

B). Total number of girls in the school, is 400.

C). Ratio of boys to girls, who play badminton, is 8:11.

a. All are true

b. Only A and B

c. Only B and C

d. None is true

e. Only A and C

10. Questions

What per cent of total number of students who play badminton in the school?

a. 24.4%

b. 16.4%

c. 22.4%

d. 18.4%

e. 20.4%

11. Questions

Study the following data carefully and answer the questions:

There are three tanks P, Q and R of different capacities and each tank has an inlet pipe and an outlet pipe connected to it.

Inlet pipe A and outlet pipe M are connected to tank P.

Inlet pipe B and outlet pipe N are connected to tank Q.

Inlet pipe C and outlet pipe O are connected to tank R.

The table given below shows the following data:

Tanks	The time, in which the inlet pipe alone can fill the tank (in minutes)	The time, in which the inlet pipe and outlet pipe together can fill the tank (in minutes)
P	T	30
Q	2t	90
R	(t + 9)	60

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Note:

1). The ratio of the efficiency of pipe M to that of pipe N is 3: 2 and the ratio of the efficiency of the pipe

M to that of pipe O is 4: 3.

The flowing rate of water from pipe M is 4.5 L per minute, which is 90% of the flowing rate of water from pipe N. If the flowing rate of water from pipe A is 'x' L per minute and that from pipe B is 'y' L per minute, then find the value of (9x-4y)/y?

a. 5.4

b. 6.8

c. 5.6

d. 7.5

e. 4.9

12. Questions

If pipes A and M work with 125% and 83 (1/3)% of their respective efficiencies and pipes B and N work with 125% and 93.75% of their respective efficiencies, then find that the time, in which pipes A and M together can fill tank P with their new efficiencies, will be what per cent of the time, in which pipes B and N together can fill tank Q with their new efficiencies?

a. 33.33%

b. 40%

c. 30%

d. 37.5%

e. 44.44%

13. Questions

If the ratio between capacities of tanks P, Q and R is 7: 6: 8 respectively, the difference between the capacities of tanks P and Q is 30 L and only pipes A, B and C are connected to tank R, then in what time tank R will be filled?

a. 8 minutes

b. 12 minutes

c. 10 minutes

d. 6 minutes

e. 15 minutes

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14. Questions

Which of the following is/are true?

X: The average time, in which pipe A alone, pipe B alone and pipe C alone can fill tanks P, Q and R respectively, is 22 minutes.

Y: If the capacity of tank R is 210 L, then the flowing rate of water from pipe O is 5.75 L.

Z: If the difference between the flowing rates of water from pipes B and N is 3 L per minute, then the capacity of tank Q is 300 L.

a. Both X and Y

b. Both X and Z

c. None is true

d. Both Y and Z

e. All are true

15. Questions

Study the following data carefully and answer the questions:

A survey is conducted to know the preferred TV channels (among A, B, and C) of the persons living in a society. Some like only one TV channel, some like only two TV channels, and remaining like all three TV channels.

The number of person who like only channel A, is 80% of those who like channel B only, which is double that of those who like channel C only. The ratio of number of persons who like both channels A and B but not C to those who like both channels A and C but not B is 5: 9. Number of persons who like all the three channels is 65 more than those who like both channels B and C but not A. The total number of persons who like channel A, is 275. The number of persons who like both channels B and C but not A, is 5 more than those who like both channels A and B but not C. One-third of the total number of persons who like channel B, like only channel B.

Find the difference between the total number of persons who like only one channel and those who like only two channels?

a. 69

b. 89

c. 49

d. 79

e. 59

16. Questions

What percent of total number of persons in society, like channel B?

a. 75%

b. 50%

c. 60%

d. 80%

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e. 40%

17. Questions

If the number of persons who like all three channels, is M% of the total number of persons who like channel A and also N% of the total number of persons in the society, then find the value of (11M – 9N) divisible by which of the following?

P). 7

Q). 3

R). 21

a. Only P and Q

b. Only R

c. Only Q

d. Only P and R

e. Only P, Q and R

18. Questions

What percent of the total number of persons in society like at-least two channels?

a. 46%

b. 54%

c. 48%

d. 52%

e. None of these

19. Questions

In the question two equations numbered I and II are given. You have to solve both the equations and mark the answer.

I). $2x^2 - 3x - A = 0$

II). $y^2 - 20y + B = 0$

One root of equation I is the largest single digit prime number and ratio of A to B is 7: 9.

a. $x < y$

b. $x \leq y$

c. $x \geq y$

d. x = y or relationship cannot be determined

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e. x > y

20. Questions

I). $2x^2 + 9x - 18 = 0$

II). $2(y^2 + 9) = Ay$

One of the roots of equations I and II are common while value of A is a positive number.

a. x < y

b. x ≤ y

c. x ≥ y

d. x = y or relationship cannot be determined

e. x > y

Explanations:

1. Questions

Answer: C

The cost price of the radio = 800

From (i)

Marked price of radio = 800 * (150/100) = 1200

Selling price of the radio = 1200 * (80/100) = 960

Profit = 960 – 800 = 160

i is correct.

From (ii)

Marked price of radio = 800 * (140/100) = 1120

Selling price of the radio = 1120 * (6/7) = 960

Profit = 960 – 800 = 160

ii is correct.

From (iii)

Marked price of radio = 800 * (130/100) = 1040

Selling price of radio = 1040 * (95/100) = 988

Profit = 988 – 800 = 188

iii is incorrect.

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Only i and ii are correct.

2. Questions

Answer: A

L.C. M of 28 and 32 = 224

Total work = 224 unit

Efficiency of A = 224/P

Efficiency of B and C together = 224/28 = 8

Efficiency of A and C together = 224/32 = 7

Let the efficiency of B be 5x and the efficiency of A be 4x.

Efficiency of C = 8 – 5x or 7 – 4x

8- 5x = 7 – 4x

x = 1

Efficiency of A = 4 * 1 = 4

A alone can complete the work in = 224/4 days = 56 days

The value of P is 56.

3. Questions

Answer: C

Surface area of a cylinder = $2\pi r^2 + 2\pi rh$

$2\pi (k^2 + k\ 42) = 4928$

$k^2 + 42k = 784$

$k^2 + (56 – 14)\ k – 784 = 0$

$k^2 + 56k – 14k – 784 = 0$

$k (k + 56) – 14 (k + 56) = 0$

$(k + 56) (k – 14) = 0$

$k = -56, 14$

The radius of the cylinder is 14 meters.

The volume of the cylinder = $\pi r^2 h$

$= \pi * 14 * 14 * 42$

$= 25872\ m^3$

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4. Questions

Answer: A

Let the speed of the boat in still water is '5x' m/s and the speed of the stream is 'y' m/s.

Since, the time taken by the boat to go 540 m downstream is 20% more than that taken by the boat to go 300 m upstream.

So,

$540/(5x+y) = 300/(5x-y) \times (120/100)$

$3/(5x+y) = 2/(5x-y)$

$15x - 3y = 10x + 2y$

$x = y$ ----------(1)

If the speed of the boat in still water were 20% more:

The new speed of boat in still water = 120% of 5x = '6x' m/s

And

$6x + y = 21$

From equation (1):

$6x + x = 21$

$x = 3, y = 3$

The speed of the boat in still water = 5 * 3 = 15 m/s

The speed of the stream = 3 m/s

Quantity I:

The downstream distance covered by the boat in 25 seconds:

(15 + 3) * 25 = 450 m

Quantity II:

The upstream distance covered by the boat in 35 seconds:

(15 – 3) * 35 = 420 m

Hence Quantity I > Quantity II

5. Questions

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Answer: B

Since, the average of present ages of A, B and C is 23 years.

So,

A + B + C = 69 ----------(1)

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Since, the ratio of A's age to B's age after 6 years will be 2: 3.

So,

$(A+6)/(B+6)= 2/3$

$3A + 18 = 2B + 12$

$2B – 3A = 6$ ----------(2)

By equation (1) * 3 + equation (2):

$3A + 3B + 3C + 2B – 3A = 207 + 6$

$5B + 3C = 213$ ----------(3)

Since, C's present age is 70% of B's present age.

So,

$C= 7B/10$

From equation (1):

$5B+ (21B/10) =213$

$B = 30$

From equation (2):

$A = 18$

From equation (1):

$C = 21$

Since, the difference between B's present age and D's present age is 10 years.

So, D's present age will be either 30 + 10 = 40 years or 30 – 10 = 20 years.

Quantity I:

A's present age = 18 years

If D's present age is 40 years:

So, the value, which will be filled in the blank = $18/40 \times 100 = 45\%$

If D's present age is 20 years:

So, the value, which will be filled in the blank = $18/20 \times 100 = 90\%$

Quantity II:

A's present age = 18 years

B's present age = 30 years

Required percentage = $18/30 \times 100 = 60\%$

Hence, the relation can't be established.

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6. Questions

Answer: D

Let the speed of the bike = '2x' m/s

So, the speed of the car = 150 of '2x' = '3x' m/s

And the speed of the Bus = (3x – 4) m/s

Since, the time taken by the car to cross the bridge is 't' seconds.

So,

3x= (240/t)----------(1)

Since, the time taken by the bike to cross the bridge is (t + 5) seconds.

So,

2x = 240/(t+5)----------(2)

From equations (1) and (2):

80/t= 120/(t+5)

2t + 10 = 3t

t = 10

From equation (1):

x = 8

Speed of the bike = 2 * 8 = 16 m/s

Speed of the car = 3 * 8 = 24 m/s

Speed of the bus = 24 – 4 = 20 m/s

Quantity I:

The new time, in which the bus will cross the bridge = 16 seconds

So, the new speed of the bus = 240/16 = 15 m/s

Since, the original speed of the bus = 20 m/s

So, the value which will be filled in the blank = 15/20 ×100 = 75%

Quantity II:

The new time, in which the bike will cross the bridge = 20 seconds

So, the new speed of the bike = 240/20 = 12 m/s

Since, the original speed of the bike = 16 m/s

So, the value which will be filled in the blank = 12/16 ×100 = 75%

Hence, Quantity I = Quantity II

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7. Questions

From the 1st pie chart:

2X + Y = 100 – (10 + 25 + 15)

2X + Y = 50 ----------(1)

From the 2nd pie chart:

(X+6)+ (Y/3)=100-(20+24+30)

3X + 18 + Y = 78

3X + Y = 60 ----------(2)

From equations (1) and (2):

X = 10, Y = 30

Since, the difference between the number of boys, who play basketball and those, who play football, is 80.

So, the number of boys, who play basketball = 80 × (10/(20-10)) = 80

The number of boys, who play football = 80 × (20/(20-10)) = 160

The number of boys, who play tennis = 80 × (25/10) = 200

The number of boys, who play cricket = 80 × (30/10) = 240

And the number of boys, who play badminton = 80 × (15/10) = 120

Since, 36% of the total number of students, who play football, are girls.

So, the number of girls, who play football = 160 × (36/64) = 90

The number of girls, who play basketball = 90 × (16/20) = 72

The number of girls, who play tennis = 90 × (24/20) = 108

The number of girls, who play cricket = 90 × (10/20) = 45

And the number of girls, who play badminton = 90 × (30/20) = 135

Answer: C

Total number of students, who play basketball = 80 + 72 = 152

Total number of students, who play badminton = 120 + 135 = 255

Required average = (152/255) × 100 = 59.6%

8. Questions

From the 1st pie chart:

2X + Y = 100 – (10 + 25 + 15)

2X + Y = 50 ----------(1)

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From the 2nd pie chart:

(X+6)+ (Y/3)=100-(20+24+30)

3X + 18 + Y = 78

3X + Y = 60 ----------(2)

From equations (1) and (2):

X = 10, Y = 30

Since, the difference between the number of boys, who play basketball and those, who play football, is 80.

So, the number of boys, who play basketball = 80 × 10/(20-10) = 80

The number of boys, who play football = 80 × 20/(20-10) = 160

The number of boys, who play tennis = 80 × (25/10) = 200

The number of boys, who play cricket = 80 × (30/10) = 240

And the number of boys, who play badminton = 80 × (15/10) = 120

Since, 36% of the total number of students, who play football, are girls.

So, the number of girls, who play football = 160 × (36/64) = 90

The number of girls, who play basketball = 90 × (16/20) = 72

The number of girls, who play tennis = 90 × (24/20) = 108

The number of girls, who play cricket = 90 × (10/20) = 45

And the number of girls, who play badminton = 90 × (30/20) = 135

Answer: B

Total number of boys, who play football, tennis, and cricket = 160 + 200 + 240 = 600

Total number of girls, who play basketball, football, and tennis = 72 + 90 + 108 = 270

Required ratio = 600:270 = 20:9

9. Questions

From the 1st pie chart:

2X + Y = 100 – (10 + 25 + 15)

2X + Y = 50 ----------(1)

From the 2nd pie chart:

(X+6)+ (Y/3)=100-(20+24+30)

3X + 18 + Y = 78

3X + Y = 60 ----------(2)

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From equations (1) and (2):

X = 10, Y = 30

Since, the difference between the number of boys, who play basketball and those, who play football, is 80.

So, the number of boys, who play basketball = 80 × 10/(20-10) = 80

The number of boys, who play football = 80 × 20/(20-10) = 160

The number of boys, who play tennis = 80 × (25/10) = 200

The number of boys, who play cricket = 80 × (30/10) = 240

And the number of boys, who play badminton = 80 × (15/10) = 120

Since, 36% of the total number of students, who play football, are girls.

So, the number of girls, who play football = 160 × (36/64) = 90

The number of girls, who play basketball = 90 × (16/20) = 72

The number of girls, who play tennis = 90 × (24/20) = 108

The number of girls, who play cricket = 90 × (10/20) = 45

And the number of girls, who play badminton = 90 × (30/20) = 135

Answer: D

From A:

Difference between number of boys and girls, who play cricket = 240 – 45 = 195

So, A is not true.

From B:

Total number of girls in the school = 72 + 90 + 108 + 45 + 135 = 450

So, B is not true.

From C:

Ratio of boys to girls, who play badminton = 120: 135 = 8: 9

So, C is not true.

Hence, none is true.

10. Questions

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From the 1st pie chart:

2X + Y = 100 – (10 + 25 + 15)

2X + Y = 50 ----------(1)

From the 2nd pie chart:

(X+6)+ (Y/3)=100-(20+24+30)

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3X + 18 + Y = 78

3X + Y = 60 ----------(2)

From equations (1) and (2):

X = 10, Y = 30

Since, the difference between the number of boys, who play basketball and those, who play football, is 80.

So, the number of boys, who play basketball = 80 × 10/(20-10) = 80

The number of boys, who play football = 80 × 20/(20-10) = 160

The number of boys, who play tennis = 80 × (25/10) = 200

The number of boys, who play cricket = 80 × (30/10) = 240

And the number of boys, who play badminton = 80 × (15/10) = 120

Since, 36% of the total number of students, who play football, are girls.

So, the number of girls, who play football = 160 × (36/64) = 90

The number of girls, who play basketball = 90 × (16/20) = 72

The number of girls, who play tennis = 90 × (24/20) = 108

The number of girls, who play cricket = 90 × (10/20) = 45

And the number of girls, who play badminton = 90 × (30/20) = 135

Answer: E

Total number of students in the school = (80 + 160 + 200 + 240 + 120) + (72 + 90 + 108 + 45 + 135) = 1250

Total number of students, who play badminton = 120 + 135 = 255

Required percentage = (255/1250) ×100 = 20.4%

11. Questions

Let the time, in which pipe M alone can empty tank P is '6m' minutes.

So, the time, in which pipe N alone can empty tank Q = 6m × 3/2 = '9m' minutes

And the time, in which pipe O alone can empty tank R = 6m × 4/3 = '8m' minutes

Since, the time, in which pipes A and M together can fill tank P is 30 minutes.

So,

(1/t) – (1/6m)= 1/30---------(1)

Since, the time, in which pipes B and N together can fill tank Q, is 90 minutes.

So,

(1/2t) – (1/9m)= 1/90----------(2)

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From equations (1) and (2):

(1/30) + (1/6m) = (2/90) + (2/9m)

(2/9m) - (1/6m) = (1/30) – (2/90)

(4-3)/18m= (3-2)/90

m = 5

From equation (1):

(1/t) – (1/30)= 1/30

t = 15

Tanks	The time, in which the inlet pipe can fill the tank (in minutes)	The time, in which outlet the pipe alone can empty the tank (in minutes)
P	Pipe A = 15	Pipe M = 6 * 5 = 30
Q	Pipe B = 2 * 15 = 30	Pipe N = 9 * 5 = 45
R	Pipe C = 15 + 9 = 24	Pipe O = 8 * 5 = 40

Answer: B

Since, the flowing rate of water from pipe M is 4.5 L per minute.

And the time, in which pipe M alone can empty tank P = 30 minutes

So, the capacity of tank P = 4.5 * 30 = 135 L

Since, the time, in which pipe A alone can fill tank P = 15 minutes

So, the flowing rate of water from pipe A = x = 135/15 = 9 L per minute

Since, the flowing rate of water from pipe M =(4.5 × 100)/90 = 5 L per minute

And the time, in which pipe N alone can empty tank Q = 45 minutes

So, the capacity of tank Q = 5 * 45 = 225 L

Since, the time, in which pipe B alone can fill tank Q = 30 minutes

So, the flowing rate of water from pipe B = y = 225/30 = 7.5 L per minute

So, the value of (9x-4y)/y = (81-30)/7.5 = 6.8

12. Questions

Let the time, in which pipe M alone can empty tank P is ‘6m’ minutes.

So, the time, in which pipe N alone can empty tank Q = 6m × 3/2 = ‘9m’ minutes

And the time, in which pipe O alone can empty tank R = 6m × 4/3 = ‘8m’ minutes

Since, the time, in which pipes A and M together can fill tank P is 30 minutes.

So,

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(1/t) – (1/6m)= 1/30---------(1)

Since, the time, in which pipes B and N together can fill tank Q, is 90 minutes.

So,

(1/2t) – (1/9m)= 1/90----------(2)

From equations (1) and (2):

(1/30) + (1/6m) = (2/90) + (2/9m)

(2/9m) - (1/6m) = (1/30) – (2/90)

(4-3)/18m= (3-2)/90

m = 5

From equation (1):

(1/t) – (1/30)= 1/30

t = 15

Tanks	The time, in which the inlet pipe can fill the tank (in minutes)	The time, in which outlet the pipe alone can empty the tank (in minutes)
P	Pipe A = 15	Pipe M = 6 * 5 = 30
Q	Pipe B = 2 * 15 = 30	Pipe N = 9 * 5 = 45
R	Pipe C = 15 + 9 = 24	Pipe O = 8 * 5 = 40

Answer: D

The time, in which pipe A alone can fill tank P with 125% of its original efficiency = 15 × 100/125 = 12 minutes

The time, in which pipe M alone can empty tank P with of its original efficiency = 30 × 300/250 = 36 minutes

Now, the part of tank P filled by pipes A and M together in 1 minute:

(1/12) – (1/36) = (3 -1)/36= 1/18

So, the time, in which pipes A and M together will fill tank P = 18 minutes

The time, in which pipe B alone can fill tank Q with 125% of its original efficiency = 30 × 100/125 = 24 minutes

The time, in which pipe N alone can empty tank Q with 93.75% of its original efficiency = 45 × 100/93.75 = 48 minutes

Now, the part of tank Q filled by pipes B and N together in 1 minute:

(1/24) – (1/48) = (2-1)/48= 1/48

So, the time, in which pipes B and N together will fill tank Q = 48 minutes

Required percentage = (18/48) ×100 = 37.5%

13. Questions

Let the time, in which pipe M alone can empty tank P is '6m' minutes.

So, the time, in which pipe N alone can empty tank Q = 6m × 3/2 = '9m' minutes

And the time, in which pipe O alone can empty tank R = 6m × 4/3 = '8m' minutes

Since, the time, in which pipes A and M together can fill tank P is 30 minutes.

So,

(1/t) – (1/6m)= 1/30---------(1)

Since, the time, in which pipes B and N together can fill tank Q, is 90 minutes.

So,

(1/2t) – (1/9m)= 1/90----------(2)

From equations (1) and (2):

(1/30) + (1/6m) = (2/90) + (2/9m)

(2/9m) - (1/6m) = (1/30) – (2/90)

(4-3)/18m= (3-2)/90

m = 5

From equation (1):

(1/t) – (1/30)= 1/30

t = 15

Tanks	The time, in which the inlet pipe can fill the tank (in minutes)	The time, in which outlet the pipe alone can empty the tank (in minutes)
P	Pipe A = 15	Pipe M = 6 * 5 = 30
Q	Pipe B = 2 * 15 = 30	Pipe N = 9 * 5 = 45
R	Pipe C = 15 + 9 = 24	Pipe O = 8 * 5 = 40

Answer: A

Since, the difference between the capacities of tanks P and Q is 30 L.

So, the capacity of tank P = 30 × (7/1) = 210 L

The capacity of tank Q = 30 × (6/1) = 180 L

And the capacity of tank R = 30 × (8/1) = 240 L

Now, the flowing rate of water from pipe A = 210/15 = 14 L per minute

The flowing rate of water from pipe B = 180/30 = 6 L per minute

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And the flowing rate of water from pipe C = 240/24 = 10 L per minute

So, the time, in which pipes A, B and C together will fill tank R:

240/(14+6+10)=8 minutes

14. Questions

Let the time, in which pipe M alone can empty tank P is '6m' minutes.

So, the time, in which pipe N alone can empty tank Q = 6m × 3/2 = '9m' minutes

And the time, in which pipe O alone can empty tank R = 6m × 4/3 = '8m' minutes

Since, the time, in which pipes A and M together can fill tank P is 30 minutes.

So,

(1/t) – (1/6m)= 1/30---------(1)

Since, the time, in which pipes B and N together can fill tank Q, is 90 minutes.

So,

(1/2t) – (1/9m)= 1/90----------(2)

From equations (1) and (2):

(1/30) + (1/6m) = (2/90) + (2/9m)

(2/9m) - (1/6m) = (1/30) – (2/90)

(4-3)/18m= (3-2)/90

m = 5

From equation (1):

(1/t) – (1/30)= 1/30

t = 15

Tanks	The time, in which the inlet pipe can fill the tank (in minutes)	The time, in which outlet the pipe alone can empty the tank (in minutes)
P	Pipe A = 15	Pipe M = 6 * 5 = 30
Q	Pipe B = 2 * 15 = 30	Pipe N = 9 * 5 = 45
R	Pipe C = 15 + 9 = 24	Pipe O = 8 * 5 = 40

Answer: C

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From X:

Average of time, in which pipe A alone, pipe B alone and pipe C alone can fill tanks P, Q and R respectively:

(15+30+24)/3=23 minutes

So, X is not true.

From Y:

If the capacity of tank R is 210 L:

And the time, in which pipe O alone can empty tank R = 40 minutes

So, the flowing rate of water from pipe O = 210/40 = 5.25 L per minute

So, Y is not true.

From Z:

If the difference between the flowing rates of water from pipes B and N is 3 L per minute:

And pipes B and N together can fill tank Q in 90 minutes.

So, the capacity of tank Q = 3 * 90 = 270 L

So, Z is not true.

Hence, none is true.

15. Questions

Let the number of persons who like only channel B = 10x

So, the number of persons who like only channel A = 80% of 10x = 8x

And the number of persons who like only channel C = 5x

Let the number of persons who like both channels A and B but not C = 5y

So, the number of persons who like both channels A and C but not B = 9y

Let the number of persons who like both channels B and C but not A = z

So, the number of persons who like all the three channels = (z + 65)

Since, the total number of persons who like channel A, is 275.

So,

8x + 5y + 9y + (z + 65) = 275

8x + 14y + z = 210 ----------(1)

Since, the number of persons who like both channels B and C but not A, is 5 more than those who like both channels A and B but not C.

So,

z – 5y = 5 ----------(2)

Since, one-third of the total number of persons who like channel B, likes only channel B.

So,

10x + 5y + z + (z + 65) = 3 × 10x

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$20x - 5y - 2z = 65$ ----------(3)

From equations (1) and (2):

$8x + 14y + 5y + 5 = 210$

$8x + 19y = 205$ ----------(4)

By equation (1) × 2 + equation (3):

$16x + 28y + 2z + 20x - 5y - 2z = 420 + 65$

$36x + 23y = 485$ ----------(5)

By equation (4) × 9 – equation (5) × 2:

$72x + 171y - 72x - 46y = 1845 - 970$

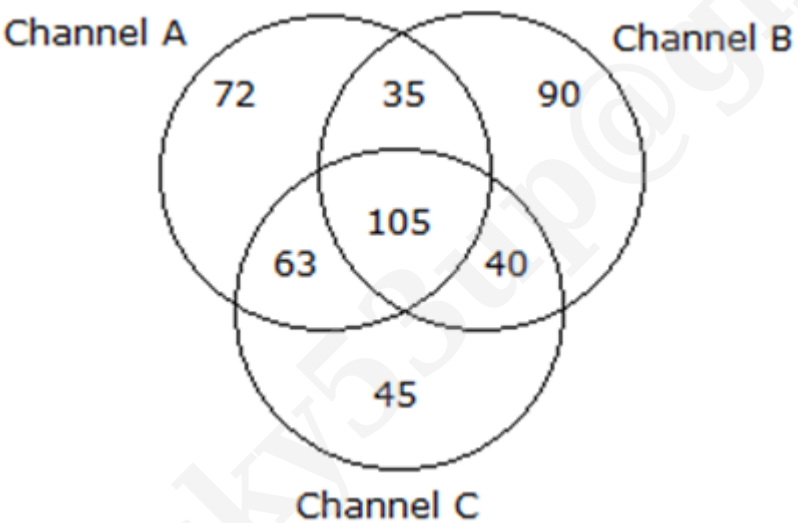
$y = 7$

From equation (2):

$z = 40$

From equation (1):

$x = 9$



Answer: A

Total number of persons who like only one channel = 72 + 90 + 45 = 207

Total number of persons who like only two channels = 35 + 40 + 63 = 138

Required difference = 207 – 138 = 69

16. Questions

Let the number of persons who like only channel B = 10x

So, the number of persons who like only channel A = 80% of 10x = 8x

And the number of persons who like only channel C = 5x

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Let the number of persons who like both channels A and B but not C = 5y

So, the number of persons who like both channels A and C but not B = 9y

Let the number of persons who like both channels B and C but not A = z

So, the number of persons who like all the three channels = (z + 65)

Since, the total number of persons who like channel A, is 275.

So,

$8x + 5y + 9y + (z + 65) = 275$

$8x + 14y + z = 210$ ----------(1)

Since, the number of persons who like both channels B and C but not A, is 5 more than those who like both channels A and B but not C.

So,

$z – 5y = 5$ ----------(2)

Since, one-third of the total number of persons who like channel B, likes only channel B.

So,

$10x + 5y + z + (z + 65) = 3 \times 10x$

$20x – 5y – 2z = 65$ ----------(3)

From equations (1) and (2):

$8x + 14y + 5y + 5 = 210$

$8x + 19y = 205$ ----------(4)

By equation (1) × 2 + equation (3):

$16x + 28y + 2z + 20x – 5y – 2z = 420 + 65$

$36x + 23y = 485$ ----------(5)

By equation (4) × 9 – equation (5) × 2:

$72x + 171y – 72x – 46y = 1845 – 970$

$y = 7$

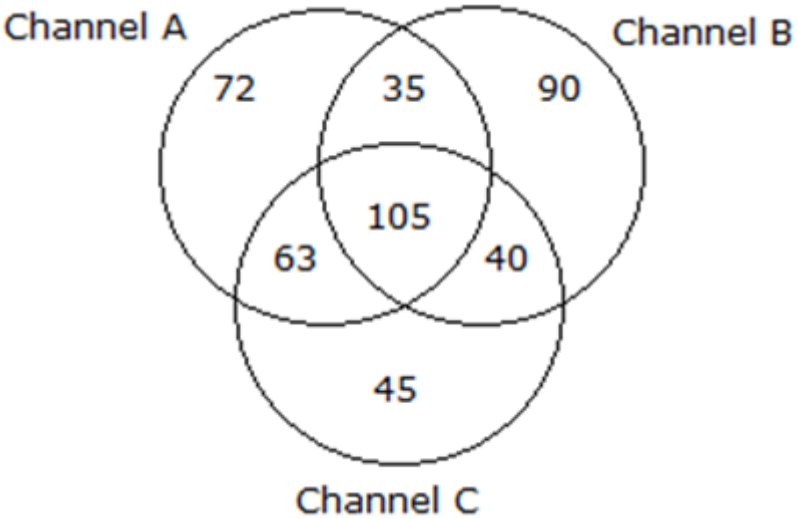
From equation (2):

$z = 40$

From equation (1):

$x = 9$

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Answer: C

Total number of persons in the society = 72 + 90 + 45 + 35 + 40 + 63 + 105 = 450

Total number of persons who like channel B = 90 + 35 + 40 + 105 = 270

Required percentage = 270/450 ×100 = 60%

17. Questions

Let the number of persons who like only channel B = 10x

So, the number of persons who like only channel A = 80% of 10x = 8x

And the number of persons who like only channel C = 5x

Let the number of persons who like both channels A and B but not C = 5y

So, the number of persons who like both channels A and C but not B = 9y

Let the number of persons who like both channels B and C but not A = z

So, the number of persons who like all the three channels = (z + 65)

Since, the total number of persons who like channel A, is 275.

So,

8x + 5y + 9y + (z + 65) = 275

8x + 14y + z = 210 ----------(1)

Since, the number of persons who like both channels B and C but not A, is 5 more than those who like both channels A and B but not C.

So,

z – 5y = 5 ----------(2)

Since, one-third of the total number of persons who like channel B, likes only channel B.

So,

10x + 5y + z + (z + 65) = 3 × 10x

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20x – 5y – 2z = 65 ----------(3)

From equations (1) and (2):

8x + 14y + 5y + 5 = 210

8x + 19y = 205 ----------(4)

By equation (1) × 2 + equation (3):

16x + 28y + 2z + 20x – 5y – 2z = 420 + 65

36x + 23y = 485 ----------(5)

By equation (4) × 9 – equation (5) × 2:

72x + 171y – 72x – 46y = 1845 – 970

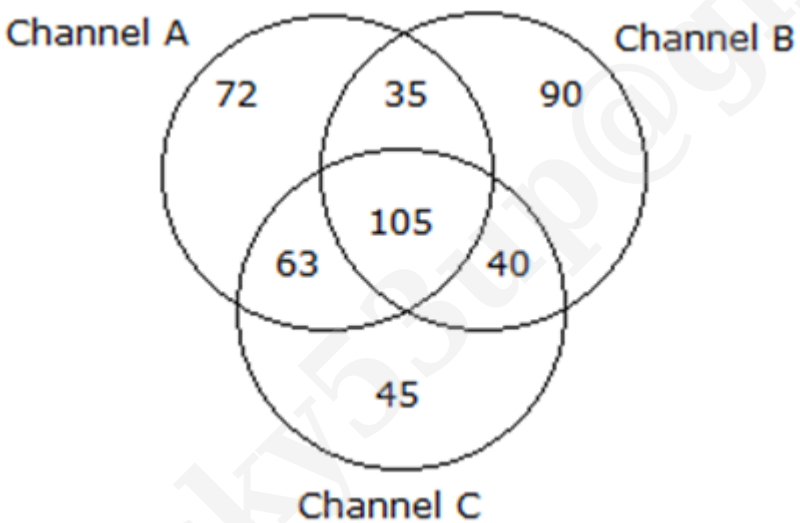
y = 7

From equation (2):

z = 40

From equation (1):

x = 9



Answer: E

Total number of persons who like channel A = 275

Total number of persons in the society = 450

The number of persons who like all the three channels = 105

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So, $M = \frac{105}{275} \times 100 = \frac{420}{11}\%$

And, $N = \frac{105}{450} \times 100 = \frac{70}{3}\%$

Now, the value of (11M – 9N) = 420 – 210 = 210

210 is divisible by 3, 7 and 21.

18. Questions

Let the number of persons who like only channel B = 10x

So, the number of persons who like only channel A = 80% of 10x = 8x

And the number of persons who like only channel C = 5x

Let the number of persons who like both channels A and B but not C = 5y

So, the number of persons who like both channels A and C but not B = 9y

Let the number of persons who like both channels B and C but not A = z

So, the number of persons who like all the three channels = (z + 65)

Since, the total number of persons who like channel A, is 275.

So,

8x + 5y + 9y + (z + 65) = 275

8x + 14y + z = 210 ----------(1)

Since, the number of persons who like both channels B and C but not A, is 5 more than those who like both channels A and B but not C.

So,

z – 5y = 5 ----------(2)

Since, one-third of the total number of persons who like channel B, likes only channel B.

So,

10x + 5y + z + (z + 65) = 3 × 10x

20x – 5y – 2z = 65 ----------(3)

From equations (1) and (2):

8x + 14y + 5y + 5 = 210

8x + 19y = 205 ----------(4)

By equation (1) × 2 + equation (3):

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$16x + 28y + 2z + 20x – 5y – 2z = 420 + 65$

$36x + 23y = 485$ ----------(5)

By equation (4) × 9 – equation (5) × 2:

$72x + 171y – 72x – 46y = 1845 – 970$

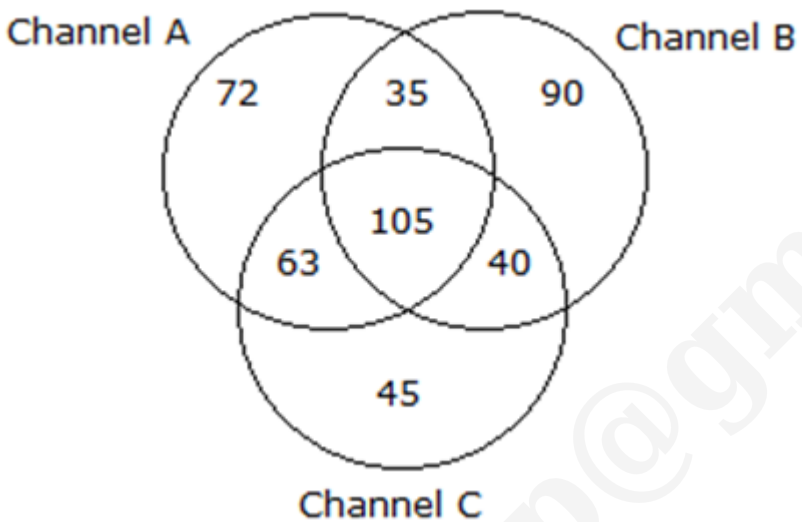
$y = 7$

From equation (2):

$z = 40$

From equation (1):

$x = 9$



Answer: B

Number of persons who like at-least two channels = Number of persons who like only two channels + Number of persons who like all the three channels

Number of persons who like at-least two channels = (35 + 40 + 63) + 105 = 243

Total number of persons in the society = 450

Required percentage = 243/450 ×100 = 54%

19. Questions

Answer: A

I: $2x^2 – 3x – A = 0$

We know that: Largest single digit prime number = 7

$2 * 7^2 – 3 * 7 – A = 0$

$98 – 21 = A$

$A = 77$

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$2x^2 – 3x – 77 = 0$

$2x^2 – 14x + 11x – 77 = 0$

$2x * (x – 7) + 11 * (x – 7) = 0$

$(x – 7) * (2x + 11) = 0$

$x = 7$ and $-11/2$

II: $y^2 – 20y + B = 0$

A: B = 7: 9

77: B = 7: 9

B = 99

$y^2 – 20y + 99 = 0$

$y^2 – 11y – 9y + 99 = 0$

$y (y – 11) – 9 (y – 11) = 0$

$(y – 11) (y – 9) = 0$

$y = 11$ and 9

When $x = 7, y = 11; x < y$

When $x = 7, y = 9; x < y$

When $x = -11/2, y = 11; x < y$

When $x = -11/2, y = 9; x < y$

Hence, $x < y$.

20. Questions

Answer: B

I: $2x^2 + 9x – 18 = 0$

$2x^2 + 12x – 3x – 18 = 0$

$2x (x + 6) – 3 (x + 6) = 0$

$(x + 6) (2x – 3) = 0$

$x = 3/2$ and -6

II: $2 (y^2 + 9) = Ay$

Case 1: When '3/2' is the common root.

$2 [(3/2)^2 + 9] = A * (3/2)$

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45/2 = 3A/2

A = 15

Case 2: When '-6' is the common root.

$2 * [(-6)^2 + 9] = A * (-6)$

90 = - 6A

A = -15 (Invalid)

Hence, $2 (y^2 + 9) = 15y$

$2y^2 + 18 – 15y = 0$

$2y^2 – 12y – 3y + 18 = 0$

$2y * (y – 6) – 3 * (y – 6) = 0$

$(y – 6) * (2y – 3) = 0$

y = 6 and 3/2

When x = 3/2, y = 6; x < y

When x = 3/2, y = 3/2; x = y

When x = -6, y = 6; x < y

When x = -6, y = 3/2; x < y

Hence, $x \leq y$.

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1. Questions

Read the following information carefully and answer the questions based on it.

A shopkeeper sold five articles – P, Q, R, S and T. Further partial information in given in table below.

Article	Cost price	% Profit	Marked price
P	-	40%	(10.8b + 3m)
Q	-	-	(12a – n)
R	-	20%	-
S	(5b – m – n – 48)	-	-
T	(4a + m/2)	-	(13a - n/2 + 6)

Note:

a). Value of 'n' is 4 times of larger root of given equation.

$K^2 - 32K + 252 = 0$

b). Article T is marked up by Rs. 1280 more than its cost price and 1280 is 28% more than profit earned on article S.

c). Discount given on article Q is 16.66%, and selling price of S is Rs. (5m + 7n + 4a + 296).

d). Value of 'm' is twice of missing value in the given sequence.

140, 136, 161, (?), 329, - 200

If cost price of P is 87.5% more than that of T, and P is sold after discount of (2L – 210) %, then find which of the following is/are is possible value of L?

I). 1.5m

II). (b – n – 8)

III). 2.4(b – a)

IV). 240

a. I and III only

b. II and IV only

c. I, II, and III only

d. I, II, III, and IV

e. None of these

2. Questions

If marked price of T is Rs. 80 less than cost price of R, which is sold after discount of Rs. 400 or (5Z/14) %. Find the value of Z.

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a. 100/7

b. 40

c. 250

d. 2/5

e. None of these

3. Questions

If Profit % given on article Q is thrice as that of Discount%, then find difference between cost price of Q and S.

a. Rs. 160

b. Rs. 80

c. Rs. 120

d. Rs. 240

e. None of these

4. Questions

If discount given on article T is (b – a) % and sold at profit of Y%, while marked price of S is Rs. 400 less than that of P, and S sold after discount of D%. Find (Y% - D %).

a. 30%

b. 20%

c. 25%

d. 40%

e. None of these

5. Questions

Study the following data carefully and answer the given questions:

The table given below shows the five different insurance companies i.e. A, B, C, D and E contain life insurance and non-life insurance schemes. The percentage distribution of total number of people taken insurance in these given 5 companies out of total number of people taken insurance in all the given companies together and also shows the number of people taken life insurance in these five companies.

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Notes:

I). Total number of people taken insurance in given 5 companies together is 7500.

II). Difference between the number of people taken non-life insurance in company A and C together and that of E is 900.

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Companies	% of total number of people taken insurance	Total number of people taken life insurance
A	(x+8)%	(y+250)
B	(2x-12)%	(0.2y+450)
C	(1.5x-8)%	(2y-750)
D	2x%	(1.2y+700)
E	(0.25x + 4)%	(y-500)

Find the value of {y/(x+4)}?

a. 12.5

b. 37.5

c. 62.5

d. 87.5

e. 100

6. Questions

Find the ratio of number of people taken non-life insurance in Company C and E together to number of people taken life insurance in Company C.

a. 11:12

b. 12:13

c. 13:14

d. 14:15

e. 16:15

7. Questions

If each person who has taken non-life insurance pay costs Rs. 80 per month, find the difference between the total cost payed by number of people taken non-life insurance in company A and Cfor one month?

a. Rs.25000

b. Rs.26000

c. Rs.27000

d. Rs.28000

e. Rs.29000

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8. Questions

In another company F, the total number of people taken insurance is 50% more than the total number of people taken life insurance in company A. Find the percentage of number of people taken life insurance in company F out of total number of people taken insurance in company F, if the ratio of number of people taken life to non-life insurancein F is 7:3.

a. 30%

b. 35%

c. 70%

d. 65%

e. 60%

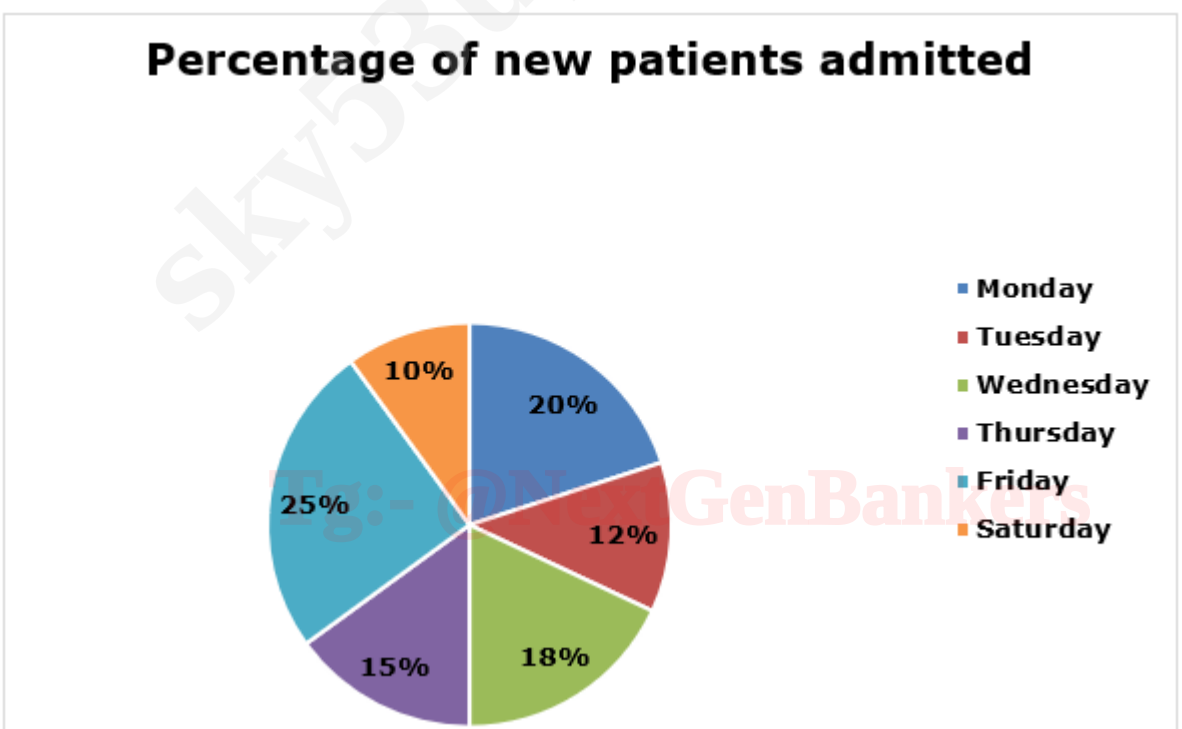
9. Questions

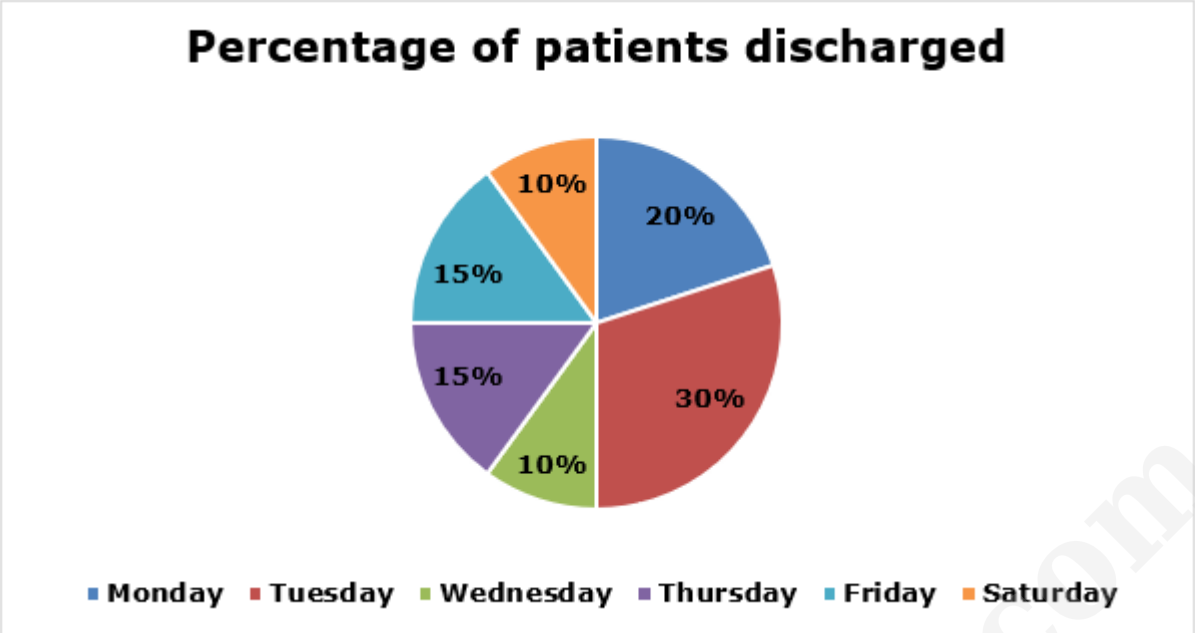
Study the following information carefully and answer the questions given below.

In the pie charts, the percentage distribution of total number of new patients admitted in a particular week and that of total number of patients discharged at the start of the day is given below. Week starts from Sunday and no one is discharged on Sunday. The total number of patients in the hospital on Sunday is 200.The total number of new patients admitted from Monday to Saturday in hospital is 2400 and the total number of patients discharged from Monday to Saturday is 1200.

Total number of patients in the hospital before discharged = Total number of patients in the hospital at the end previous day + Total number of new patients admitted on that particular day

Total number of patients in the hospital at the end of the day = Total number of patients in the hospital before discharged - Total number of patients discharged at the start of the day





Find the difference between the total number of patients in the hospital at the end of Monday and the total number of patients in the hospital at the end of Tuesday?

a. 25

b. 22

c. 72

d. 36

e. 32

10. Questions

Find the ratio between the total number of patients in the hospital at the end of Friday and the total number of patients in the hospital at the end of Saturday?

a. 27:47

b. 31:35

c. 58:73

d. 32:35

e. 35:43

11. Questions

Find the difference between the total number of new patients admitted on Wednesday and Thursday together and the total number of patients discharged on the same days?

a. 558

b. 492

c. 652

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d. 328

e. 464

12. Questions

Find the difference between the average number of new patients admitted in the hospital and the average number of patients discharged in the hospital from Monday to Saturday?

a. 200

b. 250

c. 180

d. 240

e. 220

13. Questions

A family has 4 members A, B, C and D. Average age of A and D is 4 years more than average age of B and C and average age of the family is 28 years. If A's age is 33(1/3)% of D's age and also 125% of B's age, then find B's age is what percent of C's age?

a. 30%

b. 20%

c. 33.33%

d. 25%

e. 37.5%

14. Questions

In a mixture, water is 16(2/3) % of milk. After replacing 14 L of mixture with water, the quantity of milk becomes 6 L more than that of water. What percent of the initial mixture was replaced?

a. 30%

b. 20%

c. 33.33%

d. 25%

e. 40%

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15. Questions

Ratio of A's efficiency to B's efficiency is 3: 2 and ratio of A's efficiency to C's efficiency is 3: 1. B and C together can complete a work in 10 days. If A works with 83(1/3)% of his original efficiency, B works with 125% of his original efficiency and C works with 125% of his original efficiency, then in what time A, B and C together will complete the work?

a. 5.4 days

b. 6 days

c. 3.6 days

d. 4.8 days

e. 4 days

16. Questions

Radius of a right circular cylinder is equal to that of a right circular cone, which is 7 cm and ratio of their height is 5: 6 respectively. Volume of the cylinder is 1617 cm^3 more than that of the cone. If both the cylinder and the cone are melted together to form a spherical ball, then what will the volume of that ball?

a. 3533 cm^3

b. 3773 cm^3

c. 3253 cm^3

d. 3693 cm^3

e. 3433 cm^3

17. Questions

Study the following information carefully and answer the questions.

The following passage gives the detail about the total number of people in village XYZ and the number of people who like three different types of chocolates Dairymilk, 5star and Kitkat.

The number of people who do not like any given chocolate is 25 and 5% of the total population likes all three types of chocolates. The ratio of the number of people who like 5star only to Kitkat only is 15:14 and 22% of total number of people like Dairymilk only. The number of people who like both Dairymilk and 5star only is 5 more than the number of people who like 5star and Kitkat only which is equal to the people who like Kitkat and Dairymilk only. The number of people who likes all the given three types of chocolates is equal to the number of people who do not like any of the given three chocolates which are 25% more than that of both Dairymilk and 5star only.

How many people like any one of the given chocolates only?

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a. 300

b. 400

c. 200

d. 150

e. 350

18. Questions

What is the ratio of the number of people who like any two of the given three chocolates only to the number of people who like Dairy milk only and Kitkat only together?

a. 3:2

b. 2:3

c. 1:5

d. 5:4

e. 2:7

19. Questions

Find the sum of the number of people who like Kitkat and Dairymilk only and the number of people 5star and Kitkat only.

a. 30

b. 25

c. 35

d. 40

e. 45

20. Questions

The number of people who like any two of the given three chocolates only is what percentage of the total population of the village?

a. 25%

b. 31%

c. 13%

d. 21%

e. 10%

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Explanations:

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1. Questions

First, we need to calculate the value of m and n.

$K^2 - 32K + 252 = 0$

$(K - 18)(K - 14) = 0$

K = 18, 14

So, value of n = 4 x 18 = 72

For calculating value of m,

$140 - 2^2 = 136$

$136 + 5^2 = 161$

$161 - 11^2 = 40$

$40 + 17^2 = 329$

$329 - 23^2 = -200$

So, value of m = 2 x 40 = 80

Marked price of T is Rs. 1280 more than its cost price, so

Cost price of T = (4a + 80/2) = (4a + 40)

Marked price of T = (13a – 72/2 + 6) = (13a – 30)

Now,

(13a – 30) – (4a + 40) = 1280

9a = 1350

So, value of a = 1350/9 = 150

Profit earned on article S = 1280/128 x 100 = Rs. 1000

Selling price of article S = (5 x 80 + 7 x 72 + 4 x 150 + 296) = Rs.1800

So, cost price of article S = 1800 – 1000 = Rs. 800

So, (5b – m – n – 48) = 800

5b = 800 + 48 + 72 + 80

Value of b = 1000/5 = 200

Now, we have the following known values.

Article	Cost price	% Profit	Marked price
P	-	40%	2400
Q	-	-	1728
R	-	20%	-
S	800		
T	640	-	1920

Answer: C

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According to question,

Cost price of P = 187.5% of 640 = Rs. 1200

Selling price of P = 1200 x 1.4 = 1680

Discount % = (2400 – 1680)/2400 x 100 = 30%

So, (2L – 210) = 30

Value of L = 240/2 = 120

I). 1.5m = 1.5 x 80 = 120

II). (b – n – 8) = 200 – 72 – 8) = 120

III). 2.4(b – a) = 2.4 x (200 – 150) = 120

IV). 240

Only I, II, and III follows.

Hence answer is option C

2. Questions

First, we need to calculate the value of m and n.

$K^2 – 32K + 252 = 0$

(K – 18) (K – 14) = 0

K = 18, 14

So, value of n = 4 x 18 = 72

For calculating value of m,

$140 – 2^2 = 136$

$136 + 5^2 = 161$

$161 – 11^2 = 40$

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Marked price of T = (13a – 72/2 + 6) = (13a – 30)

Now,

(13a – 30) – (4a + 40) = 1280

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9a = 1350

So, value of a = 1350/9 = 150

Profit earned on article S = 1280/128 x 100 = Rs. 1000

Selling price of article S = (5 x 80 + 7 x 72 + 4 x 150 + 296) = Rs.1800

So, cost price of article S = 1800 – 1000 = Rs. 800

So, (5b – m – n – 48) = 800

5b = 800 + 48 + 72 + 80

Value of b = 1000/5 = 200

Now, we have the following known values.

Article	Cost price	% Profit	Marked price
P	-	40%	2400
Q	-	-	1728
R	-	20%	-
S	800		
T	640	-	1920

Answer: B

Cost price of R = 80 + 1920 = Rs. 2000

MRP of R = 2000 x 1.2 + 400 = Rs. 2800

So, discount % = 400/2800 x 100 = 100/7%

5Z/14 = 100/7

Value of Z = 40

Hence answer is option B

3. Questions

First, we need to calculate the value of m and n.

$K^2 – 32K + 252 = 0$

(K – 18) (K – 14) = 0

K = 18, 14

So, value of n = 4 x 18 = 72

For calculating value of m,

$140 – 2^2 = 136$

$136 + 5^2 = 161$

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$161 - 11^2 = 40$

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$329 - 23^2 = -200$

So, value of m = 2 x 40 = 80

Marked price of T is Rs. 1280 more than its cost price, so

Cost price of T = (4a + 80/2) = (4a + 40)

Marked price of T = (13a – 72/2 + 6) = (13a – 30)

Now,

(13a – 30) – (4a + 40) = 1280

9a = 1350

So, value of a = 1350/9 = 150

Profit earned on article S = 1280/128 x 100 = Rs. 1000

Selling price of article S = (5 x 80 + 7 x 72 + 4 x 150 + 296) = Rs.1800

So, cost price of article S = 1800 – 1000 = Rs. 800

So, (5b – m – n – 48) = 800

5b = 800 + 48 + 72 + 80

Value of b = 1000/5 = 200

Now, we have the following known values.

Article	Cost price	% Profit	Marked price
P	-	40%	2400
Q	-	-	1728
R	-	20%	-
S	800		
T	640	-	1920

Answer: A

Discount % given on article Q = 50/3%

So, profit % on article Q = 3 x 50/3 = 50%

Selling price of article Q = 5/6 x 1728 = 1440

Cost price of article Q = 1440/1.5 = Rs. 960

Cost price of article S = Rs. 800

Required difference = 960 – 800 = Rs. 160

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Hence answer is option A

4. Questions

First, we need to calculate the value of m and n.

$K^2 - 32K + 252 = 0$

$(K - 18)(K - 14) = 0$

$K = 18, 14$

So, value of n = 4 x 18 = 72

For calculating value of m,

$140 - 2^2 = 136$

$136 + 5^2 = 161$

$161 - 11^2 = 40$

$40 + 17^2 = 329$

$329 - 23^2 = -200$

So, value of m = 2 x 40 = 80

Marked price of T is Rs. 1280 more than its cost price, so

Cost price of T = (4a + 80/2) = (4a + 40)

Marked price of T = (13a – 72/2 + 6) = (13a – 30)

Now,

(13a – 30) – (4a + 40) = 1280

9a = 1350

So, value of a = 1350/9 = 150

Profit earned on article S = 1280/128 x 100 = Rs. 1000

Selling price of article S = (5 x 80 + 7 x 72 + 4 x 150 + 296) = Rs.1800

So, cost price of article S = 1800 – 1000 = Rs. 800

So, (5b – m – n – 48) = 800

5b = 800 + 48 + 72 + 80

Value of b = 1000/5 = 200

Now, we have the following known values.

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Article	Cost price	% Profit	Marked price
P	-	40%	2400
Q	-	-	1728
R	-	20%	-
S	800		
T	640	-	1920

Answer: D

Discount % on T = (200 – 150) % = 50%

So, selling price of T = 1920 x 50% = Rs. 960

Profit % on T (Y%) = (960 – 640)/640 x 100 = 50%

MRP of S = 2400 – 400 = Rs. 2000

Selling price of S = Rs. 1800

Discount % on S = (2000 – 1800)/2000 x 100 = 10%

Required value = (50% - 10%) = 40%

Hence answer is option D

5. Questions

From note I,

(x + 8) + (2x - 12) + (1.5x - 8) + 2x + (0.25x + 4) = 100

6.75x = 108

x = 16

Companies	Total no. of people taken insurance	No. of people taken life insurance	No. of people taken non-life insurance
A	7500*24% = 1800	(y+250)	1800 - (y+250) = 1550 - y
B	7500*20% = 1500	(0.2y+450)	1500 -(0.2y+450) = 1050 – 0.2y
C	7500*16% = 1200	(2y-750)	1200 - (2y-750) = 1950 – 2y
D	7500*32% = 2400	(1.2y+700)	2400 - (1.2y+700) = 1700 – 1.2y
E	7500*8% = 600	(y-500)	600 - (y-500) = 1100 - y

From note II,

Case 1,

1550 – y + 1950 – 2y – (1100 – y) = 900

2y = 1500

y = 750

Case 2,

1100 – y – (1550 – y + 1950 – 2y) = 900

1100 – y – 3500 + 3y = 900

2y = 3300

y = 1650

But y = 1650 is not possible because the number of people taken life insurance in company A will become not more than 1800.

So, y = 750

In company A,

Number of people taken life insurance = y + 250 = 750 + 250 = 1000

Number of people taken non-life insurance = 1550 – y = 1550 – 750 = 800

Similarly, we can calculate other companies.

Companies	Total number of people taken insurance	Total number of people taken life insurance	Total number of people taken non-life insurance
A	1800	1000	800
B	1500	600	900
C	1200	750	450
D	2400	1600	800
E	600	250	350

Answer: B

According to question,

Required value = 750/ (16 + 4) = 750/20 = 37.5

6. Questions

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From note I,

(x + 8) + (2x - 12) + (1.5x - 8) + 2x + (0.25x + 4) = 100

6.75x = 108

x = 16

Companies	Total no. of people taken insurance	No. of people taken life insurance	No. of people taken non-life insurance
A	7500*24% = 1800	(y+250)	1800 - (y+250) = 1550 - y
B	7500*20% = 1500	(0.2y+450)	1500 -(0.2y+450) = 1050 – 0.2y
C	7500*16% = 1200	(2y-750)	1200 - (2y-750) = 1950 – 2y
D	7500*32% = 2400	(1.2y+700)	2400 - (1.2y+700) = 1700 – 1.2y
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$1100 – y – (1550 – y + 1950 – 2y) = 900$

$1100 – y – 3500 + 3y = 900$

$2y = 3300$

$y = 1650$

But y = 1650 is not possible because the number of people taken life insurance in company A will become not more than 1800.

So, y = 750

In company A,

Number of people taken life insurance = y + 250 = 750 + 250 = 1000

Number of people taken non-life insurance = 1550 – y = 1550 – 750 = 800

Similarly, we can calculate other companies.

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Companies	Total number of people taken insurance	Total number of people taken life insurance	Total number of people taken non-life insurance
A	1800	1000	800
B	1500	600	900
C	1200	750	450
D	2400	1600	800
E	600	250	350

Answer: E

Required ratio = (450 + 350):750 = 800:750 = 16:15

7. Questions

From note I,

(x + 8) + (2x - 12) + (1.5x - 8) + 2x + (0.25x + 4) = 100

6.75x = 108

x = 16

Companies	Total no. of people taken insurance	No. of people taken life insurance	No. of people taken non-life insurance
A	7500*24% = 1800	(y+250)	1800 - (y+250) = 1550 - y
B	7500*20% = 1500	(0.2y+450)	1500 -(0.2y+450) = 1050 – 0.2y
C	7500*16% = 1200	(2y-750)	1200 - (2y-750) = 1950 – 2y
D	7500*32% = 2400	(1.2y+700)	2400 - (1.2y+700) = 1700 – 1.2y
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From note II,

Case 1,

1550 – y + 1950 – 2y – (1100 – y) = 900

2y = 1500

y = 750

Case 2,

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1100 – y – (1550 – y + 1950 – 2y) = 900

1100 – y – 3500 + 3y = 900

2y = 3300

y = 1650

But y = 1650 is not possible because the number of people taken life insurance in company A will become not more than 1800.

So, y = 750

In company A,

Number of people taken life insurance = y + 250 = 750 + 250 = 1000

Number of people taken non-life insurance = 1550 – y = 1550 – 750 = 800

Similarly, we can calculate other companies.

Companies	Total number of people taken insurance	Total number of people taken life insurance	Total number of people taken non-life insurance
A	1800	1000	800
B	1500	600	900
C	1200	750	450
D	2400	1600	800
E	600	250	350

Answer: D

Required difference = (800 - 450)*80 = 350*80 = Rs. 28000

8. Questions

From note I,

(x + 8) + (2x - 12) + (1.5x - 8) + 2x + (0.25x + 4) = 100

6.75x = 108

x = 16

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Companies	Total no. of people taken insurance	No. of people taken life insurance	No. of people taken non-life insurance
A	7500*24% = 1800	(y+250)	1800 - (y+250) = 1550 - y
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Case 2,

$1100 – y – (1550 – y + 1950 – 2y) = 900$

$1100 – y – 3500 + 3y = 900$

$2y = 3300$

$y = 1650$

But y = 1650 is not possible because the number of people taken life insurance in company A will become not more than 1800.

So, y = 750

In company A,

Number of people taken life insurance = y + 250 = 750 + 250 = 1000

Number of people taken non-life insurance = 1550 – y = 1550 – 750 = 800

Similarly, we can calculate other companies.

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Companies	Total number of people taken insurance	Total number of people taken life insurance	Total number of people taken non-life insurance
A	1800	1000	800
B	1500	600	900
C	1200	750	450
D	2400	1600	800
E	600	250	350

Answer: C

Total number of people taken both insurance in F= 1000*150% = 1500

Required % = [1500*7/(1500*10)]*100 = 70%

9. Questions

Total number of patients admitted in hospital on Monday = 2400*(20/100) = 480

Total no. of patients in the hospital on Monday before discharged = 480+200 = 680

Total no. of patients discharged in the hospital on Monday = 1200*(20/100) = 240

Total no. of patients in the hospital at the end of day on Monday = 680 - 240 = 440

Similarly, we can calculate others value also.

Day	Total no. of new patients admitted	Total no. of patients before discharged	Total no. of patients discharged	Total no. of patients after discharged
Monday	480	680	240	440
Tuesday	288	728	360	368
Wednesday	432	800	120	680
Thursday	360	1040	180	860
Friday	600	1460	180	1280
Saturday	240	1520	120	1400

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Answer: C

Required difference = [440 - 368] = 72

10. Questions

Total number of patients admitted in hospital on Monday = 2400*(20/100) = 480

Total no. of patients in the hospital on Monday before discharged = 480+200 = 680

Total no. of patients discharged in the hospital on Monday = 1200*(20/100) = 240

Total no. of patients in the hospital at the end of day on Monday = 680 - 240 = 440

Similarly, we can calculate others value also.

Day	Total no. of new patients admitted	Total no. of patients before discharged	Total no. of patients discharged	Total no. of patients after discharged
Monday	480	680	240	440
Tuesday	288	728	360	368
Wednesday	432	800	120	680
Thursday	360	1040	180	860
Friday	600	1460	180	1280
Saturday	240	1520	120	1400

Answer: D

Required ratio = 1280:1400 = 32:35

11. Questions

Total number of patients admitted in hospital on Monday = 2400*(20/100) = 480

Total no. of patients in the hospital on Monday before discharged = 480+200 = 680

Total no. of patients discharged in the hospital on Monday = 1200*(20/100) = 240

Total no. of patients in the hospital at the end of day on Monday = 680 - 240 = 440

Similarly, we can calculate others value also.

Day	Total no. of new patients admitted	Total no. of patients before discharged	Total no. of patients discharged	Total no. of patients after discharged
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Tuesday	288	728	360	368
Wednesday	432	800	120	680
Thursday	360	1040	180	860
Friday	600	1460	180	1280
Saturday	240	1520	120	1400

Answer: B

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Required difference = [432+360] – [120+180] =792-300= 492

12. Questions

Total number of patients admitted in hospital on Monday = 2400*(20/100) = 480

Total no. of patients in the hospital on Monday before discharged = 480+200 = 680

Total no. of patients discharged in the hospital on Monday = 1200*(20/100) = 240

Total no. of patients in the hospital at the end of day on Monday = 680 - 240 = 440

Similarly, we can calculate others value also.

Day	Total no. of new patients admitted	Total no. of patients before discharged	Total no. of patients discharged	Total no. of patients after discharged
Monday	480	680	240	440
Tuesday	288	728	360	368
Wednesday	432	800	120	680
Thursday	360	1040	180	860
Friday	600	1460	180	1280
Saturday	240	1520	120	1400

Answer: A

Required difference = [2400]/6 – [1200]/6 = 400-200 = 200

13. Questions

Answer: A

Since, average age of the family = 28 years

So, A + B + C + D = 4 * 28

A + B + C + D = 112 ----------> (1)

Since, average age of A and D is 4 years more than average age of B and C.

So, A + D = B + C + 8

A + D – B – C = 8 ----------> (2)

From equations (1) and (2):

A + D = 60 ----------> (3)

B + C = 52 ----------> (4)

Since, A's age is 33(1/3) % of D's age.

So, A = D/3

From equation (3):

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A = 15 years

Since, A's age is 125% of B's age.

So, B = 15 * (100/125) = 12 years

And C = 52 – 12 = 40 years

Required percentage = (12/40) * 100 = 30%

14. Questions

Answer: C

Let the initial quantity of milk in the mixture = '6x' L

So, initial quantity of water in the mixture = 16(2/3) % of '6x' = 'x' L

And initial ratio of milk to water in the mixture = 6: 1

After replacing 14 L of mixture with water:

New quantity of milk in the mixture = 6x – [14 * (6/7)] = (6x – 12) L

And new quantity of water in the mixture = x – [14 * (1/7)] + 14 = (x + 12) L

From the question:

6x – 12 – x – 12 = 6

x = 6

Total quantity of mixture = 6 * (6 + 1) = 42 L

Required percentage = (14/42) * 100 = 33.33%

15. Questions

Answer: D

Ratio of A's efficiency, B's efficiency and C's efficiency respectively:

3: 2: 1

Let the time taken by B alone and C alone to complete the work is 't' days and '2t' days respectively.

So, (1/t) + (1/2t) = 1/10

(2 + 1)/2t = 1/10

t = 15

Time taken by B alone to complete the work = 15 days

Time taken by C alone to complete the work = 2 * 15 = 30 days

Time taken by A alone to complete the work = 15 * (2/3) = 10 days

Now, time taken by A alone to complete the work with 83(1/3)% efficiency:

10 * (300/250) = 12 days

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Time taken by B alone to complete the work with 125% efficiency:

15 * (100/125) = 12 days

Time taken by C alone to complete the work with 125% efficiency:

30 * (100/125) = 24 days

Work done by A, B and C together in 1 day:

(1/12) + (1/12) + (1/24) = (2 + 2 + 1)/24 = 1/4.8

Time taken by A, B and C together to complete the work with their new efficiencies = 4.8 days

16. Questions

Answer: B

Radius of the cylinder = Radius of the cone = 7 cm

Let the height of the cylinder is '5h' cm and that of the cone is '6h' cm.

From the question:

[(22/7) * 49 * 5h] – [(1/3) * (22/7) * 49 * 6h] = 1617

[22 * 7 * 5h] – [22 * 7 * 2h] = 1617

h = 3.5

Now, height of the cylinder = 5 * 3.5 = 17.5 cm

And height of the cone = 6 * 3.5 = 21 cm

If both the cylinder and the cone are melted down and a spherical ball is formed.

The volume of the spherical ball will be:

[(22/7) * 49 * 17.5] + [(1/3) * (22/7) * 49 * 21] = 2695 + 1078 = 3773 cm^3

17. Questions

Total number of of people = 25*100/5 = 500

The number of people who do not like any given chocolate = 25

Number of people who likes all three types of chocolates = 25

Number of people who likes Dairymilk only = 22*500/100 = 110

Number of people who likes both Dairymilk and 5star only = 25*100/125 = 20

Number of people who likes both 5star and Kitkat only = 20 – 5 = 15

Number of people who likes both Kitkat and Dairymilk only = 15

Number of people who like 5star only and Kitkat only = 500 – (25+25+15+15+20+110) = 290

Number of people who likes 5star only = 290**15/29 = 150

Number of people who likes Kitkat only = 290 – 150 =140

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Answer: B

Number of people who likes any one of the given chocolates only = 110+150+140 = 400

18. Questions

Total number of of people = 25*100/5 = 500

The number of people who do not like any given chocolate = 25

Number of people who likes all three types of chocolates = 25

Number of people who likes Dairymilk only = 22*500/100 = 110

Number of people who likes both Dairymilk and 5star only = 25*100/125 = 20

Number of people who likes both 5star and Kitkat only = 20 – 5 = 15

Number of people who likes both Kitkat and Dairymilk only = 15

Number of people who like 5star only and Kitkat only = 500 – (25+25+15+15+20+110) = 290

Number of people who likes 5star only = 290**15/29 = 150

Number of people who likes Kitkat only = 290 – 150 =140

Answer: C

Number of people like any two of the given three chocolates only = 20+15+15 = 50

Number of people who like Dairy milk only and Kitkat only together = 110+140 = 250

Required Ratio = 50:250 = 1:5

19. Questions

Total number of of people = 25*100/5 = 500

The number of people who do not like any given chocolate = 25

Number of people who likes all three types of chocolates = 25

Number of people who likes Dairymilk only = 22*500/100 = 110

Number of people who likes both Dairymilk and 5star only = 25*100/125 = 20

Number of people who likes both 5star and Kitkat only = 20 – 5 = 15

Number of people who likes both Kitkat and Dairymilk only = 15

Number of people who like 5star only and Kitkat only = 500 – (25+25+15+15+20+110) = 290

Number of people who likes 5star only = 290**15/29 = 150

Number of people who likes Kitkat only = 290 – 150 =140

Answer: A

Required sum = 15+15 = 30

20. Questions

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Total number of of people = 25*100/5 = 500

The number of people who do not like any given chocolate = 25

Number of people who likes all three types of chocolates = 25

Number of people who likes Dairymilk only = 22*500/100 = 110

Number of people who likes both Dairymilk and 5star only = 25*100/125 = 20

Number of people who likes both 5star and Kitkat only = 20 – 5 = 15

Number of people who likes both Kitkat and Dairymilk only = 15

Number of people who like 5star only and Kitkat only = 500 – (25+25+15+15+20+110) = 290

Number of people who likes 5star only = 290**15/29 = 150

Number of people who likes Kitkat only = 290 – 150 =140

Answer: E

Required % = 50*100/500 = 10%

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1. Questions

Study the following information carefully and answer the questions given below.

The given table shows the number of matches played by five all-rounders, the total runs scored, the Number of the match in which the batsman is out, the total runs conceded in bowling and the total number of wickets taken, batting average and bowling average of five all-rounders.

Player	Total number of matches played	Total Runs scored	The number of players was out	Batting average	Total run conceded	Total number of wickets taken	Bowling average
A	40	1200	35	____	1600	70	_____
B	60	____	50	_____	1500	90	_____
C	80	2800	____	40	2400	_____	20
D	30	____	25	40	_____	_____	30
E	70	1800	_____	30	2400	____	25

Note:

Batting average = [total runs scored/ number of times player was out]

Bowling average=[total runs conceded/ Number of wickets taken]

Ratio of the number of runs scored by players B and D is 2:1. Find the difference between the batting average of player B and batting average of player D?

a. 1

b. 5

c. 0

d. 2

e. 3

2. Questions

I= Difference in the number of times players C and E were out.

M= Difference between the number of wickets taken by C and E.

Find the relation between I and M?

a. I+2M=68

b. 2I+M=58

c. I+M+25=85

d. I+M=34

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e. 3I+2M=98

3. Questions

B scores 800 runs more than A. Find the difference between the batting average of A and B?

a. 4.65

b. 5.72

c. 8.52

d. 8.65

e. None of these

4. Questions

Ratio of runs conceded by C and D is 2:1. Find the average of wickets taken by all five players?

a. 83.2

b. 84.5

c. 88.3

d. 89.3

e. None of these

5. Questions

Ratio of runs scored by A and F is 2:3. F played 50 matches and in 10 matches he remains not out. F conceded 1400 runs and took 60 wickets. Find the batting and bowling average of F respectively?

a. 45,21.33

b. 48.23.35

c. 41,25.33

d. 44,23.33

e. None of these

6. Questions

Study the following information carefully and answer the questions given below. There are three rows, calculate the roots of the equation and compare them with other columns. Check which one matches.

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Equation	Statements
i). $5x^2-18x+9=0$	**a).** Sum of the roots is positive.
ii). $2x^2-12x+18=0$	**b).** Sum of roots is negative.
iii). $2x^2+15x+13=0$	**c).** Both roots are integers.

a. ii)-b)

b. i)-b)

c. i)-a) & iii) -b)

d. iii)-c)

e. None of these

7. Questions

Equation	Statements
i) $x^2+16x+39=0$	**a)** Sum of the roots is positive.
ii) $x^2+15x+56=0$	**b)** Sum of roots is negative.
iii) $x^2-9x+14=0$	**c)** Both the roots are positive.

a. ii)-a)

b. i)-c)

c. i)-a) & ii) -c)

d. iii)-c)

e. None of these

8. Questions

Equation	Statements
i). $x^2-56x+784=0$	**a).** Sum of the roots is positive.
ii). $x^2+10x-171=0$	**b).** Sum of roots is negative.
iii). $2x^2+12x+16=0$	**c).** Highest root is a perfect square number.

a. ii)-a)

b. i)-b)

c. ii)-c) & ii) -b)

d. iii)-c)

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e. None of these

9. Questions

Equation	Statements
i). $2x^2-19x+17=0$	**a).** Sum of the roots is positive.
ii). $x^2+16x+60=0$	**b).** Both the roots are negative.
iii). $x^2+20x+75=0$	**c).** Both the roots are positive.

a. ii)-a)

b. i)-b)

c. i)-b) & ii) -b)

d. i)-c)

e. None of these

10. Questions

Equation	Statements
i) $4x^2+2x-20=0$	**a)** Sum of the roots is positive.
ii) $3x^2-10x-13=0$	**b)** Sum of roots is negative.
iii) $10x^2-8x-2=0$	**c)** Both the roots are positive prime numbers.

a. ii)-a)

b. i)-a)

c. i)-a) & ii) -b)

d. iii)-c)

e. None of these

11. Questions

Study the following data carefully and answer the questions given below:

Two inlet pipes P and Q and an outlet pipe R is used to fill a tank. Find the time taken by all the three pipes together to fill the tank?

Statement I: Time taken by pipe P alone and pipes P and R together to fill the tank is 6 hours and 12 hours respectively.

Statement II: Time taken by pipes P and Q together to fill the tank is 24/7 hours.

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Statement III: Pipe R alone can empty the tank in 12 hours.

a. Statements I and II together are sufficient
b. Statements I and III together are sufficient
c. Statements II and III together are sufficient
d. Either statements I and II together or Statements II and III together are sufficient
e. Either statements I and III together or Statements II and III together are sufficient

12. Questions

Find the value of 'D'?

Statement I: A boat can go 'D + 15' in downstream in 2.4 hours.

Statement II: Time taken by boat to go 'D – 20' km in downstream speed is 1 hour.

Statement III: Speed of boat in still water is 4 times of the speed of stream of the river.

a. Either statements I and II together or Statements II and III together are sufficient
b. Statements I and III together are sufficient
c. Statements II and III together are sufficient
d. Either statements I and III together or Statements II and III together are sufficient
e. Statements I and II together are sufficient

13. Questions

Three persons P, Q, and R enter into a partnership. Find the ratio of profit share received by them?

Statement I: Ratio of initial investment of P, Q, and R is 4: 2: 5 respectively.

Statement II: Ratio of time period of the investment of P and R is 5: 6.

Statement III: Time period of investment of Q is 2 months more than the time period of investment of R.

a. Statements I and III together are sufficient.
b. All the statements I, II, and III together are not sufficient
c. Statements I and II together are sufficient
d. Statements II and III together are sufficient
e. All the statements I, II, and III together are sufficient

14. Questions

Priya invested some amount in two different schemes, P and Q. Scheme P offered 20%p.a. in SI and scheme Q offered 25%p.a. in CI. How much amount was invested?

Statement I: Difference of interest earned in 2 years is Rs. 4200.

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Statement II: Respective interests earned in both SI & CI after 2 years and 3 years are Rs. 13800 and Rs. 22450.

a. The data in statement I alone is sufficient to answer the question.

b. The data in statements I and II together are required to answer the question

c. The data in statement neither I nor II are sufficient to answer the question

d. The data in statement II alone is sufficient to answer the question

e. The data in statement either I or II are required to answer the question

15. Questions

Two ships M and N begin to move towards each other. Initial distance between the ships is 300km. then what is the approximate time taken by ships for collision?

Statement I: Respective ratio of upstream speed of ship M and speed of ship N in still water is 4:1. Respective ratio of speed of ship N in still water and speed of stream is 3:2.

Statement II: Average of downstream speed of boat M and downstream speed of boat N is 21kmph.

Statement III: Downstream speed of boat M is 12kmph.

a. The data in statement I alone is sufficient to answer the question.

b. The data in statements I and III together or I and II togetheris required to answer the question

c. The data in statement III alone is sufficient to answer the question.

d. The data in statements II alone is required to answer the question

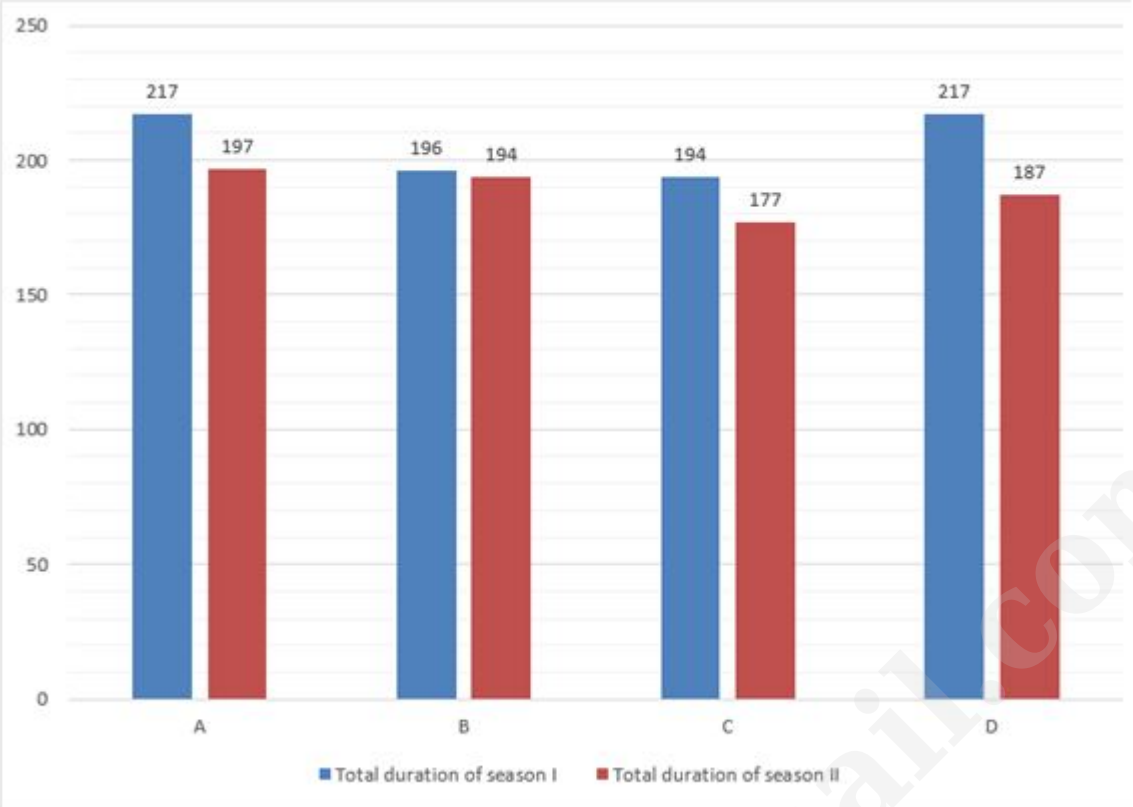
e. The data in statements III and II together are required to answer the question

16. Questions

Study the following information carefully and answer the questions given below.

The given bar graphs shows the total duration of season I and season II of four web series. Each web series has five episodes in each season. The given table shows the duration of episode 5 [E5], duration of episode1[E1],the total duration of episode 2 [E2] and episode 4 [E4] together, and the total duration of episode 1 [E1] ,episode 2[E2] and episode 3 [E3] together(in hours).

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SEASON I

Web series	Duration of E5	Duration of E1	Duration of E4 and E2 together	Duration of E1, E2 and E3 together
A	40	50	85	132
B	40	35	86	112
C	42	40	82	110
D	44	42	89	128

SEASON II

Web series	Duration of E5	Duration of E1	Duration of E4 and E2 together	Duration of E1, E2 and E3 together
A	30	40	82	127
B	32	40	82	122
C	42	30	73	97
D	40	32	70	117

Find the difference between the total duration of E4 in season I of all web series and the total duration E4 in season II of all web series?

a. 22 hours

b. 26 hours

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c. 28 hours

d. 29 hours

e. 30 hours

17. Questions

Find the sum of the total duration of E3 of the season I of all web series?

a. 142 hours

b. 158 hours

c. 162 hours

d. 149 hours

e. 168 hours

18. Questions

Find the ratio of the sum of the total duration of E2 of web series A and B in season I and the total duration of E3 of web series C and D in Season II?

a. 81:85

b. 82:77

c. 96:51

d. 88:89

e. None of these

19. Questions

Find the difference between the total duration of the last three episodes of web series C in season I and the total duration of the last three episodes of web series D in season II?

a. 1hours

b. 2hours

c. 3hours

d. 4hours

e. None of these

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20. Questions

Find the sum of the total duration of episode 3 of season II of all web series together?

a. 162hours

b. 125hours

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c. 142hours

d. 182hours

e. None of these

Explanations:

1. Questions

Answer: C

Runs scored by player D is = 25*40=1000

Runs scored by player B is =1000*2/1=2000

Batting average of player B is=2000/50=40

So, the difference is 40-40=0

2. Questions

Answer: D

The number of times C was out is =2800/40=70

The number of times E was out is =1800/30=60

I=70-60=10

Wicket taken by C is = 2400/20=120

Wicket took by E is=2400/25=96

So, M=120-96=24

So, I+M=10+24=34

3. Questions

Answer: B

Batting average of A is =1200/35=34.28

Batting average of B is =[1200+800]/50=40

So, the difference is 40-34.28=5.72

4. Questions

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Answer: A

Runs conceded by D is =2400/2=1200

Wicket taken by D is =1200/30=40

Total wicket taken by five players is =[70+90+(2400/20)+(40)+(2400/25)]=416

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So, the average is 416/5=83.2

5. Questions

Answer: E

Run score by F is =1200*3/2=1800

Batting average of F is 1800/(50-10)=1800/40=45

The bowling average of F is 1400/60=23.33

6. Questions

Answer: C

$5x^2-18x+9=0$

Or, $5x^2-15x-3x+9=0$

Or, $(x-3)(5x-3)=0$, $x=3,5/3$

$2x^2-12x+18=0$

Or, $2x^2-6x-6x+18=0$

Or, $(x-3)(2x-6)=0$, $x=3,3$

$2x^2+15x+13=0$

Or, $2x^2+13x+2x+13=0$

Or, $(2x+13)(x+1)=0$, $x=-1,-13/2$

So, option c is true.

7. Questions

Answer: D

$x^2+16x+39=0$

or, $x^2+13x+3x+39=0$

or, $(x+13)(x+3)=0$, $x=-13,-3$

$x^2+15x+56=0$

or, $x^2+7x+8x+56=0$

or, $(x+7)(x+8)=0$, $x=-7,-8$

$x^2-9x+14=0$

or, $x^2-7x-2x+14=0$

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or, (x-7)(x-2)=0, x=7,2

so, option D is true.

8. Questions

Answer: C

x^2-56x+784=0

or, x^2-2*28x+28^2=0

or, $(x-28)^2$=0, x=28, 28

x^2+10x-171=0

or, x^2+19x-9x-171=0

or, (x+19)(x-9)=0, x=-19,9

$2x^2$+12x+16=0

$2x^2$+8x+4x+16=0

Or, (x+4)(2x+4)=0, x=-4,-2

So, option C is true.

9. Questions

Answer: D

$2x^2$-19x+17=0

Or, $2x^2$-17x-2x+17=0

Or, (x-1)(2x-17)=0, x=1,17/2

x^2+16x+60=0

or, x^2+10x+6x+60=0

or, (x+10)(x+6)=0, x=-10,-6

x^2+20x+75=0

or, x^2+15x+5x+75=0

or, (x+15)(x+5)=0, x=-15,-5

so, option D is true.

10. Questions

Answer: A

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$4x^2+2x-20=0$

Or, $4x^2+10x-8x-20=0$

Or, $(2x+5)(2x-4)=0$, $x=-5/2,2$

$3x^2-10x-13=0$

Or,$3x^2-13x+3x-13=0$

Or, $(3x-13)(x+1)=0$, $x=13/3,-1$

$10x^2-8x-2=0$

Or, $10x^2-10x+2x-2=0$

Or, $(x-1)(10x+2)=0$, $x=1,-1/5$

So, option A is true.

11. Questions

Answer: D

From I and II together:

From I,

According to the question:

Time taken by P alone = 6 hours …….. (1)

1/P-1/R=1/12

From equation (1):

1/6-1/R=1/12

1/R=1/6-1/12=(2-1)/12

Pipe R alone can empty the tank = 12 hours

From II,

1/P+1/Q=7/24

Time taken by all the three pipes together to fill the tank = 1/(1/P+1/Q-1/R)=1/(7/24-1/12)=24/(7-2)

= 4.8 hours

Statements I and II together are sufficient.

From I and III together:

From I,

According to the question:

Time taken by P alone = 6 hours …….. (1)

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1/P-1/R=1/12

From equation (1):

1/6-1/R=1/12

1/R=1/6-1/12=(2-1)/12

Pipe R alone can empty the tank = 12 hours

We did not have information about pipe Q.

Hence, we cannot determine the time taken by all the three pipes together to fill the tank.

Statements I and III together are not sufficient.

From II and III together:

From II,

According to the question:

1/P+1/Q=7/24

Pipe R alone can empty the tank = 12 hours

Time taken by all the three pipes together to fill the tank = 1/(1/P+1/Q-1/R)=1/(7/24-1/12)=24/(7-2)

= 4.8 hours

Statements II and III together are sufficient.

12. Questions

Answer: E

From I and II together:

Let speed of boat in still water and speed of stream is 'x' km/hr and 'y' km/hr respectively.

(D+15)/(x+y)=2.4

(D+15)/2.4=x+y……(1)

(D-20)/(x+y)=1

D – 20 = x + y ……. (2)

From (1) and (2):

D + 15 = 2.4 * (D - 20)

D = 45 km

Value of D = 45 km

Statements I and II together are sufficient.

From I and III together:

Let speed of boat in still water and speed of stream is 'x' km/hr and 'y' km/hr respectively.

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(D+15)/(x+y)=2.4……(3)

x = 4y …… (4)

Two equations are not sufficient to calculate the value of 3 variables.

Statements I and III together are not sufficient.

From II and III together:

Let speed of boat in still water and speed of stream is '4x' km/hr and 'x' km/hr respectively.

(D-20)/(x+y)=1

D – 20 = 4x + x = 5x

Statements II and III together are not sufficient.

13. Questions

Answer: B

From I, II, and III together:

Let the initial investment of P, Q, and R is '4x', '2x', and '5x' respectively.

Let the time period of investment of P and R is '5y' and '6y' respectively.

Time period of investment of Q = (6y + 2)

Ratio of profit share of P, Q, and R = (4x * 5y): [2x * (6y + 2)]: (5x * 6y)

= 10y: (6y + 2): 15y

We cannot determine the desired ratio.

Statements I, II, and III together are not sufficient.

14. Questions

Answer: D

Let the amount invested on SI = Rs. 'a'

Let the amount invested on CI = Rs. 'b'

Scheme P offered 20% SI

Scheme Q offered 25% CI

Statement I: Difference of interest earned in 2 years = Rs. 4200

Principle amount is different for both schemes

So, Statement I alone is not sufficient to answer the question.

Statement II: Interest earned after 2 years = Rs. 13800

(20% x a x 2) + [b * $(1.25)^2$ – 1] = 13800

32a + 45b = 1104000 → 1

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Interest earned after 3 years = Rs. 22450

(20% x a x 3) + [b * $(1.25)^3$ – 1] = 22450

192a + 305b = 7184000 → 2

From both the equations,

a = Rs. 12000 and b = Rs. 16000

Total Rs. 28000 was invested

Statement II alone is sufficient to answer the question.

Hence, the answer is option d)

15. Questions

Answer: B

Let the speed of ship M = 'M' kmph

Let the speed of ship N = 'N' kmph

Let the speed of stream = 'S' kmph

Distance between two ships = 300km

Statement I: Speed of ship N in still water: speed of stream = 3:2

N/S = 3/2

Upstream speed of ship M: speed of ship N in still water = 4: 1

(M - S): N = 4: 1

So, (M – 2x): 3x = 4: 1

So, M = 14x

So, statement I alone is not sufficient to answer the question.

Statement II: Downstream speed of boat M +downstream speed of boat N = 42kmph

300 / (M + S + N + S) = 42

Time taken by ships for collision = 7.14 hours

Hence, the statement II alone is sufficient to answer the question.

Statement III: Downstream speed of boat M = 12kmph

M + S = 12

Hence, statement III alone is not sufficient to answer the question.

Combining statements, I and III

Since, S = 2x and M = 14x

M + S = 12

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14x + 2x = 12

So, x = 3/4

Thus, M = 42/4 and N = 9/4

Time taken by ships for collision = 300/(51/4) = 23.5 hours

Statements I and III together are sufficient to answer the question.

Combining statements, I and II

N/S = 3/2

N:S = 3x:2x

(M - S): N = 4: 1

So, (M – 2x): 3x = 4: 1

So, M = 14x

Downstream speed of boat M + downstream speed of boat N = 42kmph

=16x + 5x = 21x = 42 , x=2

From that 300/(28+6)= 8.8hours

Hence, the answer is option b)

16. Questions

For the season I in web series A,

Duration of E1+E2+E3+E4=217-40=177hours

Duration of E4=177-132=45hours

Duration of E2=85-45=40hours

Duration of E1+E3=132-40=92hours

Duration of E3=92-50=42hours

Similarly, we can calculate all others values of all other web series.

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	50	40	42	45	40
B	35	42	35	44	40
C	40	40	30	42	42
D	42	44	42	45	44

SEASON II

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Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	40	42	45	40	30
B	40	42	40	40	32
C	30	35	32	38	42
D	32	40	45	30	40

Answer: C

Required difference =[45+44+42+45]-[40+40+38+30]=28hours

17. Questions

For the season I in web series A,

Duration of E1+E2+E3+E4=217-40=177hours

Duration of E4=177-132=45hours

Duration of E2=85-45=40hours

Duration of E1+E3=132-40=92hours

Duration of E3=92-50=42hours

Similarly, we can calculate all others values of all other web series.

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	50	40	42	45	40
B	35	42	35	44	40
C	40	40	30	42	42
D	42	44	42	45	44

SEASON II

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	40	42	45	40	30
B	40	42	40	40	32
C	30	35	32	38	42
D	32	40	45	30	40

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Answer: D

Total sum = [42+35+30+42] =149hours

18. Questions

For the season I in web series A,

Duration of E1+E2+E3+E4=217-40=177hours

Duration of E4=177-132=45hours

Duration of E2=85-45=40hours

Duration of E1+E3=132-40=92hours

Duration of E3=92-50=42hours

Similarly, we can calculate all others values of all other web series.

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	50	40	42	45	40
B	35	42	35	44	40
C	40	40	30	42	42
D	42	44	42	45	44

SEASON II

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	40	42	45	40	30
B	40	42	40	40	32
C	30	35	32	38	42
D	32	40	45	30	40

Answer: B

Required ratio =[40+42]:[32+45]=82:77

19. Questions

For the season I in web series A,

Duration of E1+E2+E3+E4=217-40=177hours

Duration of E4=177-132=45hours

Duration of E2=85-45=40hours

Duration of E1+E3=132-40=92hours

Duration of E3=92-50=42hours

Similarly, we can calculate all others values of all other web series.

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Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	50	40	42	45	40
B	35	42	35	44	40
C	40	40	30	42	42
D	42	44	42	45	44

SEASON II

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	40	42	45	40	30
B	40	42	40	40	32
C	30	35	32	38	42
D	32	40	45	30	40

Answer: A

Required difference =[45+30+40]-[30+42+42]=1hours

20. Questions

For the season I in web series A,

Duration of E1+E2+E3+E4=217-40=177hours

Duration of E4=177-132=45hours

Duration of E2=85-45=40hours

Duration of E1+E3=132-40=92hours

Duration of E3=92-50=42hours

Similarly, we can calculate all others values of all other web series.

Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	50	40	42	45	40
B	35	42	35	44	40
C	40	40	30	42	42
D	42	44	42	45	44

SEASON II

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Web series	Duration of E1	Duration of E2	Duration of E3	Duration of E4	Duration of E5
A	40	42	45	40	30
B	40	42	40	40	32
C	30	35	32	38	42
D	32	40	45	30	40

Answer: A

Required sum =[45+40+32+45]=162hours

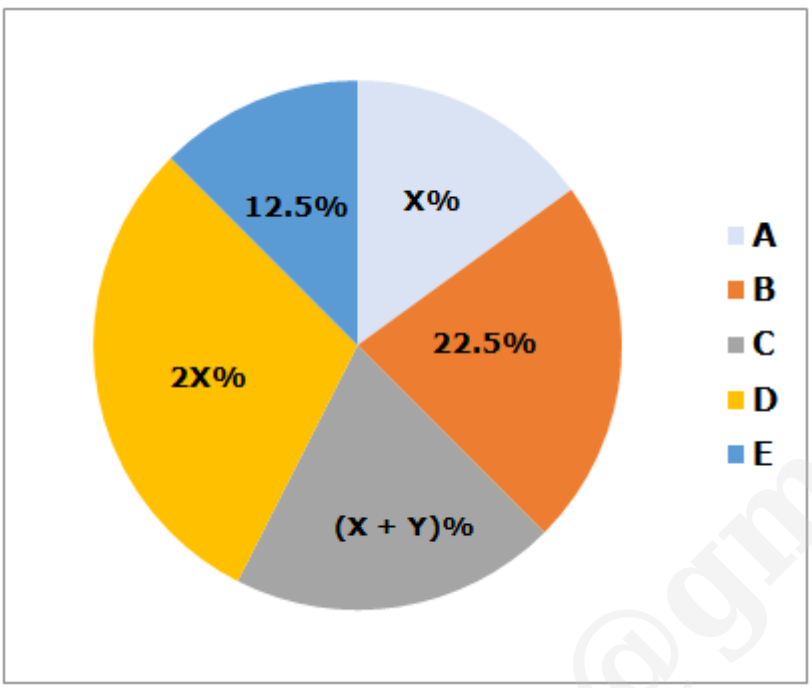
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1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the electricity load (in kWh) sanctioned in five buildings A, B, C, D and E.

The pie chart given below shows the percentage distribution of the electricity load sanctioned in each of the given buildings.



Notes:

1). The sum of the electricity loads sanctioned in buildings A and B is 45 kWh.

2). The difference between the electricity loads sanctioned in buildings B and E is 12 kWh.

3). kWh - Kilowatt per hour

On a particular day, building B used 66(2/3)% of the electricity load sanctioned in it and building D used 75% of the electricity load sanctioned in it, then find the difference between the electricity loads used by buildings B and D on that day?

a. 5 kWh

b. 9 kWh

c. 4 kWh

d. 8 kWh

e. 10 kWh

2. Questions

If there are 3 floors in building E and the electricity loads sanctioned in ground floor and 1st floor are respectively 20% and 50% of that sanctioned in building E, then find the difference between the

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electricity loads sanctioned in 1st and 2nd floors of building E?

a. 2.5 kWh
b. 3 kWh
c. 4.5 kWh
d. 3.5 kWh
e. 2 kWh

3. Questions

Which of the following statements is/are true?

P). The Ratio of X to Y is 3: 1.

Q). The value of $(X^2 - Y^2)$ is divisible by 8.

R). The value of $(X^2 + Y^2)$ is divisible by 40.

a. Only P and Q
b. Only Q and R
c. All are true
d. Only R
e. Only P and R

4. Questions

If there are 4 floors in building A and the electricity loads sanctioned in ground floor, 1st floor and 2nd floor of building A are respectively (X + Y)%, (2X – Y)% and 2(X + Y)% of that sanctioned in building A, then find the electricity load sanctioned in 3rd floor of building A?

a. 1.8 kWh
b. 3.6 kWh
c. 2 kWh
d. 1.5 kWh
e. 2.7 kWh

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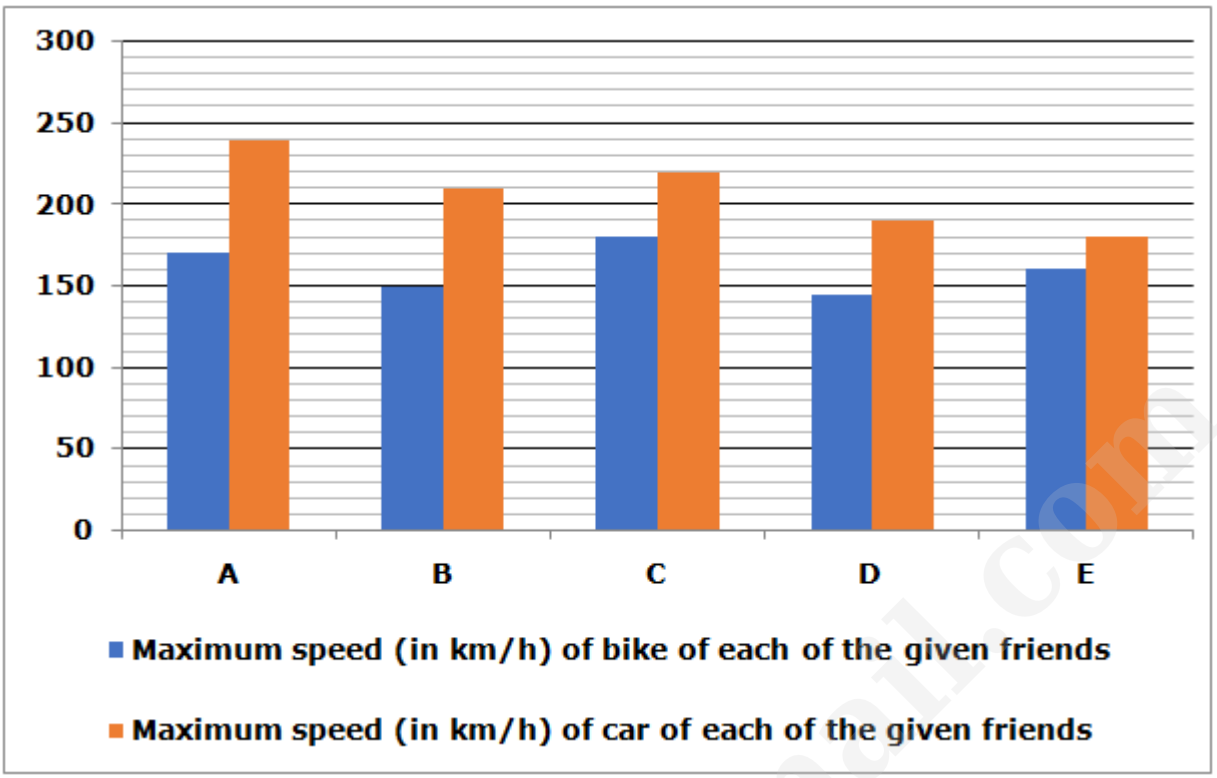
5. Questions

Study the following data carefully and answer the questions:

There are five friends A, B, C, D and E and each friend has a bike and a car.

The bar graph given below shows the maximum speeds (in km/h) of bikes and cars of each of the given

friends



A is driving his car at 30% of its maximum speed and B is following A at his bike at 50% of its maximum speed. If B crosses A in 72 seconds in the same direction, then find the distance between A's car and B's bike?

a. 48 m

b. 96 m

c. 60 m

d. 72 m

e. 88 m

6. Questions

C is riding his bike at 30% of its maximum speed, E is following C at his bike at 45% of its maximum speed and the distance between C's bike and E's bike is 120 m. If E crosses C in 't' seconds in the same direction, then find the time, in which D will cross a 435 m long tunnel at his bike at 40% of its maximum speed?

a. (t – 4) seconds

b. (t + 2) seconds

c. (t + 5) seconds

d. (t – 2) seconds

e. (t + 3) seconds

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7. Questions

If the distance covered by B by his car at 30% of its maximum speed in 14 seconds is 'd' m more than that covered by E by his car at 40% of its maximum speed in 12 seconds, then find the distance covered by D by his car at 50% of its maximum speed in 3.6 minutes?

a. (d +20.5) km

b. (d -12.3) km

c. (d +12.3) km

d. (d – 7.5) km

e. None of these

8. Questions

Each friend participated in a bike race of 960 m and A, B, C, D and E are riding their bikes at 40%, 48%, 35%, 48% and 42% of their bike's maximum speeds respectively. Find the time taken by the friend to complete the race, who won the race?

a. 48 seconds

b. 120 seconds

c. 60 seconds

d. 40 seconds

e. 80 seconds

9. Questions

Study the following data carefully and answer the questions:

The data given below is related to the price (in Rs) of shares of three companies A, B and C in five days Monday, Tuesday, Wednesday, Thursday and Friday.

The table given below shows the following data:

Day	The price of each share of company A (in Rs)	The price of each share of company B as % of that of company A	The price of each share of company C as % of that of company A
Mon	(4x + 200)	125%	85%
Tue	(3y + x)	80%	90%
Wed	(z + 200)	105%	95%
Thu	3z	90%	105%
Fri	1250	80%	120%

Notes:

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1). The difference between the price of each share of company B on Monday and that of company C on Monday, is Rs. 480.

2). The price of each share of company B on Thursday is Rs. 620 more than that on Friday.

3). The price of each share of company A is increased by Rs. 550 from Monday to Tuesday.

If the number of shares sold by company B on Wednesday is 'x' and the number of shares sold by company C on Wednesday is 'y', then find the difference between the total amount received by companies B and C on Wednesday?

a. Rs. 510y

b. Rs. 170y

c. Rs. 340y

d. Rs. 680y

e. Rs. 500y

10. Questions

Company A sold ___ the number of shares on Tuesday, (z – x) number of shares on Wednesday and (x + y – z) number of shares on Thursday. If the total amount received by company A on Tuesday, Wednesday and Thursday together is Rs.9.7 lakhs, then find the value, which can be filled in the blank?

a. (y – 240)

b. (z – 40)/2

c. (y – x)

d. (x – 10)

e. (z + 90)/3

11. Questions

If the average price of each share of company B on Tuesday, Thursday and Friday is Rs.M and the average price of each share of company C on Monday, Tuesday and Thursday is Rs.N, then find that which of the following statements is/are not true?

P). The difference between M and N is divisible by 11.

Q). The sum of M and N is divisible by 15.

R). Both M and N are co-prime with each other.

a. Only P and R

b. Only P and Q

c. Only Q

d. Only R

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e. Only Q and R

12. Questions

Find the ratio of the 2nd highest price of each share of company B on a day to the 3rd highest price of each share of company C on a day?

a. 9: 8

b. 27: 28

c. 1: 1

d. 15: 16

e. 3: 4

13. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of students of a college, who participated in three exams A, B and C. Some students cleared only one exam, some cleared only two exams and the remaining cleared all the three exams.

In the college, 16(2/3)% of the total number of students who cleared exam A only, 10% of the total number of students who cleared exam B only and 25% of the total number of students who cleared exam C only. The number of students, who cleared both exams A and B but not C, is equal to those, who cleared exam B only. The number of students, who cleared both exams A and C but not B, is equal to those, who cleared exam A only. The number of students, who cleared all the three exams, is 16 more than those, who cleared both exams B and C but not A. The total number of students, who cleared only one exam, is 188 more than those, who cleared all the three exams. The total number of students, who cleared only two exams, is 172.

The total number of students, who cleared exam A and those, who cleared exam B are respectively ___% and ___% of the total number of students in the college.

Find the values, which can be filled in the blanks in the same order?

a. 55.83, 41.67

b. 55.83, 63.33

c. 63.33, 41.67

d. 63.33, 55.83

e. 41.67, 55.83

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14. Questions

If the ratio of boys to girls, who cleared at-least two exams is 31: 27 and the ratio of boys to girls, who cleared all the three exams is 2: 3, then find that what percent of the number of girls, who cleared at-least two exams, cleared all the three exams ?

a. 30%

b. 25%

c. 33.33%

d. 40%

e. 22.5%

15. Questions

The ratio of the number of students, who cleared exam A only, to those, who cleared all the three exams, is P: Q. If the ratio of boys to girls, who cleared exam B only, is (P + Q): (Q + 2), then find the number of girls, who cleared exam B only?

a. 16

b. 30

c. 20

d. 24

e. 32

16. Questions

The number of students, who cleared exam C only, is approximately what per cent of the total number of students, who cleared exam C?

a. 45.4%

b. 39.5%

c. 49.1%

d. 35.9%

e. None of these

17. Questions

Study the following data carefully and answer the questions:

The data given below is related to the quantities (in liters) of milk and water in four mixtures A and B.

The ratio of milk to water in mixture A is 4: 1. The quantity of milk in mixture B is 25% more than that in mixture A and the quantity of water in mixture B is double of that in mixture A. The total quantity of mixture C is 80% of that of mixture A and the percentage of milk in mixture C is 75%. The quantity of milk in mixture D is equal to that in mixture B and the ratio of quantity of water in mixture D to that in mixture B is 3 : 2.

If the difference between the quantities of milk and water in mixture A is 24 L and 14 L quantity of mixture B is taken out which is replaced with the same quantity of milk, then what will be the new ratio of milk to water in mixture B?

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a. Can't be determined

b. 11: 3

c. 3: 1

d. 13: 1

e. 6: 1

18. Questions

If the difference between the quantities of milk in mixtures A and B is 16 L and mixtures B and C are mixed together, then what will be the difference between the quantities of milk and water in the new mixture?

a. 64 L

b. 96 L

c. 80 L

d. 112 L

e. 48 L

19. Questions

If the cost of mixture A is Rs. 4.8 per L more than that of mixture B, then what will be the cost of the new mixture, which is formed by mixing mixtures C and D together?

a. Rs. 41.11 per L

b. Rs. 32.5 per L

c. Rs. 45.25 per L

d. Rs. 29.29 per L

e. Rs. 37.33 per L

20. Questions

Find the ratio of the total quantity of milk in those mixtures together, in which the quantities of milk are equal, to the total quantity of water in those mixtures together, in which the quantities of water are equal?

a. 2: 1

b. 10: 3

c. 5: 3

d. 5: 1

e. 4: 1

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Explanations:

1. Questions

From the pie chart:

X + 22.5 + (X + Y) + 2X + 12.5 = 100

4X + Y = 65 ---------- (1)

Since, the difference between the electricity loads sanctioned in buildings B and E is 12 kWh.

So, the electricity load sanctioned in building B = 12 * [22.5/(22.5 – 12.5)] = 27 kWh

Since, the sum of the electricity loads sanctioned in buildings A and B = 45 kWh

So, the electricity load sanctioned in building A = 45 – 27 = 18 kWh

And, X = 22.5 * (18/27) = 15

From equation (1):

4 * 15 + Y = 65

Y = 5

The percentage distribution of the electricity load sanctioned in building A = 15%

The percentage distribution of the electricity load sanctioned in building B = 22.5%

The percentage distribution of the electricity load sanctioned in building C = 15 + 5 = 20%

The percentage distribution of the electricity load sanctioned in building D = 2 * 15 = 30%

The percentage distribution of the electricity load sanctioned in building A = 12.5%

Now, the electricity load sanctioned in building A = 18 kWh

The electricity load sanctioned in building B = 27 kWh

The electricity load sanctioned in building C = 18 * (20/15) = 24 kWh

The electricity load sanctioned in building D = 18 * (30/15) = 36 kWh

And the electricity load sanctioned in building E = 27 – 12 = 15 kWh

Answer: B

Since, the electricity load sanctioned in building B = 27 kWh

So, the electricity load used by building B = 66(2/3)% of 27

=66(2/3)% = 2/3 = 2/3*27 =18 kWh

Since, the electricity load sanctioned in building D = 36 kWh

So, the electricity load used by building D = 75% of 36

= 75%= 3/4 =3/4*36 = 27 kWh

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Required difference = 27 – 18 = 9 kWh

2. Questions

From the pie chart:

X + 22.5 + (X + Y) + 2X + 12.5 = 100

4X + Y = 65 ---------- (1)

Since, the difference between the electricity loads sanctioned in buildings B and E is 12 kWh.

So, the electricity load sanctioned in building B = 12 * [22.5/(22.5 – 12.5)] = 27 kWh

Since, the sum of the electricity loads sanctioned in buildings A and B = 45 kWh

So, the electricity load sanctioned in building A = 45 – 27 = 18 kWh

And, X = 22.5 * (18/27) = 15

From equation (1):

4 * 15 + Y = 65

Y = 5

The percentage distribution of the electricity load sanctioned in building A = 15%

The percentage distribution of the electricity load sanctioned in building B = 22.5%

The percentage distribution of the electricity load sanctioned in building C = 15 + 5 = 20%

The percentage distribution of the electricity load sanctioned in building D = 2 * 15 = 30%

The percentage distribution of the electricity load sanctioned in building A = 12.5%

Now, the electricity load sanctioned in building A = 18 kWh

The electricity load sanctioned in building B = 27 kWh

The electricity load sanctioned in building C = 18 * (20/15) = 24 kWh

The electricity load sanctioned in building D = 18 * (30/15) = 36 kWh

And the electricity load sanctioned in building E = 27 – 12 = 15 kWh

Answer: B

Since, the electricity load sanctioned in building E = 15 kWh

So, the electricity load sanctioned in ground floor of building E = 20% of 15 = 3 kWh

The electricity load sanctioned in 1st floor of building E = 50% of 15 = 7.5 kWh

And the electricity load sanctioned in 2nd floor of building E = 15 – (3 + 7.5) = 4.5 kWh

Required difference = 7.5 – 4.5 = 3 kWh

3. Questions

From the pie chart:

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$X + 22.5 + (X + Y) + 2X + 12.5 = 100$

$4X + Y = 65$ ---------- (1)

Since, the difference between the electricity loads sanctioned in buildings B and E is 12 kWh.

So, the electricity load sanctioned in building B = 12 * [22.5/(22.5 – 12.5)] = 27 kWh

Since, the sum of the electricity loads sanctioned in buildings A and B = 45 kWh

So, the electricity load sanctioned in building A = 45 – 27 = 18 kWh

And, X = 22.5 * (18/27) = 15

From equation (1):

4 * 15 + Y = 65

Y = 5

The percentage distribution of the electricity load sanctioned in building A = 15%

The percentage distribution of the electricity load sanctioned in building B = 22.5%

The percentage distribution of the electricity load sanctioned in building C = 15 + 5 = 20%

The percentage distribution of the electricity load sanctioned in building D = 2 * 15 = 30%

The percentage distribution of the electricity load sanctioned in building A = 12.5%

Now, the electricity load sanctioned in building A = 18 kWh

The electricity load sanctioned in building B = 27 kWh

The electricity load sanctioned in building C = 18 * (20/15) = 24 kWh

The electricity load sanctioned in building D = 18 * (30/15) = 36 kWh

And the electricity load sanctioned in building E = 27 – 12 = 15 kWh

Answer: A

From P:

The ratio of X to Y = 15: 5 = 3: 1

So, P is true.

From Q:

The value of $(X^2 – Y^2) = (225 – 25) = 200$

Since, 200 is divisible by 8.

= 200/8 = 25

So, Q is true.

From R:

The value of $(X^2 + Y^2) = (225 + 25) = 250$

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Since, 250 is not divisible by 40.

So, R is not true.

Hence, only P and Q are true.

4. Questions

From the pie chart:

X + 22.5 + (X + Y) + 2X + 12.5 = 100

4X + Y = 65 ---------- (1)

Since, the difference between the electricity loads sanctioned in buildings B and E is 12 kWh.

So, the electricity load sanctioned in building B = 12 * [22.5/(22.5 – 12.5)] = 27 kWh

Since, the sum of the electricity loads sanctioned in buildings A and B = 45 kWh

So, the electricity load sanctioned in building A = 45 – 27 = 18 kWh

And, X = 22.5 * (18/27) = 15

From equation (1):

4 * 15 + Y = 65

Y = 5

The percentage distribution of the electricity load sanctioned in building A = 15%

The percentage distribution of the electricity load sanctioned in building B = 22.5%

The percentage distribution of the electricity load sanctioned in building C = 15 + 5 = 20%

The percentage distribution of the electricity load sanctioned in building D = 2 * 15 = 30%

The percentage distribution of the electricity load sanctioned in building A = 12.5%

Now, the electricity load sanctioned in building A = 18 kWh

The electricity load sanctioned in building B = 27 kWh

The electricity load sanctioned in building C = 18 * (20/15) = 24 kWh

The electricity load sanctioned in building D = 18 * (30/15) = 36 kWh

And the electricity load sanctioned in building E = 27 – 12 = 15 kWh

Answer: E

Since, the electricity load sanctioned in building A = 18 kWh

So, the electricity load sanctioned in ground floor = (X + Y) % of 18 = (15 + 5) % of 18 = 3.6 kWh

The electricity load sanctioned in 1st floor = (2X – Y) % of 18 = (30 – 5) % of 18 = 4.5 kWh

The electricity load sanctioned in 2nd floor = 2(X + Y) % of 18 = 2(15 + 5) % of 18 = 7.2 kWh

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And the electricity load sanctioned in 3rd floor = 18 – (3.6 + 4.5 + 7.2) = 2.7 kWh

5. Questions

Answer: C

Since, the maximum speed of A's car = 240 km/h

So, the speed, at which A is driving is car = 30% of 240 = 72 km/h

Since, the maximum speed of B's bike = 150 km/h

So, the speed, at which B is riding his bike = 50% of 150 = 75 km/h

Let the distance between A's car and B's bike = 'P' m

Since, B crosses A in 72 seconds.

So,

(75 – 72) * (5/18) = P/72

P = 60 m

6. Questions

Answer: E

Since, the maximum speed of C's bike = 180 km/h

So, the speed, at which C is riding his bike = 30% of 180 = 54 km/h

Since, the maximum speed of E's bike = 160 km/h

So, the speed, at which E is riding his bike = 45% of 160 = 72 km/h

Since, the distance between C's bike and E's bike = 120 m

So,

(72 – 54) * (5/18) = 120/t

t = 24

Since, the maximum speed of D's bike = 145 km/h

So, the speed, at which D will cross the tunnel at his bike = 40% of 145 = 58 km/h

And the required time = 435/[58 * (5/18)] = 27 seconds = (t + 3) seconds

7. Questions

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Answer: B

Since, the maximum speed of B's car = 210 km/h

So, the distance covered by B by his car at 30% of its maximum speed in 14 seconds = 30% of 210 * (5/18) * 14 = 245 m

Since, the maximum speed of E's car = 180 km/h

So, the distance covered by E by his car at 40% of its maximum speed in 12 seconds = 40% of 180 * (5/18) * 12 = 240 m

d = 245 – 240 = 5 m = 5 * 18/5 =18 km

Since, the maximum speed of D's car = 190 km/h

So, the distance covered by D by his car at 50% of its maximum speed in 3.6 minutes = 50% of 190 * (5/18) * (3.6 * 60) = 5700 m = 5.7 km = (d – 12.3) km

8. Questions

Answer: A

Since, the maximum speed of A's bike = 170 km/h

So, the speed, at which A is riding his bike = 40% of 170 = 68 km/h

Since, the maximum speed of B's bike = 150 km/h

So, the speed, at which B is riding his bike = 48% of 150 = 72 km/h

Since, the maximum speed of C's bike = 180 km/h

So, the speed, at which C is riding his bike = 35% of 180 = 63 km/h

Since, the maximum speed of D's bike = 145 km/h

So, the speed, at which D is riding his bike = 48% of 145 = 69.6 km/h

Since, the maximum speed of E's bike = 160 km/h

So, the speed, at which E is riding his bike = 42% of 160 = 67.2 km/h

Since, the speed of B's bike is maximum, so, B won the race.

And the time, in which B will complete the race:

960/[72 * (5/18)] = 48 seconds

9. Questions

Since, the difference between the price of each share of company B on Monday and that of company C on Monday is Rs. 480.

So,

(125 – 85)% of (4x + 200) = 480

40% of (4x + 200) = 480

4x + 200 = 1200

x = 250

Since, the price of each share of company A is increased by Rs. 550 from Monday to Tuesday.

So,

3y + x = (4x + 200) + 550

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3y + 250 = (1000 + 200) + 550

y = 500

Since, the price of each share of company B on Thursday is Rs. 620 more than that on Friday.

So,

90% of 3z – 80% of 1250 = 620

27z/10 = 1620

z = 600

Day	The price of each share of company A (in Rs.)	The price of each share of company B (in Rs.)	The price of each share of company C (in Rs.)
Mon	4 * 250 + 200 = 1200	125% of 1200 = 1500	85% of 1200 = 1020
Tue	3 * 500 + 250 = 1750	80% of 1750 = 1400	90% of 1750 = 1575
Wed	600 + 200 = 800	105% of 800 = 840	95% of 800 = 760
Thu	3 * 600 = 1800	90% of 1800 = 1620	105% of 1800 = 1890
Fri	1250	80% of 1250 = 1000	120% of 1250 = 1500

Answer: C

Since, the price of each share of company B on Wednesday = Rs.840

So, the total amount received by company B on Wednesday =

840 * x = 840 * 250 = Rs.210000

Since, the price of each share of company C on Wednesday = Rs.760

So, the total amount received by company C on Wednesday=

760 * y = 760 * 500 = Rs.380000

Required difference = 380000 – 210000 = Rs.170000 = Rs.340y

10. Questions

Since, the difference between the price of each share of company B on Monday and that of company C on Monday is Rs. 480.

So,

(125 – 85)% of (4x + 200) = 480

40% of (4x + 200) = 480

4x + 200 = 1200

x = 250

Since, the price of each share of company A is increased by Rs. 550 from Monday to Tuesday.

So,

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$3y + x = (4x + 200) + 550$

$3y + 250 = (1000 + 200) + 550$

$y = 500$

Since, the price of each share of company B on Thursday is Rs. 620 more than that on Friday.

So,

90% of $3z$ – 80% of 1250 = 620

$27z/10 = 1620$

$z = 600$

Day	The price of each share of company A (in Rs.)	The price of each share of company B (in Rs.)	The price of each share of company C (in Rs.)
Mon	4 * 250 + 200 = 1200	125% of 1200 = 1500	85% of 1200 = 1020
Tue	3 * 500 + 250 = 1750	80% of 1750 = 1400	90% of 1750 = 1575
Wed	600 + 200 = 800	105% of 800 = 840	95% of 800 = 760
Thu	3 * 600 = 1800	90% of 1800 = 1620	105% of 1800 = 1890
Fri	1250	80% of 1250 = 1000	120% of 1250 = 1500

Answer: D

Let the number of shares sold by company A on Tuesday = a

Since, the number of shares sold by company A on Wednesday = $(z – x) = (600 – 250) = 350$

And the number of shares sold by company A on Thursday = $(x + y – z) = (250 + 500 – 600) = 150$

So,

$(1750 * a) + (800 * 350) + (1800 * 150) = 970000$

$1750a + 280000 + 270000 = 970000$

$1750a = 420000$

$a = 240 = (x – 10)$

11. Questions

Since, the difference between the price of each share of company B on Monday and that of company C on Monday is Rs. 480.

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So,

$(125 – 85)\%$ of $(4x + 200) = 480$

40% of $(4x + 200) = 480$

$4x + 200 = 1200$

$x = 250$

Since, the price of each share of company A is increased by Rs. 550 from Monday to Tuesday.

So,

3y + x = (4x + 200) + 550

3y + 250 = (1000 + 200) + 550

y = 500

Since, the price of each share of company B on Thursday is Rs. 620 more than that on Friday.

So,

90% of 3z – 80% of 1250 = 620

27z/10 = 1620

z = 600

Day	The price of each share of company A (in Rs.)	The price of each share of company B (in Rs.)	The price of each share of company C (in Rs.)
Mon	4 * 250 + 200 = 1200	125% of 1200 = 1500	85% of 1200 = 1020
Tue	3 * 500 + 250 = 1750	80% of 1750 = 1400	90% of 1750 = 1575
Wed	600 + 200 = 800	105% of 800 = 840	95% of 800 = 760
Thu	3 * 600 = 1800	90% of 1800 = 1620	105% of 1800 = 1890
Fri	1250	80% of 1250 = 1000	120% of 1250 = 1500

Answer: A

The average price of each share of company B on Tuesday, Thursday and Friday:

M = (1400 + 1620 + 1000)/3 = Rs.1340

The average price of each share of company C on Monday, Tuesday and Thursday:

N = (1020 + 1575 + 1890)/3 = Rs.1495

From P:

The difference between M and N = 1495 – 1340 = 155

Since, 155 is not divisible by 11.

So, P is not true.

From Q:

The sum of M and N = 1340 + 1495 = 2835

Since, 2835 is divisible by 15.

=2835/15 = 189

So, Q is true.

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From R:

Both M and N are multiple of 5.

Since, M and N are not co-prime with each other.

So, R is not true.

Hence, only P and R are not true.

12. Questions

Since, the difference between the price of each share of company B on Monday and that of company C on Monday is Rs. 480.

So,

(125 – 85)% of (4x + 200) = 480

40% of (4x + 200) = 480

4x + 200 = 1200

x = 250

Since, the price of each share of company A is increased by Rs. 550 from Monday to Tuesday.

So,

3y + x = (4x + 200) + 550

3y + 250 = (1000 + 200) + 550

y = 500

Since, the price of each share of company B on Thursday is Rs. 620 more than that on Friday.

So,

90% of 3z – 80% of 1250 = 620

27z/10 = 1620

z = 600

Day	The price of each share of company A (in Rs.)	The price of each share of company B (in Rs.)	The price of each share of company C (in Rs.)
Mon	4 * 250 + 200 = 1200	125% of 1200 = 1500	85% of 1200 = 1020
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Thu	3 * 600 = 1800	90% of 1800 = 1620	105% of 1800 = 1890
Fri	1250	80% of 1250 = 1000	120% of 1250 = 1500

Answer: C

Since, the 2nd highest price of each share of company B on a day = Rs.1500

And the 3rd highest price of each share of company C on a day = Rs.1500

So, the required ratio = 1500: 1500 = 1: 1

13. Questions

Let the total number of students in the college = 60x

So, the number of students, who cleared exam A only = 16(2/3)% of 60x = 10x

The number of students, who cleared exam B only = 10% of 60x = 6x

And the number of students, who cleared exam C only = 25% of 60x = 15x

Since, the number of students, who cleared both exams A and B but not C, is equal to those, who cleared exam B only.

So, the number of students, who cleared both exams A and B but not C = 6x

Since, the number of students, who cleared both exams A and C but not B, is equal to those, who cleared exam A only.

So, the number of students, who cleared both exams A and C but not B = 10x

Let the number of students, who cleared both exams B and C but not A = y

So, the number of students, who cleared all the three exams = (y + 16)

Since, the total number of students, who cleared only one exam, is 188 more than those, who cleared all the three exams.

So,

(10x + 6x + 15x) – (y + 16) = 188

31x – y = 204 ---------- (1)

Since, the total number of students, who cleared only two exams, is 172.

So,

6x + 10x + y = 172

16x + y = 172 ---------- (2)

From equations (1) and (2):

31x – 204 = 172 – 16x

x = 8

From equation (1):

248 – y = 204

y = 44

The total number of students in the college = 60 * 8 = 480

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The number of students, who cleared exam A only = 10 * 8 = 80

The number of students, who cleared exam B only = 6 * 8 = 48

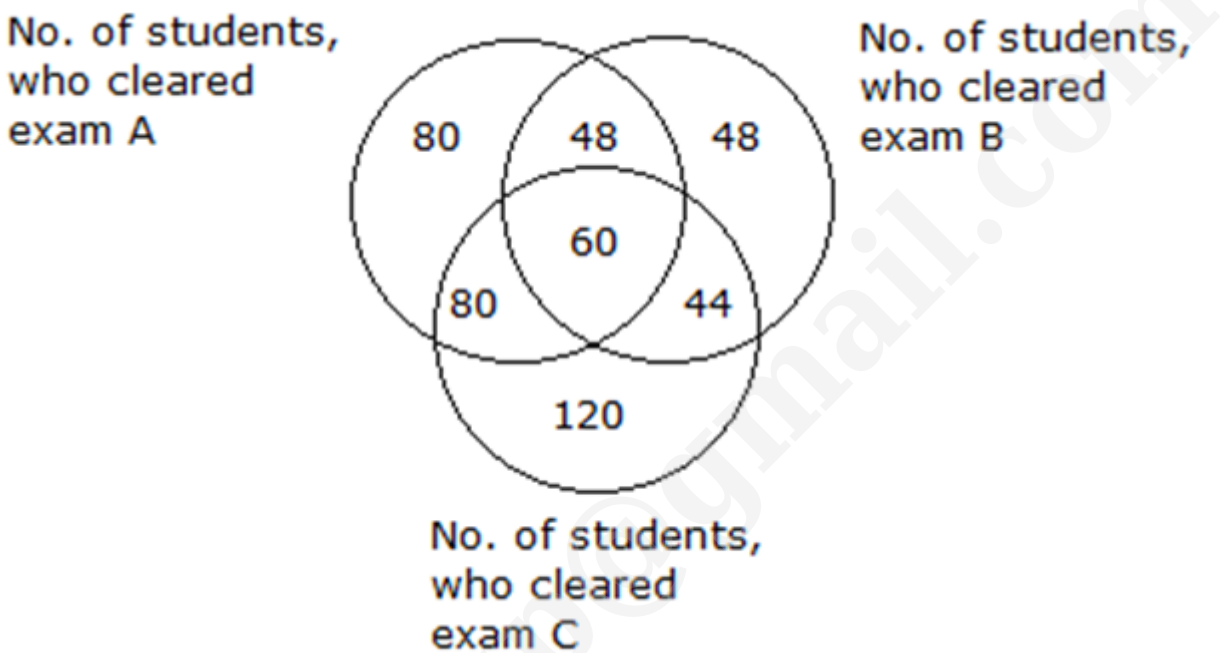
The number of students, who cleared exam C only = 15 * 8 = 120

The number of students, who cleared both exams A and B but not C = 48

The number of students, who cleared both exams A and C but not B = 80

The number of students, who cleared both exams B and C but not A = 44

The number of students, who cleared all the three exams = 44 + 16 = 60



Answer: A

Since, the total number of students, who cleared exam A:

80 + 48 + 80 + 60 = 268

So, the value, which will be filled in the 1st blank = (268/480) * 100 = 55.83%

Since, the total number of students, who cleared exam B:

48 + 48 + 44 + 60 = 200

So, the value, which will be filled in the 2nd blank = (200/480) * 100 = 41.67%

14. Questions

Let the total number of students in the college = 60x

So, the number of students, who cleared exam A only = 16(2/3)% of 60x = 10x

The number of students, who cleared exam B only = 10% of 60x = 6x

And the number of students, who cleared exam C only = 25% of 60x = 15x

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Since, the number of students, who cleared both exams A and B but not C, is equal to those, who cleared exam B only.

So, the number of students, who cleared both exams A and B but not C = 6x

Since, the number of students, who cleared both exams A and C but not B, is equal to those, who cleared exam A only.

So, the number of students, who cleared both exams A and C but not B = 10x

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So, the number of students, who cleared all the three exams = (y + 16)

Since, the total number of students, who cleared only one exam, is 188 more than those, who cleared all the three exams.

So,

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The total number of students in the college = 60 * 8 = 480

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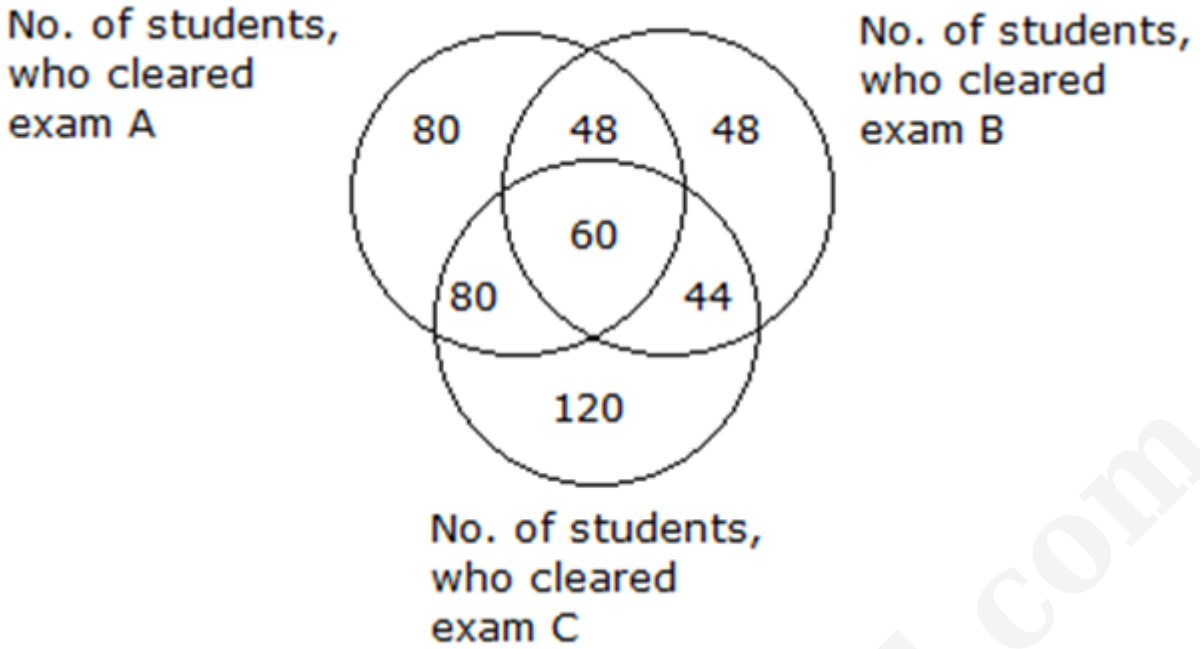
The number of students, who cleared both exams A and B but not C = 48

The number of students, who cleared both exams A and C but not B = 80

The number of students, who cleared both exams B and C but not A = 44

The number of students, who cleared all the three exams = 44 + 16 = 60

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Answer: C

Since, the total number of students, who cleared at-least two exams:

48 + 44 + 80 + 60 = 232

So, the number of girls, who cleared at-least two exams = 232 * (27/58) = 108

Since, the number of students, who cleared all the three exams = 60

So, the number of girls, who cleared all the three exams = 60 * (3/5) = 36

Required percentage = (36/108) * 100 = 33.33%

15. Questions

Let the total number of students in the college = 60x

So, the number of students, who cleared exam A only = 16(2/3)% of 60x = 10x

The number of students, who cleared exam B only = 10% of 60x = 6x

And the number of students, who cleared exam C only = 25% of 60x = 15x

Since, the number of students, who cleared both exams A and B but not C, is equal to those, who cleared exam B only.

So, the number of students, who cleared both exams A and B but not C = 6x

Since, the number of students, who cleared both exams A and C but not B, is equal to those, who cleared exam A only.

So, the number of students, who cleared both exams A and C but not B = 10x

Let the number of students, who cleared both exams B and C but not A = y

So, the number of students, who cleared all the three exams = (y + 16)

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Since, the total number of students, who cleared only one exam, is 188 more than those, who cleared all the three exams.

So,

(10x + 6x + 15x) – (y + 16) = 188

31x – y = 204 ---------- (1)

Since, the total number of students, who cleared only two exams, is 172.

So,

6x + 10x + y = 172

16x + y = 172 ---------- (2)

From equations (1) and (2):

31x – 204 = 172 – 16x

x = 8

From equation (1):

248 – y = 204

y = 44

The total number of students in the college = 60 * 8 = 480

The number of students, who cleared exam A only = 10 * 8 = 80

The number of students, who cleared exam B only = 6 * 8 = 48

The number of students, who cleared exam C only = 15 * 8 = 120

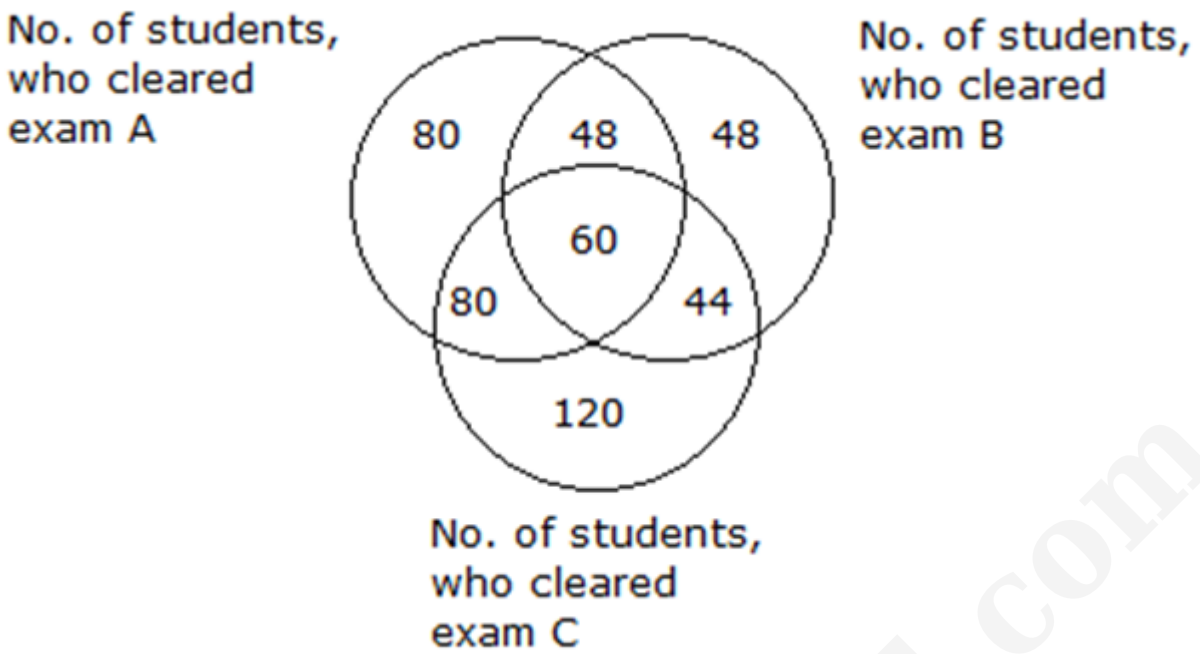
The number of students, who cleared both exams A and B but not C = 48

The number of students, who cleared both exams A and C but not B = 80

The number of students, who cleared both exams B and C but not A = 44

The number of students, who cleared all the three exams = 44 + 16 = 60

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Answer: C

Since, the number of students, who cleared exam A only = 80

And the number of students, who cleared all the three exams = 60

So, P: Q = 80: 60 = 4: 3

Since, the number of students, who cleared exam B only = 48

And the ratio of boys to girls, who cleared exam B only:

(P + Q): (Q + 2) = (4 + 3): (3 + 2) = 7: 5

So, the number of girls, who cleared exam B only = 48 * (5/12) = 20

16. Questions

Let the total number of students in the college = 60x

So, the number of students, who cleared exam A only = 16(2/3)% of 60x = 10x

The number of students, who cleared exam B only = 10% of 60x = 6x

And the number of students, who cleared exam C only = 25% of 60x = 15x

Since, the number of students, who cleared both exams A and B but not C, is equal to those, who cleared exam B only.

So, the number of students, who cleared both exams A and B but not C = 6x

Since, the number of students, who cleared both exams A and C but not B, is equal to those, who cleared exam A only.

So, the number of students, who cleared both exams A and C but not B = 10x

Let the number of students, who cleared both exams B and C but not A = y

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So, the number of students, who cleared all the three exams = (y + 16)

Since, the total number of students, who cleared only one exam, is 188 more than those, who cleared all the three exams.

So,

(10x + 6x + 15x) – (y + 16) = 188

31x – y = 204 ---------- (1)

Since, the total number of students, who cleared only two exams, is 172.

So,

6x + 10x + y = 172

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31x – 204 = 172 – 16x

x = 8

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248 – y = 204

y = 44

The total number of students in the college = 60 * 8 = 480

The number of students, who cleared exam A only = 10 * 8 = 80

The number of students, who cleared exam B only = 6 * 8 = 48

The number of students, who cleared exam C only = 15 * 8 = 120

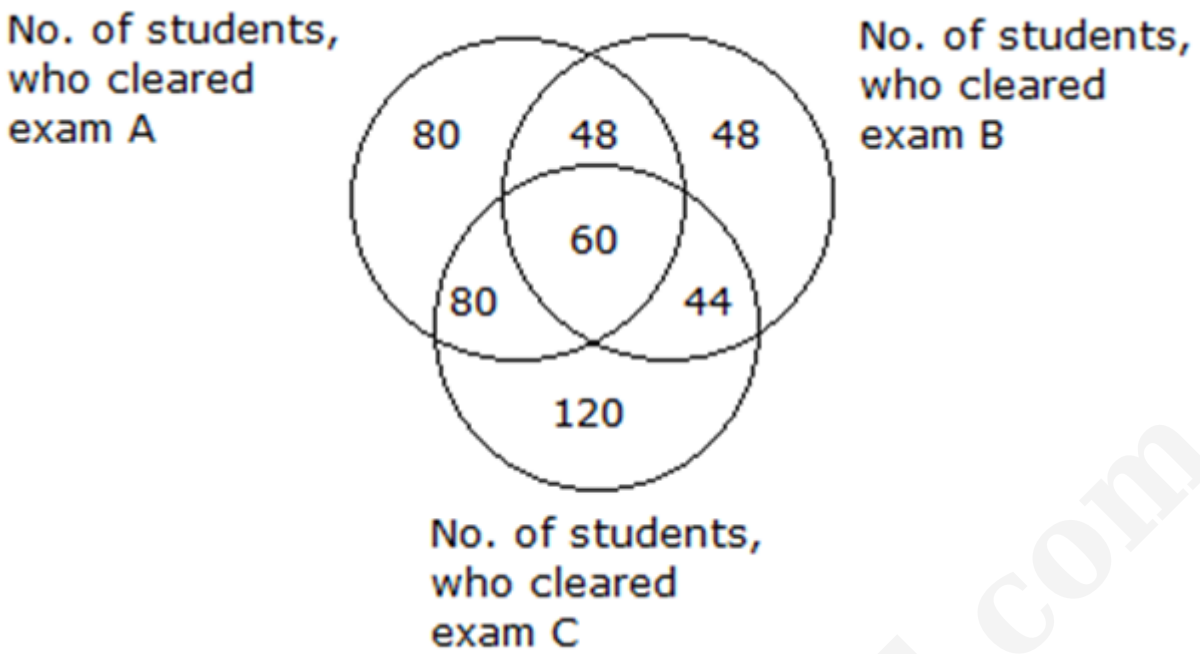
The number of students, who cleared both exams A and B but not C = 48

The number of students, who cleared both exams A and C but not B = 80

The number of students, who cleared both exams B and C but not A = 44

The number of students, who cleared all the three exams = 44 + 16 = 60

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Answer: B

Since, the number of students, who cleared exam C only = 120

And the total number of students, who cleared exam C = 120 + 80 + 44 + 60 = 304

So, the required percentage = (120/304) * 100 = 39.5%

17. Questions

Let the quantities of milk and water in mixture A be '4x' L and 'x' L respectively.

Since, the quantity of milk in mixture B is 25% more than that in mixture A and the quantity of water in mixture B is double of that in mixture A.

So, the quantity of milk in mixture B = 125% of 4x = 5x L

And the quantity of water in mixture B = 2 * x = 2x L

Since, the quantity of mixture C is 80% of that of mixture A.

So, the quantity of mixture C = 80% of (4x + x) = 4x L

The quantity of milk in mixture C = 75% of 4x = 3x L

And the quantity of water in mixture C = 4x – 3x = x L

Since, the quantity of milk in mixture D is equal to that in mixture B and the ratio of quantity of water in mixture D to that in mixture B is 3: 2.

So, the quantity of milk in mixture D = 5x L

And the quantity of water in mixture D = 2x * (3/2) = 3x L

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Mixture	Quantity of milk (in Lit)	Quantity of water (in Lit)
A	4x	x
B	5x	2x
C	3x	x
D	5x	3x

Answer: B

Since, the difference between the quantities of milk and water in mixture A = 24 L

So, 4x – x = 24

x = 8

The quantity of milk in mixture B = 5 * 8 = 40 L

The quantity of water in mixture B = 2 * 8 = 16 L

The ratio of milk to water in mixture B = 40: 16 = 5: 2

When 14 L quantity of mixture B is taken out which is replaced with the same quantity of milk:

The new quantity of milk in mixture B = 40 – 14 * (5/7) + 14 = 44 L

The new quantity of water in mixture B = 16 – 14 * (2/7) = 12 L

Required ratio = 44: 12 = 11: 3

18. Questions

Let the quantities of milk and water in mixture A be '4x' L and 'x' L respectively.

Since, the quantity of milk in mixture B is 25% more than that in mixture A and the quantity of water in mixture B is double of that in mixture A.

So, the quantity of milk in mixture B = 125% of 4x = 5x L

And the quantity of water in mixture B = 2 * x = 2x L

Since, the quantity of mixture C is 80% of that of mixture A.

So, the quantity of mixture C = 80% of (4x + x) = 4x L

The quantity of milk in mixture C = 75% of 4x = 3x L

And the quantity of water in mixture C = 4x – 3x = x L

Since, the quantity of milk in mixture D is equal to that in mixture B and the ratio of quantity of water in mixture D to that in mixture B is 3: 2.

So, the quantity of milk in mixture D = 5x L

And the quantity of water in mixture D = 2x * (3/2) = 3x L

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Mixture	Quantity of milk (in Lit)	Quantity of water (in Lit)
A	4x	x
B	5x	2x
C	3x	x
D	5x	3x

Answer: C

Since, the difference between the quantities of milk in mixtures A and B = 16 L

So, 5x – 4x = 16

x = 16

The quantity of milk in mixture B = 5 * 16 = 80 L

The quantity of water in mixture B = 2 * 16 = 32 L

The quantity of milk in mixture C = 3 * 16 = 48 L

The quantity of water in mixture C = 16 L

When mixtures B and C are mixed together:

The quantity of milk in the new mixture = 80 + 48 = 128 L

The quantity of water in the new mixture = 32 + 16 = 48 L

Required difference = 128 – 48 = 80 L

19. Questions

Let the quantities of milk and water in mixture A be '4x' L and 'x' L respectively.

Since, the quantity of milk in mixture B is 25% more than that in mixture A and the quantity of water in mixture B is double of that in mixture A.

So, the quantity of milk in mixture B = 125% of 4x = 5x L

And the quantity of water in mixture B = 2 * x = 2x L

Since, the quantity of mixture C is 80% of that of mixture A.

So, the quantity of mixture C = 80% of (4x + x) = 4x L

The quantity of milk in mixture C = 75% of 4x = 3x L

And the quantity of water in mixture C = 4x – 3x = x L

Since, the quantity of milk in mixture D is equal to that in mixture B and the ratio of quantity of water in mixture D to that in mixture B is 3: 2.

So, the quantity of milk in mixture D = 5x L

And the quantity of water in mixture D = 2x * (3/2) = 3x L

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Mixture	Quantity of milk (in Lit)	Quantity of water (in Lit)
A	4x	x
B	5x	2x
C	3x	x
D	5x	3x

Answer: E

Let the cost of pure milk = Rs. y per L

Since, the difference between the costs of mixtures A and B is Rs. 4.8 per L.

So,

$[(4x * y)/(4x + x)] – [(5x * y)/(5x + 2x)] = 4.8$

$(4y/5) – (5y/7) = 4.8$

$(28y – 25y)/35 = 4.8$

$y = 56$

When mixtures C and D are mixed together:

The quantity of milk in the new mixture = 3x + 5x = 8x L

The quantity of water in the new mixture = x + 3x = 4x L

So, the cost of the new mixture = $(8x * 56)/(8x + 4x)$ = Rs. 37.33 per L

20. Questions

Let the quantities of milk and water in mixture A be '4x' L and 'x' L respectively.

Since, the quantity of milk in mixture B is 25% more than that in mixture A and the quantity of water in mixture B is double of that in mixture A.

So, the quantity of milk in mixture B = 125% of 4x = 5x L

And the quantity of water in mixture B = 2 * x = 2x L

Since, the quantity of mixture C is 80% of that of mixture A.

So, the quantity of mixture C = 80% of (4x + x) = 4x L

The quantity of milk in mixture C = 75% of 4x = 3x L

And the quantity of water in mixture C = 4x – 3x = x L

Since, the quantity of milk in mixture D is equal to that in mixture B and the ratio of quantity of water in mixture D to that in mixture B is 3: 2.

So, the quantity of milk in mixture D = 5x L

And the quantity of water in mixture D = 2x * (3/2) = 3x L

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Mixture	Quantity of milk (in Lit)	Quantity of water (in Lit)
A	4x	x
B	5x	2x
C	3x	x
D	5x	3x

Answer: D

Since, the mixtures, in which the quantities of milk are equal, are mixtures B and D.

So, the total quantity of milk in mixtures B and D together = 5x + 5x = 10x L

Since, the mixtures, in which the quantities of water are equal, are mixtures A and C.

So, the total quantity of water in mixtures A and C together = x + x = 2x L

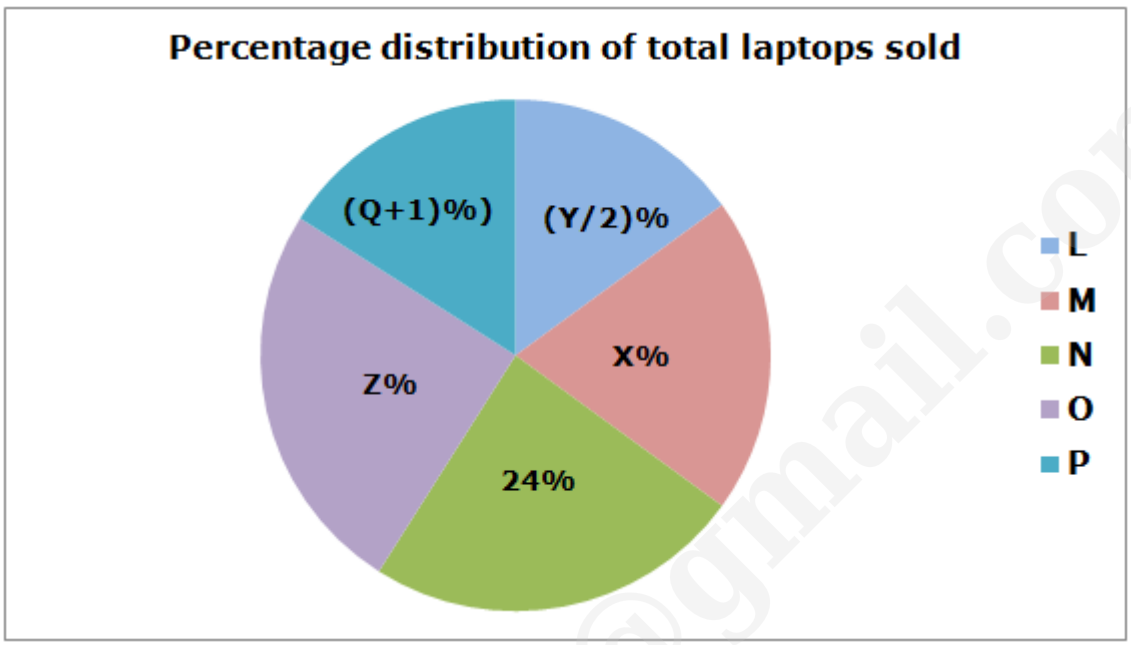
Required ratio = 10x: 2x = 5: 1

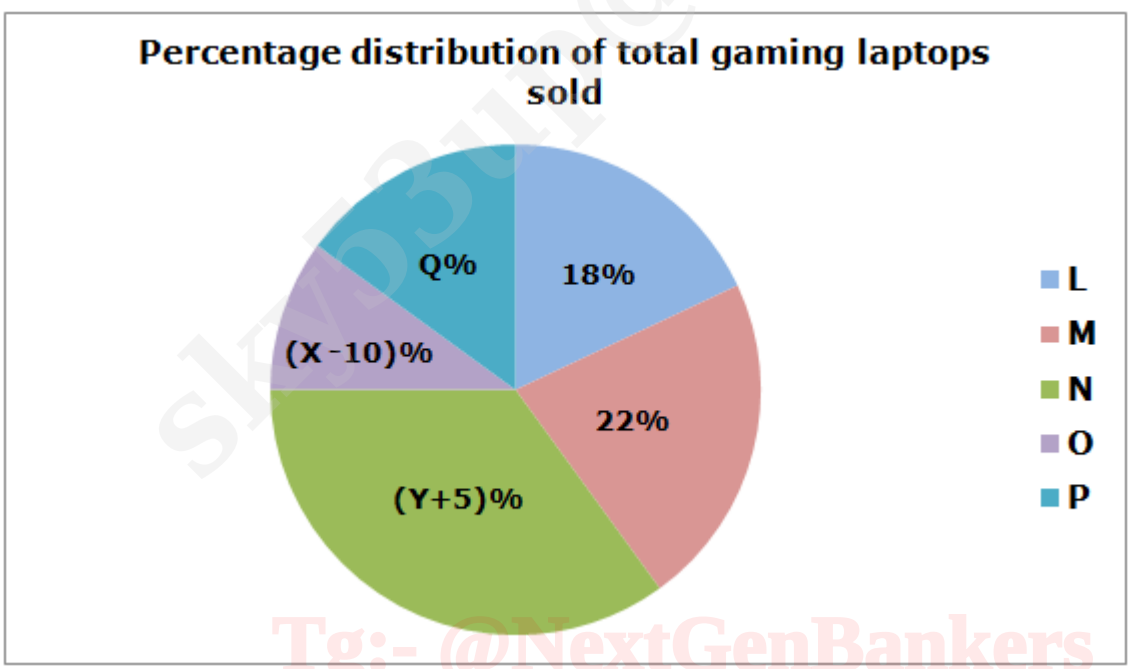
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1. Questions

Read the following information carefully and answer the questions based on it.

There are five different companies – L, M, N, O and P that sold two different types of laptops – gaming laptops and ultrabooks in 2023. The 1st pie chart given below shows the percentage distribution of total laptops sold by all companies together and 2nd pie chart given below shows the percentage distribution of gaming laptops sold by all companies together.





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Note :

A). The total number of laptops sold by all companies together is 25000.

B). X + Z = 45 and X < Z where X is a multiple of 10 and Z is a perfect square number.

C). The ratio of the total number of laptops sold by company M to the total number of gaming laptops sold

by company L is 25 : 9 respectively.

Ultrabooks sold by company O are (2a+250) and gaming laptops sold by company P are (0.3b + 600). If the total number of 12" screen ultrabooks sold by company L is (b/4) and the total number of 12" screen gaming laptops sold by company L is (a/2), then find the total number of laptops sold by company L other than 12" screen.

a. 1200

b. 1500

c. 1750

d. 1800

e. None of these

2. Questions

In 2023, the total number of laptops sold by company N is 20% more than the total number of laptops sold by company N in 2022. The ratio of the number of gaming laptops sold by company N in 2022 to 2023 is 4:7 respectively. Out of the total number of ultrabooks sold by company N in 2022, (2Y/5)% laptops were returned. Find the number of ultrabooks sold by company N in 2022 that were not returned.

a. 2460

b. 2440

c. 2660

d. 2640

e. None of these

3. Questions

Company P sold only two types of laptops– with 8GB RAM and 12GB RAM. The number of gaming laptops having 8GB RAM sold was 220 more than the gaming laptops having 12GB RAM sold while the number of ultrabooks having 8GB RAM sold was twice the number of gaming laptops having 12GB RAM sold. Find the total number of laptops having 12GB RAM sold by the company.

a. 1280

b. 1220

c. 1860

d. 640

e. None of these

4. Questions

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Find twice the difference between the average number of gaming laptops sold by company L, company N and company O together and the average number of ultrabooks sold by company O and company P together.

a. $2 \times (Q + X)^2$

b. $71 \times (X + Y)$

c. 3500

d. 3050

e. None of these

5. Questions

The total number of laptops sold by company K is 8% more than that of company M out of which 40% laptops were gaming laptops and the rest were ultrabooks. If the price of one gaming laptop is 50 thousand and the total revenue generated by company K is 2376 lakhs the find the price of one ultrabook.

a. Rs.40000

b. Rs.50000

c. Rs.55000

d. Rs.44000

e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the percentage distribution of total earphones of three brands (Redmi, Boat, and Noise) sold by five companies and 'Redmi' earphones sold by five companies. Study the data carefully and answer the following questions.

Name of company	% of earphones sold out of total earphones sold	No. of Redmi earphones sold
A	12%	750
B	X%	1650
C	21%	860
D	16%	620
E	Y%	520

Note:

1). The total number of earphones sold = Redmi + Boat + Noise

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2). The total number of earphones sold by all companies = 42000

3). The total number of earphones sold by company B is twice the total number of earphones sold by company E.

If the average number of Boat and Redmi earphones sold by company A are 1475 and the average number of Noise and Redmi earphones sold by company C are 2140 then the number of Noise earphones sold by company A is how much percentage more or less than the number of Boat earphones sold by company C (approx.)?

a. 56% more

b. 54% less

c. 52% more

d. 54% more

e. None of these

7. Questions

Quantity I: Find the average number of Redmi earphones sold by all 5 companies?

Quantity II: If Boat earphones sold by company C are 13.33% more than Redmi earphones sold by company B. Then find the difference between boat earphones sold by company C and Redmi earphones sold by company C?

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II

d. Quantity I ≥ Quantity II

e. Quantity I ≤ Quantity II

8. Questions

Find the sum of Noise and Redmi earphones sold by Company E?

Statement I: The number of Redmi earphones sold by company E is less than the number of Boat and Noise earphones sold by company E and the average number of boat and noise earphones sold by company E is 3310.

Statement II: The average number of boat earphones sold by all 5 companies is 3894 and the ratio of Boat earphones sold by company C and company E is 11: 12.

a. I only

b. II only

c. Either I or II

d. Both I and II

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e. Neither I nor II

9. Questions

There is another company F that sells the same brands of earphones and the total number of earphones sold by company F is equal to the average number of earphones sold by company B and C. The average number of Redmi and Boat earphones sold by company B is 5150 and the number of Boat earphones sold by Company F is 150 more than the number of noise earphones sold by company B. Also the difference between Boat and Noise earphones sold by company D (Noise >Boat) is 1140 and number of Noise earphones sold by company F is 37.5% more than boat earphones sold by company D. Find the number of Redmi earphones sold by company F.

a. 4210

b. 4010

c. 3910

d. 4110

e. None of these

10. Questions

If X% of the sum of Boat and Noise earphones sold by company D are defective and the ratio of the defective Boat and Noise earphones sold by company D is 10:7. And the defective Redmi earphones sold by company D are (Y – 2) % of defective boat earphones sold by company D. What is true about the non – defective earphones sold by company D?

a. It is a composite number

b. It is a prime number

c. Sum of digits of the number is a prime number

d. It is divisible by 3

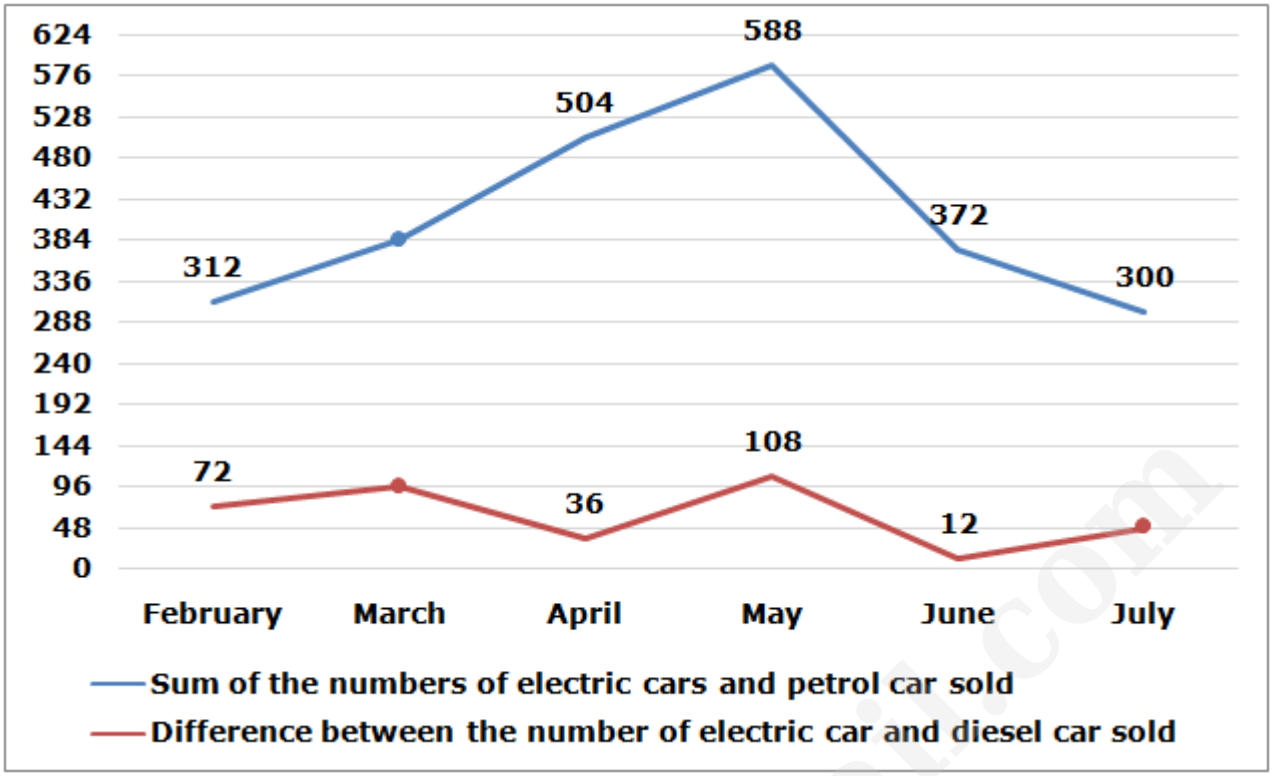
e. More than one option is correct

11. Questions

Read the following information carefully and answer the questions based on it.

There are three different cars (Electrical cars, Petrol cars and Diesel cars) sold by a showroom in 6 different months. The line graph below shows the sum of the numbers of electric cars and petrol cars sold in different months and the difference between the number of electric cars and diesel cars sold in different months by the showroom.

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The table below shows the ratio of petrol cars sold to diesel cars sold in different months.

Months	Petrol car : Diesel car
February	x:y
March	z:x
April	8:z
May	7:x
June	(z+3):7
July	4:x

Note:

1). The numbers of electric cars sold every month in more than the number of diesel cars sold in every month.

2). LCM of x, y and z is 30.

3). x, y and z is a prime number and z > x > y

The ratio of the selling price of electric cars, petrol cars and diesel cars is x:z:4y. The total amount received by the showroom owner after selling all the cars in April is 14.8176 crore. If petrol cars are sold at a profit of 40%, then find the cost price of petrol cars.

a. Rs. 120000

b. Rs. 130000

c. Rs. 140000

d. Rs. 150000

e. None of these

12. Questions

Petrol cars sold in April have three different colours of black, red and white in the ratio y:(x-y):(z-4) respectively. The selling price of each white coloured petrol car is 500% more than the selling price of each black coloured petrol car. If the difference between the amount collected by a selling a white coloured petrol cars and black coloured petrol cars is Rs. 5.184 crore, then find the selling price of black coloured petrol cars.

a. Rs. 150000

b. Rs 160000

c. Rs. 170000

d. Rs. 180000

e. None of these

13. Questions

In June, 40% of the electric cars that the showroom owner had were defective and he sold the rest of the electric cars. The defective electric cars in june are what percent more/less than the diesel cars sold in February?

a. 15%

b. 20%

c. 25%

d. 30%

e. None of these

14. Questions

Quantity I: Find the difference between the sum of electric cars sold in March, April and June together and the sum of petrol cars sold in February and May.

Quantity II: Find the average number of diesel cars sold in February, March and May.

a. Quantity I > Quantity II

b. Quantity I ≥ Quantity II

c. Quantity I < Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I = Quantity II

15. Questions

Which of the following statement(s) is/are true:

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I). The number of diesel cars sold in June is equal to the number of Electric cars sold in February.

II). The ratio of number of petrol cars sold in March to the number of electric cars sold in April is z:2x.

III). The average number of petrol cars sold in all 6 months together is greater than the average number of electric cars sold in all 6 months together.

a. Only I

b. Both II & III

c. Both I & II

d. Both I & III

e. All I, II & III

16. Questions

Read the following information carefully and answer the questions based on it.

Two persons P and Q enter into a partnership to open a shop by investing some amount in ratio a:b respectively. The difference between the initial investment of P and Q is Rs X. R who joined them after Y months had already started a cafe with S. R and S together earned a profit of Rs 9090 from the cafe, Where R invested in the cafe is Rs Z which is Rs 6750 less than S. R invested the total amount received from the partnership of Cafe in shop as an initial investment for the partnership of shop. The respective ratio of period of investment for R and S in cafe's partnership is in ratio a:c. After 6 months of joining of R in shop's partnership S joined with Rs 3000 more than the initial investment of P, and Q withdraw 1/4th of his investment. At the end of 20 months of Shop's partnership, profit earned by P is Rs 1500 less than Q. From Cafe, profit earned by R is Rs 90 less than S. In Shop's partnership, profit earned by P and S is in ratio $16b:c^3$ respectively and profit earned by R and S is in ratio $(6b + 1): 2c^2$ respectively.

Note:

1). LCM of b and c is 15 and b > c

2). 3a = 1.5(b+c)

P and R started a business by investing profit received from shop's partnership. After ___ months P withdraw Rs 600 and R withdraw Rs 180. If the profit earned by P and R after 1 year partnership is in the ratio (5abc+10):(3abc+3) respectively, then what should be come in place of the given blank.

a. b+1

b. c+2

c. a-1

d. a+b-2

e. None of these

17. Questions

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If the profit earned by Q is (4M+5)% of his initial investment and the difference between S's investment in shop and cafe partnership is Rs (16N+140), then what is the value of (N-4y-5M)?

a. 7

b. 13

c. 17

d. 19

e. None of these

18. Questions

The difference between the profit earned by S in cafe's partnership and shop's partnership is what percent of the amount invested by him in cafe's partnership?

a. 3.5%

b. 6.25%

c. 5.29%

d. 7.3%

e. None of these

19. Questions

Quantity I: Find the difference between the sum of profit earned by Q in shop's partnership and profit and by R in cafe's partnership together and the profit earned by P in shop's partnership.

Quantity II: Find the difference between P's initial investment in shop partnership and R's initial investment in cafes partnership.

a. Quantity I > Quantity II

b. Quantity I ≥ Quantity II

c. Quantity I < Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I = Quantity II

20. Questions

Which of the following statement/s is/are true:

I: X is completely divisible by Y.

II: Initial investment of Q in shop's partnership is more than initial investment of S in cafe's partnership.

III: The profit share of R in shops partnership is Rs (X - 420)

a. Only I

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b. Both II & III

c. Both I & II

d. Both I & III

e. All I, II & III

Explanations:

1. Questions

It is given that,

X + Z = 45 and X < Z where X is a multiple of 10 and Z is a perfect square number.

Since the unit digit of 45 is 5 and the unit digit of X is 0 as it is a multiple of 10.

So Z must have the unit digit 5 and is a perfect square number.

So, Z = 25(only possible case)

Therefore, X = 45 – 25 = 20

It is given that, the total number of laptops sold by all companies together is 25000.

Also it is given that, the ratio of total number of laptops sold by company M to the total number of gaming laptops sold by company L is 25 : 9 respectively.

So, (X% of 25000)/ 18% of total gaming laptops sold] = 25/9

Or, Total gaming laptops sold = (9/25) x (20% of 25000)/18% = 10000

Now from 1st pie chart we have,

(Y/2) + X + 24 + Z + (Q+1) = 100

Or, (Y/2) + 20 + 24 + 25 + (Q+1) = 100

Or, Y + 2Q = 60 -----(i)

And from 2nd pie chart we have,

18 + 22 + (Y+5) + (X – 10) + Q = 100

Or, 18 + 22 + (Y+5) + 10 + Q = 100

Or, Q + Y = 45 -----(ii)

On solving equations (i) and (ii) we get,

Q = 15 and Y = 30

From the above data we can conclude the following table:

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Company	Total laptops sold(25000)	Gaming laptops sold(10000)	Ultrabooks sold(15000)
L	15% = 3750	18% = 1800	1950
M	20% = 5000	22% = 2200	2800
N	24% = 6000	35% = 3500	2500
O	25% = 6250	10% = 1000	5250
P	16% = 4000	15% = 1500	2500

Answer: C

According to the question,

2a + 250 = 5250

Or, a = 2500

And, 0.3b + 600 = 1500

Or, b = 3000

Now, the total number of ultrabooks sold by company L = 1950

The total number of 12” screen ultrabooks sold by company L = (b/4) = (3000/4) = 750

So, the total number of ultrabooks other than 12” sold by company L = 1950 – 750 = 1200

Also, the total number of gaming laptops sold by company L = 1800

The total number of 12” screen gaming laptops sold by company L = (a/2) = (2500/2) = 1250

So, total number of gaming laptops other than 12” sold by company L = 1800 – 1250 = 550

So, The total number of laptops sold by company L other than 12” screen = 550 + 1200 = 1750

Hence, the answer is option C.

2. Questions

It is given that,

X + Z = 45 and X < Z where X is a multiple of 10 and Z is a perfect square number.

Since the unit digit of 45 is 5 and the unit digit of X is 0 as it is a multiple of 10.

So Z must have the unit digit 5 and is a perfect square number.

So, Z = 25(only possible case)

Therefore, X = 45 – 25 = 20

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Or, Total gaming laptops sold = (9/25) x (20% of 25000)/18% = 10000

Now from 1st pie chart we have,

(Y/2) + X + 24 + Z + (Q+1) = 100

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Or, Y + 2Q = 60 -----(i)

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Or, Q + Y = 45 -----(ii)

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O	25% = 6250	10% = 1000	5250
P	16% = 4000	15% = 1500	2500

Answer: D

The total number of laptops sold by company N in 2023 = 6000

So, the total number of laptops sold by company N in 2022 = (5/6) x 6000 = 5000

Now, the number of gaming laptops sold by company N in 2022 = (4/7) x 3500 = 2000

And, the number of ultrabooks sold by company N in 2022 =5000 – 2000 = 3000

The total number of Ultrabooks that were returned = (2Y/5)% of 3000 = 12% of 5000 = 360

The required value = 3000 – 360 = 2640

Hence, the answer is option D.

3. Questions

It is given that,

X + Z = 45 and X < Z where X is a multiple of 10 and Z is a perfect square number.

Since the unit digit of 45 is 5 and the unit digit of X is 0 as it is a multiple of 10.

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So Z must have the unit digit 5 and is a perfect square number.

So, Z = 25(only possible case)

Therefore, X = 45 – 25 = 20

It is given that, the total number of laptops sold by all companies together is 25000.

Also it is given that, the ratio of total number of laptops sold by company M to the total number of gaming laptops sold by company L is 25 : 9 respectively.

So, (X% of 25000)/ 18% of total gaming laptops sold] = 25/9

Or, Total gaming laptops sold = (9/25) x (20% of 25000)/18% = 10000

Now from 1st pie chart we have,

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Or, (Y/2) + 20 + 24 + 25 + (Q+1) = 100

Or, Y + 2Q = 60 -----(i)

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On solving equations (i) and (ii) we get,

Q = 15 and Y = 30

From the above data we can conclude the following table:

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O	25% = 6250	10% = 1000	5250
P	16% = 4000	15% = 1500	2500

Answer: C

The total number of gaming laptops sold by company P = 1500

The total number of ultrabooks sold by company P = 2500

It is given that,

Let, The number of gaming laptops having 12GB RAM sold =‘p’

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So, The number of gaming laptops having 8GB RAM sold ='p – 220'

Therefore, p + (p + 220) = 1500

Or, 2p = 1500 – 220 = 1280

Or, p = 640

Now, The number of ultrabooks having 8GB RAM sold ='2p' = 1280

And, The number of ultrabooks having 12GB RAM sold =2500 – 1280 = 1220

The required value = 640 + 1220 = 1860

Hence, the answer is option C.

4. Questions

It is given that,

X + Z = 45 and X < Z where X is a multiple of 10 and Z is a perfect square number.

Since the unit digit of 45 is 5 and the unit digit of X is 0 as it is a multiple of 10.

So Z must have the unit digit 5 and is a perfect square number.

So, Z = 25(only possible case)

Therefore, X = 45 – 25 = 20

It is given that, the total number of laptops sold by all companies together is 25000.

Also it is given that, the ratio of total number of laptops sold by company M to the total number of gaming laptops sold by company L is 25 : 9 respectively.

So, (X% of 25000)/ 18% of total gaming laptops sold] = 25/9

Or, Total gaming laptops sold = (9/25) x (20% of 25000)/18% = 10000

Now from 1st pie chart we have,

(Y/2) + X + 24 + Z + (Q+1) = 100

Or, (Y/2) + 20 + 24 + 25 + (Q+1) = 100

Or, Y + 2Q = 60 -----(i)

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O	25% = 6250	10% = 1000	5250
P	16% = 4000	15% = 1500	2500

Answer: B

The average number of gaming laptops sold by company L, company N and company O together = (1800 + 3500 + 1000)/3 = 2100

The average number of ultrabooks sold by company O and company P together = (5250 + 2500)/2 = 3875

The required difference = 2x(3875 – 2100) = 2 x 1775 = 3550

$2 \times (Q + X)^2 = 2 \times (15 + 20)^2 = 2450$(incorrect)

71 x (X + Y) = 71 x (20+30) = 3550(correct)

Hence, the answer is option B.

5. Questions

It is given that,

X + Z = 45 and X < Z where X is a multiple of 10 and Z is a perfect square number.

Since the unit digit of 45 is 5 and the unit digit of X is 0 as it is a multiple of 10.

So Z must have the unit digit 5 and is a perfect square number.

So, Z = 25(only possible case)

Therefore, X = 45 – 25 = 20

It is given that, the total number of laptops sold by all companies together is 25000.

Also it is given that, the ratio of total number of laptops sold by company M to the total number of gaming laptops sold by company L is 25 : 9 respectively.

So, (X% of 25000)/ 18% of total gaming laptops sold] = 25/9

Or, Total gaming laptops sold = (9/25) x (20% of 25000)/18% = 10000

Now from 1st pie chart we have,

(Y/2) + X + 24 + Z + (Q+1) = 100

Or, (Y/2) + 20 + 24 + 25 + (Q+1) = 100

Or, Y + 2Q = 60 -----(i)

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Or, 18 + 22 + (Y+5) + 10 + Q = 100

Or, Q + Y = 45 -----(ii)

On solving equations (i) and (ii) we get,

Q = 15 and Y = 30

From the above data we can conclude the following table:

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M	20% = 5000	22% = 2200	2800
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O	25% = 6250	10% = 1000	5250
P	16% = 4000	15% = 1500	2500

Answer: A

The total number of laptops sold by company K = 1108/100 x 5000 = 5400

The number of gaming laptops sold = 40% of 5400 = 2160

The number of ultrabooks sold = 60% of 5400 = 3240

The price of one gaming laptop = 50000

Total revenue from gaming laptops sold = 2160 x 50000 = 10,80,00,000

Total revenue generated = 23,76,00,000

Total revenue generated from ultrabooks = 23,76,00,000 – 10,80,00,000 = 12,96,00,000

Price of one ultrabook = 129600000/3240 = 40,000

Hence, the answer is option A.

6. Questions

The given total number of earphones sold by all companies = 42000

And the total number of earphones sold by company B is twice the total number of earphones sold by company E which means,

X = 2Y

So, 12% + X% + 21% + 16% + Y% = 100%

49 + X% + Y% = 100%

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49 + 2Y% + Y% = 100% [Putting X = 2Y]

3Y% = 51%

3Y = 51

Y = 17

And X = 2Y

X = 2(17) = X = 34

So, the number of earphones sold by A = 12% * 42000= 5040

The number of earphones sold by B = X% * 42000= 34% * 42000 = 14280

The number of earphones sold by C = 21% * 42000= 8820

The number of earphones sold by D = 16% * 42000= 6720

And, the number of earphones sold by E = Y% * 42000= 17% * 42000= 7140

So, the sum of Boat and Noise earphones sold = Total earphones sold – Redmi earphones sold.

From the above data we can conclude the following table,

Name of company	Total earphones sold	No. of Redmi earphones sold	Boat + Noise earphones sold
A	5040	750	4290
B	14280	1650	12630
C	8820	860	7960
D	6720	620	6100
E	7140	520	6620

Answer: B

According to the question,

Given,

The average number of Boat and Redmi earphones sold by company A = 1475

(Boat + Redmi earphones)/2 = 1475

The total number of Boat + Redmi earphones sold by A = 1475 * 2 = 2950

Boat + 750 = 2950

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Boat earphones sold by A = 2200

The number of Noise earphones sold by A = Total number of earphones – (Boat + Redmi earphones sold) = Noise earphones sold by A

= 5040 – 750 – 2200= 2090

Also it is given that,

The average number of Noise and Redmi earphones sold by company C= 2140

(Noise + Redmi earphones)/2 = 2140

The total number of Noise + Redmi earphones sold by C = 2140 * 2 = 4280 Noise + 860 = 4280

Noise earphones sold by C = 3420

Number of Boat earphones sold by C = Total number of earphones – (Noise + Redmi earphones sold)

= Boat earphones sold by C = 8820 – 860 – 3420= 4540

So, the number of Boat earphones sold by C - Noise earphones sold by A

= 4540 – 2090= 2450

The required percentage = 2450/4540 * 100= 53.96% = 54% (approx.)

Hence, the answer is option B.

7. Questions

The given total number of earphones sold by all companies = 42000

And the total number of earphones sold by company B is twice the total number of earphones sold by company E which means,

X = 2Y

So, 12% + X% + 21% + 16% + Y% = 100%

49 + X% + Y% = 100%

49 + 2Y% + Y% = 100% [Putting X = 2Y]

3Y% = 51%

3Y = 51

Y = 17

And X = 2Y

X = 2(17) = X = 34

So, the number of earphones sold by A = 12% * 42000= 5040

The number of earphones sold by B = X% * 42000= 34% * 42000 = 14280

The number of earphones sold by C = 21% * 42000= 8820

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And, the number of earphones sold by E = Y% * 42000= 17% * 42000= 7140

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A	5040	750	4290
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C	8820	860	7960
D	6720	620	6100
E	7140	520	6620

Answer: B

For Quantity I: Total number of Redmi earphones sold by all 5 companies = 750 + 1650 + 860 + 620 + 520 = 4400

The average number of Redmi earphones sold by all 5 companies = 4400/5= 880

So, Quantity I = 880

For Quantity II: Given: Boat earphones sold by company C are 13.33% more than Redmi earphones sold by company B

13.33% = 2/15

Boat earphones sold by company C = 1650 * 17/15= 1870

So, Boat earphones sold by company C - Redmi earphones sold by company C = 1870 – 860= 1010

So, Quantity II = 1010

Therefore, Quantity I < Quantity II.

Hence, the answer is option B.

8. Questions

The given total number of earphones sold by all companies = 42000

And the total number of earphones sold by company B is twice the total number of earphones sold by company E which means,

X = 2Y

So, 12% + X% + 21% + 16% + Y% = 100%

49 + X% + Y% = 100%

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3Y% = 51%

3Y = 51

Y = 17

And X = 2Y

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So, the sum of Boat and Noise earphones sold = Total earphones sold – Redmi earphones sold.

From the above data we can conclude the following table,

Name of company	Total earphones sold	No. of Redmi earphones sold	Boat + Noise earphones sold
A	5040	750	4290
B	14280	1650	12630
C	8820	860	7960
D	6720	620	6100
E	7140	520	6620

Answer: E

In order to find the sum of Noise and Redmi earphones sold by Company E, we need to find Boat earphones sold by company E because,

Noise + Redmi earphones sold by Company E = Total earphones by company E – Boat earphones by company E

From Statement I:

The average number of boat and noise earphones sold by E = 3310

The total number of boat and noise earphones sold by E = 3310 * 2= 6620

This statement is redundant as we can also derive it from the question.

And also it is given that the Redmi earphones sold by company E are less than the Boat and Noise earphones sold by company E.

This statement doesn't help in finding Boat earphones sold by company E.

So, this statement alone is not sufficient to answer the question.

From Statement II:

The average number of boat earphones sold by all 5 companies = 3894

The total number of boat earphones sold by all 5 companies = 3894*5= 19470

And also the ratio of Boat earphones sold by company C & company E is 11: 12.

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We do not have any value of Boat earphones sold by company C or E or sum of earphones sold by them.

So, this statement alone is not sufficient to answer the question.

On combining both statement(s), we get

The total number of Boat and Noise earphones sold by E = 6620

The total number of Boat earphones sold by all 5 companies = 19470

We cannot find the sum of Noise and Redmi earphones sold by Company E or the number of Boat earphones sold by company E.

So both statements together are not sufficient to answer this question.

Hence, the answer is option E.

9. Questions

The given total number of earphones sold by all companies = 42000

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X = 2Y

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From the above data we can conclude the following table,

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Name of company	Total earphones sold	No. of Redmi earphones sold	Boat + Noise earphones sold
A	5040	750	4290
B	14280	1650	12630
C	8820	860	7960
D	6720	620	6100
E	7140	520	6620

Answer: B

Given that the total number of earphones sold by F = the average number of earphones by B and C

The total number of earphones sold by F = (14280 + 8820)/2= 23100/2

The total number of earphones sold by F = 11550

Also, the average number of Redmi and Boat earphones sold by B = 5150

The total number of Redmi and Boat earphones sold by B = 5150 * 2= 10300

The total number of Boat earphones sold by B = 10300 – 1650= 8650

The total number of Noise earphones sold by B = 14280 – 8650 – 1650

= 3980

The total number of Boat earphones sold by F = Noise earphones sold by B + 150= 3980 + 150

The total number of Boat earphones sold by F = 4130

And, the difference between Boat and Noise earphones sold by company D = 1140-----[Noise > Boat]

Noise – Boat earphones sold by D = 1140 ……. (1)

Noise + Boat earphones sold by D = 6720 – 620

Noise + Boat earphones sold by D = 6100 …….. (2)

By adding equations 1 and 2,

2(Noise) = 1140 + 6100

2(Noise) = 7240

The total number of Noise earphones sold by D = 3620

The total number of Boat earphones sold by D = 6100 – 3620= 2480

The total number of Noise earphones sold by F = Boat earphones sold by D * (137.5%)

= 2480 * 11/8-----[37.5% = 3/8]

Noise earphones sold by F = 3410

Now, The number of Redmi earphones sold by F = Total earphones sold by F – Boat and Noise earphones sold by F

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The number of Redmi earphones sold by F = 11550 – 4130 – 3410= 4010

Hence, the answer is option B

10. Questions

The given total number of earphones sold by all companies = 42000

And the total number of earphones sold by company B is twice the total number of earphones sold by company E which means,

X = 2Y

So, 12% + X% + 21% + 16% + Y% = 100%

49 + X% + Y% = 100%

49 + 2Y% + Y% = 100% [Putting X = 2Y]

3Y% = 51%

3Y = 51

Y = 17

And X = 2Y

X = 2(17) = X = 34

So, the number of earphones sold by A = 12% * 42000= 5040

The number of earphones sold by B = X% * 42000= 34% * 42000 = 14280

The number of earphones sold by C = 21% * 42000= 8820

The number of earphones sold by D = 16% * 42000= 6720

And, the number of earphones sold by E = Y% * 42000= 17% * 42000= 7140

So, the sum of Boat and Noise earphones sold = Total earphones sold – Redmi earphones sold.

From the above data we can conclude the following table,

Name of company	Total earphones sold	No. of Redmi earphones sold	Boat + Noise earphones sold
A	5040	750	4290
B	14280	1650	12630
C	8820	860	7960
D	6720	620	6100
E	7140	520	6620

Answer: E

Given: X% of the sum of Boat and Noise earphones sold by company D are defective

We know that X = 34 Y = 17

= 34% of Boat and Noise earphones sold by D are defective

= 6100 * 34% are defective

= 2074 Boat and Noise earphones sold by D are defective

And The ratio of defective Boat and Noise earphones sold by D = 10:7

The total number of Boat earphones sold by D that are defective = 2074 * 10/17= 1220

The total number of Noise earphones sold by D that are defective = 2074 * 7/17= 854

Now, The total number of Defective Redmi earphones sold by company D are (Y – 2) % of the defective boat earphones sold by company D

The total number of Redmi earphones sold by D that are defective = 15% * 1220= 183

The total number of defective earphones sold by D = 1220 + 854 + 183= 2257

The total number of non - defective earphones sold by D = Total earphones sold by D - Defective earphones sold by D

The total number of non - defective earphones sold by D = 6720 – 2257 = 4463

Therefore,

Option a) 4463 is a prime number which has only 2 factor which is 1 and 4463

Option b) Sum of digits of 4437 = 4+4+6+3 = 19 which is also a prime number

So, options b and c are correct

Hence, more than one option is correct.

Hence, the answer is option E.

11. Questions

LCM of three numbers= {product of three number × HCF(x,y,z)}/ HCF(x,y)× HCF (y,z)× HCF (x,z)

We know that x, y & z is a prime numbers

So, HCF = 1

30 = product of xyz

xyz = 30 (is only possible when 2 is prime)

So, y = 2

xz = 15 (only possible 3 × 5)

So, x= 3, z = 5

So, we have the following table

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Month	Petrol : Diesel
February	3:2
March	5:3
April	8:5
May	7:3
June	8:7
July	4:3

Let us calculate data for February:

It is given that,

Electric + Petrol = 312

Electric – Diesel = 72

Petrol : Diesel = 3:2

Let petrol cars and diesel cars are 3a and 2a

Then, electric cars are 2a+72

2a+72 + 3a = 312

5a = 240

So, a = 48

It is given that,

Electric cars = 2×48+72 = 168

Petrol cars = 312-168 = 144

Diesel cars = 168-72 = 96

Similarly, on calculating the data for remaining months we have the following table

Months	Electric cars	Petrol cars	Diesel cars
February	168	144	96
March	204	180	108
April	216	288	180
May	252	336	144
June	180	192	168
July	156	144	108

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Answer: D

According to the question,

The ratio of electric cars, petrol cars and diesel cars is x:z:4y

x = 3, y = 2, z = 5

So, the ratio of electric cars, petrol cars and diesel cars is 3:5:8

3a×216+5a×288+8a×180 = 148176000

3528a = 148176000

So, a = 42000

Selling price of petrol cars = 5×42000 = 210000

Cost price of petrol car = 210000/140×100= 150000

Hence, the answer is option D.

12. Questions

LCM of three numbers= {product of three number × HCF(x,y,z)}/ HCF(x,y)× HCF (y,z)× HCF (x,z)

We know that x, y & z is a prime numbers

So, HCF = 1

30 = product of xyz

xyz = 30 (is only possible when 2 is prime)

So, y = 2

xz = 15 (only possible 3 × 5)

So, x= 3, z = 5

So, we have the following table

Month	Petrol : Diesel
February	3:2
March	5:3
April	8:5
May	7:3
June	8:7
July	4:3

Let us calculate data for February:

It is given that,

Electric + Petrol = 312

Electric – Diesel = 72

Petrol : Diesel = 3:2

Let petrol cars and diesel cars are 3a and 2a

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Then, electric cars are 2a+72

2a+72 + 3a = 312

5a = 240

So, a = 48

It is given that,

Electric cars = 2×48+72 = 168

Petrol cars = 312-168 = 144

Diesel cars = 168-72 = 96

Similarly, on calculating the data for remaining months we have the following table

Months	Electric cars	Petrol cars	Diesel cars
February	168	144	96
March	204	180	108
April	216	288	180
May	252	336	144
June	180	192	168
July	156	144	108

Answer: D

The ratio of black, red and white is y:(x-y):(z-4)

x = 3, y = 2, z = 5

So, The ratio of black, red and white is 2:1:1

black = 288×2/4 = 144

red = 288×1/4 = 72

white = 288×1/4 = 72

Let the selling price of black car = 1a

So, the selling price of white car = 6a

According to the question,

72×6a-144×1a= 51840000

288a = 51840000

a = 180000

Selling price of black colour petrol car is Rs 180000.

Hence, the answer is option D.

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13. Questions

LCM of three numbers= {product of three number × HCF(x,y,z)}/ HCF(x,y)× HCF (y,z)× HCF (x,z)

We know that x, y & z is a prime numbers

So, HCF = 1

30 = product of xyz

xyz = 30 (is only possible when 2 is prime)

So, y = 2

xz = 15 (only possible 3 × 5)

So, x= 3, z = 5

So, we have the following table

Month	Petrol : Diesel
February	3:2
March	5:3
April	8:5
May	7:3
June	8:7
July	4:3

Let us calculate data for February:

It is given that,

Electric + Petrol = 312

Electric – Diesel = 72

Petrol : Diesel = 3:2

Let petrol cars and diesel cars are 3a and 2a

Then, electric cars are 2a+72

2a+72 + 3a = 312

5a = 240

So, a = 48

It is given that,

Electric cars = 2×48+72 = 168

Petrol cars = 312-168 = 144

Diesel cars = 168-72 = 96

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Similarly, on calculating the data for remaining months we have the following table

Months	Electric cars	Petrol cars	Diesel cars
February	168	144	96
March	204	180	108
April	216	288	180
May	252	336	144
June	180	192	168
July	156	144	108

Answer: C

The total electric cars in June = 180/60×100 = 300

The defective electric cars = 300-180 = 120

The diesel cars in February. = 96

The required % = (120-96)/96×100 = 25%

Hence, the answer is option C.

14. Questions

LCM of three numbers= {product of three number × HCF(x,y,z)}/ HCF(x,y)× HCF (y,z)× HCF (x,z)

We know that x, y & z is a prime numbers

So, HCF = 1

30 = product of xyz

xyz = 30 (is only possible when 2 is prime)

So, y = 2

xz = 15 (only possible 3 × 5)

So, x= 3, z = 5

So, we have the following table

Month	Petrol : Diesel
February	3:2
March	5:3
April	8:5
May	7:3
June	8:7
July	4:3

Let us calculate data for February:

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It is given that,

Electric + Petrol = 312

Electric – Diesel = 72

Petrol : Diesel = 3:2

Let petrol cars and diesel cars are 3a and 2a

Then, electric cars are 2a+72

2a+72 + 3a = 312

5a = 240

So, a = 48

It is given that,

Electric cars = 2×48+72 = 168

Petrol cars = 312-168 = 144

Diesel cars = 168-72 = 96

Similarly, on calculating the data for remaining months we have the following table

Months	Electric cars	Petrol cars	Diesel cars
February	168	144	96
March	204	180	108
April	216	288	180
May	252	336	144
June	180	192	168
July	156	144	108

Answer: A

For Quantity I:

The required difference = (204+216+180) – (144+336) = 120

So, Quantity I = 120

For Quantity II :

The required average = (96+108+144)/3 = 166

So, Quantity II = 116

Quantity I> Quantity II

Hence, the answer is option A.

15. Questions

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LCM of three numbers= {product of three number × HCF(x,y,z)}/ HCF(x,y)× HCF (y,z)× HCF (x,z)

We know that x, y & z is a prime numbers

So, HCF = 1

30 = product of xyz

xyz = 30 (is only possible when 2 is prime)

So, y = 2

xz = 15 (only possible 3 × 5)

So, x= 3, z = 5

So, we have the following table

Month	Petrol : Diesel
February	3:2
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Let us calculate data for February:

It is given that,

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Electric – Diesel = 72

Petrol : Diesel = 3:2

Let petrol cars and diesel cars are 3a and 2a

Then, electric cars are 2a+72

2a+72 + 3a = 312

5a = 240

So, a = 48

It is given that,

Electric cars = 2×48+72 = 168

Petrol cars = 312-168 = 144

Diesel cars = 168-72 = 96

Similarly, on calculating the data for remaining months we have the following table

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Months	Electric cars	Petrol cars	Diesel cars
February	168	144	96
March	204	180	108
April	216	288	180
May	252	336	144
June	180	192	168
July	156	144	108

Answer: E

I). The number of diesel cars sold in June = 168

The number of Electric cars sold in February = 168

168 = 168 (True)

II). The number of petrol cars sold in March = 180

The number of electric cars sold in April = 216

The required ratio = 180:216 = 5:6 = z:2x (True)

III). The average number of petrol cars sold in all 6 months =

(144+180+288+336+192+144)/6 = 214

The average number of electric cars sold in all 6 months =

(168+204+216+252+180+156)/6 = 196

214 > 196 (True)

All I, II & III is True

Hence, the answer is option E.

16. Questions

LCM of (b, c) = 15

Only possible value of b=5 & c=3

3a = (5+3)1.5

a= 4

For Cafe

R's investment = Z

S's investment = Z+6750

Time of investment = a:c = 4:3

The profit ratio of R and S = (Z×4):{(Z+6750)×3}= 4Z: (3Z+20250)

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Total profit of R and S = 9090

R's profit share = (9090-90)/2 = 4500

S's profit share = 4500+90 = 4590

Now, 4Z: (3Z+20250) = 4500:4590

Z= 18750

S's investment = 18750+6750= 25500

For Shop

P's investment = 4x

Q's investment = 5x

S's investment = 4x+3000

R's investment = 18750+4500 = 23250

The profit ratio of P:Q:R:S =

(4x×20):[5x(y+6)+(3/4×5x)×(20-y-6)]:[(23250×(20-y)]:[(4x+3000)×(20-y-6)]

P:S= (4x×20):[(4x+3000)×(20-y-6)]=16b:c^3= 80:27

Y= (29x+42000)/(4x+3000)

R:S = [(23250×(20-y)]:[(4x+3000)×(20-y-6)] = (6b+1):$2c^2$= 31:18

Y = (14x-57000)/(x-2625)

(29x+42000)/(4x+3000) = (14x-57000)/(x-2625)

X= 6000

Y = 8

The profit share of P:Q:R:S = 160:185:93:54

The total profit earned = 1500×(160+185+93+54)/(185-160)= 29520

The profit share of P = 29520×160/492 = 9600

The profit share of Q = 29520×185/492 = 11100

The profit share of R = 29520×93/492 = 5580

The profit share of S = 29520×54/492 = 3240

Answer: A

According to the question,

Let months = k

P initial investment = 9600

R initial investment = 5580

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The profit ratio = [(9600×k)+(9600-600)(12-k)]:[5580×k+(5580-180)(12-k)

= (5abc+10):(3abc+3) = 310:183

k= 6

The required value is b+1

Hence, the answer is option A.

17. Questions

LCM of (b, c) = 15

Only possible value of b=5 & c=3

3a = (5+3)1.5

a= 4

For Cafe

R's investment = Z

S's investment = Z+6750

Time of investment = a:c = 4:3

The profit ratio of R and S = (Z×4):{(Z+6750)×3}= 4Z: (3Z+20250)

Total profit of R and S = 9090

R's profit share = (9090-90)/2 = 4500

S's profit share = 4500+90 = 4590

Now, 4Z: (3Z+20250) = 4500:4590

Z= 18750

S's investment = 18750+6750= 25500

For Shop

P's investment = 4x

Q's investment = 5x

S's investment = 4x+3000

R's investment = 18750+4500 = 23250

The profit ratio of P:Q:R:S =

(4x×20):[5x(y+6)+(3/4×5x)×(20-y-6)]:[(23250×(20-y)]:[(4x+3000)×(20-y-6)]

P:S= (4x×20):[(4x+3000)×(20-y-6)]=16b:c^3= 80:27

Y= (29x+42000)/(4x+3000)

R:S = [(23250×(20-y)]:[(4x+3000)×(20-y-6)] = (6b+1):$2c^2$= 31:18

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Y = (14x-57000)/(x-2625)

(29x+42000)/(4x+3000) = (14x-57000)/(x-2625)

X= 6000

Y = 8

The profit share of P:Q:R:S = 160:185:93:54

The total profit earned = 1500×(160+185+93+54)/(185-160)= 29520

The profit share of P = 29520×160/492 = 9600

The profit share of Q = 29520×185/492 = 11100

The profit share of R = 29520×93/492 = 5580

The profit share of S = 29520×54/492 = 3240

Answer: B

11100= (4M+5)% of 30000

M = 8

(16N+140) = 4×6000+3000-25500

N = 85

Y= 8

N-4Y-5M = 85-32-40 = 13

Hence, the answer is option B.

18. Questions

LCM of (b, c) = 15

Only possible value of b=5 & c=3

3a = (5+3)1.5

a= 4

For Cafe

R's investment = Z

S's investment = Z+6750

Time of investment = a:c = 4:3

The profit ratio of R and S = (Z×4):{(Z+6750)×3}= 4Z: (3Z+20250)

Total profit of R and S = 9090

R's profit share = (9090-90)/2 = 4500

S's profit share = 4500+90 = 4590

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Now, 4Z: (3Z+20250) = 4500:4590

Z= 18750

S's investment = 18750+6750= 25500

For Shop

P's investment = 4x

Q's investment = 5x

S's investment = 4x+3000

R's investment = 18750+4500 = 23250

The profit ratio of P:Q:R:S =

(4x×20):[5x(y+6)+(3/4×5x)×(20-y-6)]:[(23250×(20-y)]:[(4x+3000)×(20-y-6)]

P:S= (4x×20):[(4x+3000)×(20-y-6)]=16b:c^3= 80:27

Y= (29x+42000)/(4x+3000)

R:S = [(23250×(20-y)]:[(4x+3000)×(20-y-6)] = (6b+1):$2c^2$= 31:18

Y = (14x-57000)/(x-2625)

(29x+42000)/(4x+3000) = (14x-57000)/(x-2625)

X= 6000

Y = 8

The profit share of P:Q:R:S = 160:185:93:54

The total profit earned = 1500×(160+185+93+54)/(185-160)= 29520

The profit share of P = 29520×160/492 = 9600

The profit share of Q = 29520×185/492 = 11100

The profit share of R = 29520×93/492 = 5580

The profit share of S = 29520×54/492 = 3240

Answer: C

The profit share of cafe's partnership – the profit share of shop's partnership

=4590-3240 = 1350

S's initial investment in café = 25500

The required % = 1350/25500×100 = 5.29%

Hence, the answer is option C.

19. Questions

LCM of (b, c) = 15

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Only possible value of b=5 & c=3

3a = (5+3)1.5

a= 4

For Cafe

R's investment = Z

S's investment = Z+6750

Time of investment = a:c = 4:3

The profit ratio of R and S = (Z×4):{(Z+6750)×3}= 4Z: (3Z+20250)

Total profit of R and S = 9090

R's profit share = (9090-90)/2 = 4500

S's profit share = 4500+90 = 4590

Now, 4Z: (3Z+20250) = 4500:4590

Z= 18750

S's investment = 18750+6750= 25500

For Shop

P's investment = 4x

Q's investment = 5x

S's investment = 4x+3000

R's investment = 18750+4500 = 23250

The profit ratio of P:Q:R:S =

(4x×20):[5x(y+6)+(3/4×5x)×(20-y-6)]:[(23250×(20-y)]:[(4x+3000)×(20-y-6)]

P:S= (4x×20):[(4x+3000)×(20-y-6)]=16b:c^3= 80:27

Y= (29x+42000)/(4x+3000)

R:S = [(23250×(20-y)]:[(4x+3000)×(20-y-6)] = (6b+1):$2c^2$= 31:18

Y = (14x-57000)/(x-2625)

(29x+42000)/(4x+3000) = (14x-57000)/(x-2625)

X= 6000

Y = 8

The profit share of P:Q:R:S = 160:185:93:54

The total profit earned = 1500×(160+185+93+54)/(185-160)= 29520

The profit share of P = 29520×160/492 = 9600

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The profit share of Q = 29520×185/492 = 11100

The profit share of R = 29520×93/492 = 5580

The profit share of S = 29520×54/492 = 3240

Answer: A

For Quantity I :

The required difference = (11100+4500) – (9600) = 6000

Quantity I = 6000

For Quantity II :

The required average = (4×6000) - 18750 = 5250

Quantity II = 5250

Quantity I > Quantity II

Hence, the answer is option A.

20. Questions

LCM of (b, c) = 15

Only possible value of b=5 & c=3

3a = (5+3)1.5

a= 4

For Cafe

R's investment = Z

S's investment = Z+6750

Time of investment = a:c = 4:3

The profit ratio of R and S = (Z×4):{(Z+6750)×3}= 4Z: (3Z+20250)

Total profit of R and S = 9090

R's profit share = (9090-90)/2 = 4500

S's profit share = 4500+90 = 4590

Now, 4Z: (3Z+20250) = 4500:4590

Z= 18750

S's investment = 18750+6750= 25500

For Shop

P's investment = 4x

Q's investment = 5x

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S's investment = 4x+3000

R's investment = 18750+4500 = 23250

The profit ratio of P:Q:R:S =

(4x×20):[5x(y+6)+(3/4×5x)×(20-y-6)]:[(23250×(20-y)]:[(4x+3000)×(20-y-6)]

P:S= (4x×20):[(4x+3000)×(20-y-6)]=16b:c^3= 80:27

Y= (29x+42000)/(4x+3000)

R:S = [(23250×(20-y)]:[(4x+3000)×(20-y-6)] = (6b+1):$2c^2$= 31:18

Y = (14x-57000)/(x-2625)

(29x+42000)/(4x+3000) = (14x-57000)/(x-2625)

X= 6000

Y = 8

The profit share of P:Q:R:S = 160:185:93:54

The total profit earned = 1500×(160+185+93+54)/(185-160)= 29520

The profit share of P = 29520×160/492 = 9600

The profit share of Q = 29520×185/492 = 11100

The profit share of R = 29520×93/492 = 5580

The profit share of S = 29520×54/492 = 3240

Answer: E

I). 6000/8 = 750

(True)

II). Initial investment of Q = 5×6000= 30000

Initial investment of s in cafe = 25500

30000>25500 (True)

III). R's profit in shop = 5580

6000-420 = 5580

5580 = 5580

(True)

All I, II & III is True

Hence, the answer is option E.

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1. Questions

There are two series given below. Series I has one wrong term and series II starts with the wrong term of series I. Both series follow the same pattern.

Series I: 10, 18, 45, 70, 413, 534

Series II: ____, __[A]__, ____, ____, ____, __[B]__

Find the value of [{√(B-1)}+A].

a. 45

b. 37

c. 29

d. 18

e. None of these

2. Questions

There are three series given below. Series I, II and III has one missing term as P, Q and R respectively. Each series follows different pattern.

Series I: 20, 42, 72, [P], 156, 210

Series II: 11, [Q], 39, 74, 123, 186

Series III: 32, 26, 38, 14, 18, [R]

Find the correct relation between P, Q and R.

a. P > Q > R

b. P < Q < R

c. P > Q < R

d. P < Q > R

e. P = Q = R

3. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below the total number of questions attempted (correct + wrong) by a student in four different subjects A, B, C and D. It also shows the percentage of the number of correct questions attempted out of the total number of questions attempted and the total number of marks obtained by the same student in each subject.

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Subject	Total number of questions attempted (correct + wrong)	Percentage of the number of correct questions attempted	Total number of marks obtained
A	20P	85%	(16P+0.75)
B	72	15Q%	(10Q-0.50)
C	20R	20R%	60
D	45S	70%	56.25

Note:

The values of each variable is positive integer.

Each subject consists of 100 questions.

In subject B, the number of correct questions attempted is (7Q+1) more than that of wrong questions attempted.

In subject C, the number of wrong questions attempted is 48 less than that of correct questions attempted.

A student will be awarded 1 mark for each correct attempt and 0.25 marks will be deducted for each wrong attempt.

No deduction for number of skipped questions.

Total number of marks obtained = (number of marks scored from correct attempts) – (number of marks deducted for wrong attempts)

Total number of questions = number of questions attempted + number of questions skipped

Total number of attempted questions = number of correct attempts + number of wrong attempts

In subject D, there are two sections theory and practical. If the ratio of the number of correct questions attempted in theory section to that of wrong questions attempted in the practical section is 25:1. And, the number of correct questions attempted in the practical section is 48% less than that of wrong questions attempted in the theory section, then find the number of wrong questions attempted in practical section.

a. 2

b. 5

c. 10

d. 12

e. None of these

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4. Questions

In subject A, if 2 marks is awarded for each correct attempt and 0.50 mark is deducted for each wrong attempt, then find the percentage change in the total number of marks obtained as compared to the original number of total marks obtained by the student in subject A.

a. 100% decreased

b. 100% increased

c. 50% increased

d. 50% decreased

e. None of these

5. Questions

If there is another subject E, in which a student has obtained the same number of total marks as in subject D and the total number of questions attempted by him in subject E is 75, then find the difference between the number of correct and wrong attempts made by the student in subject E.

a. 40

b. 35

c. 20

d. Cannot be determined

e. None of these

6. Questions

If there is another student, whose number of correct attempts in subject B is 4 less than that of the present student and his number of wrong attempts is 2 more than that of the present student in the same subject, then find the total number of marks obtained by another student in subject B.

a. 32

b. 45

c. 26

d. 50

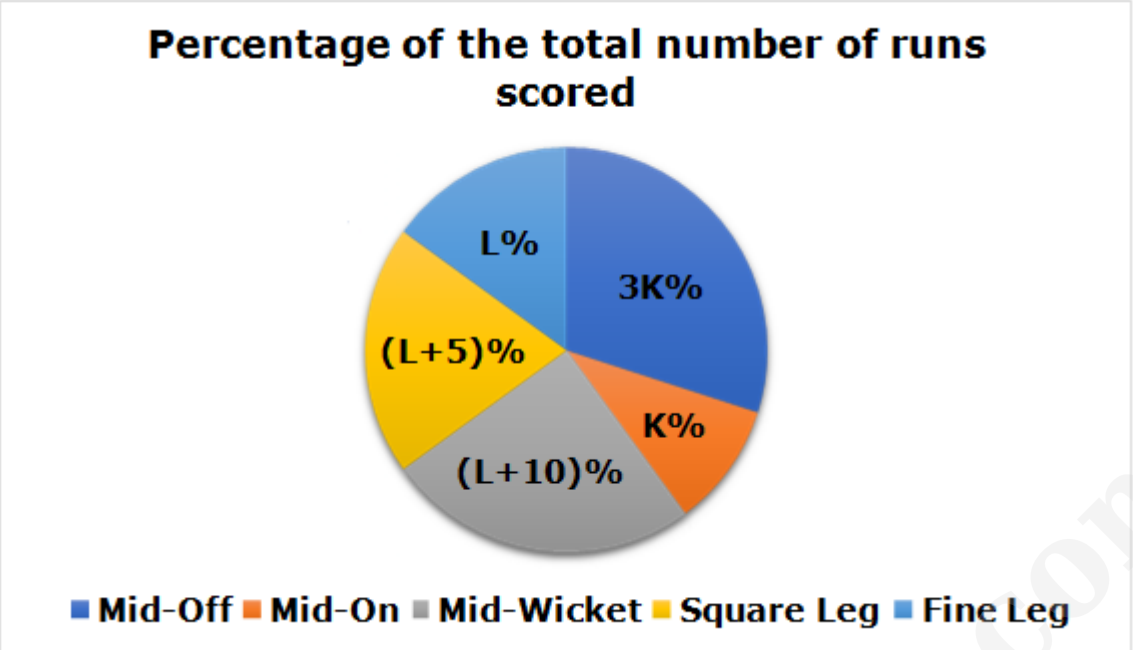
e. None of these

7. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of runs scored by a player among 11 players of a particular team in five different fielding positions i.e., mid-off, mid-on, mid-wicket, square-leg and fine leg during a cricket match.

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Note:

The ratio of the number of runs scored by the player in mid-off to that of square-leg is 3:2.

The number of runs scored in mid-wicket is (K+L+10).

A team consists a total of 11 players.

Strike rate = [{(number of runs scored)/(number of balls faced)}×100]

The number of runs scored can scored as singles (1 run), doubles (2 runs), triple (3 runs), fours (4 runs) and sixes (6 runs).

The number of runs scored in boundaries = number of fours scored + number of sixes scored

The dot ball faced by a player means zero (0) number of runs scored on that particular ball by that player.

Total number of runs scored by a team of 11 players = number of runs scored by that particular player + number of runs scored by the remaining 10 players

If the total number of scored by a player is X% of the total number of runs scored by all the 11 players together (including that player also). Then find the value of 2X, if the average number of runs scored by the remaining 10 players is 26.

a. 35

b. 70

c. 21

d. 42

e. None of these

8. Questions

If the total number of balls faced by the player is B and his overall strike rate is (B+130), then find

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the difference between the total number of runs scored and the overall strike rate of the same player. (B should be positive integer)

a. 80

b. 100

c. 60

d. Cannot be determined

e. None of these

9. Questions

Out of the total number of runs scored by the player in mid-off, the number of singles, doubles and triples scored are 13, 1 and 1 respectively. If the total number of boundaries (fours + sixes) scored is 5, then find the total number of runs scored in mid-off through sixes only.

a. 12

b. 18

c. 16

d. 14

e. None of these

10. Questions

The player has a strike rate of 210 in fine leg and the number of dot balls faced by the same player in the fine leg position is 3. Find his strike rate in the fine leg position, if the number of dot balls faced by him is excluded.

a. 320

b. 280

c. 240

d. 300

e. None of these

11. Questions

The ratio of the total number of balls faced to that of the overall strike rate of another player of the same team is 5:16 and the total number of runs scored by that player is 42.85% less than that of the present player, then find the total number of balls faced by another player.

a. 50

b. 60

c. 40

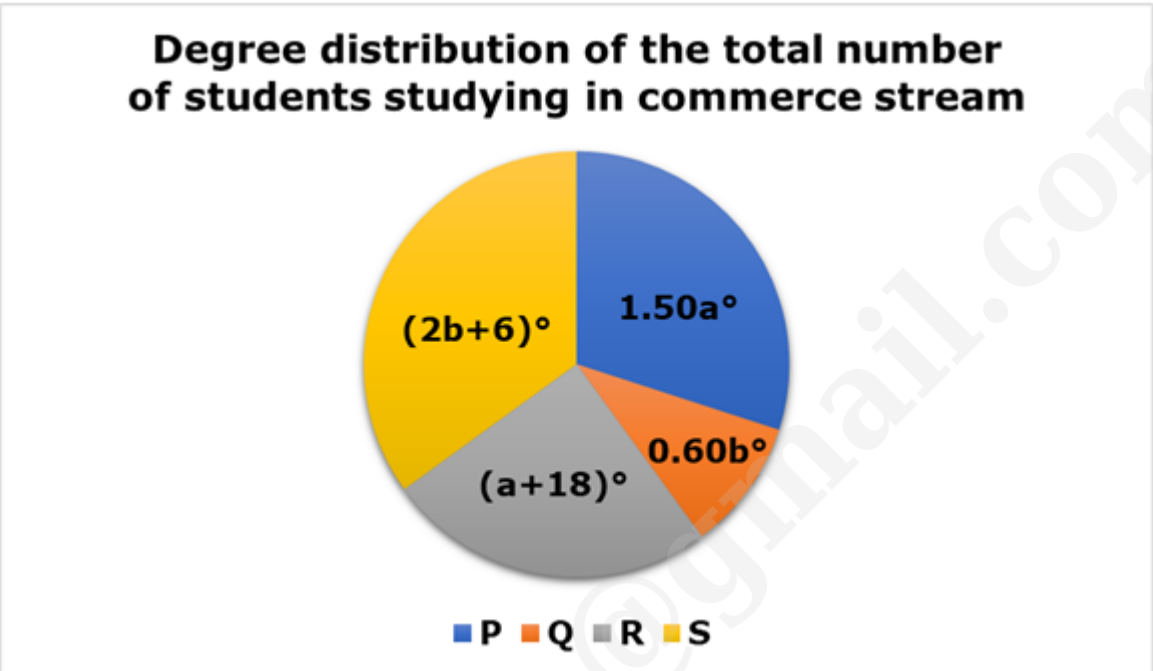
d. 30

e. None of these

12. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of students studying in commerce stream in four different colleges P, Q, R and S.



Note:

The difference between the values of a and b is the positive root of the equation $P^2 - 3P - 108 = 0$. *(a > b)*

The average number of students studying in commerce stream in college P and college Q is 40.

Total number of students studying = number of students studying science + number of students studying commerce + number of students studying on arts

Total number of students studying = number of boys studying + number of girls studying

In college P, the total number of students studying in commerce stream is X% of that of arts stream and the number of students studying in science stream is (X-48), which is 62 more than that of arts stream, then find the total number of students studying in all three streams together in college P. (X is a positive integer)

a. 184

b. 158

c. 202

d. Cannot be determined

e. None of these

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13. Questions

In college S in commerce stream, the number of boys studying is 8Y, which is Z more than that of girls studying in the same stream of same college. If the number of girls studying in commerce stream in college S is 3Z, then find the value of (Y:Z).

a. 2:1

b. 1:2

c. 3:2

d. 2:3

e. None of these

14. Questions

The total number of students studying in all the three stream (science + commerce + arts) together in college Q is 81 and the number of students studying science stream is 1 more than five times the difference between the number of students studying in commerce and arts stream, then find the number of students studying in science stream in college Q. (number of commerce students studying is more than that of arts students studying in college Q)

a. 42

b. 37

c. 23

d. 51

e. None of these

15. Questions

If there is another college T, where the number of students studying in commerce stream is K, which is 1/6th of the total number of students studying in all the five colleges together (including college T), then find the number of students studying in commerce stream in college T.

a. 40

b. 33

c. 27

d. 15

e. None of these

16. Questions

Following question have two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give an answer accordingly.

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A and B together can complete a piece of work in (40/7) days, whereas C and D together can complete the same piece of work in (60/11) days. Efficiency of A is 40% less than that of D and the ratio of the efficiency of B to that of C is 5:4.

Quantity I: The time taken by A and C to complete the work together.

Quantity II: The time taken by B and D to complete the work together.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II

17. Questions

The downstream and upstream speed of the boat are (7X+2) km/h and (X^2+2) km/h respectively. The ratio of the time taken by the boat to cover 240 km in downstream to that of 180 km in upstream is 4:5. ***(X is a positive integer)***

Quantity I: The speed of the boat in still water is what percent of that of the speed of the river?

Quantity II: The downstream distance covered by the boat in 7 hours.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II

18. Questions

The cost price of the shirt is Rs. 500, which is sold at a profit of 20%. The marked price of the jeans is Rs. 370 more than that of shirt and jeans is marked-up 40% above its cost price. Selling price of the shirt is 3/4th of the cost price of the jeans and jeans is sold at a discount of 10%.

Quantity I: Marked price of shirt.

Quantity II: Selling price of jeans.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II

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19. Questions

A person invested Rs. 40000 at the rate of 5R% per annum simple interest for R years and received Rs. 18000 as interest.

Quantity I: Find the compound interest received after 2 years, if he had invested Rs. 30000 at the (R+7)% per annum compounded annually.

Quantity II: If Rs. 8000R is invested at the rate of 18% per annum simple interest, then interest received after 4 years.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II

20. Questions

The ratio of the income of A to that of B is 7:6 and the expenditure of B is 10% less than that of A.

The savings of A is 1/3rd more than that of B. The income of A is Rs. 10000 more than the expenditure of B and the income of B is Rs. 4000 more than the expenditure of A.

Quantity I: The difference between the expenditure and savings of B.

Quantity II: The total income (expenditure + savings) of A.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II

Explanations:

1. Questions

Answer: B

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Series I: 10, 18, 45, 70, 413, 534

6th term = 534

5th term = $(534-11^2) = (534-121) = 413$

4th term = $(413-7^3) = (413-343) = 70$

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3rd term = $(70-5^2)$ = (70-25) = 45

2nd term = $(45-3^3)$ = (45-27) = 18

1st term = $(18-2^2)$ = (18-4) = 14 (10 is wrong)

So, the wrong term = 10

And, the correct term = 14

Series II:

1st term = 10

2nd term = $(10+2^2)$ = (10+4) = 14

3rd term = $(14+3^3)$ = (14+27) = 41

4th term = $(41+5^2)$ = (41+25) = 66

5th term = $(66+7^3)$ = (66+343) = 409

6th term = $(409+11^2)$ = (409+121) = 530

So, A = 14

And, B = 530

Hence, the value of,

[{√(B-1)}+A] = [{√(530-1)}+14] = [{√(529)}+14] = (23+14) = 37.

2. Questions

Answer: A

Series I: 20, 42, 72, [P], 156, 210

1st term = (4×5) = 20

2nd term = (6×7) = 42

3rd term = (8×9) = 72

4th term = (10×11) = 110

5th term = (12×13) = 156

6th term = (14×15) = 210

So, P = 110

Series II: 11, [Q], 39, 74, 123, 186

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6th term = 186

5th term = {186-(7×9)} = (186-63) = 123

4th term = {123-(7×7)} = (123-49) = 74

3rd term = {74-(7×5)} = (74-35) = 39

2nd term = {39-(7×3)} = (39-21) = 18

1st term = {18-(7×1)} = (18-7) = 11

So, Q = 18

Series III: 32, 26, 38, 14, 18, [R]

1st term = 32

2nd term = {32-(3×2)} = (32-6) = 26

3rd term = {26+(2×6)} = (26+12) = 38

4th term = {38-(3×8)} = (38-24) = 14

5th term = {14+(1×4)} = (14+4) = 18

6th term = {18-(1×8)} = (18-8) = 10

So, R = 10

Now,

P = 110

Q = 18

R = 10

Hence, P > Q > R.

3. Questions

Subject A

The total number of questions attempted = 20P

The number of correct questions attempted = 85% of 20P = 17P

The number of wrong questions attempted = 15% of 20P = 3P

So, the total number of marks scored from correct attempts = (17P×1) = 17P

And, the total number of marks deducted from wrong attempts = (3P×0.25) = 0.75P

Therefore, the total number of marks obtained = (17P-0.75P) = 16.25P

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Now,

16.25P = (16P+0.75)

0.25P = 0.75

P = 3

So, the total number of questions attempted = 20P = 60

The number of correct questions attempted = 17P = 51

The number of wrong questions attempted = 3P = 9

Therefore, the total number of marks scored from correct attempts = 17P = 51

And, the total number of marks deducted from wrong attempts = 0.75P = 2.25

Therefore, the total number of marks obtained = 16.25P = 48.75

Subject B

The total number of questions attempted = 72

The number of correct attempts = 15Q% of 72 = 10.80Q

So, the number of wrong attempts = (72-10.80Q)

Now,

10.80Q – (72-10.80Q) = (7Q+1)

3.80Q – 1 = 72 – 10.80Q

14.60Q = 73

Q = 5

Therefore, the total number of questions attempted = 72

The number of correct attempts = 10.80Q = 54

And, the number of wrong attempts = (72-10.80Q) = (72-54) = 18

The total number of marks scored from correct attempts = (54×1) = 54

And, the total number of marks deducted for wrong attempts = (18×0.25) = 4.50

So, the total marks obtained = (54-4.50) = 49.50

Subject C

The total number of questions attempted = 20R

The number of correct attempts = 20R% of 20R = $4R^2$

So, the number of wrong attempts = ($20R-4R^2$)

Now,

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$4R^2 – (20R-4R^2) = 48$

$8R^2 – 20R – 48 = 0$

$2R^2 – 5R – 12 = 0$

$2R^2 – 8R + 3R – 12 = 0$

$2R(R-4) + 3(R-4) = 0$

$(2R+3) (R-4) = 0$

$R = -1.50$ or 4

So, R = 4 (positive integer)

Therefore, the total number of questions attempted = 20R = 80

The number of correct attempts = 20R% of 20R = $4R^2 = 64$

And, the number of wrong attempts = $(20R-4R^2) = (80-64) = 16$

Now,

The total number of marks scored from correct attempts = (64×1) = 64

And, the total number of marks deducted for wrong attempts = (16×0.25) = 4

So, the total marks obtained = (64-4) = 60

Subject D

The total number of questions attempted = 45S

The number of correct attempts = 70% of 45S = 31.50S

So, the number of wrong attempts = (45S-31.50S) = 13.50S

Now,

The total number of marks scored from correct attempts = (31.50S×1) = 31.50S

And, the total number of marks deducted for wrong attempts = (13.50S×0.25) = 3.375S

So, the total marks obtained = (31.50S-3.375S) = 28.125S

28.125S = 56.25

S = 2

Therefore, the total number of questions attempted = 45S = 90

The number of correct attempts = 31.50S = 63

So, the number of wrong attempts = 13.50S = 27

And, the total number of marks scored from correct attempts = 31.50S = 63

And, the total number of marks deducted for wrong attempts = 3.375S = 6.75

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So, the total marks obtained = (63-6.75) = 56.25

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: A

According to the questions,

Subject D

Let, the total number of correct questions attempted in theory section and wrong questions attempted in practical section be 25m and m respectively.

And, the ratio of the number of correct questions attempted in practical section to that of wrong questions attempted in theory section = 52:100 = 13:25. Let, it be 13n and 25n respectively.

The total number of correct questions attempted = 63

The total number of wrong questions attempted = 27

Now,

25m + 13n = 63……..(i)

m + 25n = 27……(ii)

Solving both the equations, we get,

25m + 13n = 63

25m + 625n = 675

612n = 612

n = 1

Now, put the value of n in equation (i), we get,

25m + 13n = 63

25m + 13 = 63

25m = 50

m = 2

Hence, the number of wrong questions attempted in practical section = m = 2.

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4. Questions

Subject A

The total number of questions attempted = 20P

The number of correct questions attempted = 85% of 20P = 17P

The number of wrong questions attempted = 15% of 20P = 3P

So, the total number of marks scored from correct attempts = (17P×1) = 17P

And, the total number of marks deducted from wrong attempts = (3P×0.25) = 0.75P

Therefore, the total number of marks obtained = (17P-0.75P) = 16.25P

Now,

16.25P = (16P+0.75)

0.25P = 0.75

P = 3

So, the total number of questions attempted = 20P = 60

The number of correct questions attempted = 17P = 51

The number of wrong questions attempted = 3P = 9

Therefore, the total number of marks scored from correct attempts = 17P = 51

And, the total number of marks deducted from wrong attempts = 0.75P = 2.25

Therefore, the total number of marks obtained = 16.25P = 48.75

Subject B

The total number of questions attempted = 72

The number of correct attempts = 15Q% of 72 = 10.80Q

So, the number of wrong attempts = (72-10.80Q)

Now,

10.80Q – (72-10.80Q) = (7Q+1)

3.80Q – 1 = 72 – 10.80Q

14.60Q = 73

Q = 5

Therefore, the total number of questions attempted = 72

The number of correct attempts = 10.80Q = 54

And, the number of wrong attempts = (72-10.80Q) = (72-54) = 18

The total number of marks scored from correct attempts = (54×1) = 54

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And, the total number of marks deducted for wrong attempts = (18×0.25) = 4.50

So, the total marks obtained = (54-4.50) = 49.50

Subject C

The total number of questions attempted = 20R

The number of correct attempts = 20R% of 20R = $4R^2$

So, the number of wrong attempts = ($20R-4R^2$)

Now,

$4R^2 – (20R-4R^2) = 48$

$8R^2 – 20R – 48 = 0$

$2R^2 – 5R – 12 = 0$

$2R^2 – 8R + 3R – 12 = 0$

$2R(R-4) + 3(R-4) = 0$

$(2R+3) (R-4) = 0$

R = -1.50 or 4

So, R = 4 (positive integer)

Therefore, the total number of questions attempted = 20R = 80

The number of correct attempts = 20R% of 20R = $4R^2$ = 64

And, the number of wrong attempts = ($20R-4R^2$) = (80-64) = 16

Now,

The total number of marks scored from correct attempts = (64×1) = 64

And, the total number of marks deducted for wrong attempts = (16×0.25) = 4

So, the total marks obtained = (64-4) = 60

Subject D

The total number of questions attempted = 45S

The number of correct attempts = 70% of 45S = 31.50S

So, the number of wrong attempts = (45S-31.50S) = 13.50S

Now,

The total number of marks scored from correct attempts = (31.50S×1) = 31.50S

And, the total number of marks deducted for wrong attempts = (13.50S×0.25) = 3.375S

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So, the total marks obtained = (31.50S-3.375S) = 28.125S

28.125S = 56.25

S = 2

Therefore, the total number of questions attempted = 45S = 90

The number of correct attempts = 31.50S = 63

So, the number of wrong attempts = 13.50S = 27

And, the total number of marks scored from correct attempts = 31.50S = 63

And, the total number of marks deducted for wrong attempts = 3.375S = 6.75

So, the total marks obtained = (63-6.75) = 56.25

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: B

Subject A

The total number of correct attempts = 51

The total number of wrong attempts = 9

So, the number of marks obtained from correct attempts = (51×2) = 102

And, the number of marks deducted from wrong attempts = (9×0.50) = 4.50

Therefore, the new number of total marks obtained = (102-4.50) = 97.50

And, the original number of total marks obtained = 48.75

Hence, the required % = [{(97.50-48.75)/48.75}×100]% = {(48.75/48.75)×100}% = 100% increased.

5. Questions

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Subject A

The total number of questions attempted = 20P

The number of correct questions attempted = 85% of 20P = 17P

The number of wrong questions attempted = 15% of 20P = 3P

So, the total number of marks scored from correct attempts = (17P×1) = 17P

And, the total number of marks deducted from wrong attempts = (3P×0.25) = 0.75P

Therefore, the total number of marks obtained = (17P-0.75P) = 16.25P

Now,

16.25P = (16P+0.75)

0.25P = 0.75

P = 3

So, the total number of questions attempted = 20P = 60

The number of correct questions attempted = 17P = 51

The number of wrong questions attempted = 3P = 9

Therefore, the total number of marks scored from correct attempts = 17P = 51

And, the total number of marks deducted from wrong attempts = 0.75P = 2.25

Therefore, the total number of marks obtained = 16.25P = 48.75

Subject B

The total number of questions attempted = 72

The number of correct attempts = 15Q% of 72 = 10.80Q

So, the number of wrong attempts = (72-10.80Q)

Now,

10.80Q – (72-10.80Q) = (7Q+1)

3.80Q – 1 = 72 – 10.80Q

14.60Q = 73

Q = 5

Therefore, the total number of questions attempted = 72

The number of correct attempts = 10.80Q = 54

And, the number of wrong attempts = (72-10.80Q) = (72-54) = 18

The total number of marks scored from correct attempts = (54×1) = 54

And, the total number of marks deducted for wrong attempts = (18×0.25) = 4.50

So, the total marks obtained = (54-4.50) = 49.50

Subject C

The total number of questions attempted = 20R

The number of correct attempts = 20R% of 20R = $4R^2$

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So, the number of wrong attempts = $(20R-4R^2)$

Now,

$4R^2 – (20R-4R^2) = 48$

$8R^2 – 20R – 48 = 0$

$2R^2 – 5R – 12 = 0$

$2R^2 – 8R + 3R – 12 = 0$

$2R(R-4) + 3(R-4) = 0$

$(2R+3) (R-4) = 0$

$R = -1.50$ or 4

So, $R = 4$ (positive integer)

Therefore, the total number of questions attempted = $20R = 80$

The number of correct attempts = 20R% of $20R = 4R^2 = 64$

And, the number of wrong attempts = $(20R-4R^2) = (80-64) = 16$

Now,

The total number of marks scored from correct attempts = $(64\times1) = 64$

And, the total number of marks deducted for wrong attempts = $(16\times0.25) = 4$

So, the total marks obtained = $(64-4) = 60$

Subject D

The total number of questions attempted = $45S$

The number of correct attempts = 70% of $45S = 31.50S$

So, the number of wrong attempts = $(45S-31.50S) = 13.50S$

Now,

The total number of marks scored from correct attempts = $(31.50S\times1) = 31.50S$

And, the total number of marks deducted for wrong attempts = $(13.50S\times0.25) = 3.375S$

So, the total marks obtained = $(31.50S-3.375S) = 28.125S$

$28.125S = 56.25$

$S = 2$

Therefore, the total number of questions attempted = $45S = 90$

The number of correct attempts = $31.50S = 63$

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So, the number of wrong attempts = 13.50S = 27

And, the total number of marks scored from correct attempts = 31.50S = 63

And, the total number of marks deducted for wrong attempts = 3.375S = 6.75

So, the total marks obtained = (63-6.75) = 56.25

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: E

Subject E

The total number of questions attempted = 75

Let, the number of correct and wrong questions attempted be K and L respectively.

So, the number of marks obtained from correct attempts = (K×1) = K

And, the number of marks deducted for wrong attempts = (L×0.25) = 0.25L

Therefore, the total number of marks obtained = (K-0.25L)

Now,

K + L = 75……(i)

K – 0.25L = 56.25…….(ii)

Solving both the equations, we get,

1.25L = 18.75

L = 15

Now, put the value of L in equation (i), we get,

K + L = 75

K + 15 = 75

K = 60

Hence, the difference between the number of correct and wrong attempts made by the student in subject E = (K-L) = (60-15) = 45.

6. Questions

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Subject A

The total number of questions attempted = 20P

The number of correct questions attempted = 85% of 20P = 17P

The number of wrong questions attempted = 15% of 20P = 3P

So, the total number of marks scored from correct attempts = (17P×1) = 17P

And, the total number of marks deducted from wrong attempts = (3P×0.25) = 0.75P

Therefore, the total number of marks obtained = (17P-0.75P) = 16.25P

Now,

16.25P = (16P+0.75)

0.25P = 0.75

P = 3

So, the total number of questions attempted = 20P = 60

The number of correct questions attempted = 17P = 51

The number of wrong questions attempted = 3P = 9

Therefore, the total number of marks scored from correct attempts = 17P = 51

And, the total number of marks deducted from wrong attempts = 0.75P = 2.25

Therefore, the total number of marks obtained = 16.25P = 48.75

Subject B

The total number of questions attempted = 72

The number of correct attempts = 15Q% of 72 = 10.80Q

So, the number of wrong attempts = (72-10.80Q)

Now,

10.80Q – (72-10.80Q) = (7Q+1)

3.80Q – 1 = 72 – 10.80Q

14.60Q = 73

Q = 5

Therefore, the total number of questions attempted = 72

The number of correct attempts = 10.80Q = 54

And, the number of wrong attempts = (72-10.80Q) = (72-54) = 18

The total number of marks scored from correct attempts = (54×1) = 54

And, the total number of marks deducted for wrong attempts = (18×0.25) = 4.50

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So, the total marks obtained = (54-4.50) = 49.50

Subject C

The total number of questions attempted = 20R

The number of correct attempts = 20R% of 20R = $4R^2$

So, the number of wrong attempts = ($20R-4R^2$)

Now,

$4R^2 – (20R-4R^2) = 48$

$8R^2 – 20R – 48 = 0$

$2R^2 – 5R – 12 = 0$

$2R^2 – 8R + 3R – 12 = 0$

$2R(R-4) + 3(R-4) = 0$

$(2R+3) (R-4) = 0$

R = -1.50 or 4

So, R = 4 (positive integer)

Therefore, the total number of questions attempted = 20R = 80

The number of correct attempts = 20R% of 20R = $4R^2 = 64$

And, the number of wrong attempts = ($20R-4R^2$) = (80-64) = 16

Now,

The total number of marks scored from correct attempts = (64×1) = 64

And, the total number of marks deducted for wrong attempts = (16×0.25) = 4

So, the total marks obtained = (64-4) = 60

Subject D

The total number of questions attempted = 45S

The number of correct attempts = 70% of 45S = 31.50S

So, the number of wrong attempts = (45S-31.50S) = 13.50S

Now,

The total number of marks scored from correct attempts = (31.50S×1) = 31.50S

And, the total number of marks deducted for wrong attempts = (13.50S×0.25) = 3.375S

So, the total marks obtained = (31.50S-3.375S) = 28.125S

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28.125S = 56.25

S = 2

Therefore, the total number of questions attempted = 45S = 90

The number of correct attempts = 31.50S = 63

So, the number of wrong attempts = 13.50S = 27

And, the total number of marks scored from correct attempts = 31.50S = 63

And, the total number of marks deducted for wrong attempts = 3.375S = 6.75

So, the total marks obtained = (63-6.75) = 56.25

Subject	Total number of questions attempted	Number of correct questions attempted	Number of wrong questions attempted	Total number of marks obtained
A	60	51	9	48.75
B	72	54	18	49.50
C	80	64	16	60
D	90	63	27	56.25
Total	302	232	70	214.50

Answer: B

Subject B (another student)

The number of correct attempts = (54-4) = 50

The number of wrong attempts = (18+2) = 20

So, the number of marks scored from correct attempts = (50×1) = 50

And, the number of marks deducted from wrong attempts = (20×0.25) = 5

Hence, the total number of marks obtained by another student in subject B = (50-5) = 45.

7. Questions

From the pie chart given above,

The total value of the pie chart = 100%

3K + K + (L+10) + (L+5) + L = 100

4K + 3L = 85……..(i)

Also,

3K/(L+5) = 3/2

6K = 3L + 15

6K – 3L = 15……(ii)

Solving, both the equations, we get,

12K + 9L = 255

12K – 6L = 30

15L = 225

L = 15

Now, put the value of K in equation (i), we get,

4K + 3L = 85

4K + 45 = 85

4K = 40

K = 10

Now, the percentage distribution in the pie chart,

Mid-Off = 3K% = 30%

Mid-On = K% = 10%

Mid-Wicket = (L+10)% = (15+10)% = 25%

Square Leg = (L+5)% = (15+5)% = 20%

Fine Leg = L% = 15%

The number of runs scored in mid-wicket = (K+L+10) = (10+15+10) = 35

So,

25% = 35

1% = (35/25) = 1.40

100% = (1.40×100) = 140

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: B

According to the question,

The total number of runs scored by the player = 140

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The total number of runs scored by the remaining 10 players of the team = (26×10) = 260

So, the total number of runs scored by the whole team of 11 players = (140+26) = 400

Therefore, the percentage of the number of runs scored by the player out of the total number of runs scored by the whole team of players = {(140/400)×100}% = 35%

So, X = 35

Hence, 2X = 70.

8. Questions

From the pie chart given above,

The total value of the pie chart = 100%

3K + K + (L+10) + (L+5) + L = 100

4K + 3L = 85……..(i)

Also,

3K/(L+5) = 3/2

6K = 3L + 15

6K – 3L = 15…….(ii)

Solving, both the equations, we get,

12K + 9L = 255

12K – 6L = 30

15L = 225

L = 15

Now, put the value of K in equation (i), we get,

4K + 3L = 85

4K + 45 = 85

4K = 40

K = 10

Now, the percentage distribution in the pie chart,

Mid-Off = 3K% = 30%

Mid-On = K% = 10%

Mid-Wicket = (L+10)% = (15+10)% = 25%

Square Leg = (L+5)% = (15+5)% = 20%

Fine Leg = L% = 15%

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The number of runs scored in mid-wicket = (K+L+10) = (10+15+10) = 35

So,

25% = 35

1% = (35/25) = 1.40

100% = (1.40×100) = 140

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: C

The total number of runs scored = 140

The total number of balls faced = B

The overall strike rate = (B+130)

So,

(140/B)×100 = (B+130)

140/B = (B+130)/100

$14000 = B^2 + 130B$

$B^2 + 130B - 14000 = 0$

$B^2 + 200B - 70B - 14000 = 0$

B(B+200) – 70(B+200) = 0

(B-70) (B+200) = 0

B = 70 or -200

So, B = 70 (positive integer)

Therefore, the overall strike rate = (B+130) = (70+130) = 200

Hence, the required difference = (200-140) = 60.

9. Questions

From the pie chart given above,

The total value of the pie chart = 100%

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3K + K + (L+10) + (L+5) + L = 100

4K + 3L = 85……..(i)

Also,

3K/(L+5) = 3/2

6K = 3L + 15

6K – 3L = 15…….(ii)

Solving, both the equations, we get,

12K + 9L = 255

12K – 6L = 30

15L = 225

L = 15

Now, put the value of K in equation (i), we get,

4K + 3L = 85

4K + 45 = 85

4K = 40

K = 10

Now, the percentage distribution in the pie chart,

Mid-Off = 3K% = 30%

Mid-On = K% = 10%

Mid-Wicket = (L+10)% = (15+10)% = 25%

Square Leg = (L+5)% = (15+5)% = 20%

Fine Leg = L% = 15%

The number of runs scored in mid-wicket = (K+L+10) = (10+15+10) = 35

So,

25% = 35

1% = (35/25) = 1.40

100% = (1.40×100) = 140

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

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The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: A

Mid-Off

The total number of runs scored = 42

The number of runs scored through singles = (1×13) = 13

The number of runs scored through doubles = (2×1) = 2

The number of runs scored through triples = (3×1) = 3

So, the total number of runs scored through singles, doubles and triples (excluding boundaries) = (13+2+3) = 18

Therefore, the total number of runs scored through boundaries (fours + sixes) = (42-18) = 24

Now, let the number of fours and sixes scored be F and S respectively,

So, the number of runs scored through fours = (F×4) = 4F

And, the number of runs scored through sixes = (S×6) = 6S

Now,

F + S = 5…….(i)

4F + 6S = 24……..(ii)

Solving both the equations, we get,

4F + 4S = 20

4F + 6S = 24

2S = 4

S = 2

Now, put the value of S in equation (i), we get,

F + S = 5

F + 2 = 5

F = 3

Hence, the total number of runs scored in mid-off through sixes only = 6S = 12.

10. Questions

From the pie chart given above,

The total value of the pie chart = 100%

3K + K + (L+10) + (L+5) + L = 100

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4K + 3L = 85……..(i)

Also,

3K/(L+5) = 3/2

6K = 3L + 15

6K – 3L = 15…….(ii)

Solving, both the equations, we get,

12K + 9L = 255

12K – 6L = 30

15L = 225

L = 15

Now, put the value of K in equation (i), we get,

4K + 3L = 85

4K + 45 = 85

4K = 40

K = 10

Now, the percentage distribution in the pie chart,

Mid-Off = 3K% = 30%

Mid-On = K% = 10%

Mid-Wicket = (L+10)% = (15+10)% = 25%

Square Leg = (L+5)% = (15+5)% = 20%

Fine Leg = L% = 15%

The number of runs scored in mid-wicket = (K+L+10) = (10+15+10) = 35

So,

25% = 35

1% = (35/25) = 1.40

100% = (1.40×100) = 140

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

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The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: D

Fine Leg

The number of runs scored = 21

Strike rate = 210

Let, the number of balls faced be n.

So,

(21/n)×100 = 210

21/n = 210/100

210n = 2100

n = 10

Therefore, the total number of balls faced = 10

The number of dot balls faced = 3

So, the number of balls faced (excluding dot balls) = (10-3) = 7

Hence, the strike rate (excluding dot balls faced) = {(21/7)×100} = 300.

11. Questions

From the pie chart given above,

The total value of the pie chart = 100%

3K + K + (L+10) + (L+5) + L = 100

4K + 3L = 85……..(i)

Also,

3K/(L+5) = 3/2

6K = 3L + 15

6K – 3L = 15…….(ii)

Solving, both the equations, we get,

12K + 9L = 255

12K – 6L = 30

15L = 225

L = 15

Now, put the value of K in equation (i), we get,

4K + 3L = 85

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4K + 45 = 85

4K = 40

K = 10

Now, the percentage distribution in the pie chart,

Mid-Off = 3K% = 30%

Mid-On = K% = 10%

Mid-Wicket = (L+10)% = (15+10)% = 25%

Square Leg = (L+5)% = (15+5)% = 20%

Fine Leg = L% = 15%

The number of runs scored in mid-wicket = (K+L+10) = (10+15+10) = 35

So,

25% = 35

1% = (35/25) = 1.40

100% = (1.40×100) = 140

Therefore, the total number of runs scored in all the fielding positions together = 140

So, the number of runs scored in Mid-Off = 30% of 140 = 42

The number of runs scored in Mid-On = 10% of 140 = 14

The number of runs scored in Mid-Wicket = 25% of 140 = 35

The number of runs scored in Square Leg = 20% of 140 = 28

The number of runs scored in Fine Leg = 15% of 140 = 21

Answer: A

Another Player

The total number of runs scored = 4/7 of 140 = 80 *(Fractional value of 42.85% = 3/7)*

Let, the total number of balls faced and the overall strike rate be 5m and 16m respectively.

So,

(80/5m)×100 = 16m

16/m = 4m/25

$4m^2 = 400$

$m^2 = 100$

m = 10

Hence, the total number of balls faced by another player = 5m = 50.

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12. Questions

From the pie chart given above, we have,

The total value of the pie chart = 360°

1.50a + 0.60b + (a+18) + (2b+6) = 360

2.50a + 2.60b = 336…….(i)

From the equation given above, we have,

$P^2 – 3P – 108 = 0$

$P^2 – 12P + 9P – 108 = 0$

P(P-12) + 9(P-12) = 0

(P+9) + (P-12) = 0

P = -9 or 12

So, the difference between the values of a and b = 12 (positive root)

Now,

a – b = 12…….(ii)

Solving both the equations, we get,

2.50a + 2.60b = 336 – (i)

a – b = 12 – (ii)

2.50a + 2.60b = 336

2.50a – 2.50b = 30

5.10b = 306

b = 60

Now, put the value of b in equation (i), we get,

2.50a + 2.60b = 336

2.50a + 156 = 336

2.50a = 180

a = 72

So, the degree values in the pie chart for each college,

P = 1.50a° = 108°

Q = 0.60b° = 36°

R = (a+18)° = (72+18)° = 90°

S = (2b+6)° = (120+6)° = 126°

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The sum of the total number of students studying in commerce stream in college P and college Q = (40×2) = 80

So,

(108+36)° = 80

144° = 80

360° = {(80/144)×360} = 200

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

P = 108° of 200 = {(108/360)×200} = 60

Q = 36° of 200 = {(36/360)×200} = 20

R = 90° of 200 = {(90/360)×200} = 50

S = 126° of 200 = {(126/360)×200} = 70

Answer: C

According to the question,

College P

The total number of students studying in science stream = (X-48)

So, the total number of students studying in arts stream = (X-48-62) = (X-110)

The number of students studying in commerce stream = X% of (X-110) = $\{(X^2-110X)/100\}$

Now,

$\{(X^2-110X)/100\} = 60$

$X^2 - 110X = 6000$

$X^2 - 110X - 6000 = 0$

$X^2 - 150X + 40X - 6000 = 0$

X(X-150) + 40(X-150) = 0

(X+40) + (X-150) = 0

X = -40 or 150

So, X = 150 (positive integer)

Hence, the total number of students studying in all the three streams together = {(X-48)+60+(X-110)} = {(150-48)+60+(150-110)} = (102+60+40) = 202.

13. Questions

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From the pie chart given above, we have,

The total value of the pie chart = 360°

1.50a + 0.60b + (a+18) + (2b+6) = 360

2.50a + 2.60b = 336…….(i)

From the equation given above, we have,

$P^2 – 3P – 108 = 0$

$P^2 – 12P + 9P – 108 = 0$

P(P-12) + 9(P-12) = 0

(P+9) + (P-12) = 0

P = -9 or 12

So, the difference between the values of a and b = 12 (positive root)

Now,

a – b = 12…….(ii)

Solving both the equations, we get,

2.50a + 2.60b = 336 – (i)

a – b = 12 – (ii)

2.50a + 2.60b = 336

2.50a – 2.50b = 30

5.10b = 306

b = 60

Now, put the value of b in equation (i), we get,

2.50a + 2.60b = 336

2.50a + 156 = 336

2.50a = 180

a = 72

So, the degree values in the pie chart for each college,

P = 1.50a° = 108°

Q = 0.60b° = 36°

R = (a+18)° = (72+18)° = 90°

S = (2b+6)° = (120+6)° = 126°

The sum of the total number of students studying in commerce stream in college P and college Q = (40×2)

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= 80

So,

(108+36)° = 80

144° = 80

360° = {(80/144)×360} = 200

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

P = 108° of 200 = {(108/360)×200} = 60

Q = 36° of 200 = {(36/360)×200} = 20

R = 90° of 200 = {(90/360)×200} = 50

S = 126° of 200 = {(126/360)×200} = 70

Answer: B

College S (Commerce Stream)

The total number of students studying = 70

The number of boys studying = 8Y

And, the number of girls studying = (8Y-Z)

Now,

(8Y-Z) = 3Z

8Y = 4Z

Y:Z= 1:2.

14. Questions

From the pie chart given above, we have,

The total value of the pie chart = 360°

1.50a + 0.60b + (a+18) + (2b+6) = 360

2.50a + 2.60b = 336…….(i)

From the equation given above, we have,

$P^2 – 3P – 108 = 0$

$P^2 – 12P + 9P – 108 = 0$

P(P-12) + 9(P-12) = 0

(P+9) + (P-12) = 0

P = -9 or 12

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So, the difference between the values of a and b = 12 (positive root)

Now,

a – b = 12…….(ii)

Solving both the equations, we get,

2.50a + 2.60b = 336 – (i)

a – b = 12 – (ii)

2.50a + 2.60b = 336

2.50a – 2.50b = 30

5.10b = 306

b = 60

Now, put the value of b in equation (i), we get,

2.50a + 2.60b = 336

2.50a + 156 = 336

2.50a = 180

a = 72

So, the degree values in the pie chart for each college,

P = 1.50a° = 108°

Q = 0.60b° = 36°

R = (a+18)° = (72+18)° = 90°

S = (2b+6)° = (120+6)° = 126°

The sum of the total number of students studying in commerce stream in college P and college Q = (40×2) = 80

So,

(108+36)° = 80

144° = 80

360° = {(80/144)×360} = 200

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

P = 108° of 200 = {(108/360)×200} = 60

Q = 36° of 200 = {(36/360)×200} = 20

R = 90° of 200 = {(90/360)×200} = 50

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S = 126° of 200 = {(126/360)×200} = 70

Answer: D

College Q

The total number of students studying all the three stream together = 81

The number of commerce students studying = 20

Let, the difference between the number of commerce and arts students studying be v. (commerce-arts)

So, the number of arts students studying = (20-v)

Therefore, the number of science students studying = (5v+1)

Now,

20 + (20-v) + (5v+1) = 81

4v = 40

v = 10

Hence, the number of science students studying = (5v+1) = (50+1) = 51.

15. Questions

From the pie chart given above, we have,

The total value of the pie chart = 360°

1.50a + 0.60b + (a+18) + (2b+6) = 360

2.50a + 2.60b = 336……(i)

From the equation given above, we have,

$P^2 - 3P - 108 = 0$

$P^2 - 12P + 9P - 108 = 0$

P(P-12) + 9(P-12) = 0

(P+9) + (P-12) = 0

P = -9 or 12

So, the difference between the values of a and b = 12 (positive root)

Now,

a – b = 12……(ii)

Solving both the equations, we get,

2.50a + 2.60b = 336 – (i)

a – b = 12 – (ii)

2.50a + 2.60b = 336

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2.50a – 2.50b = 30

5.10b = 306

b = 60

Now, put the value of b in equation (i), we get,

2.50a + 2.60b = 336

2.50a + 156 = 336

2.50a = 180

a = 72

So, the degree values in the pie chart for each college,

P = 1.50a° = 108°

Q = 0.60b° = 36°

R = (a+18)° = (72+18)° = 90°

S = (2b+6)° = (120+6)° = 126°

The sum of the total number of students studying in commerce stream in college P and college Q = (40×2) = 80

So,

(108+36)° = 80

144° = 80

360° = {(80/144)×360} = 200

Therefore, the total number of students studying in commerce stream = 200

So, the number of students studying commerce stream in each college,

P = 108° of 200 = {(108/360)×200} = 60

Q = 36° of 200 = {(36/360)×200} = 20

R = 90° of 200 = {(90/360)×200} = 50

S = 126° of 200 = {(126/360)×200} = 70

Answer: A

Commerce Stream

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The number of students studying in college T = K

So, the total number of students studying in all the five colleges (including college T) together = {(K/1)×6} = 6K

Therefore, the total number of students studying in rest of the four colleges (excluding college T) together = (6K-K) = 5K

Now,

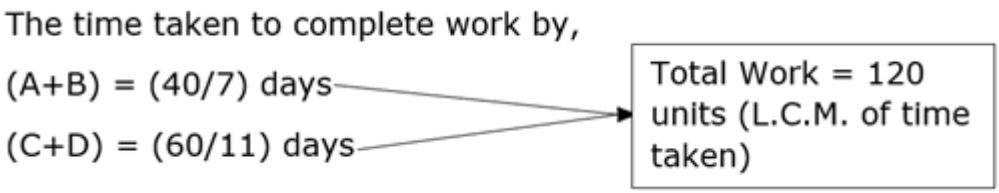
5K = 200

K = 40

Hence, the number of students studying in commerce stream in college T = K = 40.

16. Questions

Answer: A



Efficiency of,

(A+B) = {120/(40/7)} = {(120×7)/40} = 21

(C+D) = {120/(60/11)} = {(120×11)/60} = 22

Let, the efficiency of,

A = p

B = (21-p)

C = q

D = (22-q)

Now,

p/(22-q) = 60/100

p/(22-q) = 3/5

5p = 66 – 3q

5p + 3q = 66……(i)

Also,

(21-p)/q = 5/4

84 – 4p = 5q

4p + 5q = 84……(ii)

Solving both the equations, we get,

20p + 12q = 264

20p + 25q = 420

13q = 156

q = 12

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Now, put the value of q in equation (i), we get,

5p + 3q = 66

5p + 36 = 66

5p = 30

p = 6

So, the efficiency of,

A = p = 6

B = (21-p) = (21-6) = 15

C = q = 12

D = (22-q) = (22-12) = 10

Quantity I: The time taken by A and C to complete the work together = {(120)/(6+12)} = (120/18) = (20/3) = 6.67 days.

Quantity II: The time taken by B and D to complete the work together = {(120)/(15+10)} = (120/25) = (24/5) = 4.80 days.

Hence, Quantity I > Quantity II.

17. Questions

Answer: A

Speed of the boat,

In downstream = (7X+2) km/h

In upstream = (X^2+2) km/h

Now,

$\{240/(7X+2)\}/\{180/(X^2+2)\} = 4/5$

$\{1200/(7X+2)\} = \{720/(X^2+2)\}$

$1200X^2 + 2400 = 5040X + 1440$

$1200X^2 – 5040X + 960 = 0$

$5X^2 – 21X + 4 = 0$

$5X^2 – 20X – X + 4 = 0$

5X(X-4) – 1(X-4) = 0

(5X-1) (X-4) = 0

X = 0.20 or 4

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So, X = 4 (positive integer)

Therefore, the downstream speed = (7X+2) = (28+2) = 30 km/h

Upstream speed = $(X^2+2) = (4^2+2) = (16+2)$ = 18 km/h

So, the speed of the boat in still water = {(30+18)/2} = (48/2) = 24 km/h

And, the speed of the river = {(30-18)/2} = (12/2) = 6 km/h

Quantity I: The speed of the boat in still water = 24 km/h

The speed of the river = 6 km/h

Hence, the required speed = {(24/6)×100}% = 400%.

So, Quantity I = 400

Quantity II: The downstream distance covered by the boat in 7 hours = (7×30) = 210 km.

So, Quantity II = 210

Hence, Quantity I > Quantity II.

18. Questions

Answer: B

Shirt

CP = Rs. 500

SP = 120% of 500 = Rs. 600

Jeans

CP = {(600/3)×4} = Rs. 800

MP = 140% of 800 = Rs. 1120

So, the MP of the shirt = (1120-370) = Rs. 750

SP of the jeans = 90% of 1120 = Rs. 1008

Quantity I: MP of shirt = Rs. 750

Quantity II: SP of jeans = Rs. 1008

Hence, Quantity I < Quantity II.

19. Questions

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Answer: B

The amount invested = Rs. 40000

The rate of interest = 5R% p.a. (S.I.)

The time duration = R years

So, the interest earned,

$(5R \times R)\%$ of $40000 = 5R^2\%$ of $40000 = \text{Rs. } 2000R^2$

Now,

$2000R^2 = 18000$

$R^2 = 9$

$R = 3$

Quantity I

The amount invested = Rs. 30000

The rate of interest = (R+7)% = (3+7)% = 10% p.a. (C.I.)

The time duration = 2 years

So, [10+10+{(10×10)/100}]% = (20+1)% = 21%

The interest earned = 21% of 30000 = Rs. 6300.

Hence, Quantity I = 6300.

Quantity II

The amount invested = 8000R = Rs. 24000

The rate of interest = 18% p.a. (S.I.)

The time duration = 4 years

So, the interest earned = 72% of 24000 = Rs. 17280.

Hence, Quantity I = 17280.

Therefore, Quantity I < Quantity II.

20. Questions

Answer: B

Let, the income of A and B be Rs. 7P and Rs. 6P respectively.

Also, the ratio of the expenditure of A to that of B = 100:90 = 10:9. Let, it be 10Q and 9Q respectively.

So, the savings of A = Rs. (7P-10Q)

And, the savings of B = Rs. (6P-9Q)

Now,

$7P - 9Q = 10000$……(i)

$6P - 10Q = 4000$……..(ii)

Solving both the equations, we get,

$42P - 54Q = 60000$

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42P – 70Q = 28000

16Q = 32000

Q = 2000

Now, put the value of Q in equation I, we get,

7P – 9Q = 10000

7P – 18000 = 10000

7P = 28000

P = 4000

Quantity I: The expenditure of B = Rs. 9Q = Rs. 18000

And, the savings of Q = (6P-9Q) = (24000-18000) = Rs. 6000

So, the required difference = (18000-6000) = Rs. 12000

Hence, Quantity II = 12000.

Quantity II: The total income of A = 7P = Rs. 28000.

Hence, Quantity II = 28000.

Therefore, Quantity I < Quantity II.

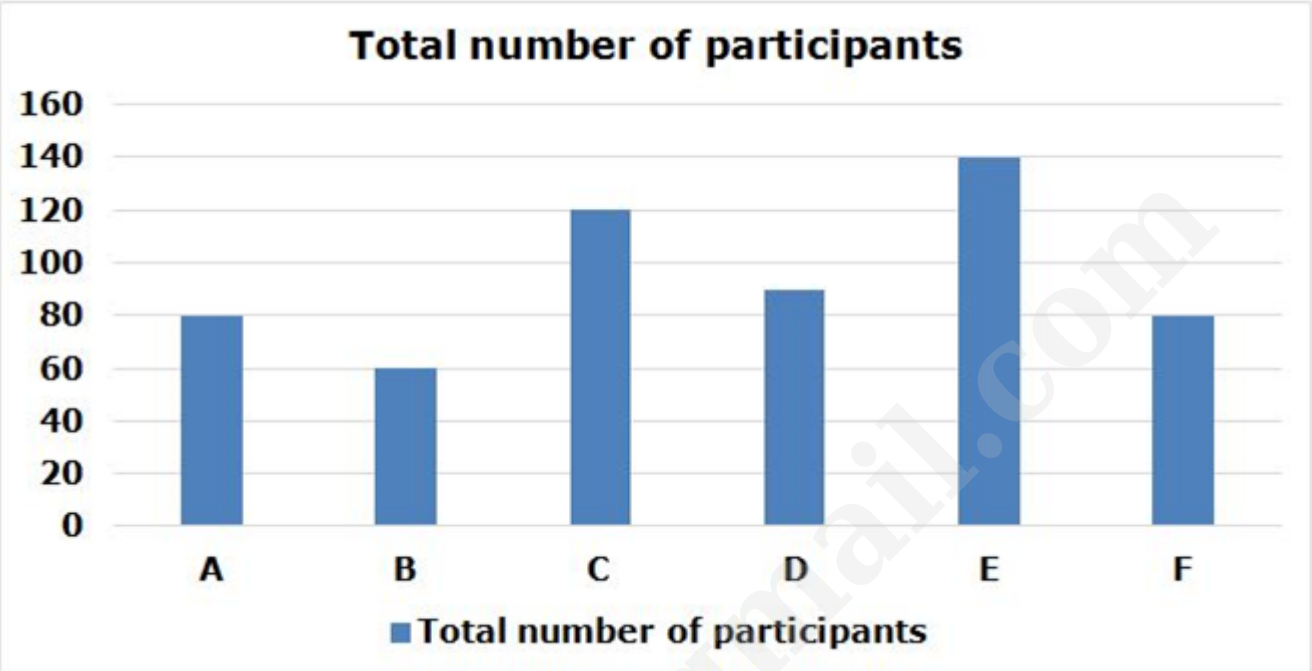
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1. Questions

Study the following information carefully and answer the questions given below.

The given bar graphs show the total number of people who participated in three different sports [High, long and triple jump] from six different complexes such as A,B,C,D,E and F.



The given table shows the percentage of the number of people who participated in the long jump out of total number of people who participated from respective complex and the number of people who participated in the high jump from those complexes

Complex	Percentage of people who participated in the long jump	Number of people who participated in the High Jump
A	10y%	_____
B	____	10x
C	25x%	_____
D	___	8y
E	20y%	____
F	____	6y

Note: The number of people who participated in the long jump from C is 60 and the ratio of the number of people who participated in the long jump from A to in the high jump from B is 6:5.

The number of people who participated in the high jump from A is 50% more than the number of people who participated in the long jump from A. The number of people who participated in the triple jump from B is 8x. The weight of each participants in the long, high, and triple jump from A is 48 kg, 54 kg, and 62 kg, respectively, and the weight of each participants in the long, high, and triple jump from B is 52 kg, 56 kg, and 58 kg, respectively. Find the approximate average weight of all the people who participated in the all sports from complexes A and B together?

a. 57.36

b. 58.24

c. 54.51

d. 49.35

e. 51.33

2. Questions

The ratio of the number of people who participated in the long jump to triple jump from C is 12:5. The ratio of the number of male to female who participated in the long jump and triple jump from C is 3:2 and 3:2 respectively. The average number of male who participated in the long jump and high jump in C is 24. The total number of male who participated from C is what percent of total number of people who participated from C?

a. 44.32%

b. 54.32%

c. 57.32%

d. 52.50%

e. 59.32%

3. Questions

The number of people who participated in the triple jump from D is 10y. The number of people who participated in the long jump from G is 25% more than the number of people who participated from D in the long jump, and the number of people who participated in the triple jump from G is 20% more than that of D. If the total number of people who participated from G is 10% more than that of D, find the difference between the total number of people who participated in the high jump from G and D?

a. 7

b. 6

c. 8

d. 9

e. None of these

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4. Questions

From D, 45% of the people participated, from F 25% of the people participated and from C, 50% of the people participated. If the total number of people who did not participate from D, C, and F together is [Ax+y] then find the value of A?

a. 233.5

b. 234.5

c. 258.3

d. 267.2

e. None of these

5. Questions

37.5% of the total number of people participated in the triple jump from F. The ratio of the number of people who participated in the long jump to discuss throw from F is 2:3. Find the number of people who participated in the high jump from F is what percent of the number of people who participated in the discus throw from F?

a. 37.5%

b. 32.5%

c. 36.5%

d. 45.3%

e. None of these

6. Questions

Study the following data carefully and answer the questions:

A shopkeeper sold 5 items an Almirah, a Chair, a Sofa, a Bed, and a Table. Cost price and marked price of each item are different and the shopkeeper sold each item at different discounts on their respective marked price.

Cost price of the Almirah is Rs.20000 more than that of the Chair and double that of the Sofa. Cost price of the Bed is 60% more than that of the Table and the Bed is marked up by Rs.4000. The Chair is marked up by 20%, the table is marked up by 8% and the marked price of the table is Rs.6000 more than that of the chair. Marked price of the Almirah is Rs.6000 more than its cost price and also Rs.10000 more than the marked price of the Bed. Average of marked prices of the Sofa and the Bed is Rs.18000.

If the Almirah is sold at a 10% discount on its marked price and the Sofa is sold at a 6.25% discount on its marked price, then find the ratio of profit received on the Almirah to that received on the Sofa.

a. 6:5

b. 3:4

c. 1:1

d. 2:3

e. 5:4

7. Questions

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If the total SP of the Bed and the Table is Rs.28660 and the Bed is sold 8% discount on its MP, then find the profit/loss percent on the Table.

a. 4.2%

b. 2.1%

c. 2.6%

d. 5.2%

e. Can't be determined

8. Questions

If the Chair is sold at two successive discounts of 5% and d% and the profit received on the Chair is 8.3%, then find the value of 'd'.

a. 10

b. 5

c. 12

d. 8

e. 15

9. Questions

Find the average of marked-up amounts of all 5 items?

a. Rs.3120

b. Rs.2880

c. Rs.3040

d. Rs.3260

e. Rs.2900

10. Questions

If the Chair and the Sofa are sold at 14% profit and 25% profit respectively, then what will be the ratio of discount percent on the marked price of the Chair to that of the Sofa?

a. 5:6

b. 2:3

c. 10:11

d. 4:5

e. 25:28

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11. Questions

Study the following information carefully and answer the questions given below.

The given table shows the expenditure on rent, grocery, transport, bill and savings of three people is given.

	A(Rs.)	B	C
Expenditure in Rent	4500	The ratio of expenditure in rent of A and B is 5:6	Rs.500 more than A
Expenditure in Grocery	6500	Expenditure in groceries is 20% less than A	The ratio of expenditure in groceries of B and C is 13:14
Expenditure in Transport	3200	Rs.800 more than A	The ratio of expenditure in transport of B and C is 8:9
Expenditure in Bill	5400	Rs.600 more than A	25% less than B
Saving	8400	The ratio of the savings of A and B is 7:6	20% less than B

Note:

Total income of a person is the sum of expenditure in rent, grocery, transport, bill, and saving together.

Expenditure of D in rent, grocery transport, and bill are 10%,20%,10%, and 20% more than that of B. If the total income of D is Rs. 32500 then find the saving of D?

a. Rs.5420

b. Rs.6820

c. Rs.8720

d. Rs.8520

e. Rs.9520

12. Questions

Find the difference between the average income of A and B together and the total saving of A, B, and C together?

a. Rs.5200

b. Rs.4520

c. Rs.6520

d. Rs.6540

e. Rs.6320

13. Questions

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Find which one is true?

I). Total expenditure in the grocery of all three together is Rs.17200.

II). Total expenditure in rent of all three together is Rs.13900.

III). Total expenditure in transport of all three together is Rs.12800.

a. Only III is true

b. None is true

c. Only II is true

d. Only II and III are true

e. None of these

14. Questions

Find the ratio of total expenditure in transport and rent of B and total expenditure in grocery and bill of C?

a. 94:101

b. 97:112

c. 98:105

d. 91:101

e. None of these

15. Questions

If the income of A and B is increased by 15% and 20% respectively. then find the new average income of A and B is approximately what percent of the income of C?

a. 130%

b. 140%

c. 160%

d. 110%

e. None of these

16. Questions

Study the information carefully and find from which of the given options we can calculate the answer to the question. [Answer may differ from each option]

Find the length of train A?

a. The Speed of train A is 20% higher than the speed of train B. The Ratio of the lengths of train A and B is 2:3. Train B covers a 140 m long bridge in 15 sec

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b. The Ratio of speed of train A and B is 4:5. Train B crosses a tree in 5.33 sec and a bridge of 180 m in 11.33 sec. The length of train B is 160 m

c. Train A crosses a tree in 5.33 sec and a bridge of 180 m in 11.33 sec. The speed of train B is 90 km/hr. The speed of train A is $6/5^{th}$ of the speed of train B

d. None

e. More than one option is correct

17. Questions

Find the age of A?

a. The Age of A is 20% more than the age of B. The Age of C is 30% more than the age of D. Sum of the age of A and D is 50 years old. The age of E is 5 more than the age of B

b. The Age of A is 10 years less than the age of D who is 9 years younger than C. The Age of B is 24 years. The age of E is 1 year younger than C

c. None

d. Sum of the age of A and B is 44 years. The average age of B and C is 27 years. The ratio of age of A and C is 2:3. Difference between the age of A and E is 9 years

e. more than one option is correct

18. Questions

Find the rate of interest?

a. Difference between compound interest and simple interest amount of the same interest rate for 2 years is 50. The principle amount is Rs.1250

b. If Rs.1640 invests for n years at some rate simple interest then the interest amount is Rs.656 and if the same amount invests for (n+1) years for 5% more interest rate then the interest amount is Rs.1230

c. Rs. p invest for 3 years compound interest at some rate of interest is Rs.11438. Rs.12500 invest for the same interest rate for some years then SI is 6250

d. None

e. more than one option is correct

19. Questions

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In each of the following questions two quantities – I and II given. You are expected to solve and find the correct relation between them.

Quantity I: Probability of not solving a question by P, Q, and R is 0.25, 0.2, and 0.5 respectively. Find the probability that at most two of them solve the question.

Quantity II: A bag contains ten red pens and six black pens, if three pens are drawn at random from the

bag. Then find the probability of getting at least one black pen.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I = Quantity II or No relation
d. Quantity I ≥ Quantity II
e. Quantity I ≤ Quantity II

20. Questions

Quantity I: M travelled from point P to Q with speed of 6 km/hr and then comes to point P with speed of 8 km/hr. N travelled from point P to Q with speed of 16 km/hr. Total journey time of M is 22 hours more than that of N. Find the distance between P and Q.

Quantity II: Find the value of 33K, where K is the unit digit of 33^{53}.

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I = Quantity II or No relation
d. Quantity I ≥ Quantity II
e. Quantity I ≤ Quantity II

Explanations:

1. Questions

The number of participants in the long jump from C is 60. So, 120*25x/100=60

Or, x=2

So, the number of people participating in the high jump from B is 10*2=20

So, the number of people participating in the long jump from A = 20*6/5=24

So, 80*10y/100=24, y=3

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Complex	Number of people who participated in Long Jumb	Number of people who participated in High Jump
A	24	_____
B	____	10*2=20
C	60	_____
D	___	8*3=24
E	140*60/100=84	____
F	____	6*3=18

Answer: C

The number of people who participated in the High Jump from A = 24*150/100=36

The number of people who participated in the triple jump from B = 8*2=16

The number of people who participated in the triple jump from A = 80-24-36=20

The number of people who participated in the long jump from B = 60-16-20=24

So, average weight

= [24*48+54*36+20*62+24*52+56*20+16*58]/140

= [1152+1944+1240+1248+1120+928]/140=54.51

2. Questions

The number of participants in the long jump from C is 60. So, 120*25x/100=60

Or, x=2

So, the number of people participating in the high jump from B is 10*2=20

So, the number of people participating in the long jump from A = 20*6/5=24

So, 80*10y/100=24, y=3

Complex	Number of people who participated in Long Jumb	Number of people who participated in High Jump
A	24	_____
B	____	10*2=20
C	60	_____
D	___	8*3=24
E	140*60/100=84	____
F	____	6*3=18

Answer: D

The total number of people who participated in the triple jump from C = 60*5/12=25

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From C -

The number of male who participated in the long jump = 60*3/5=36

The number of female who participated in the long jump = 60*2/5=24

The number of male who participated in the triple jump = 25*3/5=15

The number of female who participated in the triple jump is = 25*2/5=10

The number of male who participated in the high jump = 24*2-36=12

Required percentage = [(36+15+12)/120]*100=52.5%

3. Questions

The number of participants in the long jump from C is 60. So, 120*25x/100=60

Or, x=2

So, the number of people participating in the high jump from B is 10*2=20

So, the number of people participating in the long jump from A = 20*6/5=24

So, 80*10y/100=24, y=3

Complex	Number of people who participated in Long Jumb	Number of people who participated in High Jump
A	24	____
B	____	10*2=20
C	60	____
D	___	8*3=24
E	140*60/100=84	____
F	____	6*3=18

Answer: B

The number of people who participated in the triple jump from D = 10*3=30

The number of people who participated in the long jump from D =90-30-24=36

The number of people who participated in the long jump from G = 36*125/100=45

The number of people who participated in the triple jump from G =30*120/100=36

The number of people who participated in the high jump of G = 90*110/100-45-36=18

So, the required difference = 24-18=6

4. Questions

The number of participants in the long jump from C is 60. So, 120*25x/100=60

Or, x=2

So, the number of people participating in the high jump from B is 10*2=20

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So, the number of people participating in the long jump from A = 20*6/5=24

So, 80*10y/100=24, y=3

Complex	Number of people who participated in Long Jumb	Number of people who participated in High Jump
A	24	_____
B	____	10*2=20
C	60	_____
D	___	8*3=24
E	140*60/100=84	____
F	____	6*3=18

Answer: A

The number of people who did not participate from D = [90/45]*55=110

The number of people who did not participate from F = [80/25]*75=240

The number of people who did not participate from C =[120/50]*50=120

The total number of people who did not participate from D, F, and C together is = 110+240+120=470=Ax+y

Or, 2A+3=470

Or, A=233.5

5. Questions

The number of participants in the long jump from C is 60. So, 120*25x/100=60

Or, x=2

So, the number of people participating in the high jump from B is 10*2=20

So, the number of people participating in the long jump from A = 20*6/5=24

So, 80*10y/100=24, y=3

Complex	Number of people who participated in Long Jumb	Number of people who participated in High Jump
A	24	_____
B	____	10*2=20
C	60	_____
D	___	8*3=24
E	140*60/100=84	____
F	____	6*3=18

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Answer: A

Number of people participated in the triple jump from F =[80*37.5/100]=30

The number of people participated in the long jump from F = 80-30-18=32

The number of people participated in the discuss throw from F = 32*3/2=48

Required percentage =[18/48]*100=37.5%

6. Questions

Let CP of the Chair = 'x' rupees

So, CP of the Almirah = (x + 20000) rupees

And CP of the Sofa = [(x + 20000)/2] rupees

Let CP of the Table = '5y' rupees

So, CP of the Bed = 160% of 5y = '8y' rupees

And MP of the Bed = (8y + 4000) rupees

Since the Chair is marked up by 20%.

MP of the chair = 120% of x = '1.2x' rupees

Since MP of the table is Rs.6000 more than that of the chair.

MP of the Table = (1.2x + 6000) rupees

Since the table is marked up by 8%.

So,

1.2x + 6000 = 5y * (108/100)

1.2x + 6000 = 27y/5

6x + 30000 = 27y

2x + 10000 = 9y ----------(1)

Since MP of the Almirah is Rs.10000 more than MP of the Bed.

So, MP of the Almirah = (8y + 4000 + 10000) rupees

Since MP of the Almirah is Rs.6000 more than its CP.

So,

8y + 4000 + 10000 – x – 20000 = 6000

8y – x = 12000

x = 8y – 12000 ----------(2)

From equations (1) and (2):

2 * (8y – 12000) + 10000 = 9y

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16y – 24000 + 10000 = 9y

y = 2000, x = 4000

Items	CP of the item (in rupees)	MP of the item (in rupees)
Almirah	4000 + 20000 = 24000	24000 + 6000 = 30000
Chair	4000	120% of 4000 = 4800
Sofa	24000/2 = 12000	2 * 18000 – 20000 = 16000
Bed	8 * 2000 = 16000	16000 + 4000 = 20000
Table	5 * 2000 = 10000	4800 + 6000 = 10800

Answer: C

CP of the Almirah = Rs.24000

MP of the Almirah = Rs.30000

SP of the Almirah = 90% of 30000 = Rs.27000

Profit received on the Almirah = 27000 – 24000 = Rs.3000

CP of the Sofa = Rs.12000

MP of the Sofa = Rs.16000

SP of the Sofa = 93.75% of 16000 = Rs.15000

Profit received on the Sofa = 15000 – 12000 = Rs.3000

Required ratio = 3000: 3000 = 1: 1

7. Questions

Let CP of the Chair = ‘x’ rupees

So, CP of the Almirah = (x + 20000) rupees

And CP of the Sofa = [(x + 20000)/2] rupees

Let CP of the Table = ‘5y’ rupees

So, CP of the Bed = 160% of 5y = ‘8y’ rupees

And MP of the Bed = (8y + 4000) rupees

Since the Chair is marked up by 20%.

MP of the chair = 120% of x = ‘1.2x’ rupees

Since MP of the table is Rs.6000 more than that of the chair.

MP of the Table = (1.2x + 6000) rupees

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Since the table is marked up by 8%.

So,

1.2x + 6000 = 5y * (108/100)

1.2x + 6000 = 27y/5

6x + 30000 = 27y

2x + 10000 = 9y ----------(1)

Since MP of the Almirah is Rs.10000 more than MP of the Bed.

So, MP of the Almirah = (8y + 4000 + 10000) rupees

Since MP of the Almirah is Rs.6000 more than its CP.

So,

8y + 4000 + 10000 – x – 20000 = 6000

8y – x = 12000

x = 8y – 12000 ----------(2)

From equations (1) and (2):

2 * (8y – 12000) + 10000 = 9y

16y – 24000 + 10000 = 9y

y = 2000, x = 4000

Items	CP of the item (in rupees)	MP of the item (in rupees)
Almirah	4000 + 20000 = 24000	24000 + 6000 = 30000
Chair	4000	120% of 4000 = 4800
Sofa	24000/2 = 12000	2 * 18000 – 20000 = 16000
Bed	8 * 2000 = 16000	16000 + 4000 = 20000
Table	5 * 2000 = 10000	4800 + 6000 = 10800

Answer: C

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MP of the Bed = Rs.20000

SP of the Bed = 92% of 20000 = Rs.18400

SP of the Table = 28660 – 18400 = Rs.10260

CP of the Table = Rs.10000

Profit percent on the Table = [(10260 – 10000)/10000] * 100 = 2.6%

8. Questions

Let CP of the Chair = 'x' rupees

So, CP of the Almirah = (x + 20000) rupees

And CP of the Sofa = [(x + 20000)/2] rupees

Let CP of the Table = '5y' rupees

So, CP of the Bed = 160% of 5y = '8y' rupees

And MP of the Bed = (8y + 4000) rupees

Since the Chair is marked up by 20%.

MP of the chair = 120% of x = '1.2x' rupees

Since MP of the table is Rs.6000 more than that of the chair.

MP of the Table = (1.2x + 6000) rupees

Since the table is marked up by 8%.

So,

1.2x + 6000 = 5y * (108/100)

1.2x + 6000 = 27y/5

6x + 30000 = 27y

2x + 10000 = 9y ----------(1)

Since MP of the Almirah is Rs.10000 more than MP of the Bed.

So, MP of the Almirah = (8y + 4000 + 10000) rupees

Since MP of the Almirah is Rs.6000 more than its CP.

So,

8y + 4000 + 10000 – x – 20000 = 6000

8y – x = 12000

x = 8y – 12000 ----------(2)

From equations (1) and (2):

2 * (8y – 12000) + 10000 = 9y

16y – 24000 + 10000 = 9y

y = 2000, x = 4000

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Items	CP of the item (in rupees)	MP of the item (in rupees)
Almirah	4000 + 20000 = 24000	24000 + 6000 = 30000
Chair	4000	120% of 4000 = 4800
Sofa	24000/2 = 12000	2 * 18000 – 20000 = 16000
Bed	8 * 2000 = 16000	16000 + 4000 = 20000
Table	5 * 2000 = 10000	4800 + 6000 = 10800

Answer: B

CP of the Chair = Rs.4000

SP of the Chair = 108.3% of 4000 = Rs.4332

MP of the Chair = Rs.4800

So,

4800 * (95/100) * (100 – d)/100 = 4332

95 * (100 – d) = 9025

100 – d = 95

d = 5%

9. Questions

Let CP of the Chair = 'x' rupees

So, CP of the Almirah = (x + 20000) rupees

And CP of the Sofa = [(x + 20000)/2] rupees

Let CP of the Table = '5y' rupees

So, CP of the Bed = 160% of 5y = '8y' rupees

And MP of the Bed = (8y + 4000) rupees

Since the Chair is marked up by 20%.

MP of the chair = 120% of x = '1.2x' rupees

Since MP of the table is Rs.6000 more than that of the chair.

MP of the Table = (1.2x + 6000) rupees

Since the table is marked up by 8%.

So,

1.2x + 6000 = 5y * (108/100)

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1.2x + 6000 = 27y/5

6x + 30000 = 27y

2x + 10000 = 9y ----------(1)

Since MP of the Almirah is Rs.10000 more than MP of the Bed.

So, MP of the Almirah = (8y + 4000 + 10000) rupees

Since MP of the Almirah is Rs.6000 more than its CP.

So,

8y + 4000 + 10000 – x – 20000 = 6000

8y – x = 12000

x = 8y – 12000 ----------(2)

From equations (1) and (2):

2 * (8y – 12000) + 10000 = 9y

16y – 24000 + 10000 = 9y

y = 2000, x = 4000

Items	CP of the item (in rupees)	MP of the item (in rupees)
Almirah	4000 + 20000 = 24000	24000 + 6000 = 30000
Chair	4000	120% of 4000 = 4800
Sofa	24000/2 = 12000	2 * 18000 – 20000 = 16000
Bed	8 * 2000 = 16000	16000 + 4000 = 20000
Table	5 * 2000 = 10000	4800 + 6000 = 10800

Answer: A

Marked up the amount of the Almirah = 30000 – 24000 = Rs.6000

Marked up the amount of the Chair = 4800 – 4000 = Rs.800

Marked up the amount of the Sofa = 16000 – 12000 = Rs.4000

Marked up the amount of the Bed = 20000 – 16000 = Rs.4000

Marked up the amount of the Table = 10800 – 10000 = Rs.800

Required average = (6000 + 800 + 4000 + 4000 + 800)/5 = Rs.3120

10. Questions

Let CP of the Chair = ‘x’ rupees

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So, CP of the Almirah = (x + 20000) rupees

And CP of the Sofa = [(x + 20000)/2] rupees

Let CP of the Table = '5y' rupees

So, CP of the Bed = 160% of 5y = '8y' rupees

And MP of the Bed = (8y + 4000) rupees

Since the Chair is marked up by 20%.

MP of the chair = 120% of x = '1.2x' rupees

Since MP of the table is Rs.6000 more than that of the chair.

MP of the Table = (1.2x + 6000) rupees

Since the table is marked up by 8%.

So,

1.2x + 6000 = 5y * (108/100)

1.2x + 6000 = 27y/5

6x + 30000 = 27y

2x + 10000 = 9y ----------(1)

Since MP of the Almirah is Rs.10000 more than MP of the Bed.

So, MP of the Almirah = (8y + 4000 + 10000) rupees

Since MP of the Almirah is Rs.6000 more than its CP.

So,

8y + 4000 + 10000 – x – 20000 = 6000

8y – x = 12000

x = 8y – 12000 ----------(2)

From equations (1) and (2):

2 * (8y – 12000) + 10000 = 9y

16y – 24000 + 10000 = 9y

y = 2000, x = 4000

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Items	CP of the item (in rupees)	MP of the item (in rupees)
Almirah	4000 + 20000 = 24000	24000 + 6000 = 30000
Chair	4000	120% of 4000 = 4800
Sofa	24000/2 = 12000	2 * 18000 – 20000 = 16000
Bed	8 * 2000 = 16000	16000 + 4000 = 20000
Table	5 * 2000 = 10000	4800 + 6000 = 10800

Answer: D

CP of the Chair = Rs.4000

SP of the Chair = 114% of 4000 = Rs.4560

MP of the Chair = Rs.4800

So, discount percent on MP of the Chair = [(4800 – 4560)/4800] * 100 = 5%

CP of the Sofa = Rs.12000

SP of the Sofa = 125% of 12000 = Rs.15000

MP of the Sofa = Rs.16000

So, discount percent on MP of the Sofa = [(16000 – 15000)/16000] * 100 = 6.25%

Required ratio = 5: 6.25 = 4: 5

11. Questions

	A(Rs.)	B (Rs.)	C(Rs.)
Expenditure in Rent	4500	4500*6/5=5400	4500+500=5000
Expenditure in Grocery	6500	6500*80/100=5200	5200*14/13=5600
Expenditure in Transport	3200	3200+800=4000	4000*9/8=4500
Expenditure in Bill	5400	5400+600=6000	6000*75/100=4500
Saving	8400	8400*6/7=7200	7200*80/100=5760
Income	28000	27800	25360

Answer: C

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The total expenditure of D is

=5400*110/100+5200*120/100+4000*110/100+6000*120/100=Rs.23780

Total saving of D is =32500-23780= Rs.8720

12. Questions

	A(Rs.)	B (Rs.)	C(Rs.)
Expenditure in Rent	4500	4500*6/5=5400	4500+500=5000
Expenditure in Grocery	6500	6500*80/100=5200	5200*14/13=5600
Expenditure in Transport	3200	3200+800=4000	4000*9/8=4500
Expenditure in Bill	5400	5400+600=6000	6000*75/100=4500
Saving	8400	8400*6/7=7200	7200*80/100=5760
Income	28000	27800	25360

Answer: D

The average income of A and B is [28000+27800]/2=55800/2=27900

Total saving is = 8400+7200+5760=21360

Required difference=27900-21360=Rs.6540

13. Questions

	A(Rs.)	B (Rs.)	C(Rs.)
Expenditure in Rent	4500	4500*6/5=5400	4500+500=5000
Expenditure in Grocery	6500	6500*80/100=5200	5200*14/13=5600
Expenditure in Transport	3200	3200+800=4000	4000*9/8=4500
Expenditure in Bill	5400	5400+600=6000	6000*75/100=4500
Saving	8400	8400*6/7=7200	7200*80/100=5760
Income	28000	27800	25360

Answer: B

I). Total expenditure in the grocery of all three together is Rs.17300.

II). Total expenditure in rent of all three together is Rs.14900.

III). Total expenditure in transport of all three together is Rs.11700.

So, None is true.

14. Questions

	A(Rs.)	B (Rs.)	C(Rs.)
Expenditure in Rent	4500	4500*6/5=5400	4500+500=5000
Expenditure in Grocery	6500	6500*80/100=5200	5200*14/13=5600
Expenditure in Transport	3200	3200+800=4000	4000*9/8=4500
Expenditure in Bill	5400	5400+600=6000	6000*75/100=4500
Saving	8400	8400*6/7=7200	7200*80/100=5760
Income	28000	27800	25360

Answer: A

Required ratio = [4000+5400] : [5600+4500]=9400:10100=94:101

15. Questions

	A(Rs.)	B (Rs.)	C(Rs.)
Expenditure in Rent	4500	4500*6/5=5400	4500+500=5000
Expenditure in Grocery	6500	6500*80/100=5200	5200*14/13=5600
Expenditure in Transport	3200	3200+800=4000	4000*9/8=4500
Expenditure in Bill	5400	5400+600=6000	6000*75/100=4500
Saving	8400	8400*6/7=7200	7200*80/100=5760
Income	28000	27800	25360

Answer: A

The new average income of A and B is [28000*115/100+27800*120/100]/2=32780

Required percentage = [32780/25360]*100=129.26%=130%

16. Questions

Answer: C

In option A, we cannot calculate the length of train B so we cannot calculate the length of train A

In option B, no information is given about the relation between train A and train b.

In option C, we can get the answer.

Speed of train A is 90*6/5=108 km/hr= 108*5/18=30 m/sec

So, length/30 =5.33, length = 30*5.33=159.9 = 160 m

17. Questions

Answer: D

From D, we can get the answer.

Let the age of A and C is 2x and 3x.

So, the age of B is 44-2x

So, 44-2x+3x=27*2=54

Or, x=10

Age of A is 20 years

18. Questions

Answer: E

From a)

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We can get by formula difference of SI and CI = principle $[R/100]^2$

From b)

1640*n*r/100=656

nr = 40

And 1640*(n+1)*(r+5)/100 =1230

(n+1)(r+5) = 75

nr + r + 5 + 5n = 75

r + 5n = 75 – 45 = 30

nr = 40

40/n + 5n = 30

$5n^2$ - 30n + 40 =0

N=4,2 so r= 10,20

We can get the answer by solving the equations

So, option E is true.

19. Questions

Answer: B

Quantity I.

Required probability = 1 – [(1 – 0.25) x (1 – 0.2) x (1 – 0.5)] = 0.7

Quantity II.

Probability (at least 1 black) = 1 – probability (no black Pen)

= 1 – [$^{10}C_3$ / $^{16}C_3$]

= 1 – [(10 x 9 x 8)/ (16 x 15 x 14)]

= 1 – 3/14

= 11/14 = 0.78 (approx.)

So, Quantity I < Quantity II

Hence answer is option B

20. Questions

Answer: B

Quantity I.

Average speed of M = 2 x (6 x 8)/ (6 + 8) = 48/7 km/hr

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Speed of N = 16 km/hr

Let total time taken by M = a hours

So, total time taken by N = (a - 22) hours

Total distance travelled by M = 2 x total distance travelled by N

Now, (48/7) x a = 2 x 16 x (a - 22)

3a = 14a – 14 x 22

So, value of a = (14 x 22 / 11) = 28 hours

Time taken by N = (28 – 22) = 6 hours

Required distance = 16 x 6 = 96 km

Quantity II.

$33^{53} = 33^{4 \times 13 + 1}$

So, unit digit of 33^{53} = 3

Value of 33K = 33 x 3 = 99

So, Quantity I < Quantity II

Hence answer is option B

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1. Questions

The cost price of article A is Rs. 10,000, which is marked up X% above its cost price and sold at a discount of 20%. The cost of article B is Rs. 8000, which is sold at a profit of Y%. Article B is marked up 2Y% above its cost price. The ratio of the selling price of article A to that of article B is 6:5. The marked price of article A is Rs. 3000 more than that of article B. Find the value of (2X-3Y).

a. 20

b. 25

c. 15

d. 10

e. None of these

2. Questions

A car starts from city P and reaches city R, which is somewhere between city P and city Q in [T] hours. After that, a bike starts from city Q towards city P but stops at city R and meets with the car at the same time as taken by the car to reach city R from city P. If the speeds of the car and bike are [4V] km/h and [3V] km/h, respectively, then find the value of [V×T] if the distance between city P and city Q is 210 km.

a. 18

b. 21

c. 30

d. Cannot be determined

e. None of these

3. Questions

The ratio of the present ages of X's brother to those of his father is 1:4. X's mother is 25 years older than X. If four years ago, X's father's age was equal to the present age of X's mother and X is 2 years younger than his brother, then find the average of the present ages of all the members of the family.

a. 21 years

b. 33 years

c. 40 years

d. 18 years

e. None of these

4. Questions

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M started a business with an investment of Rs. [P]. After 5 months, N also joined the business with an investment of Rs. 4000 less than that of M. At the end of the year, the ratio of the profit share of M to that of N is 15:7. Find the initial investment made by M.

a. Rs. 27000

b. Rs. 25000

c. Rs. 21000

d. Cannot be determined

e. None of these

5. Questions

4P and 5Q are the roots of the equation $x^2 - ax + b = 0$, where both P and Q are positive integers. (4P > 5Q)

Sum of the roots of the equation is a 2-digit prime number less than 20.

One of the roots of the equation is also a prime number, which is the sum of the smallest and second smallest prime numbers.

Product of the roots of the equation is less than 50 but more than 30.

Find the value of $\sqrt{\{9a+(b/10)\}}$.

a. 11

b. 14

c. 9

d. 5

e. None of these

6. Questions

There are two quadratic equations given below.

Equation I: $a^2 + (p+1)a - 480 = 0$

Equation II: $b^2 - (q+1)b - 340 = 0$

The greater root of equation I is equal to that of equation II.

The difference between the roots of equation I is 44.

The sum of the roots of the equation I is -4.

Find the difference between the values of p and q.

a. 2

b. 1

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c. 4

d. 3

e. None of these

7. Questions

There is a quadratic equation given below.

$k^2 - mk + n = 0$

Both the roots of the equation are consecutive positive integers.

The value of n is less than 13 but more than 10.

One of the roots of the equation is the smallest composite number.

Find the value of (m×n).

a. 72

b. 68

c. 84

d. Cannot be determined

e. None of these

8. Questions

Read the following information carefully and answer the questions based on it.

There are a certain number of students participating in three different activities: singing, painting and dancing. Out of the total number of students participating in each activity some are boys while rest are girls.

The total number of students (boys and girls) participating in singing and painting are 150 and (4A-B-30) respectively.

The number of boys participating in singing is (A+20) and the number of girls participating in singing is (B-40). The number of boys and girls participating in painting are 120 and 80 respectively.

Average of the total number of students (boys + girls) participating in all the three activities is 180. The number of boys participating in dancing is 30% more than that of singing and the number of girls participating in dancing is 1/4th less than that of painting.

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Note:

The total number of students = number of students participated + number of students did not participated

The total number of students participated = number of boys participated + number of girls participated

The total number of students participated = number of students participated in singing + painting + dancing

Out of the total number of students (boys + girls) participating in singing, the ratio of the number of boys participating in rock singing to that of girls participating in jazz singing is 7:1. The number of boys participating in jazz singing is 25% less than the number of girls participating in rock singing. Find the number of boys participating in rock singing is what percent of the number of boys participated in jazz singing?

a. 266.66%

b. 233.33%

c. 212.50%

d. 237.50%

e. None of these

9. Questions

If there is another activity drawing, where the number of boys and girls participating are [15N] and [N^2-4] respectively and ratio of the number of boys to that of girls participated is 2:1, then find the total number of students (boys + girls) participating in drawing. (N is a positive integer)

a. 180

b. 192

c. 164

d. Cannot be determined

e. None of these

10. Questions

If the fees for each boy are Rs. 100 for participating in painting and the total revenue earned from all the boys and girls for participating in painting is Rs. 21,600, then find the fees for each girl for participating in painting.

a. Rs. 130

b. Rs. 150

c. Rs. 100

d. Rs. 120

e. None of these

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11. Questions

In dancing, out of the total number of boys, P% did not participate, while rest participated. If the difference between the percentage values of the number of boys who participated and did not participate is (P-5), then find the value of √(2P+11). (number of boys participated is more than that of not participated)

a. 3

b. 9

c. 27

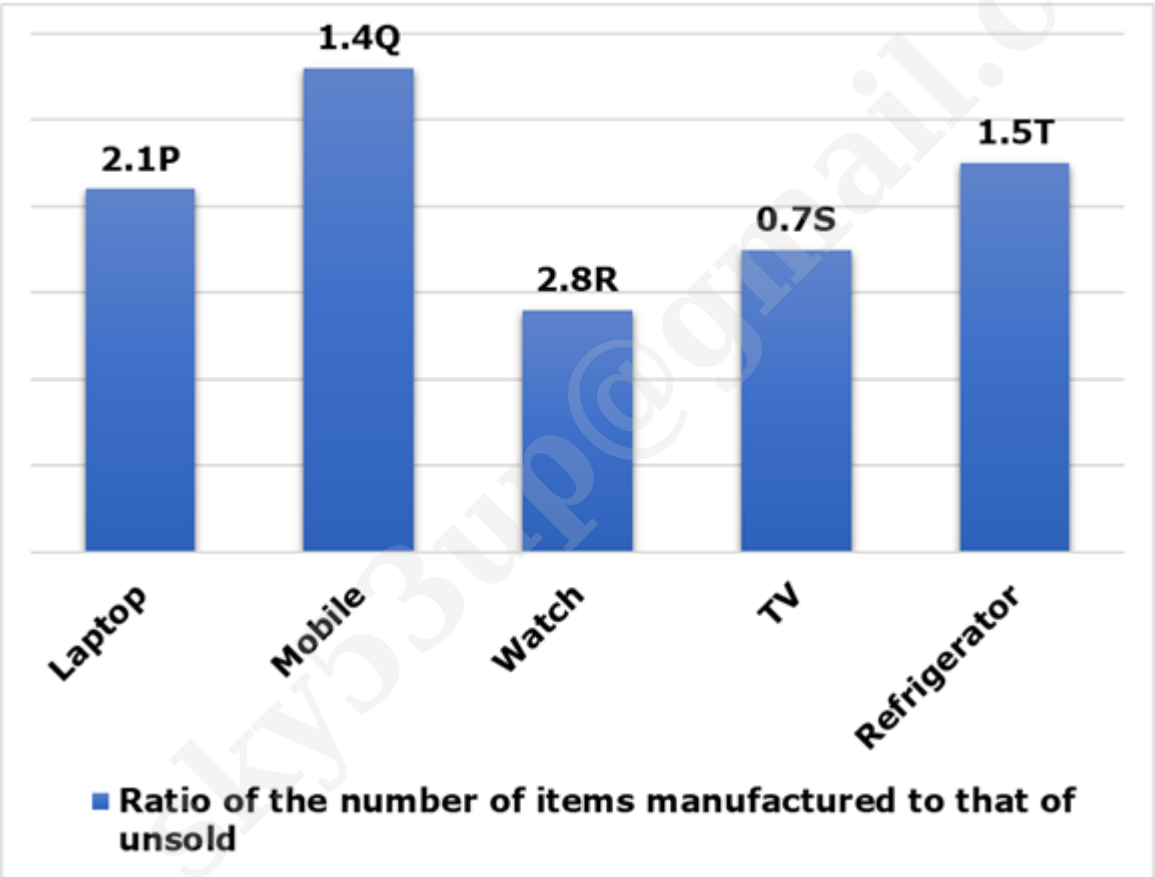
d. 81

e. None of these

12. Questions

Read the following information carefully and answer the questions based on it.

A shopkeeper sales five different electronic items i.e., laptop, mobile, watch, TV and refrigerator. The bar graph given below shows the ratio of the number of items manufactured (sold + unsold) to that of unsold by the shopkeeper for each item.



Note:

The values of all the variables are positive integers.

The values of P and Q are the roots of the equation $K^2 - 6K + 8 = 0$. *(Q>P)*

The values of S and T are prime numbers more than 2 but less than 7. *(S>T)*

The value of R is the smallest natural number.

The difference between the number of laptops sold and unsold is 330.

The number of laptops and watches unsold are equal, which is 50% more than the number of mobiles

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unsold.

Similarly, the number of TV and refrigerator unsold are also equal, which is 20% less than the number of mobiles unsold.

The total number of items manufactured = number of items sold + number of items unsold

The total number of items manufactured = number of defective items + number of non-defective items

Only non-defective items can be sold.

Not necessarily, all the non-defective items are sold.

Out of the total number of laptops manufactured, if the number of defective laptops manufactured is 100, then find the number of non-defective laptops that remain unsold.

a. 60

b. 45

c. 50

d. 35

e. None of these

13. Questions

There are two types of TVs sold, i.e., LED and LCD. The number of LED TVs sold is 4 times the number of LCD TVs sold. If the ratio of the number of black LEDs to that of white LCDs sold is 12:1 and the number of white LEDs is 1/3rd more than that of black LCDs sold, then find the total number of white TVs (LED+LCD) sold, which is what fraction of the total number of black TVs (LED+LCD) sold by the shopkeeper?

a. 1/3

b. 2/5

c. 4/7

d. 3/4

e. None of these

14. Questions

If the manufacturing cost (CP) of each mobile is Rs. 10,000 and each mobile has been sold at a profit of 20%, then find the overall profit earned/loss incurred by the shopkeeper on selling all the mobiles.

a. Rs. 75000

b. Rs. 80000

c. Rs. 60000

d. Rs. 55000

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e. None of these

15. Questions

There are two types of watches manufactured i.e., Digital and Analog. The ratio of the number of digital watches sold to that of analog watches unsold is 4:1. While, the number of analog watches sold is 30% less than the number of digital watches unsold. Find the difference between the total number of digital and analog watches manufactured.

a. 250

b. 200

c. 120

d. 180

e. None of these

16. Questions

If the number of washing machines manufactured is 60 less than the number of refrigerators manufactured and the number of washing machines sold is X% of the total number of washing machines sold. The number of washing machines sold is (X+100) more than the number of washing machines unsold. Find the value of √(X+20).

a. 10

b. 12

c. 8

d. Cannot be determined

e. None of these

17. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the total number of fast foods (pizzas + burgers + sandwiches) ordered from four different shops (A, B, C and D). It also shows the average of the number of pizzas and burgers ordered from each shop and the percentage of the number of sandwiches sold out of the total number of fast foods ordered from each shop.

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Shops	Total number of fast foods ordered	Average of number of pizzas and burgers ordered	Percentage of the number of sandwiches ordered
A	8M	96	M%
B	280	(N+27)	(N-5)%
C	(M+N+70)	M	N%
D	(4N+40)	84	30%

Note:

T he values of all the variables are positive integers.

The value of M is less than 50.

The total number of fast foods ordered = number of pizzas ordered + number of burgers ordered + number of sandwiches ordered

The total number of fast foods ordered =number of fast foods delivered + number of fast foods not delivered

The total number of fast foods delivered = number of fast foods returned + number of fast foods accepted

Out of the total number of orders (pizzas + burgers + sandwiches) in shop A, if the number of orders not delivered and the number of orders accepted out of the total number of orders delivered are equal, then find the total number of orders returned, if the ratio of the number of orders returned to the total number of orders delivered is 6:7.

a. 240

b. 200

c. 320

d. Cannot be determined

e. None of these

18. Questions

If the number of pizzas and burgers ordered from shop B is [W] and [Z] respectively and the price of each pizza is Rs. 40, whereas the price of each burger is Rs. 30 and the total revenue generated from selling all the pizzas and burgers is Rs. 5180. Find the total ratio of the number of pizzas and burgers ordered from shop B, if all the pizzas and burgers are delivered by shop B and also accepted by the customers.

a. 3:2

b. 4:7

c. 5:1

d. 1:2

e. None of these

19. Questions

In shop E, the number of pizzas, burgers and sandwiches ordered are in decreasing arithmetic progression respectively starting from 60 and the common difference between them is 20. Find the total number of fast foods (pizzas + burgers + sandwiches) ordered from shop E.

a. 110

b. 160

c. 120

d. 150

e. None of these

20. Questions

From shop D, if the difference between the number of sandwiches and burgers ordered is equal to the number of pizzas ordered, then find the number of burgers ordered from shop D, if it is more than the number of sandwiches ordered.

a. 120

b. 144

c. 112

d. 150

e. None of these

Explanations:

1. Questions

Answer: B

Article A

CP = Rs. 10000

MP = {10000+(X% of 10000)} = Rs. (10000+100X)

SP = 80% of (10000+100X) = Rs. (8000+80X)

Article B

CP = Rs. 8000

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MP = {8000+(2Y% of 8000)} = Rs. (8000+160Y)

SP = {8000+(Y% of 8000)} = Rs. (8000+80Y)

Now,

(8000+80X)/(8000+80Y) = 6/5

40000 + 400X = 48000 + 480Y

400X – 480Y = 8000…….(i)

Also,

(10000+100X) – (8000+160Y) = 3000

7000 + 100X = 8000 + 160Y

100X – 160Y = 1000…….(ii)

Solving both the equation, we get,

400X – 480Y = 8000

400X – 640Y = 4000

160Y = 4000

Y = 25

Now, put the value of Y in equation (i), we get,

400X – 480Y = 8000

400X – 12000 = 8000

400X = 20000

X = 50

Hence, the value of (2X-3Y) = (100-75) = 25.

2. Questions

Answer: C

Speed of the car = 4V km/h

Speed of the bike = 3V km/h

The total distance between city P and city Q = 210 km

The time taken by the car to reach city R from city P = T hours

And, the time taken by the bike to reach city R from city P = T hours

Now,

The total distance covered by the car = (4V×T) = 4VT km

And, the total distance covered by the bike = (210-4VT)

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So,

(210-4VT)/3V = T

210 – 4VT = 3VT

7VT = 210

VT = 30

Hence, the value of (V×T) = 30.

3. Questions

Answer: A

Let, the present ages of,

X's brother = a years

X's father = 4a years

X = b years

X's mother = (b+25) years

Now,

4a – 4 = b + 25

4a – b = 29……(i)

Also,

a – b = 2……(ii)

Solving both the equations, we get,

4a – b = 29

4a – 4b = 8

3b = 21

b = 7

Now, put the value of b in equation (i), we get,

4a – b = 29

4a – 7 = 29

4a = 36

a = 9

Now, the present ages of,

X's brother = a = 9 years

X's father = 4a = 36 years

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X = b = 7 years

X's mother = (b+25) = (7+25) = 32 years

Hence, the required average = {(9+36+7+32)/4} = (84/4) = 21 years.

4. Questions

Answer: E

Total investment made by,

M = (P×12) = Rs. 12P

N = {(P-4000)×(12-5)} = {(P-4000)×7} = Rs. (7P-28000)

Now,

12P/(7P-28000) = 15/7

84P = 105P – 420000

21P = 420000

P = 20000

Hence, the initial investment made by M = P = Rs. 20000.

5. Questions

Answer: A

Equation

$x^2 - ax + b = 0$

Sum of the roots of the equation = a

Possible values of a = 11, 13, 17 or 19 (2-digit prime numbers less than 20)

One of the roots of the equation,

Sum of the smallest prime number and second smallest prime number = (2+3) = 5

So, one of the roots of the equation = 5.

Now, let the sum of the roots (a) = 11

So, other roots of the equation = (11-5) = 6

Product of the roots (b) = (5×6) = 30 (not possible)

As, the product should be more than 30.

Now, let the sum of the roots (a) = 13

So, other roots of the equation = (13-5) = 8

Product of the roots (b) = (5×8) = 40 (possible)

As, 40 is more than 30 and less than 50.

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But, also check the sum the roots = 17

So, other root of the equation = (17-5) = 12

Product of the roots (b) = (5×12) = 60 (not possible)

As, 60 is more than 50.

Hence, the confirmed values of,

a = 13 (8+5)

b = 40

So, the greater root of the equation = 4P = 8

Therefore, P = (8/4) = 2

And, the smaller roots of the equation = 5Q = 5

Hence, Q = (5/5) = 1.

Now, the value of $\sqrt{\{9a+(b/10)\}} = \sqrt{\{117+(40/10)\}} = \sqrt{(117+4)} = \sqrt{121} = 11$.

6. Questions

Answer: B

Equation I

$a^2 + (p+1)a - 480$

Sum of the roots = -4

So,

(p+1) = 4

p = (4-1)

p = 3

Now, put the value of p in equation (i), we get,

$a^2 + (p+1)a - 480 = 0$

$a^2 + (3+1)a - 480 = 0$

$a^2 + 4a - 480 = 0$

Let, the greater root of the equation be x and smaller root be y.

Greater root = x

And, the smaller root = -y

So,

x – (-y) = 44

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$x + y = 44$……(i)

And,

$x + (-y) = -4$

$x – y = -4$……(ii)

Solving both the equations, we get,

$2y = 48$

$y = 24$

Now, put the value of y in equation (i), we get,

$x + y = 44$

$x + 24 = 44$

$x = 44 – 24$

$x = 20$

So, the greater root = x = 20

And, smaller roots = -y = -24

Equation II

$b^2 – (q+1)b – 340 = 0$

$b^2 – (q+1)b – 340 = 0$

The greater root of equation II is equal to the greater root of equation I.

So, the greater root of the equation II = 20

Product of the roots of the equation II = -340

So, let, the smaller root of the equation II be -z.

Now,

$\{20\times(-z)\} = -20z$

So,

$-20z = -340$

$z = 17$.

Therefore, the greater root = 20

And, the smaller root = -17

So, the sum of the roots = $\{20+(-17)\} = (20-17) = 3$

Now,

$(q+1) = 3$

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q = (3-1)

q = 2

So, the values of,

p = 3

q = 2

Hence, the difference between the values of p and q = (3-2) = 1.

7. Questions

Answer: C

Smallest composite number = 4

Therefore, one of the roots of the equation = 4

So, the other root could be either 3 or 5, as both the roots of the equation are consecutive positive numbers.

Now, let the other root of the equation be 3.

So, the product of the roots = (4×3) = 12 (possible)

As, 12 is more than 10 but less than 13.

But, also check the value of other root as 5

So, the product of the roots = (4×5) = 20 (not possible)

As, 20 is more than 13.

Hence, the confirmed roots of the equation = 4 and 3.

So, the product of the roots = n

n = (4×3) = 12

And, sum of the roots = m

m = (4+3) = 7

Hence, the value of (m×n) = (7×12) = 84.

8. Questions

Singing

The total number of students (boys + girls) participated = 150

The number of boys participated = (A+20)

The number of girls participated = (B-40)

So,

(A+20) + (B-40) = 150

A + B = 170……..(i)

Painting

The total number of students (boys + girls) participated = (4A-B-30)

The number of boys participated = 120

The number of girls participated = 80

So,

120 + 80 = (4A-B-30)

4A – B = 230……..(ii)

Solving both the equations, we get,

4A + 4B = 680

4A – B = 230

5B = 450

B = 90

Now, put the value of B in equation (i), we get,

A + B = 170

A + 90 = 170

A = 80

Singing

The total number of students (boys + girls) participated = 150

The number of boys participated = (A+20) = (80+20) = 100

The number of girls participated = (B-40) = (90-40) = 50

Painting

The total number of students (boys + girls) participated = (4A-B-30) = (320-90-30) = (320-120) = 200

The number of boys participated = 120

The number of girls participated = 80

Dancing

The total number of students (boys + girls) participated in all the activities together = (180×3) = 540

So, the total number of students (boys + girls) participated in dancing = {540-(150+200)} = (540-350) = 190

The number of boys participated in dancing = 130% of 100 = 130

The number of girls participated in dancing = 3/4 of 80 = 60

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Activities	Total number of students participated	Number of boys participated	Number of girls participated
Singing	150	100	50
Painting	200	120	80
Dancing	190	130	60
Total	540	350	190

Answer: B

According to the question,

Singing

The total number of boys participated = 100

The total number of girls participated = 50

Let, the number of boys participating in rock singing and girls participating in jazz singing be 7x and x respectively.

And, the ratio of the number of boys participating in jazz singing and girls participating in rock singing = 75:100 = 3:4. Let, it be 3y and 4y respectively.

Now,

7x + 3y = 100……..(i)

x + 4y = 50……(ii)

Solving both the equations, we get,

7x + 3y = 100

7x + 28y = 350

25y = 250

y = 10

Now, put the value of y in equation (i), we get,

7x + 3y = 100

7x + 30 = 100

7x = 70

x = 10

So, the number of boys participated in rock singing = 7x = 70

And, the number of boys participated in jazz singing = 3y = 30

Hence, the required % = {(70/30)×100}% = 233.33%.

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9. Questions

Singing

The total number of students (boys + girls) participated = 150

The number of boys participated = (A+20)

The number of girls participated = (B-40)

So,

(A+20) + (B-40) = 150

A + B = 170……..(i)

Painting

The total number of students (boys + girls) participated = (4A-B-30)

The number of boys participated = 120

The number of girls participated = 80

So,

120 + 80 = (4A-B-30)

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Singing

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The number of boys participated = (A+20) = (80+20) = 100

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Painting

The total number of students (boys + girls) participated = (4A-B-30) = (320-90-30) = (320-120) = 200

The number of boys participated = 120

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The number of girls participated = 80

Dancing

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So, the total number of students (boys + girls) participated in dancing = {540-(150+200)} = (540-350) = 190

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Activities	Total number of students participated	Number of boys participated	Number of girls participated
Singing	150	100	50
Painting	200	120	80
Dancing	190	130	60
Total	540	350	190

Answer: A

Drawing

The number of boys participated = 15N

The number of girls participated = (N^2-4)

Now,

$15N/(N^2-4) = 2/1$

$15N = 2N^2 - 8$

$2N^2 - 15N - 8 = 0$

$2N^2 - 16N + N - 8 = 0$

$2N(N-8) + 1(N-8) = 0$

$(2N+1)(N-8) = 0$

$N = -0.50$ or 8

So, N = 8 (positive integer)

Hence, the total number of students (boys + girls) participated in drawing = $\{15N+(N^2-4)\} = \{120+(8^2-4)\} = \{120+(64-4)\} = (120+60) = 180$.

10. Questions

Singing

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The total number of students (boys + girls) participated = 150

The number of boys participated = (A+20)

The number of girls participated = (B-40)

So,

(A+20) + (B-40) = 150

A + B = 170……..(i)

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The total number of students (boys + girls) participated = (4A-B-30)

The number of boys participated = 120

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So,

120 + 80 = (4A-B-30)

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Singing

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The number of boys participated = (A+20) = (80+20) = 100

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The total number of students (boys + girls) participated = (4A-B-30) = (320-90-30) = (320-120) = 200

The number of boys participated = 120

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Dancing

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Activities	Total number of students participated	Number of boys participated	Number of girls participated
Singing	150	100	50
Painting	200	120	80
Dancing	190	130	60
Total	540	350	190

Answer: D

Painting

The total number of boys participated = 120

So, the total revenue earned from all the boys = (120×100) = Rs. 12000

The total revenue earned from all the students (boys + girls) = Rs. 21600

Therefore, the total revenue earned from all the girls = (21600-12000) = Rs. 9600

The total number of girls participated = 80

Hence, the fees for each girl participated = (9600/80) = Rs. 120.

11. Questions

Singing

The total number of students (boys + girls) participated = 150

The number of boys participated = (A+20)

The number of girls participated = (B-40)

So,

(A+20) + (B-40) = 150

A + B = 170……..(i)

Painting

The total number of students (boys + girls) participated = (4A-B-30)

The number of boys participated = 120

The number of girls participated = 80

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So,

120 + 80 = (4A-B-30)

4A – B = 230……..(ii)

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The number of boys participated = 120

The number of girls participated = 80

Dancing

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Activities	Total number of students participated	Number of boys participated	Number of girls participated
Singing	150	100	50
Painting	200	120	80
Dancing	190	130	60
Total	540	350	190

Answer: B

Dancing

The percentage of the number of boys did not participated = P%

So, the percentage of the number of boys participated =(100-P)%

Now,

(100-P) – P = (P-5)

100 – 2P = P – 5

3P = 105

P = 35.

Hence, the value of √(2P+11) = √(70+11) = √81 = 9.

12. Questions

From the equation, we have,

$K^2 – 6K + 8 = 0$

$K^2 – 4K – 2K + 8 = 0$

K(K-4) – 2(K-4) = 0

(K-2) (K-4) = 0

K = 2 or 4

So, P = 2

And, Q = 4

Now,

The only prime numbers more than 2 but less than 7 = 3 and 5

So, the only possible values of S and T could be either 3 or 5.

So, S = 5

And, T = 3

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Now,

R = 1 (smallest natural number)

Laptop

The ratio of the number of items manufactured to that of unsold = 2.1P = (2.1×2) = 4.2 = 42/10 = 21/5 = 21:5

So, the ratio of the number of items sold to that of unsold = (21-5):5 = 16:5. Let, it be 16 units and 5 units respectively.

(16-5) units = 330

11 units = 330

1 unit = 30

Therefore, the number of laptops sold = 16 units = 480

And, the number of laptops unsold = 5 units = 150

So, the total number of laptops manufactured = (480+150) = 630

Mobiles

The ratio of the number of items manufactured to that of unsold = 1.4Q = (1.4×4) = 5.6 = 56/10 = 28/5 = 28:5

So, the ratio of the number of items sold to that of unsold = (28-5):5 = 23:5. Let it be 23 units and 5 units respectively.

The number of mobiles unsold = {(150/150)×100} = 100

5 units = 100

1 unit = 20

So, the number of mobiles sold = 23 units = 460

Therefore, the total number of mobiles manufactured = (460+100) = 560

Watch

The ratio of the number of items manufactured to that of unsold = 2.8R = (2.8×1) = 2.8 = 28/10 = 14/5 = 14:5

So, the ratio of the number of items sold to that of unsold = (14-5):5 = 9:5. Let it be 9 units and 5 units respectively.

The total number of watches unsold = 150

5 units = 150

1 unit = 30

So, the total number of watches sold = 9 units = 270

Therefore, the total number of mobiles manufactured = (270+150) = 420

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TV

The ratio of the number of items manufactured to that of unsold = 0.7S = (0.7×5) = 3.5 = 35/10 = 7/2 = 7:2

So, the ratio of the number of items sold to that of unsold = (7-2):2 = 5:2. Let it be 5 units and 2 units respectively.

The total number of TVs unsold = 80% of 100 = 80

2 units = 80

1 unit = 40

So, the total number of TVs sold = 5 units = 200

Therefore, the total number of mobiles manufactured = (200+80) = 280

Refrigerator

The ratio of the number of items manufactured to that of unsold = 1.5T = (1.5×3) = 4.5 = 45/10 = 9:2

So, the ratio of the number of items sold to that of unsold = (9-2):2 = 7:2. Let it be 7 units and 2 units respectively.

The total number of unsold = 80

2 units = 80

1 unit = 40

So, the total number of refrigerators sold = 7 units = 280

Therefore, the total number of refrigerators manufactured = (280+80) =360

Items	Total number of items manufactured	Number of items sold	Number of items unsold
Laptop	630	480	150
Mobile	560	460	100
Watch	420	270	150
TV	280	200	80
Refrigerator	360	280	80
Total	2250	1690	560

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Answer: C

According to the question,

Laptops

The total number of laptops manufactured = 630

The number of defective (unsold) laptops = 100

So, the total number of non-defective (sold + unsold) laptops = (630-100) = 530

The number of laptops (non-defective) sold = 480

Hence, the number of non-defective laptops which remain unsold = (530-480) = 50.

13. Questions

From the equation, we have,

$K^2 - 6K + 8 = 0$

$K^2 - 4K - 2K + 8 = 0$

$K(K-4) - 2(K-4) = 0$

$(K-2)(K-4) = 0$

$K = 2$ or 4

So, $P = 2$

And, $Q = 4$

Now,

The only prime numbers more than 2 but less than 7 = 3 and 5

So, the only possible values of S and T could be either 3 or 5.

So, S = 5

And, T = 3

Now,

R = 1 (smallest natural number)

Laptop

The ratio of the number of items manufactured to that of unsold = 2.1P = (2.1×2) = 4.2 = 42/10 = 21/5 = 21:5

So, the ratio of the number of items sold to that of unsold = (21-5):5 = 16:5. Let, it be 16 units and 5 units respectively.

(16-5) units = 330

11 units = 330

1 unit = 30

Therefore, the number of laptops sold = 16 units = 480

And, the number of laptops unsold = 5 units = 150

So, the total number of laptops manufactured = (480+150) = 630

Mobiles

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Therefore, the total number of mobiles manufactured = (270+150) = 420

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The total number of TVs unsold = 80% of 100 = 80

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So, the ratio of the number of items sold to that of unsold = (9-2):2 = 7:2. Let it be 7 units and 2 units respectively.

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The total number of unsold = 80

2 units = 80

1 unit = 40

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Therefore, the total number of refrigerators manufactured = (280+80) =360

Items	Total number of items manufactured	Number of items sold	Number of items unsold
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Watch	420	270	150
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Total	2250	1690	560

Answer: A

TV

The total number of TVs sold = 200

The ratio of the number of LEDs to that of LCDs sold = 4:1

So, the number of LEDs sold = 4/5 of 200 = 160

And, the number of LCDs sold = 1/5 of 200 = 40

Let, the number of black LEDs and white LCDs sold be 12a and a respectively.

And, the ratio of the number of white LEDs to that of black LCDs sold = 4:3. Let, it be 4b and 3b respectively.

Now,

12a + 4b = 160…….(i)

a + 3b = 40…….(ii)

Solving both the equations, we get,

12a + 4b = 160

12a + 36b = 480

32b = 320

b = 10

Now, put the value of b in equation (i), we get,

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12a + 4b = 160

12a + 40 = 160

12a = 120

a = 10

Therefore, the total number of white TVs (LED+LCD) sold = (4b+a) = (40+10) = 50

And, the total number of black TVs (LED+LCD) sold = (12a+3b) = (120+30) = 150

Hence, the required fraction = (50/150) = 1/3.

14. Questions

From the equation, we have,

$K^2 - 6K + 8 = 0$

$K^2 - 4K - 2K + 8 = 0$

K(K-4) – 2(K-4) = 0

(K-2) (K-4) = 0

K = 2 or 4

So, P = 2

And, Q = 4

Now,

The only prime numbers more than 2 but less than 7 = 3 and 5

So, the only possible values of S and T could be either 3 or 5.

So, S = 5

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Now,

R = 1 (smallest natural number)

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The number of mobiles unsold = {(150/150)×100} = 100

5 units = 100

1 unit = 20

So, the number of mobiles sold = 23 units = 460

Therefore, the total number of mobiles manufactured = (460+100) = 560

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Therefore, the total number of mobiles manufactured = (270+150) = 420

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So, the total number of TVs sold = 5 units = 200

Therefore, the total number of mobiles manufactured = (200+80) = 280

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The ratio of the number of items manufactured to that of unsold = 1.5T = (1.5×3) = 4.5 = 45/10 = 9:2

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TV	280	200	80
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Total	2250	1690	560

Answer: B

Mobile

The total number of mobiles manufactured = 560

CP of each mobile = Rs. 10000

So, the total CP all the mobiles manufactured = (560×10000) = Rs. 5600000

SP of each mobile = 120% of 10000 = Rs. 12000

So, the SP of all the mobiles sold = (460×12000) = Rs. 5520000

Hence, the overall loss incurred by the shopkeeper on selling the mobiles = (5600000-55200000) = Rs. 80,000.

15. Questions

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From the equation, we have,

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$K^2 - 4K - 2K + 8 = 0$

$K(K-4) - 2(K-4) = 0$

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(K-2) (K-4) = 0

K = 2 or 4

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So, the ratio of the number of items sold to that of unsold = (9-2):2 = 7:2. Let it be 7 units and 2 units respectively.

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2 units = 80

1 unit = 40

So, the total number of refrigerators sold = 7 units = 280

Therefore, the total number of refrigerators manufactured = (280+80) =360

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Items	Total number of items manufactured	Number of items sold	Number of items unsold
Laptop	630	480	150
Mobile	560	460	100
Watch	420	270	150
TV	280	200	80
Refrigerator	360	280	80
Total	2250	1690	560

Answer: D

Watches

The total number of watches sold = 270

The total number of watches unsold = 150

Now, let the number of digital watches sold and analog watches unsold be 4m and m respectively.

And, also the ratio of the number of analog watches sold to that of digital watches unsold = 70:100 = 7:10. Let it be 7n and 10n respectively.

Now,

4m + 7n = 270…….(i)

m + 10n = 150……(ii)

Solving both the equations, we get,

4m + 7n = 270

4m + 40n = 600

33n = 330

n = 10

Now, put the value of n in equation (i), we get,

4m + 7n = 270

4m + 70 = 270

4m = 200

m = 50

So, the total number of digital watches manufactured = (4m+10n) = (200+100) = 300

And, the total number of analog watches manufactured = (7n+m) = (70+50) = 120

Hence, the required difference = (300-120) = 180.

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16. Questions

From the equation, we have,

$K^2 - 6K + 8 = 0$

$K^2 - 4K - 2K + 8 = 0$

$K(K-4) - 2(K-4) = 0$

$(K-2)(K-4) = 0$

$K = 2$ or 4

So, $P = 2$

And, $Q = 4$

Now,

The only prime numbers more than 2 but less than 7 = 3 and 5

So, the only possible values of S and T could be either 3 or 5.

So, S = 5

And, T = 3

Now,

R = 1 (smallest natural number)

Laptop

The ratio of the number of items manufactured to that of unsold = 2.1P = (2.1×2) = 4.2 = 42/10 = 21/5 = 21:5

So, the ratio of the number of items sold to that of unsold = (21-5):5 = 16:5. Let, it be 16 units and 5 units respectively.

(16-5) units = 330

11 units = 330

1 unit = 30

Therefore, the number of laptops sold = 16 units = 480

And, the number of laptops unsold = 5 units = 150

So, the total number of laptops manufactured = (480+150) = 630

Mobiles

The ratio of the number of items manufactured to that of unsold = 1.4Q = (1.4×4) = 5.6 = 56/10 = 28/5 = 28:5

So, the ratio of the number of items sold to that of unsold = (28-5):5 = 23:5. Let it be 23 units and 5 units respectively.

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The number of mobiles unsold = {(150/150)×100} = 100

5 units = 100

1 unit = 20

So, the number of mobiles sold = 23 units = 460

Therefore, the total number of mobiles manufactured = (460+100) = 560

Watch

The ratio of the number of items manufactured to that of unsold = 2.8R = (2.8×1) = 2.8 = 28/10 = 14/5 = 14:5

So, the ratio of the number of items sold to that of unsold = (14-5):5 = 9:5. Let it be 9 units and 5 units respectively.

The total number of watches unsold = 150

5 units = 150

1 unit = 30

So, the total number of watches sold = 9 units = 270

Therefore, the total number of mobiles manufactured = (270+150) = 420

TV

The ratio of the number of items manufactured to that of unsold = 0.7S = (0.7×5) = 3.5 = 35/10 = 7/2 = 7:2

So, the ratio of the number of items sold to that of unsold = (7-2):2 = 5:2. Let it be 5 units and 2 units respectively.

The total number of TVs unsold = 80% of 100 = 80

2 units = 80

1 unit = 40

So, the total number of TVs sold = 5 units = 200

Therefore, the total number of mobiles manufactured = (200+80) = 280

Refrigerator

The ratio of the number of items manufactured to that of unsold = 1.5T = (1.5×3) = 4.5 = 45/10 = 9:2

So, the ratio of the number of items sold to that of unsold = (9-2):2 = 7:2. Let it be 7 units and 2 units respectively.

The total number of unsold = 80

2 units = 80

1 unit = 40

So, the total number of refrigerators sold = 7 units = 280

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Therefore, the total number of refrigerators manufactured = (280+80) =360

Items	Total number of items manufactured	Number of items sold	Number of items unsold
Laptop	630	480	150
Mobile	560	460	100
Watch	420	270	150
TV	280	200	80
Refrigerator	360	280	80
Total	2250	1690	560

Answer: A

Washing Machine

The total number of washing machines manufactured = (360-60) = 300

The number of washing machines sold = X% of 300 = 3X

The number of washing machines unsold = (300-3X)

Now,

3X – (300-3X) = (X+100)

2X – 100 = 300 – 3X

5X = 400

X = 80

Hence, the value of √(X+20) = √(80+20) = √100 = 10.

17. Questions

Shop A

The total number of fast foods ordered = 8M

The sum of the number of pizzas and burgers ordered = (96×2) = 192

The number of sandwiches ordered = M% of 8M = $(8M^2/100)$

Now,

$8M – 192 = 8M^2/100$

$800M – 19200 = 8M^2$

$8M^2 – 800M + 19200 = 0$

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$M^2 - 100M + 2400 = 0$

$M^2 - 60M - 40M + 2400 = 0$

$M(M-60) - 40(M-60) = 0$

$(M-40)(M-60) = 0$

$M = 40$ or 60

So, $M = 40$ (less than 50)

Therefore, the total number of fast foods ordered = $8M = 320$

And, the number of sandwiches ordered = $(8M^2/100) = \{(8\times40\times40)/100\} = 128$

Shop B

The total number of fast foods ordered = 280

The sum of the number of pizzas and burgers ordered = $\{(N+27)\times2\} = (2N+54)$

The number of sandwiches ordered = (N-5)% of 280 = (2.80N-14)

Now,

$280 - (2N+54) = 2.80N - 14$

$294 - 2.80N = 2N + 54$

$4.80N = 240$

$N = 50$

Therefore, the sum of the number of pizzas and burgers ordered = $(2N+54) = (100+54) = 154$

And, the number of sandwiches ordered = $(2.80N-14) = (140-14) = 126$

Shop C

The total number of fast foods ordered = $(M+N+70) = (40+50+70) = 160$

The sum of the number of pizzas and burgers ordered = $(M\times2) = 2M = 80$

The number of sandwiches ordered = 50% of 160 = 80

Shop D

The total number of fast foods ordered = $(4N+40) = (200+40) = 240$

The sum of the number of pizzas and burgers ordered = $(M\times2) = 2M = 80$

The number of sandwiches ordered = 30% of 240 = 72

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Shop	Total number of fast foods ordered	Sum of the number of pizzas and burgers ordered	Number of sandwiches ordered
A	320	192	128
B	280	154	126
C	160	80	80
D	240	168	72
Total	1000	594	406

Answer: A

According to the question,

Shop A

The total number of orders delivered = 320

Let, the number of orders not delivered be p.

So, the number of orders delivered = (320-p)

Now, the number of orders accepted = p

So, the number of orders returned = {(320-p)-p} = (320-2p)

Now,

(320-2p)/(320-p) = 6/7

2240 – 14p = 1920 – 6p

8p = 320

p = 40

Hence, the number of orders returned = (320-2p) = (320-80) = 240.

18. Questions

Shop A

The total number of fast foods ordered = 8M

The sum of the number of pizzas and burgers ordered = (96×2) = 192

The number of sandwiches ordered = M% of 8M = $(8M^2/100)$

Now,

$8M – 192 = 8M^2/100$

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$800M - 19200 = 8M^2$

$8M^2 - 800M + 19200 = 0$

$M^2 - 100M + 2400 = 0$

$M^2 - 60M - 40M + 2400 = 0$

$M(M-60) - 40(M-60) = 0$

$(M-40)(M-60) = 0$

$M = 40$ or 60

So, $M = 40$ (less than 50)

Therefore, the total number of fast foods ordered = $8M = 320$

And, the number of sandwiches ordered = $(8M^2/100) = \{(8\times40\times40)/100\} = 128$

Shop B

The total number of fast foods ordered = 280

The sum of the number of pizzas and burgers ordered = $\{(N+27)\times2\} = (2N+54)$

The number of sandwiches ordered = (N-5)% of 280 = (2.80N-14)

Now,

$280 - (2N+54) = 2.80N - 14$

$294 - 2.80N = 2N + 54$

$4.80N = 240$

$N = 50$

Therefore, the sum of the number of pizzas and burgers ordered = $(2N+54) = (100+54) = 154$

And, the number of sandwiches ordered = $(2.80N-14) = (140-14) = 126$

Shop C

The total number of fast foods ordered = $(M+N+70) = (40+50+70) = 160$

The sum of the number of pizzas and burgers ordered = $(M\times2) = 2M = 80$

The number of sandwiches ordered = 50% of 160 = 80

Shop D

The total number of fast foods ordered = $(4N+40) = (200+40) = 240$

The sum of the number of pizzas and burgers ordered = $(M\times2) = 2M = 80$

The number of sandwiches ordered = 30% of 240 = 72

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Shop	Total number of fast foods ordered	Sum of the number of pizzas and burgers ordered	Number of sandwiches ordered
A	320	192	128
B	280	154	126
C	160	80	80
D	240	168	72
Total	1000	594	406

Answer: B

Shop B

The total number of pizzas and burgers ordered = 154

The number of pizzas ordered = W

The number of burgers ordered = Z

So, the total revenue generated from selling all the pizzas = (W×40) = Rs. 40W

And, the total revenue generated from selling all the burgers = (Z×30) = Rs. 30Z

Now,

W + Z = 154……..(i)

40W + 30Z = 5180……..(ii)

Solving both the equations, we get,

40W + 40Z = 6160

40W + 30Z = 5180

10Z = 980

Z = 98

Now, put the value of Z in equation (i), we get,

W + Z = 154

W + 98 = 154

W = 56

Hence, the ratio of the number of pizzas and burgers ordered = W:Z = 56:98 = 4:7.

19. Questions

Shop A

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The total number of fast foods ordered = 8M

The sum of the number of pizzas and burgers ordered = (96×2) = 192

The number of sandwiches ordered = M% of 8M = ($8M^2$/100)

Now,

8M – 192 = $8M^2$/100

800M – 19200 = $8M^2$

$8M^2$ – 800M + 19200 = 0

M^2 – 100M + 2400 = 0

M^2 – 60M – 40M + 2400 = 0

M(M-60) – 40(M-60) = 0

(M-40) (M-60) = 0

M = 40 or 60

So, M = 40 (less than 50)

Therefore, the total number of fast foods ordered = 8M = 320

And, the number of sandwiches ordered = ($8M^2$/100) = {(8×40×40)/100} = 128

Shop B

The total number of fast foods ordered = 280

The sum of the number of pizzas and burgers ordered = {(N+27)×2} = (2N+54)

The number of sandwiches ordered = (N-5)% of 280 = (2.80N-14)

Now,

280 – (2N+54) = 2.80N – 14

294 – 2.80N = 2N + 54

4.80N = 240

N = 50

Therefore, the sum of the number of pizzas and burgers ordered = (2N+54) = (100+54) = 154

And, the number of sandwiches ordered = (2.80N-14) = (140-14) = 126

Shop C

The total number of fast foods ordered = (M+N+70) = (40+50+70) = 160

The sum of the number of pizzas and burgers ordered = (M×2) = 2M = 80

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The number of sandwiches ordered = 50% of 160 = 80

Shop D

The total number of fast foods ordered = (4N+40) = (200+40) = 240

The sum of the number of pizzas and burgers ordered = (M×2) = 2M = 80

The number of sandwiches ordered = 30% of 240 = 72

Shop	Total number of fast foods ordered	Sum of the number of pizzas and burgers ordered	Number of sandwiches ordered
A	320	192	128
B	280	154	126
C	160	80	80
D	240	168	72
Total	1000	594	406

Answer: C

Shop E

The only possible numbers of pizzas, burgers and sandwiches ordered from shop E are 60, 40 and 20 respectively.

So, the number of pizzas ordered = 60

The number of burgers ordered = 40

The number of sandwiches ordered = 20

Hence, the total number of fast foods (pizzas + burgers + sandwiches) ordered from shop E = (60+40+20) = 120.

20. Questions

Shop A

The total number of fast foods ordered = 8M

The sum of the number of pizzas and burgers ordered = (96×2) = 192

The number of sandwiches ordered = M% of 8M = ($8M^2$/100)

Now,

8M – 192 = $8M^2$/100

800M – 19200 = $8M^2$

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$8M^2 – 800M + 19200 = 0$

$M^2 – 100M + 2400 = 0$

$M^2 – 60M – 40M + 2400 = 0$

M(M-60) – 40(M-60) = 0

(M-40) (M-60) = 0

M = 40 or 60

So, M = 40 (less than 50)

Therefore, the total number of fast foods ordered = 8M = 320

And, the number of sandwiches ordered = $(8M^2/100)$ = {(8×40×40)/100} = 128

Shop B

The total number of fast foods ordered = 280

The sum of the number of pizzas and burgers ordered = {(N+27)×2} = (2N+54)

The number of sandwiches ordered = (N-5)% of 280 = (2.80N-14)

Now,

280 – (2N+54) = 2.80N – 14

294 – 2.80N = 2N + 54

4.80N = 240

N = 50

Therefore, the sum of the number of pizzas and burgers ordered = (2N+54) = (100+54) = 154

And, the number of sandwiches ordered = (2.80N-14) = (140-14) = 126

Shop C

The total number of fast foods ordered = (M+N+70) = (40+50+70) = 160

The sum of the number of pizzas and burgers ordered = (M×2) = 2M = 80

The number of sandwiches ordered = 50% of 160 = 80

Shop D

The total number of fast foods ordered = (4N+40) = (200+40) = 240

The sum of the number of pizzas and burgers ordered = (M×2) = 2M = 80

The number of sandwiches ordered = 30% of 240 = 72

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Shop	Total number of fast foods ordered	Sum of the number of pizzas and burgers ordered	Number of sandwiches ordered
A	320	192	128
B	280	154	126
C	160	80	80
D	240	168	72
Total	1000	594	406

Answer: A

Shop D

The total number of fast foods (pizzas + burgers + sandwiches) ordered = 240

The number of sandwiches ordered = 72

Let, the number of pizzas ordered be X.

So, the difference between the number of burgers and sandwiches ordered = X. (burgers-sandwiches)

Therefore, the number of burgers ordered = (72+X)

Now,

72 + X + (72+X) = 240

2X = 240 – 144

2X = 96

X = 48

Hence, the number of burgers ordered = (72+X) = (72+48) = 120.

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1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the total transaction (Debited amount + Credited amount) in two bank accounts A and B of a person in five months January, February, March, April and May.

The table given below shows the following data:

Month	Ratio of total transaction in account A to that in account B	Ratio of debited amount to credited amount in account A	Ratio of debited amount to credited amount in account B
January	4: 3	3: 5	2: 1
February	1: 2	11: 9	3: 5
March	2: 3	3: 2	3: 2
April	3: 1	4: 5	1: 3
May	3: 5	2: 3	2: 1

If the total transaction in account A in February is Rs. 8 lakhs and the total transaction in account B in March is Rs. 15 lakhs, then find that the debited amount from account B in February will be what per cent of the credited amount in account A in March?

a. 120%

b. 133.33%

c. 150%

d. 125%

e. 166.67%

2. Questions

If the ratio of the credited amount in account A in January to that in account B in January is M: N, then find that which of the following statements is/are true?

P: Value of (3M – 2N) is a prime number.

Q: Value of $(M^2 – N^2)$ is divisible by 7.

R: Value of $(M^2 + N^2)$ is a prime number.

a. All are true

b. Only P and R

c. Only Q

d. Only P and Q

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e. Only R

3. Questions

If the difference between the total transaction in accounts A and B in May is Rs. 6 lakhs, then what will be the difference between the credited amounts in accounts A and B in May?

a. Rs. 6.4 lakhs

b. Rs. 0.4 lakhs

c. Rs. 1.4 lakhs

d. Rs. 4.4 lakhs

e. Rs. 2.4 lakhs

4. Questions

If the ratio of total transaction in account A in April to that in May is 2: 1, then find the ratio of the total debited amount from accounts A and B together in April to the total debited amount from accounts A and B together in May?

a. 19: 34

b. 95: 68

c. 19: 17

d. 57: 68

e. 95: 136

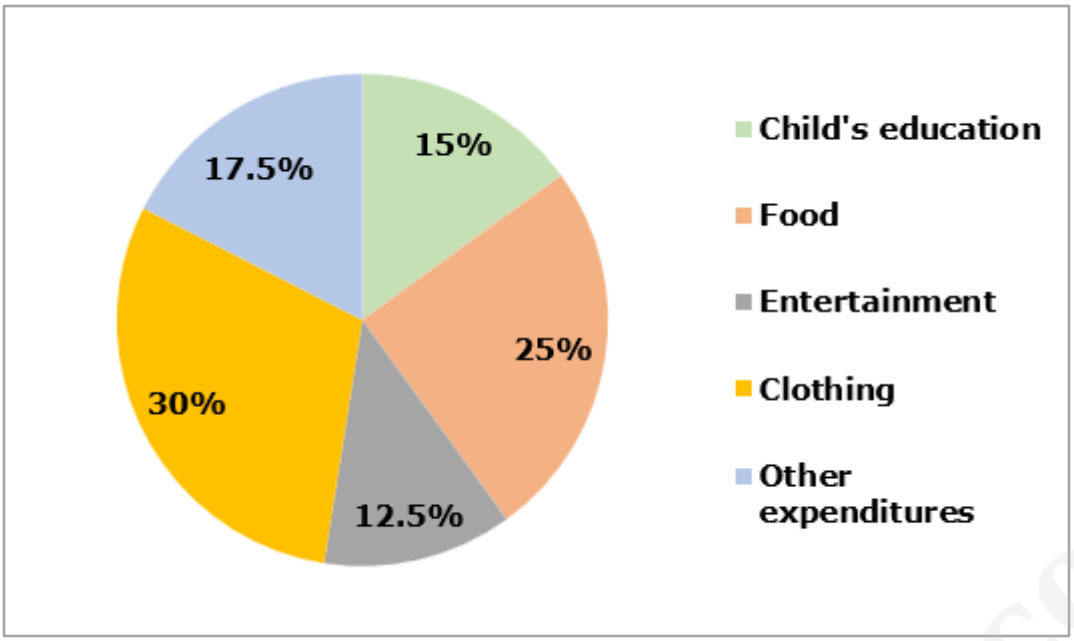
5. Questions

Study the following data carefully and answer the questions:

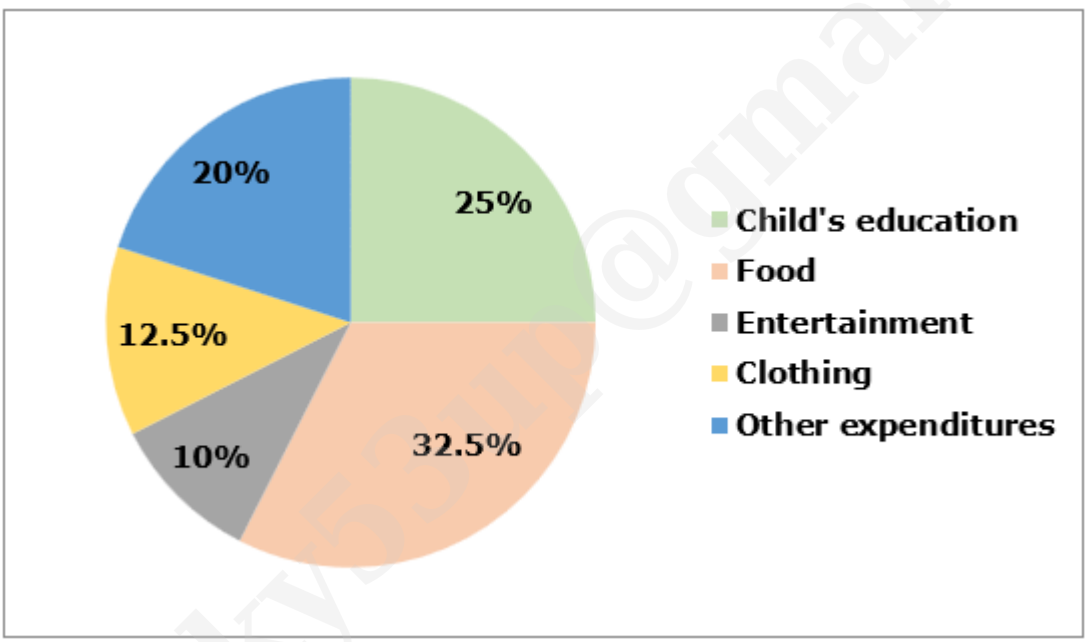
The data given below is related to the expenditures of two families on child's education, food, entertainment, clothing and other.

The pie chart given below shows the percentage distribution of the expenditures on child's education, food, entertainment, clothing and other expenditures of family A.

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The pie chart given below shows the percentage distribution of the expenditures on child's education, food, entertainment, clothing and other expenditures of family B.



Notes:

1). The ratio of the expenditure on child's education in family A to that in family B is 9:10.

If the difference between the expenditures on child's education in families A and B is Rs. 400 and the total expenditure of family B is 40% of the income of family B, then what will be the savings of family B?

a. Rs. 20000

b. Rs. 24000

c. Rs. 16000

d. Rs. 25000

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e. Rs. 18000

6. Questions

If the expenditure on clothing in family A is Rs. 3000 more than other expenditures in family A, then what will be the difference between the expenditure on clothing in family B and other expenditures in family B?

a. Rs. 2200

b. Rs. 720

c. Rs. 1620

d. Rs. 420

e. Rs. 1200

7. Questions

If the difference between the expenditure on food and that on entertainment in family A is Rs. 3000 and the expenditure on food in family A is 12.5% of the total income of family A, then what will be the savings of family A?

a. Rs. 24000

b. Rs. 16000

c. Rs. 20000

d. Rs. 12000

e. Can't be determined

8. Questions

Which of the following statements is/are true?

P: Ratio of the total expenditure on food in families A and B together to that on clothing in families A and B together is 28:23.

Q: Other expenditure in family A is 32.5 per cent more than that in family B.

R: Ratio of the expenditure on clothing in family A to that in family B is 16:5.

a. Only P and Q

b. All are true

c. Only R

d. Only P

e. Only Q and R

9. Questions

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If the ratio of the expenditure on entertainment in family B to that in family C is 2: 3, the ratio of other expenditures in family B to that in family C is 8: 7 and the difference between the expenditure on entertainment and other expenditures in family C is Rs. 350, then find the difference between the expenditure on entertainment and other expenditures in family A?

a. Rs. 1150
b. Rs. 1650
c. Rs. 1250
d. Rs. 1050
e. Rs. 1750

10. Questions

Each question given below consists of a question and two statements I and II. You have to decide whether the data provided in the statements are sufficient to answer the question.

In mixture A of milk and water, the quantity of water is 25%. Ratio of milk in mixture A to that in mixture B is 5: 4 and the ratio of water in mixture B to that in mixture A is 3: 5. Find the cost of mixture B?

Statement I: The cost of mixture A is Rs. 45 per L.

Statement II: The cost of pure milk is Rs. 48 per L.

a. Statement I alone is sufficient to answer the question but statement II alone is not sufficient
b. Statement II alone is sufficient to answer the question but statement I alone is not sufficient
c. Both the statements together are necessary to answer the questions
d. Either statement I alone or statement II alone is sufficient to answer the question
e. Both the statements together are not sufficient to answer the question

11. Questions

A shopkeeper marked up an article by Rs. X and sold at Rs. 350 profit after giving some discount on its marked price. Find the profit per cent?

Statement I: If the shopkeeper had marked up the article by 30% and sold at 20% discount on its marked price, there would have been a profit of Rs. 280.

Statement II: The discount given on the marked price of the article is Rs. 1400.

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a. Statement I alone is sufficient to answer the question but statement II alone is not sufficient
b. Statement II alone is sufficient to answer the question but statement I alone is not sufficient
c. Both the statements together are necessary to answer the questions
d. Either statement I alone or statement II alone is sufficient to answer the question
e. Both the statements together are not sufficient to answer the question

12. Questions

The difference between compound interest and simple interest on Rs. X at R% rate after 2 years is (9/400) times of Rs. X. Find the value of X?

Statement I: The difference between compound interest and simple interest on Rs. (X + 400) at R% rate after 2 years is Rs. 63.

Statement II: The simple interest received on Rs. (X – 400) at R% rate after 3 years is Rs. 900.

a. Statement I alone is sufficient to answer the question but statement II alone is not sufficient
b. Statement II alone is sufficient to answer the question but statement I alone is not sufficient
c. Both the statements together are necessary to answer the questions
d. Either statement I alone or statement II alone is sufficient to answer the question
e. Both the statements together are not sufficient to answer the question

13. Questions

Train A (200 m long) crosses a 'd' m long tunnel in 28 seconds and train B (180 m long) crosses the same tunnel in 30 seconds. Find the speed of train B?

Statement I: Train C (240 m long) crosses the same tunnel in 25 seconds.

Statement II: Train A crosses train D in 10 seconds, while running in opposite direction and the speed of train D is 18 m/s.

a. Statement I alone is sufficient to answer the question but statement II alone is not sufficient
b. Statement II alone is sufficient to answer the question but statement I alone is not sufficient
c. Both the statements together are necessary to answer the questions
d. Either statement I alone or statement II alone is sufficient to answer the question
e. Both the statements together are not sufficient to answer the question

14. Questions

Study the following data carefully and answer the questions:

There are three rooms A, B and C in a house. Their lengths (in feet) and widths (in feet) are different from each other, but their heights (in feet)are equal.

The length of room A is (x + 3) feet and also 10x% of its width. Average of the lengths of rooms A and B is (3y + 2) feet and the width of room B is (x/y) feet more than the width of room A. The height of each room is (x – y) feet, the width of room C is also (x – y) feet and the volume of room C is 960 feet^3 less than that of volume A.

Values of 'x' and 'y' can be calculated from the equations given below.

I: $x^2 + 9x – 360 = 0$ ('x' is a positive integer)

II: $7y - 2x = 5$

If the cost of installing PVC panel on wall is Rs. 25 per square feet and on roof is Rs. 40 per square feet, then find the total cost of installing PVC panel in room B? (excluding the floor)

a. Rs. 24300

b. Rs. 25100

c. Rs. 21500

d. Rs. 23400

e. Rs. 24500

15. Questions

If the cost of painting a wall is Rs. 18 per square feet, then find the difference between the total cost of painting walls of room A and the total cost of painting walls of room C?

a. Rs. 1860

b. Rs. 2410

c. Rs. 1690

d. Rs. 2250

e. Rs. 2880

16. Questions

What is the maximum number of cuboidal boxes, which can be put inside room B, if the length, the width and the height of each box are 8 feet, 6 feet and 5 feet respectively?

a. $(x - 3)$

b. $(y + 3)$

c. $(x - y)$

d. $(x + y)/4$

e. $(2x - 3y)$

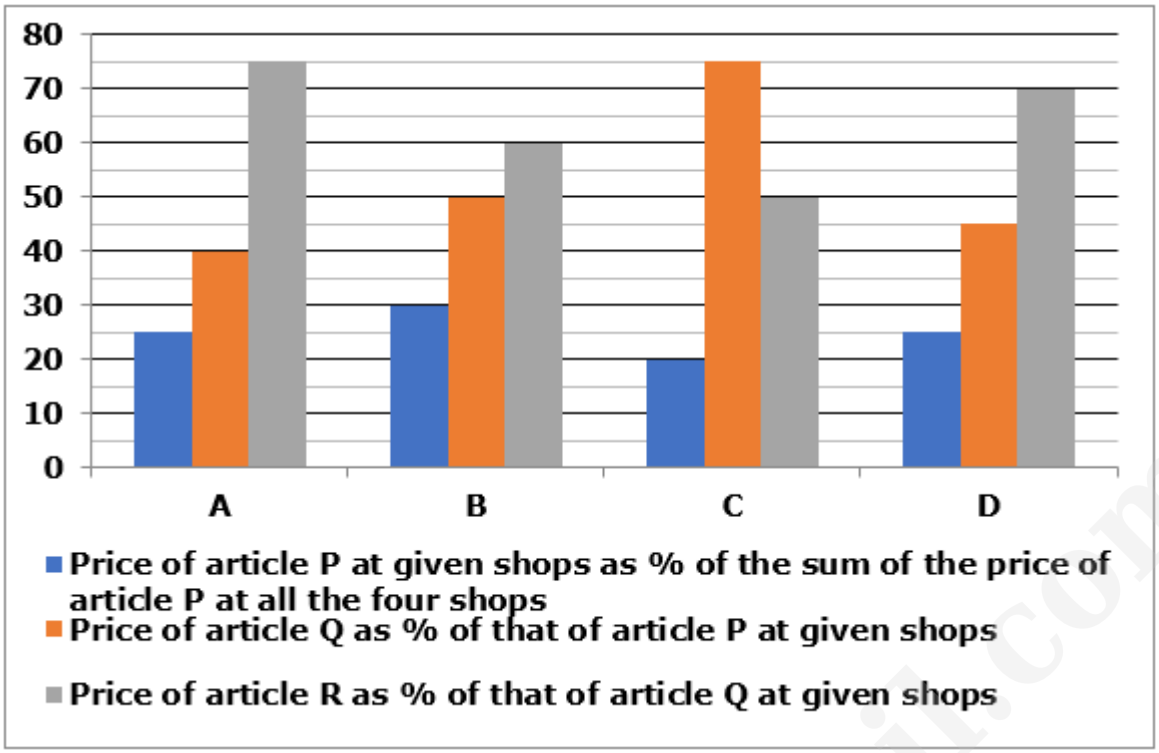
17. Questions

Study the following data carefully and answer the questions:

The data given below is relate to the prices (in Rs.) of three articles P, Q and R in four shops A, B, C and D.

The bar graph given below shows the following data:

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Note: The price of article R at shop B is Rs. 108 more than that of article R at shop A.

If shop B sold article P at 5% profit of its price, article Q at 2.5% profit of its price and article R at 7.5% profit of its price, then find the average profit received by shop B at each of the given articles?

a. Rs. 51.2

b. Rs. 81.2

c. Rs. 41.2

d. Rs. 71.2

e. Rs. 61.2

18. Questions

If the material cost of article P is equal to the price of article R at shop C and the labour cost of article P is Rs. 12 more than the price of article R at shop B, then find that at what profit per cent (at manufacturing cost) did shop C sold article P?

Manufacturing cost = Material cost + Labour cost

a. 25%

b. 12%

c. 20%

d. 14.4%

e. 24%

19. Questions

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If shops A, B, C and D sold article Q at 5%, 25, 30% and 10% discounts on the price of article Q at the respective shops, then find that which shop sold article Q at lowest price?

a. Shop A
b. Shop D
c. Shop B
d. Shop C
e. Both shops A and B

20. Questions

Which of the following statements is/are true?

X: Average price of article P at shops B, C and D is Rs. 1800.

Y: Price of article R at shop B is 20% more than that of article R at shop C.

Z: Average price of articles P, Q and R at shop D is Rs. 1059.

a. Only X and Y
b. Only X and Z
c. Only X
d. All are true
e. Only Y and Z

Explanations:

1. Questions

Answer: C

Since, the total transaction in account A in February = Rs. 8 lakhs

So, the total transaction in account B in February = 8 * (2/1) = Rs. 16 lakhs

And the debited amount from account B in February = 16 * (3/8) = Rs. 6 lakhs

Since, the total transaction in account B in March = Rs. 15 lakhs

So, the total transaction in account A in March = 15 * (2/3) = Rs. 10 lakhs

And the credited amount from account A in March = 10 * (2/5) = Rs. 4 lakhs

The required percentage = (6/4) * 100 = 150%

2. Questions

Answer: A

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Let the total transaction in accounts A and B in January be Rs. 40x and Rs. 30x respectively.

So, the credited amount in account A in January = 40x * (5/8) = Rs. 25x

And the credited amount in account B in January = 30x * (1/3) = Rs. 10x

So, M: N = 25x: 10x = 5: 2

From P:

Value of (3M – 2N) = (3 * 5 – 2 * 2) = 11

Since, 11 is a prime number.

So, P is true.

From Q:

Value of $(M^2 – N^2)$ = (25 – 4) = 21

Since, 21 is divisible by 7.

So, Q is true.

From R:

Value of $(M^2 + N^2)$ = (25 + 4) = 29

Since, 29 is a prime number.

So, R is true.

Hence, all are true.

3. Questions

Answer: B

Since, the difference between the total transactions in accounts A and B in May is Rs. 6 lakhs.

So, the total transaction in account A in May = 6 * [3/(5 – 3)] = Rs. 9 lakhs

And the total transaction in account B in May = 6 * [5/(5 – 3)] = Rs. 15 lakhs

Since, the credited amount in account A in May = 9 * (3/5) = Rs. 5.4 lakhs

And the credited amount in account B in May = 15 * (1/3) = Rs. 5 lakhs

So, the required difference = 5.4 – 5 = Rs. 0.4 lakhs

4. Questions

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Answer: E

Let the total transaction in account A in April and May be Rs. 6y and Rs. 3y respectively.

So, the total transaction in account B in April = 6y * (1/3) = Rs. 2y

And the total transaction in account B in May = 3y * (5/3) = Rs. 5y

Now, the debited amount from account A in April = 6y * (4/9) = Rs. (8y/3)

The debited amount from account B in April = 2y * (1/4) = Rs. (y/2)

The debited amount from account A in May = 3y * (2/5) = Rs. (6y/5)

The debited amount from account B in May = 5y * (2/3) = Rs. (10y/3)

Since, the debited amount from accounts A and B together in April:

(8y/3) + (y/2) = (16y + 3y)/6 = Rs. (19y/6)

And the debited amount from accounts A and B together in May:

(6y/5) + (10y/3) = (18y + 50y)/15 = Rs. (68y/15)

So, the required ratio = (19y/6): (68y/15) = 95: 136

5. Questions

Let the expenditure on child's education in family A = Rs. 45x

So, the expenditure on child's education in family B = 45x * (10/9) = Rs. 50x

Expenditure type	Expenditure in family A (in ₹)	Expenditure in family B (in ₹)
Child's education	45x	50x
Food	45x * (25/15) = 75x	50x * (32.5/25) = 65x
Entertainment	45x * (12.5/15) = 37.5x	50x * (10/25) = 20x
Clothing	45x * (30/15) = 90x	50x * (12.5/25) = 25x
Other expenditures	45x * (17.5/15) = 52.5x	50x * (20/25) = 40x

Answer: B

Since, the difference between the expenditures on child's education in families A and B = Rs. 400

So, the expenditure on child's education in family B = 400 * [50x/(50x – 45x)] = Rs. 4000

Since, the total expenditure of family B = 4000 * (100/25) = Rs. 16000

So, the income of family B = 16000 * (100/40) = Rs. 40000

And the savings of family B = 40000 – 16000 = Rs. 24000

6. Questions

Let the expenditure on child's education in family A = Rs. 45x

So, the expenditure on child's education in family B = 45x * (10/9) = Rs. 50x

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Expenditure type	Expenditure in family A (in ₹)	Expenditure in family B (in ₹)
Child's education	45x	50x
Food	45x * (25/15) = 75x	50x * (32.5/25) = 65x
Entertainment	45x * (12.5/15) = 37.5x	50x * (10/25) = 20x
Clothing	45x * (30/15) = 90x	50x * (12.5/25) = 25x
Other expenditures	45x * (17.5/15) = 52.5x	50x * (20/25) = 40x

Answer: E

Since, the expenditure on clothing in family A is Rs. 3000 more than other expenditures in family A.

So, 90x – 52.5x = 3000

x = 80

And the required difference = (40x – 25x) = (40 – 25) * 80 = Rs. 1200

7. Questions

Let the expenditure on child's education in family A = Rs. 45x

So, the expenditure on child's education in family B = 45x * (10/9) = Rs. 50x

Expenditure type	Expenditure in family A (in ₹)	Expenditure in family B (in ₹)
Child's education	45x	50x
Food	45x * (25/15) = 75x	50x * (32.5/25) = 65x
Entertainment	45x * (12.5/15) = 37.5x	50x * (10/25) = 20x
Clothing	45x * (30/15) = 90x	50x * (12.5/25) = 25x
Other expenditures	45x * (17.5/15) = 52.5x	50x * (20/25) = 40x

Answer: A

Since, the difference between the expenditure on food and that on entertainment in family A is Rs. 3000.

So, the expenditure on food in family A = 3000 * [75x/(75x – 37.5x)] = Rs. 6000

And the total expenditure of family A = 6000 * (100/25) = Rs. 24000

Since, the total income of family A = 6000 * (100/12.5) = Rs. 48000

So, the savings of family A = 48000 – 24000 = Rs. 24000

8. Questions

Let the expenditure on child's education in family A = Rs. 45x

So, the expenditure on child's education in family B = 45x * (10/9) = Rs. 50x

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Expenditure type	Expenditure in family A (in ₹)	Expenditure in family B (in ₹)
Child's education	45x	50x
Food	45x * (25/15) = 75x	50x * (32.5/25) = 65x
Entertainment	45x * (12.5/15) = 37.5x	50x * (10/25) = 20x
Clothing	45x * (30/15) = 90x	50x * (12.5/25) = 25x
Other expenditures	45x * (17.5/15) = 52.5x	50x * (20/25) = 40x

Answer: D

From P:

Ratio of the total expenditure on food in families A and B together to that on clothing in families A and B together:

(75x + 65x): (90x + 25x) = 140: 115 = 28: 23

So, P is true.

From Q:

Other expenditures in family A = 52.5x

Other expenditures in family B = 40x

Required percentage = [(52.5x – 40x)/40x] * 100 = 31.25%

So, Q is not true.

From R:

Ratio of the expenditure on clothing in family A to that in family B:

90x: 25x = 18: 5

So, R is not true.

Hence, only P is true.

9. Questions

Let the expenditure on child's education in family A = Rs. 45x

So, the expenditure on child's education in family B = 45x * (10/9) = Rs. 50x

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Expenditure type	Expenditure in family A (in ₹)	Expenditure in family B (in ₹)
Child's education	45x	50x
Food	45x * (25/15) = 75x	50x * (32.5/25) = 65x
Entertainment	45x * (12.5/15) = 37.5x	50x * (10/25) = 20x
Clothing	45x * (30/15) = 90x	50x * (12.5/25) = 25x
Other expenditures	45x * (17.5/15) = 52.5x	50x * (20/25) = 40x

Answer: D

Since, the expenditure on entertainment in family B = 20x

So, the expenditure on entertainment in family C = 20x * (3/2) = 30x

Since, the other expenditures in family B = 40x

So, the other expenditures in family C = 40x * (7/8) = 35x

Since, 35x – 30x = 350

x = 70

So, the required difference = (52.5 – 37.5) * 70 = Rs. 1050

10. Questions

Answer: D

Let the quantities of milk and water in mixture A be 75x L and 25x L.

So, the quantity of milk in mixture B = 75x * (4/5) = 60x L

And the quantity of water in mixture B = 25x * (3/5) = 15x L

From statement I alone:

Let the cost of pure milk = Rs. y per L.

Since,

(75x * y)/(75x + 25x) = 45

75y/100 = 45

y = 60

So, the cost of mixture B = (60x * 60)/(60x + 15x) = Rs. 48 per L.

So, statement I alone is sufficient to answer the question.

From statement II alone:

Since, the cost of pure milk is Rs. 48 per L.

So, the cost of mixture B = (60x * 48)/(60x + 15x) = Rs. 38.4 per L

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So, statement II alone is sufficient to answer the question.

Hence, either statement I alone or statement II alone is sufficient to answer the question.

11. Questions

Answer: A

Let the CP of the article = Rs. 100a

So, the MP of the article = Rs. (100a + X)

And the SP of the article = Rs. (100a + 350)

From statement I alone:

If the shopkeeper had marked up the article by 30% and sold at 20% discount on its marked price, there would have been a profit of Rs. 280.

Since, the CP of the article = Rs. 100a

So, the new MP of the article = 130% of 100a = Rs. 130a

And the new SP of the article = 80% of 130a = Rs. 104a

So, 104a – 100a = 280

a = 70

The CP of the article = 100 * 70 = Rs. 7000

So, the required profit per cent = (350/7000) * 100 = 5%

So, statement I alone is sufficient to answer the question.

From statement II alone:

Since, the discount given on the marked price of the article = Rs. 1400

So,

100a + X – 100a – 350 = 1400

X = 1750

Since, we can't find the value of 'a', so we can't find the CP of the article and the required profit per cent.

So, statement II alone is not sufficient to answer the question.

Hence, statement I alone is sufficient to answer the question but statement II alone is not sufficient.

12. Questions

Answer: D

Since, the difference between CI and SI on Rs. X at R% rate after 2 years is (9/400) times of Rs. X.

So,

$X * (R/100)^2 = 9X/400$

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R = 15

From statement I alone:

Since, the difference between CI and SI on Rs. (X + 400) at R% rate after 2 years is Rs. 63.

So,

$(X + 400) * (15/100)^2 = 63$

X + 400 = 2800

X = 2400

So, statement I alone is sufficient to answer the question.

From statement II alone:

Since, the SI received on Rs. (X – 400) at R% rate after 3 years is Rs. 900.

So,

[(X – 400) * 15 * 3]/100 = 900

X – 400 = 2000

X = 2400

So, statement II alone is sufficient to answer the question.

Hence, either statement I alone or statement II alone is sufficient to answer the question.

13. Questions

Answer: E

Let the speeds of trains A and B be 'a' m/s and 'b' m/s respectively.

So, a = (200 + d)/28 ---------(1)

And, b = (180 + d)/30 ---------(2)

From statement I alone:

Let the speed of train C = 'c' m/s

Since, train C (240 m long) crosses the same tunnel in 25 seconds.

So, c = (240 + d)/25 ---------(3)

Since, there are four variables and three equations, so none equation can be solved.

So, statement I alone is not sufficient to answer the question.

From statement II alone:

Let the length of train D = 'P' m

Since, train A crosses train D in 10 seconds, while running in opposite direction and the speed of train D is 18 m/s.

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So,

(a + 18) = (200 + P)/10 ----------(4)

Since, there are four variables and three equations, so none equation can be solved.

So, statement II alone is not sufficient to answer the question.

From statements I and II together:

a = (200 + d)/28 ---------(1)

b = (180 + d)/30 ---------(2)

c = (240 + d)/25 ---------(3)

(a + 18) = (200 + P)/10 ----------(4)

Since, there are five variables and four equations, so none equation can be solved.

Hence, both the statements together are not sufficient to answer the question.

14. Questions

From equation I:

$x^2 + 9x - 360 = 0$

$x^2 + 24x - 15x - 360 = 0$

x(x + 24) – 15(x + 24) = 0

x = -24, 15

Since, 'x' is positive integer.

So, x = 15

From equation II:

7y – 2x = 5

7y – 30 = 5

y = 5

The length of room A = 15 + 3 = 18 feet

The width of room A = 18 * (100/10x) = 18 * (100/150) = 12 feet

Since, the average of the lengths of rooms A and B = (3y + 2) = (3 * 5 + 2) = 17 feet

So, the length of room B = 2 * 17 – 18 = 16 feet

The width of room B = 12 + (x/y) = 12 + (15/5) = 15 feet

Since, the height of room A = (x – y) = (15 – 5) = 10 feet

So, the volume of room A = 18 * 12 * 10 = 2160 $feet^3$

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And the volume of room C = 2160 – 960 = 1200 $feet^3$

Since, the height of each room is equal.

So, the height of room B = 10 feet

And the height of room C = 10 feet

Since, the width of room C = (x – y) = (15 – 5) = 10 feet

So, the length of room C = 1200/(10 * 10) = 12 feet

Room	Length (in feet)	Width (in feet)	Height (in feet)
A	18	12	10
B	16	15	10
C	12	10	10

Answer: B

Since, the area of the roof of room B = 16 * 15 = 240 square feet

And the total area of walls of room B:

Total area = 2 (length *height + width* height)

2 * [(10 * 16) + (10 * 15)] = 620 square feet

So, the total cost of installing PVC panel in room B:

(620 * 25) + (240 * 40) = 15500 + 9600 = Rs. 25100

15. Questions

From equation I:

$x^2 + 9x – 360 = 0$

$x^2 + 24x – 15x – 360 = 0$

x(x + 24) – 15(x + 24) = 0

x = -24, 15

Since, 'x' is positive integer.

So, x = 15

From equation II:

7y – 2x = 5

7y – 30 = 5

y = 5

The length of room A = 15 + 3 = 18 feet

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The width of room A = 18 * (100/10x) = 18 * (100/150) = 12 feet

Since, the average of the lengths of rooms A and B = (3y + 2) = (3 * 5 + 2) = 17 feet

So, the length of room B = 2 * 17 – 18 = 16 feet

The width of room B = 12 + (x/y) = 12 + (15/5) = 15 feet

Since, the height of room A = (x – y) = (15 – 5) = 10 feet

So, the volume of room A = 18 * 12 * 10 = 2160 $feet^3$

And the volume of room C = 2160 – 960 = 1200 $feet^3$

Since, the height of each room is equal.

So, the height of room B = 10 feet

And the height of room C = 10 feet

Since, the width of room C = (x – y) = (15 – 5) = 10 feet

So, the length of room C = 1200/(10 * 10) = 12 feet

Room	Length (in feet)	Width (in feet)	Height (in feet)
A	18	12	10
B	16	15	10
C	12	10	10

Answer: E

Since, the total area of walls of room A:

Total area = 2 (length *height + width* height)

2 * [(18 * 10) + (12 * 10)] = 600 square feet

And the total area of walls of room C:

Total area = 2 (length *height + width* height)

2 * [(12 * 10) + (10 * 10)] = 440 square feet

So, the required difference = (600 – 440) * 18 = Rs. 2880

16. Questions

From equation I:

$x^2 + 9x – 360 = 0$

$x^2 + 24x – 15x – 360 = 0$

x(x + 24) – 15(x + 24) = 0

x = -24, 15

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Since, 'x' is positive integer.

So, x = 15

From equation II:

7y – 2x = 5

7y – 30 = 5

y = 5

The length of room A = 15 + 3 = 18 feet

The width of room A = 18 * (100/10x) = 18 * (100/150) = 12 feet

Since, the average of the lengths of rooms A and B = (3y + 2) = (3 * 5 + 2) = 17 feet

So, the length of room B = 2 * 17 – 18 = 16 feet

The width of room B = 12 + (x/y) = 12 + (15/5) = 15 feet

Since, the height of room A = (x – y) = (15 – 5) = 10 feet

So, the volume of room A = 18 * 12 * 10 = 2160 $feet^3$

And the volume of room C = 2160 – 960 = 1200 $feet^3$

Since, the height of each room is equal.

So, the height of room B = 10 feet

And the height of room C = 10 feet

Since, the width of room C = (x – y) = (15 – 5) = 10 feet

So, the length of room C = 1200/(10 * 10) = 12 feet

Room	Length (in feet)	Width (in feet)	Height (in feet)
A	18	12	10
B	16	15	10
C	12	10	10

Answer: C

Since, the volume of each cuboidal box = 8 * 6 * 5 = 240 $feet^3$

And the volume of room B = 16 * 15 * 10 = 2400 $feet^3$

So, the maximum number of cuboidal boxes, which can be put inside room B:

2400/240 = 10 = (x – y)

17. Questions

Let the sum of the price of article P at each shop = Rs. 100x

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So, the price of article P at shop A = 25% of 100x = Rs. 25x

And the price of article P at shop B = 30% of 100x = Rs. 30x

Now, the price of article R at shop A = 75% of 40% of 25x = Rs. 7.5x

And the price of article R at shop B = 60% of 50% of 30x = Rs. 9x

Since, the price of article R at shop B is Rs. 108 more than that of article R at shop A.

So,

9x – 7.5x = 108

x = 72

The sum of the price of article P at each shop = 100 * 72 = Rs. 7200

Shop	Price of article P (in ₹)	Price of article Q (in ₹)	Price of article R (in ₹)
A	25% of 7200 = 1800	40% of 1800 = 720	75% of 720 = 540
B	30% of 7200 = 2160	50% of 2160 = 1080	60% of 1080 = 648
C	20% of 7200 = 1440	75% of 1440 = 1080	50% of 1080 = 540
D	25% of 7200 = 1800	45% of 1800 = 810	70% of 810 = 567

Answer: E

Since, the price of article P at shop B = Rs. 2160

So, the profit received by shop B on article P = 5% of 2160 = Rs. 108

Since, the price of article Q at shop B = Rs. 1080

So, the profit received by shop B on article Q = 2.5% of 1080 = Rs. 27

Since, the price of article R at shop B = Rs. 648

So, the profit received by shop B on article R = 7.5% of 648 = Rs. 48.6

The required average = (108 + 27 + 48.6)/3 = Rs. 61.2

18. Questions

Let the sum of the price of article P at each shop = Rs. 100x

So, the price of article P at shop A = 25% of 100x = Rs. 25x

And the price of article P at shop B = 30% of 100x = Rs. 30x

Now, the price of article R at shop A = 75% of 40% of 25x = Rs. 7.5x

And the price of article R at shop B = 60% of 50% of 30x = Rs. 9x

Since, the price of article R at shop B is Rs. 108 more than that of article R at shop A.

So,

9x – 7.5x = 108

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A	25% of 7200 = 1800	40% of 1800 = 720	75% of 720 = 540
B	30% of 7200 = 2160	50% of 2160 = 1080	60% of 1080 = 648
C	20% of 7200 = 1440	75% of 1440 = 1080	50% of 1080 = 540
D	25% of 7200 = 1800	45% of 1800 = 810	70% of 810 = 567

Answer: C

The material cost of article P = The price of article R at shop C = Rs. 540

The labour cost of article P = The price of article R at shop B + 12 = 648 + 12 = Rs. 660

Since, the manufacturing cost of article P = 540 + 660 = Rs. 1200

And the price of article P at shop C = Rs. 1440

So, the required profit per cent = [(1440 – 1200)/1200] * 100 = 20%

19. Questions

Let the sum of the price of article P at each shop = Rs. 100x

So, the price of article P at shop A = 25% of 100x = Rs. 25x

And the price of article P at shop B = 30% of 100x = Rs. 30x

Now, the price of article R at shop A = 75% of 40% of 25x = Rs. 7.5x

And the price of article R at shop B = 60% of 50% of 30x = Rs. 9x

Since, the price of article R at shop B is Rs. 108 more than that of article R at shop A.

So,

9x – 7.5x = 108

x = 72

The sum of the price of article P at each shop = 100 * 72 = Rs. 7200

Shop	Price of article P (in ₹)	Price of article Q (in ₹)	Price of article R (in ₹)
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C	20% of 7200 = 1440	75% of 1440 = 1080	50% of 1080 = 540
D	25% of 7200 = 1800	45% of 1800 = 810	70% of 810 = 567

Answer: A

Since, the price of article Q at shop A = Rs. 720

So, the price, at which shop A sold article Q = 95% of 720 = 684

Since, the price of article Q at shop B = Rs. 1080

So, the price, at which shop B sold article Q = 75% of 1080 = 810

Since, the price of article Q at shop C = Rs. 1080

So, the price, at which shop C sold article Q = 70% of 1080 = 756

Since, the price of article Q at shop D = Rs. 810

So, the price, at which shop D sold article Q = 90% of 810 = 729

Hence, shop A sold article Q at lowest price.

20. Questions

Let the sum of the price of article P at each shop = Rs. 100x

So, the price of article P at shop A = 25% of 100x = Rs. 25x

And the price of article P at shop B = 30% of 100x = Rs. 30x

Now, the price of article R at shop A = 75% of 40% of 25x = Rs. 7.5x

And the price of article R at shop B = 60% of 50% of 30x = Rs. 9x

Since, the price of article R at shop B is Rs. 108 more than that of article R at shop A.

So,

9x – 7.5x = 108

x = 72

The sum of the price of article P at each shop = 100 * 72 = Rs. 7200

Shop	Price of article P (in ₹)	Price of article Q (in ₹)	Price of article R (in ₹)
A	25% of 7200 = 1800	40% of 1800 = 720	75% of 720 = 540
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C	20% of 7200 = 1440	75% of 1440 = 1080	50% of 1080 = 540
D	25% of 7200 = 1800	45% of 1800 = 810	70% of 810 = 567

Answer: D

From X:

Average price of article P at shops B, C and D:

(2160 + 1440 + 1800)/3 = Rs. 1800

So, X is true.

From Y:

The price of article R at shop B = Rs. 648

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The price of article R at shop C = Rs. 540

The required percentage = [(648 – 540)/540] * 100 = 20%

So, Y is true.

From Z:

Average price of articles P, Q and R at shop D:

(1800 + 810 + 567)/3 = Rs. 1059

So, Z is true.

Hence, all are true.

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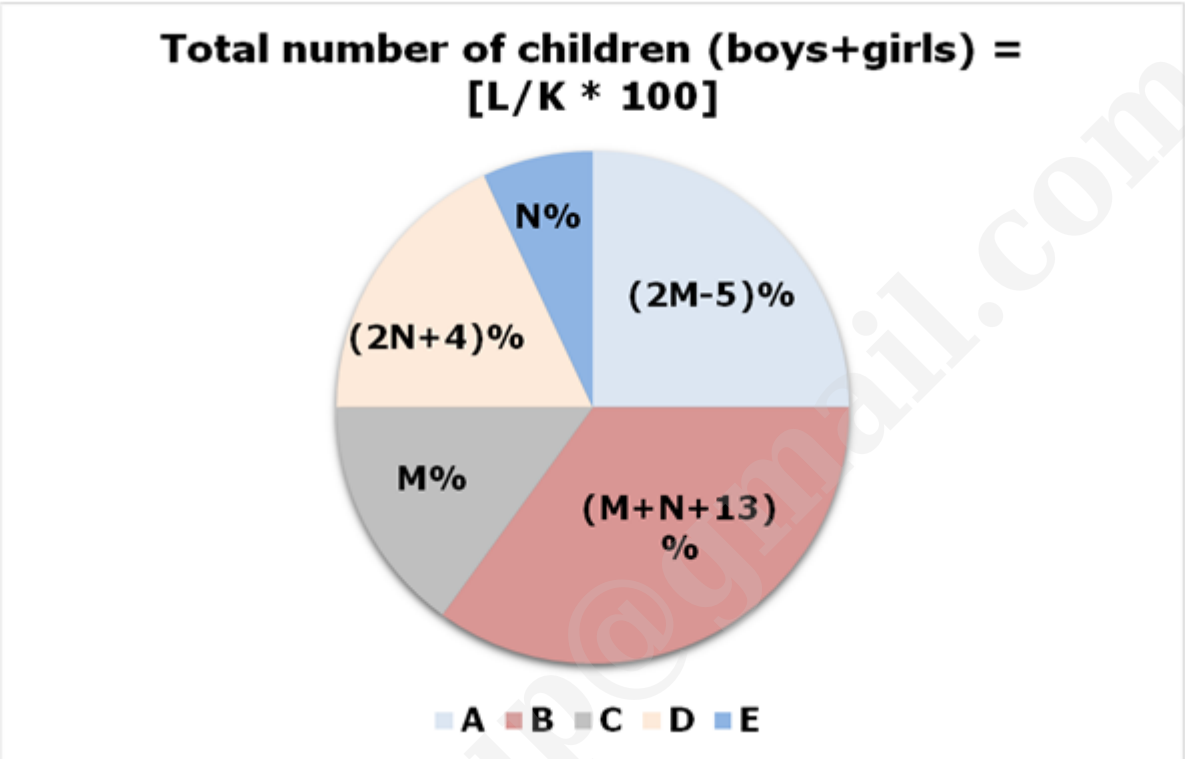
1. Questions

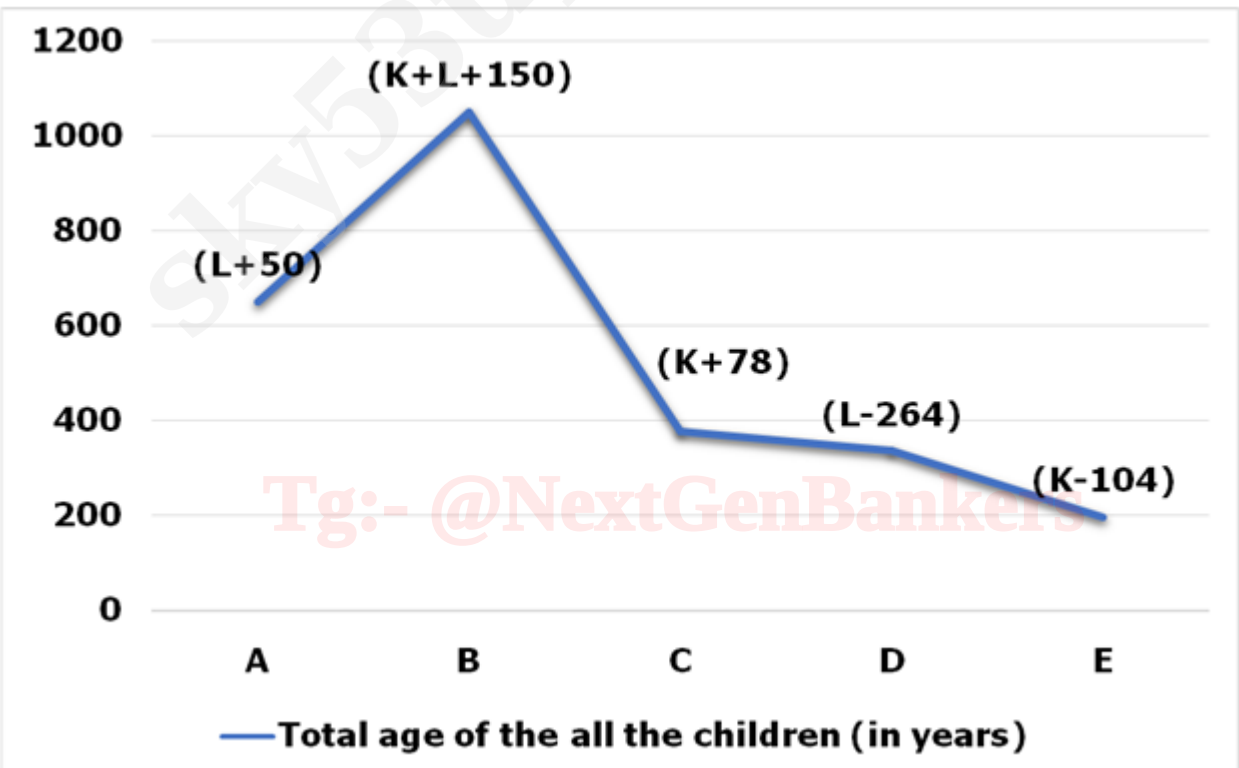
Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of children (boys + girls) living in five different towns A, B, C, D, and E.

The line graph given below shows the total age of all the children (boys + girls) living in each town.

The total number of children (boys + girls) living in all the five towns together is [L/K * 100].





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Note:

The values of M and N are positive integers.

The value of N is a prime number, which lies between both the roots of the equation $Z^2 - 13Z + 36 = 0$.

The value of M is an odd composite number.

The ratio of the values of M to that of K is 1:20.

The value of L is 100 less than the value of (10^2N).

The total number of children living in any town = the number of boys living + the number of girls living

The total age of all the children living in any town = {(the total number of children) × (the average age of each child)}

In town A, if the average age of all the boys is 15 years, which is 50% more than that of girls, then find the difference between the number of boys and girls living in town A.

a. 10 years

b. 16 years

c. 13 years

d. 19 years

e. None of these

2. Questions

In town B, the ratio of the number of boys to that of girls is 4:3 and the average age of each girl is 14 years less than that of each boy. Find the sum of the total age of all the girls living in town B.

a. 270 years

b. 250 years

c. 200 years

d. 210 years

e. None of these

3. Questions

In town C, if there are adults (males + females) are also living, which is 70% of the total number of people (children + adults) living in town C, then find the number of males living in town C, if it is $5/7^{th}$ of the total number of adults (males + females) living in the same town.

a. 20

b. 50

c. 70

d. 30

e. None of these

4. Questions

If there is another town F, where the total number of children (boys + girls) living is 11.11% more than that of town D and the average age of all the boys is 8 years, which is twice the average age of all the girls in the same town. Find the ratio of the number of boys to that of girls living in town F, if the total age of all the boys is 140 years more than that of girls in the same town.

a. 2:1

b. 4:5

c. 3:5

d. 3:2

e. None of these

5. Questions

In town E, if the product of the number of boys and girls is 49, then find the average age of all the boys, if the average age of all the girls is 14 years.

a. 14 years

b. 21 years

c. 7 years

d. Cannot be determined

e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the total number of players playing three different sports i.e., cricket, football and hockey from five different sports clubs P, Q, R, S and T. It also shows the percentage of the number of players playing football out of the total number of players playing all three sports together in each sports club. The table also shows information about the difference between the number of players playing cricket and hockey (cricket-hockey) from each sports club in the form of a statement.

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Club	The total number of players playing (cricket + football + hockey)	The percentage of the number of players playing football	The difference between the number of players playing cricket and hockey (cricket-hockey)
P	(A+20)	40%	10% of the total number of players playing all the sports
Q	210	2B%	42 less than the number of players playing football
R	(C+90)	C%	2/3rd of the number of players playing cricket
S	20D	55%	Equal to the number of players playing hockey
T	240	(6E+3)%	The square root of the difference between the number of players playing cricket and football (football-cricket)

Note:

The values of the variables are positive integers.

In each sports club, the number of players playing football is more than that of hockey.

In club P, the number of players playing hockey is 80.

The ratio of the number of players playing football to that of hockey in club Q is 5:1.

The number of players playing football in club R is 90.

The number of players playing cricket in club S is 50% more than the number of players playing hockey in club P.

In club T, the number of players playing football is 180, whereas the number of cricket players playing is 3E.

The number of players playing football is more than that of cricket in club T.

The total number of players playing = the number of players playing cricket + the number of players playing football + the number of players playing hockey

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In club P, out of the total number of players playing cricket, the ratio of the number of average runs scored by 44 players and 38 players is 5:4. If the average runs scored by the remaining number of players is 30 and the total number of runs scored by all the players playing cricket is 2760, then find the difference between the total number of runs scored by 44 players and 38 players.

a. 420

b. 260

c. 180

d. 340

e. None of these

7. Questions

If there is another sports club U, where the total number of players playing all the sports is 40 less than that of sports club T and the number of players playing cricket is X% of the total number of players playing all the sports. If the ratio of the number of players playing football to that of hockey is 13:7 and the number of players playing cricket is 35 more than that of football, then find the total number of players playing hockey in sports club U.

a. 41

b. 35

c. 23

d. Cannot be determined

e. None of these

8. Questions

The ratio of the number of players playing football above 25 years of age from club Q to that of players playing football below 25 years of age from club R is 12:7. While the number of players playing football below 25 years of age from club Q is $2/11^{th}$ less than the number of players playing football above 25 years of age from club R. Find the total number of players playing football above 25 years of age is what percent of the total number of players playing football below 25 years of age from both the clubs together?

a. 143.75%

b. 122.22%

c. 112.50%

d. 140.67%

e. None of these

9. Questions

If there is another sports club S i.e., volleyball and the number of players playing volleyball from club S is [K] more than the number of players playing hockey from the same club and is also [L] less than the number of players playing cricket from the same club, then find the value of √(K+L+4).

a. 4

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b. 16

c. 8

d. 32

e. None of these

10. Questions

In club T, if the average age of all the players playing hockey is [J] years and if the age of their coach is also included then the average age of all the players including their coach increases by 0.80 years. If the age of their coach is [2J] years, then find the value of [J^2]

a. 424

b. 400

c. 441

d. Cannot be determined

e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

A certain number of employees (males + females) work in two different banks A and B. Out of the total number of employees in each bank, some are officers whereas the rest are clerks.

Bank A: The ratio of the number of male officers working to that of female clerks working in the bank is 3:1. Whereas, the number of male clerks working in bank A is $1/7^{th}$ less than the number of female officers working in the same bank. The total number of male (officers + clerks) employees working in bank A is 150, which is 50% more than that of female (officers + clerks) employees.

Bank B: In bank B, the total number of male (officers + clerks) employees working is 140, which is 80 more than that of female (officers + clerks) employees working in the same bank. The number of male officers working in bank B is 150% more than that of female officers working in the same bank. The ratio of the number of male to female clerks working in bank B is 2:1.

Note:

The total number of employees working = the number of male employees working + the number of female employees working

The total number of employees working = the number of officers working + the number of clerks working

If out of the total number of male (officers + clerks) employees working in bank A, the ratio of the number of senior officers to that of junior clerks working is 8:7. The number of junior officers working in bank A is twice the number of senior clerks working in the same bank.

Find the correct relation between quantity I and quantity II.

Quantity I: The difference between the number of senior officers and clerks working in bank A.

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Quantity II: The average number of junior officers and clerks working in bank A.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I ≥ Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I = Quantity II

12. Questions

In bank B, out of the total number of female clerks working, the difference between the number of graduates and post graduates working is [K^2], while the number of post graduates working is [4K] and the number of graduates working is more than that of post graduates working in bank B among female clerks. Find the value of [√L], if 2K = L. (K is a positive integer)

a. 2

b. 5

c. 3

d. Cannot be determined

e. None of these

13. Questions

If there is another bank C, where the total number of employees working is the average of the total number of employees working in bank A and bank B together. If the total number of clerks working in bank C is 33.33% more than that of bank A, then find the total number of officers working in bank C.

a. 100

b. 80

c. 105

d. 75

e. None of these

14. Questions

There are three quadratic equations I, II and III given below

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Equations	Remarks
I). $X^2 - 41X + P^2 = 0$	c and d are the roots of the equation, where both c and d are exact multiples of 5 and also the difference between the roots of c and d is 5. (c > d)
II). $Y^2 - 35Y + 2Q = 0$	e and f are the roots of the equation, where both the roots are prime numbers. (e = f)
III). $Z^2 - 34Z + (12R+1) = 0$	a and b are the roots of the equation, where both a and b are the squares of a positive integer. (a > b)

Find the value of [(Q-50)/(R-P)].

a. 49

b. 25

c. 36

d. 64

e. None of these

15. Questions

There are three quadratic equations I, II and III given below with their roots as X, Y and Z but not necessarily in the same order.

Equation	Roots
I). $A^2 + A - M = 0$	**X).** 30 and 23
II). $B^2 - 2B - 3N = 0$	**Y).** 20 and -21
III). $C^2 - 53C + 2K = 0$	**Z).** -7 and 9

Find the correct pair of equations with its roots.

a. I-Y

b. II-X

c. III-Z

d. All of the above

e. None of these

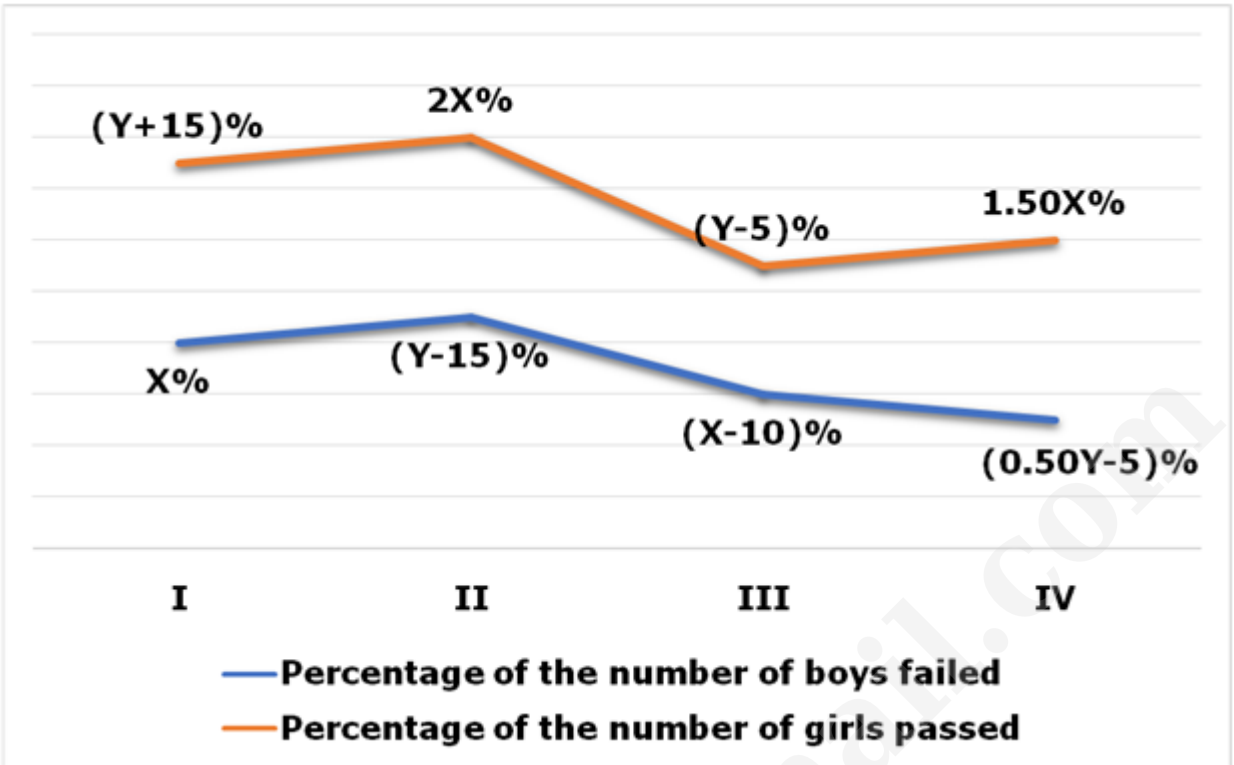
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16. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the percentage of the number of boys who failed out of the total number of boys who appeared in an examination from four different classes i.e., class I, class II, class III and class IV. It also shows the percentage of number of girls who passed out of the total number of girls who

appeared in the same examination from each class.



Note:

In class I, the ratio of the number of boys who passed to those of failed the examination is 3: 2, whereas the number of girls who failed the examination is 66.67% less than that of the number of girls who passed the examination.

The total number of students = the number of boys + the number of girls

The total number of students in the class = the number of students who appeared + the number of students who did not appear.

The total number of students who appeared in the examination = the number of students who passed + the number of students who failed

Out of the total number of students (boys + girls) who appeared in the examination from class III, 65% passed the examination. Find the number of girls who failed the examination out of the total number of girls who appeared in the examination from class III, if the total number of students (boys + girls) who appeared in the examination is 360.

a. 45

b. 54

c. 66

d. Cannot be determined

e. None of these

17. Questions

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In class II, if the number of boys who passed the examination is 88 and the total number of girls (passed + failed) who appeared in the examination is 15/23 times the total number of students (boys + girls) who appeared in the same examination from the same class, then find the total number of students (boys + girls) who appeared in the examination from class II.

a. 460

b. 340

c. 290

d. 150

e. None of these

18. Questions

If the difference between the number of boys who passed and failed the examination from class IV is 90, then find the total number of girls who passed the examination from the same class, if the number of girls who failed the examination is 55 more than that of boys who failed the examination from class IV.

a. 100

b. 180

c. 150

d. 120

e. None of these

19. Questions

If there is another class V, if the total number of students (boys + girls) who appeared is 400 and the ratio of the number of boys to that of girls who appeared in 5:3. If the number of boys who passed the examination is 6 times the number of girls who failed the examination, whereas the number of boys who failed the examination is 41.67% less than that number of girls who passed the examination. Find the total number of students (boys + girls) who passed the examination from class V.

a. 300

b. 250

c. 200

d. 350

e. None of these

20. Questions

Out of the total number of students in the class, 15% did not appear in the examination. If the number of boys who passed the examination from class I is 120, which is 85 more than the number

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of girls who failed the examination from the same class, then find the total number of students (appeared + did not appear) in class I.

a. 144

b. 256

c. 324

d. 400

e. None of these

Explanations:

1. Questions

$2M - 5 + M + N + 13 + M + 2N + 4 + N = 100$

$M + N = 22$ ------- (i)

From the given equation, we have,

$Z^2 - 13Z + 36 = 0$

$Z^2 - 9Z - 4Z + 36 = 0$

$Z(Z-9) - 4(Z-9) = 0$

$(Z-4)(Z-9) = 0$

$Z = 4$ or 9

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

Let, the value of N = 5

Now, put the value of N in equation (i), we get,

$M + N = 22$

$M + 5 = 22$

$M = 17$ (not possible)

17 is a prime number, but not an odd composite number.

Now, let the value of N = 7

Put the value of N in equation (i), we get,

$M + N = 22$

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M + 7 = 22

M = 15 (possible)

15 is an odd composite number.

Therefore, the values of,

N = 7 (confirmed)

M = 15 (confirmed)

So, K = {(15/1)×20} = 300

Now,

$L = \{(10^2N)-100\}$

L = {(100×7)-100}

L = (700-100)

L = 600

Therefore, the total number of children (boys + girls) living in each town = (L/K * 100) = (600/300 * 100) = 200

Now, the total number of children (boys + girls) living in each town,

A = (2M-5) % of 200 = (30-5)% of 200 = 25% of 200 = 50

B = (M+N+13) % of 200 = (15+7+13)% of 200 = 35% of 200 = 70

C = M% of 200 = 15% of 200 = 30

D = (2N+4) % of 200 = (14+4)% of 200 = 18% of 200 = 36

E = N% = 7% of 200 = 14

So, the total age of all the children (boys + girls) in each town,

A = (L+50) = (600+50) = 650 years

B = (K+L+150) = (300+600+150) = 1050 years

C = (K+78) = (300+78) = 378 years

D = (L-264) = (600-264) = 336 years

E = (K-104) = (300-104) = 196 years

Answer: A

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According to the question,

Town A

The total number of children (boys + girls) living = 50

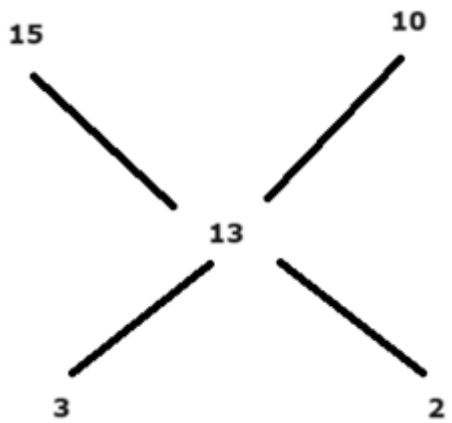
The total age of all the children (boys + girls) = 650

So, the average age of all the children (boys + girls) = (650/50) = 13

The average age of all the boys = 15 years

So, the average age of all the girls = {(15/150)×100} = 10 years

Using allegation method,



So, the ratio of the number of boys to that of girls = 3:2

Hence, the difference between the number of boys and girls living in town A = 1/5 of 50 = 10.

2. Questions

2M – 5 + M + N + 13 + M + 2N + 4 + N = 100

M + N = 22 ------- (i)

From the given equation, we have,

$Z^2 – 13Z + 36 = 0$

$Z^2 – 9Z – 4Z + 36 = 0$

Z(Z-9) – 4(Z-9) = 0

(Z-4) (Z-9) = 0

Z = 4 or 9

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

Let, the value of N = 5

Now, put the value of N in equation (i), we get,

M + N = 22

M + 5 = 22

M = 17 (not possible)

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17 is a prime number, but not an odd composite number.

Now, let the value of N = 7

Put the value of N in equation (i), we get,

M + N = 22

M + 7 = 22

M = 15 (possible)

15 is an odd composite number.

Therefore, the values of,

N = 7 (confirmed)

M = 15 (confirmed)

So, K = {(15/1)×20} = 300

Now,

L = {(10^2N)-100}

L = {(100×7)-100}

L = (700-100)

L = 600

Therefore, the total number of children (boys + girls) living in each town = (L/K * 100) = (600/300 * 100) = 200

Now, the total number of children (boys + girls) living in each town,

A = (2M-5) % of 200 = (30-5)% of 200 = 25% of 200 = 50

B = (M+N+13) % of 200 = (15+7+13)% of 200 = 35% of 200 = 70

C = M% of 200 = 15% of 200 = 30

D = (2N+4) % of 200 = (14+4)% of 200 = 18% of 200 = 36

E = N% = 7% of 200 = 14

So, the total age of all the children (boys + girls) in each town,

A = (L+50) = (600+50) = 650 years

B = (K+L+150) = (300+600+150) = 1050 years

C = (K+78) = (300+78) = 378 years

D = (L-264) = (600-264) = 336 years

E = (K-104) = (300-104) = 196 years

Answer: D

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Town B

The total number of children (boys + girls) living = 70

So, the number of boys living = 4/7 of 70 = 40

And, the number of girls living = 3/7 of 70 = 30

Let, the average age of each boy and each girl be p years and (p-14) years respectively.

So, the total age of all the boys = (40×p) = 40p years

And, the total age of all the girls = {30× (p-14)} = (30p-420) years

Now,

40p + (30p-420) = 1050

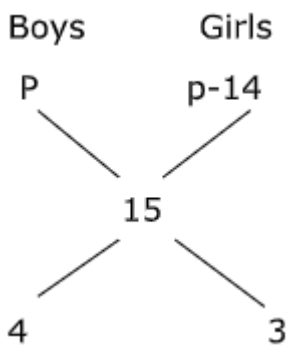
70p = 1470

p = 21

Hence, the sum of the ages of all the girls = (30p-420) = (630-420) = 210 years.

Alternate Method -

Using allegation method –



p – 15 / 15 – p +14 = 4/3

p = 21

The total age of all the girls = 30 * (p -14) =30 * (21 -14) = 30 * 7 =210 years

3. Questions

2M – 5 + M + N + 13 + M + 2N + 4 + N = 100

M + N = 22 ------- (i)

From the given equation, we have,

Z^2 – 13Z + 36 = 0

Z^2 – 9Z – 4Z + 36 = 0

Z(Z-9) – 4(Z-9) = 0

(Z-4) (Z-9) = 0

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Z = 4 or 9

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

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Put the value of N in equation (i), we get,

M + N = 22

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Therefore, the values of,

N = 7 (confirmed)

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So, K = {(15/1)×20} = 300

Now,

$L = \{(10^2N)-100\}$

L = {(100×7)-100}

L = (700-100)

L = 600

Therefore, the total number of children (boys + girls) living in each town = (L/K * 100) = (600/300 * 100) = 200

Now, the total number of children (boys + girls) living in each town,

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C = M% of 200 = 15% of 200 = 30

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D = (2N+4) % of 200 = (14+4)% of 200 = 18% of 200 = 36

E = N% = 7% of 200 = 14

So, the total age of all the children (boys + girls) in each town,

A = (L+50) = (600+50) = 650 years

B = (K+L+150) = (300+600+150) = 1050 years

C = (K+78) = (300+78) = 378 years

D = (L-264) = (600-264) = 336 years

E = (K-104) = (300-104) = 196 years

Answer: B

Town C

The percentage of the total number of people (children + adults) living = 100%

And, the percentage of the total number of adults (males + females) living = 70%

So, the percentage of the total number of children (boys + girls) living = (100-70)% = 30%

30% = 30

1% = (30/30) = 1

100% = (100×1) = 100

Therefore, the total number of people (children + adults) living = 100

So, the total number of adults (males + females) living = 70% of the 100 = 70

Hence, the number of males living in town C = 5/7 of 70 = 50.

4. Questions

2M – 5 + M + N + 13 + M + 2N + 4 + N = 100

M + N = 22 ------- (i)

From the given equation, we have,

$Z^2 – 13Z + 36 = 0$

$Z^2 – 9Z – 4Z + 36 = 0$

Z(Z-9) – 4(Z-9) = 0

(Z-4) (Z-9) = 0

Z = 4 or 9

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

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Now,

$L = \{(10^2N)-100\}$

L = {(100×7)-100}

L = (700-100)

L = 600

Therefore, the total number of children (boys + girls) living in each town = (L/K * 100) = (600/300 * 100) = 200

Now, the total number of children (boys + girls) living in each town,

A = (2M-5) % of 200 = (30-5)% of 200 = 25% of 200 = 50

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So, the total age of all the children (boys + girls) in each town,

A = (L+50) = (600+50) = 650 years

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B = (K+L+150) = (300+600+150) = 1050 years

C = (K+78) = (300+78) = 378 years

D = (L-264) = (600-264) = 336 years

E = (K-104) = (300-104) = 196 years

Answer: E

Town F

Fractional value of 11.11% = 1/9

So, the total number of children (boys + girls) living = 10/9 of 36 = 40

Let, the number of boys and girls living be s and t respectively.

Therefore, the total age of all the boys = (s×8) = 8s years

And, the total age of all the girls = {t × (8/2) × 1} = (t×4) = 4t years

Now,

s + t = 40……. (i)

8s – 4t = 140…… (ii)

Solving both the equations, we get,

8s + 8t = 320

8s – 4t = 140

12t = 180

t = 15

Now, put the value of t in equation (i), we get,

s + t = 40

s + 15 = 40

s = 25

Hence, the ratio of the number of boys to that of girls living in town F =

s:t = 25:15 = 5:3.

5. Questions

2M – 5 + M + N + 13 + M + 2N + 4 + N = 100

M + N = 22 ------- (i)

From the given equation, we have,

$Z^2 – 13Z + 36 = 0$

$Z^2 – 9Z – 4Z + 36 = 0$

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$Z(Z-9) - 4(Z-9) = 0$

$(Z-4)(Z-9) = 0$

$Z = 4$ or 9

So, the roots are = 4 and 9

Therefore, the value of N will be between 4 and 9.

So, the possible values of N = 5 or 7 (only prime numbers between 4 and 9)

Let, the value of N = 5

Now, put the value of N in equation (i), we get,

$M + N = 22$

$M + 5 = 22$

$M = 17$ (not possible)

17 is a prime number, but not an odd composite number.

Now, let the value of N = 7

Put the value of N in equation (i), we get,

$M + N = 22$

$M + 7 = 22$

$M = 15$ (possible)

15 is an odd composite number.

Therefore, the values of,

N = 7 (confirmed)

M = 15 (confirmed)

So, $K = \{(15/1)\times 20\} = 300$

Now,

$L = \{(10^2N)-100\}$

$L = \{(100\times 7)-100\}$

$L = (700-100)$

$L = 600$

Therefore, the total number of children (boys + girls) living in each town = (L/K * 100) = (600/300 * 100) = 200

Now, the total number of children (boys + girls) living in each town,

A = (2M-5) % of 200 = (30-5)% of 200 = 25% of 200 = 50

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B = (M+N+13) % of 200 = (15+7+13)% of 200 = 35% of 200 = 70

C = M% of 200 = 15% of 200 = 30

D = (2N+4) % of 200 = (14+4)% of 200 = 18% of 200 = 36

E = N% = 7% of 200 = 14

So, the total age of all the children (boys + girls) in each town,

A = (L+50) = (600+50) = 650 years

B = (K+L+150) = (300+600+150) = 1050 years

C = (K+78) = (300+78) = 378 years

D = (L-264) = (600-264) = 336 years

E = (K-104) = (300-104) = 196 years

Answer: A

Town E

The total number of children (boys + girls) living = 14

Let, the number of boys and girls living be v and (14-v) respectively.

Now,

$v \times (14-v) = 49$

$14v - v^2 = 49$

$v^2 - 14v + 49 = 0$

$v^2 - 7v - 7v + 49 = 0$

$v(v-7) - 7(v-7) = 0$

$(v-7)(v-7) = 0$

v = 7 or 7

So, v = 7

Therefore, the number of boys living = v = 7

And, the number of girls living = (14-v) = (14-7) = 7

The sum of the ages of all the girls = (14×7) = 98 years

And, the total ages of all the children (boys + girls) = 196

So, the sum of the ages of all the boys = (196-98) = 98 years

Hence, the average age of all the boys = (98/7) = 14 years.

6. Questions

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Club P

The total number of players playing all the sports = (A+20)

So, the number of players playing football = 40% of (A+20) = (0.40A+8)

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of (A+20) = (0.10A+2)

The number of players playing hockey = 80

So, the number of players playing cricket = {80 + (0.10A+2)} = (0.10A+82)

Now,

(0.40A+8) + 80 + (0.10A+82) = (A+20)

0.50A + 170 = A + 20

0.50A = 150

A = 300

So, the total number of players playing all the sports = (A+20) = (300+20) = 320

The number of players playing cricket = (0.10A+82) = (30+82) = 112

The number of players playing football = (0.40A+8) = (120+8) = 128

Club Q

The total number of players playing all the sports = 210

So, the number of players playing football = 2B% of 210 = 4.20B

The difference between the number of players playing cricket and hockey (cricket-hockey) = (4.20B-42)

The number of players playing hockey = [{(4.20B)/5} × 1] = 0.84B

So, the number of players playing cricket = {0.84B + (4.20B-42)} = (5.04B-42)

Now,

4.20B + 0.84B + (5.04B-42) = 210

10.08B – 42 = 210

10.08B = 252

B = 25

So, the number of players playing cricket = (5.04B-42) = 84

The number of players playing football = 4.20B = 105

The number of players playing hockey = 0.84B = 21

Club R

The total number of players playing all the sports = (C+90)

Tg:- @NextGenBankers

So, the number of players playing football = C% of (C+90) = $\{(C^2/100)+0.90C\} = \{(10C^2+900C)/1000\}$

Now,

$\{(10C^2+900C)/1000\} = 90$

$10C^2 + 900C = 90000$

$10C^2 + 900C – 90000 = 0$

$C^2 + 90C – 9000 = 0$

$C^2 + 150C – 60C – 9000 = 0$

$C(C+150) – 60(C+150) = 0$

$(C-60)\ (C+150) = 0$

$C = 60$ or -150

So, C = 60 (positive integer)

Therefore, the total number of players playing all the sports = (C+90) = (60+90) = 150

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = (150-90) = 60

Let, the number of players playing cricket be 3n

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = 2/3 of 3n = 2n

Therefore, the number of players playing hockey = (3n-2n) = n

Now,

(3n+n) = 60

4n = 60

n = 15

So, the number of players playing cricket = 3n = 45

And, the number of players playing hockey = n = 15

Club S

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

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Therefore, the number of players playing cricket = (m+m) = 2m

Now,

2m = 120

m = 60

So, the number of players playing hockey = m = 60

Now,

11D + 120 + 60 = 20D

9D = 180

D = 20

Hence, the total number of players playing all the sports = 20D = 400

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = (6E+3) % of 240 = (14.40E+7.20)

The number of players playing cricket = 3E

Now,

(14.40E+7.20) = 180

14.40E = 172.80

E = 12

So, the number of players playing football = (14.40E+7.20) = (172.80+7.20) = 180

And, the number of players playing cricket = 3E = 36

The difference between the number of players playing cricket and football (football-cricket) = (180-36) = 144

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = √144 = 12.

Therefore, the number of players playing hockey = (36-12) = 24

Tg:- @NextGenBankers

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
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R	150	45	90	15
S	400	120	220	60
T	240	36	180	24
Total	1320	397	723	200

Answer: D

According to the question,

Club P

The total number of players playing cricket = 112

Let, the average number of runs scored by 44 players and 38 players be 5r and 4r respectively.

So, the remaining number of players playing cricket = {112-(44+38)} = (112-82) = 30

Therefore, the total number of runs scored by 30 players = (30×30) = 900

And, the total number of runs scored by all the players playing cricket = 2760

So, the sum of the total number of runs scored by 44 players and 38 players = (2760-900) = 1860

Now, the total number of runs scored by 44 players = (44×5r) = 220r

And, the total number of runs scored by 38 players = (38×4r) = 152r

So,

220r + 152r = 1860

372r = 1860

r = 5

Therefore, the total number of runs scored by 44 players = 220r = 1100

And, the total number of runs scored by 38 players = 152r = 760

Hence, the required difference = (1100-760) = 340.

7. Questions

Club P

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Club Q

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So, the number of players playing football = 2B% of 210 = 4.20B

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The number of players playing hockey = [{(4.20B)/5} × 1] = 0.84B

So, the number of players playing cricket = {0.84B + (4.20B-42)} = (5.04B-42)

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B = 25

So, the number of players playing cricket = (5.04B-42) = 84

The number of players playing football = 4.20B = 105

The number of players playing hockey = 0.84B = 21

Club R

Tg:- @NextGenBankers

The total number of players playing all the sports = (C+90)

So, the number of players playing football = C% of (C+90) = $\{(C^2/100)+0.90C\} = \{(10C^2+900C)/1000\}$

Now,

$\{(10C^2+900C)/1000\} = 90$

$10C^2 + 900C = 90000$

$10C^2 + 900C - 90000 = 0$

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$C(C+150) - 60(C+150) = 0$

$(C-60)(C+150) = 0$

$C = 60$ or -150

So, $C = 60$ (positive integer)

Therefore, the total number of players playing all the sports = (C+90) = (60+90) = 150

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = (150-90) = 60

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Therefore, the number of players playing hockey = (3n-2n) = n

Now,

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Club S

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = (m+m) = 2m

Now,

2m = 120

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m = 60

So, the number of players playing hockey = m = 60

Now,

11D + 120 + 60 = 20D

9D = 180

D = 20

Hence, the total number of players playing all the sports = 20D = 400

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = (6E+3) % of 240 = (14.40E+7.20)

The number of players playing cricket = 3E

Now,

(14.40E+7.20) = 180

14.40E = 172.80

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So, the number of players playing football = (14.40E+7.20) = (172.80+7.20) = 180

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Answer: B

Club U

The total number of players playing all the sports = (240-40) = 200

So, the number of players playing cricket = X% of 200 = 2X

Now, let the number of players playing football and hockey be 13Y and 7Y respectively.

2X + 13Y + 7Y = 200

2X + 20Y = 200

X + 10Y = 100…… (i)

And,

2X – 13Y = 35…….. (ii)

Solving both the equations, we get,

2X + 20Y = 200

2X – 13Y = 35

33Y = 165

Y = 5

Now, put the value of Y in equation (i), we get,

X + 10Y = 100

X + 50 = 100

X = 50

Hence, the number of players playing hockey = 7Y = 35.

8. Questions

Club P

The total number of players playing all the sports = (A+20)

So, the number of players playing football = 40% of (A+20) = (0.40A+8)

The difference between the number of players playing cricket and hockey (cricket-hockey) = 10% of (A+20) = (0.10A+2)

The number of players playing hockey = 80

So, the number of players playing cricket = {80 + (0.10A+2)} = (0.10A+82)

Now,

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0.50A + 170 = A + 20

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0.50A = 150

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So, the total number of players playing all the sports = (A+20) = (300+20) = 320

The number of players playing cricket = (0.10A+82) = (30+82) = 112

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So, the number of players playing cricket = {0.84B + (4.20B-42)} = (5.04B-42)

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So, the number of players playing cricket = (5.04B-42) = 84

The number of players playing football = 4.20B = 105

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Club R

The total number of players playing all the sports = (C+90)

So, the number of players playing football = C% of (C+90) = $\{(C^2/100)+0.90C\} = \{(10C^2+900C)/1000\}$

Now,

$\{(10C^2+900C)/1000\} = 90$

$10C^2 + 900C = 90000$

$10C^2 + 900C – 90000 = 0$

$C^2 + 90C – 9000 = 0$

$C^2 + 150C – 60C – 9000 = 0$

C(C+150) – 60(C+150) = 0

(C-60) (C+150) = 0

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C = 60 or -150

So, C = 60 (positive integer)

Therefore, the total number of players playing all the sports = (C+90) = (60+90) = 150

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = (150-90) = 60

Let, the number of players playing cricket be 3n

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = 2/3 of 3n = 2n

Therefore, the number of players playing hockey = (3n-2n) = n

Now,

(3n+n) = 60

4n = 60

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So, the number of players playing cricket = 3n = 45

And, the number of players playing hockey = n = 15

Club S

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = (m+m) = 2m

Now,

2m = 120

m = 60

So, the number of players playing hockey = m = 60

Now,

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11D + 120 + 60 = 20D

9D = 180

D = 20

Hence, the total number of players playing all the sports = 20D = 400

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = (6E+3) % of 240 = (14.40E+7.20)

The number of players playing cricket = 3E

Now,

(14.40E+7.20) = 180

14.40E = 172.80

E = 12

So, the number of players playing football = (14.40E+7.20) = (172.80+7.20) = 180

And, the number of players playing cricket = 3E = 36

The difference between the number of players playing cricket and football (football-cricket) = (180-36) = 144

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Therefore, the number of players playing hockey = (36-12) = 24

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
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Total	1320	397	723	200

Answer: A

The total number of players playing football from club Q = 105

The total number of players playing football from club R = 90

Let, the total number of players playing football above 25 years of age from club Q and the number of players playing football below 25 of age years from club R be 12w and 7w respectively.

Also, let the total number of players playing football below 25 years of age from club Q and the number of players playing football above 25 years of age from club R be 9z and 11z respectively.

Now,

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$12w + 9z = 105$…… (i)

$7w + 11z = 90$…… (ii)

Solving both the equations, we get,

$84w + 63z = 735$

$84w + 132z = 1080$

$69z = 345$

$z = 5$

Now, put the value of z in equation (i), we get,

$12w + 9z = 105$

$12w + 45 = 105$

$12w = 60$

$w = 5$

Therefore, the total number of players playing football above 25 years of age = (12w+11z) = (60+55) = 115

And, the total number of players playing football below 25 years of age = (9z+7w) = (45+35) = 80

Hence, the required % = {(115/80) × 100} % = 143.75%.

9. Questions

Club P

The total number of players playing all the sports = (A+20)

So, the number of players playing football = 40% of (A+20) = (0.40A+8)

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So, the number of players playing cricket = (5.04B-42) = 84

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So, the number of players playing football = C% of (C+90) = $\{(C^2/100)+0.90C\} = \{(10C^2+900C)/1000\}$

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(C-60) (C+150) = 0

C = 60 or -150

So, C = 60 (positive integer)

Therefore, the total number of players playing all the sports = (C+90) = (60+90) = 150

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Club S

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The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

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T	240	36	180	24
Total	1320	397	723	200

Answer: C

Club S

The total number of players playing cricket = 120

And, the total number of players playing hockey = 60

So,

120-L = 60+K

K + L = 60

Hence, the value of √(K+L+4) = √(60+4) = √64 = 8.

10. Questions

Club P

The total number of players playing all the sports = (A+20)

So, the number of players playing football = 40% of (A+20) = (0.40A+8)

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The total number of players playing all the sports = (C+90)

So, the number of players playing football = C% of (C+90) = $\{(C^2/100)+0.90C\} = \{(10C^2+900C)/1000\}$

Now,

$\{(10C^2+900C)/1000\} = 90$

$10C^2 + 900C = 90000$

$10C^2 + 900C - 90000 = 0$

$C^2 + 90C - 9000 = 0$

$C^2 + 150C - 60C - 9000 = 0$

$C(C+150) - 60(C+150) = 0$

$(C-60)(C+150) = 0$

$C = 60$ or -150

So, $C = 60$ (positive integer)

Therefore, the total number of players playing all the sports = (C+90) = (60+90) = 150

The number of players playing football = 90

So, the sum of the number of players playing cricket and hockey = (150-90) = 60

Let, the number of players playing cricket be 3n

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = 2/3 of 3n = 2n

Therefore, the number of players playing hockey = (3n-2n) = n

Now,

(3n+n) = 60

4n = 60

n = 15

So, the number of players playing cricket = 3n = 45

And, the number of players playing hockey = n = 15

Club S

The total number of players playing all the sports = 20D

So, the number of players playing football = 55% of 20D = 11D

The number of players playing cricket = 150% of 80 = 120

Let, the number of players playing hockey be m

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = m

Therefore, the number of players playing cricket = (m+m) = 2m

Now,

2m = 120

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m = 60

So, the number of players playing hockey = m = 60

Now,

11D + 120 + 60 = 20D

9D = 180

D = 20

Hence, the total number of players playing all the sports = 20D = 400

Club T

The total number of players playing all the sports = 240

So, the number of players playing football = (6E+3) % of 240 = (14.40E+7.20)

The number of players playing cricket = 3E

Now,

(14.40E+7.20) = 180

14.40E = 172.80

E = 12

So, the number of players playing football = (14.40E+7.20) = (172.80+7.20) = 180

And, the number of players playing cricket = 3E = 36

The difference between the number of players playing cricket and football (football-cricket) = (180-36) = 144

So, the difference between the number of players playing cricket and hockey (cricket-hockey) = √144 = 12.

Therefore, the number of players playing hockey = (36-12) = 24

Club	The total number of players playing	The number of players playing cricket	The number of players playing football	The number of players playing hockey
P	320	112	128	80
Q	210	84	105	21
R	150	45	90	15
S	400	120	220	60
T	240	36	180	24
Total	1320	397	723	200

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Answer: B

Club T

The total number of players playing hockey = 24

The sum of the ages of all the players playing hockey = (24×J) = 24J years

The average age of all the players including their coach = (J+0.80) years

So, the sum of the ages of all the players including their coach = {(24+1) × (J+0.80)} = (25J+20)

Therefore, the age of their coach = {(25J+20)-24J}

So,

{(25J+20)-24J} = 2J

25J + 20 = 26J

J = 20

Hence, the value of $J^2 = 20^2 = 400$.

11. Questions

Bank A

Let, the number of male officers working to that of female clerks working be 3x and x respectively.

And, the ratio of the number of male clerks working to that of female officers working = 6:7. Let, it be 6y and 7y respectively.

Now, the total number of male (officers + clerks) employees working = 150

And, the total number of female (officers + clerks) employees working = {(150/150) × 100} = 100

So,

3x + 6y = 150……. (i)

x + 7y = 100…….. (ii)

Solving both the equations, we get,

3x + 6y = 150

3x + 21y = 300

15y = 150

y = 10

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Now, put the value of y in equation (i), we get,

3x + 6y = 150

3x + 60 = 150

3x = 90

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x = 30

So, the total number of male officers working = 3x = 90

The total number of male clerks working = 6y = 60

And, the total number of female officers working = 7y = 70

The total number of female clerks working = x = 30

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	90	60	150
Females	70	30	100
Total	160	90	250

Bank B

The total number of male (officers + clerks) employees working = 140

The total number of female (officers + clerks) employees working = (140-80) = 60

The ratio of the number of male officers to that of female officers working = 250:100 = 5:2. Let, it be 5m and 2m respectively.

Also, let the number of male and female clerks working be 2n and n respectively.

Now,

5m + 2n = 140……..(i)

2m + n = 60……..(ii)

Solving both the equations, we get,

10m + 4n = 280

10m + 5n = 300

n = 20

Now, put the value of n in equation (i), we get,

5m + 2n = 140

5m + 40 = 140

5m = 100

m = 20

So, the total number of male officers working = 5m = 100

The total number of male clerks working = 2n = 40

And, the total number of female officers working = 2m = 40

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The total number of female clerks working = n = 20

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	100	40	140
Females	40	20	60
Total	140	60	200

Answer: B

According to the question,

Bank A (Male Employees)

Let, the number of senior officers and junior clerks working be 8p and 7p respectively.

Also, the ratio of the number of junior officers to that of senior clerks working = 2:1. Let, it be 2q and q respectively.

The total number of officers working = 90

The total number of clerks working = 60

Now,

8p + 2q = 90…… (i)

7p + q = 60……. (ii)

Solving both the equations, we get,

56p + 14q = 630

56p + 8q = 480

6q = 150

q = 25

Now, put the value of q in equation (i), we get,

8p + 2q = 90

8p + 50 = 90

8p = 40

p = 5

Quantity I: The number of senior officers working = 8p = 40

And, the number of senior clerks working = q = 25

So, the required difference = (40-25) = 15

Hence, Quantity I = 15

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Quantity II: The number of junior officers working = 2q = 50

And, the number of junior clerks working = 7p = 35

So, the required average = {(50+35)/2} = (85/2) = 42.50

Hence, Quantity II = 42.50

Therefore, Quantity I < Quantity II.

12. Questions

Bank A

Let, the number of male officers working to that of female clerks working be 3x and x respectively.

And, the ratio of the number of male clerks working to that of female officers working = 6:7. Let, it be 6y and 7y respectively.

Now, the total number of male (officers + clerks) employees working = 150

And, the total number of female (officers + clerks) employees working = {(150/150) × 100} = 100

So,

3x + 6y = 150……. (i)

x + 7y = 100…….. (ii)

Solving both the equations, we get,

3x + 6y = 150

3x + 21y = 300

15y = 150

y = 10

Now, put the value of y in equation (i), we get,

3x + 6y = 150

3x + 60 = 150

3x = 90

x = 30

So, the total number of male officers working = 3x = 90

The total number of male clerks working = 6y = 60

And, the total number of female officers working = 7y = 70

The total number of female clerks working = x = 30

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Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	90	60	150
Females	70	30	100
Total	160	90	250

Bank B

The total number of male (officers + clerks) employees working = 140

The total number of female (officers + clerks) employees working = (140-80) = 60

The ratio of the number of male officers to that of female officers working = 250:100 = 5:2. Let, it be 5m and 2m respectively.

Also, let the number of male and female clerks working be 2n and n respectively.

Now,

5m + 2n = 140……..(i)

2m + n = 60…….(ii)

Solving both the equations, we get,

10m + 4n = 280

10m + 5n = 300

n = 20

Now, put the value of n in equation (i), we get,

5m + 2n = 140

5m + 40 = 140

5m = 100

m = 20

So, the total number of male officers working = 5m = 100

The total number of male clerks working = 2n = 40

And, the total number of female officers working = 2m = 40

The total number of female clerks working = n = 20

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Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	100	40	140
Females	40	20	60
Total	140	60	200

Answer: A

Bank B (Female Clerks)

The total number of employees working = 20

The number of post graduates working = 4K

The number of graduates working – The number of post graduates working = K^2

The number of graduates working – 4K = K^2

So, the number of graduates working = $(4K+K^2)$

Now,

$(4K+K^2) + 4K = 20$

$K^2 + 8K – 20 = 0$

$K^2 + 10K – 2K – 20 = 0$

$K(K+10) – 2(K+10) = 0$

$(K-2) (K+10) = 0$

$K = 2$ or -10

$K = 2$ (positive integer)

So,

$2K = L$

$L = 4$

Hence, the value of $\sqrt{L} = \sqrt{4} = 2$.

13. Questions

Bank A

Let, the number of male officers working to that of female clerks working be 3x and x respectively.

And, the ratio of the number of male clerks working to that of female officers working = 6:7. Let, it be 6y and 7y respectively.

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Now, the total number of male (officers + clerks) employees working = 150

And, the total number of female (officers + clerks) employees working = {(150/150) × 100} = 100

So,

3x + 6y = 150…… (i)

x + 7y = 100…….. (ii)

Solving both the equations, we get,

3x + 6y = 150

3x + 21y = 300

15y = 150

y = 10

Now, put the value of y in equation (i), we get,

3x + 6y = 150

3x + 60 = 150

3x = 90

x = 30

So, the total number of male officers working = 3x = 90

The total number of male clerks working = 6y = 60

And, the total number of female officers working = 7y = 70

The total number of female clerks working = x = 30

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	90	60	150
Females	70	30	100
Total	160	90	250

Bank B

The total number of male (officers + clerks) employees working = 140

The total number of female (officers + clerks) employees working = (140-80) = 60

The ratio of the number of male officers to that of female officers working = 250:100 = 5:2. Let, it be 5m and 2m respectively.

Also, let the number of male and female clerks working be 2n and n respectively.

Now,

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5m + 2n = 140……..(i)

2m + n = 60…….(ii)

Solving both the equations, we get,

10m + 4n = 280

10m + 5n = 300

n = 20

Now, put the value of n in equation (i), we get,

5m + 2n = 140

5m + 40 = 140

5m = 100

m = 20

So, the total number of male officers working = 5m = 100

The total number of male clerks working = 2n = 40

And, the total number of female officers working = 2m = 40

The total number of female clerks working = n = 20

Employees	The number of officers working	The number of clerks working	The total number of employees working
Males	100	40	140
Females	40	20	60
Total	140	60	200

Answer: C

Bank C

The total number of employees working = {(250+200)/2} = (450/2) = 225

The total number of clerks working = 4/3 of 90 = 120 (Fractional value of 33.33% = 1/3)

Hence, the total number of officers working in bank C = (225-120) = 105.

14. Questions

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Answer: B

Equation I

$X^2 - 41X + P^2 = 0$

The roots are = 25 and 16

The sum of the roots = (25+16) = 41

$5^2 = 25$

$4^2 = 16$

Both the roots are squares of the positive integer.

So, a and b are the roots of the equation I

Therefore, a = 25

And, b = 16

So, the product of the roots = P^2

$P^2 = (a \times b)$

$P^2 = (25 \times 16)$

$P^2 = 400$

P = 20

Equation II

$Y^2 - 35Y + 2Q = 0$

Roots of the equation = 25 and 10 (c=25, d=10, =>25-10 = 15) (it's not satisfied)

Roots of the equation = 30 and 5 (c=30, d=5 =>30-5 = 25) (it's also not satisfied)

So, Roots of the equation = 20 and 15 (c=20, d=15 =>20-15 = 5) (it's satisfied)

The sum of the roots = (20+15) = 35

Both 20 and 15 are exact multiples of 5.

So, c and d are the roots of the equation II

Therefore, c = 20

And, d = 15

So, the product of the roots = 2Q

$2Q = (c \times d)$

$2Q = (20 \times 15) = 300$

Q = 150

Equation III

$Z^2 - 34Z + (12R+1) = 0$

The roots of the equation are equal

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The sum of the roots = 34

So, both the roots = (34/2) = 17

17 is the prime number

Hence, e and f are the roots of the equation.

e = 17

f = 17

The product of the roots = (12R+1)

So,

(12R+1) = (e × f)

12R = {(17×17)-1}

12R = (289-1)

12R = 288

R = 24

Now, the value of,

[(Q-50)/(R-P)] = [(150-50) / (24-20)] = (100/4) = 25.

15. Questions

Answer: A

Equation I

$A^2 + A - M = 0$

The sum of the roots = -1

The roots of Y = 20 and -21

{20 + (-21)} = -1

So, Y is the roots of equation I

The product of the roots = {20 × (-21)} = -420

So, M = 420

Equation II

$B^2 - 2B - 3N = 0$

The sum of the roots = 2

The roots of Z = -7 and 9

(-7+9) = 2

So, Z is the roots of the equation II

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The product of the roots = (-7×9) = -63

So,

3N = 63

N = 21

Equation III

$C^2 - 53C + 2K = 0$

The sum of the roots = 53

The roots of X = 30 and 23

(30+23) = 53

So, Z is the roots of the equation III.

The product of the roots = (30×23) = 690

So,

2K = 690

K = 345

16. Questions

Class I

The percentage of the number of boys who failed the examination = X%

So, the percentage of the number of boys who passed the examination = (100-X)%

(100-X)/X = 3/2

200 – 2X = 3X

5X = 200

X = 40

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of 66.67% = 2/3)

The percentage of the number of girls who passed the examination = (Y+15)%

So, the percentage of the number of girls who failed the examination = {100-(Y+15)} % = (85-Y) %

(Y+15) / (85-Y) = 3/1

Y + 15 = 255 – 3Y

4Y = 240

Y = 60

Class I

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The percentage of the number of boys who failed the examination = X% = 40%

The percentage of the number of boys who passed the examination = (100-40) % = 60%

The percentage of the number of girls who passed the examination = (Y+15)% = (60+15) % = 75%

So, the percentage of the number of girls who failed the examination = (85-Y) % = (85-60) % = 25%

Class II

The percentage of the number of boys who failed the examination = (Y-15) % = (60-15)% = 45%

So, the percentage of the number of boys who passed the examination = (100-45) % = 55%

The percentage of the number of girls who passed the examination = 2X% = 80%

So, the percentage of the number of girls who failed the examination = (100-80) % = 20%

Class III

The percentage of the number of boys who failed the examination = (X-10) % = (40-10) % = 30%

So, the percentage of the number of boys who passed the examination = (100-30) % = 70%

The percentage of the number of girls who passed the examination = (Y-5) % = (60-5) % = 55%

So, the percentage of the number of girls who failed the examination = (100-55) % = 45%

Class IV

The percentage of the number of boys who failed the examination = (0.50Y-5)% = (30-5)% = 25%

So, the percentage of the number of boys who passed the examination = (100-25) % = 75%

The percentage of the number of girls who passed the examination = 1.50X% = 60%

So, the percentage of the number of girls who failed the examination = (100-60) % = 40%

Answer: B

According to the question,

Class III

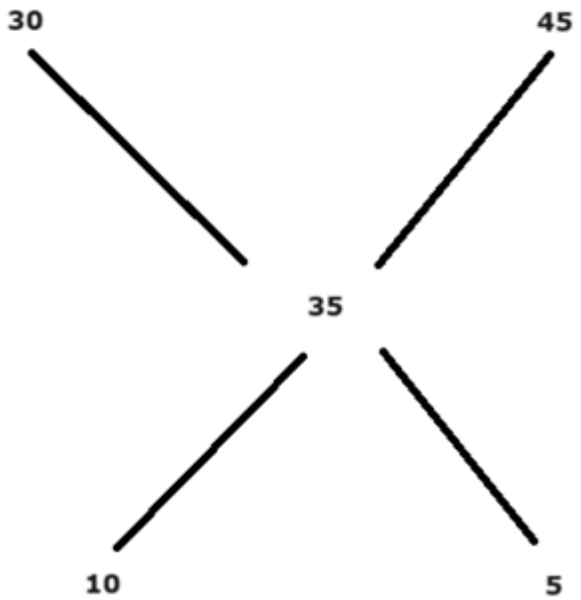
The percentage of the total number of students (boys + girls) who failed the examination = (100-65) % = 35%

The percentage of the number of boys who failed the examination = 30%

The percentage of the number of girls who failed the examination = 45%

Using allegation method,

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So, the ratio of the number of boys to that of girls who appeared in the examination = 10:5 = 2:1

Therefore, the total number of girls who appeared in the examination = 1/3 of 360 = 120

Hence, the number of girls who failed the examination = 45% of 120 = 54.

17. Questions

Class I

The percentage of the number of boys who failed the examination = X%

So, the percentage of the number of boys who passed the examination = (100-X)%

(100-X)/X = 3/2

200 – 2X = 3X

5X = 200

X = 40

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of 66.67% = 2/3)

The percentage of the number of girls who passed the examination = (Y+15)%

So, the percentage of the number of girls who failed the examination = {100-(Y+15)} % = (85-Y) %

(Y+15) / (85-Y) = 3/1

Y + 15 = 255 – 3Y

4Y = 240

Y = 60

Class I

The percentage of the number of boys who failed the examination = X% = 40%

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The percentage of the number of boys who passed the examination = (100-40) % = 60%

The percentage of the number of girls who passed the examination = (Y+15)% = (60+15) % = 75%

So, the percentage of the number of girls who failed the examination = (85-Y) % = (85-60) % = 25%

Class II

The percentage of the number of boys who failed the examination = (Y-15) % = (60-15)% = 45%

So, the percentage of the number of boys who passed the examination = (100-45) % = 55%

The percentage of the number of girls who passed the examination = 2X% = 80%

So, the percentage of the number of girls who failed the examination = (100-80) % = 20%

Class III

The percentage of the number of boys who failed the examination = (X-10) % = (40-10) % = 30%

So, the percentage of the number of boys who passed the examination = (100-30) % = 70%

The percentage of the number of girls who passed the examination = (Y-5) % = (60-5) % = 55%

So, the percentage of the number of girls who failed the examination = (100-55) % = 45%

Class IV

The percentage of the number of boys who failed the examination = (0.50Y-5)% = (30-5)% = 25%

So, the percentage of the number of boys who passed the examination = (100-25) % = 75%

The percentage of the number of girls who passed the examination = 1.50X% = 60%

So, the percentage of the number of girls who failed the examination = (100-60) % = 40%

Answer: A

Class II

The percentage of the number of boys who passed the examination = 55%

55% = 88

1% = (88/55) = 8/5

100% = {100 × (8/5)} = 160

So, the total number of boys who appeared in the examination = 160

The ratio of the total number of boys to that of girls who appeared in the examination = 8:15. Let, it be 8a and 15a respectively.

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Now,

8a = 160

a = 20

Hence, the total number of students (boys + girls) who appeared in the examination from class II = (8a+15a) = 23a = 460.

18. Questions

Class I

The percentage of the number of boys who failed the examination = X%

So, the percentage of the number of boys who passed the examination = (100-X)%

(100-X)/X = 3/2

200 – 2X = 3X

5X = 200

X = 40

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of 66.67% = 2/3)

The percentage of the number of girls who passed the examination = (Y+15)%

So, the percentage of the number of girls who failed the examination = {100-(Y+15)} % = (85-Y) %

(Y+15) / (85-Y) = 3/1

Y + 15 = 255 – 3Y

4Y = 240

Y = 60

Class I

The percentage of the number of boys who failed the examination = X% = 40%

The percentage of the number of boys who passed the examination = (100-40) % = 60%

The percentage of the number of girls who passed the examination = (Y+15)% = (60+15) % = 75%

So, the percentage of the number of girls who failed the examination = (85-Y) % = (85-60) % = 25%

Class II

The percentage of the number of boys who failed the examination = (Y-15) % = (60-15)% = 45%

So, the percentage of the number of boys who passed the examination = (100-45) % = 55%

The percentage of the number of girls who passed the examination = 2X% = 80%

So, the percentage of the number of girls who failed the examination = (100-80) % = 20%

Class III

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The percentage of the number of boys who failed the examination = (X-10) % = (40-10) % = 30%

So, the percentage of the number of boys who passed the examination = (100-30) % = 70%

The percentage of the number of girls who passed the examination = (Y-5) % = (60-5) % = 55%

So, the percentage of the number of girls who failed the examination = (100-55) % = 45%

Class IV

The percentage of the number of boys who failed the examination = (0.50Y-5)% = (30-5)% = 25%

So, the percentage of the number of boys who passed the examination = (100-25) % = 75%

The percentage of the number of girls who passed the examination = 1.50X% = 60%

So, the percentage of the number of girls who failed the examination = (100-60) % = 40%

Answer: C

Class IV

The percentage of the number of boys who failed the examination = 25%

And the percentage of the number of boys who passed the examination = 75%

So,

(75-25)% = 90

50% = 90

1% = (90/50) = 9/5

100% = {100 × (9/5)} = 180

So, the number of boys who failed the examination = 25% of 180 = 45

Therefore, the number of girls who failed the examination = (45+55) = 100

The percentage of the number of girls who failed the examination = 40%

So,

40% = 100

1% = (100/40) = (5/2)

100% = {100 × (5/2)} = 250

Therefore, the total number of girls who passed the examination = 250

Hence, the number of girls who passed the examination = 60% of 250 = 150.

19. Questions

Class I

The percentage of the number of boys who failed the examination = X%

So, the percentage of the number of boys who passed the examination = (100-X)%

(100-X)/X = 3/2

200 – 2X = 3X

5X = 200

X = 40

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The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of 66.67% = 2/3)

The percentage of the number of girls who passed the examination = (Y+15)%

So, the percentage of the number of girls who failed the examination = {100-(Y+15)} % = (85-Y) %

(Y+15) / (85-Y) = 3/1

Y + 15 = 255 – 3Y

4Y = 240

Y = 60

Class I

The percentage of the number of boys who failed the examination = X% = 40%

The percentage of the number of boys who passed the examination = (100-40) % = 60%

The percentage of the number of girls who passed the examination = (Y+15)% = (60+15) % = 75%

So, the percentage of the number of girls who failed the examination = (85-Y) % = (85-60) % = 25%

Class II

The percentage of the number of boys who failed the examination = (Y-15) % = (60-15)% = 45%

So, the percentage of the number of boys who passed the examination = (100-45) % = 55%

The percentage of the number of girls who passed the examination = 2X% = 80%

So, the percentage of the number of girls who failed the examination = (100-80) % = 20%

Class III

The percentage of the number of boys who failed the examination = (X-10) % = (40-10) % = 30%

So, the percentage of the number of boys who passed the examination = (100-30) % = 70%

The percentage of the number of girls who passed the examination = (Y-5) % = (60-5) % = 55%

So, the percentage of the number of girls who failed the examination = (100-55) % = 45%

Class IV

The percentage of the number of boys who failed the examination = (0.50Y-5)% = (30-5)% = 25%

So, the percentage of the number of boys who passed the examination = (100-25) % = 75%

The percentage of the number of girls who passed the examination = 1.50X% = 60%

So, the percentage of the number of girls who failed the examination = (100-60) % = 40%

Answer: A

Class V

The total number of students (boys + girls) who appeared in the examination = 400

So, the number of boys who appeared in the examination = 5/8 of 400 = 250

And, the number of girls who appeared in the examination = 3/8 of 400 = 150

Now, the ratio of the number of boys who passed to that of girls who failed the examination = 6:1. Let, it be 6m and m respectively.

Also, of the number of boys who failed to that of girls who passed the examination = 7:12. Let, it be 7n and 12n respectively. (Fractional value of 41.67% = 5/12)

Now,

6m + 7n = 250……(i)

And,

m + 12n = 150…….(ii)

Solving both the equations, we get,

6m + 7n = 250

6m + 72n = 900

65n = 650

n = 10

Now, put the value of n in equation (i), we get,

6m + 7n = 250

6m + 70 = 250

6m = 180

m = 30

Hence, the total number of students (boys + girls) who passed the examination = (6m+12n) = (180+120) = 300.

20. Questions

Class I

The percentage of the number of boys who failed the examination = X%

So, the percentage of the number of boys who passed the examination = (100-X)%

(100-X)/X = 3/2

200 – 2X = 3X

5X = 200

X = 40

The ratio of the number of girls who passed to those of failed the examination = 3:1. (Fractional value of 66.67% = 2/3)

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The percentage of the number of girls who passed the examination = (Y+15)%

So, the percentage of the number of girls who failed the examination = {100-(Y+15)} % = (85-Y) %

(Y+15) / (85-Y) = 3/1

Y + 15 = 255 – 3Y

4Y = 240

Y = 60

Class I

The percentage of the number of boys who failed the examination = X% = 40%

The percentage of the number of boys who passed the examination = (100-40) % = 60%

The percentage of the number of girls who passed the examination = (Y+15)% = (60+15) % = 75%

So, the percentage of the number of girls who failed the examination = (85-Y) % = (85-60) % = 25%

Class II

The percentage of the number of boys who failed the examination = (Y-15) % = (60-15)% = 45%

So, the percentage of the number of boys who passed the examination = (100-45) % = 55%

The percentage of the number of girls who passed the examination = 2X% = 80%

So, the percentage of the number of girls who failed the examination = (100-80) % = 20%

Class III

The percentage of the number of boys who failed the examination = (X-10) % = (40-10) % = 30%

So, the percentage of the number of boys who passed the examination = (100-30) % = 70%

The percentage of the number of girls who passed the examination = (Y-5) % = (60-5) % = 55%

So, the percentage of the number of girls who failed the examination = (100-55) % = 45%

Class IV

The percentage of the number of boys who failed the examination = (0.50Y-5)% = (30-5)% = 25%

So, the percentage of the number of boys who passed the examination = (100-25) % = 75%

The percentage of the number of girls who passed the examination = 1.50X% = 60%

So, the percentage of the number of girls who failed the examination = (100-60) % = 40%

Answer: D

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Class I

The percentage of the number of boys who passed the examination = 60%

60% = 120

1% = (120/60) = 2

100% = {100×2) = 200

Therefore, the total number of boys who appeared in the examination = 200

And, the percentage of the number of girls who failed the examination = 25%

25% = (120-85) = 35

1% = (35/25) = (7/5)

100% = {100 × (7/5)} = 140

Therefore, the total number of girls who appeared in the examination = 140

So, the total number of students (boys + girls) who appeared in the examination = (200+140) = 340

The percentage of the total number students who appeared in the examination = (100-15) % = 85%

85% = 340

1% = (340/85) = 4

100% = (100×4) = 400

Hence, the total number of students (appeared + did not appear) in class I = 400.

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1. Questions

Read the following information carefully and answer the questions based on it.

There are three types of mobiles i.e., 4G, 5G and 6G manufactured by two different companies P and Q. Cost price, marked price and selling price of each item may be different.

Company P: The cost price of each 4G mobile is Rs. 6000 and each 4G mobile is marked up by A% above the cost price and sold at a discount of Rs. 800. Cost price of each 5G mobile is Rs. 600A and each 5G mobile is marked up 50% above its cost price and the selling price of each 5G mobile is Rs. (B+1200). The sum of the selling price of each 4G and each 5G mobile is Rs. 22600. The profit earned on selling each 5G mobile is Rs. 4200. Cost price of each 6G mobile is Rs. 3000 more than that of each 5G mobile.

Each 6G mobile is marked up 1/3rd above its cost price and sold at a profit of Rs. 3000.

Company Q: The cost price of each 4G mobile is Rs. C. The marked price of each 4G mobile is Rs. 12000 and sold at a discount of 0.5D%. The profit earned on selling each 4G mobile is Rs. 12.50%. Cost price of each 5G mobile is Rs. 2000 more than that of each 4G mobile and each 5G mobile has been marked up by Rs. 0.5C and sold at a discount of 20%, earning a profit of Rs. (20D+200) on each 5G mobile. Cost price of each 6G mobile is Rs. 2C and each 6G mobile is marked up by 25% above its cost and sold at a discount of (D-35)%.

Note:

Profit percentage is calculated on cost price, unless mentioned in the question.

Total number of mobiles manufactured = number of 4G mobiles manufactured + number of 5G mobiles manufactured + number of 6G mobiles manufactured

Total number of mobiles manufactured = number of mobiles sold + number of mobiles unsold

Out of the total number of 4G mobiles manufactured by company P, only 2/3rd of the mobiles were sold and the rest were unsold. The number of 5G mobiles sold by the company P is 10% less than that of 4G mobiles sold by the same company. Company P sold 75% of the total 5G mobiles manufactured. The total profit earned on selling all the 4G and 5G mobiles is Rs. 418000. Find the difference between the total number of 4G and 5G mobiles manufactured.

a. 20

b. 30

c. 15

d. 45

e. None of these

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2. Questions

Company Q sold a certain number of 6G mobiles and the total profit earned from selling all the 6G mobiles is invested in a scheme, which offers compound interest at the rate of 20% per annum and receives Rs. 44000 as interest after 2 years. Find the total number of 6G mobiles sold by company Q.

a. 100

b. 180

c. 160

d. 140

e. None of these

3. Questions

If company P gives an additional discount of X% on the sale of each 6G mobile, then profit percentage on selling each 6G mobile becomes 8%, whereas cost price and marked price remain unchanged. Find the selling price of each 3G mobiles sold by company P, if the cost price of each 3G mobile is Rs. 5000 and sold at the profit of 2X%.

a. Rs. 10000

b. Rs. 8000

c. Rs. 7000

d. Cannot be determined

e. None of these

4. Questions

If there is another company R, who sold only 5G mobiles and the cost price of each 5G mobile is Rs. (B+C-3000) and marked up each 5G mobile by [(A+D)/2]% above its cost price and sold at a discount of 11.11%. Find the profit percentage earned by company R on the sale of each 5G mobile.

a. 50%

b. 45%

c. 65%

d. 20%

e. None of these

5. Questions

Read the following information carefully and answer the questions based on it.

Table given below shows the sum of the number of males and females visited a park on four different weeks of a month. It also shows the difference between the number of females and children (females-children) and the number of children visiting the same park each week.

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Week	Sum of the number of males and females visited	Difference between the number of females and children visited (females-children)	Number of children visited
First	160	10	50% of the number of males visited
Second	140	20	$3/17^{th}$ of the total number of people visited
Third	220	50	5/7 times less than the number of females visited
Fourth	120	10	Half of the average number of males and females visited

Note:

In each week, the number of females visiting the park is more than that of children.

Total number of people visited the park = number of males visited + number of females visited + number of children visited

Total number of children visited the park = number of boys visited + number of girls visited

In the first week, out of the total number of children who visited the park, the number of boys visited was X more than the number of girls who visited the same park in the same week. If the number of girls visiting the park is Y and the sum of the number of females and children (boys + girls) visiting the park is [{3×(Y+X)}+5]. Then, find the number of girls who visited the park in the first week.

a. 15

b. 35

c. 20

d. Cannot be determined

e. None of these

6. Questions

The ratio of the price of a ticket for visiting the park for each male to that of each female to that of each child is 3:2:1. The total revenue generated from the sale of the tickets from all the people (male + females + children) visited the park in the second week is Rs. 8000. Find the difference between the price of ticket for each male and the price of ticket for each child for visiting the park in the second week.

a. Rs.20

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b. Rs.40

c. Rs.50

d. Rs.60

e. None of these

7. Questions

In the first week of another month, the total number of people (males + females + children) visited the park is twice the difference between the total number of people (males + females + children) visited in the park in the third and fourth week of this month. The number of males visiting the park in the first week of another month is 80 more than that of females and the number of children who visited the park is 16.66% of that of males who visited the park in the first week of another month. Find the number of females who visited the park in the first week of another month.

a. 50

b. 36

c. 66

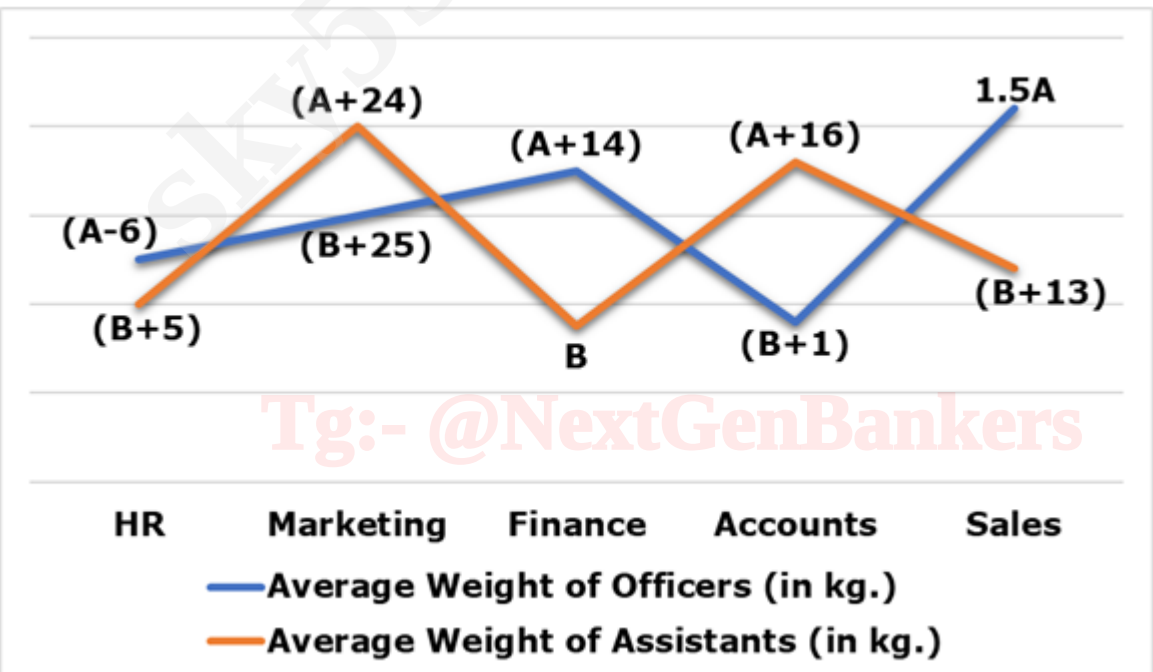
d. 40

e. None of these

8. Questions

Read the following information carefully and answer the questions based on it.

The line graph shows the average weight (in kg.) of all the officers and the average weight (in kg.) of all the assistants working in five different departments: HR, Finance, Marketing, Accounts and Sales of a company.



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Note:

The average weight of all the officers working in the HR and Marketing department is 110 kg.

The average weight of all the assistants working in the Marketing department is 40 kg more than that of the HR department.

The average weight of all the employees (officers + assistants) in department HR, Marketing, Finance, Accounts and Sales are 46 kg., 72 kg., 55 kg., 56 kg. and 69 kg respectively.

The total number of employees (officers + assistants) working in departments HR, Marketing, Finance, Accounts and Sales are 200, 250, 140, 360 and 120 in respectively.

The total number of employees working in any department = number of officers working + number of assistants working

The total number of employees working in any department at any post = number of male employees working + number of females employees working

In the HR department, the number of female officers working is the same as that of male assistants working and the number of male officers working is 80% more than that of female assistants working.

P = Difference between the total number of males (officers + assistant) and females (officers + assistants) working in department HR department

Q = Average of the total number of males (officers + assistant) and females (officers + assistants) working in department HR department

Find the value of (M-N), if [(45% of P)+M] = [(55% of Q)+N].

a. 52

b. 37

c. 43

d. 28

e. None of these

9. Questions

In the marketing department, there are (X-46) more officers joining the company and the average weight of all the officers becomes (X-5) kg. If the total weight of all the officers is increased by 20X kg. After the inclusion of X more officers, then find the number of newly joined officers in the marketing department. (X is a positive integer)

a. 30

b. 25

c. 20

d. 45

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e. None of these

10. Questions

In the finance department, the number of male assistants is twice that of female assistants. Whereas, the number of male officers is 37.50% more than that of male assistants. Average weight of each male assistant is 40 kg., while the average weight of each male officer is twice that of male assistants. Find the difference between the average weight of female assistants and female officers.

a. 32 kg

b. 20 kg

c. 31 kg

d. 24 kg

e. None of these

11. Questions

In the accounts department, there are two types of employees (officers + assistants) working i.e., junior and senior. Sum of the number of senior officers and senior assistants working is equal to the number of junior officers working, which is 60 less than the number of junior assistants working in the same department. Find the ratio of the number of junior (officers + assistants) employees to that of senior (officers + assistants) employees working in the accounts department.

a. 5:13

b. 6:11

c. 4:15

d. Cannot be determined

e. None of these

12. Questions

In the operation department of the same company, the average weight of all the officers working is 64 kg and the average weight of all the assistants working is 40 kg. If the average weight of all the employees (officers + assistants) working is 55 kg., then find the total number of male (officers + assistants) employees working in the operation department , given that the difference between the total number of officers and assistants working is 100 and ratio of the number of male to that of female employees working is 7:3.

a. 120

b. 280

c. 160

d. 250

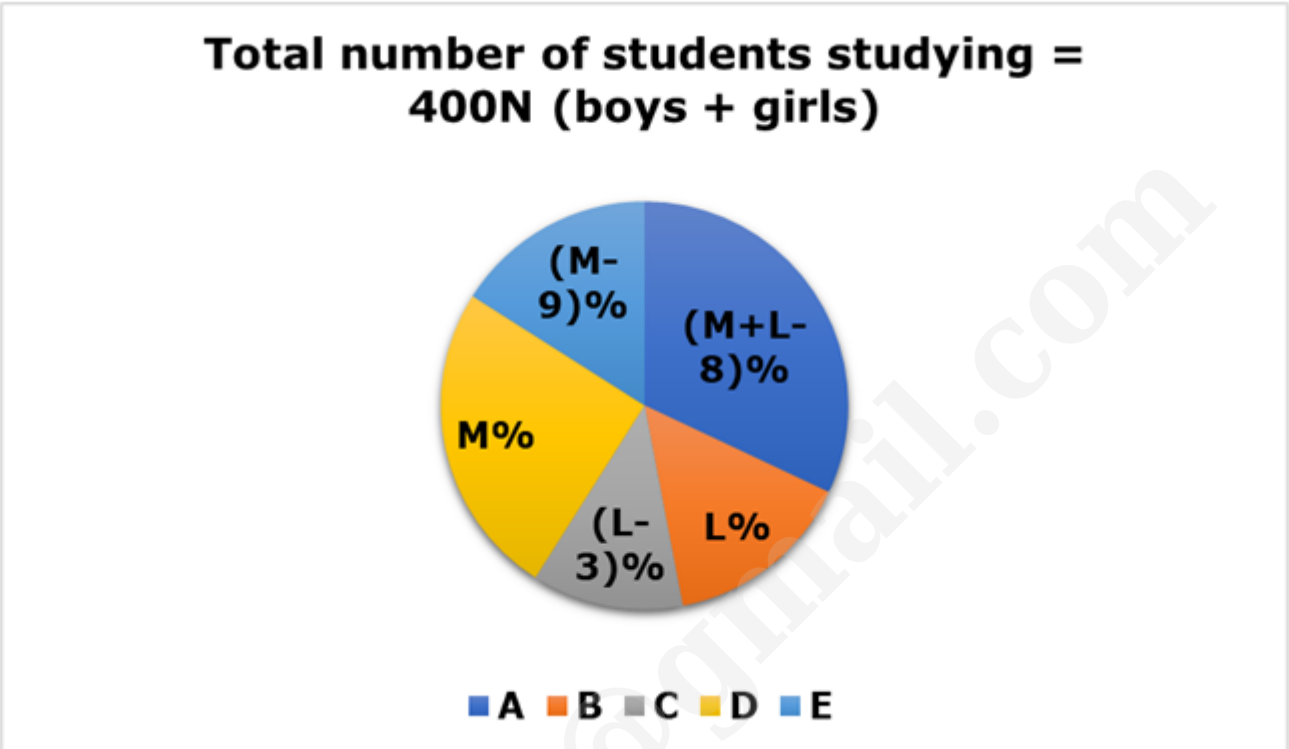
e. None of these

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13. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of students (boys + girls) studying in five different colleges A, B, C, D and E. Total number of students (boys + girls) studying in all the five colleges together is 400N.



Note:

Ratio of the total number of students (boys + girls) studying in college C to that of college E is 3:4.

Total number of students (boys + girls) studying in college A is (K^k), where K is the smallest composite number.

Total number of students studying in each college = number of boys studying + number of girls studying

The number of girls studying in college B is 11, more than twice the number of girls studying in college C. If the number of boys studying in college B is 19 less than that of college C, then find the difference between the total number of boys and girls studying in college B and college C together.

a. 5

b. 7

c. 2

d. 4

e. None of these

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14. Questions

In college D, students (boys + girls) are pursuing two courses i.e., course X and course Y. Number

of boys and girls pursuing course X are __[S]__ and __[S-42]__ respectively, and their product is 2160. While, the number of boys and girls pursuing the course Y are __[T+22]__ and __[T]__ respectively, and their product is 2280. Find the average of the values of S and T together. (S and T both are positive integers)

a. 110

b. 55

c. 60

d. 115

e. None of these

15. Questions

In college A, the ratio of the number of boys to that of girls studying is 45:19. On a particular day, the number of girls absent was 15% of the number of boys present on the same day. Whereas, the number of boys absent were 8/11 times the number of girls present on the same day. The difference between the number of girls present and absent is what percent of the difference between the number of boys present and absent on that particular day?

a. 29%

b. 21%

c. 38%

d. 34%

e. None of these

16. Questions

In college E, the fee for each boy is Rs. 15000 and the fees of each girl is Rs. 10000. The average fees received from all students (boys + girls) together is Rs. 13750. Find the ratio of the number of boys studying in college E to that of the total number of students (boys + girls) studying in college C.

a. 1:1

b. 2:5

c. 3:2

d. 4:1

e. None of these

17. Questions

If there is another college F, where the total number of students (boys + girls) studying is [(L×M)-135] and the ratio of the number of boys to that of girls studying is K:N. Find the number of girls, who are scholarship holders, if out of the total number of girls studying, 30% are non-

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scholarship holders in college F.

a. 33

b. 56

c. 45

d. Cannot be determined

e. None of these

18. Questions

The quantity of milk, water and honey is (A+5) litters, 60 liters and (B+25) liters respectively. If the 15 litres of milk and 30 liters of honey is removed from the mixture, then the quantity of milk and honey becomes equal. If the ratio of the initial quantity of milk to that of honey was 5:6, then find the initial quantity of mixture.

a. 225 liters

b. 200 liters

c. 300 liters

d. 325 liters

e. None of these

19. Questions

A person invested a certain sum of amount in scheme X at the rate of 12.50% per annum compounded annually and received Rs. P as interest after 2 years. Now, he invested Rs. (P+16000) in scheme Y, which offers compound interest at the rate of 10% per annum compounded half yearly and received Rs. 5125 as interest after 1 year. Find the amount invested by the person in scheme X.

a. Rs. 50000

b. Rs. 65000

c. Rs. 120000

d. Rs. 128000

e. None of these

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20. Questions

A bag contains total (Black + white) [2K + 3] balls out of which K are black. On picking two balls randomly from the bag, the probability of getting both white balls is 12/35. Find the value of K.

a. 2

b. 6

c. 5

d. 7

e. None of these

Explanations:

1. Questions

Company P

4G Mobiles

CP = Rs. 6000

MP = {6000+(A% of 6000)} = Rs. (6000+60A)

SP = (6000+60A-800) = Rs. (5200+60A)

5G Mobile

CP = Rs. 600A

MP = 150% of 600A = Rs. 900A

SP = Rs. (B+1200)

Profit = Rs. {(B+1200)-600A}

Now,

(B+1200) + (5200+60A) = 22600

B + 60A = 16200……(i)

Also,

(B+1200) – 600A = 4200

B – 3000 = 600A

B – 600A = 3000……..(ii)

Solving both the equations, we get,

B + 60A = 16200 – (i)

B – 600A = 3000 – (ii)

660A = 13200

A = 20

Now, put the value of A in equation (i), we get,

B + 60A = 16200

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B = 16200 – 1200

B = 15000

4G Mobile

CP = Rs. 6000

MP = (6000+60A) = (6000+1200) = Rs. 7200

SP = (5200+60A) = (5200+1200) = Rs. 6400

5G Mobile

CP = 600A = (600×20) = Rs. 12000

MP = 900A = (900×20) = Rs. 18000

SP = Rs. (B+1200) = (15000+1200) = Rs. 16200

6G Mobile

CP = (12000+3000) = Rs. 15000

MP = 4/3 of 15000 = Rs. 20000

SP = (15000+3000) = Rs. 18000

Company Q

4G Mobile

CP = Rs. C

MP = Rs. 12000

Discount = 0.5D% of 12000 = Rs. 60D

SP = Rs. (12000-60D)

Fractional value of 12.50% = 1/8

So, the ratio of the CP to that of SP of each 4G mobile = 8:9

Now,

C/(12000-60D) = 8/9

9C = 96000 – 480D

9C + 480D = 96000…….(iii)

5G Mobile

CP = Rs. (C+2000)

MP = (C+2000+0.5C) = (1.5C+2000)

SP = 80% of (1.5C+2000) = Rs. (1.2C+1600)

Profit = Rs. (20D+200)

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Now,

(1.2C+1600) – (C+2000) = (20D+200)

0.2C – 400 = 20D + 200

0.2C – 20D = 600…….(iv)

Solving both the equation, we get,

9C + 480D = 96000 – (iii)

0.2C – 20D = 600 – (iv)

1.8C + 96D = 19200

1.8C – 180D = 5400

276D = 13800

D = 50

Now, put the value of D in equation (iii), we get,

9C + 480D = 96000

9C = 96000 – 24000

9C = 72000

C = 8000

4G Mobile

CP = C = Rs. 8000

MP = Rs. 12000

SP = (12000-60D) = (12000-3000) = Rs. 9000

5G Mobile

CP = (C+2000) = (8000+2000) = Rs. 10000

MP = (1.5C+2000) = (12000+2000) = Rs. 14000

SP = (1.2C+1600) = (9600+1600) = Rs. 11200

6G Mobile

CP = 2C = (2×8000) = Rs. 16000

MP = 125% of 16000 = Rs. 20000

Discount % = (D-35)% = (50-35)% = 15%

SP = 85% of 20000 = 17000

Company P

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Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	6000	7200	6400
5G	12000	18000	16200
6G	15000	20000	18000
Total	33000	45200	40600

Company Q

Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	8000	12000	9000
5G	10000	14000	11200
6G	16000	20000	17000
Total	34000	46000	37200

Answer: B

According to the question,

Company P

Let, the total number of 4G mobiles manufactured be 300N.

So, the number of 4G mobiles sold = 2/3 of 300N = 200N

And, the number of 5G mobiles sold = 90% of 200N = 180N

Profit earned on selling each 4G mobile = (6400-6000) = Rs. 400

Profit earned on selling each 5G mobile = (16200-12000) = Rs. 4200

So, the profit earned on selling all the 4G mobiles = (400×200N) = Rs. 80000N

And, the profit earned on selling all the 5G mobiles = (4200×180N) = Rs. 756000N

Now,

80000N + 756000N = 418000

836000N = 418000

N = 0.5

So, the total number of 4G mobiles manufactured by company P = 300N = (300×0.5) = 150

And, the total number of 5G mobiles sold by company P = 180N = (180×0.5) = 90

Therefore, the total number of 5G mobiles manufactured by company P = {(90×100)/75} = 120

Hence, the required difference = (150-120) = 30.

2. Questions

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Company P

4G Mobiles

CP = Rs. 6000

MP = {6000+(A% of 6000)} = Rs. (6000+60A)

SP = (6000+60A-800) = Rs. (5200+60A)

5G Mobile

CP = Rs. 600A

MP = 150% of 600A = Rs. 900A

SP = Rs. (B+1200)

Profit = Rs. {(B+1200)-600A}

Now,

(B+1200) + (5200+60A) = 22600

B + 60A = 16200……(i)

Also,

(B+1200) – 600A = 4200

B – 3000 = 600A

B – 600A = 3000……..(ii)

Solving both the equations, we get,

B + 60A = 16200 – (i)

B – 600A = 3000 – (ii)

660A = 13200

A = 20

Now, put the value of A in equation (i), we get,

B + 60A = 16200

B = 16200 – 1200

B = 15000

4G Mobile

CP = Rs. 6000

MP = (6000+60A) = (6000+1200) = Rs. 7200

SP = (5200+60A) = (5200+1200) = Rs. 6400

5G Mobile

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CP = 600A = (600×20) = Rs. 12000

MP = 900A = (900×20) = Rs. 18000

SP = Rs. (B+1200) = (15000+1200) = Rs. 16200

6G Mobile

CP = (12000+3000) = Rs. 15000

MP = 4/3 of 15000 = Rs. 20000

SP = (15000+3000) = Rs. 18000

Company Q

4G Mobile

CP = Rs. C

MP = Rs. 12000

Discount = 0.5D% of 12000 = Rs. 60D

SP = Rs. (12000-60D)

Fractional value of 12.50% = 1/8

So, the ratio of the CP to that of SP of each 4G mobile = 8:9

Now,

C/(12000-60D) = 8/9

9C = 96000 – 480D

9C + 480D = 96000…….(iii)

5G Mobile

CP = Rs. (C+2000)

MP = (C+2000+0.5C) = (1.5C+2000)

SP = 80% of (1.5C+2000) = Rs. (1.2C+1600)

Profit = Rs. (20D+200)

Now,

(1.2C+1600) – (C+2000) = (20D+200)

0.2C – 400 = 20D + 200

0.2C – 20D = 600…….(iv)

Solving both the equation, we get,

9C + 480D = 96000 – (iii)

0.2C – 20D = 600 – (iv)

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1.8C + 96D = 19200

1.8C – 180D = 5400

276D = 13800

D = 50

Now, put the value of D in equation (iii), we get,

9C + 480D = 96000

9C = 96000 – 24000

9C = 72000

C = 8000

4G Mobile

CP = C = Rs. 8000

MP = Rs. 12000

SP = (12000-60D) = (12000-3000) = Rs. 9000

5G Mobile

CP = (C+2000) = (8000+2000) = Rs. 10000

MP = (1.5C+2000) = (12000+2000) = Rs. 14000

SP = (1.2C+1600) = (9600+1600) = Rs. 11200

6G Mobile

CP = 2C = (2×8000) = Rs. 16000

MP = 125% of 16000 = Rs. 20000

Discount % = (D-35)% = (50-35)% = 15%

SP = 85% of 20000 = 17000

Company P

Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	6000	7200	6400
5G	12000	18000	16200
6G	15000	20000	18000
Total	33000	45200	40600

Company Q

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Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	8000	12000	9000
5G	10000	14000	11200
6G	16000	20000	17000
Total	34000	46000	37200

Answer: A

Company Q

Let, the amount invested in the scheme be Rs. 100M

Scheme

Amount invested = Rs. 100M

Rate of interest = 20% p.a.

Time duration = 2 years

Type of interest = C.I.

[(20+20)+{(20×20)/100}]% = (40+4)% = 44%

So, interest earned = 44% of 100M = Rs. 44M

Now,

44M = 44000

M = (44000/44) = 1000

Therefore, the total amount invested in the scheme = 100M = (100×1000) = Rs. 100000

So, the total profit earned from selling all the 6G mobiles by company Q = Rs. 100000

Profit earned on selling each 6G mobile by company Q = (17000-16000) = Rs. 1000

Hence, total number of 6G mobiles sold by company Q = (100000/1000) = 100.

3. Questions

Company P

4G Mobiles

CP = Rs. 6000

MP = {6000+(A% of 6000)} = Rs. (6000+60A)

SP = (6000+60A-800) = Rs. (5200+60A)

5G Mobile

CP = Rs. 600A

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MP = 150% of 600A = Rs. 900A

SP = Rs. (B+1200)

Profit = Rs. {(B+1200)-600A}

Now,

(B+1200) + (5200+60A) = 22600

B + 60A = 16200……(i)

Also,

(B+1200) – 600A = 4200

B – 3000 = 600A

B – 600A = 3000……..(ii)

Solving both the equations, we get,

B + 60A = 16200 – (i)

B – 600A = 3000 – (ii)

660A = 13200

A = 20

Now, put the value of A in equation (i), we get,

B + 60A = 16200

B = 16200 – 1200

B = 15000

4G Mobile

CP = Rs. 6000

MP = (6000+60A) = (6000+1200) = Rs. 7200

SP = (5200+60A) = (5200+1200) = Rs. 6400

5G Mobile

CP = 600A = (600×20) = Rs. 12000

MP = 900A = (900×20) = Rs. 18000

SP = Rs. (B+1200) = (15000+1200) = Rs. 16200

6G Mobile

CP = (12000+3000) = Rs. 15000

MP = 4/3 of 15000 = Rs. 20000

SP = (15000+3000) = Rs. 18000

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Company Q

4G Mobile

CP = Rs. C

MP = Rs. 12000

Discount = 0.5D% of 12000 = Rs. 60D

SP = Rs. (12000-60D)

Fractional value of 12.50% = 1/8

So, the ratio of the CP to that of SP of each 4G mobile = 8:9

Now,

C/(12000-60D) = 8/9

9C = 96000 – 480D

9C + 480D = 96000…….(iii)

5G Mobile

CP = Rs. (C+2000)

MP = (C+2000+0.5C) = (1.5C+2000)

SP = 80% of (1.5C+2000) = Rs. (1.2C+1600)

Profit = Rs. (20D+200)

Now,

(1.2C+1600) – (C+2000) = (20D+200)

0.2C – 400 = 20D + 200

0.2C – 20D = 600…….(iv)

Solving both the equation, we get,

9C + 480D = 96000 – (iii)

0.2C – 20D = 600 – (iv)

1.8C + 96D = 19200

1.8C – 180D = 5400

276D = 13800

D = 50

Now, put the value of D in equation (iii), we get,

9C + 480D = 96000

9C = 96000 – 24000

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9C = 72000

C = 8000

4G Mobile

CP = C = Rs. 8000

MP = Rs. 12000

SP = (12000-60D) = (12000-3000) = Rs. 9000

5G Mobile

CP = (C+2000) = (8000+2000) = Rs. 10000

MP = (1.5C+2000) = (12000+2000) = Rs. 14000

SP = (1.2C+1600) = (9600+1600) = Rs. 11200

6G Mobile

CP = 2C = (2×8000) = Rs. 16000

MP = 125% of 16000 = Rs. 20000

Discount % = (D-35)% = (50-35)% = 15%

SP = 85% of 20000 = 17000

Company P

Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	6000	7200	6400
5G	12000	18000	16200
6G	15000	20000	18000
Total	33000	45200	40600

Company Q

Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	8000	12000	9000
5G	10000	14000	11200
6G	16000	20000	17000
Total	34000	46000	37200

Answer: E

Company P (6G Mobile)

CP = Rs. 15000

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MP = Rs. 20000

SP = Rs. 18000

After giving an additional discount, the profit percentage becomes 8%.

So,

New profit = 8% of 15000 = Rs. 1200

Therefore, new SP = (15000+1200) = Rs. 16200

And, old SP = Rs. 18000

So, the additional discount was given = (18000-16200) = Rs. 1800

Hence, the additional discount % given = {(1800/18000)×100}% = 10%

Therefore, X = 10.

Now,

CP of each 3G mobile = Rs. 5000

Profit % earned on each 3G mobile = 2X% = 20%

Hence, SP of each 3G mobile = 120% of 5000 = Rs. 6000.

4. Questions

Company P

4G Mobiles

CP = Rs. 6000

MP = {6000+(A% of 6000)} = Rs. (6000+60A)

SP = (6000+60A-800) = Rs. (5200+60A)

5G Mobile

CP = Rs. 600A

MP = 150% of 600A = Rs. 900A

SP = Rs. (B+1200)

Profit = Rs. {(B+1200)-600A}

Now,

(B+1200) + (5200+60A) = 22600

B + 60A = 16200……(i)

Also,

(B+1200) – 600A = 4200

B – 3000 = 600A

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B – 600A = 3000……..(ii)

Solving both the equations, we get,

B + 60A = 16200 – (i)

B – 600A = 3000 – (ii)

660A = 13200

A = 20

Now, put the value of A in equation (i), we get,

B + 60A = 16200

B = 16200 – 1200

B = 15000

4G Mobile

CP = Rs. 6000

MP = (6000+60A) = (6000+1200) = Rs. 7200

SP = (5200+60A) = (5200+1200) = Rs. 6400

5G Mobile

CP = 600A = (600×20) = Rs. 12000

MP = 900A = (900×20) = Rs. 18000

SP = Rs. (B+1200) = (15000+1200) = Rs. 16200

6G Mobile

CP = (12000+3000) = Rs. 15000

MP = 4/3 of 15000 = Rs. 20000

SP = (15000+3000) = Rs. 18000

Company Q

4G Mobile

CP = Rs. C

MP = Rs. 12000

Discount = 0.5D% of 12000 = Rs. 60D

SP = Rs. (12000-60D)

Fractional value of 12.50% = 1/8

So, the ratio of the CP to that of SP of each 4G mobile = 8:9

Now,

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C/(12000-60D) = 8/9

9C = 96000 – 480D

9C + 480D = 96000……(iii)

5G Mobile

CP = Rs. (C+2000)

MP = (C+2000+0.5C) = (1.5C+2000)

SP = 80% of (1.5C+2000) = Rs. (1.2C+1600)

Profit = Rs. (20D+200)

Now,

(1.2C+1600) – (C+2000) = (20D+200)

0.2C – 400 = 20D + 200

0.2C – 20D = 600……(iv)

Solving both the equation, we get,

9C + 480D = 96000 – (iii)

0.2C – 20D = 600 – (iv)

1.8C + 96D = 19200

1.8C – 180D = 5400

276D = 13800

D = 50

Now, put the value of D in equation (iii), we get,

9C + 480D = 96000

9C = 96000 – 24000

9C = 72000

C = 8000

4G Mobile

CP = C = Rs. 8000

MP = Rs. 12000

SP = (12000-60D) = (12000-3000) = Rs. 9000

5G Mobile

CP = (C+2000) = (8000+2000) = Rs. 10000

MP = (1.5C+2000) = (12000+2000) = Rs. 14000

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SP = (1.2C+1600) = (9600+1600) = Rs. 11200

6G Mobile

CP = 2C = (2×8000) = Rs. 16000

MP = 125% of 16000 = Rs. 20000

Discount % = (D-35)% = (50-35)% = 15%

SP = 85% of 20000 = 17000

Company P

Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	6000	7200	6400
5G	12000	18000	16200
6G	15000	20000	18000
Total	33000	45200	40600

Company Q

Mobile	Cost Price (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)
4G	8000	12000	9000
5G	10000	14000	11200
6G	16000	20000	17000
Total	34000	46000	37200

Answer: D

From common explanation, we got the values of,

A = 20

B = 15000

C = 8000

D = 50

Company R (5G Mobile)

CP = (B+C-3000) = (15000+8000-3000) = Rs. 20000

Mark-up % = [(A+D)/2]% = [(20+50)/2]% = (70/2)% = 35%

MP = 135% of 20000 = Rs. 27000

SP = 8/9 of 27000 = Rs. 24000 *(Fractional value of 11.11% = 1/9)*

So, the profit earned on selling each 5G mobile by company R = (24000-20000) = Rs. 4000

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Hence, required profit % = {(4000/20000)×100}% = 20%.

5. Questions

First Week

Let, the number of males who visited the park be 2a

So, the number of females who visited the park = (160-2a)

And, the number of children who visited park = (160-2a-10) = (150-2a)

Now,

2a/(150-2a) = 100/50

2a/(150-2a) = 2/1

2a = 300 – 4a

6a = 300

a = 50

Number of males who visited the park = 2a = 100

Number of females who visited the park = (160-2a) = (160-100) = 60

Number of children who visited the park = (150-2a) = (150-100) = 50

The total number of people visited the park = (100+60+50) = 210

Second Week

Let, the number of males who visited the park be b

So, the number of females who visited the park = (140-b)

And, the number of children who visited the park = (140-b-20) = (120-b)

The total number of people visited the park = {b+(140-b)+(120-b)} = (260-b)

So,

(120-b)/(260-b) = 3/17

2040 – 17b = 780 – 3b

14b = 1260

b = 90

Number of males who visited the park = b = 90

Number of females who visited the park = (140-b) = (140-90) = 50

Number of children who visited the park = (120-b) = (120-90) = 30

The total number of people visited the park = (90+50+30) = 170

Third Week

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Ratio of the number of females and children visited the park = 7:2. Let, it be 7c and 2c respectively.

So,

7c – 2c = 50

5c = 50

c = 10

Number of females visited the park = 7c = 70

Number of children visited the park = 2c = 20

Number of males visited the park = (220-70) = 150

So, the total number of people visited the park = (70+20+150) = 240

Fourth Week

Average of the number of males and females visited the park = (120/2) = 60

Number of children who visited the park = (60/2) = 30

Number of females who visited the park = (30+10) = 40

Number of males who visited the park = (120-40) = 80

So, the total number of people visited the park = (30+40+80) = 150

Week	Total number of people visited	Number of males visited	Number of females visited	Number of children visited
First	210	100	60	50
Second	170	90	50	30
Third	240	150	70	20
Fourth	150	80	40	30
Total	770	420	220	130

Answer: A

According to the question,

First Week

The total number of children who visited the park = 50

Number of girls who visited the park = Y

Number of boys who visited the park = (Y+X)

So,

2Y + X = 50……..(i)

Sum of the number females and children (boys + girls) visited the park = (60+50) = 110

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Now,

[{3×(Y+X)}+5] = 110

3Y + 3X = 105

Y + X = 35……(ii)

Solving both the equations, we get,

2Y + X = 50

2Y + 2X = 70

X = 20

Now, put the value of X in equation (i), we get,

2Y + X = 50

2Y = 50 – 20

2Y = 30

Y = 15

Hence, the number of girls who visited the park in the first week = 15.

6. Questions

First Week

Let, the number of males who visited the park be 2a

So, the number of females who visited the park = (160-2a)

And, the number of children who visited park = (160-2a-10) = (150-2a)

Now,

2a/(150-2a) = 100/50

2a/(150-2a) = 2/1

2a = 300 – 4a

6a = 300

a = 50

Number of males who visited the park = 2a = 100

Number of females who visited the park = (160-2a) = (160-100) = 60

Number of children who visited the park = (150-2a) = (150-100) = 50

The total number of people visited the park = (100+60+50) = 210

Second Week

Let, the number of males who visited the park be b

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So, the number of females who visited the park = (140-b)

And, the number of children who visited the park = (140-b-20) = (120-b)

The total number of people visited the park = {b+(140-b)+(120-b)} = (260-b)

So,

(120-b)/(260-b) = 3/17

2040 – 17b = 780 – 3b

14b = 1260

b = 90

Number of males who visited the park = b = 90

Number of females who visited the park = (140-b) = (140-90) = 50

Number of children who visited the park = (120-b) = (120-90) = 30

The total number of people visited the park = (90+50+30) = 170

Third Week

Ratio of the number of females and children visited the park = 7:2. Let, it be 7c and 2c respectively.

So,

7c – 2c = 50

5c = 50

c = 10

Number of females visited the park = 7c = 70

Number of children visited the park = 2c = 20

Number of males visited the park = (220-70) = 150

So, the total number of people visited the park = (70+20+150) = 240

Fourth Week

Average of the number of males and females visited the park = (120/2) = 60

Number of children who visited the park = (60/2) = 30

Number of females who visited the park = (30+10) = 40

Number of males who visited the park = (120-40) = 80

So, the total number of people visited the park = (30+40+80) = 150

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Week	Total number of people visited	Number of males visited	Number of females visited	Number of children visited
First	210	100	60	50
Second	170	90	50	30
Third	240	150	70	20
Fourth	150	80	40	30
Total	770	420	220	130

Answer: B

Second Week

Let, the price of a ticket for visiting the park for each male to that of each female to that of each child be 3p, 2p and p respectively.

Total revenue generated from the sale of tickets from all the males = (90×3p) = Rs. 270p

Total revenue generated from the sale of tickets from all the females = (50×2p) = Rs. 100p

Total revenue generated from the sale of tickets from all the children = (30×p) = Rs. 30p

Now,

(270p+100p+30p) = 8000

400p = 8000

p = 20

Therefore, the price of ticket for visiting the park for each male = 3*20 = Rs. 60

And, the price of ticket for visiting the park for each children = p = Rs. 20

Required difference = 60 – 20 = Rs.40

7. Questions

First Week

Let, the number of males who visited the park be 2a

So, the number of females who visited the park = (160-2a)

And, the number of children who visited park = (160-2a-10) = (150-2a)

Now,

2a/(150-2a) = 100/50

2a/(150-2a) = 2/1

2a = 300 – 4a

6a = 300

a = 50

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Number of males who visited the park = 2a = 100

Number of females who visited the park = (160-2a) = (160-100) = 60

Number of children who visited the park = (150-2a) = (150-100) = 50

The total number of people visited the park = (100+60+50) = 210

Second Week

Let, the number of males who visited the park be b

So, the number of females who visited the park = (140-b)

And, the number of children who visited the park = (140-b-20) = (120-b)

The total number of people visited the park = {b+(140-b)+(120-b)} = (260-b)

So,

(120-b)/(260-b) = 3/17

2040 – 17b = 780 – 3b

14b = 1260

b = 90

Number of males who visited the park = b = 90

Number of females who visited the park = (140-b) = (140-90) = 50

Number of children who visited the park = (120-b) = (120-90) = 30

The total number of people visited the park = (90+50+30) = 170

Third Week

Ratio of the number of females and children visited the park = 7:2. Let, it be 7c and 2c respectively.

So,

7c – 2c = 50

5c = 50

c = 10

Number of females visited the park = 7c = 70

Number of children visited the park = 2c = 20

Number of males visited the park = (220-70) = 150

So, the total number of people visited the park = (70+20+150) = 240

Fourth Week

Average of the number of males and females visited the park = (120/2) = 60

Number of children who visited the park = (60/2) = 30

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Number of females who visited the park = (30+10) = 40

Number of males who visited the park = (120-40) = 80

So, the total number of people visited the park = (30+40+80) = 150

Week	Total number of people visited	Number of males visited	Number of females visited	Number of children visited
First	210	100	60	50
Second	170	90	50	30
Third	240	150	70	20
Fourth	150	80	40	30
Total	770	420	220	130

Answer: D

Another Month (First Week)

Total number of people visited the park = {2×(240-150)} = (2×90) = 180

Fractional value of 16.66% = 1/6

Now, let the number of males visited the park be 6n

So, the number of children who visited the park = 1/6 of 6n = n

Therefore, the number of females visited the park = (6n-80)

Now,

6n + n + (6n-80) = 180

13n = 260

n = 20

Hence, the number of females visited the park in the first week of another month = (6n-80) = (120-80) = 40.

8. Questions

Sum of the average weight of all the officers working in the HR and marketing department = 110 kg.

So,

(A-6) + (B+25) = 110

A = 91 - B

A + B = 91.....(i)

Also,

(A+24) – (B+5) = 40

A – 16 = B + 5

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A – B = 21……(ii)

Solving both the equations, we get,

2B = 70

B = 35

Now, put the value of B in equation (i), we get,

A + B = 91

A = 91 – 35

A = 56

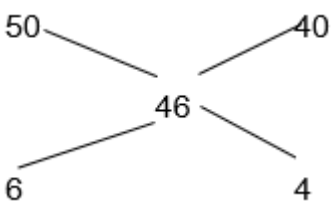
HR department

Average weight of all the officers working = (A-6) = (56-6) = 50 kg.

Average weight of all the assistants working = (B+5) = (35+5) = 40 kg.

Average weight of all the employees (officers + assistants) working = 46 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 6:4 = 3:2

Total number of employees working = 200

Number of officers working = 3/5 of 200 = 120

Number of assistants working = 2/5 of 200 = 80

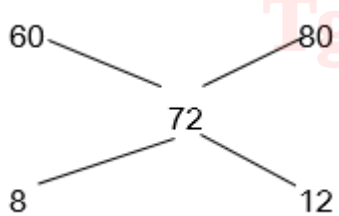
Marketing department

Average weight of all the officers working = (B+25) = (35+25) = 60 kg.

Average weight of all the assistants working = (A+24) = (56+24) = 80 kg.

Average weight of all the employees (officers + assistants) working = 72 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 8:12 = 2:3

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Total number of employees working = 250

Number of officers working = 2/5 of 250 = 100

Number of assistants working = 3/5 of 250 = 150

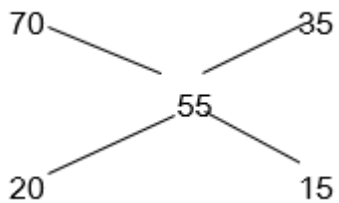
Finance department

Average weight of all the officers working = (A+14) = (56+14) = 70 kg.

Average weight of all the assistants working = 35 kg.

Average weight of all the employees (officers + assistants) working = 55 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 20:15 = 4:3

Total number of employees working = 140

Number of officers working = 4/7 of 140 = 80

Number of assistants working = 3/7 of 140 = 60

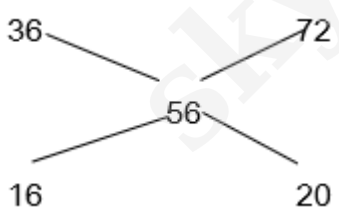
Accounts department

Average weight of all the officers working = (B+1) = (35+1) = 36 kg.

Average weight of all the assistants working = (A+16) = (56+16) = 72 kg.

Average weight of all the employees (officers + assistants) working = 56 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 16:20 = 4:5

The total number of employees working = 360

Number of officers working = 4/9 of 360 = 160

Number of assistants working = 5/9 of 360 = 200

Sales department

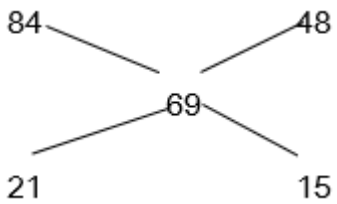
Average weight of all the officers = 1.5A = (1.5×56) = 84 kg.

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Average weight of all the assistants (B+13) = (35+13) = 48 kg.

Average weight of all the employees (officers + assistants) = 69 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 21:15 = 7:5

The total number of employees working = 120

Number of officers working = 7/12 of 120 = 70

Number of assistants working = 5/12 of 120 = 50

Department	Total number of employees working	Number of officers working	Number of assistants working	Average weight of all the officers (in kg.)	Average weight of all the assistants (in kg.)	Average weight of all the employees (in kg.)
HR	200	120	80	50	40	46
Marketing	250	100	150	60	80	72
Finance	140	80	60	70	35	55
Accounts	360	160	200	36	72	56
Sales	120	70	50	84	48	69
Total	1070	530	540	300	275	298

Answer: B

According to the question,

HR department

Total number of employees working = 200

Total number of officers working = 120

Total number of assistants working = 80

Let, the number of female officers and male assistants working be k.

So, the number of male officers working = (120-k)

And, the number of female assistants working = (80-k)

Now,

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M, (120-k)/(80-k) = 180/100

(120-k)/(80-k) = 9/5

600 – 5k = 720 – 9k

4k = 120

k = 30

Total number of male employees (officers + assistants) working = {(120-k)+k} = 120

Total number of female employees (officers + assistants) working = {k+(80-k)} = 80

P = Difference between the total number of males (officers + assistants) employees and female (officers + assistants) employees working = (120-80) = 40

Q = Average of the total number of males (officers + assistants) employees and female (officers + assistants) employees working = {(120+80)/2} = (200/2) = 100

So, P = 40

And, Q = 100

Now,

[(45% of P)+M] = [(55% of Q)+N]

[(45% of 40)+M] = [(55% of 100)+N]

18 + M = 55 + N

M – N = 37

Hence, the value of (M-N) = 37.

9. Questions

Sum of the average weight of all the officers working in the HR and marketing department = 110 kg.

So,

(A-6) + (B+25) = 110

A = 91 - B

A + B = 91……(i)

Also,

(A+24) – (B+5) = 40

A – 16 = B + 5

A – B = 21……(ii)

Solving both the equations, we get,

2B = 70

B = 35

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Now, put the value of B in equation (i), we get,

A + B = 91

A = 91 – 35

A = 56

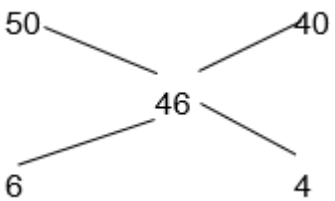
HR department

Average weight of all the officers working = (A-6) = (56-6) = 50 kg.

Average weight of all the assistants working = (B+5) = (35+5) = 40 kg.

Average weight of all the employees (officers + assistants) working = 46 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 6:4 = 3:2

Total number of employees working = 200

Number of officers working = 3/5 of 200 = 120

Number of assistants working = 2/5 of 200 = 80

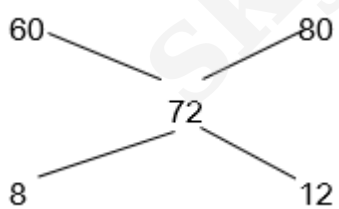
Marketing department

Average weight of all the officers working = (B+25) = (35+25) = 60 kg.

Average weight of all the assistants working = (A+24) = (56+24) = 80 kg.

Average weight of all the employees (officers + assistants) working = 72 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 8:12 = 2:3

Total number of employees working = 250

Number of officers working = 2/5 of 250 = 100

Number of assistants working = 3/5 of 250 = 150

Finance department

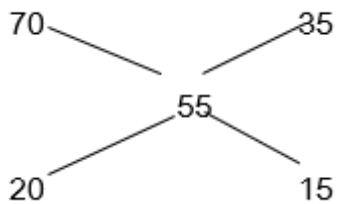
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Average weight of all the officers working = (A+14) = (56+14) = 70 kg.

Average weight of all the assistants working = 35 kg.

Average weight of all the employees (officers + assistants) working = 55 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 20:15 = 4:3

Total number of employees working = 140

Number of officers working = 4/7 of 140 = 80

Number of assistants working = 3/7 of 140 = 60

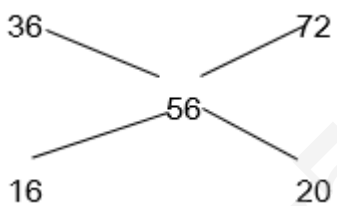
Accounts department

Average weight of all the officers working = (B+1) = (35+1) = 36 kg.

Average weight of all the assistants working = (A+16) = (56+16) = 72 kg.

Average weight of all the employees (officers + assistants) working = 56 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 16:20 = 4:5

The total number of employees working = 360

Number of officers working = 4/9 of 360 = 160

Number of assistants working = 5/9 of 360 = 200

Sales department

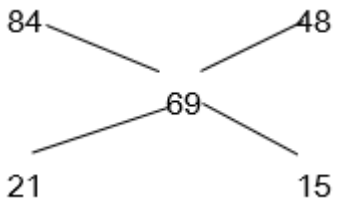
Average weight of all the officers = 1.5A = (1.5×56) = 84 kg.

Average weight of all the assistants (B+13) = (35+13) = 48 kg.

Average weight of all the employees (officers + assistants) = 69 kg.

Using allegation method,

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So, the ratio of the number of officers to that of assistants working = 21:15 = 7:5

The total number of employees working = 120

Number of officers working = 7/12 of 120 = 70

Number of assistants working = 5/12 of 120 = 50

Department	Total number of employees working	Number of officers working	Number of assistants working	Average weight of all the officers (in kg.)	Average weight of all the assistants (in kg.)	Average weight of all the employees (in kg.)
HR	200	120	80	50	40	46
Marketing	250	100	150	60	80	72
Finance	140	80	60	70	35	55
Accounts	360	160	200	36	72	56
Sales	120	70	50	84	48	69
Total	1070	530	540	300	275	298

Answer: C

Marketing department

Total number of officers working initially = 100

Sum of the weight of all the officers working initially = (100×60) = 6000 kg.

Now,

[(6000+20X)/{100+(X-46)}] = (X-5)

[(6000+20X)/(54+X)= (X-5)

$54X - 270 + X^2 - 5X = 6000 + 20X$

$X^2 = 6270 - 29X$

$X^2 + 29X - 6270 = 0$

$X^2 + 95X - 66X - 6270 = 0$

$X(X+95) - 66(X+95) = 0$

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(X-66) (X+95) = 0

X = 66 or -95

So, X = 66 (positive integer)

Hence, the newly joined officers in the marketing department = (X-46) = (66-46) = 20.

10. Questions

Sum of the average weight of all the officers working in the HR and marketing department = 110 kg.

So,

(A-6) + (B+25) = 110

A = 91 - B

A + B = 91……(i)

Also,

(A+24) – (B+5) = 40

A – 16 = B + 5

A – B = 21……(ii)

Solving both the equations, we get,

2B = 70

B = 35

Now, put the value of B in equation (i), we get,

A + B = 91

A = 91 – 35

A = 56

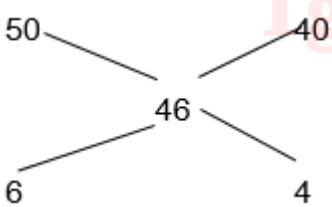
HR department

Average weight of all the officers working = (A-6) = (56-6) = 50 kg.

Average weight of all the assistants working = (B+5) = (35+5) = 40 kg.

Average weight of all the employees (officers + assistants) working = 46 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 6:4 = 3:2

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Total number of employees working = 200

Number of officers working = 3/5 of 200 = 120

Number of assistants working = 2/5 of 200 = 80

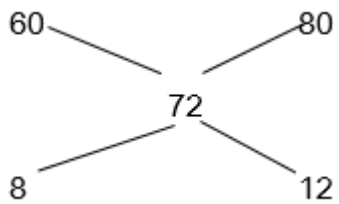
Marketing department

Average weight of all the officers working = (B+25) = (35+25) = 60 kg.

Average weight of all the assistants working = (A+24) = (56+24) = 80 kg.

Average weight of all the employees (officers + assistants) working = 72 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 8:12 = 2:3

Total number of employees working = 250

Number of officers working = 2/5 of 250 = 100

Number of assistants working = 3/5 of 250 = 150

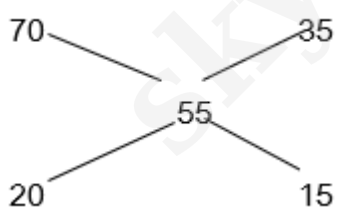
Finance department

Average weight of all the officers working = (A+14) = (56+14) = 70 kg.

Average weight of all the assistants working = 35 kg.

Average weight of all the employees (officers + assistants) working = 55 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 20:15 = 4:3

Total number of employees working = 140

Number of officers working = 4/7 of 140 = 80

Number of assistants working = 3/7 of 140 = 60

Accounts department

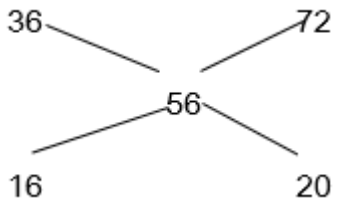
Average weight of all the officers working = (B+1) = (35+1) = 36 kg.

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Average weight of all the assistants working = (A+16) = (56+16) = 72 kg.

Average weight of all the employees (officers + assistants) working = 56 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 16:20 = 4:5

The total number of employees working = 360

Number of officers working = 4/9 of 360 = 160

Number of assistants working = 5/9 of 360 = 200

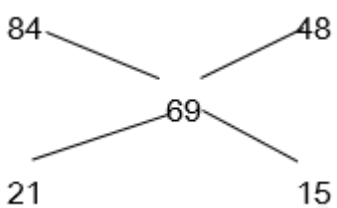
Sales department

Average weight of all the officers = 1.5A = (1.5×56) = 84 kg.

Average weight of all the assistants (B+13) = (35+13) = 48 kg.

Average weight of all the employees (officers + assistants) = 69 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 21:15 = 7:5

The total number of employees working = 120

Number of officers working = 7/12 of 120 = 70

Number of assistants working = 5/12 of 120 = 50

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Department	Total number of employees working	Number of officers working	Number of assistants working	Average weight of all the officers (in kg.)	Average weight of all the assistants (in kg.)	Average weight of all the employees (in kg.)
HR	200	120	80	50	40	46
Marketing	250	100	150	60	80	72
Finance	140	80	60	70	35	55
Accounts	360	160	200	36	72	56
Sales	120	70	50	84	48	69
Total	1070	530	540	300	275	298

Answer: E

Finance department

Total number of assistants = 60

Ratio of the number of male to that of female assistants = 2:1

Number of male assistants = 2/3 of 60 = 40

Number of female assistants = 1/3 of 60 = 20

Total number of officers = 80

Number of male officers = 11/8 of 40 = 55 *(Fractional value of 37.50% = 3/8)*

Number of female officers = 5/4 of 20 = 25

Now,

Total weight of all the assistants (male + female) = (35×60) = 2100

Total weight of all the male assistants = (40×40) = 1600

So, the total weight of all the female assistants = (2100-1600) = 500

Therefore, the average weight of all the female assistants = (500/20) = 25 kg.

Also,

Total weight of all the officers (male + female) = (70×80) = 5600

Total weight of all the male officers = {(2×40)×55} = (80×55) = 4400

So, the total weight of all the female officers = (5600-4400) = 1200

Therefore, the average weight of all the female officers = (1200/25) = 48 kg.

Hence, the difference between the average weight of all the female assistants and female officers = (48-25) = 23 kg.

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11. Questions

Sum of the average weight of all the officers working in the HR and marketing department = 110 kg.

So,

(A-6) + (B+25) = 110

A = 91 - B

A + B = 91.....(i)

Also,

(A+24) – (B+5) = 40

A – 16 = B + 5

A – B = 21.....(ii)

Solving both the equations, we get,

2B = 70

B = 35

Now, put the value of B in equation (i), we get,

A + B = 91

A = 91 – 35

A = 56

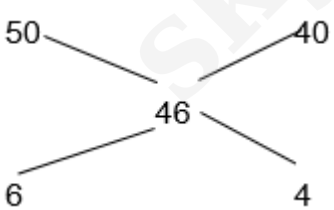
HR department

Average weight of all the officers working = (A-6) = (56-6) = 50 kg.

Average weight of all the assistants working = (B+5) = (35+5) = 40 kg.

Average weight of all the employees (officers + assistants) working = 46 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 6:4 = 3:2

Total number of employees working = 200

Number of officers working = 3/5 of 200 = 120

Number of assistants working = 2/5 of 200 = 80

Marketing department

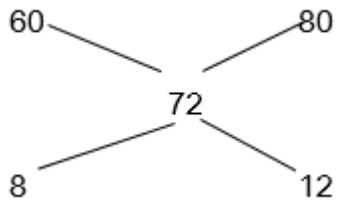
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Average weight of all the officers working = (B+25) = (35+25) = 60 kg.

Average weight of all the assistants working = (A+24) = (56+24) = 80 kg.

Average weight of all the employees (officers + assistants) working = 72 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 8:12 = 2:3

Total number of employees working = 250

Number of officers working = 2/5 of 250 = 100

Number of assistants working = 3/5 of 250 = 150

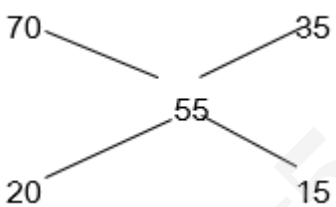
Finance department

Average weight of all the officers working = (A+14) = (56+14) = 70 kg.

Average weight of all the assistants working = 35 kg.

Average weight of all the employees (officers + assistants) working = 55 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 20:15 = 4:3

Total number of employees working = 140

Number of officers working = 4/7 of 140 = 80

Number of assistants working = 3/7 of 140 = 60

Accounts department

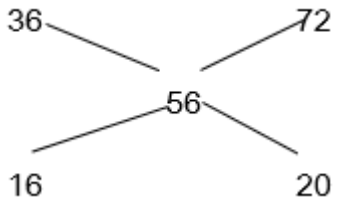
Average weight of all the officers working = (B+1) = (35+1) = 36 kg.

Average weight of all the assistants working = (A+16) = (56+16) = 72 kg.

Average weight of all the employees (officers + assistants) working = 56 kg.

Using allegation method,

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So, the ratio of the number of officers to that of assistants working = 16:20 = 4:5

The total number of employees working = 360

Number of officers working = 4/9 of 360 = 160

Number of assistants working = 5/9 of 360 = 200

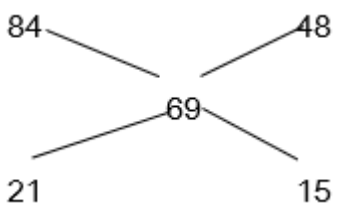
Sales department

Average weight of all the officers = 1.5A = (1.5×56) = 84 kg.

Average weight of all the assistants (B+13) = (35+13) = 48 kg.

Average weight of all the employees (officers + assistants) = 69 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 21:15 = 7:5

The total number of employees working = 120

Number of officers working = 7/12 of 120 = 70

Number of assistants working = 5/12 of 120 = 50

Department	Total number of employees working	Number of officers working	Number of assistants working	Average weight of all the officers (in kg.)	Average weight of all the assistants (in kg.)	Average weight of all the employees (in kg.)
HR	200	120	80	50	40	46
Marketing	250	100	150	60	80	72
Finance	140	80	60	70	35	55
Accounts	360	160	200	36	72	56
Sales	120	70	50	84	48	69
Total	1070	530	540	300	275	298

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Answer: A

Accounts department

Total number of officers working = 160

Total number of assistants working = 200

Let, the number of senior officers and senior assistants working be v and s respectively.

Number of junior officers working = (v+s)

Number of senior officers working = v

Number of junior assistants working = (v+s+60)

Number of senior assistants working = s

Now,

(v+s) + v = 160

2v + s = 160……..(i)

And,

(v+s+60) + s = 200

v + 2s = 140……..(ii)

Solving both the equations, we get,

2v + s = 160

2v + 4s = 280

3s = 120

s = 40

Now, put the value of s in equation (i), we get,

2v + s = 160

2v = 160 – 40

2v = 120

v = 60

So, the total number of junior officers and junior assistants working in finance department = {(v+s)+(v+s+60)} = (2v+2s+60) = (120+80+60) = 260

And, the total number of senior officers and senior assistants working in finance department = (v+s) = (60+40) = 100

Hence, required ratio = 260:100 = 13:5.

12. Questions

Sum of the average weight of all the officers working in the HR and marketing department = 110 kg.

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So,

(A-6) + (B+25) = 110

A = 91 - B

A + B = 91……(i)

Also,

(A+24) – (B+5) = 40

A – 16 = B + 5

A – B = 21……(ii)

Solving both the equations, we get,

2B = 70

B = 35

Now, put the value of B in equation (i), we get,

A + B = 91

A = 91 – 35

A = 56

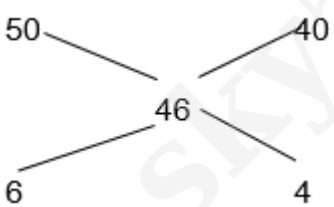
HR department

Average weight of all the officers working = (A-6) = (56-6) = 50 kg.

Average weight of all the assistants working = (B+5) = (35+5) = 40 kg.

Average weight of all the employees (officers + assistants) working = 46 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 6:4 = 3:2

Total number of employees working = 200

Number of officers working = 3/5 of 200 = 120

Number of assistants working = 2/5 of 200 = 80

Marketing department

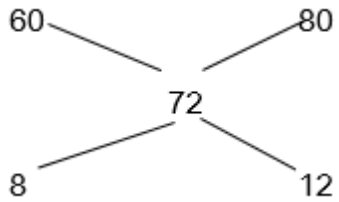
Average weight of all the officers working = (B+25) = (35+25) = 60 kg.

Average weight of all the assistants working = (A+24) = (56+24) = 80 kg.

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Average weight of all the employees (officers + assistants) working = 72 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 8:12 = 2:3

Total number of employees working = 250

Number of officers working = 2/5 of 250 = 100

Number of assistants working = 3/5 of 250 = 150

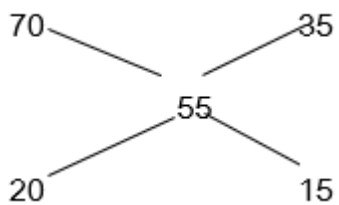
Finance department

Average weight of all the officers working = (A+14) = (56+14) = 70 kg.

Average weight of all the assistants working = 35 kg.

Average weight of all the employees (officers + assistants) working = 55 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 20:15 = 4:3

Total number of employees working = 140

Number of officers working = 4/7 of 140 = 80

Number of assistants working = 3/7 of 140 = 60

Accounts department

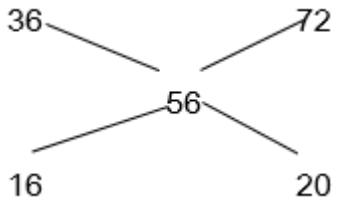
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Average weight of all the assistants working = (A+16) = (56+16) = 72 kg.

Average weight of all the employees (officers + assistants) working = 56 kg.

Using allegation method,

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So, the ratio of the number of officers to that of assistants working = 16:20 = 4:5

The total number of employees working = 360

Number of officers working = 4/9 of 360 = 160

Number of assistants working = 5/9 of 360 = 200

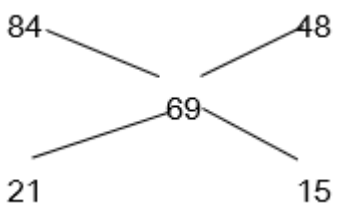
Sales department

Average weight of all the officers = 1.5A = (1.5×56) = 84 kg.

Average weight of all the assistants (B+13) = (35+13) = 48 kg.

Average weight of all the employees (officers + assistants) = 69 kg.

Using allegation method,



So, the ratio of the number of officers to that of assistants working = 21:15 = 7:5

The total number of employees working = 120

Number of officers working = 7/12 of 120 = 70

Number of assistants working = 5/12 of 120 = 50

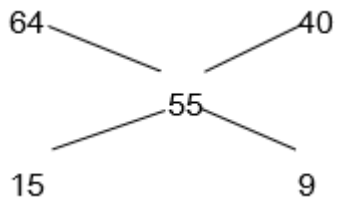
Department	Total number of employees working	Number of officers working	Number of assistants working	Average weight of all the officers (in kg.)	Average weight of all the assistants (in kg.)	Average weight of all the employees (in kg.)
HR	200	120	80	50	40	46
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Accounts	360	160	200	36	72	56
Sales	120	70	50	84	48	69
Total	1070	530	540	300	275	298

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Answer: B

Operation department

Using allegation method,



So, the ratio of the number of officers and assistants working = 15:9 = 5:3. Let it be 5u and 3u respectively.

So,

5u – 3u = 100

2u = 100

u = 50

So, the total number of employees working = (5u+3u) = 8u = (8×50) = 400

Hence, the total number of male employees (officers + assistants) working = 7/10 of 400 = 280.

13. Questions

From the pie chart, we have,

Total value of pie chart = 100%

(M+L-8) + L + (L-3) + M + (M-9) = 100

3M + 3L = 120

M + L = 40…….(i)

Also,

(L-3)/(M-9) = 3/4

4L – 12 = 3M – 27

3M – 4L = 15…….(ii)

Solving both the equations, we get,

3M + 3L = 120

3M – 4L = 15

7L = 105

L = 15

Now, put the value of L in equation (i), we get,

M + L = 40

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M = 40 – 15

M = 25

So, the percentage distribution of the total number of students (boys + girls) studying in each college is,

A = (M+L-8) = (25+15-8) = 32%

B = L = 15%

C = (L-3) = (15-3) = 12%

D = M = 25%

E = (M-9) = (25-9) = 16%

Now,

Smallest composite number = 4

So, K = 4

Total number of students (boys + girls) studying in college A = $K^k = 4^4 = 256$

So,

32% = 256

1% = (256/32) = 8

100% = (100×8) = 800

Therefore, the total number of students (boys + girls) studying in college in all the colleges together = 800

So,

400N = 800

N = (800/400) = 2

Now, the total number of students (boys + girls) studying in each college,

A = 32% of 800 = 256

B = 15% of 800 = 120

C = 12% of 800 = 96

D = 25% of 800 = 200

E = 16% of 800 = 128

Total = 800

Answer: C

Now, according to the question,

Let, the number of girls studying in college C be p.

College C

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Total number of students studying = 96

Number of girls studying = p

Number of boys studying = (96-p)

College B

Total number of students studying = 120

Number of girls studying = {(2×p)+11} = (2p+11)

Number of boys studying = {120-(2p+11)} = (109-2p)

Now,

(96-p) – (109-2p) = 19

77 – p = 109 – 2p

p = 32

So, the total number of boys in both the classes together = {(96-p)+(109-2p)} = {(96-32)+(109-64)} = (64+45) = 109

And, the total number of girls in both the classes together = {(p)+(2p+11)} = {32+(64+11)} = (32+75) = 107

Hence, the required difference = (109-107) = 2.

14. Questions

From the pie chart, we have,

Total value of pie chart = 100%

(M+L-8) + L + (L-3) + M + (M-9) = 100

3M + 3L = 120

M + L = 40…….(i)

Also,

(L-3)/(M-9) = 3/4

4L – 12 = 3M – 27

3M – 4L = 15…….(ii)

Solving both the equations, we get,

3M + 3L = 120

3M – 4L = 15

7L = 105

L = 15

Now, put the value of L in equation (i), we get,

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M + L = 40

M = 40 – 15

M = 25

So, the percentage distribution of the total number of students (boys + girls) studying in each college is,

A = (M+L-8) = (25+15-8) = 32%

B = L = 15%

C = (L-3) = (15-3) = 12%

D = M = 25%

E = (M-9) = (25-9) = 16%

Now,

Smallest composite number = 4

So, K = 4

Total number of students (boys + girls) studying in college A = $K^k = 4^4 = 256$

So,

32% = 256

1% = (256/32) = 8

100% = (100×8) = 800

Therefore, the total number of students (boys + girls) studying in college in all the colleges together = 800

So,

400N = 800

N = (800/400) = 2

Now, the total number of students (boys + girls) studying in each college,

A = 32% of 800 = 256

B = 15% of 800 = 120

C = 12% of 800 = 96

D = 25% of 800 = 200

E = 16% of 800 = 128

Total = 800

Answer: B

College D

Number of boys pursuing course X = S

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Number of girls pursuing X = (S-42)

So,

$\{S\times(S-42)\} = 2160$

$S^2 - 42S = 2160$

$S^2 - 42S - 2160 = 0$

$S^2 - 72S + 30S - 2160 = 0$

$S(S-72) + 30(S-72) = 0$

$(S+30)(S-72) = 0$

$S = -30$ or 72

Therefore, S = 72 (positive integer)

Number of boys pursuing course Y = (T+22)

Number of girls pursuing Y = T

So,

$\{(T+22)\times T\} = 2280$

$T^2 + 22T = 2280$

$T^2 + 22T - 2280 = 0$

$T^2 + 60T - 38T - 2280 = 0$

$T(T+60) - 38(T+60) = 0$

$(T-38)(T+60) = 0$

$T = 38$ or -60

Therefore, T = 38 (positive integer)

Hence, the average of the values of S and T = $\{(S+T)/2\} = \{(72+38)/2\} = (110/2) = 55$.

15. Questions

From the pie chart, we have,

Total value of pie chart = 100%

$(M+L-8) + L + (L-3) + M + (M-9) = 100$

$3M + 3L = 120$

$M + L = 40$……(i)

Also,

$(L-3)/(M-9) = 3/4$

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4L – 12 = 3M – 27

3M – 4L = 15…….(ii)

Solving both the equations, we get,

3M + 3L = 120

3M – 4L = 15

7L = 105

L = 15

Now, put the value of L in equation (i), we get,

M + L = 40

M = 40 – 15

M = 25

So, the percentage distribution of the total number of students (boys + girls) studying in each college is,

A = (M+L-8) = (25+15-8) = 32%

B = L = 15%

C = (L-3) = (15-3) = 12%

D = M = 25%

E = (M-9) = (25-9) = 16%

Now,

Smallest composite number = 4

So, K = 4

Total number of students (boys + girls) studying in college A = $K^k = 4^4 = 256$

So,

32% = 256

1% = (256/32) = 8

100% = (100×8) = 800

Therefore, the total number of students (boys + girls) studying in college in all the colleges together = 800

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Now, the total number of students (boys + girls) studying in each college,

A = 32% of 800 = 256

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B = 15% of 800 = 120

C = 12% of 800 = 96

D = 25% of 800 = 200

E = 16% of 800 = 128

Total = 800

Answer: D

College A

Total number of students studying = 256

Total number of boys studying = 45/64 of 256 = 180

Total number of girls studying = 19/64 of 256 = 76

Ratio of the number of boys present to that of girls absent = 100:15 = 20:3. Let, it be 20x and 3x respectively.

Ratio of the number of boys absent to that of girls present = 8:11. Let, it be 8y and 11y respectively.

Now,

20x + 8y = 180……..(i)

And,

3x + 11y = 76…….(ii)

Solving both the equations, we get,

60x + 24y = 540

60x + 220y = 1520

196y = 980

y = 5

Now, put the value of y in equation (i), we get,

20x + 8y = 180

20x = 180 – 40

20x = 140

x = 7

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Therefore, the difference between the number of boys present and absent = (20x-8y) = (140-40) = 100

And, the difference between the number of girls present and absent = (11y-3x) = (55-21) = 34

Hence, the required difference = {(34/100)×100}% = 34%.

16. Questions

From the pie chart, we have,

Total value of pie chart = 100%

(M+L-8) + L + (L-3) + M + (M-9) = 100

3M + 3L = 120

M + L = 40…….(i)

Also,

(L-3)/(M-9) = 3/4

4L – 12 = 3M – 27

3M – 4L = 15…….(ii)

Solving both the equations, we get,

3M + 3L = 120

3M – 4L = 15

7L = 105

L = 15

Now, put the value of L in equation (i), we get,

M + L = 40

M = 40 – 15

M = 25

So, the percentage distribution of the total number of students (boys + girls) studying in each college is,

A = (M+L-8) = (25+15-8) = 32%

B = L = 15%

C = (L-3) = (15-3) = 12%

D = M = 25%

E = (M-9) = (25-9) = 16%

Now,

Smallest composite number = 4

So, K = 4

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Total number of students (boys + girls) studying in college A = $K^k = 4^4 = 256$

So,

32% = 256

1% = (256/32) = 8

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100% = (100×8) = 800

Therefore, the total number of students (boys + girls) studying in college in all the colleges together = 800

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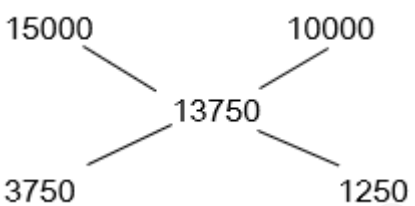
E = 16% of 800 = 128

Total = 800

Answer: A

College E

Using allegation method,



So, the ratio of the number of boys to that of girls studying in college E = 3750:1250 = 3:1

Total number of students studying in college E = 128

Therefore, the number of boys studying in college E = 3/4 of 128 = 96

And, the number of students studying in college C = 96

Hence, required ratio = 96:96 = 1:1.

17. Questions

From the pie chart, we have,

Total value of pie chart = 100%

(M+L-8) + L + (L-3) + M + (M-9) = 100

3M + 3L = 120

M + L = 40……(i)

Also,

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(L-3)/(M-9) = 3/4

4L – 12 = 3M – 27

3M – 4L = 15……(ii)

Solving both the equations, we get,

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So, the percentage distribution of the total number of students (boys + girls) studying in each college is,

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C = (L-3) = (15-3) = 12%

D = M = 25%

E = (M-9) = (25-9) = 16%

Now,

Smallest composite number = 4

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Total number of students (boys + girls) studying in college A = $K^k = 4^4 = 256$

So,

32% = 256

1% = (256/32) = 8

100% = (100×8) = 800

Therefore, the total number of students (boys + girls) studying in college in all the colleges together = 800

So,

400N = 800

N = (800/400) = 2

Now, the total number of students (boys + girls) studying in each college,

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A = 32% of 800 = 256

B = 15% of 800 = 120

C = 12% of 800 = 96

D = 25% of 800 = 200

E = 16% of 800 = 128

Total = 800

Answer: B

College F

Values we got from the common explanation,

L = 15

M = 25

K = 4

N = 2

Total number of students studying in college F = [(L×M)-135] = (375-135) = 240

Ratio of the number of boys to that of girls studying = K:N = 4:2 = 2:1.

So, the number of girls studying = 1/3 of 240 = 80

Hence, the number of girls who are scholarship holders = 70% of 80 = 56.

18. Questions

Answer: A

Quantity of milk = (A+5) liters

Quantity of water = 60 liters

Quantity of honey = (B+25) liters

Now, after removing 15 liters of milk and 30 liters of honey from the mixture,

(A+5-15) = (B+25-30)

A – 10 = B – 5

A – B = 5…….(i)

Also, initial ratio of the quantity of milk to that of honey was 5:6,

(A+5)/(B+25) = 5/6

6A + 30 = 5B + 125

6A – 5B = 95…….(ii)

Solving both the equations, we get,

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6A – 6B = 30

6A – 5B = 95

B = 65

Now, put the value of B in equation (i), we get,

A – B = 5

A = 5 + 65

A = 70

Hence, the initial quantity of mixture was = {(A+5)+60+(B+25)} = {(70+5)+60+(65+25)} = (75+60+90) = 225 liters.

19. Questions

Answer: D

Scheme Y

Total amount invested = Rs. (P+16000)

Rate of interest (p.a.) = 10%

Time duration = 1 year (half yearly basis)

So, the rate of interest = (10/2)% = 5% p.a.

And, the time duration = (1×2) = 2 years

[(5+5)+{(5×5)/100}]% = (10+0.25)% = 10.25%

Interest received = Rs. 5125

So,

10.25% = 5125

1% = 500

100% = (500×100) = Rs. 50000

Therefore, total amount invested in scheme B = Rs. 50000

So,

(P+16000) = 50000

P = (50000-16000)

P = 34000

Scheme X

Rate of interest (p.a.) = 12.50%

Time duration = 2 years (yearly basis)

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$[(12.50+12.50)+\{(12.50\times12.50)/100\}]\% = (25+1.5625)\% = 26.5625\%$

Interest received = P = Rs. 34000

So,

$26.5625\% = 34000$

$1\% = (34000/26.5625) = 1280$

$100\% = (1280\times100) = \text{Rs. } 128000$

Hence, the total amount invested in scheme A = Rs. 128000.

Alter method,

Scheme Y

Total amount invested = Rs. (P+16000)

Rate of interest (p.a.) = 10%

Time duration = 1 year (half yearly basis)

So, the rate of interest = (10/2)% = 5% p.a.

And, the time duration = (1×2) = 2 years

$CI = P\{[1+R/2/100]^{2N} - 1\}$

$(P+16000)[\{1+5/100\}^2 - 1] = 5125$

$(P+16000)(441/400 - 1) = 5125$

$(P+16000)(41/400) = 5125$

$P+16000 = 50000$

$P = 50000 - 16000$

P = Rs.34000

Scheme X

Rate of interest (p.a.) = 12.50%

Time duration = 2 years (yearly basis)

$P[(1+R/100)^n - 1] = CI$

$P[(1+125/1000)^2 - 1] = 34000$

$P[(1+1/8)^2 - 1] = 34000$

$P[81/64 - 1] = 34000$

$P(17/64) = 34000$

$P = 2000*64$

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P = Rs.128000

20. Questions

Answer: B

Total balls in bag = 2K + 3

Black balls in bag = K

So, white balls in bag = 2K + 3 – K = K + 3

Now,

$^{K+3}C_2 / {}^{2K+3}C_2 = 12/35$

[(K + 3) (K + 2)] / [(2K + 3) (2K + 2)] = 12/35

No need to solve this dicey equation, we can easily go with options.

35 = 7 x 5

There should be a multiple of 7 and 5 in denominator.

By simple putting values, in only option b, we are getting multiples of both 5 and 7, not in other option.

If K = 6, then

(9 x 8) / (14 x 15) = 12/35

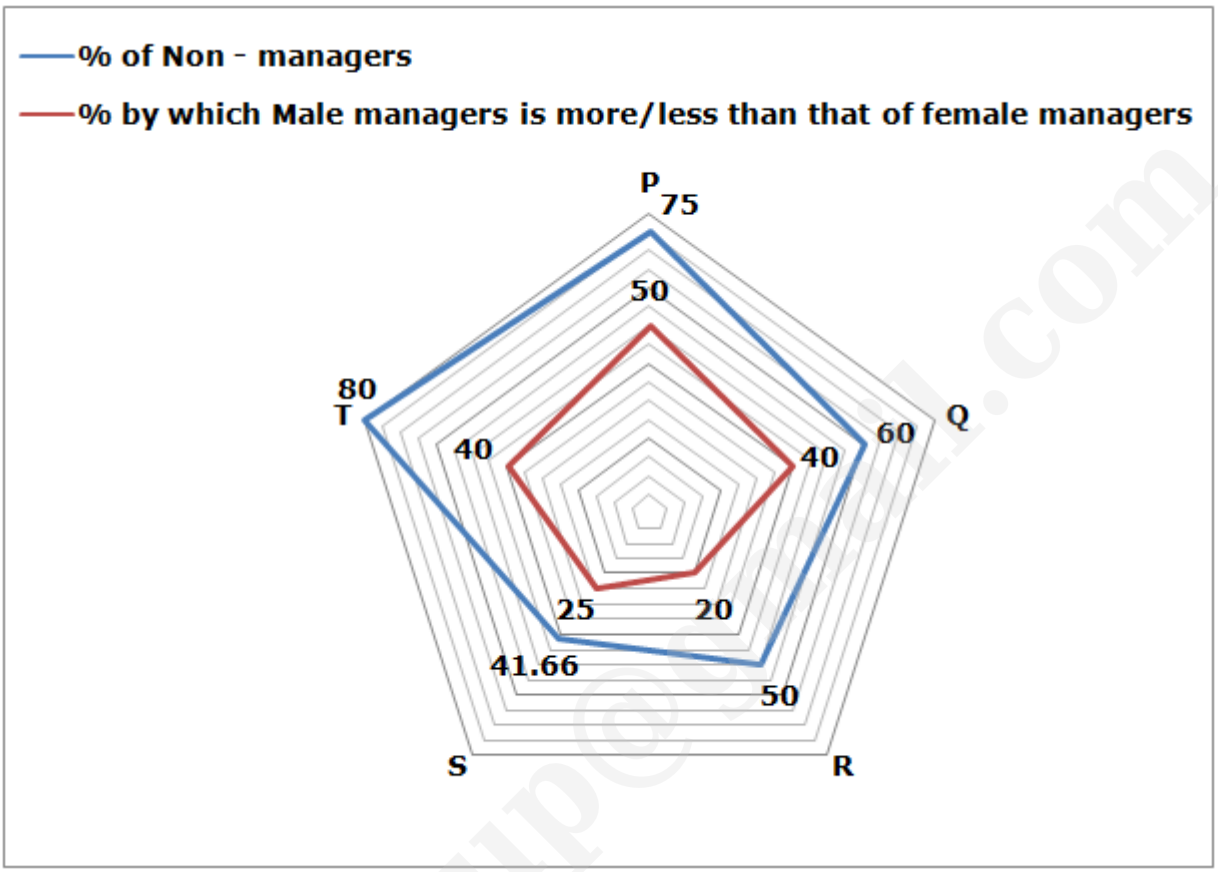
Hence answer is option B

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1. Questions

Read the following information carefully and answer the questions based on it.

The given below radar graph shows the % by which number of male managers in a company is more or less than that of female managers, and % of non – managers in a company out of total employees in that company.



Note:

a). Total number of managers in all five companies together is 1247, while number of managers in company R is same as total number of managers in company P and Q together.

b). Number of male managers < Number of female managers (companies Q and S only).

c). Number of managers in company S is 175, which is 17 less than that of company T, while number of managers in company Q is 20% more than that of company P.

In company Q, number of male employees is twice as that of females, then find number of non–manager females?

a. 110

b. 50

c. 80

d. 180

e. None of these

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2. Questions

In company R, number of female employees is 80 more than those male employees, then find total number of non–manager males is what percent of total number male employees.

a. 40%

b. 25%

c. 60%

d. 50%

e. None of these

3. Questions

Find N is how much % more or less than that of M.

I). M = difference between number of male managers in company P and female managers in company T.

II). N = Difference between female managers in company Q and male managers in company S.

a. 46.33%

b. 62.5%

c. 87.5%

d. 50%

e. None of these

4. Questions

Difference between the total number of employees in companies S and T is what percent of difference between total number of employees in companies P and Q?

a. 430%

b. 230%

c. 630%

d. 530%

e. None of these

5. Questions

Read the following information carefully and answer the questions based on it.

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Column 1	Column 2
1). $P^2 - 14P + L = 0$ a and b are roots of equation, such that sum of reciprocal of a and b is 14/45. (a > b)	**I).** one of them is either perfect square or perfect cube
2). $Q^2 + 8Q - M = 0$, c and d are roots of equation, where c – d = - 24	**II).** Difference of square of roots is exactly divisible by 4
3). $R^2 - 16R - N = 0$, e and f are roots of equation, where multiplication of reciprocal of roots is – 1/ (960). Value of f is negative. (e > f)	**III).** negative root is < - 10

Find which of the following statement(s) is/are true.

Z). b/e = 1/8

Y). unit digit of $(c - f)^8$ is 1.

K). (M + 253) is a perfect cube

a. Only Z and Y are true

b. Only K is true

c. Only Y and K are true

d. Only Z and K are true

e. None of these

6. Questions

Which of the following combination(s) is/are true.

Z). 1 – I, 2 – II, 3 – III

Y). 1 – II, 2 – I, 3 – I

K). 1 – II, 2 – III, 3 – III

a. Z and Y only

b. Z and K only

c. Y and K only

d. Z, Y, and K

e. None of these

7. Questions

Find the remainder when (L + N) divided by 13?

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a. 4

b. 6

c. 9

d. 5

e. None of these

8. Questions

Train A can cross a 120 m long tunnel in 24 seconds and train B can cross the same tunnel in ___ seconds. The speeds of trains A and B are 15 m/s and 25 m/s respectively and train B can cross train A in 10 seconds, while running in opposite direction.

Find the value, which can be filled in the blank?

a. 12.8 seconds

b. 8 seconds

c. 11.2 seconds

d. 14.4 seconds

e. 10.4 seconds

9. Questions

A person invested a total amount of Rs.5000 in two different schemes A and B. The scheme A gives simple interest at 10% rate and scheme B gives simple interest at ___% rate. Interest received from scheme B after 4 years is Rs 480 more than that of received from scheme A after 3 years. If interest received from scheme A after 2 years is Rs. 560.

Find the value, which can be filled in the blank?

a. 20%

b. 15%

c. 25%

d. 5%

e. 10%

10. Questions

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The time taken by A and B together to complete a work is 9.6 hours and the time taken by B and C together to complete the work is ___ hours and the time taken by A and C together to complete the work is 7(23/31)hours. C is 60% more efficient than B.

Find the value, which can be filled in the blank?

a. 6(2/13) hours

b. 11(7/13) hours

c. 10(10/13) hours

d. 8(6/13) hours

e. 9(3/13) hours

11. Questions

A shopkeeper marked up an article by ___% and sold it at two successive discounts of 5% and 15% on its marked price and earned a profit of Rs. 460. If the shopkeeper had sold the article at marked price, he would have received Rs. 8000.

Find the value, which can be filled in the blank?

a. 33.33%

b. 25%

c. 37.5%

d. 16.67%

e. 20%

12. Questions

Study the following data carefully and answer the given questions:

There are five classes A, B, C, D, and E in a school which consists of some boys and girls.

Probability of selecting of one boy and one girl from class A is 16/35 and ratio of boys to girls in that class is 2: 1.

Boys in class B are 9 less than the boys in class A and probability of selecting 2 boys from class B is 3/17.

Ratio of total students in class B to that in class C is 5: 4 and probability of selecting two students (either both boys or both girls) from class C is 31/63. Boys in class C is more than girls in that class.

Girls in class D is 5 more than boys in that class and probability of selecting one boy and one girl from class D is 50/99. Total students in class D are less than 50.

Average boys in all the five classes are 18 while probability of selecting two girls from class E is 1/19.

Find the sum of probability of selecting two boys from class A and probability of selecting two girls from class D.

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a. 205/811

b. 315/907

c. 657/1216

d. 756/1055

e. 856/1155

13. Questions

What is the average number of the students in all the five classes of the school together?

a. 36.4

b. 32.8

c. 33.6

d. 34

e. 30.8

14. Questions

Find the probability of selecting two students from class E such that boys are either more than or equal to girls.

a. 21/38

b. 34/57

c. 18/19

d. 6/11

e. 17/23

15. Questions

If a class F is formed by merging two classes B and D together, then find the probability of selecting one boy and one girl from class F.

a. 203/757

b. 115/316

c. 232/413

d. 315/632

e. 191/312

16. Questions

Study the following information carefully and answer the questions given below.

The table shows the number of trains that passed through five different bridges on Monday and some information about the number of trains that passed through the same bridges on Tuesday.

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Bridge	Number of trains passed on Monday	Number of trains passed on Tuesday
A	X + 100	10% more than that that on Monday
B	2X + 40	3/4th of Monday
C	2X – 30	20% more than that on Monday
D	X + 70	10% less than that on Monday
E	180	2/3rd of Monday

Note : Ratio of the number of trains passing over bridges B and C on Monday is 4:3.

The ratio of the number of local trains and express trains passed on Monday over the bridge E is 3:2 respectively and the ratio of local trains and express trains passed on Tuesday on the same bridge is 3:5 respectively. Find the difference between total local trains passed in two days together and total express trains passed in two days together?

a. 1

b. 2

c. 6

d. 4

e. 3

17. Questions

The number of trains passed on Wednesday over bridge B is 10% more than the trains passing on Monday over bridge C. Find the ratio between the total number of trains passed on Wednesday and Monday together over bridge B to that of number of trains passed on Tuesday and Monday together over bridge C?

a. 83:77

b. 97:88

c. 76:55

d. 73:66

e. 59:44

18. Questions

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Find the difference between the total number of trains passed on Monday over bridges A, B, and C together and the total number of trains passed on Tuesday over bridges B, E, and D together?

a. 258

b. 209

c. 268

d. 254

e. None of these

19. Questions

The number of trains that passed over bridge F on Monday is 5/7th of the number of trains that passed over bridge C on the same day. If the total number of trains passed over bridge F on Monday and Tuesday together is 270, then find the number of trains passed over the bridge F on Tuesday?

a. 120

b. 140

c. 160

d. 180

e. None of these

20. Questions

The total number of trains passed over bridge B in two days together is what percent of the total number of trains passed over bridge E in two days together?

a. 163.33%

b. 153.33%

c. 154.33%

d. 167.33%

e. None of these

Explanations:

1. Questions

Number of managers in company S = 175

So, number of managers in company T = 175 + 17

So, number of managers in companies (P + Q + R) = 1247 – 175 – 192 = 880

Number of managers in companies (P + Q) = Number of managers in company R = 880/2 = 440

Number of managers in company Q is 20% more than that of managers in company P.

So, number of managers in company P = 5/11 x 440 = 200

So, number of managers in company Q = 440 – 200 = 240

In company P,

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Number of managers = 200

Number of Non–managers = 75% of total employees

So, number of managers = 25% of total employees

So, total number of employees = 200/25 x 100 = 800

Number of non–managers = 75% x 800 = 600

% By which male managers is more than that of female managers = 50%

So, number of male managers = 3/5 x 200 = 120

Number of female managers = 200 – 120 = 80

Similarly, we can find the data for all companies.

Company	Total employees	Non-managers	Managers	Male managers	Female managers
P	800	600	200	120	80
Q	600	360	240	90	150
R	880	440	440	240	200
S	300	125	175	75	100
T	960	768	192	112	80

Answer: B

According to question,

In company Q,

Total number of employees = 600

Number of male employees = 2 x number of female employees

So, number of female employees = 1/3 x 600 = 200

Number of female managers = 150

So, number of non–manager females = 200 – 150 = 50

Hence answer is option B

2. Questions

Number of managers in company S = 175

So, number of managers in company T = 175 + 17

So, number of managers in companies (P + Q + R) = 1247 – 175 – 192 = 880

Number of managers in companies (P + Q) = Number of managers in company R = 880/2 = 440

Number of managers in company Q is 20% more than that of managers in company P.

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So, number of managers in company P = 5/11 x 440 = 200

So, number of managers in company Q = 440 – 200 = 240

In company P,

Number of managers = 200

Number of Non–managers = 75% of total employees

So, number of managers = 25% of total employees

So, total number of employees = 200/25 x 100 = 800

Number of non–managers = 75% x 800 = 600

% By which male managers is more than that of female managers = 50%

So, number of male managers = 3/5 x 200 = 120

Number of female managers = 200 – 120 = 80

Similarly, we can find the data for all companies.

Company	Total employees	Non-managers	Managers	Male managers	Female managers
P	800	600	200	120	80
Q	600	360	240	90	150
R	880	440	440	240	200
S	300	125	175	75	100
T	960	768	192	112	80

Answer: A

In company R,

Difference between male and female employees = 80

Total number of employees = 880

So, number of male employees = (880 – 80)/2 = 400

Number of male managers = 240

So, number of non–manager male employees = 400 – 240 = 160

Required percentage = 160/400 x 100 = 40%

Hence answer is option A

3. Questions

Number of managers in company S = 175

So, number of managers in company T = 175 + 17

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So, number of managers in companies (P + Q + R) = 1247 – 175 – 192 = 880

Number of managers in companies (P + Q) = Number of managers in company R = 880/2 = 440

Number of managers in company Q is 20% more than that of managers in company P.

So, number of managers in company P = 5/11 x 440 = 200

So, number of managers in company Q = 440 – 200 = 240

In company P,

Number of managers = 200

Number of Non–managers = 75% of total employees

So, number of managers = 25% of total employees

So, total number of employees = 200/25 x 100 = 800

Number of non–managers = 75% x 800 = 600

% By which male managers is more than that of female managers = 50%

So, number of male managers = 3/5 x 200 = 120

Number of female managers = 200 – 120 = 80

Similarly, we can find the data for all companies.

Company	Total employees	Non-managers	Managers	Male managers	Female managers
P	800	600	200	120	80
Q	600	360	240	90	150
R	880	440	440	240	200
S	300	125	175	75	100
T	960	768	192	112	80

Answer: C

Male managers in company P = 120

Female managers in company T = 80

Value of M = 120 – 80 = 40

Male managers in company S = 75

Female managers in company Q = 150

Value of N = 150 – 75 = 75

Required % change = (75 – 40)/40 x 100 = 87.5%

Hence answer is option C

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4. Questions

Number of managers in company S = 175

So, number of managers in company T = 175 + 17

So, number of managers in companies (P + Q + R) = 1247 – 175 – 192 = 880

Number of managers in companies (P + Q) = Number of managers in company R = 880/2 = 440

Number of managers in company Q is 20% more than that of managers in company P.

So, number of managers in company P = 5/11 x 440 = 200

So, number of managers in company Q = 440 – 200 = 240

In company P,

Number of managers = 200

Number of Non–managers = 75% of total employees

So, number of managers = 25% of total employees

So, total number of employees = 200/25 x 100 = 800

Number of non–managers = 75% x 800 = 600

% By which male managers is more than that of female managers = 50%

So, number of male managers = 3/5 x 200 = 120

Number of female managers = 200 – 120 = 80

Similarly, we can find the data for all companies.

Company	Total employees	Non-managers	Managers	Male managers	Female managers
P	800	600	200	120	80
Q	600	360	240	90	150
R	880	440	440	240	200
S	300	125	175	75	100
T	960	768	192	112	80

Answer: E

Difference between total employees in company S and T = 960 – 300 = 660

Difference between total employees in company P and Q = 800 – 600 = 200

Required percentage = 660/200 x 100 = 330%

Hence answer is option E

5. Questions

1). $P^2 - 14P + L = 0$

$1/a + 1/b = 14/45$

$(a + b)/ab = 14/45$

Sum of roots = - (- 14)/1 = 14

So, (a x b) = 45

Value of L = 45

45 = 9 x 5

So, a = 9, and b = 5

I). One of them is either perfect square or perfect cube [follows]

9 is perfect square.

II). Difference of square of roots is exactly divisible by 4 [follows]

$9^2 - 5^2 = 56$ (divisible by 4)

III). Negative root is < - 10 [don't follows]

There is no negative root

Combinations = 1 – I, 1 - II

2). $Q^2 + 8Q - M = 0$, c and d are roots of equation, where c – d = - 24

Sum of roots = - 8

Difference of roots = - 24

So, c = - 16

And d = 8

Value of M = - 16 x 8 = - 128

I). One of them is either perfect square or perfect cube [Follows]

8 is perfect cube

II). Difference of square of roots is exactly divisible by 4 [follows]

$(-16)^2 - 8^2 = 192$ (divisible by 4)

III). Negative root is < - 10 [follows]

- 16 < - 10

Combinations = 2 – I, 2 – II, 2 - III

3). $R^2 - 16R - N = 0$,

1/ (e x f) = - 1/960

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So, (e x f) = - 960

So, N = 960

Now,

$R^2 – 16R – 960 = 0$

(R – 40) (R + 24) = 0

So, value of R = 40, - 24

So, e = 40, and f = - 24

I). One of the is either perfect square or perfect cube [Don't follows]

II). Difference of square of roots is exactly divisible by 4 [follows]

$40^2 – (- 24)^2 = 1024$ (divisible by 4)

III). Negative root is < - 10 [Follows]

- 24 < - 10

Combinations = 3 – II, 3 – III

Answer: D

According to Question,

Z). b/e = 1/8

b/e = 5/40 = 1/8

This statement is true.

Y). unit digit of $(c – f)^8$ is 1.

(c – f) = - 16 + 24 = 8

Unit digit of $8^8 = 6$

This statement is not true

K). (M + 253) is a perfect cube

Value of M = - 128

(M + 253) = (- 128 + 253) = 125

This statement is true.

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So, Only Z and K are true.

Hence answer is option D

6. Questions

1). $P^2 – 14P + L = 0$

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$1/a + 1/b = 14/45$

$(a + b)/ab = 14/45$

Sum of roots = - (- 14)/1 = 14

So, (a x b) = 45

Value of L = 45

45 = 9 x 5

So, a = 9, and b = 5

I). One of them is either perfect square or perfect cube [follows]

9 is perfect square.

II). Difference of square of roots is exactly divisible by 4 [follows]

$9^2 – 5^2 = 56$ (divisible by 4)

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There is no negative root

Combinations = 1 – I, 1 - II

2). $Q^2 + 8Q – M = 0$, c and d are roots of equation, where c – d = - 24

Sum of roots = - 8

Difference of roots = - 24

So, c = - 16

And d = 8

Value of M = - 16 x 8 = - 128

I). One of them is either perfect square or perfect cube [Follows]

8 is perfect cube

II). Difference of square of roots is exactly divisible by 4 [follows]

$(-16)^2 – 8^2 = 192$ (divisible by 4)

III). Negative root is < - 10 [follows]

- 16 < - 10

Combinations = 2 – I, 2 – II, 2 - III

3). $R^2 – 16R – N = 0$,

1/ (e x f) = - 1/960

So, (e x f) = - 960

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So, N = 960

Now,

$R^2 - 16R - 960 = 0$

$(R - 40)(R + 24) = 0$

So, value of R = 40, - 24

So, e = 40, and f = - 24

I). One of the is either perfect square or perfect cube [Don't follows]

II). Difference of square of roots is exactly divisible by 4 [follows]

$40^2 - (-24)^2 = 1024$ (divisible by 4)

III). Negative root is < - 10 [Follows]

- 24 < - 10

Combinations = 3 – II, 3 – III

Answer: B

We have the following possible combinations

Combinations = 1 – I, 1 – II

Combinations = 2 – I, 2 – II, 2 - III

Combinations = 3 – II, 3 – III

Only Z and K follows.

Hence answer is option B

7. Questions

1). $P^2 - 14P + L = 0$

1/a + 1/b = 14/45

(a + b)/ab = 14/45

Sum of roots = - (- 14)/1 = 14

So, (a x b) = 45

Value of L = 45

45 = 9 x 5

So, a = 9, and b = 5

I). One of them is either perfect square or perfect cube [follows]

9 is perfect square.

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II). Difference of square of roots is exactly divisible by 4 [follows]

$9^2 – 5^2 = 56$ (divisible by 4)

III). Negative root is < - 10 [don't follows]

There is no negative root

Combinations = 1 – I, 1 - II

2). $Q^2 + 8Q – M = 0$, c and d are roots of equation, where c – d = - 24

Sum of roots = - 8

Difference of roots = - 24

So, c = - 16

And d = 8

Value of M = - 16 x 8 = - 128

I). One of them is either perfect square or perfect cube [Follows]

8 is perfect cube

II). Difference of square of roots is exactly divisible by 4 [follows]

$(-16)^2 – 8^2 = 192$ (divisible by 4)

III). Negative root is < - 10 [follows]

- 16 < - 10

Combinations = 2 – I, 2 – II, 2 - III

3). $R^2 – 16R – N = 0$,

1/ (e x f) = - 1/960

So, (e x f) = - 960

So, N = 960

Now,

$R^2 – 16R – 960 = 0$

(R – 40) (R + 24) = 0

So, value of R = 40, - 24

So, e = 40, and f = - 24

I). One of the is either perfect square or perfect cube [Don't follows]

II). Difference of square of roots is exactly divisible by 4 [follows]

$40^2 – (- 24)^2 = 1024$ (divisible by 4)

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III). Negative root is < - 10 [Follows]

- 24 < - 10

Combinations = 3 – II, 3 – III

Answer: A

(L + N) = 960 + 45 = 1005

Required remainder = 1005/13 = 4

Hence answer is option A

8. Questions

Answer: C

Let the length of trains A and B are 'x' m and 'y' m respectively.

Since, train A can cross a 120 m long tunnel in 24 seconds.

So, (x + 120) = 24*15

x = 240m

Since, train B can cross train A in 10 seconds, while running in opposite direction.

So, (240 + y) = 10*(15 + 25)

y = 160m

Now, time taken by train B to cross the tunnel = (160 + 120)/25 = 11.2 seconds

9. Questions

Answer: B

Since, interest received from scheme A after 2 years is Rs. 560.

Let the amount invested in scheme A = 'x' rupees

So, (x * 10 * 2)/100 = 560

x = 2800

Amount invested in scheme B = 5000 – 2800 = Rs. 2200

Interest received from scheme A after 3 years = (2800*10*3)/100 = Rs. 840

Interest received from scheme B after 4 years = 840 + 480 = Rs. 1320

Let the rate of interest in scheme B = 'y'%

So, (2200*y*4)/100 = 1320

y = 15%

10. Questions

Answer: E

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Let the time taken by A alone to complete the work = ‘x’ hours

The time taken by B alone to complete the work = ‘8y’ hours

And time taken by C alone to complete the work = 8y * (100/160) = ‘5y’ hours

So, (1/x) + (1/8y) = 1/9.6 ----------(1)

And, (1/x) + (1/5y) = 31/240 ----------(2)

From equations (1) and (2):

(1/9.6) – (1/8y) = (31/240) – (1/5y)

(31/240) – (1/9.6) = (1/5y) – (1/8y)

(31 – 25)/240 = (8 – 5)/40y

y = 3

The time taken by B alone to complete the work = 8 * 3 = 24 hours

The time taken by C alone to complete the work = 5 * 3 = 15 hours

(1/24) + (1/15) = (5 + 8)/120 = 13/120

So, the time taken by B and C together to complete the work = 9(3/13) hours

11. Questions

Answer: A

Let the CP of the article = ‘x’ rupees

Let marked up percent = ‘y’%

Marked price of the article = Rs.8000

So,

[8000 * (95/100) * (85/100)] – x = 460

(1615*4) – x= 460

x = 6000

CP of the article = Rs. 6000

MP of the article = Rs. 8000

Required percentage = [(8000 – 6000)/6000]*100 = 33(1/3) %

12. Questions

Let boys and girls in class A are ‘2a’ and ‘a’ respectively.

Total students in class A = 2a + a = 3a

Probability of selecting of one boy and one girl from class A = 16/35

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$$\frac{2a_{C_1} * a_{C_1}}{3a_{C_2}} = \frac{16}{35}$$

$$\frac{2a * a}{3a * (3a-1) \div 2} = \frac{16}{35}$$

$$\frac{a}{3(3a-1)} = \frac{4}{35}$$

35a = 36a – 12

a = 12

Boys in class A = 2a = 24

Girls in class A = a = 12

Total students in class A = 36

Boys in class B = 24 – 9 = 15

Let girls in class B = 'b'

Total students in class B = (15 + b)

Probability of selecting 2 boys from class B = 3/17

$$\frac{15_{C_2}}{(15+b)_{C_2}} = \frac{3}{17}$$

$$\frac{210}{(15+b) * (14+b)} = \frac{3}{17}$$

$$\frac{70}{(15+b) * (14+b)} = \frac{1}{17}$$

(15 + b) (14 + b) = 70 * 17

$b^2 + 29b – 980 = 0$

b= 20, -49(Negative values are neglected)

b = 20

Girls in class B = 20

Now, Total students in class B = 35

Total students in class C = (15 + 20) * (4/5) = 28

Let boys and girls in class C is 'c' and '28 – c' respectively.

Probability of selecting two students (either both boys or both girls) from class C = 31/63

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$$\frac{c_{C_2} + (28-c)_{C_2}}{28_{C_2}} = \frac{31}{63}$$

$$\frac{[c\,(c-1) \div 2] + [(28-c)(27-c) \div 2]}{14 * 27} = \frac{31}{63}$$

$$\frac{c^2 - c + 756 - 55c + c^2}{2 * 2 * 3} = 31$$

$2c^2 - 56c + 756 = 31 * 2 * 6$

$c^2 - 28c + 378 = 186$

$c^2 - 28c + 192 = 0$

$c^2 - 16c - 12c + 192 = 0$

$c = 16, 12$

We take,

$c = 16$

Because boys in class C is more than girls in that class.

Boys in class C = c = 16

Girls in class C = 28 – c = 12

Let boys and girls in class D is 'd' and 'd + 5' respectively.

Total students in class D = d + (d + 5) = (2d + 5)

Probability of selecting one boy and one girl from class D = 50/99

$$\frac{d_{C_1} * (d+5)_{C_1}}{(2d+5)_{C_2}} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(2d+4) \div 2} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(d+2)} = \frac{50}{99}$$

$99d\,(d + 5) = 50\,((2d + 5)\,(d + 2))$

$99d^2 + 495d = 100d^2 + 450d + 500$

$d^2 – 45d + 500 = 0$

$d^2 – 25d – 20d + 500 = 0$

$d = 25, 20$

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We take,

d = 20

Because total students in class D are less than 50

Boys in class D = d = 20

Girls in class D = d + 5 = 25

Total students in class D = 45

Boys in class E = 5 * 18 – 24 – 15 – 16 – 20 = 15

Let girls in class E = 'e'

Total students in class E = 'e + 15'

Probability of selecting two girls from class E = 1/19

$$\frac{e_{C_2}}{(e+15)_{C_2}} = \frac{1}{19}$$

$$\frac{e*(e-1)}{(e+15)*(e+14)} = \frac{1}{19}$$

19e (e – 1) = (e + 15) (e + 14)

$19e^2 – 19e = e^2 + 29e + 210$

$18e^2 – 48e – 210 = 0$

$3e^2 – 8e – 35 = 0$

$3e^2 – 15e + 7e – 35 = 0$

e = 5, -7/3(Negative values are neglected)

e = 5

Girls in class E = e = 5

Total students in class E = 20

Classes	Total boys	Total girls	Total students
A	24	12	24 + 12 = 36
B	15	20	15 + 20 = 35
C	16	12	16 + 12 = 28
D	20	25	20 + 25 = 45
E	15	5	15 + 5 = 20

Answer: E

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Probability of selecting two boys from class A = ${}^{24}C_2/{}^{36}C_2 = 46/105$

Probability of selecting two girls from class D = ${}^{25}C_2/{}^{45}C_2 = 10/33$

Required sum = (46/105) + (10/33)

= (506 + 350)/1155

= 856/1155

13. Questions

Let boys and girls in class A are '2a' and 'a' respectively.

Total students in class A = 2a + a = 3a

Probability of selecting of one boy and one girl from class A = 16/35

$$\frac{2a_{C_1} * a_{C_1}}{3a_{C_2}} = \frac{16}{35}$$

$$\frac{2a * a}{3a * (3a-1) \div 2} = \frac{16}{35}$$

$$\frac{a}{3(3a-1)} = \frac{4}{35}$$

35a = 36a – 12

a = 12

Boys in class A = 2a = 24

Girls in class A = a = 12

Total students in class A = 36

Boys in class B = 24 – 9 = 15

Let girls in class B = 'b'

Total students in class B = (15 + b)

Probability of selecting 2 boys from class B = 3/17

$$\frac{15_{C_2}}{(15+b)_{C_2}} = \frac{3}{17}$$

$$\frac{210}{(15+b) * (14+b)} = \frac{3}{17}$$

$$\frac{70}{(15+b) * (14+b)} = \frac{1}{17}$$

(15 + b) (14 + b) = 70 * 17

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$b^2 + 29b - 980 = 0$

b= 20, -49(Negative values are neglected)

b = 20

Girls in class B = 20

Now, Total students in class B = 35

Total students in class C = (15 + 20) * (4/5) = 28

Let boys and girls in class C is 'c' and '28 – c' respectively.

Probability of selecting two students (either both boys or both girls) from class C = 31/63

$$\frac{{}^{c}C_{2} + {}^{(28-c)}C_{2}}{{}^{28}C_{2}} = \frac{31}{63}$$

$$\frac{[c\,(c-1) \div 2] + [(28-c)(27-c) \div 2]}{14 * 27} = \frac{31}{63}$$

$$\frac{c^2 - c + 756 - 55c + c^2}{2 * 2 * 3} = 31$$

$2c^2 - 56c + 756 = 31 * 2 * 6$

$c^2 - 28c + 378 = 186$

$c^2 - 28c + 192 = 0$

$c^2 - 16c - 12c + 192 = 0$

c = 16, 12

We take,

c = 16

Because boys in class C is more than girls in that class.

Boys in class C = c = 16

Girls in class C = 28 – c = 12

Let boys and girls in class D is 'd' and 'd + 5' respectively.

Total students in class D = d + (d + 5) = (2d + 5)

Probability of selecting one boy and one girl from class D = 50/99

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$$\frac{d_{C_1} * (d+5)_{C_1}}{(2d+5)_{C_2}} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(2d+4) \div 2} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(d+2)} = \frac{50}{99}$$

$99d (d + 5) = 50 ((2d + 5) (d + 2))$

$99d^2 + 495d = 100d^2 + 450d + 500$

$d^2 – 45d + 500 = 0$

$d^2 – 25d – 20d + 500 = 0$

$d = 25, 20$

We take,

$d = 20$

Because total students in class D are less than 50

Boys in class D = d = 20

Girls in class D = d + 5 = 25

Total students in class D = 45

Boys in class E = 5 * 18 – 24 – 15 – 16 – 20 = 15

Let girls in class E = ‘e’

Total students in class E = ‘e + 15’

Probability of selecting two girls from class E = 1/19

$$\frac{e_{C_2}}{(e+15)_{C_2}} = \frac{1}{19}$$

$$\frac{e * (e-1)}{(e+15) * (e+14)} = \frac{1}{19}$$

$19e (e – 1) = (e + 15) (e + 14)$

$19e^2 – 19e = e^2 + 29e + 210$

$18e^2 – 48e – 210 = 0$

$3e^2 – 8e – 35 = 0$

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$3e^2 - 15e + 7e - 35 = 0$

e = 5, -7/3(Negative values are neglected)

e = 5

Girls in class E = e = 5

Total students in class E = 20

Classes	Total boys	Total girls	Total students
A	24	12	24 + 12 = 36
B	15	20	15 + 20 = 35
C	16	12	16 + 12 = 28
D	20	25	20 + 25 = 45
E	15	5	15 + 5 = 20

Answer: B

Total number of students in all the five classes of the school together= 36 + 35 + 28 + 45 + 20 = 164

Required average = 164/5

= 32.8

14. Questions

Let boys and girls in class A are '2a' and 'a' respectively.

Total students in class A = 2a + a = 3a

Probability of selecting of one boy and one girl from class A = 16/35

$$\frac{2a_{C_1} * a_{C_1}}{3a_{C_2}} = \frac{16}{35}$$

$$\frac{2a * a}{3a * (3a - 1) \div 2} = \frac{16}{35}$$

$$\frac{a}{3(3a - 1)} = \frac{4}{35}$$

35a = 36a – 12

a = 12

Boys in class A = 2a = 24

Girls in class A = a = 12

Total students in class A = 36

Boys in class B = 24 – 9 = 15

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Let girls in class B = 'b'

Total students in class B = (15 + b)

Probability of selecting 2 boys from class B = 3/17

$$\frac{15_{C_2}}{(15+b)_{C_2}} = \frac{3}{17}$$

$$\frac{210}{(15+b)*(14+b)} = \frac{3}{17}$$

$$\frac{70}{(15+b)*(14+b)} = \frac{1}{17}$$

(15 + b) (14 + b) = 70 * 17

$b^2 + 29b - 980 = 0$

b= 20, -49(Negative values are neglected)

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Girls in class B = 20

Now, Total students in class B = 35

Total students in class C = (15 + 20) * (4/5) = 28

Let boys and girls in class C is 'c' and '28 – c' respectively.

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$2c^2 - 56c + 756 = 31 * 2 * 6$

$c^2 - 28c + 378 = 186$

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$c^2 - 16c - 12c + 192 = 0$

c = 16, 12

We take,

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c = 16

Because boys in class C is more than girls in that class.

Boys in class C = c = 16

Girls in class C = 28 – c = 12

Let boys and girls in class D is ‘d’ and ‘d + 5’ respectively.

Total students in class D = d + (d + 5) = (2d + 5)

Probability of selecting one boy and one girl from class D = 50/99

$$\frac{d_{C_1} * (d+5)_{C_1}}{(2d+5)_{C_2}} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(2d+4) \div 2} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(d+2)} = \frac{50}{99}$$

$99d (d + 5) = 50 ((2d + 5) (d + 2))$

$99d^2 + 495d = 100d^2 + 450d + 500$

$d^2 – 45d + 500 = 0$

$d^2 – 25d – 20d + 500 = 0$

d = 25, 20

We take,

d = 20

Because total students in class D are less than 50

Boys in class D = d = 20

Girls in class D = d + 5 = 25

Total students in class D = 45

Boys in class E = 5 * 18 – 24 – 15 – 16 – 20 = 15

Let girls in class E = ‘e’

Total students in class E = ‘e + 15’

Probability of selecting two girls from class E = 1/19

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$$\frac{e_{C_2}}{(e+15)_{C_2}} = \frac{1}{19}$$

$$\frac{e*(e-1)}{(e+15)*(e+14)} = \frac{1}{19}$$

$19e(e-1) = (e+15)(e+14)$

$19e^2 - 19e = e^2 + 29e + 210$

$18e^2 - 48e - 210 = 0$

$3e^2 - 8e - 35 = 0$

$3e^2 - 15e + 7e - 35 = 0$

e = 5, -7/3(Negative values are neglected)

e = 5

Girls in class E = e = 5

Total students in class E = 20

Classes	Total boys	Total girls	Total students
A	24	12	24 + 12 = 36
B	15	20	15 + 20 = 35
C	16	12	16 + 12 = 28
D	20	25	20 + 25 = 45
E	15	5	15 + 5 = 20

Answer: C

Two students need to be selected from class E. Those two students can be either both boys or one boy and a girl.

Probability of selecting two students from class E = $({}^{15}C_2 + {}^{15}C_1 * {}^{5}C_1)/{}^{20}C_2$

= (105 + 75)/190

= 180/190

= 18/19

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15. Questions

Let boys and girls in class A are '2a' and 'a' respectively.

Total students in class A = 2a + a = 3a

Probability of selecting of one boy and one girl from class A = 16/35

$$\frac{2a_{C_1} * a_{C_1}}{3a_{C_2}} = \frac{16}{35}$$

$$\frac{2a * a}{3a * (3a-1) \div 2} = \frac{16}{35}$$

$$\frac{a}{3(3a-1)} = \frac{4}{35}$$

35a = 36a – 12

a = 12

Boys in class A = 2a = 24

Girls in class A = a = 12

Total students in class A = 36

Boys in class B = 24 – 9 = 15

Let girls in class B = 'b'

Total students in class B = (15 + b)

Probability of selecting 2 boys from class B = 3/17

$$\frac{15_{C_2}}{(15+b)_{C_2}} = \frac{3}{17}$$

$$\frac{210}{(15+b) * (14+b)} = \frac{3}{17}$$

$$\frac{70}{(15+b) * (14+b)} = \frac{1}{17}$$

(15 + b) (14 + b) = 70 * 17

$b^2 + 29b – 980 = 0$

b= 20, -49(Negative values are neglected)

b = 20

Girls in class B = 20

Now, Total students in class B = 35

Total students in class C = (15 + 20) * (4/5) = 28

Let boys and girls in class C is 'c' and '28 – c' respectively.

Probability of selecting two students (either both boys or both girls) from class C = 31/63

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$$\frac{c_{C_2} + (28-c)_{C_2}}{28_{C_2}} = \frac{31}{63}$$

$$\frac{[c\,(c-1) \div 2] + [(28-c)(27-c) \div 2]}{14 * 27} = \frac{31}{63}$$

$$\frac{c^2 - c + 756 - 55c + c^2}{2 * 2 * 3} = 31$$

$2c^2 - 56c + 756 = 31 * 2 * 6$

$c^2 - 28c + 378 = 186$

$c^2 - 28c + 192 = 0$

$c^2 - 16c - 12c + 192 = 0$

$c = 16, 12$

We take,

$c = 16$

Because boys in class C is more than girls in that class.

Boys in class C = c = 16

Girls in class C = 28 – c = 12

Let boys and girls in class D is 'd' and 'd + 5' respectively.

Total students in class D = d + (d + 5) = (2d + 5)

Probability of selecting one boy and one girl from class D = 50/99

$$\frac{d_{C_1} * (d+5)_{C_1}}{(2d+5)_{C_2}} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(2d+4) \div 2} = \frac{50}{99}$$

$$\frac{d * (d+5)}{(2d+5)(d+2)} = \frac{50}{99}$$

$99d\,(d + 5) = 50\,((2d + 5)\,(d + 2))$

$99d^2 + 495d = 100d^2 + 450d + 500$

$d^2 – 45d + 500 = 0$

$d^2 – 25d – 20d + 500 = 0$

$d = 25, 20$

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We take,

d = 20

Because total students in class D are less than 50

Boys in class D = d = 20

Girls in class D = d + 5 = 25

Total students in class D = 45

Boys in class E = 5 * 18 – 24 – 15 – 16 – 20 = 15

Let girls in class E = 'e'

Total students in class E = 'e + 15'

Probability of selecting two girls from class E = 1/19

$$\frac{e_{C_2}}{(e+15)_{C_2}} = \frac{1}{19}$$

$$\frac{e*(e-1)}{(e+15)*(e+14)} = \frac{1}{19}$$

19e (e – 1) = (e + 15) (e + 14)

$19e^2 – 19e = e^2 + 29e + 210$

$18e^2 – 48e – 210 = 0$

$3e^2 – 8e – 35 = 0$

$3e^2 – 15e + 7e – 35 = 0$

e = 5, -7/3(Negative values are neglected)

e = 5

Girls in class E = e = 5

Total students in class E = 20

Classes	Total boys	Total girls	Total students
A	24	12	24 + 12 = 36
B	15	20	15 + 20 = 35
C	16	12	16 + 12 = 28
D	20	25	20 + 25 = 45
E	15	5	15 + 5 = 20

Answer: D

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Total boys in class F = 15 + 20 = 35

Total girls in class F = 20 + 25 = 45

Total students in class F = 35 + 45 = 80

Probability of selecting one boy and one girl from class F = $({}^{35}C_1 * {}^{45}C_1)/{}^{80}C_2$

= (35 * 45)/(40 * 79)

= 315/632

16. Questions

So, (2X + 40): (2X - 30) = 4/3

Or, 6X + 120 = 8X - 120, or, 2X = 240, X = 120

Bridge	Number of trains passed on Monday	Number of trains passed on Tuesday
A	120 + 100 = 220	220 * 110/100 = 242
B	2 * 120 + 40 = 280	280 * 3/4 = 210
C	2 * 120 – 30 = 210	210 * 120/100 = 252
D	120 + 70 = 190	190 * 90/100 = 171
E	180	180 * 2/3 = 120

Answer: C

Required difference = [(180 * 3/5) + (120 * 3/8)] - [(180 * 2/5) + (120 * 5/8)] = 6

17. Questions

So, (2X + 40): (2X - 30) = 4/3

Or, 6X + 120 = 8X - 120, or, 2X = 240, X = 120

Bridge	Number of trains passed on Monday	Number of trains passed on Tuesday
A	120 + 100 = 220	220 * 110/100 = 242
B	2 * 120 + 40 = 280	280 * 3/4 = 210
C	2 * 120 – 30 = 210	210 * 120/100 = 252
D	120 + 70 = 190	190 * 90/100 = 171
E	180	180 * 2/3 = 120

Answer: D

The number of trains passing on Wednesday over the bridge B = 210 * 110/100 = 231

So, Required ratio = [231 + 280]: [210 + 252] = 73:66

18. Questions

So, (2X + 40): (2X - 30) = 4/3

Or, 6X + 120 = 8X - 120, or, 2X = 240, X = 120

Bridge	Number of trains passed on Monday	Number of trains passed on Tuesday
A	120 + 100 = 220	220 * 110/100 = 242
B	2 * 120 + 40 = 280	280 * 3/4 = 210
C	2 * 120 – 30 = 210	210 * 120/100 = 252
D	120 + 70 = 190	190 * 90/100 = 171
E	180	180 * 2/3 = 120

Answer: B

Required difference = [220 + 280 + 210] – [210 + 171 + 120] = 209

19. Questions

So, (2X + 40): (2X - 30) = 4/3

Or, 6X + 120 = 8X - 120, or, 2X = 240, X = 120

Bridge	Number of trains passed on Monday	Number of trains passed on Tuesday
A	120 + 100 = 220	220 * 110/100 = 242
B	2 * 120 + 40 = 280	280 * 3/4 = 210
C	2 * 120 – 30 = 210	210 * 120/100 = 252
D	120 + 70 = 190	190 * 90/100 = 171
E	180	180 * 2/3 = 120

Answer: A

Total number of trains passed over the bridge F on Tuesday = 270 - (5 * 210/7) = 270 – 150 = 120

20. Questions

So, (2X + 40): (2X - 30) = 4/3

Or, 6X + 120 = 8X - 120, or, 2X = 240, X = 120

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Bridge	Number of trains passed on Monday	Number of trains passed on Tuesday
A	120 + 100 = 220	220 * 110/100 = 242
B	2 * 120 + 40 = 280	280 * 3/4 = 210
C	2 * 120 – 30 = 210	210 * 120/100 = 252
D	120 + 70 = 190	190 * 90/100 = 171
E	180	180 * 2/3 = 120

Answer: A

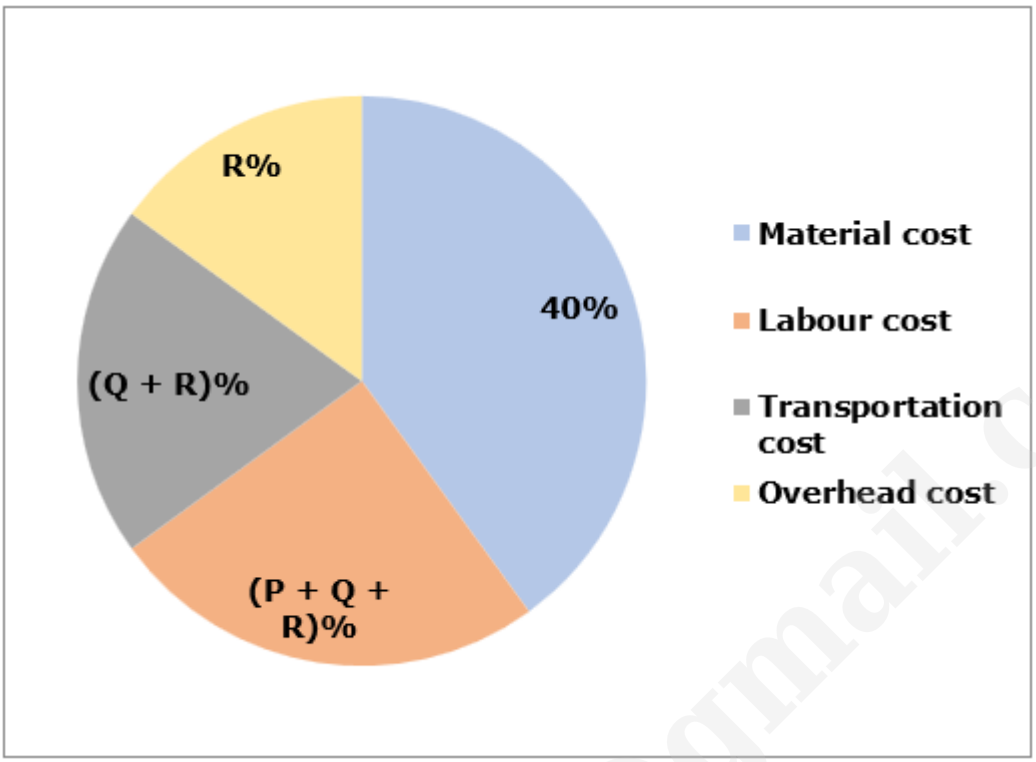
Required percentage = [(280 + 210)/ (180 + 120)] * 100 = 163.33%

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1. Questions

Study the following data carefully and answer the questions:

The pie chart given below shows the price breakup of manufacturing of a product.



Notes:

1: The ratio of Q to R is 1: 3.

2: If the material cost of the product is ₹640, then the transportation cost of the product will be ₹320.

If the difference between the material cost and labour cost of the product is ₹2PQR, then what will be the difference between the transportation cost and overhead cost of the product? (in terms of P, Q and R)

a. ₹12(4P + 4Q – R)

b. ₹10(P + Q + R)

c. ₹15(PQ – R)

d. ₹4(PR – 3Q)

e. ₹8(PQ + R)

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2. Questions

Which of the following statements is/are true?

X: Labour cost is (P + Q + R)% more than transportation cost.

Y: Ratio of material cost to overhead cost is (P + 3): (Q – 2).

Z: Overhead cost is PR% of labour cost.

a. Only X

b. Only X and Z

c. Only Y

d. Only X and Y

e. Only Y and Z

3. Questions

If the product is marked up by (PQ + R – 3P)% more than its total manufacturing cost and its marked price is ₹1500, then find the sum of the material cost and the labour cost of the product?

a. ₹1300

b. ₹650

c. ₹910

d. ₹1040

e. ₹780

4. Questions

If the difference between the initial material cost and initial overhead cost is ₹240 and all types of costs are increased by 25%, then what will be the average of new labour cost and new transportation cost?

a. ₹270

b. ₹216

c. ₹232

d. ₹235

e. None of these

5. Questions

If the overhead cost of the product is a multiple of 9 more than ₹400 and less than ₹420, then which of the following can't be the labour cost of the product?

a. ₹675

b. ₹666

c. ₹690

d. Both (a) and (b)

e. Both (b) and (c)

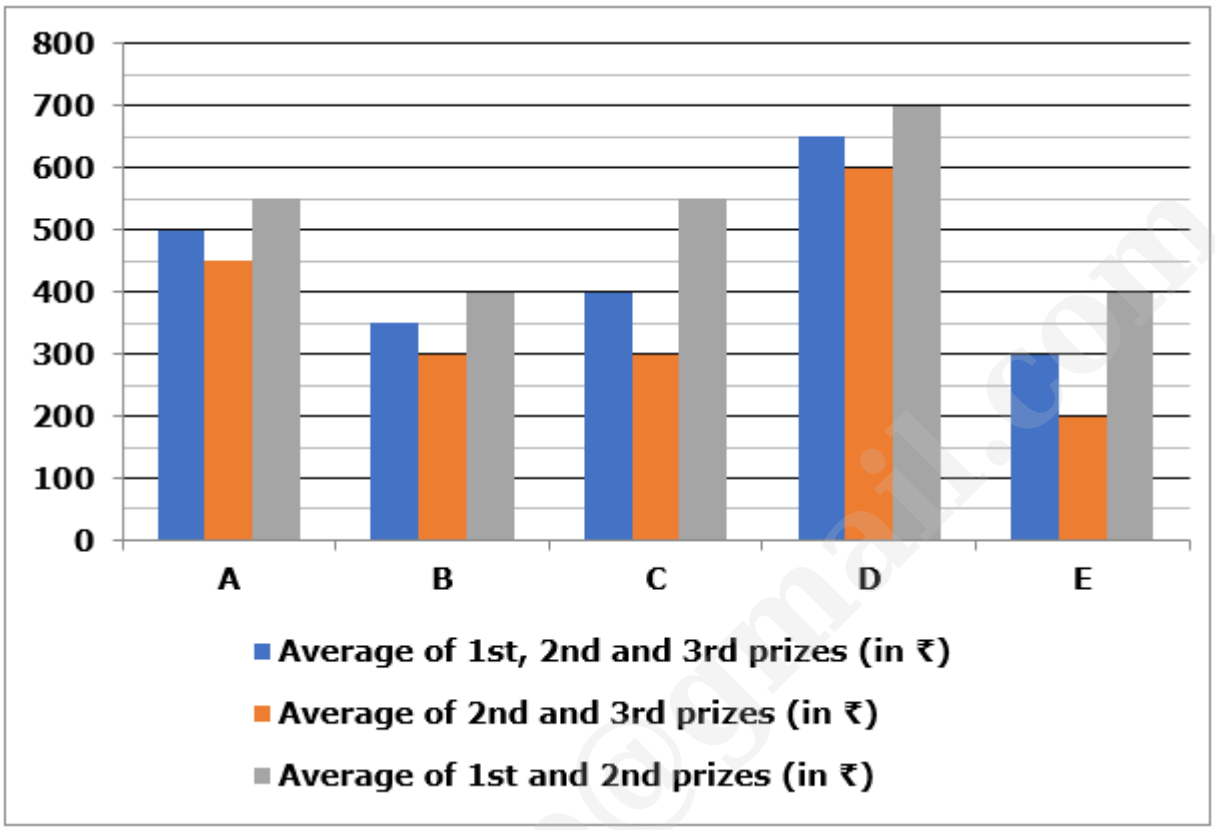
6. Questions

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Study the following data carefully and answer the questions:

The data given below is related to the 1st prize (in ₹), 2nd prize (in ₹) and 3rd prize (in ₹) in five different games A, B, C, D and E organised by a college.

The bar graph given below shows the following data:



Note: In any game, more than one person can be in 1st position, 2nd position, and 3rd position.

If the total amount distributed for the 3rd prize in game B is ₹3000 and that distributed for the 3rd prize in game C is ₹1500, then find the ratio of the number of persons at 3rd position in game B to those in game C?

a. 6: 7

b. 2: 5

c. 4: 5

d. 3: 5

e. 12: 17

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7. Questions

If the number of persons at 1st position in games A, B and C is 3, 4 and 5 respectively, then find the total amount distributed for 1st prize in games A, B and C?

a. ₹5400

b. ₹7200

c. ₹4800

d. ₹7800

e. ₹6600

8. Questions

If the sum of the 3rd prize in games D and E is M% of the sum of 1st prize in games D and E, then find that the sum of 2nd prize in games D and E is what per cent of the sum of 1st prize in games D and E?

a. (M + 18)%

b. (M – 4)%

c. (M + 12)%

d. (M + 24)%

e. (M – 20)%

9. Questions

If the total amount distributed for the 2nd prize in game B is ₹2800 and the number of persons at 2nd position in games C and D is respectively 125% and 75% of those at 2nd position in game B, then find the ratio of the total amount distributed for 2nd prize in game C to that in game D?

a. 10: 13

b. 50: 39

c. 25: 26

d. 5: 13

e. 5: 6

10. Questions

Quantity I: What is 75% of the sum of the 2nd prize in all the 5 games?

Quantity II: What is the 125% of the sum of the 3rd prize in all the 5 games?

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I ≥ Quantity II

d. Quantity I ≤ Quantity II

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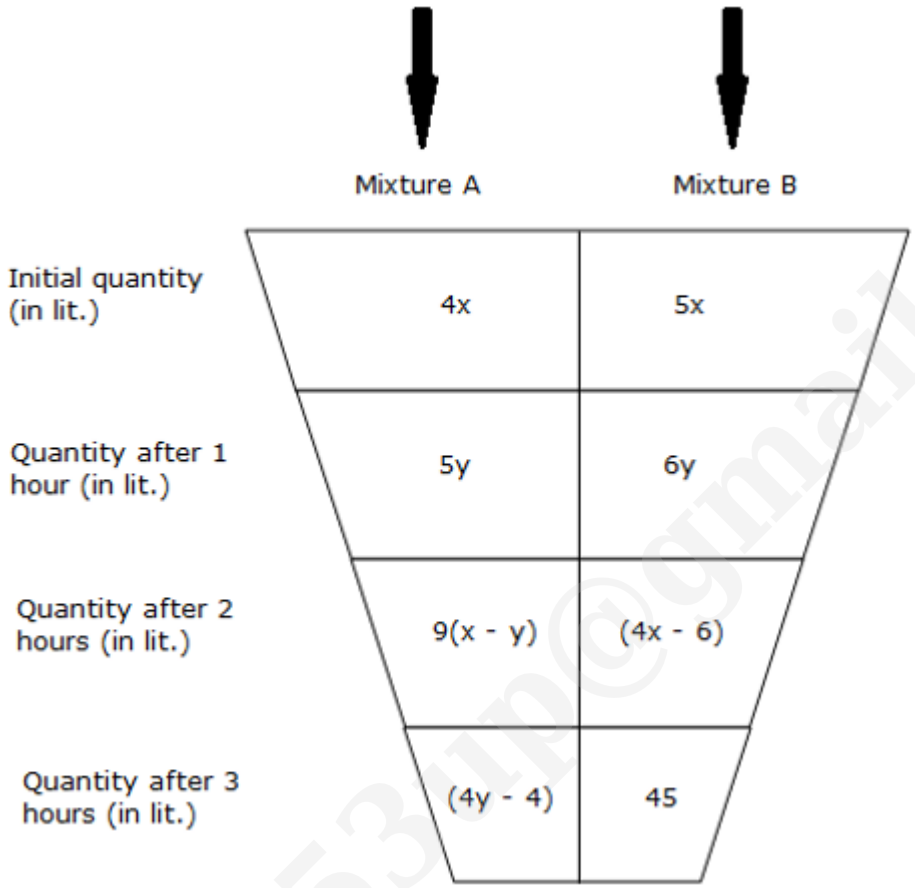
e. Quantity I = Quantity II or relation can't be established

11. Questions

Study the following data carefully and answer the questions:

There are two mixtures (A and B) of milk and water and both the mixtures are evaporated for some time (in hours).

The funnel graph given below shows the following data:



Notes:

1: The evaporation rate of water from each mixture for each hour can be different.

2: The evaporation rate of water from mixture B in the 3rd hour is 0.15 L per minute.

3: The evaporated quantity of water from mixture A in the 3rd hour is 9 L.

What is the difference between the evaporation rate (per minute) of water from mixture A in the 2nd hour and the evaporation rate of water from mixture B in the 2nd hour?

a. (1/120) L per minute

b. (1/36) L per minute

c. (1/60) L per minute

d. (1/45) L per minute

e. (1/15) L per minute

12. Questions

If the initial ratio of milk to water in mixture A is 8: 7 and the initial ratio of milk to water in mixture B is 8: 7, then what will be the ratio of the quantity of water in mixture A to that in mixture B after 1 hour?

a. 6: 5

b. 6: 7

c. 2: 3

d. 9: 10

e. 3: 5

13. Questions

If the ratio of milk to water in mixture A after 1 hour is 3: 2 and the cost of mixture A after 1 hour is ₹36 per L, then what will be the cost of mixture A after 2 hours?

a. ₹80 L

b. ₹50 L

c. ₹30 L

d. ₹60 L

e. ₹40 L

14. Questions

If the evaporation rate of mixture A in the 4th hour is 0.1 L per minute and that of mixture B in the 4th hour is 0.15 L per minute, then what will be the ratio of the quantity of mixture A to that of mixture B after 4 hours?

a. 15: 16

b. 5: 6

c. 6: 7

d. 10: 11

e. 2: 3

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15. Questions

Which of the following statements is/are true?

P: The ratio of the quantity of mixture A to that of mixture B after 2 hours is 5: 6.

Q: The ratio of the quantity of mixture A to that of mixture B after 3 hours is 4: 5.

R: The average quantity of mixtures A and B after 1 hour is 45 L.

a. Only P and Q

b. All are true S

c. Only P and R

d. Only Q

e. Only Q and R

16. Questions

In each question, some statements are given in options. Find that which of the following option/s is/are sufficient to answer the question.

The time taken by boat A to go 1.76 km upstream is 50 seconds more than that taken to come back. Find the speed of boat A in still water?

a. The time taken by boat A to go 1.485 km in the upstream direction is 2.25 minutes

b. Boat B can cover 1.08 km in still water in 1.2 minutes and the ratio of the speed of boat B in still water to the speed of the stream is 6: 1

c. The time taken by boat A to go 1.2 km in the downstream direction is 1.25 minutes

d. Either (a) alone or (c) alone

e. Either (a) alone or (b) alone or (c) alone

17. Questions

There are four friends (A, B, C and D) in a class. The average of present ages of A, B and C is 23 years and the ratio of B's present age to C's present age is 2: 1. Find the average of present ages of A and B?

a. The average of present ages of A, B, C and D is 27.75 years

b. D is 36 years old and the ratio of A's present age to D's present age will be 2: 3

c. The ratio of B's present age to D's present age is 5: 7

d. The average of present ages of B, C and D is 29 years

e. Either (b) alone or (d) alone

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18. Questions

Find the difference between compound interest and simple interest on ₹18000 at R% rate after 2 years?

a. The difference between compound interest and simple interest on ₹6000 at (R/2)% rate after 2 years is ₹33.75

b. The ratio of simple interest on ₹5000 at R% rate after 1 year to that on ₹3000 at 2R% rate after 1 year is 5: 6

c. Either option (a) alone or option (d) alone

d. The simple interest received on ₹18000 at (R + 5)% rate after 't' years will be ₹3600

e. Either option (a) alone or option (b) alone

19. Questions

The ratio of A's efficiency to B's efficiency is 3: 1 and the time taken by B alone to complete the work is 30 hours more than that taken by C alone to complete the work. In how many hours can C alone complete the work?

a. D and E together can complete the work in (40/3) hours and the ratio of A's efficiency to D's efficiency is 2: 1

b. A alone can complete 45% part of the work in 9 hours

c. The average of time in which A alone, B alone and C alone can complete the work is 36 hours and 40 minutes

d. A and D together can complete the work in (40/3) hours

e. Either (b) alone or (c) alone

20. Questions

A bike is going at 64.8 km/h, and it crosses a '3d' m long tunnel in 't' seconds. Find the distance covered by the bike in (t + 5) seconds?

a. The bike crosses a man, who is running at 5 m/s in the same direction (d – 54) m ahead of the bike, in 12 seconds

b. Value of 'd' is a multiple of 7 between 203 and 217

c. The bike can cross a pole, which is (d – 30) m far from the bike, in 10 seconds

d. The bike can cover (d + 60) long bridge in (t – 20) seconds

e. Either (a) alone or (b) alone or (c) alone or (d) alone

Explanations:

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1. Questions

From the pie chart:

40 + P + 2Q + 3R = 100

P + 2Q + 3R = 60 ----------(1)

If the material cost of the product is ₹640, then the transportation cost of the product will be ₹320.

So, the value of (Q + R) = 40 * (320/640) = 20

Since, the ratio of Q to R = 1: 3

So, the value of Q = 20 * (1/4) = 5

And the value of R = 20 – 5 = 15

From equation (1):

P + 10 + 45 = 60

P = 5

Material cost of the product = 40%

Labour cost of the product = (P + Q + R)% = (5 + 5 + 15)% = 25%

Transportation cost of the product = (Q + R)% = (5 + 15)% = 20%

Overhead cost of the product = R% = 15%

Answer: B

Since, the difference between the material cost and labour cost of the product:

2PQR = 2 * 5 * 5 * 15 = ₹750

So, the difference between the transportation cost and overhead cost of the product = 750 * [(20 – 15)/(40 – 25)] = ₹250 = ₹10(P + Q + R)

2. Questions

From the pie chart:

40 + P + 2Q + 3R = 100

P + 2Q + 3R = 60 ----------(1)

If the material cost of the product is ₹640, then the transportation cost of the product will be ₹320.

So, the value of (Q + R) = 40 * (320/640) = 20

Since, the ratio of Q to R = 1: 3

So, the value of Q = 20 * (1/4) = 5

And the value of R = 20 – 5 = 15

From equation (1):

P + 10 + 45 = 60

P = 5

Material cost of the product = 40%

Labour cost of the product = (P + Q + R)% = (5 + 5 + 15)% = 25%

Transportation cost of the product = (Q + R)% = (5 + 15)% = 20%

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Overhead cost of the product = R% = 15%

Answer: D

From X:

Labour cost = 25%

Transportation cost = 20%

Required percentage = [(25 – 20)/20] * 100 = 25%

Since, the value of (P + Q + R) = 5 + 5 + 15 = 25

So, X is true.

From Y:

Ratio of material cost to overhead cost = 40: 15 = 8: 3

Since, the value of (P + 3): (Q – 2) = (5 + 3): (5 – 2) = 8: 3

So, Y is true.

From Z:

Overhead cost = 15%

Labour cost = 25%

Required percentage = (15/25) * 100 = 60%

Since, the value of PR = 5 * 15 = 75

So, Z is not true.

Hence, only X and Y are true.

3. Questions

From the pie chart:

40 + P + 2Q + 3R = 100

P + 2Q + 3R = 60 ----------(1)

If the material cost of the product is ₹640, then the transportation cost of the product will be ₹320.

So, the value of (Q + R) = 40 * (320/640) = 20

Since, the ratio of Q to R = 1: 3

So, the value of Q = 20 * (1/4) = 5

And the value of R = 20 – 5 = 15

From equation (1):

P + 10 + 45 = 60

P = 5

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Material cost of the product = 40%

Labour cost of the product = (P + Q + R)% = (5 + 5 + 15)% = 25%

Transportation cost of the product = (Q + R)% = (5 + 15)% = 20%

Overhead cost of the product = R% = 15%

Answer: E

Since, the marked up per cent of the product = (PQ + R – 3P)% = (25 + 15 – 15)% = 25%

And the MP of the product = ₹1500

So, the manufacturing cost of the product = 1500 * (100/125) = ₹1200

And the sum of material cost and labour cost of the product = (40 + 25)% of 1200 = ₹780

4. Questions

From the pie chart:

40 + P + 2Q + 3R = 100

P + 2Q + 3R = 60 ----------(1)

If the material cost of the product is ₹640, then the transportation cost of the product will be ₹320.

So, the value of (Q + R) = 40 * (320/640) = 20

Since, the ratio of Q to R = 1: 3

So, the value of Q = 20 * (1/4) = 5

And the value of R = 20 – 5 = 15

From equation (1):

P + 10 + 45 = 60

P = 5

Material cost of the product = 40%

Labour cost of the product = (P + Q + R)% = (5 + 5 + 15)% = 25%

Transportation cost of the product = (Q + R)% = (5 + 15)% = 20%

Overhead cost of the product = R% = 15%

Answer: A

Since, the difference between initial material cost and initial overhead cost = ₹240

So, the initial labour cost = 240 * [25/(40 – 15)] = ₹240

And the initial transportation cost = 240 * [20/(40 – 15)] = ₹192

Since, all types of costs are increased by 25%.

So, the new labour cost = 125% of 240 = 300

The new transportation cost = 125% of 192 = ₹240

And the required average = (300 + 240)/2 = ₹270

5. Questions

From the pie chart:

40 + P + 2Q + 3R = 100

P + 2Q + 3R = 60 ----------(1)

If the material cost of the product is ₹640, then the transportation cost of the product will be ₹320.

So, the value of (Q + R) = 40 * (320/640) = 20

Since, the ratio of Q to R = 1: 3

So, the value of Q = 20 * (1/4) = 5

And the value of R = 20 – 5 = 15

From equation (1):

P + 10 + 45 = 60

P = 5

Material cost of the product = 40%

Labour cost of the product = (P + Q + R)% = (5 + 5 + 15)% = 25%

Transportation cost of the product = (Q + R)% = (5 + 15)% = 20%

Overhead cost of the product = R% = 15%

Answer: B

Since, the overhead cost of the product is a multiple of 9 more than ₹400 and less than ₹420.

So, the overhead cost of the product will be either ₹405 or ₹414.

If the overhead cost of the product is ₹405.

The labour cost of the product will be = 405 * (25/15) = ₹675

If the overhead cost of the product is ₹414.

The labour cost of the product will be = 414 * (25/15) = ₹690

6. Questions

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Games	Sum of 1st, 2nd and 3rd prizes (in ₹)	Sum of 2nd and 3rd prizes (in ₹)	Sum of 1st and 2nd prizes (in ₹)
A	3 * 500 = 1500	2 * 450 = 900	2 * 550 = 1100
B	3 * 350 = 1050	2 * 300 = 600	2 * 400 = 800
C	3 * 400 = 1200	2 * 300 = 600	2 * 550 = 1100
D	3 * 650 = 1950	2 * 600 = 1200	2 * 700 = 1400
E	3 * 300 = 900	2 * 200 = 400	2 * 400 = 800

Games	1st prize (in ₹)	2nd prize (in ₹)	3rd prize (in ₹)
A	1500 – 900 = 600	1100 – 600 = 500	900 – 500 = 400
B	1050 – 600 = 450	800 – 450 = 350	600 – 350 = 250
C	1200 – 600 = 600	1100 – 600 = 500	600 – 500 = 100
D	1950 – 1200 = 750	1400 – 750 = 650	1200 – 650 = 550
E	900 – 400 = 500	800 – 500 = 300	400 – 300 = 100

Answer: C

Since, the prize for 3rd position in game B = ₹250

And the total amount distributed for 3rd prize in game B = ₹3000

So, the number of persons at 3rd position in game B = 3000/250 = 12

Since, the prize for 3rd position in game C = ₹100

And the total amount distributed for 3rd prize in game C = ₹1500

So, the number of persons at 3rd position in game C = 1500/100 = 15

Required ratio = 12: 15 = 4: 5

7. Questions

Games	Sum of 1st, 2nd and 3rd prizes (in ₹)	Sum of 2nd and 3rd prizes (in ₹)	Sum of 1st and 2nd prizes (in ₹)
A	3 * 500 = 1500	2 * 450 = 900	2 * 550 = 1100
B	3 * 350 = 1050	2 * 300 = 600	2 * 400 = 800
C	3 * 400 = 1200	2 * 300 = 600	2 * 550 = 1100
D	3 * 650 = 1950	2 * 600 = 1200	2 * 700 = 1400
E	3 * 300 = 900	2 * 200 = 400	2 * 400 = 800

Games	1st prize (in ₹)	2nd prize (in ₹)	3rd prize (in ₹)
A	1500 – 900 = 600	1100 – 600 = 500	900 – 500 = 400
B	1050 – 600 = 450	800 – 450 = 350	600 – 350 = 250
C	1200 – 600 = 600	1100 – 600 = 500	600 – 500 = 100
D	1950 – 1200 = 750	1400 – 750 = 650	1200 – 650 = 550
E	900 – 400 = 500	800 – 500 = 300	400 – 300 = 100

Answer: E

The prize for 1st position in game A = ₹600

The prize for 1st position in game B = ₹450

The prize for 1st position in game C = ₹600

So, the total amount distributed for 1st prize in games A, B and C:

(600 * 3) + (450 * 4) + (600 * 5) = 1800 + 1800 + 3000 = ₹6600

8. Questions

Games	Sum of 1st, 2nd and 3rd prizes (in ₹)	Sum of 2nd and 3rd prizes (in ₹)	Sum of 1st and 2nd prizes (in ₹)
A	3 * 500 = 1500	2 * 450 = 900	2 * 550 = 1100
B	3 * 350 = 1050	2 * 300 = 600	2 * 400 = 800
C	3 * 400 = 1200	2 * 300 = 600	2 * 550 = 1100
D	3 * 650 = 1950	2 * 600 = 1200	2 * 700 = 1400
E	3 * 300 = 900	2 * 200 = 400	2 * 400 = 800

Games	1st prize (in ₹)	2nd prize (in ₹)	3rd prize (in ₹)
A	1500 – 900 = 600	1100 – 600 = 500	900 – 500 = 400
B	1050 – 600 = 450	800 – 450 = 350	600 – 350 = 250
C	1200 – 600 = 600	1100 – 600 = 500	600 – 500 = 100
D	1950 – 1200 = 750	1400 – 750 = 650	1200 – 650 = 550
E	900 – 400 = 500	800 – 500 = 300	400 – 300 = 100

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Answer: D

The sum of 1st prize in games D and E = 750 + 500 = ₹1250

The sum of 2nd prize in games D and E = 650 + 300 = ₹950

The sum of 3rd prize in games D and E = 550 + 100 = ₹650

Since, M = (650/1250) * 100 = 52%

So, the required percentage = (950/1250) * 100 = 76% = (M + 24)%

9. Questions

Games	Sum of 1st, 2nd and 3rd prizes (in ₹)	Sum of 2nd and 3rd prizes (in ₹)	Sum of 1st and 2nd prizes (in ₹)
A	3 * 500 = 1500	2 * 450 = 900	2 * 550 = 1100
B	3 * 350 = 1050	2 * 300 = 600	2 * 400 = 800
C	3 * 400 = 1200	2 * 300 = 600	2 * 550 = 1100
D	3 * 650 = 1950	2 * 600 = 1200	2 * 700 = 1400
E	3 * 300 = 900	2 * 200 = 400	2 * 400 = 800

Games	1st prize (in ₹)	2nd prize (in ₹)	3rd prize (in ₹)
A	1500 – 900 = 600	1100 – 600 = 500	900 – 500 = 400
B	1050 – 600 = 450	800 – 450 = 350	600 – 350 = 250
C	1200 – 600 = 600	1100 – 600 = 500	600 – 500 = 100
D	1950 – 1200 = 750	1400 – 750 = 650	1200 – 650 = 550
E	900 – 400 = 500	800 – 500 = 300	400 – 300 = 100

Answer: B

Since, the 2nd prize in game B = ₹350

And the total amount distributed for 2nd position in game B = ₹2800

So, the number of persons at 2nd position in game B = 2800/350 = 8

The number of persons at 2nd position in game C = 125% of 8 = 10

And the number of persons at 2nd position in game D = 75% of 8 = 6

Since, the 2nd prize in game C = ₹500

And the 2nd prize in game D = ₹650

So, the required ratio = (500 * 10): (650 * 6) = 50: 39

10. Questions

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Games	Sum of 1st, 2nd and 3rd prizes (in ₹)	Sum of 2nd and 3rd prizes (in ₹)	Sum of 1st and 2nd prizes (in ₹)
A	3 * 500 = 1500	2 * 450 = 900	2 * 550 = 1100
B	3 * 350 = 1050	2 * 300 = 600	2 * 400 = 800
C	3 * 400 = 1200	2 * 300 = 600	2 * 550 = 1100
D	3 * 650 = 1950	2 * 600 = 1200	2 * 700 = 1400
E	3 * 300 = 900	2 * 200 = 400	2 * 400 = 800

Games	1st prize (in ₹)	2nd prize (in ₹)	3rd prize (in ₹)
A	1500 – 900 = 600	1100 – 600 = 500	900 – 500 = 400
B	1050 – 600 = 450	800 – 450 = 350	600 – 350 = 250
C	1200 – 600 = 600	1100 – 600 = 500	600 – 500 = 100
D	1950 – 1200 = 750	1400 – 750 = 650	1200 – 650 = 550
E	900 – 400 = 500	800 – 500 = 300	400 – 300 = 100

Answer: B

Quantity I:

Since, the sum of 2nd prize in all the 5 games:

500 + 350 + 500 + 650 + 300 = ₹2300

So, the 75% of the sum of 2nd prize in all the 5 games = 75% of 2300 = ₹1725

Quantity II:

Since, the sum of 3rd prize in all the 5 games:

400 + 250 + 100 + 550 + 100 = ₹1400

So, the 125% of the sum of 3rd prize in all the 5 games = 125% of 1400 = ₹1750

Hence, Quantity I < Quantity II

11. Questions

Since, the evaporation rate of water from mixture B in the 3rd hour is 0.15 L per minute.

And the quantity of mixture B after 3 hours = 45 L

So, the quantity of mixture B after 2 hours:

4x – 6 = 45 + (0.15 * 60)

4x – 6 = 54

x = 15

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Since, the evaporated quantity of water from mixture A in the 3rd hour is 9 L.

So,

$9(x - y) - (4y - 4) = 9$

$9(15 - y) - (4y - 4) = 9$

$135 - 9y - 4y + 4 = 9$

$y = 10$

Time	Mixture A	Mixture B
Initial quantity (in lit.)	4 * 15 = 60	5 * 15 = 75
Quantity after 1 hour (in lit.)	5 * 10 = 50	6 * 10 = 60
Quantity after 2 hours (in lit.)	9(15 – 10) = 45	(4 * 15 – 6) = 54
Quantity after 3 hours (in lit.)	(4 * 10 – 4) = 36	45

Answer: C

The evaporation rate of water from mixture A in the 2nd hour = (50 – 45)/60 = (1/12) L per minute

The evaporation rate of water from mixture B in the 2nd hour = (60 – 54)/60 = (1/10) L per minute

Required difference = (1/10) – (1/12) = (6 – 5)/60 = (1/60) L per minute

12. Questions

Since, the evaporation rate of water from mixture B in the 3rd hour is 0.15 L per minute.

And the quantity of mixture B after 3 hours = 45 L

So, the quantity of mixture B after 2 hours:

$4x - 6 = 45 + (0.15 * 60)$

$4x - 6 = 54$

$x = 15$

Since, the evaporated quantity of water from mixture A in the 3rd hour is 9 L.

So,

$9(x - y) - (4y - 4) = 9$

$9(15 - y) - (4y - 4) = 9$

$135 - 9y - 4y + 4 = 9$

$y = 10$

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Time	Mixture A	Mixture B
Initial quantity (in lit.)	4 * 15 = 60	5 * 15 = 75
Quantity after 1 hour (in lit.)	5 * 10 = 50	6 * 10 = 60
Quantity after 2 hours (in lit.)	9(15 – 10) = 45	(4 * 15 – 6) = 54
Quantity after 3 hours (in lit.)	(4 * 10 – 4) = 36	45

Answer: D

The initial quantity of water in mixture A = 60 * (7/15) = 28 L

Since, 10 L quantity of water from mixture A is evaporated after 1 hour.

So, the quantity of water in mixture A after 1 hour = 28 – 10 = 18 L

The initial quantity of water in mixture B = 75 * (7/15) = 35 L

Since, 15 L quantity of water from mixture B is evaporated after 1 hour.

So, the quantity of water in mixture B after 1 hour = 35 – 15 = 20 L

Required ratio = 18: 20 = 9: 10

13. Questions

Since, the evaporation rate of water from mixture B in the 3rd hour is 0.15 L per minute.

And the quantity of mixture B after 3 hours = 45 L

So, the quantity of mixture B after 2 hours:

4x – 6 = 45 + (0.15 * 60)

4x – 6 = 54

x = 15

Since, the evaporated quantity of water from mixture A in the 3rd hour is 9 L.

So,

9(x – y) – (4y – 4) = 9

9(15 – y) – (4y – 4) = 9

135 – 9y – 4y + 4= 9

y = 10

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Time	Mixture A	Mixture B
Initial quantity (in lit.)	4 * 15 = 60	5 * 15 = 75
Quantity after 1 hour (in lit.)	5 * 10 = 50	6 * 10 = 60
Quantity after 2 hours (in lit.)	9(15 – 10) = 45	(4 * 15 – 6) = 54
Quantity after 3 hours (in lit.)	(4 * 10 – 4) = 36	45

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Answer: E

The quantity of milk in mixture A after 1 hour = 50 * (3/5) = 30 L

And the quantity of water in mixture A after 1 hour = 50 – 30 = 20 L

Let the cost of pure milk be ₹M per L.

So,

(30 * M)/(30 + 20) = 36

M = 60

The quantity of water evaporated from mixture A in the 1st hour = 50 – 45 = 5 L

The quantity of water in mixture A after 2 hours = 20 – 5 = 15 L

So, the cost of mixture A after 2 hours:

(30 * 60)/(30 + 15) = ₹40 L

14. Questions

Since, the evaporation rate of water from mixture B in the 3rd hour is 0.15 L per minute.

And the quantity of mixture B after 3 hours = 45 L

So, the quantity of mixture B after 2 hours:

4x – 6 = 45 + (0.15 * 60)

4x – 6 = 54

x = 15

Since, the evaporated quantity of water from mixture A in the 3rd hour is 9 L.

So,

9(x – y) – (4y – 4) = 9

9(15 – y) – (4y – 4) = 9

135 – 9y – 4y + 4= 9

y = 10

Time	Mixture A	Mixture B
Initial quantity (in lit.)	4 * 15 = 60	5 * 15 = 75
Quantity after 1 hour (in lit.)	5 * 10 = 50	6 * 10 = 60
Quantity after 2 hours (in lit.)	9(15 – 10) = 45	(4 * 15 – 6) = 54
Quantity after 3 hours (in lit.)	(4 * 10 – 4) = 36	45

Answer: B

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The quantity of mixture A after 3 hours = 36 L

The quantity of mixture A after 4 hours = 36 – (0.1 * 60) = 30 L

The quantity of mixture B after 3 hours = 45 L

The quantity of mixture B after 4 hours = 45 – (0.15 * 60) = 36 L

Required ratio = 30: 36 = 5: 6

15. Questions

Since, the evaporation rate of water from mixture B in the 3rd hour is 0.15 L per minute.

And the quantity of mixture B after 3 hours = 45 L

So, the quantity of mixture B after 2 hours:

$4x - 6 = 45 + (0.15 * 60)$

$4x - 6 = 54$

$x = 15$

Since, the evaporated quantity of water from mixture A in the 3rd hour is 9 L.

So,

$9(x - y) - (4y - 4) = 9$

$9(15 - y) - (4y - 4) = 9$

$135 - 9y - 4y + 4 = 9$

$y = 10$

Time	Mixture A	Mixture B
Initial quantity (in lit.)	4 * 15 = 60	5 * 15 = 75
Quantity after 1 hour (in lit.)	5 * 10 = 50	6 * 10 = 60
Quantity after 2 hours (in lit.)	9(15 – 10) = 45	(4 * 15 – 6) = 54
Quantity after 3 hours (in lit.)	(4 * 10 – 4) = 36	45

Answer: A

From P:

Ratio of the quantity of mixture A to that of mixture B after 2 hours is 5: 6:

45: 54 = 5: 6.

So, P is true.

From Q:

Ratio of the quantity of mixture A to that of mixture B after 3 hours is 4: 5:

36: 45 = 4: 5

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So, Q is true.

From R:

Average quantity of mixtures A and B after 1 hour:

(50 + 60)/2 = 55 L

So, R is not true.

Hence, only P and Q are true.

16. Questions

Answer: E

Let the speed of boat A in still water be 'x' m/s and the speed of stream be 'y' m/s.

So,

[1760/(x – y)] – [1760/(x + y)] = 50 ----------(1)

From option (a):

The time taken by boat A to go 1.485 km in upstream direction is 2.25 minutes.

So, the upstream speed of boat A:

(x – y) = (1.485 * 1000)/(2.25 * 60)

x – y = 11

From equation (1):

[1760/11] – [1760/(x + y)] = 50

x + y = 16

The downstream speed of boat A = 16 m/s

The speed of boat A in still water = (16 + 11)/2 = 13.5 m/s

So, option (a) is sufficient to answer the question.

From option (b):

Since, boat B can cover 1.08 km in still water in 1.2 minutes.

So, the speed of boat B in still water = (1.08 * 1000)/(1.2 * 60) = 15 m/s

And the speed of stream = y = 15 * (1/6) = 2.5 m/s

From equation (1):

[1760/(x – 2.5)] – [1760/(x + 2.5)] = 50

x = 13.5

The speed of boat A in still water = 13.5 m/s

So, option (b) is sufficient to answer the question.

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From option (c):

The time taken by boat A to go 1.2 km in downstream direction is 1.25 minutes.

So, the downstream speed of boat A:

x + y = (1.2 * 1000)/(1.25 * 60)

x + y = 16

From equation (1):

[1760/(x – y)] – [1760/16] = 50

x – y = 11

The upstream speed of boat A = 11 m/s

The speed of boat A in still water = (16 + 11)/2 = 13.5 m/s

So, option (c) is sufficient to answer the question.

Hence, either option (a) alone or option (b) alone or option (c) alone is sufficient to answer the question.

17. Questions

Answer: B

Let A's present age be 'x' years and B's and C's present ages be '2y' years and 'y' years respectively.

Also let D's present age be 'z' years

So,

x + 2y + y = 3 * 23

x + 3y = 69 ----------(1)

From option (a):

Average of present ages of A, B, C and D = 27.75 years

So, x + 3y + z = 4 * 27.75

x + 3y + z = 111

From this, we can't find the value of 'x', so we can neither find A's present age nor the average of present ages of A and B.

So, option (a) is not sufficient to answer the question.

From option (b):

D is 36 years old and the ratio of A's present age to D's present age will be 2: 3.

So, A's present age = x = 36 * (2/3) = 24 years

From equation (1):

24 + 3y = 69

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y = 15

So, B's present age = 2 * 15 = 30 years

And the average of A's and B's present ages = (24 + 30)/2 = 27 years

So, option (b) is sufficient to answer the question.

From option (c):

Ratio of B's present age to D's present age is 5: 7.

2y/z = 5/7

From this, we can't find the value of 'x', so we can neither find A's present age nor the average present ages of A and B.

So, option (c) is not sufficient to answer the question.

From option (d):

Average of present ages of B, C and D is 29 years.

2y + y + z = 3 * 29

3y + z = 87

From this, we can't find the value of 'x', so we can neither find A's present age nor the average present ages of A and B.

So, option (d) is not sufficient to answer the question.

Hence, option (b) is sufficient to answer the question.

18. Questions

Answer: A

From option (a):

The difference between CI and SI on ₹6000 at (R/2)% rate after 2 years is ₹33.75.

So,

$6000 * (R/200)^2 = 33.75$

R = 15

The difference between CI and SI on ₹18000 at R% rate after 2 years:

$18000 * (15/100)^2 = ₹405$

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So, option (a) is sufficient to answer the question.

From option (b):

The ratio of simple interest on ₹5000 at R% rate after 1 year to that on ₹3000 at 2R% rate after 1 year is 5: 6.

So,

(5000 * R)/(3000 * 2R) = 5/6

Since, the value of R can't be determined, so the required answer can't be determined.

So, option (b) is not sufficient to answer the question.

From option (d):

The simple interest received on ₹18000 at (R + 5)% rate after 't' years will be ₹3600.

So,

[18000 * (R + 5) * t]/100 = 3600

Since, the value of R can't be determined, so the required answer can't be determined.

So, option (d) is not sufficient to answer the question.

Hence, option (a) is sufficient to answer the question.

19. Questions

Answer: E

Let the time, in which A alone and B alone can complete the work be 'x' hours and '3x' hours.

So, the time, in which C alone can complete the work = (3x – 30) hours

From option (a):

D and E together can complete the work in (40/3) hours and the ratio of A's efficiency to D's efficiency is 2: 1.

Since, we can't find the time, in which D alone can complete the work, so we can't find the time, in which A alone can complete the work.

Since, we can't find the time, in which A alone can complete the work, so we can't find the time, in which C alone can complete the work.

So, option (a) is not sufficient to answer the question.

From option (b):

Since, A alone can complete 45% part of work in 9 hours.

So, the time, in which A alone can complete the work = x = 9 * (100/45) = 20 hours

The time, in which B alone can complete the work = 3 * 20 = 60 hours

And the time, in which C alone can complete the work = 60 – 30 = 30 hours

So, option (b) is sufficient to answer the question.

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From option (c):

Since, the average of time in which A alone, B alone and C alone can complete the work = 36 hours and 40 minutes = (110/3) hours

So,

$[x + 3x + (3x – 30)]/3 = 110/3$

$7x – 30 = 110$

$x = 20$

And the time, in which C alone can complete the work = $(3 * 20 – 30) = 30$ hours

So, option (c) is sufficient to answer the question.

From option (d):

A and D together can complete the work in (40/3) hours.

Since, there is no other relation between A and D, so we can't find the time, in which A alone can complete the work.

Since, we can't find the time, in which A alone can complete the work, so we can't find the time, in which C alone can complete the work.

So, option (d) is not sufficient to answer the question.

Hence, either option (b) alone or option (c) alone is sufficient to answer the question.

20. Questions

Answer: E

The speed of the bike = 64.8 km/h = 18 m/s

So,

$18 = 3d/t$ --------(1)

From option (a):

Since, the bike crosses a man, who is running at 5 m/s in the same direction (d – 54) m ahead of the bike, in 12 seconds.

So,

$18 – 5= (d – 54)/12$

$d = 210$

From equation (1):

$18 = 630/t$

$t = 35$

And the distance covered by the bike in (t + 5) seconds = $18 * (35 + 5) = 720$ m

So, option (a) is sufficient to answer the question.

From option (b):

Value of 'd' is a multiple of 7 between 203 and 217.

Since, 203, 210 and 217 are multiples of 7.

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So, the only possible value of 'd' = 210

From equation (1):

18 = 630/t

t = 35

And the distance covered by the bike in (t + 5) seconds = 18 * (35 + 5) = 720 m

So, option (b) is sufficient to answer the question.

From option (c):

The bike can cross a pole, which is (d – 30) m far from the bike, in 10 seconds.

So,

18 = (d – 30)/10

d = 210

From equation (1):

18 = 630/t

t = 35

And the distance covered by the bike in (t + 5) seconds = 18 * (35 + 5) = 720 m

So, option (c) is sufficient to answer the question.

From option (d):

The bike can cover (d + 60) long bridge in (t – 20) seconds.

So,

18 = (d + 60)/(t – 20)

From equation (1):

18 = (6t + 60)/(t – 20)

18t – 360 = 6t + 60

t = 35

And the distance covered by the bike in (t + 5) seconds = 18 * (35 + 5) = 720 m

So, option (d) is sufficient to answer the question.

Hence, either option (a) alone or option(b) alone or option(c) alone or option(d) alone is sufficient to answer the question.

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1. Questions

Read the following information carefully and answer the questions based on it.

Line graph given below shows the distribution of the total number of people who participated in three different sports i.e., cricket, football, and hockey, from five different sports clubs, A, B, C, D, and E, either in percentage or absolute value.

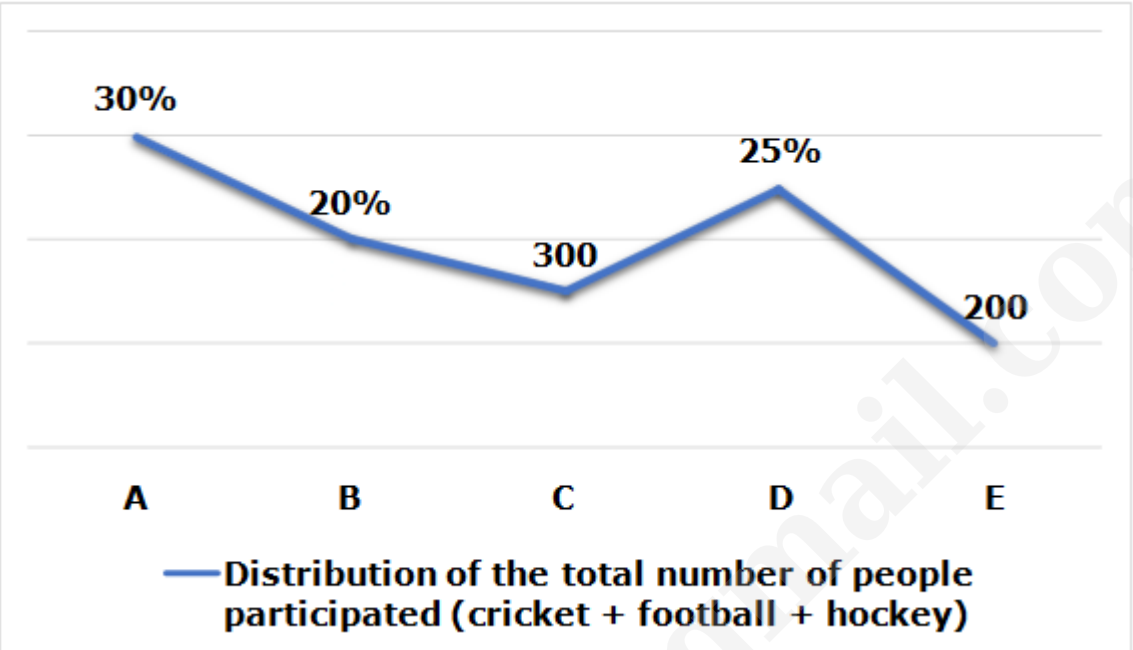


Table given below shows the percentage of the number of people who participated in cricket out of the total number of people who participated in all the sports (cricket + football + hockey) together from each club. Table also shows the number of people who participated in football from each club.

Club	Percentage of the number of people participated in cricket	Number of people participated in football
A	50%	(2P+20)
B	40%	Q
C	60%	(Q-10)
D	75%	(P+50)
E	35%	(P+Q-50)

Note:

Number of people who participated in cricket from club A is equal to the total number of people who participated in all the sports (cricket + football + hockey) together from club C.

Sum of the number of people who participated in hockey from club A and club B together is 320.

Number of people who participated in football from club D and club E are same.

Total number of people who participated in sports = number of people who participated in cricket + number of people who participated in football + number of people who participated in hockey

Total number of people in a sports club = number of people who participated + number of people who did not participate.

Out of the total number of people in club A, 40% did not participate in any sports. Out of the total number of people who participated in all the sports (cricket + football + hockey) together, the number of males and females is 50M and 25N, respectively. The number of males and females who did not participate in any sport is equal. The value of (M+N) is 17. Find the total number of males (participated + not participated) in club A.

a. 540

b. 330

c. 460

d. 610

e. None of these

2. Questions

From club B, if there are some more people who participated in another sport, volleyball, which is X% more than that of hockey from the same club, and the total number of participants from all the sports together including volleyball now increased by 40%. Then find the number of people who participated in volleyball from club E, if the number of people who participated in cricket in club E is 2X% less than that of volleyball from the same club.

a. 91

b. 89

c. 98

d. 81

e. None of these

3. Questions

If there is another club F, where the total number of people who participated in all three sports together is [3×(P+Q)]. Find the number of people who participated in football from club F.Which of the following statement(s) is/are sufficient to answer the question?

Statement I: Ratio of the number of people who participated in cricket to that of hockey from club F is 7:5.

Statement II: Number of people who participated in football from club F is 50% more than that of cricket from the same club.

Statement III: Number of people who participated in hockey from club F is 110 less than that of football from the same club.

a. Only statement II alone is sufficient, while statements I and III are not sufficient

b. Both statements I and II together are necessary, while statement III alone is not sufficient

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c. Any of the two statements I and II or II and III or I and III together is sufficient

d. All the statements I, II and III together are necessary

e. Neither of the statements is sufficient

4. Questions

Sum of the weights of all the people who participated in all the sports from club C and D together is 45000 kg. Average weight of all the people who participated in all the sports together from club C is 50 kg. If the average weight of all the people who participated in all the sports together from club D is [U] kg, then find the value of (U-T), if the ratio of the values of T to that of U is 2:3.

a. (T/2)

b. (2U-40)

c. 3T

d. (T+U-10)

e. None of these

5. Questions

Out of the total number of people who participated in all the sports together from club E,the difference between the number of males and females is twice the number of females who participated in all the sports together from the same club. Number of males who participated in all the sports together from club E is more than that of females from the same club. Find the number of males who participated in all the sports together from club E.

a. 50

b. 150

c. 100

d. Cannot be determined

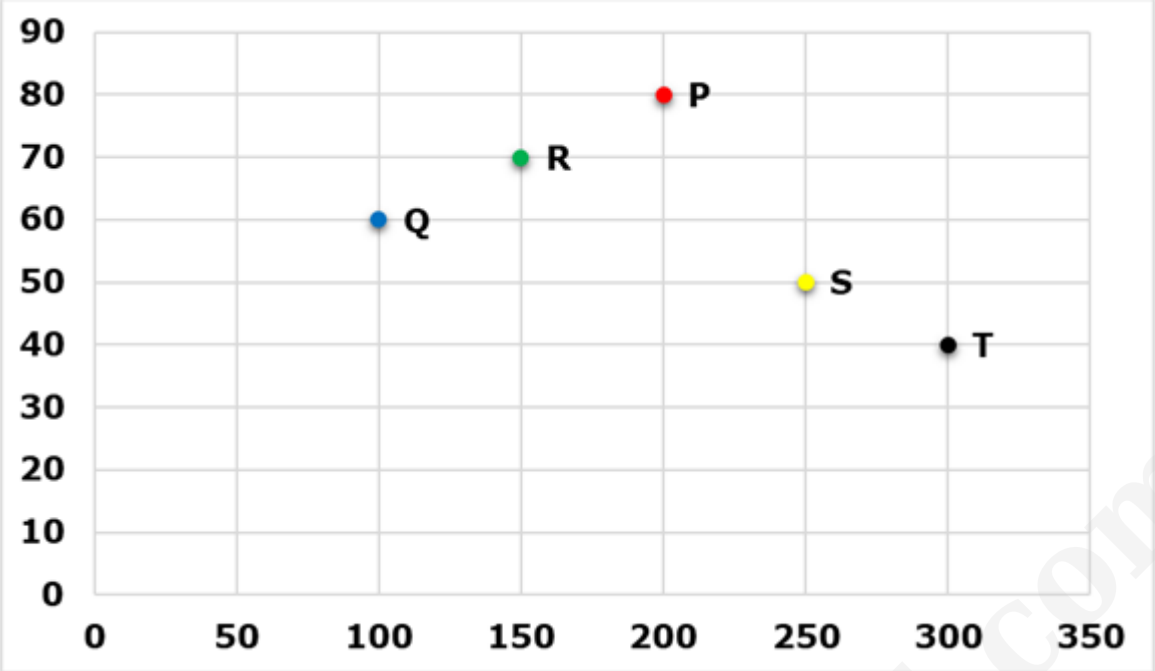
e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The scatter chart given below shows the total number of marks scored by five students, namely P, Q, R, S, and T in three subjects, i.e., A, B and C. X-axis shows the total marks scored in all the subjects (A+B+C) by each student and Y-axis shows marks scored in subject A by each student. Red, blue, green, yellow and black depictthe names of each student, i.e. P, Q, R, S and T respectively.

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Note:

Marks scored by P in subject C are 40% more than that in subject B.

Ratio of the total marks scored by Q to that of R in subject B is 6:11.

Marks scored by R in subject C are 16.66% of the total marks scored by him in all the subjects (A+B+C) together.

Marks scored by S in subject B are twice the marks scored by T in subject C.

Ratio of marks scored by S in subject C to that of T in subject B is 2:5.

Percentage of marks obtained in any subject = [{(marks scored)/(maximum marks)}×100]%

Maximum marks of each subject may be different, unless mentioned in the question.

Total marks scored = marks scored in subject A + marks scored in subject B + marks scored in subject C

Total marks scored in any subject = marks scored in theory section + marks scored in practical section

Total marks scored = fair marks scored + grace marks scored.

Ratio of marks scored by S in the theory section of subject A to that of the practical section of subject B is 4:3. Marks scored by S in the practical section of subject A are11.11% of marks scored by him in the theory section of subject B. If the total marks scored by S in all the subjects (A+B+C) together in the practical section is 55, then find the marks scored by S in theory section of subject C.

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a. 65

b. 72

c. 44

d. 51

e. None of these

7. Questions

If the maximum marks of subject A and subject C are 2K and K, respectively, and the sum of the percentage of marks scored by P in subject A and subject C is (K+10)%. Find the percentage of marks scored by P in subject B, if the maximum marks of subject B are (K+50). (K is a positive integer)

a. 37.50%

b. 33.33%

c. 62.50%

d. 83.33%

e. None of these

8. Questions

Ratio of fair marks scored by Q in subject A to that of grace marks scored by him in subject B is 11:2. Grace marks scored by Q in subject A is 1/4th of that of fair marks scored by him in subject B. Find the difference between the total fair and total grace marks scored by Q in subject A and subject B together.

a. 61

b. 56

c. 57

d. 60

e. None of these

9. Questions

If there is another student U, who scored 25% more marks than T in subject A, whereas the total marks scored by U in all the subjects (A+B+C) together is same as that of T. Difference between the marks scored by U in subject B and subject C is same as the marks scored by him in subject A. Find the ratio of the marks scored by U in subject B to that of subject C, if the number of marks scored by him in subject B is more than that of subject C.

a. 2:3

b. 3:2

c. 4:5

d. 5:4

e. None of these

10. Questions

Maximum marks for theory and practical in each subject are 50 and 20 respectively. The total

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marks scored by R in the theory section in all the subjects together is 300% more than that of practical section. Find the sum of the percentage of the marks scored by R in the theory section and in the practical sections in all the subjects together.

a. 120%

b. 100%

c. 130%

d. Cannot be determined

e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

Certain number of customers purchased two different products M, and N, from two different shops X and Y. Some customers purchased only product M, while some purchased only product N and some customers purchased both products M and N.

Number of customers purchased only product M from shop X and shop Y is 5A and 5B respectively. Number of customers purchased only product N from shop X is 20% less than the number of customers purchased only product M from the same shop. Ratio of the number of customers purchased only product M to that of only product N from shop Y is 5:3.

Number of customers who purchased both products from shop X is 15 more than the average number of customers purchased only product M and only product N from the same shop. Whereas, the number of customers purchased both products from shop Y is 5, more than the number of customers purchased only product M from the same shop.

Sum of the total number of customers who purchased only product M from shop X and the number of customers who purchased only product N from shop Y is 95. While the number of customers who purchased only product N from shop X is 35 less than the number of customers who purchased only product M from shop Y.

Note:

Total number of customers purchased products = number of customers purchased products from shop X + number of customers purchased products from shop Y

Total number of customers purchased products from shop X = number of customers who purchased only M + number of customers who purchased only N + number of customers who purchased both products

Total number of customers purchased products from shop Y = number of customers who purchased only M + number of customers who purchased only N + number of customers who purchased both products

(Each customer purchased only one item of each product)

In shop X,the selling price of each item of product M is Rs. P and the selling price of each item of product N is Rs. Q. Total revenue generated from selling all the items of product M and product N together by shop X is Rs. 19000. If the sum of the revenue generated from selling only product M and only product N is Rs. 8200. Find the difference between the selling price of each item of

product M and product N.

a. Rs. 10

b. Rs. 20

c. Rs. 30

d. Rs. 25

e. None of these

12. Questions

If there is another shop Z, where the number of customers who purchased only product M is 6A and the number of customers who purchased only product N is 2B. Number of customers who purchased both products from shop Z is R. Find the total number of customers who purchased all the products (M+N) from shop Z. (There are only two products M and N sold by shop Z)

a. (90+R)

b. (40+R)

c. (60+R)

d. (30+R)

e. None of these

13. Questions

Ratio of the number of male customers who purchased only product M from shop Y to that of female customers who purchased only product N from the same shop is 10:3. Number of female customers who purchased only product M from shop Y is 1/6th less than that of male customers who purchased only product N from the same shop. Find the total number of male customers who purchased only product M and only product N from shop Y.

a. 66

b. 49

c. 80

d. 57

e. None of these

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14. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the number of runs scored by three different players, namely A, B and C in five different matches, i.e., match 1, match 2, match 3, match 4 and match 5. It shows that the number of runs scored by A, the percentage of runs scored by B with respect to that of A and the ratio of runs scored by B to that of C in each match.

Match	Runs scored by A	Percentage of runs scored by B with respect to that of A	Ratio of runs scored by B to that of C
1	(L+5)	75%	9:(M-N)
2	35	(K-5)%	2:3
3	75	120%	(N/10):1
4	2N	(4M+5)%	5:4
5	(K+L)	80%	4:1

Note:

Runs scored by B is 45 more than that of C in match 3.

Ratio of runs scored by C in match 2 to that of match 3 is 7:15.

Runs scored by B in match 1 is half of that of match 3.

Runs scored by A in match 1 is 20% more than that of C in the same match.

Strike Rate = [{(runs scored)/(number of balls faced)}×100]

A player can score runs either in singles (1 run), doubles (2 runs), triples (3 runs), fours (4 runs) or sixes (6 runs)

Runs scored in boundary =runs scored in fours +runs scored in sixes

Total runs scored in any match = runs scored by A + runs scored by B + runs scored by C

In match 4, number of balls faced by A is equal to that of the runs scored by B in the same match. Strike rate of B is 45 more than that of A. Find the strike rate of C in match 4, if the number of balls faced by C is average of the number of balls faced by A and B in the same match.

a. 72.72

b. 88.88

c. 62.50

d. 57.14

e. None of these

15. Questions

In match 3, out of total runs scored by B,the ratio of the number of singles to that of doubles to that of triples scored by B is 10:5:6. If the number of fours and sixes scored by B are 4's and 6's respectively, then find runs scored by B in doubles.

a. 10

b. 5

c. 20

d. 12

e. None of these

16. Questions

If there is another player D, who scored 50% more runs than that of C in match 1 and the number of balls faced by him is X. Strike rate of D is (2X+5). (X is a positive integer)

Find the value of [√(S×T)].

S = Number of balls faced by D in match 1.

T = Runs scored by A in match 1.

a. Equals to the value of S

b. Same as the value of T

c. Average of the values of S and T

d. All of these

e. None of these

17. Questions

In another match 6, total runs scored by A, B and C together is 50 less than that of previous match. Ratio of runs scored by A to that of C is 7:5. Runs scored by A is 133.33% more than that of B. Find the difference between runs scored by B in match 6 and match 2.

a. 21

b. 12

c. 16

d. 25

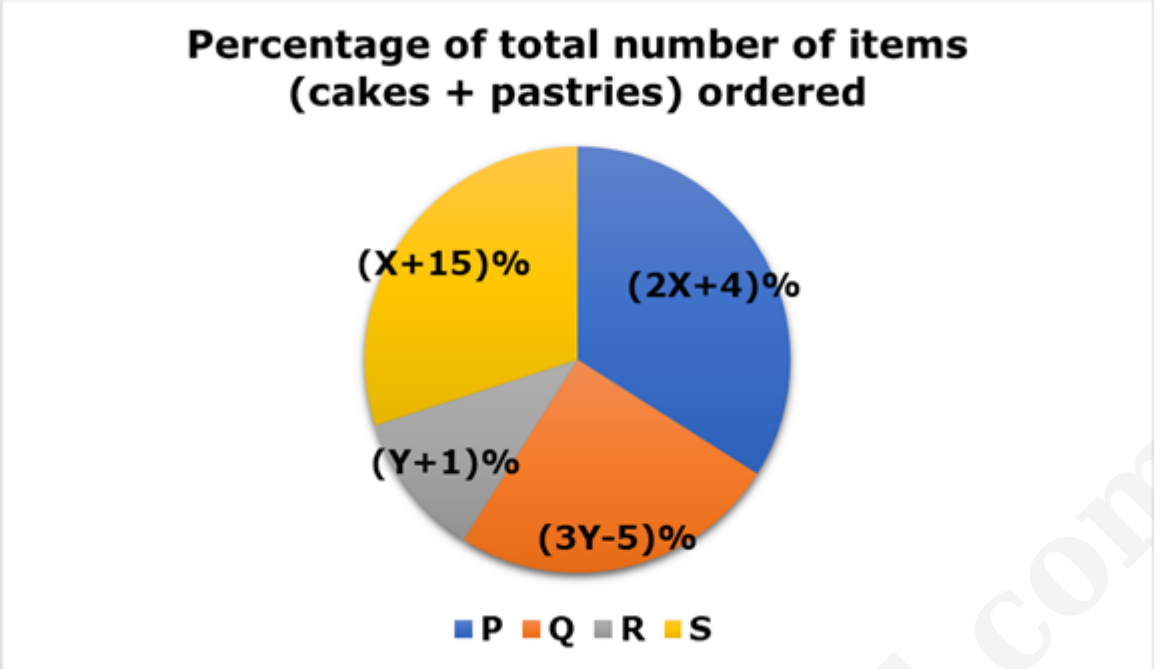
e. None of these

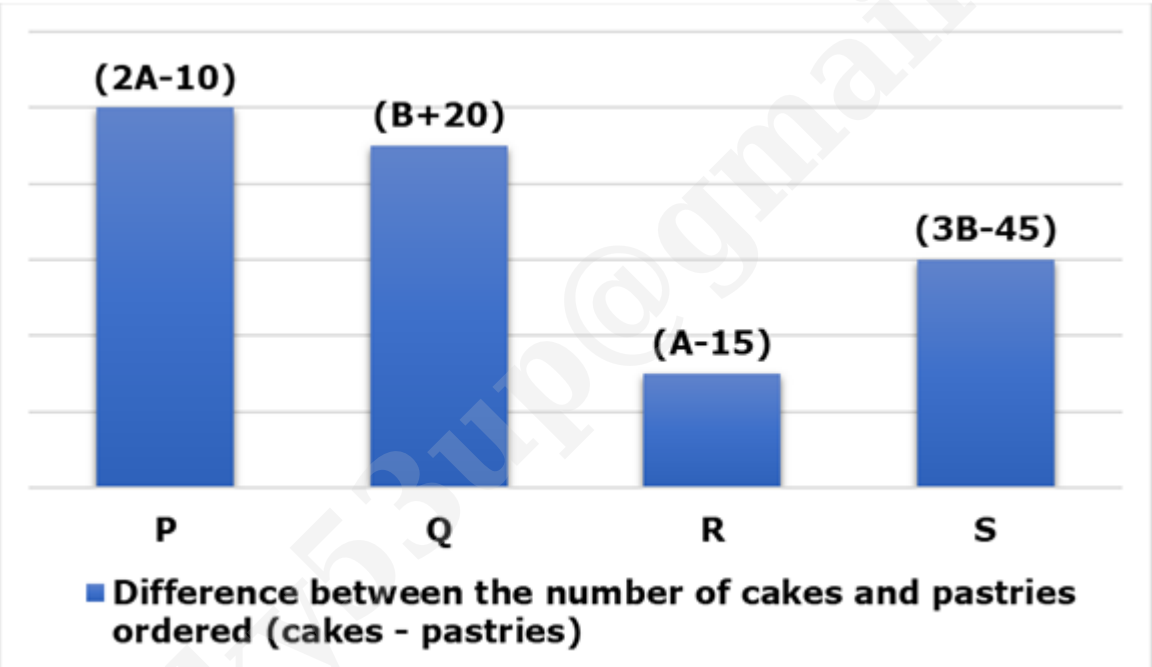
18. Questions

Read the following information carefully and answer the questions based on it.

Pie char given below shows the percentage distribution of the total number of items (cakes + pastries) ordered by four different shops P, Q, R and S. Bar graph given below shows the difference between the number of cakes and pastries (cakes – pastries) ordered by each shop.

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Note:

Values of X and A are the roots of the equation $V^2 - 45V + 450 = 0$. (A > X)

Value of B is 250% of that of Y.

Total number of items (cakes + pastries) ordered by shop R is 55.

In each shop, number of cakes ordered are more than that of pastries.

Total number of items ordered = number of items delivered + number of items not delivered

Total number of items delivered = number of items accepted + number of items returned

Total number of items (cakes + pastries) delivered by another shop T is 71.42% of the total number of items (cakes + pastries) ordered by the same shop. Out of the total number of items

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(cakes + pastries) delivered by shop T, 10% were returned by the customer. Difference between the number of items accepted by the customer out of the total delivered items (cakes + pastries) and the number of items not delivered by shop T is 50, then find the total number of items (cakes + pastries) by shop T.

a. 150
b. 120
c. 140
d. 160
e. None of these

19. Questions

Shop P ordered two different types of cakes and pastries i.e., banana and coconut. Number of banana cakes and banana pastries ordered by shop P are K and L respectively. Whereas, the number of coconut cakes and coconut pastries ordered by shop P is equal. Find twice of the difference between the number of banana cakes and banana pastries ordered by the shop P.

a. 100
b. 200
c. 50
d. 25
e. None of these

20. Questions

Out of total number of cakes ordered by shop S, (W-10)% are delivered, while out of total number of pastries ordered, (W+15)% are delivered. Ratio of the number of cakes not delivered to that of pastries not delivered by shop S is 3:1. Find the positive value of Z, if (W+21) = Z^2.

a. 3
b. 9
c. 27
d. Cannot be determined
e. None of these

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Explanations:

1. Questions

Total number of people who participated in all the sports together (cricket + football + hockey) from club

C = 300

So, the total number of people who participated in cricket from club A = 300

Percentage of the number of people who participated in cricket from club A = 50%

50% = 300

1% = (300/50) = 6

100% = (6×100) = 600

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from club A = 600

Number of people who participated in football from club A = (2P+20)

So, the number of people who participated in hockey from club A = {600-(300+2P+20)} = {600-(320+2P)} = (280-2P)

Percentage of the total number of people who participated in all the sports together (cricket + football + hockey) from club A. Out of the total number of people participated from all the clubs (cricket + football + hockey) = 30%

30% = 600

1% = (600/30) = 20

100% = (20×100) = 2000

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from all the clubs together = 2000

Now,

Total number of people who participated (cricket + football + hockey) from club B = 20% of 2000 = 400

Number of people who participated in cricket from club B = 40% of 400 = 160

Number of people who participated in football from club B = Q

So, the number of people who participated in hockey club B = {400-(160+Q)} = (240-Q)

Now,

Sum of the number of people who participated in hockey from club A and club B together = 320

So,

(280-2P) + (240-Q) = 320

520 – 2P – Q = 320

2P + Q = 200……(i)

Also,

The number of people who participated in football from club D and club E are equal

So,

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(P+50) = (P+Q-50)

Q = (50+50)

Q = 100

Now, put the value of Q in equation (i), we get,

2P + Q = 200

2P = 200 – 100

2P = 100

P = (100/2) = 50

Club A

Total number of people who participated = 600

Number of people who participated in cricket = 300

Number of people who participated in football = (2P+20) = (100+20) = 120

Number of people who participated in hockey = (280-2P) = (280-100) = 180

Club B

Total number of people who participated = 400

Number of people who participated in cricket = 160

Number of people who participated in football = Q = 100

Number of people who participated in hockey = (240-Q) = (240-100) = 140

Club C

Total number of people who participated = 300

Number of people who participated in cricket = 60% of 300 = 180

Number of people who participated in football = (Q-10) = (100-10) = 90

Number of people who participated in hockey = {300-(180+90)} = (300-270) = 30

Club D

Total number of people who participated = 25% of 2000 = 500

Number of people who participated in cricket = 75% of 500 = 375

Number of people who participated in football = (P+50) = (50+50) = 100

Number of people who participated in hockey = {500-(375+100)} = (500-475) = 25

Club E

Total number of people who participated = 200

Number of people who participated in cricket = 35% of 200 = 70

Number of people who participated in football = (P+Q-50) = (50+100-50) = (150-50) = 100

Number of people who participated in hockey = {200-(70+100)} = (200-170) = 30

Club	Total number of people who participated	Number of people who participated in cricket	Number of people who participated in football	Number of people who participated in hockey
A	600	300	120	180
B	400	160	100	140
C	300	180	90	30
D	500	375	100	25
E	200	70	100	30
Total	2000	1085	510	405

Answer: E

According to the question,

In club A,

Percentage of the total number of people (participated + not participated) from all the sports together = 100%

Percentage of thenumber of people not participated in any sport = 40%

So, the percentage of thenumber of people participated from all the sports together = (100-40)% = 60%

Total number of people participated in all the sports together = 600

60% = 600

1% = (600/60) = 10

100% = (10×100) = 1000

Therefore, the total number of people (participated + not participated) from all the sports together = 1000

So, thenumber of people not participated in any sport = 40% of 1000 = 400

Ratio of the number of males to that of females who did not participate in any sport = 1:1

Number of males not participated in any sport = 1/2 of 400 = 200

Number of females not participated in any sport = 1/2 of 400 = 200

Number of males participated in all the sports together = 50M

Number of females participated in all the sports together = 25N

So,

50M + 25N = 600……..(i)

And,

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M + N = 17

Solving both the equations, we get,

50M + 25N = 600

50M + 50N = 850

25N = 250

N = 10

Now, put the value of N in equation (i), we get,

50M + 25N = 600

50M = 600 – 250

50M = 350

M = 7

Hence, total number of males (participated + not participated) in club A = (50M+200) = (350+200) = 550.

2. Questions

Total number of people who participated in all the sports together (cricket + football + hockey) from club C = 300

So, the total number of people who participated in cricket from club A = 300

Percentage of the number of people who participated in cricket from club A = 50%

50% = 300

1% = (300/50) = 6

100% = (6×100) = 600

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from club A = 600

Number of people who participated in football from club A = (2P+20)

So, the number of people who participated in hockey from club A = {600-(300+2P+20)} = {600-(320+2P)} = (280-2P)

Percentage of the total number of people who participated in all the sports together (cricket + football + hockey) from club A. Out of the total number of people participated from all the clubs (cricket + football + hockey) = 30%

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30% = 600

1% = (600/30) = 20

100% = (20×100) = 2000

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from all the clubs together = 2000

Now,

Total number of people who participated (cricket + football + hockey) from club B = 20% of 2000 = 400

Number of people who participated in cricket from club B = 40% of 400 = 160

Number of people who participated in football from club B = Q

So, the number of people who participated in hockey club B = {400-(160+Q)} = (240-Q)

Now,

Sum of the number of people who participated in hockey from club A and club B together = 320

So,

(280-2P) + (240-Q) = 320

520 – 2P – Q = 320

2P + Q = 200……(i)

Also,

The number of people who participated in football from club D and club E are equal

So,

(P+50) = (P+Q-50)

Q = (50+50)

Q = 100

Now, put the value of Q in equation (i), we get,

2P + Q = 200

2P = 200 – 100

2P = 100

P = (100/2) = 50

Club A

Total number of people who participated = 600

Number of people who participated in cricket = 300

Number of people who participated in football = (2P+20) = (100+20) = 120

Number of people who participated in hockey = (280-2P) = (280-100) = 180

Club B

Total number of people who participated = 400

Number of people who participated in cricket = 160

Number of people who participated in football = Q = 100

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Number of people who participated in hockey = (240-Q) = (240-100) = 140

Club C

Total number of people who participated = 300

Number of people who participated in cricket = 60% of 300 = 180

Number of people who participated in football = (Q-10) = (100-10) = 90

Number of people who participated in hockey = {300-(180+90)} = (300-270) = 30

Club D

Total number of people who participated = 25% of 2000 = 500

Number of people who participated in cricket = 75% of 500 = 375

Number of people who participated in football = (P+50) = (50+50) = 100

Number of people who participated in hockey = {500-(375+100)} = (500-475) = 25

Club E

Total number of people who participated = 200

Number of people who participated in cricket = 35% of 200 = 70

Number of people who participated in football = (P+Q-50) = (50+100-50) = (150-50) = 100

Number of people who participated in hockey = {200-(70+100)} = (200-170) = 30

Club	Total number of people who participated	Number of people who participated in cricket	Number of people who participated in football	Number of people who participated in hockey
A	600	300	120	180
B	400	160	100	140
C	300	180	90	30
D	500	375	100	25
E	200	70	100	30
Total	2000	1085	510	405

Answer: C

Club B

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Total number of people who participated in cricket, football, and hockey together = 400

Total number of people who participated in cricket, football, hockey and volleyball together = 140% of 400 = 560

So, the number of people who participated in volleyball = (560-400) = 160

Number of people who participated in hockey = 140

Therefore, the percentage of the number of people who participated in volleyball is more than that of hockey by = [{(160-140)/140}]% = 14.28%

So, X% = 14.28%

Club E

Number of people who participated in cricket = 70

Fractional value of X% = 14.28% = 1/7

Therefore, the fractional value of 2X% = {(1/7)×2} = 2/7

Now, the ratio of the number of people who participated in cricket to that of volleyball = 5:7. Let, it be 5k and 7k respectively.

So, 5k = 70

k = (70/5) = 14

Therefore, 7k = (7×14) = 98

Hence, the number of people who participated in volleyball from club E = 98.

3. Questions

Total number of people who participated in all the sports together (cricket + football + hockey) from club C = 300

So, the total number of people who participated in cricket from club A = 300

Percentage of the number of people who participated in cricket from club A = 50%

50% = 300

1% = (300/50) = 6

100% = (6×100) = 600

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from club A = 600

Number of people who participated in football from club A = (2P+20)

So, the number of people who participated in hockey from club A = {600-(300+2P+20)} = {600-(320+2P)} = (280-2P)

Percentage of the total number of people who participated in all the sports together (cricket + football + hockey) from club A. Out of the total number of people participated from all the clubs (cricket + football + hockey) = 30%

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30% = 600

1% = (600/30) = 20

100% = (20×100) = 2000

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from all the clubs together = 2000

Now,

Total number of people who participated (cricket + football + hockey) from club B = 20% of 2000 = 400

Number of people who participated in cricket from club B = 40% of 400 = 160

Number of people who participated in football from club B = Q

So, the number of people who participated in hockey club B = {400-(160+Q)} = (240-Q)

Now,

Sum of the number of people who participated in hockey from club A and club B together = 320

So,

(280-2P) + (240-Q) = 320

520 – 2P – Q = 320

2P + Q = 200……(i)

Also,

The number of people who participated in football from club D and club E are equal

So,

(P+50) = (P+Q-50)

Q = (50+50)

Q = 100

Now, put the value of Q in equation (i), we get,

2P + Q = 200

2P = 200 – 100

2P = 100

P = (100/2) = 50

Club A

Total number of people who participated = 600

Number of people who participated in cricket = 300

Number of people who participated in football = (2P+20) = (100+20) = 120

Number of people who participated in hockey = (280-2P) = (280-100) = 180

Club B

Total number of people who participated = 400

Number of people who participated in cricket = 160

Number of people who participated in football = Q = 100

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Number of people who participated in hockey = (240-Q) = (240-100) = 140

Club C

Total number of people who participated = 300

Number of people who participated in cricket = 60% of 300 = 180

Number of people who participated in football = (Q-10) = (100-10) = 90

Number of people who participated in hockey = {300-(180+90)} = (300-270) = 30

Club D

Total number of people who participated = 25% of 2000 = 500

Number of people who participated in cricket = 75% of 500 = 375

Number of people who participated in football = (P+50) = (50+50) = 100

Number of people who participated in hockey = {500-(375+100)} = (500-475) = 25

Club E

Total number of people who participated = 200

Number of people who participated in cricket = 35% of 200 = 70

Number of people who participated in football = (P+Q-50) = (50+100-50) = (150-50) = 100

Number of people who participated in hockey = {200-(70+100)} = (200-170) = 30

Club	Total number of people who participated	Number of people who participated in cricket	Number of people who participated in football	Number of people who participated in hockey
A	600	300	120	180
B	400	160	100	140
C	300	180	90	30
D	500	375	100	25
E	200	70	100	30
Total	2000	1085	510	405

Answer: C

Club F

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Total number of people who participated in all the three sports together = [3×(P+Q)] = [3×(50+100)] = (3×150) = 450

Using statement I and II together

Let, the number of participants in cricket and hockey be 7y and 5y respectively.

So, the number of people who participated in football = 150% of 7y = 10.50y

Now,

7y + 10.50y + 5y = 450

22.50y = 450

y = (450/22.50)

y = 20

Hence, the number of people who participated in football from club F = 10.50y = (10.50×20) = 210.

So, statement I and II together are necessary.

Using statement II and III together

Ratio of the number of participants in football and cricket = 150:100 = 3:2. Let, it be 3z and 2z respectively.

So, the number of people who participated in hockey = (3z-110)

Now,

2z + 3z + (3z-110) = 450

8z = 560

z = 70

Hence, the number of people who participated in football from club F = 3z = (3×70) = 210.

So, statement II and III together are necessary.

Using statement I and III together

Let, the number of participants in cricket and hockey be 7s and 5s respectively.

So, the number of people who participated in football = (5s+110)

Now,

7s + (5s+110) + 5s = 450

17s = 340

s = 20

Hence, the number of people who participated in football from club F = (5s+110) = (100+110) = 210

So, statement I and III together are necessary.

Hence, the combination of any of the two statements is sufficient to answer the question.

4. Questions

Total number of people who participated in all the sports together (cricket + football + hockey) from club C = 300

So, the total number of people who participated in cricket from club A = 300

Percentage of the number of people who participated in cricket from club A = 50%

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50% = 300

1% = (300/50) = 6

100% = (6×100) = 600

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from club A = 600

Number of people who participated in football from club A = (2P+20)

So, the number of people who participated in hockey from club A = {600-(300+2P+20)} = {600-(320+2P)} = (280-2P)

Percentage of the total number of people who participated in all the sports together (cricket + football + hockey) from club A. Out of the total number of people participated from all the clubs (cricket + football + hockey) = 30%

30% = 600

1% = (600/30) = 20

100% = (20×100) = 2000

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from all the clubs together = 2000

Now,

Total number of people who participated (cricket + football + hockey) from club B = 20% of 2000 = 400

Number of people who participated in cricket from club B = 40% of 400 = 160

Number of people who participated in football from club B = Q

So, the number of people who participated in hockey club B = {400-(160+Q)} = (240-Q)

Now,

Sum of the number of people who participated in hockey from club A and club B together = 320

So,

(280-2P) + (240-Q) = 320

520 – 2P – Q = 320

2P + Q = 200……(i)

Also,

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The number of people who participated in football from club D and club E are equal

So,

(P+50) = (P+Q-50)

Q = (50+50)

Q = 100

Now, put the value of Q in equation (i), we get,

2P + Q = 200

2P = 200 – 100

2P = 100

P = (100/2) = 50

Club A

Total number of people who participated = 600

Number of people who participated in cricket = 300

Number of people who participated in football = (2P+20) = (100+20) = 120

Number of people who participated in hockey = (280-2P) = (280-100) = 180

Club B

Total number of people who participated = 400

Number of people who participated in cricket = 160

Number of people who participated in football = Q = 100

Number of people who participated in hockey = (240-Q) = (240-100) = 140

Club C

Total number of people who participated = 300

Number of people who participated in cricket = 60% of 300 = 180

Number of people who participated in football = (Q-10) = (100-10) = 90

Number of people who participated in hockey = {300-(180+90)} = (300-270) = 30

Club D

Total number of people who participated = 25% of 2000 = 500

Number of people who participated in cricket = 75% of 500 = 375

Number of people who participated in football = (P+50) = (50+50) = 100

Number of people who participated in hockey = {500-(375+100)} = (500-475) = 25

Club E

Total number of people who participated = 200

Number of people who participated in cricket = 35% of 200 = 70

Number of people who participated in football = (P+Q-50) = (50+100-50) = (150-50) = 100

Number of people who participated in hockey = {200-(70+100)} = (200-170) = 30

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Club	Total number of people who participated	Number of people who participated in cricket	Number of people who participated in football	Number of people who participated in hockey
A	600	300	120	180
B	400	160	100	140
C	300	180	90	30
D	500	375	100	25
E	200	70	100	30
Total	2000	1085	510	405

Answer: A

Total weight of all the people who participated in all the sports together from club C = (50×300) = 15000 kg

Total weight of all the people who participated in all the sports together from club D = (U×500) = 500U kg

So,

15000 + 500U = 45000

500U = 30000

U = 60

So, T = {(60×2)/3} = 40

Therefore, the value of (U-T) = (60-40) = 20.

Using option A

(T/2) = (40/2) = 20 (satisfied)

Hence, option A is the correct answer.

5. Questions

Total number of people who participated in all the sports together (cricket + football + hockey) from club C = 300

So, the total number of people who participated in cricket from club A = 300

Percentage of the number of people who participated in cricket from club A = 50%

50% = 300

1% = (300/50) = 6

100% = (6×100) = 600

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from club A = 600

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Number of people who participated in football from club A = (2P+20)

So, the number of people who participated in hockey from club A = {600-(300+2P+20)} = {600-(320+2P)} = (280-2P)

Percentage of the total number of people who participated in all the sports together (cricket + football + hockey) from club A. Out of the total number of people participated from all the clubs (cricket + football + hockey) = 30%

30% = 600

1% = (600/30) = 20

100% = (20×100) = 2000

Therefore, the total number of people who participated in all the sports together (cricket + football + hockey) from all the clubs together = 2000

Now,

Total number of people who participated (cricket + football + hockey) from club B = 20% of 2000 = 400

Number of people who participated in cricket from club B = 40% of 400 = 160

Number of people who participated in football from club B = Q

So, the number of people who participated in hockey club B = {400-(160+Q)} = (240-Q)

Now,

Sum of the number of people who participated in hockey from club A and club B together = 320

So,

(280-2P) + (240-Q) = 320

520 – 2P – Q = 320

2P + Q = 200……(i)

Also,

The number of people who participated in football from club D and club E are equal

So,

(P+50) = (P+Q-50)

Q = (50+50)

Q = 100

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Now, put the value of Q in equation (i), we get,

2P + Q = 200

2P = 200 – 100

2P = 100

P = (100/2) = 50

Club A

Total number of people who participated = 600

Number of people who participated in cricket = 300

Number of people who participated in football = (2P+20) = (100+20) = 120

Number of people who participated in hockey = (280-2P) = (280-100) = 180

Club B

Total number of people who participated = 400

Number of people who participated in cricket = 160

Number of people who participated in football = Q = 100

Number of people who participated in hockey = (240-Q) = (240-100) = 140

Club C

Total number of people who participated = 300

Number of people who participated in cricket = 60% of 300 = 180

Number of people who participated in football = (Q-10) = (100-10) = 90

Number of people who participated in hockey = {300-(180+90)} = (300-270) = 30

Club D

Total number of people who participated = 25% of 2000 = 500

Number of people who participated in cricket = 75% of 500 = 375

Number of people who participated in football = (P+50) = (50+50) = 100

Number of people who participated in hockey = {500-(375+100)} = (500-475) = 25

Club E

Total number of people who participated = 200

Number of people who participated in cricket = 35% of 200 = 70

Number of people who participated in football = (P+Q-50) = (50+100-50) = (150-50) = 100

Number of people who participated in hockey = {200-(70+100)} = (200-170) = 30

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Club	Total number of people who participated	Number of people who participated in cricket	Number of people who participated in football	Number of people who participated in hockey
A	600	300	120	180
B	400	160	100	140
C	300	180	90	30
D	500	375	100	25
E	200	70	100	30
Total	2000	1085	510	405

Answer: B

Club E

Total number of people who participated in all the sports together = 200

Let, the number of females who participated in all the sports together be r.

So, the difference between the number of males and females who participated in all the sports together = (2×r) = 2r (males – females)

Therefore, the number of males who participated in all the sports together = (r+2r) = 3r

Now,

3r + r = 200

4r = 200

r = (200/4) = 50

Hence, the number of females who participated in all the sports together = 3r = (3×50) = 150.

6. Questions

For P

Total marks scored (A+B+C) = 200

Marks scored in subject A = 80

Marks scored in subject (B+C) = (200-80) = 120

Ratio of marks scored in subject B to that of C = 100:140 = 5:7

So, marks scored in subject B = 5/12 of 120 = 50

And, marks scored in subject C = 7/12 of 120 = 70

For R

Total marks scored (A+B+C) = 150

Marks scored in subject A = 70

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Marks scored in subject C = 1/6 of 150 = 25 *(Fractional value of 16.66% = 1/6)*

So, marks scored in subject B = {150-(70+25)} = (150-95) = 55

For Q

Total marks scored (A+B+C) = 100

Marks scored in subject A = 60

Marks scored in subject B = {(55×6)/11} = 30

So, marks scored in subject C = {100-(60+30)} = (100-90) = 10

Now,

Let, marks scored by S in subject B and T in subject C be 2m and m respectively.

And, also let the marks scored by S in subject C and T in subject B be 2n and 5n respectively.

For S

Total marks scored (A+B+C) = 250

Marks scored in subject A = 50

Marks scored in subject (B+C) = (250-50) = 200

Marks scored in subject B = 2m

Marks scored in subject C = 2n

For T

Total marks scored (A+B+C) = 300

Marks scored in subject A = 40

Marks scored in subject (B+C) = (300-40) = 260

Marks scored in subject B = 5n

Marks scored in subject C = m

Now,

2m + 2n = 200……..(i)

m + 5n = 260…….(ii)

Solving both the equations, we get,

2m + 2n = 200

2m + 10n = 520

8n = 320

n = 40

Now, put the value of n in equation (i), we get,

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2m + 2n = 200

2m = 200 – 80

2m = 120

m = 60

So, marks scored by S in subject B = 2m = 120

And, marks scored by S in subject C = 2n = 80

Also,

Marks scored by T in subject B = 5n = 200

And, marks scored by T in subject C = m = 60

Student	Marks scored in subject A	Marks scored in subject B	Marks scored in subject C	Total marks scored (A+B+C)
P	80	50	70	200
Q	60	30	10	100
R	70	55	25	150
S	50	120	80	250
T	40	200	60	300
Total	300	455	245	1000

Answer: A

According to the question,

For S

Let, marks scored in the theory section of subject A and in the practical section of subject B be 4x and 3x respectively.

Fractional value of 11.11% = 1/9

So, the ratio of marks scored in the practical section of subject A and in the theory section of subject B = 1:9. Let, it be y and 9y respectively.

Now,

4x + y = 50…….(i)

3x + 9y = 120……..(ii)

Solving both equations, we get,

12x + 3y = 150

12x + 36y = 480

33y = 330

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y = 10

Now, put the value of y in equation (i), we get,

4x + y = 50

4x = 50 – 10

4x = 40

x = 10

Now,

Total marks scored in all the subjects (theory + practical) = 250

Total marks scored in the practical section = 55

So, the total marks scored in the theory section = (250-55) = 195

Hence, the marks scored by S in the theory section of subject C = {195-(40+90)} = (195-130) = 65.

7. Questions

For P

Total marks scored (A+B+C) = 200

Marks scored in subject A = 80

Marks scored in subject (B+C) = (200-80) = 120

Ratio of marks scored in subject B to that of C = 100:140 = 5:7

So, marks scored in subject B = 5/12 of 120 = 50

And, marks scored in subject C = 7/12 of 120 = 70

For R

Total marks scored (A+B+C) = 150

Marks scored in subject A = 70

Marks scored in subject C = 1/6 of 150 = 25 *(Fractional value of 16.66% = 1/6)*

So, marks scored in subject B = {150-(70+25)} = (150-95) = 55

For Q

Total marks scored (A+B+C) = 100

Marks scored in subject A = 60

Marks scored in subject B = {(55×6)/11} = 30

So, marks scored in subject C = {100-(60+30)} = (100-90) = 10

Now,

Let, marks scored by S in subject B and T in subject C be 2m and m respectively.

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And, also let the marks scored by S in subject C and T in subject B be 2n and 5n respectively.

For S

Total marks scored (A+B+C) = 250

Marks scored in subject A = 50

Marks scored in subject (B+C) = (250-50) = 200

Marks scored in subject B = 2m

Marks scored in subject C = 2n

For T

Total marks scored (A+B+C) = 300

Marks scored in subject A = 40

Marks scored in subject (B+C) = (300-40) = 260

Marks scored in subject B = 5n

Marks scored in subject C = m

Now,

2m + 2n = 200……..(i)

m + 5n = 260…….(ii)

Solving both the equations, we get,

2m + 2n = 200

2m + 10n = 520

8n = 320

n = 40

Now, put the value of n in equation (i), we get,

2m + 2n = 200

2m = 200 – 80

2m = 120

m = 60

So, marks scored by S in subject B = 2m = 120

And, marks scored by S in subject C = 2n = 80

Also,

Marks scored by T in subject B = 5n = 200

And, marks scored by T in subject C = m = 60

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Student	Marks scored in subject A	Marks scored in subject B	Marks scored in subject C	Total marks scored (A+B+C)
P	80	50	70	200
Q	60	30	10	100
R	70	55	25	150
S	50	120	80	250
T	40	200	60	300
Total	300	455	245	1000

Answer: B

For P

Marks scored in subject A = 80

Marks scored in subject C = 70

So,

The percentage of marks scored in subject A = {(80/2K)×100}%

And, the percentage of marks scored in subject C = {(70/K)×100}%

Now,

{(80/2K)×100} + {(70/K)×100} = (K+10)

4000/K + 7000/K = (K+10)

11000/K= (K+10)

K^2 + 10K = 11000

K^2 + 10K – 11000 = 0

K^2 + 110K – 100K – 11000 = 0

K(K+110) – 100(K+110) = 0

(K-100) (K+110) = 0

K = 100 or -110

So, K = 100 (positive integer)

Now,

Marks scored by P in subject B = 50

Maximum marks of subject B = (K+50) = (100+50) = 150

Hence, the percentage of marks obtained by P in subject B = {(50/150)×100}% = 33.33%.

8. Questions

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For P

Total marks scored (A+B+C) = 200

Marks scored in subject A = 80

Marks scored in subject (B+C) = (200-80) = 120

Ratio of marks scored in subject B to that of C = 100:140 = 5:7

So, marks scored in subject B = 5/12 of 120 = 50

And, marks scored in subject C = 7/12 of 120 = 70

For R

Total marks scored (A+B+C) = 150

Marks scored in subject A = 70

Marks scored in subject C = 1/6 of 150 = 25 *(Fractional value of 16.66% = 1/6)*

So, marks scored in subject B = {150-(70+25)} = (150-95) = 55

For Q

Total marks scored (A+B+C) = 100

Marks scored in subject A = 60

Marks scored in subject B = {(55×6)/11} = 30

So, marks scored in subject C = {100-(60+30)} = (100-90) = 10

Now,

Let, marks scored by S in subject B and T in subject C be 2m and m respectively.

And, also let the marks scored by S in subject C and T in subject B be 2n and 5n respectively.

For S

Total marks scored (A+B+C) = 250

Marks scored in subject A = 50

Marks scored in subject (B+C) = (250-50) = 200

Marks scored in subject B = 2m

Marks scored in subject C = 2n

For T

Total marks scored (A+B+C) = 300

Marks scored in subject A = 40

Marks scored in subject (B+C) = (300-40) = 260

Marks scored in subject B = 5n

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Marks scored in subject C = m

Now,

2m + 2n = 200……..(i)

m + 5n = 260…….(ii)

Solving both the equations, we get,

2m + 2n = 200

2m + 10n = 520

8n = 320

n = 40

Now, put the value of n in equation (i), we get,

2m + 2n = 200

2m = 200 – 80

2m = 120

m = 60

So, marks scored by S in subject B = 2m = 120

And, marks scored by S in subject C = 2n = 80

Also,

Marks scored by T in subject B = 5n = 200

And, marks scored by T in subject C = m = 60

Student	Marks scored in subject A	Marks scored in subject B	Marks scored in subject C	Total marks scored (A+B+C)
P	80	50	70	200
Q	60	30	10	100
R	70	55	25	150
S	50	120	80	250
T	40	200	60	300
Total	300	455	245	1000

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Answer: D

For Q

Total marks scored in subject A = 60

Total marks scored in subject B = 30

Let, the fair marks scored in subject A to that of grace marks scored in subject B be 11y and 2y, respectively.

Also, let the grace marks scored in subject A to that of fair marks scored in subject B be w and 4w, respectively.

Now,

11y + w = 60……..(i)

2y + 4w = 30……. (ii)

Solving both equations, we get,

22y + 2w = 120

22y + 44w = 330

42w = 210

w = 5

Now, put the value of w in equation (i), we get,

11y + w = 60

11y = 60 – 5

11y = 55

y = 5

So, the total fair marks scored in (A+B) by P = (11y+4w) = (55+20) = 75

And, the total grace marks scored in (A+B) by P = (w+2y) = (5+10) = 15

Hence, the required difference = (75-15) = 60.

9. Questions

For P

Total marks scored (A+B+C) = 200

Marks scored in subject A = 80

Marks scored in subject (B+C) = (200-80) = 120

Ratio of marks scored in subject B to that of C = 100:140 = 5:7

So, marks scored in subject B = 5/12 of 120 = 50

And, marks scored in subject C = 7/12 of 120 = 70

For R

Total marks scored (A+B+C) = 150

Marks scored in subject A = 70

Marks scored in subject C = 1/6 of 150 = 25 *(Fractional value of 16.66% = 1/6)*

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So, marks scored in subject B = {150-(70+25)} = (150-95) = 55

For Q

Total marks scored (A+B+C) = 100

Marks scored in subject A = 60

Marks scored in subject B = {(55×6)/11} = 30

So, marks scored in subject C = {100-(60+30)} = (100-90) = 10

Now,

Let, marks scored by S in subject B and T in subject C be 2m and m respectively.

And, also let the marks scored by S in subject C and T in subject B be 2n and 5n respectively.

For S

Total marks scored (A+B+C) = 250

Marks scored in subject A = 50

Marks scored in subject (B+C) = (250-50) = 200

Marks scored in subject B = 2m

Marks scored in subject C = 2n

For T

Total marks scored (A+B+C) = 300

Marks scored in subject A = 40

Marks scored in subject (B+C) = (300-40) = 260

Marks scored in subject B = 5n

Marks scored in subject C = m

Now,

2m + 2n = 200……..(i)

m + 5n = 260…….(ii)

Solving both the equations, we get,

2m + 2n = 200

2m + 10n = 520

8n = 320

n = 40

Now, put the value of n in equation (i), we get,

2m + 2n = 200

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2m = 200 – 80

2m = 120

m = 60

So, marks scored by S in subject B = 2m = 120

And, marks scored by S in subject C = 2n = 80

Also,

Marks scored by T in subject B = 5n = 200

And, marks scored by T in subject C = m = 60

Student	Marks scored in subject A	Marks scored in subject B	Marks scored in subject C	Total marks scored (A+B+C)
P	80	50	70	200
Q	60	30	10	100
R	70	55	25	150
S	50	120	80	250
T	40	200	60	300
Total	300	455	245	1000

Answer: B

For U

Total marks scoredin (A+B+C) = 300

Marks scored in subject A = 125% of 40 = 50

So, the sum of the marks scored in subject B and subject C together = (300-50) = 250

Therefore, the difference between the marks scored in subject B and subject C = 50

Now, let the marks scored in subject B and subject C be v and (v-50).

So,

2v – 50 = 250

2v = 300

v = 150

So, the marks scored in subject B = v = 150

And, the marks scored in subject C = (v-50) = (150-50) = 100

Hence, the required ratio = 150:100 = 3:2.

10. Questions

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For P

Total marks scored (A+B+C) = 200

Marks scored in subject A = 80

Marks scored in subject (B+C) = (200-80) = 120

Ratio of marks scored in subject B to that of C = 100:140 = 5:7

So, marks scored in subject B = 5/12 of 120 = 50

And, marks scored in subject C = 7/12 of 120 = 70

For R

Total marks scored (A+B+C) = 150

Marks scored in subject A = 70

Marks scored in subject C = 1/6 of 150 = 25 *(Fractional value of 16.66% = 1/6)*

So, marks scored in subject B = {150-(70+25)} = (150-95) = 55

For Q

Total marks scored (A+B+C) = 100

Marks scored in subject A = 60

Marks scored in subject B = {(55×6)/11} = 30

So, marks scored in subject C = {100-(60+30)} = (100-90) = 10

Now,

Let, marks scored by S in subject B and T in subject C be 2m and m respectively.

And, also let the marks scored by S in subject C and T in subject B be 2n and 5n respectively.

For S

Total marks scored (A+B+C) = 250

Marks scored in subject A = 50

Marks scored in subject (B+C) = (250-50) = 200

Marks scored in subject B = 2m

Marks scored in subject C = 2n

For T

Total marks scored (A+B+C) = 300

Marks scored in subject A = 40

Marks scored in subject (B+C) = (300-40) = 260

Marks scored in subject B = 5n

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Marks scored in subject C = m

Now,

2m + 2n = 200……..(i)

m + 5n = 260…….(ii)

Solving both the equations, we get,

2m + 2n = 200

2m + 10n = 520

8n = 320

n = 40

Now, put the value of n in equation (i), we get,

2m + 2n = 200

2m = 200 – 80

2m = 120

m = 60

So, marks scored by S in subject B = 2m = 120

And, marks scored by S in subject C = 2n = 80

Also,

Marks scored by T in subject B = 5n = 200

And, marks scored by T in subject C = m = 60

Student	Marks scored in subject A	Marks scored in subject B	Marks scored in subject C	Total marks scored (A+B+C)
P	80	50	70	200
Q	60	30	10	100
R	70	55	25	150
S	50	120	80	250
T	40	200	60	300
Total	300	455	245	1000

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Answer: C

For R

Total marks scored in (theory + practical) in all the subjects together = 150

Ratio of marks scored in the theory to that of practical section = 400:100 = 4:1

Marks scored in the theory section in (A+B+C) = 4/5 of 150 = 120

Marks scored in the practical section in (A+B+C) = 1/5 of 150 = 30

Maximum marks for the theory section including all the subjectstogether = (50×3) = 150

Maximum marks for the practical section including all the subjectstogether = (20×3) = 60

So, the percentage of the total marks scored in the theory section in (A+B+C) = {(120/150)×100}% = 80%

And, the percentage of the total marks scored in the practical section in (A+B+C) = {(30/60)×100}% = 50%

Hence, the required sum = (80+50)% = 130%.

11. Questions

Shop X

Number of customers purchased only product M = 5A

Number of customers purchased only product N = 80% of 5A = 4A

Number of customers purchased both the products = [{(5A+4A)/2}+15] = (4.5A+15)

Shop Y

Number of customers purchased only product M = 5B

Number of customers purchased only product N = {(5B×3)/5} = 3B

Number of customers purchased both the products = (5B+5)

Now,

3B + 5A = 95……(i)

5B – 4A = 35……..(ii)

Solving both the equation, we get,

15B + 25A = 475

15B – 12A = 105

37A = 370

A = 10

Now, put the value of A in equation (i), wet get,

3B + 5A = 95

3B = 95 – 50

3B = 45

B = 15

Shop X

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Number of customers purchased only product M = 5A = 50

Number of customers purchased only product N = 4A = 40

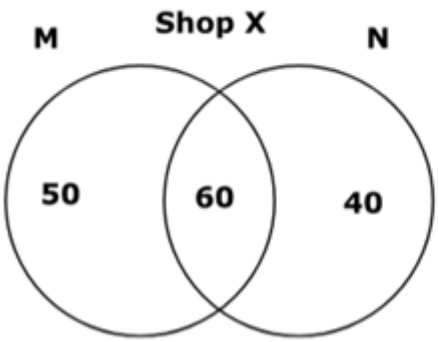
Number of customers purchased both the products = (4.5A+15) = (45+15) = 60

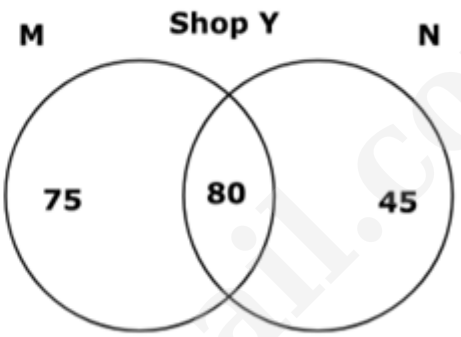
Shop Y

Number of customers purchased only product M = 5B = 75

Number of customers purchased only product N = 3B = 45

Number of customers purchased both products = (5B+5) = (75+5) = 80





Answer: B

According to the question,

In shop X,

Total number of product M sold = (50+60) = 110

Total number of product N sold = (40+60) = 100

Total number of only product M sold = 50

Total number of only product N sold = 40

Total revenue generated from selling all the items of product M = (P×110) = Rs. 110P

Total revenue generated from selling all the items of product N = (Q×100) = Rs. 100Q

Total revenue generated from selling only product M = (P×50) = Rs. 50P

Total revenue generated from selling only product N = (Q×40) = Rs. 40Q

Now,

110P + 100Q = 19000……(i)

50P + 40Q = 8200……(ii)

Solving both equations, we get,

550P + 500Q = 95000

550P + 440Q = 90200

60Q = 4800

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Q = 80

Now, put the value of Q in equation (i), we get,

110P + 100Q = 19000

110P = 19000 – 8000

110P = 11000

P = 100

Hence, the difference between the selling price of each item of product M and product N = (P-Q) = (100-80) = Rs. 20.

12. Questions

Shop X

Number of customers purchased only product M = 5A

Number of customers purchased only product N = 80% of 5A = 4A

Number of customers purchased both the products = [{(5A+4A)/2}+15] = (4.5A+15)

Shop Y

Number of customers purchased only product M = 5B

Number of customers purchased only product N = {(5B×3)/5} = 3B

Number of customers purchased both the products = (5B+5)

Now,

3B + 5A = 95……(i)

5B – 4A = 35……..(ii)

Solving both the equation, we get,

15B + 25A = 475

15B – 12A = 105

37A = 370

A = 10

Now, put the value of A in equation (i), wet get,

3B + 5A = 95

3B = 95 – 50

3B = 45

B = 15

Shop X

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Number of customers purchased only product M = 5A = 50

Number of customers purchased only product N = 4A = 40

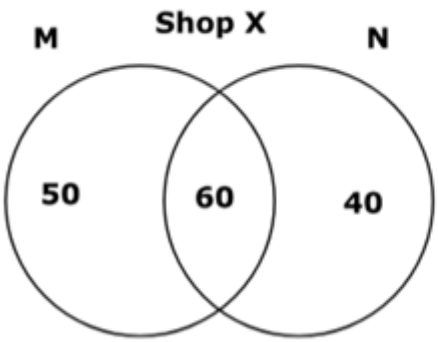
Number of customers purchased both the products = (4.5A+15) = (45+15) = 60

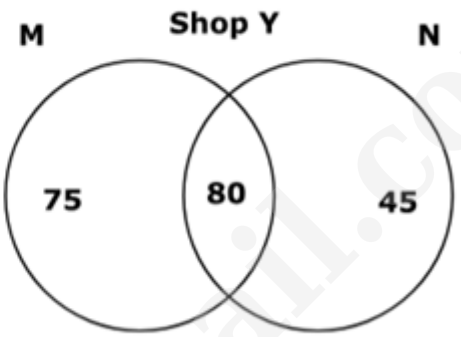
Shop Y

Number of customers purchased only product M = 5B = 75

Number of customers purchased only product N = 3B = 45

Number of customers purchased both products = (5B+5) = (75+5) = 80





Answer: A

Shop Z

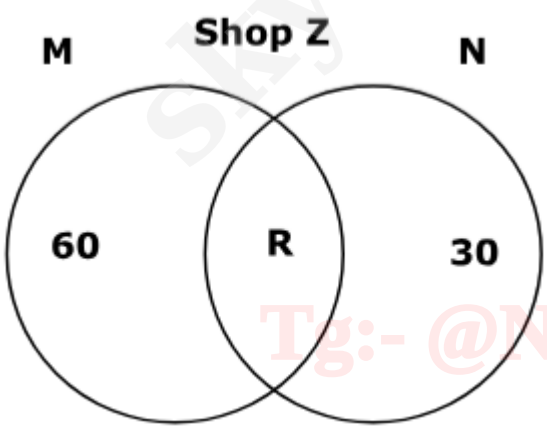
Number of customers who purchased both products = R

Number of customers who purchased only product M = 6A = 60 (A=10, from common explanation)

Number of customers who purchased only product N = 2B = 30 (B=15, from common explanation)

Hence, the total number of customers who purchased all the products (M+N) from shop Z = (60+R+30) = (90+R).

So, option A is the correct answer.



13. Questions

Shop X

Number of customers purchased only product M = 5A

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Number of customers purchased only product N = 80% of 5A = 4A

Number of customers purchased both the products = [{(5A+4A)/2}+15] = (4.5A+15)

Shop Y

Number of customers purchased only product M = 5B

Number of customers purchased only product N = {(5B×3)/5} = 3B

Number of customers purchased both the products = (5B+5)

Now,

3B + 5A = 95……(i)

5B – 4A = 35……..(ii)

Solving both the equation, we get,

15B + 25A = 475

15B – 12A = 105

37A = 370

A = 10

Now, put the value of A in equation (i), wet get,

3B + 5A = 95

3B = 95 – 50

3B = 45

B = 15

Shop X

Number of customers purchased only product M = 5A = 50

Number of customers purchased only product N = 4A = 40

Number of customers purchased both the products = (4.5A+15) = (45+15) = 60

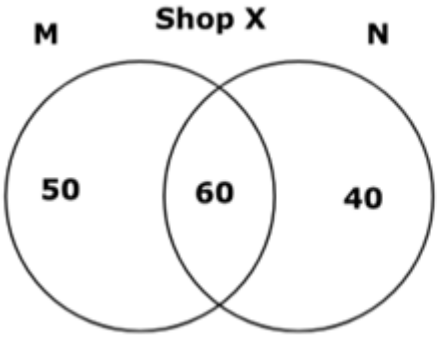
Shop Y

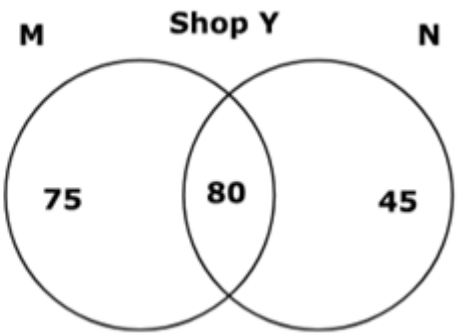
Number of customers purchased only product M = 5B = 75

Number of customers purchased only product N = 3B = 45

Number of customers purchased both products = (5B+5) = (75+5) = 80

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Answer: C

Shop Y

Total number of customers purchased only product M = 75

Total number of customers purchased only product N = 45

Let, the number of male customers who purchased only product M and female customers who purchased only product N be 10s and 3s respectively.

Ratio of the number of female customers who purchased only product M and the male customers who purchased only product N = 5:6. Let, it be 5t and 6t respectively.

So,

10s + 5t = 75……..(i)

3s + 6t = 45…….(ii)

Solving both equations, we get,

30s + 15t = 225

30s + 60t = 450

45t = 225

t = 5

Now, put the value of t in equation (i), we get,

10s + 5t = 75

10s = 75 – 25

10s = 50

s = 5

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Hence, the total number of male customers who purchased only product M and only product N from shop Y = (10s+6t) = (50+30) = 80.

14. Questions

Runs scored by A in match 3 = 75

So, runs scored by B in match 3 = 120% of 75 = 90

Therefore, runs scored by C in match 3 = (90-45) = 45

Ratio of runs scored by B to that of C in match 3 = 90:45 = 2:1

So,

(N/10):1 = 2:1

N/10 = 2

Therefore, N = (10×2) = 20

Now,

Runs scored by C in match 2 = {(45×7}/15} = 21

So, runs scored by B in match 2 = {(21×2)/3} = 14

Runs scored by A in match 2 = 35

Now,

Percentage of runs scored by B with respect to that of A in match 2 = {(14/35)×100}% = 40%

So,

(K-5)% = 40%

K = (40+5)

Therefore, K = 45

Now,

Runs scored by B in match 1 = (90/2) = 45

Percentage of runs scored by B with respect to that of A in match 1 = 75%

75% = 45

1% = (45/75) = 3/5

100% = {(3×100)/5} = 60

So, runs scored by A in match 1 = 60

Therefore,

(L+5) = 60

L = (60-5) = 55

Hence, L = 55

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Now, runs scored by C in match 1 = {(60×100)/120} = 50

Ratio of runs scored by B to that of C in match 1 = 45:50 = 9:10

So,

9:(M-N) = 9:10

(M-N) = 10

(M-20) = 10

M = (10+20)

M = 30

Now,

Match 4

Runs scored by A = 2N = 40

Runs scored by B = (4M+5)% of 40 = 125% of 40 = 50

So, runs scored by C = {(50×4)/5} = 40

Match 5

Runs scored by A = (K+L) = (45+55) = 100

Runs scored by B = 80% of 100 = 80

So, runs scored by C = {(80×1)/4} = 20

Match	Runs scored by A	Runs scored by B	Runs scored by C	Total runs scored (A+B+C)
1	60	45	50	155
2	35	14	21	70
3	75	90	45	210
4	40	50	40	130
5	100	80	20	200
Total	310	279	176	765

Answer: B

According to the question,

In match 4,

Runs scored by A = 40

Runs scored by B = 50

So, the number of balls faced by A = 50

Therefore, strike rate of A = {(40/50)×100} = 80

So,

Strike rate of B = (80+45) = 125

Now, let the number of balls faced by B be p.

So,

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{(50/p)×100} = 125

50/p = 125/100

50/p = 5/4

5p = 200

p = (200/5) = 40

Therefore, the number of balls faced by B = p = 40

Now,

The number of balls faced by C = {(50+40)/2} = (90/2) = 45

Runs scored by C = 40

Hence, the strike rate of C = {(40/45)×100} = 88.88.

15. Questions

Runs scored by A in match 3 = 75

So, runs scored by B in match 3 = 120% of 75 = 90

Therefore, runs scored by C in match 3 = (90-45) = 45

Ratio of runs scored by B to that of C in match 3 = 90:45 = 2:1

So,

(N/10):1 = 2:1

N/10 = 2

Therefore, N = (10×2) = 20

Now,

Runs scored by C in match 2 = {(45×7}/15} = 21

So, runs scored by B in match 2 = {(21×2)/3} = 14

Runs scored by A in match 2 = 35

Now,

Percentage of runs scored by B with respect to that of A in match 2 = {(14/35)×100}% = 40%

So,

(K-5)% = 40%

K = (40+5)

Therefore, K = 45

Now,

Runs scored by B in match 1 = (90/2) = 45

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Percentage of runs scored by B with respect to that of A in match 1 = 75%

75% = 45

1% = (45/75) = 3/5

100% = {(3×100)/5} = 60

So, runs scored by A in match 1 = 60

Therefore,

(L+5) = 60

L = (60-5) = 55

Hence, L = 55

Now, runs scored by C in match 1 = {(60×100)/120} = 50

Ratio of runs scored by B to that of C in match 1 = 45:50 = 9:10

So,

9:(M-N) = 9:10

(M-N) = 10

(M-20) = 10

M = (10+20)

M = 30

Now,

Match 4

Runs scored by A = 2N = 40

Runs scored by B = (4M+5)% of 40 = 125% of 40 = 50

So, runs scored by C = {(50×4)/5} = 40

Match 5

Runs scored by A = (K+L) = (45+55) = 100

Runs scored by B = 80% of 100 = 80

So, runs scored by C = {(80×1)/4} = 20

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Match	Runs scored by A	Runs scored by B	Runs scored by C	Total runs scored (A+B+C)
1	60	45	50	155
2	35	14	21	70
3	75	90	45	210
4	40	50	40	130
5	100	80	20	200
Total	310	279	176	765

Answer: A

B in match 3,

Total runs scored = 90

Let, the number of singles, doubles and triples scored by B be 10r, 5r and 6r respectively.

Now, runs scored in fours = (4×4) = 16

And, runs scored in sixes (6×6) = 36

So, the total runs scored through boundary (fours + sixes) = (16+36) = 52

Therefore, the remaining runs scored in singles, doubles and triples only = (90-52) = 38

Runs scored in singles = (1×10r) = 10r

Runs scored in doubles = (2×5r) = 10r

Runs scored in triples = (3×6r) = 18r

So,

10r + 10r + 18r = 38

38r = 38

r = 1

Hence,runs scored by B in match 3 in doubles = 10r = (10×1) = 10.

16. Questions

Runs scored by A in match 3 = 75

So, runs scored by B in match 3 = 120% of 75 = 90

Therefore, runs scored by C in match 3 = (90-45) = 45

Ratio of runs scored by B to that of C in match 3 = 90:45 = 2:1

So,

(N/10):1 = 2:1

N/10 = 2

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Therefore, N = (10×2) = 20

Now,

Runs scored by C in match 2 = {(45×7}/15} = 21

So, runs scored by B in match 2 = {(21×2)/3} = 14

Runs scored by A in match 2 = 35

Now,

Percentage of runs scored by B with respect to that of A in match 2 = {(14/35)×100}% = 40%

So,

(K-5)% = 40%

K = (40+5)

Therefore, K = 45

Now,

Runs scored by B in match 1 = (90/2) = 45

Percentage of runs scored by B with respect to that of A in match 1 = 75%

75% = 45

1% = (45/75) = 3/5

100% = {(3×100)/5} = 60

So, runs scored by A in match 1 = 60

Therefore,

(L+5) = 60

L = (60-5) = 55

Hence, L = 55

Now, runs scored by C in match 1 = {(60×100)/120} = 50

Ratio of runs scored by B to that of C in match 1 = 45:50 = 9:10

So,

9:(M-N) = 9:10

(M-N) = 10

(M-20) = 10

M = (10+20)

M = 30

Now,

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Match 4

Runs scored by A = 2N = 40

Runs scored by B = (4M+5)% of 40 = 125% of 40 = 50

So, runs scored by C = {(50×4)/5} = 40

Match 5

Runs scored by A = (K+L) = (45+55) = 100

Runs scored by B = 80% of 100 = 80

So, runs scored by C = {(80×1)/4} = 20

Match	Runs scored by A	Runs scored by B	Runs scored by C	Total runs scored (A+B+C)
1	60	45	50	155
2	35	14	21	70
3	75	90	45	210
4	40	50	40	130
5	100	80	20	200
Total	310	279	176	765

Answer: D

D in match 1,

Runs scored = 150% of 50 = 75

Number of balls faced = X

Strike rate = (2X+5)

So,

(75/X)×100 = (2X+5)

$2X^2 + 5X = 7500$

$2X^2 + 5X - 7500 = 0$

$2X^2 + 125X - 120X - 7500 = 0$

2X(X+62.50) – 120(X+62.50) = 0

(2X-120)(X+62.5) = 0

X = 60 or -62.50

So, X = 60 (positive integer)

Now,

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S = Number of balls faced by D in match 1 = X = 60

T = Runs scored by A in match 1 = 60

So,

$\sqrt{(S \times T)} = \sqrt{(60 \times 60)} = \sqrt{3600} = 60.$

Therefore, 60 = S = T

Hence, both options A and B follows. So, option D is the correct answer.

17. Questions

Runs scored by A in match 3 = 75

So, runs scored by B in match 3 = 120% of 75 = 90

Therefore, runs scored by C in match 3 = (90-45) = 45

Ratio of runs scored by B to that of C in match 3 = 90:45 = 2:1

So,

(N/10):1 = 2:1

N/10 = 2

Therefore, N = (10×2) = 20

Now,

Runs scored by C in match 2 = {(45×7}/15} = 21

So, runs scored by B in match 2 = {(21×2)/3} = 14

Runs scored by A in match 2 = 35

Now,

Percentage of runs scored by B with respect to that of A in match 2 = {(14/35)×100}% = 40%

So,

(K-5)% = 40%

K = (40+5)

Therefore, K = 45

Now,

Runs scored by B in match 1 = (90/2) = 45

Percentage of runs scored by B with respect to that of A in match 1 = 75%

75% = 45

1% = (45/75) = 3/5

100% = {(3×100)/5} = 60

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So, runs scored by A in match 1 = 60

Therefore,

(L+5) = 60

L = (60-5) = 55

Hence, L = 55

Now, runs scored by C in match 1 = {(60×100)/120} = 50

Ratio of runs scored by B to that of C in match 1 = 45:50 = 9:10

So,

9:(M-N) = 9:10

(M-N) = 10

(M-20) = 10

M = (10+20)

M = 30

Now,

Match 4

Runs scored by A = 2N = 40

Runs scored by B = (4M+5)% of 40 = 125% of 40 = 50

So, runs scored by C = {(50×4)/5} = 40

Match 5

Runs scored by A = (K+L) = (45+55) = 100

Runs scored by B = 80% of 100 = 80

So, runs scored by C = {(80×1)/4} = 20

Match	Runs scored by A	Runs scored by B	Runs scored by C	Total runs scored (A+B+C)
1	60	45	50	155
2	35	14	21	70
3	75	90	45	210
4	40	50	40	130
5	100	80	20	200
Total	310	279	176	765

Answer: C

In match 6,

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Total runs scored (A+B+C) = (200-50) = 150

Ratio of runs scored by A to that of C = 7:5

And, the ratio of runs scored by A to that of B = 7:3 (Fractional value of 133.33% = 4/3)

So, ratio of runs scored by,

C : A : B

5 : 7

7 : 3

35 : 49 : 21

Therefore, the ratio of runs scored by A to that of B to that of C in match 6 = A:B:C = 49:21:35 = 7:3:5.

So, runs scored by B in match 6 = 3/15 of 150 = 1/5 of 150 = 30

And, runs scored by B in match 2 = 14

Hence, required difference = (30-14) = 16.

18. Questions

From the pie chart, we have,

Total distribution of the pie chart = 100%

(2X+4) + (3Y-5) + (Y+1) + (X+15) = 100

3X + 4Y = 85……(i)

From the given equation, we have,

$V^2 - 45V + 450 = 0$

$V^2 - 30V - 15V + 450 = 0$

V(V-30) – 15(V-30) = 0

(V-15) (V-30) = 0

V = 15 or 30

So, the roots are 15 and 30

A > X (given)

Therefore, A = 30

And, X = 15

Now, put the value of X in equation (i), we get,

3X + 4Y = 85

45 + 4Y = 85

4Y = 40

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Y = 10

Now, we can find the value of B,

B = 250% of Y

B = 250% of 10

B = 25

Therefore, B = 25

Now,

Percentage of the total number of items (cakes + pastries) ordered by shop R = (Y+1)% = (10+1)% = 11%

Total number of items (cakes + pastries) ordered by shop R = 55

So,

11% = 55

1% = (55/11) = 5

100% = (5×100) = 500

Therefore, total number of items (cakes + pastries) ordered by all the four shops together = 500

Shop P

Total number of items (cakes + pastries) ordered = (2X+4)% of 500 = 34% of 500 = 170

Difference between the number of cakes and pastries ordered (cakes-pastries) = (2A-10) = (60-10) = 50

Let, the number of cakes and pastries ordered be c and (c-50) respectively.

2c – 50 = 170

2c = 220

c = 110

So, the number of cakes ordered = c = 110

And, the number of pastries ordered = (c-50) = (110-50) = 60

Shop Q

Total number of items (cakes + pastries) ordered = (3Y-5)% of 500 = 25% of 500 = 125

Difference between the number of cakes and pastries ordered (cakes-pastries) = (B+20) = (25+20) = 45

Let, the number of cakes and pastries ordered be d and (d-45) respectively.

2d – 45 = 125

2d = 170

d = 85

So, the number of cakes ordered = d = 85

And, the number of pastries ordered = (d-45) = (85-45) = 40

Shop R

Total number of items (cakes + pastries) ordered = (Y+1)% of 500 = 11% of 500 = 55

Difference between the number of cakes and pastries ordered (cakes-pastries) = (A-15) = (30-15) = 15

Let, the number of cakes and pastries ordered be e and (e-15) respectively.

2e – 15 = 55

2e = 70

e = 35

So, the number of cakes ordered = e = 35

And, the number of pastries ordered = (e-15) = (35-15) = 20

Shop S

Total number of items (cakes + pastries) ordered = (X+15)% of 500 = 30% of 500 = 150

Difference between the number of cakes and pastries ordered (cakes-pastries) = (3B-45) = (75-45) = 30

Let, the number of cakes and pastries ordered be f and (f-30) respectively.

2f – 30 = 150

2f = 180

f = 90

So, the number of cakes ordered = f = 90

And, the number of pastries ordered = (f-30) = (90-30) = 60

Shop	Total number of items ordered	Number of cakes ordered	Number of pastries ordered
P	170	110	60
Q	125	85	40
R	55	35	20
S	150	90	60
Total	500	320	180

Answer: C

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According to the question,

Shop T

Fractional value of 71.42% = 5/7

Let, the total number of items (cakes + pastries) ordered be 700n.

So, the total number of items (cakes + pastries) delivered = 5/7 of 700n = 500n

And, the total number of items (cakes + pastries) not delivered = 2/7 of 700n = 200n

Number of items (cakes + pastries) returned = 10% of 500n = 50n

Number of items (cakes + pastries) accepted = 90% of 500n = 450n

So,

450n – 200n = 50

250n = 50

n = (50/250) = 1/5

Hence, thetotal number of items (cakes + pastries) ordered by shop T = 700n = {(700×1)/5} = 140.

19. Questions

From the pie chart, we have,

Total distribution of the pie chart = 100%

(2X+4) + (3Y-5) + (Y+1) + (X+15) = 100

3X + 4Y = 85……(i)

From the given equation, we have,

$V^2 – 45V + 450 = 0$

$V^2 – 30V – 15V + 450 = 0$

V(V-30) – 15(V-30) = 0

(V-15) (V-30) = 0

V = 15 or 30

So, the roots are 15 and 30

A > X (given)

Therefore, A = 30

And, X = 15

Now, put the value of X in equation (i), we get,

3X + 4Y = 85

45 + 4Y = 85

4Y = 40

Y = 10

Now, we can find the value of B,

B = 250% of Y

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B = 250% of 10

B = 25

Therefore, B = 25

Now,

Percentage of the total number of items (cakes + pastries) ordered by shop R = (Y+1)% = (10+1)% = 11%

Total number of items (cakes + pastries) ordered by shop R = 55

So,

11% = 55

1% = (55/11) = 5

100% = (5×100) = 500

Therefore, total number of items (cakes + pastries) ordered by all the four shops together = 500

Shop P

Total number of items (cakes + pastries) ordered = (2X+4)% of 500 = 34% of 500 = 170

Difference between the number of cakes and pastries ordered (cakes-pastries) = (2A-10) = (60-10) = 50

Let, the number of cakes and pastries ordered be c and (c-50) respectively.

2c – 50 = 170

2c = 220

c = 110

So, the number of cakes ordered = c = 110

And, the number of pastries ordered = (c-50) = (110-50) = 60

Shop Q

Total number of items (cakes + pastries) ordered = (3Y-5)% of 500 = 25% of 500 = 125

Difference between the number of cakes and pastries ordered (cakes-pastries) = (B+20) = (25+20) = 45

Let, the number of cakes and pastries ordered be d and (d-45) respectively.

2d – 45 = 125

2d = 170

d = 85

So, the number of cakes ordered = d = 85

And, the number of pastries ordered = (d-45) = (85-45) = 40

Shop R

Total number of items (cakes + pastries) ordered = (Y+1)% of 500 = 11% of 500 = 55

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Difference between the number of cakes and pastries ordered (cakes-pastries) = (A-15) = (30-15) = 15

Let, the number of cakes and pastries ordered be e and (e-15) respectively.

2e – 15 = 55

2e = 70

e = 35

So, the number of cakes ordered = e = 35

And, the number of pastries ordered = (e-15) = (35-15) = 20

Shop S

Total number of items (cakes + pastries) ordered = (X+15)% of 500 = 30% of 500 = 150

Difference between the number of cakes and pastries ordered (cakes-pastries) = (3B-45) = (75-45) = 30

Let, the number of cakes and pastries ordered be f and (f-30) respectively.

2f – 30 = 150

2f = 180

f = 90

So, the number of cakes ordered = f = 90

And, the number of pastries ordered = (f-30) = (90-30) = 60

Shop	Total number of items ordered	Number of cakes ordered	Number of pastries ordered
P	170	110	60
Q	125	85	40
R	55	35	20
S	150	90	60
Total	500	320	180

Answer: A

Shop P

Let, the number of coconut cakes and coconut pastries ordered be J each.

Total number of cakes ordered = 110

Number of banana cakes ordered = K

Number of coconut cakes ordered = J

Total number of pastries ordered = 60

Number of banana pastries ordered = L

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Number of coconut pastries ordered = J

Now,

K + J = 110……..(i)

L + J = 60……..(ii)

Solving both the equations, we get,

K – L = 50

So, the difference between the number of banana cakes and banana pastries ordered = (K-L) = 50

Hence, twice of the difference between the number of banana cakes and banana pastries ordered = {2×(K-L)} = (2×50) = 100.

20. Questions

From the pie chart, we have,

Total distribution of the pie chart = 100%

(2X+4) + (3Y-5) + (Y+1) + (X+15) = 100

3X + 4Y = 85……(i)

From the given equation, we have,

$V^2 – 45V + 450 = 0$

$V^2 – 30V – 15V + 450 = 0$

V(V-30) – 15(V-30) = 0

(V-15) (V-30) = 0

V = 15 or 30

So, the roots are 15 and 30

A > X (given)

Therefore, A = 30

And, X = 15

Now, put the value of X in equation (i), we get,

3X + 4Y = 85

45 + 4Y = 85

4Y = 40

Y = 10

Now, we can find the value of B,

B = 250% of Y

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B = 250% of 10

B = 25

Therefore, B = 25

Now,

Percentage of the total number of items (cakes + pastries) ordered by shop R = (Y+1)% = (10+1)% = 11%

Total number of items (cakes + pastries) ordered by shop R = 55

So,

11% = 55

1% = (55/11) = 5

100% = (5×100) = 500

Therefore, total number of items (cakes + pastries) ordered by all the four shops together = 500

Shop P

Total number of items (cakes + pastries) ordered = (2X+4)% of 500 = 34% of 500 = 170

Difference between the number of cakes and pastries ordered (cakes-pastries) = (2A-10) = (60-10) = 50

Let, the number of cakes and pastries ordered be c and (c-50) respectively.

2c – 50 = 170

2c = 220

c = 110

So, the number of cakes ordered = c = 110

And, the number of pastries ordered = (c-50) = (110-50) = 60

Shop Q

Total number of items (cakes + pastries) ordered = (3Y-5)% of 500 = 25% of 500 = 125

Difference between the number of cakes and pastries ordered (cakes-pastries) = (B+20) = (25+20) = 45

Let, the number of cakes and pastries ordered be d and (d-45) respectively.

2d – 45 = 125

2d = 170

d = 85

So, the number of cakes ordered = d = 85

And, the number of pastries ordered = (d-45) = (85-45) = 40

Shop R

Total number of items (cakes + pastries) ordered = (Y+1)% of 500 = 11% of 500 = 55

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Difference between the number of cakes and pastries ordered (cakes-pastries) = (A-15) = (30-15) = 15

Let, the number of cakes and pastries ordered be e and (e-15) respectively.

2e – 15 = 55

2e = 70

e = 35

So, the number of cakes ordered = e = 35

And, the number of pastries ordered = (e-15) = (35-15) = 20

Shop S

Total number of items (cakes + pastries) ordered = (X+15)% of 500 = 30% of 500 = 150

Difference between the number of cakes and pastries ordered (cakes-pastries) = (3B-45) = (75-45) = 30

Let, the number of cakes and pastries ordered be f and (f-30) respectively.

2f – 30 = 150

2f = 180

f = 90

So, the number of cakes ordered = f = 90

And, the number of pastries ordered = (f-30) = (90-30) = 60

Shop	Total number of items ordered	Number of cakes ordered	Number of pastries ordered
P	170	110	60
Q	125	85	40
R	55	35	20
S	150	90	60
Total	500	320	180

Answer: B

Shop S

Total number of cakes ordered = 90

Number of cakes delivered = (W-10)% of 90 = (0.9W-9)

Number of cakes not delivered = {90-(0.9W-9)} = (99-0.9W)

Total number of pastries ordered = 60

Number of pastries delivered = (W+15)% of 60 = (0.6W+9)

Number of pastries not delivered = {60-(0.6W+9)} = (51-0.6W)

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Now,

(99-0.9W)/(51-0.6W) = 3/1

99 – 0.9W = 153 – 0.18W

0.9W = 54

W = (54/0.9) = 60

Now,

$Z^2 = W + 21$

$Z^2 = 60 + 21$

$Z^2 = 81$

Z = 9

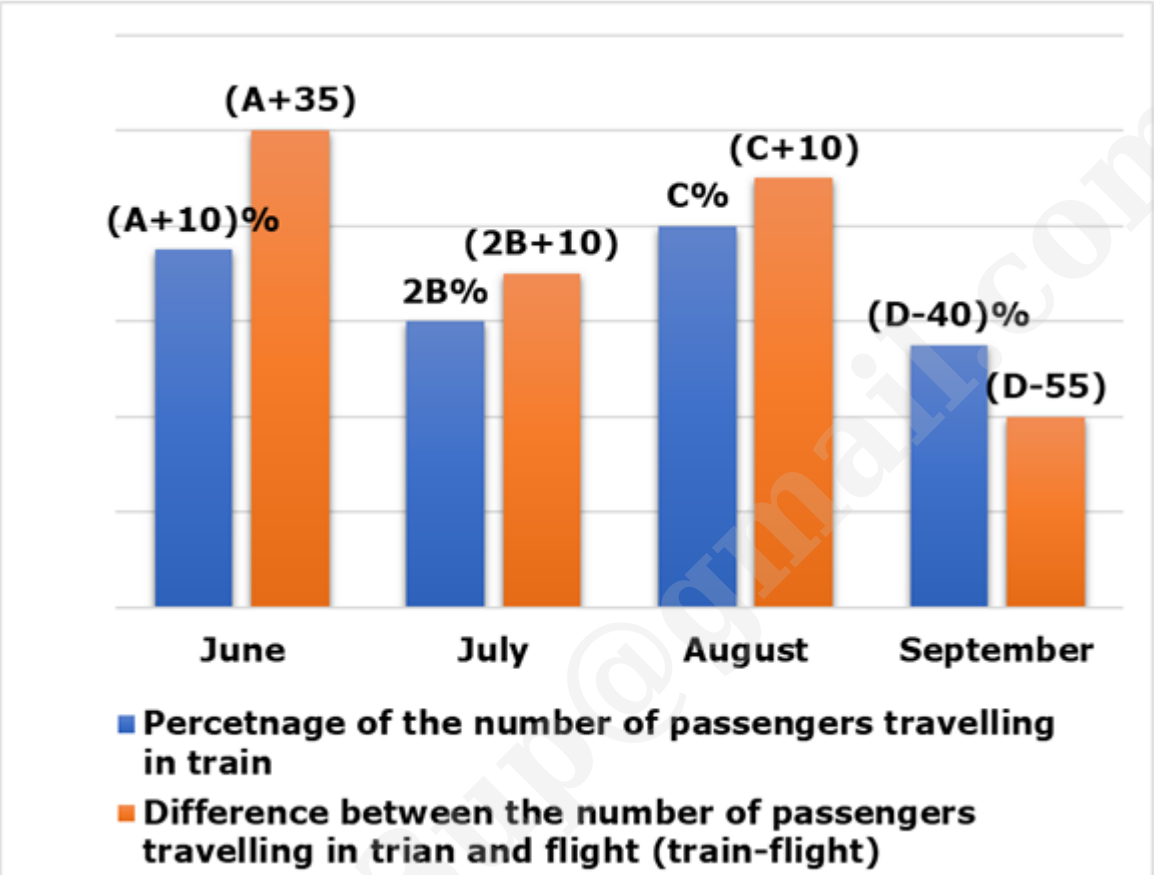
Hence, value of Z = 9.

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1. Questions

Read the following information carefully and answer the questions based on it.

The bar graph given below shows the percentage of passengers traveling by train out of the total number of passengers traveling by train and flight together in each of the four different months, i.e., June, July, August, and September of the year 2023. It also shows the difference between the number of passengers traveling by train and flight (train-flight) in each of the four months of the year 2023.



Note:

The average of the total number of passengers traveling by train and flight together in all four months of 2023 is 275.

The total number of passengers traveling by train and flight together in June 2023 is 200.

In July 2023, the total number of passengers traveling by train and flight together was 75% more than that of the previous month.

The ratio of the total number of passengers traveling by train and flight together in July 2023 to that of August 2023 is 7:3.

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In each month, the number of passengers traveling by train is more than that of flight.

The total number of passengers traveling in any month = number of passengers traveling by train + number of passengers traveling by flight

In September 2023, the ratio of the number of male passengers traveling by train to that of female passengers traveling by flight will be 3:2. The number of female passengers traveling in trains is

12.50% less than the number of male passengers traveling in flights.

Find the correct relation between M and N.

M = Average of the total number of male passengers traveling in train and flight together

N = The sum of the number of female passengers traveling in train and flight together.

a. M > N

b. M < N

c. M = N

d. M ≥ N

e. M ≤ N

2. Questions

In July 2023, out of the total number of flight tickets booked by the passengers, some passengers did not board the flight, but the amount of the price of the tickets was not refunded to those passengers. If the total number of passengers who are traveling (boarded) in the flight is 70% of the total number of booked tickets, then find the total revenue generated from those passengers who did not board the flight (a non-refundable amount) if the price of each flight ticket is Rs. 500 in July 2023.

a. Rs. 30000

b. Rs. 24000

c. Rs. 48000

d. Cannot be determined

e. None of these

3. Questions

In June 2023, there will be some local and foreign passengers traveling by train and flight. If the difference between the local and foreign passengers traveling by train is equal to the number of local passengers traveling by flight, and the ratio of the number of local passengers traveling by train to that of foreign passengers traveling by flight is 9:2, If the number of local passengers traveling by train is greater than that of foreign passengers traveling by train, then find the difference between the number of foreign passengers traveling by train and the number of the number of local passengers traveling by flight.

a. 32

b. 29

c. 30

d. 21

e. None of these

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4. Questions

If there is another month in October 2023, if the number of passengers traveling in train K% more than that of flight, which is 8K, and the total number of passengers traveling in train and flight together is 352, Find the value of (K2–3K). (K is a positive integer.)

a. 440

b. 280

c. 160

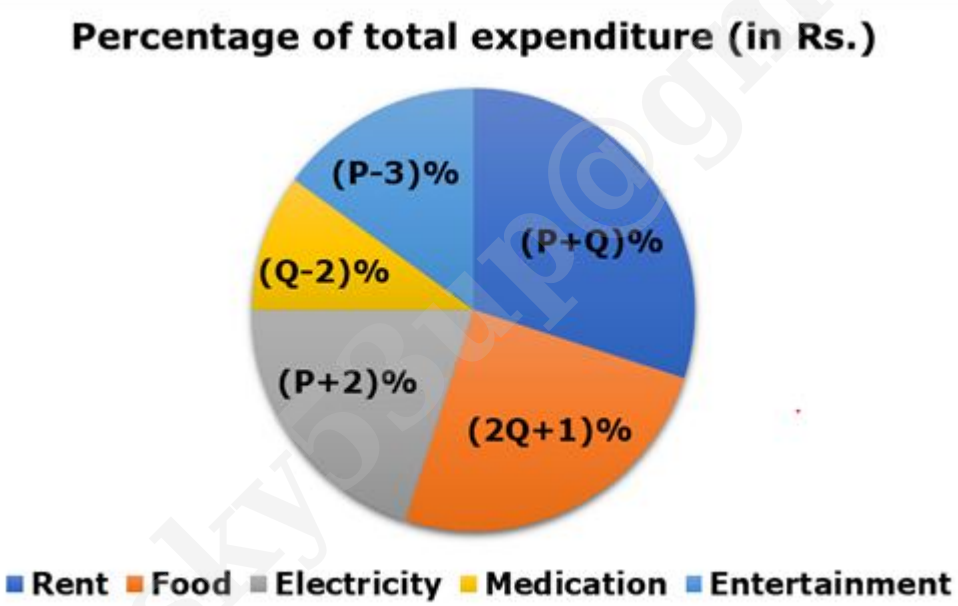
d. 320

e. None of these

5. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total expenditure (in Rs.) of a person on five different necessities, i.e., rent, food, electricity, medication, and entertainment, in the month of April 2024.



Note:

In April 2024, the expenditure on rent is 19.20% of the total income (expenditure + savings) of the person.

Out of his total income, a person saves 36%, and the rest are all his expenditures.

The difference between the expenditure on medication and entertainment is Rs. 4000.

The total income = total expenditure + total savings

The total expenditure = amount spent on rent + food + electricity + medication + entertainment

Out of his total savings in April 2024, a person kept 1/3rd of the amount with himself at home, and the remaining amount he deposited in the bank, which offers compound interest at the rate of R%

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per annum, and received a total amount of Rs. 46128 after 2 years. Find the simple interest earned if he would have invested the same amount at the rate of (R-4)% for 3 years. (The value of R should be a positive integer.)

a. Rs. 15000

b. Rs. 18000

c. Rs. 12000

d. Rs. 10000

e. None of these

6. Questions

In March 2024, if the percentage distribution of expenditure on each necessity was the same as that of April 2024, the person would save 10% of his income in March 2024 and spend the rest of the amount. In April 2024, if a person got a salary (income) hike of 25% as compared to March 2024, then find the difference between the amount spent by the person on food in the months of March and April 2024.

a. Rs. 1500

b. Rs. 2000

c. Rs. 3000

d. Rs. 2500

e. None of these

7. Questions

If there is another person who spends 12% of his total expenditure on electricity in the month of April 2024, His expenditure and savings are Rs. 400N and Rs. 500M, respectively, whereas his expenditure on electricity is Rs. 7800 less than that of his total savings. Find the value of √(N-M-20) if his income is Rs. 50000 less than that of the present person.

a. 10

b. 15

c. 5

d. 8

e. None of these

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8. Questions

Out of the total expense on medication, a person has purchased two different types of medicines, i.e., generic medicines and normal medicines. If the price of each generic and normal medicine purchased by the person is Rs. 100 and Rs. 200 and the person purchased a total of 48 medicines (generic and normal), then find the total amount spent on the purchase of generic medicines.

a. Rs. 8000

b. Rs. 6400

c. Rs. 1600

d. Rs. 2000

e. None of these

9. Questions

Read the following information carefully and answer the questions based on it.

There are three different bakery shops, X, Y, and Z, that bake cookies of two different types, i.e., chocolate cookies and sugar cookies. Out of the total number of cookies (chocolate + sugar) baked, some are sold, while the rest are unsold.

Shop X: Out of the total number of cookies baked, the number of chocolate and sugar cookies are (P+30) and (Q+20) respectively. Ratio of the total number of cookies (chocolate + sugar) sold to that of unsold is 4:1. The total number of cookies (chocolate + sugar) baked are 250.

Shop Y: The total number of cookies baked is P+2Q, while the number of chocolate and sugar cookies baked is 210 and 70, respectively. The total number of cookies (chocolate and sugar) sold is 200 more than the number of unsold cookies in the same shop.

Shop Z: The total number of cookies sold (chocolate + sugar) is equal to that of shop Y, and the total number of chocolate cookies baked is 2/3 of the total number of cookies (chocolate + sugar) baked. The total number of cookies sold (chocolate + sugar) is 80% of the total number of cookies baked (chocolate + sugar).

Note:

The total number of cookies baked = number of chocolate cookies baked + number of sugar cookies baked

The total number of cookies baked = number of cookies sold + number of cookies unsold

In shop X, out of the total number of chocolate cookies baked, the number of sold and unsold cookies is 2A and (B+10), respectively. Whereas, out of the total number of sugar cookies baked, the number of sold and unsold cookies is 4B and (A-40), respectively. Find the value of √ (A-B-4).

a. (A/10)

b. (B-14)

c. {(A×B)/200}

d. All of the above

e. None of these

10. Questions

In shop Y, out of the total number of cookies sold, the ratio of the number of chocolate cookies to

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the number of sugar cookies sold is 2:1. If the price of each sugar cookie sold is 40% less than that of each chocolate cookie sold and the total revenue earned from selling all the chocolate cookies is Rs. 560 more than that of sugar cookies, then find the difference between the price of each chocolate and each sugar cookie sold.

a. 2

b. 5

c. 4

d. 8

e. None of these

11. Questions

If there is another type of cookie baked by shop Z, i.e., peanut cookies, the difference between the number of peanut cookies sold and unsold by shop Z is T% of the total number of peanut cookies (sold and unsold) baked, which is 150. The number of peanut cookies unsold is 3T. Find the ratio of the number of peanut cookies sold to those unsold by shop Z. (The number of peanut cookies sold is more than the number of unsold ones.)

a. 2:3

b. 3:2

c. 4:1

d. 5:1

e. None of these

12. Questions

If there is another shop W, where the number of chocolate cookies baked is 150 and the product of the number of chocolate and sugar cookies baked is 18000. Find the total number of cookies (chocolate + sugar) baked by the shop W, if the sugar cookies baked is (U^2-1).

a. 300

b. 180

c. 270

d. Cannot be determined

e. None of these

13. Questions

The quantity of milk and water in container M are [3X] litters and [X] liters respectively. Whereas, the quantity of milk and water in container N are [Y+16] litters and [Y] liters respectively. If 50% and 25% from container M and container N respectively taken out of and poured into empty

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container K, then the resultant quantity of container K becomes equal to that of initial quantity of container N. If the value of [3X-Y] is 58, then find the ratio of the initial quantity of milk in container M to that of container N.

a. 12:1

b. 15:8

c. 16:7

d. 14:3

e. None of these

14. Questions

The work done by 10 men in 10 days is equal to the work done by 25 women in 8 days. If the 5 men and 4 women started working for A days, Then 2 men left and 2 new women joined the work, and together they completed the remaining work in (A–5) days. Find the time taken by (A+6) boys to complete the whole work if the efficiency of each boy is half that of each woman.

a. 25 days

b. 22 days

c. 26 days

d. 21 days

e. None of these

15. Questions

A man invested Rs. 20000 in scheme P at the rate of R% per annum simple interest and received Rs. 12000 as interest after (R-11) years. Find the compound interest earned if he had invested Rs. 1000R in scheme Q at the rate of (R-5)% per annum compounded annually after 2 years. (The value of R should be a positive integer)

a. Rs. 2275

b. Rs. 2925

c. Rs. 3150

d. Rs. 3360

e. None of these

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16. Questions

There are two boats, A and B, traveling in the same river. The speed of the river is 12 km/h, and the ratio of the speed of boat A to that of boat B in still water is 6:5. Boat A can cover the (D+90) km downstream distance in the same time as the time taken by boat B to cover the (D-60) km upstream distance. If boat B can cover 420 km of downstream distance in 10 hours, then find the time taken by boat A to cover (D+42) km of upstream distance.

a. 10 hours

b. 8 hours

c. 5 hours

d. 12 hours

e. None of these

17. Questions

For the questions given below, three statements I, II and III have been given. Analyse and answer whether the data provided in the three statements are sufficient to answer the question or not.

The train A can cross a platform of length P meters in 4 seconds at a speed of 60 m/sec. The time taken by train B to cross the same platform is 1.50 seconds more than that of train A. Find the value of √(P+1).

Statement I: The ratio of the speed of train A to that of train B is 3:2.

Statement II: The length of train B is 20 meters less than that of train A.

Statement III: The train B can crosses a pole in 3.50 seconds.

For the questions given below, three statements I, II & III have been given. Analyse and answer whether the data provided in the three statements are sufficient to answer the question or not.

a. Only statement II alone is sufficient, while statements I and III are not sufficient

b. Both statements I and III together are necessary, while statement II alone is not sufficient

c. Either statements I, II, or III alone are sufficient

d. All the statements I, II, and III together are necessary

e. Neither statement I, II, nor III is sufficient

18. Questions

X started a business with an investment of Rs. 12,000; after 5 months, Y joined him with a certain amount of investment. After some more months, Z also joined the business with 25% more investment than that of X. If the total profit at the end of the year is Rs. 274000, then find the profit share of Y.

Statement I: Investment of Y is 2/3rd of that of Z.

Statement II: Z invested his money for 8 months less than that of X.

Statement III: Profit share of Z is 14.28% less than that of Y.

a. Only statement II alone is sufficient, while statements I and III are not sufficient

b. Both statements I and III together are necessary, while statement II alone is not sufficient

c. Either statement I and II together or I and III together are sufficient

d. All the statements I, II and III together are necessary

e. Neither statement I or II nor III is sufficient

19. Questions

Pipe P and pipe Q together can fill the tank in (24/5) minutes. Whereas, pipe Q and pipe R can fill the same tank in (40/7) minutes. Find the time taken by pipe P and pipe R together to fill the tank.

Statement I: Pipe Q can alone fill 25% of the tank in 2 minutes.

Statement II: Efficiency of pipe Q is 9 more than that of pipe R.

Statement III: Ratio of the efficiency of pipe P to that of pipe Q is 2:3.

a. Only statement II alone is sufficient, while statements I and III are not sufficient

b. Both statements I and III together are necessary, while statement II alone is not sufficient

c. Either statements I, II or III alone is sufficient

d. All the statements I, II and III together are necessary

e. Neither statement I or II nor III is sufficient

20. Questions

The ratio of the number of boys and girls in class A is 3:1. Whereas, the number of boys in class B is 50% more than that of girls in the same class. Find the total number of students (boys + girls) in both the classes together.

Statement I: The total number of boys in both schools together is 240.

Statement II: The number of girls in class A is 10 less than that of class B.

Statement III: Product of the number of boys in class A to that of girls in class B is 9000.

a. Only statement II alone is sufficient, while statements I and III are not sufficient

b. Both statements I and II together are necessary, while statement III alone is not sufficient

c. Either statements I, II or III alone is sufficient

d. All the statements I, II and III together are necessary

e. Neither statement I or II nor III is sufficient

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Explanations:

1. Questions

The total number of passengers travelling in train and flight together in all the given four months of 2023 = (275×4) = 1100

June 2023

The total number of passengers traveling by train and flight together = 200

The percentage of the number of passengers traveling by train = (A+10)%

So, the percentage of the number of passengers traveling flight = {100-(A+10)}% = (90-A)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {(A+10)-(90-A)}% = (2A-80)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (A+35)

So,

(2A-80)% of 200 = (A+35)

4A – 160 = A + 35

3A = 195

A = 65

So, the number of passengers traveling by train = (A+10)% of 200 = (65+10)% of 200 = 75% of 200 = 150

Therefore, the number of passengers traveling by flight = {150-(A+35)} = {150-(65+35)} = (150-100) = 50

July 2023

The total number of passengers traveling by train and flight together = 175% of 200 = 350

The percentage of the number of passengers traveling by train = 2B%

So, the percentage of the number of passengers traveling flight = (100-2B)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {2B-(100-2B)}% = (4B-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (2B+10)

So,

(4B-100)% of 350 = (2B+10)

14B – 350 = 2B + 10

12B = 360

B = 30

So, the number of passengers traveling by train = 2B% of 350 = 60% of 350 = 210

Therefore, the number of passengers traveling by flight = {210-(2B+10)} = {210-(60+10)} = (210-70) = 140

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The total number of passengers traveling by train and flight together = {(350/7)×3} = 150

The percentage of the number of passengers traveling by train = C%

So, the percentage of the number of passengers traveling flight = (100-C)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {C-(100-C)}% = (2C-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (C+10)

So,

(2C-100)% of 150 = (C+10)

3C – 150 = C + 10

2C = 160

C = 80

So, the number of passengers traveling by train = C% of 150 = 80% of 150 = 120

Therefore, the number of passengers traveling by flight = {120-(C+10)} = {120-(80+10)} = (120-90) = 30

September 2023

The total number of passengers traveling by train and flight together = {1100-(200+350+150)} = (1100-700) = 400

The percentage of the number of passengers traveling by train = (D-40)%

So, the percentage of the number of passengers traveling flight = {100-(D-40)}% = (140-D)%

The difference between the percentage of the number of passengers travelling by train and flight (train-flight) = {(D-40)-(140-D)}% = (2D-180)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (D-55)

So,

(2D-180)% of 400 = (D-55)

8D – 720 = D – 55

7D = 665

D = 95

So, the number of passengers traveling by train = (D-40)% of 400 = (95-40)% of 400 = 55% of 400 = 220

Therefore, the number of passengers traveling by flight = {220-(D-55)} = {220-(95-55)} = (220-40) = 180

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Months	Total number of passengers traveling (train + flight)	Number of passengers traveling in train	Number of passengers traveling in flight
June	200	150	50
July	350	210	140
August	150	120	30
September	400	220	180
Total	1100	700	400

Answer: B

According to the question,

September 2023,

The total number of passengers traveling by train = 220

The total number of passengers traveling by flight = 180

Let, the number of male passengers traveling by train and female passengers traveling by flight be 3p and 2p respectively.

Also, the ratio of the number of female passengers traveling by train to that of male passengers traveling by flight = 7:8. Let it be 7q and 8q respectively. (Fractional value of 12.50% = 1/8)

Now,

3p + 7q = 220……(i)

2p + 8q = 180……(ii)

Solving both the equations, we get,

6p + 14q = 440

6p + 24q = 540

10q = 100

q = 10

Now, put the value of q in equation (i), we get,

3p + 7q = 220

3p + 70 = 220

3p = 150

p = 50

M: The number of male passengers traveling by train = 3p = 150

And, the number of male passengers traveling by flight = 8q = 80

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Hence, the average of the total number of male passengers traveling by train and flight together = {(150/80)/2} = (230/2) = 115.

So, M = 115

N: The number female passengers traveling by train = 7q = 70

And, the number of female passengers traveling by flight = 2q = 100

Hence, the sum of the number of female passengers traveling by train and flight together = (70+100) = 170.

So, N = 170

Hence, M < N.

2. Questions

The total number of passengers travelling in train and flight together in all the given four months of 2023 = (275×4) = 1100

June 2023

The total number of passengers traveling by train and flight together = 200

The percentage of the number of passengers traveling by train = (A+10)%

So, the percentage of the number of passengers traveling flight = {100-(A+10)}% = (90-A)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {(A+10)-(90-A)}% = (2A-80)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (A+35)

So,

(2A-80)% of 200 = (A+35)

4A – 160 = A + 35

3A = 195

A = 65

So, the number of passengers traveling by train = (A+10)% of 200 = (65+10)% of 200 = 75% of 200 = 150

Therefore, the number of passengers traveling by flight = {150-(A+35)} = {150-(65+35)} = (150-100) = 50

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July 2023

The total number of passengers traveling by train and flight together = 175% of 200 = 350

The percentage of the number of passengers traveling by train = 2B%

So, the percentage of the number of passengers traveling flight = (100-2B)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {2B-(100-2B)}% = (4B-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (2B+10)

So,

(4B-100)% of 350 = (2B+10)

14B – 350 = 2B + 10

12B = 360

B = 30

So, the number of passengers traveling by train = 2B% of 350 = 60% of 350 = 210

Therefore, the number of passengers traveling by flight = {210-(2B+10)} = {210-(60+10)} = (210-70) = 140

August 2023

The total number of passengers traveling by train and flight together = {(350/7)×3} = 150

The percentage of the number of passengers traveling by train = C%

So, the percentage of the number of passengers traveling flight = (100-C)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {C-(100-C)}% = (2C-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (C+10)

So,

(2C-100)% of 150 = (C+10)

3C – 150 = C + 10

2C = 160

C = 80

So, the number of passengers traveling by train = C% of 150 = 80% of 150 = 120

Therefore, the number of passengers traveling by flight = {120-(C+10)} = {120-(80+10)} = (120-90) = 30

September 2023

The total number of passengers traveling by train and flight together = {1100-(200+350+150)} = (1100-700) = 400

The percentage of the number of passengers traveling by train = (D-40)%

So, the percentage of the number of passengers traveling flight = {100-(D-40)}% = (140-D)%

The difference between the percentage of the number of passengers travelling by train and flight (train-flight) = {(D-40)-(140-D)}% = (2D-180)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (D-55)

So,

(2D-180)% of 400 = (D-55)

8D – 720 = D – 55

7D = 665

D = 95

So, the number of passengers traveling by train = (D-40)% of 400 = (95-40)% of 400 = 55% of 400 = 220

Therefore, the number of passengers traveling by flight = {220-(D-55)} = {220-(95-55)} = (220-40) = 180

Months	Total number of passengers traveling (train + flight)	Number of passengers traveling in train	Number of passengers traveling in flight
June	200	150	50
July	350	210	140
August	150	120	30
September	400	220	180
Total	1100	700	400

Answer: A

July 2023 (Flight)

Let, the total number of passengers booked (boarded + not boarded) be 100%.

And, the percentage of the number of passengers traveling (boarded) the flight = 70%

The actual number of passengers traveling (boarded) = 140

70% = 140

1% = (140/70) = 2

100% = (100×2) = 200

Therefore, the total number of tickets booked (boarded + not boarded) = 200

So, the number of passengers did not boarded = (200-140) = 60

Hence, the total revenue generated from the passengers who did not board the flight in July 2023 = (60×500) = Rs. 30000.

3. Questions

The total number of passengers travelling in train and flight together in all the given four months of 2023 = (275×4) = 1100

June 2023

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The total number of passengers traveling by train and flight together = 200

The percentage of the number of passengers traveling by train = (A+10)%

So, the percentage of the number of passengers traveling flight = {100-(A+10)}% = (90-A)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {(A+10)-(90-A)}% = (2A-80)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (A+35)

So,

(2A-80)% of 200 = (A+35)

4A – 160 = A + 35

3A = 195

A = 65

So, the number of passengers traveling by train = (A+10)% of 200 = (65+10)% of 200 = 75% of 200 = 150

Therefore, the number of passengers traveling by flight = {150-(A+35)} = {150-(65+35)} = (150-100) = 50

July 2023

The total number of passengers traveling by train and flight together = 175% of 200 = 350

The percentage of the number of passengers traveling by train = 2B%

So, the percentage of the number of passengers traveling flight = (100-2B)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {2B-(100-2B)}% = (4B-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (2B+10)

So,

(4B-100)% of 350 = (2B+10)

14B – 350 = 2B + 10

12B = 360

B = 30

So, the number of passengers traveling by train = 2B% of 350 = 60% of 350 = 210

Therefore, the number of passengers traveling by flight = {210-(2B+10)} = {210-(60+10)} = (210-70) = 140

August 2023

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The total number of passengers traveling by train and flight together = {(350/7)×3} = 150

The percentage of the number of passengers traveling by train = C%

So, the percentage of the number of passengers traveling flight = (100-C)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {C-(100-C)}% = (2C-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (C+10)

So,

(2C-100)% of 150 = (C+10)

3C – 150 = C + 10

2C = 160

C = 80

So, the number of passengers traveling by train = C% of 150 = 80% of 150 = 120

Therefore, the number of passengers traveling by flight = {120-(C+10)} = {120-(80+10)} = (120-90) = 30

September 2023

The total number of passengers traveling by train and flight together = {1100-(200+350+150)} = (1100-700) = 400

The percentage of the number of passengers traveling by train = (D-40)%

So, the percentage of the number of passengers traveling flight = {100-(D-40)}% = (140-D)%

The difference between the percentage of the number of passengers travelling by train and flight (train-flight) = {(D-40)-(140-D)}% = (2D-180)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (D-55)

So,

(2D-180)% of 400 = (D-55)

8D – 720 = D – 55

7D = 665

D = 95

So, the number of passengers traveling by train = (D-40)% of 400 = (95-40)% of 400 = 55% of 400 = 220

Therefore, the number of passengers traveling by flight = {220-(D-55)} = {220-(95-55)} = (220-40) = 180

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Months	Total number of passengers traveling (train + flight)	Number of passengers traveling in train	Number of passengers traveling in flight
June	200	150	50
July	350	210	140
August	150	120	30
September	400	220	180
Total	1100	700	400

Answer: C

July 2023,

The total number of passengers traveling by train = 150

Let, the local passengers traveling by train be X. (local-foreigner)

Also, let the number of foreign passengers traveling by train be 150 – X

Now, the total number of passengers traveling by flight = 50

So, the number of local passengers traveling by the flight = Y

And, the number of foreign passengers traveling by the flight = (50-Y)

Now,

X – (150–X) = Y

2X – Y = 150…….(i)

Also,

X/(50-Y) = 9/2

2X = 450 – 9Y

2X + 9Y = 450…….(ii)

Solving both the equations, we get,

10Y = 300

Y = 30

Now, put the value of X in equation (i), we get,

2X – Y = 150

2X – 30 = 150

2X = 180

X = 90

So, the number of foreign passengers traveling by train = 150 – 90 = 60

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And, the number of local passengers traveling by flight = Y = 30

Hence, required difference = (60-30) = 30.

4. Questions

The total number of passengers travelling in train and flight together in all the given four months of 2023 = (275×4) = 1100

June 2023

The total number of passengers traveling by train and flight together = 200

The percentage of the number of passengers traveling by train = (A+10)%

So, the percentage of the number of passengers traveling flight = {100-(A+10)}% = (90-A)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {(A+10)-(90-A)}% = (2A-80)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (A+35)

So,

(2A-80)% of 200 = (A+35)

4A – 160 = A + 35

3A = 195

A = 65

So, the number of passengers traveling by train = (A+10)% of 200 = (65+10)% of 200 = 75% of 200 = 150

Therefore, the number of passengers traveling by flight = {150-(A+35)} = {150-(65+35)} = (150-100) = 50

July 2023

The total number of passengers traveling by train and flight together = 175% of 200 = 350

The percentage of the number of passengers traveling by train = 2B%

So, the percentage of the number of passengers traveling flight = (100-2B)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {2B-(100-2B)}% = (4B-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (2B+10)

So,

(4B-100)% of 350 = (2B+10)

14B – 350 = 2B + 10

12B = 360

B = 30

So, the number of passengers traveling by train = 2B% of 350 = 60% of 350 = 210

Therefore, the number of passengers traveling by flight = {210-(2B+10)} = {210-(60+10)} = (210-70) = 140

August 2023

The total number of passengers traveling by train and flight together = {(350/7)×3} = 150

The percentage of the number of passengers traveling by train = C%

So, the percentage of the number of passengers traveling flight = (100-C)%

The difference between the percentage of the number of passengers traveling by train and flight (train-flight) = {C-(100-C)}% = (2C-100)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (C+10)

So,

(2C-100)% of 150 = (C+10)

3C – 150 = C + 10

2C = 160

C = 80

So, the number of passengers traveling by train = C% of 150 = 80% of 150 = 120

Therefore, the number of passengers traveling by flight = {120-(C+10)} = {120-(80+10)} = (120-90) = 30

September 2023

The total number of passengers traveling by train and flight together = {1100-(200+350+150)} = (1100-700) = 400

The percentage of the number of passengers traveling by train = (D-40)%

So, the percentage of the number of passengers traveling flight = {100-(D-40)}% = (140-D)%

The difference between the percentage of the number of passengers travelling by train and flight (train-flight) = {(D-40)-(140-D)}% = (2D-180)%

The actual difference between the number of passengers traveling by train and flight (train-flight) = (D-55)

So,

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(2D-180)% of 400 = (D-55)

8D – 720 = D – 55

7D = 665

D = 95

So, the number of passengers traveling by train = (D-40)% of 400 = (95-40)% of 400 = 55% of 400 = 220

Therefore, the number of passengers traveling by flight = {220-(D-55)} = {220-(95-55)} = (220-40) = 180

Months	Total number of passengers traveling (train + flight)	Number of passengers traveling in train	Number of passengers traveling in flight
June	200	150	50
July	350	210	140
August	150	120	30
September	400	220	180
Total	1100	700	400

Answer: E

October 2023

The number of passengers traveling by flight = 8K

So, the number of passengers traveling by train = {8K+(K% of 8K)} = $(0.08K^2+8K)$

Now,

$8K + (0.08K^2+8K) = 352$

$0.08K^2 + 16K – 352 = 0$

$0.01K^2 + 2K – 44 = 0$

$K^2 + 200K – 4400 = 0$

$K^2 + 220K – 20K – 4400 = 0$

$K(K+220) – 20(K+220) = 0$

$(K-20) (K+220) = 0$

$K = 20$ or -220

So, K = 20 (positive integer)

Hence, the value of $(K^2-3K) = (20^2-60) = (400-60) = 340$.

5. Questions

From the pie chart, we have,

The total value of the pie chart = 100%

$(P+Q) + (2Q+1) + (P+2) + (Q-2) + (P-3) = 100$

$3P + 4Q = 102$……(i)

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Now,

Let, the income of the person be Rs. 100K

So, the total savings of the person = 36% of 100K = Rs. 36K

Therefore, the total expenditure of the person = 64% of 100K = Rs. 64K

Now,

Amount spent on rent = 19.20% of 100K = Rs. 19.20K

Percentage of amount spent on rent out of the total expenditure = {(19.20/64)×100}% = 30%

So,

P + Q = 30…….(ii)

Solving both the equations, we get,

3P + 4Q = 102 – (i)

P + Q = 30 – (ii)

3P + 4Q = 102

3P + 3Q = 90

Q = 12

Now, put the value of Q in equation (i), we get,

3P + 4Q = 102

3P + 48 = 102

3P = 54

P = 18

So, the percentage distribution of the total expenditure,

Rent = (P+Q)% = (18+12)% = 30%

Food = (2Q+1)% = (24+1)% = 25%

Electricity = (P+2)% = (18+2)% = 20%

Medication = (Q-2)% = (12-2)% = 10%

Entertainment = (P-3)% = (18-3)% = 15%

Total = 100%

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Difference between the medication and entertainment = (15-10)% = 5%

5% = 4000

1% = (4000/5) = 800

100% = (100×800) = 80000

Therefore, the total expenditure of the person = Rs. 80000

Now,

64K = 80000

K = (80000/64) = 1250

So, the total savings of the person = 36K = Rs. 45000

Hence, the total income of the person = (80000+45000) = Rs. 125000.

Now, the expenditure on each necessity,

Rent = 30% of 80000 = Rs. 24000

Food = 25% of 80000 = Rs. 20000

Electricity = 20% of 80000 = Rs. 16000

Medication = 10% of 80000 = Rs. 8000

Entertainment = 15% of 80000 = Rs. 12000

Total = 100% of 80000 = Rs. 80000

Answer: B

According to the question,

The total savings of the person = Rs. 45000

Amount of savings kept at home = 1/3 of 45000 = Rs. 15000

So, the amount deposited in the bank = 2/3 of 45000 = Rs. 30000

Rate of interest = R% (C.I.)

Time duration = 2 years

The total amount received = 46128

Hence, the interest earned = (46128-30000) = Rs. 16128

So,

$[(R+R)+\{(R\times R)/100\}]\% = \{2R+(R^2/100)\}\% = \{(200R+R^2)/100\}\%$

Now,

Interest earned = $\{(200R+R^2)/100\}\%$ of 30000 = $(600R+3R^2)$

$(600R+3R^2) = 16128$

$3R^2 + 600R = 16128$

$3R^2 + 600R - 16128 = 0$

$R^2 + 200R - 5376 = 0$

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$R^2 + 224R - 24R - 5376 = 0$

$R(R+224) - 24(R+224) = 0$

$(R-24)(R+224) = 0$

$R = 24$ or -224

So, $R = 24$ (positive integer)

So, the rate of interest (S.I.) = $(R-4)\% = (24-4)\% = 20\%$

Therefore, the simple interest received = 20% of 30000 for 3 years

Hence, 60% of 30000 = Rs. 18000.

6. Questions

From the pie chart, we have,

The total value of the pie chart = 100%

$(P+Q) + (2Q+1) + (P+2) + (Q-2) + (P-3) = 100$

$3P + 4Q = 102$……(i)

Now,

Let, the income of the person be Rs. 100K

So, the total savings of the person = 36% of 100K = Rs. 36K

Therefore, the total expenditure of the person = 64% of 100K = Rs. 64K

Now,

Amount spent on rent = 19.20% of 100K = Rs. 19.20K

Percentage of amount spent on rent out of the total expenditure = $\{(19.20/64)\times100\}\% = 30\%$

So,

$P + Q = 30$……(ii)

Solving both the equations, we get,

$3P + 4Q = 102$ – (i)

$P + Q = 30$ – (ii)

$3P + 4Q = 102$

$3P + 3Q = 90$

$Q = 12$

Now, put the value of Q in equation (i), we get,

$3P + 4Q = 102$

$3P + 48 = 102$

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3P = 54

P = 18

So, the percentage distribution of the total expenditure,

Rent = (P+Q)% = (18+12)% = 30%

Food = (2Q+1)% = (24+1)% = 25%

Electricity = (P+2)% = (18+2)% = 20%

Medication = (Q-2)% = (12-2)% = 10%

Entertainment = (P-3)% = (18-3)% = 15%

Total = 100%

Difference between the medication and entertainment = (15-10)% = 5%

5% = 4000

1% = (4000/5) = 800

100% = (100×800) = 80000

Therefore, the total expenditure of the person = Rs. 80000

Now,

64K = 80000

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So, the total savings of the person = 36K = Rs. 45000

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Now, the expenditure on each necessity,

Rent = 30% of 80000 = Rs. 24000

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Medication = 10% of 80000 = Rs. 8000

Entertainment = 15% of 80000 = Rs. 12000

Total = 100% of 80000 = Rs. 80000

Answer: D

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March 2024

Income (salary) = {(125000/125)×100} = Rs. 100000

So, the savings = 10% of 100000 = Rs. 10000

And, the expenditure = 90% of 100000 = Rs. 90000

Now,

Amount spent on food = 25% of 90000 = Rs. 22500

And, the amount spent on food in April 2024 = Rs. 20000 (from common explanation)

Hence, the required difference = (22500-20000) = Rs. 2500

7. Questions

From the pie chart, we have,

The total value of the pie chart = 100%

(P+Q) + (2Q+1) + (P+2) + (Q-2) + (P-3) = 100

3P + 4Q = 102……(i)

Now,

Let, the income of the person be Rs. 100K

So, the total savings of the person = 36% of 100K = Rs. 36K

Therefore, the total expenditure of the person = 64% of 100K = Rs. 64K

Now,

Amount spent on rent = 19.20% of 100K = Rs. 19.20K

Percentage of amount spent on rent out of the total expenditure = {(19.20/64)×100}% = 30%

So,

P + Q = 30……(ii)

Solving both the equations, we get,

3P + 4Q = 102 – (i)

P + Q = 30 – (ii)

3P + 4Q = 102

3P + 3Q = 90

Q = 12

Now, put the value of Q in equation (i), we get,

3P + 4Q = 102

3P + 48 = 102

3P = 54

P = 18

So, the percentage distribution of the total expenditure,

Rent = (P+Q)% = (18+12)% = 30%

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Food = (2Q+1)% = (24+1)% = 25%

Electricity = (P+2)% = (18+2)% = 20%

Medication = (Q-2)% = (12-2)% = 10%

Entertainment = (P-3)% = (18-3)% = 15%

Total = 100%

Difference between the medication and entertainment = (15-10)% = 5%

5% = 4000

1% = (4000/5) = 800

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Therefore, the total expenditure of the person = Rs. 80000

Now,

64K = 80000

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So, the total savings of the person = 36K = Rs. 45000

Hence, the total income of the person = (80000+45000) = Rs. 125000.

Now, the expenditure on each necessity,

Rent = 30% of 80000 = Rs. 24000

Food = 25% of 80000 = Rs. 20000

Electricity = 20% of 80000 = Rs. 16000

Medication = 10% of 80000 = Rs. 8000

Entertainment = 15% of 80000 = Rs. 12000

Total = 100% of 80000 = Rs. 80000

Answer: A

Another person

Income = (125000-50000) = Rs. 75000

So,

500M + 400N = 75000……..(i)

Now,

Amount spent of electricity = 12% of 400N = Rs. 48N

So,

500M – 48N = 7800……..(ii)

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Solving both the equations, we get,

448N = 67200

N = 150

Now, put the value of N in equation (i), we get,

500M + 400N = 75000

500M + 60000 = 75000

500M = 15000

M = 30

Hence, the value of √(N-M-20) = √(150-30-20) = √100 = 10.

8. Questions

From the pie chart, we have,

The total value of the pie chart = 100%

(P+Q) + (2Q+1) + (P+2) + (Q-2) + (P-3) = 100

3P + 4Q = 102……(i)

Now,

Let, the income of the person be Rs. 100K

So, the total savings of the person = 36% of 100K = Rs. 36K

Therefore, the total expenditure of the person = 64% of 100K = Rs. 64K

Now,

Amount spent on rent = 19.20% of 100K = Rs. 19.20K

Percentage of amount spent on rent out of the total expenditure = {(19.20/64)×100}% = 30%

So,

P + Q = 30…….(ii)

Solving both the equations, we get,

3P + 4Q = 102 – (i)

P + Q = 30 – (ii)

3P + 4Q = 102

3P + 3Q = 90

Q = 12

Now, put the value of Q in equation (i), we get,

3P + 4Q = 102

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3P + 48 = 102

3P = 54

P = 18

So, the percentage distribution of the total expenditure,

Rent = (P+Q)% = (18+12)% = 30%

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Electricity = (P+2)% = (18+2)% = 20%

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Entertainment = (P-3)% = (18-3)% = 15%

Total = 100%

Difference between the medication and entertainment = (15-10)% = 5%

5% = 4000

1% = (4000/5) = 800

100% = (100×800) = 80000

Therefore, the total expenditure of the person = Rs. 80000

Now,

64K = 80000

K = (80000/64) = 1250

So, the total savings of the person = 36K = Rs. 45000

Hence, the total income of the person = (80000+45000) = Rs. 125000.

Now, the expenditure on each necessity,

Rent = 30% of 80000 = Rs. 24000

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Electricity = 20% of 80000 = Rs. 16000

Medication = 10% of 80000 = Rs. 8000

Entertainment = 15% of 80000 = Rs. 12000

Total = 100% of 80000 = Rs. 80000

Answer: C

The total number medicines (generic + normal) purchased = 48

The total amount spent on medication (generic medicines + normal medicines) = Rs. 8000

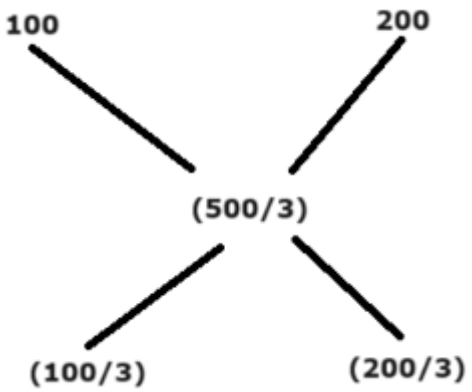
So, the price of each medicine (generic + normal) = (8000/48) = Rs. (500/3)

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Price of each generic medicine = Rs. 100

Price of each normal medicine = Rs. 200

Using Allegation Method,



So, the ratio of the number of generic medicines to that of normal medicines purchased by the person = (100/3):(200/3) = 1:2.

Therefore, the number of generic medicine purchased by the person = 1/3 of 48 = 16

Hence, the total amount spent on the purchase of generic medicines = (100×16) = Rs. 1600.

Using General Method,

Let, the number of generic and normal medicines purchased by the person be a and (48-a) respectively.

So, the total amount spent on the purchase of generic medicines = (a×100) = Rs. 100a

And, the total amount spent on the purchase of normal medicines = {(48-a)×200} = Rs. (9600-200a)

Now,

100a + (9600-200a) = 8000

9600 – 100a = 8000

100a = 1600

a = 16

Therefore, the number of generic medicine purchased by the person = a = 16

Hence, the total amount spent on the purchase of generic medicines = (100×16) = Rs. 1600.

9. Questions

Shop X

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The total number of cookies (chocolate + sugar) baked = 250

The total number of cookies (chocolate + sugar) sold = 4/5 of 250 = 200

The total number of cookies (chocolate + sugar) unsold = 1/5 of 250 = 50

The number of chocolate cookies baked = (P+30)

The number of sugar cookies baked = (Q+20)

Now,

(P+30) + (Q+20) = 250

P + Q = 200……..(i)

Shop Y

The total number of cookies (chocolate + sugar) baked = (P+2Q)

The number of chocolate cookies baked = 210

The number of sugar cookies baked = 70

Now,

P + 2Q = 210 + 70

P + 2Q = 280…….(ii)

Solving both the equations, we get,

P + Q = 200 – (i)

P + 2Q = 280 – (ii)

Q = 80

Now, put the value of Q in equation (i), we get,

P + Q = 200

P + 80 = 200

P = 120

Therefore, the total number of cookies (chocolate + sugar) baked = (P+2Q) = (120+160) = 280

Now, let the total number of cookies (chocolate + sugar) sold and unsold be n and (n-200) respectively.

2n – 200 = 280

2n = 480

n = 240

So, the total number of cookies (chocolate + sugar) sold = n = 240

And, the total number of cookies (chocolate + sugar) unsold = (n-200) = (240-200) = 40

Shop Z

The total number of cookies (chocolate + sugar) sold = 240

Percentage of the total number of cookies (chocolate + sugar) sold = 80%

80% = 240

1% = (240/80) = 3

100% = (100×3) = 300

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Therefore, the total number of cookies (chocolate + sugar) baked = 300

So, the total number of cookies (chocolate + sugar) unsold = 20% of 300 = 60

Now,

The total number of chocolate cookies baked = 2/3 of 300 = 200

And, the total number of sugar cookies baked = 1/3 of 300 = 100

Shop	Total number of cookies baked	Number of chocolate cookies baked	Number of sugar cookies baked	Number of cookies sold	Number of cookies unsold
X	250	150	100	200	50
Y	280	210	70	240	40
Z	300	200	100	240	60
Total	830	560	270	680	150

Answer: D

According to the question,

Shop X

The total number of cookies (chocolate + sugar) sold = 200

The number of chocolate cookies sold = 2A

The number of sugar cookies sold = 4B

The total number of cookies (chocolate + sugar) unsold = 50

The number of chocolate cookies unsold = (B+10)

The number of sugar cookies unsold = (A-40)

Now,

2A + 4B = 200

A + 2B = 100……(i)

Also,

(A-40) + (B+10) = 50

A + B = 80……(ii)

Solving both the equations, we get,

B = 20

Now, put the value of B in equation (i), we get,

A + 2B = 100

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A + 40 = 100

A = 60

Hence, the value of √(A-B-4) = √(60-20-4) = √36 = 6.

Checking each option,

a). (A/10) = (60/10) = 6 (satisfied)

b). (B-14) = (20-14) = 6 (satisfied)

c). {(A×B)/200} = {(60×20)/200} = (1200/200) = 6 (satisfied)

Therefore, all the options are satisfied.

So, option d) is the correct answer.

10. Questions

Shop X

The total number of cookies (chocolate + sugar) baked = 250

The total number of cookies (chocolate + sugar) sold = 4/5 of 250 = 200

The total number of cookies (chocolate + sugar) unsold = 1/5 of 250 = 50

The number of chocolate cookies baked = (P+30)

The number of sugar cookies baked = (Q+20)

Now,

(P+30) + (Q+20) = 250

P + Q = 200……..(i)

Shop Y

The total number of cookies (chocolate + sugar) baked = (P+2Q)

The number of chocolate cookies baked = 210

The number of sugar cookies baked = 70

Now,

P + 2Q = 210 + 70

P + 2Q = 280…….(ii)

Solving both the equations, we get,

P + Q = 200 – (i)

P + 2Q = 280 – (ii)

Q = 80

Now, put the value of Q in equation (i), we get,

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P + Q = 200

P + 80 = 200

P = 120

Therefore, the total number of cookies (chocolate + sugar) baked = (P+2Q) = (120+160) = 280

Now, let the total number of cookies (chocolate + sugar) sold and unsold be n and (n-200) respectively.

2n – 200 = 280

2n = 480

n = 240

So, the total number of cookies (chocolate + sugar) sold = n = 240

And, the total number of cookies (chocolate + sugar) unsold = (n-200) = (240-200) = 40

Shop Z

The total number of cookies (chocolate + sugar) sold = 240

Percentage of the total number of cookies (chocolate + sugar) sold = 80%

80% = 240

1% = (240/80) = 3

100% = (100×3) = 300

Therefore, the total number of cookies (chocolate + sugar) baked = 300

So, the total number of cookies (chocolate + sugar) unsold = 20% of 300 = 60

Now,

The total number of chocolate cookies baked = 2/3 of 300 = 200

And, the total number of sugar cookies baked = 1/3 of 300 = 100

Shop	Total number of cookies baked	Number of chocolate cookies baked	Number of sugar cookies baked	Number of cookies sold	Number of cookies unsold
X	250	150	100	200	50
Y	280	210	70	240	40
Z	300	200	100	240	60
Total	830	560	270	680	150

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Answer: A

Shop Y

The total number of cookies sold = 240

The number of chocolate cookies sold = 2/3 of 240 = 160

The number of sugar cookies sold = 1/3 of 240 = 80

The ratio of the price of each chocolate to that of each sugar cookie sold = 100:60 = 5:3. Let it be 5s and 3s respectively.

So, the total revenue earned from selling all the chocolate cookies = (5s×160) = 800s

And, the total revenue earned from selling all the sugar cookies = (3s×80) = 240s

Now,

800s – 240s = 560

560s = 560

s = 1

Hence, the difference between the price of each chocolate and each sugar cookie sold = (5s-3s) = 2s = 2.

11. Questions

Shop X

The total number of cookies (chocolate + sugar) baked = 250

The total number of cookies (chocolate + sugar) sold = 4/5 of 250 = 200

The total number of cookies (chocolate + sugar) unsold = 1/5 of 250 = 50

The number of chocolate cookies baked = (P+30)

The number of sugar cookies baked = (Q+20)

Now,

(P+30) + (Q+20) = 250

P + Q = 200……..(i)

Shop Y

The total number of cookies (chocolate + sugar) baked = (P+2Q)

The number of chocolate cookies baked = 210

The number of sugar cookies baked = 70

Now,

P + 2Q = 210 + 70

P + 2Q = 280……..(ii)

Solving both the equations, we get,

P + Q = 200 – (i)

P + 2Q = 280 – (ii)

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Q = 80

Now, put the value of Q in equation (i), we get,

P + Q = 200

P + 80 = 200

P = 120

Therefore, the total number of cookies (chocolate + sugar) baked = (P+2Q) = (120+160) = 280

Now, let the total number of cookies (chocolate + sugar) sold and unsold be n and (n-200) respectively.

2n – 200 = 280

2n = 480

n = 240

So, the total number of cookies (chocolate + sugar) sold = n = 240

And, the total number of cookies (chocolate + sugar) unsold = (n-200) = (240-200) = 40

Shop Z

The total number of cookies (chocolate + sugar) sold = 240

Percentage of the total number of cookies (chocolate + sugar) sold = 80%

80% = 240

1% = (240/80) = 3

100% = (100×3) = 300

Therefore, the total number of cookies (chocolate + sugar) baked = 300

So, the total number of cookies (chocolate + sugar) unsold = 20% of 300 = 60

Now,

The total number of chocolate cookies baked = 2/3 of 300 = 200

And, the total number of sugar cookies baked = 1/3 of 300 = 100

Shop	Total number of cookies baked	Number of chocolate cookies baked	Number of sugar cookies baked	Number of cookies sold	Number of cookies unsold
X	250	150	100	200	50
Y	280	210	70	240	40
Z	300	200	100	240	60
Total	830	560	270	680	150

Answer: B

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Shop Z

The total number of peanut cookies baked = 150

Difference between the number of peanut cookies sold and unsold = T% of 150 = 1.50T (sold-unsold)

The number of peanut cookies unsold = 3T

So, the number of peanut cookies sold = (3T+1.50T) = 4.50T

Now,

3T + 4.50T = 150

7.50T = 150

T = 20

Therefore, the number of peanut cookies sold = 4.50T = 90

And, the number of peanut cookies unsold = 3T = 60

Hence, the required ratio = 90:60 = 3:2.

12. Questions

Shop X

The total number of cookies (chocolate + sugar) baked = 250

The total number of cookies (chocolate + sugar) sold = 4/5 of 250 = 200

The total number of cookies (chocolate + sugar) unsold = 1/5 of 250 = 50

The number of chocolate cookies baked = (P+30)

The number of sugar cookies baked = (Q+20)

Now,

(P+30) + (Q+20) = 250

P + Q = 200……..(i)

Shop Y

The total number of cookies (chocolate + sugar) baked = (P+2Q)

The number of chocolate cookies baked = 210

The number of sugar cookies baked = 70

Now,

P + 2Q = 210 + 70

P + 2Q = 280……..(ii)

Solving both the equations, we get,

P + Q = 200 – (i)

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P + 2Q = 280 – (ii)

Q = 80

Now, put the value of Q in equation (i), we get,

P + Q = 200

P + 80 = 200

P = 120

Therefore, the total number of cookies (chocolate + sugar) baked = (P+2Q) = (120+160) = 280

Now, let the total number of cookies (chocolate + sugar) sold and unsold be n and (n-200) respectively.

2n – 200 = 280

2n = 480

n = 240

So, the total number of cookies (chocolate + sugar) sold = n = 240

And, the total number of cookies (chocolate + sugar) unsold = (n-200) = (240-200) = 40

Shop Z

The total number of cookies (chocolate + sugar) sold = 240

Percentage of the total number of cookies (chocolate + sugar) sold = 80%

80% = 240

1% = (240/80) = 3

100% = (100×3) = 300

Therefore, the total number of cookies (chocolate + sugar) baked = 300

So, the total number of cookies (chocolate + sugar) unsold = 20% of 300 = 60

Now,

The total number of chocolate cookies baked = 2/3 of 300 = 200

And, the total number of sugar cookies baked = 1/3 of 300 = 100

Shop	Total number of cookies baked	Number of chocolate cookies baked	Number of sugar cookies baked	Number of cookies sold	Number of cookies unsold
X	250	150	100	200	50
Y	280	210	70	240	40
Z	300	200	100	240	60
Total	830	560	270	680	150

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Answer: C

Shop W

The number of chocolate cookies baked = 150

The number of sugar cookies baked = (U^2-1)

Now,

$150 \times (U^2-1) = 18000$

$150U^2 - 150 = 18000$

$150U^2 = 18150$

$U^2 = 121$

$U = 11$

Hence, the total number of cookies baked (chocolate + sugar) by the shop Z = $\{150+(U^2-1)\}$ = $\{150+(11^2-1)\}$ = $\{150+(121-1)\}$ = (150+120) = 270.

13. Questions

Answer: B

The quantity of milk taken out of from container M = 50% of 3X = 1.50X

The quantity of water taken out of from container M = 50% of X = 0.50X

The quantity of milk taken out of from container N = 25% of (Y+16) = (0.25Y+4)

The quantity of water taken out of from container N = 25% of Y = 0.25Y

Now,

Container K (resultant mixture)

The quantity of milk = (1.50X+0.25Y+4)

The quantity of water = (0.50X+0.25Y)

So,

$2X + 0.50Y + 4 = 2Y + 16$

$2X - 1.50Y = 12$

$4X - 3Y = 24$……..(i)

And,

$3X - Y = 58$…….(ii)

Solving both the equations, we get,

$12X - 9Y = 72$

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12X – 4Y = 232

5Y = 160

Y = 32

Now, put the value of Y in equation (i), we get,

4X – 3Y = 24

4X – 96 = 24

4X = 120

X = 30

So, the ratio of the initial quantity of milk in container M to that of container N = 3X:(Y+16) = 90:(32+16) = 30:48 = 15:8.

14. Questions

Answer: A

Let, the efficiency of each man and each woman be M and W respectively.

So,

{(10×M)×10} = {(25×W)×8}

100M = 200W

M:W = 2:1.

So, the ratio of the efficiency of each man to that of each woman = 2:1.

Therefore, the total work = {(10×M)×10} = {(10×2)×10} = 200

Now, the work done by 5 men and 4 women in A days = {(5M+4W)×A} = (10A+4A) = 14A

After 2 men left and 2 new women joined, the number of men and women working becomes,

Men = (5-2) = 3

Women = (4+2) = 6

So, the remaining work was done by 3 men and 6 women in (A-5) days = {(3M+6W)×(A-5)} = {(6+6)×(A-5)} = (12A-60)

14A + (12A-60) = 200

26A = 260

A = 10

Efficiency of each boy = 1/2 of 1 = 1/2 = 0.50

So, the (A+6) boys can complete the whole work in = [200/{(A+6)×0.50}] = [200/{(10+6)×0.50}] = (200/8) = 25 days.

15. Questions

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Answer: C

Scheme P

Amount invested = Rs. 20000

Rate of interest = R% p.a.

Time duration = (R-11) years

Type of interest = S.I.

So, the interest earned = [{(R/100)×(R-11)} of 20000] = [{(R^2-11R)/100} of 20000] = ($200R^2$-2200R)

Now,

($200R^2$-2200R) = 12000

$200R^2$ - 2200R – 12000 = 0

R^2 – 11R – 60 = 0

R^2 – 15R + 4R – 60 = 0

R(R-15) + 4(R-15) = 0

(R+4) (R-15) = 0

R = -4 or 15

So, R = 15 (positive integer)

Scheme Q

Amount invested = 1000R = Rs. 15000

Rate of interest = (R-5)% = (15-5)% = 10% p.a.

Time duration = 2 years

Type of interest = C.I.

[(10+10)+{(10×10)/100}]% = (20+1)% = 21%

Hence, the compound interest earned = 21% of 15000 = Rs. 3150.

16. Questions

Answer: B

Let, the speed of boat A to that of boat B in still water be 6v km/h and 5v km/h respectively.

Boat A

Speed of the boat in still water = 6v km/h

Speed of river = 12 km/h

Downstream speed = (6v+12) km/h

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Upstream speed = (6v-12) km/h

Boat B

Speed of the boat in still water = 5v km/h

Speed of river = 12 km/h

Downstream speed = (5v+12) km/h

Upstream speed = (5v-12) km/h

Now, the downstream distance of the boat B = (420/10) = 42 km/h

(5v+12) = 42

5v = 30

v = 6

So,

(D+90)/(6v+12) = (D-60)/(5v-12)

(D+90)/(36+12) = (D-60)/(30-12)

(D+90)/48 = (D-60)/18

18D + 1620 = 48D – 2880

30D = 4500

D = 150

Hence, the time taken by the boat A to cover (D+42) km upstream distance = {(D+42)/(6v-12)} = {(150+42)/(36-12)} = (192/24) = 8 hours.

17. Questions

Answer: B

Let, the length of train A be L meters

Length of the platform = P meters

Speed of train A = 60 m/sec

So,

(L+P)/60 = 4

L + P = 240……(i)

Using statements I and II together

Speed of train B = {(60/3)×2} = 40 m/sec

Length of train B = (L-20) meters

So,

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{(L-20)+P}/40 = (4+1.50)

{(L-20)+P}/40 = 5.50

L – 20 = 220 – P

L + P = 240…….(ii)

From both the equations, we are getting the same resultants.

So, the data is inadequate.

Hence, statements I and II together are not sufficient.

Using statements I and III together

Speed of train B = {(60/3)×2} = 40 m/sec

Let, the length of train B be Q meters

So, when train B crosses a pole,

Q /40 = 3.50

Q = 140

So, the length of train B = 140 meters

Now,

(140+P)/40 = (4+1.50)

(140+P)/40 = 5.50

140 + P = 220

P = 80

Therefore, the value of √(P+1) = √(80+1) = √81 = 9.

Hence, statements I, and III together are necessary to answer the question.

18. Questions

Answer: C

The total investment made by X = (12000×12) = Rs. 144000

Z's investment = 125% of 12000 = Rs. 15000

Using statements I and II together

The investment of Y = 2/3 of 15000 = Rs. 10000

The time duration for which Y invested his money = (12-5) = 7 months

So, the total investment made by Y = (10000×7) = Rs. 70000

The time duration for which Z invested his money = (12-8) = 4 months

So, the total investment made by Z = (15000×4) = Rs. 60000

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Therefore, the ratio of the profit share of X to that Y to that of Z = 144000:70000:60000 = 72:35:30

Hence, the profit share of Y = 35/137 of 274000 = Rs. 70000.

So, statements I and II together are sufficient.

Using statements I and III together

The investment of Y = 2/3 of 15000 = Rs. 10000

The time duration for which Y invested his money = (12-5) = 7 months

So, the total investment made by Y = (10000×7) = Rs. 70000

The fractional value of 14.28% = 1/7

The ratio of the profit share of Y to that of Z = 7:6

The profit will be divided in the ratio of the total investment made by them

So, the total investment made by Z = {(70000/7)×6} = Rs. 60000

Therefore, the ratio of the profit share of X to that Y to that of Z = 144000:70000:60000 = 72:35:30

Hence, the profit share of Y = 35/137 of 274000 = Rs. 70000.

So, statements I and III together are sufficient.

Hence, either statement I and II together or I and III together are sufficient.

19. Questions

Answer: C

Time taken to fill the tank alone by pipe,



Efficiency of pipe,

(P+Q) = (120/24/5) = {(120×5)/24} = 25

(Q+R) = (120/40/7) = {(120×7)/40} = 21

(P-R) = (25-21) = 4

Let, the efficiency of pipe,

R = v

P = (v+4)

Q = (21-v)

Using statement I alone

Time taken by pipe Q to fill the whole tank = {(2/25)×100} = 8 minutes

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So, the efficiency of pipe Q = (120/8) = 15

(21-v) = 15

v = 6

So, the efficiency of pipe P = (v+4) = (6+4) = 10

and, the efficiency of pipe R = v = 6

Hence, the time taken by pipe P and pipe R together to fill the whole tank = {120/(10+6)} = (120/16) = (15/2) = 7.50 = 7 minutes 30 seconds.

Therefore, statement I alone is sufficient.

Using statement II alone

(21-v) – v = 9

12 – v = v

2v = 12

v = 6

So, the efficiency of pipe P = (v+4) = (6+4) = 10

and, the efficiency of pipe R = v = 6

Hence, the time taken by pipe P and pipe R together to fill the whole tank = {120/(10+6)} = (120/16) = (15/2) = 7.50 = 7 minutes 30 seconds.

Therefore, statement II alone is sufficient.

Using statement III alone

(v+4)/(21-v) = 2/3

3v + 12 = 42 – 2v

5v = 30

v = 6

So, the efficiency of pipe P = (v+4) = (6+4) = 10

and, the efficiency of pipe R = v = 6

Hence, the time taken by pipe P and pipe R together to fill the whole tank = {120/(10+6)} = (120/16) = (15/2) = 7.50 = 7 minutes 30 seconds.

Therefore, statement III alone is sufficient.

So, either of the statements I, II or III alone is sufficient.

20. Questions

Answer: B

Let, the number of boys and girls in school A be 3x and x respectively.

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Also, let the number of boys and girls in school B be 3y and 2y respectively.

Using statements I and II alone

3y + 3x = 240

y + x = 80……..(i)

Also,

2y – x = 10……(ii)

Solving both the equations, we get,

2y + 2x = 160

2y – x = 10

3x = 150

x = 50

Now, put the value of x in equation (i), we get,

y + x = 80

y + 50 = 80

y = 30

Hence, the total number of students (boys + girls) in both the classes together = (3x+x+3y+2y) = (4x+5y) = (200+150) = 350.

Therefore, statements I and II together are necessary.

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1. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the number of cricket balls manufactured by a company starting from Monday to till Friday of a week. A company manufactured two type of balls – One Day match ball (White) and Test match ball (Red and Pink).

Days	Total number of balls manufactured	% of white balls out of total balls manufactured	Average of (Red + Pink) balls manufactured
Monday	Z	60%	80
Till Tuesday	760	50%	(3Y + 10)
Till Wednesday	1000	(L – 25) %	225
Till Thursday	1500	60%	M
Till Friday	(4K + 500)	58%	441

Find the number of white balls manufactured on Friday?

a. 328

b. 318

c. 338

d. 308

e. None of these

2. Questions

Find the minimum possible difference between number of white balls and pink balls manufactured on Thursday?

a. 200

b. 250

c. 150

d. 350

e. None of these

3. Questions

Find the value of Y?

a. 80

b. 60

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c. 30

d. 90

e. None of these

4. Questions

Find (L/2) % of [Z + K]?

a. 640

b. 360

c. 320

d. 400

e. None of these

5. Questions

The number of red balls manufactured on Wednesday is same as that of pink balls on the same day and number of pink balls manufactured on Wednesday is less than that of any other day. Find the maximum possible sum of number of red balls manufactured on Tuesday and Thursday together?

a. 288

b. 300

c. 296

d. 278

e. 298

6. Questions

Read the following information carefully and answer the questions based on it.

Three persons – P, Q, and R starting from point M from their bikes at same time and travelled towards N, but they are using different Routes – R1, R2 and R3 respectively. Speed of Q is twice as that of P. Total time taken by P for travelling from M to N and then returned to M is 25 hours. Ratio of distance of R1: R3: R2 is 15:18:20. Total time taken by Q for travelling from M to N and then returned to M is 5 hours 25 minutes more than time taken by R to travel from M to N. Speed of Q is 24 km/h.

When R started from point M, after covering 128 km, due to engine problem, speed of his bike reduced by 18.75%. After reaching at point N, he found the mechanic to solve the engine problem, so he has to wait for 2 hours 30 minutes. Now he has to travel at K% more speed than his last speed to reach on time at point N. find approximate value of K?

a. 41%

b. 73%

c. 45%

d. 75%

e. 51%

7. Questions

Speed of R is how much % more or less than that of Q?

a. 25%

b. 50%

c. 33.33%

d. 40%

e. None of these

8. Questions

If Q is started from point M and R started form point N (both are at same route R2), find the distance between second meeting point of Q and R, and point M.

a. 160 km

b. 120 km

c. 80 km

d. 40 km

e. None of these

9. Questions

Find the sum of distance covered by P, Q and R together.

a. 265 km

b. 795 km

c. 540 km

d. 1060 km

e. None of these

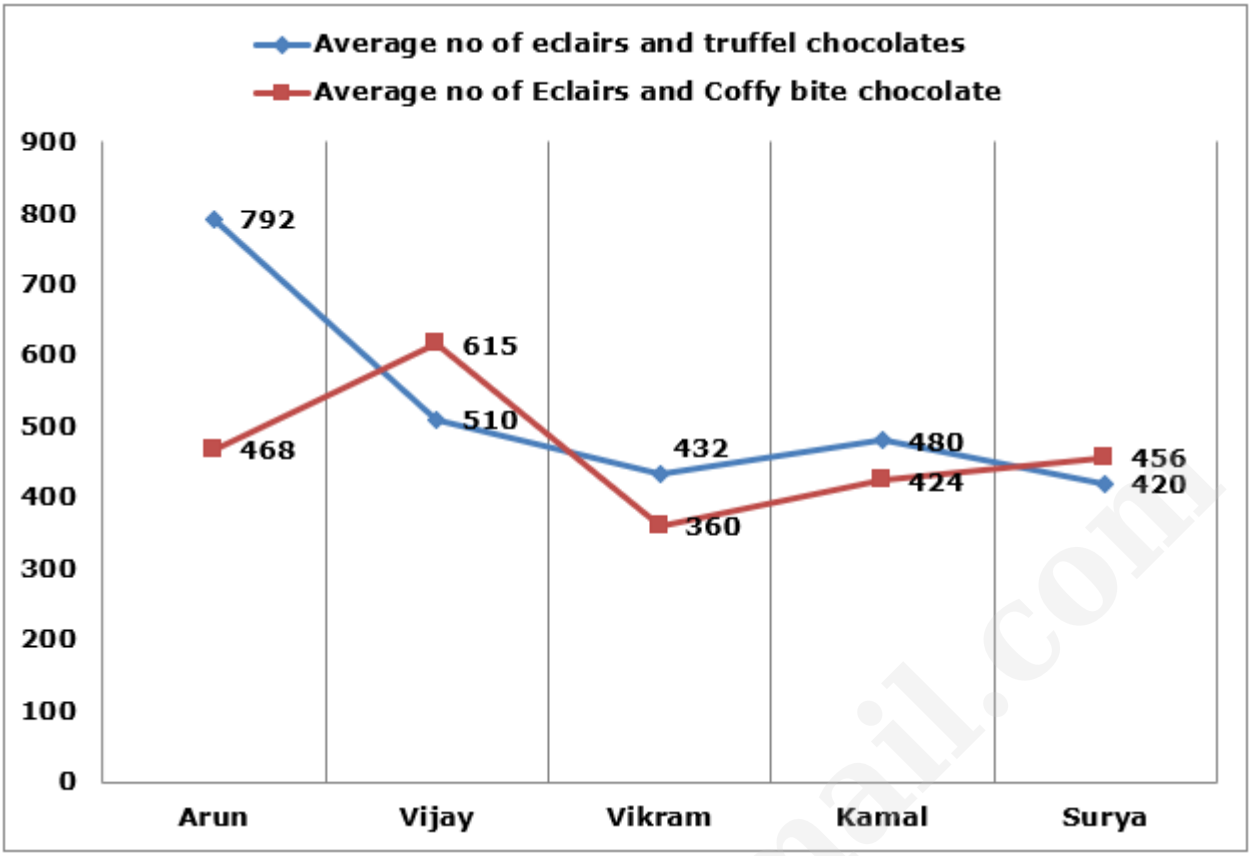
10. Questions

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Study the following information carefully and answer the questions.

Five friends namely Arun, Vijay, Vikram, Kamal and Surya have three types of chocolates (Eclairs, Truffel and Coffy bite) with them. The Line graph given below shows the average number of Eclairs and Truffel chocolates and average number of Eclairs and Coffy bite chocolates that each of them have.

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NOTES:

1). Ratio of total number of chocolates that Arun and Vijay have is 12:11, respectively.

2). Number of chocolates that Vijay has is 37.5% more than number of chocolates that Vikram has which is 208 less than number of chocolates that Kamal has.

3). Total number of chocolates that Surya has is 6% more than that of Vikram has and difference between numbers of chocolates that Vikram and Surya have is 72.

Find the number of truffel chocolates that Arun has?

a. 720

b. 576

c. 864

d. 216

e. None of these

11. Questions

Number of Eclairs chocolates that Arun has is how much percent more/less than that Vijay has?

a. 16(2/3) %

b. 20%

c. 40%

d. 25%

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e. None of these

12. Questions

Find the average number of Coffy bite chocolates that Vijay and Vikram together have?

a. 463

b. 443

c. 453

d. 483

e. None of these

13. Questions

If Kamal bought each Eclairs, Truffel and coffy bite chocolates for Rs. 5, Rs. 8 and Rs. 6, respectively, then find total amount spent by him on buying all three types of chocolates?

a. Rs. 8944

b. Rs. 8800

c. Rs. 9120

d. Rs. 9168

e. None of these

14. Questions

Find the Number of Eclairs chocolates that Surya has?

a. 480

b. 432

c. 360

d. 420

e. None of these

15. Questions

A shopkeeper sold two articles P and Q after discount of 10% and 40% respectively thus earning profit of 2Z% and (3Z – 25) %. If cost price of P is 50% more than that of Q, and marked price of both the article is same, then find the value of (Z + 15)?

a. 25

b. 40

c. 50

d. 30

e. None of these

16. Questions

P is twice as efficient as Q. Q can complete 1/3rd of job in 24 days. Q started the work and worked for 40 days. In how many days P alone does half of the remaining work?

a. 16 days

b. 12 days

c. 8 days

d. 10 days

e. None of these

17. Questions

Boat A can cover a distance of 300 km downstream in 8 hours and can cover the same distance upstream in 12 hours. Find the approximate time taken by boat B to travel the same distance downstream, if the speed of boat B is 20% more than speed of boat A?

a. 7.25 hours

b. 6.55 hours

c. 8.15 hours

d. 6.85 hours

e. None of these

18. Questions

Bag A contains 7 yellow, 5 pink and 8 violet balls and Bag B contains 6 yellow, 3 pink and 5 violet balls. Find the probability of drawing a yellow ball out of two bags?

a. 109/280

b. 93/280

c. 105/280

d. 119/280

e. None of these

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19. Questions

Simple interest earned on Rs. P at R% per annum for five years is twice of simple interest earned on Rs. 2P at (R – 15) % per annum for two years. Find the amount received after 3 years when Rs. P invested under CI for three years at (R + 10) % per annum is how much % of amount received

when Rs. 3P invested at 5R% per annum under CI for two years?

a. 25%

b. 8.33%

c. 6.25%

d. 12.5%

e. None of these

20. Questions

A and B invested in a company with an investment of Rs.35000 and Rs.40000 respectively. Out of the total profit gained, 10% and 15% of it is donated to charity and expenses for the company respectively. Find the total profit at the end of the year, if the difference between their profits are Rs.2000?

a. Rs. 30000

b. Rs. 40000

c. Rs. 35000

d. Rs. 20000

e. None of these

Explanations:

1. Questions

For Monday

Total number of (Red + Pink) balls manufactured = 80 x 2 = 160

So, total number of balls manufactured = 160 x (100/40) = 400

Number of white balls manufactured = 60% x 400 = 240

Value of Z = 400

For Tuesday

Total number of balls manufactured = 760 – 400 = 360

Number of white balls manufactured = (50% x 760) – 240 = 140

Number of (Red + Pink) balls manufactured = (50% x 760) – 160 = 220

Now,

(3Y + 10) = 380/2

3Y + 10 = 190

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Value of Y = 180/3 = 60

For Wednesday

Total number of balls manufactured = 1000 – 760 = 240

Number of (Red + Pink) manufactured = (225 x 2) – 380 = 70

Number of white balls manufactured = 240 – 70 = 170

Now,

(L – 25) % = [(1000 – 450)/1000] x 100

L – 25 = 55

Value of L = 55 + 25 = 80

For Thursday

Total number of balls manufactured = 1500 – 1000 = 500

Number of white balls manufactured = (60% x 1500) – 550 = 350

Number of (Pink + Red) balls manufactured = (40% x 1500) – 450 = 150

Value of M = (40% x 1500)/2 = 300

For Friday

Total (Red + Pink) manufactured till Friday = 441 x 2 = 882

So, total number of balls manufactured till Friday = 882 x (100/42) = 2100

Total balls manufactured on Friday = 2100 – 1500 = 600

Number of white balls manufactured = (58% x 2100) – 900 = 318

Number of (Pink + Red) manufactured = 882 – 600 = 282

Now,

4K + 500 = 2100

Value of K = 1600/4 = 400

Days	Total balls	White balls	(Red + Pink)
Monday	400	240	160
Tuesday	360	140	220
Wednesday	240	170	70
Thursday	500	350	150
Friday	600	318	282

Answer: B

According to question,

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Number of white balls manufactured on Friday = 318

Hence, the answer is option B

2. Questions

For Monday

Total number of (Red + Pink) balls manufactured = 80 x 2 = 160

So, total number of balls manufactured = 160 x (100/40) = 400

Number of white balls manufactured = 60% x 400 = 240

Value of Z = 400

For Tuesday

Total number of balls manufactured = 760 – 400 = 360

Number of white balls manufactured = (50% x 760) – 240 = 140

Number of (Red + Pink) balls manufactured = (50% x 760) – 160 = 220

Now,

(3Y + 10) = 380/2

3Y + 10 = 190

Value of Y = 180/3 = 60

For Wednesday

Total number of balls manufactured = 1000 – 760 = 240

Number of (Red + Pink) manufactured = (225 x 2) – 380 = 70

Number of white balls manufactured = 240 – 70 = 170

Now,

(L – 25) % = [(1000 – 450)/1000] x 100

L – 25 = 55

Value of L = 55 + 25 = 80

For Thursday

Total number of balls manufactured = 1500 – 1000 = 500

Number of white balls manufactured = (60% x 1500) – 550 = 350

Number of (Pink + Red) balls manufactured = (40% x 1500) – 450 = 150

Value of M = (40% x 1500)/2 = 300

For Friday

Total (Red + Pink) manufactured till Friday = 441 x 2 = 882

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So, total number of balls manufactured till Friday = 882 x (100/42) = 2100

Total balls manufactured on Friday = 2100 – 1500 = 600

Number of white balls manufactured = (58% x 2100) – 900 = 318

Number of (Pink + Red) manufactured = 882 – 600 = 282

Now,

4K + 500 = 2100

Value of K = 1600/4 = 400

Days	Total balls	White balls	(Red + Pink)
Monday	400	240	160
Tuesday	360	140	220
Wednesday	240	170	70
Thursday	500	350	150
Friday	600	318	282

Answer: A

Number of white balls manufactured on Thursday = 350

Number of (Pink + Red) manufactured on Thursday = 150

We need to find minimum value of = (White balls – Pink balls)

White balls are fixed, so for minimum required values, number of pink balls manufactured is maximum possible = 150

So, Required difference = 350 – 150 = 200

Hence, the answer is option A

3. Questions

For Monday

Total number of (Red + Pink) balls manufactured = 80 x 2 = 160

So, total number of balls manufactured = 160 x (100/40) = 400

Number of white balls manufactured = 60% x 400 = 240

Value of Z = 400

Tg:- @NextGenBankers

For Tuesday

Total number of balls manufactured = 760 – 400 = 360

Number of white balls manufactured = (50% x 760) – 240 = 140

Number of (Red + Pink) balls manufactured = (50% x 760) – 160 = 220

Now,

(3Y + 10) = 380/2

3Y + 10 = 190

Value of Y = 180/3 = 60

For Wednesday

Total number of balls manufactured = 1000 – 760 = 240

Number of (Red + Pink) manufactured = (225 x 2) – 380 = 70

Number of white balls manufactured = 240 – 70 = 170

Now,

(L – 25) % = [(1000 – 450)/1000] x 100

L – 25 = 55

Value of L = 55 + 25 = 80

For Thursday

Total number of balls manufactured = 1500 – 1000 = 500

Number of white balls manufactured = (60% x 1500) – 550 = 350

Number of (Pink + Red) balls manufactured = (40% x 1500) – 450 = 150

Value of M = (40% x 1500)/2 = 300

For Friday

Total (Red + Pink) manufactured till Friday = 441 x 2 = 882

So, total number of balls manufactured till Friday = 882 x (100/42) = 2100

Total balls manufactured on Friday = 2100 – 1500 = 600

Number of white balls manufactured = (58% x 2100) – 900 = 318

Number of (Pink + Red) manufactured = 882 – 600 = 282

Now,

4K + 500 = 2100

Value of K = 1600/4 = 400

Days	Total balls	White balls	(Red + Pink)
Monday	400	240	160
Tuesday	360	140	220
Wednesday	240	170	70
Thursday	500	350	150
Friday	600	318	282

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Answer: B

Value of Y = 60

Hence, the answer is option B

4. Questions

For Monday

Total number of (Red + Pink) balls manufactured = 80 x 2 = 160

So, total number of balls manufactured = 160 x (100/40) = 400

Number of white balls manufactured = 60% x 400 = 240

Value of Z = 400

For Tuesday

Total number of balls manufactured = 760 – 400 = 360

Number of white balls manufactured = (50% x 760) – 240 = 140

Number of (Red + Pink) balls manufactured = (50% x 760) – 160 = 220

Now,

(3Y + 10) = 380/2

3Y + 10 = 190

Value of Y = 180/3 = 60

For Wednesday

Total number of balls manufactured = 1000 – 760 = 240

Number of (Red + Pink) manufactured = (225 x 2) – 380 = 70

Number of white balls manufactured = 240 – 70 = 170

Now,

(L – 25) % = [(1000 – 450)/1000] x 100

L – 25 = 55

Value of L = 55 + 25 = 80

For Thursday

Total number of balls manufactured = 1500 – 1000 = 500

Number of white balls manufactured = (60% x 1500) – 550 = 350

Number of (Pink + Red) balls manufactured = (40% x 1500) – 450 = 150

Value of M = (40% x 1500)/2 = 300

For Friday

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Tg: @NextGenBankers

Total (Red + Pink) manufactured till Friday = 441 x 2 = 882

So, total number of balls manufactured till Friday = 882 x (100/42) = 2100

Total balls manufactured on Friday = 2100 – 1500 = 600

Number of white balls manufactured = (58% x 2100) – 900 = 318

Number of (Pink + Red) manufactured = 882 – 600 = 282

Now,

4K + 500 = 2100

Value of K = 1600/4 = 400

Days	Total balls	White balls	(Red + Pink)
Monday	400	240	160
Tuesday	360	140	220
Wednesday	240	170	70
Thursday	500	350	150
Friday	600	318	282

Answer: C

Value of L = 80

Value of Z = K = 400

So, Required value = 40% x (400 + 400) = 320

Hence, the answer is option C

5. Questions

For Monday

Total number of (Red + Pink) balls manufactured = 80 x 2 = 160

So, total number of balls manufactured = 160 x (100/40) = 400

Number of white balls manufactured = 60% x 400 = 240

Value of Z = 400

For Tuesday

Total number of balls manufactured = 760 – 400 = 360

Number of white balls manufactured = (50% x 760) – 240 = 140

Number of (Red + Pink) balls manufactured = (50% x 760) – 160 = 220

Now,

(3Y + 10) = 380/2

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3Y + 10 = 190

Value of Y = 180/3 = 60

For Wednesday

Total number of balls manufactured = 1000 – 760 = 240

Number of (Red + Pink) manufactured = (225 x 2) – 380 = 70

Number of white balls manufactured = 240 – 70 = 170

Now,

(L – 25) % = [(1000 – 450)/1000] x 100

L – 25 = 55

Value of L = 55 + 25 = 80

For Thursday

Total number of balls manufactured = 1500 – 1000 = 500

Number of white balls manufactured = (60% x 1500) – 550 = 350

Number of (Pink + Red) balls manufactured = (40% x 1500) – 450 = 150

Value of M = (40% x 1500)/2 = 300

For Friday

Total (Red + Pink) manufactured till Friday = 441 x 2 = 882

So, total number of balls manufactured till Friday = 882 x (100/42) = 2100

Total balls manufactured on Friday = 2100 – 1500 = 600

Number of white balls manufactured = (58% x 2100) – 900 = 318

Number of (Pink + Red) manufactured = 882 – 600 = 282

Now,

4K + 500 = 2100

Value of K = 1600/4 = 400

Days	Total balls	White balls	(Red + Pink)
Monday	400	240	160
Tuesday	360	140	220
Wednesday	240	170	70
Thursday	500	350	150
Friday	600	318	282

Tg:- @NextGenBankers

Answer: E

Number of pink balls manufactured on Wednesday = 70/2 = 35

So, number of pink balls manufactured on any other day, except Wednesday > 35

We need to find maximum number of red balls, so pink balls should be minimum.

So, minimum possible number of pink balls manufactured other than Wednesday = 36

So, maximum possible number of red balls manufactured on (Tuesday + Thursday) together = (220 – 36) + (150 – 36) = 298

Hence, the answer is option E

6. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: B

According to question,

Total time taken by R for back-and-forth journey = 11.25 x 2 = 22.5 hours

Due to engine problem, speed of R = 16 – 16 x 3/16 = 13 km/h

Now total time taken by R to travel from M to N = 128/16 + 52/13 = 12 hours

He has to wait for 2 hours at point N, so time left to travel from N to M for R = 22.5 - 12 – 2.5 = 8 hours

Now speed of R for N to M = 180/8 = 22.5 km/h

Required % change = (22.5 – 13)/13 x 100 = 73%

Hence answer is option B

7. Questions

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Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: C

Speed of R = 16 km/h

Speed of Q = 24 km/h

Required % change = (24 – 16)/24 x 100 = 33.33%

Hence answer is option C

8. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: D

Both Q and R, running on Route 2, distance between then initially = 200 km

When Q and R meets first time, they travelled = 200 km together

After they reached respective positions and then turns back, then they meet for the second time.

So, in second meeting, they together need to travel = 2 x 200 = 400 km

So, total distance covered = 200 + 400 = 600 km

Time is constant, so ratio of distance is the same as the ratio of speed.

Ratio of speed of Q and R = 24:16 = 3:2

Distance covered by R (N to M + M to second meeting point) = 2/5 x 600 = 240 km

So, distance between M and second meeting point = 240 – 200 = 40 km

Hence answer is option D

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9. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: E

Required total distance = 150 + 200 + 180 = 530 km

Hence answer is option E

10. Questions

Let, total number of chocolates that Arun and Vijay have is 12x and 11x, respectively.

Total number of chocolates that Vikram has = 11x/1.375 = 8x

Total number of chocolates that Kamal has = 8x + 208

Total number of chocolates that Surya has = 1.06 × 8x = 212x/25

So, 212x/25 – 8x = 72

12x = 72 × 25

x = 150

So, Total number of chocolates that Arun has = 12 × 150 = 1800

Total number of chocolates that Vijay has = 11 × 150 = 1650

Total number of chocolates that Vikram has = 8 × 150 = 1200

Total number of chocolates that Kamal has = 8x + 208 = 1408

Total number of chocolates that Surya has = 1200 × 106/100 = 1272

For Arun:

Number of coffy bite chocolates = 1800 – 792 × 2 = 216

Number of Eclairs chocolates = 468 × 2 – 216 = 720

Number of truffel chocolates = 792 × 2 – 720 = 864

Similarly, we can find number of chocolates that rest of the people has.

Tg:- @NextGenBankers

Persons	Number of Eclairs chocolates	Number of truffel chocolates	Number of Coffy Bite chocolates
Arun	720	864	216
Vijay	600	420	630
Vikram	384	480	336
Kamal	400	560	448
Surya	480	360	432

Answer: C

According to question,

Number of truffel chocolates that Arun has = 864

11. Questions

Let, total number of chocolates that Arun and Vijay have is 12x and 11x, respectively.

Total number of chocolates that Vikram has = 11x/1.375 = 8x

Total number of chocolates that Kamal has = 8x + 208

Total number of chocolates that Surya has = 1.06 × 8x = 212x/25

So, 212x/25 – 8x = 72

12x = 72 × 25

x = 150

So, Total number of chocolates that Arun has = 12 × 150 = 1800

Total number of chocolates that Vijay has = 11 × 150 = 1650

Total number of chocolates that Vikram has = 8 × 150 = 1200

Total number of chocolates that Kamal has = 8x + 208 = 1408

Total number of chocolates that Surya has = 1200 × 106/100 = 1272

For Arun:

Tg:- @NextGenBankers

Number of coffy bite chocolates = 1800 – 792 × 2 = 216

Number of Eclairs chocolates = 468 × 2 – 216 = 720

Number of truffel chocolates = 792 × 2 – 720 = 864

Similarly, we can find number of chocolates that rest of the people has.

Persons	Number of Eclairs chocolates	Number of truffel chocolates	Number of Coffy Bite chocolates
Arun	720	864	216
Vijay	600	420	630
Vikram	384	480	336
Kamal	400	560	448
Surya	480	360	432

Answer: B

Number of Eclairs chocolates that Arun has = 720

Number of Eclairs chocolates that Vijay has = 600

Required percentage = [(720 – 600)/600] × 100 = 20%

12. Questions

Let, total number of chocolates that Arun and Vijay have is 12x and 11x, respectively.

Total number of chocolates that Vikram has = 11x/1.375 = 8x

Total number of chocolates that Kamal has = 8x + 208

Total number of chocolates that Surya has = 1.06 × 8x = 212x/25

So, 212x/25 – 8x = 72

12x = 72 × 25

x = 150

So, Total number of chocolates that Arun has = 12 × 150 = 1800

Total number of chocolates that Vijay has = 11 × 150 = 1650

Total number of chocolates that Vikram has = 8 × 150 = 1200

Total number of chocolates that Kamal has = 8x + 208 = 1408

Total number of chocolates that Surya has = 1200 × 106/100 = 1272

For Arun:

Number of coffy bite chocolates = 1800 – 792 × 2 = 216

Number of Eclairs chocolates = 468 × 2 – 216 = 720

Number of truffel chocolates = 792 × 2 – 720 = 864

Similarly, we can find number of chocolates that rest of the people has.

Persons	Number of Eclairs chocolates	Number of truffel chocolates	Number of Coffy Bite chocolates
Arun	720	864	216
Vijay	600	420	630
Vikram	384	480	336
Kamal	400	560	448
Surya	480	360	432

Answer: D

Number of Coffy bite chocolates that Vijay has = 630

Number of Coffy bite chocolates that Vikram has = 336

Required Average = (630 + 336)/2 = 483

13. Questions

Let, total number of chocolates that Arun and Vijay have is 12x and 11x, respectively.

Total number of chocolates that Vikram has = 11x/1.375 = 8x

Total number of chocolates that Kamal has = 8x + 208

Total number of chocolates that Surya has = 1.06 × 8x = 212x/25

So, 212x/25 – 8x = 72

12x = 72 × 25

x = 150

So, Total number of chocolates that Arun has = 12 × 150 = 1800

Total number of chocolates that Vijay has = 11 × 150 = 1650

Total number of chocolates that Vikram has = 8 × 150 = 1200

Total number of chocolates that Kamal has = 8x + 208 = 1408

Total number of chocolates that Surya has = 1200 × 106/100 = 1272

For Arun:

Number of coffy bite chocolates = 1800 – 792 × 2 = 216

Number of Eclairs chocolates = 468 × 2 – 216 = 720

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Number of truffel chocolates = 792 × 2 – 720 = 864

Similarly, we can find number of chocolates that rest of the people has.

Persons	Number of Eclairs chocolates	Number of truffel chocolates	Number of Coffy Bite chocolates
Arun	720	864	216
Vijay	600	420	630
Vikram	384	480	336
Kamal	400	560	448
Surya	480	360	432

Answer: D

Number of Eclairs chocolates that Kamal has = 400

Number of Truffel chocolates that Kamal has = 560

Number of Coffy bite chocolates that Kamal has = 448

Required amount spent = 400 × 5 + 560 × 8 + 448 × 6 = Rs. 9168.

14. Questions

Let, total number of chocolates that Arun and Vijay have is 12x and 11x, respectively.

Total number of chocolates that Vikram has = 11x/1.375 = 8x

Total number of chocolates that Kamal has = 8x + 208

Total number of chocolates that Surya has = 1.06 × 8x = 212x/25

So, 212x/25 – 8x = 72

12x = 72 × 25

x = 150

So, Total number of chocolates that Arun has = 12 × 150 = 1800

Total number of chocolates that Vijay has = 11 × 150 = 1650

Total number of chocolates that Vikram has = 8 × 150 = 1200

Total number of chocolates that Kamal has = 8x + 208 = 1408

Total number of chocolates that Surya has = 1200 × 106/100 = 1272

For Arun:

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Number of coffy bite chocolates = 1800 – 792 × 2 = 216

Number of Eclairs chocolates = 468 × 2 – 216 = 720

Number of truffel chocolates = 792 × 2 – 720 = 864

Similarly, we can find number of chocolates that rest of the people has.

Persons	Number of Eclairs chocolates	Number of truffel chocolates	Number of Coffy Bite chocolates
Arun	720	864	216
Vijay	600	420	630
Vikram	384	480	336
Kamal	400	560	448
Surya	480	360	432

Answer: A

Number of Eclairs cookies that Surya has = 480

15. Questions

Answer: B

Marked price of both articles is same, so ratio of their selling price = 90%:60% = 3:2

Ratio of cost price of both articles = 3:2

If ratio of CP and SP is same, then profit % of both the articles must be same.

So, 2Z = 3Z – 25

Value of Z = 25

Required value = 25 + 15 = 40

Hence, the answer is option B

16. Questions

Answer: C

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Ratio of efficiency of P and Q = 2:1

Total work = 3 x 24 x 1 = 72 units

Amount of work done by Q in 40 days = 40 x 1 = 40 units

Reaming work = 72 – 40 = 32 units

Required time = 1/2 x (32/2) = 8 days

Hence answer is option C

17. Questions

Answer: D

According to the question,

Downstream speed of the boat A = 300/8 = 37.5 km/hr

Upstream speed of boat A = 300/12 = 25 km/hr

Speed of boat A = (37.5 + 25)/2 = 31.25 km/hr

Speed of stream = 31.25 – 25 = 6.25 km/hr

Speed of boat B = 31.25*(120/100) = 37.5 km/hr

Time taken by boat B = 300/ (37.50+6.25) = 300/43.75 = 6.85 hours

Hence, the answer is option D.

18. Questions

Answer: A

According to the question,

Probability of selecting one yellow ball out of two bags =1/2

Required probability = $[1/2*7c_1/20c_1] + [1/2*6c_1/14c_1] = 1/2 * (7/20+ 6/14) = 109/280$

Hence, the answer is option A.

19. Questions

Answer: D

According to the Question,

P x R% x 5 = 2 x [2P x (R – 15) % x 2]

5R = 8R – 120

3R = 120

Value of R = 40

So, ratio of amount on P and 3P = [P x 3/2 x 3/2 x 3/2]: [3P x 3 x 3] = 1:8

Required % = (1/8) x 100 = 12.5%

Hence, the answer is option D

20. Questions

Answer: B

According to the question,

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Ratio of profits received by 'A' and 'B' = 35:40 = 7:8

Difference between their profits = Rs.2000

Profit percentage for A and B = 100% - 10% - 15% = 75%

Profit percentage of A = 75% * (7/15) = 35%

Profit percentage of B = 75% * (8/15) = 40%

Difference in their profits = 40% - 35% = 5% = 2000

Total profit = 2000* (100/5) = Rs.40000.

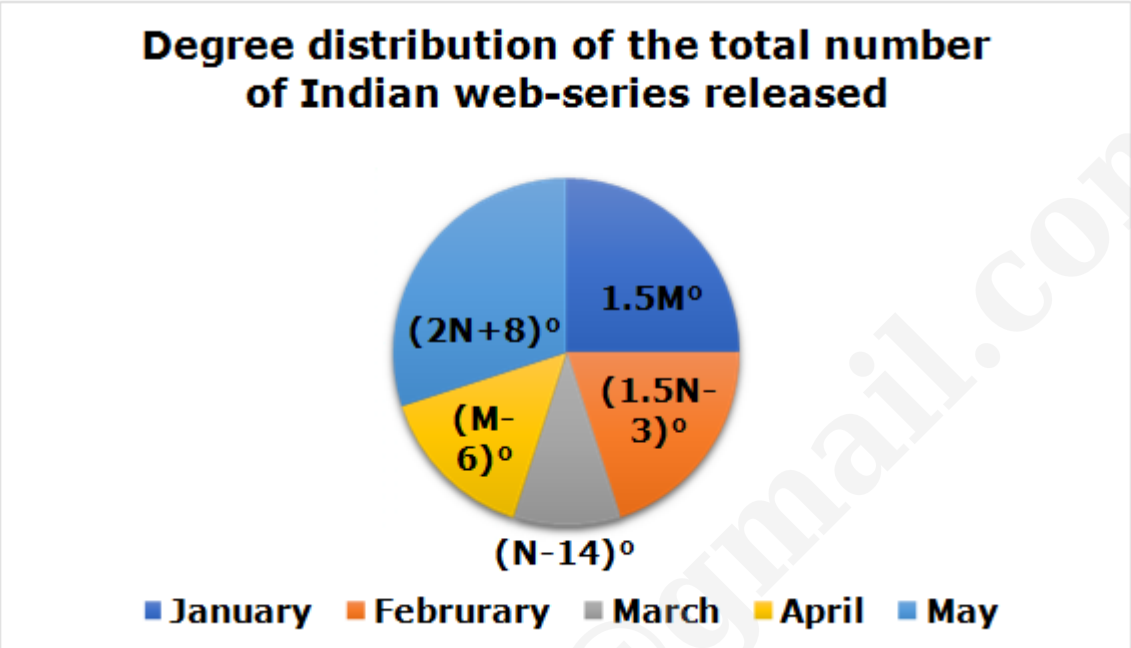
Hence, the option is B

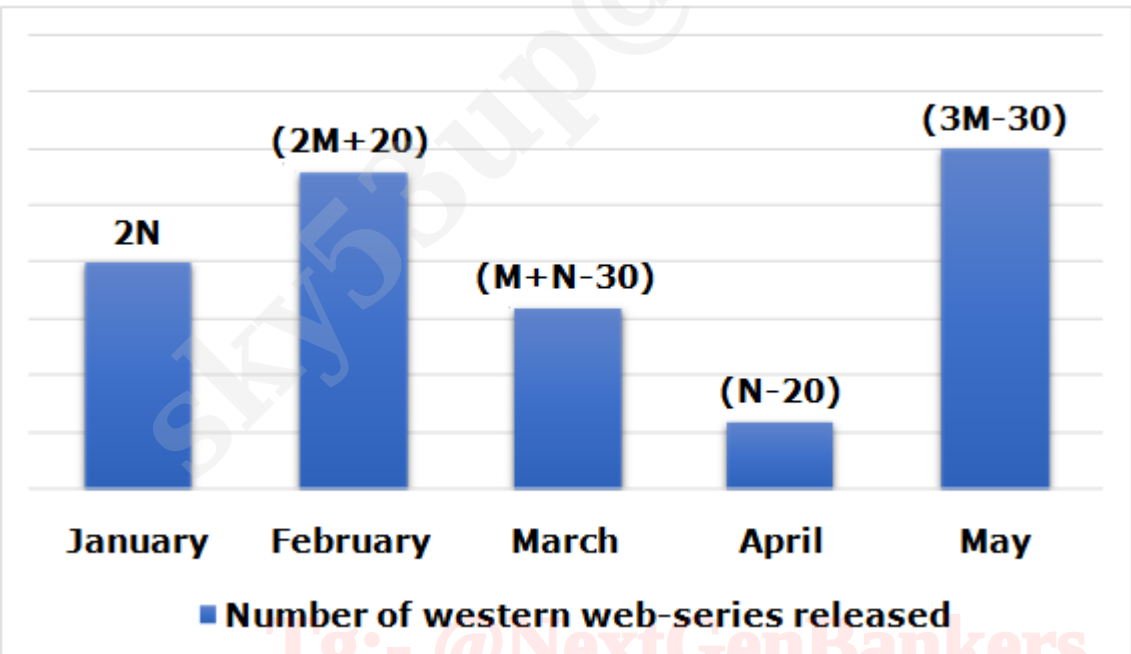
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1. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the degree distribution of the total number of Indian web-series released on an OTT platform in five different months of the year 2023 in India. The bar graph given below shows the number of western web-series released out of the total number of web-series (Indian + Western) released in India in each month.





Note:

The ratio of the total number of Indian to that of western web-series released in all the given five months together is 6:5.

The number of western web-series released in the month of March 2023 is 80.

The total number of web-series released in any month = number of Indian web-series released + number of

western web-series released

All the web-series (Indian + Western) released in India only in all the given five months of 2023.

For the questions given below, three statements I, II & III have been given. Analyse and answer whether the data provided in the three statements are sufficient to answer the question or not.

In June 2023, total number of web-series (Indian + Western) released is 30 less than that of previous month. There are only two types of Indian and western web-series released in the month of June 2023 i.e., Comedy and Thriller. Find the number of comedy western web-series released in the month of June 2023, if the total number of Indian web-series (comedy + thriller) released is 40 more than that of total number of western web-series (comedy + thriller) released.

Statement I: The ratio of the total number of comedy (Indian + Western) web-series released to that of total number of thriller (Indian + Western) web-series released in June 2023 is 1:1.

Statement II: The number of thriller western web-series released is 10 more than that of thriller Indian web-series released in June 2023.

Statement III: The number of comedy Indian web-series released is twice that of comedy western web-series released in June 2023.

a. Only statement II alone is sufficient, while statements I and III are not sufficient

b. Both statements I and II together are necessary, while statement III alone is not sufficient

c. Any of the two statements I and II or II and III or I and III together are sufficient

d. All the statements I, II and III together are necessary

e. Neither of the statements is sufficient

2. Questions

In January 2023, the ratio of the number of dubbed to that of non-dubbed Indian web-series released is 2:1. If the number of dubbed and non-dubbed western web-series released in the same month are P and Q respectively, then the ratio of the total number of dubbed (Indian + Western) to that non-dubbed (Indian + Western) web-series released becomes 17:8. Find the difference between the number of dubbed and non-dubbed western web-series released in January 2023.

a. 40

b. 24

c. 38

d. 52

e. None of these

3. Questions

In April 2023, out of the total number of Indian web-series released, the number of web-series watched and did not watched by the people are (50+L) and (60-K) respectively. Whereas, out of the total number western web-series released in the same month, number web series watched and

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didn't watched by the people are (K-20) and (L-20) respectively. Find the total number web-series (Indian + Western) watched by the people.

a. (3M-90)

b. 2N

c. (N+30)

d. M

e. None of these

4. Questions

In February 2023, the cost of purchasing the subscription of each Indian web-series is Rs. 5C for a package of 100 Indian web-series and cost of purchasing the subscription of each western web-series is Rs. 4D for a package of 100 western web-series. On purchase of each 10 Indian web-series extra, a customer has to pay an extra 10% of actual cost of each web-series. Whereas, on purchase of each 20 western web-series extra, a customer has to pay an extra 25% of actual cost of each web-series. Customer purchased all the Indian and western web-series that was released in the month of February 2023 and paid Rs. 1830 for all the Indian web-series released and Rs. 3000 for all the western web-series released. Find the C is what percent of D?

a. 50

b. 45

c. 25

d. 60

e. None of these

5. Questions

In May 2023, if out of the total number of Indian web-series released world-wide, (M+30)% was released in India and rest released in other countries and out of total number of western web-series released, (N+10)% was released in India and remaining released in other countries. Find the ratio of the total number of Indian web-series to that of western web-series released in other countries.

a. 5:2

b. 1:5

c. 3:4

d. 4:3

e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

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The table given below shows the number of chocolate ice creams sold by five different ice cream parlours J, K, L, M and N. The table also shows the percentage of the number of vanilla ice creams sold out of the total number of ice creams (chocolate + vanilla + strawberry) sold and the number of strawberry ice creams sold by each ice cream parlour.

Ice Cream Parlour	Number of chocolate ice creams sold	Percentage of the number of vanilla ice creams sold	Number of strawberry ice creams sold
J	150	35%	60 less than the number of vanilla ice cream sold
K	100	20%	Same as the number of chocolate ice creams sold
L	70	10%	28.57% of the number of chocolate ice creams sold
M	60	22.22%	$4/9^{th}$ of the total number of ice creams sold
N	216	20%	Half of the difference between the number of chocolate and vanilla ice creams sold (chocolate – vanilla)

Note:

Total number of ice creams sold = number of chocolate ice creams sold + number of vanilla ice creams sold + number of strawberry ice creams sold

Total number of ice creams made = number of ice creams sold + number of ice creams unsold

In ice cream parlour K, out of the total number of chocolate ice creams made, (P/6) ice creams were unsold, while out of the total number of vanilla ice creams made, (Q/6) ice creams were unsold and out of the total number of strawberry ice creams made, (2R/7) ice creams were unsold. The total numbers of chocolate, vanilla and strawberry ice creams made are P, Q and R respectively. Find the difference between the total number of sold (chocolate + vanilla + strawberry) and unsold (chocolate + vanilla + strawberry) ice creams by ice cream parlour K.

a. 160

b. 180

c. 150

d. 170

e. None of these

7. Questions

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In ice cream parlour M, cost of each chocolate, each vanilla and each strawberry ice cream sold is Rs. 2X, Rs. Y and Rs. X respectively. The total revenue generated from the selling all the chocolate and vanilla ice creams is Rs. 1800, whereas the total revenue generated from selling all the strawberry ice creams is Rs. 200 more than that of vanilla ice creams. Find the total revenue generated, if the ice cream parlour M had sold the ice creams (chocolate + vanilla + strawberry) combo for Rs. (X+Y).

a. Rs. 5100

b. Rs. 2000

c. Rs. 3200

d. Rs. 4700

e. None of these

8. Questions

If another ice cream parlour S, the number of vanilla ice cream sold is equal to that of ice parlour K and the number of strawberry ice cream sold in ice cream parlour S is V less than that of vanilla ice cream sold by the same ice cream parlour. Find what can be the possible number of chocolate ice creams sold by ice cream parlour S, if ice cream parlour S sold total of 5V ice creams (chocolate + vanilla + strawberry).

a. (V+20)

b. 3V

c. (6V-100)

d. V

e. None of these

9. Questions

For the questions given below, three cases I, II & III have been given. Analyse and answer whether the data provided in the three cases are satisfying the values of [S], [T] and [U] or not.

In ice cream parlour J and L, there is another verity of ice cream is sold i.e., cherry ice cream. In ice cream parlour J, number of cherry ice cream sold is ___[S]___% more than that of strawberry ice creams sold in the same ice creams parlour. Whereas, the number of cherry ice cream sold in the ice cream parlour L is ___[T]___% more than that of strawberry ice cream sold in the same ice cream parlour. Difference between the number of cherry ice creams sold in both the ice cream parlours J and L is ___[U]___.

Case I: S = 11.11, T = 25, U = 25

Case II: S = 20, T = 50, U = 24

Case III: S = 40, T = 10, U = 41

a. Only case I is satisfying

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b. Only case II is satisfying

c. Both cases II and III are satisfying

d. All the cases I, II and III are satisfying

e. Neither case I or II nor III is satisfying

10. Questions

In ice cream parlour N, the total number of ice creams (chocolate + vanilla + strawberry) sold on only three days i.e., Monday, Tuesday and Wednesday together of a certain week. The number of ice creams (chocolate + vanilla + strawberry) sold on Tuesday is 120. If the number of ice creams (chocolate + vanilla + strawberry) sold on each day is in arithmetic progression in increasing order starting from Monday, then find the number of ice creams sold on Wednesday. (Total number of ice creams sold on each day is more than 100 but less than 140).

a. Cannot be determined

b. 110

c. 120

d. 130

e. None of these

11. Questions

Following questions have three quantities as Quantity I, Quantity II and Quantity III. You have to determine the relationship between them and give an answer accordingly.

The length of train P is (A+40) meters, crosses a platform of length 80 meters in (C-2) seconds with the speed of 144 km/h. The length of train Q is (B-90) meters, crosses the same platform in (C-4) seconds with the same speed as that of train P.

Quantity I: If the sum of A and B is 450, then find the value of $[\sqrt{(B \times C)}]$.

Quantity II: Find the time taken by another train R whose length is $[6 \times (B-A)]$ meters to cross another platform, whose length is 50% more than that of given platform with the speed of train R is 21 m/sec.

Quantity III: Find the smaller root of the equation $[D^2 - \{(B-A)D\} + 600 = 0]$.

a. Quantity I = Quantity II < Quantity III

b. Quantity I > Quantity II = Quantity III

c. Quantity I < Quantity II < Quantity III

d. Quantity I > Quantity II > Quantity III

e. Quantity I = Quantity II = Quantity III

12. Questions

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Quantity I: A person has a total amount of Rs. 40000. He invests one part in scheme X at the rate of 25% per annum compounded annually for 2 years and other at the rate of 27% per annum simple interest for 3 years in scheme Y. Simple interest earned is Rs. 10440 more than that of compound interest earned. Find the difference between the amount invested in both the schemes.

Quantity II: The difference between the compound and simple interest earned on a certain sum of money at the rate 15% per annum after 2 years is Rs. 450. Find the simple interest earned on the same sum at the rate of 20% per annum after 4 years.

Quantity III: Find the value of [100×(S-R)], if (S-√1600) = R.

a. Quantity I < Quantity II > Quantity III
b. Quantity I > Quantity II = Quantity III
c. Quantity I = Quantity II < Quantity III
d. Quantity I > Quantity II < Quantity III
e. Quantity I = Quantity II = Quantity III

13. Questions

There are two schools M and N. The ratio of the number of boys to that of girls in school M is 2:1. The number of girls in both the schools are equal. The number of boys in school N is 40 more than that of girls in the same school. The total number of students (boys + girls) participated in sports competition from school M is 200 more than that of who didn't participated in the sports competition from the same school. The total number of students (boys + girls) participated in sports competition from school N is 70 less than that of school M. The total number of students (boys + girls) who didn't participated in sports competition from school N is 10 more than that of school M.

Quantity I: The difference between the total number of students (boys + girls) in both the schools.

Quantity II: The average of total number of girls in both the schools together.

Quantity III: The sum of the total number of students (boys + girls) who didn't participated in the sports competition from both the schools together.

a. Quantity I > Quantity II > Quantity III
b. Quantity I > Quantity II < Quantity III
c. Quantity I = Quantity II = Quantity III
d. Quantity I < Quantity II > Quantity III
e. Quantity I < Quantity II < Quantity III

14. Questions

Quantity I: P, Q and R invested Rs. K, Rs. 2K and Rs. (K+2000) in a business. After 6 months, P withdrew Rs. 1200, whereas R added Rs. 1000 to his initial investment. After one year, profit share of P is Rs. 4800 out of total the total profit of Rs. 27800. Find the initial investment of Q is what percent more or

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less than that of R.

Quantity II: S invested Rs. 1600 in a business for X months, while T invested Rs. 200 more than the 125% of S's investment in the same business for Y months. At the end of (X+Y-6) months, S and T received profit in the ratio of 32:55 respectively. Find the value of (X+Y-6), if difference between the values of X and Y is 2.

Quantity III: Find the value of [(M+N)/10)], if 5/6 of N = M, where (3M + 2N) = 810.

a. Quantity I < Quantity II < Quantity III
b. Quantity I > Quantity II < Quantity III
c. Quantity I < Quantity II > Quantity III
d. Quantity I > Quantity II > Quantity III
e. Quantity I = Quantity II = Quantity III

15. Questions

Pipe A and pipe B together can fill the tank in (60/7) minutes. Whereas, pipe C can fill the tank alone in 25 minutes. Pipe B and pipe D together can fill the same tank in 6 minutes. Difference between the efficiencies of A and C is 20% of that of A and D. Efficiency of A is more than that of C.

Quantity I: Find the time taken by pipe B alone to fill the tank.

Quantity II: Efficiency of C is what percent of that of D?

Quantity III: Sum of the efficiencies of B and C together.

a. Quantity I < Quantity II < Quantity III
b. Quantity I > Quantity II < Quantity III
c. Quantity I = Quantity II = Quantity III
d. Quantity I < Quantity II > Quantity III
e. Quantity I > Quantity II > Quantity III

16. Questions

Following questions have two series as series I and series II. You have to find the logic of the series and give an answer accordingly.

A and B are the missing terms of series I and series II respectively. Which of the following statement(s) is/are true?

Series I: 24, 50, A, 770, 5397, 59378

Series II: 60, 80, 122, B, 304, 460

Statement I: Both A and B are multiples of 9.

Statement II: If C is the sum of the digits of B, then the value of (C^2-2) = B.

Statement III: A is an odd composite number

a. Only statement II is true

b. Both statements I and II and true

c. Both statements II and III are true

d. All the statements I, II and III are true

e. Neither statement I or II nor III is true

17. Questions

There are two series I and II are given. Both series follows the same pattern. There is a wrong term in series I. First term of series II starts with the correct term which replaces the wrong term of series I. M is the correct term which replaces the wrong term in series I, whereas 6th term of series II is N. Which of the following equation will satisfy the values of M and N is correct manner?

Series I: 35, 120, 185, 325, 425, 535

Series II: M, ___, ___, ___, ___, N

a. (M+N) = 780

b. 5M = 2N

c. (N-450) = M

d. All of the above

e. None of these

18. Questions

Both series I and II follows same logic. If J is the resultant of the sum of LCM and HCF of K and L, then find the value of 2J.

Series I: 10, __[K]__, 135, 360, 760, 1385

Series II: __[K]__, ___, __[L]__, ___, ___, 1410

a. 1125

b. 2250

c. 1120

d. 2245

e. None of these

19. Questions

Following questions have three series as series I, series II and series III. You have to find the logic

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of the series and give an answer accordingly.

X, Y and Z are the missing terms of series I, II and III respectively. Find the value of [√Z+{√(X-1)}+{√(Y+302)}].

Series I: [X], 75, 125, 200, 300, 425

Series II: 1332, [Y], 3376, 4914, 6860, 9262

Series III: 324, 972, 486, 1458, [Z], 2187

a. 53

b. 26

c. 37

d. 84

e. None of these

20. Questions

There is one wrong term in each series I, II and III. Wrong term of series I, II and III is denoted as P, Q and R respectively. S, T and U will replace the wrong terms P, Q and R respectively as correct terms. Find the correct relation between P, Q, R, S, T and U.

Series I: 32, 46, 74, 130, 239, 466

Series II: 70, 144, 437, 1755, 8777, 52670

Series III: 124, 60, 65, 134, 544, 4368

Relation I: P < S, Q > T, R < U

Relation II: P > S, Q > T, R > U

Relation III: P < S, Q < T, R < U

a. Both relations II and III are correct

b. Only relation I is correct

c. Only relation II is correct

d. All the relations I, II and III are correct

e. Neither relation I or II nor III is correct

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Explanations:

1. Questions

From the given pie chart, we have,

Total degree distribution of the pie chart = 360°

1.5M + (1.5N-3) + (N-14) + (M-6) + (2N+8) = 360

2.5M + 4.5N = 360 + 15

2.5M + 4.5N = 375……..(i)

From the bar graph, we have,

Total number of western web-series released in March 2023 = 80

M + N – 30 = 80

M + N = 110…….(ii)

Solving both the equations, we get,

2.5M + 4.5N = 375

2.5M + 2.5N = 275

2N = 100

N = 50

Now, put the value of N in equation (i), we get,

2.5M + 4.5N = 375

2.5M = 375 – 225

2.5M = 150

M = 60

Now, the total number of western web-series released in each month,

January = 2N = 100

February = (2M+20) = (120+20) = 140

March = (M+N-30) = (110-30) = 80

April = (N-20) = (50-20) = 30

May = (3M-30) = (180-30) = 150

Total number of western web-series in all the months together = (100+140+80+30+150) = 500

So, the total number of Indian web-series in all the months together = {(500×6)/5} = 600

Total number of Indian web-series released in each month,

January = 1.5M° of 600 = 90° of 600 = {(90×600)/360} = 150

February = (1.5N-3)° of 600 = 72° of 600 = {(72×600)/360} = 120

March = (N-14)° of 600 = 36° of 600 = {(36×600)/360} = 60

April = (M-6)° of 600 = 54° of 600 = {(54×600)/360} = 90

May = (2N+8)° of 600 = 108° of 600 = {(108×600)/360} = 180

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Month	Total number of web-series (Indian + Western) released	Number of Indian web-series released	Number of western web-series released
January	250	150	100
February	260	120	140
March	140	60	80
April	120	90	30
May	330	180	150
Total	1100	600	500

Answer: C

According to the question,

In June 2023,

Total number of web-series (Indian + Western) released = (330-30) = 300

Let, the total number of Indian and western web-series released be x and (x-40)

So,

2x – 40 = 300

2x = 340

x = 170

So, the total number of Indian web-series (comedy + thriller) released = x = 170

And, the total number of western web-series (comedy + thriller) released = (x-40) = (170-40) = 130

Using statements I and II together

Total number of comedy web-series (Indian + Western) released = 1/2 of 300 = 150

Total number of thriller web-series (Indian + Western) released = 1/2 of 300 = 150

Let, the number of thriller western and Indian web-series released be a and (a-10)

2a – 10 = 150

2a = 160

a = 80

Number of thriller Indian movies released = (a-10) = (80-10) = 70

So, the number of comedy Indian movies released = (170-70) = 100

Number of thriller western movies released = a = 80

Hence, the number of comedy western movies released = (130-80) = 50

So, statements I and II together are necessary.

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Using statements II and III together

Let, the number of thriller western and Indian web-series released be b and (b-10)

So, the number of comedy Indian web-series released = {170-(b-10)} = (180-b)

And, the number of comedy western web-series released = (130-b)

(180-b)/(130-b) = 2/1

180 – b = 260 – 2b

b = 80

Hence, the number of comedy western web-series released = (130-b) = (130-80) = 50

So, statements II and III together are necessary.

Using statements I and III together

Total number of comedy web-series (Indian + Western) released = 1/2 of 300 = 150

Ratio of the number of comedy Indian web-series to that of comedy western web-series released = 2:1

Hence, the number of comedy western web-series released = 1/3 of 150 = 50

So, statements I and II together are necessary.

Therefore, combination of any of the two statements together are sufficient to answer the question.

2. Questions

From the given pie chart, we have,

Total degree distribution of the pie chart = 360°

1.5M + (1.5N-3) + (N-14) + (M-6) + (2N+8) = 360

2.5M + 4.5N = 360 + 15

2.5M + 4.5N = 375……..(i)

From the bar graph, we have,

Total number of western web-series released in March 2023 = 80

M + N – 30 = 80

M + N = 110…….(ii)

Solving both the equations, we get,

2.5M + 4.5N = 375

2.5M + 2.5N = 275

2N = 100

N = 50

Now, put the value of N in equation (i), we get,

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2.5M + 4.5N = 375

2.5M = 375 – 225

2.5M = 150

M = 60

Now, the total number of western web-series released in each month,

January = 2N = 100

February = (2M+20) = (120+20) = 140

March = (M+N-30) = (110-30) = 80

April = (N-20) = (50-20) = 30

May = (3M-30) = (180-30) = 150

Total number of western web-series in all the months together = (100+140+80+30+150) = 500

So, the total number of Indian web-series in all the months together = {(500×6)/5} = 600

Total number of Indian web-series released in each month,

January = 1.5M° of 600 = 90° of 600 = {(90×600)/360} = 150

February = (1.5N-3)° of 600 = 72° of 600 = {(72×600)/360} = 120

March = (N-14)° of 600 = 36° of 600 = {(36×600)/360} = 60

April = (M-6)° of 600 = 54° of 600 = {(54×600)/360} = 90

May = (2N+8)° of 600 = 108° of 600 = {(108×600)/360} = 180

Month	Total number of web-series (Indian + Western) released	Number of Indian web-series released	Number of western web-series released
January	250	150	100
February	260	120	140
March	140	60	80
April	120	90	30
May	330	180	150
Total	1100	600	500

Answer: A

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In January 2023,

Indian web-series

Total number of web-series released = 150

Number of dubbed web-series released = 2/3 of 150 = 100

Number of non-dubbed web-series released = 1/3 of 150 = 50

Western web-series

Total number of web-series released = 100

Number of dubbed web-series released = P

Number of non-dubbed web-series released = Q

So,

P + Q = 100……(i)

And,

(100+P)/(50+Q) = 17/8

800 + 8P = 850 + 17Q

8P – 17Q = 50…….(ii)

Solving both the equations, we get,

8P + 8Q = 800

8P – 17Q = 50

25Q = 750

Q = 30

Now, put the value of Q in equation (i), we get,

P + Q = 100

P = 100 – 30

P = 70

Hence, difference between the dubbed and non-dubbed western web-series released in January 2023 = (P-Q) = (70-30) = 40.

3. Questions

From the given pie chart, we have,

Total degree distribution of the pie chart = 360°

1.5M + (1.5N-3) + (N-14) + (M-6) + (2N+8) = 360

2.5M + 4.5N = 360 + 15

2.5M + 4.5N = 375……..(i)

From the bar graph, we have,

Total number of western web-series released in March 2023 = 80

M + N – 30 = 80

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M + N = 110……(ii)

Solving both the equations, we get,

2.5M + 4.5N = 375

2.5M + 2.5N = 275

2N = 100

N = 50

Now, put the value of N in equation (i), we get,

2.5M + 4.5N = 375

2.5M = 375 – 225

2.5M = 150

M = 60

Now, the total number of western web-series released in each month,

January = 2N = 100

February = (2M+20) = (120+20) = 140

March = (M+N-30) = (110-30) = 80

April = (N-20) = (50-20) = 30

May = (3M-30) = (180-30) = 150

Total number of western web-series in all the months together = (100+140+80+30+150) = 500

So, the total number of Indian web-series in all the months together = {(500×6)/5} = 600

Total number of Indian web-series released in each month,

January = 1.5M° of 600 = 90° of 600 = {(90×600)/360} = 150

February = (1.5N-3)° of 600 = 72° of 600 = {(72×600)/360} = 120

March = (N-14)° of 600 = 36° of 600 = {(36×600)/360} = 60

April = (M-6)° of 600 = 54° of 600 = {(54×600)/360} = 90

May = (2N+8)° of 600 = 108° of 600 = {(108×600)/360} = 180

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Month	Total number of web-series (Indian + Western) released	Number of Indian web-series released	Number of western web-series released
January	250	150	100
February	260	120	140
March	140	60	80
April	120	90	30
May	330	180	150
Total	1100	600	500

Answer: B

In April 2023,

Indian web-series

Total number of web-series released = 90

Number of web-series watched by the people = (50+L)

Number of web-series didn't watched by the people = (60-K)

Western web-series

Total number of web-series released = 30

Number of web-series watched by the people = (K-20)

Number of web-series didn't watched by the people = (L-20)

Now,

(50+L) + (60-K) = 90

110 + L = 90 + K

K – L = 20……..(i)

Also,

(K-20) + (L-20) = 30

K – 20 = 50 - L

K + L = 70……(ii)

Solving both the equations, we get,

2L = 50

L = 25

Now, put the value of L in equation (i), we get,

K – L = 20

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K = 20 + 25

K = 45

Hence, the total number of web-series (Indian + Western) watched by the people = {(50+L)+(K-20)} = {(50+25)+(45-20)} = (75+25) = 100.

So,

(3M-90) = (180-90) = 90 (not satisfied)

2N = 100 (satisfied)

(N+30) = 80 (not satisfied)

M = 60 (not satisfied)

Hence, option B is the correct answer.

4. Questions

From the given pie chart, we have,

Total degree distribution of the pie chart = 360°

1.5M + (1.5N-3) + (N-14) + (M-6) + (2N+8) = 360

2.5M + 4.5N = 360 + 15

2.5M + 4.5N = 375……..(i)

From the bar graph, we have,

Total number of western web-series released in March 2023 = 80

M + N – 30 = 80

M + N = 110…….(ii)

Solving both the equations, we get,

2.5M + 4.5N = 375

2.5M + 2.5N = 275

2N = 100

N = 50

Now, put the value of N in equation (i), we get,

2.5M + 4.5N = 375

2.5M = 375 – 225

2.5M = 150

M = 60

Now, the total number of western web-series released in each month,

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January = 2N = 100

February = (2M+20) = (120+20) = 140

March = (M+N-30) = (110-30) = 80

April = (N-20) = (50-20) = 30

May = (3M-30) = (180-30) = 150

Total number of western web-series in all the months together = (100+140+80+30+150) = 500

So, the total number of Indian web-series in all the months together = {(500×6)/5} = 600

Total number of Indian web-series released in each month,

January = 1.5M° of 600 = 90° of 600 = {(90×600)/360} = 150

February = (1.5N-3)° of 600 = 72° of 600 = {(72×600)/360} = 120

March = (N-14)° of 600 = 36° of 600 = {(36×600)/360} = 60

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May = (2N+8)° of 600 = 108° of 600 = {(108×600)/360} = 180

Month	Total number of web-series (Indian + Western) released	Number of Indian web-series released	Number of western web-series released
January	250	150	100
February	260	120	140
March	140	60	80
April	120	90	30
May	330	180	150
Total	1100	600	500

Answer: D

In February 2023,

Total number of Indian web-series released = 120

Total number of western web-series released = 140

Total number of Indian web-series purchased = 120

Total number of western web-series purchased = 140

Total amount paid by the customer on purchase of a package of 100 Indian web-series = (5C×100) = Rs. 500C

Cost of purchasing each extra 10 Indian web-series = 110% of 5C = Rs. 5.5C

But the customer purchased an extra of 20 Indian web-series.

Extra amount paid by the customer on purchase of extra 20 Indian web-series = {5.5C×20} = Rs. 110C

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Total of cost of purchasing all the 120 Indian web-series = (500C+110C) = Rs. 610C

So,

610C = 1830

C = 3

Now,

Total amount paid by the customer on purchase of a package of 100 western web-series = (4D×100) = Rs. 400D

Cost of purchasing each extra 20 western web-series = 125% of 4D = Rs. 5D

But the customer purchased an extra of 40 western web-series.

Extra amount paid by the customer on purchase of extra 40 western web-series = {(5D×40} = Rs. 200D

Total of cost of purchasing all the 120 western web-series = (400D+200D) = Rs. 600D

So,

600D = 3000

D = 5

Hence, required % = {(C/D)×100}% = {(3/5)×100}% = 60%.

5. Questions

From the given pie chart, we have,

Total degree distribution of the pie chart = 360°

1.5M + (1.5N-3) + (N-14) + (M-6) + (2N+8) = 360

2.5M + 4.5N = 360 + 15

2.5M + 4.5N = 375……..(i)

From the bar graph, we have,

Total number of western web-series released in March 2023 = 80

M + N – 30 = 80

M + N = 110…….(ii)

Solving both the equations, we get,

2.5M + 4.5N = 375

2.5M + 2.5N = 275

2N = 100

N = 50

Now, put the value of N in equation (i), we get,

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2.5M + 4.5N = 375

2.5M = 375 – 225

2.5M = 150

M = 60

Now, the total number of western web-series released in each month,

January = 2N = 100

February = (2M+20) = (120+20) = 140

March = (M+N-30) = (110-30) = 80

April = (N-20) = (50-20) = 30

May = (3M-30) = (180-30) = 150

Total number of western web-series in all the months together = (100+140+80+30+150) = 500

So, the total number of Indian web-series in all the months together = {(500×6)/5} = 600

Total number of Indian web-series released in each month,

January = 1.5M° of 600 = 90° of 600 = {(90×600)/360} = 150

February = (1.5N-3)° of 600 = 72° of 600 = {(72×600)/360} = 120

March = (N-14)° of 600 = 36° of 600 = {(36×600)/360} = 60

April = (M-6)° of 600 = 54° of 600 = {(54×600)/360} = 90

May = (2N+8)° of 600 = 108° of 600 = {(108×600)/360} = 180

Month	Total number of web-series (Indian + Western) released	Number of Indian web-series released	Number of western web-series released
January	250	150	100
February	260	120	140
March	140	60	80
April	120	90	30
May	330	180	150
Total	1100	600	500

Answer: B

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In May 2023,

Total number of Indian web-series released in India = 180

(M+30)% = 180

(60+30)% = 180 (M = 60, from common explanation)

90% = 180

1% = (180/90) = 2

100% = (2×100) = 200

So, the total number of Indian web-series released world-wide = 200

Therefore, number of Indian web-series released in other countries = (200-180) = 20

Total number of western web-series released in India = 150

(N+10)% = 150

(50+10)% = 150 (N = 50, from common explanation)

60% = 150

1% = (150/60) = 2.5

100% = (2.5×100) = 250

So, the total number of western web-series released world-wide = 250

Therefore, number of western web-series released in other countries = (250-150) = 100

Hence, required ratio = 20:100 = 1:5.

6. Questions

In ice cream parlour J

Let, the total number of ice creams sold be 100a

So, the number of vanilla ice creams sold = 35% of 100a = 35a

Number of chocolate ice creams sold = 150

Therefore, the number of strawberry ice creams sold = (35a-60)

Now,

150 + 35a + (35a-60) = 100a

70a = 100a – 90

30a = 90

a = 3

So,

The total number of ice creams sold be 100a = 300

Number of chocolate ice creams sold = 150

Number of vanilla ice creams sold = 35a = 105

Number of strawberry ice creams sold = (35a-60) = (105-60) = 45

In ice cream parlour K

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Total number of ice creams sold = 100%

Number of chocolate ice creams sold = 100

Number of strawberry ice creams sold = 100

Number of vanilla ice creams sold = 20%

So,

The percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-20)% = 80%

80% = 100 + 100

80% = 200

1% = (200/80) = 2.5

100% = (2.5×100) = 250

Therefore, the total number of ice creams sold = 250

Number of chocolate ice creams sold = 100

Number of vanilla ice creams sold = 20% of 250 = 50

Number of strawberry ice creams sold = 100

Ice cream parlour L

Number of chocolate ice cream sold = 70

Number of strawberry ice cream sold = 2/7 of 70 = 20 *(Fractional value of 28.57% = 2/7)*

Percentage of the number of vanilla ice creams sold = 10%

So, the percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-10)% = 90%

90% = 70 + 20

90% = 90

1% = (90/90) = 1

100% = (1×100) = 100

Therefore, the total number of ice creams sold = 100

Number of chocolate ice creams sold = 70

Number of vanilla ice creams sold = 10% of 100 = 10

Number of strawberry ice creams sold = 20

Ice cream parlour M

Fractional value of 22.22% = 2/9

Let, the total number of ice creams sold be 9b.

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So, the number of vanilla ice cream sold = 2/9 of 9b = 2b

And, number of strawberry ice creams sold = 4/9 of 9b = 4b

Therefore, the number of chocolate ice creams sold = {9b-(2b+4b)} = (9b-6b) = 3b

3b = 60

b = (60/3) = 20

Therefore, the total number of ice creams sold = 9b = 180

Number of chocolate ice creams sold = 60

Number of vanilla ice creams sold = 2b = 40

Number of strawberry ice creams sold = 4b = 80

Ice cream parlour N

Let, the total number of ice creams sold be 100c.

So, the number of vanilla ice creams sold = 20% of 100c = 20c

Number of chocolate ice creams sold = 216

Therefore, the number of strawberry ice creams sold = {(216-20c)/2} = (108-10c)

Now,

216 + 20c + (108-10c) = 100c

90c = 324

c = (324/90) = (18/5)

Therefore, the total number of ice creams sold = 100c = {(100×18)/5} = 360

Number of chocolate ice creams sold = 216

Number of vanilla ice creams sold = 20% of 360 = 72

Number of strawberry ice creams sold = (108-10c) = [108-{(10×18)/5}] = (108-36) = 72

Ice Cream Parlour	Total number of ice creams sold	Number of chocolate ice creams sold	Number of vanilla ice creams sold	Number of strawberry ice creams sold
J	300	150	105	45
K	250	100	50	100
L	100	70	10	20
M	180	60	40	80
N	360	216	72	72
Total	1190	596	277	317

Answer: B

According to the question,

In ice cream parlour K,

Chocolate

Total number of ice creams made = P

Number of ice creams sold = 100

Number of ice creams unsold = (P/6)

100 + (P/6) = P

P – (P/6) = 100

(6P-P)/6 = 100

5P = 600

P = 120

Vanilla

Total number of ice creams made = Q

Number of ice creams sold = 50

Number of ice creams unsold = (Q/6)

50 + (Q/6) = Q

Q – (Q/6) = 50

(6Q-Q)/6 = 50

5Q = 300

Q = 60

Strawberry

Total number of ice creams made = R

Number of ice creams sold = 100

Number of ice creams unsold = (2R/7)

100 + (2R/7) = R

R – (2R/7) = 100

(7R-2R)/7 = 100

5R = 700

R = 140

Now,

Total number of ice creams (chocolate + vanilla + strawberry) sold = (100+50+100) = 250

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Total number of ice creams (chocolate + vanilla + strawberry) unsold = {(P/6)+(Q/6)+(2R/7)} = {(120/6)+(60/6)+(280/7)} = (20+10+40) = 70

Hence, required difference = (250-70) = 180.

7. Questions

In ice cream parlour J

Let, the total number of ice creams sold be 100a

So, the number of vanilla ice creams sold = 35% of 100a = 35a

Number of chocolate ice creams sold = 150

Therefore, the number of strawberry ice creams sold = (35a-60)

Now,

150 + 35a + (35a-60) = 100a

70a = 100a – 90

30a = 90

a = 3

So,

The total number of ice creams sold be 100a = 300

Number of chocolate ice creams sold = 150

Number of vanilla ice creams sold = 35a = 105

Number of strawberry ice creams sold = (35a-60) = (105-60) = 45

In ice cream parlour K

Total number of ice creams sold = 100%

Number of chocolate ice creams sold = 100

Number of strawberry ice creams sold = 100

Number of vanilla ice creams sold = 20%

So,

The percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-20)% = 80%

80% = 100 + 100

80% = 200

1% = (200/80) = 2.5

100% = (2.5×100) = 250

Therefore, the total number of ice creams sold = 250

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Number of chocolate ice creams sold = 100

Number of vanilla ice creams sold = 20% of 250 = 50

Number of strawberry ice creams sold = 100

Ice cream parlour L

Number of chocolate ice cream sold = 70

Number of strawberry ice cream sold = 2/7 of 70 = 20 *(Fractional value of 28.57% = 2/7)*

Percentage of the number of vanilla ice creams sold = 10%

So, the percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-10)% = 90%

90% = 70 + 20

90% = 90

1% = (90/90) = 1

100% = (1×100) = 100

Therefore, the total number of ice creams sold = 100

Number of chocolate ice creams sold = 70

Number of vanilla ice creams sold = 10% of 100 = 10

Number of strawberry ice creams sold = 20

Ice cream parlour M

Fractional value of 22.22% = 2/9

Let, the total number of ice creams sold be 9b.

So, the number of vanilla ice cream sold = 2/9 of 9b = 2b

And, number of strawberry ice creams sold = 4/9 of 9b = 4b

Therefore, the number of chocolate ice creams sold = {9b-(2b+4b)} = (9b-6b) = 3b

3b = 60

b = (60/3) = 20

Therefore, the total number of ice creams sold = 9b = 180

Number of chocolate ice creams sold = 60

Number of vanilla ice creams sold = 2b = 40

Number of strawberry ice creams sold = 4b = 80

Ice cream parlour N

Let, the total number of ice creams sold be 100c.

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So, the number of vanilla ice creams sold = 20% of 100c = 20c

Number of chocolate ice creams sold = 216

Therefore, the number of strawberry ice creams sold = {(216-20c)/2} = (108-10c)

Now,

216 + 20c + (108-10c) = 100c

90c = 324

c = (324/90) = (18/5)

Therefore, the total number of ice creams sold = 100c = {(100×18)/5} = 360

Number of chocolate ice creams sold = 216

Number of vanilla ice creams sold = 20% of 360 = 72

Number of strawberry ice creams sold = (108-10c) = [108-{(10×18)/5}] = (108-36) = 72

Ice Cream Parlour	Total number of ice creams sold	Number of chocolate ice creams sold	Number of vanilla ice creams sold	Number of strawberry ice creams sold
J	300	150	105	45
K	250	100	50	100
L	100	70	10	20
M	180	60	40	80
N	360	216	72	72
Total	1190	596	277	317

Answer: E

In ice cream parlour M,

Total number of chocolate ice cream sold = 60

Total revenue generated from selling all the chocolate ice creams = (60×2X) = Rs. 120X

Total number of vanilla ice cream sold = 40

Total revenue generated from selling all the vanilla ice creams = (40×Y) = Rs. 40Y

Total number of strawberry ice cream sold = 80

Total revenue generated from selling all the strawberry ice creams = (80×X) = Rs. 80X

Now,

120X + 40Y = 1800……(i)

And,

80X – 40Y = 200……(ii)

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Solving both the equations, we get,

240X + 80Y = 3600

240X – 120Y = 600

200Y = 3000

Y = 15

Now, put the value of Y in equation (i), we get,

120X + 40Y = 1800

120X = 1800 – 600

120X = 1200

X = 10

Now, total number of ice creams (chocolate + vanilla + strawberry) sold = (60+40+80) = 180

Hence, total revenue generated from selling all the ice creams together for Rs. (X+Y) = {180×(X+Y)} = (180×25) = Rs. 4500.

8. Questions

In ice cream parlour J

Let, the total number of ice creams sold be 100a

So, the number of vanilla ice creams sold = 35% of 100a = 35a

Number of chocolate ice creams sold = 150

Therefore, the number of strawberry ice creams sold = (35a-60)

Now,

150 + 35a + (35a-60) = 100a

70a = 100a – 90

30a = 90

a = 3

So,

The total number of ice creams sold be 100a = 300

Number of chocolate ice creams sold = 150

Number of vanilla ice creams sold = 35a = 105

Number of strawberry ice creams sold = (35a-60) = (105-60) = 45

In ice cream parlour K

Total number of ice creams sold = 100%

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Number of chocolate ice creams sold = 100

Number of strawberry ice creams sold = 100

Number of vanilla ice creams sold = 20%

So,

The percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-20)% = 80%

80% = 100 + 100

80% = 200

1% = (200/80) = 2.5

100% = (2.5×100) = 250

Therefore, the total number of ice creams sold = 250

Number of chocolate ice creams sold = 100

Number of vanilla ice creams sold = 20% of 250 = 50

Number of strawberry ice creams sold = 100

Ice cream parlour L

Number of chocolate ice cream sold = 70

Number of strawberry ice cream sold = 2/7 of 70 = 20 *(Fractional value of 28.57% = 2/7)*

Percentage of the number of vanilla ice creams sold = 10%

So, the percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-10)% = 90%

90% = 70 + 20

90% = 90

1% = (90/90) = 1

100% = (1×100) = 100

Therefore, the total number of ice creams sold = 100

Number of chocolate ice creams sold = 70

Number of vanilla ice creams sold = 10% of 100 = 10

Number of strawberry ice creams sold = 20

Ice cream parlour M

Fractional value of 22.22% = 2/9

Let, the total number of ice creams sold be 9b.

So, the number of vanilla ice cream sold = 2/9 of 9b = 2b

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And, number of strawberry ice creams sold = 4/9 of 9b = 4b

Therefore, the number of chocolate ice creams sold = {9b-(2b+4b)} = (9b-6b) = 3b

3b = 60

b = (60/3) = 20

Therefore, the total number of ice creams sold = 9b = 180

Number of chocolate ice creams sold = 60

Number of vanilla ice creams sold = 2b = 40

Number of strawberry ice creams sold = 4b = 80

Ice cream parlour N

Let, the total number of ice creams sold be 100c.

So, the number of vanilla ice creams sold = 20% of 100c = 20c

Number of chocolate ice creams sold = 216

Therefore, the number of strawberry ice creams sold = {(216-20c)/2} = (108-10c)

Now,

216 + 20c + (108-10c) = 100c

90c = 324

c = (324/90) = (18/5)

Therefore, the total number of ice creams sold = 100c = {(100×18)/5} = 360

Number of chocolate ice creams sold = 216

Number of vanilla ice creams sold = 20% of 360 = 72

Number of strawberry ice creams sold = (108-10c) = [108-{(10×18)/5}] = (108-36) = 72

Ice Cream Parlour	Total number of ice creams sold	Number of chocolate ice creams sold	Number of vanilla ice creams sold	Number of strawberry ice creams sold
J	300	150	105	45
K	250	100	50	100
L	100	70	10	20
M	180	60	40	80
N	360	216	72	72
Total	1190	596	277	317

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Answer: C

In ice cream parlour S,

Total number of ice creams (chocolate + vanilla + strawberry) sold = 5V

Number of vanilla ice creams sold = 50

Number of strawberry ice creams sold = (50-V)

So, the number of chocolate ice creams sold = {5V-(50+50-V)} = {5V-(100-V)} = (6V-100)

Hence, option C is the correct answer.

9. Questions

In ice cream parlour J

Let, the total number of ice creams sold be 100a

So, the number of vanilla ice creams sold = 35% of 100a = 35a

Number of chocolate ice creams sold = 150

Therefore, the number of strawberry ice creams sold = (35a-60)

Now,

150 + 35a + (35a-60) = 100a

70a = 100a – 90

30a = 90

a = 3

So,

The total number of ice creams sold be 100a = 300

Number of chocolate ice creams sold = 150

Number of vanilla ice creams sold = 35a = 105

Number of strawberry ice creams sold = (35a-60) = (105-60) = 45

In ice cream parlour K

Total number of ice creams sold = 100%

Number of chocolate ice creams sold = 100

Number of strawberry ice creams sold = 100

Number of vanilla ice creams sold = 20%

So,

The percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-20)% = 80%

80% = 100 + 100

80% = 200

1% = (200/80) = 2.5

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100% = (2.5×100) = 250

Therefore, the total number of ice creams sold = 250

Number of chocolate ice creams sold = 100

Number of vanilla ice creams sold = 20% of 250 = 50

Number of strawberry ice creams sold = 100

Ice cream parlour L

Number of chocolate ice cream sold = 70

Number of strawberry ice cream sold = 2/7 of 70 = 20 *(Fractional value of 28.57% = 2/7)*

Percentage of the number of vanilla ice creams sold = 10%

So, the percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-10)% = 90%

90% = 70 + 20

90% = 90

1% = (90/90) = 1

100% = (1×100) = 100

Therefore, the total number of ice creams sold = 100

Number of chocolate ice creams sold = 70

Number of vanilla ice creams sold = 10% of 100 = 10

Number of strawberry ice creams sold = 20

Ice cream parlour M

Fractional value of 22.22% = 2/9

Let, the total number of ice creams sold be 9b.

So, the number of vanilla ice cream sold = 2/9 of 9b = 2b

And, number of strawberry ice creams sold = 4/9 of 9b = 4b

Therefore, the number of chocolate ice creams sold = {9b-(2b+4b)} = (9b-6b) = 3b

3b = 60

b = (60/3) = 20

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Therefore, the total number of ice creams sold = 9b = 180

Number of chocolate ice creams sold = 60

Number of vanilla ice creams sold = 2b = 40

Number of strawberry ice creams sold = 4b = 80

Ice cream parlour N

Let, the total number of ice creams sold be 100c.

So, the number of vanilla ice creams sold = 20% of 100c = 20c

Number of chocolate ice creams sold = 216

Therefore, the number of strawberry ice creams sold = {(216-20c)/2} = (108-10c)

Now,

216 + 20c + (108-10c) = 100c

90c = 324

c = (324/90) = (18/5)

Therefore, the total number of ice creams sold = 100c = {(100×18)/5} = 360

Number of chocolate ice creams sold = 216

Number of vanilla ice creams sold = 20% of 360 = 72

Number of strawberry ice creams sold = (108-10c) = [108-{(10×18)/5}] = (108-36) = 72

Ice Cream Parlour	Total number of ice creams sold	Number of chocolate ice creams sold	Number of vanilla ice creams sold	Number of strawberry ice creams sold
J	300	150	105	45
K	250	100	50	100
L	100	70	10	20
M	180	60	40	80
N	360	216	72	72
Total	1190	596	277	317

Answer: D

Using case I

In ice cream parlour J,

Number of strawberry ice creams sold = 45

So, the number of cherry ice creams sold = 10/9 of 45 = 50 *(Fractional value of 11.11% = 1/9)*

In ice cream parlour L,

Number of strawberry ice creams sold = 20

So, the number of cherry ice creams sold = 125% of 20 = 25

Therefore, the difference between the number of cherry ice creams sold by both the ice cream parlours = (50-25) = 25

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So, case I is satisfied.

Using case II

In ice cream parlour J,

Number of strawberry ice creams sold = 45

So, the number of cherry ice creams sold = 120% of 45 = 54

In ice cream parlour L,

Number of strawberry ice creams sold = 20

So, the number of cherry ice creams sold = 150% of 20 = 30

Therefore, the difference between the number of cherry ice creams sold by both the ice cream parlours = (54-30) = 24

So, case II is satisfied.

Using case III

In ice cream parlour J,

Number of strawberry ice creams sold = 45

So, the number of cherry ice creams sold = 140% of 45 = 63

In ice cream parlour L,

Number of strawberry ice creams sold = 20

So, the number of cherry ice creams sold = 110% of 20 = 22

Therefore, the difference between the number of cherry ice creams sold by both the ice cream parlours = (63-22) = 41

So, case III is satisfied.

Hence, all the cases are satisfied.

10. Questions

In ice cream parlour J

Let, the total number of ice creams sold be 100a

So, the number of vanilla ice creams sold = 35% of 100a = 35a

Number of chocolate ice creams sold = 150

Therefore, the number of strawberry ice creams sold = (35a-60)

Now,

150 + 35a + (35a-60) = 100a

70a = 100a – 90

30a = 90

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a = 3

So,

The total number of ice creams sold be 100a = 300

Number of chocolate ice creams sold = 150

Number of vanilla ice creams sold = 35a = 105

Number of strawberry ice creams sold = (35a-60) = (105-60) = 45

In ice cream parlour K

Total number of ice creams sold = 100%

Number of chocolate ice creams sold = 100

Number of strawberry ice creams sold = 100

Number of vanilla ice creams sold = 20%

So,

The percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-20)% = 80%

80% = 100 + 100

80% = 200

1% = (200/80) = 2.5

100% = (2.5×100) = 250

Therefore, the total number of ice creams sold = 250

Number of chocolate ice creams sold = 100

Number of vanilla ice creams sold = 20% of 250 = 50

Number of strawberry ice creams sold = 100

Ice cream parlour L

Number of chocolate ice cream sold = 70

Number of strawberry ice cream sold = 2/7 of 70 = 20 *(Fractional value of 28.57% = 2/7)*

Percentage of the number of vanilla ice creams sold = 10%

So, the percentage of the sum of the number of chocolate and strawberry ice creams sold = (100-10)% = 90%

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90% = 70 + 20

90% = 90

1% = (90/90) = 1

100% = (1×100) = 100

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Therefore, the total number of ice creams sold = 100

Number of chocolate ice creams sold = 70

Number of vanilla ice creams sold = 10% of 100 = 10

Number of strawberry ice creams sold = 20

Ice cream parlour M

Fractional value of 22.22% = 2/9

Let, the total number of ice creams sold be 9b.

So, the number of vanilla ice cream sold = 2/9 of 9b = 2b

And, number of strawberry ice creams sold = 4/9 of 9b = 4b

Therefore, the number of chocolate ice creams sold = {9b-(2b+4b)} = (9b-6b) = 3b

3b = 60

b = (60/3) = 20

Therefore, the total number of ice creams sold = 9b = 180

Number of chocolate ice creams sold = 60

Number of vanilla ice creams sold = 2b = 40

Number of strawberry ice creams sold = 4b = 80

Ice cream parlour N

Let, the total number of ice creams sold be 100c.

So, the number of vanilla ice creams sold = 20% of 100c = 20c

Number of chocolate ice creams sold = 216

Therefore, the number of strawberry ice creams sold = {(216-20c)/2} = (108-10c)

Now,

216 + 20c + (108-10c) = 100c

90c = 324

c = (324/90) = (18/5)

Therefore, the total number of ice creams sold = 100c = {(100×18)/5} = 360

Number of chocolate ice creams sold = 216

Number of vanilla ice creams sold = 20% of 360 = 72

Number of strawberry ice creams sold = (108-10c) = [108-{(10×18)/5}] = (108-36) = 72

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Ice Cream Parlour	Total number of ice creams sold	Number of chocolate ice creams sold	Number of vanilla ice creams sold	Number of strawberry ice creams sold
J	300	150	105	45
K	250	100	50	100
L	100	70	10	20
M	180	60	40	80
N	360	216	72	72
Total	1190	596	277	317

Answer: A

In ice cream parlour N,

Total number of ice creams (chocolate + vanilla + strawberry) sold on Monday, Tuesday and Wednesday together = 360

Total number of ice creams (chocolate + vanilla + strawberry) sold on Tuesday = 120

So, the total number of ice creams (chocolate + vanilla + strawberry) sold on Monday should be least among all the three days (Monday, Tuesday and Wednesday)

Total number of ice creams (chocolate + vanilla + strawberry) sold on Monday < Tuesday < Wednesday (In increasing order starting from Monday)

Let, the total number of ice creams (chocolate + vanilla + strawberry) sold on Monday = 110 *(more than 100 but less than 140)*

So, the total number of ice creams (chocolate + vanilla + strawberry) sold on Wednesday might be = 130 *(more than 100 but less than 140)*

Now,

Total number of ice creams (chocolate + vanilla + strawberry) sold on all the three days (Monday + Tuesday + Wednesday) together = (110+120+130) = 360

108 + 120 + 132 = 360

109 + 120 + 131 = 360 .....

There are many possible values

Hence, there are many possibilities the solution, cannot be determined.

11. Questions

Answer: B

Speed of train P, which is equal to that of train Q = {(144×5)/18} = 40 m/sec

So,

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(A+40+80)/40 = (C-2)

40C – 80 = A + 120

40C – A = 200……(i)

And,

(B-90+80)/40 = (C-4)

40C – 160 = B – 10

40C – B = 150……(ii)

Solving both the equations, we get,

B – A = 50

Quantity I: B + A = 450……(iii)

B – A = 50……(iv)

Solving both the equations, we get,

2A = 400

A = 200

Now, put the value of A in equation (iii), we get,

B + A = 450

B = 450 – 200

B = 250

Now, put the value of B in equation (ii), we get,

40C – B = 150

40C = 150 + 250

40C = 400

C = 10

So, [√(B×C)] = [√(250×10)] = √2500 = 50.

Hence, Quantity I = 50.

Quantity II: Length of train R = [6×(B-A)] = (6×50) = 300 meters

And, length of another platform = 150% of 80 = 120 meters

Speed of train R = 21 m/sec

So, the time taken by train R to cross the another platform = {(300+120)/21} = (420/21) = 20 seconds.

Therefore, Quantity II = 20.

Quantity III: [D^2 – {(B-A)D} + 600 = 0]

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$D^2 - 50D + 600 = 0$

$D^2 - 30D - 20D + 600 = 0$

$D(D-30) - 20(D-30) = 0$

$(D-20)(D-30) = 0$

$D = 20$ or 30

So, the smaller root the equation = 20.

Therefore, Quantity III = 20.

Hence, Quantity I > Quantity II = Quantity III.

12. Questions

Answer: A

Quantity I: Let, the amount invested in scheme X be Rs. 100M

Scheme X

Amount invested = Rs. 100M

Rate of interest (p.a.) = 25%

Time duration = 2 years

Type of interest = C.I.

$[(25+25)+\{(25\times25)/100\}]\% = (50+6.25)\% = 56.25\%$

Therefore, interest earned = 56.25% of 100M = Rs. 56.25M

Scheme Y

Amount invested = Rs. (40000-100M)

Rate of interest (p.a.) = 27%

Time duration = 3 years

Type of interest = S.I.

Therefore, interest earned = 81% of (40000-100M) = Rs. (32400-81M)

So,

(32400-81M) – 56.25M = 10440

137.25M = 21960

M = 160

Therefore, amount invested in scheme X = 100M = Rs. 16000

And, amount invested in scheme Y = (40000-100M) = (40000-16000) = Rs. 24000

Hence, required difference = (24000-16000) = Rs. 8000.

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So, Quantity I = 8000.

Quantity II: Let, the amount invested be Rs. 100N

Rate of interest (p.a.) = 15%

Time duration = 2 years

Compound interest earned = [(15+15)+{(15×15)/100}]% of 100N = (30+2.25)% of 100N = 32.25% of 100N = Rs. 32.25N

Simple interest earned = 30% of 100N = Rs. 30N

So,

32.25N – 30N = 450

2.25N = 450

N = 200

Therefore, amount invested = 100N = Rs. 20000.

Hence, simple interest earned on the same amount at 20% p.a. after 4 years = 80% of 20000 = Rs. 16000.

So, Quantity II = 16000.

Quantity III: (S-√1600) = R

S = R + √1600

S = R + 40

S – R = 40

Hence, [100×(S-R)] = (100×40) = 4000.

So, Quantity III = 4000

Hence, Quantity I < Quantity II > Quantity III.

13. Questions

Answer: E

Let, the total number of boys and girls in school M be 2x and x respectively.

Total number of students (boys + girls) in school M = (2x+x) = 3x

So, the total number of girls in school N = x

And, the number of boys in school N = (x+40)

Therefore, total number of students (boys + girls) in school N = (2x+40)

Now,

Let, the total number of students (boys + girls) who participated and who didn't participated in sports competition from school M be (y+200) and y respectively.

So, the total number of students (boys + girls) who participated in sports competition from school N =

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{(y+200)-70} = (y+130)

And, the total number of students (boys + girls) who didn't participated in sports competition from school N = {(2x+40)-(y+130)} = (2x-y-90)

Now,

(y+200) + y = 3x

3x – 2y = 200……..(i)

And,

(2x-y-90) – y = 10

2x – y – 90 = 10 + y

2x = 100 + 2y

2x – 2y = 100…….(ii)

Solving both the equations, we get,

6x – 4y = 400

6x – 6y = 300

2y = 100

y = 50

Now, put the value of y in equation (i), we get,

3x – 2y = 200

3x = 200 + 100

3x = 300

x = 100

Quantity I: Total number of students (boys + girls) in school M = 3x = 300

Total number of students (boys + girls) in school N = (2x+40) = (200+40) = 240

So, the difference between the total number of students in both the schools = (300-240) = 60.

Therefore, Quantity I = 60.

Quantity II: Total number of girls in school M = x = 100

Total number of girls in school N = x = 100

So, the average of the total number of girls in both the schools = {(100+100)/2} = (200/2) = 100.

Therefore, Quantity II = 100.

Quantity III: Total number of students (boys + girls) who didn't participated in the sports competition from school M = y = 50

Total number of students (boys + girls) who didn't participated in the sports competition from school N =

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(2x-y-90) = (200-50-90) = (200-140) = 60.

Hence, sum of the total number of students (boys + girls) who didn't participated in the sports competition from both the schools together = (50+60) = 110.

Therefore, Quantity III = 110.

Hence, Quantity I < Quantity II < Quantity III.

14. Questions

Answer: B

Quantity I: Investment made by,

P = {(K×6) + 6(K-1200)} = Rs. (12K-7200)

Q = (2K×12) = Rs. 24K

R = {6(K+2000) + 6(K+3000)} = (12K+30000)

Now,

Profit share of P = Rs. 4800

So, the sum of the profit share of Q and R together = (27800-4800) = Rs. 23000

(12K-7200)/(24K+12K+30000) = 4800/23000

(12K-7200)/(36K+30000) = 24/115

1380K – 828000 = 864K + 720000

516K = 1548000

K = 3000

So, the initial investment of,

Q = 2K = Rs. 6000

R = (K+2000) = (3000+2000) = Rs. 5000

Hence, required % = [{(6000-5000)/5000}×100}]% = {(1000/5000)×100}% = 20%.

So, Quantity I = 20.

Quantity II: Investment made by S = Rs. 1600

So, the investment made by T = {200+(125% of 1600)} = (200+2000) = Rs. 2200

Now,

(1600×X)/(2200×Y) = 32/55

88000X = 70400Y

X:Y = 4:5.

Let, the values of X and Y be 4z and 5z respectively.

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$5z - 4z = 2$

$z = 2$

Therefore, $X = 4z = 8$

And, $Y = 5z = 10$

Hence, $(X+Y-6) = (8+10-6) = (18-6) = 12$.

So, Quantity II = 12.

Quantity III: 5/6 of N = M

$5N = 6M$

M:N = 5:6.

Let, the values of,

$M = 5w$

$N = 6w$

Now,

$3M + 2N = 810$

$15w + 12w = 810$

$27w = 810$

$w = 30$

So, $M = 5w = 150$

And, $N = 6w = 180$

Hence, $[(M+N)/10)] = [(150+180)/10)] = (330/10) = 33$.

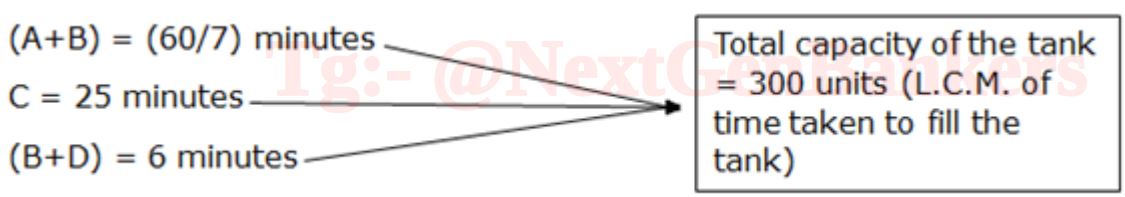
So, Quantity III = 33.

Hence, Quantity I > Quantity II < Quantity III.

15. Questions

Answer: D

Time taken to fill the tank by pipe,



Efficiency of pipe,

(A+B) = (300/60/7) = {(300×7)/60} = 35

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C = (300/25) = 12

(B+D) = (300/6) = 50

(D-A) = (50-35) = 15

So, (A-C) = 20% of 15 = 3

Therefore, A = (12+3) = 15

So, B = (35-15) = 20

D = (50-20) = 30

Quantity I: Time taken by B to fill the tank alone = (300/20) = 15 minutes.

So, Quantity I = 15.

Quantity II: Efficiency of C = 12

Efficiency of D = 30

Hence, required % = {(12/30)×100}% = 40%.

So, Quantity II = 40.

Quantity III: Efficiency of B = 20

Efficiency of C = 12

Hence, required sum = (20+12) = 32.

So, Quantity III = 32.

Hence, Quantity I < Quantity II > Quantity III.

16. Questions

Answer: C

Series I: 1st term = 24

2nd term = 24×2+2 = 50

3rd term = 50×3+3 = 153 **(A)**

4th term = 153×5+5 = 770

5th term = 770×7+7 = 5397

6th term = 5397×11+11 = 59378

So, A = 153

Pattern of the series is to multiply the given term with increasing prime number starting with 2 and add the same number.

Series II: 1st term = 60

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2nd term = $60+4+4^2$ = (60+4+16) = 80

3rd term = $80+6+6^2$ = (80+6+36) = 122

4th term = $122+8+8^2$ = (122+72) = 194 **(B)**

5th term = $192+10+10^2$ = (192+110) = 302

6th term = $302+12+12^2$ = (302+156) = 460

So, B = 194

Pattern of the series to add even number starting with 4 and then add the square of the same number.

Statement I

Only A is the multiple of 9, whereas B is not the multiple of 9.

So, statement I is not true.

Statement II

B = 194

Sum of the digits of 194 = (1+9+4) = 14

So, C = 14

Now,

$(C^2-2) = B$

$14^2 - 2 = 194$

196-2 = 194

194 = 194

So, statement II is true.

Statement III

A = 153

153 is an odd number, but not the prime number. So, A is an odd composite number.

So, statement III is true.

Hence, both the statements II and III are true.

17. Questions

Answer: A

Series I: 1st term = 35

2nd term = [35+{(3+5)×10}] = 115 **(wrong term = 120)**

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3rd term = [115+{(1+1+5)×10}] = 185

4th term = [185+{(1+8+5)×10}] = 325

5th term = [325+{(3+2+5)×10}] = 425

6th term = [425+{(4+2+5)×10}] = 535

So, the wrong term = 120

And, the correct term = 115

Therefore, M = 115

Series II: 1st term = M = 115

2nd term = [115+{(1+1+5)×10}] = 185

3rd term = [185+{(1+8+5)×10}] = 325

4th term = [325+{(3+2+5)×10}] = 425

5th term = [425+{(4+2+5)×10}] = 535

6th term = [535+{(5+3+5)×10}] = 665

Therefore, N = 665

Now,

Check option A

M + N = (115+665) = 780

So, option A is satisfied.

Check option B

5M = 2N

5M = (5×115) = 575

2N = (2×665) = 1330

575 ≠ 1330

Therefore, 5M ≠ 2N

So, option B is not satisfied.

Check option C

(N-450) = M

665 – 450 = 215

But, M = 115

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Therefore, (N-450) ≠ M

So, option C is not satisfied.

Hence, option A is correct answer.

18. Questions

Answer: B

Pattern of both the series is to subtract the square of multiples of 5 starting from 25 in each term in decreasing order from the last term.

Or, add the square of multiples of 5 starting from 5 in each term in increasing order from the first term.

Series I: 6th term = 1385

5th term = $(1385-25^2)$ = (1385-625) = 760

4th term = $(760-20^2)$ = (760-400) = 360

3rd term = $(360-15^2)$ = (360-225) = 135

2nd term = $(135-10^2)$ = (135-100) = 35 **[K]**

1st term = $(35-5^2)$ = (35-25) = 10

So, K = 35

Series II: 1st term = K = 35

2nd term = $(35+5^2)$ = (35+25) = 60

3rd term = $(60+10^2)$ = (60+100) = 160 **[L]**

4th term = $(160+15^2)$ = (160+225) = 385

5th term = $(385+20^2)$ = (385+400) = 785

6th term = $(785+25^2)$ = (785+625) = 1410

So, L = 160

Now,

HCF of K and L = HCF of 35 and 160 = 5

LCM of K and L = LCM of 35 and 160 = 1120

Therefore, J = (5+1120) = 1125

Hence, 2J = (1125×2) = 2250.

19. Questions

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Answer: D

Series I: 6th term = 425

5th term = (425-125) = 300

4th term = (300-100) = 200

3rd term = (200-75) = 125

2nd term = (125-50) = 75

1st term = (75-25) = 50 **(X)**

Hence, X = 50

So, the pattern of the series is subtracting consecutive multiples of 25 starting from 25 beginning from the last term.

Series II: 6th term = (21^3+1) = (9261+1) = 9262

5th term = (19^3+1) = (6859+1) = 6860

4th term = (17^3+1) = (4913+1) = 4914

3rd term = (15^3+1) = (3375+1) = 3376

2nd term = (13^2+1) = (2197+1) = 2198 **(Y)**

1st term = (11^3+1) = (1331+1) = 1332

Hence, Y = 2198

So, the pattern of the series is cube of odd numbers starting from 11 then add 1 to each number beginning from the last term.

Series III: 1st term = 324

2nd term = (324×3) = 972

3rd term = (972/2) = 486

4th term = (486×3) = 1458

5th term = (1458/2) = 729 **(Z)**

6th term = (729×3) = 2187

Hence, Z = 729

So, the pattern of the series is first multiply with 3 then divide by 2 alternatively beginning from the first term.

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Hence, the value of [√Z+{√(X-1)}+{√(Y+302)}] will be,

[√Z+{√(X-1)}+{√(Y+302)}] = [√729+{√(50-1)}+{√(2198+302)}] = (√729+√49+√2500) = (27+7+50) = 84.

20. Questions

Answer: B

Series I: 1st term = 32

2nd term = (32+14) = 46

3rd term = (46+28) = 74

4th term = (74+56) = 130

5th term = (130+112) = 242 (239 is wrong)

6th term = (242+224) = 466

So, P = 239 (wrong term)

And, S = 242 (correct term)

Pattern of the series is to add consecutive multiples of 14 starting from 14 from the beginning.

Series II: 1st term = 70

2nd term = (70×2+4) = 144

3rd term = (144×3+5) = 437

4th term = (437×4+6) = 1754 (1755 is wrong)

5th term = (1754×5+7) = 8777

6th term = (8777×6+8) = 52670

So, Q = 1755 (wrong term)

And, T = 1754 (correct term)

Series III: 1st term = 124

2nd term = (124×0.5+1) = 63 (60 is wrong)

3rd term = (63×1+2) = 65

4th term = (65×2+4) = 134

5th term = (134×4+8) = 544

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6th term = (544×8+16) = 4368

So, R = 60 (wrong term)

And, U = 63 (correct term)

Therefore,

P < S

Q > T

R < U

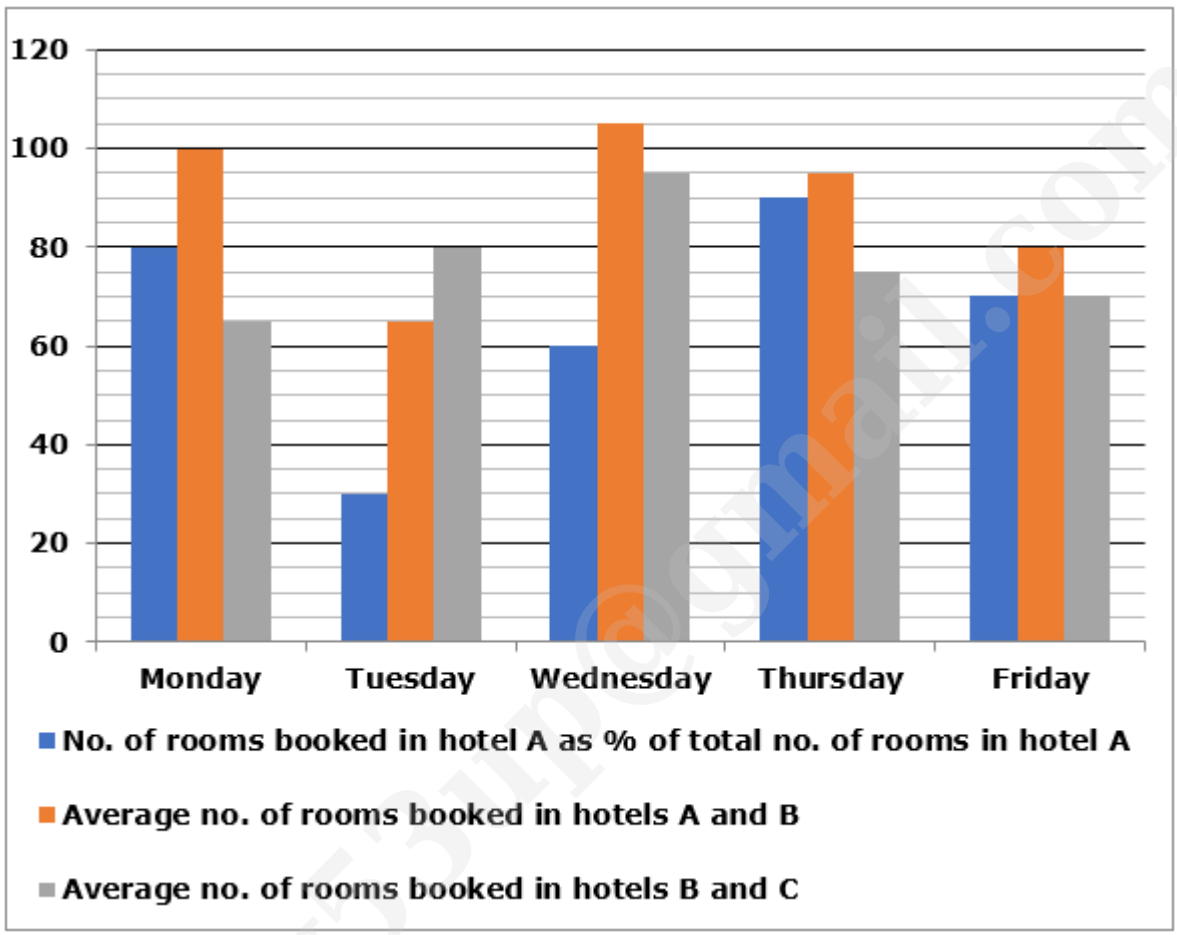
Hence, only relation I follows.

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1. Questions

Study the following data carefully and answer the questions:

The given bar graph shows the number of rooms booked in three Hotels A, B, and C on five different days: Monday, Tuesday, Wednesday, Thursday, and Friday. It also illustrates the number of rooms booked in Hotel A as a percentage out of the total number of rooms in Hotel A. Additionally, it provides the average number of rooms booked in Hotels A and B together and the average number of rooms booked in Hotels B and C together.



Note: In Hotel C, the ratio of the number of rooms booked on Monday to that booked on Tuesday, is 2: 3 respectively.

On Tuesday, the average number of persons in booked rooms of Hotel A, is 4, the average number of persons in booked rooms of Hotel B, is 2 and the average number of persons in booked rooms of Hotel C, is 3, then find the ratio between the total number of persons in booked rooms of Hotels A, B and C respectively?

a. 12: 17: 15

b. 18: 17: 15

c. 9: 8: 15

d. 6: 7: 5

e. 36: 34: 45

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2. Questions

Find the ratio between the total number of rooms booked in all three Hotels together on those days when the number of rooms booked in Hotel B is equal.

a. 19: 14

b. 9: 7

c. 38: 35

d. 57: 49

e. None of these

3. Questions

The total number of rooms in Hotels B and C respectively is 6(2/3)% and 33(1/3)% more than that of Hotel A. If M% of the total number of rooms in Hotel B was booked on Wednesday and N% of the total number of rooms in Hotel C was booked on Wednesday, then find the value of (2M+N).

a. 155

b. 185

c. 168

d. 125

e. 135

4. Questions

If the rent of 1 room in Hotels A, B, and C is Rs.1500, Rs.1750, and Rs.2400 respectively, then find the total amount received by all three Hotels together on Monday.

a. Rs.4.4 lakhs

b. Rs.3.8 lakhs

c. Rs.3.4 lakhs

d. Rs.4 lakhs

e. Rs.4.8 lakhs

5. Questions

The table given below shows the following data:

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Days	The ratio of the number of AC rooms to non-AC rooms booked in Hotel C
Monday	3: 7
Tuesday	2: 3
Wednesday	2: 5
Thursday	7: 12
Friday	5: 12

Find the average number of AC rooms booked in Hotel C on each of the given days.

a. 35

b. 20

c. 25

d. 15

e. 30

6. Questions

A is running at 21.6 km/hr and B is running 'd' meters behind A at 27 km/hr. After 48 seconds, B will be (d – 8) meters ahead of A. Find the approximate time at which B crossed A.

a. 41 seconds

b. 27 seconds

c. 47 seconds

d. 35 seconds

e. 59 seconds

7. Questions

In a mixture, the ratio of milk to water is 2:1 respectively, and the cost of pure milk is Rs. 48 per liter. When 15 liters of the mixture is replaced with 5 liters of milk, the cost of the mixture is increased by Rs. 1.6 per liter. Find the initial quantity of the mixture.

a. 30 liters

b. 48 liters

c. 36 liters

d. 72 liters

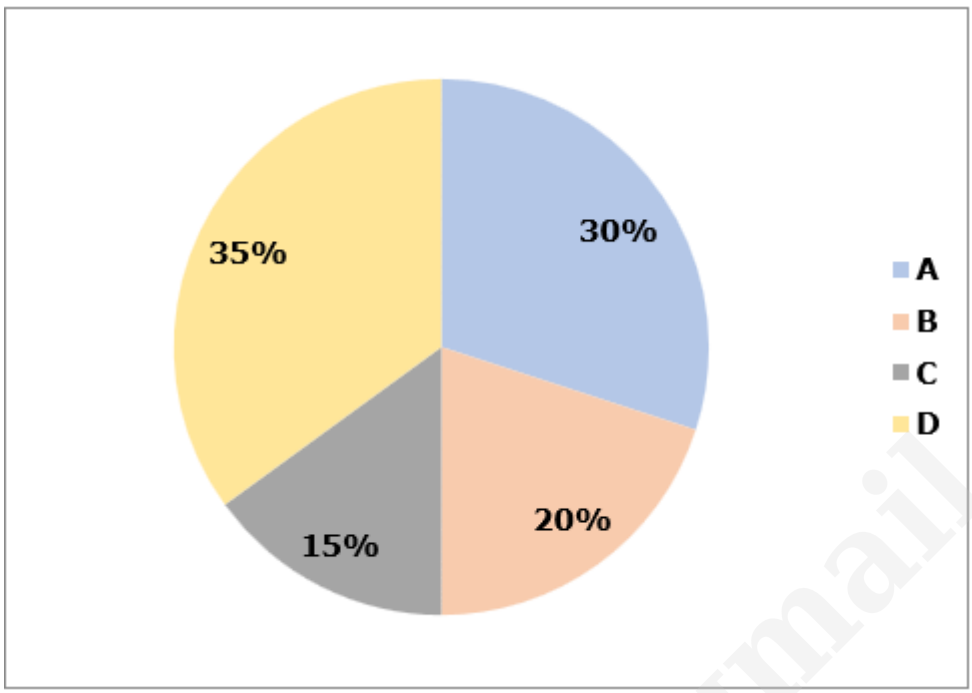
e. 60 liters

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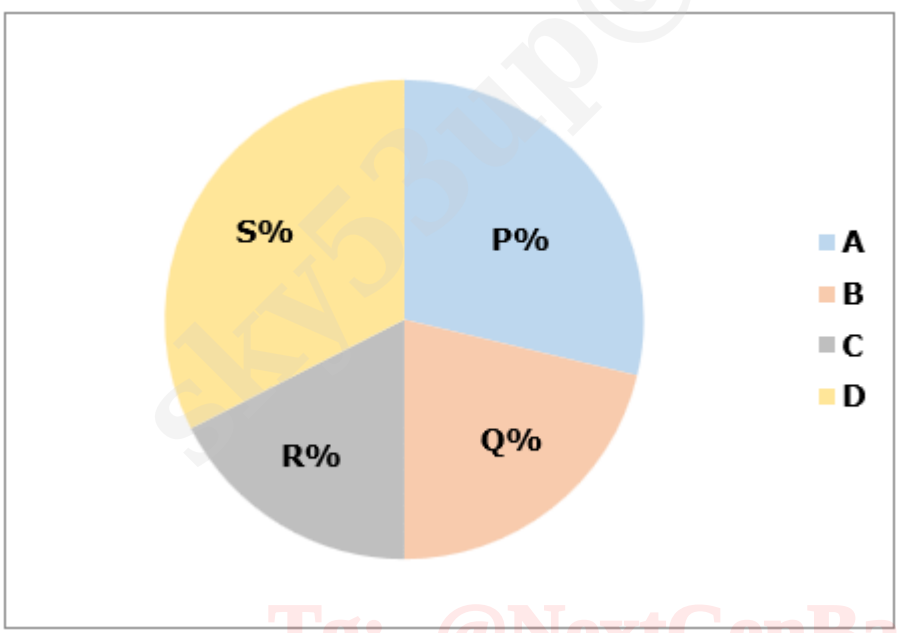
8. Questions

Study the following data carefully and answer the questions:

The given pie chart shows the percentage distribution of the present ages of four different persons namely A, B, C and D.



The given pie chart shows the percentage distribution of the ages after 5 years of four different persons namely A, B, C and D.



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Notes:

i). The difference between the present ages of C and D is 12 years.

Find the ratio of A's age after (P – 10.75) years to B's age after (Q – 9.25) years.

a. 9: 8

b. 12: 11

c. 4: 3

d. 6: 5

e. 3: 2

9. Questions

Find the average of A's age after 6 years, B's age after 9 years, C's age after 3 years and D's age after 12 years.

a. (P – 5.25) years

b. (Q + 0.25) years

c. (R + 6) years

d. (S – 10) years

e. None of these

10. Questions

If the ratio of A's age to B's age after 2(P – Q) years, is M: N respectively, then find the ratio of C's present age to D's present age. (In terms of M and N)

a. (N/3): (M – 4)

b. (3N): (M + 15)

c. (N/2): (M – 3)

d. (N/4): (M – 2)

e. (2N): (M + 10)

11. Questions

Which of the following statements is/are true?

Statement X: The difference between P and R is divisible by 1.5.

Statement Y: The Sum of Q and S is divisible by 1.75.

Statement Z: B's age after 5 years is approximately 21.4% more than that of C's age.

a. Only statement Z

b. Only statements X and Y

c. None is true

d. Only statements Y and Z

e. Only statement Y

12. Questions

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Study the following data carefully and answer the questions:

Two series I and II are given below:

I: 569, 400, 279, (15A – B), 205, 196, 192

II: 840, 420, 168, 56, A, 4

Which of the following statements is/are true?

Statement P: A and B are co-prime with each other.

Statement Q: The value of $(A^2 – B^2)$ is divisible by 9.

Statement R: The value of $(A^2 + B^2)$ is divisible by 17.

a. None is true

b. Only statements P and Q

c. Only statements P and R

d. Only statements Q and R

e. Only statement R

13. Questions

If a series is formed, where the 1st term is B, the 2nd term is A, the 3rd term is 4(A – B), the 4th term is (2A + 2), and the 5th term is (5B – 4), then find the 6th term of the series.

a. 56

b. 64

c. 68

d. 52

e. 60

14. Questions

Study the following data carefully and answer the questions:

A company held three meetings in a week. The data given below shows the total number of employees of the company who attended these meetings. Some employees attended only one meeting, some attended only two meetings, and the remaining attended all three meetings.

The number of employees who attended only the 2nd meeting is half of those who attended only the 1st meeting and also 75% of those who attended only the 3rd meeting.

The number of employees who attended both the 2nd and 3rd meetings but not the 1st meeting is half of those who attended both the 1st and 2nd meetings but not the 3rd meeting and also 75% of those who attended both the 1st and 3rd meetings but not the 2nd meeting.

The number of employees who attended all three meetings is 32. The number of employees who attended

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only one meeting is 104 more than those who attended only two meetings. The total number of employees who attended the 1st meeting is 144.

Find the ratio of the total number of employees, who attended the 2nd meeting, to the total number of employees, who didn't attend the 2nd meeting?

a. 26: 37
b. 13: 17
c. 11: 27
d. 16: 47
e. 2: 3

15. Questions

If the ratio of male to female employees, who attended both the 1st and 2nd meetings but not the 3rd meeting, is 5: 7 respectively, the ratio of male to female employees, who attended both the 2nd and 3rd meetings but not the 1st meeting, is 3: 1 respectively and the ratio of male to female employees, who attended both the 1st and 3rd meetings but not the 2nd meeting, is 3: 5 respectively, then find the difference between the number of male and female employees, who attended only two meetings?

a. 2
b. 5
c. 1
d. 7
e. 3

16. Questions

The total number of employees who attended the 3rd meeting is what percentage of the total number of employees in the company.

a. 36%
b. 54%
c. 18%
d. 60%
e. 45%

17. Questions

Find the difference between the total number of employees who attended only one meeting, to the total number of employees who attended at least two meetings.

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a. 72

b. 64

c. 39

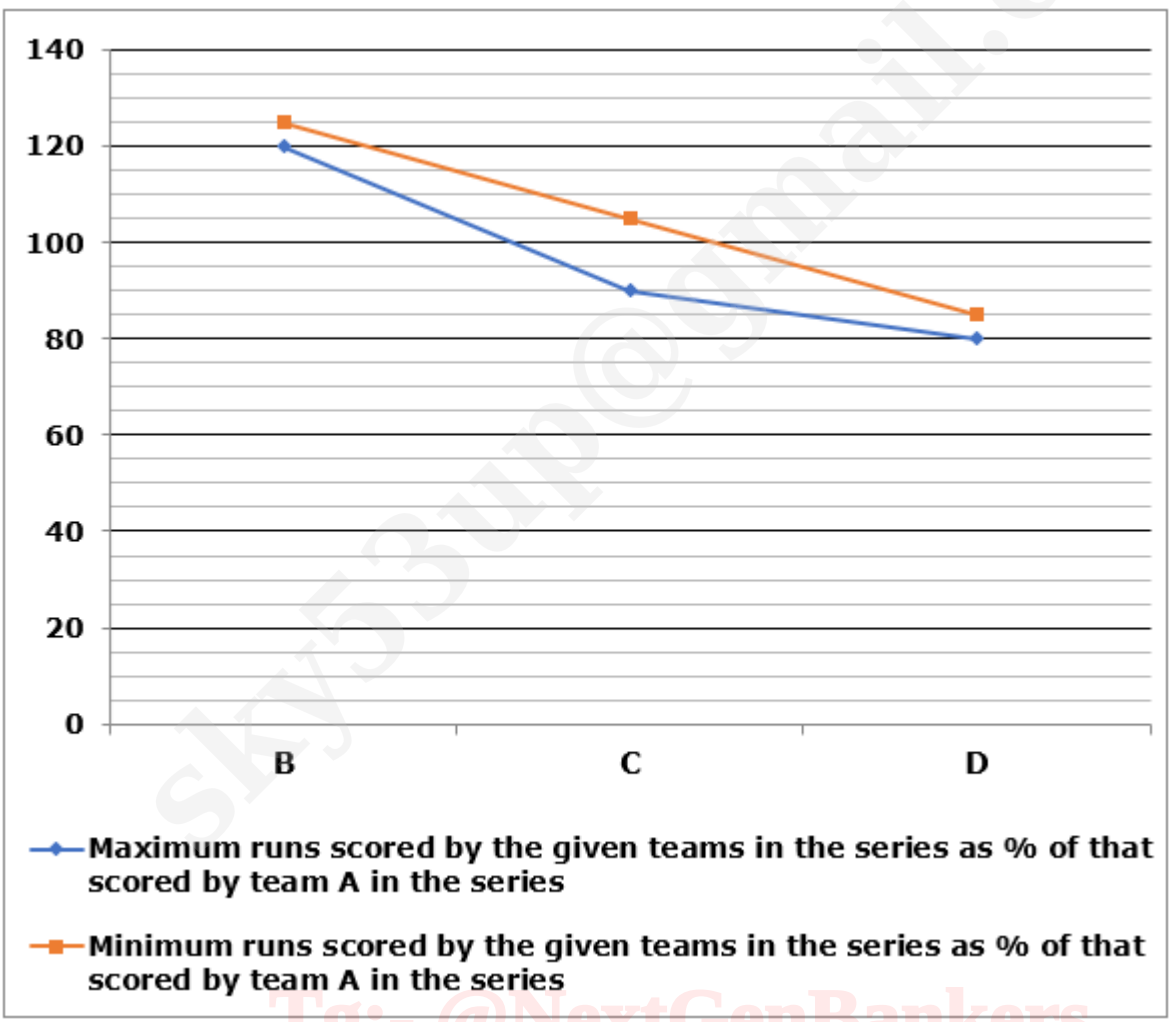
d. 45

e. 98

18. Questions

Study the following data carefully and answer the questions:

The given line graph shows the maximum runs and minimum runs scored by four different Teams, namely A, B, C, and D, in a cricket match series. It also shows the maximum runs scored by a Team in a series as a percentage of that scored by Team A in the series and the minimum runs scored by the given Teams in the series as a percentage of that scored by Team A in the series.



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Notes:

1: The ratio of maximum runs scored by Team A to minimum runs scored by Team A in the series is 5: 4.

2: The average of minimum runs scored by each Team in the series is 249.

If the total runs scored by Team A in the remaining matches of the series is 60 more than the sum of the maximum and minimum runs scored by Team A, and the total runs scored by Team B in the

remaining matches of the series is 60 less than the sum of the maximum and minimum runs scored by Team B, then find the ratio of the total runs scored by Team A to Team B in the series.

a. 38: 41

b. 6: 7

c. 57: 61

d. 19: 21

e. 2: 3

19. Questions

If Team C hit 15 sixes and 20 fours in the match in which it scored maximum runs and hit 8 sixes and 16 fours in the match, in which it scored minimum runs, then find the total runs scored by Team C by running in both matches together?

a. 210

b. 224

c. 214

d. 254

e. 240

20. Questions

The sum of the minimum runs scored by each Team is approximately what percentage of the sum of the maximum runs scored by each Team?

a. 65%

b. 85%

c. 70%

d. 90%

e. 75%

Explanations:

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1. Questions

Let the total number of rooms in Hotel A = 100a

So, the number of rooms booked in Hotel A on Monday = 80% of 100a = 80a

The number of rooms booked in Hotel A on Tuesday = 30% of 100a = 30a

The number of rooms booked in Hotel B on Monday = (2 * 100) – 80a

= (200 – 80a)

The number of rooms booked in Hotel B on Tuesday = (2 * 65) – 30a

= (130 – 30a)

The number of rooms booked in Hotel C on Monday = (2 * 65) – (200 – 80a)

= (80a – 70)

And the number of rooms booked in Hotel C on Tuesday

= (2 * 80) – (130 – 30a) = (30a + 30)

Since, in Hotel C, the ratio of the number of rooms booked on Monday to that booked on Tuesday is 2: 3.

So,

(80a – 70)/(30a + 30) = 2/3

(8a – 7)/(3a + 3) = 2/3

24a – 21 = 6a + 6

18a = 27

a = 1.5

The total number of rooms in Hotel A = 100 * 1.5 = 150

Days	Number of rooms booked in Hotel A	Number of rooms booked in Hotel B	Number of rooms booked in Hotel C
Monday	80% of 150 = 120	(2 * 100) – 120 = 80	(2 * 65) – 80 = 50
Tuesday	30% of 150 = 45	(2 * 65) – 45 = 85	(2 * 80) – 85 = 75
Wednesday	60% of 150 = 90	(2 * 105) – 90 = 120	(2 * 95) – 120 = 70
Thursday	90% of 150 = 135	(2 * 95) – 135 = 55	(2 * 75) – 55 = 95
Friday	70% of 150 = 105	(2 * 80) – 105 = 55	(2 * 70) – 55 = 85

Answer: E

According to the question,

On Tuesday:

Since, the average number of persons in booked rooms of Hotel A = 4

So, the total number of persons in booked rooms of Hotel A = 4 * 45 = 180

Since, the average number of persons in booked rooms of Hotel B = 2

So, the total number of persons in booked rooms at Hotel B = 2 * 85 = 170

Since, the average number of persons in booked rooms of Hotel C = 3

So, the total number of persons in booked rooms of Hotel C = 3 * 75 = 225

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Required ratio = 180: 170: 225 = 36: 34: 45

2. Questions

Let the total number of rooms in Hotel A = 100a

So, the number of rooms booked in Hotel A on Monday = 80% of 100a = 80a

The number of rooms booked in Hotel A on Tuesday = 30% of 100a = 30a

The number of rooms booked in Hotel B on Monday = (2 * 100) – 80a

= (200 – 80a)

The number of rooms booked in Hotel B on Tuesday = (2 * 65) – 30a

= (130 – 30a)

The number of rooms booked in Hotel C on Monday = (2 * 65) – (200 – 80a)

= (80a – 70)

And the number of rooms booked in Hotel C on Tuesday

= (2 * 80) – (130 – 30a) = (30a + 30)

Since, in Hotel C, the ratio of the number of rooms booked on Monday to that booked on Tuesday is 2: 3.

So,

(80a – 70)/(30a + 30) = 2/3

(8a – 7)/(3a + 3) = 2/3

24a – 21 = 6a + 6

18a = 27

a = 1.5

The total number of rooms in Hotel A = 100 * 1.5 = 150

Days	Number of rooms booked in Hotel A	Number of rooms booked in Hotel B	Number of rooms booked in Hotel C
Monday	80% of 150 = 120	(2 * 100) – 120 = 80	(2 * 65) – 80 = 50
Tuesday	30% of 150 = 45	(2 * 65) – 45 = 85	(2 * 80) – 85 = 75
Wednesday	60% of 150 = 90	(2 * 105) – 90 = 120	(2 * 95) – 120 = 70
Thursday	90% of 150 = 135	(2 * 95) – 135 = 55	(2 * 75) – 55 = 95
Friday	70% of 150 = 105	(2 * 80) – 105 = 55	(2 * 70) – 55 = 85

Answer: D

The days, on which the number of rooms booked in Hotel B is equal, are Thursday and Friday.

Since, the total number of rooms booked in all three Hotels together on Thursday = 135 + 55 + 95 = 285

And the total number of rooms booked in all three Hotels together on Friday

= 105 + 55 + 85 = 245

So, the required ratio = 285: 245 = 57: 49

3. Questions

Let the total number of rooms in Hotel A = 100a

So, the number of rooms booked in Hotel A on Monday = 80% of 100a = 80a

The number of rooms booked in Hotel A on Tuesday = 30% of 100a = 30a

The number of rooms booked in Hotel B on Monday = (2 * 100) – 80a

= (200 – 80a)

The number of rooms booked in Hotel B on Tuesday = (2 * 65) – 30a

= (130 – 30a)

The number of rooms booked in Hotel C on Monday = (2 * 65) – (200 – 80a)

= (80a – 70)

And the number of rooms booked in Hotel C on Tuesday

= (2 * 80) – (130 – 30a) = (30a + 30)

Since, in Hotel C, the ratio of the number of rooms booked on Monday to that booked on Tuesday is 2: 3.

So,

(80a – 70)/(30a + 30) = 2/3

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24a – 21 = 6a + 6

18a = 27

a = 1.5

The total number of rooms in Hotel A = 100 * 1.5 = 150

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Monday	80% of 150 = 120	(2 * 100) – 120 = 80	(2 * 65) – 80 = 50
Tuesday	30% of 150 = 45	(2 * 65) – 45 = 85	(2 * 80) – 85 = 75
Wednesday	60% of 150 = 90	(2 * 105) – 90 = 120	(2 * 95) – 120 = 70
Thursday	90% of 150 = 135	(2 * 95) – 135 = 55	(2 * 75) – 55 = 95
Friday	70% of 150 = 105	(2 * 80) – 105 = 55	(2 * 70) – 55 = 85

Answer: B

Since, the total number of rooms in Hotel A = 150

So, the total number of rooms in Hotel B = 106(2/3)% of 150 = 160

And the total number of rooms in Hotel C = 133(1/3)% of 150 = 200

Since, the number of rooms booked in Hotel B on Wednesday = 120

So, M = (120/160) * 100 = 75%

Since, the number of rooms booked in Hotel C on Wednesday = 70

So, N = (70/200) * 100 = 35%

Required value = (2M + N) = (2*75) + 35 = 185.

4. Questions

Let the total number of rooms in Hotel A = 100a

So, the number of rooms booked in Hotel A on Monday = 80% of 100a = 80a

The number of rooms booked in Hotel A on Tuesday = 30% of 100a = 30a

The number of rooms booked in Hotel B on Monday = (2 * 100) – 80a

= (200 – 80a)

The number of rooms booked in Hotel B on Tuesday = (2 * 65) – 30a

= (130 – 30a)

The number of rooms booked in Hotel C on Monday = (2 * 65) – (200 – 80a)

= (80a – 70)

And the number of rooms booked in Hotel C on Tuesday

= (2 * 80) – (130 – 30a) = (30a + 30)

Since, in Hotel C, the ratio of the number of rooms booked on Monday to that booked on Tuesday is 2: 3.

So,

(80a – 70)/(30a + 30) = 2/3

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24a – 21 = 6a + 6

18a = 27

a = 1.5

The total number of rooms in Hotel A = 100 * 1.5 = 150

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Days	Number of rooms booked in Hotel A	Number of rooms booked in Hotel B	Number of rooms booked in Hotel C
Monday	80% of 150 = 120	(2 * 100) – 120 = 80	(2 * 65) – 80 = 50
Tuesday	30% of 150 = 45	(2 * 65) – 45 = 85	(2 * 80) – 85 = 75
Wednesday	60% of 150 = 90	(2 * 105) – 90 = 120	(2 * 95) – 120 = 70
Thursday	90% of 150 = 135	(2 * 95) – 135 = 55	(2 * 75) – 55 = 95
Friday	70% of 150 = 105	(2 * 80) – 105 = 55	(2 * 70) – 55 = 85

Answer: A

Since, the number of rooms booked in Hotel A on Monday = 120

So, the total amount received by Hotel A on Monday = 120 * 1500 = Rs.180000

Since, the number of rooms booked in Hotel B on Monday = 80

So, the total amount received by Hotel B on Monday = 80 * 1750 = Rs.140000

Since, the number of rooms booked in Hotel C on Monday = 50

So, the total amount received by Hotel C on Monday = 50 * 2400 = Rs.120000

Required sum = 180000 + 140000 + 120000 = Rs.440000 = Rs.4.4 lakhs

5. Questions

Let the total number of rooms in Hotel A = 100a

So, the number of rooms booked in Hotel A on Monday = 80% of 100a = 80a

The number of rooms booked in Hotel A on Tuesday = 30% of 100a = 30a

The number of rooms booked in Hotel B on Monday = (2 * 100) – 80a

= (200 – 80a)

The number of rooms booked in Hotel B on Tuesday = (2 * 65) – 30a

= (130 – 30a)

The number of rooms booked in Hotel C on Monday = (2 * 65) – (200 – 80a)

= (80a – 70)

And the number of rooms booked in Hotel C on Tuesday

= (2 * 80) – (130 – 30a) = (30a + 30)

Since, in Hotel C, the ratio of the number of rooms booked on Monday to that booked on Tuesday is 2: 3.

So,

(80a – 70)/(30a + 30) = 2/3

(8a – 7)/(3a + 3) = 2/3

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24a – 21 = 6a + 6

18a = 27

a = 1.5

The total number of rooms in Hotel A = 100 * 1.5 = 150

Days	Number of rooms booked in Hotel A	Number of rooms booked in Hotel B	Number of rooms booked in Hotel C
Monday	80% of 150 = 120	(2 * 100) – 120 = 80	(2 * 65) – 80 = 50
Tuesday	30% of 150 = 45	(2 * 65) – 45 = 85	(2 * 80) – 85 = 75
Wednesday	60% of 150 = 90	(2 * 105) – 90 = 120	(2 * 95) – 120 = 70
Thursday	90% of 150 = 135	(2 * 95) – 135 = 55	(2 * 75) – 55 = 95
Friday	70% of 150 = 105	(2 * 80) – 105 = 55	(2 * 70) – 55 = 85

Answer: C

Number of AC rooms booked in Hotel C on Monday = 50 * (3/10) = 15

Number of AC rooms booked in Hotel C on Tuesday = 75 * (2/5) = 30

Number of AC rooms booked in Hotel C on Wednesday = 70 * (2/7) = 20

Number of AC rooms booked in Hotel C on Thursday = 95 * (7/19) = 35

Number of AC rooms booked in Hotel C on Friday = 85 * (5/17) = 25

Required average = (15 + 30 + 20 + 35 + 25)/5 = 25

6. Questions

Answer: B

The speed of A = 21.6 km/h = 21.6*(5/18) = 6 m/s

The speed of B = 27 km/h = 27*(5/18) = 7.5 m/s

Distance covered by A in 48 seconds = 6 * 48 = 288 meters

Distance covered by B in 48 seconds = 7.5 * 48 = 360 meters

So,

d + d – 8 + 288 = 360

2d = 80

D = 40

Since, the initial distance between A and B = 40 meters

So, the time, in which B crossed A = 40/(7.5 – 6) = 26.66 seconds

= 27 seconds (approx.)

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7. Questions

Answer: E

Let the initial quantities of milk and water in the mixture be '2x' L liters and 'x' liters respectively.

And the cost of pure milk = Rs.48 per liter

So, the initial cost of the mixture = (2x * 48)/(2x + x) = Rs.32 per liter

After replacing 15 liters mixture with 5 liters milk:

The new quantity of milk = 2x – 15 * (2/3) + 5 = (2x – 5) liters

The new quantity of water = x – 15 * (1/3) = (x – 5) liters

Since, the new cost of the mixture = 32 + 1.6 = Rs.33.6 per liters

So,

[(2x – 5) * 48]/[(2x – 5) + (x – 5)] = 33.6

96x – 240 = 100.8x – 336

x = 20

And the initial quantity of the mixture = (2 + 1) * 20 = 60 liters

8. Questions

Since the difference between the present ages of C and D is 12 years.

The total present age of all = [12/(35% - 15%)]*100 = 60 years

The present age of A = (30/100)*60 = 18 years

The present age of B = (20/100)*60 = 12 years

The present age of C = (15/100)*60 = 9 years

The present age of D = (35/100)*60 = 21 years

Now,

A's age after 5 years = 18 + 5 = 23 years

B's age after 5 years = 12 + 5 = 17 years

C's age after 5 years = 9 + 5 = 14 years

D's age after 5 years = 21 + 5 = 26 years

The total of age of all after 5 years = 23 + 17 + 14 + 26 = 80 years

So,

P = (23/80) * 100 = 28.75%

Q = (17/80) * 100 = 21.25%

R = (14/80) * 100 = 17.5%

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S = (26/80) * 100 = 32.5%

Answer: E

According to the equation,

Since, the present age of A = 18 years

So, A's age after (P – 10.75) years = 18 + (28.75 – 10.75) = 36 years

Since, the present age of B = 12 years

So, B's age after (Q – 9.25) years = 12 + (21.25 – 9.25) = 24 years

Required ratio = 36: 24 = 3: 2

9. Questions

Since the difference between the present ages of C and D is 12 years.

The total present age of all = [12/(35% - 15%)]*100 = 60 years

The present age of A = (30/100)*60 = 18 years

The present age of B = (20/100)*60 = 12 years

The present age of C = (15/100)*60 = 9 years

The present age of D = (35/100)*60 = 21 years

Now,

A's age after 5 years = 18 + 5 = 23 years

B's age after 5 years = 12 + 5 = 17 years

C's age after 5 years = 9 + 5 = 14 years

D's age after 5 years = 21 + 5 = 26 years

The total of age of all after 5 years = 23 + 17 + 14 + 26 = 80 years

So,

P = (23/80) * 100 = 28.75%

Q = (17/80) * 100 = 21.25%

R = (14/80) * 100 = 17.5%

S = (26/80) * 100 = 32.5%

Answer: D

A's age after 6 years = 18 + 6 = 24 years

B's age after 9 years = 12 + 9 = 21 years

C's age after 3 years = 9 + 3 = 12 years

D's age after 12 years = 21 + 12 = 33 years

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Required average = (24 + 21 + 12 + 33)/4 = 22.5 years = (S – 10) years

10. Questions

Since the difference between the present ages of C and D is 12 years.

The total present age of all = [12/(35% - 15%)]*100 = 60 years

The present age of A = (30/100)*60 = 18 years

The present age of B = (20/100)*60 = 12 years

The present age of C = (15/100)*60 = 9 years

The present age of D = (35/100)*60 = 21 years

Now,

A's age after 5 years = 18 + 5 = 23 years

B's age after 5 years = 12 + 5 = 17 years

C's age after 5 years = 9 + 5 = 14 years

D's age after 5 years = 21 + 5 = 26 years

The total of age of all after 5 years = 23 + 17 + 14 + 26 = 80 years

So,

P = (23/80) * 100 = 28.75%

Q = (17/80) * 100 = 21.25%

R = (14/80) * 100 = 17.5%

S = (26/80) * 100 = 32.5%

Answer: A

Since, the present age of A = 18 years

So, A's age after 2(P – Q) years = 18 + 2(28.75 – 21.25) = 33 years

Since, the present age of B = 12 years

So, B's age after 2(P – Q) years = 12 + 2(28.75 – 21.25) = 27 years

And, M: N = 33: 27 = 11: 9

Since, the present age of C = 9 years

The present age of D= 21 years

So, the required ratio = 9: 21 = 3: 7 = (N/3): (M – 4)

11. Questions

Since the difference between the present ages of C and D is 12 years.

The total present age of all = [12/(35% - 15%)]*100 = 60 years

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The present age of A = (30/100)*60 = 18 years

The present age of B = (20/100)*60 = 12 years

The present age of C = (15/100)*60 = 9 years

The present age of D = (35/100)*60 = 21 years

Now,

A's age after 5 years = 18 + 5 = 23 years

B's age after 5 years = 12 + 5 = 17 years

C's age after 5 years = 9 + 5 = 14 years

D's age after 5 years = 21 + 5 = 26 years

The total of age of all after 5 years = 23 + 17 + 14 + 26 = 80 years

So,

P = (23/80) * 100 = 28.75%

Q = (17/80) * 100 = 21.25%

R = (14/80) * 100 = 17.5%

S = (26/80) * 100 = 32.5%

Answer: A

Using statement X:

Difference of P and R = 28.75 – 17.5 = 11.25

Since 11.25 is not divisible by 1.5.

So, the statement X is not true.

Using statement Y:

Sum of Q and S = 21.25 + 32.5 = 53.75

Since 53.75 is not divisible by 1.75.

So, the statement Y is not true.

Using statement Z:

Since, B's age after 5 years = 17 years

And C's age after 5 years = 14 years

So, the required percentage = [(17 – 14)/14] * 100 = 21.4% (approx.)

So, the statement Z is true.

Hence, Only statement Z is true.

12. Questions

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Logic of the Series I:

$569 - 13^2 = 400$

$400 - 11^2 = 279$

$279 - 7^2 = \mathbf{230}$

$230 - 5^2 = 205$

$205 - 3^2 = 196$

$196 - 2^2 = 192$

So,

15A – B = 230 ----------(1)

Logic of the Series II:

840 ÷ 2 = 420

420 ÷ 2.5 = 168

168 ÷ 3 = 56

56 ÷ 3.5 = **16**

16 ÷ 4 = 4

So, A = 16

From equation (1):

15 * 16 – B = 230

B = 10

Answer: A

According to the equation,

From the statement P:

Since, both A and B are multiple of 2.

So, A and B are not co-prime with each other.

So, the statement P is not true.

From the statement Q:

Value of $(A^2 – B^2) = (16^2 - 10^2) = 156$

Since 156 is not divisible by 9.

So, the statement Q is not true.

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From the statement R:

Value of $(A^2 + B^2) = (16^2 + 10^2) = 356$

Since 356 is not divisible by 17.

So, the statement R is not true.

Hence, none of the statements is true.

13. Questions

Logic of the Series I:

$569 - 13^2 = 400$

$400 - 11^2 = 279$

$279 - 7^2 =$ **230**

$230 - 5^2 = 205$

$205 - 3^2 = 196$

$196 - 2^2 = 192$

So,

15A – B = 230 ----------(1)

Logic of the Series II:

840 ÷ 2 = 420

420 ÷ 2.5 = 168

168 ÷ 3 = 56

56 ÷ 3.5 = **16**

16 ÷ 4 = 4

So, A = 16

From equation (1):

15 * 16 – B = 230

B = 10

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Answer: E

The 1st term of the series = B = 10

The 2nd term of the series = A = 16

The 3rd term of the series = 4(A – B) = 24

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The 4th term of the series = (2A + 2) = 34

The 5th term of the series = (5B – 4) = 46

Logic in the series:

10 + 6 = 16

16 + 8 = 24

24 + 10 = 34

34 + 12 = 46

So, the 6th term of the series = 46 + 14 = 60

14. Questions

Let's we assume that,

The number of employees, who attended only the 3rd meeting = 4x

So, the number of employees, who attended only the 2nd meeting = 75% of 4x = 3x

And the number of employees, who attended only the 1st meeting = 2 * 3x = 6x

Let the number of employees, who attended both the 1st and 3rd meetings but not 2nd meeting = 4y

So, the number of employees, who attended both the 2nd and 3rd meetings but not 1st meeting = 75% of 4y = 3y

And the number of employees, who attended both the 1st and 2nd meetings but not 3rd meeting = 2 * 3y = 6y

The number of employees, who attended all the three meetings = 32

The total number of employees, who attended only one meeting:

4x + 3x + 6x = 13x

The total number of employees, who attended only two meetings:

4y + 3y + 6y = 13y

The total number of employees, who attended the 1st meeting:

6x + 4y + 6y + 32 = 6x + 10y + 32

Since, the number of employees, who attended only one meeting, is 104 more than those, who attended only two meetings.

So, 13x – 13y = 104

x – y = 8 ----------(1)

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Since, the total number of employees, who attended the 1st meeting = 144

So, 6x + 10y + 32 = 144

3x + 5y = 56 ----------(2)

Multiplying equation (1) by 3 then and solving both equations we get,

3x + 5(x – 8) = 56

3x + 5x – 40 = 56

x = 12

Putting the value of x into equation (1), then we get the value of y,

x – y = 8

12 – y = 8

y = 4

The number of employees, who attended only the 1st meeting = 6 * 12 = 72

The number of employees, who attended only the 2nd meeting = 3 * 12 = 36

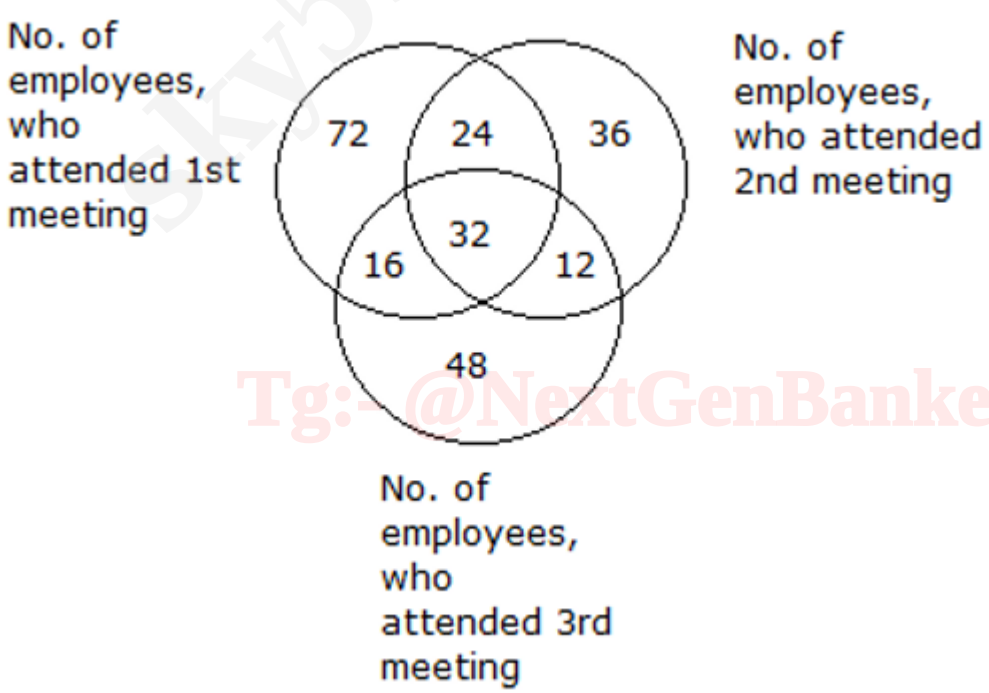
The number of employees, who attended only the 3rd meeting = 4 * 12 = 48

The number of employees, who attended both the 1st and 2nd meetings but not the 3rd meeting = 6 * 4 = 24

The number of employees, who attended both the 2nd and 3rd meetings but not the 1st meeting = 3 * 4 = 12

The number of employees, who attended both the 1st and 3rd meetings but not the 2nd meeting = 4 * 4 = 16

The number of employees, who attended all the three meetings = 32



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Answer: B

According to the question,

The total number of employees, who attended the 2nd meeting:

36 + 24 + 12 + 32 = 104

The total number of employees, who didn't attend the 2nd meeting:

72 + 16 + 48 = 136

Required ratio = 104: 136 = 13: 17

15. Questions

Let's we assume that,

The number of employees, who attended only the 3rd meeting = 4x

So, the number of employees, who attended only the 2nd meeting = 75% of 4x = 3x

And the number of employees, who attended only the 1st meeting = 2 * 3x = 6x

Let the number of employees, who attended both the 1st and 3rd meetings but not 2nd meeting = 4y

So, the number of employees, who attended both the 2nd and 3rd meetings but not 1st meeting = 75% of 4y = 3y

And the number of employees, who attended both the 1st and 2nd meetings but not 3rd meeting = 2 * 3y = 6y

The number of employees, who attended all the three meetings = 32

The total number of employees, who attended only one meeting:

4x + 3x + 6x = 13x

The total number of employees, who attended only two meetings:

4y + 3y + 6y = 13y

The total number of employees, who attended the 1st meeting:

6x + 4y + 6y + 32 = 6x + 10y + 32

Since, the number of employees, who attended only one meeting, is 104 more than those, who attended only two meetings.

So, 13x – 13y = 104

x – y = 8 ----------(1)

Since, the total number of employees, who attended the 1st meeting = 144

So, 6x + 10y + 32 = 144

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$3x + 5y = 56$ ----------(2)

Multiplying equation (1) by 3 then and solving both equations we get,

$3x + 5(x - 8) = 56$

$3x + 5x - 40 = 56$

$x = 12$

Putting the value of x into equation (1), then we get the value of y,

$x - y = 8$

$12 - y = 8$

$y = 4$

The number of employees, who attended only the 1st meeting = 6 * 12 = 72

The number of employees, who attended only the 2nd meeting = 3 * 12 = 36

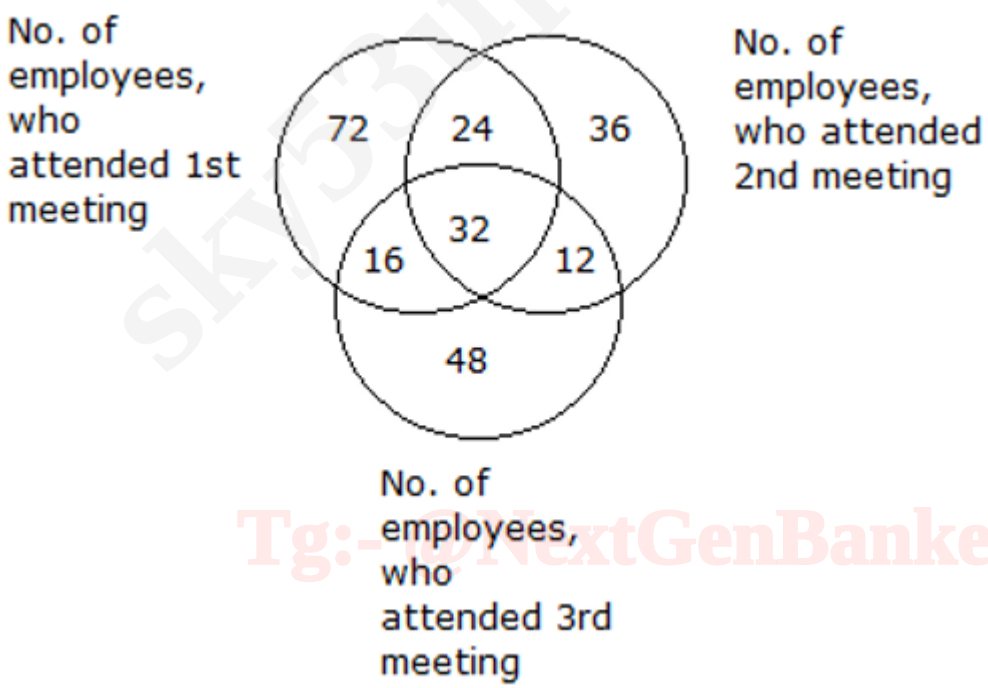
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The number of employees, who attended both the 1st and 2nd meetings but not the 3rd meeting = 6 * 4 = 24

The number of employees, who attended both the 2nd and 3rd meetings but not the 1st meeting = 3 * 4 = 12

The number of employees, who attended both the 1st and 3rd meetings but not the 2nd meeting = 4 * 4 = 16

The number of employees, who attended all the three meetings = 32



Answer: A

The number of male employees, who attended both the 1st and 2nd meetings but not the 3rd meeting = 24 *

(5/12) = 10

The number of female employees, who attended both the 1st and 2nd meetings but not the 3rd meeting = 24 – 10 = 14

The number of male employees, who attended both the 2nd and 3rd meetings but not the 1st meeting = 12 * (3/4) = 9

The number of female employees, who attended both the 2nd and 3rd meetings but not 1st meeting = 12 – 9 = 3

The number of male employees, who attended both the 1st and 3rd meetings but not the 2nd meeting = 16 * (3/8) = 6

The number of female employees, who attended both the 1st and 3rd meetings but not the 2nd meeting = 16 – 6 = 10

Required difference = (14 + 3 + 10) – (10 + 9 + 6) = 2

16. Questions

Let's we assume that,

The number of employees, who attended only the 3rd meeting = 4x

So, the number of employees, who attended only the 2nd meeting = 75% of 4x = 3x

And the number of employees, who attended only the 1st meeting = 2 * 3x = 6x

Let the number of employees, who attended both the 1st and 3rd meetings but not 2nd meeting = 4y

So, the number of employees, who attended both the 2nd and 3rd meetings but not 1st meeting = 75% of 4y = 3y

And the number of employees, who attended both the 1st and 2nd meetings but not 3rd meeting = 2 * 3y = 6y

The number of employees, who attended all the three meetings = 32

The total number of employees, who attended only one meeting:

4x + 3x + 6x = 13x

The total number of employees, who attended only two meetings:

4y + 3y + 6y = 13y

The total number of employees, who attended the 1st meeting:

6x + 4y + 6y + 32 = 6x + 10y + 32

Since, the number of employees, who attended only one meeting, is 104 more than those, who attended only two meetings.

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So, 13x – 13y = 104

x – y = 8 ----------(1)

Since, the total number of employees, who attended the 1st meeting = 144

So, 6x + 10y + 32 = 144

3x + 5y = 56 ----------(2)

Multiplying equation (1) by 3 then and solving both equations we get,

3x + 5(x – 8) = 56

3x + 5x – 40 = 56

x = 12

Putting the value of x into equation (1), then we get the value of y,

x – y = 8

12 – y = 8

y = 4

The number of employees, who attended only the 1st meeting = 6 * 12 = 72

The number of employees, who attended only the 2nd meeting = 3 * 12 = 36

The number of employees, who attended only the 3rd meeting = 4 * 12 = 48

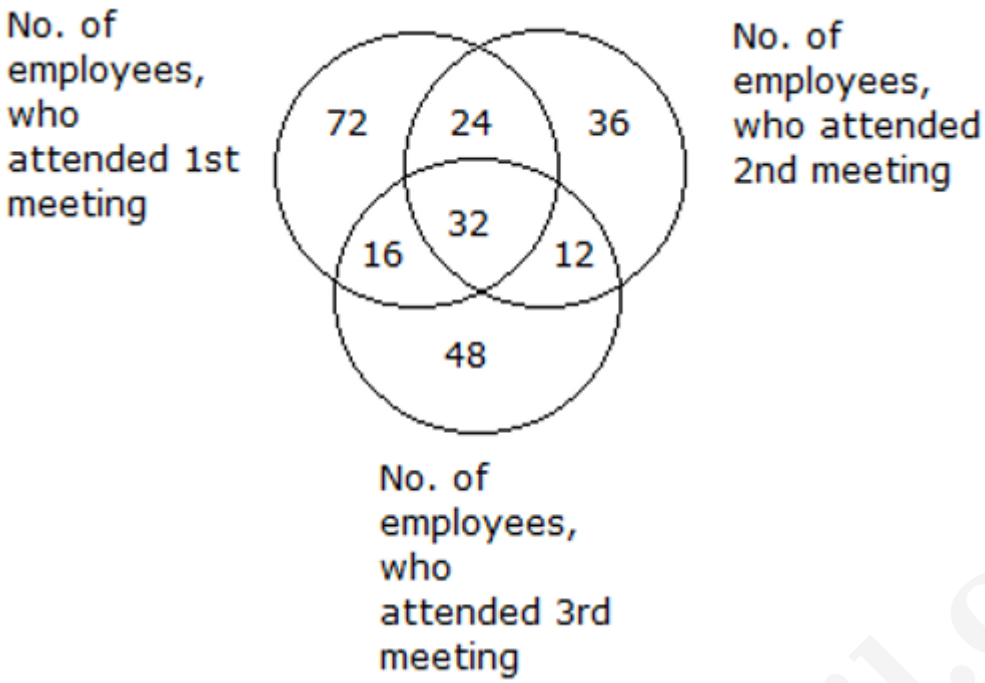
The number of employees, who attended both the 1st and 2nd meetings but not the 3rd meeting = 6 * 4 = 24

The number of employees, who attended both the 2nd and 3rd meetings but not the 1st meeting = 3 * 4 = 12

The number of employees, who attended both the 1st and 3rd meetings but not the 2nd meeting = 4 * 4 = 16

The number of employees, who attended all the three meetings = 32

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Answer: E

The total number of employees in the company:

72 + 36 + 48 + 24 + 12 + 16 = 32 = 240

The total number of employees, who attended the 3rd meeting:

48 + 16 + 12 + 32 = 108

Required percentage = (108/240) * 100 = 45%

17. Questions

Let's we assume that,

The number of employees, who attended only the 3rd meeting = 4x

So, the number of employees, who attended only the 2nd meeting = 75% of 4x = 3x

And the number of employees, who attended only the 1st meeting = 2 * 3x = 6x

Let the number of employees, who attended both the 1st and 3rd meetings but not 2nd meeting = 4y

So, the number of employees, who attended both the 2nd and 3rd meetings but not 1st meeting = 75% of 4y = 3y

And the number of employees, who attended both the 1st and 2nd meetings but not 3rd meeting = 2 * 3y = 6y

The number of employees, who attended all the three meetings = 32

The total number of employees, who attended only one meeting:

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$4x + 3x + 6x = 13x$

The total number of employees, who attended only two meetings:

$4y + 3y + 6y = 13y$

The total number of employees, who attended the 1st meeting:

$6x + 4y + 6y + 32 = 6x + 10y + 32$

Since, the number of employees, who attended only one meeting, is 104 more than those, who attended only two meetings.

So, $13x - 13y = 104$

$x - y = 8$ ----------(1)

Since, the total number of employees, who attended the 1st meeting = 144

So, $6x + 10y + 32 = 144$

$3x + 5y = 56$ ----------(2)

Multiplying equation (1) by 3 then and solving both equations we get,

$3x + 5(x - 8) = 56$

$3x + 5x - 40 = 56$

$x = 12$

Putting the value of x into equation (1), then we get the value of y,

$x - y = 8$

$12 - y = 8$

$y = 4$

The number of employees, who attended only the 1st meeting = $6 * 12 = 72$

The number of employees, who attended only the 2nd meeting = $3 * 12 = 36$

The number of employees, who attended only the 3rd meeting = $4 * 12 = 48$

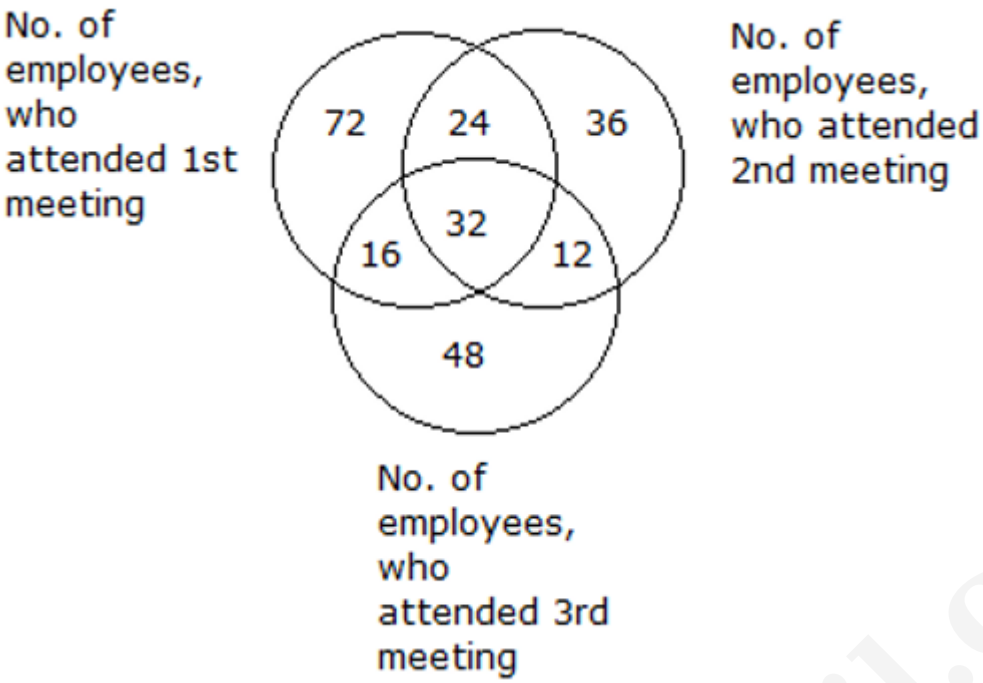
The number of employees, who attended both the 1st and 2nd meetings but not the 3rd meeting = $6 * 4 = 24$

The number of employees, who attended both the 2nd and 3rd meetings but not the 1st meeting = $3 * 4 = 12$

The number of employees, who attended both the 1st and 3rd meetings but not the 2nd meeting = $4 * 4 = 16$

The number of employees, who attended all the three meetings = 32

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Answer: A

The total number of employees, who attended only one meeting:

72 + 36 + 48 = 156

The total number of employees, who attended at least two meetings:

24 + 12 + 16 + 32 = 84

Required difference = (156 – 84) = 72

18. Questions

Let the minimum runs scored by Team A = 100x

So, the minimum runs scored by Teams B, C and D will be 125x, 105x and 85x respectively.

Since, the average of minimum runs scored by each Team, is 249.

So,

(100x + 125x + 105x + 85x)/4 = 249

415x = 996

x = 2.4

Minimum runs scored by Team A = 100 * 2.4 = 240

Maximum runs scored by Team A = 240 * (5/4) = 300

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Team	Maximum runs scored	Minimum runs scored
A	300	240
B	120% of 300 = 360	125% of 240 = 300
C	90% of 300 = 270	105% of 240 = 252
D	80% of 300 = 240	85% of 240 = 204

Answer: D

The sum of maximum and minimum runs scored by Team A = 300 + 240 = 540

So, the runs scored by Team A in the remaining matches = 540 + 60 = 600

And the total runs scored by Team A in the series = 540 + 600 = 1140

The sum of maximum and minimum runs scored by Team B = 360 + 300 = 660

So, the runs scored by Team B in the remaining matches = 660 – 60 = 600

And the total runs scored by Team B in the series = 660 + 600 = 1260

Required ratio = 1140: 1260 = 19: 21

19. Questions

Let the minimum runs scored by Team A = 100x

So, the minimum runs scored by Teams B, C and D will be 125x, 105x and 85x respectively.

Since, the average of minimum runs scored by each Team, is 249.

So,

(100x + 125x + 105x + 85x)/4 = 249

415x = 996

x = 2.4

Minimum runs scored by Team A = 100 * 2.4 = 240

Maximum runs scored by Team A = 240 * (5/4) = 300

Team	Maximum runs scored	Minimum runs scored
A	300	240
B	120% of 300 = 360	125% of 240 = 300
C	90% of 300 = 270	105% of 240 = 252
D	80% of 300 = 240	85% of 240 = 204

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Answer: E

The maximum runs scored by Team C in a match = 270

The runs scored by Team C by hitting boundaries in the match, in which it scored maximum run = (15 * 6) + (20 * 4) = 90 + 80 = 170

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So, the runs scored by Team C by running in the match, in which it scored maximum runs = 270 – 170 = 100

The minimum runs scored by Team C in a match = 252

The runs scored by Team C by hitting boundaries in the match, in which it scored minimum run = (8 * 6) + (16 * 4) = 48 + 64 = 112

So, the runs scored by Team C by running in the match, in which it scored minimum runs = 252 – 112 = 140

Required sum = 100 + 140 = 240

20. Questions

Let the minimum runs scored by Team A = 100x

So, the minimum runs scored by Teams B, C and D will be 125x, 105x and 85x respectively.

Since, the average of minimum runs scored by each Team, is 249.

So,

(100x + 125x + 105x + 85x)/4 = 249

415x = 996

x = 2.4

Minimum runs scored by Team A = 100 * 2.4 = 240

Maximum runs scored by Team A = 240 * (5/4) = 300

Team	Maximum runs scored	Minimum runs scored
A	300	240
B	120% of 300 = 360	125% of 240 = 300
C	90% of 300 = 270	105% of 240 = 252
D	80% of 300 = 240	85% of 240 = 204

Answer: B

The sum of the minimum runs scored by each Team = 240 + 300 + 252 + 204 = 996

The sum of the maximum runs scored by each Team = 300 + 360 + 270 + 240 = 1170

Required percentage = (996/1170) * 100 = 85% (approx.)

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1. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the cumulative percentage distribution of the total number of mobiles (iPhone + Android + Symbian) sold by five different shops P, Q, R, S, and T. The Table also shows the average number of Android and Symbian mobiles sold and the difference between the number of Android and Symbian (android-symbian) mobiles sold by each shop.

Shop	Cumulative percentage of the total number of mobiles sold	The average number of Android and Symbian mobiles sold	Difference between the number of Android and Symbian mobiles sold (android-Symbian)
P	20%	(2A-60)	20
Q	30%	(B+5)	20
R	55%	(A-5)	40
S	85%	(3B+15)	50
T	100%	(A/2)	10

Note:

The difference between the total number of mobiles (iPhone + android + Symbian) sold by shop R and T is 90.

The sum of the total number of Android and Symbian mobiles sold by Shop P and Shop Q together is 130.

The sum of the number of Android and Symbian mobiles sold by shop S is 60 more than that of shop R.

In each shop, the number of Android mobiles sold is more than that of Symbian mobiles.

Total number of mobiles sold by any shop = number of iPhone mobiles sold + number of Android mobiles sold + number of Symbian mobiles sold

Total number of mobiles manufactured = number of mobiles sold + number of mobiles unsold

Find the relation between L, M, and N.

In shop P, out of the total number of mobiles manufactured (iPhone + Android + Symbian), 40% remain unsold. The ratio of the number of manufactured iPhones to that of manufactured Android to that of manufactured Symbian mobiles is 17:9:4. Whereas, the number of unsold iPhone mobiles is 75% more than that of unsold Android mobiles, which is four times the number of unsold Symbian mobiles sold.

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L = The average of the number of sold and unsold iPhone mobiles.

M =The difference between the number of sold and unsold Android mobiles.

N = The sum of the number of sold and unsold Symbian mobiles.

a. L > M > N

b. L < M < N

c. L > M < N

d. L < M > N

e. L = M = N

2. Questions

In another shop U, the total number of mobiles (iPhone + android + Symbian) sold is 65 more than that of shop T. While the number of iPhones, android and Symbian mobiles sold are (2A+10), 3B and [{(A+B)/2}-5] respectively. Find the total number of mobiles (iPhone + Android + Symbian) manufactured by Shop U, if out of the total number of mobiles manufactured 88% of iPhone, 75% of Android mobiles, and 50% of Symbian mobiles are sold.

a. 254

b. 265

c. 282

d. 271

e. None of these

3. Questions

In shop Q, there are three types of iPhone mobiles sold i.e., iPhone Max, phone smart, and iPhone Pro. The number of iPhone Max, iPhone Smart, and iPhone Pro mobiles sold are E^2, 2E, and E respectively. Out of the total number of Android mobiles sold by shop Q, 4E% are 4G mobiles, while the rest are 5G mobiles. Then find the difference between the number 4G and 5G android mobiles sold by shop Q. (Value of E is a positive integer)

a. 21

b. 40

c. 32

d. Cannot be determined

e. None of these

4. Questions

In shop R, the price of each iPhone, Android, and Symbian mobile sold is Rs. F thousand, Rs. G thousand and Rs. 0.5G thousand respectively. The total revenue generated from selling all the iPhone and Android mobiles is Rs. 9350 thousand. If the shopkeeper had sold 20 iPhones and 30 Android mobiles, then the total revenue generated would have been Rs. 7150 thousand less than the of actual revenue generated. Find the total revenue generated from selling all the Symbian mobiles by shop R.

a. Rs. 300 thousand

b. Rs. 240 thousand

c. Rs. 410 thousand

d. Rs. 550 thousand

e. None of these

5. Questions

For the questions given below, three fillers I, II & III have been given. Analyse and answer whether the data provided in the three fillers are filling the blanks correctly in the same order or not.

In shop S, out of the total number of Android mobiles manufactured, 20% are defective and hence could not be sold. Out of the total number of non-defective Android mobiles, _____% are sold. The total number of Android mobiles manufactured by shop S is _____.

Filler I: 50, 250

Filler II: 25, 500

Filler III: 10, 1000

a. Only filler I is correct

b. Only filler III is correct

c. Both fillers I and II are correct

d. All the fillers I, II and III are correct

e. Neither filler I or II nor III is correct

6. Questions

Read the following information carefully and answer the questions based on it.

There are five different vehicles car, bus, bike, truck, and cab. They travel different distances with different speeds and the time taken by them to travel a certain distance is also different.

The time taken by the car to travel 250 km with the speed of (S-10) km/h is (S/12) hours. The same car can cross a man running in the opposite direction with a speed of $1/5^{th}$ of that of a car and covers (D+40) km in (D/50) hours. The bus covers a distance of (D-60) km with a speed of (V+30) km/h in (V/20) hours. The bike can cross the same man whom the car has crossed running in the opposite direction with the speed of the bike is 4T km/h in T hours, covering (7V+20) km, running in the same direction. The truck travels a distance of (X+60) km with a speed of (X-20) km/h in (T/5) hours. Cab travels (Y+30) km distance with a speed of (Y-90) km/h in (Y/50) hours.

Note:

Values of S, D, V, T, X, and Y are positive integers.

A train whose length is 300 meters, crosses a platform of length of length P meters in 4 seconds with the speed of A m/sec. Another train, whose length is 40 meters less than that of the first train, crosses the same platform in 2 seconds with a speed of 80 m/sec more than that of the first train.

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Find the time taken by the bus to cover (P+350) km, if its speed is (A+50) km/h.

a. 2 hours

b. 4 hours

c. 3 hours

d. 5 hours

e. None of these

7. Questions

The distance between city B and city C is 3Z km. The bike starts from City B towards City C at 9:00 am, but stops at 11:30 am after reaching City E, which is somewhere between City B and City C. Now, the bike is stopped and waiting for the car to reach City E from City C. The time taken by the car to reach city E from city C is 4 hours. Find the time taken by bike and car to cover (2Z-20) km, travelling in the opposite direction to each other.

a. 2 hours

b. 1 hour

c. 6 hours

d. 9 hours

e. None of these

8. Questions

For the questions given below, three statements I, II & III have been given. Analyse and answer whether the data provided in the three statements are true or not.

The time taken by a train, whose length is (X+Y) meters to cross a tunnel whose length is 2V meters with the speed of (S+6) m/sec is M seconds. Whereas, the time taken by the truck to cover (U-W) km distance is M hours. The cab can cover (U-60) km distance in 9 hours.

Statement I: The ratio of the values of U to that of W is 3:1.

Statement II: The bus can cover (W+10) km in 5 hours.

Statement III: The value of W is half of that of X.

a. Only statement III is true

b. Only statement I is true

c. Both the statements I and II are true

d. All the statements I, II and III are true

e. Neither statement I or II nor III is true

9. Questions

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The time taken by the bus to travel (Q+50) km is the same as the time taken by the cab to travel (R-200) km. Also, the time taken by the bus to travel (R-10) km is the same as the time taken by the cab to travel (Q+120) km. Find the distance travelled by car in 10 hours.

a. (2R-500) km

b. (Q+100) km

c. (R-Q) km

d. All of the above

e. None of these

10. Questions

The sum of the distance travelled by car in F hours and bus in G hours is 1340 km. Whereas, the time taken by the truck to cover (4F+5G) km with 25% more speed than its usual speed is (F/10) hours. Find the difference between the roots of the equation $[L^2 - \{(F+G)L\} + FG] = 0$

a. 12

b. 6

c. 8

d. 17

e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

Three quadratic equations I, II and III are given below.

Equations	Remarks
I). $X^2 - 13X + A = 0$	p and q are the roots of the equation, where one of the roots is a prime number less than 11 but more than 5. (p > q)
II). $Y^2 + 3Y - B = 0$	r and s are the roots of the equation, where the difference between them is 13. (r > s)
III). $Z^2 - 2Z - C = 0$	t and u are the roots of the equation, where one root is twice and other root is equal to that of r and s in any order. (t > u)

Find the correct relation between A, B and C.

Relation I: (A-82) = B

Relation II: 2B = C

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Relation III: (C+120) = A

a. Only relation I is correct

b. Both the relations I and II are correct

c. Both the relations II and III are correct

d. All the relations are correct

e. None of these

12. Questions

Which of the following match follows correctly?

Value of X	**L:** The square of twice the greater root is 100
Value of Y	**M:** The product of roots is 10 times the cube of -2
Value of Z	**N:** The difference between the roots is 1

I: X-M, Y-N, Z-L

II: X-N, Y-L, Z-M

III: X-L, Y-M, Z-N

a. Only I follow

b. Both I and III follow

c. Only II follows

d. All I, II, and III follow

e. Neither I nor II nor III follows

13. Questions

Find the value of [{√(A-17)}+(B-C)].

a. 50

b. 34

c. 22

d. 45

e. None of these

14. Questions

Find the smaller root of the equation $k^2 - (p+t)\,k + 14r = 0$.

a. t

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b. u

c. r

d. s

e. None of these

15. Questions

Which of the following equations have roots (-r) and t?

a. $d^2 - 5d - 50 = 0$

b. $d^2 + 5d - 50 = 0$

c. $d^2 - 15d + 50 = 0$

d. Cannot be determined

e. None of these

16. Questions

A man invested Rs. 3000X in a scheme at the rate of 15% per annum simple interest for 3 years. The total amount received from the scheme is reinvested in another scheme, which offers compound interest at the rate of 20% per annum for 2 years. The difference between the simple and compound interest earned is equal to Rs. 282Y. Find the ratio of 2X:3Y.

a. 1:2

b. 1:3

c. 3:4

d. 2:5

e. None of these

17. Questions

A, B and C can complete a piece of work in (P-2) days, (P+3) days and (P-7) days respectively. If A and B together work for 4 days, then 2/3rd of the total work gets completed. Find the time taken by C alone to complete the work, if the total work is 30 units.

a. 5 days

b. 6 days

c. 10 days

d. 12 days

e. None of these

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18. Questions

Find the value of $\sqrt{(K-L-4^2)}$.

The cost price of the calculator is Rs. (M+200), which is marked up 50% above its cost price and sold at a profit of 20%. The marked price of the watch is Rs. (N+400), which is sold at a discount of 10%. The watch was marked up 20% above its cost price. The marked price of the calculator is Rs. 900 less than that of the watch. Whereas, the sum of the selling price of both articles is Rs. 3360.

K = Cost price of watch.

L = Selling price of calculator.

a. 29

b. 24

c. 21

d. 28

e. None of these

19. Questions

A mixture contains 180 ml of pure milk. If (Q-40) ml of mixture is replaced with the same quantity of water, then the ratio of the quantity of milk to that of water in the resultant mixture becomes 7:2. Had (Q+10) ml of pure milk was removed and replaced with (0.5Q-10) ml of water, then the total quantity of milk and water in the resultant mixture becomes 2R ml. Find the value of (4Q-3R).

a. 120

b. 150

c. 140

d. Cannot be determined

e. None of these

20. Questions

The time taken by boat A to cover 28 km in upstream is 2 hours, which is the same as the time taken by boat B to cover 72 km in downstream. The speed of boat A in still water is 33.33% less than that of boat B. Both the boats are rowing in the same river. The difference between the downstream speed of boat A and the upstream speed of boat B is _____.

a. 2 km/h

b. 3 km/h

c. 6 km/h

d. 7 km/h

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e. None of these

Explanations:

1. Questions

Percentage distribution of the total number of mobiles (iPhone + Android + Symbian) sold by each shop,

P = 20%

Q = (30-20)% = 10%

R = (55-30)% = 25%

S = (85-55)% = 30%

T = (100-85)% = 15%

Difference between the total number of mobiles (iPhone + android + Symbian) sold by shop R and T = 90

So,

(25-15)% = 90

10% = 90

1% = (90/10) = 9

100% = (9×100) = 900

Now, total number of mobiles (iphone + android + symbian) sold by each shop,

P = 20% of 900 = 180

Q = 10% of 900 = 90

R = 25% of 900 = 225

S = 30% of 900 = 270

T = 15% of 900 = 135

Now,

The sum of the total number of Android and Symbian mobiles sold by shop P = {2×(2A-60)} = (4A-120)

The sum of the total number of Android and Symbian mobiles sold by shop Q = {2×(B+5)} = (2B+10)

So,

(2B+10) + (4A-120) = 130

2B + 4A = 240……..(i)

Also,

The sum of the total number of Android and Symbian mobiles sold by shop R = {2×(A-5)} = (2A-10)

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The sum of the total number of Android and Symbian mobiles sold by shop S = {2×(3B+15)} = (6B+30)

So,

(6B+30) – (2A-10) = 60

6B – 30 = 2A – 10

6B – 2A = 20……..(ii)

Solving both the equation, we get,

2B + 4A = 240

6B – 2A = 20

6B + 12A = 720

6B – 2A = 20

14A = 700

A = 50

Now, put the value of A in equation (i), we get,

2B + 4A = 240

2B = 240 – 200

2B = 40

B = 20

Now,

Shop P

Total number of mobiles (iPhone + android + Symbian) sold = 180

Average number of android and Symbian mobiles sold = (2A-60) = (100-60) = 40

The sum of the number of Android and Symbian mobiles sold = (40×2) = 80

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be v and (v-20)

2v – 20 = 80

2v = 100

v = 50

Number of android mobiles sold = v = 50

Number of Symbian mobiles sold = (v-20) = (50-20) = 30

Number of iPhone mobiles sold = {180-(50+30)} = (180-80) = 100

Shop Q

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Total number of mobiles (iPhone + android + Symbian) sold = 90

Average number of android and Symbian mobiles sold = (B+5) = (20+5) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be w and (w-20)

2w– 20 = 50

2w = 70

w = 35

Number of android mobiles sold = w = 35

Number of Symbian mobiles sold = (w-20) = (35-20) = 15

Number of iPhone mobiles sold = {90-(35+15)} = (90-50) = 40

Shop R

Total number of mobiles (iPhone + android + Symbian) sold = 225

Average number of android and Symbian mobiles sold = (A-5) = (50-5) = 45

The sum of the number of Android and Symbian mobiles sold = (45×2) = 90

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be x and (x-40)

2x – 40 = 90

2x = 130

x = 65

Number of android mobiles sold = x = 65

Number of Symbian mobiles sold = (x-40) = (65-40) = 25

Number of iPhone mobiles sold = {225-(65+25)} = (225-90) = 135

Shop S

Total number of mobiles (iPhone + android + Symbian) sold = 270

Average number of android and Symbian mobiles sold = (3B+15) = (60+15) = 75

The sum of the number of Android and Symbian mobiles sold = (75×2) = 150

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be y and (y-50)

2y – 50 = 150

2y = 200

y = 100

Number of android mobiles sold = y = 100

Number of Symbian mobiles sold = (y-50) = (100-50) = 50

Number of iPhone mobiles sold = {270-(100+50)} = (270-150) = 120

Shop T

Total number of mobiles (iPhone + android + Symbian) sold = 135

Average number of android and Symbian mobiles sold = (A/2) = (50/2) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of Android and Symbian (android-symbian) mobiles sold = 10

Let, the number of Android and Symbian mobiles sold be z and (z-10)

2z – 10 = 50

2z = 60

z = 30

Number of android mobiles sold = z = 30

Number of Symbian mobiles sold = (z-10) = (30-10) = 20

Number of iPhone mobiles sold = {135-(30+20)} = (135-50) = 85

Shop	Total number of mobiles sold	Number of iPhone mobiles sold	Number of Android mobiles sold	Number of Symbian mobiles sold
P	180	100	50	30
Q	90	40	35	15
R	225	135	65	25
S	270	120	100	50
T	135	85	30	20
Total	900	480	280	140

Answer: C

According to the question,

Shop P,

Let, the total number of mobiles (iPhone + Android + Symbian) manufactured be 100K.

Now,

Total number of mobiles (iphone + andoird + symbian) unsold = 40% of 100K = 40K

Total number of mobiles (iphone + andoird + symbian) sold = 60% of 100K = 60K

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Total number of mobiles (iphone + andoird + symbian) sold = 180

So,

60K = 180

K = 3

Total number of mobiles (iphone + andoird + symbian) manufactured = 100K = (100×3) = 300

Total number of mobiles (iPhone + android + Symbian) sold = 180

Number of iPhone mobiles sold = 100

Number of Android mobiles sold = 50

Number of Symbian mobiles sold = 30

Total number of mobiles (iphone + andoird + symbian) unsold = (300-180) = 120

Now,

Let, the number of manufactured iPhones, manufactured Android, and manufactured Symbian mobiles be 17C, 9C, and 4C respectively.

The ratio of the number of unsold iPhones to that of unsold Androids to that of unsold Symbian mobiles = 175:100:25 = 7:4:1. Let, it be 7D, 4D, and D respectively.

Now,

17C – 7D = 100……..(i)

Also,

9C – 4D = 50……(ii)

Solving both equations, we get,

153C – 63D = 900

153C – 68D = 850

5D = 50

D = 10

Now, put the value of D in equation (i), we get,

17C – 7D = 100

17C = 100 + 70

17C = 170

C = 10

Now,

L: Number of iPhone mobiles sold = 100

Number of iPhone mobiles unsold = 7D = (7×10) = 70

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Average of the number of sold and unsold iPhone mobiles = {(100+70)/2} = (170/2) = 85.

So, L = 85

M: Number of Android mobiles sold = 50

Number of android mobiles unsold = 4D = (4×10) = 40

Difference between the number of sold and unsold android mobiles = (50-40) = 10

So, M = 10

N: Number of Symbian mobiles sold = 30

Number of Symbian mobiles unsold = D = 10

The sum of the number of sold and unsold Symbian mobiles = (30+10) = 40

So, N = 40

Therefore, L > M < N.

2. Questions

Percentage distribution of the total number of mobiles (iPhone + Android + Symbian) sold by each shop,

P = 20%

Q = (30-20)% = 10%

R = (55-30)% = 25%

S = (85-55)% = 30%

T = (100-85)% = 15%

Difference between the total number of mobiles (iPhone + android + Symbian) sold by shop R and T = 90

So,

(25-15)% = 90

10% = 90

1% = (90/10) = 9

100% = (9×100) = 900

Now, total number of mobiles (iphone + android + symbian) sold by each shop,

P = 20% of 900 = 180

Q = 10% of 900 = 90

R = 25% of 900 = 225

S = 30% of 900 = 270

T = 15% of 900 = 135

Now,

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The sum of the total number of Android and Symbian mobiles sold by shop P = {2×(2A-60)} = (4A-120)

The sum of the total number of Android and Symbian mobiles sold by shop Q = {2×(B+5)} = (2B+10)

So,

(2B+10) + (4A-120) = 130

2B + 4A = 240……..(i)

Also,

The sum of the total number of Android and Symbian mobiles sold by shop R = {2×(A-5)} = (2A-10)

The sum of the total number of Android and Symbian mobiles sold by shop S = {2×(3B+15)} = (6B+30)

So,

(6B+30) – (2A-10) = 60

6B – 30 = 2A – 10

6B – 2A = 20……..(ii)

Solving both the equation, we get,

2B + 4A = 240

6B – 2A = 20

6B + 12A = 720

6B – 2A = 20

14A = 700

A = 50

Now, put the value of A in equation (i), we get,

2B + 4A = 240

2B = 240 – 200

2B = 40

B = 20

Now,

Shop P

Total number of mobiles (iPhone + android + Symbian) sold = 180

Average number of android and Symbian mobiles sold = (2A-60) = (100-60) = 40

The sum of the number of Android and Symbian mobiles sold = (40×2) = 80

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be v and (v-20)

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2v – 20 = 80

2v = 100

v = 50

Number of android mobiles sold = v = 50

Number of Symbian mobiles sold = (v-20) = (50-20) = 30

Number of iPhone mobiles sold = {180-(50+30)} = (180-80) = 100

Shop Q

Total number of mobiles (iPhone + android + Symbian) sold = 90

Average number of android and Symbian mobiles sold = (B+5) = (20+5) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be w and (w-20)

2w– 20 = 50

2w = 70

w = 35

Number of android mobiles sold = w = 35

Number of Symbian mobiles sold = (w-20) = (35-20) = 15

Number of iPhone mobiles sold = {90-(35+15)} = (90-50) = 40

Shop R

Total number of mobiles (iPhone + android + Symbian) sold = 225

Average number of android and Symbian mobiles sold = (A-5) = (50-5) = 45

The sum of the number of Android and Symbian mobiles sold = (45×2) = 90

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be x and (x-40)

2x – 40 = 90

2x = 130

x = 65

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Number of android mobiles sold = x = 65

Number of Symbian mobiles sold = (x-40) = (65-40) = 25

Number of iPhone mobiles sold = {225-(65+25)} = (225-90) = 135

Shop S

Total number of mobiles (iPhone + android + Symbian) sold = 270

Average number of android and Symbian mobiles sold = (3B+15) = (60+15) = 75

The sum of the number of Android and Symbian mobiles sold = (75×2) = 150

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be y and (y-50)

2y – 50 = 150

2y = 200

y = 100

Number of android mobiles sold = y = 100

Number of Symbian mobiles sold = (y-50) = (100-50) = 50

Number of iPhone mobiles sold = {270-(100+50)} = (270-150) = 120

Shop T

Total number of mobiles (iPhone + android + Symbian) sold = 135

Average number of android and Symbian mobiles sold = (A/2) = (50/2) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of Android and Symbian (android-symbian) mobiles sold = 10

Let, the number of Android and Symbian mobiles sold be z and (z-10)

2z – 10 = 50

2z = 60

z = 30

Number of android mobiles sold = z = 30

Number of Symbian mobiles sold = (z-10) = (30-10) = 20

Number of iPhone mobiles sold = {135-(30+20)} = (135-50) = 85

Shop	Total number of mobiles sold	Number of iPhone mobiles sold	Number of Android mobiles sold	Number of Symbian mobiles sold
P	180	100	50	30
Q	90	40	35	15
R	225	135	65	25
S	270	120	100	50
T	135	85	30	20
Total	900	480	280	140

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Answer: B

In shop U,

Total number of mobiles (iPhone + android + Symbian) sold = (135+65) = 200

Number of iPhone mobiles sold = (2A+10) = (2×50+10) = 110

Number of android mobiles sold = 3B = (3×20) = 60

Number of Symbian mobiles sold = [{(A+B)/2}-5] = [{(50+20)/2}-5] = 30

Now,

The number of iPhone mobiles sold is 88% of the total manufactured,

So,

88% = 110

1% = (110/88) = (5/4)

100% = {(5/4)×100} = 125

Therefore, the total number of iPhone mobiles manufactured = 125

The number of Android mobiles sold is 75% of the total manufactured,

So,

75% = 60

1% = (60/75) = (4/5)

100% = {(4/5)×100} = 80

Therefore, the total number of Android mobiles manufactured = 80

The number of Symbian mobiles sold is 50% of the total manufactured,

So,

50% = 30

1% = (30/50) = (3/5)

100% = {(3/5)×100} = 60

Therefore, the total number of Symbian mobiles manufactured = 60

Hence, the total number of mobiles (iPhone + Android + Symbian) manufactured = (125+80+60) = 265.

3. Questions

Percentage distribution of the total number of mobiles (iPhone + Android + Symbian) sold by each shop,

P = 20%

Q = (30-20)% = 10%

R = (55-30)% = 25%

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S = (85-55)% = 30%

T = (100-85)% = 15%

Difference between the total number of mobiles (iPhone + android + Symbian) sold by shop R and T = 90

So,

(25-15)% = 90

10% = 90

1% = (90/10) = 9

100% = (9×100) = 900

Now, total number of mobiles (iphone + android + symbian) sold by each shop,

P = 20% of 900 = 180

Q = 10% of 900 = 90

R = 25% of 900 = 225

S = 30% of 900 = 270

T = 15% of 900 = 135

Now,

The sum of the total number of Android and Symbian mobiles sold by shop P = {2×(2A-60)} = (4A-120)

The sum of the total number of Android and Symbian mobiles sold by shop Q = {2×(B+5)} = (2B+10)

So,

(2B+10) + (4A-120) = 130

2B + 4A = 240……..(i)

Also,

The sum of the total number of Android and Symbian mobiles sold by shop R = {2×(A-5)} = (2A-10)

The sum of the total number of Android and Symbian mobiles sold by shop S = {2×(3B+15)} = (6B+30)

So,

(6B+30) – (2A-10) = 60

6B – 30 = 2A – 10

6B – 2A = 20……..(ii)

Solving both the equation, we get,

2B + 4A = 240

6B – 2A = 20

6B + 12A = 720

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6B – 2A = 20

14A = 700

A = 50

Now, put the value of A in equation (i), we get,

2B + 4A = 240

2B = 240 – 200

2B = 40

B = 20

Now,

Shop P

Total number of mobiles (iPhone + android + Symbian) sold = 180

Average number of android and Symbian mobiles sold = (2A-60) = (100-60) = 40

The sum of the number of Android and Symbian mobiles sold = (40×2) = 80

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be v and (v-20)

2v – 20 = 80

2v = 100

v = 50

Number of android mobiles sold = v = 50

Number of Symbian mobiles sold = (v-20) = (50-20) = 30

Number of iPhone mobiles sold = {180-(50+30)} = (180-80) = 100

Shop Q

Total number of mobiles (iPhone + android + Symbian) sold = 90

Average number of android and Symbian mobiles sold = (B+5) = (20+5) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be w and (w-20)

2w– 20 = 50

2w = 70

w = 35

Number of android mobiles sold = w = 35

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Number of Symbian mobiles sold = (w-20) = (35-20) = 15

Number of iPhone mobiles sold = {90-(35+15)} = (90-50) = 40

Shop R

Total number of mobiles (iPhone + android + Symbian) sold = 225

Average number of android and Symbian mobiles sold = (A-5) = (50-5) = 45

The sum of the number of Android and Symbian mobiles sold = (45×2) = 90

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be x and (x-40)

2x – 40 = 90

2x = 130

x = 65

Number of android mobiles sold = x = 65

Number of Symbian mobiles sold = (x-40) = (65-40) = 25

Number of iPhone mobiles sold = {225-(65+25)} = (225-90) = 135

Shop S

Total number of mobiles (iPhone + android + Symbian) sold = 270

Average number of android and Symbian mobiles sold = (3B+15) = (60+15) = 75

The sum of the number of Android and Symbian mobiles sold = (75×2) = 150

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be y and (y-50)

2y – 50 = 150

2y = 200

y = 100

Number of android mobiles sold = y = 100

Number of Symbian mobiles sold = (y-50) = (100-50) = 50

Number of iPhone mobiles sold = {270-(100+50)} = (270-150) = 120

Shop T

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Total number of mobiles (iPhone + android + Symbian) sold = 135

Average number of android and Symbian mobiles sold = (A/2) = (50/2) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of Android and Symbian (android-symbian) mobiles sold = 10

Let, the number of Android and Symbian mobiles sold be z and (z-10)

$2z - 10 = 50$

$2z = 60$

$z = 30$

Number of android mobiles sold = z = 30

Number of Symbian mobiles sold = (z-10) = (30-10) = 20

Number of iPhone mobiles sold = {135-(30+20)} = (135-50) = 85

Shop	Total number of mobiles sold	Number of iPhone mobiles sold	Number of Android mobiles sold	Number of Symbian mobiles sold
P	180	100	50	30
Q	90	40	35	15
R	225	135	65	25
S	270	120	100	50
T	135	85	30	20
Total	900	480	280	140

Answer: A

In shop Q,

Total number of iPhone mobiles sold = 40

Number of iPhone max mobiles sold = E^2

Number of iPhone smart mobiles sold = 2E

Number of iPhone pro mobiles sold = E

So,

$E^2 + 2E + E = 40$

$E^2 + 3E = 40$

$E^2 + 3E - 40 = 0$

$E^2 + 8E - 5E - 40 = 0$

$E(E+8) - 5(E+8) = 0$

$(E-5)(E+8) = 0$

$E = 5$ or -8

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So, E = 5 (positive integer)

Total number of Android mobiles sold = 35

Number of 4G mobiles sold = 4E% of 35 = (4×5)% of 35 = 20% of 35 = 7

Number of 5G mobiles sold = 80% of 35 = 28

Hence, the difference between the number of 4G and 5G mobiles sold = (28-7) = 21.

4. Questions

Percentage distribution of the total number of mobiles (iPhone + Android + Symbian) sold by each shop,

P = 20%

Q = (30-20)% = 10%

R = (55-30)% = 25%

S = (85-55)% = 30%

T = (100-85)% = 15%

Difference between the total number of mobiles (iPhone + android + Symbian) sold by shop R and T = 90

So,

(25-15)% = 90

10% = 90

1% = (90/10) = 9

100% = (9×100) = 900

Now, total number of mobiles (iphone + android + symbian) sold by each shop,

P = 20% of 900 = 180

Q = 10% of 900 = 90

R = 25% of 900 = 225

S = 30% of 900 = 270

T = 15% of 900 = 135

Now,

The sum of the total number of Android and Symbian mobiles sold by shop P = {2×(2A-60)} = (4A-120)

The sum of the total number of Android and Symbian mobiles sold by shop Q = {2×(B+5)} = (2B+10)

So,

(2B+10) + (4A-120) = 130

2B + 4A = 240……..(i)

Also,

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The sum of the total number of Android and Symbian mobiles sold by shop R = {2×(A-5)} = (2A-10)

The sum of the total number of Android and Symbian mobiles sold by shop S = {2×(3B+15)} = (6B+30)

So,

(6B+30) – (2A-10) = 60

6B – 30 = 2A – 10

6B – 2A = 20……..(ii)

Solving both the equation, we get,

2B + 4A = 240

6B – 2A = 20

6B + 12A = 720

6B – 2A = 20

14A = 700

A = 50

Now, put the value of A in equation (i), we get,

2B + 4A = 240

2B = 240 – 200

2B = 40

B = 20

Now,

Shop P

Total number of mobiles (iPhone + android + Symbian) sold = 180

Average number of android and Symbian mobiles sold = (2A-60) = (100-60) = 40

The sum of the number of Android and Symbian mobiles sold = (40×2) = 80

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be v and (v-20)

2v – 20 = 80

2v = 100

v = 50

Number of android mobiles sold = v = 50

Number of Symbian mobiles sold = (v-20) = (50-20) = 30

Number of iPhone mobiles sold = {180-(50+30)} = (180-80) = 100

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Shop Q

Total number of mobiles (iPhone + android + Symbian) sold = 90

Average number of android and Symbian mobiles sold = (B+5) = (20+5) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be w and (w-20)

2w– 20 = 50

2w = 70

w = 35

Number of android mobiles sold = w = 35

Number of Symbian mobiles sold = (w-20) = (35-20) = 15

Number of iPhone mobiles sold = {90-(35+15)} = (90-50) = 40

Shop R

Total number of mobiles (iPhone + android + Symbian) sold = 225

Average number of android and Symbian mobiles sold = (A-5) = (50-5) = 45

The sum of the number of Android and Symbian mobiles sold = (45×2) = 90

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be x and (x-40)

2x – 40 = 90

2x = 130

x = 65

Number of android mobiles sold = x = 65

Number of Symbian mobiles sold = (x-40) = (65-40) = 25

Number of iPhone mobiles sold = {225-(65+25)} = (225-90) = 135

Shop S

Total number of mobiles (iPhone + android + Symbian) sold = 270

Average number of android and Symbian mobiles sold = (3B+15) = (60+15) = 75

The sum of the number of Android and Symbian mobiles sold = (75×2) = 150

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be y and (y-50)

2y – 50 = 150

2y = 200

y = 100

Number of android mobiles sold = y = 100

Number of Symbian mobiles sold = (y-50) = (100-50) = 50

Number of iPhone mobiles sold = {270-(100+50)} = (270-150) = 120

Shop T

Total number of mobiles (iPhone + android + Symbian) sold = 135

Average number of android and Symbian mobiles sold = (A/2) = (50/2) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of Android and Symbian (android-symbian) mobiles sold = 10

Let, the number of Android and Symbian mobiles sold be z and (z-10)

2z – 10 = 50

2z = 60

z = 30

Number of android mobiles sold = z = 30

Number of Symbian mobiles sold = (z-10) = (30-10) = 20

Number of iPhone mobiles sold = {135-(30+20)} = (135-50) = 85

Shop	Total number of mobiles sold	Number of iPhone mobiles sold	Number of Android mobiles sold	Number of Symbian mobiles sold
P	180	100	50	30
Q	90	40	35	15
R	225	135	65	25
S	270	120	100	50
T	135	85	30	20
Total	900	480	280	140

Answer: E

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In shop R

Total number of iPhone mobiles sold = 135

Total number of Android mobiles sold = 65

Total selling price of all the iPhone mobiles sold = (135×F) = Rs. 135F

The total selling price of all the Android mobiles sold = (65×G) = Rs. 65G

Also,

Total selling price of 20 iPhone mobiles sold = (20×F) = Rs. 20F

Total selling price of all 30 Android mobiles sold = (30×G) = Rs. 30G

Now,

135F + 65G = 9350……..(i)

And,

20F + 30G = 9350 – 7150

20F + 30G = 2200…….(ii)

Solving both the equations, we get,

540F + 260G = 37400

540F + 810G = 59400

550G = 22000

G = 40

Now, put the value of G in equation (i),

135F + 65G = 9350

135F = 9350 – 2600

135F = 6750

F = 50

Hence, the total revenue generated from selling all the Symbian mobiles by shop R = (0.5G×25) thousand = (20×25) thousand = Rs. 500 thousand.

5. Questions

Percentage distribution of the total number of mobiles (iPhone + Android + Symbian) sold by each shop,

P = 20%

Q = (30-20)% = 10%

R = (55-30)% = 25%

S = (85-55)% = 30%

T = (100-85)% = 15%

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Difference between the total number of mobiles (iPhone + android + Symbian) sold by shop R and T = 90

So,

(25-15)% = 90

10% = 90

1% = (90/10) = 9

100% = (9×100) = 900

Now, total number of mobiles (iphone + android + symbian) sold by each shop,

P = 20% of 900 = 180

Q = 10% of 900 = 90

R = 25% of 900 = 225

S = 30% of 900 = 270

T = 15% of 900 = 135

Now,

The sum of the total number of Android and Symbian mobiles sold by shop P = {2×(2A-60)} = (4A-120)

The sum of the total number of Android and Symbian mobiles sold by shop Q = {2×(B+5)} = (2B+10)

So,

(2B+10) + (4A-120) = 130

2B + 4A = 240……..(i)

Also,

The sum of the total number of Android and Symbian mobiles sold by shop R = {2×(A-5)} = (2A-10)

The sum of the total number of Android and Symbian mobiles sold by shop S = {2×(3B+15)} = (6B+30)

So,

(6B+30) – (2A-10) = 60

6B – 30 = 2A – 10

6B – 2A = 20……..(ii)

Solving both the equation, we get,

2B + 4A = 240

6B – 2A = 20

6B + 12A = 720

6B – 2A = 20

14A = 700

A = 50

Now, put the value of A in equation (i), we get,

2B + 4A = 240

2B = 240 – 200

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2B = 40

B = 20

Now,

Shop P

Total number of mobiles (iPhone + android + Symbian) sold = 180

Average number of android and Symbian mobiles sold = (2A-60) = (100-60) = 40

The sum of the number of Android and Symbian mobiles sold = (40×2) = 80

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be v and (v-20)

2v – 20 = 80

2v = 100

v = 50

Number of android mobiles sold = v = 50

Number of Symbian mobiles sold = (v-20) = (50-20) = 30

Number of iPhone mobiles sold = {180-(50+30)} = (180-80) = 100

Shop Q

Total number of mobiles (iPhone + android + Symbian) sold = 90

Average number of android and Symbian mobiles sold = (B+5) = (20+5) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of android and Symbian (android-symbian) mobiles sold = 20

Let, the number of Android and Symbian mobiles sold be w and (w-20)

2w– 20 = 50

2w = 70

w = 35

Number of android mobiles sold = w = 35

Number of Symbian mobiles sold = (w-20) = (35-20) = 15

Number of iPhone mobiles sold = {90-(35+15)} = (90-50) = 40

Shop R

Total number of mobiles (iPhone + android + Symbian) sold = 225

Average number of android and Symbian mobiles sold = (A-5) = (50-5) = 45

The sum of the number of Android and Symbian mobiles sold = (45×2) = 90

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Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be x and (x-40)

2x – 40 = 90

2x = 130

x = 65

Number of android mobiles sold = x = 65

Number of Symbian mobiles sold = (x-40) = (65-40) = 25

Number of iPhone mobiles sold = {225-(65+25)} = (225-90) = 135

Shop S

Total number of mobiles (iPhone + android + Symbian) sold = 270

Average number of android and Symbian mobiles sold = (3B+15) = (60+15) = 75

The sum of the number of Android and Symbian mobiles sold = (75×2) = 150

Difference between the number of android and Symbian (android-symbian) mobiles sold = 40

Let, the number of Android and Symbian mobiles sold be y and (y-50)

2y – 50 = 150

2y = 200

y = 100

Number of android mobiles sold = y = 100

Number of Symbian mobiles sold = (y-50) = (100-50) = 50

Number of iPhone mobiles sold = {270-(100+50)} = (270-150) = 120

Shop T

Total number of mobiles (iPhone + android + Symbian) sold = 135

Average number of android and Symbian mobiles sold = (A/2) = (50/2) = 25

The sum of the number of Android and Symbian mobiles sold = (25×2) = 50

Difference between the number of Android and Symbian (android-symbian) mobiles sold = 10

Let, the number of Android and Symbian mobiles sold be z and (z-10)

2z – 10 = 50

2z = 60

z = 30

Number of android mobiles sold = z = 30

Number of Symbian mobiles sold = (z-10) = (30-10) = 20

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Number of iPhone mobiles sold = {135-(30+20)} = (135-50) = 85

Shop	Total number of mobiles sold	Number of iPhone mobiles sold	Number of Android mobiles sold	Number of Symbian mobiles sold
P	180	100	50	30
Q	90	40	35	15
R	225	135	65	25
S	270	120	100	50
T	135	85	30	20
Total	900	480	280	140

Answer: C

In shop S

Total number of android mobiles sold = 100 (non-defective)

Using filler I

50% = 100 (non-defective sold)

1% = (100/50) = 2

100% = 200

Therefore, the total number of non-defective Android mobiles = 200

Out of the total number of Android mobiles manufactured, 20% are defective, which means the rest 80% are non-defective

So,

80% = 200

1% = (200/80) = 2.5

100% = 250

Hence, the total number of Android mobiles manufactured by shop S = 250

So, filler I is correct.

Using filler II

25% = 100 (non-defective sold)

1% = (100/25) = 4

100% = 400

Therefore, the total number of non-defective Android mobiles = 400

Out of the total number of Android mobiles manufactured, 20% are defective, which means the rest 80%

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are non-defective

So,

80% = 400

1% = (400/80) = 5

100% = 500

Hence, the total number of Android mobiles manufactured by shop S = 500

So, filler II is correct.

Using filler III

10% = 100 (non-defective sold)

1% = (100/10) = 10

100% = 1000

Therefore, the total number of non-defective Android mobiles = 1000

Out of the total number of Android mobiles manufactured, 20% are defective, which means the rest 80% are non-defective

So,

80% = 1000

1% = (1000/80) = 12.5

100% = 1250

Hence, the total number of Android mobiles manufactured by shop S = 1250

So, filler III is not correct.

Therefore, only fillers I and II are correctly filling the blanks in the same order.

6. Questions

For Car,

Total distance travelled = 250 km

Speed = (S-10) km/h

Time taken = (S/12) hours

So,

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250/(S-10) = (S/12)

$3000 = S^2 - 10S$

$S^2 - 10S - 3000 = 0$

$S^2 - 60S + 50S - 3000 = 0$

$S(S-60) + 50(S-60) = 0$

$(S+50)(S-60) = 0$

$S = -50$ or 60

So, $S = 60$ (positive integer)

Now,

Speed of car = $(S-10) = (60-10) = 50$ km/h

So, speed of the man = 1/5 of 50 = 10 km/h

Now,

$(D+40)/(50+10) = (D/50)$ (Relative speeds of the car and the man will be added in case of opposite direction)

$(D+40)/60 = (D/50)$

$50D + 2000 = 60D$

$10D = 2000$

$D = 200$

For Bus,

Total distance travelled = $(D-60)$ km

Speed = $(V+30)$ km/h

Time taken = $(V/20)$ hours

So,

$(D-60)/(V+30) = (V/20)$

$(200-60)/(V+30) = (V/20)$

$140/(V+30) = (V/20)$

$2800 = V^2 + 30V$

$V^2 + 30V - 2800 = 0$

$V^2 + 70V - 40V - 2800 = 0$

$V(V+70) - 40(V+70) = 0$

$(V-40)(V+70) = 0$

$V = 40$ or -70

So, $V = 40$ (positive integer)

For Bike,

Total distance travelled = $(7V+20)$ km

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Speed of bike = 4T km/h

Speed of the man = 10 km/h

Time taken = T hours

So,

(7V+20)/(4T-10) = T (Relative speeds of the car and the man will be subtracted in case of same direction)

(280+20)/(4T-10) = T

$4T^2 – 10T = 300$

$4T^2 – 10T – 300 = 0$

$2T^2 – 5T – 150 = 0$

$2T^2 – 20T + 15T – 150 = 0$

2T(T-10) + 15(T-10) = 0

(2T+15) (T-10) = 0

T = -7.5 or 10

So, T = 10 (positive integer)

For Truck,

Total distance travelled = (X+60) km

Speed = (X-20) km/h

Time taken = (T/5) hours

So,

(X+60)/(X-20) = (T/5)

(X+60)/(X-20) = (10/5)

(X+60)/(X-20) = 2

2X – 40 = X + 60

2X – X = 60 + 40

X = 100

For Cab,

Total distance travelled = (Y+30) km

Speed = (Y-90) km/h

Time taken = (Y/50) hours

So,

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$(Y+30)/(Y-90) = (Y/50)$

$50Y + 1500 = Y^2 - 90Y$

$Y^2 - 140Y - 1500 = 0$

$Y^2 - 150Y + 10Y - 1500 = 0$

$Y(Y-150) + 10(Y-150) = 0$

$(Y+10)(Y-150) = 0$

$Y = -10$ or 150

So, $Y = 150$ (positive integer)

Vehicle/Person	Distance travelled (in km)	Speed (in km/h)	Time taken (in hours)
Car	250	50	5
(Car+Man) (in the opposite direction)	240	(50+10) = 60	4
Bus	140	70	2
(Bike+Man) (in same direction)	300	(40-10) = 30	10
Truck	160	80	2
Cab	180	60	3

Answer: C

According to the question,

Length of the platform = P meter

First Train,

Length = 300 meter

Speed = A m/sec

Time taken to cross the platform = 4 seconds

So,

$(300+P)/A = 4$

$300 + P = 4A$

$4A - P = 300$……..(i)

Second Train

Length = (300-40) = 260 meter

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Speed = (A+80) m/sec

Time taken to cross the platform = 2 seconds

So,

(260+P)/(A+80) = 2

2A + 160 = 260 + P

2A – P = 100……...(ii)

Solving both the equations, we get,

4A – P = 300

4A – 2P = 200

P = 100

Now, put the value of P in equation (i), we get,

4A – P = 300

4A = 300 + 100

4A = 400

A = 100

Hence, the time is taken by the bus to cover (P+350) km with the speed of (A+50) km/h = (P+350)/(A+50) = (450/150) = 3 hours.

7. Questions

For Car,

Total distance travelled = 250 km

Speed = (S-10) km/h

Time taken = (S/12) hours

So,

250/(S-10) = (S/12)

$3000 = S^2 – 10S$

$S^2 – 10S – 3000 = 0$

$S^2 – 60S + 50S – 3000 = 0$

S(S-60) + 50(S-60) = 0

(S+50) (S-60) = 0

S = -50 or 60

So, S = 60 (positive integer)

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Now,

Speed of car = (S-10) = (60-10) = 50 km/h

So, speed of the man = 1/5 of 50 = 10 km/h

Now,

(D+40)/(50+10) = (D/50) (Relative speeds of the car and the man will be added in case of opposite direction)

(D+40)/60 = (D/50)

50D + 2000 = 60D

10D = 2000

D = 200

For Bus,

Total distance travelled = (D-60) km

Speed = (V+30) km/h

Time taken = (V/20) hours

So,

(D-60)/(V+30) = (V/20)

(200-60)/(V+30) = (V/20)

140/(V+30) = (V/20)

$2800 = V^2 + 30V$

$V^2 + 30V - 2800 = 0$

$V^2 + 70V - 40V - 2800 = 0$

V(V+70) – 40(V+70) = 0

(V-40) (V+70) = 0

V = 40 or -70

So, V = 40 (positive integer)

For Bike,

Total distance travelled = (7V+20) km

Speed of bike = 4T km/h

Speed of the man = 10 km/h

Time taken = T hours

So,

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(7V+20)/(4T-10) = T (Relative speeds of the car and the man will be subtracted in case of same direction)

(280+20)/(4T-10) = T

$4T^2 – 10T = 300$

$4T^2 – 10T – 300 = 0$

$2T^2 – 5T – 150 = 0$

$2T^2 – 20T + 15T – 150 = 0$

2T(T-10) + 15(T-10) = 0

(2T+15) (T-10) = 0

T = -7.5 or 10

So, T = 10 (positive integer)

For Truck,

Total distance travelled = (X+60) km

Speed = (X-20) km/h

Time taken = (T/5) hours

So,

(X+60)/(X-20) = (T/5)

(X+60)/(X-20) = (10/5)

(X+60)/(X-20) = 2

2X – 40 = X + 60

2X – X = 60 + 40

X = 100

For Cab,

Total distance travelled = (Y+30) km

Speed = (Y-90) km/h

Time taken = (Y/50) hours

So,

(Y+30)/(Y-90) = (Y/50)

$50Y + 1500 = Y^2 – 90Y$

$Y^2 – 140Y – 1500 = 0$

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$Y^2 - 150Y + 10Y - 1500 = 0$

$Y(Y-150) + 10(Y-150) = 0$

$(Y+10)(Y-150) = 0$

$Y = -10$ or 150

So, $Y = 150$ (positive integer)

Vehicle/Person	Distance travelled (in km)	Speed (in km/h)	Time taken (in hours)
Car	250	50	5
(Car+Man) (in the opposite direction)	240	(50+10) = 60	4
Bus	140	70	2
(Bike+Man) (in same direction)	300	(40-10) = 30	10
Truck	160	80	2
Cab	180	60	3

Answer: A

Total distance between city B and city C = 3Z km

The bike starts from City B at 9:00 am and stops at 11:30 am at City E

So, the total time duration that bike has travelled = (9:00 am – 11:30 am) = 2 hours 30 minutes = 2.50 hours

Speed of the bike = 40 km/h

So, distance covered by the bike = (40×2.50) = 100 km

Therefore, the distance between city B and city E = 100 km

Now, remaining distance between city E and city C = (3Z-100) (As city E is somewhere between city B and city C)

Now,

Speed of the car = 50 km/h

And, the time taken by the car to cover the distance from city C to city E = 4 hours

So,

$(3Z-100)/50 = 4$

$3Z - 100 = 200$

$3Z = 300$

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Z = 100

Hence, time is taken by bike and car to cover (2Z-20) km in opposite direction = {(200-20)/(40+50)} = (180/90) = 2 hours. (Relative speeds of the bike and the car will be added in case of opposite directions)

8. Questions

For Car,

Total distance travelled = 250 km

Speed = (S-10) km/h

Time taken = (S/12) hours

So,

250/(S-10) = (S/12)

$3000 = S^2 - 10S$

$S^2 - 10S - 3000 = 0$

$S^2 - 60S + 50S - 3000 = 0$

S(S-60) + 50(S-60) = 0

(S+50) (S-60) = 0

S = -50 or 60

So, S = 60 (positive integer)

Now,

Speed of car = (S-10) = (60-10) = 50 km/h

So, speed of the man = 1/5 of 50 = 10 km/h

Now,

(D+40)/(50+10) = (D/50) (Relative speeds of the car and the man will be added in case of opposite direction)

(D+40)/60 = (D/50)

50D + 2000 = 60D

10D = 2000

D = 200

For Bus,

Total distance travelled = (D-60) km

Speed = (V+30) km/h

Time taken = (V/20) hours

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So,

(D-60)/(V+30) = (V/20)

(200-60)/(V+30) = (V/20)

140/(V+30) = (V/20)

$2800 = V^2 + 30V$

$V^2 + 30V - 2800 = 0$

$V^2 + 70V - 40V - 2800 = 0$

V(V+70) – 40(V+70) = 0

(V-40) (V+70) = 0

V = 40 or -70

So, V = 40 (positive integer)

For Bike,

Total distance travelled = (7V+20) km

Speed of bike = 4T km/h

Speed of the man = 10 km/h

Time taken = T hours

So,

(7V+20)/(4T-10) = T (Relative speeds of the car and the man will be subtracted in case of same direction)

(280+20)/(4T-10) = T

$4T^2 - 10T = 300$

$4T^2 - 10T - 300 = 0$

$2T^2 - 5T - 150 = 0$

$2T^2 - 20T + 15T - 150 = 0$

2T(T-10) + 15(T-10) = 0

(2T+15) (T-10) = 0

T = -7.5 or 10

So, T = 10 (positive integer)

For Truck,

Total distance travelled = (X+60) km

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Speed = (X-20) km/h

Time taken = (T/5) hours

So,

(X+60)/(X-20) = (T/5)

(X+60)/(X-20) = (10/5)

(X+60)/(X-20) = 2

2X – 40 = X + 60

2X – X = 60 + 40

X = 100

For Cab,

Total distance travelled = (Y+30) km

Speed = (Y-90) km/h

Time taken = (Y/50) hours

So,

(Y+30)/(Y-90) = (Y/50)

$50Y + 1500 = Y^2 – 90Y$

$Y^2 – 140Y – 1500 = 0$

$Y^2 – 150Y + 10Y – 1500 = 0$

Y(Y-150) + 10(Y-150) = 0

(Y+10) (Y-150) = 0

Y = -10 or 150

So, Y = 150 (positive integer)

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Vehicle/Person	Distance travelled (in km)	Speed (in km/h)	Time taken (in hours)
Car	250	50	5
(Car+Man) (in the opposite direction)	240	(50+10) = 60	4
Bus	140	70	2
(Bike+Man) (in same direction)	300	(40-10) = 30	10
Truck	160	80	2
Cab	180	60	3

Answer: B

Length of the tunnel = 2V = (2×40) = 80 meters

For Train,

Length = (X+Y) = (100+150) = 250 meters

Speed = (S+6) = (60+6) = 66 m/sec

Time taken by the train to cross the tunnel = {(250+80)/66} = 5 seconds

So, M = 5

Now,

The truck can cover (U-W) km distance in M hours,

So,

(U-W)/80 = M

(U-W)/80 = 5

U – W = 400……(i)

Now, time is taken by the cab to cover (U-60) km = 9 hours

So,

(U-60)/60 = 9

U – 60 = 540

U = 600

Now, put the value of U in equation (i), we get,

U – W = 400

600 = 400 + W

W = 600 – 400

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W = 200

Statement I

The ratio of the values of U to that of W = 600:200 = 3:1

So, statement I is true.

Statement II

Time taken by the bus to cover (W+10) km = {(W+10)/70} = (210/70) = 3 hours

So, statement II is not true.

Statement III

X = 100

W = 200

Therefore, the value of W is twice that of X = (200/100) = 2

So, statement III is not true.

Hence, only statement I is true.

9. Questions

For Car,

Total distance travelled = 250 km

Speed = (S-10) km/h

Time taken = (S/12) hours

So,

250/(S-10) = (S/12)

$3000 = S^2 - 10S$

$S^2 - 10S - 3000 = 0$

$S^2 - 60S + 50S - 3000 = 0$

S(S-60) + 50(S-60) = 0

(S+50) (S-60) = 0

S = -50 or 60

So, S = 60 (positive integer)

Now,

Speed of car = (S-10) = (60-10) = 50 km/h

So, speed of the man = 1/5 of 50 = 10 km/h

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Now,

(D+40)/(50+10) = (D/50) (Relative speeds of the car and the man will be added in case of opposite direction)

(D+40)/60 = (D/50)

50D + 2000 = 60D

10D = 2000

D = 200

For Bus,

Total distance travelled = (D-60) km

Speed = (V+30) km/h

Time taken = (V/20) hours

So,

(D-60)/(V+30) = (V/20)

(200-60)/(V+30) = (V/20)

140/(V+30) = (V/20)

$2800 = V^2 + 30V$

$V^2 + 30V - 2800 = 0$

$V^2 + 70V - 40V - 2800 = 0$

V(V+70) – 40(V+70) = 0

(V-40) (V+70) = 0

V = 40 or -70

So, V = 40 (positive integer)

For Bike,

Total distance travelled = (7V+20) km

Speed of bike = 4T km/h

Speed of the man = 10 km/h

Time taken = T hours

So,

(7V+20)/(4T-10) = T (Relative speeds of the car and the man will be subtracted in case of same direction)

(280+20)/(4T-10) = T

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$4T^2 - 10T = 300$

$4T^2 - 10T - 300 = 0$

$2T^2 - 5T - 150 = 0$

$2T^2 - 20T + 15T - 150 = 0$

$2T(T-10) + 15(T-10) = 0$

$(2T+15)(T-10) = 0$

$T = -7.5$ or 10

So, $T = 10$ (positive integer)

For Truck,

Total distance travelled = (X+60) km

Speed = (X-20) km/h

Time taken = (T/5) hours

So,

$(X+60)/(X-20) = (T/5)$

$(X+60)/(X-20) = (10/5)$

$(X+60)/(X-20) = 2$

$2X - 40 = X + 60$

$2X - X = 60 + 40$

$X = 100$

For Cab,

Total distance travelled = (Y+30) km

Speed = (Y-90) km/h

Time taken = (Y/50) hours

So,

$(Y+30)/(Y-90) = (Y/50)$

$50Y + 1500 = Y^2 - 90Y$

$Y^2 - 140Y - 1500 = 0$

$Y^2 - 150Y + 10Y - 1500 = 0$

$Y(Y-150) + 10(Y-150) = 0$

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(Y+10) (Y-150) = 0

Y = -10 or 150

So, Y = 150 (positive integer)

Vehicle/Person	Distance travelled (in km)	Speed (in km/h)	Time taken (in hours)
Car	250	50	5
(Car+Man) (in the opposite direction)	240	(50+10) = 60	4
Bus	140	70	2
(Bike+Man) (in same direction)	300	(40-10) = 30	10
Truck	160	80	2
Cab	180	60	3

Answer: A

(Q+50)/70 = (R-200)/60

60Q + 3000 = 70R – 14000

70R – 60Q = 17000……(i)

Also,

(R-10)/70 = (Q+120)/60

60R – 600 = 70Q + 8400

60R – 70Q = 9000……..(ii)

Solving both the equations, we get,

420R – 360Q = 102000

420R – 490Q = 63000

130Q = 39000

Q = 300

Now, put the value of R in equation (i), we get,

70R – 60Q = 17000

70R = 17000 + 18000

70R = 35000

R = 500

So, the distance travelled by the car in 10 hours = (50×10) = 500 km.

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(2R-500) = (1000-500) = 500.

Hence, option A is the correct answer.

10. Questions

For Car,

Total distance travelled = 250 km

Speed = (S-10) km/h

Time taken = (S/12) hours

So,

250/(S-10) = (S/12)

$3000 = S^2 - 10S$

$S^2 - 10S - 3000 = 0$

$S^2 - 60S + 50S - 3000 = 0$

S(S-60) + 50(S-60) = 0

(S+50) (S-60) = 0

S = -50 or 60

So, S = 60 (positive integer)

Now,

Speed of car = (S-10) = (60-10) = 50 km/h

So, speed of the man = 1/5 of 50 = 10 km/h

Now,

(D+40)/(50+10) = (D/50) (Relative speeds of the car and the man will be added in case of opposite direction)

(D+40)/60 = (D/50)

50D + 2000 = 60D

10D = 2000

D = 200

For Bus,

Total distance travelled = (D-60) km

Speed = (V+30) km/h

Time taken = (V/20) hours

So,

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(D-60)/(V+30) = (V/20)

(200-60)/(V+30) = (V/20)

140/(V+30) = (V/20)

$2800 = V^2 + 30V$

$V^2 + 30V - 2800 = 0$

$V^2 + 70V - 40V - 2800 = 0$

V(V+70) – 40(V+70) = 0

(V-40) (V+70) = 0

V = 40 or -70

So, V = 40 (positive integer)

For Bike,

Total distance travelled = (7V+20) km

Speed of bike = 4T km/h

Speed of the man = 10 km/h

Time taken = T hours

So,

(7V+20)/(4T-10) = T (Relative speeds of the car and the man will be subtracted in case of same direction)

(280+20)/(4T-10) = T

$4T^2 - 10T = 300$

$4T^2 - 10T - 300 = 0$

$2T^2 - 5T - 150 = 0$

$2T^2 - 20T + 15T - 150 = 0$

2T(T-10) + 15(T-10) = 0

(2T+15) (T-10) = 0

T = -7.5 or 10

So, T = 10 (positive integer)

For Truck,

Total distance travelled = (X+60) km

Speed = (X-20) km/h

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Time taken = (T/5) hours

So,

(X+60)/(X-20) = (T/5)

(X+60)/(X-20) = (10/5)

(X+60)/(X-20) = 2

2X – 40 = X + 60

2X – X = 60 + 40

X = 100

For Cab,

Total distance travelled = (Y+30) km

Speed = (Y-90) km/h

Time taken = (Y/50) hours

So,

(Y+30)/(Y-90) = (Y/50)

$50Y + 1500 = Y^2 – 90Y$

$Y^2 – 140Y – 1500 = 0$

$Y^2 – 150Y + 10Y – 1500 = 0$

Y(Y-150) + 10(Y-150) = 0

(Y+10) (Y-150) = 0

Y = -10 or 150

So, Y = 150 (positive integer)

Vehicle/Person	Distance travelled (in km)	Speed (in km/h)	Time taken (in hours)
Car	250	50	5
(Car+Man) (in the opposite direction)	240	(50+10) = 60	4
Bus	140	70	2
(Bike+Man) (in same direction)	300	(40-10) = 30	10
Truck	160	80	2
Cab	180	60	3

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Answer: E

Distance travelled by car in F hours = (F×50) = 50F

Distance travelled by the car in G hours = (G×70) = 70G

So,

50F + 70G = 1340……(i)

Increased speed of the truck = 125% of 80 = 100 km/h

So,

(4F+5G)/100 = F/10

40F + 50G = 100F

60F = 50G

F:G = 5:6

Let, the values of,

F = 5h

G = 6h

Now, put these values in equation (i), we get,

50F + 70G = 1340

250h + 420h = 1340

670h = 1340

h = 2

Therefore, the values of,

F = 5h = 10

G = 6h = 12

Now,

$[L^2 – \{(F+G)L\} + FG] = 0$

$L^2 – 22L + 120 = 0$

$L^2 – 22L + 120 = 0$

$L^2 – 12L – 10L + 120 = 0$

L(L-12) – 10(L-12) = 0

(L-10) (L-12) = 0

L = 10 or 12

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So, the roots of the equation = 10 and 12

Hence, the difference between the roots of the equation = (12-10) = 2.

11. Questions

Equation I: $X^2 - 13X + A = 0$

The sum of the roots = 13

p + q = 13……(i)

One of the roots is the prime number, which is less than 11 but more than 5.

So, only possible value = 7

Therefore, other root = (13-7) = 6

p > q (given)

So, p = 7

And, q = 6

Now,

A = (p × q)

A = (7 × 6)

A = 42

Equation II: $Y^2 + 3Y - B = 0$

The sum of the roots = -3

Difference between the roots = 13

r > s (given)

So,

r + s = -3……(i)

r – s = 13……..(ii)

Solving both the equations, we get,

2s = -16

s = -8

Now, put the value of s in equation (i), we get,

r + s = -3

r = {-3–(-8)}

r = (-3+8)

r = 5

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Therefore,

B = (r × s)

B = {5×(-8)}

B = -40

Equation III: $Z^2 - 2Z - C = 0$

The sum of the roots = 2

t > u (given)

Let,

t = 2s

t = {2×(-8)}

t = -16

So, u = 5

Which means, t < u (not possible)

Now, let the value of,

t = 2r

t = (2×5)

t = 10

So, u = -8

Which mean t > u (possible)

Now,

t + u = 2

10 + (-8) = 2

10 – 8 = 2 (satisfied)

So, t = 10

And, u = -8

Therefore, C = (t × u)

C = {10×(-8)}

C = -80

Answer: B

According to the question,

A = 42

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B = -40

C = -80

Relation I: (A-82) = B

42 – 82 = -40

-40 = -40

So, relation I is correct.

Relation II: 2B = C

{2×(-40)} = -80

-80 = -80

So, relation II is correct.

Relation III: (C+120) = A

(-80+120) = 40

But, A = 42

So, relation III is not correct.

Hence, only relations I and II are correct.

12. Questions

Equation I: $X^2 - 13X + A = 0$

The sum of the roots = 13

p + q = 13……(i)

One of the roots is the prime number, which is less than 11 but more than 5.

So, only possible value = 7

Therefore, other root = (13-7) = 6

p > q (given)

So, p = 7

And, q = 6

Now,

A = (p × q)

A = (7 × 6)

A = 42

Equation II: $Y^2 + 3Y - B = 0$

The sum of the roots = -3

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Difference between the roots = 13

r > s (given)

So,

r + s = -3……(i)

r – s = 13……..(ii)

Solving both the equations, we get,

2s = -16

s = -8

Now, put the value of s in equation (i), we get,

r + s = -3

r = {-3–(-8)}

r = (-3+8)

r = 5

Therefore,

B = (r × s)

B = {5×(-8)}

B = -40

Equation III: $Z^2 – 2Z – C = 0$

The sum of the roots = 2

t > u (given)

Let,

t = 2s

t = {2×(-8)}

t = -16

So, u = 5

Which means, t < u (not possible)

Now, let the value of,

t = 2r

t = (2×5)

t = 10

So, u = -8

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Which mean t > u (possible)

Now,

t + u = 2

10 + (-8) = 2

10 – 8 = 2 (satisfied)

So, t = 10

And, u = -8

Therefore, C = (t × u)

C = {10×(-8)}

C = -80

Answer: C

Value of X,

Roots are = 7 and 6

Difference between the roots = (7-6) = 1

So, only N follows.

Value of Y,

Roots are = 5 and -8

Square of twice the greater root = $(5\times2)^2 = 10^2 = 100$

So, only L follows.

Value of Z,

Roots are = 10 and -8

Product of roots = {10×(-8)} = -80

So, only M follows.

Hence, only II follows.

13. Questions

Equation I: $X^2 - 13X + A = 0$

The sum of the roots = 13

p + q = 13……(i)

One of the roots is the prime number, which is less than 11 but more than 5.

So, only possible value = 7

Therefore, other root = (13-7) = 6

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$p > q$ (given)

So, $p = 7$

And, $q = 6$

Now,

$A = (p \times q)$

$A = (7 \times 6)$

$A = 42$

Equation II: $Y^2 + 3Y - B = 0$

The sum of the roots = -3

Difference between the roots = 13

$r > s$ (given)

So,

$r + s = -3$……(i)

$r - s = 13$……..(ii)

Solving both the equations, we get,

$2s = -16$

$s = -8$

Now, put the value of s in equation (i), we get,

$r + s = -3$

$r = \{-3-(-8)\}$

$r = (-3+8)$

$r = 5$

Therefore,

$B = (r \times s)$

$B = \{5\times(-8)\}$

$B = -40$

Equation III: $Z^2 - 2Z - C = 0$

The sum of the roots = 2

$t > u$ (given)

Let,

$t = 2s$

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t = {2×(-8)}

t = -16

So, u = 5

Which means, t < u (not possible)

Now, let the value of,

t = 2r

t = (2×5)

t = 10

So, u = -8

Which mean t > u (possible)

Now,

t + u = 2

10 + (-8) = 2

10 – 8 = 2 (satisfied)

So, t = 10

And, u = -8

Therefore, C = (t × u)

C = {10×(-8)}

C = -80

Answer: D

A = 42, B = -40 and C = -80

So,

[{√(A-17)}+(B-C)] = [{√(42-17)}+{-40-(-80)}].

{(√25)+(-40+80)} = (5+40) = 45.

14. Questions

Equation I: $X^2 - 13X + A = 0$

The sum of the roots = 13

p + q = 13……(i)

One of the roots is the prime number, which is less than 11 but more than 5.

So, only possible value = 7

Therefore, other root = (13-7) = 6

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$p > q$ (given)

So, $p = 7$

And, $q = 6$

Now,

$A = (p \times q)$

$A = (7 \times 6)$

$A = 42$

Equation II: $Y^2 + 3Y - B = 0$

The sum of the roots = -3

Difference between the roots = 13

$r > s$ (given)

So,

$r + s = -3$……(i)

$r - s = 13$……..(ii)

Solving both the equations, we get,

$2s = -16$

$s = -8$

Now, put the value of s in equation (i), we get,

$r + s = -3$

$r = \{-3-(-8)\}$

$r = (-3+8)$

$r = 5$

Therefore,

$B = (r \times s)$

$B = \{5\times(-8)\}$

$B = -40$

Equation III: $Z^2 - 2Z - C = 0$

The sum of the roots = 2

$t > u$ (given)

Let,

$t = 2s$

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$t = \{2\times(-8)\}$

$t = -16$

So, $u = 5$

Which means, $t < u$ (not possible)

Now, let the value of,

$t = 2r$

$t = (2\times5)$

$t = 10$

So, $u = -8$

Which mean $t > u$ (possible)

Now,

$t + u = 2$

$10 + (-8) = 2$

$10 – 8 = 2$ (satisfied)

So, $t = 10$

And, $u = -8$

Therefore, $C = (t \times u)$

$C = \{10\times(-8)\}$

$C = -80$

Answer: E

From the common explanation, we have,

$p = 5$

$q = 6$

$r = 5$

$s = -8$

$t = 10$

$u = -8$

So,

$k^2 – (p+t)k + 14r = 0$

$k^2 – (7+10)k + (14\times5) = 0$

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$k^2 - 17k + 70 = 0$

$k^2 - 10k - 7k + 70 = 0$

$k(k-10) - 7(k-10) = 0$

$(k-7)(k-10) = 0$

$k = 7$ or 10

So, the roots are = 7 and 10

Hence, the smaller root = 7 = p.

15. Questions

Equation I: $X^2 - 13X + A = 0$

The sum of the roots = 13

$p + q = 13$……(i)

One of the roots is the prime number, which is less than 11 but more than 5.

So, only possible value = 7

Therefore, other root = (13-7) = 6

$p > q$ (given)

So, $p = 7$

And, $q = 6$

Now,

$A = (p \times q)$

$A = (7 \times 6)$

$A = 42$

Equation II: $Y^2 + 3Y - B = 0$

The sum of the roots = -3

Difference between the roots = 13

$r > s$ (given)

So,

$r + s = -3$……(i)

$r - s = 13$……..(ii)

Solving both the equations, we get,

$2s = -16$

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s = -8

Now, put the value of s in equation (i), we get,

r + s = -3

r = {-3–(-8)}

r = (-3+8)

r = 5

Therefore,

B = (r × s)

B = {5×(-8)}

B = -40

Equation III: $Z^2 – 2Z – C = 0$

The sum of the roots = 2

t > u (given)

Let,

t = 2s

t = {2×(-8)}

t = -16

So, u = 5

Which means, t < u (not possible)

Now, let the value of,

t = 2r

t = (2×5)

t = 10

So, u = -8

Which mean t > u (possible)

Now,

t + u = 2

10 + (-8) = 2

10 – 8 = 2 (satisfied)

So, t = 10

And, u = -8

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Therefore, C = (t × u)

C = {10×(-8)}

C = -80

Answer: A

From the common explanation, we have,

r = 5

t = 10

So, the roots should be = -5 and 10

Using option A,

$d^2 – 5d – 50 = 0$

$d^2 – 10d + 5d – 50 = 0$

d(d-10) + 5(d-10) = 0

(d+5) (d-10) = 0

d = -5 or 10

Therefore, roots are = -5 and 10 (satisfied)

Hence, option A is the correct answer.

16. Questions

Answer: B

First scheme

Amount invested = Rs. 3000X

Rate of interest = 15% p.a.

Time duration = 3 years

Type of interest = S.I.

Interest earned = 45% of 3000X = Rs. 1350X

Amount received = (3000X+1350X) = Rs. 4350X

Second scheme

Amount invested = Rs. 4350X

Rate of interest = 20% p.a.

Time duration = 2 years

Type of interest = C.I.

[(20+20)+{(20×20)/100}]% = (40+4)% = 44%

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Interest earned = 44% of 4350X = Rs. 1914X

The difference between the simple and compound interest earned = (1914X-1350X) = Rs. 564X

564X = 282Y

X:Y = 1:2

Hence, 2X:3Y = 2:6 = 1:3.

17. Questions

Answer: A

Total work = 30 units

Efficiency of,

A = {30/(P-2)}

B = {30/(P+3)}

C = {30/(P-7)}

Now,

4×[{30/(P-2)} + {30/(P+3)}] = 2/3 of 30

[{(30P+90)+(30P-60)}/{(P^2 + P – 6)}] = 20/4

60P + 30 = $5P^2$ + 5P – 30

$5P^2$ – 55P – 60 = 0

P^2 – 11P – 12 = 0

P^2 -12P + P – 12 = 0

P(P-12) + 1(P-12) = 0

(P+1) (P-12) = 0

P = -1 or 12

So, P = 12 (positive integer)

Therefore, efficiency of C = {30/(P-7)} = {30/(12-7)} = (30/5) = 6

Hence, the time taken by C alone to complete the work = (30/6) = 5 days.

18. Questions

Answer: D

Calculator

CP = Rs. (M+200)

MP = 150% of (M+200) = Rs. (1.5M+300)

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SP = 120% of (M+200) = Rs. (1.2M+240)

Watch

MP = Rs. (N+400)

SP = 90% of (N+400) = Rs. (0.9N+360)

Now,

(N+400) – (1.5M+300) = 900

N – 500 = 1.5M + 300

N – 1.5M = 800……..(i),

Also,

(0.9N+360) + (1.2M+240) = 3360

0.9N + 1.2M = 2760……(ii),

Solving both the equations, we get,

9N – 13.5M = 7200

9N + 12M = 27600

25.5M = 20400

M = 800

Now, put the value of M in equation (i), we get,

N – 1.5M = 800

N = 800 + 1200

N = 2000

K: Marked price of watch = (N+400) = (2000+400) = Rs. 2400

The ratio of the marked price and cost price of the watch = 120:100 = 6:5 (the Watch was marked up 20% above its cost price)

So, the cost price of watch = {(2400×5)/6} = Rs. 2000

Therefore, K = 2000

L: Selling price of the calculator = (1.2M+240) = (960+240) = Rs. 1200

Therefore, L = 1200

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So,

$\sqrt{(K-L-4^2)} = \sqrt{(2000-1200-16)} = \sqrt{(2000-1216)} = \sqrt{784} = 28.$

19. Questions

Answer: C

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The initial quantity of pure milk in the mixture = 180 ml

Now, after (Q-40) ml is replaced with the same quantity of water,

Resultant quantity of milk in the mixture = {180-(Q-40)} = (220-Q) ml

The resultant quantity of water in the mixture = (Q-40) ml

So,

(220-Q)/(Q-40) = 7/2

440 – 2Q = 7Q – 280

9Q = 720

Q = 80

Had (Q+10) ml of milk replaced with (0.5Q-10) ml of water,

Resultant quantity of milk in the mixture = {180-(Q+10)} = {180-(80+10)} = (180-90) = 90 ml

Resultant quantity of water in the mixture = (0.5Q-10) = (40-10) = 30 ml

Therefore, the total quantity of milk and water in the resultant mixture = (90+30) = 120 ml

So,

2R = 120

R = 60

Hence, (4Q-3R) = (320-180) = 140.

20. Questions

Answer: A

The ratio of the speed of boat A to that of boat B in still water = 2:3. Let, it be 2v km/h and 3v km/h respectively. (Fractional value of 33.33% = 1/3)

Also, let the speed of the river be s km/h.

So,

Boat A

Speed of boat in still water = 2v km/h

Speed of river = s km/h

Downstream speed = (2v+s) km/h

Upstream speed = (2v-s) km/h

Boat B

Speed of boat in still water = 3v km/h

Speed of river = s km/h

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Downstream speed = (3v+s) km/h

Upstream speed = (3v-s) km/h

Now,

28/(2v-s) = 2

4v – 2s = 28……..(i),

Also,

72/(3v+s) = 2

6v + 2s = 72…….(ii)

Solving both the equations, we get,

12v – 6s = 84

12v + 4s = 144

10s = 60

s = 6

Now, put the value of s in equation (i), we get,

4v – 2s = 28

4v = 28 + 12

4v = 40

v = 10

Hence, the difference between the downstream speed of boat A and the upstream speed of boat B = {(2v+s)-(3v-s)} = {(20+6)-(30-6)} = (26-24) = 2 km/h.

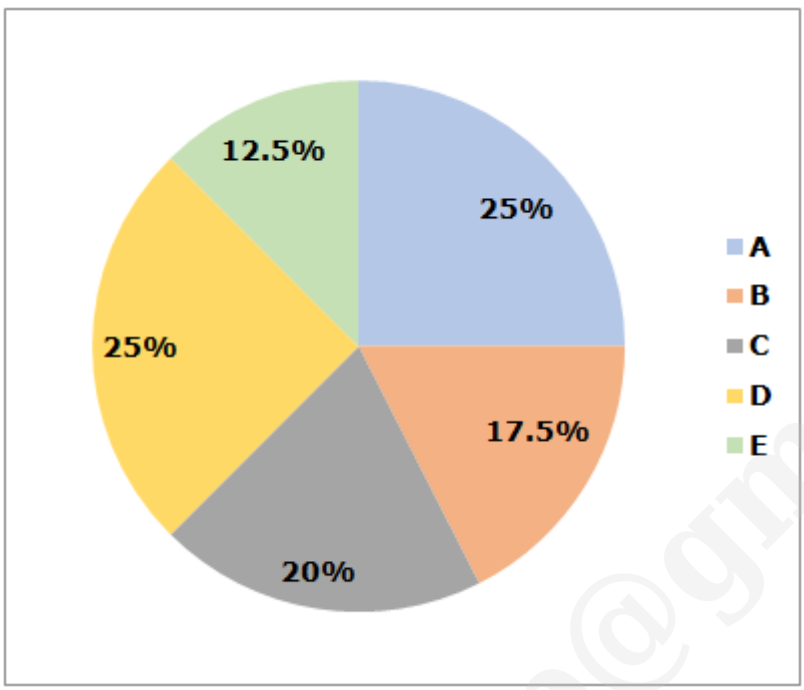
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1. Questions

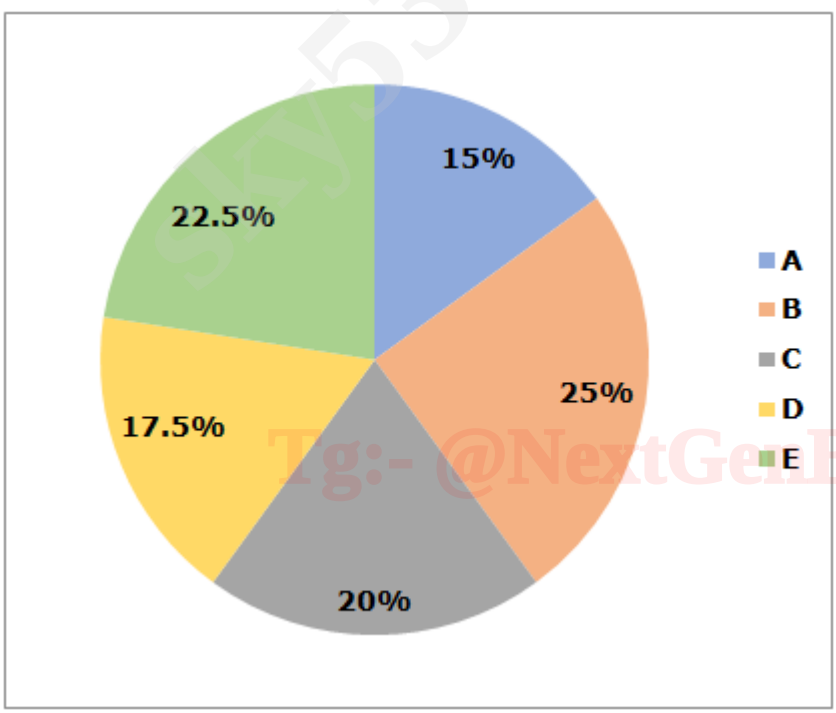
Study the following data carefully and answer the questions:

The data given below is related to the number of points scored by five archery players (A, B, C, D, and E) in three attempts (1st, 2nd, and 3rd) in an archery match.

Pie chart given below shows the percentage distribution of the number of points scored by each player in 1st attempt.



The pie chart given below shows the percentage distribution of the number of points scored by each player in 2nd attempt.



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Notes:

1: The ratio of the points scored by A in 2nd attempt to those scored in 3rd attempt is 3:4.

2: The ratio of the points scored by B in 2nd attempt to those scored in 3rd attempt is 5:3.

3: The total points scored by A and B in the match are 24 and 23, respectively.

4: Each player is given three attempts only, and each player can score a maximum of 10 points in each attempt.

If the difference between the total points scored by A and C is four times the difference between the total points scored by A and B, then find the points scored by C in 3rd attempt?

a. 9

b. 4

c. 5

d. 8

e. 6

2. Questions

If the total points scored by D in 1st and 2nd attempts together is 112.5% more than those scored in 3rd attempt, then find that the points scored by D in 3rd attempt is what per cent of those scored by B in 3rd attempt?

a. 75%

b. 120%

c. 80%

d. 100%

e. 133.33%

3. Questions

The maximum points scored by a player in 2nd attempt is what per cent of the total points scored by those players together in 2nd attempt, who scored equal points in 1st attempt?

a. 76.9%

b. 72.6%

c. 96.7%

d. 66.9%

e. 69.6%

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4. Questions

If the total points scored by B in the match are 15% more than those scored by E in the match, then find that which of the following statements is/are true?

P: The ratio of points scored by A to E in 3rd attempt is 4: 3.

Q: Points scored by E in 3rd attempt are 66(2/3)% of those scored in 2nd attempt.

R: Total points scored by A, B and E together in 3rd attempt are less than 21.

a. Only P and R

b. Only Q

c. Only P and Q

d. All are true

e. Only R

5. Questions

If the ratio of points scored by A to F in 1st attempt is 5: 3, the ratio of points scored by A to F in 2nd attempt is 2: 3 and the ratio of points scored by A to F in 3rd attempt is 8: 5, then find out what percentage of the maximum points in the match F scored in the match ?

a. 50%

b. 66.67%

c. 75%

d. 60%

e. 82.5%

6. Questions

Read the following information carefully and answer the following questions carefully.

Ranjan had 'a' litres of alcohol. He took q litres of alcohol and replaced with same amount of water. He again took (q + 24) litres and replaced with same amount of water such that the resultant mixture contains (q^2 – 5124) litres alcohol and rest is water. Raman has two mixtures X and Y of alcohol and water, only. Mixture X contains 3q litres water and Mixture Y contains 2q litres of alcohol, while the quantity of water in mixture Y is (a - 62)% more than the quantity of alcohol in mixture X. Total quantity of mixture Y is 25% more than the total quantity of mixture X. Raghu took 50% of mixture X and whole of mixture Y and mixed them to form another mixture Z. He sold mixture Z to Ravi who then added 3q litres of alcohol and (3q - 120) litres of water to mixture Z such that the ratio of alcohol to water in the mixture becomes 7:8. Sumathi has alcohol equal to the new quantity of mixture Z and she took 180 litres of alcohol and replaced with water also took 40% of the mixture and replaced with water such that the final mixture contains 'm' litres of alcohol where 'q' is calculated from the equation q^2 – 67q - 360 = 0 where 'q' is a positive integer.

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What is the ratio of value of 'a' to that of 'q'?

a. 8:3
b. 7:5
c. 5:4
d. 9:5
e. none of these

7. Questions

Find water in Mixture Y is how much percent more/less than that of alcohol in the same Mixture?

a. 83.67%
b. 91.67%
c. 89.33%
d. 97.67%
e. none of these

8. Questions

If mixture U and Mixture V are mixed in the ratio 7:5 to form final mixture L. Cost price of alcohol is Rs.50 per litre while the final mixture of L is sold at Rs.25 per litre, then what is the profit amount earned where the final quantity of mixture is equal to alcohol quantity of mixture Y and also the respective ratio of alcohol and water ofmixture U and mixture V is equal to the ratio of alcohol and water in Mixture X and final Mixture of Z?

a. Rs.800
b. Rs.900
c. Rs.700
d. Rs.1000
e. none of these

9. Questions

Find the value of (m-a) * (3q-120)?

a. 25450
b. 26480
c. 24640
d. 23040
e. none of these

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10. Questions

Find the difference between the quantity of water of sumathi has in her final mixture and the quantity of water in Mixture Y?

a. 188 litres

b. 176 litres

c. 190 litres

d. 212 litres

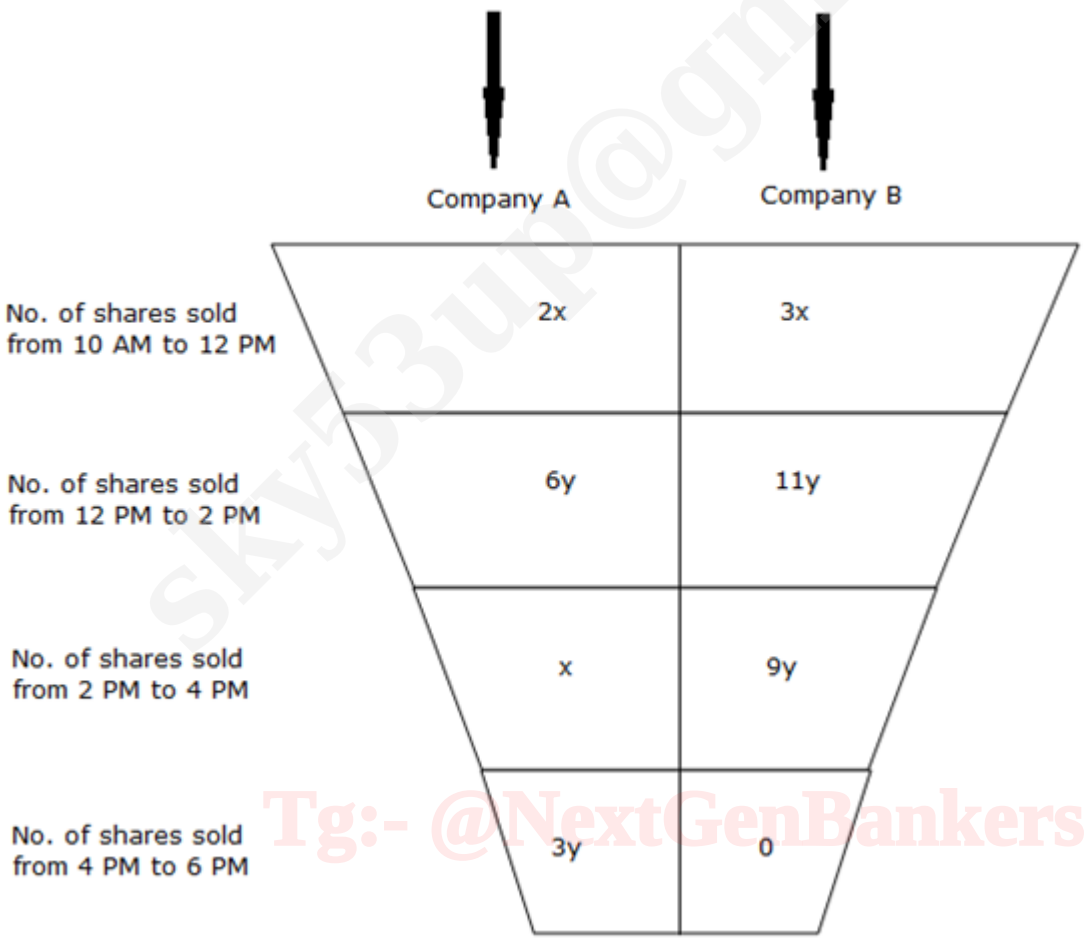
e. none of these

11. Questions

Study the following data carefully and answer the questions:

Two companies A and B launched certain number of shares in the market to sell on a particular day and the shares were sold from 10 a.m. to 6 p.m.

The funnel graph given below shows the number of shares sold by two companies A and B from 10 a.m. to 6 p.m. on that particular day.



Notes:

1: All shares of company B were sold till 4 p.m. and some shares of company A remain unsold after 6 p.m.

2: The total number of shares sold by company B is 4,400 more than that sold by company A.

3: The number of shares sold by company B from 2 p.m. to 4 p.m. is 2100 more than that sold by company A from 2 p.m. to 4 p.m.

If company A sold 81% of the total number of shares till 6 p.m., then find the number of shares of company A remain unsold after 6 p.m.?

a. (2x – 3y)

b. (xy)/4

c. 2(x – y)

d. (x + y)

e. None of these

12. Questions

Company A received a total amount of Rs.4.5 lakhs by selling its shares from 10 a.m. to 12 p.m. If company A reduces the price of its each share by 16(2/3)% after 12 p.m., then find the amount received by company A by selling its shares from 12 p.m. to 2 p.m.?

a. Rs.3.6 lakhs

b. Rs.3 lakhs

c. Rs.4 lakhs

d. Rs.2.7 lakhs

e. Rs.2.4 lakhs

13. Questions

If company A sold M% of its shares from 2 p.m. to 4 p.m. and company B sold N% of its shares from 2 p.m. to 4 p.m., then find the ratio of M to N? (Consider the data of question 1)

a. 15: 28

b. 25: 48

c. 5: 8

d. 10: 17

e. 3: 5

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14. Questions

Company B received a total amount of Rs. 8.1 lakhs by selling its shares from 10 p.m. to 12 p.m. If company B reduced the price of its each share by 16(2/3)% for next two hours and again it reduced the price of its each share by 20% for next two hours, then find the total amount received by company B selling its all shares?

a. Rs.15.05 lakhs

b. Rs.18.06 lakhs

c. Rs.16.08 lakhs

d. Rs.17.01 lakhs

e. Rs.19.02 lakhs

15. Questions

If the number of shares of company A, which were unsold on that particular day, were sold on next day at Rs.[(x – y)/4] per share, then find the amount received by company A by selling those shares on next day? (Consider the data from question 1)

a. Rs. 552200

b. Rs. 525200

c. Rs. 522500

d. Rs. 555500

e. Rs. 555200

16. Questions

In each question, three equations given in column M and their roots are given in column N. Find the correct combination of equations with their roots.

Column M	Column N
A: $x^2 + 21x - 100 = 0$	**P:** 3/2
B: $2y^2 - 53y + 75 = 0$	**Q:** -25
C: $2z^2 + 5z - 12 = 0$	**R:** -4

a. A-Q, B-P, C-P, C-R

b. A-R, B-P, C-P, C-R

c. A-Q, B-Q, C-P, C-R

d. A-Q, B-P, C-Q, C-R

e. A-P, B-R, C-P, C-Q

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17. Questions

Column M	Column N
A: $6x^2 - 23x + 21 = 0$	**P:** 7/2
B: $2y^2 - 17y + 35 = 0$	**Q:** 3/2
C: $2z^2 - 19z + 35 = 0$	**R:** 5/2

a. A-Q, B-R, C-P, C-R

b. A-P, B-Q, C-R

c. A-Q, B-P, B-Q, C-R

d. A-P, B-P, B-R, C-Q

e. A-Q, B-P, C-R

18. Questions

Column M	Column N
A: $x^2 - 47x + 420 = 0$	**P:** 35
B: $5y^2 - 52y + 96 = 0$	**Q:** 8
C: $z^2 - 43z + 280 = 0$	**R:** 12

a. A-P, A-Q, B-Q, C-P, C-R

b. A-P, A-R, B-Q, C-P, C-Q

c. A-P, A-R, B-P, C-P, C-Q

d. A-P, A-R, B-Q, C-P, C-R

e. A-P, A-Q, B-R, C-P, C-Q

19. Questions

Column M	Column N
A: $3x^2 - 59x + 90 = 0$	**P:** 9
B: $5y^2 - 28y + 36 = 0$	**Q:** 2
C: $2z^2 - 23z + 45 = 0$	**R:** 18

Find that which of following combination given in options can't be correct?

a. A-P

b. B-R

c. C-Q

d. Both (a) and (b)

e. All (a), (b) and (c)

20. Questions

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Column M	Column N
A: $x^2 - 13x - 114 = 0$	P: -19
B: $y^2 + 13y - 114 = 0$	Q: -9
C: $z^2 - 7z - 144 = 0$	R: -6

a. A-P, B-Q, C-R

b. A-R, B-P, C-Q

c. A-Q, B-P, C-R

d. A-P, B-R, C-Q

e. A-R, B-Q, C-P

Explanations:

1. Questions

Let the points scored by A, B, C, D and E in 1st attempt be 25x, 17.5x, 20x, 25x and 12.5x, respectively.

Also let the points scored by A, B, C, D and E in 2nd attempt be 15y, 25y, 20y, 17.5y and 22.5y respectively.

So, the points scored by A in 3rd attempt = 15y * (4/3) = 20y

And the points scored by B in 3rd attempt = 25y * (3/5) = 15y

Since, the total points scored by A in the match = 24

So,

25x + 15y + 20y = 24

25x + 35y = 24 ----------(1)

Since, the total points scored by B in the match = 23

So,

17.5x + 25y + 15y = 23

17.5x + 40y = 23

y = (23 – 17.5x)/40 ---------- (2)

From equations (1) and (2):

25x + 35 * [(23 – 17.5x)/40] = 24

1000x + 805 – 612.5x = 960

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387.5x = 155

x = 0.4

From equation (2):

y = (23 – 17.5 * 0.4)/40

y = 0.4

Player	Point scored in 1st attempt	Points scored in 2nd attempt	Points scored in 3rd attempt
A	25 * 0.4 = 10	15 * 0.4 = 6	6 * (4/3) = 8
B	17.5 * 0.4 = 7	25 * 0.4 = 10	10 * (3/5) = 6
C	20 * 0.4 = 8	20 * 0.4 = 8	__
D	25 * 0.4 = 10	17.5 * 0.4 = 7	__
E	12.5 * 0.4 = 5	22.5 * 0.4 = 9	__

Answer: B

Since, the difference between the total points scored by A and B = 24 – 23 = 1

So, the difference between the total points scored by A and C = 1 * 4 = 4

And the total points scored by C will be either = 24 + 4 = 28 or 24 – 4 = 20

If the total points scored by C = 28

So, the points scored by C in 3rd attempt would be = 28 – (8 + 8) = 12

If the total points scored by C = 20

So, the points scored by C in 3rd attempt would be = 20 – (8 + 8) = 4

Since, each player can score maximum 10 points in each attempt.

So, the points scored by C in 3rd attempt = 4

2. Questions

Let the points scored by A, B, C, D and E in 1st attempt be 25x, 17.5x, 20x, 25x and 12.5x, respectively.

Also let the points scored by A, B, C, D and E in 2nd attempt be 15y, 25y, 20y, 17.5y and 22.5y respectively.

So, the points scored by A in 3rd attempt = 15y * (4/3) = 20y

And the points scored by B in 3rd attempt = 25y * (3/5) = 15y

Since, the total points scored by A in the match = 24

So,

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25x + 15y + 20y = 24

25x + 35y = 24 ----------(1)

Since, the total points scored by B in the match = 23

So,

17.5x + 25y + 15y = 23

17.5x + 40y = 23

y = (23 – 17.5x)/40 ---------- (2)

From equations (1) and (2):

25x + 35 * [(23 – 17.5x)/40] = 24

1000x + 805 – 612.5x = 960

387.5x = 155

x = 0.4

From equation (2):

y = (23 – 17.5 * 0.4)/40

y = 0.4

Player	Point scored in 1st attempt	Points scored in 2nd attempt	Points scored in 3rd attempt
A	25 * 0.4 = 10	15 * 0.4 = 6	6 * (4/3) = 8
B	17.5 * 0.4 = 7	25 * 0.4 = 10	10 * (3/5) = 6
C	20 * 0.4 = 8	20 * 0.4 = 8	__
D	25 * 0.4 = 10	17.5 * 0.4 = 7	__
E	12.5 * 0.4 = 5	22.5 * 0.4 = 9	__

Answer: E

Since, the total points scored by D in 1st and 2nd attempts together = 10 + 7 = 17

So, the points scored by D in 3rd attempt = 17 * (100/212.5) = 8

Since, the points scored by B in 3rd attempt = 6

So, the required percentage = (8/6) * 100 = 133.33%

3. Questions

Let the points scored by A, B, C, D and E in 1st attempt be 25x, 17.5x, 20x, 25x and 12.5x, respectively.

Also let the points scored by A, B, C, D and E in 2nd attempt be 15y, 25y, 20y, 17.5y and 22.5y

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respectively.

So, the points scored by A in 3rd attempt = 15y * (4/3) = 20y

And the points scored by B in 3rd attempt = 25y * (3/5) = 15y

Since, the total points scored by A in the match = 24

So,

25x + 15y + 20y = 24

25x + 35y = 24 ----------(1)

Since, the total points scored by B in the match = 23

So,

17.5x + 25y + 15y = 23

17.5x + 40y = 23

y = (23 – 17.5x)/40 ---------- (2)

From equations (1) and (2):

25x + 35 * [(23 – 17.5x)/40] = 24

1000x + 805 – 612.5x = 960

387.5x = 155

x = 0.4

From equation (2):

y = (23 – 17.5 * 0.4)/40

y = 0.4

Player	Point scored in 1st attempt	Points scored in 2nd attempt	Points scored in 3rd attempt
A	25 * 0.4 = 10	15 * 0.4 = 6	6 * (4/3) = 8
B	17.5 * 0.4 = 7	25 * 0.4 = 10	10 * (3/5) = 6
C	20 * 0.4 = 8	20 * 0.4 = 8	__
D	25 * 0.4 = 10	17.5 * 0.4 = 7	__
E	12.5 * 0.4 = 5	22.5 * 0.4 = 9	__

Answer: A

The maximum points scored by a player in 2nd attempt = 10

Since, the players, who scored equal points in 1st attempt, are A and D.

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So, the total points scored by A and D together in 2nd attempt = 6 + 7 = 13

And the required percentage = (10/13) * 100 = 76.9% (approx.)

4. Questions

Let the points scored by A, B, C, D and E in 1st attempt be 25x, 17.5x, 20x, 25x and 12.5x, respectively.

Also let the points scored by A, B, C, D and E in 2nd attempt be 15y, 25y, 20y, 17.5y and 22.5y respectively.

So, the points scored by A in 3rd attempt = 15y * (4/3) = 20y

And the points scored by B in 3rd attempt = 25y * (3/5) = 15y

Since, the total points scored by A in the match = 24

So,

25x + 15y + 20y = 24

25x + 35y = 24 ----------(1)

Since, the total points scored by B in the match = 23

So,

17.5x + 25y + 15y = 23

17.5x + 40y = 23

y = (23 – 17.5x)/40 ---------- (2)

From equations (1) and (2):

25x + 35 * [(23 – 17.5x)/40] = 24

1000x + 805 – 612.5x = 960

387.5x = 155

x = 0.4

From equation (2):

y = (23 – 17.5 * 0.4)/40

y = 0.4

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Player	Point scored in 1st attempt	Points scored in 2nd attempt	Points scored in 3rd attempt
A	25 * 0.4 = 10	15 * 0.4 = 6	6 * (4/3) = 8
B	17.5 * 0.4 = 7	25 * 0.4 = 10	10 * (3/5) = 6
C	20 * 0.4 = 8	20 * 0.4 = 8	__
D	25 * 0.4 = 10	17.5 * 0.4 = 7	__
E	12.5 * 0.4 = 5	22.5 * 0.4 = 9	__

Answer: D

The total points scored by B in the match = 23

So, the total points scored by E in the match = 23 * (100/115) = 20

And the points scored by E in 3rd attempt = 20 – (5 + 9) = 6

From P:

Ratio of points scored by A to E in 3rd attempt = 8: 6 = 4: 3

So, P is true.

From Q:

Points scored by E in 2nd attempt = 9

Points scored by E in 3rd attempt = 6

Required percentage = (6/9) * 100 = 66(2/3)%

So, Q is true.

From R:

The total points scored by A, B and E together in 3rd attempt = 8 + 6 + 6 = 20

Since, 20 is less than 21.

So, R is true.

Hence, all are true.

5. Questions

Let the points scored by A, B, C, D and E in 1st attempt be 25x, 17.5x, 20x, 25x and 12.5x, respectively.

Also let the points scored by A, B, C, D and E in 2nd attempt be 15y, 25y, 20y, 17.5y and 22.5y respectively.

So, the points scored by A in 3rd attempt = 15y * (4/3) = 20y

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And the points scored by B in 3rd attempt = 25y * (3/5) = 15y

Since, the total points scored by A in the match = 24

So,

25x + 15y + 20y = 24

25x + 35y = 24 ----------(1)

Since, the total points scored by B in the match = 23

So,

17.5x + 25y + 15y = 23

17.5x + 40y = 23

y = (23 – 17.5x)/40 ---------- (2)

From equations (1) and (2):

25x + 35 * [(23 – 17.5x)/40] = 24

1000x + 805 – 612.5x = 960

387.5x = 155

x = 0.4

From equation (2):

y = (23 – 17.5 * 0.4)/40

y = 0.4

Player	Point scored in 1st attempt	Points scored in 2nd attempt	Points scored in 3rd attempt
A	25 * 0.4 = 10	15 * 0.4 = 6	6 * (4/3) = 8
B	17.5 * 0.4 = 7	25 * 0.4 = 10	10 * (3/5) = 6
C	20 * 0.4 = 8	20 * 0.4 = 8	__
D	25 * 0.4 = 10	17.5 * 0.4 = 7	__
E	12.5 * 0.4 = 5	22.5 * 0.4 = 9	__

Answer: B

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Since, the points scored by A in 1st attempt = 10

So, the points scored by F in 1st attempt = 10 * (3/5) = 6

Since, the points scored by A in 2nd attempt = 6

So, the points scored by F in 2nd attempt = 6 * (3/2) = 9

Since, the points scored by A in 3rd attempt = 8

So, the points scored by F in 3rd attempt = 8 * (5/8) = 5

Since, the total points scored by F in the match = 6 + 9 + 5 = 20

And the maximum points of the match = 3 * 10 = 30

So, the required percentage = (20/30) * 100 = 66.67%

6. Questions

q^2 - 67q - 360 = 0

= q^2 – 72q + 5q - 360 = 0

= (q - 72) * (q + 5) = 0

Where q = 72

By using the formula,

Final quantity = initial quantity $(1 – \text{amount replaced/total quantity})^{\text{n times}}$

Final quantity = 72 * 72 – 5124 = 60 litres

60 = a * (a - 72/a) * (a – 96/a)

60a = a^2 – 168a + 6912

a^2 – 228a + 6912 = 0

(a – 192) * (a - 36) = 0

a = 192, where 36 is not possible

Initial quantity of alcohol = 192 litres

Alcohol and water in mixture X = 10x and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 23x litres respectively.

10x + 216 = 4

23x + 144 = 5

50x + 1080 = 92x + 576

x = 12

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Alcohol and water in mixture X = 120 and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 276 litres respectively.

Raghu took 50% of mixture X and whole of mixture Y and mixed them to form another mixture Z,

60 + 144: 108 + 276 = 204:384

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Alcohol and water added by Ravi to mixture Z,

204 + 3 * 72 = 420 litres

384 + 3 * 72 – 120 = 480 litres

Sumathi has 420 + 480 = 900 litres of alcohol

Final quantity of alcohol 'm'= 900 * (900 - 180)/900 * 3/5 = 432 litres

Answer: A

From the common explanation,

The value of a = 192

The value of q = 72

Respective ratio = 192:72 = 8:3

7. Questions

q^2 - 67q - 360 = 0

= q^2 – 72q + 5q - 360 = 0

= (q - 72) * (q + 5) = 0

Where q = 72

By using the formula,

Final quantity = initial quantity $(1 – \text{amount replaced/total quantity})^{\text{n times}}$

Final quantity = 72 * 72 – 5124 = 60 litres

60 = a * (a - 72/a) * (a – 96/a)

60a = a^2 – 168a + 6912

a^2 – 228a + 6912 = 0

(a – 192) * (a - 36) = 0

a = 192, where 36 is not possible

Initial quantity of alcohol = 192 litres

Alcohol and water in mixture X = 10x and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 23x litres respectively.

10x + 216 = 4

23x + 144 = 5

50x + 1080 = 92x + 576

x = 12

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60 + 144: 108 + 276 = 204:384

Alcohol and water added by Ravi to mixture Z,

204 + 3 * 72 = 420 litres

384 + 3 * 72 – 120 = 480 litres

Sumathi has 420 + 480 = 900 litres of alcohol

Final quantity of alcohol 'm'= 900 * (900 - 180)/900 * 3/5 = 432 litres

Answer: B

From the common explanation,

Quantity of alcohol in mixture Y = 144 litres

Quantity of water in Mixture Y = 276 litres

Respective percentage = (276 - 144) * 100/144 = 91.67%

8. Questions

$q^2 - 67q - 360 = 0$

$= q^2 – 72q + 5q - 360 = 0$

$= (q - 72) * (q + 5) = 0$

Where q = 72

By using the formula,

Final quantity = initial quantity $(1 – \text{amount replaced/total quantity})^{n \text{ times}}$

Final quantity = 72 * 72 – 5124 = 60 litres

$60 = a * (a - 72/a) * (a – 96/a)$

$60a = a^2 – 168a + 6912$

$a^2 – 228a + 6912 = 0$

$(a – 192) * (a - 36) = 0$

a = 192, where 36 is not possible

Initial quantity of alcohol = 192 litres

Alcohol and water in mixture X = 10x and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 23x litres respectively.

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10x + 216 = 4

23x + 144 = 5

50x + 1080 = 92x + 576

x = 12

Alcohol and water in mixture X = 120 and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 276 litres respectively.

Raghu took 50% of mixture X and whole of mixture Y and mixed them to form another mixture Z,

60 + 144: 108 + 276 = 204:384

Alcohol and water added by Ravi to mixture Z,

204 + 3 * 72 = 420 litres

384 + 3 * 72 – 120 = 480 litres

Sumathi has 420 + 480 = 900 litres of alcohol

Final quantity of alcohol 'm'= 900 * (900 - 180)/900 * 3/5 = 432 litres

Answer: C

From the common explanation,

Ratio of alcohol and water in Mixture X = 5:9 (U)

Ratio of alcohol and water in Mixture Z = 7:8 (V)

By using allegation,

5/14 7/15

L

7 5

12L = 5/2 + 7/3

L = 29/72

So, ratio of alcohol and water in Mixture L = 29:43

Final mixture is equal to alcohol value of mixture Y,

Quantity of alcohol in mixture Y = 144 litres

Quantity of alcohol in Mixture L = 144 * 29/72 = 58 litres

Quantity of water in Mixture L = 144 * 43/72 = 86 litres

Respective profit = 144 * 25 – 58 * 50 = Rs.700

9. Questions

q^2 - 67q - 360 = 0

$= q^2 – 72q + 5q - 360 = 0$

$= (q - 72) * (q + 5) = 0$

Where q = 72

By using the formula,

Final quantity = initial quantity $(1 – \text{amount replaced/total quantity})^{\text{n times}}$

Final quantity = 72 * 72 – 5124 = 60 litres

$60 = a * (a - 72/a) * (a – 96/a)$

$60a = a^2 – 168a + 6912$

$a^2 – 228a + 6912 = 0$

$(a – 192) * (a - 36) = 0$

a = 192, where 36 is not possible

Initial quantity of alcohol = 192 litres

Alcohol and water in mixture X = 10x and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 23x litres respectively.

10x + 216 = 4

23x + 144 = 5

50x + 1080 = 92x + 576

x = 12

Alcohol and water in mixture X = 120 and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 276 litres respectively.

Raghu took 50% of mixture X and whole of mixture Y and mixed them to form another mixture Z,

60 + 144: 108 + 276 = 204:384

Alcohol and water added by Ravi to mixture Z,

204 + 3 * 72 = 420 litres

384 + 3 * 72 – 120 = 480 litres

Sumathi has 420 + 480 = 900 litres of alcohol

Final quantity of alcohol ‘m’= 900 * (900 - 180)/900 * 3/5 = 432 litres

Answer: D

From the common explanation,

The value of m= 432

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The value of a = 192

The value of q = 72

(m – a) * (3q - 120) = (432 - 192) * (3 * 72 – 120) = 23040

10. Questions

q^2 - 67q - 360 = 0

= q^2 – 72q + 5q - 360 = 0

= (q - 72) * (q + 5) = 0

Where q = 72

By using the formula,

Final quantity = initial quantity $(1 – \text{amount replaced/total quantity})^{\text{n times}}$

Final quantity = 72 * 72 – 5124 = 60 litres

60 = a * (a - 72/a) * (a – 96/a)

60a = a^2 – 168a + 6912

a^2 – 228a + 6912 = 0

(a – 192) * (a - 36) = 0

a = 192, where 36 is not possible

Initial quantity of alcohol = 192 litres

Alcohol and water in mixture X = 10x and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 23x litres respectively.

10x + 216 = 4

23x + 144 = 5

50x + 1080 = 92x + 576

x = 12

Alcohol and water in mixture X = 120 and 216 litres respectively.

Alcohol and water in mixture Y = 144 and 276 litres respectively.

Raghu took 50% of mixture X and whole of mixture Y and mixed them to form another mixture Z,

60 + 144: 108 + 276 = 204:384

Alcohol and water added by Ravi to mixture Z,

204 + 3 * 72 = 420 litres

384 + 3 * 72 – 120 = 480 litres

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Sumathi has 420 + 480 = 900 litres of alcohol

Final quantity of alcohol 'm'= 900 * (900 - 180)/900 * 3/5 = 432 litres

Answer: E

From the common explanation,

Quantity of water of sumathi has in her final mixture = (900 - 432) = 468 litres

Quantity of water in Mixture Y = 276 litres

Required difference = 192 litres

11. Questions

The total number of shares sold by company A:

2x + 6y + x + 3y = 3x + 9y

The total number of shares sold by company B:

3x + 11y + 9y = 3x + 20y

Since, the total number of shares sold by company B is 4400 more than that sold by company A.

So,

3x + 20y – 3x – 9y = 4400

y = 400

Since, the number of shares sold by company B from 2 p.m. to 4 p.m. is 2100 more than that sold by company A from 2 p.m. to 4 p.m.

So,

9y – x = 2100

9 * 400 – x = 2100

x = 1500

Time period	No. of shares sold by company A	No. of shares sold by company B
From 10 a.m. to 12 p.m.	2 * 1500 = 3000	3 * 1500 = 4500
From 12 p.m. to 2 p.m.	6 * 400 = 2400	11 * 400 = 4400
From 2 p.m. to 4 p.m.	1500	9 * 400 = 3600
From 4 p.m. to 6 p.m.	3 * 400 = 1200	

Answer: D

Since, the total number of shares sold by company A till 6 p.m.:

3000 + 2400 + 1500 + 1200 = 8100

So, the total number of shares launched by company A = 8100 * (100/81) = 10000

And the number of shares of company A, remain unsold after 6 p.m.:

10000 – 8100 = 1900 = (x + y)

12. Questions

The total number of shares sold by company A:

2x + 6y + x + 3y = 3x + 9y

The total number of shares sold by company B:

3x + 11y + 9y = 3x + 20y

Since, the total number of shares sold by company B is 4400 more than that sold by company A.

So,

3x + 20y – 3x – 9y = 4400

y = 400

Since, the number of shares sold by company B from 2 p.m. to 4 p.m. is 2100 more than that sold by company A from 2 p.m. to 4 p.m.

So,

9y – x = 2100

9 * 400 – x = 2100

x = 1500

Time period	No. of shares sold by company A	No. of shares sold by company B
From 10 a.m. to 12 p.m.	2 * 1500 = 3000	3 * 1500 = 4500
From 12 p.m. to 2 p.m.	6 * 400 = 2400	11 * 400 = 4400
From 2 p.m. to 4 p.m.	1500	9 * 400 = 3600
From 4 p.m. to 6 p.m.	3 * 400 = 1200	

Answer: B

The number of shares sold by company A from 10 a.m. to 12 p.m. = 3000

So, the price of shares of company A from 10 a.m. to 12 p.m. = 450000/3000 = Rs.150 per share

When company A reduces the price of its each share by 16(2/3)%:

The price of shares of company A from 12 p.m. to 2 p.m. = 83(1/3)% of 150 = Rs.125

Since, the number of shares sold by company A from 12 p.m. to 2 p.m. = 2400

So, the amount received by company A by selling its shares from 12 p.m. to 2 p.m.:

2400 * 125 = Rs.3 lakhs

13. Questions

The total number of shares sold by company A:

2x + 6y + x + 3y = 3x + 9y

The total number of shares sold by company B:

3x + 11y + 9y = 3x + 20y

Since, the total number of shares sold by company B is 4400 more than that sold by company A.

So,

3x + 20y – 3x – 9y = 4400

y = 400

Since, the number of shares sold by company B from 2 p.m. to 4 p.m. is 2100 more than that sold by company A from 2 p.m. to 4 p.m.

So,

9y – x = 2100

9 * 400 – x = 2100

x = 1500

Time period	No. of shares sold by company A	No. of shares sold by company B
From 10 a.m. to 12 p.m.	2 * 1500 = 3000	3 * 1500 = 4500
From 12 p.m. to 2 p.m.	6 * 400 = 2400	11 * 400 = 4400
From 2 p.m. to 4 p.m.	1500	9 * 400 = 3600
From 4 p.m. to 6 p.m.	3 * 400 = 1200	

Answer: B

From question 1:

The total number of shares of company A = 10000

The number of shares sold by company A from 2 p.m. to 4 p.m. = 1500

So, M = (1500/10000) * 100 = 15%

The total number of shares of company B = 4500 + 4400 + 3600 = 12500

The number of shares sold by company B from 2 p.m. to 4 p.m. = 3600

So, N = (3600/12500) * 100 = 28.8%

Required ratio = 15: 28.8 = 25: 48

14. Questions

The total number of shares sold by company A:

2x + 6y + x + 3y = 3x + 9y

The total number of shares sold by company B:

3x + 11y + 9y = 3x + 20y

Since, the total number of shares sold by company B is 4400 more than that sold by company A.

So,

3x + 20y – 3x – 9y = 4400

y = 400

Since, the number of shares sold by company B from 2 p.m. to 4 p.m. is 2100 more than that sold by company A from 2 p.m. to 4 p.m.

So,

9y – x = 2100

9 * 400 – x = 2100

x = 1500

Time period	No. of shares sold by company A	No. of shares sold by company B
From 10 a.m. to 12 p.m.	2 * 1500 = 3000	3 * 1500 = 4500
From 12 p.m. to 2 p.m.	6 * 400 = 2400	11 * 400 = 4400
From 2 p.m. to 4 p.m.	1500	9 * 400 = 3600
From 4 p.m. to 6 p.m.	3 * 400 = 1200	

Answer: E

Since, the number of shares sold by company B from 10 AM to 12 PM = 4500

So, the price of shares of company B from 10 AM to 12 PM = 810000/4500 = Rs. 180 per share

The price of shares of company B from 12 PM to 2 PM = 83(1/3)% of 180 = Rs. 150 per share

And the price of shares of company B from 2 PM to 4 PM = 80% of 150 = Rs. 120 per share

Since, the number of shares sold by company B from 12 PM to 2 PM = 4400

And the number of shares sold by company B from 2 PM to 4 PM = 3600

So, the total amount received by company B by selling its all shares:

810000 + (4400 * 150) + (3600 * 120)

810000 + 660000 + 432000 = 1902000 = Rs. 19.02 lakhs

15. Questions

The total number of shares sold by company A:

2x + 6y + x + 3y = 3x + 9y

The total number of shares sold by company B:

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$3x + 11y + 9y = 3x + 20y$

Since, the total number of shares sold by company B is 4400 more than that sold by company A.

So,

$3x + 20y – 3x – 9y = 4400$

$y = 400$

Since, the number of shares sold by company B from 2 p.m. to 4 p.m. is 2100 more than that sold by company A from 2 p.m. to 4 p.m.

So,

$9y – x = 2100$

$9 * 400 – x = 2100$

$x = 1500$

Time period	No. of shares sold by company A	No. of shares sold by company B
From 10 a.m. to 12 p.m.	2 * 1500 = 3000	3 * 1500 = 4500
From 12 p.m. to 2 p.m.	6 * 400 = 2400	11 * 400 = 4400
From 2 p.m. to 4 p.m.	1500	9 * 400 = 3600
From 4 p.m. to 6 p.m.	3 * 400 = 1200	

Answer: C

From question 1:

The number of shares of company A, which were unsold on that particular day = 1900

And the price, at which those share were sold on next day:

$(x – y)/4 = (1500 – 400)/4$ = Rs.275 per share

So, the total amount received by company A by selling those shares on next day = 1900 * 275 = Rs.522500

16. Questions

Answer: A

From equation A:

$x^2 + 21x – 100 = 0$

$x^2 + 25x – 4x – 100 = 0$

$x(x + 25) – 4(x + 25) = 0$

$x = -25, 4$

From equation B:

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$2y^2 - 53y + 75 = 0$

$2y^2 - 50y - 3y + 75 = 0$

$2y(y - 25) - 3(y - 25) = 0$

$y = 25, 3/2$

From equation C:

$2z^2 + 5z - 12 = 0$

$2z^2 + 8z - 3z - 12 = 0$

$2z(z + 4) - 3(z + 4) = 0$

$z = -4, 3/2$

The correct combination: A-Q, B-P, C-P, C-R.

17. Questions

Answer: E

From equation A:

$6x^2 - 23x + 21 = 0$

$6x^2 - 9x - 14x + 21 = 0$

$3x(2x - 3) - 7(2x - 3) = 0$

$x = 3/2, 7/3$

From equation B:

$2y^2 - 17y + 35 = 0$

$2y^2 - 10y - 7y + 35 = 0$

$2y(y - 5) - 7(y - 5) = 0$

$y = 5, 7/2$

From equation C:

$2z^2 - 19z + 35 = 0$

$2z^2 - 14z - 5z + 35 = 0$

$2z(z - 7) - 5(z - 7) = 0$

$z = 7, 5/2$

The correct combination: A-Q, B-P, C-R.

18. Questions

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Answer: B

From equation A:

$x^2 - 47x + 420 = 0$

$x^2 - 35x - 12x + 420 = 0$

$x(x - 35) - 12(x - 35) = 0$

$x = 35, 12$

From equation B:

$5y^2 - 52y + 96 = 0$

$5y^2 - 40y - 12y + 96 = 0$

$5y(y - 8) - 12(y - 8) = 0$

$y = 8, 12/5$

From equation C:

$z^2 - 43x + 280 = 0$

$z^2 - 35z - 8z + 280 = 0$

$z(z - 35) - 8(z - 35) = 0$

$z = 35, 8$

The correct combination: A-P, A-R, B-Q, C-P, C-Q.

19. Questions

Answer: E

From equation A:

$3x^2 - 59x + 90 = 0$

$3x^2 - 54x - 5x + 90 = 0$

$3x(x - 18) - 5(x - 18) = 0$

$x = 18, 5/3$

From equation B:

$5y^2 - 28y + 36 = 0$

$5y^2 - 10y - 18y + 36 = 0$

$5y(y - 2) - 18(y - 2) = 0$

$y = 2, 18/5$

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From equation C:

$2z^2 - 23z + 45 = 0$

$2z^2 - 18z - 5z + 45 = 0$

$2z(z - 9) - 5(z - 9) = 0$

$z = 9, 5/2$

The incorrect combination: A-R, B-Q, C-P.

20. Questions

Answer: B

From equation A:

$x^2 - 13x - 114 = 0$

$x^2 - 19x + 6x - 114 = 0$

$x(x - 19) + 6(x - 19) = 0$

$x = 19, -6$

From equation B:

$y^2 + 13y - 114 = 0$

$y^2 + 19y - 6y - 114 = 0$

$y(y + 19) - 6(y + 19) = 0$

$y = -19, 6$

From equation C:

$z^2 - 7z - 144 = 0$

$z^2 - 16z + 9z - 144 = 0$

$z(z - 16) + 9(z - 16) = 0$

$z = 16, -9$

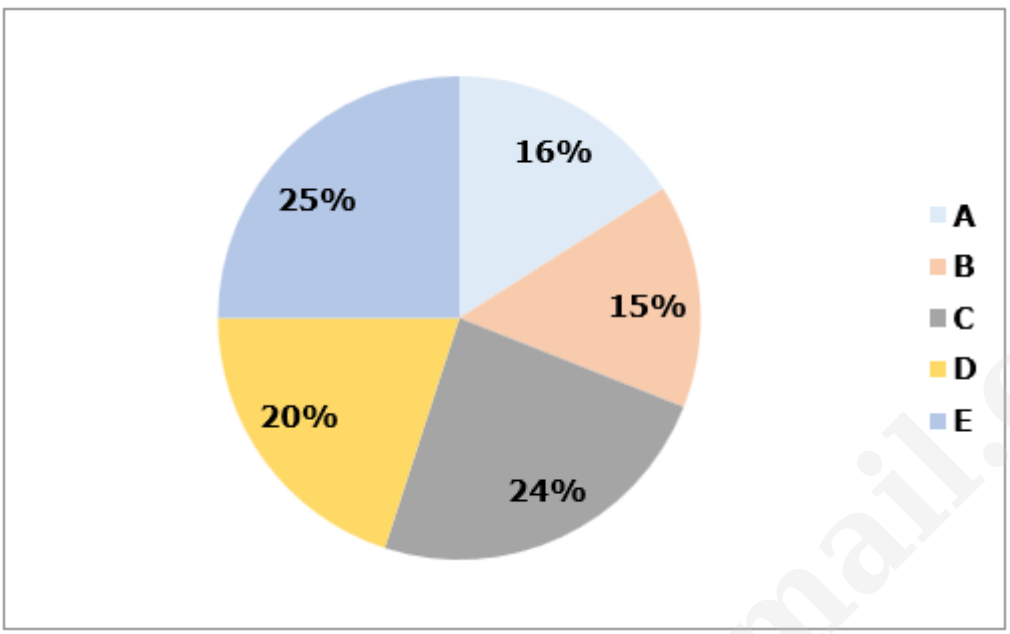
The correct combination: A-R, B-P, C-Q.

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1. Questions

Study the following data carefully and answer the questions:

The pie chart given below shows the percentage distribution of the difference between the number of votes received in 2021 and that received in 2022 by each of the given five candidates A, B, C, D and E.



Notes:

1). The number of votes received by A and C in 2021 is 45000 and 40000 respectively.

2). The ratio of the number of votes received by A in 2022 to that received by C in 2022 is 189: 254.

3). The number of votes received by A in 2021 > The number of votes received by A in 2022

4). The number of votes received by C in 2021 < The number of votes received by C in 2022

The difference between the number of votes received by A and C in 2021 is approximately what per cent of the difference between the number of votes received by A and C in 2022?

a. Can't be determined

b. 38.5%

c. 25%

d. 34.5%

e. 30%

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2. Questions

If the number of votes received by D in 2021 is 94% of that received by A in 2021 and the average of the number of votes received by B and C in 2021 is 48375, then find that the number of votes received by D in 2022 will be what per cent more than that received by B in 2022?

Condition 1- The number of votes received by B in 2021 > The number of votes received by B in 2022

Condition 2- The number of votes received by D in 2021 < The number of votes received by D in 2022

a. 6.5%

b. 9.1%

c. 1.3%

d. 3.9%

e. 2.6%

3. Questions

If the average of the number of votes received by A and E in 2022 is 41400, then choose the correct option.

Quantity I: 80% of the maximum possible number of votes received by E in 2021.

Quantity II: 140% of the minimum possible number of votes received by E in 2021.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I ≥ Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I = Quantity II or relation can't be established

4. Questions

If the average of the number of votes received by A, C and F in 2021 is 43000 and the average of the number of votes received by A, C and F in 2022 is 40000, then find the difference between the number of votes received by F in 2021 and that received by him in 2022?

a. 12200

b. 11800

c. 11600

d. 12600

e. 11400

5. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of parcels received by three post offices A, B and C on five days Mon, Tue, Wed, Thu and Fri.

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Day	The number of parcels received by post office A	The number of parcels received by post office B as % of that received by post office A	The average number of parcels received by post offices A and C
Mon	2x	5y%	150
Tue	5y	140%	(4y + x/2)
Wed	3x	75%	(xy)/10
Thu	7y	x%	160
Fri	200	90%	170

Notes:

1). The number of parcels received by post office C on Mon is equal to that received by C on Fri.

2). The average number of parcels received by post office A on each of the given days is 180.

If the average number of parcels received by post office A on Mon, Tue and Wed is M and the average number of parcels received by post office B on Mon, Tue and Wed is N, then find the value of 5.5(N – M)? (In terms of 'x' and 'y')

a. (2x – 5y)

b. (x – y)

c. (x + y)/5

d. (y + x/4)

e. (x + 2y)/2

6. Questions

Which of the following statements is/are not true?

P). The difference between the number of parcels received by A and C on Thu is 'y'.

Q). The average number of parcels received by post office B on Tue, Wed and Thu is (2x + 5).

R). The ratio between the number of parcels received by post offices A, B and C on Wed is 12: 9: 8 respectively.

a. Only P

b. Only Q and R

c. None is true

d. Only P and Q

e. Only Q

7. Questions

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The total number of parcels received by all the three post offices together on Thu is what per cent of the total number of parcels received by all the three post offices together on Mon?

a. (x + y – 23)%

b. (x + y – 11)%

c. (x + y – 15)%

d. (x + y – 19)%

e. (x + y – 13)%

8. Questions

Find the ratio of the total number of parcels received by post office A on all the five days together to that received by post office B on all the five days together?

a. (x – 44): (y + 10)

b. (x – 34): (y + 5)

c. (x – 54): (y + 15)

d. (x – 24): (y + 20)

e. None of these

9. Questions

If the average number of parcels received by post office A on Fri and Sat is 170, the average number of parcels received by post office B on Fri and Sat is 165 and the average number of parcels received by post office C on Fri and Sat is ___, then the ratio of the total number of parcels received by all the three post offices together on Fri to that received by all the three post offices together on Sat will be 4: 3.

Find the value, which can be filled in the blank?

a. 150

b. 110

c. 160

d. 120

e. 130

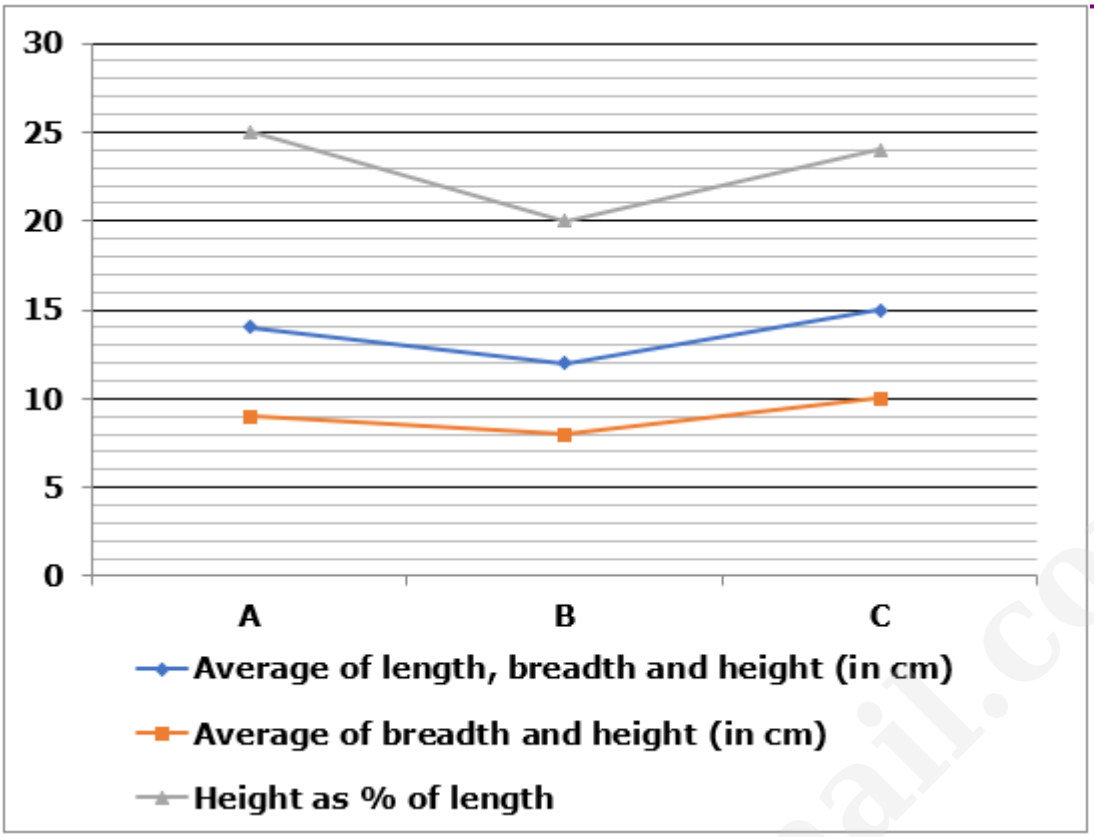
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10. Questions

Study the following data carefully and answer the questions:

The data given below is related to length (in cm), breadth (in cm) and height (in cm) of three solid cuboids A, B and C.

The line graph given below shows the following data:



If the ratio of length to breadth of cuboid A is P: Q and the ratio of length to breadth of cuboid B is R: S, then find the ratio of breadth to height of cuboid C?

a. (P + R): S

b. (R + S – Q): P

c. (PS + Q): (R – 1)

d. None of these

e. (PR – S): Q

11. Questions

If the cost of painting a cuboid is ₹15 per cm^2, then find the difference between the total costs of painting cuboids B and C?

a. ₹ 6630

b. ₹ 6210

c. ₹ 6660

d. ₹ 6480

e. ₹ 6360

12. Questions

Cuboid A is melted down to form 'x' number of solid cubes and the side of each cube is 'y' cm. If the ratio of 'x' to 'y' is 4: 3, then find the value of 'x'?

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a. 12

b. 16

c. 8

d. 20

e. 4

13. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of employees of a company, who like movies of three genres (action, comedy and horror). Some like movies of only one genre, some like movies of only two genres and remaining like movies of all the three genres.

25% of the total number of employees in the company likes action movies only. The ratio of the number of employees, who like action movies only, to those, who like both action and comedy movies but not horror movies, is 10: 1 and the ratio of the number of employees, who like action movies only, to those, who like movies of all the three genres, is 5: 2. The number of employees, who like both action and horror movies but not comedy movies, is 30. The ratio between the number of employees, who like only comedy movies, those, who like both comedy and horror movies but not action movies and those, who like only horror movies, is 5: 4: 3 respectively. A total of 50% of the total number of employees of the company likes action movies.

If the ratio of male to female employees, who like only action movies, is 8: 7, the ratio of male to female employees, who like only comedy movies, is 3: 2 and the ratio of male to female employees, who like only horror movies, is 3: 2, then find that approximately what per cent of the total number of employees, who like movies of only one genre, are female?

a. 32.8%

b. 58.8%

c. 48.8%

d. 42.8%

e. 38.8%

14. Questions

What percent of the total number of employees of the company like horror movies?

a. 44.67%

b. 51.67%

c. 55.67%

d. 41.67%

e. 59.67%

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15. Questions

If the ratio of male to female employees, who like movies of only two genres, is 8: 11 and the ratio of the total number of male to female employees in the company is 2: 1, then find that what per cent of the total number of female employees in the company like movies of only two genres?

a. 55%

b. 40%

c. 50%

d. 44%

e. 34%

16. Questions

Find the ratio of the total number of employees, who like movies of only one genre, to those, who like movies of only two genres?

a. 14: 9

b. 28: 19

c. 7: 4

d. 7: 2

e. 35: 19

17. Questions

If the ratio between the number of employees, who like only action movies, those, who like only comedy movies and those, who like only horror movies, is P: Q: R, then find that which of the following statements is/are true?

X). The value of $(P^2 - Q^2)$ is divisible by R.

Y). The value of $(P * Q * R) < 45$.

Z). The value of $(P^2 + Q^2 + R^2)$ is divisible by 9.

a. Only X

b. Only X and Y

c. Only X and Z

d. None is true

e. Only Y

18. Questions

Study the following data carefully and answer the questions:

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Three series I, II and III are given below and each series follows the same pattern:

I). 298, _, _, 104, 68, _

II). 341, 260, 196, _, _, 86

III). 325, (P + Q), (P – Q), 131

The 6th term of series I is what per cent of the 6th term of series II?

a. 33.33%

b. 43%

c. 25%

d. 50%

e. 172%

19. Questions

Find the average of the 2nd and 3rd term of series I?

a. (P + 3Q)/2

b. (P – Q + 5)

c. (2P + Q)/2

d. (2P + 2Q)/3

e. (P – Q + 10)

20. Questions

Which of the following statements is/are true?

X). The difference between the 4th and 5th terms of series II is (Q + 4).

Y). The 5th term of series III is [(P/2) – 10].

Z). Q is 15% of P.

a. Only X

b. Only X and Y

c. Only X and Z

d. None is true

e. Only Y

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Explanations:

1. Questions

Let the difference between the number of votes received by A in 2021 and 2022 = 16x

So, the difference between the number of votes received by C in 2021 and 2022 = 24x

Since, the number of votes received by A in 2021 = 45000

So, the number of votes received by A in 2022 = 45000 – 16x

Since, the number of votes received by C in 2021 = 40000

So, the number of votes received by C in 2022 = 40000 + 24x

Since, the ratio of the number of votes received by A in 2022 to that received by C in 2022 is 189: 254.

So,

(45000 – 16x)/(40000 + 24x) = 189/254

(5625 – 2x)/(5000 + 3x) = 189/254

1428750 – 508x = 945000 + 567x

483750 = 1075x

x = 450

Candidates	Difference between the no. of votes received in 2021 and 2022
A	16 * 450 = 7200
B	15 * 450 = 6750
C	24 * 450 = 10800
D	20 * 450 = 9000
E	25 * 450 = 11250

Answer: B

Since, the number of votes received by A in 2021 = 45000

And the number of votes received by C in 2021 = 40000

So, the difference between the number of votes received by A and C in 2021:

45000 – 40000 = 5000

Since, the number of votes received by A in 2022 = 45000 – 7200 = 37800

And the number of votes received by C in 2022 = 40000 + 10800 = 50800

So, the difference between the number of votes received by A and C in 2022:

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50800 – 37800 = 13000

The required percentage = (5000/13000) * 100 = 38.5% (approx.)

Another method:

The number of votes received by A in 2021 is 45000.

The number of votes received by A in 2022 is 45000 X 84 /100= 37800

The ratio of the number of votes received by A and C in 2022 is 189:254

Since, 37800 = 189x

X = 200

The number of votes received by C in 2022 is 254 x 200 = 50800

So, the difference between the number of votes received by A and C in 2021:

45000 – 40000 = 5000

So, the difference between the number of votes received by A and C in 2022:

50800 – 37800 = 13000

The required percentage = (5000/13000) * 100 = 38.5% (approx).

2. Questions

Let the difference between the number of votes received by A in 2021 and 2022 = 16x

So, the difference between the number of votes received by C in 2021 and 2022 = 24x

Since, the number of votes received by A in 2021 = 45000

So, the number of votes received by A in 2022 = 45000 – 16x

Since, the number of votes received by C in 2021 = 40000

So, the number of votes received by C in 2022 = 40000 + 24x

Since, the ratio of the number of votes received by A in 2022 to that received by C in 2022 is 189: 254.

So,

(45000 – 16x)/(40000 + 24x) = 189/254

(5625 – 2x)/(5000 + 3x) = 189/254

1428750 – 508x = 945000 + 567x

483750 = 1075x

x = 450

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Candidates	Difference between the no. of votes received in 2021 and 2022
A	16 * 450 = 7200
B	15 * 450 = 6750
C	24 * 450 = 10800
D	20 * 450 = 9000
E	25 * 450 = 11250

Answer: E

Since, the number of votes received by A in 2021 = 45000

So, the number of votes received by D in 2021 = 94% of 45000 = 42300

And the number of votes received by D in 2022 = 42300 + 9000 = 51300

Since, the number of votes received by C in 2021 = 40000

So, the number of votes received by B in 2021 = 2 * 48375 – 40000 = 56750

And the number of votes received by B in 2022 = 56750 – 6750 = 50000

The required percentage = [(51300 – 50000)/50000] * 100 = 2.6%

3. Questions

Let the difference between the number of votes received by A in 2021 and 2022 = 16x

So, the difference between the number of votes received by C in 2021 and 2022 = 24x

Since, the number of votes received by A in 2021 = 45000

So, the number of votes received by A in 2022 = 45000 – 16x

Since, the number of votes received by C in 2021 = 40000

So, the number of votes received by C in 2022 = 40000 + 24x

Since, the ratio of the number of votes received by A in 2022 to that received by C in 2022 is 189: 254.

So,

(45000 – 16x)/(40000 + 24x) = 189/254

(5625 – 2x)/(5000 + 3x) = 189/254

1428750 – 508x = 945000 + 567x

483750 = 1075x

x = 450

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Candidates	Difference between the no. of votes received in 2021 and 2022
A	16 * 450 = 7200
B	15 * 450 = 6750
C	24 * 450 = 10800
D	20 * 450 = 9000
E	25 * 450 = 11250

Answer: B

Since, the average of the number of votes received by A and E in 2022 is 41400.

And the number of votes received by A in 2022 = 45000 – 7200 = 37800

So, the number of votes received by E in 2022 = 2 * 41400 – 37800 = 45000

Quantity I:

Since, the maximum possible number of votes received by E in 2021:

45000 + 11250 = 56250

So, the required value = 80% of 56250 = 45000

Quantity II:

Since, the minimum possible number of votes received by E in 2021:

45000 – 11250 = 33750

So, the required value = 140% of 33750 = 47250

Hence, Quantity I < Quantity II

4. Questions

Let the difference between the number of votes received by A in 2021 and 2022 = 16x

So, the difference between the number of votes received by C in 2021 and 2022 = 24x

Since, the number of votes received by A in 2021 = 45000

So, the number of votes received by A in 2022 = 45000 – 16x

Since, the number of votes received by C in 2021 = 40000

So, the number of votes received by C in 2022 = 40000 + 24x

Since, the ratio of the number of votes received by A in 2022 to that received by C in 2022 is 189: 254.

So,

(45000 – 16x)/(40000 + 24x) = 189/254

(5625 – 2x)/(5000 + 3x) = 189/254

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1428750 – 508x = 945000 + 567x

483750 = 1075x

x = 450

Candidates	Difference between the no. of votes received in 2021 and 2022
A	16 * 450 = 7200
B	15 * 450 = 6750
C	24 * 450 = 10800
D	20 * 450 = 9000
E	25 * 450 = 11250

Answer: D

Since, the average of the number of votes received by A, C and F in 2021 = 43000

So, the number of votes received by F in 2021:

3 * 43000 – (45000 + 40000) = 44000

Since, the average of the number of votes received by A, C and F in 2022 = 40000

So, the number of votes received by F in 2022:

3 * 40000 – (37800 + 50800) = 31400

The required difference = 44000 – 31400 = 12600

5. Questions

Since, the number of parcels received by post office C on Fri = (2 * 170) – 200 = 140

And the number of parcels received by post office C on Mon = 140

So,

(2 * 150) – 2x = 140

x = 80

Since, the average number of parcels received by post office A on each of the given days is 180.

So,

2x + 5y + 3x + 7y + 200 = 5 * 180

160 + 12y + 240 + 200 = 900

y = 25

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Day	The number of parcels received by post office A	The number of parcels received by post office B as % of that received by post office A	The average number of parcels received by post offices A and C
Mon	2 * 80 = 160	5 * 25 = 125%	150
Tue	5 * 25 = 125	140%	(4 * 25 + 80/2) = 140
Wed	3 * 80 = 240	75%	(80 * 25)/10 = 200
Thu	7 * 25 = 175	80%	160
Fri	200	90%	170

Day	The number of parcels received by post office A	The number of parcels received by post office B	The number of parcels received by post office C
Mon	160	125% of 160 = 200	2 * 150 – 160 = 140
Tue	125	140% of 125 = 175	2 * 140 – 125 = 155
Wed	240	75% of 240 = 180	2 * 200 – 240 = 160
Thu	175	80% of 175 = 140	2 * 160 – 175 = 145
Fri	200	90% of 200 = 180	2 * 170 – 200 = 140

Answer: B

Since, the average number of parcels received by post office A on Mon, Tue and Wed = M = (160 + 125 + 240)/3 = 175

And the average number of parcels received by post office B on Mon, Tue and Wed = N = (200 + 175 + 180)/3 = 185

So, the value of 5.5(N – M) = 5.5(185 – 175) = 5.5*10 = 55 = (x – y) = (80 – 25) = 55

6. Questions

Since, the number of parcels received by post office C on Fri = (2 * 170) – 200 = 140

And the number of parcels received by post office C on Mon = 140

So,

(2 * 150) – 2x = 140

x = 80

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Since, the average number of parcels received by post office A on each of the given days is 180.

So,

2x + 5y + 3x + 7y + 200 = 5 * 180

160 + 12y + 240 + 200 = 900

y = 25

Day	The number of parcels received by post office A	The number of parcels received by post office B as % of that received by post office A	The average number of parcels received by post offices A and C
Mon	2 * 80 = 160	5 * 25 = 125%	150
Tue	5 * 25 = 125	140%	(4 * 25 + 80/2) = 140
Wed	3 * 80 = 240	75%	(80 * 25)/10 = 200
Thu	7 * 25 = 175	80%	160
Fri	200	90%	170

Day	The number of parcels received by post office A	The number of parcels received by post office B	The number of parcels received by post office C
Mon	160	125% of 160 = 200	2 * 150 – 160 = 140
Tue	125	140% of 125 = 175	2 * 140 – 125 = 155
Wed	240	75% of 240 = 180	2 * 200 – 240 = 160
Thu	175	80% of 175 = 140	2 * 160 – 175 = 145
Fri	200	90% of 200 = 180	2 * 170 – 200 = 140

Answer: A

From P:

The difference between the number of parcels received by A and C on Thu:

(175 – 145) = 30

Since, the value of ‘y’ = 25

So, P is not true.

From Q:

The average number of parcels received by post office B on Tue, Wed and Thu:

(175 + 180 + 140)/3 = 165

Since, the value of (2x + 5) = (2 * 80 + 5) = 165

So, Q is true.

From R:

The ratio between the number of parcels received by post offices A, B and C on Wed:

240: 180: 160 = 12: 9: 8

So, R is true.

Hence, only P is not true.

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7. Questions

Since, the number of parcels received by post office C on Fri = (2 * 170) – 200 = 140

And the number of parcels received by post office C on Mon = 140

So,

(2 * 150) – 2x = 140

x = 80

Since, the average number of parcels received by post office A on each of the given days is 180.

So,

2x + 5y + 3x + 7y + 200 = 5 * 180

160 + 12y + 240 + 200 = 900

y = 25

Day	The number of parcels received by post office A	The number of parcels received by post office B as % of that received by post office A	The average number of parcels received by post offices A and C
Mon	2 * 80 = 160	5 * 25 = 125%	150
Tue	5 * 25 = 125	140%	(4 * 25 + 80/2) = 140
Wed	3 * 80 = 240	75%	(80 * 25)/10 = 200
Thu	7 * 25 = 175	80%	160
Fri	200	90%	170

Day	The number of parcels received by post office A	The number of parcels received by post office B	The number of parcels received by post office C
Mon	160	125% of 160 = 200	2 * 150 – 160 = 140
Tue	125	140% of 125 = 175	2 * 140 – 125 = 155
Wed	240	75% of 240 = 180	2 * 200 – 240 = 160
Thu	175	80% of 175 = 140	2 * 160 – 175 = 145
Fri	200	90% of 200 = 180	2 * 170 – 200 = 140

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Answer: E

Since, the total number of parcels received by all the three post offices together on Thu = 175 + 140 + 145 = 460

And the total number of parcels received by all the three post offices together on Mon = 160 + 200 + 140 = 500

So, the required percentage = (460/500) * 100 = 92% = (x + y – 13)%

8. Questions

Since, the number of parcels received by post office C on Fri = (2 * 170) – 200 = 140

And the number of parcels received by post office C on Mon = 140

So,

(2 * 150) – 2x = 140

x = 80

Since, the average number of parcels received by post office A on each of the given days is 180.

So,

2x + 5y + 3x + 7y + 200 = 5 * 180

160 + 12y + 240 + 200 = 900

y = 25

Day	The number of parcels received by post office A	The number of parcels received by post office B as % of that received by post office A	The average number of parcels received by post offices A and C
Mon	2 * 80 = 160	5 * 25 = 125%	150
Tue	5 * 25 = 125	140%	(4 * 25 + 80/2) = 140
Wed	3 * 80 = 240	75%	(80 * 25)/10 = 200
Thu	7 * 25 = 175	80%	160
Fri	200	90%	170

Day	The number of parcels received by post office A	The number of parcels received by post office B	The number of parcels received by post office C
Mon	160	125% of 160 = 200	2 * 150 – 160 = 140
Tue	125	140% of 125 = 175	2 * 140 – 125 = 155
Wed	240	75% of 240 = 180	2 * 200 – 240 = 160
Thu	175	80% of 175 = 140	2 * 160 – 175 = 145
Fri	200	90% of 200 = 180	2 * 170 – 200 = 140

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Answer: A

Since, the total number of parcels received by post office A on all the five days together = 160 + 125 + 240 + 175 + 200 = 900

And the total number of parcels received by post office B on all the five days together = 200 + 175 + 180 + 140 + 180 = 875

So, the required ratio = 900: 875 = 36: 35 = (x – 44): (y + 10)

9. Questions

Since, the number of parcels received by post office C on Fri = (2 * 170) – 200 = 140

And the number of parcels received by post office C on Mon = 140

So,

(2 * 150) – 2x = 140

x = 80

Since, the average number of parcels received by post office A on each of the given days is 180.

So,

2x + 5y + 3x + 7y + 200 = 5 * 180

160 + 12y + 240 + 200 = 900

y = 25

Day	The number of parcels received by post office A	The number of parcels received by post office B as % of that received by post office A	The average number of parcels received by post offices A and C
Mon	2 * 80 = 160	5 * 25 = 125%	150
Tue	5 * 25 = 125	140%	(4 * 25 + 80/2) = 140
Wed	3 * 80 = 240	75%	(80 * 25)/10 = 200
Thu	7 * 25 = 175	80%	160
Fri	200	90%	170

Day	The number of parcels received by post office A	The number of parcels received by post office B	The number of parcels received by post office C
Mon	160	125% of 160 = 200	2 * 150 – 160 = 140
Tue	125	140% of 125 = 175	2 * 140 – 125 = 155
Wed	240	75% of 240 = 180	2 * 200 – 240 = 160
Thu	175	80% of 175 = 140	2 * 160 – 175 = 145
Fri	200	90% of 200 = 180	2 * 170 – 200 = 140

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Answer: D

Since, the average number of parcels received by post office A on Fri and Sat is 170.

So, the number of parcels received by post office A on Sat = 2 * 170 – 200 = 140

Since, the average number of parcels received by post office B on Fri and Sat is 165.

So, the number of parcels received by post office B on Sat = 2 * 165 – 180 = 150

Since, the total number of parcels received by all the three post offices together on Fri = 200 + 180 + 140 = 520

So, the total number of parcels received by all the three post offices together on Sat = 520 * (3/4) = 390

And the number of parcels received by post office C on Sat:

390 – (140 + 150) = 100

So, the average number of parcels received by post office C on Fri and Sat is (100+140)/2 = 120

10. Questions

Cuboid A:

Since, the sum of length, breadth and height = 3 * 14 = 42 cm

And the sum of breadth and height = 2 * 9 = 18 cm

So, the length = 42 – 18 = 24 cm

The height = 25% of 24 = 6 cm

And the breadth = 42 – (24 + 6) = 12 cm

Similarly, we can calculate for other cuboids as well:

Cuboid	Length (in cm)	Breadth (in cm)	Height (in cm)
A	24	12	6
B	20	12	4
C	25	14	6

Answer: A

Since, the ratio of length to breadth of cuboid A = P: Q = 24: 12 = 2: 1

And the ratio of length to breadth of cuboid B = R: S = 20: 12 = 5: 3

So, the ratio of breadth to height of cuboid C = 14: 6 = 7: 3 = (P + R): S

11. Questions

Cuboid A:

Since, the sum of length, breadth and height = 3 * 14 = 42 cm

And the sum of breadth and height = 2 * 9 = 18 cm

So, the length = 42 – 18 = 24 cm

The height = 25% of 24 = 6 cm

And the breadth = 42 – (24 + 6) = 12 cm

Similarly, we can calculate for other cuboids as well:

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Cuboid	Length (in cm)	Breadth (in cm)	Height (in cm)
A	24	12	6
B	20	12	4
C	25	14	6

Answer: D

Since, the total surface area of cuboid B = 2 * (20 * 12 + 12 * 4 + 20 * 4) = 736 cm^2

And the total surface area of cuboid C = 2 * (25 * 14 + 14 * 6 + 25 * 6) = 1168 cm^2

So, the required difference = 15 * (1168 – 736) = ₹ 6480

12. Questions

Cuboid A:

Since, the sum of length, breadth and height = 3 * 14 = 42 cm

And the sum of breadth and height = 2 * 9 = 18 cm

So, the length = 42 – 18 = 24 cm

The height = 25% of 24 = 6 cm

And the breadth = 42 – (24 + 6) = 12 cm

Similarly, we can calculate for other cuboids as well:

Cuboid	Length (in cm)	Breadth (in cm)	Height (in cm)
A	24	12	6
B	20	12	4
C	25	14	6

Answer: C

The volume of cuboid A = 24 * 12 * 6 = 1728 cm^3

So,

$x * y^3 = 1728$ ---------(1)

Since, the ratio of 'x' to 'y' = x: y = 4: 3

So,

$x = 4y/3$ ---------- (2)

From equations (1) and (2):

$(4y/3) * y^3 = 1728$

$y^4 = 1296$

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y = 6

From equation (2):

x = 8

13. Questions

Let the total number of employees in the company = 4x

So, the number of employees, who like only action movies = 25% of 4x = x

The number of employees, who like both action and comedy movies but not horror movies = x * (1/10) = 0.1x

And the number of employees, who like movies of all the three genres = x * (2/5) = 0.4x

The number of employees, who like both action and horror movies but not comedy movies, is 30

Since, a total of 50% of the total number of employees in the company likes action movies.

So,

x + 0.1x + 0.4x + 30 = 50% of 4x

1.5x + 30 = 2x

x = 60

Since, the total number of employees in the company = 4 * 60 = 240

So, the number of employees, who like only action movies = 25% of 240 = 60

The number of employees, who like both action and comedy movies but not horror movies = 0.1 * 60 = 6

And the number of employees, who like movies of all the three genres = 0.4 * 60 = 24

So, the total number of employees, who like action movies = 50% of 240 = 120

And the total number of employees, who don't like action movies = 240 – 120 = 120

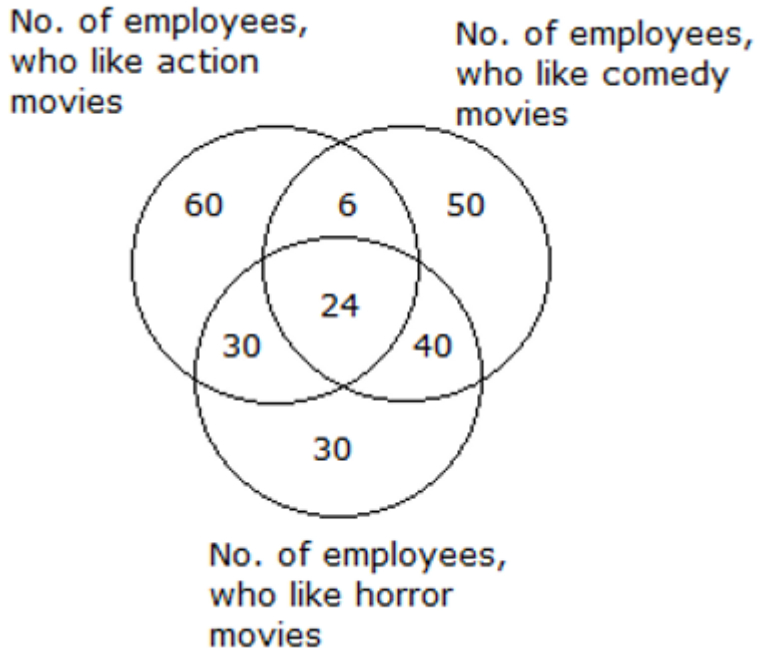
Since, the ratio between the number of employees, who like only comedy movies, those, who like both comedy and horror movies but not action movies and those, who like only horror movies, is 5: 4: 3 respectively

So, the number of employees, who like only comedy movies = 120 * (5/12) = 50

The number of employees, who like both comedy and horror movies but not action movies = 120 * (4/12) = 40

And the number of employees, who like only horror movies = 120 * (3/12) = 30

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Answer: D

The number of male employees, who like only action movies = 60 * (8/15) = 32

The number of female employees, who like only action movies = 60 – 32 = 28

The number of male employees, who like only comedy movies = 50 * (3/5) = 30

The number of female employees, who like only comedy movies = 50 – 30 = 20

The number of male employees, who like only horror movies = 30 * (3/5) = 18

The number of female employees, who like only comedy movies = 30 – 18 = 12

Since, the total number of employees, who like movies of only one genre = 60 + 50 + 30 = 140

And the total number of female employees, who like movies of only one genre = 28 + 20 + 12 = 60

So, the required percentage = (60/140) * 100 = 42.8% (approx)

14. Questions

Let the total number of employees in the company = 4x

So, the number of employees, who like only action movies = 25% of 4x = x

The number of employees, who like both action and comedy movies but not horror movies = x * (1/10) = 0.1x

And the number of employees, who like movies of all the three genres = x * (2/5) = 0.4x

The number of employees, who like both action and horror movies but not comedy movies, is 30

Since, a total of 50% of the total number of employees in the company likes action movies.

So,

x + 0.1x + 0.4x + 30 = 50% of 4x

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1.5x + 30 = 2x

x = 60

Since, the total number of employees in the company = 4 * 60 = 240

So, the number of employees, who like only action movies = 25% of 240 = 60

The number of employees, who like both action and comedy movies but not horror movies = 0.1 * 60 = 6

And the number of employees, who like movies of all the three genres = 0.4 * 60 = 24

So, the total number of employees, who like action movies = 50% of 240 = 120

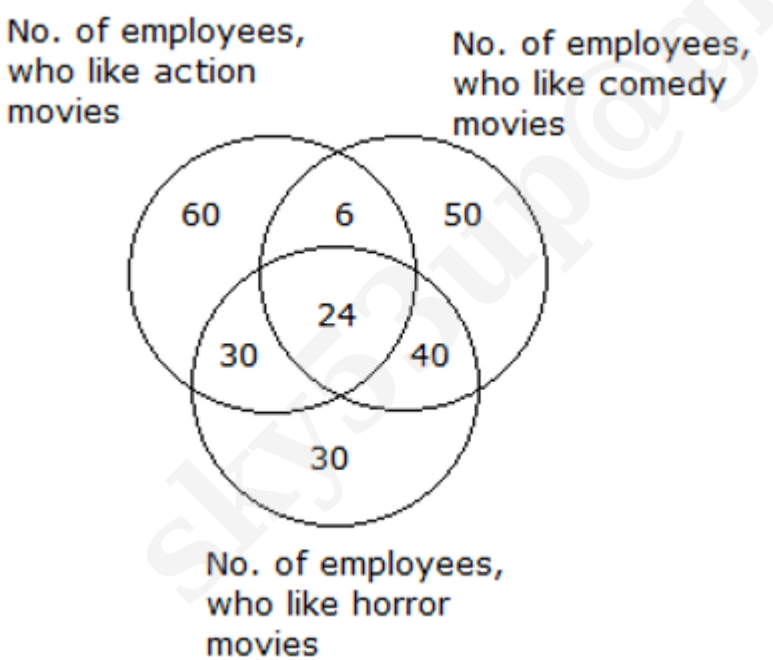
And the total number of employees, who don't like action movies = 240 – 120 = 120

Since, the ratio between the number of employees, who like only comedy movies, those, who like both comedy and horror movies but not action movies and those, who like only horror movies, is 5: 4: 3 respectively

So, the number of employees, who like only comedy movies = 120 * (5/12) = 50

The number of employees, who like both comedy and horror movies but not action movies = 120 * (4/12) = 40

And the number of employees, who like only horror movies = 120 * (3/12) = 30



Answer: B

Since, the total number of employees in the company = 240

And the total number of employees, who like horror movies:

30 + 30 + 40 + 24 = 124

So, the required percentage = (124/240) * 100 = 51.67%

15. Questions

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Let the total number of employees in the company = 4x

So, the number of employees, who like only action movies = 25% of 4x = x

The number of employees, who like both action and comedy movies but not horror movies = x * (1/10) = 0.1x

And the number of employees, who like movies of all the three genres = x * (2/5) = 0.4x

The number of employees, who like both action and horror movies but not comedy movies, is 30

Since, a total of 50% of the total number of employees in the company likes action movies.

So,

x + 0.1x + 0.4x + 30 = 50% of 4x

1.5x + 30 = 2x

x = 60

Since, the total number of employees in the company = 4 * 60 = 240

So, the number of employees, who like only action movies = 25% of 240 = 60

The number of employees, who like both action and comedy movies but not horror movies = 0.1 * 60 = 6

And the number of employees, who like movies of all the three genres = 0.4 * 60 = 24

So, the total number of employees, who like action movies = 50% of 240 = 120

And the total number of employees, who don't like action movies = 240 – 120 = 120

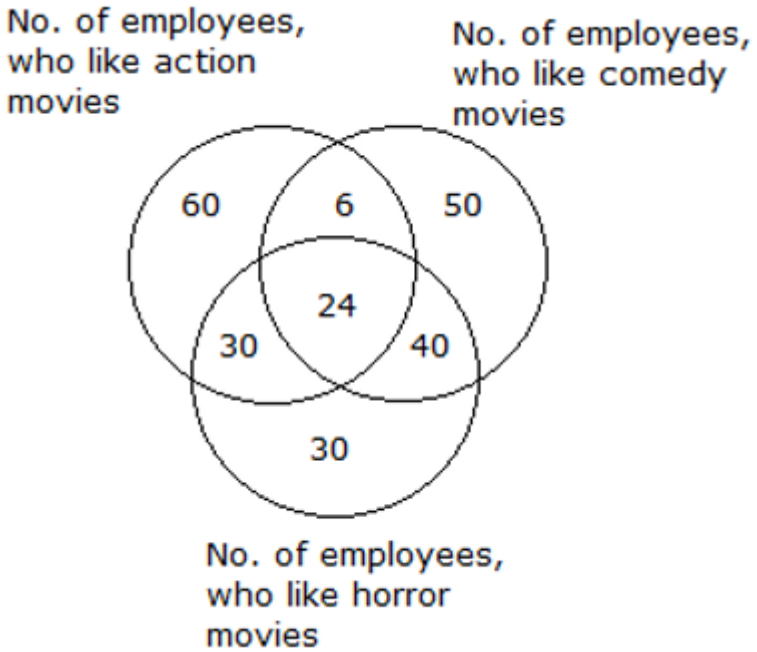
Since, the ratio between the number of employees, who like only comedy movies, those, who like both comedy and horror movies but not action movies and those, who like only horror movies, is 5: 4: 3 respectively

So, the number of employees, who like only comedy movies = 120 * (5/12) = 50

The number of employees, who like both comedy and horror movies but not action movies = 120 * (4/12) = 40

And the number of employees, who like only horror movies = 120 * (3/12) = 30

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Answer: A

Since, the total number of employees, who like movies of only two genres:

6 + 40 + 30 = 76

So, the total number of female employees, who like movies of only two genres:

76 * (11/19) = 44

Since, the total number of employees in the company = 240

And the total number of female employees in the company = 240 * (1/3) = 80

So, the required percentage = (44/80) * 100 = 55%

16. Questions

Let the total number of employees in the company = 4x

So, the number of employees, who like only action movies = 25% of 4x = x

The number of employees, who like both action and comedy movies but not horror movies = x * (1/10) = 0.1x

And the number of employees, who like movies of all the three genres = x * (2/5) = 0.4x

The number of employees, who like both action and horror movies but not comedy movies, is 30

Since, a total of 50% of the total number of employees in the company likes action movies.

So,

x + 0.1x + 0.4x + 30 = 50% of 4x

1.5x + 30 = 2x

x = 60

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Since, the total number of employees in the company = 4 * 60 = 240

So, the number of employees, who like only action movies = 25% of 240 = 60

The number of employees, who like both action and comedy movies but not horror movies = 0.1 * 60 = 6

And the number of employees, who like movies of all the three genres = 0.4 * 60 = 24

So, the total number of employees, who like action movies = 50% of 240 = 120

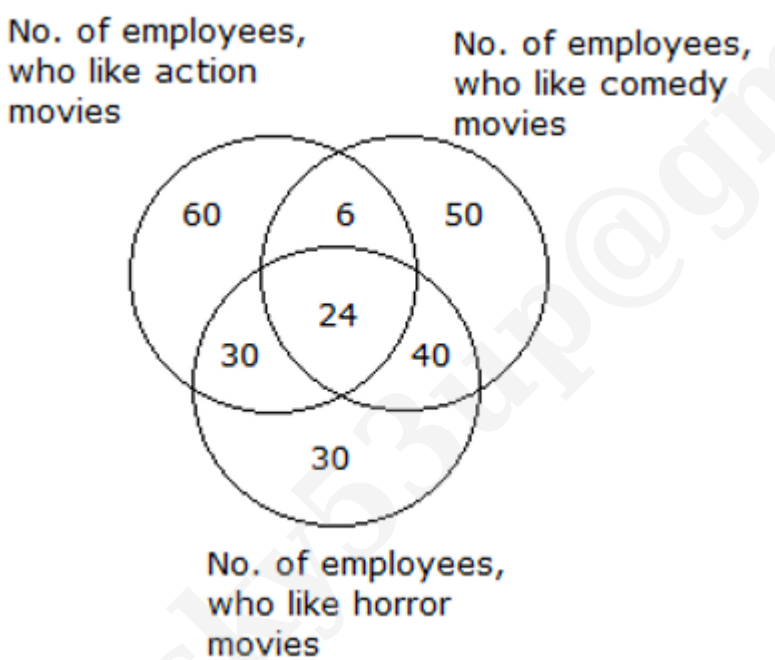
And the total number of employees, who don't like action movies = 240 – 120 = 120

Since, the ratio between the number of employees, who like only comedy movies, those, who like both comedy and horror movies but not action movies and those, who like only horror movies, is 5: 4: 3 respectively

So, the number of employees, who like only comedy movies = 120 * (5/12) = 50

The number of employees, who like both comedy and horror movies but not action movies = 120 * (4/12) = 40

And the number of employees, who like only horror movies = 120 * (3/12) = 30



Answer: E

The total number of employees, who like movies of only one genre:

60 + 50 + 30 = 140

The total number of employees, who like movies of only two genres:

6 + 30 + 40 = 76

The required ratio = 140: 76 = 35: 19

17. Questions

Let the total number of employees in the company = 4x

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So, the number of employees, who like only action movies = 25% of 4x = x

The number of employees, who like both action and comedy movies but not horror movies = x * (1/10) = 0.1x

And the number of employees, who like movies of all the three genres = x * (2/5) = 0.4x

The number of employees, who like both action and horror movies but not comedy movies, is 30

Since, a total of 50% of the total number of employees in the company likes action movies.

So,

x + 0.1x + 0.4x + 30 = 50% of 4x

1.5x + 30 = 2x

x = 60

Since, the total number of employees in the company = 4 * 60 = 240

So, the number of employees, who like only action movies = 25% of 240 = 60

The number of employees, who like both action and comedy movies but not horror movies = 0.1 * 60 = 6

And the number of employees, who like movies of all the three genres = 0.4 * 60 = 24

So, the total number of employees, who like action movies = 50% of 240 = 120

And the total number of employees, who don't like action movies = 240 – 120 = 120

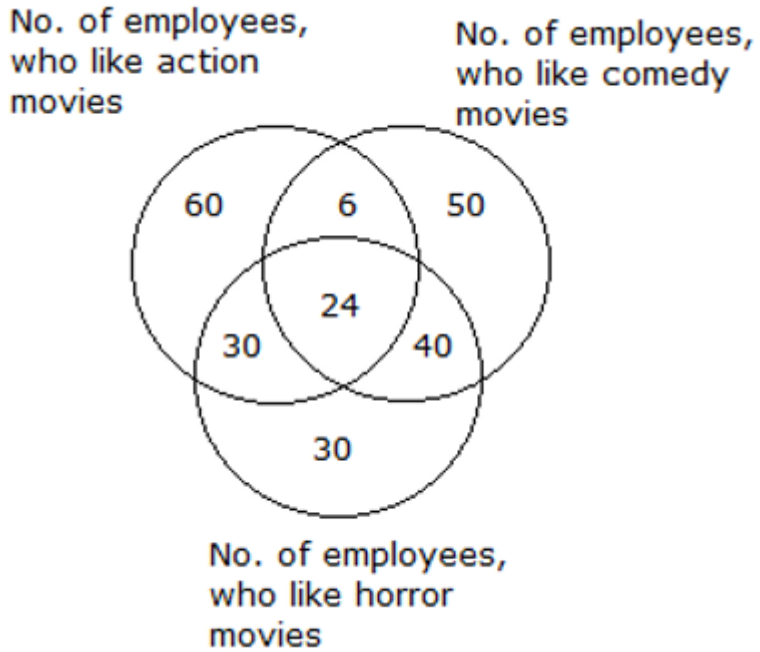
Since, the ratio between the number of employees, who like only comedy movies, those, who like both comedy and horror movies but not action movies and those, who like only horror movies, is 5: 4: 3 respectively

So, the number of employees, who like only comedy movies = 120 * (5/12) = 50

The number of employees, who like both comedy and horror movies but not action movies = 120 * (4/12) = 40

And the number of employees, who like only horror movies = 120 * (3/12) = 30

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Answer: D

The ratio between the number of employees, who like only action movies, those, who like only comedy movies and those, who like only horror movies:

P: Q: R = 60: 50: 30 = 6: 5: 3

From X:

Value of $(P^2 – Q^2) = (36 – 25) = 11$

Since, the value of R = 3

And 11 is not divisible by 3.

So, X is not true.

From Y:

The value of (P * Q * R) = 6 * 5 * 3 = 90

Since, 90 > 45

So, Y is not true.

From Z:

Since, the value of $(P^2 + Q^2 + R^2) = (36 + 25 + 9) = 70$

And 70 is not divisible by 9.

So, Z is not true.

Hence, none is true.

18. Questions

Series I:

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$298 - 9^2 = 217$

$217 - 8^2 = 153$

$153 - 7^2 = 104$

$104 - 6^2 = 68$

$68 - 5^2 = 43$

Series II:

$341 - 9^2 = 260$

$260 - 8^2 = 196$

$196 - 7^2 = 147$

$147 - 6^2 = 111$

$111 - 5^2 = 86$

Series III:

$325 - 9^2 = 244$

$244 - 8^2 = 180$

$180 - 7^2 = 131$

Since, P + Q = 244

And, P – Q = 180

So, P = (244 + 180)/2 = 212

And, Q = (244 – 180)/2 = 32

Answer: D

The 6th term of series I = 43

The 6th term of series II = 86

So, the required percentage = (43/86) * 100 = 50%

19. Questions

Series I:

$298 - 9^2 = 217$

$217 - 8^2 = 153$

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$153 - 7^2 = 104$

$104 - 6^2 = 68$

$68 - 5^2 = 43$

Series II:

$341 - 9^2 = 260$

$260 - 8^2 = 196$

$196 - 7^2 = 147$

$147 - 6^2 = 111$

$111 - 5^2 = 86$

Series III:

$325 - 9^2 = 244$

$244 - 8^2 = 180$

$180 - 7^2 = 131$

Since, P + Q = 244

And, P – Q = 180

So, P = (244 + 180)/2 = 212

And, Q = (244 – 180)/2 = 32

Answer: B

The 2nd term of series I = 217

The 3rd term of series I = 153

The required average = (217 + 153)/2 = 185 = (P – Q + 5).

20. Questions

Series I:

$298 - 9^2 = 217$

$217 - 8^2 = 153$

$153 - 7^2 = 104$

$104 - 6^2 = 68$

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$68 - 5^2 = 43$

Series II:

$341 - 9^2 = 260$

$260 - 8^2 = 196$

$196 - 7^2 = 147$

$147 - 6^2 = 111$

$111 - 5^2 = 86$

Series III:

$325 - 9^2 = 244$

$244 - 8^2 = 180$

$180 - 7^2 = 131$

Since, P + Q = 244

And, P – Q = 180

So, P = (244 + 180)/2 = 212

And, Q = (244 – 180)/2 = 32

Answer: C

From X:

The difference between the 4th and 5th terms of series II:

147 – 111 = 36 = (Q + 4)

So, X is true.

From Y:

The 5th term of series III = $131 - 6^2 = 95$

Since, the value of [(P/2) – 10] = [(212/2) – 10] = 96

So, Y is not true.

From Z:

Since, Q is (32/212) * 100 = 15.09% (approx.) of P.

So, Z is true.

Hence, only X and Z are true.

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1. Questions

Study the following information carefully and answer the questions given below.

The given table below shows the number of male visitors in five different malls likely i.e., A, B, C, D and E. And also some information about the number of females and children visiting the malls.

Malls	Number of males	Information
A	Square of 16.	The number of females is 25% less than the number of males, and the number of children is 1/8th of the number of males and females together.
B	24% of 576 +116.76	The number of females is 20% more than the number of males, and the number of children is 1/6th of the number of females.
C	Difference of squares of 29 and 23.	The average of males and females is 315 and the number of children visiting the mall is a square of 8.
D	280	The number of children visiting the mall is 60% less than the number of males. The number of females in D is missing the number of the below series. 45, 81,145, x, 389
E	320	The ratio of males and children visiting the mall is 4:1, and the number of females visiting the mall is 360% higher than the number of children visiting the mall.

Out of the total number of males who visit mall A, 87.5% are married. The number of married females visits in A is 350% more than the number of unmarried males in A. Find the difference between the total number of unmarried males and females together in A and the total number of children in A.

a. 15

b. 16

c. 24

d. 21

e. 18

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2. Questions

Out of the total number of male visits in B, 20% come to watch movies, and 25% of the rest come for shopping. The number of females visiting B for shopping is three times the number of males visiting B for shopping. Out of the total number of females who visit for shopping, 66.66% buy shoes in B. Find the number of females who visit B to buy shoes.

a. 109

b. 91

c. 108

d. 102

e. 112

3. Questions

Find which of the following is true.

I). The average number of children visiting D and E together is 93.

II). The total number of people visiting C is 694.

III). The total number of children in all malls together is 444.

a. All true

b. All false

c. Only II true

d. Only II and III true

e. None of these

4. Questions

The number of male visits in F is 20% more than the number of male visits in E and the number of females visiting F is 25% more than the number of females visiting E. The total number of people visiting F is 948, find the difference between the number of children in E and F.

a. 26

b. 22

c. 29

d. 24

e. None of these

5. Questions

The number of male visits in E is r% more than the number of male visits in A. The number of children in D is x% more than the number of children in C. If the number of children in G is [r+x]% more than the number of children in C, then find the number of children in G.

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a. 128

b. 142

c. 124

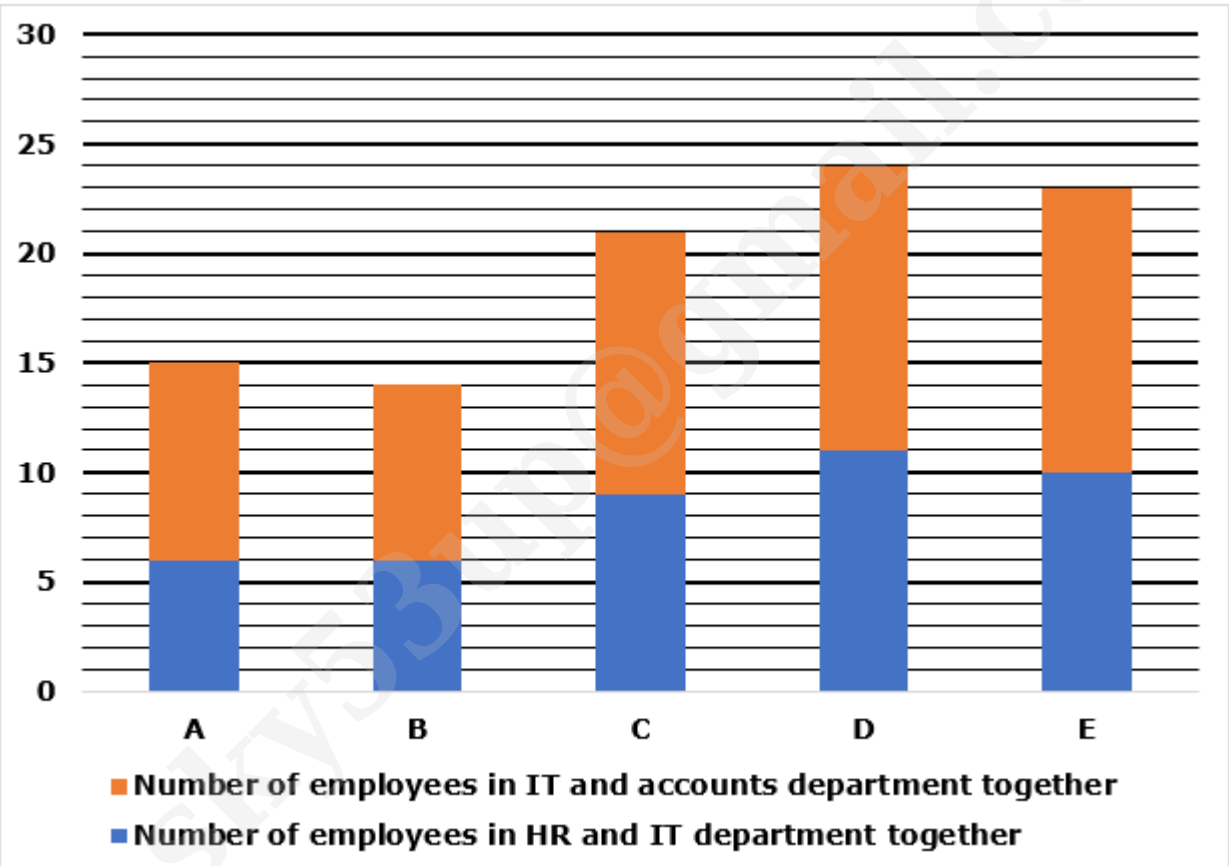
d. 162

e. None of these

6. Questions

Study the following information carefully and answer the questions given below.

The given bar graph shows the ratio of the number of employees in the HR and IT departments together and the number of employees in the IT and accounts departments together of five different companies [A, B, C, D and E].



If the number of employees in the HR department of company A is 20. The number of employees in the accounts department of the same company is 150% higher than the number of employees in the HR department of company A. The ratio of male and female employees in company A in each department is 5:3. The average weight of male and female employees in the IT department is 72 and 62 kg respectively. Find the average weight of all the employees in the IT department.

a. 65.66 kg

b. 69.32 kg

c. 68.25 kg

d. 66.75 kg

e. 67.25 kg

7. Questions

The number of employees in the IT department of company B is 100 more than the number of employees in the HR department of the same company. The number of employees in the accounts department of company B is 50 less than the number of employees in the IT department. The ratio of males and females in each department is 3:2. Find the difference between the total number of males and females in company B.

a. 46

b. 48

c. 55

d. 45

e. 65

8. Questions

In company C, the ratio of the number of employees in the HR and accounts department is 4:7. The number of employees in the IT department is 12 more than the total number of employees in the HR department of company C. Out of the total number of employees in the IT department of C, 40% are married. Then find the number of unmarried employees in the IT department is what percent of the total number of employees is in C?

a. 12.35%

b. 18.75%

c. 16.66%

d. 22.22%

e. None of these

9. Questions

In company D, the number of employees in the IT department is 62.5% higher than the number of employees in the HR department. The number of employees in the accounts department is 18 more than the number of employees in the HR department, and the number of employees in the accounts department is 27 less than the number of employees in the IT department. Find the total number of employees in Company D.

a. 367

b. 254

c. 267

d. 279

e. None of these

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10. Questions

The ratio of the number of employees in the IT and accounts departments of E is 6:7. The total number of employees in the HR and IT departments of E is 110. Find the difference between the number of employees in the IT and HR departments of E.

a. 22

b. 23

c. 29

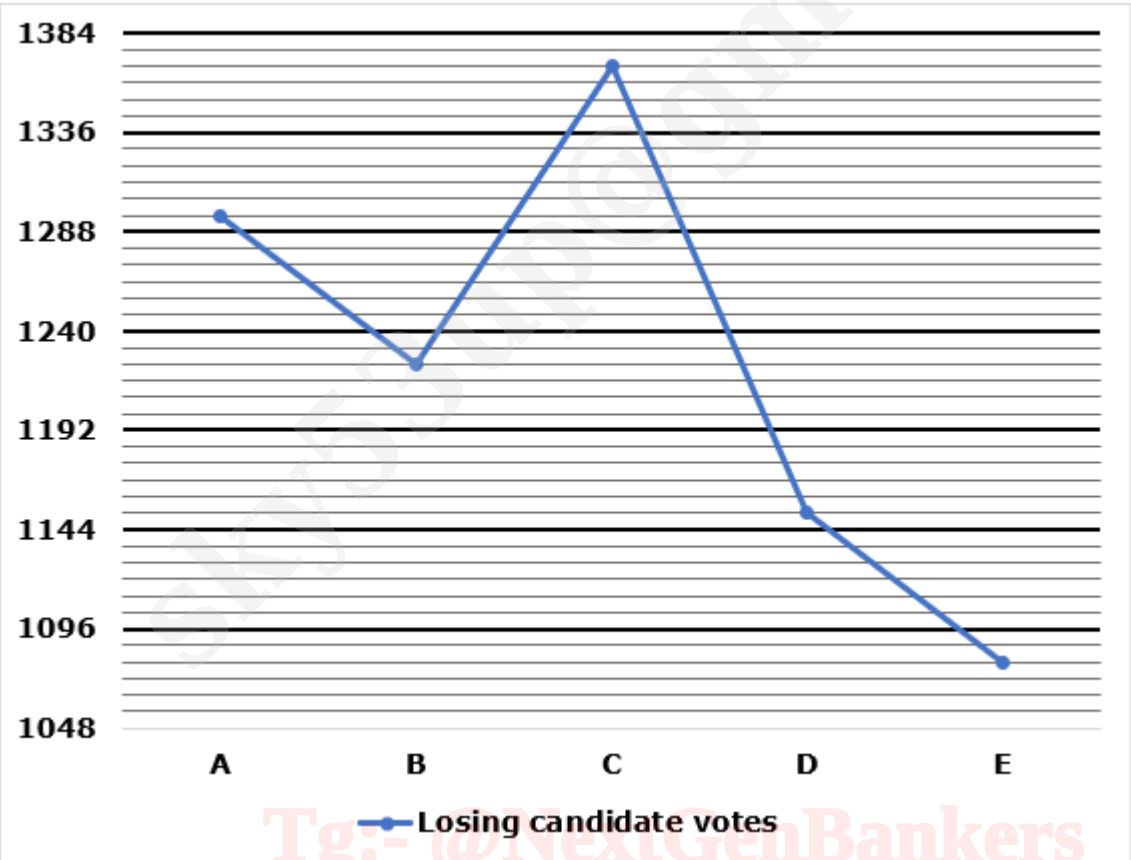
d. 19

e. None of these

11. Questions

Study the following information carefully and answer the questions given below.

The line graph shows the vote of the losing candidate of five different villages [A, B, C, D and E]. The election was conducted in five different villages and only two candidates contested in election in each village.



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Note: – In each village out of the total population, 80% are registered voter, out of the total registered voter 60% cast their votes, and out of the total cast votes 80% are valid votes. The ratio between the number of votes obtained by winning and losing candidates' is 5:3.

The difference between the total population of villages A and B is x and the difference between the invalid votes in D and E is y. Then find the total number of votes received by the winning

candidates in C and D together.

a. 7x+8y+12

b. 8x+4y+8

c. 8x+6y+22

d. 9x+7y+23

e. More than one option is true

12. Questions

The ratio of male to female population in village C is 3:2 and the ratio of male to female population in village D is 5:3. Find the difference between the total number of male population in villages C and D together and the total number of female population in villages C and D together.

a. 3600

b. 3200

c. 3300

d. 3900

e. 4900

13. Questions

The total population of village F is 20% more than the total population of village E. The ratio of males and females in village F is 3:2. Out of the total males 20% are illiterate and out of the total females 25% are illiterate. Find the total literate population in village F.

a. 2580

b. 7020

c. 6580

d. 9560

e. None of these

14. Questions

Find the difference between the total number of votes obtained by the winning and losing candidates in village G.If the total population in village G is 25% less than that of village D.

(**Note:** The number of Register voters, cast votes and valid votes is the same as other villages)

a. 576

b. 956

c. 856

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d. 876

e. None of these

15. Questions

The difference between the total number of votes obtained by the losing and winning candidates' votes of all villages together is x and the total invalid votes of all villages is y. Then find which of the following is true.

a. 3x=2y

b. 2x+3y=9850

c. x=y+1

d. $(3x-2y)/8=(23^2 - 19)$

e. None of these

16. Questions

Study the following information carefully and answer the questions given below.

The height of a cone is 3/2 times the height of a cylinder. The radius of a cylinder and a sphere are the same. The ratio of height and breadth of the cuboid is 11:7. The height of the cylinder is equal to the length of the cuboid. The radius of a cone is equal to the breadth of a cuboid. The ratio of length and height of the cuboid is 9:11. Radius of the sphere is 16.66% more than the length of the cuboid. The volume of cuboid is 5544 m^3.

Find which of the following is true.

I). The total surface area of the cylinder is 5148 m^2.

II). The volume of the sphere is 38608 m^3.

III). The volume of cone is 5544 m^3.

a. Only I true

b. Only I and II true

c. Only I and III true

d. Only II true

e. Only III true

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17. Questions

The ratio of the radius of a given cylinder and another cylinder is 3:4 and the ratio of the height of the given cylinder and another cylinder is 3:7, find the difference between the volume of the two cylinders.

a. 76540

b. 79820

c. 79640

d. 78540

e. 77640

18. Questions

The length of the rectangle is _____ more than that of the cuboid and the breadth of the rectangle is ____ more than that of the cuboid. The area of the rectangle is _____.

Find which of the following is true.

a. 7, 3, 665

b. 4, 3, 374

c. 5, 2, 356

d. 7, 5, 445

e. more than one option is true

19. Questions

A boat covers 182 km downstream and 108 km upstream in 13 hours. The same boat covers 130 km downstream and 72 km upstream in 9 hours. The boat covered d km distance in 4 hours of still water, find the total distance traveled by boat in 3 hours downstream and 2 hours upstream.

a. d+26

b. 3d-124

c. 0.5d+85

d. 2d-61

e. More than one option is true

20. Questions

A, B and C start a business with Rs. 12000, Rs.16000 and Rs.18000 respectively. After 4 months A invests Rs.6000 more and B invests Rs.2000 more. After 2 months later C withdraws Rs.4000. After a year total profit is Rs.20350. With profit share B buy a watch and sell at (r+15)% profit. If the selling price is Rs.8580. Then find which of the following is the sum of the share of profit of A and C.

a. 2150r

b. 2340r

c. 2190r

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d. 2640r

e. None of these

Explanations:

1. Questions

Malls	Number of males	Number of females and children
A	Square of 16= 16*16=256	The number of females is 256*75/100=192. The number of children is 1/8*[256+192]=56
B	24% of 576 +116.76 =24*576/100+116.76 =255	The number of females is 255*120/100=306 The number of children is 1/6*306=51
C	Difference of squares of 29 and 23 =841-529=312	The number of females is 315*2-312=318. The number of children visiting the mall is 8*8=64
D	280	The number of children visiting the mall =280*40/100=112. 4581145245389 +36 +64 +100 +144 The number of females visiting Mall D is 245.
E	320	The number of children visiting the mall is 320/4=80. The number of females visiting the mall is 80*460/100=368

Answer: C

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Required sum,

Total number of unmarried males = 256 × (12.5/100)=32

Total number of unmarried females = 192-(32*450/100) =48

So, x=[32+48]-56=24.

Hence option C.

2. Questions

Malls	Number of males	Number of females and children
A	Square of 16= 16*16=256	The number of females is 256*75/100=192. The number of children is 1/8*[256+192]=56
B	24% of 576 +116.76 =24*576/100+116.76 =255	The number of females is 255*120/100=306 The number of children is 1/6*306=51
C	Difference of squares of 29 and 23 =841-529=312	The number of females is 315*2-312=318. The number of children visiting the mall is 8*8=64
D	280	The number of children visiting the mall =280*40/100=112. 4581145245389 +36 +64 +100 +144 The number of females visiting Mall D is 245.
E	320	The number of children visiting the mall is 320/4=80. The number of females visiting the mall is 80*460/100=368

Answer: D

The number of males visiting B for shopping [255*80/100]*25/100=51

The number of females visiting B for shopping 51*3=153

The number of females visiting B to buy shoes 153*66.66/100=102

Hence option D

3. Questions

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Malls	Number of males	Number of females and children
A	Square of 16= 16*16=256	The number of females is 256*75/100=192. The number of children is 1/8*[256+192]=56
B	24% of 576 +116.76 =24*576/100+116.76 =255	The number of females is 255*120/100=306 The number of children is 1/6*306=51
C	Difference of squares of 29 and 23 =841-529=312	The number of females is 315*2-312=318. The number of children visiting the mall is 8*8=64
D	280	The number of children visiting the mall =280*40/100=112. 4581145245389 +36 +64 +100 +144 The number of females visiting Mall D is 245.
E	320	The number of children visiting the mall is 320/4=80. The number of females visiting the mall is 80*460/100=368

Answer: C

I). The average number of children visiting D and E together is 93

=[112+80]/2=96 (False)

II). The total number of people visiting C is 694

= 312+318+64=694 (True)

III). The total number of children in all malls together is 444

56+51+64+112+80=363 (False)

Hence option C

4. Questions

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Malls	Number of males	Number of females and children
A	Square of 16= 16*16=256	The number of females is 256*75/100=192. The number of children is 1/8*[256+192]=56
B	24% of 576 +116.76 =24*576/100+116.76 =255	The number of females is 255*120/100=306 The number of children is 1/6*306=51
C	Difference of squares of 29 and 23 =841-529=312	The number of females is 315*2-312=318. The number of children visiting the mall is 8*8=64
D	280	The number of children visiting the mall =280*40/100=112. 4581145245389 +36 +64 +100 +144 The number of females visiting Mall D is 245.
E	320	The number of children visiting the mall is 320/4=80. The number of females visiting the mall is 80*460/100=368

Answer: D

The number of children visiting F is = 948-[320*120/100] - [368*125/100]=948-384-460=104

So, the required difference = 104-80=24

Hence option D

5. Questions

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Malls	Number of males	Number of females and children
A	Square of 16= 16*16=256	The number of females is 256*75/100=192. The number of children is 1/8*[256+192]=56
B	24% of 576 +116.76 =24*576/100+116.76 =255	The number of females is 255*120/100=306 The number of children is 1/6*306=51
C	Difference of squares of 29 and 23 =841-529=312	The number of females is 315*2-312=318. The number of children visiting the mall is 8*8=64
D	280	The number of children visiting the mall =280*40/100=112. 4581145245389 +36 +64 +100 +144 The number of females visiting Mall D is 245.
E	320	The number of children visiting the mall is 320/4=80. The number of females visiting the mall is 80*460/100=368

Answer: A

So, r=[320-256]×100/256=25

So, x=[112-64]×100/64=75

So, the number of children visiting in G is = 64×(200/100)=128

Hence option A.

6. Questions

Answer: C

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Number of employees in HR department of company A is 20.

The number of employees in the accounts department of company A is 20*250/100=50

Let the number of employees in the IT department of company A be x.

So, [x+20]/[50+x]=6/9

Or, 9x+180=300+6x

Or, 3x=120,

x=[120/3]=40

So, the number of males in A = 40*5/8=25

the number of females in A = 40-25=15

So, Required average age = [25*72+15*62]/40=2730/40=68.25 kg

Hence option C

7. Questions

Answer: D

Let the number of employees in the IT department of company B be x.

The number of employees in the HR department of company B is x-100.

The number of employees in the Accounts department of company B is x-50.

So, [x+x-100]/[x+x-50]=6/8

Or, 16x-800=12x-300

Or, 4x=500, x=500/4=125

The number of employees in the IT department is 125,

The number of employees in the HR department is 125-100=25

The number of employees in the accounts department is 125-50=75

So, the required difference

= [125+25+75]×(3/5)-[125+25+75]×(2/5)=135-90=45

Hence option D

8. Questions

Answer: B

Let the number of employees in the HR and accounts departments be 4x and 7x respectively.

So, the Number of employeesin the IT department is 4x+12.

So, [4x+4x+12]/[4x+12+7x]=9/12

Or, 96x+144=99x+108

Or, 3x=36

Or, x=[36/3]=12

The number of employees in the IT department is 4x+12=4*12+12=60

So, the number of unmarried employees in the IT department is = 60*60/100=36

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So, required percentage =[36/(48+60+84)]*100=18.75%

Hence option B

9. Questions

Answer: D

Number of employees in IT department is 62.5% more than the number of employees in HR department.

So, Let the number of employees in the HR department =8x

The number of employees in the IT department =8x*162.5/100=13x.

The number of employees in the accounts department is 8x+18.

The number of employees in the accounts department is 13x-27

So, 8x+18=13x-27

Or, 5x=45, x=9

So, the total number of employees in D is = (29×9)+18=279

Hence option D

10. Questions

Answer: A

The Total number of employees in the IT and accounts department is =110*13/10=143

The Number of employees in the IT department is 143*6/13=66

The Number of employees in the HR department is 110-66=44

So, difference = 66-44=22

Hence option A.

11. Questions

Village A,

The total number of votes received by thelosing candidates = 1296

The total number of votes received by the winning candidates = 1296*5/3=2160

So, the total number of valid votes = 2160+1296=3456

The total cast vote is 3456*100/80=4320

The total number of registered voters is 4320*100/60=7200

The total number of populations in the village is 7200*100/80=9000

Similarly, we can calculate others' values also.

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Village	Losing candidates votes	Winning candidates votes	Total valid votes	Total invalid votes	Total cast votes	Total register voters	Total population
A	1296	2160	3456	864	4320	7200	9000
B	1224	2040	3264	816	4080	6800	8500
C	1368	2280	3648	912	4560	7600	9500
D	1152	1920	3072	768	3840	6400	8000
E	1080	1800	2880	720	3600	6000	7500

Answer: B

Required sum,

The difference between the total population of villages A and B,

x = 9000-8500=500,

The difference between the invalid votes in D and E,

y=768-720=48

The total number of votes received by the winning candidates in C and D together= 2280+1920=4200

Option A:

x = 500, y = 48

7x+8y+12 = 4200

3500+384+12 = 4200

3896 = 4200 (False)

Option B:

x = 500, y = 48

8x+4y+8 = 4200

4000+192+8 = 4200

4200 = 4200 (True)

Option C:

x = 500, y = 48

8x+6y+22 = 4200

4000+288+22 = 4200

4310 = 4200 (False)

Option D:

x = 500, y = 48

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9x+7y+23 = 4200

4500+336+23 = 4200

4859 = 4200 (False)

Hence option B.

12. Questions

Village A,

The total number of votes received by thelosing candidates = 1296

The total number of votes received by the winning candidates = 1296*5/3=2160

So, the total number of valid votes = 2160+1296=3456

The total cast vote is 3456*100/80=4320

The total number of registered voters is 4320*100/60=7200

The total number of populations in the village is 7200*100/80=9000

Similarly, we can calculate others' values also.

Village	Losing candidates votes	Winning candidates votes	Total valid votes	Total invalid votes	Total cast votes	Total register voters	Total population
A	1296	2160	3456	864	4320	7200	9000
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C	1368	2280	3648	912	4560	7600	9500
D	1152	1920	3072	768	3840	6400	8000
E	1080	1800	2880	720	3600	6000	7500

Answer: D

Required difference is

= [9500*3/5+8000*5/8]-[9500*2/5+8000*3/8] =10700-6800=3900

Hence, option D.

13. Questions

Village A,

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The total number of votes received by thelosing candidates = 1296

The total number of votes received by the winning candidates = 1296*5/3=2160

So, the total number of valid votes = 2160+1296=3456

The total cast vote is 3456*100/80=4320

The total number of registered voters is 4320*100/60=7200

The total number of populations in the village is 7200*100/80=9000

Similarly, we can calculate others' values also.

Village	Losing candidates votes	Winning candidates votes	Total valid votes	Total invalid votes	Total cast votes	Total register voters	Total population
A	1296	2160	3456	864	4320	7200	9000
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C	1368	2280	3648	912	4560	7600	9500
D	1152	1920	3072	768	3840	6400	8000
E	1080	1800	2880	720	3600	6000	7500

Answer: B

The total population in village F is 7500*120/100=9000

The total number of literate Population in the village

=[9000×(3/5)]×(80/100)+[9000*2/5]*75/100 = 4320+2700=7020

Hence option B.

14. Questions

Village A,

The total number of votes received by thelosing candidates = 1296

The total number of votes received by the winning candidates = 1296*5/3=2160

So, the total number of valid votes = 2160+1296=3456

The total cast vote is 3456*100/80=4320

The total number of registered voters is 4320*100/60=7200

The total number of populations in the village is 7200*100/80=9000

Similarly, we can calculate others' values also.

Village	Losing candidates votes	Winning candidates votes	Total valid votes	Total invalid votes	Total cast votes	Total register voters	Total population
A	1296	2160	3456	864	4320	7200	9000
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C	1368	2280	3648	912	4560	7600	9500
D	1152	1920	3072	768	3840	6400	8000
E	1080	1800	2880	720	3600	6000	7500

Answer: A

The Total population in village G is =8000*75/100=6000

So, the required difference = 6000×[80/100]×[60/100]×[80/100]×[2/8] = 576

Hence, option A

15. Questions

Village A,

The total number of votes received by thelosing candidates = 1296

The total number of votes received by the winning candidates = 1296*5/3=2160

So, the total number of valid votes = 2160+1296=3456

The total cast vote is 3456*100/80=4320

The total number of registered voters is 4320*100/60=7200

The total number of populations in the village is 7200*100/80=9000

Similarly, we can calculate others' values also.

Village	Losing candidates votes	Winning candidates votes	Total valid votes	Total invalid votes	Total cast votes	Total register voters	Total population
A	1296	2160	3456	864	4320	7200	9000
B	1224	2040	3264	816	4080	6800	8500
C	1368	2280	3648	912	4560	7600	9500
D	1152	1920	3072	768	3840	6400	8000
E	1080	1800	2880	720	3600	6000	7500

Answer: D

Required difference

x=[2160+2040+2280+1920+1800]-[1296+1224+1368+1152+1080]

x =10200-6120=4080

So, y =864+816+912+768+720=4080

Option A:

3*4080 = 2*4080 (False)

Option B:

2*4080+3*4080 = 9850

20400 = 9850 (False)

Option C:

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4080 = 4080+1

4080=4081 (False)

Option D:

[(3*4080)-(2*4080)]/8 = (23*23) – 19

(12240 – 8160)/8 = 529 – 19

4080/8 = 510

4080 = 4080 (True)

Hence option D.

16. Questions

Let the length and height of the cuboid be 9x and 11x respectively.

The breadth of the cuboid is 7x.

So, [9x*11x*7x]=5544

Or, $693x^3=5544$

Or, $x^3=[5544/693]=8$

So, x=2

Length and height of the cuboid are 9*2=18 and 11*2=22 respectively.

The breadth of the cuboid is 7*2=14.

The height of the cylinder is 18 m.

The height of the cone is 18*3/2=27.

Radius of sphere is = 18*116.66/100=21

The radius of the cylinder is = 21

The radius of the cone is = 14

Answer: C

According to the question,

I). Total surface area of a cylinder= $2\pi r(r+h) = 2*(22/7)*21*(21+18)=5148\ m^2$.(True)

II). The volume of the sphere = $(4/3)\pi r^3 = [4/3]*[22/7]*[21*21*21]=38808\ m^3$.(False)

III). The volume of a cone = $(1/3)\pi r^2h = [1/3]*[22/7]*14*14*27=5544\ m^3$.(True)

Hence option C.

17. Questions

Let the length and height of the cuboid be 9x and 11x respectively.

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The breadth of the cuboid is 7x.

So, [9x*11x*7x]=5544

Or, $693x^3=5544$

Or, $x^3=[5544/693]=8$

So, x=2

Length and height of the cuboid are 9*2=18 and 11*2=22 respectively.

The breadth of the cuboid is 7*2=14.

The height of the cylinder is 18 m.

The height of the cone is 18*3/2=27.

Radius of sphere is = 18*116.66/100=21

The radius of the cylinder is = 21

The radius of the cone is = 14

Answer: D

The radius of the given cylinder is = 21

The height of the given cylinder is = 18

The radius of the new cylinder is = 21*4/3=28

The height of the new cylinder is = 18*7/3=42

The volume of the cylinder = $\pi r^2 h$

The volume of the new cylinder= [22/7]*[28*28]*42=103488

The volume of the given cylinder=[22/7]*[21*21]*18=24948

So, difference=103488-24948=78540

Hence, Option D

18. Questions

Let the length and height of the cuboid be 9x and 11x respectively.

The breadth of the cuboid is 7x.

So, [9x*11x*7x]=5544

Or, $693x^3=5544$

Or, $x^3=[5544/693]=8$

So, x=2

Length and height of the cuboid are 9*2=18 and 11*2=22 respectively.

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The breadth of the cuboid is 7*2=14.

The height of the cylinder is 18 m.

The height of the cone is 18*3/2=27.

Radius of sphere is = 18*116.66/100=21

The radius of the cylinder is = 21

The radius of the cone is = 14

Answer: B

From the Option A,

The length of a rectangle is = [18+7]=25

The breadth of a rectangle is [14+3]=17

So, the area of the rectangle = 25*17=425 m^2 (False)

From the Option B,

The length of a rectangle = [18+4]=22

The breadth of a rectangle = [14+3]=17

So, the area of the rectangle = 22*17=374 m^2 (True)

For the Option C,

The length of a rectangle = [18+5]=23

The breadth of a rectangle = [14+2]=16

So, the area of the rectangle = 23*16=368 m^2 (False)

For the Option D,

The length of a rectangle is = [18+7]=25

The breadth of a rectangle is [14+5]=19

So, the area of the rectangle = 25*19=475 m^2(False)

Hence, option B

19. Questions

Answer: A

Let,The speed of the boat = x km/hr

The speed of the stream = y km/hr.

So, {182/[x+y] + 108/[x-y]}=13 ----(i)

And also, {130/[x+y] + 72/[x-y]}=9 ---- (ii)

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By solving the above equation, x=22 and y=4.

So, d=22*4=88

So, The total distance cover by boat =[22+4]*3+[22-4]*2 = 78+36=114.

114 = d+26.

Hence, option A.

20. Questions

Answer: D

Investment ratio of A, B, and C

=[12000*12+6000*8]:[16000*12+2000*8]:[18000*12-4000*6]

=192000:208000:192000=12:13:12.

Profit share of B is = 20350*13/[37]=7150

So, profit percentage of watch =[8580-7150]*100/7150=20%

So, [r+15]=20, r=20-15=5

The sum of the share of A and C is 20350*24/37=13200=2640r

Hence option D.

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1. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the number of passengers (Sleeper + AC class) in four different trains – P, Q, R and S

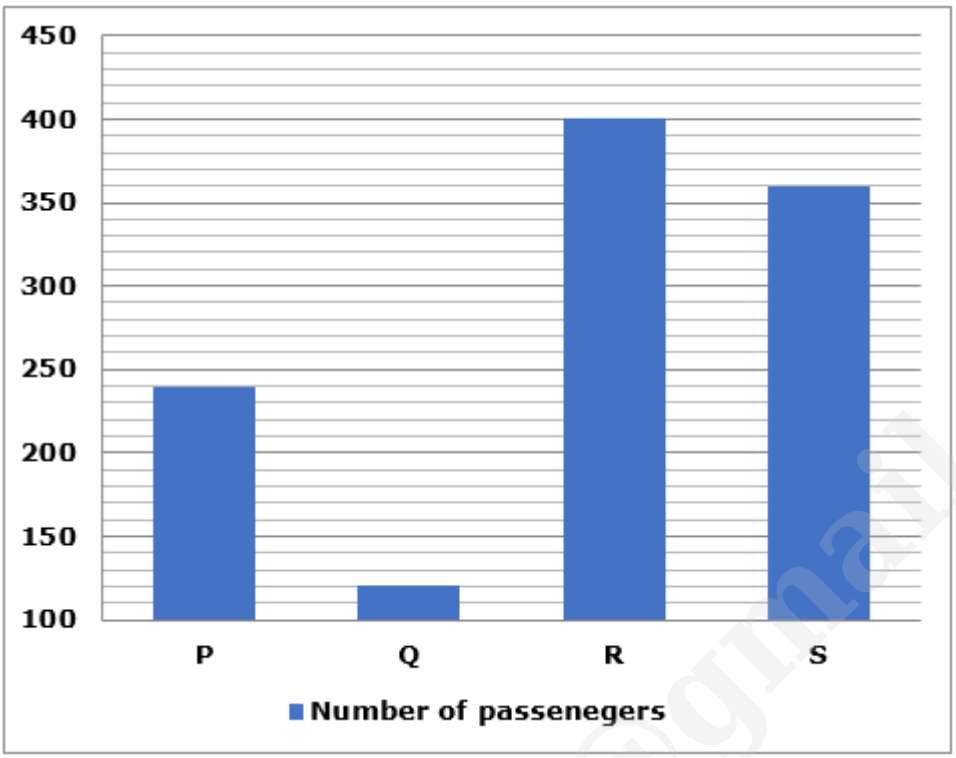


Table given below shows the % of sleeper class passengers in a train out of total passengers in that train, and number of female passengers in AC class.

Name of train	% of Sleeper class passengers	Number of females in AC class.
P	Z%	(Y + 16)
Q	33.33%	4Y/5
R	(Z – 20)%	2.5Z
S	-	84

Note: Total number of female passengers in AC class is 286, and difference between Z and Y (Z > Y) is 40.

For S, the number of passengers in sleeper class and in AC class is same. Out of total male passengers, 1/3rd is going to Delhi rest is going to Bihar. Find the number of male passengers going to Bihar, if the ratio of male and female passengers in sleeper class is Z: (Y + 20) respectively.

a. 123

b. 132

c. 136

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d. 108

e. None of these

2. Questions

For train P, 25% of sleeper class passenger and 75% of AC class passengers booked their tickets online rest offline. Total number of passengers booked their tickets online is (M + 22)% of male passengers in AC class of train P and R together. Find difference between Y and M.

a. 30

b. 20

c. 10

d. 40

e. None of these

3. Questions

The difference between Total number of male and female in AC class of all trains together is 1.8Y. Number of total female passenger sleeper class in train R is 82.

A = Number of male passenger in train R

B = Total number of passengers in sleeper class of train S.

a. A > B

b. A < B

c. A = B or No relation

d. A ≤ B

e. A ≥ B

4. Questions

For train Q, respective ratio of male passengers in AC class and females in sleeper class is 7:16 respectively. Out of total male passengers in sleeper class, 25% are aged > 30 years. Number of male passengers in sleeper class whose age is exactly 30 years is M% of number of male passengers in AC class of train P. Find maximum possible value of M.

a. 16.66%

b. 15%

c. 12.5%

d. 20%

e. None of these

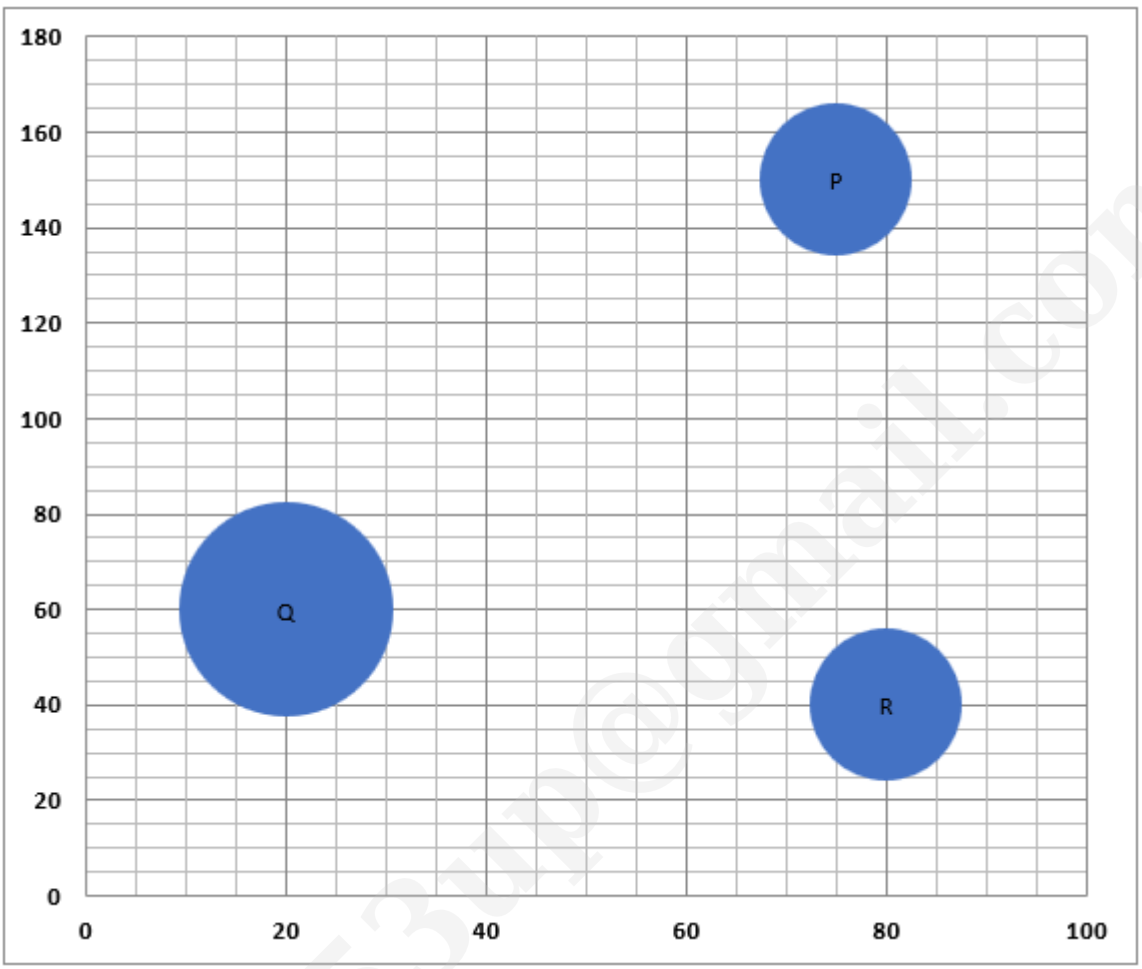
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5. Questions

Read the following information carefully and answer the questions based on it.

There are three companies – P, Q, and R. Each company has three departments – A, B, and C.

The chart given below shows the % of employees in department A (X axis) out of total employees in that company, and % of employees in department B out of employees in department C (Y axis).



The table given below shows the difference between number of employees in department A and C of each company.

Name of company	Difference between students in department A and department C
P	(Z + 90)
Q	(Y – 80)
R	460

Note:

a). Total employees in company = Department A + Department B + Department C

b). $K^2 – 220K + 4000 = 0$, where A and B are roots of equation such that (A > B).

Z = 15B, and Y = A

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Out of total employees in department C of company P and company R 40% and 65% are male employees. Find number of male employees in department C of company P and R together?

a. 99

b. 101

c. 89

d. 91

e. None of these

6. Questions

For another company S, number of employees in company is 2/7 more than that of company R. out of total employees in company 1/3rd are in department A and number of employees in department B is twice as that of the employees in department B of company Q, rest employees are in department C. Find number employees in department C.

a. 360

b. 240

c. 420

d. 320

e. None of these

7. Questions

D = Difference between employees in department A and B of company P

E = Difference between employees in department B and C of company Q

Find 2E is how much % more or less than that of D.

a. 45.45%

b. 55.55%

c. 36.36%

d. 44.44%

e. None of these

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8. Questions

For company Q, number of females in department B is same as males in department A. Number of females in department C is four times as that of in department A. Find number of male employees in department C if number of male and female employees in department B are same.

a. 80

b. 60

c. 140

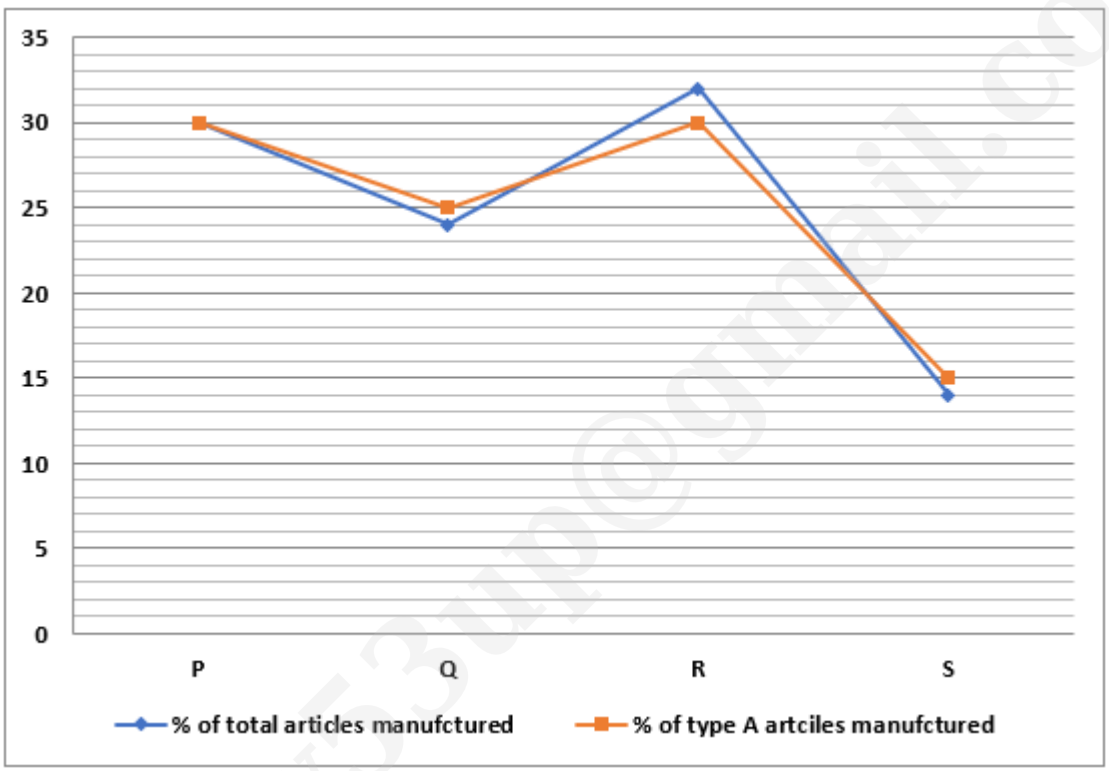
d. 120

e. None of these

9. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the % of articles manufactured (A + B) by a company out of total number of articles manufactured by all companies together, and % of article B manufactured out of total articles A manufactured by all companies – P, Q, R, and S together.



Note: Difference between article B manufactured by company P and R is 200.

The number of type B articles manufactured by S is 200, and out of total type B articles manufactured by P 25% were sold. If total articles remains unsold by company P is 2450. Find number of type A articles sold by company P.

a. 400

b. 300

c. 450

d. 600

e. None of these

10. Questions

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Out of total articles manufactured by P, 80% are of type A and total articles manufactured by company Q is six times as that of type B articles manufactured. Find difference between type B articles manufactured by company P and Q.

a. 400
b. 600
c. 200
d. 240
e. None of these

11. Questions

The difference between number of type B articles manufactured by R and S is 600, and number of type B articles manufactured by P is (Z/2). Find the value Z.

a. 600
b. 800
c. 400
d. 1200
e. None of these

12. Questions

Type B articles manufactured by company P is 50% more than that of company Q. Find the difference between total articles manufactured by company Q and company S?

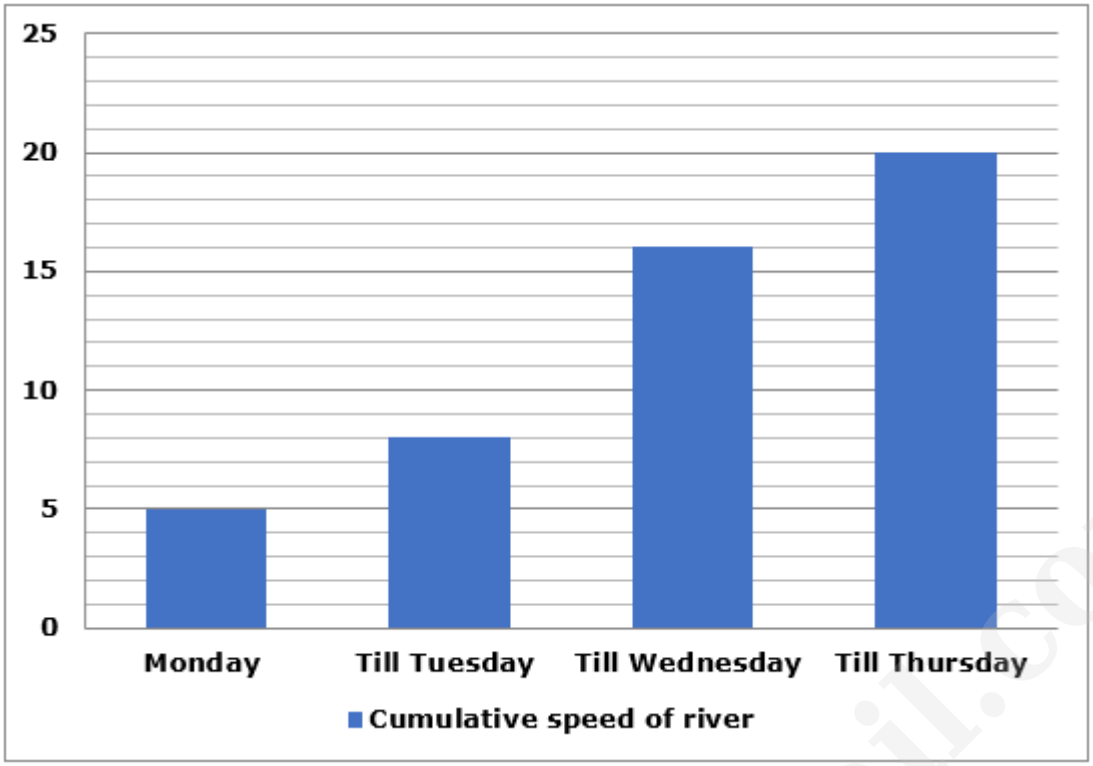
a. 1200
b. 1000
c. 1600
d. 800
e. None of these

13. Questions

Read the following information carefully and answer the questions based on it.

A boat travels certain distance on four days of a week – Monday to Thursday. Each day speed of boat in still water is different and speed of river is also different. Each day boat travels some distance upstream and some distance downstream.

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Days	Total distance covered	Downstream Distance: Upstream Distance
Monday	150	2: Z
Tuesday	39	3Y:4
Wednesday	160	3: (Y – 2)
Thursday	44	6:5

Note:

a). Z and Y are natural numbers, and on each day downstream distance travelled by boat is more than that of upstream distance.

b). On any day upstream distance travelled by boat is not less than 10 km.

Downstream travelling time of boat on Monday is same as that of upstream travelling time on same day. Downstream speed of boat on Monday is same as that on Wednesday. Find the upstream travelling time of boat on Wednesday.

a. 8 hours

b. 10 hours

c. 6 hours

d. 12 hours

e. None of these

14. Questions

On Tuesday, the respective ratio of downstream and upstream travelling time is (Z +2): 4

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respectively. Find speed of boat in still water of boat on Tuesday?

a. 6 km/h

b. 3 km/h

c. 9 km/h

d. 12 km/h

e. None of these

15. Questions

The Upstream travelling time on Thursday is (2Y – 3) hours more than downstream time on same day. Downstream travelling time on Thursday is 50% as that on Monday. Find the upstream speed of boat on Monday.

a. 15 km/h

b. 10 km/h

c. 20 km/h

d. 12.5 km/h

e. None of these

16. Questions

On Friday, boats travel downstream distance which is same as the sum of upstream distance on Wednesday and Thursday together. Speed of boat in still water is twice as that on Tuesday. If downstream travelling time of boat on friday is 5 hours which is 3 hours less than upstream travelling time on Tuesday. Find speed of river on Friday.

a. 6 km/h

b. 2.5 km/h

c. 2 km/h

d. 4 km/h

e. None of these

17. Questions

Read the following information carefully and answer the questions based on it.

A survey is conducted in class of students for the movies they have watched among – P, Q, and R. $1/4^{th}$ of total surveyed people did not watch any movie among the three.

- Students who watched all three movies is 40% less than students watched only Q. And the number of students who watched all three movies is equal to 9 .
- Students watched only R is Z which is more than those watched both Q and R but not P. Those watched both P and R but not Q is twice as that of those who watched only R.

- Number of students those who watched both P and Q but not R is Y, while those who watched only P is 3 more than those who like both P and R but not Q.
- Number of students who watched both Q and R but not P (at least one) is 30 less than thrice of those who watched both P and Q but not R.
- Number of students those who watched only P + number of students who watched both P and Q but not R = three more than twice the number of students who watched only Q

If M is the total number of people surveyed then find the value of (M + 60)?

a. 150

b. 176

c. 180

d. 184

e. None of these

18. Questions

A = Number of students those who watched both P and Q but not R

B = Number of students who watched only Q

Find difference between A and B

a. 3

b. 6

c. 4

d. 2

e. None of these

19. Questions

Number of students who watched movie P is 2A, while the number of students who did not watch any movie is B.

I). A > B

II). A + B = 2B

III). Those who watched all three movies = 30% of B

a. I only

b. II and III only

c. I and III only

d. III only

e. None of these

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20. Questions

The number of students who watched both P and Q but not R is what % of half of the total number of people surveyed?

a. 10%

b. 13.33%

c. 26.67%

d. 20%

e. None of these

Explanations:

1. Questions

Total number of female passengers in AC class = 286

Now,

Y + 16 + 4Y/5 + 2.5Z + 84 = 286

9Y/5 + 2.5Z = 186……. (1)

Also, (Z – Y) = 40……... (2)

On solving both equations, we get

Z = 60 and Y = 20

Train	Total	Sleeper	AC	AC (F)	AC (M)
P	240	144	96	36	60
Q	120	40	80	16	64
R	400	160	240	150	90
S	360	-	-	84	

Answer: C

According to question,

For S

Number of passengers in AC class = Sleeper class = 360/2 = 180

Number of male passengers in sleeper class = 180 x 60/(60 + 40) = 108

Number of male passengers in AC class = 180 – 84 = 96

Number of male passengers going to Bihar = 2/3 x (108 + 96) = 136

Hence answer is option C

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2. Questions

Total number of female passengers in AC class = 286

Now,

Y + 16 + 4Y/5 + 2.5Z + 84 = 286

9Y/5 + 2.5Z = 186……. (1)

Also, (Z – Y) = 40……... (2)

On solving both equations, we get

Z = 60 and Y = 20

Train	Total	Sleeper	AC	AC (F)	AC (M)
P	240	144	96	36	60
Q	120	40	80	16	64
R	400	160	240	150	90
S	360	-	-	84	

Answer: A

Number of passengers in train P, those booked tickets online = 25% x 144 + 75% x 96 = 108

Number of male passengers in AC class of train P and R = 60 + 90 = 150

(M + 22)% = 108/150 x 100

M = 72 – 22 = 50

Required difference = 50 – 20 = 30

Hence answer is option A

3. Questions

Total number of female passengers in AC class = 286

Now,

Y + 16 + 4Y/5 + 2.5Z + 84 = 286

9Y/5 + 2.5Z = 186……. (1)

Also, (Z – Y) = 40……... (2)

On solving both equations, we get

Z = 60 and Y = 20

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Train	Total	Sleeper	AC	AC (F)	AC (M)
P	240	144	96	36	60
Q	120	40	80	16	64
R	400	160	240	150	90
S	360	-	-	84	

Answer: D

For train S,

Difference between male and female passengers in AC class = 1.8Y = 1.8 x 20 = 36

So, male passengers in AC class of train trains together = 286 + 36 OR 286 – 36 = 322 or 250

So, male passengers in AC class of train S = (322 – 214) OR (250 – 214) = 108 OR 36

So, total passengers in AC class of train S = 84 + 108 or 84 + 36 = 192 or 120

So, passengers in sleeper class of train S = 360 – 192 OR 360 – 120 = 168 OR 240 (B)

For Train R

Total number of female passengers = 150 + 82 = 232

So, male passengers = 400 – 232 = 168 (A)

A ≤ B

Hence answer is option D

4. Questions

Total number of female passengers in AC class = 286

Now,

Y + 16 + 4Y/5 + 2.5Z + 84 = 286

9Y/5 + 2.5Z = 186……. (1)

Also, (Z – Y) = 40……... (2)

On solving both equations, we get

Z = 60 and Y = 20

Train	Total	Sleeper	AC	AC (F)	AC (M)
P	240	144	96	36	60
Q	120	40	80	16	64
R	400	160	240	150	90
S	360	-	-	84	

Answer: B

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For train Q,

Number of female passengers in Sleeper class = 7/16 x 64 = 28

So, number of male passengers in sleeper class = 40 – 28 = 12

Maximum possible number of male passengers in AC class = 12 x 75% = 9

Required maximum value (M)% = 9/60 x 100 = 15%

Hence answer is option B

5. Questions

$K^2 – 220K + 4000 = 0$

(K – 200) (K – 20) = 0

K = 20, 200

So, Z = 15 x 20 = 300

Y = 200

For P

Number of employees in department A = 75% x total students

So, number of employees in department B = 150% x employees in department C

% of employees in department C = 2/5 x 25% = 10%

Now,

(75 – 10) % x total employees = 390

So, total employees = 390/65 x 100 = 600

Employees in department A = 75% x 600 = 450

Employees in department C = 450 – 390 = 60

Employees in department B = 600 – 450 – 60 = 90

Similarly, we can find the data for all companies

Name of COMPANY	Total employees	Department A	Department B	Department C
P	600	450	90	60
Q	400	80	120	200
R	700	560	40	100
Total	1700	1090	250	360

Answer: C

According to question,

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Number of male employees in department C of company P = 60 x 40% = 24

Number of male employees in department C of company R = 65% x 100 = 65

Required sum = 65 + 24 = 89

Hence answer is option C

6. Questions

$K^2 - 220K + 4000 = 0$

(K – 200) (K – 20) = 0

K = 20, 200

So, Z = 15 x 20 = 300

Y = 200

For P

Number of employees in department A = 75% x total students

So, number of employees in department B = 150% x employees in department C

% of employees in department C = 2/5 x 25% = 10%

Now,

(75 – 10) % x total employees = 390

So, total employees = 390/65 x 100 = 600

Employees in department A = 75% x 600 = 450

Employees in department C = 450 – 390 = 60

Employees in department B = 600 – 450 – 60 = 90

Similarly, we can find the data for all companies

Name of COMPANY	Total employees	Department A	Department B	Department C
P	600	450	90	60
Q	400	80	120	200
R	700	560	40	100
Total	1700	1090	250	360

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Answer: A

For company S

Total number of employees = 9/7 x 700 = 900

Number of employees in department A = 1/3 x 900 = 300

Number of employees in department B = 120 x 2 = 240

Number of employees in department C = 900 – 300 – 240 = 360

Hence answer is option A

7. Questions

$K^2 – 220K + 4000 = 0$

(K – 200) (K – 20) = 0

K = 20, 200

So, Z = 15 x 20 = 300

Y = 200

For P

Number of employees in department A = 75% x total students

So, number of employees in department B = 150% x employees in department C

% of employees in department C = 2/5 x 25% = 10%

Now,

(75 – 10) % x total employees = 390

So, total employees = 390/65 x 100 = 600

Employees in department A = 75% x 600 = 450

Employees in department C = 450 – 390 = 60

Employees in department B = 600 – 450 – 60 = 90

Similarly, we can find the data for all companies

Name of COMPANY	Total employees	Department A	Department B	Department C
P	600	450	90	60
Q	400	80	120	200
R	700	560	40	100
Total	1700	1090	250	360

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Answer: B

Value of D = 450 – 90 = 360

Value of E = 200 – 120 = 80

Required % change = (360 – 160)/360 x 100 = 55.55%

Hence answer is option B

8. Questions

$K^2 - 220K + 4000 = 0$

$(K - 200)(K - 20) = 0$

$K = 20, 200$

So, $Z = 15 \times 20 = 300$

$Y = 200$

For P

Number of employees in department A = 75% x total students

So, number of employees in department B = 150% x employees in department C

% of employees in department C = 2/5 x 25% = 10%

Now,

(75 – 10) % x total employees = 390

So, total employees = 390/65 x 100 = 600

Employees in department A = 75% x 600 = 450

Employees in department C = 450 – 390 = 60

Employees in department B = 600 – 450 – 60 = 90

Similarly, we can find the data for all companies

Name of COMPANY	Total employees	Department A	Department B	Department C
P	600	450	90	60
Q	400	80	120	200
R	700	560	40	100
Total	1700	1090	250	360

Answer: D

Number of females in department B = 120/2 = 60

So, number of males in department A = 60

So, number of females in department A = 80 – 60 = 20

So, number of females in department C = 20 x 4 = 80

So, number of males in department C = 200 – 80 = 120

Hence answer is option D

9. Questions

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Let total number of articles manufactured by all companies together = 100a

Let total number of articles of type A manufactured by all companies together = 100b

Number of type B articles manufactured by company P = 30% x 100a – 30% x 100b = 30a – 30b

Number of type B articles manufactured by company R = 32% x a – 30% x b = 32a – 30b

Now,

(32a – 30b) – (30a – 30b) = 200

2a = 200

Value of a = 100

Total number of articles manufactured by al companies together = 100 x 100 = 10000

Name of company	Total number of articles manufactured
P	3000
Q	2400
R	3200
S	1400

Answer: A

According to question,

Number of type A, articles manufactured by S = 1400 – 200 = 1200

So, number of type A, articles manufactured by P = 1200/15 x 30 = 2400

So, number of type B articles manufactured by P = 3000 – 2400 = 600

Number of articles sold by P = 3000 – 2450 = 550

So, number of type A, articles sold by P = 550 – 25% x 600 = 400

Hence answer is option A

10. Questions

Let total number of articles manufactured by all companies together = 100a

Let total number of articles of type A manufactured by all companies together = 100b

Number of type B articles manufactured by company P = 30% x 100a – 30% x 100b = 30a – 30b

Number of type B articles manufactured by company R = 32% x a – 30% x b = 32a – 30b

Now,

(32a – 30b) – (30a – 30b) = 200

2a = 200

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Value of a = 100

Total number of articles manufactured by al companies together = 100 x 100 = 10000

Name of company	Total number of articles manufactured
P	3000
Q	2400
R	3200
S	1400

Answer: C

Number of type B articles manufactured by company P = 20% x 3000 = 600

Number of type B articles manufactured by company Q = 1/6 x 2400 = 400

Required difference = 600 – 400 = 200

Hence answer is option C

11. Questions

Let total number of articles manufactured by all companies together = 100a

Let total number of articles of type A manufactured by all companies together = 100b

Number of type B articles manufactured by company P = 30% x 100a – 30% x 100b = 30a – 30b

Number of type B articles manufactured by company R = 32% x a – 30% x b = 32a – 30b

Now,

(32a – 30b) – (30a – 30b) = 200

2a = 200

Value of a = 100

Total number of articles manufactured by al companies together = 100 x 100 = 10000

Name of company	Total number of articles manufactured
P	3000
Q	2400
R	3200
S	1400

Answer: D

Number of type B articles manufactured by R = 3200 – 30b

Number of type B articles manufactured by S = 1400 – 15b

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Now,

(3200 – 30b) – (1400 – 15b) = 600

1200 = 15b

Value of b = 80

So, number of type B articles manufactured by P = 3000 – 30 x 80 = 600

Value of Z = 600 x 2 = 1200

Hence answer is option D

12. Questions

Let total number of articles manufactured by all companies together = 100a

Let total number of articles of type A manufactured by all companies together = 100b

Number of type B articles manufactured by company P = 30% x 100a – 30% x 100b = 30a – 30b

Number of type B articles manufactured by company R = 32% x a – 30% x b = 32a – 30b

Now,

(32a – 30b) – (30a – 30b) = 200

2a = 200

Value of a = 100

Total number of articles manufactured by al companies together = 100 x 100 = 10000

Name of company	Total number of articles manufactured
P	3000
Q	2400
R	3200
S	1400

Answer: B

Difference between total articles manufactured by Q and S = 2400 – 1400 = 1000

Hence answer is option B

13. Questions

Speed of river on Monday = 5 km/h

Speed of river Tuesday = 8 -5 = 3 km/h

Speed of river on Wednesday = 16 – 8 = 8 km/h

Speed of river on Thursday = 20 – 16 = 4 km/h

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For Monday

Downstream distance > Upstream distance

Ratio of downstream distance and upstream distance = 2: Z

So, value of Z = 1

So, downstream distance travelled = 150 x 2/3 = 100 km

Upstream distance travelled = 150 – 100 = 50 km

For Tuesday

Total distance travelled = 39 km

Ratio of distance on Wednesday = 3: (Y – 2)

So, Y > 2, so value of Y = 3

Ratio of distance on Tuesday = 9:4

Upstream distance = 4/13 x 39 = 12 km

If Y = 4, so upstream distance = 39 x 1/4 = 9.75 (not possible)

So, Y = 3, and Downstream distance = 39 – 12 = 27 km

For Wednesday

Total distance = 160 km

Ratio of distance = 3: (3 – 2) = 3:1

Downstream distance = 3/4 x 160 = 120 km

Upstream distance = 160 – 120 = 40 km

For Thursday

Total distance = 44 km

Downstream distance = 44 x 6/11 = 24 km

Upstream distance = 44 – 24 = 20 km

Answer: B

According to question,

Let speed of boat in still water on Monday = a

Now,

(a + 5)/ (a – 5) = 2/1

2a – 10 = a + 5

Value of a = 15

So, downstream speed of boat on Wednesday = 20 km /h

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So, upstream speed of boat on Wednesday = 20 – 8 – 8 = 4 km/h

So, required time = 40/4 = 10 hours

Hence answer is option B

14. Questions

Speed of river on Monday = 5 km/h

Speed of river Tuesday = 8 -5 = 3 km/h

Speed of river on Wednesday = 16 – 8 = 8 km/h

Speed of river on Thursday = 20 – 16 = 4 km/h

For Monday

Downstream distance > Upstream distance

Ratio of downstream distance and upstream distance = 2: Z

So, value of Z = 1

So, downstream distance travelled = 150 x 2/3 = 100 km

Upstream distance travelled = 150 – 100 = 50 km

For Tuesday

Total distance travelled = 39 km

Ratio of distance on Wednesday = 3: (Y – 2)

So, Y > 2, so value of Y = 3

Ratio of distance on Tuesday = 9:4

Upstream distance = 4/13 x 39 = 12 km

If Y = 4, so upstream distance = 39 x 1/4 = 9.75 (not possible)

So, Y = 3, and Downstream distance = 39 – 12 = 27 km

For Wednesday

Total distance = 160 km

Ratio of distance = 3: (3 – 2) = 3:1

Downstream distance = 3/4 x 160 = 120 km

Upstream distance = 160 – 120 = 40 km

For Thursday

Total distance = 44 km

Downstream distance = 44 x 6/11 = 24 km

Upstream distance = 44 – 24 = 20 km

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Answer: A

On Tuesday

Ratio of downstream and upstream travelling time = (1 + 2): 2 x 2 = 3:4

Ratio of upstream and downstream distance = 9:4

Ratio of upstream speed and downstream speed = 9/3: 4/2 = 3:1

So, ratio of speed of boat in still water and speed of stream = 2:1

So, speed of boat in still water = 2/1 x 3 = 6 km/h

Hence answer is option A

15. Questions

Speed of river on Monday = 5 km/h

Speed of river Tuesday = 8 -5 = 3 km/h

Speed of river on Wednesday = 16 – 8 = 8 km/h

Speed of river on Thursday = 20 – 16 = 4 km/h

For Monday

Downstream distance > Upstream distance

Ratio of downstream distance and upstream distance = 2: Z

So, value of Z = 1

So, downstream distance travelled = 150 x 2/3 = 100 km

Upstream distance travelled = 150 – 100 = 50 km

For Tuesday

Total distance travelled = 39 km

Ratio of distance on Wednesday = 3: (Y – 2)

So, Y > 2, so value of Y = 3

Ratio of distance on Tuesday = 9:4

Upstream distance = 4/13 x 39 = 12 km

If Y = 4, so upstream distance = 39 x 1/4 = 9.75 (not possible)

So, Y = 3, and Downstream distance = 39 – 12 = 27 km

For Wednesday

Total distance = 160 km

Ratio of distance = 3: (3 – 2) = 3:1

Downstream distance = 3/4 x 160 = 120 km

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Upstream distance = 160 – 120 = 40 km

For Thursday

Total distance = 44 km

Downstream distance = 44 x 6/11 = 24 km

Upstream distance = 44 – 24 = 20 km

Answer: A

For Thursday

Let speed of boat in still water = a km/h

Now,

$20/ (a – 4) – 24/(a + 4) = 3$

$3a^2 + 4a – 224 = 0$

$3a^2 – 24a + 28a – 224 = 0$

$3a (a – 8) + 28(a – 8) = 0$

$(3a + 28) (a – 8) = 0$

So, value of a = 8

Downstream speed of boat = 8 + 4 = 12 km/h

So, downstream traveling time of boat on Thursday = 24/12 = 2 hours

So, downstream travelling time of boat on Monday = 2 x 2 = 4 hours

So, downstream speed of boat on Monday = 100/4 = 25 km/h

So, upstream speed on Monday = 25 – 5 – 5 = 15 km/h

Hence answer is option A

16. Questions

Speed of river on Monday = 5 km/h

Speed of river Tuesday = 8 -5 = 3 km/h

Speed of river on Wednesday = 16 – 8 = 8 km/h

Speed of river on Thursday = 20 – 16 = 4 km/h

For Monday

Downstream distance > Upstream distance

Ratio of downstream distance and upstream distance = 2: Z

So, value of Z = 1

So, downstream distance travelled = 150 x 2/3 = 100 km

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Upstream distance travelled = 150 – 100 = 50 km

For Tuesday

Total distance travelled = 39 km

Ratio of distance on Wednesday = 3: (Y – 2)

So, Y > 2, so value of Y = 3

Ratio of distance on Tuesday = 9:4

Upstream distance = 4/13 x 39 = 12 km

If Y = 4, so upstream distance = 39 x 1/4 = 9.75 (not possible)

So, Y = 3, and Downstream distance = 39 – 12 = 27 km

For Wednesday

Total distance = 160 km

Ratio of distance = 3: (3 – 2) = 3:1

Downstream distance = 3/4 x 160 = 120 km

Upstream distance = 160 – 120 = 40 km

For Thursday

Total distance = 44 km

Downstream distance = 44 x 6/11 = 24 km

Upstream distance = 44 – 24 = 20 km

Answer: E

Upstream travelling time on Tuesday = 5 + 3 = 8 hours

Upstream speed on Tuesday = 12/8 = 1.5 km/h

So, speed of boat in still water on Friday = 2 x (1.5 + 3) = 9 km/h

Downstream distance on Friday = 40 + 20 = 60 km

So, speed of river on Friday = 60/5 – 9 = 3 km/h

Hence answer is option E

17. Questions

Students those watched only R = Z

So, students who watched both P and R but not Q = 2 x Z = 2Z

So, the number of students who watched only P = 2Z + 3

Number of students those watched all three movies = 9

So, the number of students those who watched only Q = 9/60 x 100 = 15

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Number of students who watched both P and Q but not R = Y

So, the number of students who watched both Q and R but not P = 3 x Y – 30 = 3Y – 30

Now,

2Z + 3 + Y = 2 x 15 + 3

2Z + Y = 30

Z = 15 – Y/2……. (1)

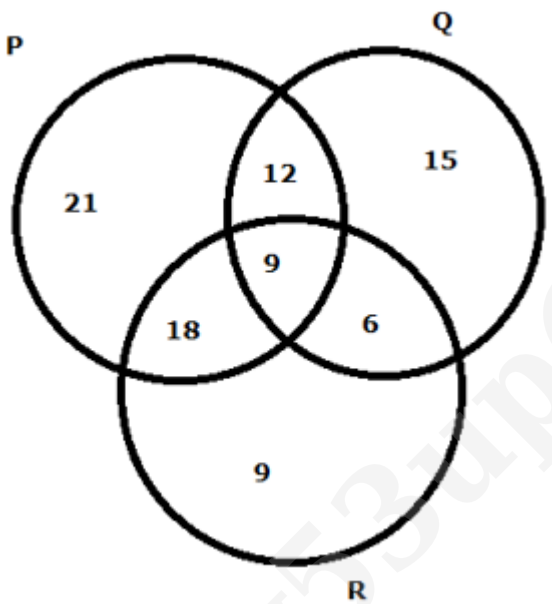
Also, Z > 3Y – 30

Y must be even number > 10

So, if Y = 12, then Z = 9 (possible)

If Y = 14, then Z = 8 (not possible because Z must be more than 3Y – 30).

So, we have only one pair of value.



Total number of surveyed people = 90/75 x 100 = 120

Answer: C

According to question,

Value of M = 120

So, required value = 120 + 60 = 180

Hence answer is option C

18. Questions

Students those watched only R = Z

So, students who watched both P and R but not Q = 2 x Z = 2Z

So, the number of students who watched only P = 2Z + 3

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Number of students those watched all three movies = 9

So, the number of students those who watched only Q = 9/60 x 100 = 15

Number of students who watched both P and Q but not R = Y

So, the number of students who watched both Q and R but not P = 3 x Y – 30 = 3Y – 30

Now,

2Z + 3 + Y = 2 x 15 + 3

2Z + Y = 30

Z = 15 – Y/2……. (1)

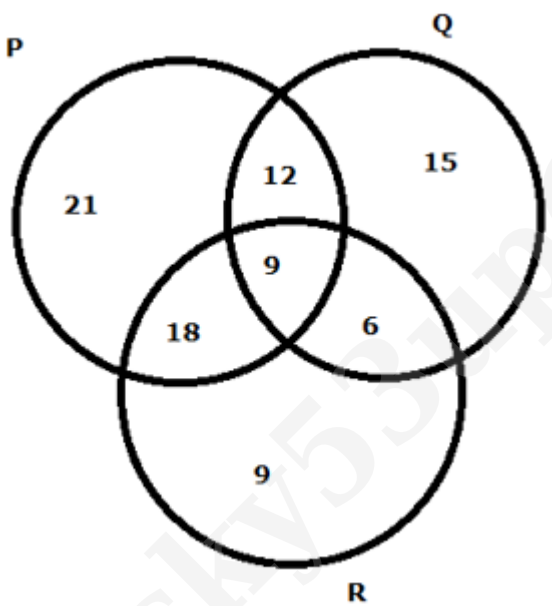
Also, Z > 3Y – 30

Y must be even number > 10

So, if Y = 12, then Z = 9 (possible)

If Y = 14, then Z = 8 (not possible because Z must be more than 3Y – 30).

So, we have only one pair of value.



Total number of surveyed people = 90/75 x 100 = 120

Answer: A

Value of A = 12

Value of B = 15

Required difference = 15 – 12 = 3

Hence answer is option A

19. Questions

Students those watched only R = Z

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So, students who watched both P and R but not Q = 2 x Z = 2Z

So, the number of students who watched only P = 2Z + 3

Number of students those watched all three movies = 9

So, the number of students those who watched only Q = 9/60 x 100 = 15

Number of students who watched both P and Q but not R = Y

So, the number of students who watched both Q and R but not P = 3 x Y – 30 = 3Y – 30

Now,

2Z + 3 + Y = 2 x 15 + 3

2Z + Y = 30

Z = 15 – Y/2……. (1)

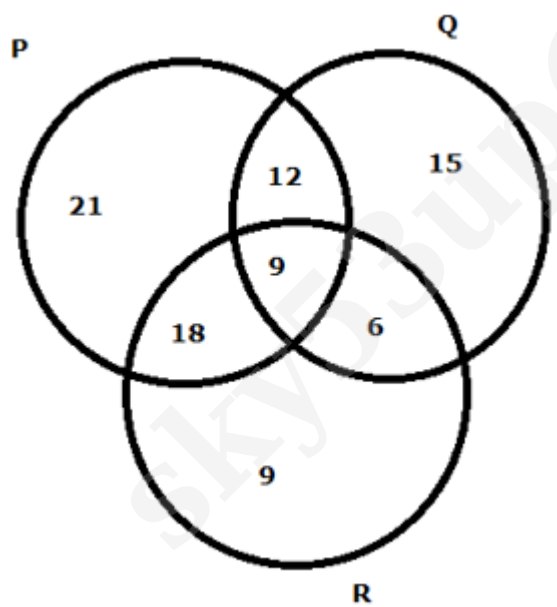
Also, Z > 3Y – 30

Y must be even number > 10

So, if Y = 12, then Z = 9 (possible)

If Y = 14, then Z = 8 (not possible because Z must be more than 3Y – 30).

So, we have only one pair of value.



Total number of surveyed people = 90/75 x 100 = 120

Answer: B

2A = 60

So, A = 30

B = 120 x 25% = 30

I). A > B

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30 = 30

This statement is false.

II). A + B = 2B

30 + 30 = 2 x 30

60 = 60

This statement is true.

III). Those who watched all three movies = 30% of B

Those who watched all three movies = 9 = 30% x 30 = 9

This statement is true.

So, only II and III are true.

Hence answer is option B

20. Questions

Students those watched only R = Z

So, students who watched both P and R but not Q = 2 x Z = 2Z

So, the number of students who watched only P = 2Z + 3

Number of students those watched all three movies = 9

So, the number of students those who watched only Q = 9/60 x 100 = 15

Number of students who watched both P and Q but not R = Y

So, the number of students who watched both Q and R but not P = 3 x Y – 30 = 3Y – 30

Now,

2Z + 3 + Y = 2 x 15 + 3

2Z + Y = 30

Z = 15 – Y/2……. (1)

Also, Z > 3Y – 30

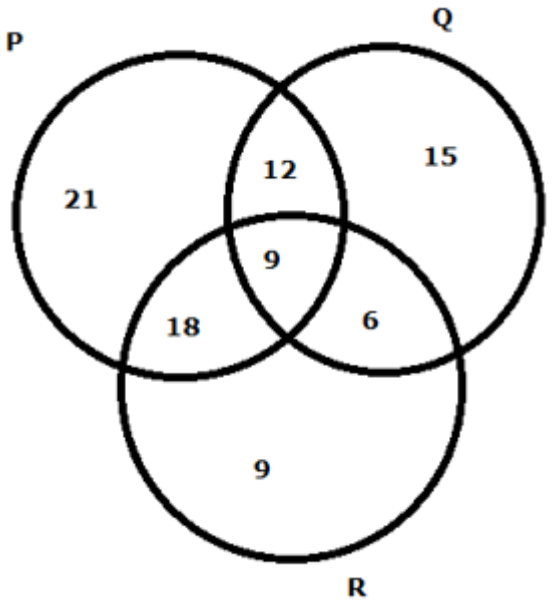
Y must be even number > 10

So, if Y = 12, then Z = 9 (possible)

If Y = 14, then Z = 8 (not possible because Z must be more than 3Y – 30).

So, we have only one pair of value.

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Total number of surveyed people = 90/75 x 100 = 120

Answer: D

Required % = [12 / (120/2)] x 100 = 20%

Hence answer is option D

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1. Questions

Study the following information carefully and answer the questions given below.

There are five moving bodies [M, N, O, P and Q]. u, v, s, a and t are different parameters of moving bodies. The values of u, v, s, a and t of five moving bodies are given in the table. Some values are missing.

Moving Body	Value of u	Value of v	Value of s	Value of a	Value of t
M	18	__	___	7	____
N	_____	97	___	____	10
O	20	____	____	9	____
P	___	____	756	____	____
Q	_____	140	____	____	7

Note:

I). $s = u * t + (1/2) * a * t^2$

II). $v = u + a * t$

III). $v^2 = u^2 + 2as$

----> Values of the variable is neither zero nor negative.

The value of s of moving body M is 46 times the value of t of the same moving body. Find which of the following statements is true?

I). The sum of digits of the value of s of moving body M is 17.

II). The square value of t of moving body M is a perfect square number and perfect cube number.

III). If the speed of a car is equal to the value of v of moving body M, then the car covers 168 km in 2 hours.

a. Only I and III true

b. Only I true

c. Only I and II true

d. Only III true

e. Only II true

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2. Questions

The ratio of value of 'a' and 't' of moving body N is 4:5. The value of 's' of moving body O is 194.5 more than the value of 's' of moving body N. The age of ram is equal to the value of t of moving body O. Ram's father's age is four times the Ram's age. The ratio of ages of Ram's father and mother after 4 years is 12:11. Find the present age of Ram's mother?

a. 45

b. 49

c. 41

d. 40

e. 39

3. Questions

In moving body P, the ratio of the value of 't' and 'a' is 3:4 and the ratio of the value of 'u' and 't' is 10:3 and the value of 'v' is 138. The difference between the value of u and v of moving body P is 'i'. j is a prime number between 30 to 40.

I). [i + j] is divisible by 5.

II). [i + j] is one more than the value of v of moving body P.

III). Difference between the value of u of moving body P and value of j is a prime number.

IV). [i + j] is one more than the square of 12.

a. If I is true then II is also true.

b. If I is false then II is true.

c. If is I true then III is false.

d. If III is true then IV is false.

e. None of these

4. Questions

Value of 'a' is twice the value of t of moving body Q. Find the value of s of moving body Q.

a. 637

b. 682

c. 696

d. 688

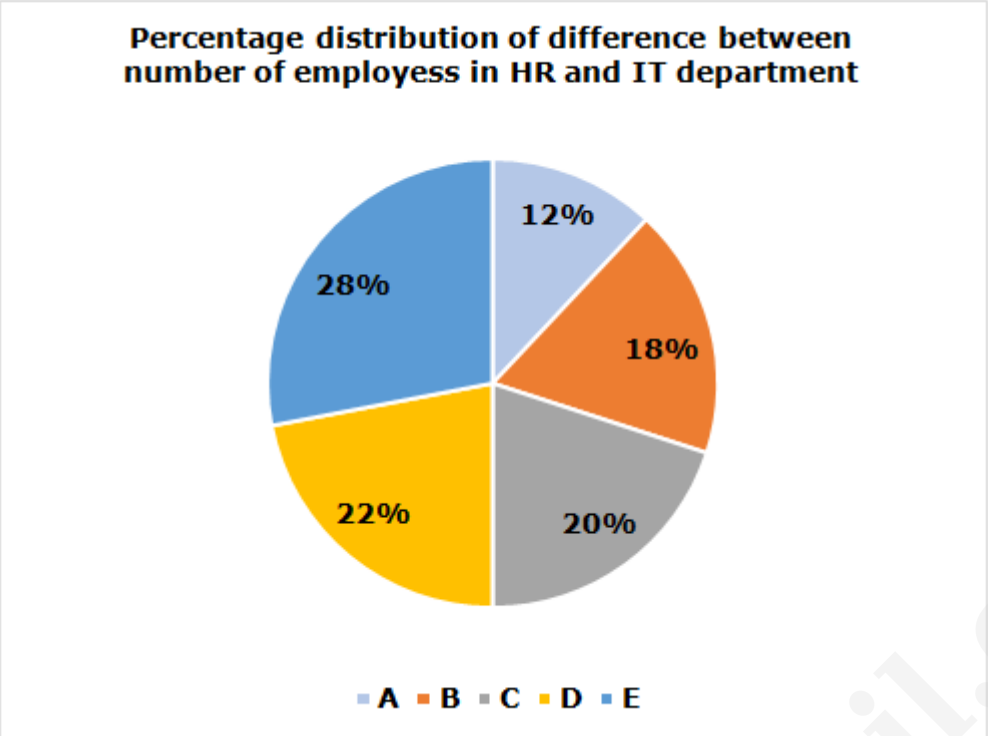
e. None of these

5. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage distribution of the difference between the number of employees in HR and IT departments of five different companies [A, B, C, D and E].

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The given table shows the percentage of employees in the HR department as well as the number of employees in IT department and the percentage of the number of employees in the accounts department more than the number of employees in the IT department.

Company	The percentage of employees in the HR department as the number of employees in the IT department	Percentage of the number of employees in the accounts department is more than the number of employees in the IT department.
A	76%	20%
B	60%	40%
C	68%	20%
D	60%	10%
E	60%	25%

Note:

The difference between the number of employees in accounts and the IT department of company A is 20.

The total number of employees in the accounts, marketing, and sales departments of company A is 310. The difference between the number of employees in the marketing and sales departments of company A is 30. The total number of employees in the IT and sales departments is not more than 200. Find the approximate percentage of the total number of employees in the HR and marketing departments of company A, out of the total number of employees in all the departments together in company A?

a. 31%

b. 28%

c. 38%

d. 48%

e. 56%

6. Questions

The difference between the number of males in the HR and IT departments of company B is 26. The ratio of the number of females in the HR and IT departments of company B is 3:4. The ratio of the number of females in IT to the number of males in the accounts department of company B is 5:9. Find the ratio between the number of females in HR and accounts department of company B?

[If the number of employees is more than one department compared to another department, then the number of males is also more for one department compared to another department]

a. 4:5

b. 6:7

c. 3:7

d. 5:9

e. 8:7

7. Questions

The total number of employees in the HR and IT departments of company F is 20% more than the total number of employees in the HR and IT departments of company C. The ratio of the number of employees in the HR and IT departments is 1:2. The number of employees in the accounts department of company F is 25% less than the number of employees in the IT department of the same company. Find the total number of employees in the accounts department of companies C and F together?

a. 258

b. 296

c. 286

d. 276

e. 236

8. Questions

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The number of employees from the HR department participating in cultural events at companies D and E is 33.33% and 25% respectively. The sum of the total number of HR department employees from companies D and E participating in the event is I and the difference between the number of employees in the IT department of companies D and E is J and the difference between the number of employees in the accounts department of companies D and E is K. Find which of the following is true?

a. I, J and K are non-prime numbers.

b. I is the prime number, and J and K are only divisible 3.

c. The sum of I, J and K is the prime number

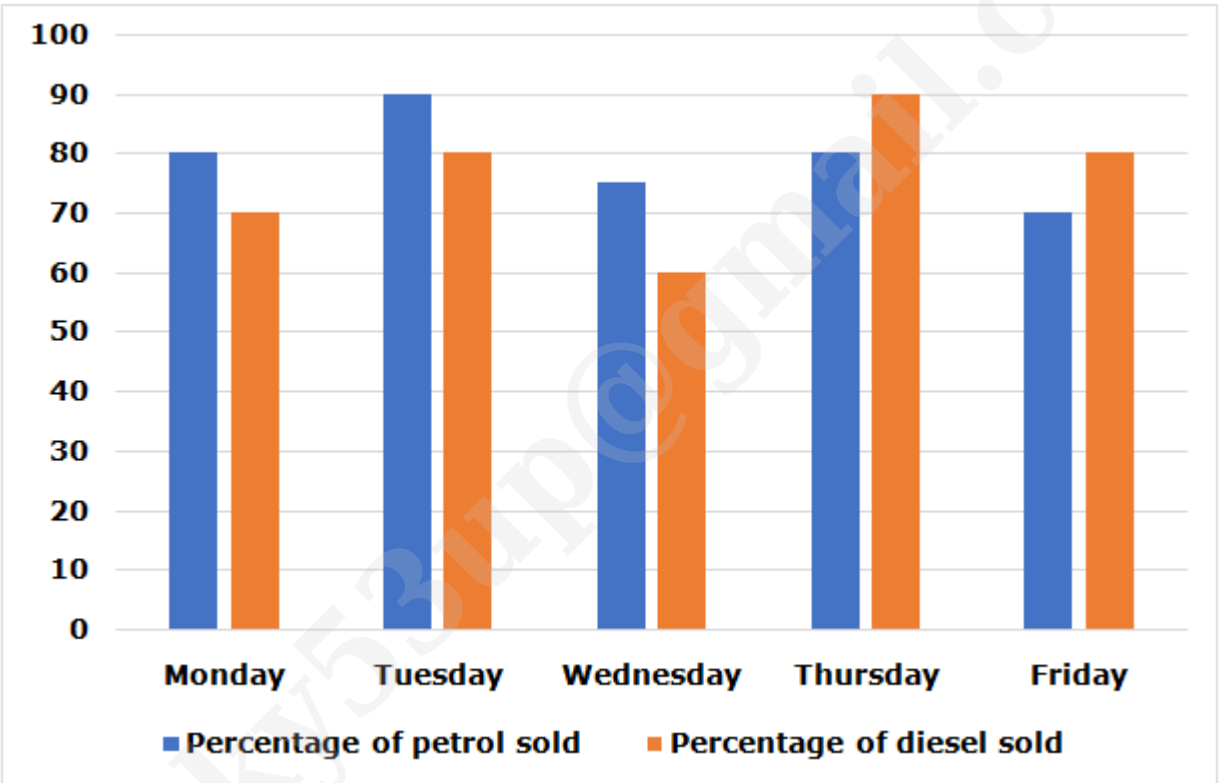
d. I is prime number, and J and K are divisible by 6.

e. More than one option true

9. Questions

Study the following information carefully and answer the questions given below.

There are six fuel dispensers [three petrol and three diesel] in fuel stations. The given bar graphs show the percentage of fuel sold from each petrol and diesel dispenser out of the total capacity of each dispenser on five different days of the week [Monday, Tuesday, Wednesday, Thursday, and Friday].



Note:

The total capacity of each dispenser (petrol or diesel) is 120 litres. At the end of each day, the vacant portion of each dispenser is filled with fuel.

The total quantity of a mixture of milk and water in container A is equal to the remaining quantity of petrol at the end of Monday, and the total quantity of a mixture of milk and water in container B is equal to the remaining quantity of diesel at the end of Monday. The ratio of milk in Containers A and B is 3:4 and the ratio of water in containers B and A is 7:3. Find the difference between the total amount of milk and water in both the container together?

a. 120

b. 160

c. 100

d. 140

e. 130

10. Questions

The total quantity of fuel left on Tuesday is equal to the cost of two items. When one item is sold at 20% profit and others at 50% profit then, the overall profit percentage is 36.11%. Find the selling price of the item whose cost price is lowest when sold at 25% profit?

a. Rs.55

b. Rs.72.5

c. Rs.66

d. Rs.62.5

e. Rs.75

11. Questions

Find which of the following statements is true?

I). The average quantity of diesel left on Thursday and Friday together is 48.

II). The sum of fuel left on Thursday is 108.

III). The difference in the quantity of fuel sold on Friday is 30.

a. All true

b. Only II true

c. All false

d. Only III true

e. None of these

12. Questions

The quantity of petrol sold on Saturday is 25% more than the quantity of petrol sold on Friday, and the quantity of diesel sold on Saturday is 25% more than the same on Friday. The price per litre of petrol and diesel is Rs. 105 and Rs. 95 respectively. Find the difference between the total revenue earned by the fuel station in these two days?

a. Rs.13455

b. Rs.25855

c. Rs.16544

d. Rs.13655

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e. None of these

13. Questions

Study the following information carefully and answer the questions given below.

There are different numbers of male and female teachers working in five different colleges [A, B, C, D, and E]. The difference between the number of female teachers in colleges A and B is ____(I). The ratio of male and female teachers in college A is 13:10. The difference between the number of male and female teachers in college B is ____ (II). Only in colleges B and D, the number of female teachers is higher than the number of male teachers. The number of male teachers in college B is 10 less than the number of male teachers in college A. The number of male teachers in E is five more than the number of male teachers in College C. The difference between the number of male and female teachers in College E is 13. The total number of female teachers in College A is 16.66% less than that of College C. The number of male teachers in College C is 33.33% more than the number of female teachers in College C. The difference between the number of male and female teachers in College D is 5. The total number of male teachers in colleges C and D together is ___(III)____. The total number of teachers in College D is 145. The number of female teachers in College C is 20% less than the number of female teachers in College D. The ratio between the number of male and female teachers in College B is 11:14. The total number of teachers in College E is ___(IV)____.

The number of male teachers in college F is _____% more than the number of male teachers in college C and the number of female teachers in college F is _____% more than the number of female teachers in college C. The total number of teachers in college F is _____. Find which of the values satisfy the blank?

a. 20, 10, 152

b. 20, 30, 174

c. 10, 20, 162

d. 30, 10, 170

e. More than one option is true

14. Questions

Out of the total male teachers in College C, 40% are lecturers, and the rest are assistant professors. The total number of lecturers in College C is 60. The number of female lecturers in College D is 25% more than that of College C. The number of male assistant professors in College D is 20% more than the number of female assistant professors in the same college. Find the number of male assistant professors in college D?

a. Thrice of the square of 3.

b. Double of square of 5.

c. Two more than the square of 6.

d. One less than the square of 7.

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e. Half of the square of 8.

15. Questions

The number of lab attendants in college A is 20% of the total number of teachers in the college. The number of lab attendants in College C is 30% of the total number of teachers in the college. The difference between the number of lab attendees in college A and C is x. The average age of all the teachers of College E is 2x. Find the total age of all the teachers of college E?

a. 5874

b. 5966

c. 6588

d. 8456

e. None of these

16. Questions

Find the sum of the values of all four blanks?

a. 342

b. 352

c. 362

d. 332

e. None of these

17. Questions

The following questions contain two statements as statement I and statement II. You have to determine which statement/s is/are necessary to answer the question and give the answer as,

Find the time taken by C alone to complete the work?

Statement I: A, B, and C complete the work together in 8 days. If A works alone for 5 days, then C finishes the rest of the work in 18days.

Statement II: The efficiency of A and B together is twice the efficiency of C alone. B alone completed the work in 30 days.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question

b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question

c. The data either in statement I alone or in statement II alone is sufficient to answer the question

d. The data given in both statements I and II together are not sufficient to answer the question

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e. The data given in both statements I and II together are necessary to answer the question.

18. Questions

Find the sum of the ages of A and B together?

Statement I: The ratio between the present age of A to B is 3:4. The sum of the ages of C after 5 years and B after 3 years is 83 years.

Statement II: The age of A before 10 years is half the age of the present age of B. The sum of the ages of B after 5 years and A after 3 years is 78 years.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question

b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question

c. The data either in statement I alone or in statement II alone is sufficient to answer the question

d. The data given in both statements I and II together are not sufficient to answer the question

e. The data given in both statements I and II together are necessary to answer the question.

19. Questions

Find the number of students who like photography?

Statement I: The number of students who like only painting and photography but not gardening is 20. The number of students who like only gardening and photography but not painting is 30. The number of students who like only gardening is 25.

Statement II: The number of students who like only photography is twice the number of students who like both photography and painting, not gardening. The number of students who like only painting is twice the number of students who like painting and gardening but not photography, which is equal to 15.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question

b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question

c. The data either in statement I alone or in statement II alone is sufficient to answer the question

d. The data given in both statements I and II together are not sufficient to answer the question

e. The data given in both statements I and II together are necessary to answer the question

20. Questions

Find the distance between P and Q?

Statement I: Boat A and Boat B meet after 3 hours when they start the journey from P and Q towards Q and P respectively, at the same time. Boat A goes with the stream, and Boat B goes against the stream and they cover an equal distance. The speed of the stream is 3 km/hr.

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Statement II: Boat A covers 84 km downstream in another river in 3 hours, and boat B covers 90 km downstream in the above river (mentioned in statement I) in 3 hours.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question
b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question
c. The data either in statement I alone or in statement II alone is sufficient to answer the question
d. The data given in both statements I and II together are not sufficient to answer the question
e. The data given in both statements I and II together are necessary to answer the question.

Explanations:

1. Questions

Answer: C

The value of u is 18 and the value of a is 7 of moving body M.

So, v=18 + 7 * t

So, $v^2 = 18^2 + 2 * 7 *s$

Or, $[18 + 7t]^2 = 18^2 + 2*7*s$

Or, $324 + 2* 18 * 7t + 49t^2= 324 + 2 * 7 * 46 t$

Or, $49t^2 – 392t = 0$

Or, $t^2 – 8t =0$

Or, t*(t-8) =0, so, t = 8 because t cannot be zero.

So, s = 8 * 46 = 368

So, v= 18 + 7*8 = 74

I). The sum of digits of the value of s of moving body M is 3 + 6 + 8 = 17.

II). The Square of value of t of moving body M is 8*8 = 64 is a perfect square number and perfect cube number.

III). If the speed of a car is equal to the value of v of moving body M, then the car covers 74*2 = 148 km in 2 hours.

2. Questions

Answer: D

The value of a of moving body N is 10 * 4 / 5 = 8

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Value of u of moving body N is = 97 – 8 * 10 = 17

So, value of s of moving body N is = 17 * 10+ (1/2) * 8 * 10 *10 = 570

The value of s of moving body O is 570 + 194 .5 = 764 .5

So, v^2 = 20 * 20 + 2 * 9 *764.5 = 14161 = 119 * 119

So, v = 119

So, t = [119 – 20] / 9 = 11

So, the age of Ram's father is 11*4 = 44

So, present age of Ram's mother is [44+4]*11/12 - 4 = 44 - 4 = 40

3. Questions

Answer: B

Let the value of t and a of moving body P be 3x and 4x respectively.

So, the value of u is 3x*10/3 = 10x.

So, $v = 10x+12x^2$

So, $138 = 10x + 12x^2$

Or, $6x^2+5x-69=0$

Or, $6x^2+23x – 18x -69 =0$

Or, x(6x+23)-3(6x+23)=0

Or, (x-3)(6x+23)=0

So, x = 3 , x = -23/6

So, the value of t, a and u of moving body P is 9, 12 and 30 respectively.

So, i=138-30=108

The value of j is either 31 or 37.

I). [i + j] is divisible is by 5.

Possible when j = 37.

II). [i + j] is one more than the value of v of moving body P.

Possible when j = 31

III). Difference between the value of u of moving body P and value j is a prime number.

Possible when j = 37

IV). [i + j] is one more than the square of 12.

Possible when j = 37

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4. Questions

Answer: A

s = 42 * 7 + (1/2) * 14* 49 = 637

5. Questions

If the number of employees in the IT department of company A is 100% and so, the number of employees in the Accounts department of company A is [100 * 120/100 = 120%, so, the percentage difference between the number of employees in accounts and IT department of company A is 20%.

So, 20% = 20

So, 1% = 1.

So, the number of employees in the IT department of company A is 100 and the number of employees in the accounts department of company A is 120.

So, the total number of employees in the HR department of company A is 100 * 76/100 = 76.

So, the difference between the number of employees in the HR and IT department of company A is 100 – 76 = 24

So, the total value of pie chart is 24 *100 /12 = 200

So, the difference between the number of employees in the HR and IT department of company B is 200* 18 /100 = 36

So, 40% = 36

So, 100% = [36 /40] *100 = 90

So, Number of employees in the IT department of company B is 90.

The number of employees in the HR department of company B is 90 – 36 = 54

The number of employees in the accounts department of company B is 90 * 140/ 100 = 126

Similarly, we can calculate others' values also.

Company	Number of employees in the HR department	Number of employees in the IT department	Number of employees in the Accounts department
A	76	100	120
B	54	90	126
C	85	125	150
D	66	110	121
E	84	140	175

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Answer: C

Let the number of employees in the marketing and sales departments of company A be x and y

respectively.

So, x + y = 310 – 120 = 190

So, Employees in Marketing department is either [190 + 30]/2 = 110 or [190 -30] /2 = 80

So, the Employees in the sales department is either 80 or 110.

Now, the total number of employees in the Sales and IT department of company A is not more than 200.

So, Number of employees in the sales department of company A is 80.

The number of employees in the marketing department of company A is 110.

So, required percentage = [186/486] *100 = 38.27% = 38%

6. Questions

If the number of employees in the IT department of company A is 100% and so, the number of employees in the Accounts department of company A is [100 * 120/100 = 120%, so, the percentage difference between the number of employees in accounts and IT department of company A is 20%.

So, 20% = 20

So, 1% = 1.

So, the number of employees in the IT department of company A is 100 and the number of employees in the accounts department of company A is 120.

So, the total number of employees in the HR department of company A is 100 * 76/100 = 76.

So, the difference between the number of employees in the HR and IT department of company A is 100 – 76 = 24

So, the total value of pie chart is 24 *100 /12 = 200

So, the difference between the number of employees in the HR and IT department of company B is 200* 18 /100 = 36

So, 40% = 36

So, 100% = [36 /40] *100 = 90

So, Number of employees in the IT department of company B is 90.

The number of employees in the HR department of company B is 90 – 36 = 54

The number of employees in the accounts department of company B is 90 * 140/ 100 = 126

Similarly, we can calculate others' values also.

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Company	Number of employees in the HR department	Number of employees in the IT department	Number of employees in the Accounts department
A	76	100	120
B	54	90	126
C	85	125	150
D	66	110	121
E	84	140	175

Answer: D

Let the number of females in the HR and IT departments of company B be 3x and 4x.

Let the number of males in the HR department be y and the number of males in the IT department be y + 26.

So, 3x + y = 54 and 4x + y +26 = 90, 4x + y = 64

By solving above equations, x = 10 and y = 24

The number of females in the HR and IT departments of company B is 3*10=30 and 4*10 = 40 respectively.

The number of males in the HR department is 24 and the number of males in the IT department is 24 + 26 = 50.

The number of male employees in the accounts department of company B is 40*9/5 = 72

The number of female employees in the accounts department of company B is 126 – 72 = 54

So, required ratio is = 30 : 54 =5:9

7. Questions

If the number of employees in the IT department of company A is 100% and so, the number of employees in the Accounts department of company A is [100 * 120/100 = 120%, so, the percentage difference between the number of employees in accounts and IT department of company A is 20%.

So, 20% = 20

So, 1% = 1.

So, the number of employees in the IT department of company A is 100 and the number of employees in the accounts department of company A is 120.

So, the total number of employees in the HR department of company A is 100 * 76/100 = 76.

So, the difference between the number of employees in the HR and IT department of company A is 100 – 76 = 24

So, the total value of pie chart is 24 *100 /12 = 200

So, the difference between the number of employees in the HR and IT department of company B is 200*

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18 /100 = 36

So, 40% = 36

So, 100% = [36 /40] *100 = 90

So, Number of employees in the IT department of company B is 90.

The number of employees in the HR department of company B is 90 – 36 = 54

The number of employees in the accounts department of company B is 90 * 140/ 100 = 126

Similarly, we can calculate others' values also.

Company	Number of employees in the HR department	Number of employees in the IT department	Number of employees in the Accounts department
A	76	100	120
B	54	90	126
C	85	125	150
D	66	110	121
E	84	140	175

Answer: D

The number of employees in the HR and IT departments of company F is

= [85 +125] * 120 /100 = 252

The number of employees in the accounts department of company F is

= [252*2 /3] * 75/100= 126

So, The total number of employees in the accounts department of company C and F is 150 + 126 = 276

8. Questions

If the number of employees in the IT department of company A is 100% and so, the number of employees in the Accounts department of company A is [100 * 120/100 = 120%, so, the percentage difference between the number of employees in accounts and IT department of company A is 20%.

So, 20% = 20

So, 1% = 1.

So, the number of employees in the IT department of company A is 100 and the number of employees in the accounts department of company A is 120.

So, the total number of employees in the HR department of company A is 100 * 76/100 = 76.

So, the difference between the number of employees in the HR and IT department of company A is 100 – 76 = 24

So, the total value of pie chart is 24 *100 /12 = 200

So, the difference between the number of employees in the HR and IT department of company B is 200* 18 /100 = 36

So, 40% = 36

So, 100% = [36 /40] *100 = 90

So, Number of employees in the IT department of company B is 90.

The number of employees in the HR department of company B is 90 – 36 = 54

The number of employees in the accounts department of company B is 90 * 140/ 100 = 126

Similarly, we can calculate others' values also.

Company	Number of employees in the HR department	Number of employees in the IT department	Number of employees in the Accounts department
A	76	100	120
B	54	90	126
C	85	125	150
D	66	110	121
E	84	140	175

Answer: E

So, I is = 66 * 33.33/100 + 84 *25/100 = 22 + 21 = 43

So, J = 140 – 110 = 30

So, K = 175 – 121 = 54

So, options C and D are true.

c). The sum of I, J and K is the prime number

d). I is a prime number, and J and K are divisible by 6.

9. Questions

The total capacity of each dispenser is 120 litres.

Total quantity of petrol sold on Monday is [120 *80 /100] *3 = 96 *3 = 288 liters

The total quantity of petrol in the dispensers at the end of Monday is 120 * 3 – 288 = 72 litres

Total quantity of diesel sold on Monday is [120 * 70/100] *3 = 252 liters

The total quantity of diesel in the dispensers at the end of Monday is 120*3 – 252 = 108 litres

Similarly, we can calculate others' values also.

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Day	The total quantity of petrol sold (litres)	Left quantity of petrol at end of the day (liters)	The total quantity of diesel sold (liters)	Left quantity of diesel at the end of the day (liters)
Monday	288	72	252	108
Tuesday	324	36	288	72
Wednesday	270	90	216	144
Thursday	288	72	324	36
Friday	252	108	288	72

Answer: C

Let milk in A and B be 3x and 4x respectively.

Water in A and B is 3y and 7y respectively.

So, we can say, 3x+3y=72 and 4x+7y=108

By solving the above equations, we get x = 20 and y = 4 .

So, the total amount of milk in containers A and B together is 7*20 = 140

So, the total amount of water in containers A and B together is 10 * 4 = 40

So, difference is = 140 – 40 = 100

10. Questions

The total capacity of each dispenser is 120 litres.

Total quantity of petrol sold on Monday is [120 *80 /100] *3 = 96 *3 = 288 liters

The total quantity of petrol in the dispensers at the end of Monday is 120 * 3 – 288 = 72 litres

Total quantity of diesel sold on Monday is [120 * 70/100] *3 = 252 liters

The total quantity of diesel in the dispensers at the end of Monday is 120*3 – 252 = 108 litres

Similarly, we can calculate others' values also.

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Day	The total quantity of petrol sold (litres)	Left quantity of petrol at end of the day (liters)	The total quantity of diesel sold (liters)	Left quantity of diesel at the end of the day (liters)
Monday	288	72	252	108
Tuesday	324	36	288	72
Wednesday	270	90	216	144
Thursday	288	72	324	36
Friday	252	108	288	72

Answer: D

So, the total cost price of two items is = 36 +72 = 108

So, by allegation method

20 50

36.11

13.89 16.11

So, the ratio of the cost price of the item is 1389:1611

So, the Cost price of the lowest item is 108 * 1389 / 3000 = 50

So, required selling price is 50 * 125/100 = Rs. 62.5

11. Questions

The total capacity of each dispenser is 120 litres.

Total quantity of petrol sold on Monday is [120 *80 /100] *3 = 96 *3 = 288 liters

The total quantity of petrol in the dispensers at the end of Monday is 120 * 3 – 288 = 72 litres

Total quantity of diesel sold on Monday is [120 * 70/100] *3 = 252 liters

The total quantity of diesel in the dispensers at the end of Monday is 120*3 – 252 = 108 litres

Similarly, we can calculate others' values also.

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Day	The total quantity of petrol sold (litres)	Left quantity of petrol at end of the day (liters)	The total quantity of diesel sold (liters)	Left quantity of diesel at the end of the day (liters)
Monday	288	72	252	108
Tuesday	324	36	288	72
Wednesday	270	90	216	144
Thursday	288	72	324	36
Friday	252	108	288	72

Answer: B

I). The average quantity of diesel left on Thursday and Friday together is [72+36]/2=108/2=54.

II). The Sum of fuel left on Thursday is 72 + 36 =108.

III). Difference in the quantity of fuel sold on Friday is 288 – 252 = 36.

12. Questions

The total capacity of each dispenser is 120 litres.

Total quantity of petrol sold on Monday is [120 *80 /100] *3 = 96 *3 = 288 liters

The total quantity of petrol in the dispensers at the end of Monday is 120 * 3 – 288 = 72 litres

Total quantity of diesel sold on Monday is [120 * 70/100] *3 = 252 liters

The total quantity of diesel in the dispensers at the end of Monday is 120*3 – 252 = 108 litres

Similarly, we can calculate others' values also.

Day	The total quantity of petrol sold (litres)	Left quantity of petrol at end of the day (liters)	The total quantity of diesel sold (liters)	Left quantity of diesel at the end of the day (liters)
Monday	288	72	252	108
Tuesday	324	36	288	72
Wednesday	270	90	216	144
Thursday	288	72	324	36
Friday	252	108	288	72

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Answer: A

The total quantity of petrol sold on Saturday is 252 * 125/100 = 315

The total quantity of diesel sold on Saturday is 288 * 125/100 = 360

So, the required difference

= [315*105 +360*95] – [252*105 + 288*95] = 13455

13. Questions

The number of male teachers in college D is [145–5]/2 = 70.

The number of female teachers in college D is 70 + 5 = 75.

The number of female teachers in college C is 75 * 80 / 100 = 60.

The number of male teachers in college C is 60 * 133.33 / 100 = 80.

The number of male teachers in college E is 80 + 5 = 85.

The number of female teachers in college E is 85 - 13 = 72.

The number of female teachers in college A is 60*83.34/100=50.

The number of male teachers in college A is 50 * 13/10 = 65.

The number of male teachers in college B is 65 -10 = 55.

The number of female teachers in college B is 55 * 14/11 = 70.

The value of the first blank is 70 - 50 = 20.

The value of the second blank is 70 - 55 = 15.

The value of the third blank is 80 + 70 = 150.

The value of the 4th blank is = 85 + 72 = 157.

Answer: E

For the value of B,

The total number of teachers in college F is

= 80*120/100 + 60*130/100 =96 + 78 = 174

So, it is true. Similarly, we can check others' values also.

14. Questions

The number of male teachers in college D is [145–5]/2 = 70.

The number of female teachers in college D is 70 + 5 = 75.

The number of female teachers in college C is 75 * 80 / 100 = 60.

The number of male teachers in college C is 60 * 133.33 / 100 = 80.

The number of male teachers in college E is 80 + 5 = 85.

The number of female teachers in college E is 85 - 13 = 72.

The number of female teachers in college A is 60*83.34/100=50.

The number of male teachers in college A is 50 * 13/10 = 65.

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The number of male teachers in college B is 65 -10 = 55.

The number of female teachers in college B is 55 * 14/11 = 70.

The value of the first blank is 70 - 50 = 20.

The value of the second blank is 70 - 55 = 15.

The value of the third blank is 80 + 70 = 150.

The value of the 4th blank is = 85 + 72 = 157.

Answer: D

The number of female lecturers in C is 60 - 80*40/100 = 28

The number of female lecturers in D is 28*125/100= 35

The number of males asst prof in D is [75-35] * 120/100 = 48

15. Questions

The number of male teachers in college D is [145–5]/2 = 70.

The number of female teachers in college D is 70 + 5 = 75.

The number of female teachers in college C is 75 * 80 / 100 = 60.

The number of male teachers in college C is 60 * 133.33 / 100 = 80.

The number of male teachers in college E is 80 + 5 = 85.

The number of female teachers in college E is 85 - 13 = 72.

The number of female teachers in college A is 60*83.34/100=50.

The number of male teachers in college A is 50 * 13/10 = 65.

The number of male teachers in college B is 65 -10 = 55.

The number of female teachers in college B is 55 * 14/11 = 70.

The value of the first blank is 70 - 50 = 20.

The value of the second blank is 70 - 55 = 15.

The value of the third blank is 80 + 70 = 150.

The value of the 4th blank is = 85 + 72 = 157.

Answer: B

So, x = [80 + 60] *30/100 - [65+50] * 20/100 = 42 – 23 = 19

So, total age is [85+72] * 2 * 19 = 5966

16. Questions

The number of male teachers in college D is [145–5]/2 = 70.

The number of female teachers in college D is 70 + 5 = 75.

The number of female teachers in college C is 75 * 80 / 100 = 60.

The number of male teachers in college C is 60 * 133.33 / 100 = 80.

The number of male teachers in college E is 80 + 5 = 85.

The number of female teachers in college E is 85 - 13 = 72.

The number of female teachers in college A is 60*83.34/100=50.

The number of male teachers in college A is 50 * 13/10 = 65.

The number of male teachers in college B is 65 -10 = 55.

The number of female teachers in college B is 55 * 14/11 = 70.

The value of the first blank is 70 - 50 = 20.

The value of the second blank is 70 - 55 = 15.

The value of the third blank is 80 + 70 = 150.

The value of the 4th blank is = 85 + 72 = 157.

Answer: A

The value of the first blank is 20.

The value of the second blank is 15.

The value of the third blank is 150.

The value of the fourth blank is 157.

So, sum is 20 + 15 +150 +157 = 342

17. Questions

Answer: E

From I, Let the efficiency of A, B and C be x, y and z.

So, total work is 8*[x + y + z]

Or also we can say, total work 5x + 18z.

From II,

2z=[x + y], or z=[x + y]/2

Total work is 30y.

So, from I and II,

8x+8y+8z=30y

Or, 8x+8z=22y

And also, 5x+18z=30y

Or, 5x+18z=30y

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Or, 5x + 9 [x+y] = 30y

Or, 5x +9x =21y

Or, 14x = 21y

Or, x/y = 3/2

So, A alone complete the work in 30 * 2 /3 = 20 days

So, 2z=3y/2 + y = 5y/2

Or, z/y = 5/4

So, Z alone complete the work in 30 * 4/5=24 days

So, The data given in both statements I and II together are necessary to answer the question.

18. Questions

Answer: B

From I, Ages of A and B are 3x and 4x.

Let, the age of C be y.

So, y+5+4x+3=83

From II,

Let the age of A be m and, the age of B is n.

So, m-10=n/2

Or, 2m-n=20

And also, n+5+m+3=78

Or, m+n=70

By solving the above equations, we can easily get the age of A and B.

So, statement II alone is sufficient.

19. Questions

Answer: D

From I,

The number of students who like only painting and photography but not gardening is 20.

Number of students who like only gardening and photography but not painting is 30.

Number of students who like only gardening is 25.

From II,

The number of students who like only painting is double of the number of students who like painting and gardening but not photography which is equal to 15.

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So, Number of students who like only painting is 15* 2= 30

From both I and II,

The number of students who like only photography is double of the number of students who like both photography and painting not gardening is 20*2=40

From I and II we cannot calculate the number of students who like all three, so we cannot calculate the number of students who like photography.

20. Questions

Answer: E

From I,

Let the speed of boats A and B be x km/hr and y km/hr respectively.

Distance between P and Q is [x+3]*3 + [y-3]*3

Or, [x+3]*3 = [y-3]*3

Or, x – y = -6

From II,

Let the speed of the stream of another river is z km/hr.

So, [x+z]*3 = 84 and [y+3]*3 = 90

So, [x+z]=28,

And y = 30 -3 = 27

So, from I and II,

Boat B cover [27-3]*3=72 km in 3 hours

So, the total distance between P and Q is 72*2=144 km

So, The data given in both statements I and II together are necessary to answer the question.

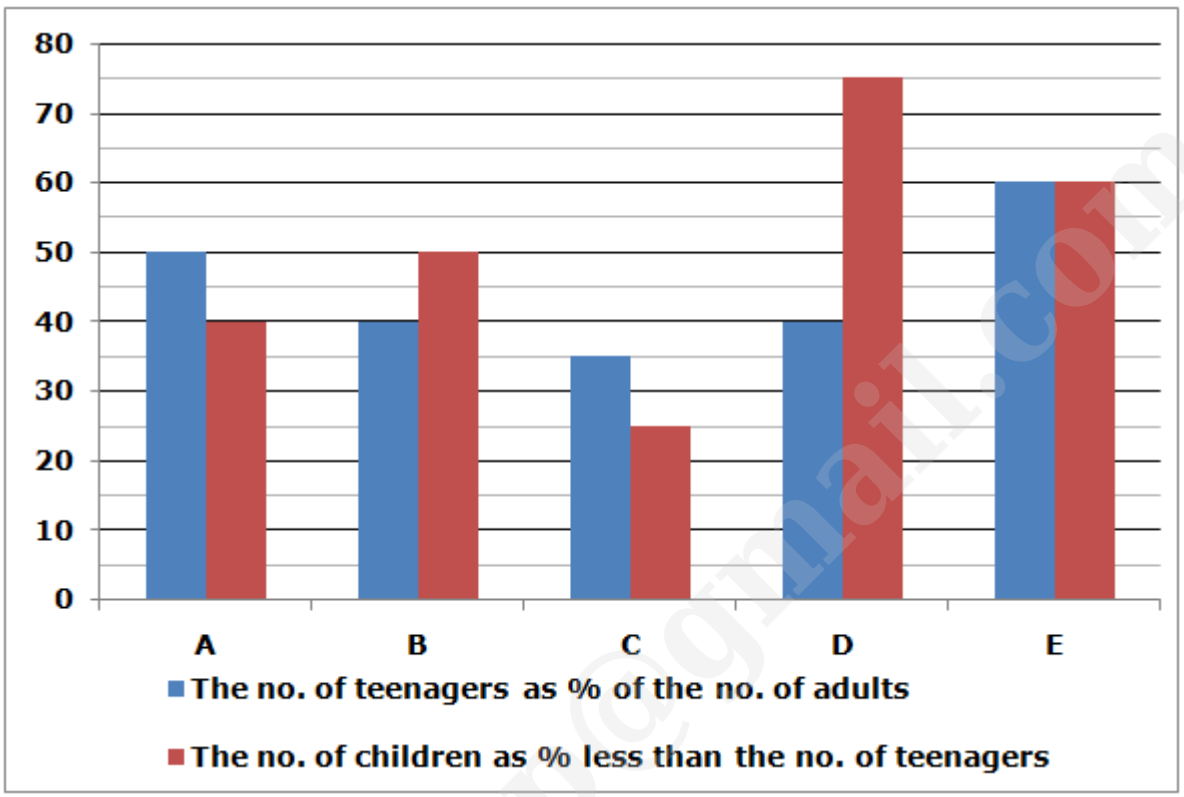
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1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of persons (children + teenagers + adults) in five buildings A, B, C, D and E.

The bar graph given below shows the following data:



If the number of children in building C is 21 and the ratio of the number of adults in building C to those in building D is 8:9, then what will be the average of the number of teenagers in buildings C and D?

a. 34

b. 30

c. 35

d. 31

e. 32

2. Questions

If the ratio of the number of adults in building B to those in building E is 3:5 and the difference between the number of teenagers in buildings B and E is 27, then what will be the difference between the number of children in buildings B and E?

a. 12

b. 5

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c. 15

d. 9

e. 8

3. Questions

In which building, the percentage of children is maximum?

a. A

b. B

c. C

d. D

e. E

4. Questions

If the average of the number of adults, the number of teenagers, and the number of children in building A is 42, then find the difference between the number of adults and teenagers in building A.

a. 35

b. 21

c. 28

d. 42

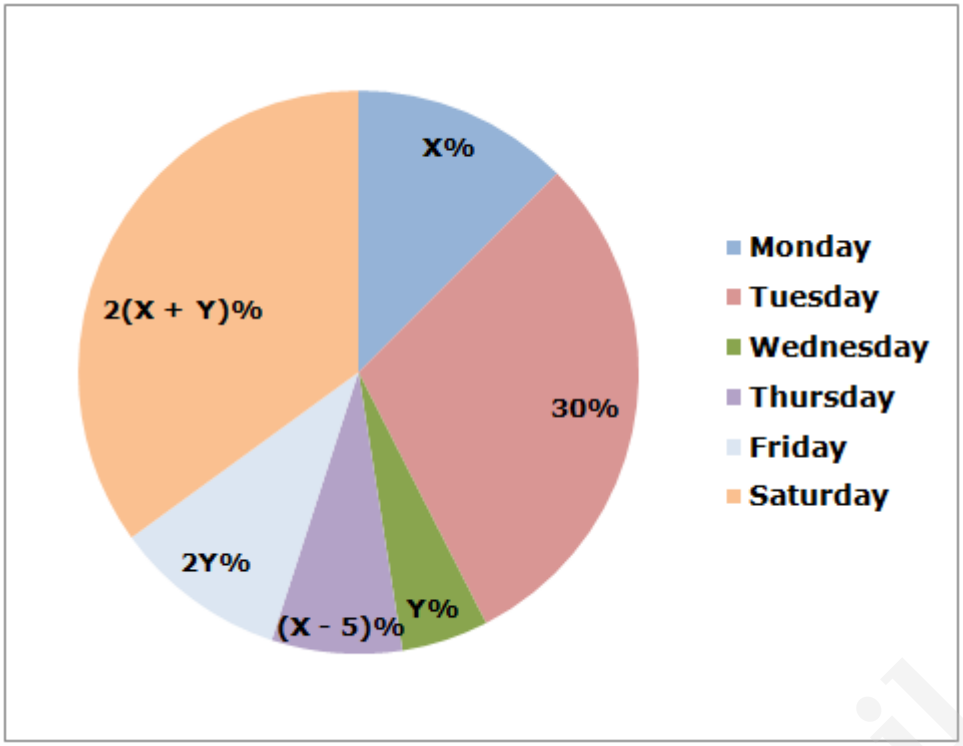
e. 14

5. Questions

Study the following data carefully and answer the questions:

The pie chart given below shows the percentage distribution of the total number of students in a class who submitted their projects in six days, from Monday to Saturday.

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Note:

If the number of students, who submitted their projects on Wednesday is 12, then the number of students, who submitted their projects on Saturday, will be 84.

If the number of students who submitted their projects on Monday is 30, the ratio of boys to girls who submitted their projects on Friday is 2:1, and the ratio of boys to girls who submitted their projects on Saturday is 2:5, then find the average number of girls who submitted their projects on Friday and Saturday.

a. 32

b. 28

c. 35

d. 29

e. 34

6. Questions

If the number of boys who submitted their projects on Tuesday is 48, which is (4X + 6Y)% of the number of girls who submitted their projects on Tuesday, and the number of boys who submitted their projects on Wednesday is 2 more than the number of girls who submitted their projects on Wednesday, then find the ratio of the number of girls who submitted their projects on Tuesday to the number of girls who submitted their projects on Wednesday.

a. 10: 3

b. 15: 2

c. 6: 1

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d. 12: 5

e. 5: 1

7. Questions

If the number of students who submitted their projects on Thursday is (2X – 1), then find the number of students who submitted their projects on Saturday.

a. (10X – 2Y – 3)

b. (10X – Y)

c. (6X + 7Y + 3)

d. (8X + 2Y)

e. None of these

8. Questions

Which of the following statements is/are true?

P: The number of students who submitted their projects on Thursday is 2X% of those who submitted their projects on Tuesday.

Q: The ratio of the number of students, who submitted their projects on Monday , to those, who submitted their projects on Friday, is 5: 4.

R: (4X – 7Y)% of the total number of students in the class submitted their projects on Wednesday.

a. Only Q

b. All are true

c. Only P and Q

d. Only P and R

e. Only R

9. Questions

Study the following data carefully and answer the questions:

The data given below is related to maximum temperature (in °C), minimum temperature (in °C), and wind speed (in km/h) in two states (A and B) on four days: Monday, Tuesday, Wednesday, and Thursday.

The table given below shows the following data:

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Day	Maximum temperature as % of minimum temperature in state A	Maximum temperature as % of minimum temperature in state B
Monday	180%	125%
Tuesday	140%	140%
Wednesday	150%	120%
Thursday	175%	130%

Notes:

1: On each day, wind speed (in km/h) in state A is half of the difference between the maximum and minimum temperature (in °C).

2: On each day, wind speed (in km/h) in state B is double of the difference between the maximum and minimum temperature (in °C).

If the wind speed in state A on Tuesday is 6 km/h and the minimum temperature in state A on Tuesday is 25% more than that of state B on Monday, then find the wind speed in state B on Monday.

a. 14 km/h

b. 8 km/h

c. 12 km/h

d. 16 km/h

e. 10 km/h

10. Questions

If the wind speed in state B on Wednesday is 8 km/h and the average of maximum temperatures in state B on Tuesday and Wednesday is 29.5°C , then find the average of wind speeds in state B on Tuesday and Wednesday.

a. 12 km/h

b. 14 km/h

c. 16 km/h

d. 18 km/h

e. 20 km/h

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11. Questions

If the maximum temperature of state A on Thursday is 10 °C more than that of state B on Thursday and the minimum temperature of state A on Thursday is 2 °C less than that of state B on Thursday, then find the ratio of wind speed in state A on Thursday to that in state B on Thursday.

a. 7: 12

b. 35: 48

c. 5: 6

d. 21: 32

e. 11: 18

12. Questions

If the ratio of the maximum temperature of state A on Monday to that on Wednesday is 15:16, then find that the wind speed in state A on Wednesday is what percent of that in state A on Monday.

a. 80%

b. 72%

c. 88%

d. 90%

e. None of these

13. Questions

Study the following data carefully and answer the questions:

The data given below is related to the total number of persons from two societies, A and B, who attended two functions, P and Q. Some attended function P only, some attended function Q only, and the some attended both functions.

The number of persons in society A who attended function P only is equal to those in society B who attended function P only. The number of people in society A who attended function Q only is double that of those in society B who attended function Q only. The total number of persons in society B who attended function Q is 30, which is 37.5% of those in society A who attended function Q. The total number of persons in society B is 75, and the total number of persons in society A who attended only one function is 65.

If M% of the total number of persons in society A attended function P only and N% of the total number of persons in society B attended function P only, then what percentage of the total number of persons in society A attended function Q only?

a. (N – M – 14)%

b. (N – M – 8)%

c. (N – M – 10)%

d. (N – M – 12)%

e. (N – M – 6)%

14. Questions

If the total number of females in society A who attended function P is equal to the total number of

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persons in society B who attended function P and the number of males in society B who attended function P is 40, then find the total number of females in societies A and B together who attended function P.

a. 80

b. 75

c. 95

d. 90

e. 85

15. Questions

What percent of the total number of persons in both societies attended both functions P and Q?

a. 24%

b. 32%

c. 20%

d. 50%

16. Questions

If the ratio of males to females of society A, who attended both the functions, is 3: 2, the ratio of males to females of society A, who attended function Q only, is 7: 3, then ratio of males to females of society B, who attended both the functions, is 1: 3 and the ratio of males to females of society B, who attended function Q only, is 1: 1, then find the ratio of the total number of males of both the societies together, who attended function Q, to the total number of females of both the societies together, who attended function Q?

a. 1: 1

b. 8: 3

c. 6: 5

d. 1: 3

e. Can't be determined

17. Questions

Study the following data carefully and answer the questions:

Three series I, II and III are given below:

I: 324, 260, 211, 175, 150, 134, ?

II: ?, 124, 173, 198, 207, 211

III: 5, 21, 53, 117, ?, 501

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The missing terms in series I, II and III are P, Q and R respectively.

Which of the following statements is/are true?

X: Difference of P and R is divisible by Q.

Y: Value of (R + 100)/Q is divisible by P.

Z: R is divisible by [(P/5) + Q^2].

a. Only X

b. Only X and Y

c. Only X and Z

d. Only Y and Z

e. Only Y

18. Questions

If a series is formed, whose 1st term is Q^2, 2nd term is (Q^2 + 7), 3rd term is (P/5), 4th term is [(P/5) + 11] and 5th term is (R/5), then find the 6th term of the series.

a. Q^4

b. (R – P + 1)

c. (P – 25)

d. (P + Q)/2

e. [(P/5) + 8Q]

19. Questions

Study the following data carefully and answer the questions:

Two series I and II are given below, with the missing term P in series I and the wrong term Q in series II.

I: 11, 19, 37, 69, 119, ?, 289

II: 19, 140, 40, 121, 57, 116, 70

If a new series is formed, whose 1st term is (P – Q) and it follows the same pattern as in series I, then find that the 4th term of the new series will be divisible by which of the following?

a. 19

b. 13

c. 23

d. 17

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e. 15

20. Questions

If a new series is formed, whose 1st term is (P + Q), and it follows the same pattern as in series II, then find the average of the first four terms of the new series.

a. 348

b. 398

c. 378

d. 358

e. 368

Explanations:

1. Questions

Let the number of adults in buildings A, B, C, D and E be 100a, 100b, 100c, 100d and 100e respectively.

Building	No. of adults	No. of teenagers	No. of children
A	100a	50% of 100a = 50a	60% of 50a = 30a
B	100b	40% of 100b = 40b	50% of 40b = 20b
C	100c	35% of 100c = 35c	75% of 35c = 26.25c
D	100d	40% of 100d = 40d	25% of 40d = 10d
E	100e	60% of 100e = 60e	40% of 60e = 24e

Answer: E

Since, the number of children in building C = 21

So, the number of teenagers in building C = 21 * 100/(100 – 25) = 28

And the number of adults in building C = 28 * (100/35) = 80

Since, the number of adults in building D = 80 * (9/8) = 90

So, the number of teenagers in building D = 40% of 90 = 36

Required average = (28 + 36)/2 = 32

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2. Questions

Let the number of adults in buildings A, B, C, D and E be 100a, 100b, 100c, 100d and 100e respectively.

Building	No. of adults	No. of teenagers	No. of children
A	100a	50% of 100a = 50a	60% of 50a = 30a
B	100b	40% of 100b = 40b	50% of 40b = 20b
C	100c	35% of 100c = 35c	75% of 35c = 26.25c
D	100d	40% of 100d = 40d	25% of 40d = 10d
E	100e	60% of 100e = 60e	40% of 60e = 24e

Answer: D

Let the number of adults in buildings B and E be 30x and 50x respectively.

So, the number of teenagers in building B = 40% of 30x = 12x

And the number of teenagers in building E = 60% of 50x = 30x

Since,

30x – 12x = 27

So,

x = 1.5

Since, the number of teenagers in building B = 12 * 1.5 = 18

So, the number of children in building B = (100 – 50)% of 18 = 9

Since, the number of teenagers in building E = 30 * 1.5 = 45

So, the number of children in building E = (100 – 60)% of 45 = 18

Required difference = 18 – 9 = 9

3. Questions

Let the number of adults in buildings A, B, C, D and E be 100a, 100b, 100c, 100d and 100e respectively.

Building	No. of adults	No. of teenagers	No. of children
A	100a	50% of 100a = 50a	60% of 50a = 30a
B	100b	40% of 100b = 40b	50% of 40b = 20b
C	100c	35% of 100c = 35c	75% of 35c = 26.25c
D	100d	40% of 100d = 40d	25% of 40d = 10d
E	100e	60% of 100e = 60e	40% of 60e = 24e

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Answer: A

Let the number of adults in buildings A, B, C, D and E be 100a, 100b, 100c, 100d and 100e respectively.
Percentage of children in building A = [30a/(100a + 50a + 30a)] * 100 = 16.67%

Percentage of children in building B = [20b/(100b + 40b + 20b)] * 100 = 12.5%

Percentage of children in building C = [26.25c/(100c + 35c + 26.25c)] * 100 = 16.3% (approx.)

Percentage of children in building D = [10d/(100d + 40d + 10d)] * 100 = 6.67%

Percentage of children in building E = [24e/(100e + 60e + 24e)] * 100 = 13% (approx.)

So, the percentage of children is maximum in building A.

4. Questions

Let the number of adults in buildings A, B, C, D and E be 100a, 100b, 100c, 100d and 100e respectively.

Building	No. of adults	No. of teenagers	No. of children
A	100a	50% of 100a = 50a	60% of 50a = 30a
B	100b	40% of 100b = 40b	50% of 40b = 20b
C	100c	35% of 100c = 35c	75% of 35c = 26.25c
D	100d	40% of 100d = 40d	25% of 40d = 10d
E	100e	60% of 100e = 60e	40% of 60e = 24e

Answer: A

Let the number of adults in building A = 100a

So, the number of teenagers in building A = 50% of 100a = 50a

And the number of children in building A = 60% of 50a = 30a

Since, the average of the number of adults, the number of teenagers and the number of children in building A is 42.

So,

100a + 50a + 30a = 3 * 42

a = 0.7

And the required difference = (100 – 50) * 0.7 = 35

5. Questions

From the pie chart:

X + 30 + Y + (X – 5) + 2Y + 2(X + Y) = 100

4X + 5Y + 25 = 100

4X + 5Y = 75 ----------(1)

If the number of students who submitted their projects on Wednesday is 12, then the number of students who submitted their projects on Saturday will be 84.

Since, the ratio of the students, who submitted their projects on Wednesday, to those, who submitted their projects on Saturday = 12: 84 = 1: 7

So,

Y/2(X + Y) = 1/7

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7Y = 2X + 2Y

5Y = 2X ----------(2)

From equations (1) and (2):

4X + 2X = 75

X = 12.5 and Y = 5

Day	% distribution of the no. of students, who submitted their projects on given days
Monday	12.5%
Tuesday	30%
Wednesday	5%
Thursday	12.5 – 5 = 7.5%
Friday	2 * 5 = 10%
Saturday	2(12.5 + 5) = 35%

Answer: E

Since, the number of students, who submitted their projects on Mon = 30

So, the number of students, who submitted their projects on Fri = 30 * (10/12.5) = 24

And the number of students, who submitted their projects on Sat = 30 * (35/12.5) = 84

Since, the number of girls, who submitted their projects on Fri = 24 * (1/3) = 8

And the number of girls, who submitted their projects on Sat = 84 * (5/7) = 60

So, the required average = (8 + 60)/2 = 34

6. Questions

From the pie chart:

X + 30 + Y + (X – 5) + 2Y + 2(X + Y) = 100

4X + 5Y + 25 = 100

4X + 5Y = 75 ----------(1)

If the number of students who submitted their projects on Wednesday is 12, then the number of students who submitted their projects on Saturday will be 84.

Since, the ratio of the students, who submitted their projects on Wednesday, to those, who submitted their projects on Saturday = 12: 84 = 1: 7

So,

Y/2(X + Y) = 1/7

7Y = 2X + 2Y

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5Y = 2X ----------(2)

From equations (1) and (2):

4X + 2X = 75

X = 12.5 and Y = 5

Day	% distribution of the no. of students, who submitted their projects on given days
Monday	12.5%
Tuesday	30%
Wednesday	5%
Thursday	12.5 – 5 = 7.5%
Friday	2 * 5 = 10%
Saturday	2(12.5 + 5) = 35%

Answer: B

Since, the number of boys, who submitted their projects on Tuesday = 48

So, the number of girls, who submitted their projects on Tuesday :

48 * 100/(4X + 6Y) = 48 * 100/(50 + 30) = 60

And the number of students, who submitted their projects on Tuesday = 48 + 60 = 108

Since, the number of students, who submitted their projects on Wednesday = 108 * (5/30) = 18

So, the number of girls, who submitted their projects on Wednesday = (18 – 2)/2 = 8

And the required ratio = 60: 8 = 15: 2

7. Questions

From the pie chart:

X + 30 + Y + (X – 5) + 2Y + 2(X + Y) = 100

4X + 5Y + 25 = 100

4X + 5Y = 75 ----------(1)

If the number of students who submitted their projects on Wednesday is 12, then the number of students who submitted their projects on Saturday will be 84.

Since, the ratio of the students, who submitted their projects on Wednesday, to those, who submitted their projects on Saturday = 12: 84 = 1: 7

So,

Y/2(X + Y) = 1/7

7Y = 2X + 2Y

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5Y = 2X ----------(2)

From equations (1) and (2):

4X + 2X = 75

X = 12.5 and Y = 5

Day	% distribution of the no. of students, who submitted their projects on given days
Monday	12.5%
Tuesday	30%
Wednesday	5%
Thursday	12.5 – 5 = 7.5%
Friday	2 * 5 = 10%
Saturday	2(12.5 + 5) = 35%

Answer: A

Since, the number of students, who submitted their projects on Thu:

(2X – 1) = (2 * 12.5 – 1) = 24

So, the number of students, who submitted their projects on Sat:

24 * (35/7.5) = 112 = (10X – 2Y – 3)

8. Questions

From the pie chart:

X + 30 + Y + (X – 5) + 2Y + 2(X + Y) = 100

4X + 5Y + 25 = 100

4X + 5Y = 75 ----------(1)

If the number of students who submitted their projects on Wednesday is 12, then the number of students who submitted their projects on Saturday will be 84.

Since, the ratio of the students, who submitted their projects on Wednesday, to those, who submitted their projects on Saturday = 12: 84 = 1: 7

So,

Y/2(X + Y) = 1/7

7Y = 2X + 2Y

5Y = 2X ----------(2)

From equations (1) and (2):

4X + 2X = 75

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X = 12.5 and Y = 5

Day	% distribution of the no. of students, who submitted their projects on given days
Monday	12.5%
Tuesday	30%
Wednesday	5%
Thursday	12.5 – 5 = 7.5%
Friday	2 * 5 = 10%
Saturday	2(12.5 + 5) = 35%

Answer: C

From P:

Since, the ratio of the number of students, who submitted their projects on Tue, to those, who submitted their projects on Thu = 30: 7.5 = 4: 1

So, the required percentage = (1/4) * 100 = 25% = 2X%

So, P is true.

From Q:

Ratio of the number of students, who submitted their projects on Mon, to those, who submitted their projects on Fri = 12.5: 10 = 5: 4

So, Q is true.

From R:

Since, the percentage distribution of the number of students, who submitted their projects on Wed = 5%

And, (4X – 7Y) = (4 * 12.5 – 7 * 5) = 15

So, R is not true.

Hence, only P and Q are true.

9. Questions

Day	Ratio of maximum temperature to minimum temperature in state A	Ratio of maximum temperature to minimum temperature in state B
Monday	180: 100 = 9: 5	125: 100 = 5: 4
Tuesday	140: 100 = 7: 5	140: 100 = 7: 5
Wednesday	150: 100 = 3: 2	120: 100 = 6: 5
Thursday	175: 100 = 7: 4	130: 100 = 13: 10

Answer: C

Since, the wind speed in state A on Tue = 6 km/h

So, the difference between maximum and minimum temperatures in state A on Tue = 2 * 6 = 12 km/h

And the minimum temperature in state A on Tuesday = 12 * 5/(7 – 5) = 30°C

Now, the minimum temperature in state B on Monday = 30 * (100/125) = 24°C

The maximum temperature in state B on Monday = 125% of 24 = 30°C

And the wind speed in state B on Monday = 2 * (30 – 24) = 12 km/h

10. Questions

Day	Ratio of maximum temperature to minimum temperature in state A	Ratio of maximum temperature to minimum temperature in state B
Monday	180: 100 = 9: 5	125: 100 = 5: 4
Tuesday	140: 100 = 7: 5	140: 100 = 7: 5
Wednesday	150: 100 = 3: 2	120: 100 = 6: 5
Thursday	175: 100 = 7: 4	130: 100 = 13: 10

Answer: B

Since, the wind speed in state B on Wednesday = 8 km/h

So, the difference between the maximum and minimum temperatures of state B on Wednesday = 8/2 = 4 km/h

The maximum temperature of state B on Wednesday = 4 * 6/(6 – 5) = 24°C

And the minimum temperature of state B on Wednesday = 24 – 4 = 20°C

Now, the maximum temperature of state B on Tuesday = (2 * 29.5) – 24 = 35°C

The minimum temperature of state B on Tuesday = 35 * (5/7) = 25°C

And the wind speed of state B on Tuesday = 2 * (35 – 25) = 20 km/h

Required average = (8 + 20)/2 = 14 km/h

11. Questions

Day	Ratio of maximum temperature to minimum temperature in state A	Ratio of maximum temperature to minimum temperature in state B
Monday	180: 100 = 9: 5	125: 100 = 5: 4
Tuesday	140: 100 = 7: 5	140: 100 = 7: 5
Wednesday	150: 100 = 3: 2	120: 100 = 6: 5
Thursday	175: 100 = 7: 4	130: 100 = 13: 10

Answer: A

Let the minimum temperatures of states A and B on Thursday be 100a°C and 100b°C respectively.

So, the maximum temperature of state A on Thursday = 175% of 100a = 175a°C

And the maximum temperature of state B on Thursday = 130% of 100b = 130b°C

175a – 130b = 10

35a – 26b = 2 ----------(1)

100b – 100a = 2

50b – 50a = 1 ----------(2)

By equation (1) * 10 + equation (2) * 7

350a – 260b + 350b – 350a = 20 + 7

b = 0.3

From equation (2):

15 – 50a = 1

a = 0.28

Since, the difference between maximum and minimum temperatures of state A on Thursday= (175 – 100) * 0.28 = 21°C

So, the wind speed of state A on Thursday = 21/2 = 10.5 km/h

Since, the difference between maximum and minimum temperatures of state B on Thursday= (130 – 100) * 0.3 = 9°C

So, the wind speed of state B on Thu = 2 * 9 = 18 km/h

Required ratio = 10.5: 18 = 7: 12

12. Questions

Day	Ratio of maximum temperature to minimum temperature in state A	Ratio of maximum temperature to minimum temperature in state B
Monday	180: 100 = 9: 5	125: 100 = 5: 4
Tuesday	140: 100 = 7: 5	140: 100 = 7: 5
Wednesday	150: 100 = 3: 2	120: 100 = 6: 5
Thursday	175: 100 = 7: 4	130: 100 = 13: 10

Answer: A

Let the minimum temperatures of state A on Monday and Wednesday be 100x°C and 100y°C.

So, the maximum temperature of state A on Monday = 180% of 100x = 180x°C

And the maximum temperature of state A on Wednesday = 150% of 100y = 150y°C

Since, the ratio of maximum temperature of state A on Monday to that on Wednesday is 15: 16.

So,

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180x/150y = 15/16

x/y = 25/32

The wind speed in state A on Monday = (180x – 100x)/2 = 40x km/h

The wind speed in state A on Wednesday = (150y – 100y)/2 = 25y km/h = 25 * (32x/25) = 32x km/h

Required percentage = (32x/40x) * 100 = 80%

13. Questions

Let the number of persons of society A, who attended function P only = the number of persons of society B, who attended function P only = x

Also let the number of persons of society B, who attended function Q only = y

So, the number of persons of society A, who attended function Q only = 2y

Since, the total number of persons of society B, who attended function Q = 30

So, the total number of persons of society A, who attended function Q:

30 * (100/37.5) = 80

Since, the total number of persons in society B = 75

So,

x + 30 = 75

x = 45

Since, the total number of persons of society A, who attended only one function, is 65.

So,

x + 2y = 65

45 + 2y = 65

y = 10

In society A:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 2 * 10 = 20

The number of persons, who attended both the functions = 80 – 20 = 60

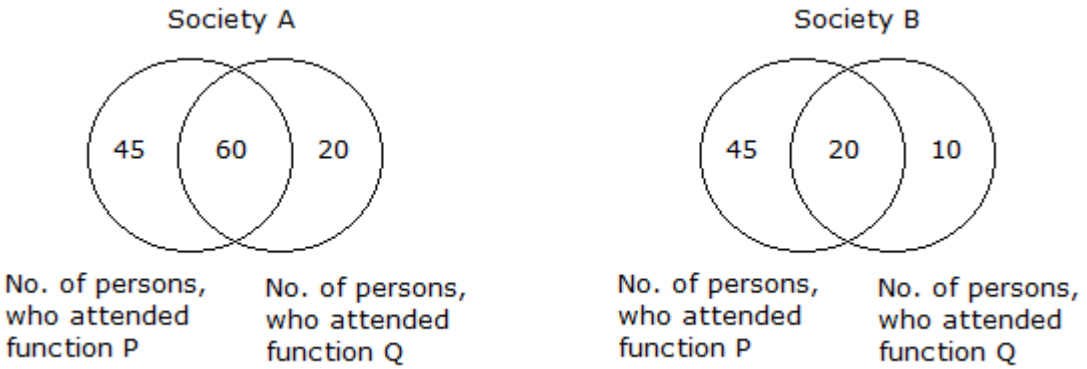
In society B:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 10

The number of persons, who attended both the functions = 30 – 10 = 20

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Answer: B

Since, the total number of persons in society A = 45 + 60 + 20 = 125

And the number of persons of society A, who attended function P only = 45

So, M = (45/125) * 100 = 36%

Since, the total number of persons in society B = 45 + 20 + 10 = 75

And the number of persons of society B, who attended function P only = 45

So, N = (45/75) * 100 = 60%

Since, the number of persons of society A, who attended function Q only = 20

So, the required percentage = (20/125) * 100 = 16% = (N – M – 8)%

14. Questions

Let the number of persons of society A, who attended function P only = the number of persons of society B, who attended function P only = x

Also let the number of persons of society B, who attended function Q only = y

So, the number of persons of society A, who attended function Q only = 2y

Since, the total number of persons of society B, who attended function Q = 30

So, the total number of persons of society A, who attended function Q:

30 * (100/37.5) = 80

Since, the total number of persons in society B = 75

So,

x + 30 = 75

x = 45

Since, the total number of persons of society A, who attended only one function, is 65.

So,

x + 2y = 65

45 + 2y = 65

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y = 10

In society A:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 2 * 10 = 20

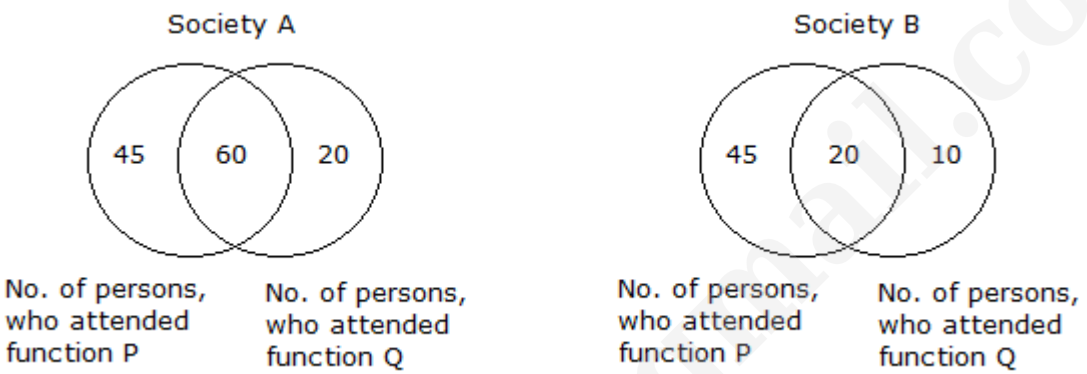
The number of persons, who attended both the functions = 80 – 20 = 60

In society B:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 10

The number of persons, who attended both the functions = 30 – 10 = 20



Answer: D

Since, the total number of persons of society B, who attended function P = 45 + 20 = 65

So, the total number of females of society A, who attended function P = 65

Since, the total number of persons of society A, who attended function P = 45 + 60 = 105

So, the total number of males of society A, who attended function P = 105 – 65 = 40

The total number of males of society B, who attended function P = 40

And the total number of females of society B, who attended function P = 65 – 40 = 25

Required sum = 65 + 25 = 90

15. Questions

Let the number of persons of society A, who attended function P only = the number of persons of society B, who attended function P only = x

Also let the number of persons of society B, who attended function Q only = y

So, the number of persons of society A, who attended function Q only = 2y

Since, the total number of persons of society B, who attended function Q = 30

So, the total number of persons of society A, who attended function Q:

30 * (100/37.5) = 80

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Since, the total number of persons in society B = 75

So,

x + 30 = 75

x = 45

Since, the total number of persons of society A, who attended only one function, is 65.

So,

x + 2y = 65

45 + 2y = 65

y = 10

In society A:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 2 * 10 = 20

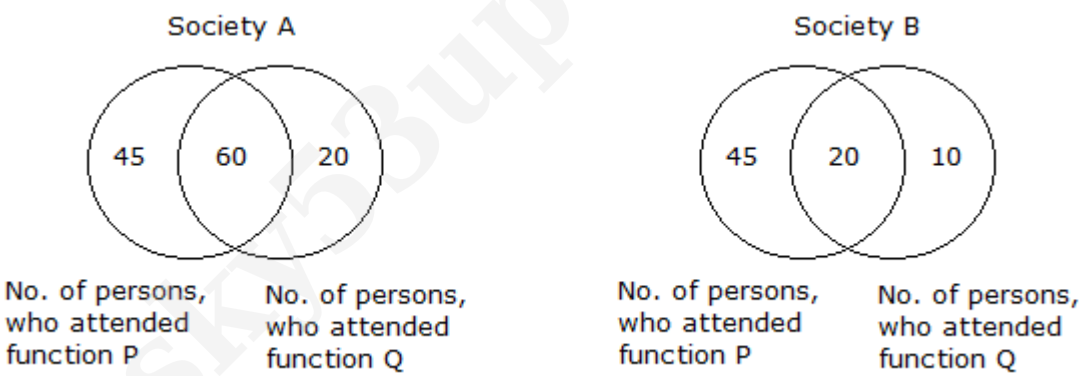
The number of persons, who attended both the functions = 80 – 20 = 60

In society B:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 10

The number of persons, who attended both the functions = 30 – 10 = 20



Answer: B

Since, the total number of persons in both the societies together:

(45 + 60 + 20) + (45 + 20 + 10) = 200

And the total number of persons of both the societies together, who attended both the functions P and Q = 60 + 20 = 80

So, the required percentage = (80/200) * 100 = 40%

16. Questions

Let the number of persons of society A, who attended function P only = the number of persons of society

B, who attended function P only = x

Also let the number of persons of society B, who attended function Q only = y

So, the number of persons of society A, who attended function Q only = 2y

Since, the total number of persons of society B, who attended function Q = 30

So, the total number of persons of society A, who attended function Q:

30 * (100/37.5) = 80

Since, the total number of persons in society B = 75

So,

x + 30 = 75

x = 45

Since, the total number of persons of society A, who attended only one function, is 65.

So,

x + 2y = 65

45 + 2y = 65

y = 10

In society A:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 2 * 10 = 20

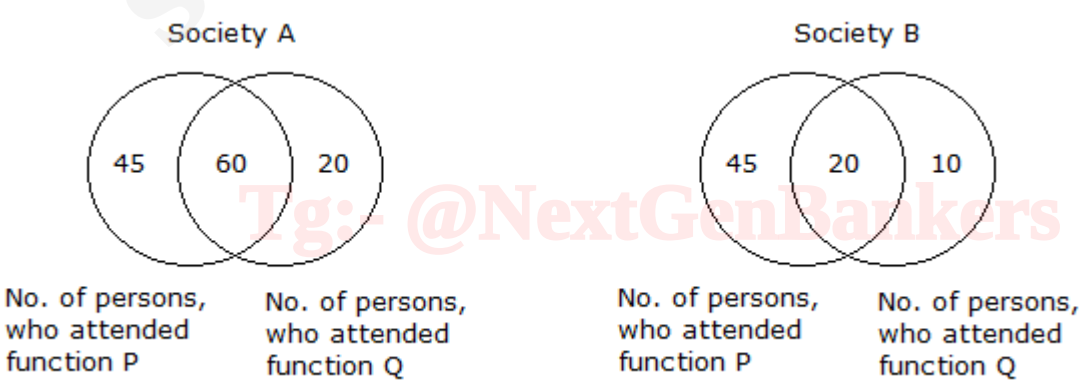
The number of persons, who attended both the functions = 80 – 20 = 60

In society B:

The number of persons, who attended function P only = 45

The number of persons, who attended function Q only = 10

The number of persons, who attended both the functions = 30 – 10 = 20



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Answer: C

The number of males of society A who attended both the functions = 60 * (3/5) = 36

The number of females of society A who attended both the functions = 60 – 36 = 24

The number of males of society A who attended function Q only = 20 * (7/10) = 14

The number of females of society A who attended function Q only = 20 – 14 = 6

The number of males of society B who attended both the functions = 20 * (1/4) = 5

The number of females of society B who attended both the functions = 20 – 5 = 15

The number of males of society B who attended function Q only = 10 * (1/2) = 5

The number of females of society B who attended function Q only = 10 – 5 = 5

The total number of males of societies A and B together, who attended function Q = 36 + 14 + 5 + 5 = 60

The total number of females of societies A and B together, who attended function Q = 24 + 6 + 15 + 5 = 50

Required ratio = 60: 50 = 6: 5

17. Questions

Series I:

$324 - 8^2 = 260$

$260 - 7^2 = 211$

$211 - 6^2 = 175$

$175 - 5^2 = 150$

$150 - 4^2 = 134$

$134 - 3^2 = 125$

So, the missing term of series I = P = 125

Series II:

$3 + 11^2 = 124$

$124 + 7^2 = 173$

$173 + 5^2 = 198$

$198 + 3^2 = 207$

$207 + 2^2 = 211$

So, the missing term of series II = Q = 3

Series III:

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5 + 16 = 21

21 + 32 = 53

53 + 64 = 117

117 + 128 = 245

245 + 256 = 501

So, the missing term of series III = R = 245

Answer: A

From X:

Difference of P and R = 245 – 125 = 120

Since, the value of Q = 3

And 120 is divisible by 3.

So, X is true.

From Y:

Value of (R + 100)/Q = (245 + 100)/3 = 115

Since, the value of P = 125

And 115 is not divisible by 125.

So, Y is not true.

From Z:

Value of $[(P/5) + Q^2] = [(125/5) + 3^3] = 34$

Since, the value of Q = 245

And 245 is not divisible by 34.

So, Z is not true.

Hence, only X is true.

18. Questions

Series I:

$324 - 8^2 = 260$

$260 - 7^2 = 211$

$211 - 6^2 = 175$

$175 - 5^2 = 150$

$150 - 4^2 = 134$

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$134 - 3^2 = 125$

So, the missing term of series I = P = 125

Series II:

$3 + 11^2 = 124$

$124 + 7^2 = 173$

$173 + 5^2 = 198$

$198 + 3^2 = 207$

$207 + 2^2 = 211$

So, the missing term of series II = Q = 3

Series III:

5 + 16 = 21

21 + 32 = 53

53 + 64 = 117

117 + 128 = 245

245 + 256 = 501

So, the missing term of series III = R = 245

Answer: D

The 1st term of the new series = $Q^2 = 3^2 = 9$

The 2nd term of the new series = $(Q^2 + 7) = (9 + 7) = 16$

The 3rd term of the new series = (P/5) + (125/5) = 25

The 4th term of the new series = [(P/5) + 11] = (25 + 11) = 36

The 5th term of the new series = (R/5) = (245/5) = 49

The series: 9, 16, 25, 36, 49

Logic in the series: $3^2, 4^2, 5^2, 6^2, 7^2$

So, the 6th term of the new series = $8^2 = 64 = (P + Q)/2$

19. Questions

Series I:

11 + 2 * 4 = 19

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19 + 2 * 9 = 37

37 + 2 * 16 = 69

69 + 2 * 25 = 119

119 + 2 * 36 = **191**

191 + 2 * 49 = 289

So, the missing term of series I = P = 191

Series II:

19 + 121 = 140

140 – 100 = 40

40 + 81 = 121

121 – 64 = 57

57 + 49 = **106(116)**

106 – 36 = 70

So, the wrong term of series II = Q = 116

Answer: A

The 1st term of the new series = (P – Q) = (191 – 116) = 75

The 2nd term of the new series = 75 + 2 * 4 = 83

The 3rd term of the new series = 83 + 2 * 9 = 101

The 4th term of the new series = 101 + 2 * 16 = 133

So, 133 is divisible by 19 only.

20. Questions

Series I:

11 + 2 * 4 = 19

19 + 2 * 9 = 37

37 + 2 * 16 = 69

69 + 2 * 25 = 119

119 + 2 * 36 = **191**

191 + 2 * 49 = 289

So, the missing term of series I = P = 191

Series II:

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19 + 121 = 140

140 – 100 = 40

40 + 81 = 121

121 – 64 = 57

57 + 49 = **106(116)**

106 – 36 = 70

So, the wrong term of series II = Q = 116

Answer: E

The 1st term of the new series = (P + Q) = (191 + 116) = 307

The 2nd term of the new series = 307 + 121 = 428

The 3rd term of the new series = 428 – 100 = 328

The 4th term of the new series = 328 + 81 = 409

Required average = (307 + 428 + 328 + 409)/4 = 368

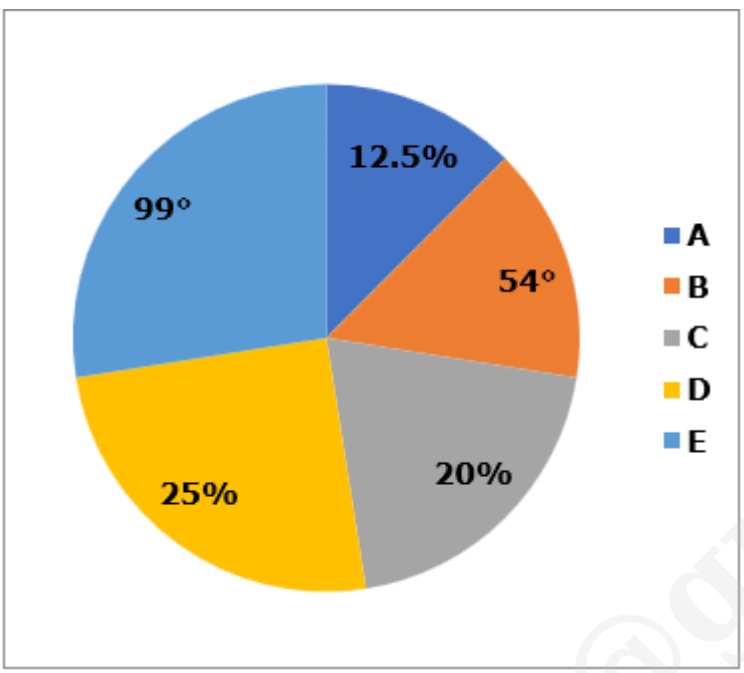
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1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of candidates selected for five different posts A, B, C, D and E.

The pie chart given below shows the percentage distribution and the degree distribution of the number of candidates selected for each of the given posts.



Notes:

1: The number of candidates selected for post A is 18% of the number of candidates applied for post A.

2: The number of candidates selected for post B is 43.2% of the number of candidates applied for post B.

3: The difference between the number of candidates, who applied for posts A and B, is 250.

If the number of male candidates selected for post A is 18 more than the number of female candidates selected for post A and the number of male candidates selected for post B is 36 less than the number of female candidates selected for post B, then find the ratio of the total number of male candidates to the total number of female candidates selected for posts A and B together?

a. 9:10

b. 18:19

c. 5:6

d. 6:7

e. 15:16

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2. Questions

The number of candidates selected for post E is 41.25% of the number of candidates applied for post E. If the number of male candidates applied for post E is 300, out of which 39% were selected

for post E, then find that what per cent of the number of female candidates, who applied for post E, was selected for post E?

a. 54%

b. 45%

c. 36%

d. 66%

e. 48%

3. Questions

48% of the number of candidates, who applied for post C, was selected for post C and 45% of the number of candidates, who applied for post D, was selected for post D. If the number of male candidates, who applied for posts C and D, are 125 and 175 respectively, then find the average number of female candidates, who applied for posts C and D?

a. 200

b. 225

c. 175

d. 150

e. 250

4. Questions

If the difference between the number of candidates, who selected for posts E and F, is three times of the difference between the number of candidates, who selected for posts A and B and the number of candidates selected for post F, is not a multiple of 72, then find the number of candidates selected for post F?

a. 234

b. 180

c. 162

d. 252

e. 216

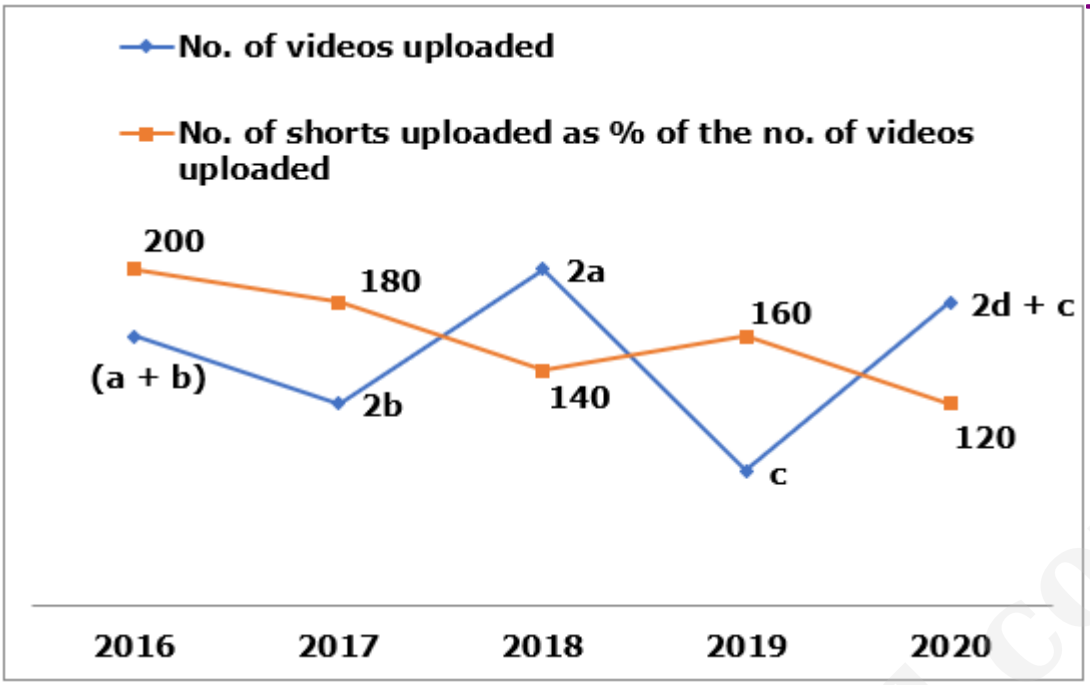
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5. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of videos (long videos) and the number of shorts (short videos) uploaded by a YouTube channel in five different years 2016, 2017, 2018, 2019 and 2020.

The line graph given below shows the number of videos uploaded and also shows the number of shorts uploaded as per cent of the number of videos uploaded in each of the given year.



Notes:

1: The number of videos uploaded in 2016 is 40 more than that uploaded in 2017.

2: The number of shorts uploaded in 2018 is 280.

3: Difference between the number of videos uploaded in 2019 and 2020, is 'a'.

4: Ratio of 'c' to 'd' is 8: 5 respectively.

Total number of shorts uploaded in 2017 and 2018 together is what per cent more than the total number of videos uploaded in 2017 and 2018 together?

a. [(a/4) + (b/3)]%

b. [(a + c)/3]%

c. [(a + d)/3]%

d. [(c/2) + (b/4)]%

e. [(a + 2b + c)/4]%

6. Questions

If the average number of likes on each short uploaded in 2019 is 2 lakhs and the average number of likes on each short uploaded in 2020 is 1.5 lakhs, then find the ratio of total number of likes on all the shorts together uploaded in 2019 to that on all the shorts together uploaded in 2020?

a. 32:45

b. 64:81

c. 16:27

d. 128:151

e. 8:13

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7. Questions

Find the ratio between the number of videos uploaded in those two years, in which the number of shorts uploaded are equal?

a. 4:5

b. 2:3

c. 5:2

d. 4:9

e. 2:1

8. Questions

If the number of videos uploaded in 2021 is 2.5(c – d)% more than that uploaded in 2020 and the number of shorts uploaded in 2021 is 1.25(a – b)% more than that uploaded in 2019, then find that the number of shorts uploaded in 2021 is approximately what per cent of the number of videos uploaded in 2021?

a. 69%

b. 51%

c. 55%

d. 65%

e. 61%

9. Questions

Study the following data carefully and answer the questions:

The table given below shows the number of shirts and the number of jeans sold by three shopkeepers A, B and C on five days Mon, Tue, Wed, Thu and Fri.

Days	Shopkeeper A		Shopkeeper B		Shopkeeper C	
	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold
Mon	45	__	25	36	__	24
Tue	72	60	60	__	32	__
Wed	__	48	42	21	27	__
Thu	50	__	__	30	51	34
Fri	41	50	55	__	22	40

Note:

1: Average number of shirts sold by A on each of the given days is 48.

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2: Average number of shirts sold by B on each of the given days is 40.

3: Average number of shirts sold by C on each of the given days is 36.

If the total number of shirts sold by A, B and C together on Wed, is 22 less than the total number of jeans sold by A, B and C together on Wed and C sold jeans at ₹750 per piece, then find the total amount received by C by selling jeans on Wed?

a. ₹ 37500

b. ₹ 45000

c. ₹ 35700

d. ₹ 40500

e. ₹ 53700

10. Questions

If the average number of jeans sold by C on Tue, Thu and Fri is 30 and the total number of jeans sold by B and C together on Tue is 16(2/3)% more than that sold by A on Tue, then find the ratio of the number of shirts to the number of jeans sold by B on Tue respectively?

a. 6:5

b. 12:11

c. 3:2

d. 15:14

e. 10:9

11. Questions

If the average number of jeans sold by A, B and C on Thu is 36 and the total number of shirts sold by A on Mon, Wed and Thu together is 1 less than the total number of jeans sold by him on Mon, Wed and Thu together, then find the average number of jeans sold by A, B and C on Mon?

a. 32

b. 30

c. 28

d. 29

e. 31

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12. Questions

If the difference between the number of shirts sold by B on Wed and Thu is 8 times of the difference between the number of jeans sold by B on Thu and Fri and the average of the number of jeans sold by B on Wed and Fri is a multiple of 9, then find the number of jeans sold by B on Fri?

a. Can't be determined

b. 27

c. 33

d. 36

e. 30

13. Questions

Study the following data carefully and answer the questions:

Data given below is related to the cost price and marked price of four articles A, B, C and D.

The ratio of the cost price of article A to that of article C is 6: 5 respectively and the ratio of the cost price of article B to that of article D is 9: 10 respectively. Article A is marked up by 50% and article B is marked up by 33(1/3)%. Article C is marked up by ₹600 and article D is also marked up by ₹600. The cost price of article B is equal to the marked price of article A and the marked prices articles B and D are equal.

Assume that article C is marked up by 2M%. If article A was sold at M% discount on its marked price and article B was sold at (M – 5)% discount on its marked price, then what would have been the total profit received on articles A and B together?

a. ₹ 960

b. ₹ 720

c. ₹ 840

d. ₹ 560

e. ₹ 800

14. Questions

If article C was sold at ₹N discount on its marked price and article D was sold at ₹2N discount on its marked price, then there would be a total profit of 20% on articles C and D together. Find the value of N?

a. 120

b. 150

c. 80

d. 90

e. 100

15. Questions

Which of the following statements is/are true?

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P: The total cost price of articles A and B together is equal to that of articles C and D together?

Q: The ratio of the marked price of article A to that of article C is 9: 7 respectively.

R: The average marked price of each of the given article is ₹3200.

a. Only P and Q

b. All are true

c. Only Q and R

d. Only Q

e. Only P and R

16. Questions

If each of the given articles was sold at two successive discounts of 5% and ₹150 on their marked prices, then what would have been the total profit per cent on all the four articles together?

a. 25%

b. 33.33%

c. 20%

d. 16.67%

e. 15%

17. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of persons, who watched three movies A, B and C. Some persons watched only one movie, some watched only two movies and remaining watched all the three movies.

The number of persons, who watched movie A only, is 8X, the number of persons, who watched movie B only, is 5X and the number of persons, who watched movie C only, is (X + Y). The number of persons, who watched both movies B and C but not A, is 3Y, which is 50% more than those, who watched both movies A and B but not C and also 75% of those, who watched both movies A and C but not B. The number of persons, who watched all the three movies, is 37.5% of those, who watched movie A only.

Note: Values of X and Y can be different for each question.

If the total number of persons, who watched movie A, is 1000 and the number of persons, who watched both movies B and C but not A, is 225, then what will be the total number of persons, who watched movie B?

a. 775

b. 750

c. 900

d. 625

e. 825

18. Questions

If the ratio of X to Y is 1:1 respectively, then what will be the ratio of the total number of persons, who watched only one movie, to those, who watched only two movies?

a. 6:5

b. 5:3

c. 5:4

d. 2:1

e. 15:13

19. Questions

If the number of persons, who watched movie C only, is 65 and the ratio of the number of persons, who watched both movies A and B but not C, to those, who watched all the three movies, is 5: 12 respectively, then what will be the number of persons, who watched all the three movies?

a. 150

b. 108

c. 180

d. 144

e. 120

20. Questions

If the ratio of the total number of persons, who watched only one movies, to those, who watched at-least two movies, is 74: 51 respectively, then what will be the ratio of the total number of persons, who watched only one movie, to those, who watched all the three movies?

a. 37:5

b. 74:5

c. Can't be determined

d. 74:15

e. 37:15

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Explanations:

1. Questions

Percentage distribution of the number of candidates selected for post A = 12.5%

Percentage distribution of the number of candidates selected for post B = (54/360) * 100 = 15%

Percentage distribution of the number of candidates selected for post C = 20%

Percentage distribution of the number of candidates selected for post D = 25%

Percentage distribution of the number of candidates selected for post E = (99/360) * 100 = 27.5%

Let the number of candidates selected for posts A, B, C, D and E be 12.5x, 15x, 20x, 25x and 27.5x respectively.

Since, the number of candidates selected for post A is 18% of the number of candidates applied for post A.

So, the number of candidates applied for post A = 12.5x * (100/18) = (625x/9)

Since, the number of candidates selected for post B is 43.2% of the number of candidates applied for post B.

So, the number of candidates applied for post B = 15x * (100/43.2) = (625x/18)

Since, the difference between the number of candidates, who applied for posts A and B, is 250.

So,

(625x/9) – (625x/18) = 250

1250x – 625x = 4500

x = 7.2

Post	No. of candidates selected for the post
A	12.5 * 7.2 = 90
B	15 * 7.2 = 108
C	20 * 7.2 = 144
D	25 * 7.2 = 180
E	27.5 * 7.2 = 198

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Answer: C

Since, the total number of candidates selected for post A = 90

So, the number of male candidates selected for post A = (90 + 18)/2 = 54

And the number of female candidates selected for post A = 90 – 54 = 36

Since, the total number of candidates selected for post B = 108

So, the number of male candidates selected for post B = (108 – 36)/2 = 36

And the number of female candidates selected for post B = 108 – 36 = 72

Since, the total number of male candidates selected for posts A and B = 54 + 36 = 90

And the total number of female candidates selected for posts A and B = 36 + 72 = 108

So, the required ratio = 90:108 = 5:6

2. Questions

Percentage distribution of the number of candidates selected for post A = 12.5%

Percentage distribution of the number of candidates selected for post B = (54/360) * 100 = 15%

Percentage distribution of the number of candidates selected for post C = 20%

Percentage distribution of the number of candidates selected for post D = 25%

Percentage distribution of the number of candidates selected for post E = (99/360) * 100 = 27.5%

Let the number of candidates selected for posts A, B, C, D and E be 12.5x, 15x, 20x, 25x and 27.5x respectively.

Since, the number of candidates selected for post A is 18% of the number of candidates applied for post A.

So, the number of candidates applied for post A = 12.5x * (100/18) = (625x/9)

Since, the number of candidates selected for post B is 43.2% of the number of candidates applied for post B.

So, the number of candidates applied for post B = 15x * (100/43.2) = (625x/18)

Since, the difference between the number of candidates, who applied for posts A and B, is 250.

So,

(625x/9) – (625x/18) = 250

1250x – 625x = 4500

x = 7.2

Post	No. of candidates selected for the post
A	12.5 * 7.2 = 90
B	15 * 7.2 = 108
C	20 * 7.2 = 144
D	25 * 7.2 = 180
E	27.5 * 7.2 = 198

Answer: B

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Since, the number of candidates selected for post E = 198

So, the number of candidates applied for post E = 198 * (100/41.25) = 480

Since, the number of male candidates applied for post E = 300

So, the number of female candidates applied for post E = 480 – 300 = 180

Since, the number of male candidates selected for post E = 39% of 300 = 117

So, the number of female candidates selected for post E = 198 – 117 = 81

Required percentage = (81/180) * 100 = 45%

3. Questions

Percentage distribution of the number of candidates selected for post A = 12.5%

Percentage distribution of the number of candidates selected for post B = (54/360) * 100 = 15%

Percentage distribution of the number of candidates selected for post C = 20%

Percentage distribution of the number of candidates selected for post D = 25%

Percentage distribution of the number of candidates selected for post E = (99/360) * 100 = 27.5%

Let the number of candidates selected for posts A, B, C, D and E be 12.5x, 15x, 20x, 25x and 27.5x respectively.

Since, the number of candidates selected for post A is 18% of the number of candidates applied for post A.

So, the number of candidates applied for post A = 12.5x * (100/18) = (625x/9)

Since, the number of candidates selected for post B is 43.2% of the number of candidates applied for post B.

So, the number of candidates applied for post B = 15x * (100/43.2) = (625x/18)

Since, the difference between the number of candidates, who applied for posts A and B, is 250.

So,

(625x/9) – (625x/18) = 250

1250x – 625x = 4500

x = 7.2

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Post	No. of candidates selected for the post
A	12.5 * 7.2 = 90
B	15 * 7.2 = 108
C	20 * 7.2 = 144
D	25 * 7.2 = 180
E	27.5 * 7.2 = 198

Answer: A

Since, the number of candidates selected for post C = 144

So, the number of candidates applied for post C = 144 * (100/48) = 300

And the number of female candidates applied for post C = 300 – 125 = 175

Since, the number of candidates selected for post D = 180

So, the number of candidates applied for post D = 180 * (100/45) = 400

And the number of female candidates applied for post D = 400 – 175 = 225

Required average = (175 + 225)/2 = 200

4. Questions

Percentage distribution of the number of candidates selected for post A = 12.5%

Percentage distribution of the number of candidates selected for post B = (54/360) * 100 = 15%

Percentage distribution of the number of candidates selected for post C = 20%

Percentage distribution of the number of candidates selected for post D = 25%

Percentage distribution of the number of candidates selected for post E = (99/360) * 100 = 27.5%

Let the number of candidates selected for posts A, B, C, D and E be 12.5x, 15x, 20x, 25x and 27.5x respectively.

Since, the number of candidates selected for post A is 18% of the number of candidates applied for post A.

So, the number of candidates applied for post A = 12.5x * (100/18) = (625x/9)

Since, the number of candidates selected for post B is 43.2% of the number of candidates applied for post B.

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So, the number of candidates applied for post B = 15x * (100/43.2) = (625x/18)

Since, the difference between the number of candidates, who applied for posts A and B, is 250.

So,

(625x/9) – (625x/18) = 250

1250x – 625x = 4500

x = 7.2

Post	No. of candidates selected for the post
A	12.5 * 7.2 = 90
B	15 * 7.2 = 108
C	20 * 7.2 = 144
D	25 * 7.2 = 180
E	27.5 * 7.2 = 198

Answer: D

Since, the difference between the number of candidates selected for posts A and B = 108 – 90 = 18

So, the difference between the number of candidates selected for posts E and F = 3 * 18 = 54

Since, the number of candidates selected for post E = 198

So, the number of candidates selected for post F will be either 198 + 54 = 252 or 198 – 54 = 144

Since, the number of candidates selected for post F, is not a multiple of 72.

And 144 is a multiple of 72.

So, the number of candidates selected for post F = 252

5. Questions

Since, the number of videos uploaded in 2016 is 40 more than that uploaded in 2017.

So,

(a + b) – 2b = 40

a – b = 40 ----------(1)

Since, the number of shorts uploaded in 2018 is 280.

So,

2a * (140/100) = 280

a = 100

From equation (1):

b = 60

Since, the difference between the number of videos uploaded in 2019 and 2020, is ‘a’.

So,

(2d + c) – c = a

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2d = 100

d = 50

Since, the ratio of 'c' to 'd' = 8:5

So, c = 50 * (8/5) = 80

Year	No. of videos uploaded	No. of shorts uploaded
2016	100 + 60 = 160	200% of 160 = 320
2017	2 * 60 = 120	180% of 120 = 216
2018	2 * 100 = 200	140% of 200 = 280
2019	80	160% of 80 = 128
2020	2 * 50 + 80 = 180	120% of 180 = 216

Answer: D

Total number of videos uploaded in 2017 and 2018 together = 120 + 200 = 320

Total number of shorts uploaded in 2017 and 2018 together = 216 + 280 = 496

Required percentage = $\frac{496-320}{320} \times 100 = 55\%$

From option (a):

[(a/4) + (b/3)] = [(100/4) + (60/3)] = (25 + 20) = 45%

So, option (a) is not correct.

From option (b):

[(a + c)/3] = [(100 + 80)/3] = 60%

So, option (b) is not correct.

From option (c):

[(a + d)/3] = [(100 + 50)/3] = 50%

So, option (c) is not correct.

From option (d):

[(c/2) + (b/4)] = [(80/2) + (60/4)] = (40 + 15) = 55%

So, option (d) is correct.

From option (e):

[(a + 2b + c)/4] = [(100 + 120 + 80)/4] = 75%

So, option (e) is not correct.

Hence, option (d) is correct.

6. Questions

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Since, the number of videos uploaded in 2016 is 40 more than that uploaded in 2017.

So,

(a + b) – 2b = 40

a – b = 40 ----------(1)

Since, the number of shorts uploaded in 2018 is 280.

So,

2a * (140/100) = 280

a = 100

From equation (1):

b = 60

Since, the difference between the number of videos uploaded in 2019 and 2020, is 'a'.

So,

(2d + c) – c = a

2d = 100

d = 50

Since, the ratio of 'c' to 'd' = 8:5

So, c = 50 * (8/5) = 80

Year	No. of videos uploaded	No. of shorts uploaded
2016	100 + 60 = 160	200% of 160 = 320
2017	2 * 60 = 120	180% of 120 = 216
2018	2 * 100 = 200	140% of 200 = 280
2019	80	160% of 80 = 128
2020	2 * 50 + 80 = 180	120% of 180 = 216

Answer: B

Since, the total number of shorts uploaded in 2019 = 128

And the average number of likes on each short uploaded in 2019 = 2 lakhs

So, the total number of likes on all the shorts together uploaded in 2019:

(128 * 2) lakhs

Since, the total number of shorts uploaded in 2020 = 216

And the average number of likes on each short uploaded in 2020 = 1.5 lakhs

So, the total number of likes on all the shorts together uploaded in 2020:

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(216 * 1.5) lakhs

Required ratio = (128 * 2): (216 * 1.5) = 256: 324 = 64: 81

7. Questions

Since, the number of videos uploaded in 2016 is 40 more than that uploaded in 2017.

So,

(a + b) – 2b = 40

a – b = 40 ----------(1)

Since, the number of shorts uploaded in 2018 is 280.

So,

2a * (140/100) = 280

a = 100

From equation (1):

b = 60

Since, the difference between the number of videos uploaded in 2019 and 2020, is 'a'.

So,

(2d + c) – c = a

2d = 100

d = 50

Since, the ratio of 'c' to 'd' = 8:5

So, c = 50 * (8/5) = 80

Year	No. of videos uploaded	No. of shorts uploaded
2016	100 + 60 = 160	200% of 160 = 320
2017	2 * 60 = 120	180% of 120 = 216
2018	2 * 100 = 200	140% of 200 = 280
2019	80	160% of 80 = 128
2020	2 * 50 + 80 = 180	120% of 180 = 216

Answer: B

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The years, in which the number of shorts uploaded are equal, are 2017 and 2020.

The number of videos uploaded in 2017 = 120

The number of videos uploaded in 2020 = 180

Required ratio = 120:180 = 2:3

8. Questions

Since, the number of videos uploaded in 2016 is 40 more than that uploaded in 2017.

So,

(a + b) – 2b = 40

a – b = 40 ----------(1)

Since, the number of shorts uploaded in 2018 is 280.

So,

2a * (140/100) = 280

a = 100

From equation (1):

b = 60

Since, the difference between the number of videos uploaded in 2019 and 2020, is 'a'.

So,

(2d + c) – c = a

2d = 100

d = 50

Since, the ratio of 'c' to 'd' = 8:5

So, c = 50 * (8/5) = 80

Year	No. of videos uploaded	No. of shorts uploaded
2016	100 + 60 = 160	200% of 160 = 320
2017	2 * 60 = 120	180% of 120 = 216
2018	2 * 100 = 200	140% of 200 = 280
2019	80	160% of 80 = 128
2020	2 * 50 + 80 = 180	120% of 180 = 216

Answer: E

Since, the number of videos uploaded in 2020 = 180

So, the number of videos uploaded in 2021:

[100 + 2.5(c – d)]% of 180 = [(100 + 2.5(80 – 50)]% of 180 = 315

Since, the number of shorts uploaded in 2019 = 128

So, the number of shorts uploaded in 2021:

[100 + 1.25(a – b)]% of 128 = [100 + 1.25(100 – 60)]% of 128 = 192

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Required percentage = (192/315) * 100 = 61% (approx.)

9. Questions

Since, the average number of shirts sold by A on each of the given days = 48

So, the number of shirts sold by A on Wed:

(5 * 48) – (45 + 72 + 50 + 41) = 32

Since, the average number of shirts sold by B on each of the given days = 40

So, the number of shirts sold by B on Thu:

(5 * 40) – (25 + 60 + 42 + 55) = 18

Since, the average number of shirts sold by C on each of the given days = 36

So, the number of shirts sold by C on Mon:

(5 * 36) – (32 + 27 + 51 + 22) = 48

Days	Shopkeeper A		Shopkeeper B		Shopkeeper C	
	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold
Mon	45	__	25	36	48	24
Tue	72	60	60	__	32	__
Wed	32	48	42	21	27	__
Thu	50	__	18	30	51	34
Fri	41	50	55	__	22	40

Answer: D

Since, the total number of shirts sold by A, B and C together on Wed:

32 + 42 + 27 = 101

So, the total number of jeans sold by A, B and C together on Wed = 101 + 22 = 123

Since, the number of jeans sold by A on Wed = 48

And the number of jeans sold by B on Wed = 21

So, the number of jeans sold by C on Wed = 123 – (48 + 21) = 54

Since, the cost, at which C sold jeans = ₹750 per piece

So, the amount received by C by selling jeans on Wed = 54 * 750 = ₹40500

10. Questions

Since, the average number of shirts sold by A on each of the given days = 48

So, the number of shirts sold by A on Wed:

(5 * 48) – (45 + 72 + 50 + 41) = 32

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Since, the average number of shirts sold by B on each of the given days = 40

So, the number of shirts sold by B on Thu:

(5 * 40) – (25 + 60 + 42 + 55) = 18

Since, the average number of shirts sold by C on each of the given days = 36

So, the number of shirts sold by C on Mon:

(5 * 36) – (32 + 27 + 51 + 22) = 48

Days	Shopkeeper A		Shopkeeper B		Shopkeeper C	
	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold
Mon	45	__	25	36	48	24
Tue	72	60	60	__	32	__
Wed	32	48	42	21	27	__
Thu	50	__	18	30	51	34
Fri	41	50	55	__	22	40

Answer: E

Since, the average number of jeans sold by C on Tue, Thu and Fri = 30

So, the number of jeans sold by C on Tue = (3 * 30) – (34 + 40) = 16

Since, the total number of jeans sold by B and C together on Tue is 16(2/3)% more than that sold by A on Tue.

And the number of jeans sold by A on Tue = 60

So, the number of jeans sold by B and C together on Tue = 116(2/3)% of 60 = 70

And the number of jeans sold by B on Tue = 70 – 16 = 54

Since, the number of shirts sold by B on Tue = 60

So, the required ratio = 60:54 = 10:9

11. Questions

Since, the average number of shirts sold by A on each of the given days = 48

So, the number of shirts sold by A on Wed:

(5 * 48) – (45 + 72 + 50 + 41) = 32

Since, the average number of shirts sold by B on each of the given days = 40

So, the number of shirts sold by B on Thu:

(5 * 40) – (25 + 60 + 42 + 55) = 18

Since, the average number of shirts sold by C on each of the given days = 36

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So, the number of shirts sold by C on Mon:

(5 * 36) – (32 + 27 + 51 + 22) = 48

Days	Shopkeeper A		Shopkeeper B		Shopkeeper C	
	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold
Mon	45	__	25	36	48	24
Tue	72	60	60	__	32	__
Wed	32	48	42	21	27	__
Thu	50	__	18	30	51	34
Fri	41	50	55	__	22	40

Answer: A

Since, the average number of jeans sold by A, B and C on Thu = 36

So, the number of jeans sold by A on Thu = (3 * 36) – (30 + 34) = 44

Since, the total number of shirts sold by A on Mon, Wed and Thu together:

45 + 32 + 50 = 127

So, the total number of jeans sold by A on Mon, Wed and Thu together:

127 + 1 = 128

And the number of jeans sold by A on Mon = 128 – (48 + 44) = 36

Since, the number of jeans sold by B on Mon = 36

And the number of jeans sold by C on Mon = 24

So, the required average = (36 + 36 + 24)/3 = 32

12. Questions

Since, the average number of shirts sold by A on each of the given days = 48

So, the number of shirts sold by A on Wed:

(5 * 48) – (45 + 72 + 50 + 41) = 32

Since, the average number of shirts sold by B on each of the given days = 40

So, the number of shirts sold by B on Thu:

(5 * 40) – (25 + 60 + 42 + 55) = 18

Since, the average number of shirts sold by C on each of the given days = 36

So, the number of shirts sold by C on Mon:

(5 * 36) – (32 + 27 + 51 + 22) = 48

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Days	Shopkeeper A		Shopkeeper B		Shopkeeper C	
	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold	No. of shirts sold	No. of jeans sold
Mon	45	__	25	36	48	24
Tue	72	60	60	__	32	__
Wed	32	48	42	21	27	__
Thu	50	__	18	30	51	34
Fri	41	50	55	__	22	40

Answer: C

Since, the difference between the number of shirts sold by B on Wed and Thu = 42 – 18 = 24

So, the difference between the number of jeans sold by B on Thu and Fri = 24/8 = 3

Since, the number of jeans sold by B on Thu = 30

So, the number of jeans sold by B on Fri will be either 30 + 3 = 33 or 30 – 3 = 27

Since, the number of jeans sold by B on Wed = 21

So, the average of the number of jeans sold by B on Wed and Fri:

Case I: (21 + 33)/2 = 27

Case II: (21 + 27)/2 = 24

Since, the average of the number of jeans sold by B on Wed and Fri is a multiple of 9.

So, case II can be neglected.

And the number of jeans sold by B on Fri = 33

13. Questions

Let the CPs of articles A and C be ₹6x and ₹5x respectively.

Also let the CPs of articles B and D be ₹9y and ₹10y respectively.

So, the MP of article A = 150% of 6x = ₹9x

The MP of article B = 133(1/3)% of 9y = ₹12y

The MP of article C = ₹(5x + 600)

And the MP of article D = ₹(10y + 600)

Since, the CP of article B is equal to the MP of article A.

So, 9y = 9x

x = y ----------(1)

Since, the MP of articles B and D is equal.

So,

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12y = 10y + 600

y = 300

From equation (1):

x = 300

Article	Cost price (in ₹)	Marked price (in ₹)
A	6 * 300 = 1800	150% of 1800 = 2700
B	9 * 300 = 2700	133(1/3)% of 2700 = 3600
C	5 * 300 = 1500	1500 + 600 = 2100
D	10 * 300 = 3000	3000 + 600 = 3600

Answer: B

Since, the CP of article C = ₹1500

And the MP of article C = ₹2100

So, $2M = \frac{2100-1500}{1500} \times 100 = 40$

M = 20

The CP of article A = ₹1800

The MP of article A = ₹2700

If article A was sold at M% discount on its MP:

So, the SP of article A would have been:

(100 – M)% of 2700 = (100 – 20)% of 2700 = ₹2160

And the profit received on article A would have been = 2160 – 1800 = ₹360

The CP of article B = ₹2700

The MP of article B = ₹3600

If article B was sold at (M – 5)% discount on its MP:

So, the SP of article B would have been:

(100 – M + 5)% of 3600 = (100 – 20 + 5)% of 3600 = 3060

And the profit received on article B would have been = 3060 – 2700 = ₹360

Required profit amount = 360 + 360 = ₹720

14. Questions

Let the CPs of articles A and C be ₹6x and ₹5x respectively.

Also let the CPs of articles B and D be ₹9y and ₹10y respectively.

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So, the MP of article A = 150% of 6x = ₹9x

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B	9 * 300 = 2700	133(1/3)% of 2700 = 3600
C	5 * 300 = 1500	1500 + 600 = 2100
D	10 * 300 = 3000	3000 + 600 = 3600

Answer: E

Since, the total CP of articles C and D together = 1500 + 3000 = ₹4500

So, the total SP of articles C and D together = 120% of 4500 = ₹5400

Since, the total MP of articles C and D together = 2100 + 3600 = ₹5700

So,

N + 2N = 5700 – 5400

N = 100

15. Questions

Let the CPs of articles A and C be ₹6x and ₹5x respectively.

Also let the CPs of articles B and D be ₹9y and ₹10y respectively.

So, the MP of article A = 150% of 6x = ₹9x

The MP of article B = 133(1/3)% of 9y = ₹12y

The MP of article C = ₹(5x + 600)

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And the MP of article D = ₹(10y + 600)

Since, the CP of article B is equal to the MP of article A.

So, 9y = 9x

x = y ----------(1)

Since, the MP of articles B and D is equal.

So,

12y = 10y + 600

y = 300

From equation (1):

x = 300

Article	Cost price (in ₹)	Marked price (in ₹)
A	6 * 300 = 1800	150% of 1800 = 2700
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C	5 * 300 = 1500	1500 + 600 = 2100
D	10 * 300 = 3000	3000 + 600 = 3600

Answer: A

From P:

The total CP of articles A and B together = The total CP of articles C and D together

1800 + 2700 = 1500 + 3000

4500 = 4500

So, P is true.

From Q:

The ratio of the MP of article A to that of article C = 2700: 2100 = 9: 7

So, Q is true.

From R:

The average marked price of each of the given article:

$$\frac{2700 + 3600 + 2100 + 3600}{4} = ₹3000$$

So, R is not true.

Hence, only P and Q are true.

16. Questions

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Let the CPs of articles A and C be ₹6x and ₹5x respectively.

Also let the CPs of articles B and D be ₹9y and ₹10y respectively.

So, the MP of article A = 150% of 6x = ₹9x

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Since, the CP of article B is equal to the MP of article A.

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Since, the MP of articles B and D is equal.

So,

12y = 10y + 600

y = 300

From equation (1):

x = 300

Article	Cost price (in ₹)	Marked price (in ₹)
A	6 * 300 = 1800	150% of 1800 = 2700
B	9 * 300 = 2700	133(1/3)% of 2700 = 3600
C	5 * 300 = 1500	1500 + 600 = 2100
D	10 * 300 = 3000	3000 + 600 = 3600

Answer: C

Since, the total MP of all the four articles together:

2700 + 3600 + 2100 + 3600 = ₹12000

So, the total SP of all the four articles together:

95% of 12000 – (4 * 150) = ₹10800

Since, the total CP of all the four articles together:

1800 + 2700 + 1500 + 3000 = ₹9000

So, the required profit per cent = $\frac{10800-9000}{9000} \times 100 = 20\%$

17. Questions

The number of persons, who watched movie A only = 8X

The number of persons, who watched movie B only = 5X

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The number of persons, who watched movie C only = (X + Y)

The number of persons, who watched both movies B and C but not A = 3Y

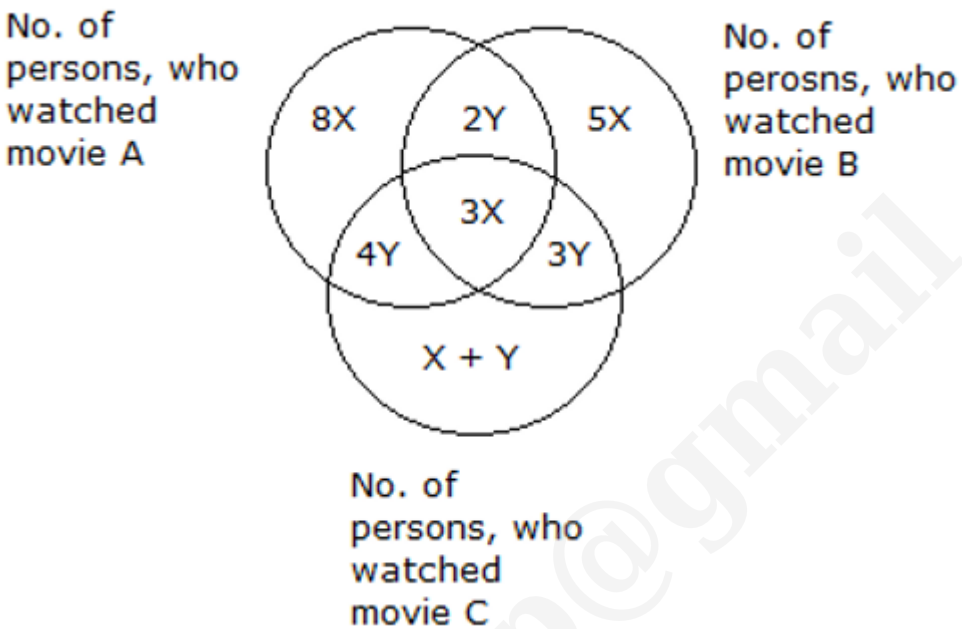
The number of persons, who watched both movies A and B but not C:

3Y * (100/150) = 2Y

The number of persons, who watched both movies A and C but not B:

3Y * (100/75) = 4Y

The number of persons, who watched all the three movies = 37.5% of 8X = 3X



Answer: A

Since, the number of persons, who watched both movies B and C but not A = 225

So, 3Y = 225

Y = 75

Since, the total number of persons, who watched movie A = 1000

So, 8X + 2Y + 4Y + 3X = 1000

11X + (6 * 75) = 1000

X = 50

So, the total number of persons, who watched movie B:

5X + 2Y + 3Y + 3X = 8X + 5Y = (8 * 50) + (5 * 75) = 775

18. Questions

The number of persons, who watched movie A only = 8X

The number of persons, who watched movie B only = 5X

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The number of persons, who watched movie C only = (X + Y)

The number of persons, who watched both movies B and C but not A = 3Y

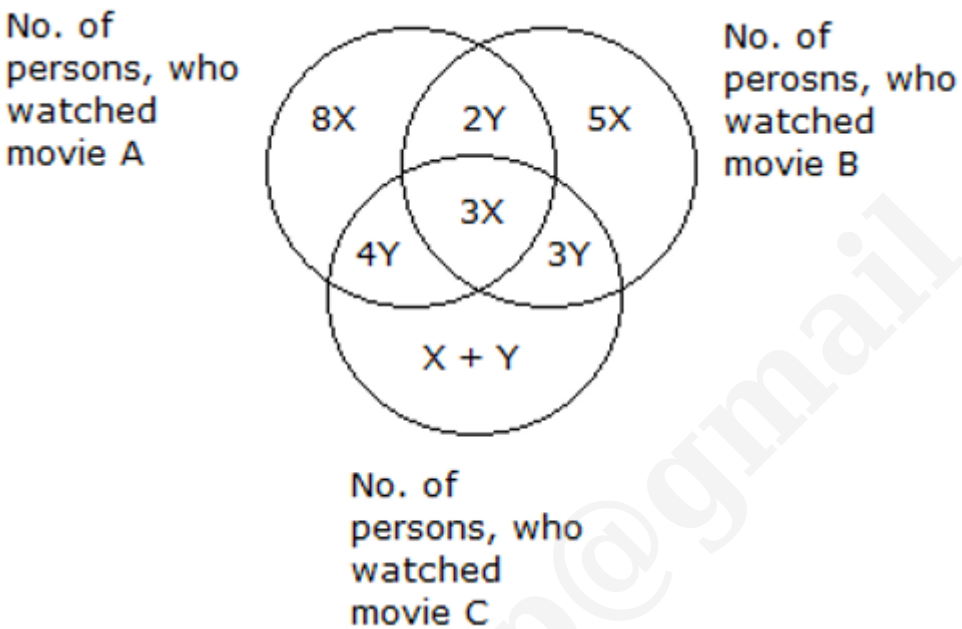
The number of persons, who watched both movies A and B but not C:

3Y * (100/150) = 2Y

The number of persons, who watched both movies A and C but not B:

3Y * (100/75) = 4Y

The number of persons, who watched all the three movies = 37.5% of 8X = 3X



Answer: B

Since, the ratio of X to Y = X: Y = 1: 1

So, X = Y

The total number of persons, who watched only one movie:

8X + 5X + (X + Y) = (14X + Y) = (14Y + Y) = 15Y

The total number of persons, who watched only two movies:

2Y + 3Y + 4Y = 9Y

Required ratio = 15Y: 9Y = 5: 3

19. Questions

The number of persons, who watched movie A only = 8X

The number of persons, who watched movie B only = 5X

The number of persons, who watched movie C only = (X + Y)

The number of persons, who watched both movies B and C but not A = 3Y

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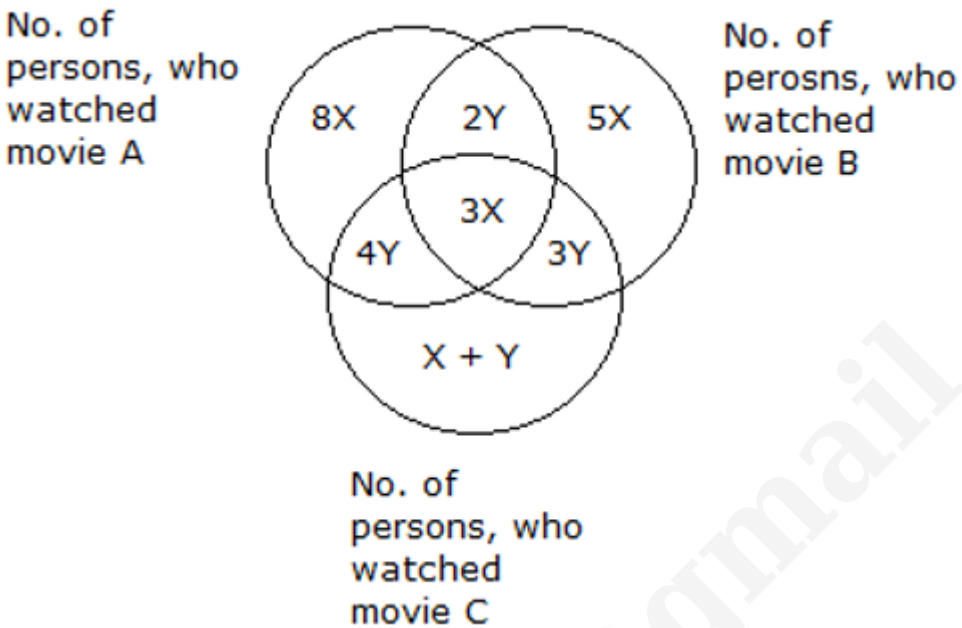
The number of persons, who watched both movies A and B but not C:

3Y * (100/150) = 2Y

The number of persons, who watched both movies A and C but not B:

3Y * (100/75) = 4Y

The number of persons, who watched all the three movies = 37.5% of 8X = 3X



Answer: E

Since, the number of persons, who watched movie C only = 65

So,

X + Y = 65 ---------(1)

Since, the ratio of the number of persons, who watched both movies A and B but not C, to those, who watched all the three movies, is 5: 12.

So,

$$\frac{2Y}{3X} = \frac{5}{12}$$

5X = 8Y

From equation (1):

X + (5X/8) = 65

X = 40

So, the number of persons, who watched all the three movies = 3 * 40 = 120

20. Questions

The number of persons, who watched movie A only = 8X

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The number of persons, who watched movie B only = 5X

The number of persons, who watched movie C only = (X + Y)

The number of persons, who watched both movies B and C but not A = 3Y

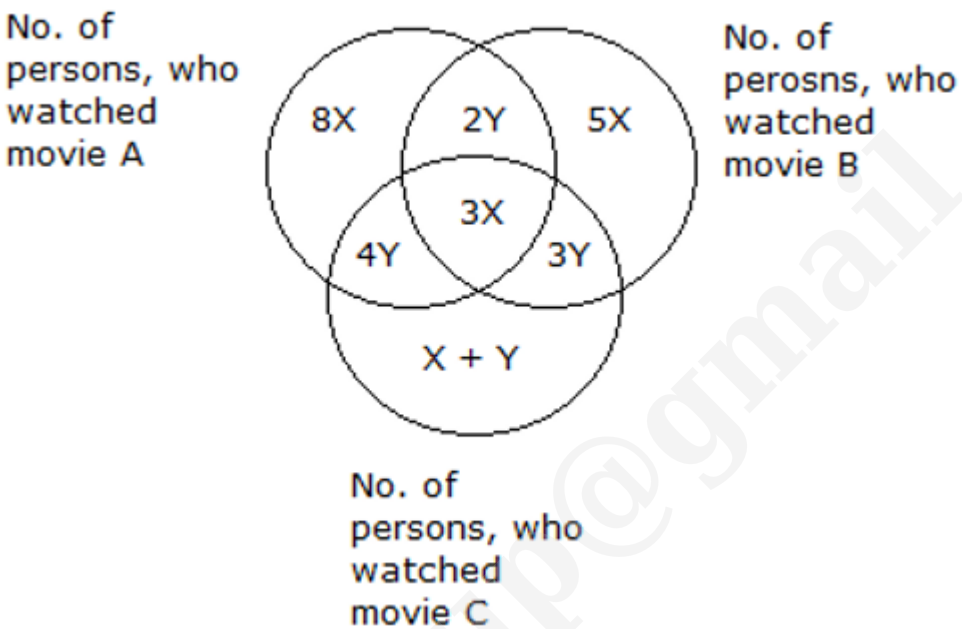
The number of persons, who watched both movies A and B but not C:

3Y * (100/150) = 2Y

The number of persons, who watched both movies A and C but not B:

3Y * (100/75) = 4Y

The number of persons, who watched all the three movies = 37.5% of 8X = 3X



Answer: D

The total number of persons, who watched only one movie:

8X + 5X + (X + Y) = 14X + Y

The total number of persons, who watched at-least two movies = (The number of persons, who watched only two movies) + (The number of persons, who watched all the three movies)

The total number of persons, who watched at-least two movies:

(2Y + 3Y + 4Y) + 3X = (3X + 9Y)

So,

$$\frac{14X + Y}{3X + 9Y} = \frac{74}{51}$$

714X + 51Y = 222X + 666Y

492X = 615Y

4X = 5Y

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X: Y = 5: 4

Since, the total number of persons, who watched only one movie:

14X + Y = 14X + (4X/5) = (74X/5)

And the number of persons, who watched all the three movies = 3X

So, the required ratio = (74X/5): 3X = 74: 15

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1. Questions

Read the following information carefully and answer the questions based on it.

There are three colleges – P, Q, and R, and each college has two streams A and B.

College P: Number of boys is thrice as that of girls, while number of boys in stream A is 400 which is 25% more than that of total students in stream B. Out of total girls 40% are in stream A.

College Q: Number of students in stream B is 1/3rd as that of boys in stream A, while total boys in both streams together is 20% more than that of boys in stream A. Total number of girls and number of girls in stream A is in 7:5 ratio, and number of boys in stream A is 600.

College R: 60% of total girls are in A, while ratio of total boys and total girls is 7:5 respectively. 9/14 of total number of boys is in stream A.

If total number of students in College P, Q, and R is in arithmetic progression in same order, then find the number of girls in stream B of college R.

a. 300
b. 200
c. 240
d. 250
e. None of these

2. Questions

Number of girls in stream A of college P is how much more than that of girls in stream B of same college.

a. 40
b. 20
c. 30
d. 60
e. None of these

3. Questions

Number of students in stream A of college R is how much % more or less than that of students in stream B of same college.

a. 60%
b. 62.5%
c. 40%
d. 66.66%

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e. None of these

4. Questions

Quantity I: 1.2 x Number of girls in stream B of college P

Quantity II: Sum of the number of boys in stream B of college Q and girls in same stream of college P.

a. Quantity I > Quantity II

b. Quantity I = Quantity II

c. Quantity I ≥ Quantity II

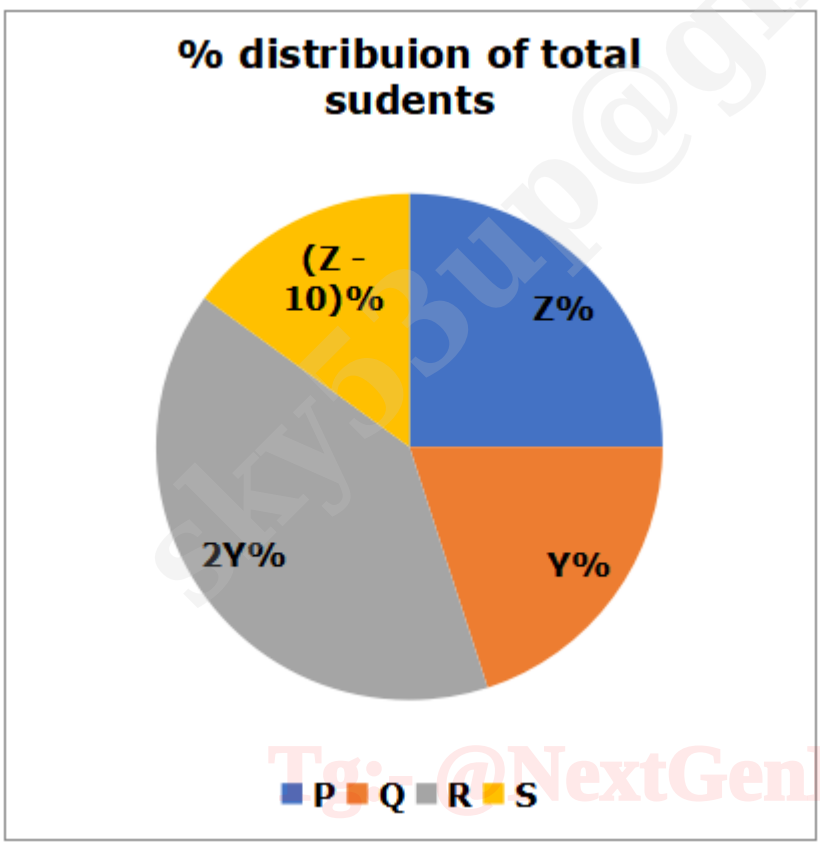
d. Quantity I ≤ Quantity II

e. Quantity I < Quantity II

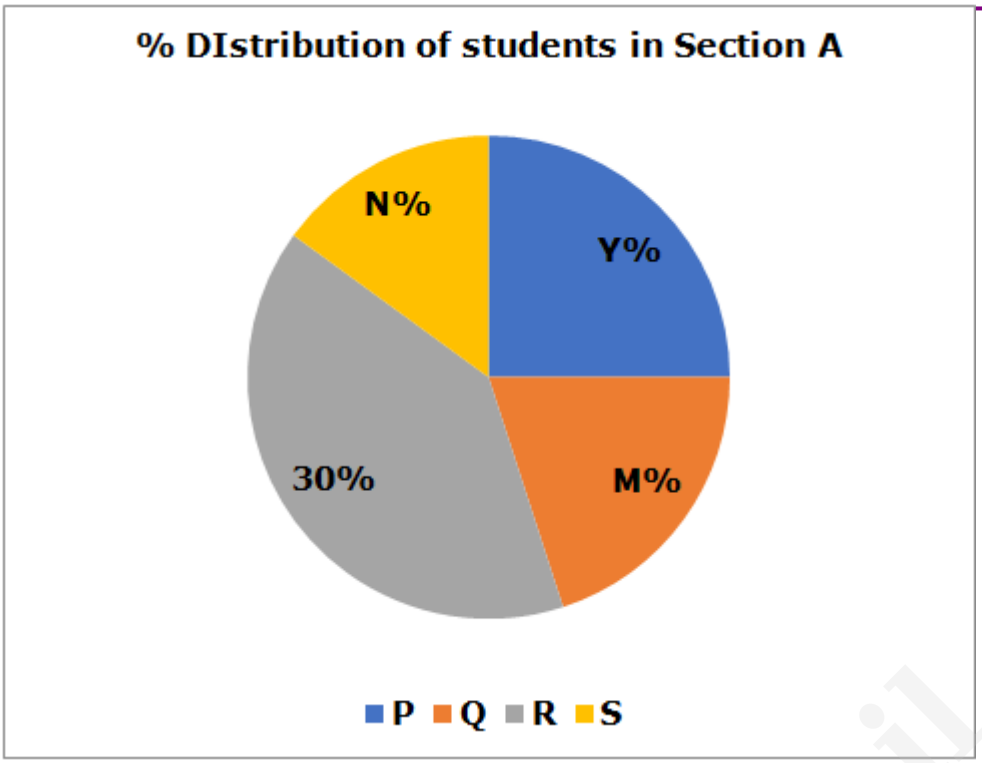
5. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the % distribution of students in four different classes of a school and % distribution of number of students in section A of four classes – P, Q, R, and S. Each class has two sections – A and B



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Note:

a). Difference between the total number of students in class P and S is 40, which is 1/6th of total students in section B.

b). Students in both section of class Q is same, while number of students in section A of class Q and S is same.

For class P, the total number of students in section A is same as number of boys in section B, while number of boys in section A is 50% of girls in section B. Find number of girls in section A of school P.

a. 18

b. 20

c. 14

d. 16

e. None of these

6. Questions

For R, out of total students 1/4th are girls and out of total boys (L – 10) % are in section B. Find the value of L, if number of girls in section A is 25% of total students in that section.

a. 75

b. 80

c. 65

d. 60

e. None of these

7. Questions

Find the difference between the number of students in section A in class Q and the number of students in section B in class S.

a. 25

b. 30

c. 15

d. 20

e. None of these

8. Questions

If the total number of students in class T is 75% more than that of class Q and the number of students in section A of class T is (Z + 20), then find the number of students in section B of class T.

a. 75

b. 80

c. 65

d. 85

e. None of these

9. Questions

Read the following information carefully and answer the questions based on it.

The table given below shows the number of workers – Male and female workers working on three different projects.

Name of project	Male workers	Number of days	Female workers	Number of days
P	(4Z + 4)	20	(2Y – 20)	(7Z + 13)
Q	(Y – 4)	(Y + 10)	1.5Y	(2Y – 8)
R	M	48	(M/2 + 19)	40

Note:

a). Amount of work done for each project is same, and efficiency of one man is twice as that of a woman.

b). All men are equal efficient and all women are equal efficient.

If (3N – 4) male workers assigned to complete the project P, it will take 15 days to complete the

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project. Find the value of N?

a. 8

b. 6

c. 12

d. 10

e. None of these

10. Questions

Quantity I. Find total number of male workers working in all three projects.

Quantity II. Find the missing term in the series given below

20, 22, 26, 34, (?)

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I ≤ Quantity II

d. Quantity I ≥ Quantity II

e. Quantity I = Quantity II

11. Questions

Find the value of (Y + 2Z + M).

a. 40

b. 50

c. 60

d. 70

e. None of these

12. Questions

If 2K more male workers working on project Q (as compared to given workers, hence work is completed in (M + 4) days earlier. Find the value of K.

a. 14

b. 7

c. 16

d. 8

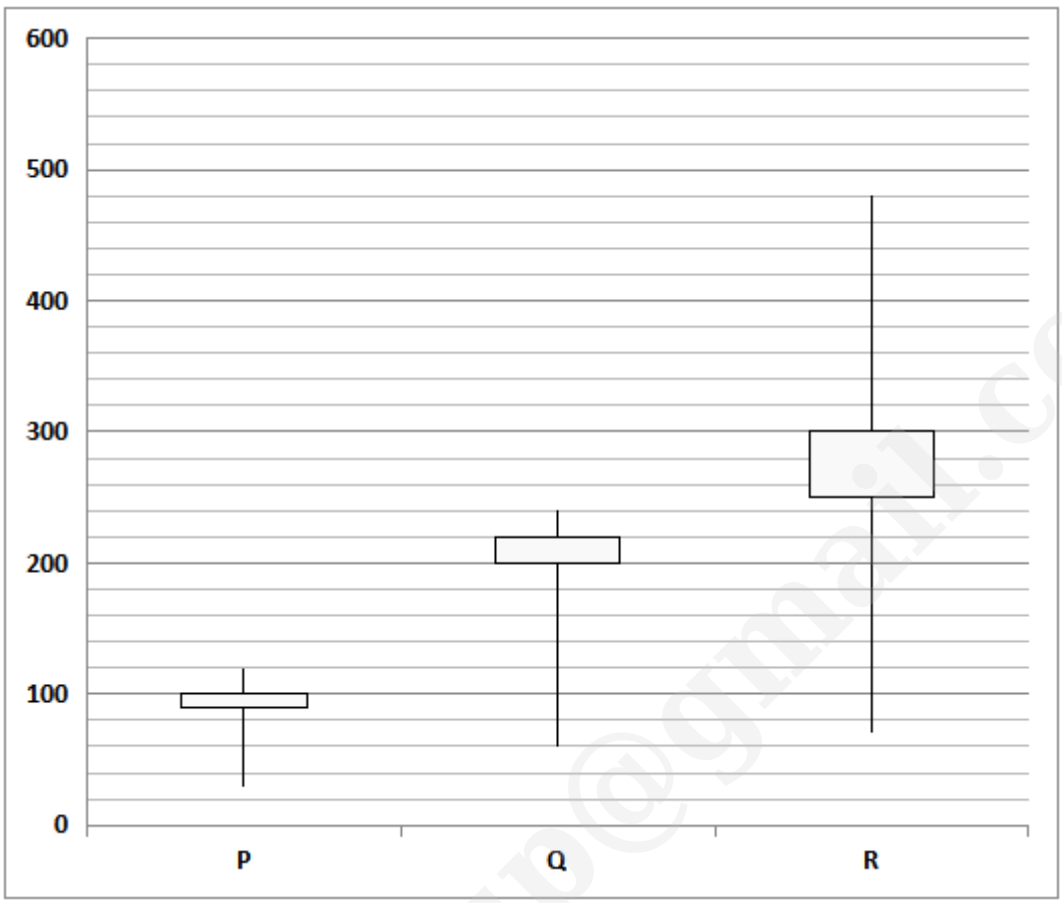
e. None of these

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13. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the number people visited three malls – P, Q, and R on three different days of a week – Monday, Tuesday, and Wednesday.



Note:

a). Sum of the value at top of box and bottom of line represents the sum of people visited mall on Wednesday and Monday together.

b). Top of line represents the number of people visited on Monday and Tuesday together, while bottom of the box represents number of people visited on Tuesday and Wednesday together.

For mall P, out of total people who visited mall 40% are males rest are females. Out of total males Z% males visited on Monday and rest Y% males visited on Wednesday. Z is a multiple of 25, and both Z and Y are natural numbers.

A = Find 75% of number of females visited on Tuesday.

B = Out of Y and Z, value of one of the variables is 25, and difference between the number of females visited on Monday and Wednesday.

a. A ≥ B

b. A < B

c. A = B or no relation

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d. A > B

e. A ≤ B

14. Questions

For mall Q, number of females visited on Monday is 30Z, which is four times the number of males on Tuesday. The number of males visited on Monday and females visited on Tuesday is A and B respectively. Z is a natural number

Which of the following can be the possible value(s) of difference between the value of A and B.

I). 10

II). 15

III). 35

a. I only

b. II and III only

c. I and III only

d. I, II and III

e. None of these

15. Questions

For mall R, the difference between the number of people visited on any two days is A, where A is maximum possible. So, value of (A – 30) is/are ___.

I). 2/3 x people visited mall R on Monday

II). 2.5 times people visited mall P on Tuesday

III). 1.25 times of people visited mall Q on Monday

a. II only

b. I and III only

c. I only

d. I, II, and III

e. None of these

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16. Questions

The number of people visited P on Tuesday and Q on Wednesday is in the ratio of A: B respectively, where A and B are co primes. So, find the value of [(2A + 1)/ (B – 1)].

a. 1.5

b. 0.33

c. 1.2

d. 1.33

e. None of these

17. Questions

For R, it was found that there is an error in calculating the number of people visited on Tuesday and Wednesday. 10% of the actual number of people those who visited on Tuesday, they are counted on Wednesday. Number of people visited (actual) mall R on Wednesday is how much % more or less than people visited mall Q on Tuesday.

a. 12.25%

b. 37.5%

c. 12.5%

d. 32.5%

e. None of these

18. Questions

Read the following information carefully and answer the questions based on it.

Three quadratic equations – I, II, and III given below.

Equations	Remarks
I). $Z^2 - 28Z + P = 0$	where a and b are roots of equations such that their difference is 4. a > b
II). $Y^2 - 17Y - Q = 0$	where c and d are roots of equation, such that sum of squares of roots is 689.
III). $M^2 + 5M - R = 0$	where e and f are roots of equation, such that ($e^2 + 2f = 89$), e < f

Which of the following statement is/are Must be true.

A). a: f = 4:1

B). on adding 5 to difference between the value of e and f, resultant is perfect square.

C). when (b + d) is divided by 3 then remainder is 1.

a. I only

b. I and III only

c. II and II only

d. I, II and III

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e. None of these

19. Questions

which of the following match is/are possible.

Value of Z	**A).** One of the roots is perfect square
Value of Y	**B).** |N| and |L| are co primes, where A and B are roots of equation
Value of M	**C).** Product of roots is > - 50

I). Z – A, Y – C, M - B

II). Z – C, Y – B, M - A

III). Z – C, Y – A, M - C

a. I and II only

b. II and III only

c. I and III only

d. I, II, and III

e. None of these

20. Questions

Find the value of (P - Q + 2R)?

a. 64

b. 72

c. 56

d. 84

e. None of these

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Explanations:

1. Questions

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For college P

Let total students = 100a

Total boys = 75% x 100a = 75a

Total girls = 25% x 100a = 25a

Boys in stream A = 400

So, total students in B = 400 /125 x 100 = 320

Boys in stream B = 75a – 400

Girls in stream B = 25a x 60% = 15a

Now,

(75a – 400) + 15a = 320

90a = 720

So, value of a = 8

Now we can find all the related data to college P

For college Q

Boys in stream A = 600

So, total students in B = 600 x 1/3 = 200

Total boys = 600 x 1.2 = 720

So, boys in stream B = 720 – 600 = 120

So, girls in stream B = 200 – 120 = 80

So, girls in stream A = [80 / (7 – 5)] x 5 = 200

Total students in A = 600 + 200 = 800

For R

Let total number of boys = 14b

Total number of girls = 14b x 5/7 = 10b

So, boys in A = 9/14 x 14b = 9b

Boys in B = 14b – 9b = 5b

Girls in A = 60% x 10b = 6b

Girls in B = 10b – 6b = 4b

Answer: B

According to question,

Number of students in P, Q, and R is in AP in same order.

So, number of students in R = (1000 – 800) + 1000 = 1200

So, (14b + 10b) = 1200

Value of b = 50

So, number of girls in stream B of college R = 4b = 50 x 4 = 200

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Hence answer is option B

2. Questions

For college P

Let total students = 100a

Total boys = 75% x 100a = 75a

Total girls = 25% x 100a = 25a

Boys in stream A = 400

So, total students in B = 400 /125 x 100 = 320

Boys in stream B = 75a – 400

Girls in stream B = 25a x 60% = 15a

Now,

(75a – 400) + 15a = 320

90a = 720

So, value of a = 8

Now we can find all the related data to college P

For college Q

Boys in stream A = 600

So, total students in B = 600 x 1/3 = 200

Total boys = 600 x 1.2 = 720

So, boys in stream B = 720 – 600 = 120

So, girls in stream B = 200 – 120 = 80

So, girls in stream A = [80 / (7 – 5)] x 5 = 200

Total students in A = 600 + 200 = 800

For R

Let total number of boys = 14b

Total number of girls = 14b x 5/7 = 10b

So, boys in A = 9/14 x 14b = 9b

Boys in B = 14b – 9b = 5b

Girls in A = 60% x 10b = 6b

Girls in B = 10b – 6b = 4b

Answer: A

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For P

Girls in stream A = 10a

Girls in stream B = 15a

Required difference = 15a – 10a = 5a = 5 x 8 = 40

Hence answer is option A

3. Questions

For college P

Let total students = 100a

Total boys = 75% x 100a = 75a

Total girls = 25% x 100a = 25a

Boys in stream A = 400

So, total students in B = 400 /125 x 100 = 320

Boys in stream B = 75a – 400

Girls in stream B = 25a x 60% = 15a

Now,

(75a – 400) + 15a = 320

90a = 720

So, value of a = 8

Now we can find all the related data to college P

For college Q

Boys in stream A = 600

So, total students in B = 600 x 1/3 = 200

Total boys = 600 x 1.2 = 720

So, boys in stream B = 720 – 600 = 120

So, girls in stream B = 200 – 120 = 80

So, girls in stream A = [80 / (7 – 5)] x 5 = 200

Total students in A = 600 + 200 = 800

For R

Let total number of boys = 14b

Total number of girls = 14b x 5/7 = 10b

So, boys in A = 9/14 x 14b = 9b

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Boys in B = 14b – 9b = 5b

Girls in A = 60% x 10b = 6b

Girls in B = 10b – 6b = 4b

Answer: D

Total students in A = (9b + 6b) = 15b

Total students in B = (5b + 4b) = 9b

Required % change = (15b – 9b)/9b x 100 = 66.66%

Hence answer is option D

4. Questions

For college P

Let total students = 100a

Total boys = 75% x 100a = 75a

Total girls = 25% x 100a = 25a

Boys in stream A = 400

So, total students in B = 400 /125 x 100 = 320

Boys in stream B = 75a – 400

Girls in stream B = 25a x 60% = 15a

Now,

(75a – 400) + 15a = 320

90a = 720

So, value of a = 8

Now we can find all the related data to college P

For college Q

Boys in stream A = 600

So, total students in B = 600 x 1/3 = 200

Total boys = 600 x 1.2 = 720

So, boys in stream B = 720 – 600 = 120

So, girls in stream B = 200 – 120 = 80

So, girls in stream A = [80 / (7 – 5)] x 5 = 200

Total students in A = 600 + 200 = 800

For R

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Let total number of boys = 14b

Total number of girls = 14b x 5/7 = 10b

So, boys in A = 9/14 x 14b = 9b

Boys in B = 14b – 9b = 5b

Girls in A = 60% x 10b = 6b

Girls in B = 10b – 6b = 4b

Answer: E

Quantity I = 1.2 x 120 = 144

Quantity II = (120 + 120) = 240

So, quantity I < Quantity II

Hence answer is option E

5. Questions

Difference between students in class P and S = 40

[Z – (Z – 10)] % x total students = 40

10% x total students = 40

So, total students in all classes together = 40/10 x 100 = 400

Total number of students in section B = 240

So, total number of section A = 400 – 240 = 160

Students in section A of class Q and S is same, so M = N

Now,

Z + Y + 2Y + Z – 10 = 100

2Z + 3Y = 110……. (1)

Number of students in section A of class Q = Number of students in section B of class Q

400 x Y% = 2 x M% x 160

5Y = 4M

Y = 4M/5…….... (2)

Now,

4M/5 + M + 30 + M = 100

14M/5 = 70

So, value of M = 25

So, value of Y = 4 x 25/5 = 20

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From (1), we get

2Z + 3 x 20 = 110

Value of Z = 25

Now we can find all the data related to all given classes.

Name of class	Total students	Students in section A	Students in section B
P	100	32	68
Q	80	40	40
R	160	48	112
S	60	40	20

Answer: C

According to question,

For P

Number of boys in B = 32

So, number of girls in B = 68 – 32 = 36

So, number of boys in A = 50% x 36 = 18

So, number of girls in A = 32 – 18 = 14

Hence answer is option C

6. Questions

Difference between students in class P and S = 40

[Z – (Z – 10)] % x total students = 40

10% x total students = 40

So, total students in all classes together = 40/10 x 100 = 400

Total number of students in section B = 240

So, total number of section A = 400 – 240 = 160

Students in section A of class Q and S is same, so M = N

Now,

Z + Y + 2Y + Z – 10 = 100

2Z + 3Y = 110……. (1)

Number of students in section A of class Q = Number of students in section B of class Q

400 x Y% = 2 x M% x 160

5Y = 4M

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Y = 4M/5…….. (2)

Now,

4M/5 + M + 30 + M = 100

14M/5 = 70

So, value of M = 25

So, value of Y = 4 x 25/5 = 20

From (1), we get

2Z + 3 x 20 = 110

Value of Z = 25

Now we can find all the data related to all given classes.

Name of class	Total students	Students in section A	Students in section B
P	100	32	68
Q	80	40	40
R	160	48	112
S	60	40	20

Answer: B

For R

Total girls = 25% x total students

Girls in section A = 25% x total students in that section

So, girls in section B must be = 25% x total girls in section B

So, boys in section B = 75% x total students in section B

Boys in section B = 75/100 * 112 = 84

Total number of boys = 3/4 * 160 = 120

(L – 10) % = 84/120 * 100

L = 70 + 10 = 80

Hence answer is option B

7. Questions

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Difference between students in class P and S = 40

[Z – (Z – 10)] % x total students = 40

10% x total students = 40

So, total students in all classes together = 40/10 x 100 = 400

Total number of students in section B = 240

So, total number of section A = 400 – 240 = 160

Students in section A of class Q and S is same, so M = N

Now,

Z + Y + 2Y + Z – 10 = 100

2Z + 3Y = 110……. (1)

Number of students in section A of class Q = Number of students in section B of class Q

400 x Y% = 2 x M% x 160

5Y = 4M

Y = 4M/5…….... (2)

Now,

4M/5 + M + 30 + M = 100

14M/5 = 70

So, value of M = 25

So, value of Y = 4 x 25/5 = 20

From (1), we get

2Z + 3 x 20 = 110

Value of Z = 25

Now we can find all the data related to all given classes.

Name of class	Total students	Students in section A	Students in section B
P	100	32	68
Q	80	40	40
R	160	48	112
S	60	40	20

Answer: D

Required difference = 40 – 20 = 20

8. Questions

Difference between students in class P and S = 40

[Z – (Z – 10)] % x total students = 40

10% x total students = 40

So, total students in all classes together = 40/10 x 100 = 400

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Total number of students in section B = 240

So, total number of section A = 400 – 240 = 160

Students in section A of class Q and S is same, so M = N

Now,

Z + Y + 2Y + Z – 10 = 100

2Z + 3Y = 110……. (1)

Number of students in section A of class Q = Number of students in section B of class Q

400 x Y% = 2 x M% x 160

5Y = 4M

Y = 4M/5…….... (2)

Now,

4M/5 + M + 30 + M = 100

14M/5 = 70

So, value of M = 25

So, value of Y = 4 x 25/5 = 20

From (1), we get

2Z + 3 x 20 = 110

Value of Z = 25

Now we can find all the data related to all given classes.

Name of class	Total students	Students in section A	Students in section B
P	100	32	68
Q	80	40	40
R	160	48	112
S	60	40	20

Answer: E

Total number of students in class T = 175/100 * 80 = 140

Number of students in section A of class T = 25 + 20 = 45

Number of students in section B of class T = 140 – 45 = 95

9. Questions

For Q

(Y – 4) x (Y + 10) x 2 = 1.5Y x (2Y – 8)

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$2Y^2 + 12Y - 80 = 3Y^2 - 12Y$

$Y^2 - 24Y + 80 = 0$

$(Y - 4)(Y - 20) = 0$

$Y = 4, 20$

So, possible value of Y = 20

For P

$(4Z + 4) \times 20 \times 2 = (2Y - 20) \times (7Z + 13)$

$8Z + 8 = 7Z + 13$

$Z = 5$

For R

$M \times 48 \times 2 = (M/2 + 19) \times 40$

$48M = 10M + 380$

$38M = 380$

$M = 10$

Answer: C

According to question,

Total work = $(3N - 4) \times 2 \times 15 = (4Z + 4) \times 20 \times 2$

$(3N - 4) = 24 \times 20 / 15$

$3N = 36$

Value of N = 12

Hence answer is option C

10. Questions

For Q

$(Y - 4) \times (Y + 10) \times 2 = 1.5Y \times (2Y - 8)$

$2Y^2 + 12Y - 80 = 3Y^2 - 12Y$

$Y^2 - 24Y + 80 = 0$

$(Y - 4)(Y - 20) = 0$

$Y = 4, 20$

So, possible value of Y = 20

For P

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$(4Z + 4) \times 20 \times 2 = (2Y - 20) \times (7Z + 13)$

$8Z + 8 = 7Z + 13$

$Z = 5$

For R

$M \times 48 \times 2 = (M/2 + 19) \times 40$

$48M = 10M + 380$

$38M = 380$

$M = 10$

Answer: E

Quantity I.

Required sum = 24 + 16 + 10 = 50

Quantity II.

20 + 2 = 22

22 + 4 = 26

26 + 8 = 34

34 + 16 = 50

So, Quantity I = Quantity II

Hence answer is option E

11. Questions

For Q

$(Y - 4) \times (Y + 10) \times 2 = 1.5Y \times (2Y - 8)$

$2Y^2 + 12Y - 80 = 3Y^2 - 12Y$

$Y^2 - 24Y + 80 = 0$

$(Y - 4)(Y - 20) = 0$

$Y = 4, 20$

So, possible value of Y = 20

For P

$(4Z + 4) \times 20 \times 2 = (2Y - 20) \times (7Z + 13)$

$8Z + 8 = 7Z + 13$

$Z = 5$

For R

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M x 48 x 2 = (M/2 + 19) x 40

48M = 10M + 380

38M = 380

M = 10

Answer: A

Required value = 20 + 2 x 5 + 10 = 40

Hence answer is option A

12. Questions

For Q

(Y – 4) x (Y + 10) x 2 = 1.5Y x (2Y – 8)

$2Y^2 + 12Y – 80 = 3Y^2 – 12Y$

$Y^2 – 24Y + 80 = 0$

(Y – 4) (Y – 20) = 0

Y = 4, 20

So, possible value of Y = 20

For P

(4Z + 4) x 20 x 2 = (2Y – 20) x (7Z + 13)

8Z + 8 = 7Z + 13

Z = 5

For R

M x 48 x 2 = (M/2 + 19) x 40

48M = 10M + 380

38M = 380

M = 10

Answer: B

16 x 30 x 2 = (16 + 2K) x 2 (30 – 14)

16 + 2K = 30

2K = 14

K = 7

Hence answer is option B

13. Questions

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For mall P

Let people visited on Monday, Tuesday, and Wednesday = a, b, and c

So, (a + b) = 120……... (1)

(b + c) = 90…… (3)

(c + a) = 100 + 30 = 130……. (2)

On adding all three equations, we get

2 x (a + b + c) = 340

(a + b + c) = 170

So, value of a = 170 – 90 = 80

Value of b = 170 – 130 = 40

Value of c = 170 – 120 = 50

Similarly, we can find the data for all malls

Name of mall	Monday	Tuesday	Wednesday	Total
P	80	40	50	170
Q	160	80	120	360
R	300	180	70	550
Total	540	300	240	1080

Answer: D

According to question,

For mall P

Number of males = 40% x 170 = 68

Number of females = 60% x 170 = 102

Out of total males Z% visited on Monday, and rest Y% visited on Wednesday. So that means no males visited on Tuesday.

So, number of females visited on Tuesday = 40

Z is a multiple of 25.

If Z = 25, then Y = 75

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So, number of males visited on Wednesday = 75% x 68 = 51 (not possible, because total population on Wednesday is 50).

So, value of Z = 50 or 75

So, number of males visited on Monday = 34 or 51

So, number of males visited on Wednesday = 34 or 17

So, number of females on Monday = 46 or 29

So, number of females visited on Wednesday = **16 or 33**

So, difference between females on Monday and Wednesday = 30 or 4

Value of A = 75% x 40 = 30

B = Out of Y and Z, value of one of the variables is 25, and difference between the number of females visited on Monday and Wednesday.

Value of B = 33 – 29 = 4

So, A > B

Hence answer is option D

14. Questions

For mall P

Let people visited on Monday, Tuesday, and Wednesday = a, b, and c

So, (a + b) = 120……... (1)

(b + c) = 90…… (3)

(c + a) = 100 + 30 = 130……. (2)

On adding all three equations, we get

2 x (a + b + c) = 340

(a + b + c) = 170

So, value of a = 170 – 90 = 80

Value of b = 170 – 130 = 40

Value of c = 170 – 120 = 50

Similarly, we can find the data for all malls

Name of mall	Monday	Tuesday	Wednesday	Total
P	80	40	50	170
Q	160	80	120	360
R	300	180	70	550
Total	540	300	240	1080

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Answer: C

For mall Q

People on Monday = 160

People on Tuesday = 80

The ratio of number of females on Monday = 4 x number of Males on Tuesday

So, 30Z must be divisible by 4,

So, possible value of Z = 2 and 4

So, possible value of A = (160 – 60) and (160 – 120) = 100 or 40

So, possible values of B = (80 – 15) and (80 – 30) = 65 or 50

Required difference = 35 or 10

So, only I and III follows.

Hence answer is option C

15. Questions

For mall P

Let people visited on Monday, Tuesday, and Wednesday = a, b, and c

So, (a + b) = 120……... (1)

(b + c) = 90…… (3)

(c + a) = 100 + 30 = 130……. (2)

On adding all three equations, we get

2 x (a + b + c) = 340

(a + b + c) = 170

So, value of a = 170 – 90 = 80

Value of b = 170 – 130 = 40

Value of c = 170 – 120 = 50

Similarly, we can find the data for all malls

Name of mall	Monday	Tuesday	Wednesday	Total
P	80	40	50	170
Q	160	80	120	360
R	300	180	70	550
Total	540	300	240	1080

Answer: B

For R

Difference between Monday and Tuesday = 300 – 180 = 120

Difference between Tuesday and Wednesday = 180 – 70 = 110

Difference between Wednesday and Monday = 70 – 300 = 230

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So, maximum possible value of (A – 30) = 230 – 30 = 200

I). 2/3 x people visited mall R on Monday = 2/3 x 300 = 200

II). 2.5 times of people visited mall P on Tuesday = 2.5 x 40 = 120

III). 1.25 times of people visited mall Q on Monday = 160 x 5/4 = 200

So, only I and III follows.

Hence answer is option B

16. Questions

For mall P

Let people visited on Monday, Tuesday, and Wednesday = a, b, and c

So, (a + b) = 120……... (1)

(b + c) = 90…… (3)

(c + a) = 100 + 30 = 130……. (2)

On adding all three equations, we get

2 x (a + b + c) = 340

(a + b + c) = 170

So, value of a = 170 – 90 = 80

Value of b = 170 – 130 = 40

Value of c = 170 – 120 = 50

Similarly, we can find the data for all malls

Name of mall	Monday	Tuesday	Wednesday	Total
P	80	40	50	170
Q	160	80	120	360
R	300	180	70	550
Total	540	300	240	1080

Answer: A

Value of A: B = 40:120 = 1:3

Required value = (2 x 1 + 1)/ (3 – 1) = 1.5

Hence answer is option A

17. Questions

For mall P

Let people visited on Monday, Tuesday, and Wednesday = a, b, and c

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So, (a + b) = 120……... (1)

(b + c) = 90…… (3)

(c + a) = 100 + 30 = 130……. (2)

On adding all three equations, we get

2 x (a + b + c) = 340

(a + b + c) = 170

So, value of a = 170 – 90 = 80

Value of b = 170 – 130 = 40

Value of c = 170 – 120 = 50

Similarly, we can find the data for all malls

Name of mall	Monday	Tuesday	Wednesday	Total
P	80	40	50	170
Q	160	80	120	360
R	300	180	70	550
Total	540	300	240	1080

Answer: B

For mall R

Actual number of people visited on Tuesday = 180/90 x 100 = 200

Actual number of people visited on Wednesday = 70 – 20 = 50

Now,

Required % change = (80 – 50)/80 x 100 = 37.5%

Hence answer is option B

18. Questions

I). $Z^2 - 28Z + P = 0$

(a + b) = 28……. (1)

(a – b) = 4

So, value of a = (28 + 4)/2 = 16

Value of b = (28 – 4)/2 = 12

So, value of P = 16 x 12 = 192

II). $Y^2 - 17Y - Q = 0$

(c + d) = 17

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On making square of both sides

$C^2 + d^2 + 2cd = 289$……... (3)

Also, $c^2 + d^2 = 689$……... (4)

From (3) and (4), we get

$- 2cd = 400$

So $(c \times d) = - 200$

So, value of Q = 200

III). $M^2 + 5M – R = 0$

$(e + f) = - 5$……. (5)

Also, $e^2 + 2f = 89$…… (6)

On solving both equations we get

$e^2 – 2e = 99$

$e^2 – 2e – 99 = 0$

$(e – 11) (e + 9) = 0$

So, values of e = 11 and – 9

If e = 11 then f = - 16 (but e < f)

So, e = - 9 and f = 4

Answer: B

According to question,

A). a: f = 4:1

Required ratio = 16:4 = 4:1

This statement is true.

B). On adding 5 to difference between e and f resultant is perfect square.

Required value = [4 – (-9)] + 5 = 18 (non perfect square)

This statement is not true.

C). (b + d) divided by 3 remainder is 1.

Value of d = 25 or – 8

(b + d) = 12 – 8 OR 12 + 25 = 4 or 37

Required remainder = 4/3 = 1 OR 37/3 = 1

This statement is true.

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So, only I and III is true

Hence answer is option B

19. Questions

I). $Z^2 - 28Z + P = 0$

$(a + b) = 28$……. (1)

$(a – b) = 4$

So, value of $a = (28 + 4)/2 = 16$

Value of $b = (28 – 4)/2 = 12$

So, value of $P = 16 \times 12 = 192$

II). $Y^2 - 17Y – Q = 0$

$(c + d) = 17$

On making square of both sides

$C^2 + d^2 + 2cd = 289$……... (3)

Also, $c^2 + d^2 = 689$……... (4)

From (3) and (4), we get

$- 2cd = 400$

So $(c \times d) = - 200$

So, value of $Q = 200$

III). $M^2 + 5M – R = 0$

$(e + f) = - 5$……. (5)

Also, $e^2 + 2f = 89$…… (6)

On solving both equations we get

$e^2 – 2e = 99$

$e^2 – 2e – 99 = 0$

$(e – 11) (e + 9) = 0$

So, values of e = 11 and – 9

If e = 11 then f = - 16 (but e < f)

So, e = - 9 and f = 4

Answer: B

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For Z

Roots are 16 and 12

A and C follows

For Y

Roots are 25 and – 8

So, A and B follows

For M

Roots are -9 and 4

So, A, B, and C follows

So, only II and III are true

Hence answer is option B

20. Questions

I). $Z^2 - 28Z + P = 0$

$(a + b) = 28$……. (1)

$(a – b) = 4$

So, value of $a = (28 + 4)/2 = 16$

Value of $b = (28 – 4)/2 = 12$

So, value of $P = 16 \times 12 = 192$

II). $Y^2 - 17Y – Q = 0$

$(c + d) = 17$

On making square of both sides

$C^2 + d^2 + 2cd = 289$……... (3)

Also, $c^2 + d^2 = 689$……... (4)

From (3) and (4), we get

$- 2cd = 400$

So $(c \times d) = - 200$

So, value of $Q = 200$

III). $M^2 + 5M – R = 0$

$(e + f) = - 5$……. (5)

Also, $e^2 + 2f = 89$…… (6)

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On solving both equations we get

$e^2 - 2e = 99$

$e^2 - 2e - 99 = 0$

$(e - 11)(e + 9) = 0$

So, values of e = 11 and – 9

If e = 11 then f = - 16 (but e < f)

So, e = - 9 and f = 4

Answer: A

Required value = 192 – 200 + 2 x 36 = 64

Hence answer is option A

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1. Questions

A person buys the same item from two different shops at different prices. The cost price of the item in the first shop is Rs.35 and the Cost price of the item in the second shop is Rs.60. The person marked up both items at 60% above the cost price and sold at 10% discount. The amount of profit in this whole transaction is Rs. x. Find which of the following is true?

I). The difference between the selling price of items is 36.

II). The profit amount of first shopkeeper's item is 15.4.

III). If cost price of another item is Rs. 5x. When it is sold at 20% profit its selling price is 250.8

a. All false

b. Only I true

c. All true

d. Only II true

e. Only III true

2. Questions

Boat A covers M km downstream in 3 hours. Boat B covers N km upstream in 4 hours. Boat C covers [M+N] Km upstream distance in t hours. The ratio of the speed of boats A and B is 5:4 respectively. Ratio of the upstream speed of boat A and the speed of boat C in still water is 11:14 respectively. The speed of the stream of the river is 3 km/hr and Boat C covers 50 km upstream distance in 2 hours. Find the value of 't'.

a. 12.08

b. 9.66

c. 7.65

d. 6.08

e. 4.54

3. Questions

A train crosses a platform, whose length is 120 m in t sec. The same train crosses a pole in [t-4.8] sec. Another train of length 200 m crosses a platform of 240 m length in [t+10] sec. The speed of the second train is 20 m/sec. Find the length of the platform which was crossed by the first train in [t+30] sec?

a. 825

b. 870

c. 876

d. 998

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e. None of these

4. Questions

The ratio of age of A after 5 years and age of D after 10 years is 5:8 respectively. Ratio of age of A after 4 years and age of D after 2 years 3:4 respectively. Ratio of age of B after 3 years and age of D after 4 years is 19:17 respectively. Ratio of age of C after 5 years and age of B after 1 year is 5:4 respectively. Find the average ages of A, B, C and D?

a. 31.25

b. 32.35

c. 35.66

d. 29.55

e. None of these

5. Questions

Study the following information carefully and answer the questions given below.

There are five different series. Some are wrong number series and some are missing number series.

Series-I 785,704,632,596,569

Series-II 45,196,1005,6066, ?

Series-III 85,170,180,996,988,9880

Series-IV 78,73,136,393, ?

Series-V 75,75,77,231,435,1175

X, Y and Z are the wrong numbers of series I, III and V and M and N are missing numbers of series II and IV.

Find the sum of the correct numbers in the wrong number series.

a. 1245

b. 1362

c. 1042

d. 1472

e. 1654

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6. Questions

Find which of the following statement/s is/are true.

I). The sum of all the digits of M is divisible by 7.

II). N is divisible by 3.

III). Z is divisible by 5 but not divisible by 3.

a. Only I true
b. Only I false
c. All true
d. All false
e. Only III true

7. Questions

The below series follows the same logic as series I. Find the value 'g'.

932, i, j, k, g, l

a. 744
b. 734
c. 764
d. 716
e. 799

8. Questions

Find the value of [7X+8Y+9Z].

a. 9845
b. 9887
c. 9865
d. 9779
e. None of these

9. Questions

Study the following information carefully and answer the questions given below.

The given table shows the average weight (kg) of boys and girls, and the difference between the number of boys and girls of five different classes [A, B, C, D and E].

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Class	Average weight of boys (kg)	Average weight of girls (kg)	Difference between the number of boys and girls
A	65	45	4
B	65	42	6
C	67	43	4
D	62	49	12
E	70	48	4

Note: – The average weight of all the students of class A, B, C, D and E is in the AP series where 1st term of the AP series is equal to the average weight of class A, 2nd term is equal to B and So on. The last term of the AP series is 58 and the common difference of the AP series is the smallest odd number**.**

In class A, _____ new boys of weight 62 kg and 5 new girls of weight 48 kg joined. In class B, 6 new boys of weight 66 kg and _____ new girls of weight 45 kg joined. Now the difference in total weight of class A and B is _____.

Find which of the following is true?

a. 6,8,354

b. 9,6,258

c. 7,5,317

d. 9,11,325

e. More than one option is true

10. Questions

If the number of boys and girls in class C and F is same, the average age of boys in class F is (t+2) and the average age of girls in class F is t years. Then the total age of class F is 1204 years. Find which of the following is true?

I). The number of boys in classes C, D and E together is (3t+6).

II). The number of girls in classes A, C and E together is (3t-10).

III). The sum of boys and girls in class D is (2t+4).

a. Only I and II

b. Only III

c. Only III and II

d. Only I and III

e. Only I

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11. Questions

The ratio of number of boys and girls in class F is 3:4 respectively. The average number of girls in classes E and F is 26. The average age of boys and girls in class F is 62 years and 45 years respectively. Find the total weight of all the students of class E and F together.

a. 5847

b. 5114

c. 5264

d. 5964

e. None of these

12. Questions

All the students of classes D and E appeared in an exam. 20% of girls and 25% of boys of class D qualify the exam and 20% of boys and 25% of girls of class E qualify the exam. The ratio between the total number of boys and the total number of girls who qualify the exams is m:n respectively. Find the value of [7m+13n].

a. 107

b. 112

c. 156

d. 147

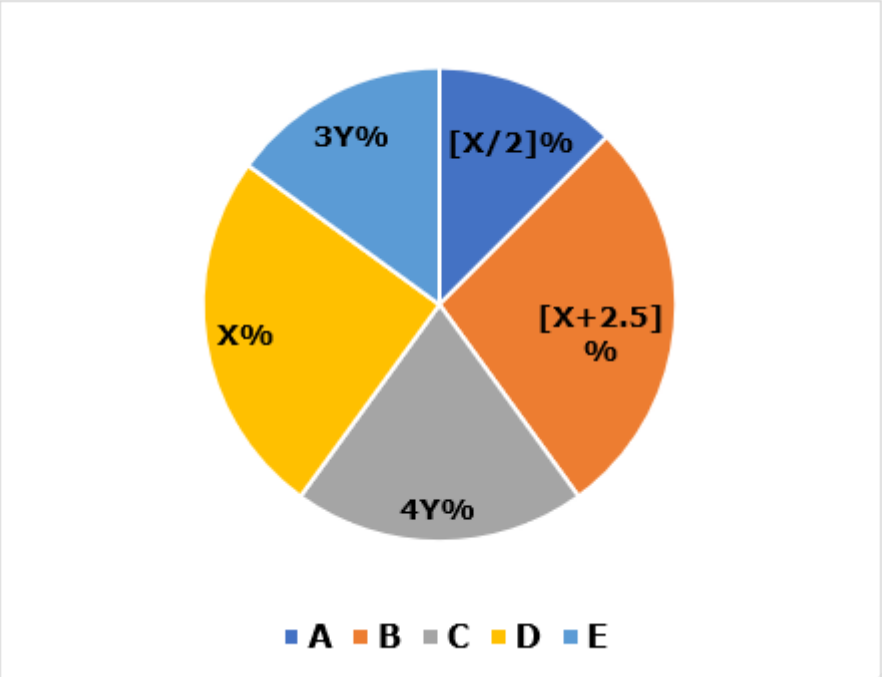
e. None of these

13. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage distribution of number of PDFs in five different memory cards [A, B, C, D and E].

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The given table shows the average number of PDFs and DOCs and the average amount of space occupied by PDFs, PPT and DOCs.

Memory card	The average number of PDFs and DOCs	Average amount of space occupied by PDFs	Average amount of space occupied by DOCs	Average amount of space occupied by PPTs
A	41	1 Mb	45 Kb	12 Mb
B	68	30 Mb	69 Kb	22 Mb
C	55	10 Mb	72 Kb	36 Mb
D	67	36 Mb	88 Kb	58 Mb
E	53	15 Mb	96 Kb	56 Mb

Note: – X+Y=30, the Memory size of each memory card is 8 Gb. 1Gb>1Mb>1Kb. The total number of PDFs in memory card is 320. The ratio of number of DOCs and PPTs in all memory cards is 2:3 respectively

1Gb=1024 Mb, 1 Mb= 1024 Kb

The approximate difference between the amount of free space (In Mb) in memory cards A and B is Z. Find which of the following is true?

I). 2X+3Y+Z=3896

II). 3X+4Y+2Z=6465

III). 8X+6Y+Z=3456

a. only False

b. all true

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c. all false

d. only I true

e. only III false

14. Questions

In free storage space of C there are t number of JPEG files is loaded and in storage free space of D there are n number of JPEG files is loaded. Find the approximate value of [t+n]? [Average size of JPEG file is 2.5 Mb]

a. 2490

b. 2960

c. 2130

d. 2270

e. 2740

15. Questions

The ratio of the number of PDFs, DOCs and PPTs in E and F is 2:3, 2:3 and 3:2 respectively. If the average amount of space occupied by F is same as that of E for PDFs, Docs, PPTs then find the approximate amount of free space (In GB) in F?

a. 4.56 Gb

b. 3.76 Gb

c. 7.56 Gb

d. 4.86 Gb

e. None of these

16. Questions

Find the approximate difference (In GB) between the amount of space occupied by PDF and PPT files in all memory cards together?

a. 7.29 Gb

b. 9.63 Gb

c. 8.88 Gb

d. 16.36 Gb

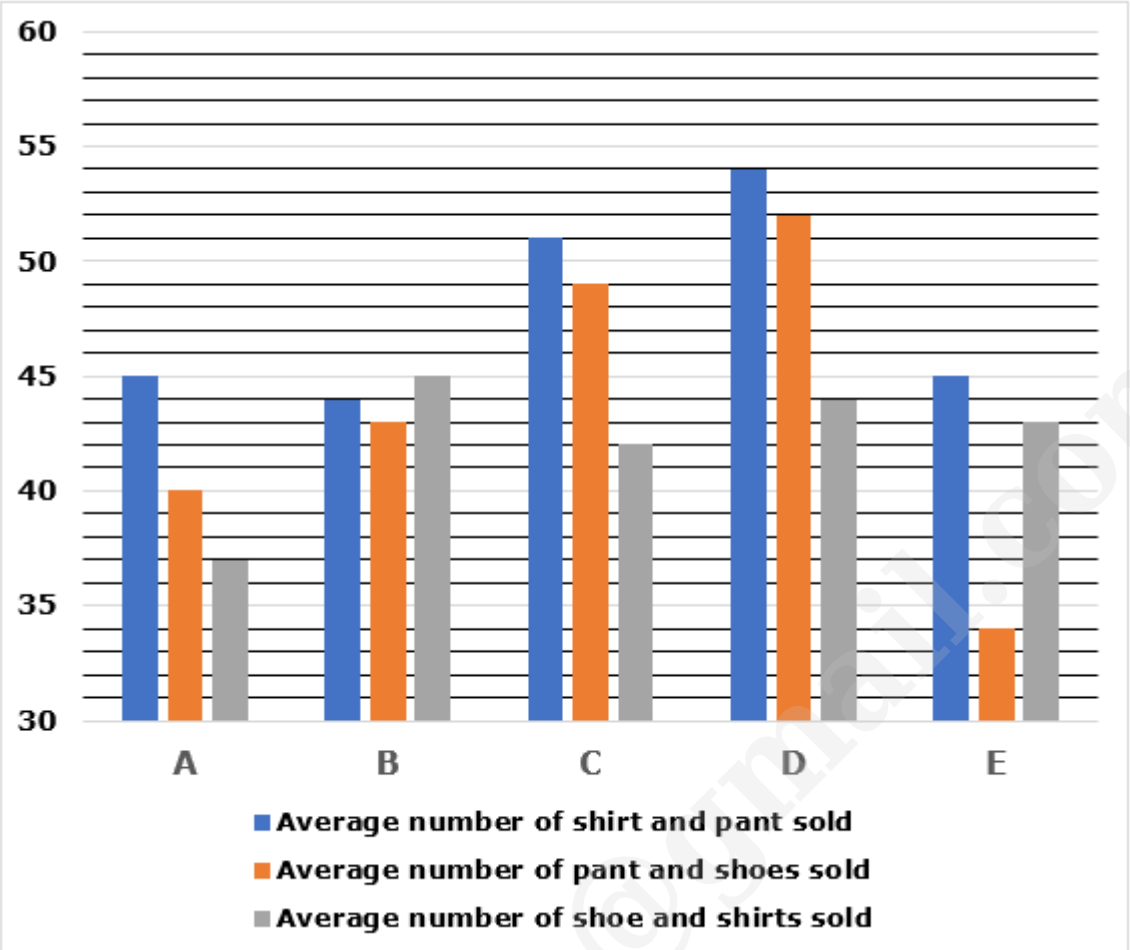
e. None of these

17. Questions

Study the following information carefully and answer the questions given below.

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The bar graph shows the average number of shirts and pants, the average number of pants and shoes and the average number of shoes and shirts sold in five different shops [A, B, C, D and E].



Out of the total pants sold in shops A and E, 25% are formal rest are casual. The ratio of the number of casual and formal pants sold in shops B, C and D is 2:1, 17:12 and 1:1 respectively. The difference between the total number of formal and the total number of causal pants sold in all shops together is x. Find the value of [7x].

a. 90

b. 92

c. 462

d. 588

e. 678

18. Questions

Find which of the following statement/s is/are true.

I). Average number of shirts sold in shops A, B and C together is 44.

II). Average number of items sold in shop D is 50.

III). Average number of shoes sold in shops C, D and E is 38.

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a. Only I true

b. Only II true

c. All false

d. All true

e. Only I and II are true

19. Questions

In shops A and B, Out of the total shirts sold, 25% are red shirts. In shops C and D, out of the total shirts sold, 30% are red shirts. In shop E, out of the total shirts sold 11.11% are red shirts. Out of the total red shirts sold in all shops together, 20% are formal. Find the total number of red formal shirts sold in all shops together?

a. 19

b. 11

c. 16

d. 18

e. None of these

20. Questions

The ratio of men's and women's shoes sold in all shops together is 3:2 respectively. Out of the total men's shoes sold in all shops together 50% are formal and out of the total women's shoes sold in all shops together 25% are formal. Find the total number of formal shoes sold in all shops together?

a. 76

b. 85

c. 82

d. 89

e. None of these

Explanations:

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1. Questions

Answer: C

Selling price of item from 1st shop = 35*[160/100]*[90/100]=50.4

Selling price of item from 2nd shop = 60*160/100*90/100=86.4

So, profit amount = [50.4+86.4]-[35+60]=41.8=x

I). The difference between the selling price of the item is 86.4-50.4=36.

II). The profit amount of first shopkeeper's item is 15.4.

III). The cost price of the item is 5x=5*41.8=209. 209*120/100=250.8

2. Questions

Answer: D

Upstream speed of boat C is 50/2=25 km/hr

So, speed of boat C in still water is 25+3=28 km/hr

Upstream speed of boat A is 28*11/14=22 km/hr

So, speed of boat A is 22+3= 25 km/hr

The speed of boat B is 25*4/5=20 km/hr.

M is = [25+3]*3=84

N is =[20-3]*4=68

So, t = [M+N]/25=[84+68]/25=6.08

3. Questions

Answer: B

The speed of the second train is 20 m/sec.

So, [200+240]/20 = t+10

Or, 22=t+10 or, t=12

So, the length of the first train is L and the speed of the first train is s.

So, (L+120)/s=12

And also, L/s=7.2

By solving the above equations,

L+120/L=12/7.2

Or, 12L=7.2L+864

Or, L=864/4.8=180

So, S= 180/7.2= 25 m/sec

So, [180+x]=42*25

Or, 180+x=1050

Or, x=1050-180=870

4. Questions

Answer: A

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Let the age of A and D be x and z

[x+5]:[z+10]= 5 : 8

So, 8x+40=5z+50

Or, 8x-5z=10

Again [x+4]:[z+2]=3 : 4

4x+16=3z+6

Or, 3z-4x=10

By solving the above equations we get x=20 and z=30.

Age of A is 20 years and age of D is 30 years.

Age of B after 3 years = 34*19/17=38

So, the Age of B is 38-3=35 years.

Age of C after 5 years 36*5/4=45 years

So, average = [20+35+40+30]/4=31.25

5. Questions

Series-I 785,704,632,596,569

785-9*9=704, 704-9*7=641, 641-9*5=596, 596-9*3=569

So, X=632

Series-II 45,196,1005,6066, ?

45*4+4*4=196, 196*5+25=1005…. 6066*7+49=42511=M

Series-III 85,170,180,996,988,9880

85*2=170, 170-4=166, 166*6=996, 996-8=988, 988*10=9880

So, Y=180

Series-IV 78,73,136,393, ?

78*1-5=73, 73*2-10=136, 136*3-15=393, 393*4-20=1552

N=1552

Series-V 75,75,77,231,435,1175

75*1=75, 75+2=77, 77*3=231, 231+4=235, 235*5=1175

So, Z=435

Answer: C

Required value = 641+166+235=1042

6. Questions

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Series-I 785,704,632,596,569

785-9*9=704, 704-9*7=641, 641-9*5=596, 596-9*3=569

So, X=632

Series-II 45,196,1005,6066, ?

45*4+4*4=196, 196*5+25=1005…. 6066*7+49=42511=M

Series-III 85,170,180,996,988,9880

85*2=170, 170-4=166, 166*6=996, 996-8=988, 988*10=9880

So, Y=180

Series-IV 78,73,136,393, ?

78*1-5=73, 73*2-10=136, 136*3-15=393, 393*4-20=1552

N=1552

Series-V 75,75,77,231,435,1175

75*1=75, 75+2=77, 77*3=231, 231+4=235, 235*5=1175

So, Z=435

Answer: D

The sum of the digits of M = 4+5+2+1+1=13, so not divisible by 7.

N is not divisible by 3.

Z is divisible by both 5 and 3.

7. Questions

Series-I 785,704,632,596,569

785-9*9=704, 704-9*7=641, 641-9*5=596, 596-9*3=569

So, X=632

Series-II 45,196,1005,6066, ?

45*4+4*4=196, 196*5+25=1005…. 6066*7+49=42511=M

Series-III 85,170,180,996,988,9880

85*2=170, 170-4=166, 166*6=996, 996-8=988, 988*10=9880

So, Y=180

Series-IV 78,73,136,393, ?

78*1-5=73, 73*2-10=136, 136*3-15=393, 393*4-20=1552

N=1552

Series-V 75,75,77,231,435,1175

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75*1=75, 75+2=77, 77*3=231, 231+4=235, 235*5=1175

So, Z=435

Answer: D

So, i=932-81=851, j=851-63=788, k=788-45=743, g=743-27=716

8. Questions

Series-I 785,704,632,596,569

785-9*9=704, 704-9*7=641, 641-9*5=596, 596-9*3=569

So, X=632

Series-II 45,196,1005,6066, ?

45*4+4*4=196, 196*5+25=1005…. 6066*7+49=42511=M

Series-III 85,170,180,996,988,9880

85*2=170, 170-4=166, 166*6=996, 996-8=988, 988*10=9880

So, Y=180

Series-IV 78,73,136,393, ?

78*1-5=73, 73*2-10=136, 136*3-15=393, 393*4-20=1552

N=1552

Series-V 75,75,77,231,435,1175

75*1=75, 75+2=77, 77*3=231, 231+4=235, 235*5=1175

So, Z=435

Answer: D

7X+8Y+9Z=7*632+8*180+9*435=9779

9. Questions

The last term of the AP series is 58.

The common difference is 1.

So, t_5=a+[n-1]*1=58 [a is 1st term]

Or, a=58-4*1=54

So, the Average age of all students of classes A, B, C, D and E is 54, [54+1]=55, [54+2]=56, [54+3]=57, [54+4]=58

Now, the Average weight of boys and girls of class A is 65 and 45 kg respectively.

So, By allegation method,

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65 45

54

9 11

So, the Ratio of boys and girls is 9:11

The difference between the number of boys and girls in A is 4.

So, 2 unit = 4, 1 unit = 2.

Number of boys in class A is 9*2=18

Number of girls in class A is 11*2=22

Similarly, we can calculate the others' values also.

Class	Number of boys	Number of girls
A	18	22
B	26	20
C	26	22
D	32	20
E	20	24

Answer: C

For the value of C,

New difference of total age of class A and B

= [42*20+65*26+6*66+5*45] - [22*45+65*18+7*62+5*48]

= 3151-2834=317

10. Questions

The last term of the AP series is 58.

The common difference is 1.

So, t_5=a+[n-1]*1=58 [a is 1st term]

Or, a=58-4*1=54

So, the Average age of all students of classes A, B, C, D and E is 54, [54+1]=55, [54+2]=56, [54+3]=57, [54+4]=58

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Now, the Average weight of boys and girls of class A is 65 and 45 kg respectively.

So, By allegation method,

Boys Girls

65 45

54

9 11

So, the Ratio of boys and girls is 9:11

The difference between the number of boys and girls in A is 4.

So, 2 unit = 4, 1 unit = 2.

Number of boys in class A is 9*2=18

Number of girls in class A is 11*2=22

Similarly, we can calculate the others' values also.

Class	Number of boys	Number of girls
A	18	22
B	26	20
C	26	22
D	32	20
E	20	24

Answer: D

So, 26*[t+2]+22*t=1204

Or, 48t=1204-26*2

Or, t= [1152]/48=24

I). The number of boys in classes C, D and E together is =26+32+20=3t+6.

II). The number of girls in classes A, C and E together is = 22+22+24=68.

III). The sum of boys and girls in class D is 20+32=52=2t+4.

11. Questions

The last term of the AP series is 58.

The common difference is 1.

So, t_5=a+[n-1]*1=58 [a is 1st term]

Or, a=58-4*1=54

So, the Average age of all students of classes A, B, C, D and E is 54, [54+1]=55, [54+2]=56, [54+3]=57, [54+4]=58

Now, the Average weight of boys and girls of class A is 65 and 45 kg respectively.

So, By allegation method,

Boys Girls

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Tg:- @NextGenBankers

65 45

54

9 11

So, the Ratio of boys and girls is 9:11

The difference between the number of boys and girls in A is 4.

So, 2 unit = 4, 1 unit = 2.

Number of boys in class A is 9*2=18

Number of girls in class A is 11*2=22

Similarly, we can calculate the others' values also.

Class	Number of boys	Number of girls
A	18	22
B	26	20
C	26	22
D	32	20
E	20	24

Answer: B

Number of girls in class F is = 26*2-24=28

Number of boys in class F is = 28*3/4=21

So, the total weight of all students in class F is = 28*45+21*62+58*44=5114

12. Questions

The last term of the AP series is 58.

The common difference is 1.

So, t_5=a+[n-1]*1=58 [a is 1st term]

Or, a=58-4*1=54

So, the Average age of all students of classes A, B, C, D and E is 54, [54+1]=55, [54+2]=56, [54+3]=57, [54+4]=58

Now, the Average weight of boys and girls of class A is 65 and 45 kg respectively.

So, By allegation method,

Boys Girls

65 45

54

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9 11

So, the Ratio of boys and girls is 9:11

The difference between the number of boys and girls in A is 4.

So, 2 unit = 4, 1 unit = 2.

Number of boys in class A is 9*2=18

Number of girls in class A is 11*2=22

Similarly, we can calculate the others' values also.

Class	Number of boys	Number of girls
A	18	22
B	26	20
C	26	22
D	32	20
E	20	24

Answer: A

Required ratio

= [32*25/100+20*20/100]: [20*20/100+24*25/100]

= [8+4]:[4+6]=12:10=6:5

So, required value=7m+13n=7*6+5*13=42+65=107

13. Questions

ATQ, X+Y=30 ----(I)

And Also, [X/2]+[X+2.5]+4Y+X+3Y=100

Or, 5X+14Y=195 ------(II)

By solving the above equation, we get X=25 and Y=5.

So, the number of PDFs in A is 320*12.5/100=40

The number of DOCs in A is 41*2-40=42

The number of PPTs in A is 42*3/2=63

Similarly, we can calculate others values also.

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Memory cards	Number of PDFs	Number of DOCs	Number of PPTs
A	40	42	63
B	88	48	72
C	64	46	69
D	80	54	81
E	48	58	87

Answer: C

Free space in A is = 8*1024-40*1-[42*45]/1024-63*12=7394.15=7394

Free Space in B is = 8*1024-88*30-[48*69]/1024-72*22=3964.76=3965

So, Z=7394-3965=3429

I). 2X+3Y+Z=3494

II). 3X+4Y+2Z=6953

III). 8X+6Y+Z=3659

14. Questions

ATQ, X+Y=30 ----(I)

And Also, [X/2]+[X+2.5]+4Y+X+3Y=100

Or, 5X+14Y=195 ------(II)

By solving the above equation, we get X=25 and Y=5.

So, the number of PDFs in A is 320*12.5/100=40

The number of DOCs in A is 41*2-40=42

The number of PPTs in A is 42*3/2=63

Similarly, we can calculate others values also.

Memory cards	Number of PDFs	Number of DOCs	Number of PPTs
A	40	42	63
B	88	48	72
C	64	46	69
D	80	54	81
E	48	58	87

Answer: D

Free space in C = 1024*8-64*10-[46*72/1024] - [69*36] = 5064.76=5065

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So, t=5065/2.5=2026

Free space in D = 1024*8-80*36-[54*88/1024]-58*81=609.35=610

So, n=610/2.5=244

So, t+n=2026+244=2270

15. Questions

ATQ, X+Y=30 ----(I)

And Also, [X/2]+[X+2.5]+4Y+X+3Y=100

Or, 5X+14Y=195 ------(II)

By solving the above equation, we get X=25 and Y=5.

So, the number of PDFs in A is 320*12.5/100=40

The number of DOCs in A is 41*2-40=42

The number of PPTs in A is 42*3/2=63

Similarly, we can calculate others values also.

Memory cards	Number of PDFs	Number of DOCs	Number of PPTs
A	40	42	63
B	88	48	72
C	64	46	69
D	80	54	81
E	48	58	87

Answer: B

The number of PDFs in F is 48*3/2=72

Number of DOCs in F is = 58*3/2=87

The number of PPTs in F is = 87*2/3=58

So, Free space = 8*1024-72*15-[87*96/1024]-58*56

= 3855.8 = 3856 Mb = [3856/1024] = 3.76 Gb

16. Questions

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ATQ, X+Y=30 ----(I)

And Also, [X/2]+[X+2.5]+4Y+X+3Y=100

Or, 5X+14Y=195 ------(II)

By solving the above equation, we get X=25 and Y=5.

So, the number of PDFs in A is 320*12.5/100=40

The number of DOCs in A is 41*2-40=42

The number of PPTs in A is 42*3/2=63

Similarly, we can calculate others values also.

Memory cards	Number of PDFs	Number of DOCs	Number of PPTs
A	40	42	63
B	88	48	72
C	64	46	69
D	80	54	81
E	48	58	87

Answer: A

Required difference

= [63*12+72*22+36*69+81*58+56*87] - [1*40+88*30+10*64+80*36+48*15]=14394-6920=7474 Mb =7.29 Gb

17. Questions

For Shop A,

Total number of shirts and pants sold is 45*2=90

The total number of pants and shoes sold is 40*2=80

The total number of shoes and shirts sold is 37*2=74

So, the number of shirts sold [90+80+74]/2-80=122-80=42

Number of Pants sold [122-74] =48

Number of shoes sold [122-90] =32

Similarly, we can calculate others' values also.

Shop	Number of shirts sold	Number of Pants sold	Number of shoes sold
A	42	48	32
B	46	42	44
C	44	58	40
D	46	62	42
E	54	36	32

Answer: C

Tg:-@NextGenBankers

Total causal pants sold in all shops together

= [48*75/100+36*75/100+42*2/3+58*17/29+62/2]

= 36+27+28+34+31=156

Total number of formal pants sold in all shops together

= (48+42+58+62+36) - 156=246 – 156 = 90

So, the difference is = 156-90 = 66 =x

So, 7x=66*7=462

18. Questions

For Shop A,

Total number of shirts and pants sold is 45*2=90

The total number of pants and shoes sold is 40*2=80

The total number of shoes and shirts sold is 37*2=74

So, the number of shirts sold [90+80+74]/2-80=122-80=42

Number of Pants sold [122-74] =48

Number of shoes sold [122-90] =32

Similarly, we can calculate others' values also.

Shop	Number of shirts sold	Number of Pants sold	Number of shoes sold
A	42	48	32
B	46	42	44
C	44	58	40
D	46	62	42
E	54	36	32

Answer: D

I). Average number of shirts sold in shops A, B and C together is [42+46+44]/3=44.

II). Average number of items sold in shop D is [46+62+42]/3=50.

III). Average number of shoes sold in shops C, D and E is [40+42+32]/3=38.

19. Questions

For Shop A,

Total number of shirts and pants sold is 45*2=90

The total number of pants and shoes sold is 40*2=80

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The total number of shoes and shirts sold is 37*2=74

So, the number of shirts sold [90+80+74]/2-80=122-80=42

Number of Pants sold [122-74] =48

Number of shoes sold [122-90] =32

Similarly, we can calculate others' values also.

Shop	Number of shirts sold	Number of Pants sold	Number of shoes sold
A	42	48	32
B	46	42	44
C	44	58	40
D	46	62	42
E	54	36	32

Answer: B

Total red formal shirts sold in all shops together

= [[42+46] *25/100+[44+46]*30/100+54/9]*20/100

= [22+27+6] *20/100=55*20/100=11

20. Questions

For Shop A,

Total number of shirts and pants sold is 45*2=90

The total number of pants and shoes sold is 40*2=80

The total number of shoes and shirts sold is 37*2=74

So, the number of shirts sold [90+80+74]/2-80=122-80=42

Number of Pants sold [122-74] =48

Number of shoes sold [122-90] =32

Similarly, we can calculate others' values also.

Shop	Number of shirts sold	Number of Pants sold	Number of shoes sold
A	42	48	32
B	46	42	44
C	44	58	40
D	46	62	42
E	54	36	32

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Answer: A

The total number of shoes sold in all shops together is 32+44+40+42+32=190

Total formal shoes sold in all shops together

[190*3/5]*50/100+[190*2/5]*25/100=57+19=76

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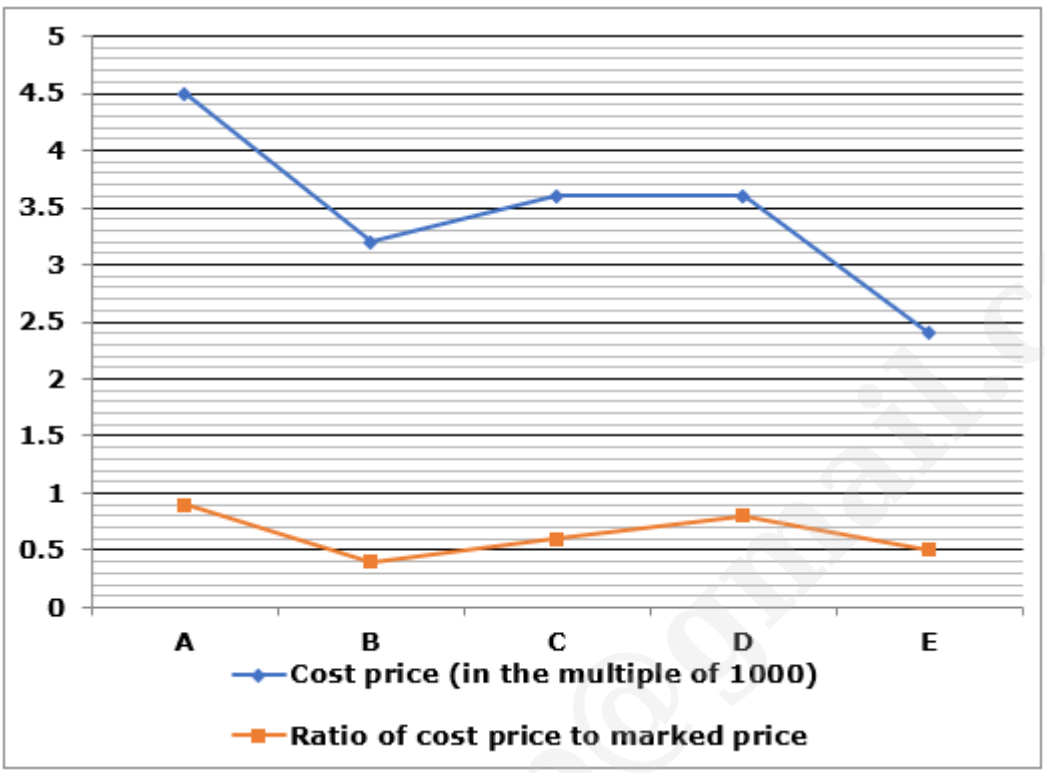
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1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the cost price and marked price of 5 articles A, B, C, D and E.

The line graph given below shows the cost price of each article (in ₹ and in the multiple of 1000) and the ratio of cost price to marked price of each article.



Note: Articles A, B, C, D and E are marked up by P%, Q%, R%, S% and T% respectively.

If article B were sold at (18P – Q)% discount on its marked price and article C were sold at [(T – S)/3]% discount on its marked price, then what would have been the ratio of profit amount received on article B to that received on article C?

a. 2:3

b. 8:9

c. 4:5

d. 18:19

e. 10:13

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2. Questions

Which of the following statements is/are not true?

X: Article B is marked up by 2(T – S)%.

Y: Value of (P + R) is divisible by 3.

Z: Article E is marked up by (Q – 2S)%.

a. Only Y

b. Only Z

c. Only X and Z

d. Only X

e. Only X and Y

3. Questions

If article D was sold at (3R/10)% discount on its marked price and article E was sold at (9P/4)% discount on its marked price, then what would have been the total profit per cent on article D and E together?

a. 25%

b. 33.33%

c. 20%

d. 15%

e. 12.5%

4. Questions

The cost price of article F is (Q – S)% of the cost price of article E and the ratio of the cost price to marked price of article F is P: R respectively. If article F is sold at (Q + S – T)% discount on its marked price, then what will be the profit received on article F?

a. ₹ 1200

b. ₹ 1800

c. ₹ 2400

d. ₹ 2100

e. ₹ 1500

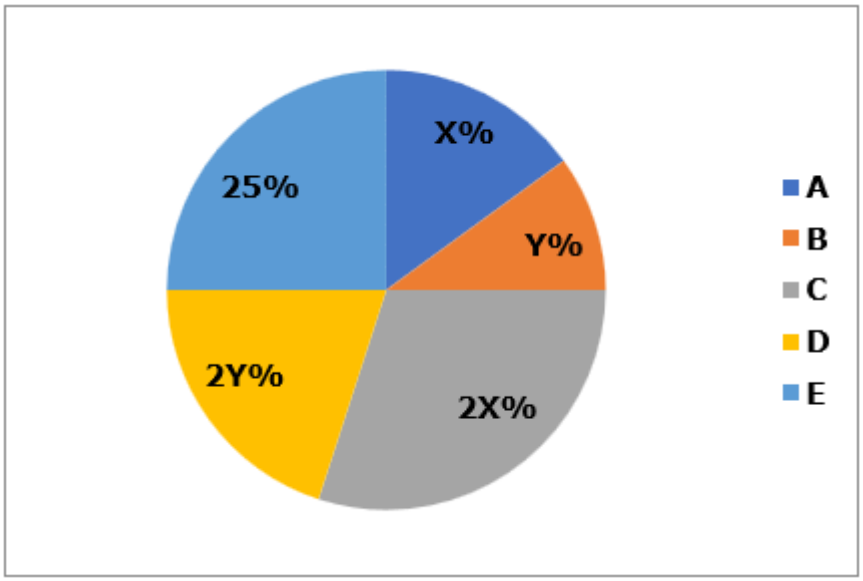
5. Questions

Study the following data carefully and answer the questions:

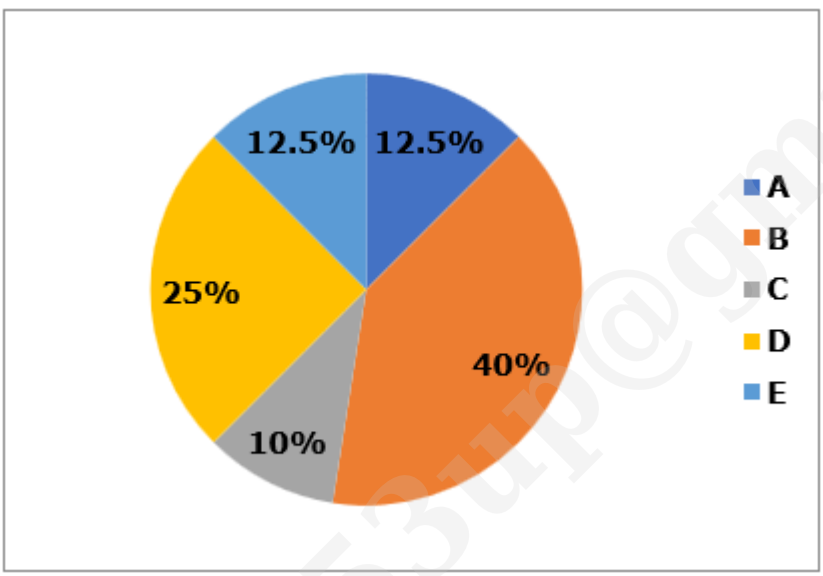
The data given below is related to the total quantity (in kg) of fruits sold by five street vendors A, B, C, D and E on Mon and Tue.

The pie chart given below shows the percentage distribution of the total quantity (in kg) of fruits sold by each street vendor on Mon.

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The pie chart given below shows the percentage distribution of the total quantity (in kg) of fruits sold by each street vendor on Tue.



Notes:

1: On Mon, difference between the quantity of fruits sold by A and C is 18 kg and the difference between the quantity of fruits sold by B and D is 12 kg.

2: Total quantity of fruits sold by A on Mon and Tue together is 28 kg.

On Mon, ratio of the quantity of fresh fruits to the quantity of stale fruits sold by C, is 2: p respectively. On Tue, quantity of fresh fruits sold by C, is 4 kg more than the quantity of stale fruits sold by him. If the total quantity of stale fruits sold by C on Mon and Tue together is 14 kg, then find the value of 'p'?

a. 13

b. Can't be determined

c. 2

d. 7

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e. 1

6. Questions

On Mon, C and D sold their fruits at ₹50 per kg and ₹60 per kg respectively. On Tue, C and D sold their fruits at ₹45 per kg and ₹75 per kg respectively. Find the ratio of the total amount received by C by selling fruits on Mon and Tue together to that received by D by selling fruits on Mon and Tue together?

a. 16:25

b. 6:11

c. 36:49

d. 18:29

e. 9:14

7. Questions

If the quantity of fruits sold by F on Mon is $(X^2 - Y^2)\%$ of that sold by E on Mon and the quantity of fruits sold by F on Tue is $(XY)\%$ of that sold by E on Tue, then find the total quantity of fruits sold by F on Mon and Tue together?

a. 45.5 kg

b. 42.5 kg

c. 54.5 kg

d. 48.5 kg

e. 52.5 kg

8. Questions

The total quantity of fruits sold by those street vendors whose sum of the quantity of fruits sold on both days together is equal , is what per cent of the total quantity of fruits sold by all the street vendors on both the days together?

a. $(4X + Y - 4)\%$

b. $15(X - Y)\%$

c. $(5X - Y + 4)\%$

d. $10(X - Y)\%$

e. $(6X - 2Y)\%$

9. Questions

Study the following data carefully and answer the questions:

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The data given below is related to the number of questions attempted by four candidates A, B, C and D in an exam consists of Maths, Reasoning and English.

The table given below shows the cumulative data about the number of questions attempted by each candidate in each subject.

Candidates	No. of questions attempted in Maths	No. of questions attempted in Reasoning as % of that attempted in Maths	No. of questions attempted in English as % of that attempted in Maths
A	32	62.5%	50%
A + B	50	78%	80%
A + B + C	75	76%	88%
A + B + C + D	90	90%	P%

Note: Each question consists of 1 mark and there is a negative mark of 0.25 marks for each wrong answer.

If the ratio of the total number of questions attempted by C to that attempted by D in the exam, is 23: 17, then find the value of P?

a. 86.67%

b. 83.33%

c. 90%

d. 93.75%

e. 80%

10. Questions

If the number of questions answered wrong by A is 36% of the number of questions answered correct by A, which is also equal to the number of questions answered correct by B, then find the difference between the marks obtained by A and B?

a. 1.25

b. 2.25

c. 1.5

d. 1.75

e. 2.5

11. Questions

If the value of P was 88(8/9), then the total number of questions attempted by D would have been approximately what per cent of the total number of questions attempted by A in the exam?

a. 85%

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b. 67%

c. 78%

d. 72%

e. 64%

12. Questions

Number of questions answered wrong by C in Maths, Reasoning and English are 5, 3 and 8 respectively. Find the ratio between the marks obtained by C in Maths, Reasoning and English respectively?

a. 5:3:4

b. 75:57:64

c. 25:19:24

d. 15:9:14

e. 9:7:64

13. Questions

The time taken by a train to cross a 300 m long platform is 5 seconds more than that taken by the same train to cross a 195 m long platform. If the speed of the train were 4 m/s more than its original speed, it would have crossed a pole in 9 seconds. Find the time taken by the train to cross a 153 m long tunnel?

a. 25 seconds

b. 16 seconds

c. 22 seconds

d. 20 seconds

e. 18 seconds

14. Questions

A and B together can complete 75% part of a work in 10.5 hours, B and C together can complete the remaining work in 1.5 hours and the ratio of B's efficiency to C's efficiency is 1: 6. If the difference between the time, in which A alone and D alone can complete the work, is 3 hours, then the time, in which D alone can complete the work, will fall in which of the following ranges ?

P: 12 and 25

Q: 17 and 27

R: 10 and 22

a. Only P

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b. Only R

c. Only P and Q

d. Only Q

e. Only P and R

15. Questions

Two boats A and B are coming towards each other, and the stream is flowing against boat B. Boats A and B are 450 m away from each other and cross each other in ___ seconds. Speed of stream is 3 m/s, downstream speed of boat A is equal to upstream speed of boat B and the ratio of upstream speed of boat A to downstream speed of boat B is 3: 7. Find the value, which can be filled in the blank?

a. 15

b. 9

c. 30

d. 25

e. Can't be determined

16. Questions

A, B and C started a business by investing in the ratio of 10: 15: 8 respectively. After 5 months, A withdrew 20% of his initial investment, B withdrew 33(1/3)% of his initial investment and C invested 50% more than his initial investment. At the end of 15 months, the total profit received by A, B and C together is ₹116250.

Find that which of the following statements is/are true?

P: A's profit share is 81.25% of C's profit share.

Q: Difference between B's profit share and C's profit share is ₹3750.

R: Average of A's profit share and B's profit share is ₹38125.

a. Only P

b. Only Q and R

c. Only R

d. Only Q

e. All are true

17. Questions

In each of the following questions, read the given statement and compare quantity I and quantity II on its basis. (only quantity is to be compared)

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The simple interest received on ₹15000 at (R + 4)% rate after 'T' years is ₹3600 and the simple interest received on ₹18000 at R% rate after 'T' years is ₹2880.

Quantity I: Value of R is what per cent less than the value of (R + 4)?

Quantity II: Value of T is what per cent of R?

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I ≥ Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I = Quantity II or relation can't be established

18. Questions

The present age of A and B is 24 years and 35 years respectively. Ratio of A's age after 'x' years to B's age after (x + y) years will be 5: 8 and 'y' is 116(2/3)% of 'x'.

Quantity I: If the difference between the present ages of A and C is 'x' years, then what is the possible present age of C (in years)?

Quantity II: If D's present age is (x – 1) years less than B's present age, then find D's present age (in years)?

a. Quantity I = Quantity II or relation can't be established

b. Quantity I < Quantity II

c. Quantity I > Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I ≥ Quantity II

19. Questions

Quantity I: Boat A went 2.16 km in downstream direction in 2.25 minutes and came back in 3.6 minutes. If the speed of boat B in still water is 2 m/s more than that of boat A, then find the total time (in seconds) taken by boat B to go 2.97 km in downstream direction and come back?

Quantity II: A shopkeeper marked up an article by 25% and sold at 40% discount on its marked price. If the shopkeeper had sold the article at 20% discount on its cost price, he would have received ₹21 more that original selling price. Find the cost price of the article (in ₹)?

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a. Quantity I ≥ Quantity II

b. Quantity I ≤ Quantity II

c. Quantity I = Quantity II or relation can't be established

d. Quantity I < Quantity II

e. Quantity I > Quantity II

20. Questions

A mixture contains 20 L milk and 12 L water. When 'X' L mixture is taken out and 14 L milk is added to the mixture, then the ratio of milk to water becomes 4: 1 respectively.

Quantity I: Find the value of 'X'?

Quantity II: If the cost of pure milk is ₹24 per L, then find the cost of the initial mixture (in ₹ per L)?

a. Quantity I = Quantity II or relation can't be established

b. Quantity I < Quantity II

c. Quantity I > Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I ≥ Quantity II

Explanations:

1. Questions

Article	Cost price (in ₹)	Ratio of cost price to marked price	Marked price (in ₹)
A	4500	9: 10	4500 * (10/9) = 5000
B	3200	4: 10	3200 * (10/4) = 8000
C	3600	6: 10	3600 * (10/6) = 6000
D	3600	8: 10	3600 * (10/8) = 4500
E	2400	5: 10	2400 * (10/5) = 4800

Since, article A is marked up by P%.

So, P% = $\frac{10-9}{9} \times 100$ = (100/9)%

Since, article B is marked up by Q%.

So, Q% = $\frac{10-4}{4} \times 100$ = 150%

Since, article C is marked up by R%.

So, R% = $\frac{10-6}{6} \times 100$ = (200/3)%

Since, article D is marked up by S%.

So, S% = $\frac{10-8}{8} \times 100$ = 25%

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Since, article E is marked up by T%.

So, T% = $\frac{10-5}{5} \times 100$ = 100%

Answer: B

The CP of article B = ₹3200

The MP of article B = ₹8000

If article B were sold at (18P – Q)% discount on its MP:

The SP of article B would have been:

[100 – (18P – Q)]% of 8000 = (100 – 50)% of 8000 = ₹4000

And the profit amount received on article B would have been = 4000 – 3200 = ₹800

The CP of article C = ₹3600

The MP of article C = ₹6000

If article C were sold at [(T – S)/3]% discount on its MP:

The SP of article C would have been:

[100 – {(T – S)/3}]% of 6000 = (100 – 25)% of 6000 = ₹4500

And the profit amount received on article C would have been = 4500 – 3600 = ₹900

Required ratio = 800:900 = 8:9

2. Questions

Article	Cost price (in ₹)	Ratio of cost price to marked price	Marked price (in ₹)
A	4500	9: 10	4500 * (10/9) = 5000
B	3200	4: 10	3200 * (10/4) = 8000
C	3600	6: 10	3600 * (10/6) = 6000
D	3600	8: 10	3600 * (10/8) = 4500
E	2400	5: 10	2400 * (10/5) = 4800

Since, article A is marked up by P%.

So, P% = $\frac{10-9}{9} \times 100$ = (100/9)%

Since, article B is marked up by Q%.

So, Q% = $\frac{10-4}{4} \times 100$ = 150%

Since, article C is marked up by R%.

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So, R% = $\frac{10-6}{6} \times 100$ = (200/3)%

Since, article D is marked up by S%.

So, S% = $\frac{10-8}{8} \times 100$ = 25%

Since, article E is marked up by T%.

So, T% = $\frac{10-5}{5} \times 100$ = 100%

Answer: A

From X:

Since, the value of 2(T – S) = 2(100 – 25) = 150

And article B is marked up by 150%.

So, X is true.

From Y:

The value of (P + R) = $\frac{100}{9} + \frac{200}{3} = \frac{100+600}{9} = \frac{700}{9}$

Since, (700/9) is not divisible by 3.

So, Y is not true.

From Z:

Since, the value of (Q – 2S) = (150 – 50) = 100

And article E is marked up by 100%.

So, Z is true.

Hence, only Y is not true.

3. Questions

Article	Cost price (in ₹)	Ratio of cost price to marked price	Marked price (in ₹)
A	4500	9: 10	4500 * (10/9) = 5000
B	3200	4: 10	3200 * (10/4) = 8000
C	3600	6: 10	3600 * (10/6) = 6000
D	3600	8: 10	3600 * (10/8) = 4500
E	2400	5: 10	2400 * (10/5) = 4800

Since, article A is marked up by P%.

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So, P% = $\frac{10-9}{9} \times 100$ = (100/9)%

Since, article B is marked up by Q%.

So, Q% = $\frac{10-4}{4} \times 100$ = 150%

Since, article C is marked up by R%.

So, R% = $\frac{10-6}{6} \times 100$ = (200/3)%

Since, article D is marked up by S%.

So, S% = $\frac{10-8}{8} \times 100$ = 25%

Since, article E is marked up by T%.

So, T% = $\frac{10-5}{5} \times 100$ = 100%

Answer: C

The CP of article D = ₹3600

The MP of article D = ₹4500

If article D were sold at (3R/10)% discount on its MP:

The SP of article D would have been:

[100 – (3R/10)]% of 4500 = [100 – 20]% of 4500 = ₹3600

The CP of article E = ₹2400

The MP of article E = ₹4800

If article E were sold at (9P/4)% discount on its MP:

The SP of article E would have been:

[100 – (9P/4)]% of 4800 = [100 – 25]% of 4800 = ₹3600

Since, the total CP of articles D and E together = 3600 + 2400 = ₹6000

And the total SP of articles D and E together = 3600 + 3600 = ₹7200

So, the required profit per cent = $\frac{7200-6000}{6000} \times 100 = 20\%$

4. Questions

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Article	Cost price (in ₹)	Ratio of cost price to marked price	Marked price (in ₹)
A	4500	9: 10	4500 * (10/9) = 5000
B	3200	4: 10	3200 * (10/4) = 8000
C	3600	6: 10	3600 * (10/6) = 6000
D	3600	8: 10	3600 * (10/8) = 4500
E	2400	5: 10	2400 * (10/5) = 4800

Since, article A is marked up by P%.

So, P% = $\frac{10-9}{9} \times 100$ = (100/9)%

Since, article B is marked up by Q%.

So, Q% = $\frac{10-4}{4} \times 100$ = 150%

Since, article C is marked up by R%.

So, R% = $\frac{10-6}{6} \times 100$ = (200/3)%

Since, article D is marked up by S%.

So, S% = $\frac{10-8}{8} \times 100$ = 25%

Since, article E is marked up by T%.

So, T% = $\frac{10-5}{5} \times 100$ = 100%

Answer: E

Since, the CP of article E = ₹2400

So, the CP of article F = (Q – S)% of 2400 = (150 – 25)% of 2400 = ₹3000

Since, the ratio of CP to MP of article F is P: R.

And, P: R = (100/9): (200/3) = 300: 1800 = 1: 6

So, the MP of article F = 3000 * (6/1) = ₹18000

If article F is sold at (Q + S – T)% discount on its MP:

The SP of article F will be:

[100 – (Q + S – T)]% of 18000 = [100 – (150 + 25 – 100)]% of 18000 = ₹4500

So, the profit received on article F = 4500 – 3000 = ₹1500

5. Questions

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From the 1st pie chart:

The ratio of the quantity of fruits sold by A to that sold by C on Mon = X: 2X = 1: 2

Since, the difference between the quantities of fruits sold by A and C on Mon = 18

So, the quantity of fruits sold by A on Mon = $18 \times \frac{1}{2-1}$ = 18 kg

And the quantity of fruits sold by C on Mon = 18 + 18 = 36 kg

The ratio of the quantity of fruits sold by B to that sold by D on Mon = Y:2Y = 1:2

Since, the difference between the quantity of fruits sold by B and D on Mon = 12

So, the quantity of fruits sold by B on Mon = $12 \times \frac{1}{2-1}$ = 12 kg

The quantity of fruits sold by D on Mon = 12 + 12 = 24 kg

And the quantity of fruits sold by E on Mon:

$$(18 + 36 + 12 + 24) \times \frac{25}{75} = 30 \text{ kg}$$

Since, the total quantity of fruits sold by all the 5 street vendors together on Mon = 18 + 36 + 12 + 24 + 30 = 120 kg

So, X = (18/120) * 100 = 15%

And, Y = (12/120) * 100 = 10%

Since, the total quantity of fruits sold by A on Mon and Tue together is 28 kg.

So, the quantity of fruits sold by A on Tue = 28 – 18 = 10 kg

The quantity of fruits sold by B on Tue = 10 * (40/12.5) = 32 kg

The quantity of fruits sold by C on Tue = 10 * (10/12.5) = 8 kg

The quantity of fruits sold by D on Tue = 10 * (25/12.5) = 20 kg

The quantity of fruits sold by E on Tue = 10 * (12.5/12.5) = 10 kg

And the total quantity of fruits sold by all the 5 street vendors together on Tue = 10 + 32 + 8 + 20 + 10 = 80 kg

Street vendor	Quantity of fruits sold on Mon (in kg)	Quantity of fruits sold on Tue (in kg)
A	18	10
B	12	32
C	36	8
D	24	20
E	30	10

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Answer: E

Since, the total quantity of fruits sold by C on Tue = 8 kg

So, the quantity of stale fruits sold by C on Tue = (8 – 4)/2 = 2 kg

And the quantity of stale fruits sold by C on Mon = 14 – 2 = 12 kg

Since, the total quantity of fruits sold by C on Mon = 36 kg

So, the quantity of fresh fruits sold by C on Mon = 36 – 12 = 24 kg

Since, the ratio of the quantity of fresh fruits to the quantity of stale fruits sold by C on Mon = 2:p = 24:12 = 2:1

So, the value of 'p'= 1

6. Questions

From the 1st pie chart:

The ratio of the quantity of fruits sold by A to that sold by C on Mon = X: 2X = 1: 2

Since, the difference between the quantities of fruits sold by A and C on Mon = 18

So, the quantity of fruits sold by A on Mon = $18 \times \frac{1}{2-1}$ = 18 kg

And the quantity of fruits sold by C on Mon = 18 + 18 = 36 kg

The ratio of the quantity of fruits sold by B to that sold by D on Mon = Y:2Y = 1:2

Since, the difference between the quantity of fruits sold by B and D on Mon = 12

So, the quantity of fruits sold by B on Mon = $12 \times \frac{1}{2-1}$ = 12 kg

The quantity of fruits sold by D on Mon = 12 + 12 = 24 kg

And the quantity of fruits sold by E on Mon:

$$(18 + 36 + 12 + 24) \times \frac{25}{75} = 30 \text{ kg}$$

Since, the total quantity of fruits sold by all the 5 street vendors together on Mon = 18 + 36 + 12 + 24 + 30 = 120 kg

So, X = (18/120) * 100 = 15%

And, Y = (12/120) * 100 = 10%

Since, the total quantity of fruits sold by A on Mon and Tue together is 28 kg.

So, the quantity of fruits sold by A on Tue = 28 – 18 = 10 kg

The quantity of fruits sold by B on Tue = 10 * (40/12.5) = 32 kg

The quantity of fruits sold by C on Tue = 10 * (10/12.5) = 8 kg

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The quantity of fruits sold by D on Tue = 10 * (25/12.5) = 20 kg

The quantity of fruits sold by E on Tue = 10 * (12.5/12.5) = 10 kg

And the total quantity of fruits sold by all the 5 street vendors together on Tue = 10 + 32 + 8 + 20 + 10 = 80 kg

Street vendor	Quantity of fruits sold on Mon (in kg)	Quantity of fruits sold on Tue (in kg)
A	18	10
B	12	32
C	36	8
D	24	20
E	30	10

Answer: C

Since, C sold their fruits on Mon and Tue at ₹50 per kg and ₹45 per kg respectively.

And the quantities of fruits sold by C on Mon and Tue are 36 kg and 8 kg respectively.

So, the total amount received by C by selling fruits on Mon and Tue together:

(36 * 50) + (8 * 45) = 1800 + 360 = ₹2160

Since, D sold their fruits on Mon and Tue at ₹60 per kg and ₹75 per kg respectively.

And the quantities of fruits sold by D on Mon and Tue are 24 kg and 20kg respectively.

So, the total amount received by D by selling fruits on Mon and Tue together:

(24 * 60) + (20 * 75) = 1440 + 1500 = ₹2940

Required ratio = 2160: 2940 = 36: 49

7. Questions

From the 1st pie chart:

The ratio of the quantity of fruits sold by A to that sold by C on Mon = X: 2X = 1: 2

Since, the difference between the quantities of fruits sold by A and C on Mon = 18

So, the quantity of fruits sold by A on Mon = $18 \times \frac{1}{2-1}$ = 18 kg

And the quantity of fruits sold by C on Mon = 18 + 18 = 36 kg

The ratio of the quantity of fruits sold by B to that sold by D on Mon = Y:2Y = 1:2

Since, the difference between the quantity of fruits sold by B and D on Mon = 12

So, the quantity of fruits sold by B on Mon = $12 \times \frac{1}{2-1}$ = 12 kg

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The quantity of fruits sold by D on Mon = 12 + 12 = 24 kg

And the quantity of fruits sold by E on Mon:

$$(18 + 36 + 12 + 24) \times \frac{25}{75} = 30 \text{ kg}$$

Since, the total quantity of fruits sold by all the 5 street vendors together on Mon = 18 + 36 + 12 + 24 + 30 = 120 kg

So, X = (18/120) * 100 = 15%

And, Y = (12/120) * 100 = 10%

Since, the total quantity of fruits sold by A on Mon and Tue together is 28 kg.

So, the quantity of fruits sold by A on Tue = 28 – 18 = 10 kg

The quantity of fruits sold by B on Tue = 10 * (40/12.5) = 32 kg

The quantity of fruits sold by C on Tue = 10 * (10/12.5) = 8 kg

The quantity of fruits sold by D on Tue = 10 * (25/12.5) = 20 kg

The quantity of fruits sold by E on Tue = 10 * (12.5/12.5) = 10 kg

And the total quantity of fruits sold by all the 5 street vendors together on Tue = 10 + 32 + 8 + 20 + 10 = 80 kg

Street vendor	Quantity of fruits sold on Mon (in kg)	Quantity of fruits sold on Tue (in kg)
A	18	10
B	12	32
C	36	8
D	24	20
E	30	10

Answer: E

Since, the quantity of fruits sold by E on Mon = 30 kg

So, the quantity of fruits sold by F on Mon:

$(X^2 – Y^2)\%$ of 30 = $(15^2 – 10^2)\%$ of 30 = 37.5 kg

Since, the quantity of fruits sold by E on Tue = 10 kg

So, the quantity of fruits sold by F on Tue:

(XY)% of 10 = (15 * 10)% of 10 = 15 kg

And the total quantity of fruits sold by F on Mon and Tue together:

37.5 + 15 = 52.5 kg

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8. Questions

From the 1st pie chart:

The ratio of the quantity of fruits sold by A to that sold by C on Mon = X: 2X = 1: 2

Since, the difference between the quantities of fruits sold by A and C on Mon = 18

So, the quantity of fruits sold by A on Mon = $18 \times \frac{1}{2-1}$ = 18 kg

And the quantity of fruits sold by C on Mon = 18 + 18 = 36 kg

The ratio of the quantity of fruits sold by B to that sold by D on Mon = Y:2Y = 1:2

Since, the difference between the quantity of fruits sold by B and D on Mon = 12

So, the quantity of fruits sold by B on Mon = $12 \times \frac{1}{2-1}$ = 12 kg

The quantity of fruits sold by D on Mon = 12 + 12 = 24 kg

And the quantity of fruits sold by E on Mon:

$$(18 + 36 + 12 + 24) \times \frac{25}{75} = 30 \text{ kg}$$

Since, the total quantity of fruits sold by all the 5 street vendors together on Mon = 18 + 36 + 12 + 24 + 30 = 120 kg

So, X = (18/120) * 100 = 15%

And, Y = (12/120) * 100 = 10%

Since, the total quantity of fruits sold by A on Mon and Tue together is 28 kg.

So, the quantity of fruits sold by A on Tue = 28 – 18 = 10 kg

The quantity of fruits sold by B on Tue = 10 * (40/12.5) = 32 kg

The quantity of fruits sold by C on Tue = 10 * (10/12.5) = 8 kg

The quantity of fruits sold by D on Tue = 10 * (25/12.5) = 20 kg

The quantity of fruits sold by E on Tue = 10 * (12.5/12.5) = 10 kg

And the total quantity of fruits sold by all the 5 street vendors together on Tue = 10 + 32 + 8 + 20 + 10 = 80 kg

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Street vendor	Quantity of fruits sold on Mon (in kg)	Quantity of fruits sold on Tue (in kg)
A	18	10
B	12	32
C	36	8
D	24	20
E	30	10

Answer: A

Total quantity of fruits sold by A on Mon and Tue together = 18 + 10 = 28 kg

Total quantity of fruits sold by B on Mon and Tue together = 12 + 32 = 44 kg

Total quantity of fruits sold by C on Mon and Tue together = 36 + 8 = 44 kg

Total quantity of fruits sold by D on Mon and Tue together = 24 + 20 = 44 kg

Total quantity of fruits sold by E on Mon and Tue together = 30 + 10 = 40 kg

Since, the street vendors, who sold equal quantity of fruits on both the days together, are B, C and D.

The total quantity of fruits sold by B, C and D on both the days together:

44 + 44 + 44 = 132 kg

And the total quantity of fruits sold by all the street vendors on both the days together = 120 + 80 = 200 kg

So, the required percentage = (132/200) * 100 = 66% = (4X + Y – 4)%

9. Questions

Candidates	No. of questions attempted in Maths	No. of questions attempted in Reasoning	No. of questions attempted in English
A	32	62.5% of 32 = 20	50% of 32 = 16
A + B	50	78% of 50 = 39	80% of 50 = 40
A + B + C	75	76% of 75 = 57	88% of 75 = 66
A + B + C + D	90	90% of 90 = 81	P% of 90 = 9P/10

Candidate	No. of questions attempted in Maths	No. of questions attempted in reasoning	No. of questions attempted in English
A	32	20	16
B	50 – 32 = 18	39 – 20 = 19	40 – 16 = 24
C	75 – 50 = 25	57 – 39 = 18	66 – 40 = 26
D	90 – 75 = 15	81 – 57 = 24	(9P/10) – 66

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Answer: A

Since, the total number of questions attempted by C in the exam:

25 + 18 + 26 = 69

So, the total number of questions attempted by D in the exam:

69 * (17/23) = 51

And the number of questions attempted by D in English = 51 – (15 + 24) = 12

(9P/10) – 66 = 12

P= 78*10/9

P = 86.67

10. Questions

Candidates	No. of questions attempted in Maths	No. of questions attempted in Reasoning	No. of questions attempted in English
A	32	62.5% of 32 = 20	50% of 32 = 16
A + B	50	78% of 50 = 39	80% of 50 = 40
A + B + C	75	76% of 75 = 57	88% of 75 = 66
A + B + C + D	90	90% of 90 = 81	P% of 90 = 9P/10

Candidate	No. of questions attempted in Maths	No. of questions attempted in reasoning	No. of questions attempted in English
A	32	20	16
B	50 – 32 = 18	39 – 20 = 19	40 – 16 = 24
C	75 – 50 = 25	57 – 39 = 18	66 – 40 = 26
D	90 – 75 = 15	81 – 57 = 24	(9P/10) – 66

Answer: D

Since, the ratio of the number of questions answered correct to the number of questions answered wrong by A = 100: 36 = 25: 9

And the total number of question attempted by A = 68

So, the number of questions answered correct by A = 68 * (25/34) = 50

And the number of questions answered wrong by A = 68 – 50 = 18

Since, the number of questions answered correct by B is equal to that answered correct by A.

So, the number of questions answered correct by B = 50

Since, the total number of questions attempted by B = 61

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So, the number of questions answered wrong by B = 61 – 50 = 11

Since, the total marks obtained by A in the exam:

(50 * 1) – (18 * 0.25) = 45.5

And the total marks obtained by B in the exam:

(50 * 1) – (11 * 0.25) = 47.25

So, the required difference = 47.25 – 45.5 = 1.75

11. Questions

Candidates	No. of questions attempted in Maths	No. of questions attempted in Reasoning	No. of questions attempted in English
A	32	62.5% of 32 = 20	50% of 32 = 16
A + B	50	78% of 50 = 39	80% of 50 = 40
A + B + C	75	76% of 75 = 57	88% of 75 = 66
A + B + C + D	90	90% of 90 = 81	P% of 90 = 9P/10

Candidate	No. of questions attempted in Maths	No. of questions attempted in reasoning	No. of questions attempted in English
A	32	20	16
B	50 – 32 = 18	39 – 20 = 19	40 – 16 = 24
C	75 – 50 = 25	57 – 39 = 18	66 – 40 = 26
D	90 – 75 = 15	81 – 57 = 24	(9P/10) – 66

Answer: C

If the value of P was 88(8/9):

The total number of questions attempted by all the four candidates together in English would have been = 88(8/9)% of 90 = 80

The number of questions attempted by D in English would have been = 80 – 66 = 14

And the total number of questions attempted by D in the exam would have been = 15 + 24 + 14 = 53

Since, the total number of questions attempted by A in the exam = 32 + 20 + 16 = 68

So, the required percentage = (53/68) * 100 = 78% (approx.)

12. Questions

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Candidates	No. of questions attempted in Maths	No. of questions attempted in Reasoning	No. of questions attempted in English
A	32	62.5% of 32 = 20	50% of 32 = 16
A + B	50	78% of 50 = 39	80% of 50 = 40
A + B + C	75	76% of 75 = 57	88% of 75 = 66
A + B + C + D	90	90% of 90 = 81	P% of 90 = 9P/10

Candidate	No. of questions attempted in Maths	No. of questions attempted in reasoning	No. of questions attempted in English
A	32	20	16
B	50 – 32 = 18	39 – 20 = 19	40 – 16 = 24
C	75 – 50 = 25	57 – 39 = 18	66 – 40 = 26
D	90 – 75 = 15	81 – 57 = 24	(9P/10) – 66

Answer: B

Since, the number of questions attempted by C in Maths = 25

So, the marks obtained by C in Maths = [(25 – 5) * 1] – [5 * 0.25] = 18.75

Since, the number of questions attempted by C in Reasoning = 18

So, the marks obtained by C in Reasoning = [(18 – 3) * 1] – [3 * 0.25] = 14.25

Since, the number of questions attempted by C in English = 26

So, the marks obtained by C in English = [(26 – 8) * 1] – [8 * 0.25] = 16

Required ratio = 18.75:14.25:16 = 75:57:64

13. Questions

Answer: E

Let the speed of the train be ‘x’ m/s and the length of the train be ‘d’ m.

Since, the time taken by the train to cross a 300 m long platform is 5 seconds more than that taken to cross a 195 m long platform.

So,

$$\frac{d + 300}{x} - \frac{d + 195}{x} = 5$$

d + 300 – d – 195 = 5x

x = 21 m/s

If the speed of the train were 4 m/s more than its original speed, it would have crosses a pole in 9 seconds.

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So, (x + 4) = d/9

d = 225

Now, the time taken by the train to cross a 153 m long tunnel:

(225 + 153)/21 = 18 seconds

14. Questions

Answer: C

Since, A and B together can complete 75% work in 10.5 hours.

So, the time, in which A and B together will complete the whole work:

10.5 * (100/75) = 14 hours

$$\frac{1}{A}+\frac{1}{B}=\frac{1}{14}-------(1)$$

Since, B and C together can complete 25% work in 1.5 hours.

So, the time, in which B and C together will complete the whole work:

1.5 * (100/25) = 6 hours

$$\frac{1}{B}+\frac{1}{C}=\frac{1}{6}-------(2)$$

Since, the ratio of B's efficiency to C's efficiency = 1: 6

So, the ratio of the time, in which B alone to the time, in which C alone can complete the work = B: C = 6: 1

From equation (2):

$$\frac{1}{6C}+\frac{1}{C}=\frac{1}{6}$$

C = 7

The time, in which C alone can complete the work = 7 hours

The time, in which B alone can complete the work = 7 * (6/1) = 42 hours

From equation (1):

$$\frac{1}{A}=\frac{1}{14}-\frac{1}{42}$$

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A = 21

The time, in which A alone can complete the work = 21 hours

Since, the difference between the time, in which A alone and D alone can complete the work, is 3 hours.

So, the time, in which D alone can complete the work, will be either 21 + 3 = 24 hours or 21 – 3 = 18

hours

So, the time, in which D alone can complete the work, will fall in ranges P and Q only.

15. Questions

Answer: A

Let the speeds of boats A and B in still water be 'a' m/s and 'b' m/s respectively.

Since, the downstream speed of boat A is equal to upstream speed of boat B.

So, (a + 3) = (b – 3)

b – a = 6 -------(1)

Since, the ratio of upstream speed of boat A to downstream speed of boat B = 3: 7

So,

$$\frac{a-3}{b+3}=\frac{3}{7}$$

7a – 21 = 3b + 9

7a – 3b = 30 --------(2)

From equations (1) and (2):

7a – 3(a + 6) = 30

7a – 3a – 18 = 30

a = 12

From equation (1):

b = 18

Since, boat A is going downstream and boat B is going upstream.

The downstream speed of boat A = 12 + 3 = 15 m/s

And the upstream speed of boat B = 18 – 3 = 15 m/s

So, the required time = 450/(15 + 15) = 15 seconds

16. Questions

Answer: E

Let the initial investments of A, B and C be 10x, 15x and 8x respectively.

Ratio between the profit shares of A, B and C respectively after 15 months:

[(10x * 5) + (8x * 10)]: [(15x * 5) + (10x * 10)]: [(8x * 5) + (12x * 10)]

[50x + 80x]: [75x + 100x]: [40x + 120x]

130x: 175x: 160x = 26: 35: 32

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Since, the total profit received by A, B and C after 15 months = ₹116250

So, A's profit share = $116250 \times \frac{26}{26+35+32}$ = ₹32500

So, B's profit share = $116250 \times \frac{35}{26+35+32}$ = ₹43750

So, C's profit share = $116250 \times \frac{32}{26+35+32}$ = ₹40000

From P:

Since, A's profit share = ₹32500

And C's profit share = ₹40000

So, required percentage = $\frac{32500}{40000} \times 100$ = 81.25%

So, P is true.

From Q:

Difference between B's profit share and C's profit share:

43750 – 40000 = ₹3750

So, Q is true.

From R:

Average of A's profit share and B's profit share:

$$\frac{32500 + 43750}{2} = ₹38125$$

So, R is true.

Hence, all are true.

17. Questions

Answer: A

Since, the SI received on ₹15000 at (R + 4)% rate after 'T' years is ₹3600.

So,

$$\frac{15000 \times T \times (R+4)}{100} = 3600 - - - - - - - - (1)$$

Since, the SI received on ₹18000 at R% rate after 'T' years is ₹2880.

So,

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$$\frac{18000 \times T \times R}{100} = 2880$$

TR = 16 ----------(2)

From equations (1) and (2):

$$\frac{15000 \times (TR + 4T)}{100} = 3600$$

$$\frac{15000 \times (16 + 4T)}{100} = 3600$$

4T = 8

T = 2

From equation (2):

R = 8

Quantity I:

Value of R = 8

Value of (R + 4) = 8 + 4 = 12

Required percentage = $\frac{12-8}{12} \times 100$ = 33.33%

Quantity II:

Value of R = 8

Value of T = 2

Required percentage = (2/8) * 100 = 25%

Hence, Quantity I > Quantity II

18. Questions

Answer: D

$$\frac{24 + x}{35 + (x + y)} = \frac{5}{8}$$

192 + 8x = 175 + 5x + 5y

5y – 3x = 17 ----------(1)

Since, ratio of 'x' to 'y' = x: y = 100: (350/3) = 6: 7

So, (x/y) = (6/7)

y = 7x/6 ----------(2)

From equations (1) and (2):

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$(35x/6) - 3x = 17$

$35x - 18x = 17 * 6$

$x = 6$

From equation (2):

$y = 7$

Quantity I:

Since, the difference between A's present age C's present age = 'x' years = 6 years

And A's present age = 24 years

So, C's age will be either 24 – 6 = 18 years or 24 + 6 = 30 years.

Quantity II:

Since, D's present age is (x – 1) years less than B's present age.

And B's present age = 35 years

So, D's present age = 35 – (x – 1) = 35 – (6 – 1) = 30 years

Hence, Quantity I ≤ Quantity II

19. Questions

Answer: D

Quantity I:

Since, the downstream speed of boat A = $\frac{2.16 \times 1000}{2.25 \times 60}$ = 16 m/s

And the upstream speed of boat A = $\frac{2.16 \times 1000}{3.6 \times 60}$ = 10 m/s

So, the speed of boat A in still water = (16 + 10)/2 = 13 m/s

And the speed of stream = (16 – 10)/2 = 3 m/s

Since, the speed of boat B in still water is 2 m/s more than that of boat A in still water.

So, the speed of boat B in still water = 13 + 2 = 15 m/s

The downstream speed of boat B = 15 + 3 = 18 m/s

And the upstream speed of boat B = 15 – 3 = 12 m/s

Now, the required time:

$$\frac{2.97 \times 1000}{18} + \frac{2.97 \times 1000}{12} = 165 + 247.5 = 412.5 \text{ seconds}$$

Quantity II:

Let the CP of the article = ₹100x

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So, the MP of the article = 125% of 100x = ₹125x

And the original SP of the article = 60% of 125x = ₹75x

If the article were sold at 20% discount on its cost price:

80% of 100x – 75x = 21

80x – 75x = 21

x = 4.2

So, the CP of the article = 100 * 4.2 = ₹420

Hence, Quantity I < Quantity II

20. Questions

Answer: C

Initial ratio of milk to water = 20: 12 = 5: 3

Quantity of milk in 'X' L mixture = X * (5/8) = (5X/8) L

Quantity of water in 'X' L mixture = X * (3/8) = (3X/8) L

So,

$$\frac{20-\frac{5X}{8}+14}{12-\frac{3X}{8}}=\frac{4}{1}$$

$$\frac{272-5X}{96-3X}=\frac{4}{1}$$

272 – 5X = 384 – 12X

X = 16

Quantity I:

Value of 'X' = 16

Quantity II:

Since, the cost of pure milk = ₹24 per L

So, the cost of initial mixture:

$$\frac{20\times 24}{20+12}=₹15 \text{ per L}$$

Hence, Quantity I > Quantity II

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1. Questions

Study the following information carefully and answer the questions given below.

The ratio of the speed of boats A and B is 8:9. The speed of boat A is 11.11% less than that of boat C. The ratio of the speed of streams in rivers Q to S is 2:1. The speed of the stream of River R is 25% more than the speed of the stream of River P. Boat A covered 70 km downstream and 48 km upstream in River P in 7.5 hours. The ratio of the speed of boats C and D is 3:5. Boat A covered 100 km downstream and 30 km upstream in River P in 7.5 hours. The speed of river Q is 20% more than the speed ofriver R.

If speed of A is increased by _____% and Speed of D is increased by _____%. Total distance covered by Boat A and D in 6 hours downstream in river _____ is _____.

Find which of the following is true?

a. 25,20,Q,224

b. 25,40,S,256

c. 25,30,R,414

d. 25,10, Q,332

e. More than one option is true

2. Questions

Boat B and C start their journey at 7 am and 11 am from M and N towards N and M respectively in river Q. B goes against the stream and C goes with the stream. After covering an equal distance they meet each other. Find the distance between M to N?

a. 240 km

b. 188 km

c. 144 km

d. 192 km

e. 162 km

3. Questions

The ratio between the speed of boat D and E is 6:7. The total distance covered by boat E in 3 hours downstream and 3 hours upstream in river R is equal to

a. Boat C cover a distance in 11 hours in river S in downstream

b. Boat A covers a distance in 10 hours in river R in downstream

c. Boat B covers a distance in 10 hours in river Q in downstream

d. Boat D covers a distance in 12.5 hours in river P in downstream

e. More than one option is true

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4. Questions

Find the difference between the distance covered by boat C in river P in 6.5 hours downstream and the distance covered by boat B in river Q in 4.5 hours upstream.

a. 89 km

b. 47 km

c. 42 km

d. 41 km

e. None of these

5. Questions

Study the following information carefully and answer the questions given below.

Inlet pipe A takes 9 hours more to fill the tank than Outlet pipe X to empty the whole tank. The ratio of time taken by inlet pipe C to fill the tank and outlet pipe Z to empty the whole tank is 1:2. Ratio of efficiency of outlet pipe Y to Z is 3:2. Quantity of tank filled when pipe A is opened for 10 hours then pipe X isopened for 5 hours is equal to the Quantity of tank filled when pipe B is opened for 4 hours and pipe Y is opened for 5 hours. The ratio of efficiency of pipes B to C is 5:4. Pipe Z takes double the time to empty the full than the time taken by pipe X to empty the full tank.

Pipe A is opened for [2t+6] hours and then pipe A closed and Pipe X is opened for t hours. Then Pipe X is closed and Pipe B and Y are opened together and filled the tank in 20 hours. Find the value of 't'.

a. 7

b. 8

c. 5

d. 9

e. 10

6. Questions

Find which of the given options will fill maximum amount of tank?

I). Pipe A is opened for 9 hours and pipe Z is opened for 2 hours.

II). Pipe B is opened for 5 hours and pipe Y is opened for 2.5 hours.

III). Pipe C is opened for 7 hours and pipe X is opened for 3 hours.

a. Both I and III is the same

b. Both II and III is the same

c. Both I and II are the same

d. Only II

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e. Only I

7. Questions

Pipe A is opened for t hours, then pipe C is opened for 2t hours and then pipe Z is opened for 3t hours and then pipes B and Y fill the remaining tank in 7.5 hours. Find the value of 't'.

a. 4

b. 3

c. 9

d. 10

e. 6

8. Questions

Inlet pipes C and D together fill the tank in 10 hours and Pipe E and X together fill the tank in 30 hours. Find the time taken by Pipe D and E together to fill the tank.

a. 6.5 hours

b. 7.5 hours

c. 4.5 hours

d. 9.5 hours

e. None of these

9. Questions

Study the following information carefully and answer the questions given below.

The given table shows the total number of adults and total number of children, the ratio of male and female out of total adults and the ratio of boys and girls out of total children in five different villages [A,B,C,D and E].

Village	Total number of adults	Total number of children	The ratio of males and females out of the total adult	The ratio of boys and girls out of total children
A	750	_____	_____	2:3
B	_____	120	5:3	_____
C	840	160	4:3	3:5
D	_____	150	5:3	_____
E	720	_____	_____	2:1

The total number of children in A is 25% more than the total number of children in B. The number of females in A is five times of number of boys in A. The ratio of males in A and B is 9:8. The

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number of females in B is 3.2 times of number of boys in B. Find the number of girls in B.

a. 62

b. 75

c. 45

d. 84

e. 95

10. Questions

The number of adults in E is 8 times the number of children in E. The number of females in E is 9 times the number of girls in E. The average age of males, females, boys and girls is 45, 42, 9 and 7 years respectively. Find the approximate average age all the persons in E?

a. 32 years

b. 33 years

c. 28 years

d. 40 years

e. 29 years

11. Questions

The number of adults in B is x more than the number of children in B. The difference between number of male and female in B is 160. Find the possible value of the number of males in B.

a. 2x-120

b. x-120

c. 3x-270

d. x+290

e. None of these

12. Questions

The number of females in C and D out of total adults is the same. The number of girls in D is 10% of the total number of males. The difference between the boys and girls is x. Find the value of $(7x^2+32)$.

a. 6332

b. 6325

c. 6542

d. 6554

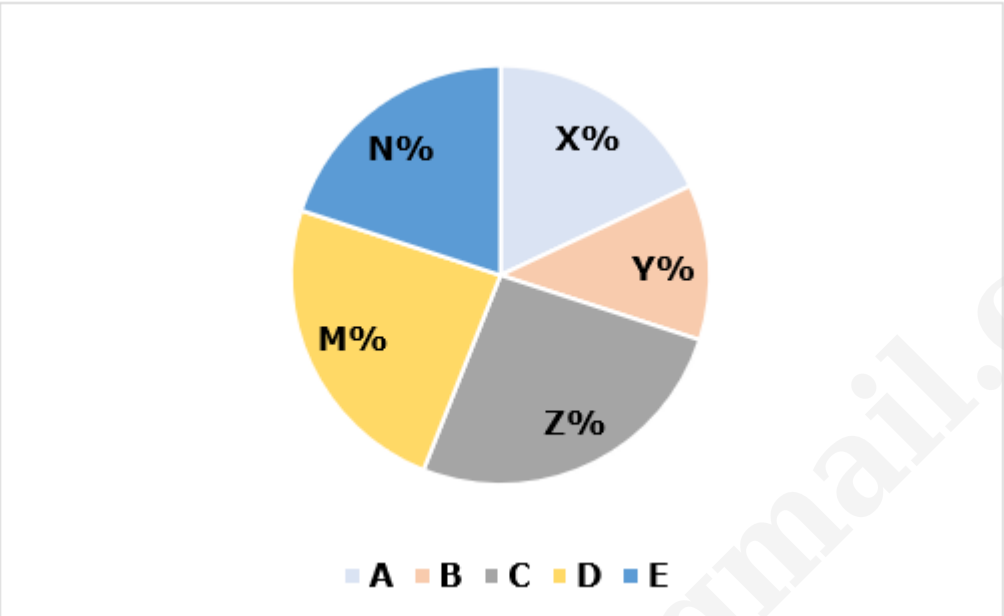
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e. None of these

13. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage distribution of the number of videos posted in five different education channels [A, B, C, D and E].



Note: The total number of video posted in a channel is equal to the sum of the number of math videos and the number of English videos posted on the channel. The ratio of the number of math and English videos posted on all the given channel is 4:3. Total number of videos posted on all the channels together is 700.X:Y=3:2, M=2Y, Z+N=46, Channel C and Dtogether make 180°. The value of Y is 12.

Duration of each math video is 48 minutes and the duration of each English video is 30 minutes. A student watches all math videos at 1.25x speed and all English videos at 1.5x speed. Find the total time taken (in hours) to watch all the videos on the channel A by the student.

a. 62.32

b. 78.04

c. 64.08

d. 96.06

e. 66.04

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14. Questions

The number of English videos on Channel F is _____% more than the number of English videos on Channel E and the number of Math video on Channel F is _____% more than the number of Math videos in channel E. Total number of video in channel F is _____.

Find which of the following is true?

I). 30,40,190

II). 40,30,188

III). 45,35,195

a. Only I true
b. Only I false
c. All false
d. All true
e. Only III false

15. Questions

The difference between the number of Maths and English videos posted on A and B is 2Q and 2R respectively. Find which of the following is true?

I). 5X+7Q=143

II). 5M+8R+7N=208

III). 6Z+2R+3X=235

a. Only III true
b. All false
c. Only I false
d. Only II true
e. None of these

16. Questions

The ratio of prelims and mains level video on C is 7:6. The ratio of math and English prelims level video is 4:3. Find the difference between mains Math and English video.

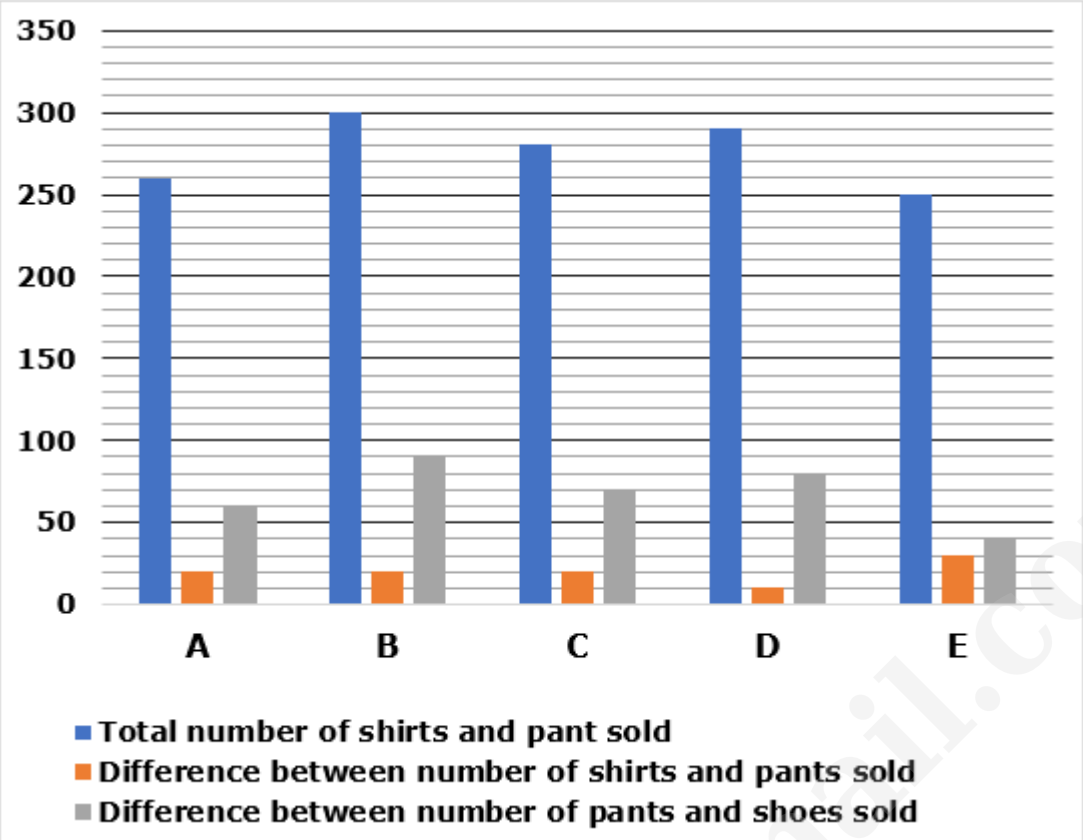
a. 12
b. 14
c. 16
d. 18
e. None of these

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17. Questions

Study the following information carefully and answer the questions given below.

The given bar graphs shows the total number of shirts and pants sold, difference between shirts and pants sold and the difference between the number of shirts and shoes sold in five different shops [A,B,C,D and E].



Notes: – The number of shirts sold in any shop is not more than 150 and number of pants sold in any shop is not less than 130. The number of shoes sold is less than the number of others two items, in all the shops. Number of shirts sold in C is equal to the number of pants sold in D.

Find which of the following statements is true.

I). The difference between the number of shoes and shirts sold in shop A is 40.

II). Average number of shoes sold in all shops is 76.

III). The sum of items sold in shop E is 350.

a. All true

b. All false

c. Only III

d. Only II

e. Only I

18. Questions

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The ratio of formal and casual shirts sold in shop A is 3:2 and the ratio of formal and causal shirts sold in shop B is 4:3. The difference between the formal shirts sold in shop A and B is x and the difference between the casual shirts sold in shops A and B is y. Find the value of (5x+7y)

a. 134

b. 154

c. 164

d. 124

e. 184

19. Questions

The number of watches sold in shop C is 15% of the number of shoes sold in C and the number of watches sold in shop D is 20% of the number of shoes sold in D. If the average price of the watches sold in shop C and D is Rs.1250, then find the total revenue earned by shops C and D together.

a. Rs.25420

b. Rs.28750

c. Rs.26520

d. Rs.26400

e. None of these

20. Questions

Average number of shirts, pants and shoes sold in shops E and F is 130, 135 and 95 respectively. Find the difference between the total number of items sold in shops D and F.

a. 10

b. 14

c. 16

d. 18

e. None of these

Explanations:

1. Questions

Let the speeds of boats A and B be 8x and 9x, respectively.

The speed of boat C is 8x*9/8=9x.

The speed of boat D is 9x*5/3=15x.

Let the speed of the stream of river P be 4y.

So, the speed of the stream of river R is 4y*5/4=5y.

Let the speed of the stream of the river Q be 5y*6/5=6y.

The speed of the stream of the river S is 6y/2=3y.

So, 70/[8x+4y]+48/[8x-4y]=7.5 and 100/[8x+4y]+30/[8x-4y]= 7.5

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By solving the above equations, we get x = 2 and y = 1.

speed of boats A and B is 8*2 = 16 km/hr and 9*2 = 18 km/hr, respectively.

The speed of boat C is 9*2 = 18 km/hr.

The speed of boat D is =15*2 = 30 km/hr.

Let the speed of the stream of river P be 4*1 = 4 km/hr.

So, the speed of the stream of the river R is 5y = 5*1 = 5 km/h.

Let the speed of the stream of the river Q be 6y = 6*1 = 6 km/hr.

The speed of the stream of river S is 6/2 = 3 km/hr.

Answer: C

For the option value of option C,

Speed of boat A is 16*125/100=20

Speed of boat D is 30*130/100=39

So, required distance =[39+5]*6+[20+5]*6=414 km

So its true.

2. Questions

Let the speeds of boats A and B be 8x and 9x, respectively.

The speed of boat C is 8x*9/8=9x.

The speed of boat D is 9x*5/3=15x.

Let the speed of the stream of river P be 4y.

So, the speed of the stream of river R is 4y*5/4=5y.

Let the speed of the stream of the river Q be 5y*6/5=6y.

The speed of the stream of the river S is 6y/2=3y.

So, 70/[8x+4y]+48/[8x-4y]=7.5 and 100/[8x+4y]+30/[8x-4y]= 7.5

By solving the above equations, we get x = 2 and y = 1.

speed of boats A and B is 8*2 = 16 km/hr and 9*2 = 18 km/hr, respectively.

The speed of boat C is 9*2 = 18 km/hr.

The speed of boat D is =15*2 = 30 km/hr.

Let the speed of the stream of river P be 4*1 = 4 km/hr.

So, the speed of the stream of the river R is 5y = 5*1 = 5 km/h.

Let the speed of the stream of the river Q be 6y = 6*1 = 6 km/hr.

The speed of the stream of river S is 6/2 = 3 km/hr.

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Answer: D

Upstream speed of boat B is 18-6=12

The downstream speed of boat C is 18+6=24

So, we can say, 12*[x+4]=24*x

x=4

So, in 4 hours boat C cover in downstream is 24*4=96

So, distance between M to N is 96*2=192 km

3. Questions

Let the speeds of boats A and B be 8x and 9x, respectively.

The speed of boat C is 8x*9/8=9x.

The speed of boat D is 9x*5/3=15x.

Let the speed of the stream of river P be 4y.

So, the speed of the stream of river R is 4y*5/4=5y.

Let the speed of the stream of the river Q be 5y*6/5=6y.

The speed of the stream of the river S is 6y/2=3y.

So, 70/[8x+4y]+48/[8x-4y]=7.5 and 100/[8x+4y]+30/[8x-4y]= 7.5

By solving the above equations, we get x = 2 and y = 1.

speed of boats A and B is 8*2 = 16 km/hr and 9*2 = 18 km/hr, respectively.

The speed of boat C is 9*2 = 18 km/hr.

The speed of boat D is =15*2 = 30 km/hr.

Let the speed of the stream of river P be 4*1 = 4 km/hr.

So, the speed of the stream of the river R is 5y = 5*1 = 5 km/h.

Let the speed of the stream of the river Q be 6y = 6*1 = 6 km/hr.

The speed of the stream of river S is 6/2 = 3 km/hr.

Answer: B

Speed of boat E is =30*7/6=35

So, total distance cover by Boat E is [35+5]*3+[35-5]*3=210

So, boat A covers distance in 10 hours in river R is [16+5]*10=210

4. Questions

Let the speeds of boats A and B be 8x and 9x, respectively.

The speed of boat C is 8x*9/8=9x.

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The speed of boat D is 9x*5/3=15x.

Let the speed of the stream of river P be 4y.

So, the speed of the stream of river R is 4y*5/4=5y.

Let the speed of the stream of the river Q be 5y*6/5=6y.

The speed of the stream of the river S is 6y/2=3y.

So, 70/[8x+4y]+48/[8x-4y]=7.5 and 100/[8x+4y]+30/[8x-4y]= 7.5

By solving the above equations, we get x = 2 and y = 1.

speed of boats A and B is 8*2 = 16 km/hr and 9*2 = 18 km/hr, respectively.

The speed of boat C is 9*2 = 18 km/hr.

The speed of boat D is =15*2 = 30 km/hr.

Let the speed of the stream of river P be 4*1 = 4 km/hr.

So, the speed of the stream of the river R is 5y = 5*1 = 5 km/h.

Let the speed of the stream of the river Q be 6y = 6*1 = 6 km/hr.

The speed of the stream of river S is 6/2 = 3 km/hr.

Answer: A

So, the required difference

=[18+4]*6.5-[18-6]*4.5=143-54=89km

5. Questions

The ratio of efficiency of pipes Y and Z is 3:2.

The time ratio of pipes Y and Z is 2:3.

Outlet pipe Y takes 2x hours and outlet pipe Z takes 3x hours to empty the tank.

So, Inlet Pipe C takes 3x/2=1.5x hours to fill the tank.

Outlet pipe X will empty the tank in 3x/2=1.5x hours.

Inlet pipe A fill the tank in [1.5x+9] hours.

Inlet pipe B fills the tank with 1.5x*4/5=1.2x.

So, we can say,

[10/(1.5x+9)]-5/1.5x=4/1.2x-5/2x

Or, 10/(1.5x+9)=[200+200-150]/60x=250/60x

Or, 12x=7.5x+45

Or, x=45/4.5=10

So, outlet pipe Y takes 2*10=20 hours, and outlet pipe Z takes 3*10=30 hours to empty the tank.

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So, Inlet Pipe C takes 1.5*10=15 hours to fill the tank.

Outlet pipe X will empty the tank in 1.5* 10 = 15 hours.

Inlet pipe A fills the tank in 15+ 9 = 24 hours.

Inlet pipe B fills the tank in 1.2* 10 = 12 hours.

The total capacity of tank = LCM of 24, 12, 15, 20, and 30 is 120.

The efficiency of inlet pipes A, B, and C is 5, 10, and 8, respectively.

The efficiency of outlet pipes X, Y, and Z is [-8], [-6], and [-4], respectively.

Answer: C

So, we can say,

5*[2t+6]-8*t+4*20=120

Or, 2t=40-30=10

Or, t=5

6. Questions

The ratio of efficiency of pipes Y and Z is 3:2.

The time ratio of pipes Y and Z is 2:3.

Outlet pipe Y takes 2x hours and outlet pipe Z takes 3x hours to empty the tank.

So, Inlet Pipe C takes 3x/2=1.5x hours to fill the tank.

Outlet pipe X will empty the tank in 3x/2=1.5x hours.

Inlet pipe A fill the tank in [1.5x+9] hours.

Inlet pipe B fills the tank with 1.5x*4/5=1.2x.

So, we can say,

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Or, 10/(1.5x+9)=[200+200-150]/60x=250/60x

Or, 12x=7.5x+45

Or, x=45/4.5=10

So, outlet pipe Y takes 2*10=20 hours, and outlet pipe Z takes 3*10=30 hours to empty the tank.

So, Inlet Pipe C takes 1.5*10=15 hours to fill the tank.

Outlet pipe X will empty the tank in 1.5* 10 = 15 hours.

Inlet pipe A fills the tank in 15+ 9 = 24 hours.

Inlet pipe B fills the tank in 1.2* 10 = 12 hours.

The total capacity of tank = LCM of 24, 12, 15, 20, and 30 is 120.

The efficiency of inlet pipes A, B, and C is 5, 10, and 8, respectively.

The efficiency of outlet pipes X, Y, and Z is [-8], [-6], and [-4], respectively.

Answer: E

I). Pipe A is opened for 9 hours and pipe Z is opened for 2 hours.

Total quantity of tank filled is =5*9-4*2=37

II). Pipe B is opened for 5 hours and pipe Y is opened for 2.5 hours.

Total quantity of tank filled is =5*10-6*2.5=35

III). Pipe C is opened for 7 hours and pipe X is opened for 3 hours.

Total quantity of tank filled is =8*7-8*3=32

7. Questions

The ratio of efficiency of pipes Y and Z is 3:2.

The time ratio of pipes Y and Z is 2:3.

Outlet pipe Y takes 2x hours and outlet pipe Z takes 3x hours to empty the tank.

So, Inlet Pipe C takes 3x/2=1.5x hours to fill the tank.

Outlet pipe X will empty the tank in 3x/2=1.5x hours.

Inlet pipe A fill the tank in [1.5x+9] hours.

Inlet pipe B fills the tank with 1.5x*4/5=1.2x.

So, we can say,

[10/(1.5x+9)]-5/1.5x=4/1.2x-5/2x

Or, 10/(1.5x+9)=[200+200-150]/60x=250/60x

Or, 12x=7.5x+45

Or, x=45/4.5=10

So, outlet pipe Y takes 2*10=20 hours, and outlet pipe Z takes 3*10=30 hours to empty the tank.

So, Inlet Pipe C takes 1.5*10=15 hours to fill the tank.

Outlet pipe X will empty the tank in 1.5* 10 = 15 hours.

Inlet pipe A fills the tank in 15+ 9 = 24 hours.

Inlet pipe B fills the tank in 1.2* 10 = 12 hours.

The total capacity of tank = LCM of 24, 12, 15, 20, and 30 is 120.

The efficiency of inlet pipes A, B, and C is 5, 10, and 8, respectively.

The efficiency of outlet pipes X, Y, and Z is [-8], [-6], and [-4], respectively.

Answer: D

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So, 5*t+8*2t-4*3t+[10-6]*7.5=120

Or,9t=90

Or, t=90/9=10

8. Questions

The ratio of efficiency of pipes Y and Z is 3:2.

The time ratio of pipes Y and Z is 2:3.

Outlet pipe Y takes 2x hours and outlet pipe Z takes 3x hours to empty the tank.

So, Inlet Pipe C takes 3x/2=1.5x hours to fill the tank.

Outlet pipe X will empty the tank in 3x/2=1.5x hours.

Inlet pipe A fill the tank in [1.5x+9] hours.

Inlet pipe B fills the tank with 1.5x*4/5=1.2x.

So, we can say,

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Or, 10/(1.5x+9)=[200+200-150]/60x=250/60x

Or, 12x=7.5x+45

Or, x=45/4.5=10

So, outlet pipe Y takes 2*10=20 hours, and outlet pipe Z takes 3*10=30 hours to empty the tank.

So, Inlet Pipe C takes 1.5*10=15 hours to fill the tank.

Outlet pipe X will empty the tank in 1.5* 10 = 15 hours.

Inlet pipe A fills the tank in 15+ 9 = 24 hours.

Inlet pipe B fills the tank in 1.2* 10 = 12 hours.

The total capacity of tank = LCM of 24, 12, 15, 20, and 30 is 120.

The efficiency of inlet pipes A, B, and C is 5, 10, and 8, respectively.

The efficiency of outlet pipes X, Y, and Z is [-8], [-6], and [-4], respectively.

Answer: B

Let the efficiency of pipes D and E is m and n respectively.

So, [8+m]*10=120

Or, m=4

[n-8]*30=120

Or, n=12

So, pipe D and E together fill the tank in =[120/(12+4)]=7.5 hours

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9. Questions

Male in C is = 840*4/7=480

Female in C is 840-480=360

Boys in C is 160*3/8=60

Girls in C is 160-60=100

Answer: C

The total number of children in A is 120*5/4=150

Number of boys in A is 150*2/5=60

So, number of females in A is 60*5=300

Number of males in A is 750-300=450

Number of males in B is 450*8/9=400

Number of females in B is 400*5/3=240

Number of boys in B is 240/3.2=75

So, number of girls in B is = 120-75=45

10. Questions

Male in C is = 840*4/7=480

Female in C is 840-480=360

Boys in C is 160*3/8=60

Girls in C is 160-60=100

Answer: D

Number of children in E is 720/8=90

Number of girls in E is 90*1/3=30

Number of boys in E is 90-30=60

Number of females in E is 30*9=270

Number of males in E is 720-270=450

So, the average age in E is

=[450*45+270*42+60*9+30*7]/810=39.92=40

11. Questions

Male in C is = 840*4/7=480

Female in C is 840-480=360

Boys in C is 160*3/8=60

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Girls in C is 160-60=100

Answer: B

So, we can say [x+120]*5/8 – [x+120]*3/8 = 160

Or, 5x+600-3x-360=1280

Or, 2x=1040

Or, x=1040/2=520

So, the number of males 640*5/8=400

12. Questions

Male in C is = 840*4/7=480

Female in C is 840-480=360

Boys in C is 160*3/8=60

Girls in C is 160-60=100

Answer: A

Number of males in D is 360*5/3=600

Number of girls in D is 600*10/100=60

Number of boys in D is 150-60=90

So, the difference in D is 90-60=30

So, 7*30*30+32=6332

13. Questions

The total number of videos posted on a channel is equal to the sum of the number of math videos and the number of English videos posted on the channel. The ratio of the number of math and English videos posted on the channel is 4:3. Total number of videos posted on all the channels together is 700.

AQT,

X:Y=3:2,

M=2Y, Y=12

So, M=12*2=24

Or, X=12*3/2=18

Channel C and D together make 180°. Means Channel C and D together posted 50% video.

So, M+Z=50

Or, Z=50-24=26

So, N=46-26=20

The number of videos posted on channel A is 700*18/100=126

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So, the number of Math videos posted on Channel A is 126*4/7=72

The number of English videos posted on Channel A is 126-72=54

Similarly, we can calculate others' values also.

Channel	Total video posted	Total Math's video posted	Total English's video posted
A	126	72	54
B	84	48	36
C	182	104	78
D	168	96	72
E	140	80	60

Answer: C

So, total time taken

= [48*72]/1.25 + [54*30]/1.5=2764.8+1080=3844.8

So, total time taken = [3844.8]/60=64.08 hours

14. Questions

The total number of videos posted on a channel is equal to the sum of the number of math videos and the number of English videos posted on the channel. The ratio of the number of math and English videos posted on the channel is 4:3. Total number of videos posted on all the channels together is 700.

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A	126	72	54
B	84	48	36
C	182	104	78
D	168	96	72
E	140	80	60

Answer: D

For value III,

The total video posted on F is

=60*145/100 + 80*135/100 = 195

So, it is true.

Similarly, we can calculate others' values also.

15. Questions

The total number of videos posted on a channel is equal to the sum of the number of math videos and the number of English videos posted on the channel. The ratio of the number of math and English videos posted on the channel is 4:3. Total number of videos posted on all the channels together is 700.

AQT,

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So, M=12*2=24

Or, X=12*3/2=18

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The number of English videos posted on Channel A is 126-72=54

Similarly, we can calculate others' values also.

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Channel	Total video posted	Total Math's video posted	Total English's video posted
A	126	72	54
B	84	48	36
C	182	104	78
D	168	96	72
E	140	80	60

Answer: B

The difference between the number of Math and English videos posted on A and B is 2Q and 2R respectively.

So, Q=[72-54]/2=18/2=9 and R=[48-36]=12/2=6

I). 5X+7Q=5*18+7*9=153

II). 5M+8R+7N=5*24+8*6+7*20=308

III). 6Z+2R+3X=6*26+2*6+3*18=222

16. Questions

The total number of videos posted on a channel is equal to the sum of the number of math videos and the number of English videos posted on the channel. The ratio of the number of math and English videos posted on the channel is 4:3. Total number of videos posted on all the channels together is 700.

AQT,

X:Y=3:2,

M=2Y, Y=12

So, M=12*2=24

Or, X=12*3/2=18

Channel C and D together make 180°. Means Channel C and D together posted 50% video.

So, M+Z=50

Or, Z=50-24=26

So, N=46-26=20

The number of videos posted on channel A is 700*18/100=126

So, the number of Math videos posted on Channel A is 126*4/7=72

The number of English videos posted on Channel A is 126-72=54

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A	126	72	54
B	84	48	36
C	182	104	78
D	168	96	72
E	140	80	60

Answer: A

Total Prelims level video in C is = 182*7/13=98

English prelims levelvideo in C is 98*3/7=42

Math prelims level video in C is 98*4/7=56

So, the difference is [104-56]-[78-42]=48-36=12

17. Questions

The number of shirts or pants sold in shop A is [260+20]/2=140 or [260-20]/2=120.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop A is 140, and the number of shirts sold in shop A is 120.

The number of shoes sold in shop A is 140-60=80.

The number of shirts or pants sold in shop B is [300+20]/2=160 or [300-20]/2=140.

The number of shirts sold in any shop is not more than 150.

So, the number of pants sold in shop B is 160, and the number of shirts sold in shop B is 140.

The number of shoes sold in shop B is 160-90=70.

The number of shirts or pants sold in shop E is [250+30]/2=140 or [250-30]/2=110.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop E is 140, and the number of shirts sold in shop E is 110.

The number of shoes sold in shop E is 140–40= 100.

The number of shirts and pants sold in C is [280+20]/2=150 or [280-20]/2=130.

The number of shirts and pants sold in D is [290+10]/2=150 or [290-10]/2=140.

Now, the number of shirts sold in C is equal to the number of pants sold in D.

So, the number of shirts sold in C is 150, or the number of pants sold in D.

The number of shirts sold in D is 140.

The number of pants sold in C is 130.

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Number of shoes sold in C is 130-70=60

Number of shoes sold in D is 150-80=70

Answer: A

I). The difference between the number of shoes and shirts sold in shop A is 40=120-80=40.

II). Average number of shoes sold in all shops is =[80+70+60+70+100]/5=76.

III). The sum of items sold in shop E is 110+140+100=350.

18. Questions

The number of shirts or pants sold in shop A is [260+20]/2=140 or [260-20]/2=120.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop A is 140, and the number of shirts sold in shop A is 120.

The number of shoes sold in shop A is 140-60=80.

The number of shirts or pants sold in shop B is [300+20]/2=160 or [300-20]/2=140.

The number of shirts sold in any shop is not more than 150.

So, the number of pants sold in shop B is 160, and the number of shirts sold in shop B is 140.

The number of shoes sold in shop B is 160-90=70.

The number of shirts or pants sold in shop E is [250+30]/2=140 or [250-30]/2=110.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop E is 140, and the number of shirts sold in shop E is 110.

The number of shoes sold in shop E is 140–40= 100.

The number of shirts and pants sold in C is [280+20]/2=150 or [280-20]/2=130.

The number of shirts and pants sold in D is [290+10]/2=150 or [290-10]/2=140.

Now, the number of shirts sold in C is equal to the number of pants sold in D.

So, the number of shirts sold in C is 150, or the number of pants sold in D.

The number of shirts sold in D is 140.

The number of pants sold in C is 130.

Number of shoes sold in C is 130-70=60

Number of shoes sold in D is 150-80=70

Tg: @NextGenBankers

Answer: D

The difference between formal shirts sold in shops A and B is = 140*4/7- 120*3/5 = 80-72=8=x

The difference between causal shirts sold in shops A and B is = 140*3/7 – 120*2/5=60-48=12=y

So, 5x+7y=5*8+7*12=40+84=124

19. Questions

The number of shirts or pants sold in shop A is [260+20]/2=140 or [260-20]/2=120.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop A is 140, and the number of shirts sold in shop A is 120.

The number of shoes sold in shop A is 140-60=80.

The number of shirts or pants sold in shop B is [300+20]/2=160 or [300-20]/2=140.

The number of shirts sold in any shop is not more than 150.

So, the number of pants sold in shop B is 160, and the number of shirts sold in shop B is 140.

The number of shoes sold in shop B is 160-90=70.

The number of shirts or pants sold in shop E is [250+30]/2=140 or [250-30]/2=110.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop E is 140, and the number of shirts sold in shop E is 110.

The number of shoes sold in shop E is 140–40= 100.

The number of shirts and pants sold in C is [280+20]/2=150 or [280-20]/2=130.

The number of shirts and pants sold in D is [290+10]/2=150 or [290-10]/2=140.

Now, the number of shirts sold in C is equal to the number of pants sold in D.

So, the number of shirts sold in C is 150, or the number of pants sold in D.

The number of shirts sold in D is 140.

The number of pants sold in C is 130.

Number of shoes sold in C is 130-70=60

Number of shoes sold in D is 150-80=70

Answer: B

Total revenue earned by shops C and D together is

= [(60*15/100) + (70*20/100)]*1250 = [9+14]*1250=28750

20. Questions

The number of shirts or pants sold in shop A is [260+20]/2=140 or [260-20]/2=120.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop A is 140, and the number of shirts sold in shop A is 120.

The number of shoes sold in shop A is 140-60=80.

The number of shirts or pants sold in shop B is [300+20]/2=160 or [300-20]/2=140.

The number of shirts sold in any shop is not more than 150.

So, the number of pants sold in shop B is 160, and the number of shirts sold in shop B is 140.

The number of shoes sold in shop B is 160-90=70.

The number of shirts or pants sold in shop E is [250+30]/2=140 or [250-30]/2=110.

The number of pants sold in any shop should not be less than 130.

So, the number of pants sold in shop E is 140, and the number of shirts sold in shop E is 110.

The number of shoes sold in shop E is 140–40= 100.

The number of shirts and pants sold in C is [280+20]/2=150 or [280-20]/2=130.

The number of shirts and pants sold in D is [290+10]/2=150 or [290-10]/2=140.

Now, the number of shirts sold in C is equal to the number of pants sold in D.

So, the number of shirts sold in C is 150, or the number of pants sold in D.

The number of shirts sold in D is 140.

The number of pants sold in C is 130.

Number of shoes sold in C is 130-70=60

Number of shoes sold in D is 150-80=70

Answer: A

Total number of items sold in F =130*2-110+135*2-140+95*2-100=150+130+90=370

So, the required difference = 370-150-140-70=10

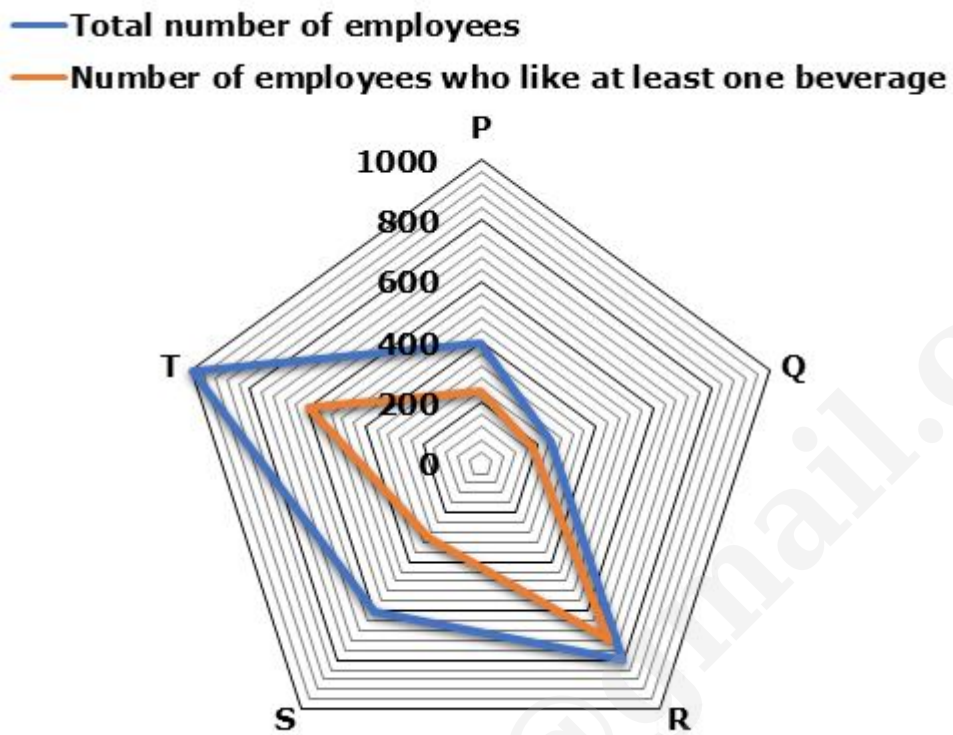
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1. Questions

Study the following information carefully and answer the questions based on it.

A survey was conducted among all employees in five different companies about their liking among three beverages – A, B, and C. An employee can like more than one beverage among the three beverages.



Note: The number of employees who like at least one beverage is in a multiple of 10.

For company Q, the number of employees who like all three beverages is four times of those who like exactly one beverage, while those who like exactly two beverages is the same as those who don't like any of the given beverages. The number of people who like exactly one beverage is (Z – 12). Find the value of Z.

a. 52

b. 24

c. 36

d. 28

e. None of these

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2. Questions

The number of employees in company S who likes at least one beverage is 66.66% more than that of male employees who likes at least one beverage. The number of female employees those likes at least one beverage in S is 50% of the number of female employees in R those likes at least one beverage. Out of the total employees in company R those like at least one beverage is (Z/3) % are males. Find the value of Z.

a. 180

b. 120

c. 200

d. 160

e. None of these

3. Questions

The average number of employees in all companies together those who do not like any beverage is A, while the total number of employees in all companies together is 5B. Find the difference between 2A and B?

a. 198

b. 408

c. 228

d. 208

e. None of these

4. Questions

For P, the number of employees who like only A is thrice that of only B (Z), while those who like only C is 35% more than those who like only B. Find the number of employees who like exactly two beverage, if the number of employees who like all three beverages is 25% of the total number of employees in company P.

a. 74

b. 33

c. 43

d. 37

e. None of these

5. Questions

The number of employees in company R who don't likes any beverage is 1/6th of the number of employees in company U who like at least one beverage. The number of employees in company U those who like at least one beverage is 75% of the total number of employees in that company. Out of the total employees who like at least one beverage 4/5th are females. Find the difference between the number of employees who do not like any beverage and the number of male employees who like at least one beverage in company U.

a. 96

b. 60

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c. 80

d. 64

e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

The data given below gives information about the number of students in three different schools.

- The total number of students in school Q is half of the total students in all schools together.
- The total number of students in school R is the same as the number of boys in school Q, and the number of boys in school P and R is the same.
- The total number of boys in all three schools together is Z% more than that of the total number of girls.
- Out of the total number of girls in all three schools together, (Y) % are in school P.

Note:

$(3Z - Y)/ (4Y - Z) = 5/2$..... (1)

$(Z^2 - M^2) = [(Z - M) \times (M + 40)^2] / (Z + M)$ ..... (2)

The difference between the number of girls in schools Q and R is [5(Z + Y)].

A = Number of girls in school P

B = Number of boys in school Q

Find the value of (A + B).

a. 900

b. 600

c. 750

d. 1200

e. None of these

7. Questions

The average number of girls in schools P and Q together is 400, and the number of students in another school S is (Z + 10) % more than that of school R. Find the difference between the sum of the number of boys in schools P and R together and the total number of students in school S.

a. 100

b. 200

c. 400

d. 300

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e. None of these

8. Questions

Quantity I: 50% of the girls in school Q are preparing for CAT. Find the number of girls in school Q those who are not preparing for CAT is how much % more or less than the number of boys in school R.

Quantity II: Find the value of (Y + 5).

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I ≥ Quantity II

9. Questions

The total number of students in all three given schools together is less than 2(Z + 10), then find the maximum possible number of students in school P.

a. 24

b. 48

c. 18

d. 20

e. None of these

10. Questions

Which of the following statement is/are must be true.

I). The total number of students in school P is greater than the total number of students in school R.

II). The number of girls in schools P and R is the same.

III). The number of girls in school Q is half of the total number of students in the same school.

a. I and II only

b. II and III only

c. I and III only

d. I, II, and III

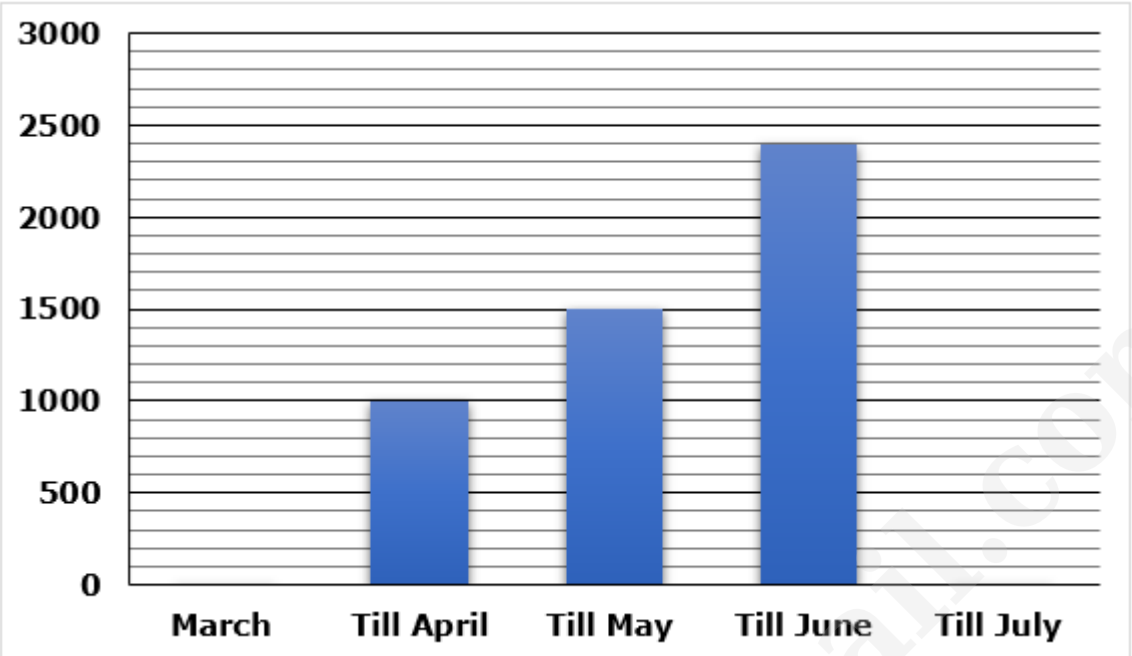
e. None of these

11. Questions

Study the following information carefully and answer the questions based on it.

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A company manufactured three types of articles – A, B, and C. The bar graph given below gives partial information about the cumulative data for the total number of units manufactured from – March to July in 2023.



The table chart given below shows the information about the % of Article A manufactured and 1/4th of the sum of the number of type B and type C articles manufactured.

Months	% of type A articles manufactured	25% of sum of (B and C)
March	20%	80
Till April	38%	-
Till May	-	220
Till June	(275/6) %	-
Till July	45.67%	-

Note: The total number of type B and type C articles manufactured is 1630.

In July, out of the total units manufactured of type A article, 1/3rd were sold and the rest remained unsold. The average selling price of each unit is Rs. 200. Find the revenue generated in the month of July from the sale of type A articles.

a. Rs. 24000

b. Rs. 40000

c. Rs. 12000

d. Rs. 18000

e. None of these

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12. Questions

For June, the number of type C articles manufactured are 120. Out of the total number of articles manufactured, 200 were sold.

Z = The number of type A articles manufactured

Y = The number of type B articles remains unsold

Find the maximum value of (Z – Y)?

a. 400

b. 380

c. 480

d. 420

e. None of these

13. Questions

The sum of the number of type B and type C articles in months of March and April together is (7Z + 60). Find the value of Z.

a. 80

b. 70

c. 60

d. 100

e. None of these

14. Questions

The number of type A articles manufactured till May is [(8Z + 28)/3] % of the total number of articles manufactured in the month of March, April, and May together. The number of type B articles manufactured in March is 40 more than that of type C articles manufactured. Find the difference between Z and Y, if Y is the number of type B articles manufactured in March.

a. 178

b. 128

c. 168

d. 148

e. None of these

15. Questions

Find the difference between the number of type A articles manufactured in the month of April and June?

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a. 180

b. 150

c. 160

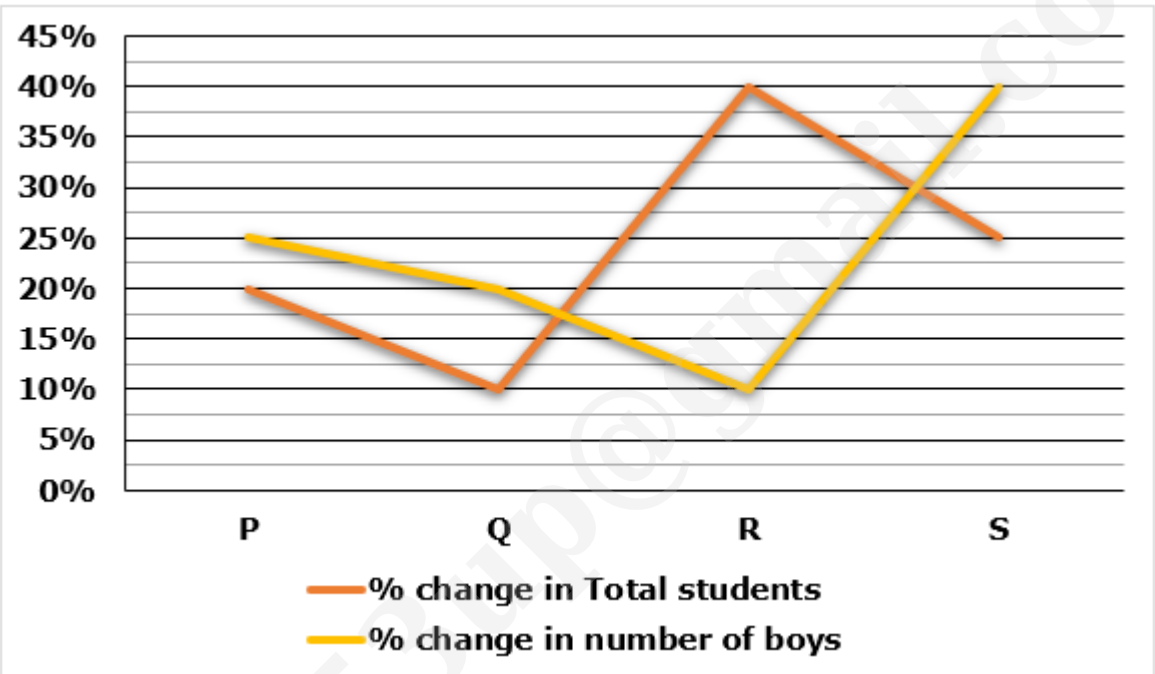
d. 60

e. None of these

16. Questions

Study the following information carefully and answer the questions based on it.

The line graph given below shows the % change in total students in 2020 as compared to 2019, and % change in the number of boys in 2020 as compared to 2019.



Name of school	Average number of girls in both years together
P	90
Q	255
R	380
S	82.5

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Note:

a). The ratio of the number of boys in P in 2019 to the number of girls in Q in the same year is 4:15.

b). The total number of boys in school Q and school R in 2020 is 680, while the number of boys in school Q in 2019 is 4/9 times the total number of students in the same school in 2020.

c). The total number of students in each school in 2020 is less than that of in 2019, while the number of

boys in each school in 2019 is less than that in 2020.

The total number of students in 2020 in school R is 375 more than that of school S in the same year. Find the number of boys in school S in 2019?

a. 240

b. 210

c. 150

d. 180

e. None of these

17. Questions

In 2020, for school P, in an examination, the number of students who passed the exam is 35% of the total number of boys in school Q in 2020, out of which 64 are boys. Find the number of girls who failed the exam?

a. 24

b. 20

c. 36

d. 40

e. None of these

18. Questions

The difference between the number of girls in 2019 and 2020 in school R is A, while the number of boys in school R in 2020 is B. Then find which of the following statements is /are true?

I). A = B

II). A + B > 900

III). (2A – B) = 88% of students in school Q in 2019

a. II only

b. I and III only

c. I and II only

d. I, II, and III

e. None of these

19. Questions

The number of girls in school S in 2020 is 30. Find the total number of students in school S in 2019 and 2020 together if the number of boys in 2020 is twice that of girls in 2019.

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a. 700

b. 840

c. 400

d. 805

e. None of these

20. Questions

Out of the total number of girls in P, Q, and R together in 2020, 100 are in class 8th and the rest are in class 9th. $1/3^{rd}$ of the total number of girls in class 9th in 2020 in (P + Q + R) together passed in the exam. Find the number of girls who failed the exam in 2020 in all these three schools together?

a. 110

b. 276

c. 231

d. 330

e. None of these

Explanations:

1. Questions

Answer: C

For Q,

The number of employees who do not like any beverages = 240 – 180 = 60

Those who like all three = 4 x who like exactly one beverage

Now,

180 = like exactly one + likes exactly two + Exactly three

180 = likes exactly one + 60 + 4 x exactly one

120 = 5x exactly one

So, exactly one = 120/5 = 24

(Z – 12) = 24

Value of Z = 36

Hence answer is option C

2. Questions

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Answer: A

Number of employees who likes at least one in S = 300

Number of males in S likes at least one in S= 3/5 x 300 = 180

Number of females in S likes at least one in S = 300 – 180 = 120

So, the number of females who likes at least one in R = 120 x 2 = 240

Number of males in R likes at least one beverage = 720 – 240 = 480

Total number of employees in R = 800

Required % = 480/800 x 100 = 60%

So, value of Z = 60 x 3 = 180

Hence answer is option A

3. Questions

Answer: D

Value of A = (160 + 60 + 80 + 300 + 400)/5 = 200

Value of B = (400 + 240 + 800 + 600 + 1000)/5 = 608

Required difference = 608 – 2 x 200 = 208

Hence answer is option D

4. Questions

Answer: B

For P

Those who like at least one = 240

Those who like all three = 25% x 400 = 100

Only B = Z

So, only A = 3 x Z = 3Z

So, only C = 27/20 x Z = 27Z/20

Now,

(3Z + Z + 27Z/20) + Exactly two + 100 = 240

107Z/20 + Exactly two = 140

If Z = 20, exactly two = 33

If Z = 40 exactly two become negative

So, the required number of employees = 33

Hence answer is option B

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5. Questions

Answer: D

For U,

Number of employees who like at least one beverage = 6 x (800 – 720) = 480

Total employees in company U = 480/75 x 100 = 640

Employees who do not like any beverage = 640 – 480 = 160

Male employees who like at least one beverage = 1/5 x 480 = 96

Required difference = 160 – 96 = 64

Hence answer is option D

6. Questions

(3Z – Y)/ (4Y – Z) = 5/2……. (1)

6Z – 2Y = 20Y – 5Z

11Z = 22Y

Z = 2Y

Now,

$(Z^2 – M^2) = [(Z – M) \times (M + 40)^2] / (Z + M)$ …… (2)

$(Z – M) \times (Z + M)^2 = (Z – M) \times (M + 40)^2$

Z + M = M + 40

Value of Z = 40

So, value of Y = 40/2 = 20

Let the total number of girls in all schools together = 5a

So, the total number of boys in all schools together = 1.4 x 5a = 7a

Total number of students in school Q = 1/2 x (7a + 5a) = 6a

Let the total number of students in school R = b

So, the number of boys in school Q = b

So, the number of girls in school Q = (6a – b)

Number of girls in school P = 20% x 5a = a

Let the number of girls in school R = c

So, the number of boys in school R = (b – c)

So, the number of boys in school R = (b – c)

Now,

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(b – c) + b + (b – c) = 7a

3b – 2c = 7a……. (3)

Also,

(6a – b) + c + a = 5a

2a = b – c

Or

4a = 2b – 2c……... (4)

On adding both (3) and (4), we get

3b – 2c + 4a = 7a + 2b – 2c

So, b = 3a

So, 3 x 3a – 2c = 7a

2a = 2c

So, a = c

For school Q

Total students = 6a

Number of boys = 3a

Number of girls = 6a – 3a = 3a

For school R

Total number of students = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

For school P

Total number of students = 12a – 6a – 3a = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

Answer: B

According to the question,

(3a – a) = 5 x (40 + 20)

2a = 300

Value of a = 150

Value of A = 150

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Value of B = 150 x 3 = 450

Required sum = 450 + 150 = 600

Hence answer is option B

7. Questions

(3Z – Y)/ (4Y – Z) = 5/2……. (1)

6Z – 2Y = 20Y – 5Z

11Z = 22Y

Z = 2Y

Now,

$(Z^2 – M^2) = [(Z – M) \times (M + 40)^2] / (Z + M)$ …… (2)

$(Z – M) \times (Z + M)^2 = (Z – M) \times (M + 40)^2$

Z + M = M + 40

Value of Z = 40

So, value of Y = 40/2 = 20

Let the total number of girls in all schools together = 5a

So, the total number of boys in all schools together = 1.4 x 5a = 7a

Total number of students in school Q = 1/2 x (7a + 5a) = 6a

Let the total number of students in school R = b

So, the number of boys in school Q = b

So, the number of girls in school Q = (6a – b)

Number of girls in school P = 20% x 5a = a

Let the number of girls in school R = c

So, the number of boys in school R = (b – c)

So, the number of boys in school R = (b – c)

Now,

(b – c) + b + (b – c) = 7a

3b – 2c = 7a……. (3)

Also,

(6a – b) + c + a = 5a

2a = b – c

Or

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4a = 2b – 2c…….... (4)

On adding both (3) and (4), we get

3b – 2c + 4a = 7a + 2b – 2c

So, b = 3a

So, 3 x 3a – 2c = 7a

2a = 2c

So, a = c

For school Q

Total students = 6a

Number of boys = 3a

Number of girls = 6a – 3a = 3a

For school R

Total number of students = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

For school P

Total number of students = 12a – 6a – 3a = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

Answer: A

The number of students in school S = 1.5 x 3a = 4.5a

Total number of boys in school P and R together = 2a + 2a = 4a

Now,

(3a + a)/2 = 400

2a = 400

Value of a = 200

Require difference = 4.5a – 4a = 0.5a = 0.5 x 200 = 100

Hence answer is option A

8. Questions

(3Z – Y)/ (4Y – Z) = 5/2……. (1)

6Z – 2Y = 20Y – 5Z

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$11Z = 22Y$

$Z = 2Y$

Now,

$(Z^2 - M^2) = [(Z - M) \times (M + 40)^2] / (Z + M)$ …… (2)

$(Z - M) \times (Z + M)^2 = (Z - M) \times (M + 40)^2$

$Z + M = M + 40$

Value of $Z = 40$

So, value of $Y = 40/2 = 20$

Let the total number of girls in all schools together = 5a

So, the total number of boys in all schools together = $1.4 \times 5a = 7a$

Total number of students in school Q = $1/2 \times (7a + 5a) = 6a$

Let the total number of students in school R = b

So, the number of boys in school Q = b

So, the number of girls in school Q = (6a – b)

Number of girls in school P = $20\% \times 5a = a$

Let the number of girls in school R = c

So, the number of boys in school R = (b – c)

So, the number of boys in school R = (b – c)

Now,

$(b - c) + b + (b - c) = 7a$

$3b - 2c = 7a$……. (3)

Also,

$(6a - b) + c + a = 5a$

$2a = b - c$

Or

$4a = 2b - 2c$…….... (4)

On adding both (3) and (4), we get

$3b - 2c + 4a = 7a + 2b - 2c$

So, $b = 3a$

So, $3 \times 3a - 2c = 7a$

$2a = 2c$

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So, a = c

For school Q

Total students = 6a

Number of boys = 3a

Number of girls = 6a – 3a = 3a

For school R

Total number of students = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

For school P

Total number of students = 12a – 6a – 3a = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

Answer: C

Quantity I.

Number of girls in school Q not preparing for CAT = 50% x 3a = 1.5a

Number of boys in school R = 2a

Required % change = (2a – 1.5a)/2a x 100 = 25%

Quantity II.

Required value = (20 + 5) = 25%

So, Quantity I = Quantity II

Hence answer is option C

9. Questions

(3Z – Y)/ (4Y – Z) = 5/2……. (1)

6Z – 2Y = 20Y – 5Z

11Z = 22Y

Z = 2Y

Now,

$(Z^2 – M^2) = [(Z – M) \times (M + 40)^2] / (Z + M)$ …… (2)

$(Z – M) \times (Z + M)^2 = (Z – M) \times (M + 40)^2$

Z + M = M + 40

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Value of Z = 40

So, value of Y = 40/2 = 20

Let the total number of girls in all schools together = 5a

So, the total number of boys in all schools together = 1.4 x 5a = 7a

Total number of students in school Q = 1/2 x (7a + 5a) = 6a

Let the total number of students in school R = b

So, the number of boys in school Q = b

So, the number of girls in school Q = (6a – b)

Number of girls in school P = 20% x 5a = a

Let the number of girls in school R = c

So, the number of boys in school R = (b – c)

So, the number of boys in school R = (b – c)

Now,

(b – c) + b + (b – c) = 7a

3b – 2c = 7a……. (3)

Also,

(6a – b) + c + a = 5a

2a = b – c

Or

4a = 2b – 2c…….... (4)

On adding both (3) and (4), we get

3b – 2c + 4a = 7a + 2b – 2c

So, b = 3a

So, 3 x 3a – 2c = 7a

2a = 2c

So, a = c

For school Q

Total students = 6a

Number of boys = 3a

Number of girls = 6a – 3a = 3a

For school R

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Total number of students = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

For school P

Total number of students = 12a – 6a – 3a = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

Answer: A

Total number of students in all three schools = 12a

12a < 2(40 + 10)

12a < 100

Value of a < 100/12 = 8.333

So, the maximum possible value of a = 8

So, the maximum possible number of students in school P = 3a = 24

Hence answer is option A

10. Questions

(3Z – Y)/ (4Y – Z) = 5/2……. (1)

6Z – 2Y = 20Y – 5Z

11Z = 22Y

Z = 2Y

Now,

$(Z^2 – M^2) = [(Z – M) \times (M + 40)^2] / (Z + M)$ …… (2)

$(Z – M) \times (Z + M)^2 = (Z – M) \times (M + 40)^2$

Z + M = M + 40

Value of Z = 40

So, value of Y = 40/2 = 20

Let the total number of girls in all schools together = 5a

So, the total number of boys in all schools together = 1.4 x 5a = 7a

Total number of students in school Q = 1/2 x (7a + 5a) = 6a

Let the total number of students in school R = b

So, the number of boys in school Q = b

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So, the number of girls in school Q = (6a – b)

Number of girls in school P = 20% x 5a = a

Let the number of girls in school R = c

So, the number of boys in school R = (b – c)

So, the number of boys in school R = (b – c)

Now,

(b – c) + b + (b – c) = 7a

3b – 2c = 7a……. (3)

Also,

(6a – b) + c + a = 5a

2a = b – c

Or

4a = 2b – 2c……... (4)

On adding both (3) and (4), we get

3b – 2c + 4a = 7a + 2b – 2c

So, b = 3a

So, 3 x 3a – 2c = 7a

2a = 2c

So, a = c

For school Q

Total students = 6a

Number of boys = 3a

Number of girls = 6a – 3a = 3a

For school R

Total number of students = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

For school P

Total number of students = 12a – 6a – 3a = 3a

Number of boys = (3a – a) = 2a

Number of girls = 3a – 2a = a

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Answer: B

I). The total number of students in school P > The total number of students in school R.

The total number of students in school P = The total number of students in school R = 3a

This statement is false.

II). The number of girls in school P and R is the same.

The number of girls in school P = The number of girls in school R = a

This statement is true.

III). The number of girls in school Q is half of the total number of students in the same school.

Required ratio = 3a/6a = 1/2

This statement is true.

Only II and III are true

Hence answer is option B

11. Questions

For March

The number of articles (B + C) manufactured = 80 x 4 = 320

So, the total Articles manufactured = 320/80 x 100 = 400

So, type A articles manufactured = 400 – 320 = 80

For April

Total number of articles manufactured = 1000 – 400 = 600

The number of type A articles manufactured = 1000 x 38% - 80 = 300

The number of (B + C) articles manufactured = 600 - 300 = 300

For May

Total number of articles manufactured = 1500 - 1000 = 500

The number of (B + C) articles manufactured = 220 x 4 – 300 – 320 = 260

The number of type A articles manufactured = 500 – 260 = 240

For June

Total number of articles manufactured = 2400 – 1500 = 900

The number of type A articles manufactured = 2400 x (395/6) % - 1100 = 480

The number of (B + C) articles manufactured = 900 – 480 = 420

For July

The number of type A articles till July = 45.67% x (total A till July) = 137/300 x (Total A till July)

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Tg: @NextGenBankers

The number of (B + C) articles till July = 1630

So, the total number of type A articles till July = 1630/163 x 137 = 1370

So, the number of Article A manufactured in July = 1370 – (80 + 300 + 240 + 480) = 270

The total number of articles = (1370 + 1630) – 2400 = 600

So, the number of (B + C) articles= 600 – 270 = 330

Name of Months	Total articles manufactured	Type A
March	400	80
April	600	300
May	500	240
June	900	480
July	600	270

Answer: D

For July

Total units of type A articles manufactured = 270

The Number of type A articles sold = 1/3 x 270 = 90

Required revenue = 90 x 200 = 18000

Hence answer is option D

12. Questions

For March

The number of articles (B + C) manufactured = 80 x 4 = 320

So, the total Articles manufactured = 320/80 x 100 = 400

So, type A articles manufactured = 400 – 320 = 80

For April

Total number of articles manufactured = 1000 – 400 = 600

The number of type A articles manufactured = 1000 x 38% - 80 = 300

The number of (B + C) articles manufactured = 600 - 300 = 300

For May

Total number of articles manufactured = 1500 - 1000 = 500

The number of (B + C) articles manufactured = 220 x 4 – 300 – 320 = 260

The number of type A articles manufactured = 500 – 260 = 240

For June

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Total number of articles manufactured = 2400 – 1500 = 900

The number of type A articles manufactured = 2400 x (395/6) % - 1100 = 480

The number of (B + C) articles manufactured = 900 – 480 = 420

For July

The number of type A articles till July = 45.67% x (total A till July) = 137/300 x (Total A till July)

The number of (B + C) articles till July = 1630

So, the total number of type A articles till July = 1630/163 x 137 = 1370

So, the number of Article A manufactured in July = 1370 – (80 + 300 + 240 + 480) = 270

The total number of articles = (1370 + 1630) – 2400 = 600

So, the number of (B + C) articles= 600 – 270 = 330

Name of Months	Total articles manufactured	Type A
March	400	80
April	600	300
May	500	240
June	900	480
July	600	270

Answer: B

For R

Value of Z = 480

The Number of type C articles manufactured = 120

So, the number of type B articles manufactured = 420 – 120 = 300

We need to find the maximum value of (Z – Y), so the value of Y must be the minimum possible.

So, all sold articles are of type B, so Y = 300 – 200 = 100

Required value = (480 – 100) = 380

Hence answer is option B

13. Questions

For March

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The number of articles (B + C) manufactured = 80 x 4 = 320

So, the total Articles manufactured = 320/80 x 100 = 400

So, type A articles manufactured = 400 – 320 = 80

For April

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Total number of articles manufactured = 1000 – 400 = 600

The number of type A articles manufactured = 1000 x 38% - 80 = 300

The number of (B + C) articles manufactured = 600 - 300 = 300

For May

Total number of articles manufactured = 1500 - 1000 = 500

The number of (B + C) articles manufactured = 220 x 4 – 300 – 320 = 260

The number of type A articles manufactured = 500 – 260 = 240

For June

Total number of articles manufactured = 2400 – 1500 = 900

The number of type A articles manufactured = 2400 x (395/6) % - 1100 = 480

The number of (B + C) articles manufactured = 900 – 480 = 420

For July

The number of type A articles till July = 45.67% x (total A till July) = 137/300 x (Total A till July)

The number of (B + C) articles till July = 1630

So, the total number of type A articles till July = 1630/163 x 137 = 1370

So, the number of Article A manufactured in July = 1370 – (80 + 300 + 240 + 480) = 270

The total number of articles = (1370 + 1630) – 2400 = 600

So, the number of (B + C) articles= 600 – 270 = 330

Name of Months	Total articles manufactured	Type A
March	400	80
April	600	300
May	500	240
June	900	480
July	600	270

Answer: A

(7Z + 60) = 620

So, the value of Z = 80

Hence answer is option A

14. Questions

For March

The number of articles (B + C) manufactured = 80 x 4 = 320

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So, the total Articles manufactured = 320/80 x 100 = 400

So, type A articles manufactured = 400 – 320 = 80

For April

Total number of articles manufactured = 1000 – 400 = 600

The number of type A articles manufactured = 1000 x 38% - 80 = 300

The number of (B + C) articles manufactured = 600 - 300 = 300

For May

Total number of articles manufactured = 1500 - 1000 = 500

The number of (B + C) articles manufactured = 220 x 4 – 300 – 320 = 260

The number of type A articles manufactured = 500 – 260 = 240

For June

Total number of articles manufactured = 2400 – 1500 = 900

The number of type A articles manufactured = 2400 x (395/6) % - 1100 = 480

The number of (B + C) articles manufactured = 900 – 480 = 420

For July

The number of type A articles till July = 45.67% x (total A till July) = 137/300 x (Total A till July)

The number of (B + C) articles till July = 1630

So, the total number of type A articles till July = 1630/163 x 137 = 1370

So, the number of Article A manufactured in July = 1370 – (80 + 300 + 240 + 480) = 270

The total number of articles = (1370 + 1630) – 2400 = 600

So, the number of (B + C) articles= 600 – 270 = 330

Name of Months	Total articles manufactured	Type A
March	400	80
April	600	300
May	500	240
June	900	480
July	600	270

Tg:- @NextGenBankers

Answer: C

According to the question,

(620/1500) x 100 = (8Z + 28)/3

Value of Z = 12

Number of type B articles manufactured in March = (320 + 40)/2 = 180

Required difference = 180 – 12 = 168

Hence answer is option C

15. Questions

For March

The number of articles (B + C) manufactured = 80 x 4 = 320

So, the total Articles manufactured = 320/80 x 100 = 400

So, type A articles manufactured = 400 – 320 = 80

For April

Total number of articles manufactured = 1000 – 400 = 600

The number of type A articles manufactured = 1000 x 38% - 80 = 300

The number of (B + C) articles manufactured = 600 - 300 = 300

For May

Total number of articles manufactured = 1500 - 1000 = 500

The number of (B + C) articles manufactured = 220 x 4 – 300 – 320 = 260

The number of type A articles manufactured = 500 – 260 = 240

For June

Total number of articles manufactured = 2400 – 1500 = 900

The number of type A articles manufactured = 2400 x (395/6) % - 1100 = 480

The number of (B + C) articles manufactured = 900 – 480 = 420

For July

The number of type A articles till July = 45.67% x (total A till July) = 137/300 x (Total A till July)

The number of (B + C) articles till July = 1630

So, the total number of type A articles till July = 1630/163 x 137 = 1370

So, the number of Article A manufactured in July = 1370 – (80 + 300 + 240 + 480) = 270

The total number of articles = (1370 + 1630) – 2400 = 600

So, the number of (B + C) articles= 600 – 270 = 330

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Name of Months	Total articles manufactured	Type A
March	400	80
April	600	300
May	500	240
June	900	480
July	600	270

Answer: A

Required difference = 480 – 300 = 180

Hence answer is option A

16. Questions

For Q

Let total students in 2019 = 10a

So, students in 2020 = 10a x 90% = 9a

So, boys in 2019 = 4/9 x 9a = 4a

So, girls in 2019 = 10a – 4a = 6a

So, boys in 2020 = 4a x 1.2 = 4.8a

So, girls in 2020 = 9a – 4.8a = 4.2a

Now,

(6a + 4.2a) = 255 x 2

10.2a = 510

So, the value of a = 50

For P,

Boys in P in 2019 = 4/15 x 6 x 50 = 80

So, boys in P in 2020 = 80 x 1.25 = 100

Total number of students in (2019 + 2020) = 80 + 100 + 2 x 90 = 360

Number of students in 2020 = 80% x 2019 =

So, the total students in 2019 = 360/180 x 100 = 200

Total students in 2020 = 200 x 0.8 = 160

Girls in 2019 = 200 – 80 = 120

Girls in 2020 = 180 – 120 = 60

For R,

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Number of boys in R in 2020 = 680 – 4.8 x 50 = 440

Number of boys in R in 2019 = 440/110 x 100 = 400

Total students in (2019 + 2020) = 840 + 760 = 1600

Ratio of the number of students in 2019 and 2020 = 100:60 = 5:3

So, students in 2019 = 5/8 x 1600 = 1000

Students in 2020 = 1600 – 1000 = 600

Answer: C

According to question,

Number of students in 2020 in S = 600 – 375 = 225

So, number of students in 2019 = 225/75 x 100 = 300

Total number of boys (2019 + 2020) = 300 + 225 – 2 x 82.5 = 360

Ratio of the number of boys in 2019 and 2020 = 5:7

So, boys in 2019 = 5/12 x 360 = 150

Hence answer is option C

17. Questions

For Q

Let total students in 2019 = 10a

So, students in 2020 = 10a x 90% = 9a

So, boys in 2019 = 4/9 x 9a = 4a

So, girls in 2019 = 10a – 4a = 6a

So, boys in 2020 = 4a x 1.2 = 4.8a

So, girls in 2020 = 9a – 4.8a = 4.2a

Now,

(6a + 4.2a) = 255 x 2

10.2a = 510

So, the value of a = 50

For P,

Boys in P in 2019 = 4/15 x 6 x 50 = 80

So, boys in P in 2020 = 80 x 1.25 = 100

Total number of students in (2019 + 2020) = 80 + 100 + 2 x 90 = 360

Number of students in 2020 = 80% x 2019 =

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So, the total students in 2019 = 360/180 x 100 = 200

Total students in 2020 = 200 x 0.8 = 160

Girls in 2019 = 200 – 80 = 120

Girls in 2020 = 180 – 120 = 60

For R,

Number of boys in R in 2020 = 680 – 4.8 x 50 = 440

Number of boys in R in 2019 = 440/110 x 100 = 400

Total students in (2019 + 2020) = 840 + 760 = 1600

Ratio of the number of students in 2019 and 2020 = 100:60 = 5:3

So, students in 2019 = 5/8 x 1600 = 1000

Students in 2020 = 1600 – 1000 = 600

Answer: D

For P (2020)

Number of students pass in exam = 35% x 240 = 84

Girls pass the exam = 84 – 64 = 20

So, the number of girls who failed in exam is = 60 – 20 = 40

Hence answer is option D

18. Questions

For Q

Let total students in 2019 = 10a

So, students in 2020 = 10a x 90% = 9a

So, boys in 2019 = 4/9 x 9a = 4a

So, girls in 2019 = 10a – 4a = 6a

So, boys in 2020 = 4a x 1.2 = 4.8a

So, girls in 2020 = 9a – 4.8a = 4.2a

Now,

(6a + 4.2a) = 255 x 2

10.2a = 510

So, the value of a = 50

For P,

Boys in P in 2019 = 4/15 x 6 x 50 = 80

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So, boys in P in 2020 = 80 x 1.25 = 100

Total number of students in (2019 + 2020) = 80 + 100 + 2 x 90 = 360

Number of students in 2020 = 80% x 2019 =

So, the total students in 2019 = 360/180 x 100 = 200

Total students in 2020 = 200 x 0.8 = 160

Girls in 2019 = 200 – 80 = 120

Girls in 2020 = 180 – 120 = 60

For R,

Number of boys in R in 2020 = 680 – 4.8 x 50 = 440

Number of boys in R in 2019 = 440/110 x 100 = 400

Total students in (2019 + 2020) = 840 + 760 = 1600

Ratio of the number of students in 2019 and 2020 = 100:60 = 5:3

So, students in 2019 = 5/8 x 1600 = 1000

Students in 2020 = 1600 – 1000 = 600

Answer: B

The difference between the number of girls in 2019 and 2020 in school R = 600 – 160 = 440 (A)

The number of boys in school R in 2020 = 440 (B)

I). A = B [True]

II). A + B > 900

A + B = 880 < 900

This statement is false

III). (2A – B) = 88% of students in school Q in 2019

Value of (2A – B) = A = 440

88% x students in school Q in 2019 = 88% x 500 = 440

This statement is true.

So, only I and III are true.

Hence answer is option B

19. Questions

For Q

Let total students in 2019 = 10a

So, students in 2020 = 10a x 90% = 9a

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So, boys in 2019 = 4/9 x 9a = 4a

So, girls in 2019 = 10a – 4a = 6a

So, boys in 2020 = 4a x 1.2 = 4.8a

So, girls in 2020 = 9a – 4.8a = 4.2a

Now,

(6a + 4.2a) = 255 x 2

10.2a = 510

So, the value of a = 50

For P,

Boys in P in 2019 = 4/15 x 6 x 50 = 80

So, boys in P in 2020 = 80 x 1.25 = 100

Total number of students in (2019 + 2020) = 80 + 100 + 2 x 90 = 360

Number of students in 2020 = 80% x 2019 =

So, the total students in 2019 = 360/180 x 100 = 200

Total students in 2020 = 200 x 0.8 = 160

Girls in 2019 = 200 – 80 = 120

Girls in 2020 = 180 – 120 = 60

For R,

Number of boys in R in 2020 = 680 – 4.8 x 50 = 440

Number of boys in R in 2019 = 440/110 x 100 = 400

Total students in (2019 + 2020) = 840 + 760 = 1600

Ratio of the number of students in 2019 and 2020 = 100:60 = 5:3

So, students in 2019 = 5/8 x 1600 = 1000

Students in 2020 = 1600 – 1000 = 600

Answer: A

For S,

Number of girls in (2019 + 2020) = 82.5 x 2 = 165

Number of girls in 2019 = 165 – 30 = 135

Number of boys in 2020 = 2 x 135 = 270

So, the total students in 2020 = 270 + 30 = 300

So, the total students in (2019 + 2020) = 300 x 7/3 = 700

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Hence answer is option A

20. Questions

For Q

Let total students in 2019 = 10a

So, students in 2020 = 10a x 90% = 9a

So, boys in 2019 = 4/9 x 9a = 4a

So, girls in 2019 = 10a – 4a = 6a

So, boys in 2020 = 4a x 1.2 = 4.8a

So, girls in 2020 = 9a – 4.8a = 4.2a

Now,

(6a + 4.2a) = 255 x 2

10.2a = 510

So, the value of a = 50

For P,

Boys in P in 2019 = 4/15 x 6 x 50 = 80

So, boys in P in 2020 = 80 x 1.25 = 100

Total number of students in (2019 + 2020) = 80 + 100 + 2 x 90 = 360

Number of students in 2020 = 80% x 2019 =

So, the total students in 2019 = 360/180 x 100 = 200

Total students in 2020 = 200 x 0.8 = 160

Girls in 2019 = 200 – 80 = 120

Girls in 2020 = 180 – 120 = 60

For R,

Number of boys in R in 2020 = 680 – 4.8 x 50 = 440

Number of boys in R in 2019 = 440/110 x 100 = 400

Total students in (2019 + 2020) = 840 + 760 = 1600

Ratio of the number of students in 2019 and 2020 = 100:60 = 5:3

So, students in 2019 = 5/8 x 1600 = 1000

Students in 2020 = 1600 – 1000 = 600

Answer: E

For (P + Q + R) together in 2020

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Total number of girls = 60 + 210 + 160 = 430

Number of girls in class 9th = 430 – 100 = 330

Number of girls failed in class 9th = 330 x 2/3 = 220

Hence answer is option E

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1. Questions

Study the following information carefully and answer the questions given below.

The table chart shows the number of cars [petrol + diesel + electric] sold, the percentage of diesel cars soldin that showroom and the percentage of petrol cars sold more or less than the number of diesel cars sold, in five showrooms [A, B, C, D and E] in the year 2023.

Showroom	Total number of cars sold	Percentage of diesel cars sold	percentage of petrol cars sold more or less than the number of diesel cars sold
A	6x	5j%	+25%
B	3x	5i%	+30%
C	6y	10m%	+40%
D	4x	[i*j]%	-25%
E	10y	40%	+30%

Note- x=16i, y=25m, $(i-5)^2+(j-4)^2+(m-2)^2=0$

Find which of the following is true?

I). The difference between the number of electric cars sold in showrooms A and B is 142.

II). The number of petrol cars sold in showroom E is 6.5 times of the number of electric cars sold in the showroom.

III). i+2j+3m=29

a. Only II and III

b. Only I and II

c. Only II

d. Only III

e. All false

2. Questions

In the year 2022, the number of cars sold in showroom B is 20% more than the number of cars sold in 2023. The percentage of electric cars sold in showroom B in 2022 is 5i. Find the difference between the total number of petrol and diesel cars sold in showroom B in 2022 and 2023?

a. x-3

b. 32m

c. 22x

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d. 19.5j

e. y+24

3. Questions

Find the possible difference between the total number of diesel cars and the total number of petrol cars sold in all showrooms together in the year 2023?

I). 22i,

II). 25j+10,

III). 2y+10

a. All false

b. All true

c. Only I true

d. Only I false

e. None of these

4. Questions

The difference between the number of diesel cars sold in showrooms A and B is _____, The difference between the number of electric cars sold in showrooms D and E is _____ and the number of diesel cars sold in C is _____% of the number of diesel cars sold in E.

Find which of the following is true?

a. 36,168,30

b. 48, 164,30

c. 16m, 36j, 6m

d. 18m, 42j,6i

e. More than one option is true

5. Questions

The number of CNG cars sold in showroomD is 25% less than the number of electric cars sold in the same showroom. The average selling price of CNG cars is 11.5 lakhs. Find the total revenue earned by showroom D by selling CNG cars?

a. 1794 lakhs

b. 1568 lakhs

c. 1684 lakhs

d. 1954 lakhs

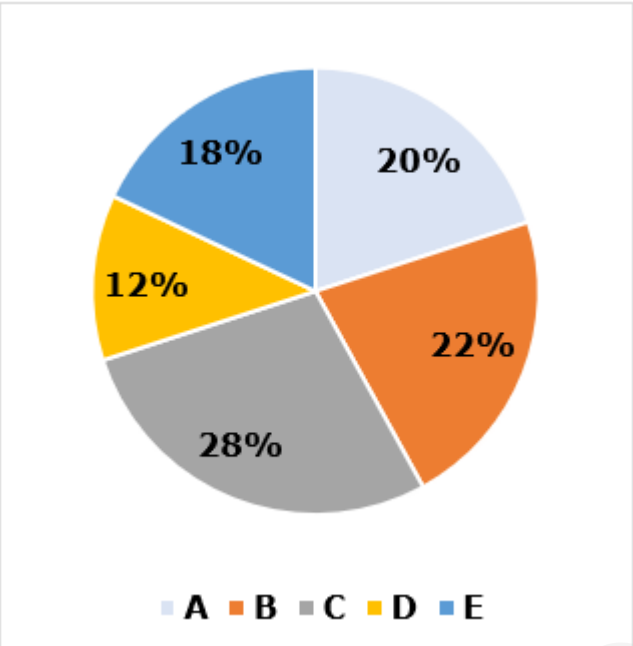
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e. None of these

6. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage of the number students who appeared in an exam from five schools [A,B,C,D and E].



The given table shows the ratio of the number of boys and girls who appeared in the exam, the ratio of the number of boys who passed and failed in the exam, ratio of the number of girls who passed and failed in the exam from five schools.

Schools	The ratio of the number of boys and girls who appeared in the exam	The ratio of the number of boys who passed and failed the exam	Ratio of the number of girls who passed and failed in the exam
A	3:2	7:2	7:1
B	6:5	8:1	3:1
C	3:4	5:1	7:5
D	2:1	11:1	7:1
E	4:5	5:1	5:1

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Note – The total number of students who appeared in the exam from all schools together is 2400.

Quantity: I Difference between the number of students who failed from school C and D is x. Find the value 9x?

Quantity: II The difference between the number of students who passed from schools A and B is y. Find the value 37y?

a. Quantity: I < Quantity: II

b. Quantity: I ≥ Quantity: II

c. Quantity: II ≥ Quantity: I

d. Quantity: I > Quantity: II

e. Quantity I = Quantity II or relation can't be established

7. Questions

Find which of the following is true?

I). The number of boys who passed from A and B is 152 more than the number of girls who passed from the same school.

II). The number of boys who passed from D and E is 32 more than the number of girls who passed from the same school.

III). The number of boys who failed from D and C is 128 more than the number of girls who failed from the same school.

a. Only I false

b. Only II false

c. All true

d. All false

e. Only III false

8. Questions

The number of students who appeared in the exam from school F is 20% more than the number of students who appeared in the exam from school A. The ratio of boys and girls who appeared in the exam from school F is 2:1. Find the difference between the number of boys and girls who appeared in the exam from school F?

a. 182

b. 192

c. 188

d. 164

e. None of these

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9. Questions

The average weight of boys who appeared in the exam from E is 58 kg and the average weight of girls who appeared in the exam from E is 42 kg. Find the approximate average age of all the students who appeared in the exam from E?

a. 49 kg

b. 47 kg

c. 51 kg

d. 46 kg

e. None of these

10. Questions

Quantity: I Number of boys who passed from A is approximately what percent of the number of girls who failed from B?

Quantity: II Number of boys who passed from B is approximately what percent of the number of girls who failed from E?

a. Quantity I < Quantity II

b. Quantity I ≥ Quantity II

c. Quantity II ≥ Quantity I

d. Quantity I > Quantity II

e. Quantity I = Quantity II or relation can't be established

11. Questions

Read the following information carefully and answer the questions based on it.

Three persons – P, Q, and R starting from point M from their bikes at same time and travelled towards N, but they are using different Routes – R1, R2 and R3 respectively. Speed of Q is twice as that of P. Total time taken by P for travelling from M to N and then returned to M is 25 hours. Ratio of distance of R1: R3: R2 is 15:18:20. Total time taken by Q for travelling from M to N and then returned to M is 5 hours 25 minutes more than time taken by R to travel from M to N. Speed of Q is 24 km/h.

When R started from point M, after covering 128 km, due to engine problem, speed of his bike reduced by 18.75%. After reaching at point N, he found the mechanic to solve the engine problem, so he has to wait for 2 hours 30 minutes. Now he has to travel at K% more speed than his last speed to reach on time at point M, find approximate value of K?

a. 41%

b. 73%

c. 45%

d. 75%

e. 51%

12. Questions

The speed of R is how much % more or less than that of Q?

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a. 25%

b. 50%

c. 33.33%

d. 40%

e. None of these

13. Questions

If Q is started from point M and R started form point N (both are at same route R2), find the distance between second meeting point of Q and R, and point M.

a. 160 km

b. 120 km

c. 80 km

d. 40 km

e. None of these

14. Questions

Find the sum of distance covered by P, Q and R together.

a. 265 km

b. 795 km

c. 540 km

d. 1060 km

e. None of these

15. Questions

A covers certain distance in 25 hours and the speed of A is 25% more than the speed of R, find the distance covered by A.

a. 300 km

b. 775 km

c. 550 km

d. 1000 km

e. None of these

16. Questions

A and B can complete a work alone in _____ and ____ days respectively. They start the work and

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after ____ days they left the work and the remaining work is finished by C in ____ days. C is double the efficient than A.

Find which of the following satisfies the blank in the same sequence?

I). 20,30, 6,5

II). 36,24,12,3

III). 24,16,4,7

a. All false
b. Only I false
c. All true
d. Only II false
e. Only III false

17. Questions

_____% of income of A is _____% of income of B. If the income of A increases by 2000 and the Income of B increases by 4000 then the ratio of income of A and B is _____. The total income of A and B is _____.

Find which of the following is true?

I).40,50,7:8,54000

II). 45,75,2:1,64000

III). 50,40,3:2,45000

a. Only III true
b. Only II false
c. Only I false
d. All false
e. All true

18. Questions

The ratio of cost price and selling price is _____. The marked price of the item is Rs. _____ and the discount percentage is 20%. The difference between the discount amount and the profit amount is _____.

Find which of the following is true?

a. 25:36, 360, 20
b. 50:52, 520, 88
c. 4:5, 375, 25

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d. 3:5,250, 34

e. More than one option is true

19. Questions

A invests Rs._____ for 3 years at a 20% rate of simple interest. After 3 years A divided the total amount in a ratio of _____ and invests in schemes M and N respectively. Scheme M and N offer 20% compound interest. The difference between the interest amount received by A after 2 years is ____.

Find which of the following is true?

a. 1800, 3:1, 633.6

b. 1500,3:2, 263.33

c. 1200, 5:3, 544.6

d. 3200, 5:3, 452.33

e. None of these

20. Questions

The ratio of speed of boat A and B in still water is _____. The speed of the stream of the river is 4 km/hr. Boat B covers _____ km upstream in the river in 4 hours. Boat A covers _____ downstream in the river in 7 hours.

a. 5:3, 104, 452

b. 7:8, 80, 224

c. 3:2, 56, 256

d. 4:5,105, 258

e. 5:4, 64, 203

Explanations:

1. Questions

x=16i, y=25m,

$(i-5)^2+(j-4)^2+(m-2)^2=0$

We know that if the sum of the squares of two or more numbers then each number is zero.

So,(i-5)=0, i=5, similarly, j-4=0, j=4, m-2=0, m=2

So, x=16*5=80 and y=25*2=50

So, number of cars sold in A is 6*80=480

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Number of diesel cars sold in A is 480*[5*4]/100=96

Number of petrol cars sold in A is 96*125/100=120

Number of electric cars sold in A is 480-96-120=264

Similarly, we can calculate the others' values also.

Showrooms	Total number of cars sold	Total number of diesel cars sold	Total number of petrol cars sold	Total number of electric cars sold
A	480	96	120	264
B	240	60	78	102
C	300	60	84	156
D	320	64	48	208
E	500	200	260	40

Answer: C

I). Difference between the number of electric cars sold in showrooms A and B is [264-102] =162

II). The number of petrol cars sold in showroom E is =260 which is 40*6.5 times of number of electric cars sold in the showroom.

III). i+2j+3m=5+2*4+3*2=5+8+6=19

2. Questions

x=16i, y=25m,

$(i-5)^2+(j-4)^2+(m-2)^2=0$

We know that if the sum of the squares of two or more numbers then each number is zero.

So,(i-5)=0, i=5, similarly, j-4=0, j=4, m-2=0, m=2

So, x=16*5=80 and y=25*2=50

So, number of cars sold in A is 6*80=480

Number of diesel cars sold in A is 480*[5*4]/100=96

Number of petrol cars sold in A is 96*125/100=120

Number of electric cars sold in A is 480-96-120=264

Similarly, we can calculate the others' values also.

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Showrooms	Total number of cars sold	Total number of diesel cars sold	Total number of petrol cars sold	Total number of electric cars sold
A	480	96	120	264
B	240	60	78	102
C	300	60	84	156
D	320	64	48	208
E	500	200	260	40

Answer: D

The number of cars sold in 2022 from showroom B is 240*120/100=288

Number of diesel and petrol cars sold in 2022 is = 288*(100-5*5)/100=(288*75)/100=216

Number of Diesel and petrol cars sold in 2023 is 60+78=138

So, the difference is 216-138=78

3. Questions

x=16i, y=25m,

$(i-5)^2+(j-4)^2+(m-2)^2=0$

We know that if the sum of the squares of two or more numbers then each number is zero.

So,(i-5)=0, i=5, similarly, j-4=0, j=4, m-2=0, m=2

So, x=16*5=80 and y=25*2=50

So, number of cars sold in A is 6*80=480

Number of diesel cars sold in A is 480*[5*4]/100=96

Number of petrol cars sold in A is 96*125/100=120

Number of electric cars sold in A is 480-96-120=264

Similarly, we can calculate the others' values also.

Showrooms	Total number of cars sold	Total number of diesel cars sold	Total number of petrol cars sold	Total number of electric cars sold
A	480	96	120	264
B	240	60	78	102
C	300	60	84	156
D	320	64	48	208
E	500	200	260	40

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Answer: B

Required difference

= [260+48+84+78+120] - [200+64+60+60+96]

= 590-480=110=22i=25j+10=2y+10

4. Questions

x=16i, y=25m,

$(i-5)^2+(j-4)^2+(m-2)^2=0$

We know that if the sum of the squares of two or more numbers then each number is zero.

So,(i-5)=0, i=5, similarly, j-4=0, j=4, m-2=0, m=2

So, x=16*5=80 and y=25*2=50

So, number of cars sold in A is 6*80=480

Number of diesel cars sold in A is 480*[5*4]/100=96

Number of petrol cars sold in A is 96*125/100=120

Number of electric cars sold in A is 480-96-120=264

Similarly, we can calculate the others' values also.

Showrooms	Total number of cars sold	Total number of diesel cars sold	Total number of petrol cars sold	Total number of electric cars sold
A	480	96	120	264
B	240	60	78	102
C	300	60	84	156
D	320	64	48	208
E	500	200	260	40

Answer: E

The difference between the number of petrol cars sold in showrooms A and B is =[96-60]=36

The difference between the number of electric cars sold in showrooms D and E is =208-40=168

The number of diesel cars sold in C is =[60/200]*100=30% of the number of diesel cars sold in E.

5. Questions

x=16i, y=25m,

$(i-5)^2+(j-4)^2+(m-2)^2=0$

We know that if the sum of the squares of two or more numbers then each number is zero.

So,(i-5)=0, i=5, similarly, j-4=0, j=4, m-2=0, m=2

So, x=16*5=80 and y=25*2=50

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So, number of cars sold in A is 6*80=480

Number of diesel cars sold in A is 480*[5*4]/100=96

Number of petrol cars sold in A is 96*125/100=120

Number of electric cars sold in A is 480-96-120=264

Similarly, we can calculate the others' values also.

Showrooms	Total number of cars sold	Total number of diesel cars sold	Total number of petrol cars sold	Total number of electric cars sold
A	480	96	120	264
B	240	60	78	102
C	300	60	84	156
D	320	64	48	208
E	500	200	260	40

Answer: A

Number of CNG cars sold in showrooms D is 208*(75/100)=156

The total revenue earned by showrooms is 156*11.5=1794 lakhs

6. Questions

Number of students who appeared from A is 2400*20/100=480

Number of boys who appeared from A is 480*3/5=288

Number of girls who appeared from A is 480-288=192

Number of boys who passed the exam from A is 288*7/9=224

Number of boys who failed the exam from A is 288-224=64

Number of girls who passed the exam from A is 192*7/8=168

Number of girls who failed the exam from A is 192-168=24

Similarly, we can calculate others' values also.

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School	The total number of students who appeared in the exam	Boys		Girls	
		Passed	Failed	Passed	Failed
A	480	224	64	168	24
B	528	256	32	180	60
C	672	240	48	224	160
D	288	176	16	84	12
E	432	160	32	200	40

Answer: A

Quantity: I Difference between the number of students who failed from schools C and D is x.

So, x= [160+48]-[16+12]=180

Find the value 9x=9*180=1620

Quantity: II The Difference between the number of students who passed from schools A and B is y.

So, y=[256+180]-[224+168]=44

So, 37*44=1628

Quantity I < Quantity II

7. Questions

Number of students who appeared from A is 2400*20/100=480

Number of boys who appeared from A is 480*3/5=288

Number of girls who appeared from A is 480-288=192

Number of boys who passed the exam from A is 288*7/9=224

Number of boys who failed the exam from A is 288-224=64

Number of girls who passed the exam from A is 192*7/8=168

Number of girls who failed the exam from A is 192-168=24

Similarly, we can calculate others' values also.

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School	The total number of students who appeared in the exam	Boys		Girls	
		Passed	Failed	Passed	Failed
A	480	224	64	168	24
B	528	256	32	180	60
C	672	240	48	224	160
D	288	176	16	84	12
E	432	160	32	200	40

Answer: D

I). Number of boys who passed from A and B is [224+256]-[168+180]=132 more than the number of girls who passed from the same school.

II). Number of boys who passed from D and E is [176+160]-[84+200]=52 more than the number of girls who passed from the same school.

III). Number of boys who failed from D and C is [160+12]-[48+16]=108 more than the number of girls who failed from the same school.

8. Questions

Number of students who appeared from A is 2400*20/100=480

Number of boys who appeared from A is 480*3/5=288

Number of girls who appeared from A is 480-288=192

Number of boys who passed the exam from A is 288*7/9=224

Number of boys who failed the exam from A is 288-224=64

Number of girls who passed the exam from A is 192*7/8=168

Number of girls who failed the exam from A is 192-168=24

Similarly, we can calculate others' values also.

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School	The total number of students who appeared in the exam	Boys		Girls	
		Passed	Failed	Passed	Failed
A	480	224	64	168	24
B	528	256	32	180	60
C	672	240	48	224	160
D	288	176	16	84	12
E	432	160	32	200	40

Answer: B

Number of students who appeared in the exam is = 480*120/100=576

The difference in the number of boys and girls who appeared in the exam from F is = 576/3=192

9. Questions

Number of students who appeared from A is 2400*20/100=480

Number of boys who appeared from A is 480*3/5=288

Number of girls who appeared from A is 480-288=192

Number of boys who passed the exam from A is 288*7/9=224

Number of boys who failed the exam from A is 288-224=64

Number of girls who passed the exam from A is 192*7/8=168

Number of girls who failed the exam from A is 192-168=24

Similarly, we can calculate others' values also.

School	The total number of students who appeared in the exam	Boys		Girls	
		Passed	Failed	Passed	Failed
A	480	224	64	168	24
B	528	256	32	180	60
C	672	240	48	224	160
D	288	176	16	84	12
E	432	160	32	200	40

Answer: A

The average weight of all the students from E is

= [58*192+240*42]/432=21216/432=49.11=49 kg

10. Questions

Number of students who appeared from A is 2400*20/100=480

Number of boys who appeared from A is 480*3/5=288

Number of girls who appeared from A is 480-288=192

Number of boys who passed the exam from A is 288*7/9=224

Number of boys who failed the exam from A is 288-224=64

Number of girls who passed the exam from A is 192*7/8=168

Number of girls who failed the exam from A is 192-168=24

Similarly, we can calculate others' values also.

School	The total number of students who appeared in the exam	Boys		Girls	
		Passed	Failed	Passed	Failed
A	480	224	64	168	24
B	528	256	32	180	60
C	672	240	48	224	160
D	288	176	16	84	12
E	432	160	32	200	40

Answer: A

Quantity: I Number of boys passed from A is approximately=[224/60]*100=373.33=373% of number of girls failed from B

Quantity: II Number of boys passed from B is approximately=[256/40]*100=640% of number girls failed from E

Quantity I < Quantity II

11. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

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Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: B

According to question,

Total time taken by R for back-and-forth journey = 11.25 x 2 = 22.5 hours

Due to engine problem, speed of R = 16 – 16 x 3/16 = 13 km/h

Now total time taken by R to travel from M to N = 128/16 + 52/13 = 12 hours

He has to wait for 2 hours at point N, so time left to travel from N to M for R = 22.5 - 12 – 2.5 = 8 hours

Now speed of R for N to M = 180/8 = 22.5 km/h

Required % change = (22.5 – 13)/13 x 100 = 73%

Hence answer is option B

12. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: C

Speed of R = 16 km/h

Speed of Q = 24 km/h

Required % change = (24 – 16)/24 x 100 = 33.33%

Hence answer is option C

13. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

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So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: D

Both Q and R, running on Route 2, distance between then initially = 200 km

When Q and R meets first time, they travelled = 200 km together

After they reached respective positions and then turns back, then they meet for the second time.

So, in second meeting, they together need to travel = 2 x 200 = 400 km

So, total distance covered = 200 + 400 = 600 km

Time is constant, so ratio of distance is the same as the ratio of speed.

Ratio of speed of Q and R = 24:16 = 3:2

Distance covered by R (N to M + M to second meeting point) = 2/5 x 600 = 240 km

So, distance between M and second meeting point = 240 – 200 = 40 km

Hence answer is option D

14. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

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Answer: E

Required total distance = 150 + 200 + 180 = 530 km

Hence answer is option E

15. Questions

Speed of Q = 24 km/h

So, speed of P = 24/2 = 12 km/hr

Total distance covered by P = 12 x 25 = 300 km

So, distance of route R1 = 300/2 = 150 km

Distance of route R3 = 150/15 x 18 = 180 km

Distance of route R2 = 150/15 x 20 = 200 km

Time taken by Q to travel from M to N and N to M = 2 x 200/24 = 50/3 hours

Time taken by R to travel from M to N = 50/3 – 65/12 = 135/12 hours = 11.25 hours

Speed of R = 180/11.25 = 16 km/h

Answer: E

Speed of A = 125/100 * 16 = 20 kmph

Required distance = 20 * 25 = 500 km

16. Questions

Answer: C

For I,

A and B can do work alone in 20 and 30 days respectively.

C is 100% more efficient than A.

So, C takes 10 days.

Total work = LCM 20,30 and 10=60

So, the Efficiency of A,B and C is 3, 2 and 6 respectively.

So, If A and B together work for 6 days then the remaining work is 60-5*6=30

So, the remaining work C is complete by 30/6=5 days.

So, it's true.

Similarly, we can check others' values also.

17. Questions

Answer: D

For I,

40% of income of A = 50% income of B

The ratio of income of A and B is 5:4.

So, the income of A is 54000*5/9=30000

The income of B is 54000-30000=24000

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So, [30000+2000]:[24000+4000]=32000:28000=8:7

So, it's false.

Similarly, we can check others' values also.

18. Questions

Answer: B

For the value of B,

The marked price is 520.

So, the selling price is 520*80/100=416

So, the cost price is 416*50/52=400

So, profit is 416-400=16

Discount is 520*20/100=104

So, difference 104-16=88

So, it's true

19. Questions

Answer: A

For A,

Total amount after 3 years = 1800+1800*3*20/100=2880

So, the amount invested in scheme M is 2880*3/4=2160

The amount invested in scheme N is 2880-2160=720

So, difference 2160*44/100 – 720*44/100=633.6

So, it's true

20. Questions

Answer: E

For the value of E,

The ratio of speed of Boat A and B in still water is 5:4.

So, the speed of boat B in upstream is 64/4=16 km/hr

So, the speed of boat B in still water is 16+4=20 km/hr

So, the speed of boat A in still water is 20*5/4=25 km/hr

Boat A cover in 7 hours downstream = 29*7=203 km

So, it's true

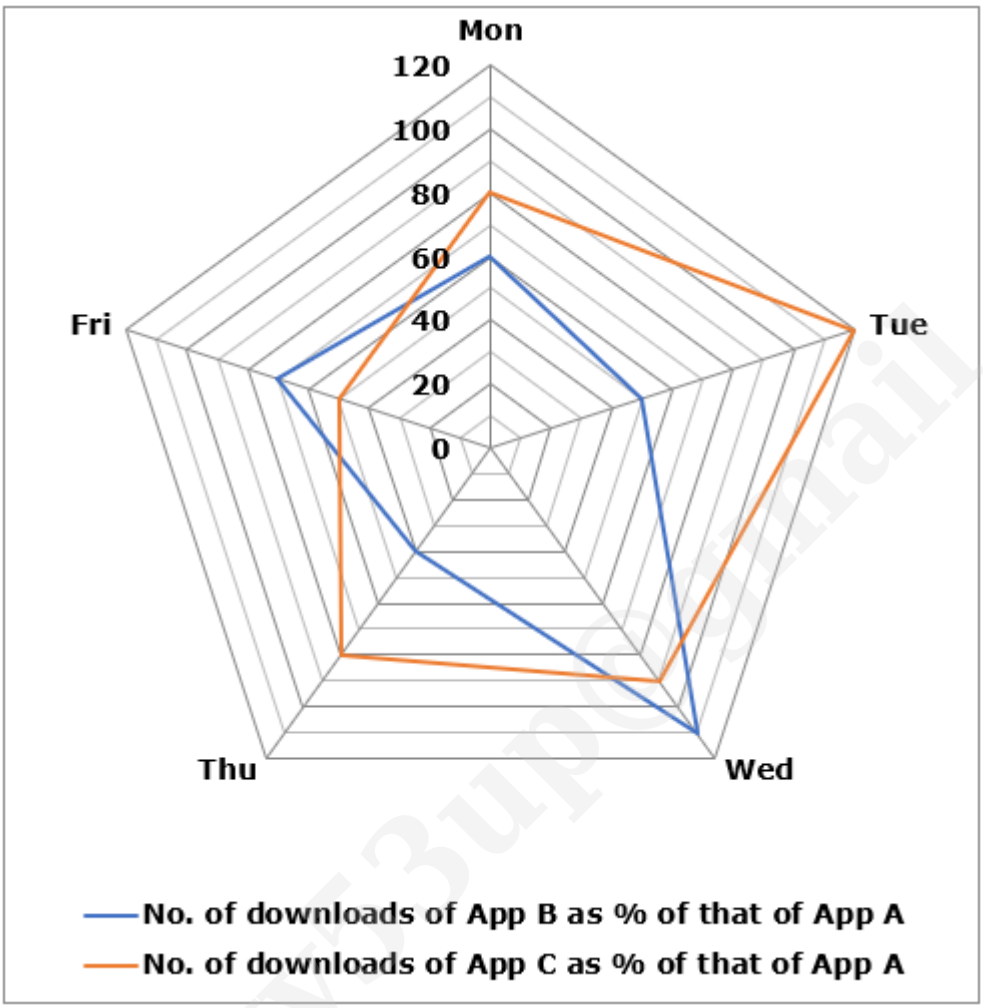
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1. Questions

Study the following data carefully and answer the questions:

Data given below is related to the number of downloads of three mobile applications (Apps) A, B and C on five days Mon, Tue, Wed, Thu and Fri, which were launched on Mon.

Radar graph given below shows the number of downloads of Apps B and C as per cent of the number of downloads of App A on each day.



Notes:

1: The ratio of the number of downloads of App C on Mon to that on Thu is 4: 5 respectively.

2: The number of downloads of App B on Mon is equal to that on Tue and also 9000 less than that on Wed.

3: The number of downloads of App B on Wed is 7000 less than that of App C on Thu.

4: The ratio of the number of downloads of App C on Wed to that on Fri is 3: 2.

5: The number of downloads of an App = The number of persons, who downloaded the App

If the number of downloads of App A on Thu is a% more/less than that on Mon, the number of downloads of App B on Fri is b% more/less than that on Mon and the number of downloads of App C on Tue is c% more/less than that on Mon, then find the ratio between a, b and c respectively?

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a. 25: 5: 88

b. 5: 1: 10

c. 5: 1: 20

d. 5: 1: 16

e. 5: 4: 16

2. Questions

The table given below shows the following data:

Day	Ratio of male to female, who downloaded App B
Mon	5: 3
Tue	1: 2
Wed	7: 4
Thu	1: 3
Fri	3: 4

Find the difference between the total number of males and the total number of females, who downloaded App B from Mon to Fri?

a. 5600

b. 6600

c. 5500

d. 6000

e. None of these

3. Questions

If the difference between the total number of downloads of Apps A and B till Wed is 'M' and the difference between the total number of downloads of Apps B and C till Wed is 'N', then find out which of the following is/are true?

P: Difference between M and N is divisible by 7.

Q: Average of M and N is divisible by 3.

R: Value of (M/N) is less than 1.2.

a. Only P

b. Only P and Q

c. All are true

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d. Only Q and R

e. Only Q

4. Questions

On Mon, the total number of persons, who downloaded App B, also downloaded Apps A and C, but the total number of persons, who downloaded App A, didn't download App C and the total number of persons, who downloaded App C, didn't download App A. Find the ratio of the number of persons, who downloaded App A only to those, who downloaded App C only, on Mon?

a. 2:1

b. 16:7

c. 8:3

d. 4:3

e. 4:1

5. Questions

If the total number of downloads of App A is increased by 25% from Fri to Sat, the total number of downloads of App B is increased by 15% from Fri to Sat and the total number of downloads of App C is increased by 30% from Fri to Sat, then find the average of the number of downloads of Apps A, B and C on Sat?

a. 42470

b. 32460

c. 44570

d. 45270

e. 43270

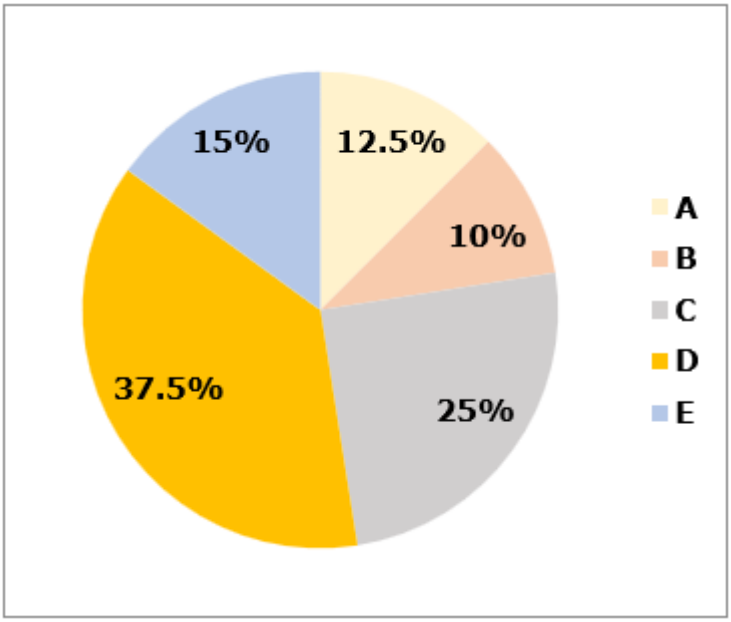
6. Questions

Study the following data carefully and answer the questions:

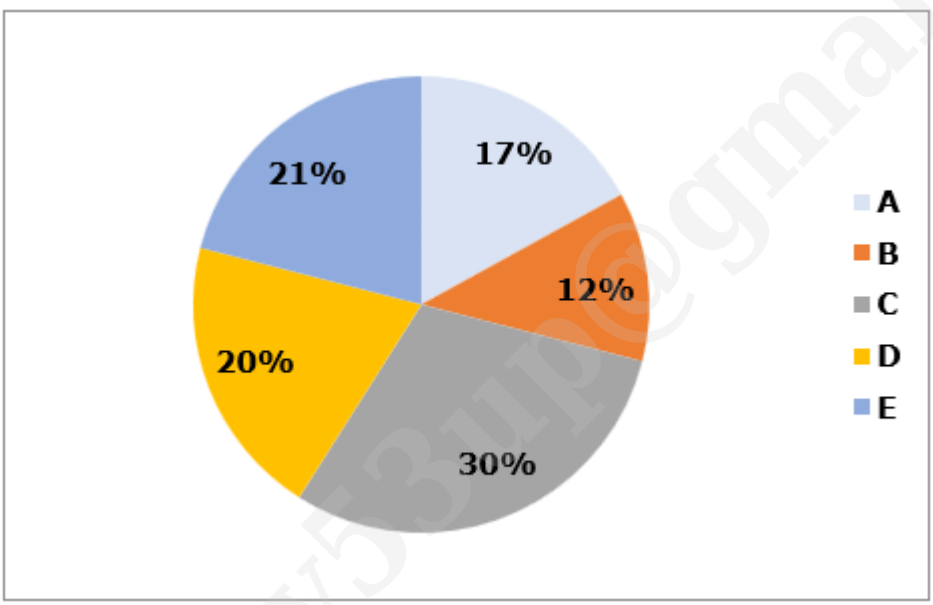
A certain amount (in the form of 500 rupee notes and 200 rupee notes) is distributed among five workers A, B, C, D and E, after they completed a work.

Pie chart given below shows the percentage distribution of total number of notes distributed among A, B, C, D and E.

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Pie chart given below shows the percentage distribution of the number of 200 rupee notes distributed among A, B, C, D and E.



Notes:

1: The number of 500 rupee notes received by B is 20.

2: The total amount received by C is ₹55000.

If the number of 500 rupee notes received by A is M% of that received by B and the number of 500 rupee notes received by C is N% of that received by D, then find the number of 500 rupee notes received by E (in terms of M and N)?

a. (M + N – 95)

b. (M – N – 25)

c. (M – 2N – 10)

d. (M – 3N + 10)

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e. None of these

7. Questions

If each worker is paid in the ratio of the part of work completed by them, then find the ratio between the part of work completed by A, B and D respectively?

a. 49: 44: 240

b. 7: 4: 40

c. 35: 24: 60

d. 63: 44: 120

e. 5: 4: 12

8. Questions

If the average of the total number of notes received by D, E and F is 165 and the average of the number of 200 rupee notes received by D, E and F is 80, then find that the total amount received by F is what per cent of that received by D?

a. 27.5%

b. 33.33%

c. 22.5%

d. 20%

e. 16.67%

9. Questions

If the number of 200 rupee notes received by each of the given workers is described in a circle and the number of 500 rupee notes received by each of the given workers is described in another circle, then find the ratio of the degree distribution of the number of 200 rupee notes received by E to that of the number of 500 rupee notes received by E?

a. 63:25

b. 7:3

c. 49:20

d. 4:3

e. 21:5

10. Questions

Study the following data carefully and answer the questions:

A, B and C were hired to complete a work and they completed the work in such a way that, A and B started the work, after some time, A left and C joined the work with B and after some time, B left and A

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joined the work with C and the work is completed.

A and B started the work together and completed 35% part of work in 2.1 hours. After that, A left the work and C joined B and they together completed 40% part of work in 2.7 hours. After that, B left the work and A joined C and they together completed the remaining work in 2.7 hours.

Find the ratio of the total part of work completed by A to the total part of work completed by C?

a. 4:3

b. 16:9

c. 5:2

d. 20:11

e. 2:1

11. Questions

If A, B and C had worked together, then in what time they would have been completed the 55% part of work?

a. 2 hours 14 minutes

b. 2 hours 49 minutes

c. 2 hours 35 minutes

d. 2 hours 28 minutes

e. 2 hours 42 minutes

12. Questions

If they are paid in the ratio of their efficiencies and the difference between the amount paid to A and B is ₹5250, then find the difference between the amount paid to B and C?

a. ₹3500

b. ₹7000

c. ₹2100

d. ₹4200

e. ₹5100

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13. Questions

Study the following data carefully and answer the questions:

Data given below is related to the number of boys and girls, who applied for two posts A and B. Some boys applied for post A only, some boys applied for post B only and remaining applied for both the posts A and B and the similar is for the girls.

40% of the total number of boys applied for post A only and the number of boys, who applied for post B

only, is 9 more than those boys, who applied for both the posts A and B.

40% of the total number of girls applied for post A only and the number of girls, who applied for post B only, is 3 more than those girls, who applied for both the posts A and B.

Total number of boys and girls is 60. The number of boys, who applied for both the posts, is equal to the number of girls, who applied for both the posts. The total number of boys, who applied for post A, and girls who applied for post A is 20 and 16 respectively.

What per cent of the total number of boys and girls together, who applied for post B, applied for both the posts A and B?

a. 40%

b. 33.33%

c. 30%

d. 25%

e. 37.5%

14. Questions

If 50% of the boys, who applied for post A only, selected for post A, 60% of the boys, who applied for post B only, selected for post B, 33(1/3)% of the boys, who applied for both the posts A and B, selected for post A only and 50% of the boys, who applied for both the posts, selected for post B only, then find the difference between the total number of boys, who selected for post A and those, who selected for post B?

a. 5

b. 2

c. 3

d. 4

e. 6

15. Questions

If 80% of the total number of girls, who applied for post B, selected for post B, out of which 66(2/3)% are those, who were applied for post B only, then find that what per cent of the number of girls, who applied for post B only, selected for post B?

a. 80%

b. 93.75%

c. 72%

d. 64%

e. 88.89%

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16. Questions

Train A, which is running at ___ km/h, is being followed by train B, which is running 27 km/h faster than train A. The length of train A is 10% more than that of train B and train B crosses train A in 84 seconds. If train A crosses a 120 m long tunnel in 20 seconds, then find the value, which can be filled in the blank?

a. 63

b. 108

c. 72

d. 90

e. 81

17. Questions

A shopkeeper marked up an article by ₹___ and sold at 12% profit after giving ___% discount on its marked price. If the article were sold at 20% discount on its cost price, the shopkeeper would have received ₹400 less than the original selling price.

Find that which of the following can be filled in the blanks in the same order?

P: 350, 12.5

Q: 500, 20

R: 750, 25

a. Only P and Q

b. Only R

c. Only Q

d. Only P

e. Only Q and R

18. Questions

A and B together can complete a work in 3.6 hours and B's efficiency is 150% of A's efficiency and ___% of C's efficiency. If A and C together can complete the work in more than 3 hours, then find that which of the following can't be filled in the blank?

P: 60

Q: 50

R: 75

a. Only P and Q

b. Only R

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c. Only Q

d. None can be filled

e. Only Q and R

19. Questions

The speed of boat A in still water is 8 m/s more than the speed of the stream and it takes ___ seconds to cover 1.05 km in downstream direction. Boat B is going upstream at 12 m/s. After travelling for 1.5 minutes, boat B started coming back and took 1 minute to come back to the starting point.

Find the value, which can be filled in the blank?

a. 80

b. 50

c. 75

d. 90

e. 66

20. Questions

Difference between compound interest and simple interest on ₹1250 at R% rate after 2 years is ₹R, the interest received on ₹X at (R + 4)% rate after 1 year is ₹___ and the simple interest received on ₹(X – 450) at (R/2)% rate after 2 years is ₹104.

Find the value, which can be filled in the blank?

a. 270

b. 300

c. 180

d. 210

e. 240

Explanations:

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1. Questions

Let the number of downloads of App C on Mon and Thu be '4x' and '5x' respectively.

So, the number of downloads of App B on Mon = 60% of [4x * (100/80)] = 3x

Since, the number of downloads of App B on Mon is equal to that on Tue and also 9000 less than that on Wed.

So, the number of downloads of App B on Tue = 3x

And the number of downloads of App B on Wed = (3x + 9000)

Since, the number of downloads of App B on Wed is 7000 less than that of App C on Thu.

So,

5x – (3x + 9000) = 7000

x = 8000

Since, the ratio of the number of downloads of App C on Wed to that on Fri is 3: 2.

And the number of downloads of App C on Wed:

90% of [(3x + 9000) * (100/110)] = 90% of [33000 * (100/110)] = 27000

So, the number of downloads of App C on Fri = 27000 * (2/3) = 18000

The number of downloads of App C on Mon = 4 * 8000 = 32000

The number of downloads of App C on Tue = 120% of [3x * (100/50)] = 57600

The number of downloads of App C on Wed = 27000

The number of downloads of App C on Thu = 5 * 8000 = 40000

The number of downloads of App C on Fri = 18000

Day	No. of downloads of App A	No. of downloads of App B	No. of downloads of App C
Mon	32000 * (100/80) = 40000	3 * 8000 = 24000	32000
Tue	57600 * (100/120) = 48000	3 * 8000 = 24000	57600
Wed	27000 * (100/90) = 30000	24000 + 9000 = 33000	27000
Thu	40000 * (100/80) = 50000	40% 50000 = 20000	40000
Fri	18000 * (100/50) = 36000	70% of 36000 = 25200	18000

Answer: D

Since, the number of downloads of App A on Mon = 40000

And the number of downloads of App A on Thu = 50000

So, a = $\frac{50000-40000}{40000} \times 100$ = 25%

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Since, the number of downloads of App B on Mon = 24000

And the number of downloads of App B on Fri = 252000

So, b = $\frac{25200-24000}{24000} \times 100$ = 5%

Since, the number of downloads of App C on Mon = 32000

And the number of downloads of App C on Tue = 57600

So, c = $\frac{57600-32000}{32000} \times 100$ = 80%

Required ratio = a: b: c = 25: 5: 80 = 5: 1: 16

2. Questions

Let the number of downloads of App C on Mon and Thu be '4x' and '5x' respectively.

So, the number of downloads of App B on Mon = 60% of [4x * (100/80)] = 3x

Since, the number of downloads of App B on Mon is equal to that on Tue and also 9000 less than that on Wed.

So, the number of downloads of App B on Tue = 3x

And the number of downloads of App B on Wed = (3x + 9000)

Since, the number of downloads of App B on Wed is 7000 less than that of App C on Thu.

So,

5x – (3x + 9000) = 7000

x = 8000

Since, the ratio of the number of downloads of App C on Wed to that on Fri is 3: 2.

And the number of downloads of App C on Wed:

90% of [(3x + 9000) * (100/110)] = 90% of [33000 * (100/110)] = 27000

So, the number of downloads of App C on Fri = 27000 * (2/3) = 18000

The number of downloads of App C on Mon = 4 * 8000 = 32000

The number of downloads of App C on Tue = 120% of [3x * (100/50)] = 57600

The number of downloads of App C on Wed = 27000

The number of downloads of App C on Thu = 5 * 8000 = 40000

The number of downloads of App C on Fri = 18000

Day	No. of downloads of App A	No. of downloads of App B	No. of downloads of App C
Mon	32000 * (100/80) = 40000	3 * 8000 = 24000	32000
Tue	57600 * (100/120) = 48000	3 * 8000 = 24000	57600
Wed	27000 * (100/90) = 30000	24000 + 9000 = 33000	27000
Thu	40000 * (100/80) = 50000	40% 50000 = 20000	40000
Fri	18000 * (100/50) = 36000	70% of 36000 = 25200	18000

Answer: B

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Day	The no. of male, who downloaded App B	The no. of females, who downloaded App B
Mon	24000 * (5/8) = 15000	24000 – 15000 = 9000
Tue	24000 * (1/3) = 8000	24000 – 8000 = 16000
Wed	33000 * (7/11) = 21000	33000 – 21000 = 12000
Thu	20000 * (1/4) = 5000	20000 – 5000 = 15000
Fri	25200 * (3/7) = 10800	25200 – 10800 = 14400

The total number of males, who downloaded App B from Mon to Fri:

15000 + 8000 + 21000 + 5000 + 10800 = 59800

The total number of females, who downloaded App B from Mon to Fri:

9000 + 16000 + 12000 + 15000 + 14400 = 66400

Required difference = 66400 – 59800 = 6600

3. Questions

Let the number of downloads of App C on Mon and Thu be '4x' and '5x' respectively.

So, the number of downloads of App B on Mon = 60% of [4x * (100/80)] = 3x

Since, the number of downloads of App B on Mon is equal to that on Tue and also 9000 less than that on Wed.

So, the number of downloads of App B on Tue = 3x

And the number of downloads of App B on Wed = (3x + 9000)

Since, the number of downloads of App B on Wed is 7000 less than that of App C on Thu.

So,

5x – (3x + 9000) = 7000

x = 8000

Since, the ratio of the number of downloads of App C on Wed to that on Fri is 3: 2.

And the number of downloads of App C on Wed:

90% of [(3x + 9000) * (100/110)] = 90% of [33000 * (100/110)] = 27000

So, the number of downloads of App C on Fri = 27000 * (2/3) = 18000

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The number of downloads of App C on Mon = 4 * 8000 = 32000

The number of downloads of App C on Tue = 120% of [3x * (100/50)] = 57600

The number of downloads of App C on Wed = 27000

The number of downloads of App C on Thu = 5 * 8000 = 40000

The number of downloads of App C on Fri = 18000

Day	No. of downloads of App A	No. of downloads of App B	No. of downloads of App C
Mon	32000 * (100/80) = 40000	3 * 8000 = 24000	32000
Tue	57600 * (100/120) = 48000	3 * 8000 = 24000	57600
Wed	27000 * (100/90) = 30000	24000 + 9000 = 33000	27000
Thu	40000 * (100/80) = 50000	40% 50000 = 20000	40000
Fri	18000 * (100/50) = 36000	70% of 36000 = 25200	18000

Answer: C

Since, the total number of downloads of App A till Wed:

40000 + 48000 + 30000 = 118000

The total number of downloads of App B till Wed:

24000 + 24000 + 33000 = 81000

And the total number of downloads of App C till Wed:

32000 + 57600 + 27000 = 116600

So, M = 118000 – 81000 = 37000

And, N = 116600 – 81000 = 35600

From P:

Difference of M and N = 37000 – 35600 = 1400

Since, 1400 is divisible by 7.

So, P is true.

From Q:

Average of M and N = (37000 + 35600)/2 = 36300

Since, 36300 is divisible by 3.

So, Q is true.

From R:

Value of (M/N) = 37000/35600 = 1.04 (approx.)

Since, the value of (M/N) is less than 1.2.

So, R is true.

Hence, all are true.

4. Questions

Let the number of downloads of App C on Mon and Thu be ‘4x’ and ‘5x’ respectively.

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So, the number of downloads of App B on Mon = 60% of [4x * (100/80)] = 3x

Since, the number of downloads of App B on Mon is equal to that on Tue and also 9000 less than that on Wed.

So, the number of downloads of App B on Tue = 3x

And the number of downloads of App B on Wed = (3x + 9000)

Since, the number of downloads of App B on Wed is 7000 less than that of App C on Thu.

So,

5x – (3x + 9000) = 7000

x = 8000

Since, the ratio of the number of downloads of App C on Wed to that on Fri is 3: 2.

And the number of downloads of App C on Wed:

90% of [(3x + 9000) * (100/110)] = 90% of [33000 * (100/110)] = 27000

So, the number of downloads of App C on Fri = 27000 * (2/3) = 18000

The number of downloads of App C on Mon = 4 * 8000 = 32000

The number of downloads of App C on Tue = 120% of [3x * (100/50)] = 57600

The number of downloads of App C on Wed = 27000

The number of downloads of App C on Thu = 5 * 8000 = 40000

The number of downloads of App C on Fri = 18000

Day	No. of downloads of App A	No. of downloads of App B	No. of downloads of App C
Mon	32000 * (100/80) = 40000	3 * 8000 = 24000	32000
Tue	57600 * (100/120) = 48000	3 * 8000 = 24000	57600
Wed	27000 * (100/90) = 30000	24000 + 9000 = 33000	27000
Thu	40000 * (100/80) = 50000	40% 50000 = 20000	40000
Fri	18000 * (100/50) = 36000	70% of 36000 = 25200	18000

Answer: A

Since, the total number of persons, who downloaded App A, on Mon = 40000

And the total number of persons, who downloaded App B as well as App A, on Mon = 24000

So, the total number of persons, who downloaded App A only, on Mon = 40000 – 24000 = 16000

Since, the total number of persons, who downloaded App C, on Mon = 32000

And the total number of persons, who downloaded App B as well as App C, on Mon = 24000

So, the total number of persons, who downloaded App C only, on Mon = 32000 – 24000 = 8000

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Required ratio = 16000:8000 = 2:1

5. Questions

Let the number of downloads of App C on Mon and Thu be '4x' and '5x' respectively.

So, the number of downloads of App B on Mon = 60% of [4x * (100/80)] = 3x

Since, the number of downloads of App B on Mon is equal to that on Tue and also 9000 less than that on Wed.

So, the number of downloads of App B on Tue = 3x

And the number of downloads of App B on Wed = (3x + 9000)

Since, the number of downloads of App B on Wed is 7000 less than that of App C on Thu.

So,

5x – (3x + 9000) = 7000

x = 8000

Since, the ratio of the number of downloads of App C on Wed to that on Fri is 3: 2.

And the number of downloads of App C on Wed:

90% of [(3x + 9000) * (100/110)] = 90% of [33000 * (100/110)] = 27000

So, the number of downloads of App C on Fri = 27000 * (2/3) = 18000

The number of downloads of App C on Mon = 4 * 8000 = 32000

The number of downloads of App C on Tue = 120% of [3x * (100/50)] = 57600

The number of downloads of App C on Wed = 27000

The number of downloads of App C on Thu = 5 * 8000 = 40000

The number of downloads of App C on Fri = 18000

Day	No. of downloads of App A	No. of downloads of App B	No. of downloads of App C
Mon	32000 * (100/80) = 40000	3 * 8000 = 24000	32000
Tue	57600 * (100/120) = 48000	3 * 8000 = 24000	57600
Wed	27000 * (100/90) = 30000	24000 + 9000 = 33000	27000
Thu	40000 * (100/80) = 50000	40% 50000 = 20000	40000
Fri	18000 * (100/50) = 36000	70% of 36000 = 25200	18000

Answer: B

the number of downloads of App A on Sat = 125% of 36000 = 45000

the number of downloads of App B on Sat = 115% of 25200 = 28980

the number of downloads of App C on Sat = 130% of 18000 = 23400

Required average = (45000 + 28980 + 23400)/3 = 32460

6. Questions

Let the total number of notes received by A, B, C, D and E be 12.5x, 10x, 25x, 37.5x and 15x respectively.

Also let the number of 200 rupee notes received by A, B, C, D and E respectively be 17y, 12y, 30y, 20y and 21y respectively.

Since, the number of 500 rupee notes received by B is 20.

So,

10x – 12y = 20

5x – 6y = 10 ----------(1)

Since, the total amount received by C is ₹55000.

So,

(25x – 30y) * 500 + 30y * 200 = 55000

125x – 150y + 60y = 550

25x – 18y = 110 ----------(2)

From equations (1) and (2):

25x – 3 * (5x – 10) = 110

25x – 15x + 30 = 110

x = 8

From equation (1):

40 – 6y = 10

y = 5

Worker	Total no. of notes received	No. of 200 - rupee notes received	No. of 500 - rupee notes received
A	12.5 * 8 = 100	17 * 5 = 85	100 – 85 = 15
B	10 * 8 = 80	12 * 5 = 60	80 – 60 = 20
C	25 * 8 = 200	30 * 5 = 150	200 – 150 = 50
D	37.5 * 8 = 300	20 * 5 = 100	300 – 100 = 200
E	15 * 8 = 120	21 * 5 = 105	120 – 105 = 15

Answer: C

Since, the number of 500 rupee notes received by A = 15

And the number of 500 rupee notes received by B = 20

So, M = (15/20) * 100 = 75%

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Since, the number of 500 rupee notes received by C = 50

And the number of 500 rupee notes received by D = 200

So, N = (50/200) * 100 = 25%

And the number of 500 rupee notes received by E = 15 = (M – 2N – 10)

7. Questions

Let the total number of notes received by A, B, C, D and E be 12.5x, 10x, 25x, 37.5x and 15x respectively.

Also let the number of 200 rupee notes received by A, B, C, D and E respectively be 17y, 12y, 30y, 20y and 21y respectively.

Since, the number of 500 rupee notes received by B is 20.

So,

10x – 12y = 20

5x – 6y = 10 ----------(1)

Since, the total amount received by C is ₹55000.

So,

(25x – 30y) * 500 + 30y * 200 = 55000

125x – 150y + 60y = 550

25x – 18y = 110 ----------(2)

From equations (1) and (2):

25x – 3 * (5x – 10) = 110

25x – 15x + 30 = 110

x = 8

From equation (1):

40 – 6y = 10

y = 5

Worker	Total no. of notes received	No. of 200 - rupee notes received	No. of 500 - rupee notes received
A	12.5 * 8 = 100	17 * 5 = 85	100 – 85 = 15
B	10 * 8 = 80	12 * 5 = 60	80 – 60 = 20
C	25 * 8 = 200	30 * 5 = 150	200 – 150 = 50
D	37.5 * 8 = 300	20 * 5 = 100	300 – 100 = 200
E	15 * 8 = 120	21 * 5 = 105	120 – 105 = 15

Answer: A

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Total amount paid to A = (15 * 500) + (85 * 200) = ₹24500

Total amount paid to B = (20 * 500) + (60 * 200) = ₹22000

Total amount paid to D = (200 * 500) + (100 * 200) = ₹120000

So, the ratio between the part of work completed by A, B and C respectively:

24500: 22000: 120000 = 49: 44: 240

8. Questions

Let the total number of notes received by A, B, C, D and E be 12.5x, 10x, 25x, 37.5x and 15x respectively.

Also let the number of 200 rupee notes received by A, B, C, D and E respectively be 17y, 12y, 30y, 20y and 21y respectively.

Since, the number of 500 rupee notes received by B is 20.

So,

10x – 12y = 20

5x – 6y = 10 ----------(1)

Since, the total amount received by C is ₹55000.

So,

(25x – 30y) * 500 + 30y * 200 = 55000

125x – 150y + 60y = 550

25x – 18y = 110 ----------(2)

From equations (1) and (2):

25x – 3 * (5x – 10) = 110

25x – 15x + 30 = 110

x = 8

From equation (1):

40 – 6y = 10

y = 5

Worker	Total no. of notes received	No. of 200 - rupee notes received	No. of 500 - rupee notes received
A	12.5 * 8 = 100	17 * 5 = 85	100 – 85 = 15
B	10 * 8 = 80	12 * 5 = 60	80 – 60 = 20
C	25 * 8 = 200	30 * 5 = 150	200 – 150 = 50
D	37.5 * 8 = 300	20 * 5 = 100	300 – 100 = 200
E	15 * 8 = 120	21 * 5 = 105	120 – 105 = 15

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Answer: C

Since, the average of the total number of notes received by D, E and F = 165

So, the total number of notes received by F = (3 * 165) – (300 + 120) = 75

Since, the average of the number of 200 rupee notes received by D, E and F = 80

So, the number of 200 rupee notes received by F = (3 * 80) – (100 + 105) = 35

Since, the total amount received by F = (75 – 35) * 500 + 35 * 200 = ₹27000

And the total amount received by D = (200 * 500) + (100 * 200) = ₹120000

So, the required percentage = (27000/120000) * 100 = 22.5%

9. Questions

Let the total number of notes received by A, B, C, D and E be 12.5x, 10x, 25x, 37.5x and 15x respectively.

Also let the number of 200 rupee notes received by A, B, C, D and E respectively be 17y, 12y, 30y, 20y and 21y respectively.

Since, the number of 500 rupee notes received by B is 20.

So,

10x – 12y = 20

5x – 6y = 10 ----------(1)

Since, the total amount received by C is ₹55000.

So,

(25x – 30y) * 500 + 30y * 200 = 55000

125x – 150y + 60y = 550

25x – 18y = 110 ----------(2)

From equations (1) and (2):

25x – 3 * (5x – 10) = 110

25x – 15x + 30 = 110

x = 8

From equation (1):

40 – 6y = 10

y = 5

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Worker	Total no. of notes received	No. of 200 - rupee notes received	No. of 500 - rupee notes received
A	12.5 * 8 = 100	17 * 5 = 85	100 – 85 = 15
B	10 * 8 = 80	12 * 5 = 60	80 – 60 = 20
C	25 * 8 = 200	30 * 5 = 150	200 – 150 = 50
D	37.5 * 8 = 300	20 * 5 = 100	300 – 100 = 200
E	15 * 8 = 120	21 * 5 = 105	120 – 105 = 15

Answer: E

The total number of 200 rupee notes received by each of the given workers:

85 + 60 + 150 + 100 + 105 = 500

So, the degree distribution of the number of 200 rupee notes received by E:

$$360 \times \frac{105}{500} = 75.6°$$

The total number of 500 rupee notes received by each of the given workers:

15 + 20 + 50 + 200 + 15 = 300

So, the degree distribution of the number of 500 rupee notes received by E:

$$360 \times \frac{15}{300} = 18°$$

Required ratio = 75.6:18 = 21:5

10. Questions

Since, A and B together completed 35% part of work in 2.1 hours.

So, the time, in which A and B together can finish the whole work = 2.1 * (100/35) = 6 hours

So,

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{6} - - - - - - - - (1)$$

Since, B and C together completed 40% part of work in 2.7 hours.

So, the time, in which B and C together can finish the whole work = 2.7 * (100/40) = 6.75 hours

So,

$$\frac{1}{B} + \frac{1}{C} = \frac{1}{6.75} = \frac{4}{27} - - - - - - - - (2)$$

Since, A and C together completed 25% part of work in 2.7 hours.

So, the time, in which A and C together can finish the whole work = 2.7 * (100/25) = 10.8 hours

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So,

$$\frac{1}{A}+\frac{1}{C}=\frac{1}{10.8}=\frac{5}{54}---------(3)$$

By equations (1) + (2) + (3):

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{1}{6}+\frac{4}{27}+\frac{5}{54}$$

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{9+8+5}{54}$$

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{22}{54}$$

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}=\frac{11}{54}--------(4)$$

By equation (4) – (1):

$$\frac{1}{C}=\frac{11}{54}-\frac{1}{6}$$

C = 27

By equation (4) – (2):

$$\frac{1}{A}=\frac{11}{54}-\frac{4}{27}$$

A = 18

By equation (4) – (3):

$$\frac{1}{B}=\frac{11}{54}-\frac{5}{54}$$

B = 9

The time, in which A alone can finish the work = 18 hours

The time, in which B alone can finish the work = 9 hours

The time, in which C alone can finish the work = 27 hours

Answer: A

Since, the ratio of A's efficiency to B's efficiency = 9: 18 = 1: 2

And A and B together completed 35% part.

So, the part of work completed by A while working with B = (35/3)%

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And the part of work completed by B while working with A = $35 \times \frac{2}{3} = \frac{70}{3}\%$

Since, the ratio of B's efficiency to C's efficiency = 27: 9 = 3: 1

And B and C together completed 40% part.

So, the part of work completed by B while working with C = 40 * (3/4) = 30%

And the part of work completed by C while working with B = 40 – 30 = 10%

Since, the ratio of A's efficiency to C's efficiency = 27: 18 = 3: 2

And A and C together completed 25% part.

So, the part of work completed by A while working with C = 25 * (3/5) = 15%

And the part of work completed by C while working with A = 25 – 15 = 10%

Since, the total part of work completed by A = (35/3) + 15 = (80/3)%

And the total part of work complete by C = 10 + 10 = 20%

So, the required ratio = (80/3): 20 = 4:3

11. Questions

Since, A and B together completed 35% part of work in 2.1 hours.

So, the time, in which A and B together can finish the whole work = 2.1 * (100/35) = 6 hours

So,

$$\frac{1}{A} + \frac{1}{B} = \frac{1}{6} - - - - - - - - - (1)$$

Since, B and C together completed 40% part of work in 2.7 hours.

So, the time, in which B and C together can finish the whole work = 2.7 * (100/40) = 6.75 hours

So,

$$\frac{1}{B} + \frac{1}{C} = \frac{1}{6.75} = \frac{4}{27} - - - - - - - - - (2)$$

Since, A and C together completed 25% part of work in 2.7 hours.

So, the time, in which A and C together can finish the whole work = 2.7 * (100/25) = 10.8 hours

So,

$$\frac{1}{A} + \frac{1}{C} = \frac{1}{10.8} = \frac{5}{54} - - - - - - - - - (3)$$

By equations (1) + (2) + (3):

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$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{1}{6}+\frac{4}{27}+\frac{5}{54}$$

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{9+8+5}{54}$$

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{22}{54}$$

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}=\frac{11}{54}--------(4)$$

By equation (4) – (1):

$$\frac{1}{C}=\frac{11}{54}-\frac{1}{6}$$

C = 27

By equation (4) – (2):

$$\frac{1}{A}=\frac{11}{54}-\frac{4}{27}$$

A = 18

By equation (4) – (3):

$$\frac{1}{B}=\frac{11}{54}-\frac{5}{54}$$

B = 9

The time, in which A alone can finish the work = 18 hours

The time, in which B alone can finish the work = 9 hours

The time, in which C alone can finish the work = 27 hours

Answer: E

If A, B and C had worked together:

The part of work completed by them together in 1 hour would have been:

$$\frac{1}{18}+\frac{1}{9}+\frac{1}{27}=\frac{3+6+2}{54}=\frac{11}{54}$$

So, the time, in which A, B and C together would have been completed the 55% part of work = 55% of (54/11) = 2.7 hours = 2 hours 42 minutes

12. Questions

Since, A and B together completed 35% part of work in 2.1 hours.

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So, the time, in which A and B together can finish the whole work = 2.1 * (100/35) = 6 hours

So,

$$\frac{1}{A}+\frac{1}{B}=\frac{1}{6}---------(1)$$

Since, B and C together completed 40% part of work in 2.7 hours.

So, the time, in which B and C together can finish the whole work = 2.7 * (100/40) = 6.75 hours

So,

$$\frac{1}{B}+\frac{1}{C}=\frac{1}{6.75}=\frac{4}{27}---------(2)$$

Since, A and C together completed 25% part of work in 2.7 hours.

So, the time, in which A and C together can finish the whole work = 2.7 * (100/25) = 10.8 hours

So,

$$\frac{1}{A}+\frac{1}{C}=\frac{1}{10.8}=\frac{5}{54}---------(3)$$

By equations (1) + (2) + (3):

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{1}{6}+\frac{4}{27}+\frac{5}{54}$$

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{9+8+5}{54}$$

$$\frac{2}{A}+\frac{2}{B}+\frac{2}{C}=\frac{22}{54}$$

$$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}=\frac{11}{54}--------(4)$$

By equation (4) – (1):

$$\frac{1}{C}=\frac{11}{54}-\frac{1}{6}$$

C = 27

By equation (4) – (2):

$$\frac{1}{A}=\frac{11}{54}-\frac{4}{27}$$

A = 18

By equation (4) – (3):

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$$\frac{1}{B} = \frac{11}{54} - \frac{5}{54}$$

B = 9

The time, in which A alone can finish the work = 18 hours

The time, in which B alone can finish the work = 9 hours

The time, in which C alone can finish the work = 27 hours

Answer: B

Let the total work = LCM of 18, 9 and 27 = 54 units

So, the ratio between the efficiencies of A, B and C:

(54/18): (54/9): (54/27) = 3: 6: 2

Since, the difference between the amount paid to A and B is ₹5250.

So, the difference between the amount paid to B and C:

$$5250 \times \frac{6-2}{6-3} = ₹7000$$

13. Questions

Let the total number of boys = 5x

So, the number of boys, who applied for post A only = 40% of 5x = 2x

Let the number of boys, who applied for both the posts A and B = y

So, the number of boys, who applied for post B only = (y + 9)

Since, the total number of boys and girls = 60

So, the total number of girls = (60 – 5x)

And the number of girls, who applied for post A only = 40% of (60 – 5x) = (24 – 2x)

Since, the number of boys, who applied for both the posts, is equal to the number of girls, who applied for both the posts.

So, the number of girls, who applied for both the posts = y

And the number of girls, who applied for post B only = (y + 3)

Since, the total number of boys, who applied for post A = 20

So,

2x + y = 20 ----------(1)

Since, the total number of girls, who applied for post A = 16

So,

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(24 – 2x) + y = 16

2x – y = 8 ----------(2)

From equations (1) and (2):

x = 7 and y = 6

The number of boys, who applied for post A only = 2 * 7 = 14

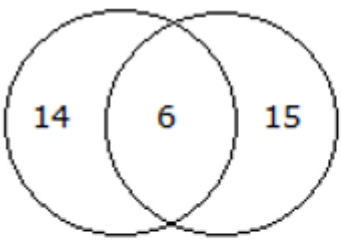
The number of boys, who applied for post B only = 6 + 9 = 15

The number of boys, who applied for both the posts A and B = 6

The number of girls, who applied for post A only = 24 – 2 * 7 = 10

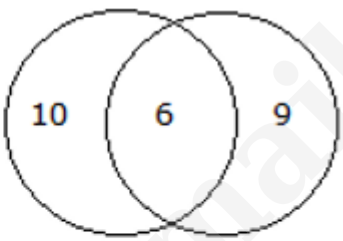
The number of girls, who applied for post B only = 6 + 3 = 9

The number of girls, who applied for both the posts A and B = 6



No. of boys, who applied for post A

No. of boys, who applied for post B



No. of girls, who applied for post A

No. of girls, who applied for post B

Answer: B

The total number of boys and girls together, who applied for post B:

(6 + 15) + (6 + 9) = 36

The total number of boys and girls together, who applied for both the posts A and B = 6 + 6 = 12

Required percentage = (12/36) * 100 = 33.33%

14. Questions

Let the total number of boys = 5x

So, the number of boys, who applied for post A only = 40% of 5x = 2x

Let the number of boys, who applied for both the posts A and B = y

So, the number of boys, who applied for post B only = (y + 9)

Since, the total number of boys and girls = 60

So, the total number of girls = (60 – 5x)

And the number of girls, who applied for post A only = 40% of (60 – 5x) = (24 – 2x)

Since, the number of boys, who applied for both the posts, is equal to the number of girls, who applied for

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both the posts.

So, the number of girls, who applied for both the posts = y

And the number of girls, who applied for post B only = (y + 3)

Since, the total number of boys, who applied for post A = 20

So,

2x + y = 20 ----------(1)

Since, the total number of girls, who applied for post A = 16

So,

(24 – 2x) + y = 16

2x – y = 8 ----------(2)

From equations (1) and (2):

x = 7 and y = 6

The number of boys, who applied for post A only = 2 * 7 = 14

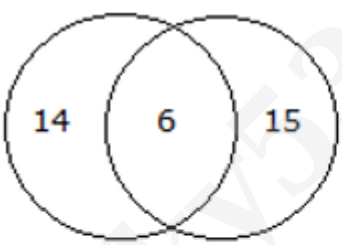
The number of boys, who applied for post B only = 6 + 9 = 15

The number of boys, who applied for both the posts A and B = 6

The number of girls, who applied for post A only = 24 – 2 * 7 = 10

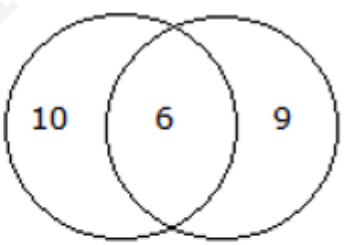
The number of girls, who applied for post B only = 6 + 3 = 9

The number of girls, who applied for both the posts A and B = 6



No. of boys, who applied for post A

No. of boys, who applied for post B



No. of girls, who applied for post A

No. of girls, who applied for post B

Answer: C

The total number of boys, who selected for post A:

[50% of 14 + 33(1/3)% of 6] = 7 + 2 = 9

The total number of boys, who selected for post B:

(60% of 15 + 50% of 6) = 9 + 3 = 12

Required difference = 12 – 9 = 3

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15. Questions

Let the total number of boys = 5x

So, the number of boys, who applied for post A only = 40% of 5x = 2x

Let the number of boys, who applied for both the posts A and B = y

So, the number of boys, who applied for post B only = (y + 9)

Since, the total number of boys and girls = 60

So, the total number of girls = (60 – 5x)

And the number of girls, who applied for post A only = 40% of (60 – 5x) = (24 – 2x)

Since, the number of boys, who applied for both the posts, is equal to the number of girls, who applied for both the posts.

So, the number of girls, who applied for both the posts = y

And the number of girls, who applied for post B only = (y + 3)

Since, the total number of boys, who applied for post A = 20

So,

2x + y = 20 ----------(1)

Since, the total number of girls, who applied for post A = 16

So,

(24 – 2x) + y = 16

2x – y = 8 ----------(2)

From equations (1) and (2):

x = 7 and y = 6

The number of boys, who applied for post A only = 2 * 7 = 14

The number of boys, who applied for post B only = 6 + 9 = 15

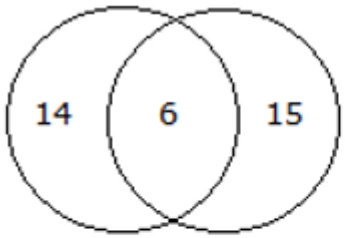
The number of boys, who applied for both the posts A and B = 6

The number of girls, who applied for post A only = 24 – 2 * 7 = 10

The number of girls, who applied for post B only = 6 + 3 = 9

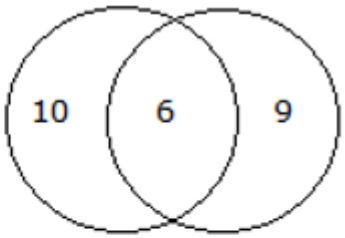
The number of girls, who applied for both the posts A and B = 6

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No. of boys, who applied for post A

No. of boys, who applied for post B



No. of girls, who applied for post A

No. of girls, who applied for post B

Answer: E

The total number of girls, who applied for post B = 6 + 9 = 15

The total number of girls selected for post B = 80% of 15 = 12

The number of such girls, who were applied for post B only and selected for post B = 66(2/3)% of 12 = 8

The total number of girls, who applied for post B only = 9

Required percentage = (8/9) * 100 = 88.89%

16. Questions

Answer: E

Let the speed of train A = 'x' m/s

So, the speed of train B = x + 27 * (5/18) = (x + 7.5) m/s

Let the length of train B = '10d' m

So, the length of train A = 110% of '10d' = '11d' m

Since, train B crosses train A in 84 seconds.

So,

$$(x + 7.5) - x = \frac{11d + 10d}{84}$$

d = 30

The length of train A = 11 * 30 = 330 m

Since, train A crosses a 120 m long tunnel in 20 seconds.

So,

$$x = \frac{330 + 120}{20}$$

x = 22.5 m/s

The speed of train A = 22.5 * (18/5) = 81 km/h

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17. Questions

Answer: A

Let the CP of the article be ₹100x.

So, the original SP of the article = 112% of 100x = ₹112x

If the article were sold at 20% discount on its cost price, the shopkeeper would have received ₹400 less than its original SP.

So,

80% of 100x = 112x – 400

80x = 112x – 400

x = 12.5

The CP of the article = 100 * 12.5 = ₹1250

The original SP of the article = 112% of 1250 = ₹1400

From P:

When the marked up amount is ₹350 and discount % is 12.5%.

The MP of the article will be = 1250 + 350 = ₹1600

And the SP of article will be = 87.5% of 1600 = ₹1400

So, the values in P can be filled in the blanks in the same order.

From Q:

When the marked up amount is ₹500 and discount % is 20%.

The MP of the article will be = 1250 + 500 = ₹1750

And the SP of article will be = 80% of 1750 = ₹1400

So, the values in Q can be filled in the blanks in the same order.

From R:

When the marked up amount is ₹750 and discount % is 25%.

The MP of the article will be = 1250 + 750 = ₹2000

And the SP of article will be = 75% of 2000 = ₹1500

So, the values in R can't be filled in the blanks.

Hence, the values in P and Q can be filled.

18. Questions

Answer: D

Let the time, in which A alone can complete the work = '3x' hours

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Since, B's efficiency is 150% of A's efficiency.

So, the time, in which B alone can complete the work = 3x * (100/150) = '2x' hours

Since, A and B together can complete the work in 3.6 hours.

So,

$$\frac{1}{3x}+\frac{1}{2x}=\frac{1}{3.6}$$

$$\frac{2+3}{6x}=\frac{1}{3.6}$$

x = 3

The time, in which A alone can complete the work = 3 * 3 = 9 hours

The time, in which B alone can complete the work = 2 * 3 = 6 hours

From P:

If B's efficiency is 60% of C's efficiency:

So, the time, in which C alone can complete the work = 60% of 6 = 3.6 hours

Since, the part of work completed by A and C together in 1 hour:

$$\frac{1}{9}+\frac{1}{3.6}=\frac{2+5}{18}=\frac{7}{18}$$

So, the time, in which A and C together will complete the work = 18/7 = 2.6 hours (approx.)

So, P can't be filled.

From Q:

If B's efficiency is 50% of C's efficiency:

So, the time, in which C alone can complete the work = 50% of 6 = 3 hours

Since, the part of work completed by A and C together in 1 hour:

$$\frac{1}{9}+\frac{1}{3}=\frac{1+3}{9}=\frac{4}{9}$$

So, the time, in which A and C together will complete the work = 9/4 = 2.25 hours

So, Q can't be filled.

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From R:

If B's efficiency is 75% of C's efficiency:

So, the time, in which C alone can complete the work = 75% of 6 = 4.5 hours

Since, the part of work completed by A and C together in 1 hour:

$$\frac{1}{9}+\frac{1}{4.5}=\frac{1+2}{9}=\frac{1}{3}$$

So, the time, in which A and C together will complete the work = 3 hours

So, R can't be filled.

Hence, none can be filled in the blank.

19. Questions

Answer: C

Since, the distance covered by boat B in upstream direction in 1.5 minutes:

12 * (1.5 * 60) = 1080 m

So, the downstream speed of boat B = 1080/60 = 18 m/s

And the speed of the stream = (18 – 12)/2 = 3 m/s

Since, the speed of boat A in still water is 8 m/s more than the speed of the stream.

So, the speed of boat A in still water = 3 + 8 = 11 m/s

The downstream speed of boat A = 11 + 3 = 14 m/s

And the time, in which boat A can cover 1.05 km in downstream direction:

$$\frac{1.05\times 1000}{14}=75 \text{ seconds}$$

20. Questions

Answer: D

Since, the difference between CI and SI in ₹1250 at R% rate after 2 years is ₹R.

So,

$$1250\times\left(\frac{R}{100}\right)^2=R$$

R = 8%

Since, the SI received on ₹(X – 450) at (R/2)% rate after 2 years is ₹104.

So,

$$\frac{(X-450)\times R\times 2}{100\times 2}=104$$

$$\frac{(X-450)\times 8\times 2}{100\times 2}=104$$

X = 1750

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Now, the interest received on ₹X at (R + 4)% rate after 1 year:

$$\frac{X \times (R+4)}{100} = \frac{1750 \times (8+4)}{100} = ₹210$$

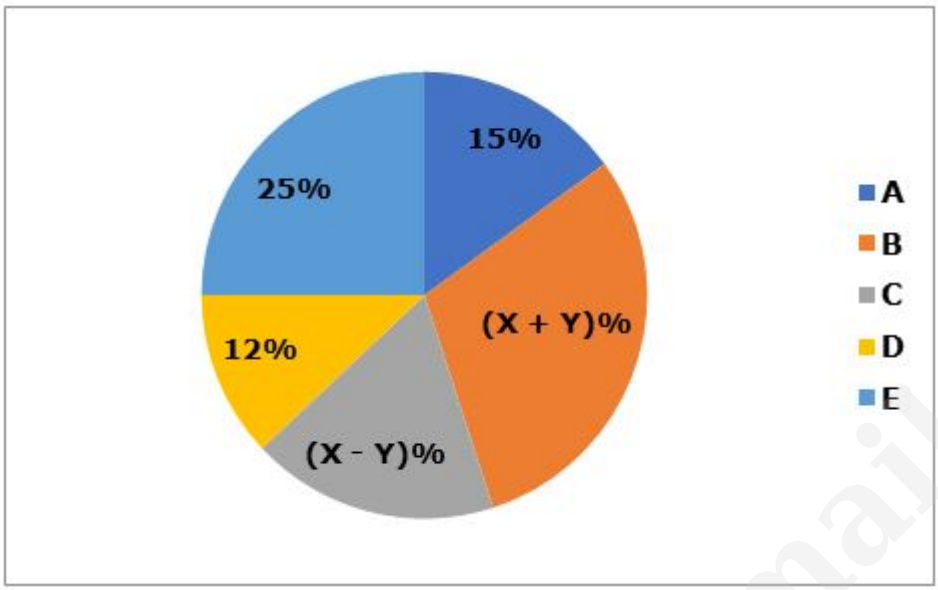
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1. Questions

Study the following data carefully and answer the questions:

The data given below is related to the sale (in ₹) of five shops A, B, C, D and E on Monday and Tuesday.

The pie chart given below shows the percentage distribution of sale of each shop on Monday.



The table given below shows the ratio of sale on Monday to that on Tuesday for each shop.

Shop	Ratio of sale on Monday to that on Tuesday
A	3: 2
B	6: 5
C	4: 5
D	12: 5
E	6: 5

Note:

1). B's sale on Monday is double of A's sale on Monday.

2). A's sale on Monday is ₹125XY.

If B's sale on Tuesday is ₹250 per article, C's sale on Tuesday is ₹300 per article and D's sale on Tuesday is ₹40 per article, then what is the ratio between the number of articles sold by shops B, C and D respectively on Tuesday?

a. 12:9:20

b. 4:3:5

c. 6:4:9

d. 30:21:50

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e. Can't be determined

2. Questions

If the ratio of A's sale on Monday to that on Wednesday is 6: 5 and the ratio of B's sale on Monday to that on Wednesday is 3: 2, then find the average of A's total sale on Tuesday and Wednesday together and B's total sale on Tuesday and Wednesday together?

a. ₹ 41400

b. ₹ 36300

c. ₹ 40500

d. ₹ 42300

e. ₹ 39900

3. Questions

If the sale of each shop on Tuesday is described in a circle, then what will be the ratio of the percentage distribution of C's sale on Monday to the degree distribution of C's sale on Tuesday?

a. 5:27

b. 9:37

c. 3:7

d. 12:47

e. 1:24

4. Questions

Table given below shows the percent increment/decrement in the sale of each shop from Tuesday to Wednesday:

Shop	% increment/decrement in the sale from Tuesday to Wednesday
A	+25%
B	-20%
C	-33(1/3)%
D	+300%
E	-24%

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By what per cent the total sale of all the five shops together is increased/decreased from Monday to Wednesday?

a. 12.5%

b. 20%

c. 11.11%

d. 10%

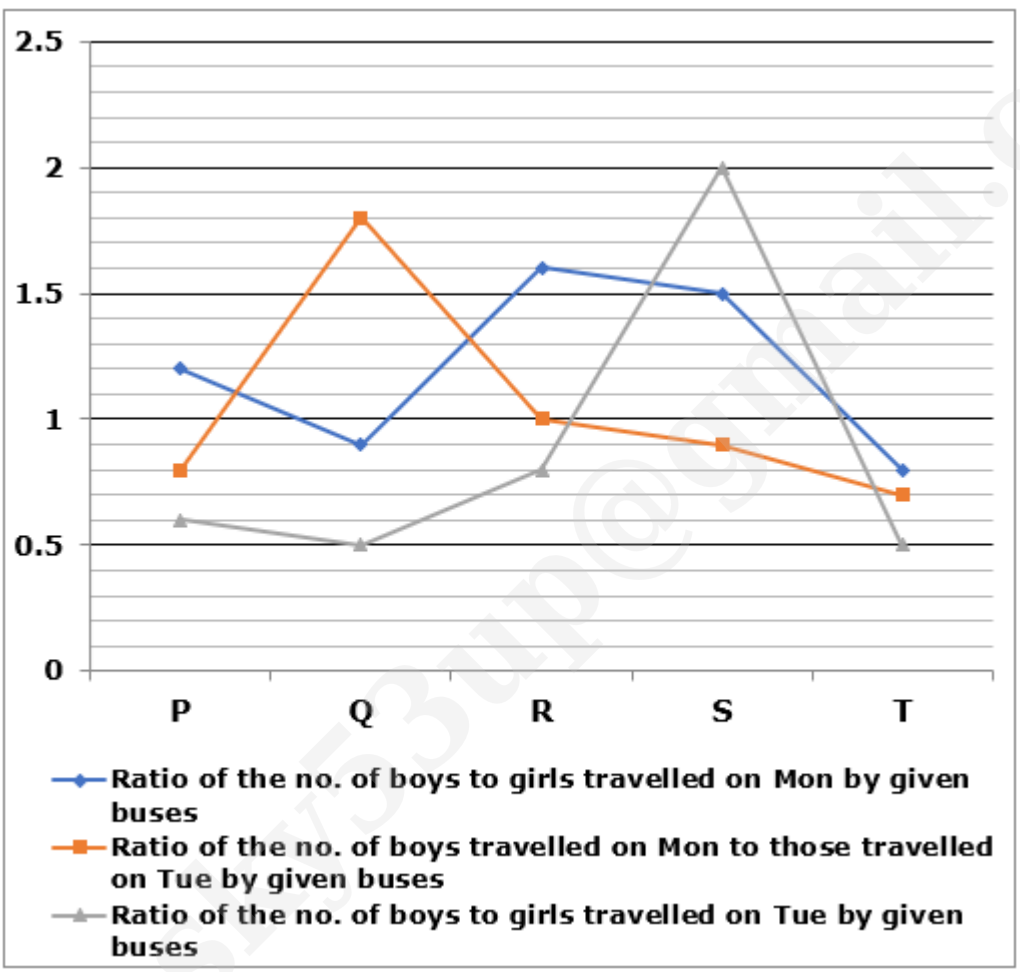
e. 16.67%

5. Questions

Study the following data carefully and answer the questions:

Data given below is related to the number of boys and girls travelled by 5 buses P, Q, R, S and T on Mon and Tue.

Line graph given below shows the following data:



Note:

1). On Mon, the number of boys travelled by bus Q is equal to those travelled by bus S and also 17 more than those travelled by bus T.

2). On Mon, the ratio of the number of boys travelled by bus P to those travelled by bus R is 6: 5.

3). On Mon, the average number of boys travelled by buses P, Q and S is 46.

4). On Mon, the number of girls travelled by bus Q is 10 more than those travelled by bus P.

If the ratio of the total number of boys and girls travelled by bus R on Mon and Tue together to those travelled by bus S on Mon and Tue together is M: N, then find the total number of girls travelled by all the five buses together on Tue?

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a. 6(M + N)

b. 5(M + N)

c. 8(M + N)

d. 2(M + N)

e. 4(M + N)

6. Questions

Which of the following statements is/are true?

A: Average of the total number of boys travelled by bus P on Mon and Tue together and the total number of girls travelled by bus P on Mon and Tue together is 134.

B: Ratio of the total number of boys and girls together travelled by bus S on Mon to those travelled by bus S on Tue is 1:1.

C: The total number of boys and girls together travelled by bus P on Mon is 55% of those travelled by bus P on Tue.

a. All are true

b. Only A and B

c. Only B

d. Only C

e. Only B and C

7. Questions

If the number of girls, travelled by each bus on Mon, is described in a circle, then what will be the difference between the degree distribution of the number of girls travelled by buses Q and T on Mon?

a. 20°

b. 45°

c. 30°

d. 25°

e. 36°

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8. Questions

The total number of boys and girls together travelled by the bus on Tue, in which the minimum number of boys travelled on Tue, is what per cent of the total number of boys and girls together travelled by the bus on Tue, in which the 2nd maximum number of girls travelled on Tue?

a. 62.5%

b. 75%

c. 80%

d. 66.67%

e. 60%

9. Questions

Study the following data carefully and answer the questions:

A person invested three difference amounts in three schemes A, B and C and rates of interest in schemes A, B and C is R%, 8% and 15% respectively. The amount invested in scheme A is ₹2500 less than that invested in scheme B and the ratio of the amount invested in scheme B to that invested in scheme C is 5: 8. The simple interest received from scheme A at R% rate after 2 years is ₹504 more than the compound interest received from scheme B at 8% rate after 2 years. The difference between compound interest and simple interest received from scheme C at 15% rate after 2 years is ₹540.

If the rates of interest in schemes A and C were interchanged, then what would have been the average of the simple interest received from scheme A after 3 years and the simple interest received from scheme C after 2 years?

a. ₹ 5522.5

b. ₹ 5962.5

c. ₹ 5252.5

d. ₹ 5852.5

e. ₹ 5692.5

10. Questions

If the simple interest received from scheme A after 1 year was ₹X, that received from scheme B after 1 year was ₹Y and that received from scheme C after 1 year was ₹Z and ₹[2X + 5Y + (5Z/3)] was invested in scheme D at (R/2)% rate of interest, then what would have been the compound interest received from scheme D after 2 years?

a. ₹ 2184

b. ₹ 1664

c. ₹ 1854

d. ₹ 1545

e. ₹ 2024

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11. Questions

Which of the following statements is/are true?

P: Difference between the total amount invested in schemes A and B together and the amount invested in

scheme C is ₹3500.

Q: Difference between compound interest and simple interest received from scheme B after 2 years would be ₹96.

R: Ratio of the amount invested in scheme A to that invested in scheme C is 25: 48 respectively.

a. Only P and Q

b. Only R

c. Only Q and R

d. Only P

e. All are true

12. Questions

If the amounts, invested in schemes A and B, were interchanged, then what would have been the difference between compound interest received from scheme A after 2 years and that received from scheme B after 2 years?

a. ₹1236

b. ₹1736

c. ₹1536

d. ₹1136

e. ₹1936

13. Questions

Study the following data carefully and answer the questions:

Data given below is related to the number of students in a class, who have their birthdays in the following months: Jan, Feb, Mar, Apr and May.

Number of students, who have their birthdays in Jan, is (A + B). The number of students, who have their birthdays in Feb, is 3A. The number of students, who have their birthdays in Mar, is 5(A – B). The number of students, who have their birthdays in Apr, is (x – 2) but more than the number of students, who have their birthdays in Feb. The number of students, who have their birthdays in May, is (x – A)/2, but not the multiple of 4.

The values of x, A and B can be determined from the equation given below:

$x^2 - (AB - 4)x + 10A(B + 1) = 0$

Note:

1). Both A and B are single digit consecutive natural numbers, where A is a multiple of 4, B is not less than 6 and also not multiple of 3.

If the ratio of boys to girls, who have their birthdays in Feb, is 5: 3 respectively, the ratio of boys to

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girls, who have their birthdays in Apr, is 3: 2 respectively and the ratio of boys to girls, who have their birthdays in May is 1: 1 respectively, then find the average number of girls, who have their birthdays in Feb, Apr and May?

a. 6

b. 11

c. 7

d. 10

e. 8

14. Questions

If the number of students, who have their birthdays in Apr, is (A + B – 2.5)% of the total number of students in the class, then find that what per cent of the total number of students in class have their birthdays in May?

a. [(A + B)/4]%

b. 3(A – B)%

c. [(A + B)/6]%

d. [(A + B)/2]%

e. 5(A – B)%

15. Questions

If the numbers of students, who have their birthdays in the given months, are described in a circle, then what will be the difference between the degree distribution of the number of students, who have their birthdays in Feb and Mar?

a. 76°

b. 85.5°

c. 72.2°

d. 91.2°

e. 57°

16. Questions

If the number of students, who have their birthdays in June, is 9 times of x% of (A + B), then find the average number of students, who have their birthdays in Apr, May and June?

a. 21

b. Can't be determined

c. 18

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d. 36

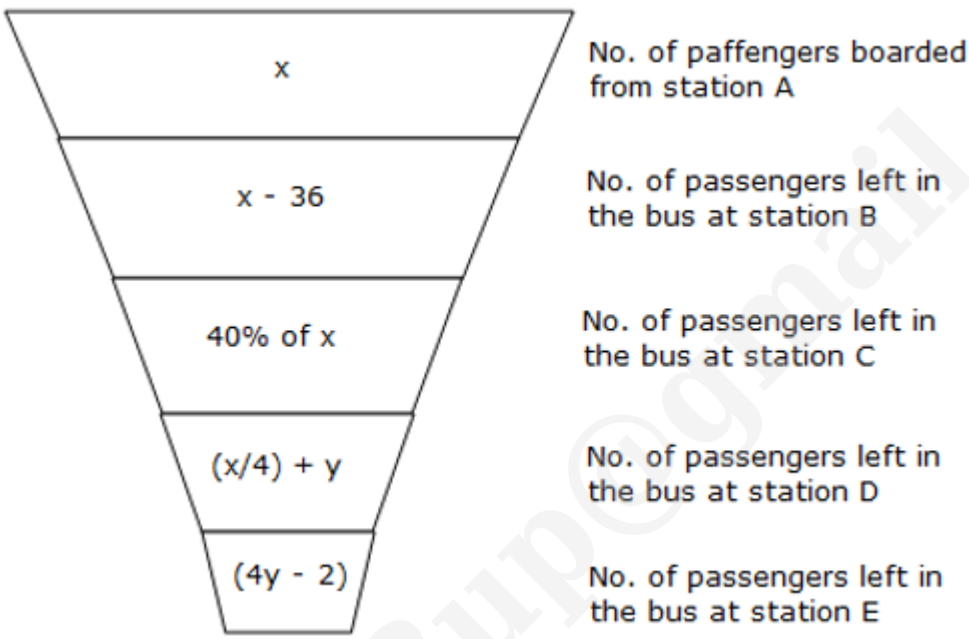
e. 25

17. Questions

Study the following data carefully and answer the questions:

A bus started from station A and went to station F via stations B, C, D and E respectively. Passengers boarded the bus only from station A. Some passengers got off the bus at station B, some got off the bus at station C, some got off the bus at station D, some got off the bus at station E and remaining got off the bus at station F.

Funnel graph given below shows the following data:



Notes:

1). 'No. of passengers left in the bus at a station' means 'the remaining no. of passengers in the bus after some passengers got off the bus at the same station'.

2). Ratio of the number of passengers, who got off the bus at station B, to those, who got off the bus at station C, is 3: 5 respectively.

3). The number of passengers, who left in the bus at station D, is 27.5% of those, who boarded from station A.

The number of males left in the bus at station B is 84 and the ratio of the number of males to females, who left in the bus at station B is M: N. If the ratio of the number of males to females, who got off the bus at station B is (M/3): (N/2), then find the number of males boarded from station A?

a. 96

b. 42

c. 84

d. 105

e. 63

18. Questions

Quantity I: Approximately what per cent of the number of passengers, who left in the bus at station B, got off the bus at station C?

Quantity II: The number of passengers, who left in the bus at station E, is what per cent of the number of passengers, who got off the bus at station E?

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I ≥ Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I = Quantity II or relation can't be established

19. Questions

The total number of passengers, who got off the bus at stations B and E together, is what per cent of those, who got off the bus at station C and D together?

a. 66%

b. 82.5%

c. 93.5%

d. 55%

e. 71.5%

20. Questions

If the number of passengers left in the bus at station B is (5P + Q + 7.5)% of those, who boarded from station A and the number of passengers who got off the bus at station F is (P + 0.75)% of those, who boarded from station A, then find the value of $(Q - P)^2$?

a. 484

b. 324

c. 576

d. 676

e. 400

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Explanations:

1. Questions

Since, B's sale on Monday is double of A's sale on Monday.

And per cent distribution of A's sale on Monday = 15%

So,

X + Y = 2 * 15

X + Y = 30 ---------(1)

Since,

15 + (X + Y) + (X – Y) + 12 + 25 = 100

15 + 30 + (X – Y) + 12 + 25 = 100

X – Y = 18 ----------(2)

By equations (1) + (2):

X + Y + X – Y = 30 + 18

X = 24

From equation (1):

24 + Y = 30

Y = 6

Since, A's sale on Monday is ₹125XY.

So, A's sale on Monday = 125 * 24 * 6 = ₹18000

And total sale of all the five shops together on Monday = 18000 * (100/15) = ₹120000

Since, B's sale on Monday is double of A's sale on Monday.

So, B's sale on Monday = 2 * 18000 = ₹36000

Since, per cent distribution of C's sale on Monday = 18%

So, C's sale on Monday = 18% of 120000 = ₹21600

D's sale on Monday = 12% of 120000 = ₹14400

And E's sale on Monday = 25% of 120000 = ₹30000

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Shop	Sale on Monday (in ₹)	Sale on Tuesday (in ₹)
A	18000	18000 * (2/3) = ₹12000
B	36000	36000 * (5/6) = ₹30000
C	21600	21600 * (5/4) = ₹27000
D	14400	14400 * (5/12) = ₹6000
E	30000	30000 * (5/6) = ₹25000

Answer: B

Since, the total sale of shop B on Tuesday = ₹30000

And per article sale of shop B on Tuesday = ₹250

So, the number of articles sold by shop B on Tuesday = 30000/250 = 120

Since, the total sale of shop C on Tuesday = ₹27000

And per article sale of shop C on Tuesday = ₹300

So, the number of articles sold by shop C on Tuesday = 27000/300 = 90

Since, the total sale of shop D on Tuesday = ₹6000

And per article sale of shop D on Tuesday = ₹40

So, the number of articles sold by shop D on Tuesday = 6000/40 = 150

Required ratio = 120:90:150 = 4:3:5

2. Questions

Since, B's sale on Monday is double of A's sale on Monday.

And per cent distribution of A's sale on Monday = 15%

So,

X + Y = 2 * 15

X + Y = 30 ---------(1)

Since,

15 + (X + Y) + (X – Y) + 12 + 25 = 100

15 + 30 + (X – Y) + 12 + 25 = 100

X – Y = 18 ----------(2)

By equations (1) + (2):

X + Y + X – Y = 30 + 18

X = 24

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From equation (1):

24 + Y = 30

Y = 6

Since, A's sale on Monday is ₹125XY.

So, A's sale on Monday = 125 * 24 * 6 = ₹18000

And total sale of all the five shops together on Monday = 18000 * (100/15) = ₹120000

Since, B's sale on Monday is double of A's sale on Monday.

So, B's sale on Monday = 2 * 18000 = ₹36000

Since, per cent distribution of C's sale on Monday = 18%

So, C's sale on Monday = 18% of 120000 = ₹21600

D's sale on Monday = 12% of 120000 = ₹14400

And E's sale on Monday = 25% of 120000 = ₹30000

Shop	Sale on Monday (in ₹)	Sale on Tuesday (in ₹)
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C	21600	21600 * (5/4) = ₹27000
D	14400	14400 * (5/12) = ₹6000
E	30000	30000 * (5/6) = ₹25000

Answer: C

Since, A's sale on Monday = ₹18000

So, A's sale on Wednesday = 18000 * (5/6) = ₹15000

Since, B's sale on Monday = ₹36000

So, B's sale on Wednesday = 36000 * (2/3) = ₹24000

Since, A's total sale on Tuesday and Wednesday together = 12000 + 15000 = ₹27000

And B's total sale on Tuesday and Wednesday together = 30000 + 24000 = ₹54000

So, required average = $\frac{27000+54000}{2}$ = ₹40500

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3. Questions

Since, B's sale on Monday is double of A's sale on Monday.

And per cent distribution of A's sale on Monday = 15%

So,

X + Y = 2 * 15

X + Y = 30 ---------(1)

Since,

15 + (X + Y) + (X – Y) + 12 + 25 = 100

15 + 30 + (X – Y) + 12 + 25 = 100

X – Y = 18 ----------(2)

By equations (1) + (2):

X + Y + X – Y = 30 + 18

X = 24

From equation (1):

24 + Y = 30

Y = 6

Since, A's sale on Monday is ₹125XY.

So, A's sale on Monday = 125 * 24 * 6 = ₹18000

And total sale of all the five shops together on Monday = 18000 * (100/15) = ₹120000

Since, B's sale on Monday is double of A's sale on Monday.

So, B's sale on Monday = 2 * 18000 = ₹36000

Since, per cent distribution of C's sale on Monday = 18%

So, C's sale on Monday = 18% of 120000 = ₹21600

D's sale on Monday = 12% of 120000 = ₹14400

And E's sale on Monday = 25% of 120000 = ₹30000

Shop	Sale on Monday (in ₹)	Sale on Tuesday (in ₹)
A	18000	18000 * (2/3) = ₹12000
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C	21600	21600 * (5/4) = ₹27000
D	14400	14400 * (5/12) = ₹6000
E	30000	30000 * (5/6) = ₹25000

Answer: A

Since, the total sale of each shop on Tuesday:

12000 + 30000 + 27000 + 6000 + 25000 = ₹100000

So, the degree distribution of C's sale on Tuesday:

$$360° \times \frac{27000}{100000} = 97.2°$$

Since, the percentage distribution of C's sale on Monday = 18%

So, the required ratio = 18:97.2 = 5:27

4. Questions

Since, B's sale on Monday is double of A's sale on Monday.

And per cent distribution of A's sale on Monday = 15%

So,

X + Y = 2 * 15

X + Y = 30 ---------(1)

Since,

15 + (X + Y) + (X – Y) + 12 + 25 = 100

15 + 30 + (X – Y) + 12 + 25 = 100

X – Y = 18 ----------(2)

By equations (1) + (2):

X + Y + X – Y = 30 + 18

X = 24

From equation (1):

24 + Y = 30

Y = 6

Since, A's sale on Monday is ₹125XY.

So, A's sale on Monday = 125 * 24 * 6 = ₹18000

And total sale of all the five shops together on Monday = 18000 * (100/15) = ₹120000

Since, B's sale on Monday is double of A's sale on Monday.

So, B's sale on Monday = 2 * 18000 = ₹36000

Since, per cent distribution of C's sale on Monday = 18%

So, C's sale on Monday = 18% of 120000 = ₹21600

D's sale on Monday = 12% of 120000 = ₹14400

And E's sale on Monday = 25% of 120000 = ₹30000

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Shop	Sale on Monday (in ₹)	Sale on Tuesday (in ₹)
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B	36000	36000 * (5/6) = ₹30000
C	21600	21600 * (5/4) = ₹27000
D	14400	14400 * (5/12) = ₹6000
E	30000	30000 * (5/6) = ₹25000

Answer: E

Since, the sale of shop A on Tuesday = ₹12000

So, the sale of shop A on Wednesday = 125% of 12000 = ₹15000

Since, the sale of shop B on Tuesday = ₹30000

So, the sale of shop B on Wednesday = 80% of 30000 = ₹24000

Since, the sale of shop C on Tuesday = ₹27000

So, the sale of shop C on Wednesday = 66(2/3)% of 27000 = ₹18000

Since, the sale of shop D on Tuesday = ₹6000

So, the sale of shop D on Wednesday = 400% of 6000 = ₹24000

Since, the sale of shop E on Tuesday = ₹25000

So, the sale of shop E on Wednesday = 76% of 25000 = ₹19000

Since, the total sale of all the 5 shops together on Wednesday:

15000 + 24000 + 18000 + 24000 + 19000 = ₹100000

And the total sale of all the 5 shops together on Monday = ₹120000

So, the required percentage = $\frac{120000-100000}{120000} \times 100 = 16.67\%$

5. Questions

Let the number of boys travelled by bus Q on Mon = The number of boys travelled by bus S on Mon = x

So, the number of boys travelled by bus T on Mon = (x – 17)

Let the number of boys travelled by buses P and R on Mon be 6y and 5y respectively.

The number of girls travelled by bus P on Mon = 6y/1.2 = 5y

The number of girls travelled by bus Q on Mon = x/0.9 = 10x/9

Since, the average number of boys travelled by buses P, Q and S on Mon is 46.

So,

6y + x + x = 3 * 46

3y + x = 69 ----------(1)

Since, the number of girls travelled by bus Q on Mon is 10 more than those travelled by bus P on Mon.

So,

(10x/9) – 5y = 10

10x – 45y = 90

2x – 9y = 18 ----------(2)

From equations (1) and (2):

2(69 – 3y) – 9y = 18

138 – 6y – 9y = 18

y = 8

From equation (1):

24 + x = 69

x = 45

Bus	No. of boys travelled on Mon	No. of girls travelled on Mon	No. of boys travelled on Tue	No. of girls travelled on Tue
P	6 * 8 = 48	48/1.2 = 40	48/0.8 = 60	60/0.6 = 100
Q	45	45/0.9 = 50	45/1.8 = 25	25/0.5 = 50
R	5 * 8 = 40	40/1.6 = 25	40/1.0 = 40	40/0.8 = 50
S	45	45/1.5 = 30	45/0.9 = 50	50/2.0 = 25
T	45 – 17 = 28	28/0.8 = 35	28/0.7 = 40	40/0.5 = 80

Answer: B

Since, the total number of boys and girls travelled by bus R on Mon and Tue together = (40 + 25) + (40 + 50) = 155

And the total number of boys and girls travelled by bus S on Mon and Tue together = (45 + 30) + (50 + 25) = 150

So, M:N = 155:150 = 31:30

And the total number of girls travelled by all the five buses together on Tue:

100 + 50 + 50 + 25 + 80 = 305 = 5(M + N)

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6. Questions

Let the number of boys travelled by bus Q on Mon = The number of boys travelled by bus S on Mon = x

So, the number of boys travelled by bus T on Mon = (x – 17)

Let the number of boys travelled by buses P and R on Mon be 6y and 5y respectively.

The number of girls travelled by bus P on Mon = 6y/1.2 = 5y

The number of girls travelled by bus Q on Mon = x/0.9 = 10x/9

Since, the average number of boys travelled by buses P, Q and S on Mon is 46.

So,

6y + x + x = 3 * 46

3y + x = 69 ----------(1)

Since, the number of girls travelled by bus Q on Mon is 10 more than those travelled by bus P on Mon.

So,

(10x/9) – 5y = 10

10x – 45y = 90

2x – 9y = 18 ----------(2)

From equations (1) and (2):

2(69 – 3y) – 9y = 18

138 – 6y – 9y = 18

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From equation (1):

24 + x = 69

x = 45

Bus	No. of boys travelled on Mon	No. of girls travelled on Mon	No. of boys travelled on Tue	No. of girls travelled on Tue
P	6 * 8 = 48	48/1.2 = 40	48/0.8 = 60	60/0.6 = 100
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R	5 * 8 = 40	40/1.6 = 25	40/1.0 = 40	40/0.8 = 50
S	45	45/1.5 = 30	45/0.9 = 50	50/2.0 = 25
T	45 – 17 = 28	28/0.8 = 35	28/0.7 = 40	40/0.5 = 80

Answer: E

From A:

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The total number of boys travelled by bus P on Mon and Tue together:

48 + 60 = 108

The total number of girls travelled by bus P on Mon and Tue together:

40 + 100 = 140

Required average = (108 + 140)/2 = 124

So, A is not true.

From B:

The total number of boys and girls together travelled by bus S on Mon:

45 + 30 = 75

The total number of boys and girls together travelled by bus S on Tue:

50 + 25 = 75

Required ratio = 75: 75 = 1: 1

So, B is true.

From C:

The total number of boys and girls together travelled by bus P on Mon:

48 + 40 = 88

The total number of boys and girls together travelled by bus P on Tue:

60 + 100 = 160

Required percentage = (88/160) * 100 = 55%

So, C is true.

Hence, only B and C are true.

7. Questions

Let the number of boys travelled by bus Q on Mon = The number of boys travelled by bus S on Mon = x

So, the number of boys travelled by bus T on Mon = (x – 17)

Let the number of boys travelled by buses P and R on Mon be 6y and 5y respectively.

The number of girls travelled by bus P on Mon = 6y/1.2 = 5y

The number of girls travelled by bus Q on Mon = x/0.9 = 10x/9

Since, the average number of boys travelled by buses P, Q and S on Mon is 46.

So,

6y + x + x = 3 * 46

3y + x = 69 ----------(1)

Since, the number of girls travelled by bus Q on Mon is 10 more than those travelled by bus P on Mon.

So,

(10x/9) – 5y = 10

10x – 45y = 90

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2x – 9y = 18 ----------(2)

From equations (1) and (2):

2(69 – 3y) – 9y = 18

138 – 6y – 9y = 18

y = 8

From equation (1):

24 + x = 69

x = 45

Bus	No. of boys travelled on Mon	No. of girls travelled on Mon	No. of boys travelled on Tue	No. of girls travelled on Tue
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S	45	45/1.5 = 30	45/0.9 = 50	50/2.0 = 25
T	45 – 17 = 28	28/0.8 = 35	28/0.7 = 40	40/0.5 = 80

Answer: C

The total number of girls travelled by all the five buses together on Mon:

40 + 50 + 25 + 30 + 35 = 180

The degree distribution of the number of girls travelled by bus Q on Mon:

360 * (50/180) = 100°

The degree distribution of the number of girls travelled by but T on Mon:

360 * (35/180) = 70°

Required difference = 100 – 70 = 30°

8. Questions

Let the number of boys travelled by bus Q on Mon = The number of boys travelled by bus S on Mon = x

So, the number of boys travelled by bus T on Mon = (x – 17)

Let the number of boys travelled by buses P and R on Mon be 6y and 5y respectively.

The number of girls travelled by bus P on Mon = 6y/1.2 = 5y

The number of girls travelled by bus Q on Mon = x/0.9 = 10x/9

Since, the average number of boys travelled by buses P, Q and S on Mon is 46.

So,

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$6y + x + x = 3 * 46$

$3y + x = 69$ ----------(1)

Since, the number of girls travelled by bus Q on Mon is 10 more than those travelled by bus P on Mon.

So,

$(10x/9) – 5y = 10$

$10x – 45y = 90$

$2x – 9y = 18$ ----------(2)

From equations (1) and (2):

$2(69 – 3y) – 9y = 18$

$138 – 6y – 9y = 18$

$y = 8$

From equation (1):

$24 + x = 69$

$x = 45$

Bus	No. of boys travelled on Mon	No. of girls travelled on Mon	No. of boys travelled on Tue	No. of girls travelled on Tue
P	6 * 8 = 48	48/1.2 = 40	48/0.8 = 60	60/0.6 = 100
Q	45	45/0.9 = 50	45/1.8 = 25	25/0.5 = 50
R	5 * 8 = 40	40/1.6 = 25	40/1.0 = 40	40/0.8 = 50
S	45	45/1.5 = 30	45/0.9 = 50	50/2.0 = 25
T	45 – 17 = 28	28/0.8 = 35	28/0.7 = 40	40/0.5 = 80

Answer: A

Since, the bus, in which the minimum number of boys travelled on Tue, is bus Q.

So, the total number of boys and girls together travelled by bus Q on Tue:

$25 + 50 = 75$

Since, the bus, in which the 2nd maximum number of girls travelled on Tue, is bus T.

So, the total number of boy and girls together travelled by bus T on Tue:

$40 + 80 = 120$

Required percentage = $(75/120) * 100 = 62.5\%$

9. Questions

Let the amount invested in schemes B and C be ₹5x and ₹8x respectively.

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So, the amount invested in scheme A = ₹(5x – 2500)

Since, the SI received from scheme A at R% rate after 2 years is ₹504 more than the CI received from B at 8% rate after 2 years:

$$\frac{(5x-2500)\times R\times 2}{100}-\left[5x\left(1+\frac{8}{100}\right)^2-5x\right]=504$$

$$\frac{10xR-5000R}{100}-\frac{729x}{125}+5x=504$$

$$\frac{xR-500R}{10}-\frac{729x}{125}+5x=504$$

$$\frac{25xR-12500R-1458x+1250x}{250}=504$$

25xR – 12500R – 208x = 126000 ----------(1)

Since, the difference between CI and SI received from scheme C at 15% rate after 2 years is ₹540.

So,

$$8x\times\left(\frac{15}{100}\right)^2=540$$

x = 3000

From equation (1):

25 * 3000 * R – 12500R – 208 * 3000 = 126000

75000R – 12500R – 624000 = 126000

R = 12%

The amount invested in scheme A = 5 * 3000 – 2500 = ₹12500

The rate of interest in scheme A = 12%

The amount invested in scheme B = 5 * 3000 = ₹15000

The rate of interest in scheme B = 8%

The amount invested in scheme C = 8 * 3000 = ₹24000

The rate of interest in scheme C = 15%

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Answer: E

The amount invested in scheme A = ₹12500

The amount invested in scheme C = ₹24000

The original rate of interest in scheme A = 12%

The original rate of interest in scheme C = 15%

If the rates of interest in schemes A and C were interchanged:

So, the SI received from scheme A after 3 years would have been:

$$\frac{12500 \times 15 \times 3}{100} = ₹5625$$

And the SI received from scheme C after 2 years would have been:

$$\frac{24000 \times 12 \times 2}{100} = ₹5760$$

Required average = $\frac{5625+5760}{2}$ = ₹ 5692.5

10. Questions

Let the amount invested in schemes B and C be ₹5x and ₹8x respectively.

So, the amount invested in scheme A = ₹(5x – 2500)

Since, the SI received from scheme A at R% rate after 2 years is ₹504 more than the CI received from B at 8% rate after 2 years:

$$\frac{(5x-2500) \times R \times 2}{100} - \left[5x\left(1+\frac{8}{100}\right)^2 - 5x\right] = 504$$

$$\frac{10xR-5000R}{100} - \frac{729x}{125} + 5x = 504$$

$$\frac{xR-500R}{10} - \frac{729x}{125} + 5x = 504$$

$$\frac{25xR-12500R-1458x+1250x}{250} = 504$$

25xR – 12500R – 208x = 126000 ----------(1)

Since, the difference between CI and SI received from scheme C at 15% rate after 2 years is ₹540.

So,

$$8x \times \left(\frac{15}{100}\right)^2 = 540$$

x = 3000

From equation (1):

25 * 3000 * R – 12500R – 208 * 3000 = 126000

75000R – 12500R – 624000 = 126000

R = 12%

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The amount invested in scheme A = 5 * 3000 – 2500 = ₹12500

The rate of interest in scheme A = 12%

The amount invested in scheme B = 5 * 3000 = ₹15000

The rate of interest in scheme B = 8%

The amount invested in scheme C = 8 * 3000 = ₹24000

The rate of interest in scheme C = 15%

Answer: C

The SI received by the person from scheme A after 1 year would be:

$$X = \frac{12500 \times 12}{100} = ₹1500$$

The SI received by the person from scheme B after 1 year would be:

$$Y = \frac{15000 \times 8}{100} = ₹1200$$

The SI received by the person from scheme C after 1 year would be:

$$Z = \frac{24000 \times 15}{100} = ₹3600$$

Now, the amount invested in scheme D:

[2X + 5Y + (5Z/3)] = [3000 + 6000 + 6000] = ₹15000

And the rate of interest in scheme D = (R/2)% = 12/2 = 6%

So, the CI received from scheme D after 2 years would be:

$$15000 \times \left(1 + \frac{6}{100}\right)^2 - 15000 = ₹1854$$

11. Questions

Let the amount invested in schemes B and C be ₹5x and ₹8x respectively.

So, the amount invested in scheme A = ₹(5x – 2500)

Since, the SI received from scheme A at R% rate after 2 years is ₹504 more than the CI received from B at 8% rate after 2 years:

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$$\frac{(5x-2500)\times R\times 2}{100}-\left[5x\left(1+\frac{8}{100}\right)^2-5x\right]=504$$

$$\frac{10xR-5000R}{100}-\frac{729x}{125}+5x=504$$

$$\frac{xR-500R}{10}-\frac{729x}{125}+5x=504$$

$$\frac{25xR-12500R-1458x+1250x}{250}=504$$

25xR – 12500R – 208x = 126000 ----------(1)

Since, the difference between CI and SI received from scheme C at 15% rate after 2 years is ₹540.

So,

$$8x\times\left(\frac{15}{100}\right)^2=540$$

x = 3000

From equation (1):

25 * 3000 * R – 12500R – 208 * 3000 = 126000

75000R – 12500R – 624000 = 126000

R = 12%

The amount invested in scheme A = 5 * 3000 – 2500 = ₹12500

The rate of interest in scheme A = 12%

The amount invested in scheme B = 5 * 3000 = ₹15000

The rate of interest in scheme B = 8%

The amount invested in scheme C = 8 * 3000 = ₹24000

The rate of interest in scheme C = 15%

Answer: E

From P:

Difference between the total amount invested in schemes A and B together and the amount invested in scheme C:

(12500 + 15000) – (24000) = ₹3500

So, P is true.

From Q:

The amount invested in scheme B = ₹15000

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Rate of interest in scheme B = 8%

So, the difference between CI and SI received from scheme B after 2 years would be:

$$15000 \times \left(\frac{8}{100}\right)^2 = ₹96$$

So, Q is true.

From R:

Ratio of the amount invested in scheme A to that invested in scheme C:

12500:24000 = 25:48

So, R is true.

Hence, all are true.

12. Questions

Let the amount invested in schemes B and C be ₹5x and ₹8x respectively.

So, the amount invested in scheme A = ₹(5x – 2500)

Since, the SI received from scheme A at R% rate after 2 years is ₹504 more than the CI received from B at 8% rate after 2 years:

$$\frac{(5x - 2500) \times R \times 2}{100} - \left[5x\left(1 + \frac{8}{100}\right)^2 - 5x\right] = 504$$

$$\frac{10xR - 5000R}{100} - \frac{729x}{125} + 5x = 504$$

$$\frac{xR - 500R}{10} - \frac{729x}{125} + 5x = 504$$

$$\frac{25xR - 12500R - 1458x + 1250x}{250} = 504$$

25xR – 12500R – 208x = 126000 ----------(1)

Since, the difference between CI and SI received from scheme C at 15% rate after 2 years is ₹540.

So,

$$8x \times \left(\frac{15}{100}\right)^2 = 540$$

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x = 3000

From equation (1):

25 * 3000 * R – 12500R – 208 * 3000 = 126000

75000R – 12500R – 624000 = 126000

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R = 12%

The amount invested in scheme A = 5 * 3000 – 2500 = ₹12500

The rate of interest in scheme A = 12%

The amount invested in scheme B = 5 * 3000 = ₹15000

The rate of interest in scheme B = 8%

The amount invested in scheme C = 8 * 3000 = ₹24000

The rate of interest in scheme C = 15%

Answer: B

The amount invested in scheme A = ₹12500

The rate of interest in scheme A = 12%

The amount invested in scheme B = ₹15000

The rate of interest in scheme B = 8%

If the amounts, invested in schemes A and B, were interchanged:

The CI received from scheme A after 2 years would be:

$$15000 \times \left(1 + \frac{12}{100}\right)^2 - 15000 = ₹3816$$

The CI received from scheme B after 2 years would be:

$$12500 \times \left(1 + \frac{8}{100}\right)^2 - 12500 = ₹2080$$

And the required difference would be = 3816 – 2080 = ₹1736

13. Questions

Since, both A and B are single digit consecutive natural numbers, where A is a multiple of 4.

So, A can be either 4 or 8.

Case I: If A is 4, then B will be either 3 or 5.

Case II: If A is 8, then B will be either 7 or 9.

Since, B is not less than 6.

So, case I can be neglected.

Since, B is not multiple of 3.

So, B will be 7 and A will be 8.

The number of students, who have their birthdays in Jan = (A + B) = 8 + 7 = 15

The number of students, who have their birthdays in Feb = 3 * 8 = 24

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The number of students, who have their birthdays in Mar = 5(A – B) = 5(8 – 7) = 5

Now, from the equation:

$x^2 – (AB – 4)x + 10A(B + 1) = 0$

$x^2 – (8 * 7 – 4)x + 10 * 8 * (7 + 1) = 0$

$x^2 – 52x + 640 = 0$

$x^2 – 20x – 32x + 640 = 0$

$x(x – 20) – 32(x – 20) = 0$

$x = 20, 32$

Since, the number of students, who have their birthdays in Apr, is (x – 2).

So, the number of students, who have their birthdays in Apr, will be either (20 – 2) = 18 or (32 – 2) = 30

Since, the number of students, who have their birthdays in Apr, is more than those, who have their birthdays in Feb.

So, the number of students, who have their birthdays in Apr = 30

Since, the number of students, who have their birthdays in May = (x – A)/2

So, the number of students, who have their birthdays in May, will be either (20 – 8)/2 = 6 or (32 – 8)/2 = 12

Since, the number of students, who have their birthdays in May, is not a multiple of 4.

So, the number of students, who have their birthdays in May = 6

Answer: E

Since, the number of students, who have their birthdays in Feb = 24

So, the number of girls, who have their birthdays in Feb = 24 * (3/8) = 9

Since, the number of students, who have their birthdays in Apr = 30

So, the number of girls, who have their birthdays in Apr = 30 * (2/5) = 12

Since, the number of students, who have their birthdays in May = 6

So, the number of girls, who have their birthdays in May = 6 * (1/2) = 3

Required average = $\frac{9+12+3}{3}$ = 8

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14. Questions

Since, both A and B are single digit consecutive natural numbers, where A is a multiple of 4.

So, A can be either 4 or 8.

Case I: If A is 4, then B will be either 3 or 5.

Case II: If A is 8, then B will be either 7 or 9.

Since, B is not less than 6.

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Since, B is not multiple of 3.

So, B will be 7 and A will be 8.

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The number of students, who have their birthdays in Feb = 3 * 8 = 24

The number of students, who have their birthdays in Mar = 5(A – B) = 5(8 – 7) = 5

Now, from the equation:

$x^2 – (AB – 4)x + 10A(B + 1) = 0$

$x^2 – (8 * 7 – 4)x + 10 * 8 * (7 + 1) = 0$

$x^2 – 52x + 640 = 0$

$x^2 – 20x – 32x + 640 = 0$

$x(x – 20) – 32(x – 20) = 0$

$x = 20, 32$

Since, the number of students, who have their birthdays in Apr, is (x – 2).

So, the number of students, who have their birthdays in Apr, will be either (20 – 2) = 18 or (32 – 2) = 30

Since, the number of students, who have their birthdays in Apr, is more than those, who have their birthdays in Feb.

So, the number of students, who have their birthdays in Apr = 30

Since, the number of students, who have their birthdays in May = (x – A)/2

So, the number of students, who have their birthdays in May, will be either (20 – 8)/2 = 6 or (32 – 8)/2 = 12

Since, the number of students, who have their birthdays in May, is not a multiple of 4.

So, the number of students, who have their birthdays in May = 6

Answer: C

Since, the number of students, who have their birthdays in Apr, is (A + B – 2.5)% of the total number of students in the class.

And the number of students, who have their birthdays in Apr = 30

So, the total number of students in the class:

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$$30 \times \frac{100}{A + B - 2.5} = 30 \times \frac{100}{8 + 7 - 2.5} = 240$$

Since, the number of students, who have their birthdays in May = 6

So, the required percentage = (6/240) * 100 = 2.5% = [(A + B)/6]%

15. Questions

Since, both A and B are single digit consecutive natural numbers, where A is a multiple of 4.

So, A can be either 4 or 8.

Case I: If A is 4, then B will be either 3 or 5.

Case II: If A is 8, then B will be either 7 or 9.

Since, B is not less than 6.

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Since, B is not multiple of 3.

So, B will be 7 and A will be 8.

The number of students, who have their birthdays in Jan = (A + B) = 8 + 7 = 15

The number of students, who have their birthdays in Feb = 3 * 8 = 24

The number of students, who have their birthdays in Mar = 5(A – B) = 5(8 – 7) = 5

Now, from the equation:

$x^2 - (AB - 4)x + 10A(B + 1) = 0$

$x^2 - (8 * 7 - 4)x + 10 * 8 * (7 + 1) = 0$

$x^2 - 52x + 640 = 0$

$x^2 - 20x - 32x + 640 = 0$

$x(x - 20) - 32(x - 20) = 0$

$x = 20, 32$

Since, the number of students, who have their birthdays in Apr, is (x – 2).

So, the number of students, who have their birthdays in Apr, will be either (20 – 2) = 18 or (32 – 2) = 30

Since, the number of students, who have their birthdays in Apr, is more than those, who have their birthdays in Feb.

So, the number of students, who have their birthdays in Apr = 30

Since, the number of students, who have their birthdays in May = (x – A)/2

So, the number of students, who have their birthdays in May, will be either (20 – 8)/2 = 6 or (32 – 8)/2 =

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12

Since, the number of students, who have their birthdays in May, is not a multiple of 4.

So, the number of students, who have their birthdays in May = 6

Answer: B

The total number of students, who have their birthdays in the given months:

15 + 24 + 5 + 30 + 6 = 80

So, the degree distribution of the number of students, who have their birthdays in

$$\text{Feb} = 360 \times \frac{24}{80} = 108°$$

And the degree distribution of the number of students, who have their birthdays in

$$\text{Mar} = 360 \times \frac{5}{80} = 22.5°$$

Required difference = 108° – 22.5° = 85.5°

16. Questions

Since, both A and B are single digit consecutive natural numbers, where A is a multiple of 4.

So, A can be either 4 or 8.

Case I: If A is 4, then B will be either 3 or 5.

Case II: If A is 8, then B will be either 7 or 9.

Since, B is not less than 6.

So, case I can be neglected.

Since, B is not multiple of 3.

So, B will be 7 and A will be 8.

The number of students, who have their birthdays in Jan = (A + B) = 8 + 7 = 15

The number of students, who have their birthdays in Feb = 3 * 8 = 24

The number of students, who have their birthdays in Mar = 5(A – B) = 5(8 – 7) = 5

Now, from the equation:

$x^2 – (AB – 4)x + 10A(B + 1) = 0$

$x^2 – (8 * 7 – 4)x + 10 * 8 * (7 + 1) = 0$

$x^2 – 52x + 640 = 0$

$x^2 – 20x – 32x + 640 = 0$

$x(x – 20) – 32(x – 20) = 0$

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x = 20, 32

Since, the number of students, who have their birthdays in Apr, is (x – 2).

So, the number of students, who have their birthdays in Apr, will be either (20 – 2) = 18 or (32 – 2) = 30

Since, the number of students, who have their birthdays in Apr, is more than those, who have their birthdays in Feb.

So, the number of students, who have their birthdays in Apr = 30

Since, the number of students, who have their birthdays in May = (x – A)/2

So, the number of students, who have their birthdays in May, will be either (20 – 8)/2 = 6 or (32 – 8)/2 = 12

Since, the number of students, who have their birthdays in May, is not a multiple of 4.

So, the number of students, who have their birthdays in May = 6

Answer: A

The number of students, who have their birthdays in June, is 9 times of x% of (A + B).

Case I: The number of students, who have their birthdays in June = 9 * 20% of (8 + 7) = 27

Case II: The number of students, who have their birthdays in June = 9 * 32% of (8 + 7) = 43.2

Since, case II is not possible.

So, the number of students, who have their birthdays in June = 27

Average number of students who have birthday in apr, may, june = (30+6+27)/3 = 63/3 = 21

17. Questions

The number of passengers, who got off the bus at station B = x – (x – 36) = 36

The number of passengers, who got off the bus at station C:

(x – 36) – 40% of x = (x – 36) – (2x/5) = (3x – 180)/5

Since, the ratio of the number of passengers, who got off the bus at station B, to those, who got off the bus at station C, is 3: 5.

So,

$$\frac{36}{\frac{(3x-180)}{5}} = \frac{3}{5}$$

$$\frac{12 \times 5}{3x-180} = \frac{1}{5}$$

3x – 180 = 300

x = 160

The number of passengers boarded from station A = 160

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The number of passengers left in the bus at station B = 160 – 36 = 124

The number of passengers left in the bus at station C = 40% of 160 = 64

Since, the number of passengers, who left in the bus at station D, is 27.5% of those, who boarded from station A.

So,

$$\frac{x}{4} + y = 27.5\% \text{ of } x$$

$$\frac{160}{4} + y = 27.5\% \text{ of } 160$$

40 + y = 44

y = 4

The number of passengers left in the bus at station D = 27.5% of 160 = 44

The number of passengers left in the bus at station E = 4y – 2 = 14

Answer: D

Since, the number of males left in the bus at station B = 84

And the number of passengers left in the bus at station B = 124

So, the number of females left in the bus at station B = 124 – 84 = 40

And, M: N = 84: 40 = 21: 10

Since, the ratio of the number of males to females, who got off the bus at station B = (M/3): (N/2) = (21/3): (10/2) = 7: 5

And the number of passengers, who got off the bus at station B = 36

So, the number of males, who got off the bus at station B = 36 * (7/12) = 21

And the number of males boarded from station A = 84 + 21 = 105

18. Questions

The number of passengers, who got off the bus at station B = x – (x – 36) = 36

The number of passengers, who got off the bus at station C:

(x – 36) – 40% of x = (x – 36) – (2x/5) = (3x – 180)/5

Since, the ratio of the number of passengers, who got off the bus at station B, to those, who got off the bus at station C, is 3: 5.

So,

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$$\frac{36}{\frac{(3x-180)}{5}} = \frac{3}{5}$$

$$\frac{12 \times 5}{3x-180} = \frac{1}{5}$$

3x – 180 = 300

x = 160

The number of passengers boarded from station A = 160

The number of passengers left in the bus at station B = 160 – 36 = 124

The number of passengers left in the bus at station C = 40% of 160 = 64

Since, the number of passengers, who left in the bus at station D, is 27.5% of those, who boarded from station A.

So,

$$\frac{x}{4} + y = 27.5\% \text{ of } x$$

$$\frac{160}{4} + y = 27.5\% \text{ of } 160$$

40 + y = 44

y = 4

The number of passengers left in the bus at station D = 27.5% of 160 = 44

The number of passengers left in the bus at station E = 4y – 2 = 14

Answer: A

Quantity I:

The number of passengers left in the bus at station B = 124

The number of passengers got off the bus at station C = 124 – 64 = 60

Required percentage = (60/124) * 100 = 48.4% (approx.)

Quantity II:

The number of passengers left in the bus at station E = 14

The number of passengers got off the bus at station E = 44 – 14 = 30

Required percentage = (14/30) * 100 = 46.67%

Hence, Quantity I > Quantity II

19. Questions

The number of passengers, who got off the bus at station B = x – (x – 36) = 36

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The number of passengers, who got off the bus at station C:

(x – 36) – 40% of x = (x – 36) – (2x/5) = (3x – 180)/5

Since, the ratio of the number of passengers, who got off the bus at station B, to those, who got off the bus at station C, is 3: 5.

So,

$$\frac{36}{\frac{(3x-180)}{5}} = \frac{3}{5}$$

$$\frac{12 \times 5}{3x-180} = \frac{1}{5}$$

3x – 180 = 300

x = 160

The number of passengers boarded from station A = 160

The number of passengers left in the bus at station B = 160 – 36 = 124

The number of passengers left in the bus at station C = 40% of 160 = 64

Since, the number of passengers, who left in the bus at station D, is 27.5% of those, who boarded from station A.

So,

$$\frac{x}{4} + y = 27.5\% \text{ of } x$$

$$\frac{160}{4} + y = 27.5\% \text{ of } 160$$

40 + y = 44

y = 4

The number of passengers left in the bus at station D = 27.5% of 160 = 44

The number of passengers left in the bus at station E = 4y – 2 = 14

Answer: B

The total number of passengers, who got off the bus at stations B and E together = (160 – 124) + (44 – 14) = 66

The total number of passengers, who got off the bus at stations C and D together = (124 – 64) + (64 – 44) = 80

Required percentage = (66/80) * 100 = 82.5%

20. Questions

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The number of passengers, who got off the bus at station B = x – (x – 36) = 36

The number of passengers, who got off the bus at station C:

(x – 36) – 40% of x = (x – 36) – (2x/5) = (3x – 180)/5

Since, the ratio of the number of passengers, who got off the bus at station B, to those, who got off the bus at station C, is 3: 5.

So,

$$\frac{36}{\frac{(3x-180)}{5}} = \frac{3}{5}$$

$$\frac{12 \times 5}{3x-180} = \frac{1}{5}$$

3x – 180 = 300

x = 160

The number of passengers boarded from station A = 160

The number of passengers left in the bus at station B = 160 – 36 = 124

The number of passengers left in the bus at station C = 40% of 160 = 64

Since, the number of passengers, who left in the bus at station D, is 27.5% of those, who boarded from station A.

So,

$$\frac{x}{4} + y = 27.5\% \text{ of } x$$

$$\frac{160}{4} + y = 27.5\% \text{ of } 160$$

40 + y = 44

y = 4

The number of passengers left in the bus at station D = 27.5% of 160 = 44

The number of passengers left in the bus at station E = 4y – 2 = 14

Answer: A

The number of passengers boarded from station A = 160

The number of passengers left in the bus at station B = 124

The number of passengers who got off the bus at station F = 14

So,

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$$(5P + Q + 7.5) = \frac{124}{160} \times 100$$

5P + Q + 7.5 = 77.5 ----------(1)

And,

$$(P + 0.75) = \frac{14}{160} \times 100$$

P = 8

From equation (1):

40 + Q + 7.5 = 77.5

Q = 30

So, the value of $(Q - P)^2 = (30 - 8)^2 = 484$

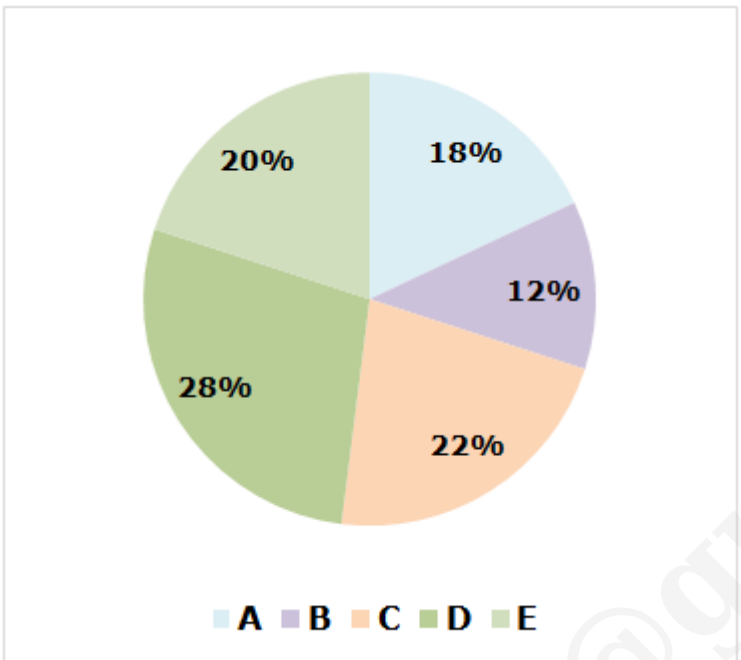
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1. Questions

Read the data carefully and answer the following questions.

The pie chart given below shows the percentage distribution of the difference between the number of males and females in five different offices [A, B, C, D and E].

Note: Only in offices A, D and E is the number of female employees more than the number of male employees in the same office.



Note: The females in office E is 55.55% of the total number of employees in office E. The number of female employees in office D is 25% more than that in office E. The number of males in office A is 10% less than that in office E. The number of female employees in offices B and C is the same and the number of male employees in office C is 174. The number of male employees in office E is 160.

The ratio of married males to females in Office A is 20:27. The ratio of unmarried males to females in Office A is 8:9. Find the difference between the difference of married and unmarried males and the difference of married and unmarried females.

a. 12

b. 14

c. 20

d. 16

e. 13

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2. Questions

Out of the total number of females in office D, (x+20)% use public transport, and the rest use private transport. Out of the total number of females in office E, x% use public transport, and the rest use private transport. The ratio of the number of females who use public transport in offices D and E is 15:8. If in office E, 60% of employees use public transport, then find the number of males

who use private transport in office E?

a. 22

b. 23

c. 29

d. 24

e. 21

3. Questions

The difference between males in offices D and E is 2x. The difference between females in offices A and C is 5y. The number of males in Office B has increased by 5x, and the number of females in Office B has increased by 7y. Now, the difference between males and females in office B is z. Find the value of 7z+32=?

a. 325

b. 305

c. 362

d. 398

e. None of these

4. Questions

The average age of males in Office A is 32 years, and the average age of females in Office A is 28 years. The average age of the male and female officers in Office C is 27 years and 36 years, respectively. The approximate average age of all the employees of offices A and C together is x years. Find the number of females in office B when it increased by x%.

a. 169

b. 159

c. 147

d. 151

e. None of these

5. Questions

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Out of the total number of males in office A, 25% lived in rented houses, and the rest lived in their own houses. Out of the total number of females in office A, _____% lived in rented houses, and the rest lived in their own houses. The total number of people living in their own house in office A is _____.

Find which of the following is true?

I). 40%, 195

II). 20%,204

III). 60%,224

a. Only I true

b. Only II true

c. All true

d. All false

e. Only I and II true

6. Questions

Read the following information carefully and answer the questions based on it.

There are three apartments – P, Q, and R in East Delhi, each apartment has a certain number of flats – 2 BHK and 3 BHK. Number of 3BHK flats in apartment Q is 50% more than that of 2BHK in Q, while the respective ratio of 2 BHK and 3 BHK flats in apartment P is 8: 7 respectively. The total number of flats in all apartments together is 405, out of which 120 are in apartment R. The number of 2 BHK flats in apartment P is 20% more than that of Q. The number of 3BHK flats in R is 40% less than that of 2BHK in R.

Find the average number of 2 BHK flats in all the three apartments together.

a. 66

b. 72

c. 68

d. 69

e. None of these

7. Questions

For Q, the average number of people living in each 2 BHK flat is 4, while in 3 BHK flat is 5, find the total number of people living in apartment Q.

a. 660

b. 690

c. 600

d. 540

e. None of these

8. Questions

The number of 2 BHK flats in apartment P and R together is approximately how much percentage more or less than that of the number of 3 BHK flats in apartment Q and R together?

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a. 12% more

b. 6% Less

c. 9% more

d. 10% Less

e. 7% more

9. Questions

If there is another department S, the total number of 2 BHK flats in P, Q and S together is 280, while the number of 3BHK flats in P and S together is 174. Find the respective ratio of the number of 3BHK to 2 BHK flats in S.

a. 3:4

b. 4:3

c. 2:3

d. 3:2

e. None of these

10. Questions

Find the difference between 2 BHK and 3 BHK flats in all departments together.

a. 12

b. 11

c. 19

d. 8

e. None of these

11. Questions

Study the following data carefully and answer the questions:

In five different bottles (B_1, B_2, B_3, B_4 and B_5), each contains mixture of honey and sugar syrup in different proportions. The table given below shows the quantity of honey and sugar syrup (in litres) in each bottle. Some of them are missed.

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Bottles	Quantity of Honey	Quantity of Sugar Syrup
B_1	120	75
B_2	9a	4a
B_3	3b	5b
B_4	8c	3c
B_5	5d	9d

Notes:

i). The total mixture of B_1 is 3 litres more than that of B_3

ii). The total mixture of B_1 is 3 litres less than that of B_4

Find the difference between the quantity of honey in bottle B_3 and the quantity of sugar syrup in bottle B_4.

a. 15 litres

b. 17 litres

c. 19 litres

d. 18 litres

e. 21 litres

12. Questions

If 65 litres mixture is taken out from Bottle B_2 and replaced with 17 litres of sugar syrup, the ratio of honey to sugar syrup in the bottle becomes 7:5. Find the initial quantity of the mixture in bottle B_2.

a. 111 litres

b. 134 litres

c. 156 litres

d. 178 litres

e. 147 litres

13. Questions

If 40% of the mixture from bottle B_1 is taken out and sold at the rate of pure honey. Now 30 litres of honey added to the mixture and then the resultant mixture is sold at a rate of 40% lower than the rate of pure honey. Find the profit % in this transaction?(The cost of sugar syrup is neglected)

a. 9%

b. 10.8%

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c. 11.11%

d. 12.12%

e. 13.5%

14. Questions

If z litres mixture are taken out from bottle B_4 and replaced with 44 litres of sugar syrup, resulting in a ratio of sugar syrup to honey becomes 5:6, then find the value of z.

a. 22

b. 44

c. 66

d. 88

e. None of these

15. Questions

In Bottle B_3, half of the mixture is taken out and poured into bottle B_5. Find the difference of the initial quantities of honey and sugar syrup in bottle B_5, if the ratio of the honey to sugar syrup in the resultant mixture in bottle B_5 is 17:30.

a. 65 litres

b. 70 litres

c. 75 litres

d. 78 litres

e. 80 litres

16. Questions

Study the following information carefully and answer the given questions.

Train A travels from Chennai to Mumbai, while train B travels from Mumbai to Chennai. Find the time taken by train A to cover a distance 25% more than the original distance.

I). Train B leaves Mumbai at 11.00 AM and train A leaves Chennai at 9.00 AM and they meet at 3.00 PM.

II). The total distance travelled by train B is 480km when trains A and B meet, and the ratio of the time taken by train A to train B to travel the total distance is 3:2.

III). The speed of train A is 33.33% less than that of train B, and they meet in 4hrs, after train B starts travelling. Train A travels for 't' hours before the train B starts traveling.

a. Data in statement III alone is sufficient to answer the question and data in statement II or statement I is not sufficient to answer the question

b. Data in statement II alone is sufficient to answer the question and data in statement I or statement III

is not sufficient to answer the question

c. All the given three statements are necessary to answer the question

d. Data in statement I alone is sufficient to answer the question and data in statement II or statement III is not sufficient to answer the question

e. Data in statement I and statement II is necessary to answer the question

17. Questions

Find the ratio of the number of shirts folded by a boy, a woman and a man in one hour.

I). 10 boys can fold 2000 shirts in a day and each boy is working 8 hours per day

II). 4 women can fold 1200 shirts in a day and each woman is working 10 hours per day

III). 8 men can fold 3200 shirts in a day and each man is working 8 hours a day

a. Data in statement III alone is sufficient to answer the question and data in statement II or statement I is not sufficient to answer the question

b. Data in statement II alone is sufficient to answer the question and data in statement I or statement III is not sufficient to answer the question

c. All the given three statements are necessary to answer the question

d. Data in statement I alone is sufficient to answer the question and data in statement II or statement III is not sufficient to answer the question

e. Data in statement I and statement II is necessary to answer the question

18. Questions

Hari's family have five members – his father, mother, wife Pooja and his son Sundhar. Find the average age of the family 4 years ago.

I). The ratio between the present ages of Hari and his mother is 7:13 and the difference between the present age of his father and mother is 10 years.

II). Before Sundhar was born, the average age of the family was 39 years.

III). The ratio between the present ages of Pooja and Hari's mother is 6:13, and Hari is 4 years elder than Pooja.

a. Data in all the given three statements are not sufficient to answer the question

b. Data in statement II alone is sufficient to answer the question and data in statement I or statement III is not sufficient to answer the question

c. All the given three statements are necessary to answer the question

d. Data in statement I alone is sufficient to answer the question and data in statement II or statement III is not sufficient to answer the question

e. Data in statement I and statement II is necessary to answer the question

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19. Questions

Find the difference volume of cuboid A and cuboid B.

Statement I: The ratio of breadth of cuboid A to cuboid B is 2:3, and the ratio of the height to length of cuboid A is 3:2. The total surface area of cuboid B is 158 cm^2 and the length and height of cuboid B is 5 cm and 8 cm respectively.

Statement II: The ratio of height, length and breadth of cuboid A and cuboid B is 3:4, 4:5 and 2:3 respectively. The total surface area of cuboid A is 88 cm^2.

a. Only Statement I alone.

b. Only Statement II alone.

c. Both Statements I and II together

d. Neither Statement I nor II is sufficient

e. Either Statement I or II.

20. Questions

A bag contains red, blue, green and yellow balls in the ratio 2:4:3:5 respectively. Find the total number of balls in the bag?

Statement I: The probability of picking a blue ball from the bag is 2/7.

Statement II: The probability of picking two red balls together from the bag is 1/63.

Statement III: The probability of picking one blue and one green ball together from the bag is 8/63.

Statement IV: The probability of picking one green and one yellow ball together from the bag is 10/63.

a. The data either in statement I alone or in statement IV alone is sufficient to answer the question.

b. The data either in statement I alone or in statement III alone is sufficient to answer the question.

c. The data either in statement II alone or in statement III alone is sufficient to answer the question.

d. The data either in statement II alone or in statement III or in statement IV alone is sufficient to answer the question.

e. The data in any 4 statements alone is sufficient to answer the question.

Explanations:

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1. Questions

Number of female employees in office E is =[160/44.44]*55.55=200

So, the difference between males and females in office E is 200-160=40

So, the total value of the pie chart = [40/20]*100=200.

The difference between males and females in office C is 200*22/100=44.

So, the females in office C = 174-44=130

Number of females in office D = 200*125/100=250

Number of males in office D = 250-[200*28/100]=194

Number of males in office A = 160*90/100=144

Number of females in office A = 144+200*18/100=180

Number of females in office B is 130

So, the number of males in office B is 130+200*12/100=154

Answer: C

Let, the married males and the married females in office A be 20x and 27x, respectively.

Unmarried males and unmarried females in office A be 8y and 9y, respectively.

So, 20x+8y= 144 and 27x+9y= 180

By solving the above equations, x=4 and y=8

So, married males and married females in office A is 20*4=80 and 27*4=108, respectively.

Unmarried males and unmarried females in office A is 8*8=64 and 9*8=72 respectively.

So, the required difference = [108-72] – [80-64]= 36-16=20

2. Questions

Number of female employees in office E is =[160/44.44]*55.55=200

So, the difference between males and females in office E is 200-160=40

So, the total value of the pie chart = [40/20]*100=200.

The difference between males and females in office C is 200*22/100=44.

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Number of females in office D = 200*125/100=250

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Number of males in office A = 160*90/100=144

Number of females in office A = 144+200*18/100=180

Number of females in office B is 130

So, the number of males in office B is 130+200*12/100=154

Answer: D

So, 250*[x+20]/200*x=15/8

Or, 10x+200=15x

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Or, x=200/5=40

So, the number of females who use public transport in office E is 200*40/100=80

The number of males who use public transport in office E is 360*60/100-80=136

The number of males who use private transport in office E is 160-136=24

3. Questions

Number of female employees in office E is =[160/44.44]*55.55=200

So, the difference between males and females in office E is 200-160=40

So, the total value of the pie chart = [40/20]*100=200.

The difference between males and females in office C is 200*22/100=44.

So, the females in office C = 174-44=130

Number of females in office D = 200*125/100=250

Number of males in office D = 250-[200*28/100]=194

Number of males in office A = 160*90/100=144

Number of females in office A = 144+200*18/100=180

Number of females in office B is 130

So, the number of males in office B is 130+200*12/100=154

Answer: B

The difference between males in offices D and E is 2x=194-160=34

So, x=34/2=17.

The difference between males and females in offices A and C is 5y=180-130=50, y=50/5=10.

The number of males in office B is increased by 5x, Now, males in office B is =154+5*17=154+85=239

The number of females in office B is increased by 7y. Now, females is =130+7*10=200 Now, the difference between males and females in office B is z.

So, z=239-200=39 so, the value of 7z+32=7*39+32=273+32=305

4. Questions

Number of female employees in office E is =[160/44.44]*55.55=200

So, the difference between males and females in office E is 200-160=40

So, the total value of the pie chart = [40/20]*100=200.

The difference between males and females in office C is 200*22/100=44.

So, the females in office C = 174-44=130

Number of females in office D = 200*125/100=250

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Number of males in office D = 250-[200*28/100]=194

Number of males in office A = 160*90/100=144

Number of females in office A = 144+200*18/100=180

Number of females in office B is 130

So, the number of males in office B is 130+200*12/100=154

Answer: A

So, the total age of all the employees of offices A and C is

=144*32+180*28+174*27+130*36

=19026

So, the average age is = [19026/628]=30.29=30 years

The required number of females is 130*130/100=169

5. Questions

Number of female employees in office E is =[160/44.44]*55.55=200

So, the difference between males and females in office E is 200-160=40

So, the total value of the pie chart = [40/20]*100=200.

The difference between males and females in office C is 200*22/100=44.

So, the females in office C = 174-44=130

Number of females in office D = 200*125/100=250

Number of males in office D = 250-[200*28/100]=194

Number of males in office A = 160*90/100=144

Number of females in office A = 144+200*18/100=180

Number of females in office B is 130

So, the number of males in office B is 130+200*12/100=154

Answer: D

For the value I,

The total number of people who lived in their own house

= 144*75/100 + 180*60/100=216

So, it's false.

Similarly, we can check other values as well.

6. Questions

Total number of flats in all apartments = 405

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Number of flats in R = 120

Respective ratio of 2BHK to 3 BHK flats in R = 5: 3

So, 2BHK flats in R = 5/8 x 120 = 75

Number of 3BHK flats in R = 3/8 x 120 = 45

Respective ratio of 2 BHK to 3 BHK flats in P = 8: 7

Respective ratio of 2 BHK to 3 BHK flats in Q = 2: 3

Respective ratio of 2 BHK flats in P to Q = 6: 5

By concept of the ratio

Respective ratio of 2 BHK to 3 BHK flats in P = [8: 7] x 3 = 24: 21

Respective ratio of 2 BHK to 3 BHK flats in Q = [2: 3] x 2 x 5 = 20: 30

Now,

= (24 + 21 + 20 + 30) units = 405 – 120 = 285

= 95 units = 285

So, 1 unit = 3

Now we can find all the data related to all given apartments.

Flats	P	Q	R	Total
2 BHK	72	60	75	207
3 BHK	63	90	45	198
Total	135	150	120	405

Answer: D

According to the question,

Average number of 2 BHK flats in all three apartments together = 207/3 = 69

Hence answer is option D

7. Questions

Total number of flats in all apartments = 405

Number of flats in R = 120

Respective ratio of 2BHK to 3 BHK flats in R = 5: 3

So, 2BHK flats in R = 5/8 x 120 = 75

Number of 3BHK flats in R = 3/8 x 120 = 45

Respective ratio of 2 BHK to 3 BHK flats in P = 8: 7

Respective ratio of 2 BHK to 3 BHK flats in Q = 2: 3

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Respective ratio of 2 BHK flats in P to Q = 6: 5

By concept of the ratio

Respective ratio of 2 BHK to 3 BHK flats in P = [8: 7] x 3 = 24: 21

Respective ratio of 2 BHK to 3 BHK flats in Q = [2: 3] x 2 x 5 = 20: 30

Now,

= (24 + 21 + 20 + 30) units = 405 – 120 = 285

= 95 units = 285

So, 1 unit = 3

Now we can find all the data related to all given apartments.

Flats	P	Q	R	Total
2 BHK	72	60	75	207
3 BHK	63	90	45	198
Total	135	150	120	405

Answer: B

For Q,

Total number of people in 2 BHK = 60 x 4 = 240

Total number of people in 3 BHK = 90 x 5 = 450

Required sum = 240 + 450 = 690

Hence answer is option B

8. Questions

Total number of flats in all apartments = 405

Number of flats in R = 120

Respective ratio of 2BHK to 3 BHK flats in R = 5: 3

So, 2BHK flats in R = 5/8 x 120 = 75

Number of 3BHK flats in R = 3/8 x 120 = 45

Respective ratio of 2 BHK to 3 BHK flats in P = 8: 7

Respective ratio of 2 BHK to 3 BHK flats in Q = 2: 3

Respective ratio of 2 BHK flats in P to Q = 6: 5

By concept of the ratio

Respective ratio of 2 BHK to 3 BHK flats in P = [8: 7] x 3 = 24: 21

Respective ratio of 2 BHK to 3 BHK flats in Q = [2: 3] x 2 x 5 = 20: 30

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Now,

= (24 + 21 + 20 + 30) units = 405 – 120 = 285

= 95 units = 285

So, 1 unit = 3

Now we can find all the data related to all given apartments.

Flats	P	Q	R	Total
2 BHK	72	60	75	207
3 BHK	63	90	45	198
Total	135	150	120	405

Answer: C

Number of 2 BHK flats in apartment P and R together = 72 + 75 = 147

Number of 2 BHK flats in apartment Q and R together = 90 + 45 = 135

Required % change = (147 – 135)/135 x 100 = 9% (approx)

Hence answer is option C

9. Questions

Total number of flats in all apartments = 405

Number of flats in R = 120

Respective ratio of 2BHK to 3 BHK flats in R = 5: 3

So, 2BHK flats in R = 5/8 x 120 = 75

Number of 3BHK flats in R = 3/8 x 120 = 45

Respective ratio of 2 BHK to 3 BHK flats in P = 8: 7

Respective ratio of 2 BHK to 3 BHK flats in Q = 2: 3

Respective ratio of 2 BHK flats in P to Q = 6: 5

By concept of the ratio

Respective ratio of 2 BHK to 3 BHK flats in P = [8: 7] x 3 = 24: 21

Respective ratio of 2 BHK to 3 BHK flats in Q = [2: 3] x 2 x 5 = 20: 30

Now,

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= (24 + 21 + 20 + 30) units = 405 – 120 = 285

= 95 units = 285

So, 1 unit = 3

Now we can find all the data related to all given apartments.

Flats	P	Q	R	Total
2 BHK	72	60	75	207
3 BHK	63	90	45	198
Total	135	150	120	405

Answer: A

Number of 2 BHK flats in S = 280 – 72 – 60 = 148

Number of 3 BHK flats in S = 174 – 63 = 111

Required ratio = 111:148 = 3:4

Hence answer is option A

10. Questions

Total number of flats in all apartments = 405

Number of flats in R = 120

Respective ratio of 2BHK to 3 BHK flats in R = 5: 3

So, 2BHK flats in R = 5/8 x 120 = 75

Number of 3BHK flats in R = 3/8 x 120 = 45

Respective ratio of 2 BHK to 3 BHK flats in P = 8: 7

Respective ratio of 2 BHK to 3 BHK flats in Q = 2: 3

Respective ratio of 2 BHK flats in P to Q = 6: 5

By concept of the ratio

Respective ratio of 2 BHK to 3 BHK flats in P = [8: 7] x 3 = 24: 21

Respective ratio of 2 BHK to 3 BHK flats in Q = [2: 3] x 2 x 5 = 20: 30

Now,

= (24 + 21 + 20 + 30) units = 405 – 120 = 285

= 95 units = 285

So, 1 unit = 3

Now we can find all the data related to all given apartments.

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Flats	P	Q	R	Total
2 BHK	72	60	75	207
3 BHK	63	90	45	198
Total	135	150	120	405

Answer: E

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Total number of 2 BHK flats = 207

Total number of 3 BHK flats = 198

Required difference = 207 – 198 = 9

Hence answer is option E

11. Questions

The total mixture of B_1 = 120+75 = 195

The total mixture of B_3 = 195 – 3 = 192

The total mixture of B_4 = 195+3 = 198

Answer: D

In bottle B_3,

The quantity of honey = 192*(3/8) =72

The quantity of sugar syrup =192*(5/8) =120

In bottle B_4,

The quantity of honey = 198*(8/11) =144

The quantity of sugar syrup = 198*(3/11) = 54

Required difference = 72 – 54 = 18 litres

12. Questions

The total mixture of B_1 = 120+75 = 195

The total mixture of B_3 = 195 – 3 = 192

The total mixture of B_4 = 195+3 = 198

Answer: C

[9a – 65*(9/13)]/ [4a – 65*(4/13) + 17] = 7/5

(9a – 45)*5 = (4a – 3)*7

45a – 28a = 225 -21

17a = 204

a = 12

Initial quantity of bottle B_2 = 13a = 13*12 = 156 litres

13. Questions

The total mixture of B_1 = 120+75 = 195

The total mixture of B_3 = 195 – 3 = 192

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The total mixture of B_4 = 195+3 = 198

Answer: B

Let the cost price of 1 litre pure honey = Rs. 100

Total cost price of the mixture = 100*(120+30) = Rs. 15000

Selling price of 40% mixture = 100*(120+75)*40% = Rs. 7800

Selling price of resultant mixture = 60% * 100*(120+75-78+30) = Rs. 8820

Profit % = [(7800+8820 - 15000)/15000]*100 = 10.8%

14. Questions

The total mixture of B_1 = 120+75 = 195

The total mixture of B_3 = 195 – 3 = 192

The total mixture of B_4 = 195+3 = 198

Answer: C

[144 – (8z/11)]/ [54 – (3z/11) + 44] = 6/5

[144 – (8z/11)]*5 = [98 – (3z/11)]*6

7920 – 40z = 6468 – 18z

22z = 1452

z = 66 litres

15. Questions

The total mixture of B_1 = 120+75 = 195

The total mixture of B_3 = 195 – 3 = 192

The total mixture of B_4 = 195+3 = 198

Answer: E

[5d + (72/2)]/ [9d + (120/2)] = 17/30

30*[5d + 36] = 17*[9d + 60]

150d + 1080 = 153d + 1020

3d = 60

d = 20

Required difference = 9d – 5d = 4d = 4*20 = 80 litres

16. Questions

Answer: E

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By combining statement I and II

Total distance = 480

Given time ratio A:B = 3:2

So speed of train A is '2x' and speed of train B is '3x'

Let the train A travelled a distance be 4x before the train B started to travel,

(480 – 4x) / 5x = 4

x = 20

Speed of train A = 40km/hr

Required answer = 480*(125/100)*40= 15h

17. Questions

Answer: C

Combining I, II and III,

No. of shirts fold by a boy in one hour = 2000/(10*8) = 25

No. of shirts fold by a woman in one hour = 1200 / (4*10) = 30

No. of shirts fold by a man in one hour = 3200 / (8 * 8) = 50

Boy: woman: man = 25:30:50 = 5:6:10

18. Questions

Answer: A

Cannot find the average age of a family because insufficient information about the age of Sundhar and also we don't know that his father is the eldest person or his mother is the eldest person in the family.

19. Questions

Answer: C

Statement I –

Let Breadth of cuboid B is 3U

So, 3U * 5 + 5 * 8 + 8 * 3U = 158/2 = 79

Or, 39U = 79 – 40

Or, U = 1

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Length, breadth, height of cuboid B is 5, 3, 8 respectively and breadth of cuboid A is 2 and height and length of cuboid A is 3v and 2v respectively.

So, we can calculate the volume of cuboid B i.e., 5 * 3 * 8 = 120 but cannot calculate the volume of cuboid A.

Statement II –

Total surface area of cuboid A is 88 cm^2. But we cannot calculate individual volume from it.

From both statements,

1st statement we get, breadth of cuboid A is 2 and height and length of cuboid A is 3v and 2v respectively

So, we can say, $2 * (2 * 3v + 3v * 2v + 2v * 2) = 88\ cm^2$

From this we get v = 2.

So, volume cuboid A = $(2 * 2) * 2 * (3 * 2) = 48\ cm^3$

Required difference = $120 - 48 = 72\ cm^3$

So, using two statements we get the answer.

20. Questions

Answer: D

Let the total no. of balls = 14x

Statement I,

4x/14x = 2/7

2/7 = 2/7

Statement I alone is not sufficient to answer the question.

Statement II,

$^{2x}C_2/^{14x}C_2 = 1/63$

$[(2x) * (2x - 1)]/ [(14x) * (14x - 1)] = 1/63$

$[4x^2 - 2x] * 63 = 196x^2 - 14x$

$252x^2 - 126x = 196x^2 - 14x$

$56x^2 - 112x = 0$

$56x (x - 2) = 0$

x = 0, x = 2

Statement II alone is sufficient to answer the question.

Similarly, we can calculate other statements.

Hint: minimum 2 balls selection of probability can the find the values.

Hence, the correct answer is option d.

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1. Questions

Study the following data carefully and answer the questions:

A farmer had three different shaped fields: rectangular field, square field and circular field and the area (in meter^2) of each field is different.

Ratio of the length of rectangular field to the side of square field is 3: 2 and the radius of circular field is 4 meters more than the breadth of rectangular field. Perimeter of the rectangular field is 56 meters. The time taken by the farmer to complete one round (at circumference) of the circular field is 10 seconds more than the time taken by him to complete one round (at perimeter) of the square field and the farmer runs at the speed of 14.4 km/h.

If the average of the area of the rectangular field and the area of the square field is (M + N) m^2 and the ratio of M to N is 20: 7, then find the average of the area of square field and the area of the circular field?

a. [10N – (M/4)] m^2

b. (2M + 5N) m^2

c. (3M + N) m^2

d. (3N + M) m^2

e. (5N + M + 50) m^2

2. Questions

If the cost of fencing the sides of rectangular field and square field is Rs. 125 per meter and the cost of fencing the side of circular field is Rs. 150 per meter, then find the difference between the total cost of fencing rectangular and square field together and the cost of fencing circular field?

a. Rs. 250

b. Rs. 200

c. Rs. 300

d. Rs. 150

e. Rs. 130

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3. Questions

If the farmer made a 3 meters wide path around the rectangular field, then find the area of the path?

a. 214 m^2

b. 254 m^2

c. 224 m^2

d. 204 m^2

e. 234 m^2

4. Questions

If a square field of maximum area is cut inside the circular field, then what will be the ratio of the side of the square field, which the farmer had already, to the side of the square field, which is inside the circular field?

a. 12: 14√7

b. 12: 7√2

c. 4: 7√2

d. 3√2 : 7

e. 3: 2√7

5. Questions

Which of the following statements is/are not true?

P: The area of circular field is maximum.

Q: Ratio of the perimeter of the rectangular field to that of the square field is 7: 6.

R: Average of the breadth of the rectangular field, the side of the square field and the radius of the circular field is 12 m.

a. All are true

b. Only P and Q

c. Only Q and R

d. Only P and R

e. Only P

6. Questions

Study the following data carefully and answer the questions:

Data given below is related to the weights (in kg) of six members of a family and the six members are son, daughter, father, mother, grandfather, and grandmother.

Bar graph given below shows the weights of grandmother, father, mother, son and daughter as per cent of the weight of grandfather on 1st Jan of a particular year.

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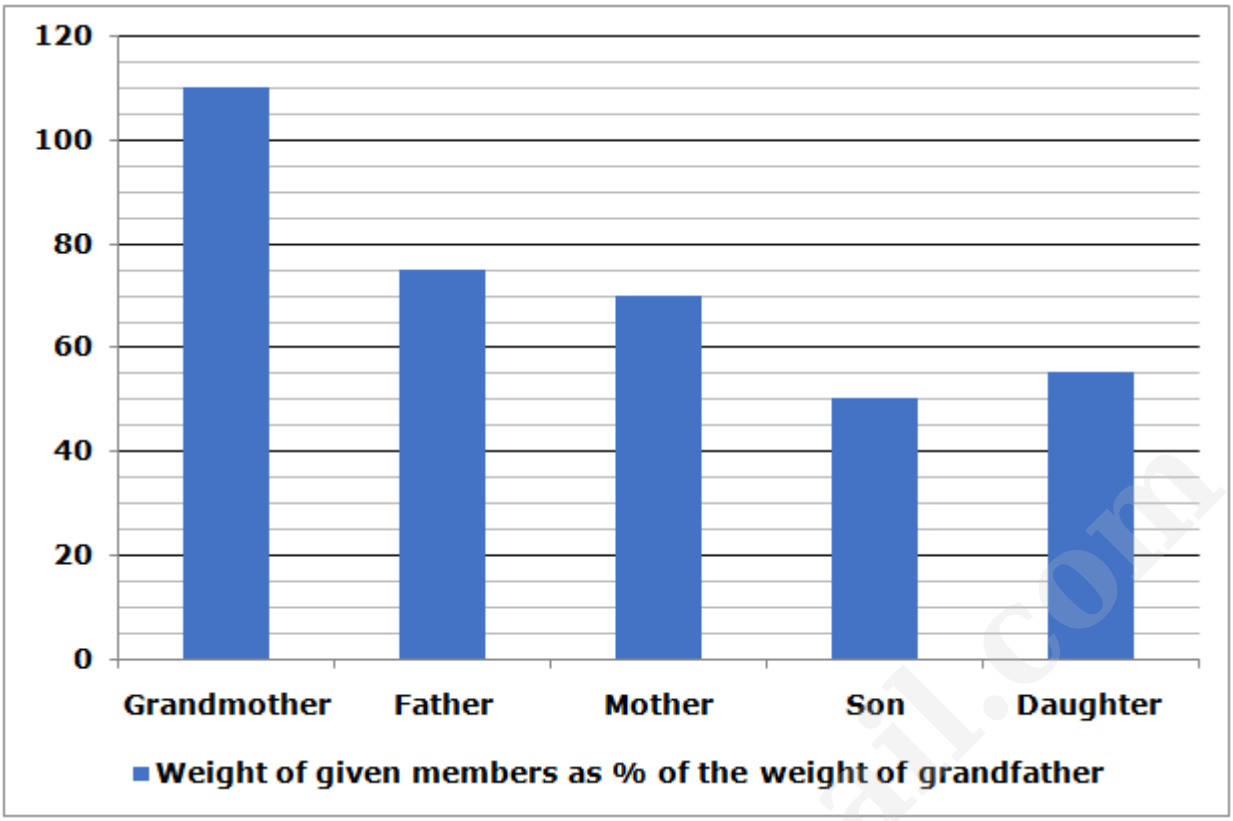


Table given below shows the increasing/decreasing rate of weight of each family member for the next 6 months.

Family member	Increasing/decreasing rate of weight for next 6 months
Grandfather	-0.75 kg/month
Grandmother	-1.2 kg/month
Father	+0.25 kg/month
Mother	-0.4 kg/month
Son	+0.9 kg/month
Daughter	-0.5 kg/month

Note:

On 30th June of that particular year, the total weight of grandfather and grandmother is 41.2 kg more than the total weight of father and mother.

If father's weight increases at the rate of 0.75 kg per month from 1st July to 31st Dec and son's weight increases at the rate of 0.65 kg per month from 1st July to 31st October, then find the ratio of father's weight on 31st Dec to son's weight on 31st Oct?

a. 11: 8

b. 22: 15

c. 3: 2

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d. 33: 28

e. 2: 1

7. Questions

Which of the following statements is/are not true?

P: Average weight of grandfather, grandmother and father on 1st Jan is 76 kg.

Q: Ratio of son's weight to daughter's weight on 30th June is 27: 25.

R: Grandfather's weight is decreased by 5(5/8)% from 1st Jan to 30th June.

a. Only P and Q

b. Only Q

c. Only R

d. All are true

e. Only P

8. Questions

If father's weight is increased by M% from 1st Jan to 30th June and son's weight is increased by N% from 1st Jan to 30th June, then find the ratio of M to N?

a. 1: 4

b. 5: 9

c. 4: 27

d. 2: 5

e. 5: 27

9. Questions

If grandmother's weight is increased at the rate of 0.3 kg per months from 1st July to 31st Dec, then find the average decrement per month in grandmother's weight from 1st Jan to 31st Dec?

a. 0.4 kg per month

b. 0.48 kg per month

c. 0.45 kg per month

d. 0.42 kg per month

e. 0.46 kg per month

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10. Questions

Find the difference between the average weight of the family on 1st Jan and the average weight of the family on 30th June?

a. 1.7 kg

b. 4.7 kg

c. 3.7 kg

d. 5.7 kg

e. 2.7 kg

11. Questions

Average age of P, Q, and R is 'x' years while it is 2 years less than the present age of P. P is twice as old as Q as he is younger than R.

Which of the following statement(s) can be determined?

I: Value of 'x'.

II: Difference between ages of Q and R.

III: Sum of ages of P and Q

a. Only statements I and II

b. Only statements II and III

c. Only statement II

d. All I, II, and III

e. Only III

12. Questions

A car is 'D' km behind a bus, speed of bus is 45 km/h. If the car can catch the bus in 40 min and reaches its destination ('6D' km far from the starting point) in 1.6 hours. Find the time taken by bus to cover a distance of 'D + 16' km.

a. 48 min

b. 54 min

c. 36 min

d. 1 hour

e. 1.2 hours

13. Questions

Two persons P and Q started a business with their investment in the ratio of 2: 3 and after 6

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months Q left the business and R joins with P with initial investment of Rs.12500. Out of total annual profit Rs.18000 from the business, share of R is Rs.7500. Which of the following is correctly matched?

Column I	Column II
A: Initial investment of P.	**P:** Rs.5000
B: Profit share of Q.	**Q:** Rs.4500
C: Difference between initial investments of P and R.	**R:** Rs.7500

a. Only A-P

b. All are correctly matched

c. Only B-Q

d. Only A-P and B-Q

e. Only A-P and C-R

14. Questions

Read the data carefully and answer the following questions.

Cost price of items A and B are 'X' and '2X – 500' respectively. Item A is marked 20% above cost price and sold after giving a discount of 'a' rupees while item B is marked 30% above cost price and sold after giving a discount of Rs.234. Total profit amount earned is Rs.92 while the selling prices of item B is Rs.180 more than the selling price of item A.

Find the profit percent earned after selling both the items A and B together.

a. 6.25%

b. 7.5%

c. 4.5%

d. 5.25%

e. 5.75%

15. Questions

If p% of X = a, then which of the following cannot be the factors of value of 'p + 6'?

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a. 2

b. 3

c. 4

d. 6

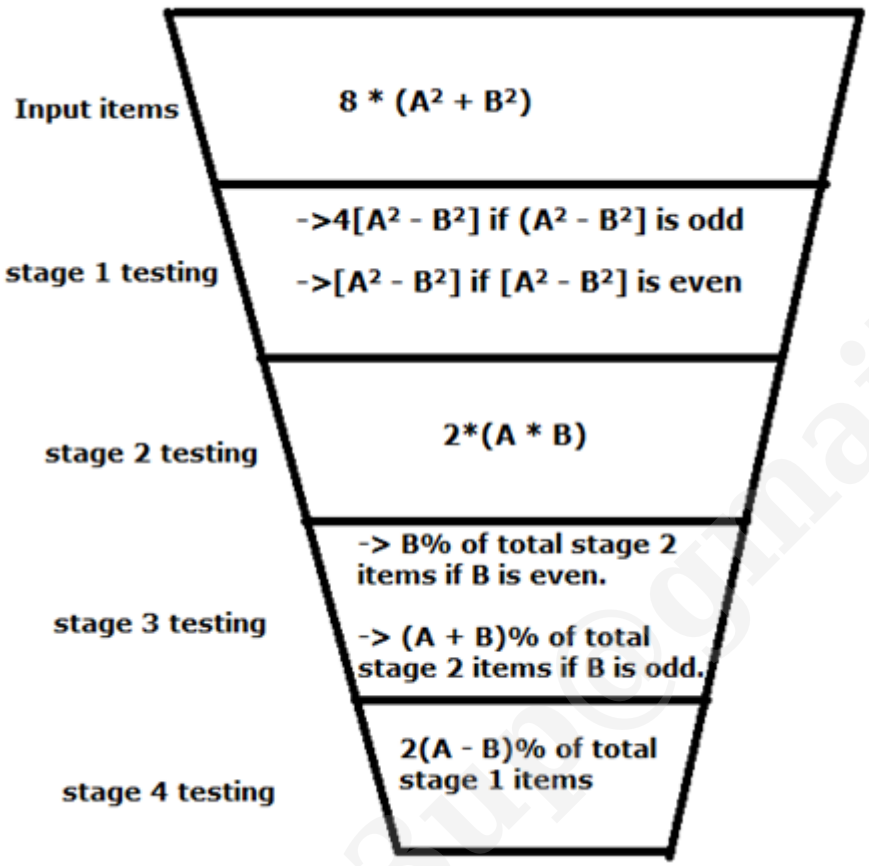
e. 9

16. Questions

Read the data carefully and answer the following questions.

In a manufacturing factory, 4 stages of testing are performed for the manufactured items such that the customers are getting quality product.

Funnel graph given below shows the data about number of input items which are being tested and four stages of testing of different days of a week.



Note:

1). At each testing stage, data shows the number of items discarded at that testing stage. Ex: At stage 2, number of items that are discarded is 2*(A*B).

2). Both 'A' and 'B' are 2-digits multiples of 5 and A is an odd number.

If total number of items discarded after stage 2 of testing on Friday was 500, then total items received after stage 3 of testing on that day was approximately what percent of total number of input items?

a. 45%

b. 50%

c. 40%

d. 75%

e. 60%

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17. Questions

If total number of items discarded on Saturday after stage 4 testing is 1280 and both A and B are odd numbers such that (A > B), then find the total number of quality items received after stage 4 testing on that day.

a. 2310

b. 2130

c. 2200

d. 2110

e. 2020

18. Questions

If total number of input items to be tested on Monday and Tuesday are 9000 and 13000 respectively, then find the difference between the total number of output items received after stage 3 on Monday and the total number of output items received after stage 2 on Tuesday.

a. 4240

b. 3860

c. 4520

d. 4840

e. 5120

19. Questions

If A = 35 and B < A also B is an even integer and also number of items received after stage 1 of testing is near to the 10000 number, then find the total number of items discarded after all the testing stages on Wednesday.

a. 8880

b. 9130

c. 9270

d. 10300

e. 9730

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20. Questions

If both A and B have the same values such that (A + B <80), then find the approximate average of number of items that are received on Monday after stage 2 of testing when taking all the possibilities together.

a. 9260

b. 9460

c. 9880

d. 9680

e. 9080

Explanations:

1. Questions

Since, the ratio of the length of rectangular field to the side of square field is 3: 2.

So, let the length of rectangular field and the side of square field be '3x' m and '2x' m respectively.

Since, the radius of circular field is 4 meters more than the breadth of rectangular field.

So, let the breadth of the rectangular field be 'y' m.

And the radius of circular field = (y + 4) m

Since, the perimeter of the rectangular field = 56 m

So,

2(3x + y) = 56

3x + y = 28 ----------(1)

The perimeter of the square field = 4 * 2x = 8x m

The circumference of the circular field = [2 * (22/7) * (y + 4)] m

Since, the time taken by the farmer to complete one round (at circumference) of the circular field is 10 seconds more than the time taken by him to complete one round (at perimeter) of the square field.

And the speed of the farmer = 14.4 km/h = 4 m/s

So,

$$\frac{2 \times 22 \times (y+4)}{7 \times 4} - \frac{8x}{4} = 10$$

$$\frac{11 \times (y+4)}{7} - 2x = 10$$

11y + 44 – 14x = 70

11y – 14x = 26 ----------(2)

From equations (1) and (2):

11(28 – 3x) – 14x = 26

308 – 33x – 14x = 26

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x = 6

From equation (1):

3 * 6 + y = 28

y = 10

The length of rectangular field = 3 * 6 = 18 m

The breadth of rectangular field = 10 m

The side of square field = 2 * 6 = 12 m

The radius of circular field = 10 + 4 = 14 m

Answer: E

The area of the rectangular field = 18 * 10 = 180 m^2

The area of the square field = 12 * 12 = 144 m^2

The area of circular field = (22/7) * 14 * 14 = 616 m^2

Since, the average of the areas of rectangular field and square field is (M + N) m^2.

So,

M + N = (180 + 144)/2

M + N = 162 ---------(1)

Since, the ratio of M to N is 20: 7.

So, M/N = 20/7 ----------(2)

From equations (1) and (2):

(20N/7) + N = 162

N = 42

From equation (1):

M = 120

Now, the average of the areas of square field and circular field:

$$\frac{144 + 616}{2} = 380\ m^2 = (5N + M + 50)m^2$$

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2. Questions

Since, the ratio of the length of rectangular field to the side of square field is 3: 2.

So, let the length of rectangular field and the side of square field be '3x' m and '2x' m respectively.

Since, the radius of circular field is 4 meters more than the breadth of rectangular field.

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So, let the breadth of the rectangular field be 'y' m.

And the radius of circular field = (y + 4) m

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So,

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308 – 33x – 14x = 26

x = 6

From equation (1):

3 * 6 + y = 28

y = 10

The length of rectangular field = 3 * 6 = 18 m

The breadth of rectangular field = 10 m

The side of square field = 2 * 6 = 12 m

The radius of circular field = 10 + 4 = 14 m

Answer: B

The perimeter of rectangular field = 56 m

The perimeter of square field = 4 * 12 = 48 m

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The circumference of circular field = 2 * (22/7) * 14 = 88 m

Since, the total cost of fencing the sides of rectangular and square fields together = 125 * (56 + 48) = Rs. 13000

And the cost of fencing the side of circular field = 150 * 88 = Rs.13200

So, the required difference = 13200 – 13000 = Rs.200

3. Questions

Since, the ratio of the length of rectangular field to the side of square field is 3: 2.

So, let the length of rectangular field and the side of square field be '3x' m and '2x' m respectively.

Since, the radius of circular field is 4 meters more than the breadth of rectangular field.

So, let the breadth of the rectangular field be 'y' m.

And the radius of circular field = (y + 4) m

Since, the perimeter of the rectangular field = 56 m

So,

2(3x + y) = 56

3x + y = 28 ----------(1)

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The circumference of the circular field = [2 * (22/7) * (y + 4)] m

Since, the time taken by the farmer to complete one round (at circumference) of the circular field is 10 seconds more than the time taken by him to complete one round (at perimeter) of the square field.

And the speed of the farmer = 14.4 km/h = 4 m/s

So,

$$\frac{2 \times 22 \times (y+4)}{7 \times 4} - \frac{8x}{4} = 10$$

$$\frac{11 \times (y+4)}{7} - 2x = 10$$

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11(28 – 3x) – 14x = 26

308 – 33x – 14x = 26

x = 6

From equation (1):

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$y = 10$

The length of rectangular field = $3 * 6 = 18$ m

The breadth of rectangular field = 10 m

The side of square field = $2 * 6 = 12$ m

The radius of circular field = $10 + 4 = 14$ m

Answer: D

The length of the rectangular field = 18 m

The breadth of the rectangular field = 10 m

The area of the rectangular field = $18 * 10 = 180$ m^2

The area of the path = The area of the field with path – The area of the field

Since, the width of the path is 3 m.

So, the area of the field with path = $(18 + 3 + 3) * (10 + 3 + 3) = 384$ m^2

And the area of the path = $384 – 180 = 204$ m^2

4. Questions

Since, the ratio of the length of rectangular field to the side of square field is 3: 2.

So, let the length of rectangular field and the side of square field be '3x' m and '2x' m respectively.

Since, the radius of circular field is 4 meters more than the breadth of rectangular field.

So, let the breadth of the rectangular field be 'y' m.

And the radius of circular field = (y + 4) m

Since, the perimeter of the rectangular field = 56 m

So,

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Since, the time taken by the farmer to complete one round (at circumference) of the circular field is 10 seconds more than the time taken by him to complete one round (at perimeter) of the square field.

And the speed of the farmer = 14.4 km/h = 4 m/s

So,

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$$\frac{2 \times 22 \times (y+4)}{7 \times 4} - \frac{8x}{4} = 10$$

$$\frac{11 \times (y+4)}{7} - 2x = 10$$

11y + 44 – 14x = 70

11y – 14x = 26 ----------(2)

From equations (1) and (2):

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308 – 33x – 14x = 26

x = 6

From equation (1):

3 * 6 + y = 28

y = 10

The length of rectangular field = 3 * 6 = 18 m

The breadth of rectangular field = 10 m

The side of square field = 2 * 6 = 12 m

The radius of circular field = 10 + 4 = 14 m

Answer: D

The side of the square field, which the farmer had already = 12 m

Since, the radius of the circular field = 14 m

Let the side of the square field, which is inside the circular field is 'x'

$\sqrt{(x^2 + x^2)} = \sqrt{28*28}$ m

$\sqrt{2}$ x = 28

x = $28/\sqrt{2}$ m

Required ratio = 12: $28/\sqrt{2} = 3\sqrt{2} : 7$

5. Questions

Since, the ratio of the length of rectangular field to the side of square field is 3: 2.

So, let the length of rectangular field and the side of square field be '3x' m and '2x' m respectively.

Since, the radius of circular field is 4 meters more than the breadth of rectangular field.

So, let the breadth of the rectangular field be 'y' m.

And the radius of circular field = (y + 4) m

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Since, the perimeter of the rectangular field = 56 m

So,

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The circumference of the circular field = [2 * (22/7) * (y + 4)] m

Since, the time taken by the farmer to complete one round (at circumference) of the circular field is 10 seconds more than the time taken by him to complete one round (at perimeter) of the square field.

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So,

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The length of rectangular field = 3 * 6 = 18 m

The breadth of rectangular field = 10 m

The side of square field = 2 * 6 = 12 m

The radius of circular field = 10 + 4 = 14 m

Answer: A

From P:

The area of the rectangular field = 18 * 10 = 180 m^2

The area of the square field = 12 * 12 = 144 m^2

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Tg:- @NextGenBankers

The area of circular field = (22/7) * 14 * 14 = 616 m^2

Since, the area of the circular field is maximum.

So, P is true.

From Q:

The perimeter of the rectangular field = 56 m

The perimeter of the square field = 4 * 12 = 48 m

Ratio of the perimeter of the rectangular field to that of the square field:

56: 48 = 7: 6

So, Q is true.

From R:

Average of the breadth of the rectangular field, the side of the square field and the radius of the circular field:

(10+12+14)/3=12 m

So, R is true.

Hence, all are true.

6. Questions

On 1st Jan:

Let the present weight of grandfather be '100x' kg.

So, the present weight of grandmother = 110% of 100x = '110x' kg

The present weight of father = 75% of 100x = '75x' kg

The present weight of mother = 70% of 100x = '70x' kg

The present weight of son = 50% of 100x = '50x' kg

And the present weight of daughter = 55% of 100x = '55x' kg

On 30th June:

Grandfather's weight after 6 months = (100x – 6 * 0.75) = (100x – 4.5) kg

Grandmother's weight after 6 mother = (110x – 6 * 1.2) = (110x – 7.2) kg

Father's weight after 6 months = (75x + 6 * 0.25) = (75x + 1.5) kg

Mother's weight after 6 months = (70x – 6 * 0.4) = (70x – 2.4) kg

Since, after 6 months, the total weight of grandfather and grandmother is 41.2 kg more than the total weight of father and mother.

So,

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$(100x - 4.5) + (110x - 7.2) = (75x + 1.5) + (70x - 2.4) + 41.2$

$210x - 11.7 = 145x + 40.3$

$65x = 52$

$x = 0.8$

Family members	Weight (in kg) on 1st Jan	Weight (in kg) on 30th June
Grandfather	100 * 0.8 = 80	80 – 6 * 0.75 = 75.5
Grandmother	110 * 0.8 = 88	88 – 6 * 1.2 = 80.8
Father	75 * 0.8 = 60	60 + 6 * 0.25 = 61.5
Mother	70 * 0.8 = 56	56 – 6 * 0.4 = 53.6
Son	50 * 0.8 = 40	40 + 6 * 0.9 = 45.4
Daughter	55 * 0.8 = 44	44 – 6 * 0.5 = 41

Answer: A

Since, father's weight on 30th June = 61.5 kg

So, father's weight on 31st Dec = 61.5 + 6 * 0.75 = 66 kg

Since, son's weight on 30th June = 45.4 kg

So, son's weight on 31st Oct = 45.4 + 4 * 0.65 = 48 kg

Required ratio = 66: 48 = 11: 8

7. Questions

On 1st Jan:

Let the present weight of grandfather be '100x' kg.

So, the present weight of grandmother = 110% of 100x = '110x' kg

The present weight of father = 75% of 100x = '75x' kg

The present weight of mother = 70% of 100x = '70x' kg

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So,

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Son	50 * 0.8 = 40	40 + 6 * 0.9 = 45.4
Daughter	55 * 0.8 = 44	44 – 6 * 0.5 = 41

Answer: B

From P:

Average weight of grandfather, grandmother and father on 1st Jan:

(80+88+60)/3=76 kg

So, P is true.

From Q:

Ratio of son's weight to daughter's weight on 30th June:

45.4: 41 = 227: 205

So, Q is not true.

From R:

Grandfather's weight on 1st Jan = 80 kg

Grandfather's weight on 30th June = 75.5 kg

Required percentage = (80-75.5)/80 ×100=5 (5/8)%

So, R is true.

Hence, only Q is not true.

8. Questions

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On 1st Jan:

Let the present weight of grandfather be '100x' kg.

So, the present weight of grandmother = 110% of 100x = '110x' kg

The present weight of father = 75% of 100x = '75x' kg

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Father's weight after 6 months = (75x + 6 * 0.25) = (75x + 1.5) kg

Mother's weight after 6 months = (70x – 6 * 0.4) = (70x – 2.4) kg

Since, after 6 months, the total weight of grandfather and grandmother is 41.2 kg more than the total weight of father and mother.

So,

(100x – 4.5) + (110x – 7.2) = (75x + 1.5) + (70x – 2.4) + 41.2

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Family members	Weight (in kg) on 1st Jan	Weight (in kg) on 30th June
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Father	75 * 0.8 = 60	60 + 6 * 0.25 = 61.5
Mother	70 * 0.8 = 56	56 – 6 * 0.4 = 53.6
Son	50 * 0.8 = 40	40 + 6 * 0.9 = 45.4
Daughter	55 * 0.8 = 44	44 – 6 * 0.5 = 41

Answer: E

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Father's weight on 1st Jan = 60 kg

Father's weight on 30th June = 61.5 kg

So, M = (61.5-60)/60 ×100 = 2.5%

Son's weight on 1st Jan = 40 kg

Son's weight on 30th June = 45.4 kg

So, N = (45.4-40)/40 ×100 = 13.5%

Required ratio = M: N = 2.5: 13.5 = 5: 27

9. Questions

On 1st Jan:

Let the present weight of grandfather be '100x' kg.

So, the present weight of grandmother = 110% of 100x = '110x' kg

The present weight of father = 75% of 100x = '75x' kg

The present weight of mother = 70% of 100x = '70x' kg

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And the present weight of daughter = 55% of 100x = '55x' kg

On 30th June:

Grandfather's weight after 6 months = (100x – 6 * 0.75) = (100x – 4.5) kg

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Father's weight after 6 months = (75x + 6 * 0.25) = (75x + 1.5) kg

Mother's weight after 6 months = (70x – 6 * 0.4) = (70x – 2.4) kg

Since, after 6 months, the total weight of grandfather and grandmother is 41.2 kg more than the total weight of father and mother.

So,

(100x – 4.5) + (110x – 7.2) = (75x + 1.5) + (70x – 2.4) + 41.2

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Family members	Weight (in kg) on 1st Jan	Weight (in kg) on 30th June
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Son	50 * 0.8 = 40	40 + 6 * 0.9 = 45.4
Daughter	55 * 0.8 = 44	44 – 6 * 0.5 = 41

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Answer: C

Since, grandmother's weight on 1st Jan = 88 kg

And grandmother's weight on 30th June = 80.8 kg

So, grandmother's weight on 31st Dec = 80.8 + 6 * 0.3 = 82.6 kg

And the average decrement per month in grandmother's weight from 1st Jan to 31st Dec = (88 – 82.6)/12 = 0.45 kg per month

10. Questions

On 1st Jan:

Let the present weight of grandfather be '100x' kg.

So, the present weight of grandmother = 110% of 100x = '110x' kg

The present weight of father = 75% of 100x = '75x' kg

The present weight of mother = 70% of 100x = '70x' kg

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Since, after 6 months, the total weight of grandfather and grandmother is 41.2 kg more than the total weight of father and mother.

So,

(100x – 4.5) + (110x – 7.2) = (75x + 1.5) + (70x – 2.4) + 41.2

210x – 11.7 = 145x + 40.3

65x = 52

x = 0.8

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Family members	Weight (in kg) on 1st Jan	Weight (in kg) on 30th June
Grandfather	100 * 0.8 = 80	80 – 6 * 0.75 = 75.5
Grandmother	110 * 0.8 = 88	88 – 6 * 1.2 = 80.8
Father	75 * 0.8 = 60	60 + 6 * 0.25 = 61.5
Mother	70 * 0.8 = 56	56 – 6 * 0.4 = 53.6
Son	50 * 0.8 = 40	40 + 6 * 0.9 = 45.4
Daughter	55 * 0.8 = 44	44 – 6 * 0.5 = 41

Answer: A

Since, the average weight of the family on 1st Jan:

(80+88+60+56+40+44)/6=61.33 kg

And the average weight of the family on 30th June:

(75.5+80.8+61.5+53.6+45.4+41)/6=59.63 kg

Required difference = 61.33 – 59.63 = 1.7 kg

11. Questions

Answer: C

Average age of P, Q, and R = 'x' years

Present age of P = (x + 2) years

Let the difference between the ages of P and R is 'a' year, then the difference between the ages of P and Q will be '2a' years.

Present age of Q = (x + 2) – 2a

Present age of R = (x + 2) + a

According to the question:

(x + 2) + [(x + 2) – 2a] + [(x + 2) + a] = 3x

3x + 6 – a = 3x

a = 6

But we cannot determine the value of 'x'. So, the age of all the persons also cannot be determined.

I: Value of 'x' cannot be determined.

II: Difference between present ages of Q and R = [(x + 2) + a] – [(x + 2) – 2a] = 3a = 18 years

Statement II can be determined.

III: Since the age of any person cannot be determined. So, sum of ages of P and Q also cannot be determined.

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Hence, only statement II can be determined.

12. Questions

Answer: A

Let speed of car = 'S' km/h

Relative speed of car and bus = (S – 45) km/h

Relative distance covered by car with respect to bus = 'D' km

According to the question:

D = (S – 45) * 40/60

3D = 2S – 90 …… (1)

6D = S * 1.6 ……. (2)

From (1) and (2):

2(2S – 90) = 1.6S

4S – 180 = 1.6S

2.4S = 180

S = 75

From equation (2):

6D = 75 * 1.6

D = 20

Time taken by bus to cover 'D + 16' km = 36/45 * 60

= 48 min

13. Questions

Answer: B

Let initial investments of P and Q is '2x' and '3x' respectively.

Initial investment of R = Rs.12500

Ratio of profit share of P, Q, and R = (2x * 12): (3x * 6): (12500 * 6) = 4x: 3x: 12500

Profit share of R = 7500

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$$18000 * \frac{12500}{4x + 3x + 12500} = 7500$$

$$\frac{12500}{4x + 3x + 12500} = \frac{5}{12}$$

$$\frac{2500}{4x + 3x + 12500} = \frac{1}{12}$$

30000 = 7x + 12500

7x = 17500

x = 2500

Ratio of profit share of P, Q, and R = 4x: 3x: 12500 = 4: 3: 5

Initial investment of P = 2x = 5000

Profit share of Q = 18000 * 3/12 = Rs.4500

Difference between initial investment of P and R = 12500 – 5000 = Rs.7500

Hence, all are correctly matched.

14. Questions

Cost price of item A = X

Marked price of item A = 120% of X = 1.2X

Selling price of item A = (1.2X – a)

Cost price of item B = 2X – 500

Marked price of item B = 130% of (2X – 500) = (2.6X – 650)

Selling price of item B = (2.6X – 650) – 234 = (2.6X – 884)

According to the question:

(2.6X – 884) – (1.2X – a) = 180

1.4X + a = 1064 …… (1)

[(2.6X – 884) + (1.2X – a)] – [X + (2X – 500)] = 92

0.8X – a = 476

a = 0.8X – 476 …….. (2)

From (1) and (2):

1.4X + (0.8X – 476) = 1064

2.2X = 1540

X = 700

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From equation (2):

a = 0.8X – 476 = 560 – 476

a = 84

Answer: E

Cost price of item A = X = Rs.700

Cost price of item B = 2X – 500 = Rs.900

Total cost price of items A and B together = 700 + 900 = Rs.1600

Profit amount earned after selling both the items together = Rs.92

Required profit percent = 92/1600 * 100

= 5.75%

15. Questions

Cost price of item A = X

Marked price of item A = 120% of X = 1.2X

Selling price of item A = (1.2X – a)

Cost price of item B = 2X – 500

Marked price of item B = 130% of (2X – 500) = (2.6X – 650)

Selling price of item B = (2.6X – 650) – 234 = (2.6X – 884)

According to the question:

(2.6X – 884) – (1.2X – a) = 180

1.4X + a = 1064 ……. (1)

[(2.6X – 884) + (1.2X – a)] – [X + (2X – 500)] = 92

0.8X – a = 476

a = 0.8X – 476 …….. (2)

From (1) and (2):

1.4X + (0.8X – 476) = 1064

2.2X = 1540

X = 700

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From equation (2):

a = 0.8X – 476 = 560 – 476

a = 84

Answer: C

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p% of X = a

p% of 700 = 84

p/100 * 700 = 84

p * 7 = 84

p = 12

p + 6 = 12 + 6 = 18

18 can be written as: $1 * 2 * 3^2$.

Factors of 18 = 1, 2, 3, 6, 9, and 18

Hence, 4 is not a factor of 18.

16. Questions

Answer: B

Total number of items discarded after stage 2 of testing on Friday = 500

2 * (A * B) = 500

A * B = 250

Possible values of (A, B) when A is 2-digit odd multiple of 5 = (25, 10)

A = 25 and B = 10 (Even)

$A^2 + B^2 = 625 + 100 = 725$

$A^2 – B^2 = 625 – 100 = 525$ (Odd)

Total number of input items on Friday = $8 * (A^2 + B^2) = 8 * 725 = 5800$

Total number of items received after stage 1 of testing = 5800 – 4 * 525 = 3700

Total number of items received after stage 2 of testing = 3700 – 500 = 3200

Total number of items received after stage 3 of testing = 3200 – 10% of 3200 = 2880

Req. % = 2880*100/5800 = 49.65%

We know that half of total number of input items on Friday will be 2900.

Hence, our required approximate percent will also be equal to 50%.

17. Questions

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Answer: D

Since both A and B are odd numbers (A > B), then possible values of (A, B) = (25, 15), (35, 15), (35, 25), (45, 15), ……..

When A = 25 and B = 15

$A^2 – B^2 = 25^2 – 15^2 = 625 – 225 = 400$ (Even)

Total number of input items = $8(A^2 + B^2) = 8 * (25^2 + 15^2) = 6800$

Total number of items after stage 1 of testing = 6800 – 400 = 6400

Total number of items discarded after stage 4 of testing = (2 * 10)% of 6400 = 1280

So, this is a valid case.

When A = 35 and B = 15

$A^2 – B^2 = 35^2 – 15^2 = 1225 – 225 = 1000$ (Even)

Total number of input items = $8(A^2 + B^2) = 8 * (35^2 + 15^2) = 11600$

Total number of items after stage 1 of testing = 11600 – 1000 = 10600

Total number of items discarded after stage 4 of testing = (2 * 10)% of 10600 = 2120

So, this is not a valid case.

When A = 35 and B = 25

$A^2 – B^2 = 35^2 – 25^2 = 1225 – 625 = 600$ (Even)

Total number of input items = $8(A^2 + B^2) = 8 * (35^2 + 25^2) = 14800$

Total number of items after stage 1 of testing = 14800 – 600 = 14200

Total number of items discarded after stage 4 of testing = (2 * 10)% of 14200 = 2840

So, this is not a valid case. Also, we can assume that no further cases will be the valid cases.

Hence, A = 25 and B = 15.

Total number of items after stage 2 of testing = 6400 – 2 * 25 * 15 =5650

Total number of items after stage 3 of testing = 5650 – (25 + 15)% of 5650 = 60% of 5650 = 3390

Total number of items after stage 4 of testing = 3390 – 1280 = 2110

18. Questions

Answer: C

Monday:

Total number of input items = $8(A^2 + B^2) = 9000 \Rightarrow A^2 + B^2 = 1125$

Since A is a 2-digit odd multiple of 5, then possible values of A = 15, 25, 35, ……

When A = 15, B = 30

When A = 25, $B^2 = 500$ (Invalid)

When A = 35, $B^2 = -100$ (Invalid)

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$|A^2 - B^2| = |30^2 - 15^2| = 675$ [Odd]

Total number of items after stage 1 = 9000 – 4 * 675 = 6300

Total number of items after stage 2 = 6300 – 2 * 15 * 30 = 5400

Total number of items after stage 3 = 5400 – 30% of 5400 = 3780

Tuesday:

Total number of input items = $8(A^2 + B^2) = 13000 \Rightarrow A^2 + B^2 = 1625$

Since A is a 2-digit odd multiple of 5, then possible values of A = 15, 25, 35, ……

When A = 15, $B^2 = 1400$ (Invalid)

When A = 25, $B^2 = 1000$ (Invalid)

When A = 35, B = 20

When A = 45, $B^2 = -400$

$|A^2 - B^2| = |20^2 - 35^2| = 825$ [Odd]

Total number of items after stage 1 = 13000 – 4 * 825 = 9700

Total number of items after stage 2 = 9700 – 2 * 35 * 20 = 8300

Required difference = 8300 – 3780

= 4520

19. Questions

Answer: C

Since A = 35 and B < A also B is an even integer. So, possible values of B are 30, 20, and 10.

When B = 30, $B^2 = 900$ and $A^2 - B^2 = 1225 - 900 = 325$ (Odd)

Total number of input items = $8(A^2 + B^2) = 8(35^2 + 30^2) = 17000$

Total number of items after stage 1 of testing = 17000 –4 * 325 = 15700

When B = 20, $B^2 = 400$ and $A^2 - B^2 = 1225 - 400 = 825$ (Odd)

Total number of input items = $8(A^2 + B^2) = 8(35^2 + 20^2) = 13000$

Total number of items after stage 1 of testing = 13000 – 4 * 825 = 9700

When B = 10, $B^2 = 100$ and $A^2 - B^2 = 1225 - 100 = 1125$ (Odd)

Total number of input items = $8(A^2 + B^2) = 8(35^2 + 10^2) = 10600$

Total number of items after stage 1 of testing = 10600 – 4 * 1125 = 6100

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Out of all the 3 possibilities, number of items received after stage 1 of testing will become close to 10000 number when B = 20.

So, A = 35 and B = 20

Total number of items after stage 2 of testing = 9700 – 2 * 35 * 20 = 8300

Total number of items after stage 3 of testing = 8300 – 20% of 8300 = 6640

Total number of items after stage 4 of testing = 6640 – (2 * 15)% of 9700 = 6640 – 2910 = 3730

Hence, total number of discarded items on Wednesday after all the four stages of testing = Total number of input items – Total number of output items

= 13000 –3730

= 9270

20. Questions

Answer: D

Since both A and B have the same values. Also, A + B < 80

2A or 2B < 80

A or B < 40

Which means,

A = B = 15

Total number of input items = $8(A^2 + B^2) = 8(15^2 + 15^2) = 3600$

Total number of items after stage 1 of testing = 3600 – 0 = 3600 [$A^2 – B^2 = 0$]

Total number of items after stage 2 of testing = 3600 – 2 * 15 * 15 = 3150

A = B = 25

Total number of input items = $8(A^2 + B^2) = 8(25^2 + 25^2) = 10000$

Total number of items after stage 1 of testing = 10000 – 0 = 10000 [$A^2 – B^2 = 0$]

Total number of items after stage 2 of testing = 10000 – 2 * 25 * 25 = 8750

A = B = 35

Total number of input items = $8(A^2 + B^2) = 8(35^2 + 35^2) = 19600$

Total number of items after stage 1 of testing = 19600 – 0 = 19600 [$A^2 – B^2 = 0$]

Total number of items after stage 2 of testing = 19600 – 2 * 35 * 35 = 17150

Required average = (3150 + 8750 + 17150)/3

= 29050/3

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= 9683.33

= 9680

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1. Questions

Read the data carefully and answer the following questions.

A company manufactured tables and chairs. There are (4M + 8) tables and (N+4) chairs and each of these types of article has two varieties, i.e.,Green and Blue.

The company marked up the green tables at (C + 5) % above its cost price while the blue tables at 2C% above its cost price. The discount given in the green tables is C % while in the blue tables is (C +5) %.

Out of the total number of chairs manufactured, 3C% are green chairs, and the rest are blue chairs. He labelled each green chair at Rs. 30 more than its cost price while the blue chairs at 33.33% above its cost price. The discounts of 11.11% and 16.66% are offered on green chairs and blue chairs, respectively.

Out of the total number of tables manufactured, 2.5R% are green tables, while the rest are blue tables. The manufacturing cost of each Table and Chair is Rs. (17M – N) and Rs. (2N+8), respectively**.** The total selling price of Chairs is Rs. 680 which is Rs. 150 less than that of Tables. Consider the nearest whole value for decimal values.

Note: 6M + 7N = 95, where both M and N are prime numbers.

Find the total selling price of blue tables (approx.)?

a. Rs. 360

b. Rs. 1000

c. Rs. 900

d. Rs. 630

e. Rs. 450

2. Questions

If another article Pen whose marked price is 30% more than the total discount offered on a green chair and the discount offered on a pen is equal to 1/10th of the total Marked price of a green table, then find the cost price of 30pens ?(Note: The profit earned by selling each pen is equal to the total number of the table)

a. Rs.1100

b. Rs.990

c. Rs.880

d. Rs.1500

e. Rs.760

3. Questions

Find which of the following is/are definitely true.

I). The difference between the total number of chairs and tables is 5 less than R.

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II). The ratio between the marked price of the green table and the selling price of the blue chair is 5:4

III). The profit earned by selling all tables is 40 more than the total profit earned by selling all chairs

a. Only I and II are true
b. Only II and III are true
c. Only I is true
d. All I, II, and III are true
e. None of these

4. Questions

Study the following information carefully and answer the questions given below.

The ratio of time taken by Bus A and the time taken by Bus C to cover a certain distance is 3:4. The distance covered by Bus B in (t+4) hours is (d+175) km, while the distance covered by Bus D in (t-2) hours is (d/2+20) km. The ratio of the speed of buses B and D is 5:6. When buses A and B start their journey simultaneously towards each other from a distance of 620 km, they meet each other after 4 hours. Similarly, when buses C and D start their journey simultaneously towards each other from a distance of 600 km, they meet each other after 4 hours.

Find the possible sum of time taken by Bus C and Bus D to cover the distance of [2d-160] km.

I). 4t+3.33

II). 5t+1.33

III). t+13.33

a. All true
b. All false
c. Only I true
d. Only I false
e. None of these

5. Questions

Bus A and C start their journey towards each other at the same time. After t+2 hours, they meet each other. Find the time taken by Bus D to cover the initial distance between Bus A and C.

a. 10.88 hours
b. 12.55 hours
c. 16.52 hours
d. 19.6 hours
e. None of these

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6. Questions

Find which of the following is true

a. d km distance covered by A in 6.25 hours

b. d km distance covered by B in 5.66 hours

c. [d-250] distance covered by C in 4.16 hours

d. d km distance covered by D in 5.55 hours

e. More than one option is true

7. Questions

Study the following information carefully and answer the questions given below.

A person divided Rs.120000 among his two sons (A and B) and two daughters (C and D) equally. All of them invested some amount in scheme P and the remaining amount in scheme Q. Scheme P provides simple interest and scheme Q provides compound interest. A invests 60% of his amount in scheme P. B invests an equal amount in scheme P and Q. The interest rate is the same for all the persons in each scheme. C invested 33.33% of his amount in scheme Q, while D invested 45% of his amount in scheme Q. The difference between the interest amount received from scheme P for A and D is Rs.900 after 3 years. The sum of the interest amount received by C from schemes P and Q after 2 years is Rs.13625.

The interest amount received from scheme Q after 2 years by A is again invested in scheme P for 6 years. Find the interest amount received from scheme P after 6 years.

a. Rs. 9400

b. Rs. 9600

c. Rs. 8100

d. Rs. 8400

e. Rs. 6400

8. Questions

The difference between the interest amount received by A from scheme P and Q after 2 years is 'm'. Find the value of 7m.

a. 3250

b. 3150

c. 3620

d. 3420

e. None of these

9. Questions

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Find the difference between the interest amount received by B and D from scheme Q after 3 years.

a. 1429.69

b. 1568.99

c. 1875.33

d. 1986.33

e. None of these

10. Questions

Find the average amount of the interest received by all persons from scheme P after 4 years.

a. 13900

b. 16540

c. 19600

d. 18450

e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

P invested a total of Rs. Z in two businesses (Rs. L in food business and rest in garments business). In food business, P has two partners Q and R, while in Garments business his partners are M and N. In food business, Q started the business with initial investment of Rs. 12000, after four months P joined, and after another eight months R joined with investment of Rs. 16000. In Garments business, M started the business with initial investment of Rs. 5000, after two months P joined him, after 7 months from starting the business N joined him with initial investment of Rs. 6000. Time period of investment in the food business is of two years while in the garment business is of one year. Profit share of P in garments business is 4/13 times as that of total profit earned from garments business.

Total investment made by P is Rs. 28000, and Total profit from food business is Rs. 1.5 lakhs, then find the profit share of P in food business.

a. Rs. 60000

b. Rs. 30000

c. Rs. 45000

d. Rs. 75000

e. None of these

12. Questions

If total investment made by P in business is Rs. 12000, total profit from garments business is Rs. 91000, then find profit share of M in garments business.

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a. Rs. 42000

b. Rs. 28000

c. Rs. 21000

d. Rs. 49000

e. None of these

13. Questions

Find the difference between the value of Z and L.

a. Rs. 4000

b. Rs. 8000

c. Rs. 2000

d. Rs. 9000

e. None of these

14. Questions

Study the following information carefully and answer the questions given below.

The speed of boat D in still water is 33.33% more than the speed of boat A. The ratio of the speed of river P and Q is 2:3. The speed of boat D is six times the speed of river P. The ratio of the speed of boats B and C in still water is 13:11. The ratio of the speed of river Q and R is 3:4. The downstream speed of boat A in river R is equal to the speed of boat B in still water.

The downstream speed of the boat D in river R is 32 km /hr. Boat C covered 140 km downstream in river Q in 't' hours. Find the distance covered by boat A in 't' hours in still water.

a. 85 km

b. 95 km

c. 90 km

d. 75 km

e. None of these

15. Questions

The upstream speed of boat A in river P is 14 km/hr. Find the distance covered by boats B and C in river Q downstream in 5.5 hours.

a. 320

b. 390

c. 330

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d. 360

e. None of these

16. Questions

The speed of boat D in still water is 36 km/hr. Boat A covers d km distance in still water in 2 hours. Find the distance covered by boat B in still water in 5.5 hours.

a. 5d-25.2

b. 6d-39.5

c. 3d+52.5

d. d+35.5

e. More than one option true

17. Questions

Study the following information carefully and answer the questions given below.

Work P: The ratio of efficiency of A and B is 5:4 respectively. A and C together can complete the work in 15 days. B and C together can complete the work in [10*x]/7 days. When A's efficiency becomes twice his original efficiency, then A alone can complete the work in [y-6] days.

Work Q: The ratio of efficiency of A and B is 3:2. The ratio of efficiency of B and C is 3:5. C alone can complete the work in 18 days. A and B together can complete the work in x days.

Work R: B and C take the same number of days to complete the work alone as they take to complete work P. A and B together can complete the work in y days.

A starts work P alone. After [y/2 + 1] days, A left the work then the rest of the work completed by B and C together in ____ days

I). x-2

II). y/2 + 1

III). [x+y]/2

a. Only A

b. Only B

c. Both A and B

d. Both B and C

e. Only C

18. Questions

Find which of the following takes the minimum number of days?

I). B and C together can complete work R with twice their original efficiency.

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II). A and C together can complete work Q.

III). A and B together can complete work P.

a. Only III

b. Only I

c. Only II

d. Only I and II

e. None of these

19. Questions

Find the approximate difference between the total time taken by A, B and C together to complete work P and R.

a. 2

b. 6

c. 8

d. 9

e. 0

20. Questions

A starts work R alone. After z days, A left the work, and B finished the rest of the work in [z+5] days. Find the value (x+y+z).

a. 45

b. 49

c. 65

d. 82

e. None of these

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Explanations:

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1. Questions

First, we need to find the value of M and N

6M + 7N = 95, where M and N are prime numbers

N = (95 – 6M)/7

If M = 2, N = 8.875 (not possible)

If M = 3, N = 11 (Possible)

If M = 5, N = 9.285 (not possible)

If M = 7, N = 7.571 (not possible)

If M = 11, N = 4.142 (not possible)

If M = 13, N = 2.428 (Not possible)

No other value is possible, which gives M and N both primes.

So, the number of tables manufactured by company = 4 x 3 + 8 = 20

Number of chairs manufactured by the company = 11 + 4 = 15

Cost price of each table = (17 x 3 – 11) = Rs. 40

Cost price of each chair = (2 x 11 + 8) = Rs. 30

For chairs

Total number of chairs = 15

Cost price of each chair = Rs. 30

Number of green chairs = 3C% x 15 = 9C/20

Number of blue chairs = (15 – 9C/20)

Marked price of green chairs = 9C/20 x (30 + 30) = 27C

Marked price of blue chairs = 133.33% x 30 x (15 – 9C/20) = (600 – 18C)

Selling price of green chairs = 8/9 x 27C = 24C

Selling price of blue chairs = 5/6 x (600 – 18C) = (500 – 15C)

Total selling price of chairs = Rs. 680

Now,

24C + (500 – 15C) = 680

9C = 180

Value of C = 20

Now we can find all the data related to Chairs

For Tables

Total number of tables = 20

Cost price of each table = Rs. 40

Number of green tables = 2.5R% x 20 = 0.5R

Number of blue tables = (20 – 0.5R)

Mark up% on green tables = (C + 5)% = (20 + 5)% = 25%

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So, the marked price of green tables = 40 x 1.25 x 0.5R = 25R

Discount given on green tables = C% = 20%

So, the selling price of green tables = 80% x 25R = 20R

Number of blue tables = (20 – 0.5R)

Mark up % on blue tables = 2C% = 2 x 20 = 40%

So, the marked price of blue tables = 40 x 1.4 x (20 – 0.5R) = (1120 – 28R)

Discount given on blue tables = (C +5)% = (20 +5)% = 25%

So, the selling price of blue tables = 75% x (1120 – 28R) = Rs. (840 – 21R)

Selling price of tables = 680 + 150 = 830

Now,

20R + (840 – 21R) = 830

Value of R = 10

Now we can find all the data related to tables

	Cost price	Selling price	Marked price	Discount	Profit
Green Chair	270	480	540	60	210
Blue chair	180	200	240	40	20
Green table	200	200	250	50	0
Blue table	600	630	840	210	30

Answer: D

According to the Question,

Selling price of blue tables = (840– 21R) = (840 – 21x 10) = Rs. 630

Hence, the answer is option D

2. Questions

First, we need to find the value of M and N

6M + 7N = 95, where M and N are prime numbers

N = (95 – 6M)/7

If M = 2, N = 8.875 (not possible)

If M = 3, N = 11 (Possible)

If M = 5, N = 9.285 (not possible)

If M = 7, N = 7.571 (not possible)

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No other value is possible, which gives M and N both primes.

So, the number of tables manufactured by company = 4 x 3 + 8 = 20

Number of chairs manufactured by the company = 11 + 4 = 15

Cost price of each table = (17 x 3 – 11) = Rs. 40

Cost price of each chair = (2 x 11 + 8) = Rs. 30

For chairs

Total number of chairs = 15

Cost price of each chair = Rs. 30

Number of green chairs = 3C% x 15 = 9C/20

Number of blue chairs = (15 – 9C/20)

Marked price of green chairs = 9C/20 x (30 + 30) = 27C

Marked price of blue chairs = 133.33% x 30 x (15 – 9C/20) = (600 – 18C)

Selling price of green chairs = 8/9 x 27C = 24C

Selling price of blue chairs = 5/6 x (600 – 18C) = (500 – 15C)

Total selling price of chairs = Rs. 680

Now,

24C + (500 – 15C) = 680

9C = 180

Value of C = 20

Now we can find all the data related to Chairs

For Tables

Total number of tables = 20

Cost price of each table = Rs. 40

Number of green tables = 2.5R% x 20 = 0.5R

Number of blue tables = (20 – 0.5R)

Mark up% on green tables = (C + 5)% = (20 + 5)% = 25%

So, the marked price of green tables = 40 x 1.25 x 0.5R = 25R

Discount given on green tables = C% = 20%

So, the selling price of green tables = 80% x 25R = 20R

Number of blue tables = (20 – 0.5R)

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Mark up % on blue tables = 2C% = 2 x 20 = 40%

So, the marked price of blue tables = 40 x 1.4 x (20 – 0.5R) = (1120 – 28R)

Discount given on blue tables = (C +5)% = (20 +5)% = 25%

So, the selling price of blue tables = 75% x (1120 – 28R) = Rs. (840 – 21R)

Selling price of tables = 680 + 150 = 830

Now,

20R + (840 – 21R) = 830

Value of R = 10

Now we can find all the data related to tables

	Cost price	Selling price	Marked price	Discount	Profit
Green Chair	270	480	540	60	210
Blue chair	180	200	240	40	20
Green table	200	200	250	50	0
Blue table	600	630	840	210	30

Answer: B

Marked price of pen = 60*130/100 = 78

Discount offered on a pen = (1/10)* 250=Rs. 25

Selling price of a pen = Rs.53

Profit = Rs.20

Cost price of 30 pens = 30*33 = Rs.990

3. Questions

First, we need to find the value of M and N

6M + 7N = 95, where M and N are prime numbers

N = (95 – 6M)/7

If M = 2, N = 8.875 (not possible)

If M = 3, N = 11 (Possible)

If M = 5, N = 9.285 (not possible)

If M = 7, N = 7.571 (not possible)

If M = 11, N = 4.142 (not possible)

If M = 13, N = 2.428 (Not possible)

No other value is possible, which gives M and N both primes.

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Number of chairs manufactured by the company = 11 + 4 = 15

Cost price of each table = (17 x 3 – 11) = Rs. 40

Cost price of each chair = (2 x 11 + 8) = Rs. 30

For chairs

Total number of chairs = 15

Cost price of each chair = Rs. 30

Number of green chairs = 3C% x 15 = 9C/20

Number of blue chairs = (15 – 9C/20)

Marked price of green chairs = 9C/20 x (30 + 30) = 27C

Marked price of blue chairs = 133.33% x 30 x (15 – 9C/20) = (600 – 18C)

Selling price of green chairs = 8/9 x 27C = 24C

Selling price of blue chairs = 5/6 x (600 – 18C) = (500 – 15C)

Total selling price of chairs = Rs. 680

Now,

24C + (500 – 15C) = 680

9C = 180

Value of C = 20

Now we can find all the data related to Chairs

For Tables

Total number of tables = 20

Cost price of each table = Rs. 40

Number of green tables = 2.5R% x 20 = 0.5R

Number of blue tables = (20 – 0.5R)

Mark up% on green tables = (C + 5)% = (20 + 5)% = 25%

So, the marked price of green tables = 40 x 1.25 x 0.5R = 25R

Discount given on green tables = C% = 20%

So, the selling price of green tables = 80% x 25R = 20R

Number of blue tables = (20 – 0.5R)

Mark up % on blue tables = 2C% = 2 x 20 = 40%

So, the marked price of blue tables = 40 x 1.4 x (20 – 0.5R) = (1120 – 28R)

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Discount given on blue tables = (C +5)% = (20 +5)% = 25%

So, the selling price of blue tables = 75% x (1120 – 28R) = Rs. (840 – 21R)

Selling price of tables = 680 + 150 = 830

Now,

20R + (840 – 21R) = 830

Value of R = 10

Now we can find all the data related to tables

	Cost price	Selling price	Marked price	Discount	Profit
Green Chair	270	480	540	60	210
Blue chair	180	200	240	40	20
Green table	200	200	250	50	0
Blue table	600	630	840	210	30

Answer: A

I: R=10

Difference between the total number of chairs and tables = 20-15 =5 which is 5 less than R

This statement is true.

II: Ratio between Marked price of green table to selling price of blue chair = 250:200 = 5:4

This statement is true.

III: Profit earned by selling all the tables is 40 more than the total profit earned by selling all the chairs

This statement is not true.

Only statement I and II is true.

Hence, the option A is correct answer.

4. Questions

Let the speed of buses A and C be 4x and 3x, respectively.

The speed of buses B and D is 5y and 6y, respectively.

So, we can say, 4x*4+5y*4=620, 4x+5y=155.

And also, 3x*4+6y*4=600, 3x+6y=150.

By solving the above equation, x = 20 and y = 15.

So, the speed of buses A and C is 4*20 = 80 and 3*20 = 60 km/hr, respectively.

The speeds of buses B and D is 5*15=75 and 6*15=90 km/hr, respectively.

So, [d+175]/[t+4]=75

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So, d-75t=125,

and also, [d/2+20]/[t-2]=90.

or, d-180t=-400

By solving the above equations, we get d = 500 and t = 5.

Answer: C

So, distance = 2*500-160=840

So, required time = 840/60+840/90=23.33=4*t+3.33

So, only I true

5. Questions

Let the speed of buses A and C be 4x and 3x, respectively.

The speed of buses B and D is 5y and 6y, respectively.

So, we can say, 4x*4+5y*4=620, 4x+5y=155.

And also, 3x*4+6y*4=600, 3x+6y=150.

By solving the above equation, x = 20 and y = 15.

So, the speed of buses A and C is 4*20 = 80 and 3*20 = 60 km/hr, respectively.

The speeds of buses B and D is 5*15=75 and 6*15=90 km/hr, respectively.

So, [d+175]/[t+4]=75

So, d-75t=125,

and also, [d/2+20]/[t-2]=90.

or, d-180t=-400

By solving the above equations, we get d = 500 and t = 5.

Answer: A

Initial distance is [5+2]*80+[5+2]*60=560+420=980

So, required time = 980/90=10.88 hours

6. Questions

Let the speed of buses A and C be 4x and 3x, respectively.

The speed of buses B and D is 5y and 6y, respectively.

So, we can say, 4x*4+5y*4=620, 4x+5y=155.

And also, 3x*4+6y*4=600, 3x+6y=150.

By solving the above equation, x = 20 and y = 15.

So, the speed of buses A and C is 4*20 = 80 and 3*20 = 60 km/hr, respectively.

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The speeds of buses B and D is 5*15=75 and 6*15=90 km/hr, respectively.

So, [d+175]/[t+4]=75

So, d-75t=125,

and also, [d/2+20]/[t-2]=90.

or, d-180t=-400

By solving the above equations, we get d = 500 and t = 5.

Answer: E

500 km distance covered by A in [500/80]=6.25 hours.

[500-250] km distance covered by C in [250/60]=4.16 hours.

500 km distance covered by train D in 500/90 = 5.55 hours.

7. Questions

The amount for each son and daughter is 120000/4=30000.

A invests 30000*60/1000=18000 in scheme P.

A invests 30000-18000=12000 in scheme Q.

B invests 30000/2=15000 in each scheme P and Q.

In scheme Q, C invests 30000*33.33/100=10000.

In scheme P, C invests 30000-10000=20000.

In scheme Q, D invests 30000*45/100=13500.

In scheme P, D invests 30000-13500=16500.

18000*r*3/100- 16500*r*3/100=900 [r is the interest rate provided in P]

Or, 15r=300

Or, r=300/15=20%

Now, in scheme P, interest received by C is 20000*20*2/100=8000.

In scheme Q, interest received by C is 13625-8000=5625

So, let the interest rate in scheme Q is n%.

So, $10000*[1+n/100]^2-10000=5625$

Or, $[1+n/100]^2=[15625/10000]=[625/400]=[25/20]^2$

Or, n/100=5/20

Or, n=25%

Answer: C

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Interest received from Q is = $12000*[1+25/100]^2-12000=6750$

So, interest in new investment is = 6750*6*20/100=Rs.8100

8. Questions

The amount for each son and daughter is 120000/4=30000.

A invests 30000*60/1000=18000 in scheme P.

A invests 30000-18000=12000 in scheme Q.

B invests 30000/2=15000 in each scheme P and Q.

In scheme Q, C invests 30000*33.33/100=10000.

In scheme P, C invests 30000-10000=20000.

In scheme Q, D invests 30000*45/100=13500.

In scheme P, D invests 30000-13500=16500.

18000*r*3/100- 16500*r*3/100=900 [r is the interest rate provided in P]

Or, 15r=300

Or, r=300/15=20%

Now, in scheme P, interest received by C is 20000*20*2/100=8000.

In scheme Q, interest received by C is 13625-8000=5625

So, let the interest rate in scheme Q is n%.

So, $10000*[1+n/100]^2-10000=5625$

Or, $[1+n/100]^2=[15625/10000]=[625/400]=[25/20]^2$

Or, n/100=5/20

Or, n=25%

Answer: B

Interest received from P

=18000*20*2/100=7200

Interest received from Q

=$12000*[1+25/100]^2-12000=18750-12000=6750$

So, difference = m = 7200-6750=450

So, 7m=7*450=3150

9. Questions

The amount for each son and daughter is 120000/4=30000.

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A invests 30000*60/1000=18000 in scheme P.

A invests 30000-18000=12000 in scheme Q.

B invests 30000/2=15000 in each scheme P and Q.

In scheme Q, C invests 30000*33.33/100=10000.

In scheme P, C invests 30000-10000=20000.

In scheme Q, D invests 30000*45/100=13500.

In scheme P, D invests 30000-13500=16500.

18000*r*3/100- 16500*r*3/100=900 [r is the interest rate provided in P]

Or, 15r=300

Or, r=300/15=20%

Now, in scheme P, interest received by C is 20000*20*2/100=8000.

In scheme Q, interest received by C is 13625-8000=5625

So, let the interest rate in scheme Q is n%.

So, $10000*[1+n/100]^2-10000=5625$

Or, $[1+n/100]^2=[15625/10000]=[625/400]=[25/20]^2$

Or, n/100=5/20

Or, n=25%

Answer: A

Interest received by B= $15000*[1+25/100]^3-15000=29296.875-15000=14296.875$

Interest received by D =$13500*[1+25/100]^3-13500=26367.1875-13500=12867.1875$

So, difference = 14296.875-12867.1875=1429.69

10. Questions

The amount for each son and daughter is 120000/4=30000.

A invests 30000*60/1000=18000 in scheme P.

A invests 30000-18000=12000 in scheme Q.

B invests 30000/2=15000 in each scheme P and Q.

In scheme Q, C invests 30000*33.33/100=10000.

In scheme P, C invests 30000-10000=20000.

In scheme Q, D invests 30000*45/100=13500.

In scheme P, D invests 30000-13500=16500.

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18000*r*3/100- 16500*r*3/100=900 [r is the interest rate provided in P]

Or, 15r=300

Or, r=300/15=20%

Now, in scheme P, interest received by C is 20000*20*2/100=8000.

In scheme Q, interest received by C is 13625-8000=5625

So, let the interest rate in scheme Q is n%.

So, $10000*[1+n/100]^2-10000=5625$

Or, $[1+n/100]^2=[15625/10000]=[625/400]=[25/20]^2$

Or, n/100=5/20

Or, n=25%

Answer: A

Total interest received from scheme P

=18000*20*4/100 + 15000*20*4/100 + 20000*20*4/100 + 16500*20*4/100

=14400+12000+16000+13200=55600

Average = 55600/4 = 13900

11. Questions

Ratio of profit share of P, Q, and R in food business = L x 20: (12000 x 24): (16000 x 12)

= 5L: 72:48 [L is in thousands]

Ratio of profit share of P, M, and N in garments business = (Z – L) x 10: 5000 x 12: 6000 x 5 = (Z – L): 6000:3000

Now,

(Z – L) / (Z – L + 6000 + 3000) = 4/13

13 x (Z – L) = 4 x (Z – L) + 36000

(Z – L) = 4000…….... (1)

Answer: D

According to Question,

Z = 28000

So, L = 28000 – 4000 = Rs. 24000

Now,

Ratio of profit share in B1 = 5 x 24: 72: 48 = 5:3:2

So, profit share of P = 5/10 x 1.5 lakhs = Rs. 75000

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Hence answer is option D

12. Questions

Ratio of profit share of P, Q, and R in food business = L x 20: (12000 x 24): (16000 x 12)

= 5L: 72:48 [L is in thousands]

Ratio of profit share of P, M, and N in garments business = (Z – L) x 10: 5000 x 12: 6000 x 5 = (Z – L): 6000:3000

Now,

(Z – L) / (Z – L + 6000 + 3000) = 4/13

13 x (Z – L) = 4 x (Z – L) + 36000

(Z – L) = 4000…….... (1)

Answer: A

Garments business,

Z = Rs. 12000

So, L = Rs. 12000 – 4000 = Rs. 8000

Amount invested by P in garments = Rs. 12000 – 8000 = Rs. 4000

Ratio of profit share = 4:6:3

Profit share of M = 6/13 x 91000 = Rs. 42000

Hence answer is option A

13. Questions

Ratio of profit share of P, Q, and R in food business = L x 20: (12000 x 24): (16000 x 12)

= 5L: 72:48 [L is in thousands]

Ratio of profit share of P, M, and N in garments business = (Z – L) x 10: 5000 x 12: 6000 x 5 = (Z – L): 6000:3000

Now,

(Z – L) / (Z – L + 6000 + 3000) = 4/13

13 x (Z – L) = 4 x (Z – L) + 36000

(Z – L) = 4000…….... (1)

Answer: A

14. Questions

Let the speed of streams of rivers P and Q be 2x and 3x, respectively.

The speed of boat D is 2x*6=12x.

The speed of boat A is 12x*100/133.33=9x.

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The speed of the stream of river R is 3x*4/3=4x.

Downstream of boat A in river R is = [9x+4x] = 13x.

The speed of boat B in still water is 13x.

So, the speed of boat C in still water is 13x*11/13=11x.

Answer: C

The downstream speed of boat D in river R is 12x+4x=16x

So, 16x=32

Or, x=32/16=2

So, downstream of boat C in river Q is [11x+3x]=14*2=28

So, t=140/28=5

So, required distance = 9*2*5=90 km

15. Questions

Let the speed of streams of rivers P and Q be 2x and 3x, respectively.

The speed of boat D is 2x*6=12x.

The speed of boat A is 12x*100/133.33=9x.

The speed of the stream of river R is 3x*4/3=4x.

Downstream of boat A in river R is = [9x+4x] = 13x.

The speed of boat B in still water is 13x.

So, the speed of boat C in still water is 13x*11/13=11x.

Answer: C

The upstream speed of boat A in river P is [9x-2x]=7x

So,7x=14, x=14/7=2 km/hr.

So, the downstream speed of boats B and C in river Q is 26+6=32 and 22+6=28 km/hr respectively.

So, required distance = [32+28]*5.5=330 km

16. Questions

Let the speed of streams of rivers P and Q be 2x and 3x, respectively.

The speed of boat D is 2x*6=12x.

The speed of boat A is 12x*100/133.33=9x.

The speed of the stream of river R is 3x*4/3=4x.

Downstream of boat A in river R is = [9x+4x] = 13x.

The speed of boat B in still water is 13x.

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So, the speed of boat C in still water is 13x*11/13=11x.

Answer: C

So, 12x=36, x=3

So, the speed of boat A is 9*3=27 km/hr

So, d=27*2=54 km

So, the speed of boat B is 13*3=39

So, required distance = 39*5.5=214.5 km=3d+52.5

17. Questions

For work Q,

C alone can complete the work in 18 days.

So, B alone can complete the work in 18*5/3=30 days.

A alone can complete the work in 30*2/3=20 days. 0

Total work = LCM of 18, 30 and 20= 180

The efficiency of A, B and C for work Q is [180/20, 180/30, 180/18]=9,6 and 10 respectively.

So, x=[180/15]=12.

For work P,

The ratio of efficiency between A and B is 5:4.

Let the time taken by A and B be 4m and 5m respectively.

A and C together can complete the work in 15 days. B and C together can complete the work in [10*x]/7 days= 10*12/7=120/7 days.

C's one day work [(1/15)-(1/4m)]

C's one day work [7/120 – 1/5m]

So, [1/15-1/4m]=[7/120 – 1/5m]

Or, 1/20m=1/120

Or, m=6

So, the time taken by A and B is 4*6=24 and 5*6=30 respectively.

C's one day work = 1/15-1/24=3/120=1/40

So, C alone complete the work in 40 days.

Total work = LCM of 24, 30 and 40 =120

The efficiency of A, B and C in work P is 5, 4 and 3 respectively.

When A doubled his original efficiency, then A alone can complete the work in 120/10=12 =[y-6] days.

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So, y=18.

For work R,

B and C take the same number of days to complete the work alone as they takes to complete work P.

So, B and C take 30 and 40 days respectively.

A and B together completed the work in y=18 days.

So, A's one day work = 1/18-1/30=(5-3)/90=1/45

So, A alone completed the work alone in 45 days.

Total work = LCM of 45, 30 and 40 = 360

So, the efficiency of A, B and C is 8, 12 and 9 respectively.

Answer: D

In [y/2 + 1] days A did = 10*5=50 unit work.

Rest work is 120-50=70

So, B and C together complete the work in [70/7]=10 days =x-2=y/2 + 1

18. Questions

For work Q,

C alone can complete the work in 18 days.

So, B alone can complete the work in 18*5/3=30 days.

A alone can complete the work in 30*2/3=20 days. 0

Total work = LCM of 18, 30 and 20= 180

The efficiency of A, B and C for work Q is [180/20, 180/30, 180/18]=9,6 and 10 respectively.

So, x=[180/15]=12.

For work P,

The ratio of efficiency between A and B is 5:4.

Let the time taken by A and B be 4m and 5m respectively.

A and C together can complete the work in 15 days. B and C together can complete the work in [10*x]/7 days= 10*12/7=120/7 days.

C's one day work [(1/15)-(1/4m)]

C's one day work [7/120 – 1/5m]

So, [1/15-1/4m]=[7/120 – 1/5m]

Or, 1/20m=1/120

Or, m=6

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So, the time taken by A and B is 4*6=24 and 5*6=30 respectively.

C's one day work = 1/15-1/24=3/120=1/40

So, C alone complete the work in 40 days.

Total work = LCM of 24, 30 and 40 =120

The efficiency of A, B and C in work P is 5, 4 and 3 respectively.

When A doubled his original efficiency, then A alone can complete the work in 120/10=12 =[y-6] days.

So, y=18.

For work R,

B and C take the same number of days to complete the work alone as they takes to complete work P.

So, B and C take 30 and 40 days respectively.

A and B together completed the work in y=18 days.

So, A's one day work = 1/18-1/30=(5-3)/90=1/45

So, A alone completed the work alone in 45 days.

Total work = LCM of 45, 30 and 40 = 360

So, the efficiency of A, B and C is 8, 12 and 9 respectively.

Answer: B

I). B and C work for work R with two times of their original efficiency.

[360/(12*2+9*2)]=360/42=8.57

II). A and C work in work Q.

=[180/19]=9.47

III). A and B work in work P.

=[120/9]=13.33

19. Questions

For work Q,

C alone can complete the work in 18 days.

So, B alone can complete the work in 18*5/3=30 days.

A alone can complete the work in 30*2/3=20 days. 0

Total work = LCM of 18, 30 and 20= 180

The efficiency of A, B and C for work Q is [180/20, 180/30, 180/18]=9,6 and 10 respectively.

So, x=[180/15]=12.

For work P,

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The ratio of efficiency between A and B is 5:4.

Let the time taken by A and B be 4m and 5m respectively.

A and C together can complete the work in 15 days. B and C together can complete the work in [10*x]/7 days= 10*12/7=120/7 days.

C's one day work [(1/15)-(1/4m)]

C's one day work [7/120 – 1/5m]

So, [1/15-1/4m]=[7/120 – 1/5m]

Or, 1/20m=1/120

Or, m=6

So, the time taken by A and B is 4*6=24 and 5*6=30 respectively.

C's one day work = 1/15-1/24=3/120=1/40

So, C alone complete the work in 40 days.

Total work = LCM of 24, 30 and 40 =120

The efficiency of A, B and C in work P is 5, 4 and 3 respectively.

When A doubled his original efficiency, then A alone can complete the work in 120/10=12 =[y-6] days.

So, y=18.

For work R,

B and C take the same number of days to complete the work alone as they takes to complete work P.

So, B and C take 30 and 40 days respectively.

A and B together completed the work in y=18 days.

So, A's one day work = 1/18-1/30=(5-3)/90=1/45

So, A alone completed the work alone in 45 days.

Total work = LCM of 45, 30 and 40 = 360

So, the efficiency of A, B and C is 8, 12 and 9 respectively.

Answer: A

Total time taken by A, B and C together to complete work P

= [120/(5+4+3)]=10

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Total time taken by A, B and C together to complete work R

= [360/(8+12+9)]=12.41=12

So, the difference is 12-10=2

20. Questions

For work Q,

C alone can complete the work in 18 days.

So, B alone can complete the work in 18*5/3=30 days.

A alone can complete the work in 30*2/3=20 days. 0

Total work = LCM of 18, 30 and 20= 180

The efficiency of A, B and C for work Q is [180/20, 180/30, 180/18]=9,6 and 10 respectively.

So, x=[180/15]=12.

For work P,

The ratio of efficiency between A and B is 5:4.

Let the time taken by A and B be 4m and 5m respectively.

A and C together can complete the work in 15 days. B and C together can complete the work in [10*x]/7 days= 10*12/7=120/7 days.

C's one day work [(1/15)-(1/4m)]

C's one day work [7/120 – 1/5m]

So, [1/15-1/4m]=[7/120 – 1/5m]

Or, 1/20m=1/120

Or, m=6

So, the time taken by A and B is 4*6=24 and 5*6=30 respectively.

C's one day work = 1/15-1/24=3/120=1/40

So, C alone complete the work in 40 days.

Total work = LCM of 24, 30 and 40 =120

The efficiency of A, B and C in work P is 5, 4 and 3 respectively.

When A doubled his original efficiency, then A alone can complete the work in 120/10=12 =[y-6] days.

So, y=18.

For work R,

B and C take the same number of days to complete the work alone as they takes to complete work P.

So, B and C take 30 and 40 days respectively.

A and B together completed the work in y=18 days.

So, A's one day work = 1/18-1/30=(5-3)/90=1/45

So, A alone completed the work alone in 45 days.

Total work = LCM of 45, 30 and 40 = 360

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So, the efficiency of A, B and C is 8, 12 and 9 respectively.

Answer: A

So, z*8+[z+5] *12=360

Or, z=300/20=15

So, x+y+z=18+12+15 = 45

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1. Questions

Study the following data carefully and answer the questions:

In a shop, ratio of the cost price of article A to that of article B is 5: 6. The shopkeeper marked up article A by 60%, article B by 50% and sold articles A and B at ₹8000 and ₹9000 discount on their respective marked prices. The total profit, received by the shopkeeper by selling both the articles, is invested in a scheme and the shopkeeper received ₹2625 simple interest from the scheme at the rate of 12.5% after 3 years.

What is the approximate average of the profit received on article A, the profit received on article B and the simple interest received from the scheme?

a. ₹3309

b. ₹3107

c. ₹3208

d. ₹3410

e. ₹3006

2. Questions

If the discount given on articles A and B were interchanged, then what would have been the ratio of the profit percentage on article A to the profit percentage on article B?

a. Can't be determined

b. 1: 5

c. 5: 12

d. 9: 10

e. 3: 5

3. Questions

If the original profit percentage on article A is M% and the original profit percentage on article B is N%, then find the value of (M* N)/(M – N)?

a. 20

b. 50

c. 16.67

d. 25

e. 33.33

4. Questions

Study the following data carefully and answer the questions:

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There are two mixtures A and B. Mixture A contains mango juice and milk and mixture B contains milk and water. In mixture A, the quantity of milk is 11 L less than that of mango juice and also 1 L more than the quantity of milk in mixture B. The ratio of milk to water in mixture B is 3: 2, the cost of the mixture A is ₹70.8 per L and the cost of mixture B ₹36 per L. The ratio of the cost of mango juice to the cost of the pure milk is 5: 4.

Which of the following statements is/are not true?

P: Total quantity of mixture B is 40% of the total quantity of mixture A.

Q: Ratio of mango juice in mixture A to milk in mixture B is 3: 1.

R: Average quantity of milk in mixtures A and B is 13 L.

a. All are true

b. Only Q

c. Only R

d. Only P and R

e. Only Q and R

5. Questions

When mixture B is evaporated (only the whole quantity of water evaporates) and then mixed with mixture A, then what will be the approximate cost of the new mixture?

a. ₹68.7 per L

b. ₹74.7 per L

c. ₹55.7 per L

d. ₹42.7 per L

e. ₹62.7 per L

6. Questions

If 5 L milk is added to mixture A, then what will be the ratio of initial cost of mixture A to the new cost of mixture A?

a. 118: 115

b. 128: 125

c. 108: 105

d. 98: 95

e. 138: 135

7. Questions

Study the following data carefully and answer the questions:

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Data given below is related to the production (in units) of articles A and B in 6 months from Jan to June.

Bar graph given below shows the average production of article A in Jan and Feb, average production of article A in Feb and Mar, average production of article A in Mar and Apr, average production of article A in Apr and May and average production of article A in May and June.

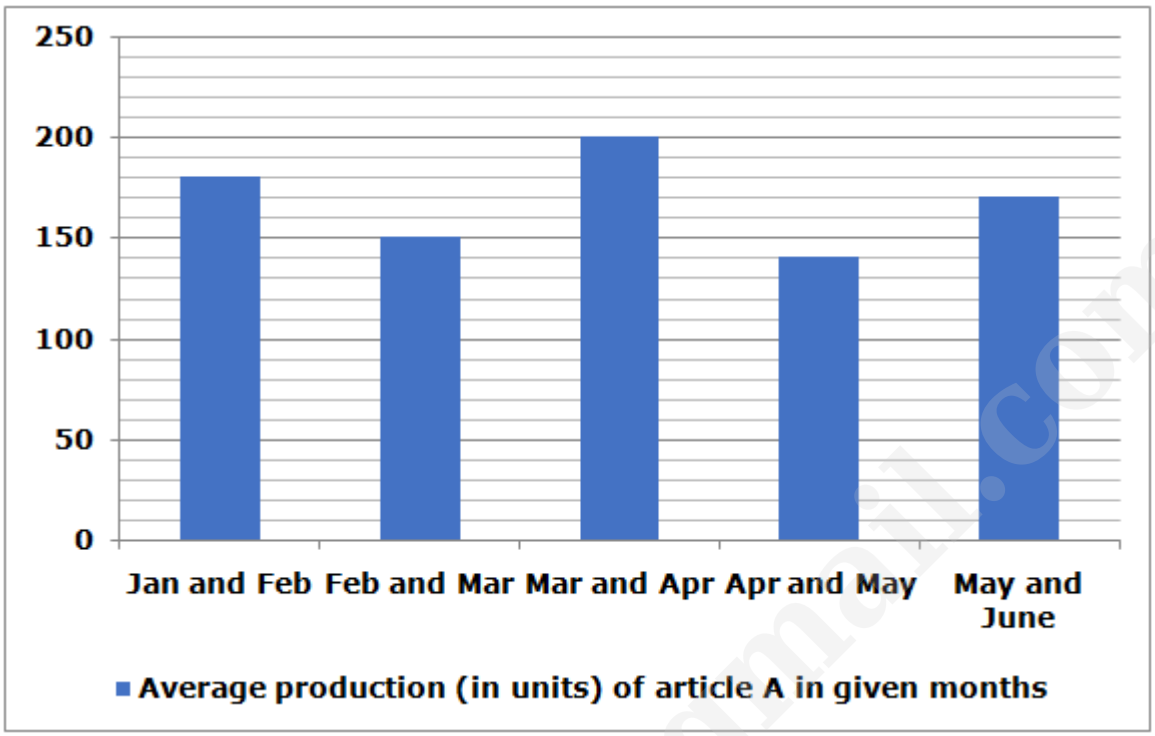


Table given below shows the average production (in units) of articles A and B in each of the given months.

Month	Average production (in units) of articles A and B
Jan	185
Feb	135
Mar	160
Apr	165
May	105
June	230

Note: Production of article B in Jan is 54(1/6)% of the production of article A in Jan.

If the ratio of the production of article A in Mar to the production of article B in Mar is p: q and the ratio of the production of article A in Apr to the production of article B in Apr is m: n, then find the value of [pq/(m + n)]?

a. 21

b. 33

c. 18

d. 30

e. 24

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8. Questions

If the production cost of article A is ₹150 per unit and the production cost of article B is ₹200per unit, then find the total production cost of article A in June is what per cent more than the total production cost of article B in June?

a. 12.5%

b. 16.67%

c. 10%

d. 20%

e. 6.67%

9. Questions

In which of the following options, the percentage increase/decrease in the production of article B is maximum?

a. From Jan to Feb

b. From Feb to Mar

c. From Mar to Apr

d. From Apr to May

e. From May to June

10. Questions

If a series is formed, whose 1st term is the production of article B in Apr, 2nd term is the production of article A in Feb, 3rd term is the production of article B in Mar, 4th term is the production of article A in Mar and 5th term is the total production of article B in Feb and Apr together, then find the 6th term of the series?

a. 400

b. 480

c. 450

d. 420

e. 460

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11. Questions

If the production of article A is increased by 25% from June to July and the production of article B is decreased by 25% from June to July, then find the ratio of total production of articles A and B together in June to that on July?

a. 23: 27

b. 115: 123

c. 92: 97

d. 46: 47

e. 138: 143

12. Questions

Persons A and B can do a piece of work in 'x' days and 'x + 8' days respectively while C alone can do the work in 'y' days. Together they can finish the work in 5 days. If x < y < (x + 8), then how many values of (x, y) are possible.

a. 1

b. 5

c. 2

d. 4

e. 3

13. Questions

In a 90 litres mixture of orange juice, water. The quantity of water is 11 (1/9)%. Find the ratio of orange juice to water in the final mixture when 'x' litters of the mixture is taken out and replaced with same quantity of the water.

Note: 'x' is 2-digits perfect cube number and 2 more than perfect square of an integer.

a. 28: 17

b. 3: 2

c. 5: 4

d. 11: 7

e. 19: 11

14. Questions

In a garden, shaped like a right-angled triangle, there are curved arc flower areas at each corner of the garden, all with the same radius. The rest of the garden is for vegetation. If the circles along the shorter side touch each other, and the longest side of the garden is 26 meters long, while there a 14-meter gap between circles on the middle side of the garden, find the area of that part of the garden which is used for vegetation. (Use $\pi = 3.14$)

a. 92.25 m^2

b. 82.75 m^2

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c. 78.5 m^2

d. 80.75 m^2

e. 72.5 m^2

15. Questions

Cost price of item A is ₹400 more than the cost price of item B. If items A and B are marked up 25% and 50% of their respective cost prices. If difference between marked price of both the items is ₹100, then find the profit amount earned after selling both the items when both the items are sold at 20% profit.

a. ₹720

b. ₹1040

c. Either ₹1040 or ₹720

d. ₹800

e. Either ₹720 or ₹800

16. Questions

Study the following data carefully and answer the questions:

Table given below shows the data regarding the number of classes taken by five teachers A, B, C, D, and E for an institute and it also shows the average fee (in Rs.) per class charged by these teachers from the institute.

Teachers	Total number of classes taken	Average fee (in Rs.) per class
A	20% of total classes taken by all the five teachers together.	750
B	Total number of classed taken by A and B together is 35.	Average fees per class charged by teachers B and C together is Rs.700.
C	Twice the number of classes taken by B.	Average fees per class charged by teachers B and D together is Rs.950.
D	10 more than the number of classes taken by B and 5 less than the number of classes taken by C.	----
E	----	Total fees charged by E is Rs.5000.

Note:

1). Average fees per class charged by all the five teachers together is Rs.715.

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2). Profit earned by institute = Total fees collected by the institutes – Total fees charged by the teachers

The institute charges 'm' rupees from each boy and 'm – 6' rupees from each girl for all the classes taken by all the five teachers together such that 'm' is the smallest 3-digits integer which have exactly 3 factors (excluding 1). If institutes earn a total profit of Rs.16200 and ratio of total boys to girls is 9: 8, then find the total number of students in the institute.

a. 750

b. 960

c. 800

d. 850

e. 900

17. Questions

Find the average fee per class charged by teachers B, D, and E together.

a. 860

b. 920

c. 840

d. 820

e. 880

18. Questions

If the values that will fill the missing value of column II (Total number of classes taken) and column III (Average fee (in ₹) per class by D) are P and Q respectively, then find the value of:

$$\sqrt{\frac{\sqrt{P*(P+Q)}}{10}}$$

a. 5

b. 4

c. 7

d. 6

e. 3

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19. Questions

If in first 12 classes of teacher C, average views per class was 2400 (A) and in the remaining classes, average views per class was 2000 (B) and average views per class in all the classes taken by him was 2200.

Which of the following value(s) marked as A and B will be replaced by the values given below such that the question becomes correct.

Statement I: A = 2200

Statement II: B = 2200

Statement III: A = 2500

a. Only statements II and III together

b. Either statement III alone or statements I and II together

c. Only statement I alone.

d. Only statement III alone.

e. Either statement I alone or statements II and III together

20. Questions

If teachers A and E both took 'x' number of classes each for another institute such that the number of classes taken by A in both the institutes together is 40% more than that by E, and their fee per class is same in both the institutes, then total fee collected by both the teachers from institute B is how many times of the value of '5x'?

a. 300

b. 400

c. 250

d. 200

e. 350

Explanations:

1. Questions

Let the total profit received by the shopkeeper by selling both the articles be ₹x.

Since, the shopkeeper invested the profit in a scheme and received ₹2625 simple interest from the scheme at 12.5% rate after 3 years.

So,

$(x \times 12.5 \times 3)/100=2625$

$x = 7000$

Let the CPs of articles A and B be ₹5y and ₹6y respectively.

So, the MP of article A = 160% of 5y = ₹8y

The MP of article B = 150% of 6y = ₹9y

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The SP of article A = ₹(8y – 8000)

And the SP of article B = ₹(9y – 9000)

Since, the profit received from article A = (8y – 8000) – 5y = ₹(3y – 8000)

And the profit received from article B = (9y – 9000) – 6y = ₹(3y – 9000)

So,

(3y – 8000) + (3y – 9000) = 7000

6y = 24000

y = 4000

The CP of article A = 5 * 4000 = ₹20000

The MP of article A = 160% of 20000 = ₹32000

The SP of article A = 32000 – 8000 = ₹24000

The CP of article B = 6 * 4000 = ₹24000

The MP of article B = 150% of 24000 = ₹36000

The SP of article B = 36000 – 9000 = ₹27000

Answer: C

The profit received on article A = 24000 – 20000 = ₹4000

The profit received on article B = 27000 – 24000 = ₹3000

The SI received from the scheme = ₹2625

Required average = (4000+3000+2625)/3 = ₹3208 (approx.)

2. Questions

Let the total profit received by the shopkeeper by selling both the articles be ₹x.

Since, the shopkeeper invested the profit in a scheme and received ₹2625 simple interest from the scheme at 12.5% rate after 3 years.

So,

(x ×12.5 ×3)/100=2625

x = 7000

Let the CPs of articles A and B be ₹5y and ₹6y respectively.

So, the MP of article A = 160% of 5y = ₹8y

The MP of article B = 150% of 6y = ₹9y

The SP of article A = ₹(8y – 8000)

And the SP of article B = ₹(9y – 9000)

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Since, the profit received from article A = (8y – 8000) – 5y = ₹(3y – 8000)

And the profit received from article B = (9y – 9000) – 6y = ₹(3y – 9000)

So,

(3y – 8000) + (3y – 9000) = 7000

6y = 24000

y = 4000

The CP of article A = 5 * 4000 = ₹20000

The MP of article A = 160% of 20000 = ₹32000

The SP of article A = 32000 – 8000 = ₹24000

The CP of article B = 6 * 4000 = ₹24000

The MP of article B = 150% of 24000 = ₹36000

The SP of article B = 36000 – 9000 = ₹27000

Answer: D

The CP of article A = 5 * 4000 = ₹20000

The MP of article A = 160% of 20000 = ₹32000

The CP of article B = 6 * 4000 = ₹24000

The MP of article B = 150% of 24000 = ₹36000

The discount given on article A = ₹8000

The discount given on article B = ₹9000

If the given discount were interchanged:

The SP of article A would have been = 32000 – 9000 = ₹23000

And the profit percentage on article A would have been:

(23000-20000)/20000 ×100=15%

The SP of article B would have been = 36000 – 8000 = ₹28000

And the profit percentage on article B would have been:

$$\frac{28000-24000}{24000} \times 100 = \frac{50}{3}\%$$

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Required ratio = 15: (50/3) = 9: 10

3. Questions

Let the total profit received by the shopkeeper by selling both the articles be ₹x.

Since, the shopkeeper invested the profit in a scheme and received ₹2625 simple interest from the scheme at 12.5% rate after 3 years.

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So,

$(x \times 12.5 \times 3)/100=2625$

$x = 7000$

Let the CPs of articles A and B be ₹5y and ₹6y respectively.

So, the MP of article A = 160% of 5y = ₹8y

The MP of article B = 150% of 6y = ₹9y

The SP of article A = ₹(8y – 8000)

And the SP of article B = ₹(9y – 9000)

Since, the profit received from article A = (8y – 8000) – 5y = ₹(3y – 8000)

And the profit received from article B = (9y – 9000) – 6y = ₹(3y – 9000)

So,

(3y – 8000) + (3y – 9000) = 7000

6y = 24000

y = 4000

The CP of article A = 5 * 4000 = ₹20000

The MP of article A = 160% of 20000 = ₹32000

The SP of article A = 32000 – 8000 = ₹24000

The CP of article B = 6 * 4000 = ₹24000

The MP of article B = 150% of 24000 = ₹36000

The SP of article B = 36000 – 9000 = ₹27000

Answer: E

Since, the CP of article A = ₹20000

And the SP of article A = ₹24000

So, M = $(24000-20000)/20000 \times 100 = 20\%$

Since, the CP of article B = ₹24000

And the SP of article B = ₹27000

So, N = $(27000-24000)/24000 \times 100 = 12.5\%$

The value of (M * N)/(M – N) = (20 * 12.5)/(20 – 12.5) = 33.33

4. Questions

Since, the ratio of milk to water in mixture B is 3: 2.

Let the quantities of milk and water in mixture B be '3x' L and '2x' L respectively.

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So, the quantity of milk in mixture A = (3x + 1) L

And the quantity of mango juice in mixture A = (3x + 1 + 11) = (3x + 12) L

Since, the ratio of the cost of mango juice to the cost of pure milk is 5: 4.

So, let the costs of mango juice and pure milk be ₹5y per L and ₹4y per L respectively.

Since, the cost of mixture A is ₹70.8 per L.

So,

$$\frac{(3x+12) \times 5y + (3x+1) \times 4y}{(6x+13)} = 70.8 - - - - - - - - - - (1)$$

Since, the cost of mixture B is ₹36 per L.

So,

$$\frac{3x \times 4y}{3x+2x} = 36$$

y = 15

From equation (1):

$$\frac{(3x+12) \times 5 \times 15 + (3x+1) \times 4 \times 15}{(6x+13)} = 70.8$$

225x + 900 + 180x + 60 = 424.8x + 920.4

19.8x = 39.6

x = 2

The quantity of mango juice in mixture A = 3 * 2 + 12 = 18 L

The quantity of milk in mixture A = 3 * 2 + 1 = 7 L

The quantity of milk in mixture B = 3 * 2 = 6 L

The quantity of water in mixture B = 2 * 2 = 4 L

The cost of mango juice = 5 * 15 = ₹75 per L

The cost of pure milk = 4 * 15 = ₹60 per L

Answer: C

From P:

Total quantity of mixture A = 18 + 7 = 25 L

Total quantity of mixture B = 6 + 4 = 10 L

Required percentage = (10/25) * 100 = 40%

So, P is true.

From Q:

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Ratio of mango juice in mixture A to milk in mixture B = 18: 6 = 3: 1

So, Q is true.

From R:

Average quantity of milk in mixtures A and B = (7 + 6)/2 = 6.5 L

So, R is not true.

Hence, only R is not true.

5. Questions

Since, the ratio of milk to water in mixture B is 3: 2.

Let the quantities of milk and water in mixture B be '3x' L and '2x' L respectively.

So, the quantity of milk in mixture A = (3x + 1) L

And the quantity of mango juice in mixture A = (3x + 1 + 11) = (3x + 12) L

Since, the ratio of the cost of mango juice to the cost of pure milk is 5: 4.

So, let the costs of mango juice and pure milk be ₹5y per L and ₹4y per L respectively.

Since, the cost of mixture A is ₹70.8 per L.

So,

$$\frac{(3x+12)\times 5y+(3x+1)\times 4y}{(6x+13)}=70.8 \text{ ---------- (1)}$$

Since, the cost of mixture B is ₹36 per L.

So,

$$\frac{3x\times 4y}{3x+2x}=36$$

y = 15

From equation (1):

$$\frac{(3x+12)\times 5\times 15+(3x+1)\times 4\times 15}{(6x+13)}=70.8$$

225x + 900 + 180x + 60 = 424.8x + 920.4

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x = 2

The quantity of mango juice in mixture A = 3 * 2 + 12 = 18 L

The quantity of milk in mixture A = 3 * 2 + 1 = 7 L

The quantity of milk in mixture B = 3 * 2 = 6 L

The quantity of water in mixture B = 2 * 2 = 4 L

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The cost of mango juice = 5 * 15 = ₹75 per L

The cost of pure milk = 4 * 15 = ₹60 per L

Answer: A

The quantities of milk and water in mixture B are 6 L and 4 L respectively.

When mixture B is evaporated until the whole quantity of water evaporates:

The quantity of mixture B = The quantity of milk in mixture B = 6 L

The quantities of mango juice and milk in mixture A are 18 L and 7 L respectively.

The quantity of mango juice in the new mixture = 18 L

The quantity of milk in the new mixture = 7 + 6 = 13 L

The cost of mango juice = ₹75 per L

The cost of pure milk = ₹60 per l

So, the cost of the new mixture:

$$\frac{18 \times 75 + 13 \times 60}{18 + 13} = ₹68.7 \text{ per L (approx)}$$

6. Questions

Since, the ratio of milk to water in mixture B is 3: 2.

Let the quantities of milk and water in mixture B be '3x' L and '2x' L respectively.

So, the quantity of milk in mixture A = (3x + 1) L

And the quantity of mango juice in mixture A = (3x + 1 + 11) = (3x + 12) L

Since, the ratio of the cost of mango juice to the cost of pure milk is 5: 4.

So, let the costs of mango juice and pure milk be ₹5y per L and ₹4y per L respectively.

Since, the cost of mixture A is ₹70.8 per L.

So,

$$\frac{(3x+12) \times 5y + (3x+1) \times 4y}{(6x+13)} = 70.8 - - - - - - - - - - (1)$$

Since, the cost of mixture B is ₹36 per L.

So,

$$\frac{3x \times 4y}{3x + 2x} = 36$$

y = 15

From equation (1):

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$$\frac{(3x+12)\times 5\times 15+(3x+1)\times 4\times 15}{(6x+13)}=70.8$$

225x + 900 + 180x + 60 = 424.8x + 920.4

19.8x = 39.6

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The quantity of mango juice in mixture A = 3 * 2 + 12 = 18 L

The quantity of milk in mixture A = 3 * 2 + 1 = 7 L

The quantity of milk in mixture B = 3 * 2 = 6 L

The quantity of water in mixture B = 2 * 2 = 4 L

The cost of mango juice = 5 * 15 = ₹75 per L

The cost of pure milk = 4 * 15 = ₹60 per L

Answer: A

The quantity of mango juice in mixture A = 18 L

The initial quantity of milk in mixture A = 7 L

The new quantity of milk in mixture A = 7 + 5 = 12 L

The cost of mango juice = ₹75 per L

The cost of pure milk = ₹60 per L

So, the new cost of mixture A:

$$\frac{18\times 75+12\times 60}{18+12}=₹69 \text{ per L}$$

Since, the initial cost of mixture A = ₹70.8 per L

So, the required ratio = 70.8: 69 = 118: 115

7. Questions

Let the production of article A in Jan be ‘x’ units.

So, the production of article B in Jan = 54(1/6)% of x = (13x/24) units

Since, the average production of articles A and B n Jan is 185 units.

So,

x + (13x/24) = 2 * 185

37x = 8880

x = 240

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Month	Production of article A (in units)	Production of article B (in units)
Jan	240	2 * 185 – 240 = 130
Feb	2 * 180 – 240 = 120	2 * 135 – 120 = 150
Mar	2 * 150 – 120 = 180	2 * 160 – 180 = 140
Apr	2 * 200 – 180 = 220	2 * 165 – 220 = 110
May	2 * 140 – 220 = 60	2 * 105 – 60 = 150
June	2 * 170 – 60 = 280	2 * 230 – 280 = 180

Answer: A

The ratio of the production of article A in Mar to the production of article B in Mar = p: q = 180: 140 = 9: 7

The ratio of the production of article A in Apr to the production of article B in Apr = m: n = 220: 110 = 2: 1

So, the value of [pq/(m + n)] = (9 * 7)/(2 + 1) = 21

8. Questions

Let the production of article A in Jan be 'x' units.

So, the production of article B in Jan = 54(1/6)% of x = (13x/24) units

Since, the average production of articles A and B n Jan is 185 units.

So,

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Apr	2 * 200 – 180 = 220	2 * 165 – 220 = 110
May	2 * 140 – 220 = 60	2 * 105 – 60 = 150
June	2 * 170 – 60 = 280	2 * 230 – 280 = 180

Answer: B

Since, the production cost of article A = ₹150 per unit

So, the total production cost of article A in June = 150 * 280 = ₹42000

Since, the production cost of article B = ₹200 per unit

So, the total production cost of article B in June = 200 * 180 = ₹36000

Required percentage = (42000-36000)/36000 ×100 = 16.67%

9. Questions

Let the production of article A in Jan be 'x' units.

So, the production of article B in Jan = 54(1/6)% of x = (13x/24) units

Since, the average production of articles A and B n Jan is 185 units.

So,

x + (13x/24) = 2 * 185

37x = 8880

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Month	Production of article A (in units)	Production of article B (in units)
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Apr	2 * 200 – 180 = 220	2 * 165 – 220 = 110
May	2 * 140 – 220 = 60	2 * 105 – 60 = 150
June	2 * 170 – 60 = 280	2 * 230 – 280 = 180

Answer: D

The percentage increase in the production of article B from Jan to Feb:

(150-130)/130 ×100=15.4% (approx)

The percentage decrease in the production of article B from Feb to Mar:

(150-140)/150 ×100=6.67%

The percentage decrease in the production of article B from Mar to Apr:

(140-110)/140 ×100=21.4% (approx)

The percentage increase in the production of article B from Apr to May:

(150-110)/110 ×100=36.4% (approx)

The percentage increase in the production of article B from May to June:

(180-150)/150 ×100=20%

So, the percentage increase/decrease in the production of article B is maximum from Apr to May.

10. Questions

Let the production of article A in Jan be 'x' units.

So, the production of article B in Jan = 54(1/6)% of x = (13x/24) units

Since, the average production of articles A and B n Jan is 185 units.

So,

x + (13x/24) = 2 * 185

37x = 8880

x = 240

Month	Production of article A (in units)	Production of article B (in units)
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Apr	2 * 200 – 180 = 220	2 * 165 – 220 = 110
May	2 * 140 – 220 = 60	2 * 105 – 60 = 150
June	2 * 170 – 60 = 280	2 * 230 – 280 = 180

Answer: D

The 1st term of the series = 110

The 2nd term of the series = 120

The 3rd term of the series = 140

The 4th term of the series = 180

The 5th term of the series = 150 + 110 = 260

Logic in the series:

110 + 10 = 120

120 + 20 = 140

140 + 40 = 180

180 + 80 = 260

So, the 6th term of the series = 260 + 160 = 420

11. Questions

Let the production of article A in Jan be ‘x’ units.

So, the production of article B in Jan = 54(1/6)% of x = (13x/24) units

Since, the average production of articles A and B n Jan is 185 units.

So,

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x + (13x/24) = 2 * 185

37x = 8880

x = 240

Month	Production of article A (in units)	Production of article B (in units)
Jan	240	2 * 185 – 240 = 130
Feb	2 * 180 – 240 = 120	2 * 135 – 120 = 150
Mar	2 * 150 – 120 = 180	2 * 160 – 180 = 140
Apr	2 * 200 – 180 = 220	2 * 165 – 220 = 110
May	2 * 140 – 220 = 60	2 * 105 – 60 = 150
June	2 * 170 – 60 = 280	2 * 230 – 280 = 180

Answer: C

Since, the production of article A in June = 280 units

So, the production of article A in July = 125% of 280 = 350

Since, the production of article B in June = 180 units

So, the production of article B in July = 75% of 180 = 135

Required ratio = (280 + 180): (350 + 135) = 460: 485 = 92: 97

12. Questions

Answer: E

According to the question:

$$\frac{1}{x}+\frac{1}{x+8}+\frac{1}{y}=\frac{1}{5}$$

$$\frac{1}{y}=\frac{1}{5}-\frac{1}{x}-\frac{1}{x+8}$$

$$\frac{1}{y}=\frac{x(x+8)-5(x+8)-5x}{5x(x+8)}$$

$$\frac{1}{y}=\frac{x^2-2x-40}{5x(x+8)}$$

$$y=\frac{5x(x+8)}{x^2-2x-40}$$

Given:

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$x < y$

$$x < \frac{5x(x+8)}{x^2-2x-40}$$

$x^2 - 2x - 40 < 5x + 40$

$x^2 - 7x < 80$

$x(x-7) < 80$ ..... (1)

Also,

$Y < (x + 8)$

$$\frac{5x(x+8)}{x^2-2x-40} < (x+8)$$

$5x < x^2 - 2x - 40$

$x^2 - 7x > 40$

$x(x-7) > 40$ ..... (2)

After combining both the equations together:

$40 < x(x-7) < 80$

Product of two numbers lie between 40 and 80 and also value of 'x' must be non-zero positive numbers as number of days cannot be negative.

Also, $(x - 7)$ must be greater than zero because $x(x - 7) > 40$.

$x - 7 > 0 \Rightarrow x > 7$

Let x = 8, then: $40 < x(x-7) < 80 \Rightarrow 40 < 8 < 80$ [Equality not holding true]

Let x = 9, then: $40 < x(x-7) < 80 \Rightarrow 40 < 18 < 80$ [Equality not holding true]

Let x = 10, then: $40 < x(x-7) < 80 \Rightarrow 40 < 30 < 80$ [Equality not holding true]

Let x = 11, then: $40 < x(x-7) < 80 \Rightarrow 40 < 44 < 80$ [Equality holding true]

Let x = 12, then: $40 < x(x-7) < 80 \Rightarrow 40 < 60 < 80$ [Equality holding true]

Let x = 13, then: $40 < x(x-7) < 80 \Rightarrow 40 < 78 < 80$ [Equality holding true]

Let x = 14, then: $40 < x(x-7) < 80 \Rightarrow 40 < 98 < 80$ [Equality not holding true]

So, possible values of x are 11, 12, and 13.

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Possible values of y are 1045/59, 15, and 1365/103.

Hence, total 3 values of (x, y) are possible.

13. Questions

Answer: A

Possible values of perfect square of integers to make 'x' is a 2-digit number = 9, 16, 25, 36, 49, 64, and 81.

Possible value of 'x' = 11, 18, 27, 38, 51, 66, and 83.

Out of the above possible values of 'x' only 27 is a perfect cube.

So, x = 27

Quantity of water in the initial mixture = $11\frac{1}{9}\%$ of 90 = 10 L

Quantity of orange juice in the initial mixture = 90 – 10 = 80 L

Ratio of orange juice and water in the initial mixture = 80: 10 = 8: 1

Quantity of orange juice when 'x' L of the mixture is taken out = 80 – 27 * 8/9 = 80 – 24 = 56 L

Quantity of water when 'x' L of the mixture is taken out and replace with same quantity of water = 10 – 27 * 1/9 + 27 = 10 – 3 + 27 = 34 L

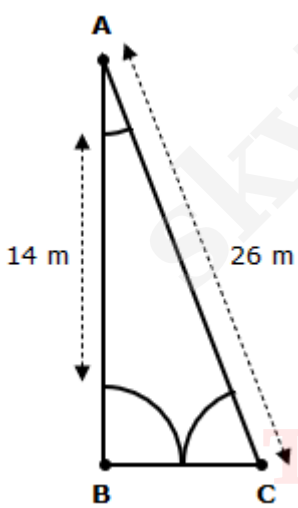
Required ratio = 56: 34

= 28: 17

14. Questions

Answer: D

The garden and area used for flower and vegetation will look like the one given below:



Let radius of each circular section = 'r' m

Now,

BC = r + r = 2r

AB = r + 14 + r = (2r + 14) m

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AC = 26 m

In a right-angled triangle:

$AC^2 = AB^2 + BC^2$

$26^2 = (2r + 14)^2 + (2r)^2$

$676 = 4r^2 + 196 + 56r + 4r^2$

$8r^2 + 56r – 480 = 0$

$r^2 + 7r – 60 = 0$

$r^2 + 12r – 5r – 60 = 0$

$r(r + 12) – 5(r + 12) = 0$

$(r + 12) (r – 5) = 0$

r = 5 (As 'r' cannot take negative values.)

AB = 24 m

BC = 10 m

Total angle formed by all the three circular sections together = 180° = Sum of all the interior angles of a triangle.

A circle makes 360° of angle. Which means area of circle with angle 180° will be half of the area of circle with angle 360°.

So, area used for vegetation = Area of triangle – Area of three circular sections

$= (1/2) * 24 * 10 - (1/2) * \pi * 5^2$

= 120 – 39.25

$= 80.75\ m^2$

15. Questions

Answer: C

Let cost price of item A and B is 'x + 400' and 'x' respectively.

Marked price of item A = 125% of (x + 400) = (1.25x + 500)

Marked price of item B = 150% of x = 1.5x

Case 1:

1.5x – (1.25x + 500) = 100

0.25x = 600

x = 2400

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Total cost price of both the items together = (x + 400) + x = ₹5200

Required profit amount = 20% of 5200 = ₹1040

Case 2:

(1.25x + 500) – 1.5x = 100

0.25x = 400

x = 1600

Total cost price of both the items together = (x + 400) + x = ₹3600

Required profit amount = 20% of 3600 = ₹720

Hence, profit amount can be both ₹1040 and ₹720.

16. Questions

Let number of classes taken by teachers A and B are '35 – x' and 'x' respectively.

Number of classes taken by C = 2x

Number of classes taken by D = x + 10 = 2x – 5

x = 15

Number of classed taken by A = 35 – x = 20

Number of classed taken by B = x = 15

Number of classed taken by C = 2x = 30

Number of classed taken by D = x + 10 = 25

Total number of classes taken by all the five teachers together = 20 * 100/20 = 100

Number of classed taken by E = 100 – 20 – 15 – 30 – 25 = 10

Total fees charged by A = 20 * 750 = Rs.15000

Average fee per class charged by E = 5000/10 = Rs.500

Let average fee per class charged by B, C, and D are 'b', 'c', and 'd' respectively.

According to the question:

$$\frac{15b + 30c}{15 + 30} = 700$$

15b + 30c = 31500 …… (1)

$$\frac{15b + 25d}{15 + 25} = 950$$

15b + 25d = 38000 …….. (2)

$$\frac{15000 + 15b + 30c + 25d + 5000}{100} = 715$$

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15b + 30c + 25d = 51500 …….. (3)

From (1) and (3):

25d = 51500 – 31500

25d = 20000

d = 800

From equation (2):

15b + 20000 = 38000

15b = 18000

b = 1200

From equation (1):

18000 + 30c = 31500

30c = 13500

c = 450

Teachers	Total number of classes taken	Average fee (in Rs.) per class
A	20	750
B	15	1200
C	30	450
D	25	800
E	10	500

Answer: D

Smallest 3-digits integer is 100 and smallest 3-digits which have exactly 3 factors (excluding 1) is 106(2, 53, 106).

So, m = 106

m – 6 = 100

Let number of boys and girls in the institute are '9p' and '8p' respectively.

Total cost of the videos for the teachers = 715 * 100 = Rs.71500

Total fess collected from the students by the institute = 9p * 106 + 8p * 100 = 71500 + 16200

954p + 800p = 87700

1754p = 87700

p = 50

Total number of students in the institute = 9p + 8p = 17p

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= 850

17. Questions

Let number of classes taken by teachers A and B are '35 – x' and 'x' respectively.

Number of classes taken by C = 2x

Number of classes taken by D = x + 10 = 2x – 5

x = 15

Number of classed taken by A = 35 – x = 20

Number of classed taken by B = x = 15

Number of classed taken by C = 2x = 30

Number of classed taken by D = x + 10 = 25

Total number of classes taken by all the five teachers together = 20 * 100/20 = 100

Number of classed taken by E = 100 – 20 – 15 – 30 – 25 = 10

Total fees charged by A = 20 * 750 = Rs.15000

Average fee per class charged by E = 5000/10 = Rs.500

Let average fee per class charged by B, C, and D are 'b', 'c', and 'd' respectively.

According to the question:

$$\frac{15b + 30c}{15 + 30} = 700$$

15b + 30c = 31500 …… (1)

$$\frac{15b + 25d}{15 + 25} = 950$$

15b + 25d = 38000 …….. (2)

$$\frac{15000 + 15b + 30c + 25d + 5000}{100} = 715$$

15b + 30c + 25d = 51500 …….. (3)

From (1) and (3):

25d = 51500 – 31500

25d = 20000

d = 800

From equation (2):

15b + 20000 = 38000

15b = 18000

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b = 1200

From equation (1):

18000 + 30c = 31500

30c = 13500

c = 450

Teachers	Total number of classes taken	Average fee (in Rs.) per class
A	20	750
B	15	1200
C	30	450
D	25	800
E	10	500

Answer: A

Total fee charged by teachers B, D, and E together = 15 * 1200 + 25 * 800 + 5000 = 18000 + 20000 + 5000 = 43000

Total classes taken by teachers B, D, and E together = 15 + 25 + 10 = 50

Required average = 43000/50

= 860

18. Questions

Let number of classes taken by teachers A and B are '35 – x' and 'x' respectively.

Number of classes taken by C = 2x

Number of classes taken by D = x + 10 = 2x – 5

x = 15

Number of classed taken by A = 35 – x = 20

Number of classed taken by B = x = 15

Number of classed taken by C = 2x = 30

Number of classed taken by D = x + 10 = 25

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Total number of classes taken by all the five teachers together = 20 * 100/20 = 100

Number of classed taken by E = 100 – 20 – 15 – 30 – 25 = 10

Total fees charged by A = 20 * 750 = Rs.15000

Average fee per class charged by E = 5000/10 = Rs.500

Let average fee per class charged by B, C, and D are 'b', 'c', and 'd' respectively.

According to the question:

$$\frac{15b + 30c}{15 + 30} = 700$$

15b + 30c = 31500 …… (1)

$$\frac{15b + 25d}{15 + 25} = 950$$

15b + 25d = 38000 …….. (2)

$$\frac{15000 + 15b + 30c + 25d + 5000}{100} = 715$$

15b + 30c + 25d = 51500 …….. (3)

From (1) and (3):

25d = 51500 – 31500

25d = 20000

d = 800

From equation (2):

15b + 20000 = 38000

15b = 18000

b = 1200

From equation (1):

18000 + 30c = 31500

30c = 13500

c = 450

Teachers	Total number of classes taken	Average fee (in Rs.) per class
A	20	750
B	15	1200
C	30	450
D	25	800
E	10	500

Answer: E

The value that will fill the missing value of column II (Total number of classes taken) = P = 10

The value that will fill the missing value of column III (Average fee (in Rs.) per class by D) = Q = 800

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Now,

$$\sqrt{\frac{\sqrt{P*(P+Q)}}{10}}$$

$$=\sqrt{\frac{\sqrt{10*(10+800)}}{10}}$$

$$=\sqrt{\frac{\sqrt{10*810}}{10}}$$

$$=\sqrt{\frac{\sqrt{8100}}{10}}$$

$$=\sqrt{\frac{90}{10}}$$

$$=\sqrt{9}$$

$$=3$$

19. Questions

Let number of classes taken by teachers A and B are '35 – x' and 'x' respectively.

Number of classes taken by C = 2x

Number of classes taken by D = x + 10 = 2x – 5

x = 15

Number of classed taken by A = 35 – x = 20

Number of classed taken by B = x = 15

Number of classed taken by C = 2x = 30

Number of classed taken by D = x + 10 = 25

Total number of classes taken by all the five teachers together = 20 * 100/20 = 100

Number of classed taken by E = 100 – 20 – 15 – 30 – 25 = 10

Total fees charged by A = 20 * 750 = Rs.15000

Average fee per class charged by E = 5000/10 = Rs.500

Let average fee per class charged by B, C, and D are 'b', 'c', and 'd' respectively.

According to the question:

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$$\frac{15b + 30c}{15 + 30} = 700$$

15b + 30c = 31500 …… (1)

$$\frac{15b + 25d}{15 + 25} = 950$$

15b + 25d = 38000 …….. (2)

$$\frac{15000 + 15b + 30c + 25d + 5000}{100} = 715$$

15b + 30c + 25d = 51500 …….. (3)

From (1) and (3):

25d = 51500 – 31500

25d = 20000

d = 800

From equation (2):

15b + 20000 = 38000

15b = 18000

b = 1200

From equation (1):

18000 + 30c = 31500

30c = 13500

c = 450

Teachers	Total number of classes taken	Average fee (in Rs.) per class
A	20	750
B	15	1200
C	30	450
D	25	800
E	10	500

Answer: B

Remaining classed = 30 – 12 = 18

Let the values marked as A and B are ‘x’ and ‘y’ respectively.

Total views per class in all the 30 classes together = 2200 * 30 = 12x + 18y

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12x + 18y = 66000

2x + 3y = 11000

From statement I alone:

A = 2200 and B = 2000 (Given in question):

= 2 * 2200 + 3 * 2000 = 10400 [Not true]

From statement II alone:

A = 2400 and B = 2200 (Given in question):

= 2 * 2400 + 3 * 2200 = 11400 [Not true]

From statement III alone:

A = 2500 and B = 2000 (Given in question):

= 2 * 2500 + 3 * 2000 = 11000 [True]

From statements I and II together:

A = 2200 and B = 2200 (Given in question):

= 2 * 2200 + 3 * 2200 = 11000 [True]

From statements II and III together:

A = 2500 and B = 2200 (Given in question):

= 2 * 2500 + 3 * 2200 = 11600 [Not true]

So, either only value of A as 2500 or values of both A and B and 2200 and 2200 respectively can replaces the values marked in the question.

Hence, either statement III alone or statements I and II together.

20. Questions

Let number of classes taken by teachers A and B are '35 – x' and 'x' respectively.

Number of classes taken by C = 2x

Number of classes taken by D = x + 10 = 2x – 5

x = 15

Number of classed taken by A = 35 – x = 20

Number of classed taken by B = x = 15

Number of classed taken by C = 2x = 30

Number of classed taken by D = x + 10 = 25

Total number of classes taken by all the five teachers together = 20 * 100/20 = 100

Number of classed taken by E = 100 – 20 – 15 – 30 – 25 = 10

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Total fees charged by A = 20 * 750 = Rs.15000

Average fee per class charged by E = 5000/10 = Rs.500

Let average fee per class charged by B, C, and D are 'b', 'c', and 'd' respectively.

According to the question:

$$\frac{15b + 30c}{15 + 30} = 700$$

15b + 30c = 31500 …… (1)

$$\frac{15b + 25d}{15 + 25} = 950$$

15b + 25d = 38000 …….. (2)

$$\frac{15000 + 15b + 30c + 25d + 5000}{100} = 715$$

15b + 30c + 25d = 51500 …….. (3)

From (1) and (3):

25d = 51500 – 31500

25d = 20000

d = 800

From equation (2):

15b + 20000 = 38000

15b = 18000

b = 1200

From equation (1):

18000 + 30c = 31500

30c = 13500

c = 450

Teachers	Total number of classes taken	Average fee (in Rs.) per class
A	20	750
B	15	1200
C	30	450
D	25	800
E	10	500

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Answer: C

Total number of classes taken by A in both the institutes together = (20 + x)

Total number of classes taken by E in both the institutes together = (10 + x)

According to the question:

20 + x = 140% of (10 + x)

20 + x = 14 + 1.4x

0.4x = 6

x = 15

Number of classes taken by A and E at institute B is 15 each.

Total fee collected by them from another institute = 15 * 750 + 15 * 500 = 15 * (750 + 500) = (15 * 1250)

Required answer = (15 * 1250)/5x

= 1250/5

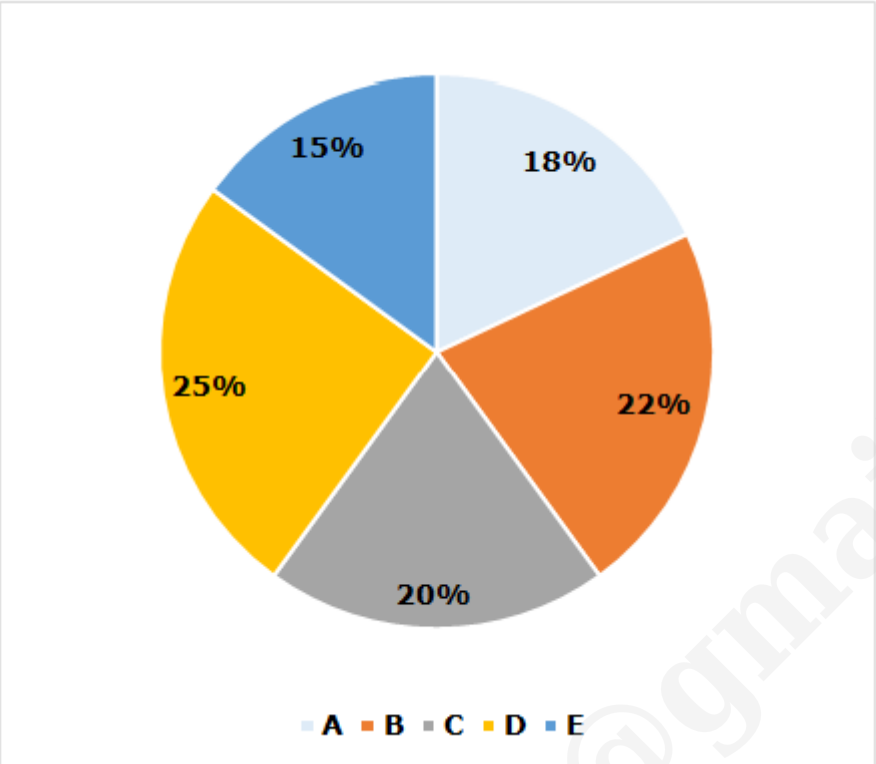
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1. Questions

Study the following information carefully and answer the questions given below.

The pie chart shows the percentage distribution of the number of sandwiches [Veg + Non-veg] sold in five different shops [A,B,C,D and E].



Note:

Total number of veg sandwiches = Total number of corn sandwiches + Total number of paneer sandwiches

Total number of non-veg sandwiches = Total number of chicken sandwiches + Total number of egg sandwiches.

In shops, A and B, 50% of veg sandwiches were sold out of the total number of sandwiches sold, while in shops C, D and E in 60% of non-veg sandwiches were sold out of the total number of sandwiches sold. In each shop, out of the total veg sandwiches, 25% are corn sandwiches and out of the total non-veg sandwiches, 25% are egg sandwiches. The number of chicken sandwiches sold in shop E is 54.

The difference between the total number of chicken sandwiches sold in shops C and D together and the total number of corn sandwiches sold in shops A and B together is x. Find the possible number of egg sandwiches sold in shops C and E together?

a. [x+32]/4

b. x-92

c. 0.5*x-19

d. 0.8*x-32

e. More than one option is true

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2. Questions

The price of corn, paneer, chicken and egg sandwiches sold in shop B is Rs.35, Rs.40, Rs. 55 and Rs.48 respectively. Find the total revenue earned by shop B.

a. Rs.8526

b. Rs.6850

c. Rs.8946

d. Rs.8096

e. Rs.8876

3. Questions

The total number of sandwiches sold in shop F is 50% more than the number of sandwiches sold in shop C. The ratio of veg and non-veg sandwiches sold in shop F is 3:2. The difference between the number of veg and non-veg sandwiches sold in shop F is equal to

I). 20 more than the number of egg sandwiches sold in shop E.

II). 4 less than the number of paneer sandwiches sold in shop D.

III). Average number of paneer sandwiches sold in shops C and B.

a. Only I

b. All false

c. Only I and III

d. All true

e. None of these

4. Questions

The number of cheese sandwiches sold in shop A is 25% more than the number of non-veg sandwiches sold in shop A. The number of cheese sandwiches sold in shop B is x more than that in shop A, where x is the sum of the number of egg sandwiches sold in shops A and B together. Find the number of cheese sandwiches sold in shop B.

a. 130

b. 120

c. 160

d. 190

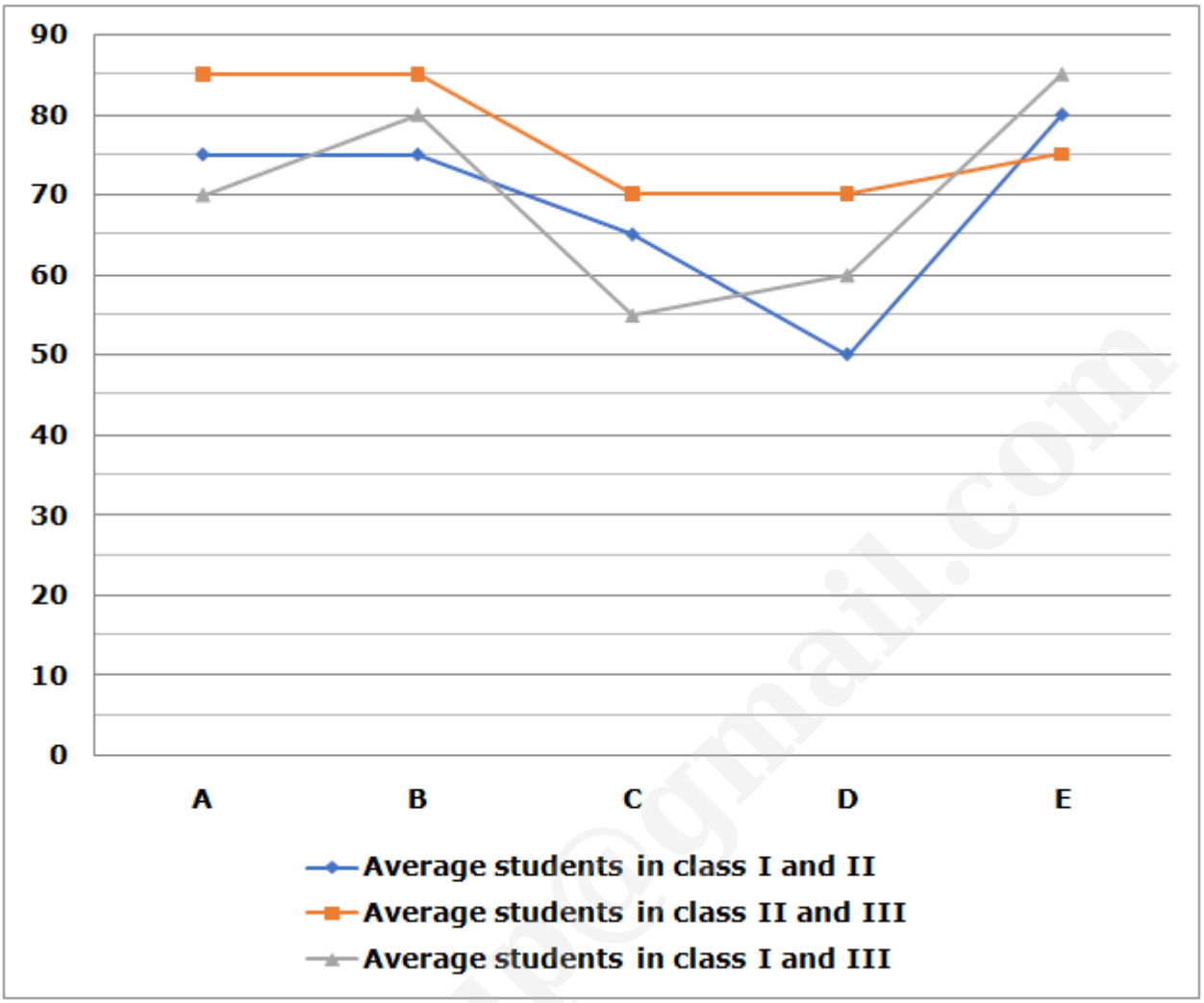
e. None of these

5. Questions

Study the following information carefully and answer the questions given below.

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The line graph shows the average number of students in classes I and II, classes II and III and classes III and I of five different schools A, B, C, D and E.



The ratio of the number of boys to girls in classes I, II and III of school A is 3:2, 5:4 and 5:3 respectively. The sum of the number of boys in class I of schools A and B is 57. The sum of the number of girls in class II of schools A and B is 74. The ratio of the number of boys to girls in class III of school B is 3:2. Find the difference of difference between the number of boys and girls in school A and school B.

a. 32

b. 36

c. 40

d. 33

e. 39

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6. Questions

The average number of boys in class I of school C and D is 27. The ratio of the number of girls of class I of school C and D is 5:4. The number of boys in class II of school C is 50% more than the number of boys of class I of school C. The difference of between the number of girls in class II of

school C and boys in class II of school C is x. Find the value of [7x+8].

a. 28

b. 26

c. 24

d. 78

e. 23

7. Questions

The average number of students in class I of all schools is 'm', the average number of students in class II of all schools is 'n' and the average number of students in class III of all schools is 'p'. The number of students in school F is [2m+3n-p]. Find the number of students in classes I and II of school F, if 50% of students of school F in class III.

a. 155

b. 169

c. 158

d. 137

e. 154

8. Questions

The number of students in class IV of school A is 2.5 times the difference between the number of students in class I and II of the same school and the number of students in class IV of school B is 3.5 times the difference between the number of students in class I and III of the same school. Find the total number of students in class IV of school A and B.

a. 154

b. 145

c. 156

d. 150

e. None of these

9. Questions

Study the following information carefully and answer the questions given below.

The table given below shows the ratio of the number of formal and casual shoes sold, the ratio of men's and women's formal shoes sold, and the ratio of men's and women's casual shoes sold in five different shops.

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	The ratio of the number of formal and causal shoes sold,	Ratio men's and women's formal shoes sold	Ratio of men's and women's casual shoes
A	5:3	3:2	3:1
B	3:5	5:4	3:2
C	5:4	2:3	5:3
D	3:4	3:4	3:2
E	3:5	5:3	1:3

Note:

I). Total number of shoes sold = Total number of formal shoes sold + Total number of casual shoes sold.

Total number of formal shoes sold = Total number of formal men shoes sold + Total number of formal women shoes sold

Total number of casual shoes sold = Total number of casual men shoes sold + Total number of casual women shoes sold

II). The ratio of the number of shoes sold in shops A to E and A to B is 1:2 and 2:3 respectively. There were 160 more shoes sold in shop E than in shop B. 60 more shoes were sold in shop C than in shop B. Furthermore, 10 less shoes sold in shop B than in shop D.

The difference between the number of men's formal shoes sold in shops A and B together and the number of women's casual shoes sold in shops C and D together is 'x'. Find the possible value of the number of formal shoes sold in shops A, C and E together.

I). 39x+38

II). 59x-258

III). 45x-70

a. Only I and III

b. Only III

c. Only I and II

d. Only II

e. Only II and III

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10. Questions

The ratio of the number of men's formal shoes sold in shops B and C together to the number of women's casual shoes sold in C and E together is m:n. Find the value of (7m+9n)

a. 456

b. 475

c. 685

d. 505

e. 495

11. Questions

The total number of women's shoes sold in shops A and B together is _____ more or less than the total number of women's shoes sold in shops C and D together. The total number of men's shoes sold in shops C and D together is _____ more or less than the total number of men's shoes sold in shops A and E together.

a. 192, 85

b. 192, 68

c. 105, 96

d. 115, 88

e. None of these

12. Questions

The difference between the average number of men's formal shoes sold in all shops together and the average number of women's formal shoes sold in all shops together is x. Find which of the following equation/relation is true.

a. $x^2+6x+6=78$

b. $x^2-9x+24=6$

c. $x^2-8x+4=6$

d. $x^2-11x+14=6$

e. More than one option is true

13. Questions

Study the following information carefully and answer the questions given below.

The cost price of item M is Rs.6a. The speed of boat X is b km/hr, and the speed of boat Y is 0.5d km/hr. The cost price of item N is Rs.8c. The speed of the stream of river P is 5 km/hr.When item M is marked e% more than its cost price and sold at a 10% discount, then the profit percentage is 35%. Boat X covered 285 km in river P in 9.5 hours in downstream. The speed of boat Y is 20% less than the speed of boat X. The selling price of item N is Rs.60.8 more than the selling price of item M when item M is sold at 20% profit and item N is sold at 20% loss. The sum of the value of a and c is 52.

Find which of the following relation is true?

I). 5a+6b+9c-4d=379

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II). 6a+7d+8e=810

III). 6a+11c+9e=932

a. All true

b. Only I false

c. All false

d. Only I and II true

e. Only II and III true

14. Questions

Find the difference between the time taken by boat X and Y to covers [8e+5a] km distance in still water.

a. [d/4]

b. [c-a]

c. [d-b]

d. [e-d]/2

e. [d-c]/2

15. Questions

R and S can complete a work alone in c days and d days respectively. R alone started the work and after 20 days, R left the work and then S finished the remaining work. Find the possible value of the total number of days taken to complete the whole work.

a. 32 days

b. 35 days

c. 42 days

d. 38 days

e. None of these

16. Questions

A is the wrong number and B is correct number of series I and II respectively. Find the relation between A and B?

Series I: 75,153,309,620,1245

Series II: 144,288,740, 2160, 7560

a. 3A+7B=6900

b. 5A+6B=258

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c. 7B-5A=22

d. 9A+5B=258

e. More than one option is true

17. Questions

Find the sum of the missing numbers of series I and II?

Series I: 850,170,340, 68, ?

Series II: 97.02, 95.06, 93.12, 91.2, ?

a. 253.2

b. 245.3

c. 225.3

d. 265.5

e. 295.5

18. Questions

The following questions contain two statements statement I and statement II. You have to determine which statement/s is/are necessary to answer the question and give the answer as,

Find the speed of train A?

Statement I: Train A and Train B crosses each other when moving in opposite directions in 8 seconds. The difference between the length of train A and B is 40 m and the difference between the speed of trains A and B is 5 m/s.

Statement II: The sum of the length of train A and train B is 360 m. The sum of the speed of train A and B is 45 m/s. Train B crosses a 140 m-long bridge in 12 seconds.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question

b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question

c. The data either in statement I alone or in statement II alone is sufficient to answer the question

d. The data given in both statements I and II together are not sufficient to answer the question

e. The data given in both statements I and II together are necessary to answer the question.

19. Questions

Find the present age of B.

Statement I: The ratio of the present ages of A and B is 5:6. The sum of the present ages of B and C is 62 years. D is 10 years younger than A, who is 7 years younger than C.

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Statement II: The present age of D is 16.66% more than the present age of B. The sum of the present ages of A and C is 57 years. The ratio of the present ages of A and D is 5:7.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question
b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question
c. The data either in statement I alone or in statement II alone is sufficient to answer the question
d. The data given in both statements I and II together are not sufficient to answer the question
e. The data given in both statements I and II together are necessary to answer the question.

20. Questions

Find the income of A.

Statement I: A's income is equal to the $3/4^{th}$ of B's income. The income of D is Rs.10000 less than the average income of B and C. A's savings is equal to 25% of his income.

Statement II: D's income is five times the savings of A. D's saving is half of A's expenditure. C's expenditure is Rs.38000 and B's saving is Rs.20000.

a. The data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question
b. The data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question
c. The data either in statement I alone or in statement II alone is sufficient to answer the question
d. The data given in both statements I and II together are not sufficient to answer the question
e. The data given in both statements I and II together are necessary to answer the question.

Explanations:

1. Questions

The number of chicken sandwiches sold in shop E is 54.

Number of egg sandwiches sold in shop E is 54*25/75=18.

So, the total non veg sandwiches sold in shop E is 54+18=72.

Number of veg sandwiches sold in shop E is = 72*40/60=48.

Total sandwiches sold in shop E together is = 48+72=120.

Total sandwiches sold in all shops together is = [120/15]*100=800.

The total number of sandwiches sold in shop A is 800*18/100=144.

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Number of veg sandwiches sold in shop A is = 144*50/100=72.

Number of non veg sandwiches sold in shop A is = 72.

Number of corn sandwiches sold in shop A is = 72*25/100=18

Number of Paneer sandwiches sold in shop A is =72-18=54.

Number of chicken sandwiches sold in shop A is = 72*75/100=54

Number of Egg sandwiches sold in shop A is = 72*25/100=18

Similarly, we can calculate others' values also.

Shop	Number of sandwiches sold	Number of veg sandwiches sold		Number of non veg sandwiches sold	
		Number of corn sandwiches sold	Number of Paneer sandwiches sold	Number of chicken sandwiches sold	Number of egg sandwiches sold
A	144	18	54	54	18
B	176	22	66	66	22
C	160	16	48	72	24
D	200	20	60	90	30
E	120	12	36	54	18

Answer: C

so, x= [72+90] –[18+22]= 122

so, egg sandwiches sold in C and E together is = 24+18=42= [0.5*x-19]

2. Questions

The number of chicken sandwiches sold in shop E is 54.

Number of egg sandwiches sold in shop E is 54*25/75=18.

So, the total non veg sandwiches sold in shop E is 54+18=72.

Number of veg sandwiches sold in shop E is = 72*40/60=48.

Total sandwiches sold in shop E together is = 48+72=120.

Total sandwiches sold in all shops together is = [120/15]*100=800.

The total number of sandwiches sold in shop A is 800*18/100=144.

Number of veg sandwiches sold in shop A is = 144*50/100=72.

Number of non veg sandwiches sold in shop A is = 72.

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Number of corn sandwiches sold in shop A is = 72*25/100=18

Number of Paneer sandwiches sold in shop A is =72-18=54.

Number of chicken sandwiches sold in shop A is = 72*75/100=54

Number of Egg sandwiches sold in shop A is = 72*25/100=18

Similarly, we can calculate others' values also.

Shop	Number of sandwiches sold	Number of veg sandwiches sold		Number of non veg sandwiches sold	
		Number of corn sandwiches sold	Number of Paneer sandwiches sold	Number of chicken sandwiches sold	Number of egg sandwiches sold
A	144	18	54	54	18
B	176	22	66	66	22
C	160	16	48	72	24
D	200	20	60	90	30
E	120	12	36	54	18

Answer: D

The total revenue earned by Shop B is

=22*35+66*40+66*55+22*48=Rs.8096

3. Questions

The number of chicken sandwiches sold in shop E is 54.

Number of egg sandwiches sold in shop E is 54*25/75=18.

So, the total non veg sandwiches sold in shop E is 54+18=72.

Number of veg sandwiches sold in shop E is = 72*40/60=48.

Total sandwiches sold in shop E together is = 48+72=120.

Total sandwiches sold in all shops together is = [120/15]*100=800.

The total number of sandwiches sold in shop A is 800*18/100=144.

Number of veg sandwiches sold in shop A is = 144*50/100=72.

Number of non veg sandwiches sold in shop A is = 72.

Number of corn sandwiches sold in shop A is = 72*25/100=18

Number of Paneer sandwiches sold in shop A is =72-18=54.

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Number of chicken sandwiches sold in shop A is = 72*75/100=54

Number of Egg sandwiches sold in shop A is = 72*25/100=18

Similarly, we can calculate others' values also.

Shop	Number of sandwiches sold	Number of veg sandwiches sold		Number of non veg sandwiches sold	
		Number of corn sandwiches sold	Number of Paneer sandwiches sold	Number of chicken sandwiches sold	Number of egg sandwiches sold
A	144	18	54	54	18
B	176	22	66	66	22
C	160	16	48	72	24
D	200	20	60	90	30
E	120	12	36	54	18

Answer: B

The number of Sandwiches sold in shop F is 160*150/100=240

The difference between veg and non veg sandwiches sold in shop F is 240/5=48

I). 48-18 = 30 more than the number of egg sandwiches sold in shop E.

II). 60-48=12 less than the number of paneer sandwiches sold in shop D.

III). The average number of paneer sandwiches sold in shops C and B is [66+48]/2=57

4. Questions

The number of chicken sandwiches sold in shop E is 54.

Number of egg sandwiches sold in shop E is 54*25/75=18.

So, the total non veg sandwiches sold in shop E is 54+18=72.

Number of veg sandwiches sold in shop E is = 72*40/60=48.

Total sandwiches sold in shop E together is = 48+72=120.

Total sandwiches sold in all shops together is = [120/15]*100=800.

The total number of sandwiches sold in shop A is 800*18/100=144.

Number of veg sandwiches sold in shop A is = 144*50/100=72.

Number of non veg sandwiches sold in shop A is = 72.

Number of corn sandwiches sold in shop A is = 72*25/100=18

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Number of Paneer sandwiches sold in shop A is =72-18=54.

Number of chicken sandwiches sold in shop A is = 72*75/100=54

Number of Egg sandwiches sold in shop A is = 72*25/100=18

Similarly, we can calculate others' values also.

Shop	Number of sandwiches sold	Number of veg sandwiches sold		Number of non veg sandwiches sold	
		Number of corn sandwiches sold	Number of Paneer sandwiches sold	Number of chicken sandwiches sold	Number of egg sandwiches sold
A	144	18	54	54	18
B	176	22	66	66	22
C	160	16	48	72	24
D	200	20	60	90	30
E	120	12	36	54	18

Answer: A

Number of cheese sandwiches sold in shop A is 72*125/100=90

The number of cheese sandwiches sold in shop B is 90+18+22=130

5. Questions

Total number of students in class I and II of school A is 75*2=150

Total number of students in classes II and III of school A is 85*2=170

Total number of students in class III and I of school A is 70*2=140

So, the total number of students in school A is [150+140+170]/2=230

Students in class I of school A is 230-170=60

Students in class II of school A is 230-140=90

Students in class III of school A is 230-150=80

Similarly, we can solve other schools also.

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School	Number of students in class I	Number of students in class II	Number of students in class III
A	60	90	80
B	70	80	90
C	50	80	60
D	40	60	80
E	90	70	80

Answer: C

The difference between boys and girls in school A is

= [60*3/5+90*5/9+80*5/8]-[60*2/5+90*4/9+80*3/8]

=136-94=42

Boys in class I of school B is = 57-60*3/5=21

So, girls in class I of school B is 70-21=49

Girls in class II of school B is = 74-90*4/9=34

Boys in class II of school B is 80-34=46

Boys in class III of school B is = 90*3/5=54

Girls in class III of school B is = 90-54=36

So, difference of boys and girls in school B is = [21+46+54]- [49+34+36]=2

So, the difference is = 42-2=40

6. Questions

Total number of students in class I and II of school A is 75*2=150

Total number of students in classes II and III of school A is 85*2=170

Total number of students in class III and I of school A is 70*2=140

So, the total number of students in school A is [150+140+170]/2=230

Students in class I of school A is 230-170=60

Students in class II of school A is 230-140=90

Students in class III of school A is 230-150=80

Similarly, we can solve other schools also.

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School	Number of students in class I	Number of students in class II	Number of students in class III
A	60	90	80
B	70	80	90
C	50	80	60
D	40	60	80
E	90	70	80

Answer: D

Total boys in class I of school C and D is 27*2=54

So, the total number of girls in class I of schools C and D is = 90-54=36

So, girls in C is 36*5/9=20

Girls in D is 36-20=16

Boys in C in class I is = 50-20=30

Boys in class II of C is = 30*150/100=45

Girls in class II of C is = 80-45=35

X=10

So, 7x+8=70+8=78

7. Questions

Total number of students in class I and II of school A is 75*2=150

Total number of students in classes II and III of school A is 85*2=170

Total number of students in class III and I of school A is 70*2=140

So, the total number of students in school A is [150+140+170]/2=230

Students in class I of school A is 230-170=60

Students in class II of school A is 230-140=90

Students in class III of school A is 230-150=80

Similarly, we can solve other schools also.

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School	Number of students in class I	Number of students in class II	Number of students in class III
A	60	90	80
B	70	80	90
C	50	80	60
D	40	60	80
E	90	70	80

Answer: D

So, m=[60+70+50+40+90]/5=62

So, n=[90+80+80+60+70]/5=76

So, p=[80+90+60+80+80]/5=78

The number of students in school F is 62*2+3*76-78=274

Number of students in class I and II together in F

= 274*50/100=137

8. Questions

Total number of students in class I and II of school A is 75*2=150

Total number of students in classes II and III of school A is 85*2=170

Total number of students in class III and I of school A is 70*2=140

So, the total number of students in school A is [150+140+170]/2=230

Students in class I of school A is 230-170=60

Students in class II of school A is 230-140=90

Students in class III of school A is 230-150=80

Similarly, we can solve other schools also.

School	Number of students in class I	Number of students in class II	Number of students in class III
A	60	90	80
B	70	80	90
C	50	80	60
D	40	60	80
E	90	70	80

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Answer: B

The Number of students in class IV of school A is = [30*2.5]=75

The Number of students in class IV of school B is = 20*3.5=70

So, the required sum is 75+70=145

9. Questions

The ratio of number of shoes sold in shops A, E and B is 2:4:3.

The difference in the number of shoes sold in shops E and B is 160.

So, 1 unit = 160.

The number of shoes sold in shops A, E and B is 160*2=320, 160*4=640, and 160*3=480 respectively.

Number of shoes sold in shop C is 480+60=540.

Number of shoes sold in shop D is 480+10=490

Number of formal shoes sold in shop A is = 320*5/8=200

Number of casual shoes sold in shop A is 320-200=120

Number of Men's formal shoes sold in shop A is 200*3/5=120

Number of Women's formal shoes sold in shop A is 200-120=80

Number of men's causal shoes sold in shop A is 120*3/4=90

Number of women's causal shows sold in shop A is 120-90=30

Similarly, we can calculate others' values also.

Shop	Total number of shoes sold	Number of formal shoes sold	Number of causal shoes sold	Number of men's formal shoes sold	Number of women's formal shoes sold	Number of men's casual shoes sold	Number of women's casual shoes sold
A	320	200	120	120	80	90	30
B	480	180	300	100	80	180	120
C	540	300	240	120	180	150	90
D	490	210	280	90	120	168	112
E	640	240	400	150	90	100	300

Answer: A

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So, x=[120+100]-[90+112]=18

So, total formal shoes sold in shops A, C and E together = 200+300+240=740=39x+38, 45x-70,

10. Questions

The ratio of number of shoes sold in shops A, E and B is 2:4:3.

The difference in the number of shoes sold in shops E and B is 160.

So, 1 unit = 160.

The number of shoes sold in shops A, E and B is 160*2=320, 160*4=640, and 160*3=480 respectively.

Number of shoes sold in shop C is 480+60=540.

Number of shoes sold in shop D is 480+10=490

Number of formal shoes sold in shop A is = 320*5/8=200

Number of casual shoes sold in shop A is 320-200=120

Number of Men's formal shoes sold in shop A is 200*3/5=120

Number of Women's formal shoes sold in shop A is 200-120=80

Number of men's causal shoes sold in shop A is 120*3/4=90

Number of women's causal shows sold in shop A is 120-90=30

Similarly, we can calculate others' values also.

Shop	Total number of shoes sold	Number of formal shoes sold	Number of causal shoes sold	Number of men's formal shoes sold	Number of women's formal shoes sold	Number of men's casual shoes sold	Number of women's casual shoes sold
A	320	200	120	120	80	90	30
B	480	180	300	100	80	180	120
C	540	300	240	120	180	150	90
D	490	210	280	90	120	168	112
E	640	240	400	150	90	100	300

Answer: D

So, m:n= [100+120]:[300+90]=220:390=22:39

So, 7*22+9*39=505

11. Questions

The ratio of number of shoes sold in shops A, E and B is 2:4:3.

The difference in the number of shoes sold in shops E and B is 160.

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So, 1 unit = 160.

The number of shoes sold in shops A, E and B is 160*2=320, 160*4=640, and 160*3=480 respectively.

Number of shoes sold in shop C is 480+60=540.

Number of shoes sold in shop D is 480+10=490

Number of formal shoes sold in shop A is = 320*5/8=200

Number of casual shoes sold in shop A is 320-200=120

Number of Men's formal shoes sold in shop A is 200*3/5=120

Number of Women's formal shoes sold in shop A is 200-120=80

Number of men's causal shoes sold in shop A is 120*3/4=90

Number of women's causal shows sold in shop A is 120-90=30

Similarly, we can calculate others' values also.

Shop	Total number of shoes sold	Number of formal shoes sold	Number of causal shoes sold	Number of men's formal shoes sold	Number of women's formal shoes sold	Number of men's casual shoes sold	Number of women's casual shoes sold
A	320	200	120	120	80	90	30
B	480	180	300	100	80	180	120
C	540	300	240	120	180	150	90
D	490	210	280	90	120	168	112
E	640	240	400	150	90	100	300

Answer: B

The total number of women's shoes sold in shops A and B together is _____ [80+30+80+120]-[180+90+120+112]=192 less than the total women's shoes sold in shops C and D together.

The total number of men's shoes sold in shops C and D together is [120+150+90+168]-[120+90+150+100]=68 more than the total men's shoes sold in shops A and E together.

12. Questions

The ratio of number of shoes sold in shops A, E and B is 2:4:3.

The difference in the number of shoes sold in shops E and B is 160.

So, 1 unit = 160.

The number of shoes sold in shops A, E and B is 160*2=320, 160*4=640, and 160*3=480 respectively.

Number of shoes sold in shop C is 480+60=540.

Number of shoes sold in shop D is 480+10=490

Number of formal shoes sold in shop A is = 320*5/8=200

Number of casual shoes sold in shop A is 320-200=120

Number of Men's formal shoes sold in shop A is 200*3/5=120

Number of Women's formal shoes sold in shop A is 200-120=80

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Number of men's causal shoes sold in shop A is 120*3/4=90

Number of women's causal shows sold in shop A is 120-90=30

Similarly, we can calculate others' values also.

Shop	Total number of shoes sold	Number of formal shoes sold	Number of causal shoes sold	Number of men's formal shoes sold	Number of women's formal shoes sold	Number of men's casual shoes sold	Number of women's casual shoes sold
A	320	200	120	120	80	90	30
B	480	180	300	100	80	180	120
C	540	300	240	120	180	150	90
D	490	210	280	90	120	168	112
E	640	240	400	150	90	100	300

Answer: E

So, x=[120+100+120+90+150]/5-[80+80+180+120+90]/5

=116-110=6

So, x^2+6x+6=78

13. Questions

So, ATQ, [8c*80/100-6a*120/100]=60.8

Or, 6.4c-7.2a=60.8

And a+c=52

By solving the above equation, we get a=20 and c=32

Cost price of item M =6*20=120

Cost price of item N =32*8=256

Selling price of item M, when it is sold at 35% profit is = 120*135/100=162

The marked price is 162*100/90=180

So, e=[180-120]*100/120=50

The downstream speed of boat X in river P is = 285/9.5=30

So, the speed of boat X is 30-5=25 km/hr =b

Speed of boat Y is 25*80/100=20=0.5d

So, d=40

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Answer: C

I). 5a+6b+9c-4d=5*20+6*25+9*32-4*40=378

II). 6a+7d+8e=6*20+7*40+8*50=800

III). 6a+11c+9e=6*20+11*32+9*50=922

14. Questions

So, ATQ, [8c*80/100-6a*120/100]=60.8

Or, 6.4c-7.2a=60.8

And a+c=52

By solving the above equation, we get a=20 and c=32

Cost price of item M =6*20=120

Cost price of item N =32*8=256

Selling price of item M, when it is sold at 35% profit is = 120*135/100=162

The marked price is 162*100/90=180

So, e=[180-120]*100/120=50

The downstream speed of boat X in river P is = 285/9.5=30

So, the speed of boat X is 30-5=25 km/hr =b

Speed of boat Y is 25*80/100=20=0.5d

So, d=40

Answer: D

Time taken by boat X = [8*50+5*20]/25=20

Time taken by boat Y= [8*50+5*20]/20=25

So, difference of time=25-20=5=[e-d]/2

15. Questions

So, ATQ, [8c*80/100-6a*120/100]=60.8

Or, 6.4c-7.2a=60.8

And a+c=52

By solving the above equation, we get a=20 and c=32

Cost price of item M =6*20=120

Cost price of item N =32*8=256

Selling price of item M, when it is sold at 35% profit is = 120*135/100=162

The marked price is 162*100/90=180

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So, e=[180-120]*100/120=50

The downstream speed of boat X in river P is = 285/9.5=30

So, the speed of boat X is 30-5=25 km/hr =b

Speed of boat Y is 25*80/100=20=0.5d

So, d=40

Answer: B

Total work = LCM of 32 and 40=160

The efficiency of R is 5 and the efficiency of S is 4.

After 20 days, R completed 5*20=100 unit

So, S complete remaining work in =[60/4]=15

So, total work completed in = 20+15=35 days

16. Questions

Answer: A

Series I – 75,153,309,620,1245

75*2+3=153, 153*2+3=309, 309*2+3=621, 621*2+3=1245

So, A=620

Series II -144,288,740,2160,7560

144*2=288, 288*2.5=720, 720*3=2160, 2160*3.5=7560

So, B=720

So, 3A+5B=3*620+7*720=6900

17. Questions

Answer: C

Series I –850 170 340 68 136

/5 *2 /5 *2

Series II –97.02 95.06 93.12 91.2 89.3

9.9*9.8 9.8*9.7 9.7*9.6 9.6*9.5 9.5*9.4

So, the sum is = 136+89.3=225.3

18. Questions

Answer: E

From statement I,

If the length of train A is l1 and speed of train A is S1 and length of train B is l2 and the speed of train B is

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S2

So, (l1+l2)/S1+S2=8

The difference of length of trains A and B is 40.

The difference in speed of trains A and B is 5.

From statement II,

L1+L2=360

S1+S2=45

From I and II

So, Length of trains are [360+40]/2=200 and [360-40]/2=160

Speed of trains are [45+5]/2=25 and [45-5]/2=20

Train B crosses a 140-long bridge in 12 sec.

The possible length and speed of the train B are 160 m and 25 m/sec.

[140+160]/25=12

So, the speed of train A is 20 m/sec

19. Questions

Answer: A

From statement I,

Let the age of A is 5x and age of B is 6x.

So, the age of D is 5x-10 years

Age of C is 5x+7 years

So, 11x+7=62

Or, x=55/11=5

So, age of B is 30 years

From II,

No direct values is given so we cannot calculate the answer from II.

20. Questions

Answer: D

From statement I,

Let A's income is 3x, so, B's income is 4x.

A's saving is 3x*25/100=75x/100

From II,

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C’s expenditure is Rs.38000 and B’s saving is Rs.20000.

From I and II.

D's income is five times the savings of A.

So, D’s income is = 5*[75x/100]=75x/20

D’s saving is half of A’s expenditure.

D’s saving is = [225x/100]/2

But we don't know the total income of B and C. So we cannot calculate the answer.

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1. Questions

Study the following data carefully and answer the questions:

Data given below is related to the quantity (in litres) of olive oil, coconut oil and mustard oil sold by a shopkeeper in a particular month. Cost (in Rs. per litre) of each type of oil is different.

Total quantity of olive oil sold is (x + 2) L. Ratio of quantity of olive oil sold to the quantity of coconut oil sold is 8: 5 and the cost of coconut oil is Rs.10 per L more than that of olive oil. Ratio of total amount received by selling olive oil to that received by selling coconut oil is 40: 27 respectively. The quantity of mustard oil is (x – 14) L more than that of olive oil. Total amount received by selling mustard oil is Rs.7200 and the cost of mustard oil is 20% more than that of olive oil.

Note: all three kinds of oil are sold at cost price

If the quantity of peanut oil sold by the shopkeeper is [(x + 10)% of 3x] L and the cost of peanut oil is (x + 10)% more than that of olive oil, then find the average amount received by the shopkeeper by selling coconut oil, mustard oil and peanut oil?

a. Rs.4500

b. Rs.3600

c. Rs.4800

d. Rs.4200

e. Rs.5400

2. Questions

If the total amount received by the shopkeeper by selling coconut oil is M% of that received by selling mustard oil, then find that the total amount received by the shopkeeper by selling mustard oil is what per cent more than that received by selling olive oil?

a. 2M%

b. (2M + 5)%

c. 4M%

d. (4M – 30)%

e. (2M – 15)%

3. Questions

If the shopkeeper had sold olive oil at Rs.50 per L more and coconut oil at Rs.27 per L more, then what would have been the ratio of profit amount received by selling olive oil to that received by selling coconut oil? (By selling whole quantity of each type of oil)

a. 40: 9

b. 25: 6

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c. 80: 27

d. 20: 3

e. 32: 17

4. Questions

Which of the following statements is/are true?

P: Cost of coconut oil is 8% more than that of olive oil.

Q: Sold quantity of olive oil is 33(1/3)% of that of mustard oil.

R: Ratio of sold quantity of coconut oil to that of mustard oil is 5: 12.

a. Only P and R

b. Only P

c. Only P and Q

d. All are true

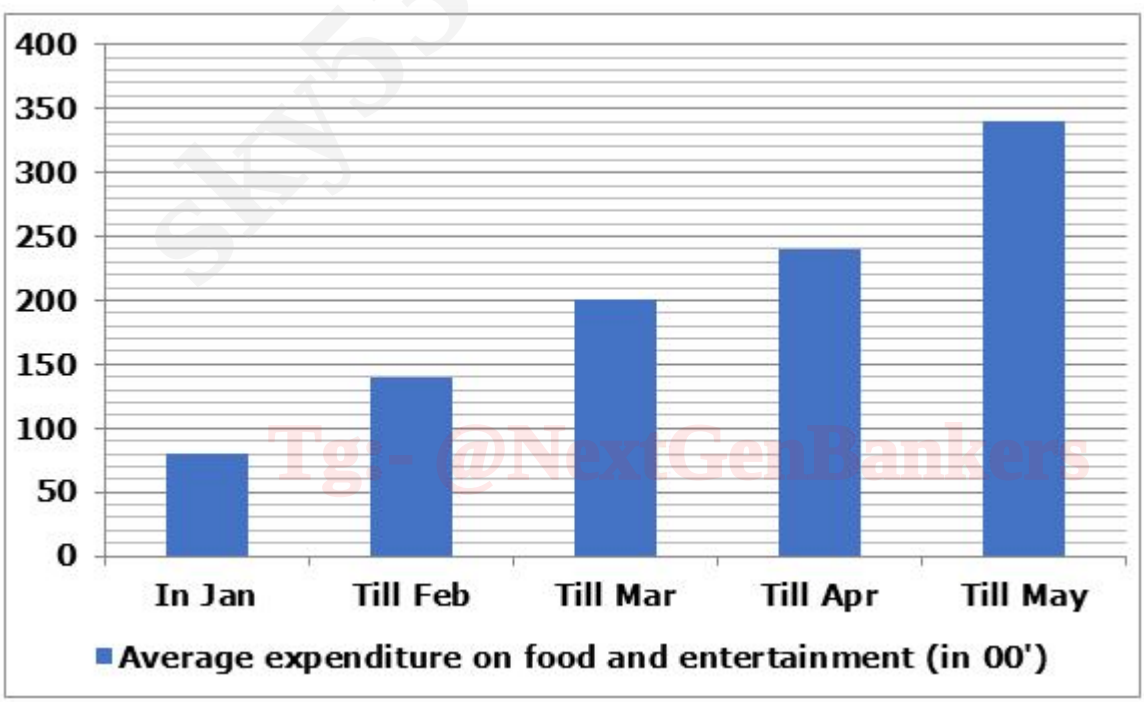
e. Only Q and R

5. Questions

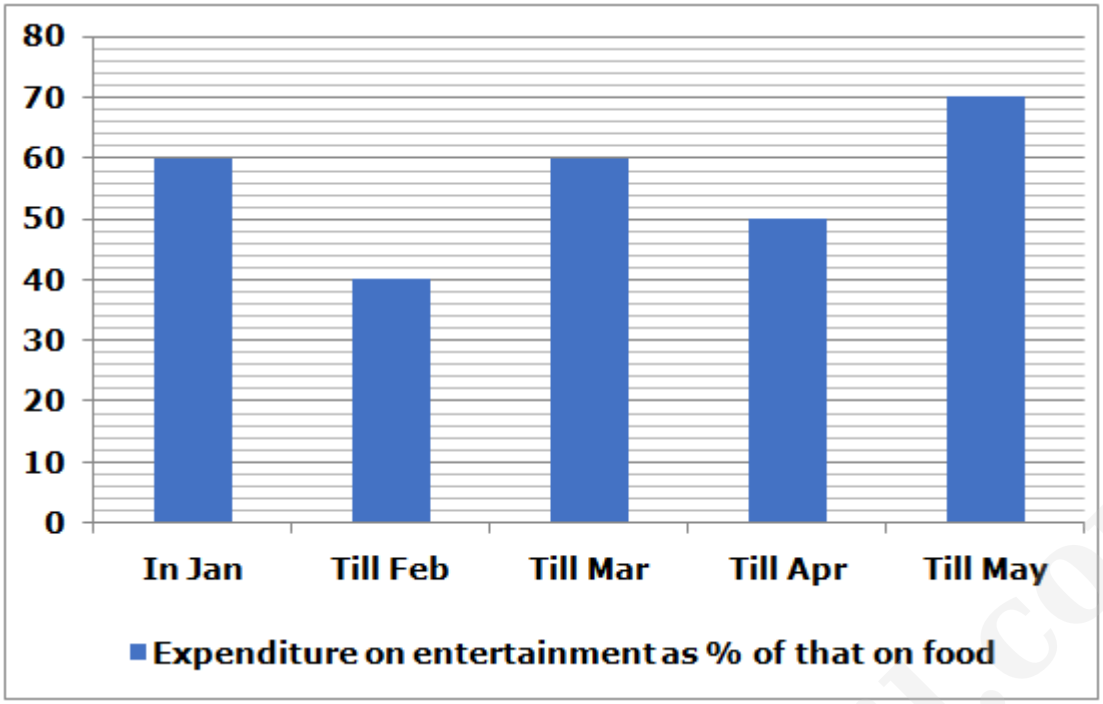
Study the following data carefully and answer the questions:

Data given below is related to the expenditure of a person on food and entertainment in five months Jan, Feb, Mar, Apr and May.

Bar graph given below shows the average of expenditure (in 00' rupees) of a person on food and entertainment.



Bar graph given below shows the expenditure on entertainment as per cent of that on food.



If the expenditure on food in Jan is 30(10/33)% of his salary in Jan and the expenditure on food in Feb is 27(7/9)% of his salary in Feb, then find that by what per cent the salary of the person increased/decreased from Jan to Feb?

a. 9(1/11)%

b. 11(1/9)%

c. 16(2/3)%

d. 3(1/3)%

e. 14(2/7)%

6. Questions

If the expenditure on entertainment in Mar is M% more than that in food in Mar and the expenditure on entertainment in May is N% more than that of food in May, then find the ratio of M to N?

a. 4: 5

b. 2: 5

c. 1: 2

d. 8: 5

e. 2: 3

7. Questions

The expenditure on entertainment in Mar is 17.5% of the salary in Mar and the salary is increased by 20% from Mar to Apr. If the total expenditure in Apr is 43.75% of the salary in Apr, then find that the expenditure on food in Apr is what per cent of the total expenditure in Apr?

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a. 20%

b. 27.5%

c. 30%

d. 25%

e. 33.33%

8. Questions

Which of the following statements is/are true?

P: Average expenditure on entertainment in Feb, Mar and May is Rs.8000.

Q: In Feb, expenditure on entertainment is 20% of that on food.

R: Average expenditure on food and entertainment in Apr is Rs.4000.

a. Only Q and R

b. Only Q

c. Only R

d. None is true

e. Only P

9. Questions

If the expenditure on entertainment in May is 40% of the total expenditure in May and the savings in May is Rs.10000 less than the total expenditure in May, then find the percentage of savings in May?

a. Can't be determined

b. 40%

c. 50%

d. 20%

e. 25%

10. Questions

Study the following data carefully and answer the questions:

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Five friends A, B, C, D and E are racing and each of them is running at a different speed (in km/h).

Table given below shows some statements about the speeds of each friend.

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Statement I:	B is running 5.4 km/h faster than A.
Statement II:	Ratio of C's speed to D's speed is 3: 10 respectively.
Statement III:	E's speed is 14 times of 10% of D's speed.
Statement IV:	Distance covered by A in 75 seconds is 15 m more than that covered by D in 42 seconds.
Statement V:	Average speed of B and C is equal to A's speed.

Match the following data given below in the table:

M: A's speed	**R:** Less than 3 m/s.
N: B's speed	**S:** 4 m/s less than E's speed.
O: C's speed	**T:** Between 4.5 m/s and 6.5 m/s.
P: D's speed	**U:** Multiple of 0.9.
Q: E's speed	**V:** 2 m/s more than D's speed.

a. M – R, N – S, O – T, P – U, Q – V

b. M – S, N – U, O – R, P – T, Q – V

c. M – S, N – U, O – T, P – R, Q – V

d. M – V, N – U, O – R, P – T, Q – S

e. M – S, N – R, O – U, P – T, Q – V

11. Questions

Which of the following can be determined from statements II and III alone?

P: Difference between C's speed and D's speed.

Q: Average of D's speed and E's speed.

R: Ratio of C's speed to E's speed.

a. None can be determined

b. Only P and R

c. Only Q

d. Only R

e. Only P and Q

12. Questions

Find the difference between the distance covered by the person in 36 seconds, who finished the race in 3rd position and the distance covered by the person in 72 seconds, who finished the race at last?

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a. 44 m

b. 64 m

c. 34 m

d. 74 m

e. 54 m

13. Questions

If after running for 12 seconds, A's increased his speed by 1.5 m/s, E decreased his speed by 0.5 m/s and E completed the race in the total time of 60 seconds, then find the distance between A and E, when E completed the race?

a. 120 m

b. 144 m

c. 110 m

d. 136 m

e. 130 m

14. Questions

Study the following data carefully and answer the questions:

Some 50 rupee notes, 100 rupee notes and 200 rupee notes are distributed among certain number of workers. Some received only one denomination notes, some received only two denominations note and remaining received all the three denominations notes

25% of the total number of workers received only 50 rupee notes. Number of workers, who received only 100 rupee notes, is 6 less than those, who received only 50 rupee notes. Number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes, is equal to those, who received only 200 rupee notes and also 6 less than those, who received both 100 and 200 rupee notes but not 50 rupee notes. Ratio of the number of workers, who received only 50 rupee notes, to those, who received all the three denominations notes, is 5: 3. The number of workers, who received both 50 and 200 rupee notes but not 100 rupee notes, is 12. Total number of persons, who received 50 rupee notes, is 70 and the total number of workers is 120.

Note: Each worker received only one note of one denomination.

If the ratio of the number of workers, who received only 50 rupee notes, to those, who received only 200 rupee notes is M: N, then find the ratio of the number of workers, who received only 50 rupee notes, to those, who received only 100 rupee notes?

a. (M + 3): (N + 2)

b. (M + 1): (N + 2)

c. (M + 2): (N + 3)

d. (M + 3): (N + 1)

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e. (M + 2): (N + 2)

15. Questions

Find the difference between the total value of 50 rupee notes and the value of 100 rupee notes, which were distributed among the workers?

a. Rs.4300

b. Rs.3000

c. Rs.5300

d. Rs.3300

e. Rs.2300

16. Questions

The total number of workers, who received only two denominations notes, is what per cent of the total number of workers, who received only one denomination note?

a. 59(3/8)%

b. 76(3/8)%

c. 19(3/8)%

d. 57(3/8)%

e. 61(3/8)%

17. Questions

The total value of 200 rupee notes is approximately what per cent of the total value of notes of all the three denominations together?

a. 56%

b. 52.1%

c. 28.5%

d. 42%

e. 84.3%

18. Questions

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In each question, two quantities I and II are given. Read the given statement and compare the quantity I and quantity II on its basis. (Only quantity is to be considered)

Quantity I: A boat can cover 30.6 km in still water in 30 minutes and 18 km downstream in 15 minutes. Find the value of 'p' if speed of the stream is p% of the speed of the boat in still water?

Quantity II: A's present age is half of B's present age and C is 2 years older than A. B's age after 4 years

will be double of C's present age. Find A's present age (in years)?

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II or relation can't be established

19. Questions

Difference between compound interest and simple interest received on Rs.X at 15% rate after 3 years is Rs.170.1.

Quantity I: What will be the simple interest (in Rs.) on Rs. (X + 100) at 2% rate after 2 years?

Quantity II: What will be difference between compound interest and simple interest (in Rs.) on Rs. (X + 2600) at 16% rate after 2 years?

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II or relation can't be established

20. Questions

An article is marked up by Rs.45 and then sold at 15% discount on its marked price. If the article was sold at Rs.27 discount on its marked price, the profit per cent would be 8%.

Quantity I: Find the numerical value of original profit per cent on the article?

Quantity II: If the article was sold at 18% discount on its marked price, then what would have been the profit/loss (in Rs.)?

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II or relation can't be established

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Explanations:

1. Questions

Let the sold quantities of olive oil and coconut oil be '8a' L and '5a' L respectively.

Also let the cost of olive oil be ₹b per L.

So, the cost of coconut oil = ₹(b + 10) per L

The total amount received by selling olive oil = ₹8ab

And the total amount received by selling coconut oil = ₹5a(b + 10)

Since, the ratio of total amount received by selling olive oil to that received by selling coconut oil is 40: 27

So,

$8ab/(5a(b+10))= 40/27$

$27b = 25b + 250$

$b = 125$

The cost of olive oil = ₹125 per L

The cost of coconut oil = 125 + 10 = ₹135 per L

Since, the cost of mustard oil is 20% more than that of olive oil.

So, the cost of mustard oil = 120% of 125 = ₹150 per L

Since, the total amount received by selling mustard oil is ₹7200.

So, the sold quantity of mustard oil = 7200/150 = 48 L

Since, the sold quantity of mustard oil is (x – 14) L more than that of olive oil.

And the sold quantity of olive oil = (x + 2) L

So,

$(x + 2) + (x – 14) = 48$

$x = 30$

So, the sold quantity of olive oil = 30 + 2 = 32 L

And the sold quantity of coconut oil = 32 * (5/8) = 20 L

Type of oil	Quantity of oil sold (in litre)	Cost of oil (in Rs. per litre)	Total amount received
Olive oil	32	125	32 * 125 = 4000
Coconut oil	20	135	20 * 135 = 2700
Mustard oil	48	150	7200

Answer: E

The quantity of peanut oil sold by the shopkeeper:

[(x + 10)% of 3x] = [(30 + 10)% of (3 * 30)] = 36 L

The cost of peanut oil:

(100 + x + 10)% of 125 = (100 + 30 + 10)% of 125 = ₹175 per L

The amount received by the shopkeeper by selling peanut oil = 36 * 175 = ₹6300

Since, the amount received by the shopkeeper by selling coconut oil = ₹2700

And the amount received by the shopkeeper by selling mustard oil = ₹7200

So, the required average:

(2700+7200+6300)/3= ₹5400

2. Questions

Let the sold quantities of olive oil and coconut oil be '8a' L and '5a' L respectively.

Also let the cost of olive oil be ₹b per L.

So, the cost of coconut oil = ₹(b + 10) per L

The total amount received by selling olive oil = ₹8ab

And the total amount received by selling coconut oil = ₹5a(b + 10)

Since, the ratio of total amount received by selling olive oil to that received by selling coconut oil is 40: 27

So,

8ab/(5a(b+10))= 40/27

27b = 25b + 250

b = 125

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The cost of coconut oil = 125 + 10 = ₹135 per L

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So, the cost of mustard oil = 120% of 125 = ₹150 per L

Since, the total amount received by selling mustard oil is ₹7200.

So, the sold quantity of mustard oil = 7200/150 = 48 L

Since, the sold quantity of mustard oil is (x – 14) L more than that of olive oil.

And the sold quantity of olive oil = (x + 2) L

So,

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(x + 2) + (x – 14) = 48

x = 30

So, the sold quantity of olive oil = 30 + 2 = 32 L

And the sold quantity of coconut oil = 32 * (5/8) = 20 L

Type of oil	Quantity of oil sold (in litre)	Cost of oil (in Rs. per litre)	Total amount received
Olive oil	32	125	32 * 125 = 4000
Coconut oil	20	135	20 * 135 = 2700
Mustard oil	48	150	7200

Answer: B

Since, the total amount received by selling coconut oil = ₹2700

And the total amount received by selling mustard oil = ₹7200

So, M = (2700/7200) * 100 = 37.5%

Since, the total amount received by selling olive oil = ₹4000

So, the required percentage:

(7200-4000)/4000 ×100=80%=(2M+5)%

3. Questions

Let the sold quantities of olive oil and coconut oil be '8a' L and '5a' L respectively.

Also let the cost of olive oil be ₹b per L.

So, the cost of coconut oil = ₹(b + 10) per L

The total amount received by selling olive oil = ₹8ab

And the total amount received by selling coconut oil = ₹5a(b + 10)

Since, the ratio of total amount received by selling olive oil to that received by selling coconut oil is 40: 27

So,

8ab/(5a(b+10))= 40/27

27b = 25b + 250

b = 125

The cost of olive oil = ₹125 per L

The cost of coconut oil = 125 + 10 = ₹135 per L

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So, the cost of mustard oil = 120% of 125 = ₹150 per L

Since, the total amount received by selling mustard oil is ₹7200.

So, the sold quantity of mustard oil = 7200/150 = 48 L

Since, the sold quantity of mustard oil is (x – 14) L more than that of olive oil.

And the sold quantity of olive oil = (x + 2) L

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So,

(x + 2) + (x – 14) = 48

x = 30

So, the sold quantity of olive oil = 30 + 2 = 32 L

And the sold quantity of coconut oil = 32 * (5/8) = 20 L

Type of oil	Quantity of oil sold (in litre)	Cost of oil (in Rs. per litre)	Total amount received
Olive oil	32	125	32 * 125 = 4000
Coconut oil	20	135	20 * 135 = 2700
Mustard oil	48	150	7200

Answer: C

Since, the quantity of olive oil sold by the shopkeeper = 32 L

If the shopkeeper had sold olive oil at ₹50 per L more:

The profit amount received by selling olive oil would have been:

32 * 50 = ₹1600

Since, the quantity of coconut oil sold by the shopkeeper = 20 L

If the shopkeeper had sold coconut oil at ₹27 per L more:

The profit amount received by selling coconut oil would have been:

20 * 27 = ₹540

Required ratio = 1600: 540 = 80: 27

4. Questions

Let the sold quantities of olive oil and coconut oil be ‘8a’ L and ‘5a’ L respectively.

Also let the cost of olive oil be ₹b per L.

So, the cost of coconut oil = ₹(b + 10) per L

The total amount received by selling olive oil = ₹8ab

And the total amount received by selling coconut oil = ₹5a(b + 10)

Since, the ratio of total amount received by selling olive oil to that received by selling coconut oil is 40: 27

So,

8ab/(5a(b+10))= 40/27

27b = 25b + 250

b = 125

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The cost of olive oil = ₹125 per L

The cost of coconut oil = 125 + 10 = ₹135 per L

Since, the cost of mustard oil is 20% more than that of olive oil.

So, the cost of mustard oil = 120% of 125 = ₹150 per L

Since, the total amount received by selling mustard oil is ₹7200.

So, the sold quantity of mustard oil = 7200/150 = 48 L

Since, the sold quantity of mustard oil is (x – 14) L more than that of olive oil.

And the sold quantity of olive oil = (x + 2) L

So,

(x + 2) + (x – 14) = 48

x = 30

So, the sold quantity of olive oil = 30 + 2 = 32 L

And the sold quantity of coconut oil = 32 * (5/8) = 20 L

Type of oil	Quantity of oil sold (in litre)	Cost of oil (in Rs. per litre)	Total amount received
Olive oil	32	125	32 * 125 = 4000
Coconut oil	20	135	20 * 135 = 2700
Mustard oil	48	150	7200

Answer: A

From P:

Cost of olive oil = ₹125 per L

Cost of coconut oil = ₹135 per L

Required percentage = (135-125)/125 ×100 = 8%

So, P is true.

From Q:

Sold quantity of olive oil = 32 L

Sold quantity of mustard oil = 48 L

Required percentage = 32/48 ×100 = 66(2/3)%

So, Q is not true.

From R:

Sold quantity of coconut oil = 20 L

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Sold quantity of mustard oil = 48 L

Required ratio = 20: 48 = 5: 12

So, R is true.

Hence, only P and R are true.

5. Questions

Let the expenditure on food in Jan be Rs.5x.

So, the expenditure on entertainment in Jan = 60% of 5x = Rs.3x

Since, the total expenditure on food and entertainment together in Jan = 2 * 8000 = Rs.16000

So, 5x + 3x = 16000

x = 2000

The expenditure on food in Jan = 5 * 2000 = Rs.10000

The expenditure on entertainment in Jan = 3 * 2000 = Rs.6000

Similarly, we can calculate the expenditure of the person on food and entertainment till Feb, till Mar, till Apr and till May.

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
Till Feb	20000	8000
Till Mar	25000	15000
Till Apr	32000	16000
Till May	40000	28000

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
In Feb	20000 – 10000 = 10000	8000 – 6000 = 2000
In Mar	25000 – 20000 = 5000	15000 – 8000 = 7000
In Apr	32000 – 25000 = 7000	16000 – 15000 = 1000
In May	40000 – 32000 = 8000	28000 – 16000 = 12000

Answer: A

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Since, the expenditure on food in Jan = Rs.10000

So, the salary in Jan = 10000 × 3300/1000 = Rs.33000

Since, the expenditure on food in Feb = Rs.10000

So, the salary in Feb = 10000 × 900/250 = Rs.36000

Required percentage = (36000-33000)/33000 ×100=9 (1/11)%

6. Questions

Let the expenditure on food in Jan be Rs.5x.

So, the expenditure on entertainment in Jan = 60% of 5x = Rs.3x

Since, the total expenditure on food and entertainment together in Jan = 2 * 8000 = Rs.16000

So, 5x + 3x = 16000

x = 2000

The expenditure on food in Jan = 5 * 2000 = Rs.10000

The expenditure on entertainment in Jan = 3 * 2000 = Rs.6000

Similarly, we can calculate the expenditure of the person on food and entertainment till Feb, till Mar, till Apr and till May.

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Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
In Feb	20000 – 10000 = 10000	8000 – 6000 = 2000
In Mar	25000 – 20000 = 5000	15000 – 8000 = 7000
In Apr	32000 – 25000 = 7000	16000 – 15000 = 1000
In May	40000 – 32000 = 8000	28000 – 16000 = 12000

Answer: A

The expenditure on food in Mar = Rs.5000

The expenditure on entertainment in Mar = Rs.7000

So, M = (7000-5000)/5000 ×100 = 40%

The expenditure on food in May = Rs.8000

The expenditure on entertainment in May = Rs.12000

So, N = (12000-8000)/8000 ×100 = 50%

Required ratio = M: N = 40: 50 = 4: 5

7. Questions

Let the expenditure on food in Jan be Rs.5x.

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So, the expenditure on entertainment in Jan = 60% of 5x = Rs.3x

Since, the total expenditure on food and entertainment together in Jan = 2 * 8000 = Rs.16000

So, 5x + 3x = 16000

x = 2000

The expenditure on food in Jan = 5 * 2000 = Rs.10000

The expenditure on entertainment in Jan = 3 * 2000 = Rs.6000

Similarly, we can calculate the expenditure of the person on food and entertainment till Feb, till Mar, till Apr and till May.

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
Till Feb	20000	8000
Till Mar	25000	15000
Till Apr	32000	16000
Till May	40000	28000

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
In Feb	20000 – 10000 = 10000	8000 – 6000 = 2000
In Mar	25000 – 20000 = 5000	15000 – 8000 = 7000
In Apr	32000 – 25000 = 7000	16000 – 15000 = 1000
In May	40000 – 32000 = 8000	28000 – 16000 = 12000

Answer: E

Since, the expenditure on entertainment in Mar = Rs.7000

So, the salary in Mar = 7000 * (100/17.5) = Rs.40000

And the salary in Apr = 120% of 40000 = Rs.48000

Since, the total expenditure in Apr = 43.75% of 48000 = Rs.21000

And the expenditure on food in Apr = Rs.7000

So, the required ratio = (7000/21000) * 100 = 33.33%

8. Questions

Let the expenditure on food in Jan be Rs.5x.

So, the expenditure on entertainment in Jan = 60% of 5x = Rs.3x

Since, the total expenditure on food and entertainment together in Jan = 2 * 8000 = Rs.16000

So, 5x + 3x = 16000

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x = 2000

The expenditure on food in Jan = 5 * 2000 = Rs.10000

The expenditure on entertainment in Jan = 3 * 2000 = Rs.6000

Similarly, we can calculate the expenditure of the person on food and entertainment till Feb, till Mar, till Apr and till May.

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
Till Feb	20000	8000
Till Mar	25000	15000
Till Apr	32000	16000
Till May	40000	28000

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
In Feb	20000 – 10000 = 10000	8000 – 6000 = 2000
In Mar	25000 – 20000 = 5000	15000 – 8000 = 7000
In Apr	32000 – 25000 = 7000	16000 – 15000 = 1000
In May	40000 – 32000 = 8000	28000 – 16000 = 12000

Answer: A

From P:

Average expenditure on entertainment in Feb, Mar and May:

(2000+7000+12000)/3= Rs.7000

So, P is not true.

From Q:

Expenditure on food in Feb = Rs.10000

Expenditure on entertainment in Feb = Rs.2000

Required percentage = (2000/10000) * 100 = 20%

So, Q is true.

From R:

Average expenditure on food and entertainment in Apr:

(7000+1000)/2= Rs. 4000

So, R is true.

Hence, only Q and R are true.

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9. Questions

Let the expenditure on food in Jan be Rs.5x.

So, the expenditure on entertainment in Jan = 60% of 5x = Rs.3x

Since, the total expenditure on food and entertainment together in Jan = 2 * 8000 = Rs.16000

So, 5x + 3x = 16000

x = 2000

The expenditure on food in Jan = 5 * 2000 = Rs.10000

The expenditure on entertainment in Jan = 3 * 2000 = Rs.6000

Similarly, we can calculate the expenditure of the person on food and entertainment till Feb, till Mar, till Apr and till May.

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
Till Feb	20000	8000
Till Mar	25000	15000
Till Apr	32000	16000
Till May	40000	28000

Month	Expenditure on food (in ₹)	Expenditure on entertainment (in ₹)
In Jan	10000	6000
In Feb	20000 – 10000 = 10000	8000 – 6000 = 2000
In Mar	25000 – 20000 = 5000	15000 – 8000 = 7000
In Apr	32000 – 25000 = 7000	16000 – 15000 = 1000
In May	40000 – 32000 = 8000	28000 – 16000 = 12000

Answer: B

Since, the expenditure on entertainment in May = Rs.12000

So, the total expenditure in May = 12000 * (100/40) = Rs.30000

And the total savings in May = 30000 – 10000 = Rs.20000

Since, the total salary in May = 30000 + 20000 = Rs.50000

So, the percentage of savings in May = (20000/50000) * 100 = 40%

10. Questions

From statement I:

Let A's speed be 'x' m/s.

So, B's speed = x + 5.4 * (5/18) = (x + 1.5) m/s

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From statement II:

Let C's and D's speed be '3y' m/s and '10y' m/s respectively.

From statement III:

E's speed is 14 times of 10% of D's speed.

So, E's speed = 14 * (10% of 10y) = '14y' m/s

From statement IV:

Distance covered by A in 75 seconds is 15 m more than that covered by D in 42 seconds.

So,

(x * 75) – (10y * 42) = 15

5x – 28y = 1 ----------(1)

From statement V:

Average speed of B and C is equal to A's speed.

x + 1.5 + 3y = 2x

x – 3y = 1.5 ----------(2)

From equations (1) and (2):

5(3y + 1.5) – 28y = 1

15y + 7.5 – 28y = 1

y = 0.5

From equation (2):

x – 1.5 = 1.5

x = 3

Now, A's speed = 3 m/s

B' speed = 3 + 1.5 = 4.5 m/s

C's speed = 3 * 0.5 = 1.5 m/s

D's speed = 10 * 0.5 = 5 m/s

And E's speed = 14 * 0.5 = 7 m/s

Answer: B

The speed, which is less than 3 m/s, is 1.5 m/s, which is C's speed.

So, O – R

The speed, which is 4 m/s less than E's speed, is 3 m/s, which is A's speed.

So, M – S

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The speed, which is between 4.5 m/s and 6.5 m/s, is 5 m/s, which is D's speed.

So, P – T

The speed, which is multiple of 0.9, is 4.5 m/s, which is B's speed.

So, N – U

The speed, which is 2 m/s more than D's speed, is 7 m/s, which is E's speed.

So, Q – V

The correct combination: M – S, N – U, O – R, P – T, Q – V

11. Questions

From statement I:

Let A's speed be 'x' m/s.

So, B's speed = x + 5.4 * (5/18) = (x + 1.5) m/s

From statement II:

Let C's and D's speed be '3y' m/s and '10y' m/s respectively.

From statement III:

E's speed is 14 times of 10% of D's speed.

So, E's speed = 14 * (10% of 10y) = '14y' m/s

From statement IV:

Distance covered by A in 75 seconds is 15 m more than that covered by D in 42 seconds.

So,

(x * 75) – (10y * 42) = 15

5x – 28y = 1 ----------(1)

From statement V:

Average speed of B and C is equal to A's speed.

x + 1.5 + 3y = 2x

x – 3y = 1.5 ----------(2)

From equations (1) and (2):

5(3y + 1.5) – 28y = 1

15y + 7.5 – 28y = 1

y = 0.5

From equation (2):

x – 1.5 = 1.5

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x = 3

Now, A's speed = 3 m/s

B' speed = 3 + 1.5 = 4.5 m/s

C's speed = 3 * 0.5 = 1.5 m/s

D's speed = 10 * 0.5 = 5 m/s

And E's speed = 14 * 0.5 = 7 m/s

Answer: D

From statement II:

Let C's and D's speed be '3y' m/s and '10y' m/s respectively.

From statement III:

E's speed is 14 times of 10% of D's speed.

So, E's speed = 14 * (10% of 10y) = '14y' m/s

From P:

Since, we can't find the speeds of C and D.

So, P can't be determined.

From Q:

Since, we can't find the speeds of D and E.

So, Q can't be determined.

From R:

Ratio of C's speed to E's speed = 3y: 14y = 3: 14

Hence, only R can be determined.

12. Questions

From statement I:

Let A's speed be 'x' m/s.

So, B's speed = x + 5.4 * (5/18) = (x + 1.5) m/s

From statement II:

Let C's and D's speed be '3y' m/s and '10y' m/s respectively.

From statement III:

E's speed is 14 times of 10% of D's speed.

So, E's speed = 14 * (10% of 10y) = '14y' m/s

From statement IV:

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Distance covered by A in 75 seconds is 15 m more than that covered by D in 42 seconds.

So,

(x * 75) – (10y * 42) = 15

5x – 28y = 1 ----------(1)

From statement V:

Average speed of B and C is equal to A's speed.

x + 1.5 + 3y = 2x

x – 3y = 1.5 ----------(2)

From equations (1) and (2):

5(3y + 1.5) – 28y = 1

15y + 7.5 – 28y = 1

y = 0.5

From equation (2):

x – 1.5 = 1.5

x = 3

Now, A's speed = 3 m/s

B' speed = 3 + 1.5 = 4.5 m/s

C's speed = 3 * 0.5 = 1.5 m/s

D's speed = 10 * 0.5 = 5 m/s

And E's speed = 14 * 0.5 = 7 m/s

Answer: E

Since, B's speed is 3rd maximum speed and he will finish the race in 3rd position.

So, the distance covered by B in 36 seconds = 4.5 * 36 = 162 m

Since, C's speed is minimum speed and he will finish the race last.

So, the distance covered by C in 72 seconds = 1.5 * 72 = 108 m

Required difference = 162 – 108 = 54 m

13. Questions

From statement I:

Let A's speed be 'x' m/s.

So, B's speed = x + 5.4 * (5/18) = (x + 1.5) m/s

From statement II:

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Let C's and D's speed be '3y' m/s and '10y' m/s respectively.

From statement III:

E's speed is 14 times of 10% of D's speed.

So, E's speed = 14 * (10% of 10y) = '14y' m/s

From statement IV:

Distance covered by A in 75 seconds is 15 m more than that covered by D in 42 seconds.

So,

(x * 75) – (10y * 42) = 15

5x – 28y = 1 ----------(1)

From statement V:

Average speed of B and C is equal to A's speed.

x + 1.5 + 3y = 2x

x – 3y = 1.5 ----------(2)

From equations (1) and (2):

5(3y + 1.5) – 28y = 1

15y + 7.5 – 28y = 1

y = 0.5

From equation (2):

x – 1.5 = 1.5

x = 3

Now, A's speed = 3 m/s

B' speed = 3 + 1.5 = 4.5 m/s

C's speed = 3 * 0.5 = 1.5 m/s

D's speed = 10 * 0.5 = 5 m/s

And E's speed = 14 * 0.5 = 7 m/s

Answer: B

E's initial speed = 7 m/s

E's speed after 12 seconds = 7 – 0.5 = 6.5 m/s

Since, E completed the race in 60 seconds.

So, the total distance covered by E for finishing the race:

(7 * 12) + [6.5 * (60 – 12)] = 84 + 312 = 396 m

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A's initial speed = 3 m/s

A's speed after 12 seconds = 3 + 1.5 = 4.5 m/s

The distance covered by A in 60 seconds:

(3 * 12) + [4.5 * (60 – 12)] = 252 m

So, the distance between A and E, when E completed the race = 396 – 252 = 144 m

14. Questions

Since, the total number of workers is 120.

So, the number of workers, who received only 50 rupee notes = 25% of 120 = 30

The number of workers, who received only 100 rupee notes = 30 – 6 = 24

And the number of workers, who received all the three denominations notes = 30 * (3/5) = 18

Let the number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = The number of workers, who received only 200 rupee notes = x

Since, the number of workers, who received both 50 and 200 rupee notes but not 100 rupee notes, is 12.

And total number of persons, who received 50 rupee notes, is 70.

So,

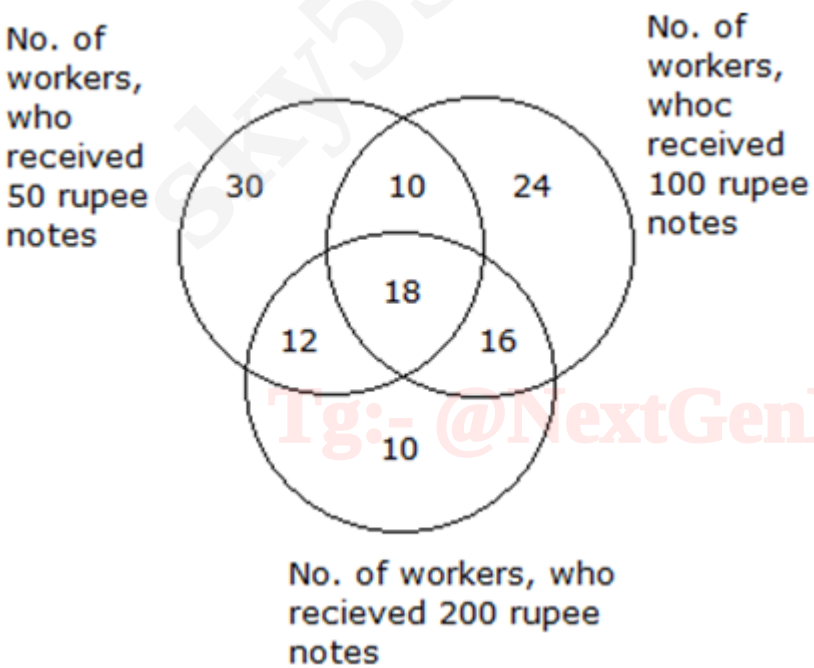
30 + 18 + x + 12 = 70

x = 10

The number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = 10

The number of workers, who received only 200 rupee notes = 10

The number of workers, who received both 100 and 200 rupee notes but not 50 rupee notes = 10 + 6 = 16



Answer: C

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Since, the number of workers, who received only 50 rupee notes = 30

And the number of workers, who received only 200 rupee notes = 10

So, M: N = 30: 10 = 3: 1

Since, the number of workers, who received only 100 rupee notes = 24

So, the required ratio = 30: 24 = 5: 4 = (M + 2): (N + 3)

15. Questions

Since, the total number of workers is 120.

So, the number of workers, who received only 50 rupee notes = 25% of 120 = 30

The number of workers, who received only 100 rupee notes = 30 – 6 = 24

And the number of workers, who received all the three denominations notes = 30 * (3/5) = 18

Let the number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = The number of workers, who received only 200 rupee notes = x

Since, the number of workers, who received both 50 and 200 rupee notes but not 100 rupee notes, is 12.

And total number of persons, who received 50 rupee notes, is 70.

So,

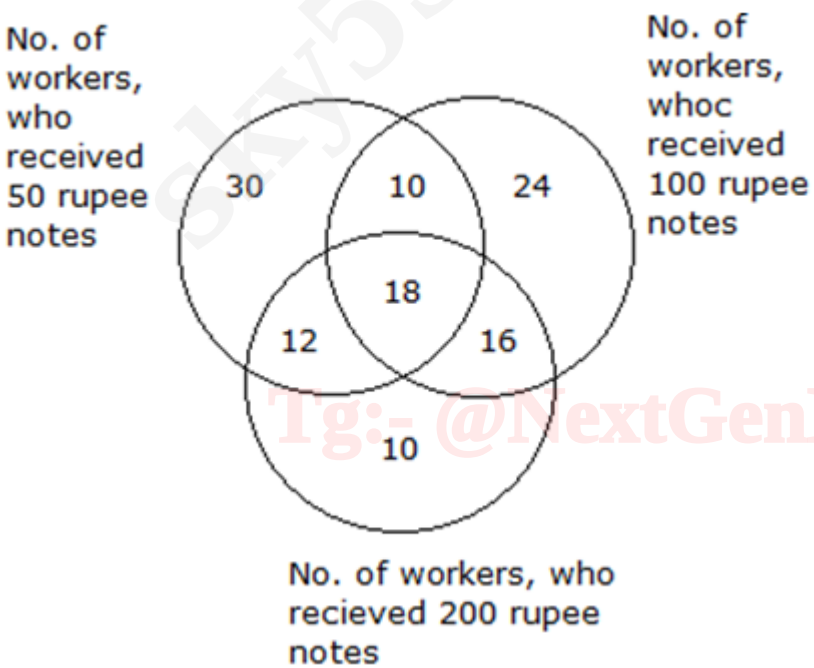
30 + 18 + x + 12 = 70

x = 10

The number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = 10

The number of workers, who received only 200 rupee notes = 10

The number of workers, who received both 100 and 200 rupee notes but not 50 rupee notes = 10 + 6 = 16



Answer: D

The total number of workers, who received 50 rupee notes:

30 + 10 + 12 + 18 = 70

Since, each worker received only one note of one denomination.

So, the total value of 50 rupee notes = 70 * 50 = Rs.3500

The total number of workers, who received 100 rupee notes:

24 + 10 + 16 + 18 = 68

Since, each worker received only one note of one denomination.

So, the total value of 100 rupee notes = 68 * 100 = Rs.6800

Required difference = 6800 – 3500 = Rs.3300

16. Questions

Since, the total number of workers is 120.

So, the number of workers, who received only 50 rupee notes = 25% of 120 = 30

The number of workers, who received only 100 rupee notes = 30 – 6 = 24

And the number of workers, who received all the three denominations notes = 30 * (3/5) = 18

Let the number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = The number of workers, who received only 200 rupee notes = x

Since, the number of workers, who received both 50 and 200 rupee notes but not 100 rupee notes, is 12.

And total number of persons, who received 50 rupee notes, is 70.

So,

30 + 18 + x + 12 = 70

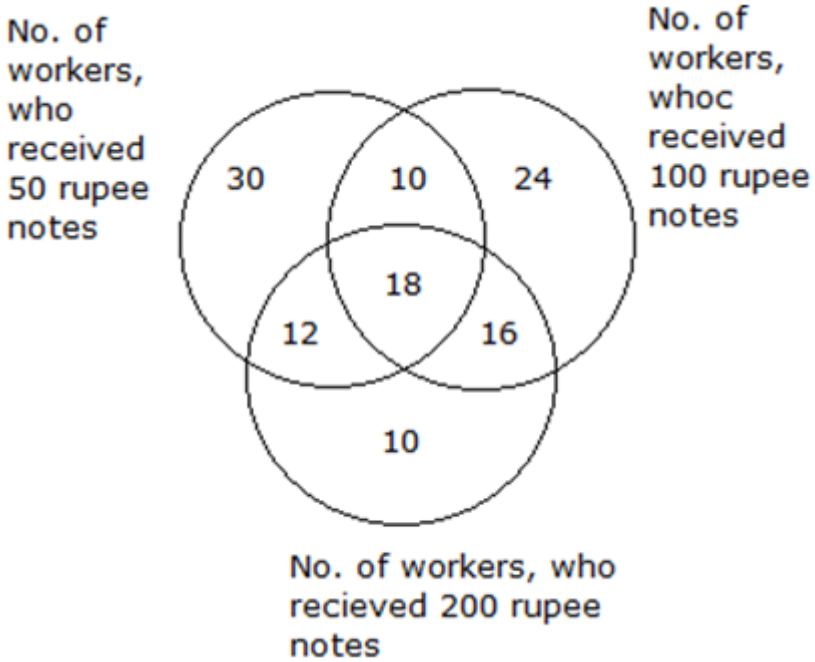
x = 10

The number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = 10

The number of workers, who received only 200 rupee notes = 10

The number of workers, who received both 100 and 200 rupee notes but not 50 rupee notes = 10 + 6 = 16

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Answer: A

The total number of workers, who received only two denominations notes:

10 + 16 + 12 = 38

The total number of workers, who received only one denomination note:

30 + 24 + 10 = 64

Required percentage = (38/64) * 100 = 59(3/8)%

17. Questions

Since, the total number of workers is 120.

So, the number of workers, who received only 50 rupee notes = 25% of 120 = 30

The number of workers, who received only 100 rupee notes = 30 – 6 = 24

And the number of workers, who received all the three denominations notes = 30 * (3/5) = 18

Let the number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = The number of workers, who received only 200 rupee notes = x

Since, the number of workers, who received both 50 and 200 rupee notes but not 100 rupee notes, is 12.

And total number of persons, who received 50 rupee notes, is 70.

So,

30 + 18 + x + 12 = 70

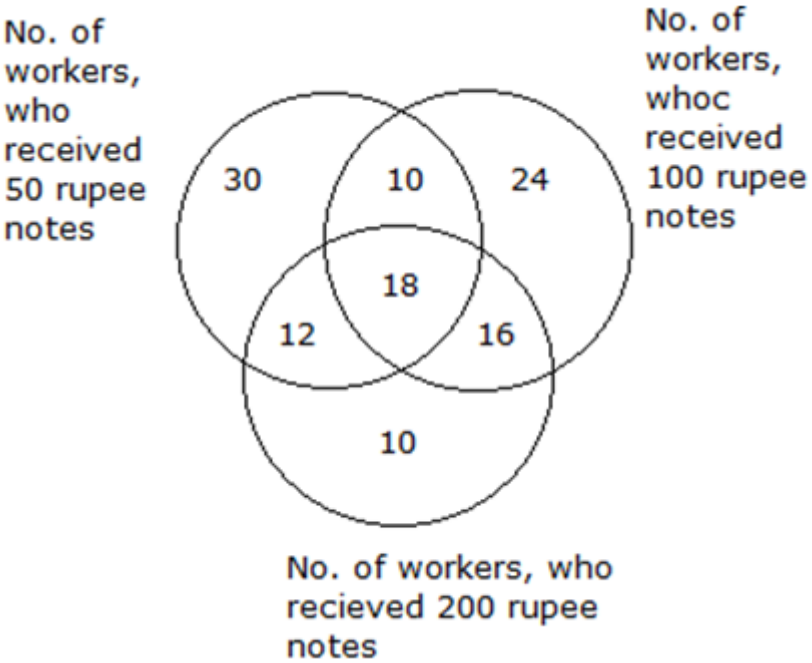
x = 10

The number of workers, who received both 50 and 100 rupee notes but not 200 rupee notes = 10

The number of workers, who received only 200 rupee notes = 10

The number of workers, who received both 100 and 200 rupee notes but not 50 rupee notes = 10 + 6 = 16

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Answer: B

The total number of workers, who received 50 rupee notes:

30 + 10 + 12 + 18 = 70

Since, each worker received only one note of one denomination.

So, the total value of 50 rupee notes = 70 * 50 = Rs.3500

The total number of workers, who received 100 rupee notes:

24 + 10 + 16 + 18 = 68

Since, each worker received only one note of one denomination.

So, the total value of 100 rupee notes = 68 * 100 = Rs.6800

The total number of workers, who received 200 rupee notes:

10 + 12 + 16 + 18 = 56

Since, each worker received only one note of one denomination.

So, the total value of 200 rupee notes = 56 * 200 = Rs.11200

Since, the total value of notes of all the three denominations together:

3500 + 6800 + 11200 = Rs.21500

So, the required percentage = (11200/21500) * 100 = 52.1% (approx.)

18. Questions

Answer: E

Quantity I:

Since, the boat can cover 30.6 km in still water in 30 minutes.

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So, the speed of boat in still water = (30.6 ×60)/30 = 61.2 km/h

Since, the boat can cover 18 km downstream in 15 minutes.

So, the downstream speed of the boat = (18 ×60)/15 = 72 km/h

And the speed of the stream = 72 – 61.2 = 10.8 km/h

Required percentage = (10.8/61.2) * 100 = 17.6% (approx.)

Quantity II:

Let B's present age be '2x' years.

So, A's present age = 2x/2 = 'x' years

And C's present age = (x + 2) years

Since, B's age after 4 years will be double of C's present age.

So,

2x + 4 = 2(x + 2)

Since, the equation can't be solved, so A's present age can't be determined.

Hence, relation can't be established.

19. Questions

Answer: B

Since, the difference between CI and SI on Rs.X at 15% rate after 3 years is Rs.170.1.

So,

$X \times (15/100)^2 \times ((300+15)/100)=170.1$

X = 2400

Quantity I:

The SI received on Rs. (X + 100) at 2% rate after 2 years:

((2400+100)×2 ×2)/100= Rs.100

Quantity II:

The difference between CI and SI received on Rs. (X + 2600) at 16% rate after 2 years:

$(2400+2600)\times (16/100)^2$= Rs.128

Hence, Quantity I < Quantity II

20. Questions

Answer: B

Let the original CP of the article be Rs.100x.

So, the original MP of the article = Rs. (100x + 45)

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And the original SP of the article = 85% of (100x + 45)

If the article was sold Rs.27 discount on its marked price:

So,

100x + 45 – 27 = 108% of 100x

100x + 18 = 108x

x = 2.25

The original CP of the article = 100 * 2.25 = Rs.225

The original MP of the article = 225 + 45 = Rs.270

The original SP of the article = 85% of 270 = Rs.229.5

Quantity I:

The original profit per cent = (229.5-225)/225 ×100 = 2%

Quantity II:

The original CP of the article = Rs.225

The original MP of the article = Rs.270

If the article was sold at 18% discount on its MP:

The SP of the article would have been = 82% of 270 = Rs.221.4

And the loss would have been = 225 – 221.4 = Rs.3.6

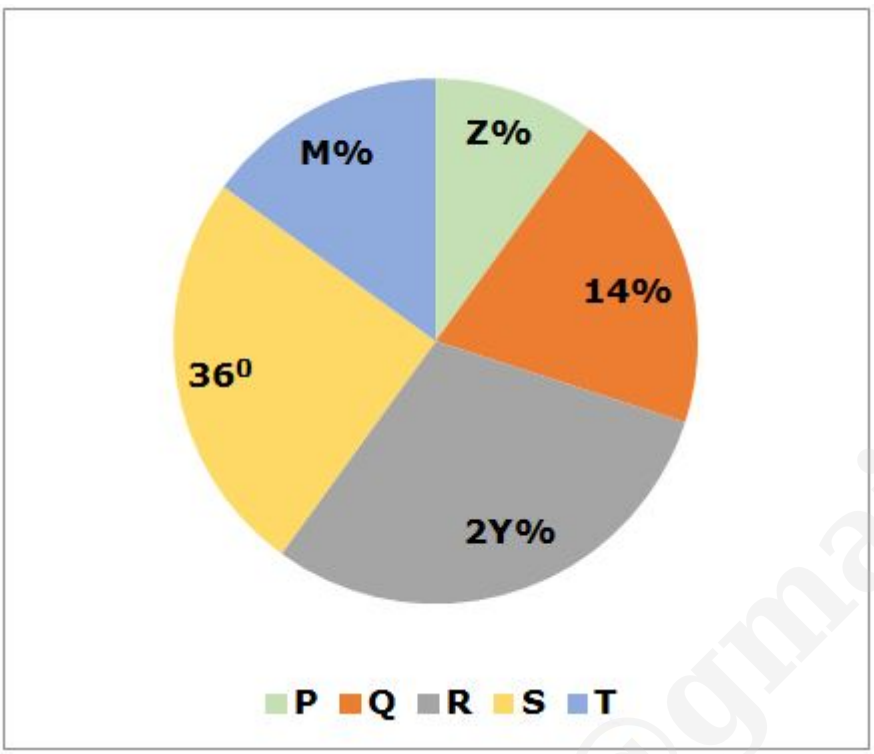
Hence, Quantity I < Quantity II

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1. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the % or degree distribution of number of articles sold by five different companies – P, Q, R, S, and T.



a). The total number of articles sold by all companies together are 800, while difference between number of articles sold by P and R is 136.

b). The number of articles sold by company T is multiple of 40, and all variables are natural numbers, such that M > Z.

c). The number of articles sold by company R is more than any other company.

The number of articles sold by company U is (3Y – 1)% more than that of company T, and out of total articles manufactured by company U, 3M% were sold and the rest were unsold. Find the difference between the sold articles of company P and company the unsold articles of company U.

a. 52

b. 48

c. 20

d. 36

e. None of these

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2. Questions

On the first day company T sold (Z – 1)/2 articles, and after the first day, on each day company sold one article more than the previous day. So, in this way company T sold all articles on the Ath day.

Company R sold 11 articles per day, find after A days what % of articles sold by company R. (approx.).

a. 53%

b. 55%

c. 60%

d. 59%

e. 57%

3. Questions

Each company sold two types of articles – A and B. The respective ratio of type A and type B articles sold by company S is 4:1 respectively, and number of type B articles sold by company T is five times of the number of type B articles sold by company S. Find average number of Type A articles sold by company S and T together.

a. 94

b. 92

c. 88

d. 96

e. None of these

4. Questions

Which of the following relation is/are true.

I). $Z = Y$

II). $(2Z + Y) < 2M$

III). $(2M + 2Y - 3) = 9^2$

a. I and III only

b. I and II only

c. I, II, and III

d. II and III only

e. None of these

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5. Questions

Out of the total articles sold by P, Q, and R together, 200 articles were sold at an average price of Rs. B and the rest were sold at an average price of Rs. A. The total revenue generated by the sale of articles by these three companies together is Rs. 11200. Find the value of B, if B is 20% more than that of A?

a. 24

b. 20

c. 30

d. 18

e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

There are three schools – P, Q, and R. Number of boys in school P and Q are Z and Y respectively, while number of girls in Q is M. Number of boys in school P is 33.33% of total students in school Q, and boys in school Q is 50% as that of in school R. Number of boys in school P is less than that of girls, and this statement is just opposite for rest two schools. Difference between boys and girls in school Q is 30.

The total students in school P are 200, and number of girls in schools P and Q is in the ratio of 32:9 respectively, then find number of boys in school R?

a. 180

b. 120

c. 150

d. 90

e. None of these

7. Questions

The number of students in schools P, Q, and R are in arithmetic progression in the same order and the total number of girls in schools P and R together is 90. Find the total number of students in all three schools together.

a. 540

b. 560

c. 480

d. 580

e. Cannot be determined

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8. Questions

The number of girls in school R is 90, while number of boys in school P is 25% as that of girls in same school. Find the total number of students in school P, if the number of students in school R is twice as that of Q?

a. 240

b. 200

c. 180

d. 280

e. None of these

9. Questions

Read the following information carefully and answer the questions based on it.

There are two vessel P and Q contains mixtures of milk and water, (completely filled).

Vessel P: Volume of Mixture of M ml and ratio of milk and water is in 2Z: (Z + 3) respectively.

Vessel Q: Mixture in vessel is 2M ml and water is 50% more than milk in vessel P. Respective ratio of water and milk is (Z – 1):Z respectively.

The amount of milk in vessel P initially is 40 ml. On adding 15Z ml of milk in vessel Q, now the amount of water in vessel Q is what % of the amount of milk in vessel Q.

a. 40%

b. 37.5%

c. 33.33%

d. 60%

e. None of these

10. Questions

180 ml from vessel P and K ml from vessel Q are poured out to form a new mixture, such that the amount of milk in new mixture is same as that of water. Find 30Z% of K.

a. 60

b. 36

c. 54

d. 24

e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

The table given below gives partial information about the number of students in five different colleges – P, Q, R, S and T in three different streams – A, B and C.

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Name of college	Average number of students in stream A and B together	Number of students in stream A	Number of students in stream C	% of students in stream B out of total students in college
P	420	9Z	360	25%
Q	200	80	400	2Y%
R	-	30Y	-	M%
S	840	-	-	20%
T	-	N	-	-

Note:

a). The total number of students in R and S together is 4400, out of which 5/11 are in college R.

b). The number of students in stream A and B together in college R is 80Y, which is 10Y more than that in college T.

For college T, the number of students in stream C is 1/8 times of the total number of students. The number of boys in stream B is the same as that of girls, which is five times the number of boys in stream C. The total number of students in stream B is same as number of boys in stream A. If the total number of girls in college T is 640, find the value of N/2.

a. 800

b. 400

c. 200

d. 300

e. None of these

12. Questions

For college U, the number of students in stream C is (Y + 20)% less than the sum of the number of students in stream B of college P and S together. The difference between boys and girls in stream C of college U is (M + 18), and number of girls in stream C of college U is ____ as compared to 50% of students in stream C of college R.

a. >

b. <

c. = or cannot be determined

d. ≥

e. ≤

13. Questions

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For college P, 60% of the total students passed in semester exams, out of which (Z/4 + 10)% are from stream B. The number of students who failed in the semester exam from stream B is what % of the number of students in stream B of college T, if the value of N is 600.

a. (Y - 5)%
b. (M – 24)%
c. (2Y – 4)%
d. 40%
e. None of these

14. Questions

Quantity I: The average number of students in stream A of college P, Q, and R together.

Quantity II: 7Z

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I = Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I ≥ Quantity II

15. Questions

For college T, the number of students in stream A is 50% of the total students, while the number of students in stream B of college T is the same as the number of students in stream A of college R. The number of students in stream B is K% of the total students. Find (2K)% of the students in stream C?

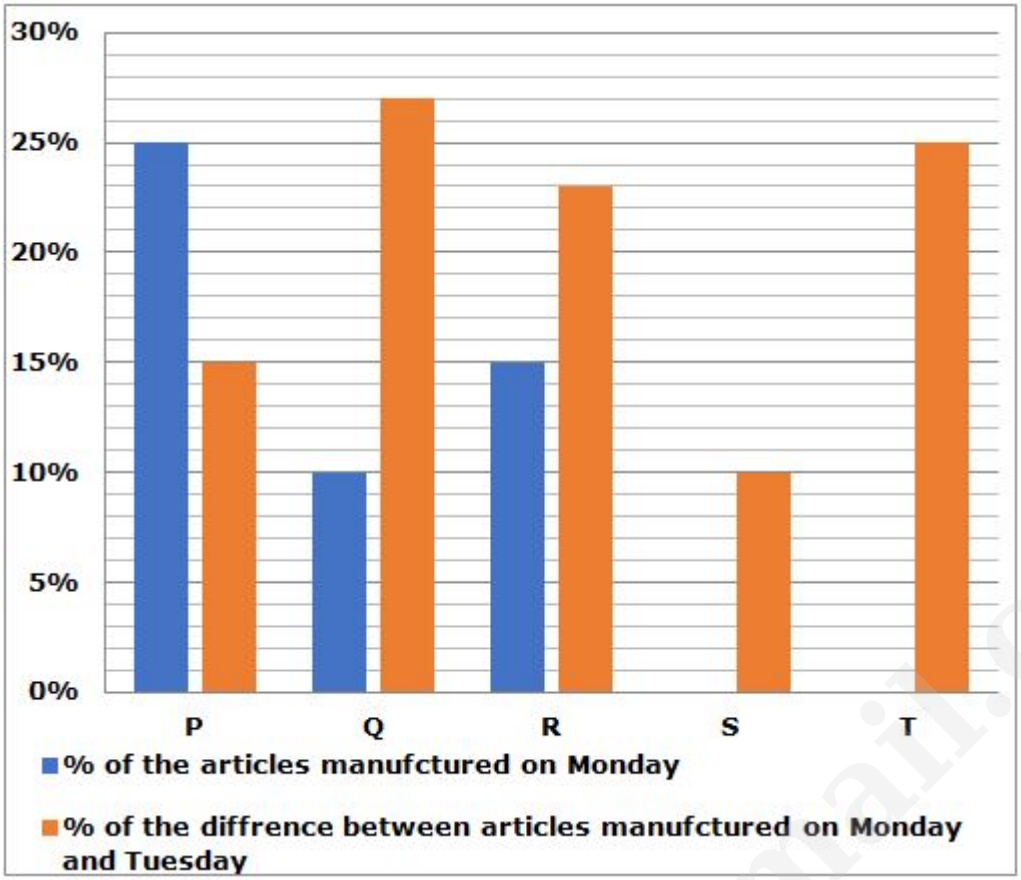
a. 75
b. 80
c. 150
d. 160
e. None of these

16. Questions

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Read the following information carefully and answer the questions based on it.

There are five companies – P, Q, R, S and T. Each company manufactured products on two different days – Monday and Tuesday. The chart given below shows the % of articles manufactured by a company on Monday out of the total articles manufactured by all companies together on Monday, and % of the difference between the articles manufactured on Monday and Tuesday out of the sum of difference between Monday and Tuesday by all companies together.



Note:

a). The sum of difference between articles manufactured on Monday and Tuesday of all companies together is 400, and the total number of articles manufactured by company R is 308

b). The number of articles manufactured on Monday by company S is 50% more than that of company T.

The difference between the number of articles manufactured by company U and T on Monday is 16, and number of articles manufactured by company U on Tuesday is 50% more than that on Monday. Out of the total articles manufactured on Tuesday 40% remain unsold. Find the average number of articles manufactured by company R and U together on Tuesday.

a. 196

b. 220

c. 174

d. Both (a) and (b)

e. None of these

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17. Questions

The total number of products manufactured by company P is more than that of company S. The average price per unit of article sold by each company on Tuesday is Rs. 40. Find the revenue generated by company P, Q, and S on Tuesday, if all manufactured articles on Tuesday were sold.

a. Rs. 20960

b. Rs. 23840

c. Rs. 19520

d. Rs. 19360

e. Cannot be determined

18. Questions

Out of the total products manufactured on Monday and Tuesday by company S, respective ratio of sold and unsold products is 3:1, and 9:2 respectively. The difference between the number of products sold by company T and company S is 94. Find the number of products unsold by company T.

a. 216

b. 212

c. 196

d. 176

e. Can't be determined

19. Questions

The total number of products manufactured by company P on Tuesday is ___ than that of company Q.

a. >

b. <

c. ≥

d. ≤

e. CND

20. Questions

Which of the following can be uniquely determined.

I). The total number of products manufactured by Company Q.

II). The total number of products unsold by company S, if the number of products sold are 400.

III). The difference between the products manufactured by P and Q on Tuesday.

a. I only

b. I and II only

c. II and III only

d. I, II, and III

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e. None of these

Explanations:

1. Questions

Difference between articles sold by P and R = 136/800 x 100 = 17%

2Y – Z = 17

2Y = Z + 17…….... (1)

Also,

M + Z + 2Y = 100 – 14 – 10

M + Z + Z + 17 = 76

M + 2Z = 59

Z = (59 – M)/2……. (2)

Number of articles sold by T is multiple of 40, that means M% is multiple of = 40/800 x 100 = 5%

Also, M > Z and M must be odd.

If M = (5, 15) then Z = (27, 22) [not possible]

If M = 25, then Z = 17, 2Y = 34 (possible)

If M = 35, then Z = 12, 2Y = 29 [not possible, because Y should be a natural number]

So, we have only one possible pair, now we can find all the related data

Name of the company	Number of articles sold
P	136
Q	112
R	272
S	80
T	200

Answer: D

For company U,

Number of articles sold = 200 x (100 + 3Y – 1)% = 200 x 150% = 300

% of sold articles = 3M% = 75%

so, % of unsold articles = 25%

so, number of unsold articles = 300/3 = 100

Required difference = 136 – 100 = 36

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Hence answer is option D

2. Questions

Difference between articles sold by P and R = 136/800 x 100 = 17%

2Y – Z = 17

2Y = Z + 17…….... (1)

Also,

M + Z + 2Y = 100 – 14 – 10

M + Z + Z + 17 = 76

M + 2Z = 59

Z = (59 – M)/2……. (2)

Number of articles sold by T is multiple of 40, that means M% is multiple of = 40/800 x 100 = 5%

Also, M > Z and M must be odd.

If M = (5, 15) then Z = (27, 22) [not possible]

If M = 25, then Z = 17, 2Y = 34 (possible)

If M = 35, then Z = 12, 2Y = 29 [not possible, because Y should be a natural number]

So, we have only one possible pair, now we can find all the related data

Name of the company	Number of articles sold
P	136
Q	112
R	272
S	80
T	200

Answer: E

Total articles sold by company T = 200

On day one, number of articles sold = 8

Number of articles sold on 2nd day = 9

Number of articles sold on 3rd day = 10

Number of articles sold in 11 days = 8 + (8 + 1) + (8 + 2)…… + 8 + 10) = 8 + 8 x [1 + 2 + 3……. + 10] = 143

In 13 days, number of articles sold = 143 + 19 + 20 = 182

So, on 14th day total number of articles sold. So, value of A = 14.

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Total number of articles sold by company R in A days = 14 x 11 = 154

Required % = 154/272 x 100 = 57% (approx.)

Hence answer is option E

3. Questions

Difference between articles sold by P and R = 136/800 x 100 = 17%

2Y – Z = 17

2Y = Z + 17…….... (1)

Also,

M + Z + 2Y = 100 – 14 – 10

M + Z + Z + 17 = 76

M + 2Z = 59

Z = (59 – M)/2……. (2)

Number of articles sold by T is multiple of 40, that means M% is multiple of = 40/800 x 100 = 5%

Also, M > Z and M must be odd.

If M = (5, 15) then Z = (27, 22) [not possible]

If M = 25, then Z = 17, 2Y = 34 (possible)

If M = 35, then Z = 12, 2Y = 29 [not possible, because Y should be a natural number]

So, we have only one possible pair, now we can find all the related data

Name of the company	Number of articles sold
P	136
Q	112
R	272
S	80
T	200

Answer: B

Number of type A articles sold by S = 4/5 x 80 = 64

Number of type B articles sold by S = 80 – 64 = 16

Number of type B articles sold by T = 16 x 5 = 80

So, number of type A articles sold by T = 200 – 80 = 120

Required average = (64 + 120)/2 = 92

Hence answer is option B

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4. Questions

Difference between articles sold by P and R = 136/800 x 100 = 17%

2Y – Z = 17

2Y = Z + 17…….... (1)

Also,

M + Z + 2Y = 100 – 14 – 10

M + Z + Z + 17 = 76

M + 2Z = 59

Z = (59 – M)/2……. (2)

Number of articles sold by T is multiple of 40, that means M% is multiple of = 40/800 x 100 = 5%

Also, M > Z and M must be odd.

If M = (5, 15) then Z = (27, 22) [not possible]

If M = 25, then Z = 17, 2Y = 34 (possible)

If M = 35, then Z = 12, 2Y = 29 [not possible, because Y should be a natural number]

So, we have only one possible pair, now we can find all the related data

Name of the company	Number of articles sold
P	136
Q	112
R	272
S	80
T	200

Answer: A

I). Z = Y

Value of Z = Y = 17

This statement is true

II). (2Z + Y) < 2M

Value of (2Z + Y) = 2 x 17 + 17 = 51

Value of 2M = 2 x 25 = 50

This relation is false

III). $(2M + 2Y – 3) = 9^2$

Required value = $(2 \times 25 + 34 – 3) = 81 = 9^2$

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Hence answer is option A

5. Questions

Difference between articles sold by P and R = 136/800 x 100 = 17%

2Y – Z = 17

2Y = Z + 17…….... (1)

Also,

M + Z + 2Y = 100 – 14 – 10

M + Z + Z + 17 = 76

M + 2Z = 59

Z = (59 – M)/2……. (2)

Number of articles sold by T is multiple of 40, that means M% is multiple of = 40/800 x 100 = 5%

Also, M > Z and M must be odd.

If M = (5, 15) then Z = (27, 22) [not possible]

If M = 25, then Z = 17, 2Y = 34 (possible)

If M = 35, then Z = 12, 2Y = 29 [not possible, because Y should be a natural number]

So, we have only one possible pair, now we can find all the related data

Name of the company	Number of articles sold
P	136
Q	112
R	272
S	80
T	200

Answer: A

Total number of articles sold by (P + Q + R) together = 136 + 112 + 272 = 520

B = 6A/5

Now,

200 x 6A/5 + 320 x A = 11200

560A = 11200

So, value of A = 20

So, value of B = 20 x 1.2 = 24

Hence answer is option A

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6. Questions

Number of boys in P = Z

So, total students in Q = 3 x Z = 3Z

Number of boys in Q = Y

So, number of boys in R = 2 x Y = 2Y

Number of girls in Q = M

Number of boys in Q > number of girls in Q

So, Y – M = 30……. (1)

Also,

3Z = Y + M……. (2)

Answer: C

According to question,

Ratio of girls in P and Q = 32:9 = 32a and 9a

So,

200 = Z + 32a……. (3)

3Z = Y + 9a……... (2)

From (2) and (3), we get

600 + 3Z = 3Z + 96a + Y + 9a

600 = Y + 105a

OR 600 = 30 + 9a + 105a

114a = 570

Value of a = 5

So, M = 45, and Y = 45 + 30 = 75

So, number of boys in R = 2Y = 2 x 75 = 150

Hence answer is option C

7. Questions

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Number of boys in P = Z

So, total students in Q = 3 x Z = 3Z

Number of boys in Q = Y

So, number of boys in R = 2 x Y = 2Y

Number of girls in Q = M

Number of boys in Q > number of girls in Q

So, Y – M = 30……. (1)

Also,

3Z = Y + M……. (2)

Answer: A

According to question,

Total number of boys in P and R together = Z + 2Y

Total students in P, Q, and R in arithmetic progression in same order

So, total students in (P + R) = 2 x 3Z = 6Z

Girls in (P + R) = 90 = 6Z – Z – 2Y =

5Z – 2Y = 90…… (1)

Total students in Q = 3Z = Y + M = (Y + Y – 30)

3Z + 30 = 2Y……... (2)

Form (1) and (2), we get

5Z – 3Z – 30 = 90

Z = 60

So, total students = 3Z + 6Z = 9Z = 9 x 60 = 540

Hence answer is option A

8. Questions

Number of boys in P = Z

So, total students in Q = 3 x Z = 3Z

Number of boys in Q = Y

So, number of boys in R = 2 x Y = 2Y

Number of girls in Q = M

Number of boys in Q > number of girls in Q

So, Y – M = 30……. (1)

Also,

3Z = Y + M……. (2)

Answer: B

Total students in Q and R = 1:2

Boys in Q and R = 1:2

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So, girls in Q and R must be in = 1:2

So, (Y – 30) = 90/2 = 45

Y = 75

So, Z = (75 + 45)/3 = 40

Ratio of boys and girls in P = 1:4 = Z, 4Z

So, total students in P = 5Z = 5 x 40 = 200

Hence answer is option B

9. Questions

Amount of water in Q = 3/2 x amount of milk in P

Amount of water in Q = = (Z – 1)/(2Z – 1) x 2M

Amount of milk in P = 2Z / (3Z + 3) x M

Now,

3/2 x [2Z/3 x (Z + 1) x M] = (Z – 1)/(2Z – 1) x 2M

[Z / (Z + 1)] = [(Z – 1) / (2Z – 1)] x 2

$2Z^2 – Z = 2Z^2 – 2$

Value of Z = 2

So, ratio of milk and water in P = 4:5

Ratio of milk and water in Q = 2:1

Answer: A

According to question,

Amount of milk in P = 40 ml

So, amount of water in Q = 40 x 1.5 = 60 ml

On adding 30 ml milk in Q, amount of milk = 120 + 30 = 150

Required % = 60/150 x 100 = 40%

Hence answer is option A

10. Questions

Amount of water in Q = 3/2 x amount of milk in P

Amount of water in Q = = (Z – 1)/(2Z – 1) x 2M

Amount of milk in P = 2Z / (3Z + 3) x M

Now,

3/2 x [2Z/3 x (Z + 1) x M] = (Z – 1)/(2Z – 1) x 2M

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$[Z / (Z + 1)] = [(Z – 1) / (2Z – 1)] \times 2$

$2Z^2 – Z = 2Z^2 – 2$

Value of Z = 2

So, ratio of milk and water in P = 4:5

Ratio of milk and water in Q = 2:1

Answer: B

4/9 x 180 + 2/3 x K = 5/9 x 180 + 1/3 x K

20 = K/3

K = 60

So, required value = 60% x 60 = 36

Hence answer is option B

11. Questions

For P

Total students in (A + B) = 2 x 420 = 840

Students in C = 360

So, total students = 840 + 360 = 1200

Students in B = 25% x 1200 = 300

So, students in A = 840 – 300 = 540

So, 9Z = 540

Value of Z = 60

For Q

Students in (A + B) = 200 x 2 = 400

Students in A = 80

Students in B = 400 – 80 = 320

Students in C = 400

Total students in College = 400 + 400 = 800

2Y% = 320/800 x 100 = 40%

So, value of Y = 20

For R

Total number of students = 4400 x 5/11 = 2000

Number of students in (A + B) = 80 x 20 = 1600

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Number of students in A = 30 x Y = 30 x 20 = 600

Number of students in B = 1600 – 600 = 1000

So, number of students in C = 2000 – 1600 = 400

Value of M% = 1000/2000 x 100 = 50%

For S

Total students in S = 4400 – 2000 = 2400

Students in (A + B) = 840 x 2 = 1680

So, students in C = 2400 – 1680 = 720

Students in B = 20% x 2400 = 480

So, students in A = 1680 – 480 = 1200

For T

Number of students in (A + B) = 80Y – 10Y = 70Y = 70 x 20 = 1400

Name of college	Total students	Stream A	Stream B	Stream C
P	1200	540	300	360
Q	800	80	320	400
R	2000	600	1000	400
S	2400	1200	480	720
T	-	N	-	-

Answer: B

According to question,

For college T

Number of students in C = 1/8 x total students

So, Number of students in (A + B) = 7/8 x total students = 1400

So, total students = 1600

Number of students in C = 200

Let number of boys in B = number of girls in B = a

So, number of boys in A = (a + a) = 2a

So, number of Boys in C = a/5

Total number of boys in college = (a/5 + 2a + a) = 16a/5

16a/5 = 1600 – 640 = 960

So, value of a = 960 x 5/16 = 300

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So, total number of students in B = 2 x a = 600

So, students in A = 1400 – 600 = 800

So, value of N/2 = 800/2 = 400

Hence answer is option B

12. Questions

For P

Total students in (A + B) = 2 x 420 = 840

Students in C = 360

So, total students = 840 + 360 = 1200

Students in B = 25% x 1200 = 300

So, students in A = 840 – 300 = 540

So, 9Z = 540

Value of Z = 60

For Q

Students in (A + B) = 200 x 2 = 400

Students in A = 80

Students in B = 400 – 80 = 320

Students in C = 400

Total students in College = 400 + 400 = 800

2Y% = 320/800 x 100 = 40%

So, value of Y = 20

For R

Total number of students = 4400 x 5/11 = 2000

Number of students in (A + B) = 80 x 20 = 1600

Number of students in A = 30 x Y = 30 x 20 = 600

Number of students in B = 1600 – 600 = 1000

So, number of students in C = 2000 – 1600 = 400

Value of M% = 1000/2000 x 100 = 50%

For S

Total students in S = 4400 – 2000 = 2400

Students in (A + B) = 840 x 2 = 1680

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So, students in C = 2400 – 1680 = 720

Students in B = 20% x 2400 = 480

So, students in A = 1680 – 480 = 1200

For T

Number of students in (A + B) = 80Y – 10Y = 70Y = 70 x 20 = 1400

Name of college	Total students	Stream A	Stream B	Stream C
P	1200	540	300	360
Q	800	80	320	400
R	2000	600	1000	400
S	2400	1200	480	720
T	-	N	-	-

Answer: D

For college U, Stream C

Number of students (Boys + girls) = (100 – Y – 20)% x [300 + 480] = 468

Difference between boys and girls = (M + 18) = 68

So, number of girls = (468 + 68)/2 OR (468 – 68)/2 = 268 or 200

50% of students in stream C of college R = 400 x (1/2) = 200

So, (268 or 200) ≥ 200

Hence answer is option D

13. Questions

For P

Total students in (A + B) = 2 x 420 = 840

Students in C = 360

So, total students = 840 + 360 = 1200

Students in B = 25% x 1200 = 300

So, students in A = 840 – 300 = 540

So, 9Z = 540

Value of Z = 60

For Q

Students in (A + B) = 200 x 2 = 400

Students in A = 80

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Students in B = 400 – 80 = 320

Students in C = 400

Total students in College = 400 + 400 = 800

2Y% = 320/800 x 100 = 40%

So, value of Y = 20

For R

Total number of students = 4400 x 5/11 = 2000

Number of students in (A + B) = 80 x 20 = 1600

Number of students in A = 30 x Y = 30 x 20 = 600

Number of students in B = 1600 – 600 = 1000

So, number of students in C = 2000 – 1600 = 400

Value of M% = 1000/2000 x 100 = 50%

For S

Total students in S = 4400 – 2000 = 2400

Students in (A + B) = 840 x 2 = 1680

So, students in C = 2400 – 1680 = 720

Students in B = 20% x 2400 = 480

So, students in A = 1680 – 480 = 1200

For T

Number of students in (A + B) = 80Y – 10Y = 70Y = 70 x 20 = 1400

Name of college	Total students	Stream A	Stream B	Stream C
P	1200	540	300	360
Q	800	80	320	400
R	2000	600	1000	400
S	2400	1200	480	720
T	-	N	-	-

Answer: A

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For college P

Number of students passed in semester exam in stream B = 1200 x 60% x 25% = 180

So, number of students failed in stream B = 300 – 180 = 120

Number of students in stream B of college T = 1400 – 600 = 800

So, Required % = 120/800 x 100 = 15% = (Y - 5)%

Hence answer is option A

14. Questions

For P

Total students in (A + B) = 2 x 420 = 840

Students in C = 360

So, total students = 840 + 360 = 1200

Students in B = 25% x 1200 = 300

So, students in A = 840 – 300 = 540

So, 9Z = 540

Value of Z = 60

For Q

Students in (A + B) = 200 x 2 = 400

Students in A = 80

Students in B = 400 – 80 = 320

Students in C = 400

Total students in College = 400 + 400 = 800

2Y% = 320/800 x 100 = 40%

So, value of Y = 20

For R

Total number of students = 4400 x 5/11 = 2000

Number of students in (A + B) = 80 x 20 = 1600

Number of students in A = 30 x Y = 30 x 20 = 600

Number of students in B = 1600 – 600 = 1000

So, number of students in C = 2000 – 1600 = 400

Value of M% = 1000/2000 x 100 = 50%

For S

Total students in S = 4400 – 2000 = 2400

Students in (A + B) = 840 x 2 = 1680

So, students in C = 2400 – 1680 = 720

Students in B = 20% x 2400 = 480

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So, students in A = 1680 – 480 = 1200

For T

Number of students in (A + B) = 80Y – 10Y = 70Y = 70 x 20 = 1400

Name of college	Total students	Stream A	Stream B	Stream C
P	1200	540	300	360
Q	800	80	320	400
R	2000	600	1000	400
S	2400	1200	480	720
T	-	N	-	-

Answer: B

Quantity I = (540 + 80 + 600)/3 = 406.67

Quantity II = 7Z = 7 x 60 = 420

So, Quantity I < Quantity II

Hence answer is option B

15. Questions

For P

Total students in (A + B) = 2 x 420 = 840

Students in C = 360

So, total students = 840 + 360 = 1200

Students in B = 25% x 1200 = 300

So, students in A = 840 – 300 = 540

So, 9Z = 540

Value of Z = 60

For Q

Students in (A + B) = 200 x 2 = 400

Students in A = 80

Students in B = 400 – 80 = 320

Students in C = 400

Total students in College = 400 + 400 = 800

2Y% = 320/800 x 100 = 40%

So, value of Y = 20

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For R

Total number of students = 4400 x 5/11 = 2000

Number of students in (A + B) = 80 x 20 = 1600

Number of students in A = 30 x Y = 30 x 20 = 600

Number of students in B = 1600 – 600 = 1000

So, number of students in C = 2000 – 1600 = 400

Value of M% = 1000/2000 x 100 = 50%

For S

Total students in S = 4400 – 2000 = 2400

Students in (A + B) = 840 x 2 = 1680

So, students in C = 2400 – 1680 = 720

Students in B = 20% x 2400 = 480

So, students in A = 1680 – 480 = 1200

For T

Number of students in (A + B) = 80Y – 10Y = 70Y = 70 x 20 = 1400

Name of college	Total students	Stream A	Stream B	Stream C
P	1200	540	300	360
Q	800	80	320	400
R	2000	600	1000	400
S	2400	1200	480	720
T	-	N	-	-

Answer: C

For College T

Number of students in stream B = 600

So, number of students in stream A = 1400 – 600 = 800

So, total number of students = 800 x 2 = 1600

So, number of students in stream C = 1600 – 1400 = 200

K = 600/1600 x 100 = 37.5%

So, required value = 2 x 37.5% x 200 = 150

Hence answer is option C

16. Questions

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For R

Total number of articles manufactured by company R = 308

Difference between articles manufactured on Monday and Tuesday = 23% x 400 = 92

Number of articles manufactured on Monday = (308 + 92)/2 OR (308 - 92)/2 = 200 or 108

So, 15% x (total products manufactured on Monday by all companies together = 200 or 108

So, total products manufactured by all companies together on Monday = 200/15 x 100 = 1333.33 (not possible)

So, total products manufactured by all companies together on Monday = 108/15 x 100 = 720

So, total products manufactured on Monday = 720

Name of company	Monday	Difference between Monday and Tuesday	Tuesday
P	180	60	120 or 240
Q	72	108	180
R	108	92	200
S	216	40	256 or 176
T	144	100	244 or 44

Answer: B

According to question,

For company U,

Number of articles manufactured on Monday = 144 + 16 OR 144 – 16 = 160 or 128

Number of articles manufactured on Tuesday = 160 x 1.5 or 128 x 1.5 = 240 or 192

Number of articles remains unsold on Tuesday = 40% = 2/5 x manufactured on Tuesday.

So, articles on Tuesday cannot be 192.

So, articles manufactured on Tuesday = 240

Articles manufactured by R on Tuesday = 200

Required average = (200 + 240)/2 = 220

Hence answer is option B

17. Questions

For R

Total number of articles manufactured by company R = 308

Difference between articles manufactured on Monday and Tuesday = 23% x 400 = 92

Number of articles manufactured on Monday = (308 + 92)/2 OR (308 - 92)/2 = 200 or 108

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So, 15% x (total products manufactured on Monday by all companies together = 200 or 108

So, total products manufactured by all companies together on Monday = 200/15 x 100 = 1333.33 (not possible)

So, total products manufactured by all companies together on Monday = 108/15 x 100 = 720

So, total products manufactured on Monday = 720

Name of company	Monday	Difference between Monday and Tuesday	Tuesday
P	180	60	120 or 240
Q	72	108	180
R	108	92	200
S	216	40	256 or 176
T	144	100	244 or 44

Answer: B

Total articles manufactured by P = 180 + 120 or 180 + 240 = 300 or 420

Total articles manufactured by S = 216 + 256 Or 216 + 176 = 472 or 392

Articles of P > S,

So, articles manufactured by P and S on Tuesday = 240 and 176 respectively

Required revenue = (240 + 176 + 180) x 40 = Rs. 23840

Hence answer is option B

18. Questions

For R

Total number of articles manufactured by company R = 308

Difference between articles manufactured on Monday and Tuesday = 23% x 400 = 92

Number of articles manufactured on Monday = (308 + 92)/2 OR (308 - 92)/2 = 200 or 108

So, 15% x (total products manufactured on Monday by all companies together = 200 or 108

So, total products manufactured by all companies together on Monday = 200/15 x 100 = 1333.33 (not possible)

So, total products manufactured by all companies together on Monday = 108/15 x 100 = 720

So, total products manufactured on Monday = 720

Name of company	Monday	Difference between Monday and Tuesday	Tuesday
P	180	60	120 or 240
Q	72	108	180
R	108	92	200
S	216	40	256 or 176
T	144	100	244 or 44

Answer: D

For company S

Products sold on Monday = 216 x 3/4 = 162

Products manufactured on Tuesday must be multiple of 11, so products manufactured on Tuesday = 176

Products sold on Tuesday = 9/11 x 176 = 144

Total products sold = 162 + 144 = 306

Total products manufactured by T = 388 or 188

Total products sold by T = (306 – 94) Or (306 + 94) = 212 or 400

So, total products manufactured by T must be = 388

So, products unsold by T = 388 – 212 = 176

Hence answer is option D

19. Questions

For R

Total number of articles manufactured by company R = 308

Difference between articles manufactured on Monday and Tuesday = 23% x 400 = 92

Number of articles manufactured on Monday = (308 + 92)/2 OR (308 - 92)/2 = 200 or 108

So, 15% x (total products manufactured on Monday by all companies together = 200 or 108

So, total products manufactured by all companies together on Monday = 200/15 x 100 = 1333.33 (not possible)

So, total products manufactured by all companies together on Monday = 108/15 x 100 = 720

So, total products manufactured on Monday = 720

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Name of company	Monday	Difference between Monday and Tuesday	Tuesday
P	180	60	120 or 240
Q	72	108	180
R	108	92	200
S	216	40	256 or 176
T	144	100	244 or 44

Answer: E

Products manufactured by P on Tuesday = 120 or 240

Products manufactured by Q on Tuesday = 180

So, 120 < 180 and 240 > 180

So, we cannot find the relation.

Hence answer is option E

20. Questions

For R

Total number of articles manufactured by company R = 308

Difference between articles manufactured on Monday and Tuesday = 23% x 400 = 92

Number of articles manufactured on Monday = (308 + 92)/2 OR (308 - 92)/2 = 200 or 108

So, 15% x (total products manufactured on Monday by all companies together = 200 or 108

So, total products manufactured by all companies together on Monday = 200/15 x 100 = 1333.33 (not possible)

So, total products manufactured by all companies together on Monday = 108/15 x 100 = 720

So, total products manufactured on Monday = 720

Name of company	Monday	Difference between Monday and Tuesday	Tuesday
P	180	60	120 or 240
Q	72	108	180
R	108	92	200
S	216	40	256 or 176
T	144	100	244 or 44

Answer: D

I). Total number of products manufactured by Company Q = 252

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II). Total products unsold by company S, if number of products sold are 400.

Total products manufactured by company S should not be less than 400. So total products manufactured by S = 216 + 256 = 472

So, unsold = 472 – 400 = 72

III). Difference between products manufactured by P and Q on Tuesday.

Require difference = 180 – 120 or 240 – 180 = 60

So, all statements can be uniquely determined

Hence answer is option D

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1. Questions

Study the following information carefully and answer the questions given below.

In each of the five different schools (A, B, C, D, and E), there are three departments: science, commerce, and arts, each department with different numbers of boys and girls.

The given table charts shows the percentage of students in the science department out of the total number of students in each school, the ratio of the number of students in the commerce and arts departments, the difference between the number of boys and girls in the science department, the percentage of the number of girls out of the total number of students in the commerce department, and the ratio of the number of boys and girls in the arts department. Some values are missing and approximate values should always be used.

SCHOOL	The percentage of students in the science department out of total number of students in each school	The ratio of the number of students in the Commerce and Arts Departments	The difference between boys and girls in the science department	The percentage of girls out of the total number of students in the commerce department	The ratio of boys and girls in the Arts department
A	___	4:3	40	_____	1:2
B	41.66%	____	_____	37.5%	7:5
C	___	1:2	10	25%	____
D	40.90%	7:6	_____	50%	____
E	_____	____	20	____	4:3

Note:

The percentage distribution of total students in schools A, B, C, D, and E is 23.63%, 21.81%, 16.36%, 20%, and 18.18%, respectively. In school D, boys and girls in each department are equal. In each school, the number of boys in the science department is either greater or equal to the number of girls. The number of girls in arts department in school D is 30.

The number of girls in the arts department in school A is 33.33% more than the number of girls in the same department of school D. Find the number of girls in the commerce department of school A if the number of boys in the commerce department of school A is 25% less than the number of boys in science department in the same school.

a. 30

b. 25

c. 20

d. 15

e. 12

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2. Questions

The number of girls in the commerce department in school D is 16.66% more than that of same department in school B. The difference between the boys and girls in the arts department in school B is half of the difference between the boys and girls in the science department in school B. Find the number of boys in the science department in school B.

a. 75

b. 65

c. 55

d. 60

e. 30

3. Questions

The number of boys in the commerce and arts department is equal in school C. In school C, the number of girls in the science department is half of the number of girls in the arts department. Find the total number of students in the science departments in schools C and D.

a. 140

b. 150

c. 160

d. 190

e. None of these

4. Questions

The difference between the number of students in the science and commerce department of D is x and the difference between the number of students in the arts and commerce department of D is y. Find the value of (7x+9y).

a. 230

b. 240

c. 250

d. 260

e. None of these

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5. Questions

The number of girls in the arts department in schools D and E is the same, and the total number of girls in school E is 80. In school E, the ratio of the number of girls in the science to commerce department is 3:2. Find the ratio of the number of boys in the science to commerce department in school E.

a. 5:3

b. 3:4

c. 6:7

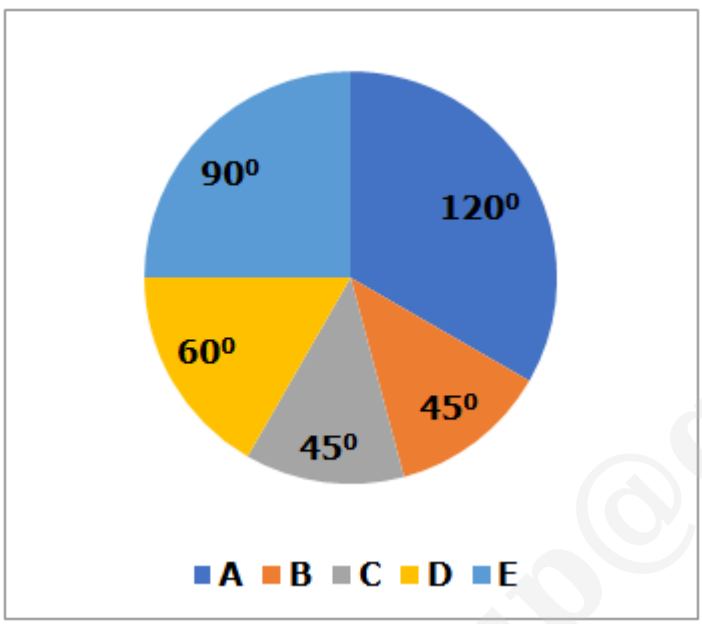
d. 7:4

e. None of these

6. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the degree distribution of the number of unsold furniture in five different shops [A, B, C, D and E] in January.



Note:

The total number of furniture manufactured = The total number of sold furniture + The total number of unsold furniture.

In each shop, 75% of furniture is sold. The number of sold furniture in shop E is 90.

The number of furniture manufactured by shop A in February is 20% more than the number of furniture manufactured by the same shop in January. The ratio of the number of sold furniture in January and February is 5:7. Find the difference between the number of unsold furniture in January and February.

a. 22

b. 18

c. 16

d. 29

e. 26

7. Questions

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Tg:- @NextGenBankers

The ratio of the number of chairs, tables and beds manufactured by shop B in January is 3:2:1 respectively. 60% of tables and 50% of beds were sold in shop B. Find the number of chairs sold in shop B.

a. 22

b. 26

c. 32

d. 28

e. 30

8. Questions

The difference between the number of furniture manufactured in shops C and D is 'x' and the difference between the number of furniture manufactured in shops D and E is 'y'.

Find which of the following equation is true.

a. 5x+9y=460

b. 5x+7x=320

c. 6x+9y=480

d. 7x+9y=500

e. More than one option is true

9. Questions

The average manufacturing cost of each furniture in shop D is Rs._____ and the average selling price of each furniture in shop D is Rs._____. The profit percentage of shop D is_____%.

Find which of the following is true.

a. 2400, 3200, 6

b. 2500, 3600, 8

c. 2000, 3200, 10

d. 2500, 4000, 12

e. More than one option is true

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10. Questions

The ratio of the number of furniture manufactured in shops E and F is _____ and the ratio of the number of furniture manufactured in shops D and G is _____. The total number of furniture manufactured in shops F and G is _____.

I). 3:4, 4:5, 260

II). 4:5, 4:3, 210

III). 4:7, 5:6, 320

a. Only I and II are true

b. Only II and III true

c. Only II false

d. Only I true

e. None of these

11. Questions

Study the following information carefully and answer the questions given below.

A survey is conducted in a class. The ratio between the number of students who like only football and the number of students who like only cricket is 9:8. The number of students who like both football and cricket but not hockey is five times the number of students who like all three sports. The number of students who like only football is three times the number of students who like only hockey. The ratio of the number of students who like both football and hockey but not cricket and the number of students who like both hockey and cricket but not football is 1:2. The number of students who like both cricket and hockey but not football is 20% less than the number of students who like both football and cricket but not hockey. The number of students who like only cricket is double the number of students who like both cricket and hockey but not football. The number of students who like both cricket and football but not hockey is a perfect square between 20 and 30.

The number of students who like only basketball is 20% more than the number of students who like only football and the number of students who only baseball is 20% more than the number of students who like only hockey. Find the total number of students who like only baseball and basketball.

a. 78

b. 76

c. 72

d. 88

e. 96

12. Questions

The difference between the number of students who like football and the number of students who like hockey is 'x' and the difference between the number of students who like cricket and the number of students who like hockey is 'y'. Find the value of (11x+19y).

a. 1149

b. 1568

c. 1976

d. 1145

e. 1354

13. Questions

The ratio of the number of boys and girls who like only football is 7:2 and the ratio of the number of boys and girls who like only cricket is 5:3. The difference between the number of boys who like only football and the number of girls who like only cricket is m. Find the value 9m.

a. 140

b. 160

c. 120

d. 180

e. None of these

14. Questions

The ratio of the number of students who like ODI, test and T20 cricket out of the total number of students who like cricket is 3:4:2 respectively. Find the difference between the number of students like ODI and T20 cricket.

a. 10

b. 12

c. 18

d. 19

e. None of these

15. Questions

Quantity I: The total number of students who like both football and cricket but not hockey is what percent of the number of students who like both hockey and cricket but not football?

Quantity I: The total number of students who like only football is what percent of the total number of students who like only cricket?

a. Quantity I < Quantity II

b. Quantity I ≥ Quantity II

c. Quantity II ≥ Quantity I

d. Quantity I > Quantity II

e. Quantity I = Quantity II or relation can't be established

16. Questions

Study the following information carefully and answer the questions given below.

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The ratio of the cost price of item A and B is 1:2. The selling price of item C is equal to the cost price of item B. Item A is marked up by 60% above its cost price. The ratio of the cost price of item B and D is 8:9. Item B is marked up by 50% above its cost price. The ratio of the selling price of item B and C is 5:4. The ratio of the selling price of item A and D is 8:17. The sum of the marked price of items A and C is Rs. 480. The sum of the selling price of items B and D is Rs. 640. The discount percentage for items A and C is the same.

The marked price of item C is Rs.288. Find the profit percentage of item A.

a. 25%

b. 22.5%

c. 33.33%

d. 37.5%

e. 16.66%

17. Questions

The selling price of item B is Rs.300, and item D is marked up by 40%. Find the marked price of item D.

a. Rs.258

b. Rs.268

c. Rs.248

d. Rs.378

e. Rs.298

18. Questions

The ratio of the cost price of items A and C is 2:3. Find the cost price of item C, if the discount amount of item B is Rs.60.

a. Rs.160

b. Rs.180

c. Rs.200

d. Rs.240

e. None of these

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19. Questions

The marked price of item D is Rs.378. Find the difference between the profit percentage of items B and D.

a. 12.5%

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b. 10.5%

c. 5.5%

d. 0.92%

e. 1.25%

20. Questions

The profit amount of item A is Rs.40. Find the difference between the discount amounts of item A and B.

a. 28

b. 22

c. 38

d. 32

e. Cannot be determined

Explanations:

1. Questions

The number of girls in the arts department in school D = 30.

The number of boys in the arts department in school D = 30.

So, the total number of students in the arts department in school D = 60.

So, the number of students in the commerce department in school D = 60*7/6=70.

The number of students in the science department in school D = [130*40.90/59.09] = 90.

The number of boys in the science department in school D = 90/2 = 45.

The number of girls in the science department in school D = 90/2 = 45.

The number of boys in the commerce department in school D = 70/2 = 35.

The number of girls in the commerce department in school D = 70/2 = 35.

The total number of students in all schools together = [220/20]*100=1100.

So, the number of students in school A = 1100 * 23.63 / 100 = 260.

Similarly, the total number of students in schools B, C, and E is 240, 180, and 200, respectively.

Answer: C

The number of girls in the arts department in school A = 30*133.33/100=40.

The number of boys in the arts department in school A = 40/2=20.

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The number of students in the commerce department in school A = [40+ 20] * 4/3 = 80.

The number of students in the science department in school A= (260-60-80)=120.

The number of boys in the science department in school A = [120+ 40]/2 = 80.

The number of boys in the commerce department in school A = 80 * 75 / 100 = 60.

The number of girls in the commerce department in school A = 80-60=20.

2. Questions

The number of girls in the arts department in school D = 30.

The number of boys in the arts department in school D = 30.

So, the total number of students in the arts department in school D = 60.

So, the number of students in the commerce department in school D = 60*7/6=70.

The number of students in the science department in school D = [130*40.90/59.09] = 90.

The number of boys in the science department in school D = 90/2 = 45.

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The number of boys in the commerce department in school D = 70/2 = 35.

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The total number of students in all schools together = [220/20]*100=1100.

So, the number of students in school A = 1100 * 23.63 / 100 = 260.

Similarly, the total number of students in schools B, C, and E is 240, 180, and 200, respectively.

Answer: D

In school B

The number of girls in the commerce department in school B = 35*100/116.66=30.

The number of boys in the commerce department in school B = [30/37.5]*62.5=50.

The number of students in the science department = 240 * 41.66 / 100 = 100.

The number of students in the arts department = 240-100-80=60.

The difference between boys and girls in the art department = 60*2/12=10.

The difference between boys and girls in the science department= 10*2=20.

The number of boys in the science department = [100 + 20]/2 = 60.

3. Questions

The number of girls in the arts department in school D = 30.

The number of boys in the arts department in school D = 30.

So, the total number of students in the arts department in school D = 60.

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So, the number of students in the commerce department in school D = 60*7/6=70.

The number of students in the science department in school D = [130*40.90/59.09] = 90.

The number of boys in the science department in school D = 90/2 = 45.

The number of girls in the science department in school D = 90/2 = 45.

The number of boys in the commerce department in school D = 70/2 = 35.

The number of girls in the commerce department in school D = 70/2 = 35.

The total number of students in all schools together = [220/20]*100=1100.

So, the number of students in school A = 1100 * 23.63 / 100 = 260.

Similarly, the total number of students in schools B, C, and E is 240, 180, and 200, respectively.

Answer: B

In school C

Let the number of students in the commerce and arts departments in school C be 100x and 200x, respectively.

The number of girls in the Commerce Department in school C = 25x.

The number of boys in the commerce department in school C = 100x-25x=75x= Number of boys in the arts department of C

The number of girls in the arts department in school C = 200x-75x=125x.

The number of girls in the science department in school C =125x/2=62.5x.

The number of boys in the science department in school C = 62.5x+10.

So, 300x+62.5x+62.5x+10=180

Or, 425x=170, x=170/425=0.4

The total number of students in the science department in school C = 125*0.4+ 10 = 60.

So, the required sum is 60+90=150.

4. Questions

The number of girls in the arts department in school D = 30.

The number of boys in the arts department in school D = 30.

So, the total number of students in the arts department in school D = 60.

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So, the number of students in the commerce department in school D = 60*7/6=70.

The number of students in the science department in school D = [130*40.90/59.09] = 90.

The number of boys in the science department in school D = 90/2 = 45.

The number of girls in the science department in school D = 90/2 = 45.

The number of boys in the commerce department in school D = 70/2 = 35.

The number of girls in the commerce department in school D = 70/2 = 35.

The total number of students in all schools together = [220/20]*100=1100.

So, the number of students in school A = 1100 * 23.63 / 100 = 260.

Similarly, the total number of students in schools B, C, and E is 240, 180, and 200, respectively.

Answer: A

So, x=90-70=20

So, y=70-60=10

So, 7*20+9*10=140+90=230

5. Questions

The number of girls in the arts department in school D = 30.

The number of boys in the arts department in school D = 30.

So, the total number of students in the arts department in school D = 60.

So, the number of students in the commerce department in school D = 60*7/6=70.

The number of students in the science department in school D = [130*40.90/59.09] = 90.

The number of boys in the science department in school D = 90/2 = 45.

The number of girls in the science department in school D = 90/2 = 45.

The number of boys in the commerce department in school D = 70/2 = 35.

The number of girls in the commerce department in school D = 70/2 = 35.

The total number of students in all schools together = [220/20]*100=1100.

So, the number of students in school A = 1100 * 23.63 / 100 = 260.

Similarly, the total number of students in schools B, C, and E is 240, 180, and 200, respectively.

Answer: A

The number of boys in the arts department in school E = 30*4/3=40.

Let the number of girls in the science and commerce departments in school E be 3x and 2x, respectively.

So, 3x+2x+30=80

Or, x=50/5=10

The number of girls in the science department in school E = 3*10=30.

The number of boys in the science department in school E = 30+20=50.

The total number of students in the commerce department in school E = 200-80-70=50

The number of boys in the commerce department in school E = 50-20=30.

So, the required ratio = 50:30=5:3.

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6. Questions

Number of furniture sold in shop E = 90.

The number of unsold furniture in shop E = 90*25/75=30.

So, the total number of furniture unsold in all shops together = [30/90]*360=120

So, Number of furniture unsold in shop A = 120*120/360=40

So, Number of furniture sold in shop A = 40*75/25=120

Similarly, we can calculate others' values also.

Shop	Number of unsold furniture	Number of sold furniture
A	40	120
B	15	45
C	15	45
D	20	60
E	30	90

Answer: C

The number of furniture manufactured in February = [40+120]*120/100=192

The number of unsold furniture in February = 192-[120*7/5]=192-168=24

So, the required difference = 40-24=16

7. Questions

Number of furniture sold in shop E = 90.

The number of unsold furniture in shop E = 90*25/75=30.

So, the total number of furniture unsold in all shops together = [30/90]*360=120

So, Number of furniture unsold in shop A = 120*120/360=40

So, Number of furniture sold in shop A = 40*75/25=120

Similarly, we can calculate others' values also.

Shop	Number of unsold furniture	Number of sold furniture
A	40	120
B	15	45
C	15	45
D	20	60
E	30	90

Answer: D

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Number of chairs manufactured in shop B = 60*3/6=30

Number of tables manufactured in shop B = 60*2/6=20

Number of beds manufactured in shop B = 60*1/6=10

Number of unsold chairs in shop B = 15-[20*40/100+10*50/100]=2

So, the number of chairs sold in B = 30-2=28

8. Questions

Number of furniture sold in shop E = 90.

The number of unsold furniture in shop E = 90*25/75=30.

So, the total number of furniture unsold in all shops together = [30/90]*360=120

So, Number of furniture unsold in shop A = 120*120/360=40

So, Number of furniture sold in shop A = 40*75/25=120

Similarly, we can calculate others' values also.

Shop	Number of unsold furniture	Number of sold furniture
A	40	120
B	15	45
C	15	45
D	20	60
E	30	90

Answer: E

So, x=80-60=20

And y=120-80=40

So, 5x+9y=100+360=460

And also, 7x+9y=140+360=500 and 6x +9y = 120+360=480

9. Questions

Number of furniture sold in shop E = 90.

The number of unsold furniture in shop E = 90*25/75=30.

So, the total number of furniture unsold in all shops together = [30/90]*360=120

So, Number of furniture unsold in shop A = 120*120/360=40

So, Number of furniture sold in shop A = 40*75/25=120

Similarly, we can calculate others' values also.

Shop	Number of unsold furniture	Number of sold furniture
A	40	120
B	15	45
C	15	45
D	20	60
E	30	90

Answer: B

For the value of B,

Profit percentage = [3600*60-2500*80]*100/[2500*80]=8%

10. Questions

Number of furniture sold in shop E = 90.

The number of unsold furniture in shop E = 90*25/75=30.

So, the total number of furniture unsold in all shops together = [30/90]*360=120

So, Number of furniture unsold in shop A = 120*120/360=40

So, Number of furniture sold in shop A = 40*75/25=120

Similarly, we can calculate others' values also.

Shop	Number of unsold furniture	Number of sold furniture
A	40	120
B	15	45
C	15	45
D	20	60
E	30	90

Answer: A

For the value I,

Total furniture manufactured in shop F and G

= 120*4/3+ 80*5/4=260

So, it's true

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Similarly, we can check others' values also.

11. Questions

The number of students who like both cricket and football but not hockey is a perfect square between 20 and 30.

So, the number of students who like both cricket and football but not hockey is 25.

The number of students who like all the sports = 25/5=5.

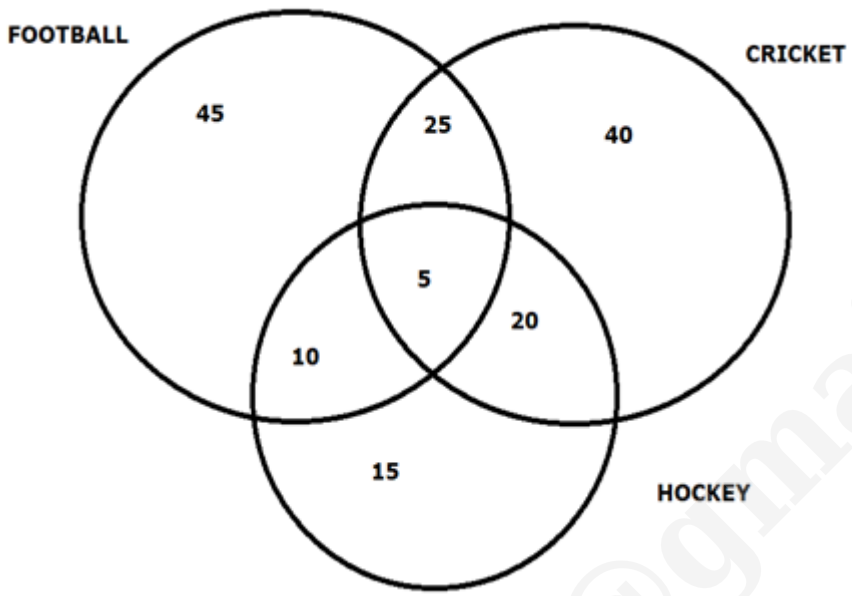
The number of students who like both cricket and hockey but not football = 25* 80 / 100 = 20.

The number of students who like both football and hockey but not cricket = 20* 1/2 = 10.

The number of students who like only cricket = 20* 2 = 40.

The number of students who like only football = 40*9/8=45.

The number of students who like only hockey = 45/3=15.



Answer: C

The total number of students who like only basketball and baseball

= 45*120/100+15*120/100=54+18=72

12. Questions

The number of students who like both cricket and football but not hockey is a perfect square between 20 and 30.

So, the number of students who like both cricket and football but not hockey is 25.

The number of students who like all the sports = 25/5=5.

The number of students who like both cricket and hockey but not football = 25* 80 / 100 = 20.

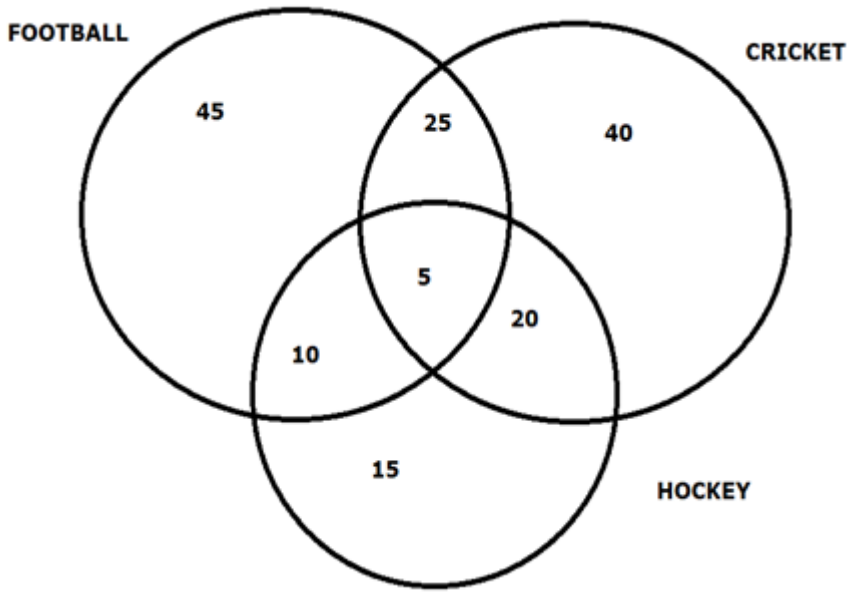
The number of students who like both football and hockey but not cricket = 20* 1/2 = 10.

The number of students who like only cricket = 20* 2 = 40.

The number of students who like only football = 40*9/8=45.

The number of students who like only hockey = 45/3=15.

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Answer: D

So, x=[45+25+5+10]-[15+10+5+20]=35

So, y=[40+25+5+20]-[15+10+5+20]=40

So, 11x+19y=11*35+19*40=1145

13. Questions

The number of students who like both cricket and football but not hockey is a perfect square between 20 and 30.

So, the number of students who like both cricket and football but not hockey is 25.

The number of students who like all the sports = 25/5=5.

The number of students who like both cricket and hockey but not football = 25* 80 / 100 = 20.

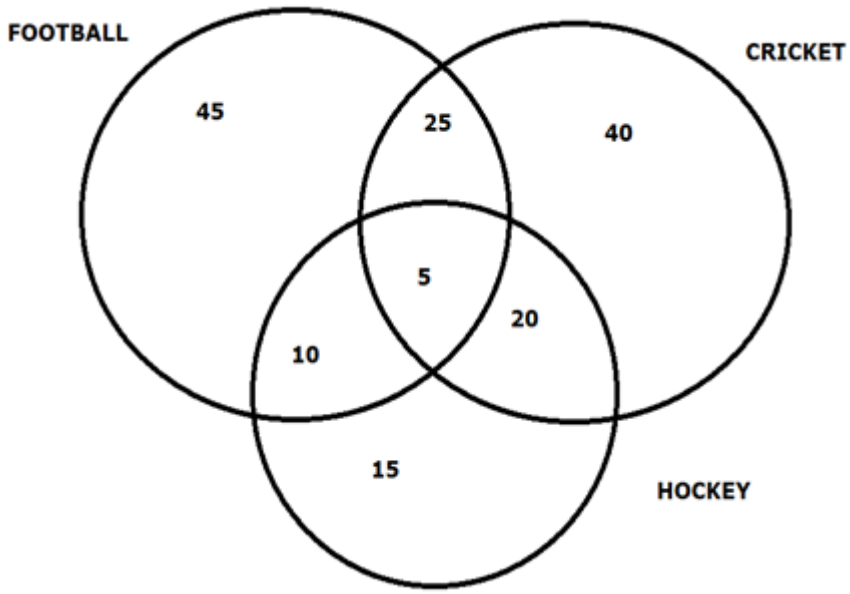
The number of students who like both football and hockey but not cricket = 20* 1/2 = 10.

The number of students who like only cricket = 20* 2 = 40.

The number of students who like only football = 40*9/8=45.

The number of students who like only hockey = 45/3=15.

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Answer: D

So, required difference = 45*7/9 – 40*3/8=35-15=20=m

So, 9m=9*20=180

14. Questions

The number of students who like both cricket and football but not hockey is a perfect square between 20 and 30.

So, the number of students who like both cricket and football but not hockey is 25.

The number of students who like all the sports = 25/5=5.

The number of students who like both cricket and hockey but not football = 25* 80 / 100 = 20.

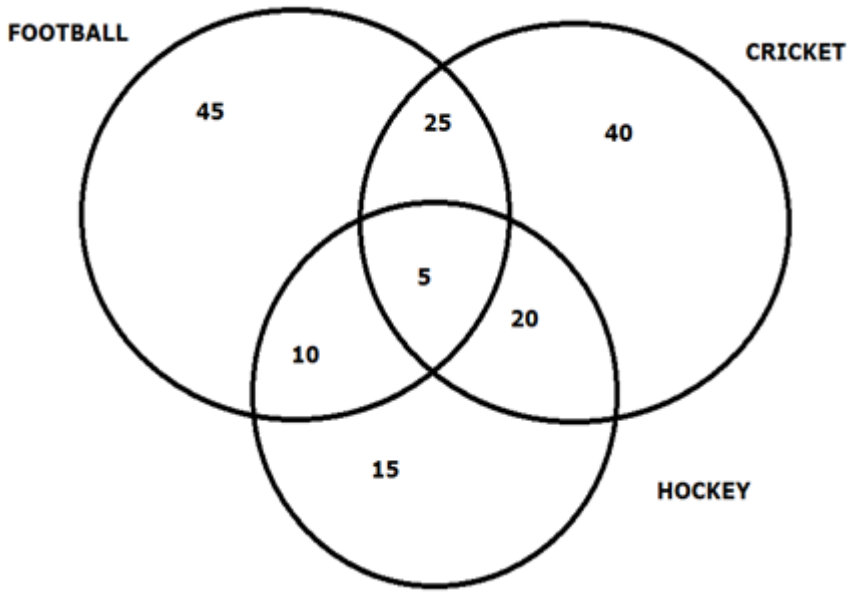
The number of students who like both football and hockey but not cricket = 20* 1/2 = 10.

The number of students who like only cricket = 20* 2 = 40.

The number of students who like only football = 40*9/8=45.

The number of students who like only hockey = 45/3=15.

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Answer: A

The total number of students who like cricket = 25+5+20+40=90

So, the required difference = 90*1/9=10

15. Questions

The number of students who like both cricket and football but not hockey is a perfect square between 20 and 30.

So, the number of students who like both cricket and football but not hockey is 25.

The number of students who like all the sports = 25/5=5.

The number of students who like both cricket and hockey but not football = 25* 80 / 100 = 20.

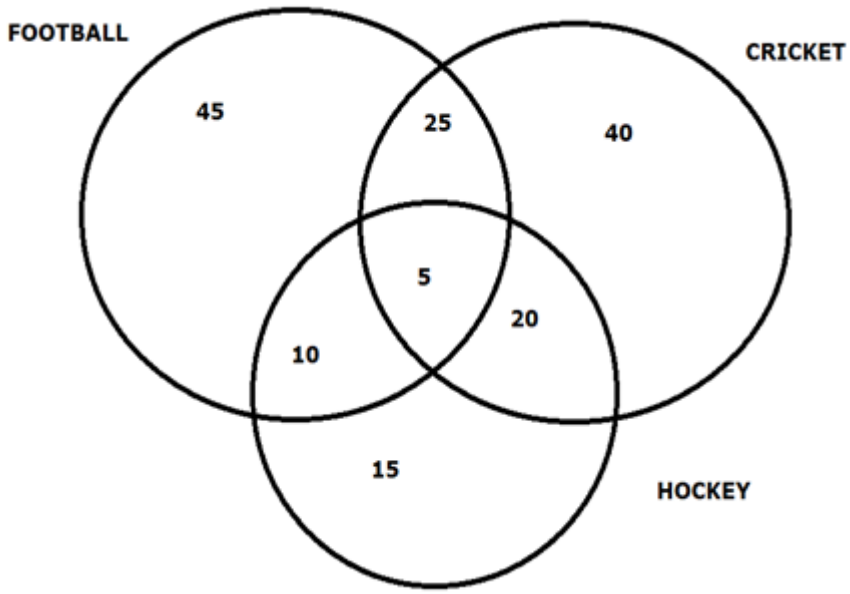
The number of students who like both football and hockey but not cricket = 20* 1/2 = 10.

The number of students who like only cricket = 20* 2 = 40.

The number of students who like only football = 40*9/8=45.

The number of students who like only hockey = 45/3=15.

Tg:- @NextGenBankers



Answer: D

Quantity I

Required percentage = [25/20]*100=125%

Quantity II

Required percentage = [45/40]*100=112.5%

Quantity I > Quantity II

16. Questions

Let the cost of item A be 100x and the cost of item B be 200x.

The selling price of item C = 200x.

The marked price of item B = 200x*150/100=300x.

The marked price of item A = 100x*160/100=160x.

The cost of item D = 200x*9/8=225x.

The selling price of item B = 200x*5/4=250x.

The marked price of item C = 480–160x.

The selling price of item D = 640-250x.

The selling price of item A = [640-250x]*8/17.

Answer: C

The marked price of item C = 480-160x=288

Or, 160x=192

Or, x=192/160=1.2

So, the selling price of item A = [640-250*1.2]*8/17=160

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The cost price of item A =100*1.2=120

So, profit percentage = [40/120]*100=33.33%

17. Questions

Let the cost of item A be 100x and the cost of item B be 200x.

The selling price of item C = 200x.

The marked price of item B = 200x*150/100=300x.

The marked price of item A = 100x*160/100=160x.

The cost of item D = 200x*9/8=225x.

The selling price of item B = 200x*5/4=250x.

The marked price of item C = 480–160x.

The selling price of item D = 640-250x.

The selling price of item A = [640-250x]*8/17.

Answer: D

So, 250x=300

Or, x=[300/250]=1.2

So, the cost price of D = 225x=225*1.2=270

So, marked price = 270*140/100=378

18. Questions

Let the cost of item A be 100x and the cost of item B be 200x.

The selling price of item C = 200x.

The marked price of item B = 200x*150/100=300x.

The marked price of item A = 100x*160/100=160x.

The cost of item D = 200x*9/8=225x.

The selling price of item B = 200x*5/4=250x.

The marked price of item C = 480–160x.

The selling price of item D = 640-250x.

The selling price of item A = [640-250x]*8/17.

Answer: B

So, 300x-250x=60

Or, x=60/50=1.2

So, the cost price of item A is 100*1.2=120

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So, the cost price of item C is 120*3/2=180

19. Questions

Let the cost of item A be 100x and the cost of item B be 200x.

The selling price of item C = 200x.

The marked price of item B = 200x*150/100=300x.

The marked price of item A = 100x*160/100=160x.

The cost of item D = 200x*9/8=225x.

The selling price of item B = 200x*5/4=250x.

The marked price of item C = 480–160x.

The selling price of item D = 640-250x.

The selling price of item A = [640-250x]*8/17.

Answer: D

[160x - [640 – 250x] * 8/17]/160x = [(480 – 160x) – 200x]/[480 – 160x]

[160x - [640 – 250x] * 8/17]/x = [(480 – 360x)]/[3-x]

[2720x – 5120 + 2000x]/17x = (480 – 360x)/(3 – x)

(590x – 640) * (3 – x) = (60 – 45x) * 17x

$1770x – 590x^2 – 1920 + 640x = 1020x – 765x^2$

$175x^2 + 1390x – 1920 = 0$

$35x^2 + 278x – 384 = 0$

x = 42/35, -320/35

x = 6/5

Profit % of B = (250x – 200x)/200x * 100 = 25%

Cp of D = 225 * 6/5 = 270

Sp of D = 640 – 250 * 6/5 = 340

Profit % of D = (340 – 270)/270 * 100 = 25.92%

Required Difference = 25.92% - 25 = 0.92%

20. Questions

Let the cost of item A be 100x and the cost of item B be 200x.

The selling price of item C = 200x.

The marked price of item B = 200x*150/100=300x.

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The marked price of item A = 100x*160/100=160x.

The cost of item D = 200x*9/8=225x.

The selling price of item B = 200x*5/4=250x.

The marked price of item C = 480–160x.

The selling price of item D = 640-250x.

The selling price of item A = [640-250x]*8/17.

Answer: A

[640-250x]*8/17-100x=40

Or, 640*8-2000x-1700x=680

Or, 3700x=4440

Or, x=4440/3700=1.2

So, the discount amount of item A = [160*1.2-100*1.2-40]=32

Discount of item B = [300x-250x]=50x=50*1.2=60

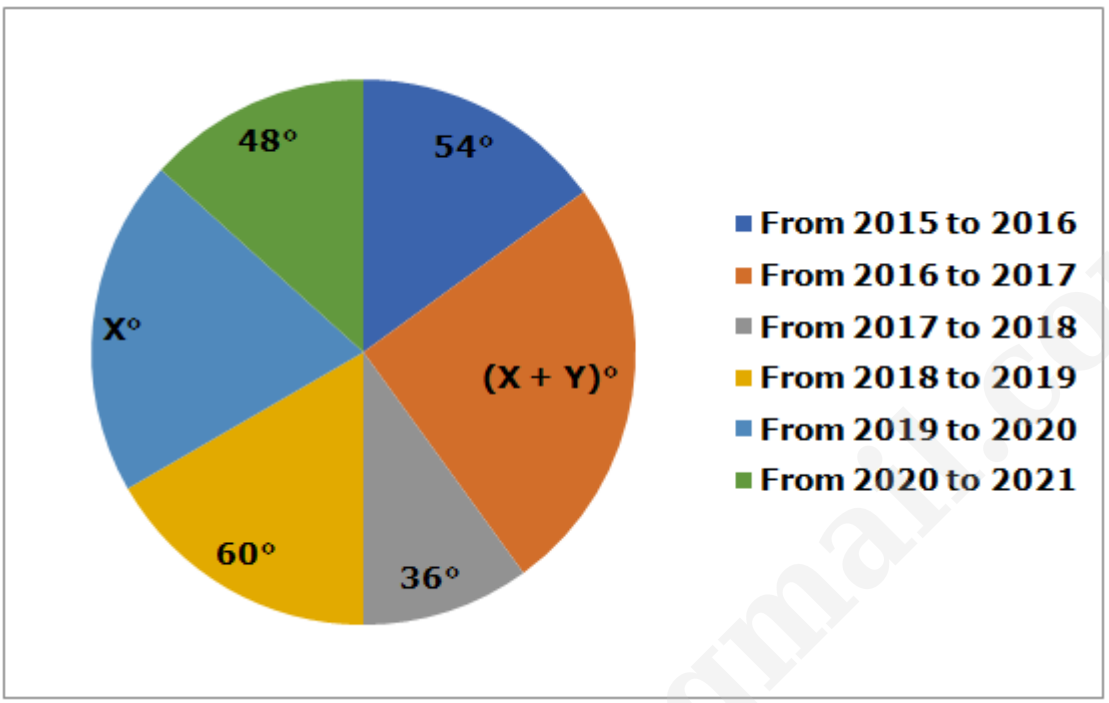
So, the difference = 60-32=28

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1. Questions

Study the following data carefully and answer the questions:

Pie chart given below shows the degree distribution of increment in monthly salary (in Rs) of a person from the year 2015 to 2021.



Notes:

1). The sum of the increment in monthly salary from 2016 to 2017 and that from 2019 to 2020 is Rs 13500.

2). The increment in monthly salary from 2016 to 2017 is Rs 3000 more than that from 2015 to 2016.

If in 2017, ratio of monthly expenditure to monthly saving is 9: 11 and difference between the monthly expenditure and monthly saving is Rs 3000, then what will be the ratio of monthly salary in 2016 to that in 2018?

a. 25: 33

b. 9: 11

c. 5: 9

d. 15: 22

e. 3: 5

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2. Questions

If the monthly salary in 2015 is Rs. 25000, the increment in monthly salary from 2015 to 2018 will be M% and the increment in monthly salary from 2018 to 2021 will be N%. Find that which of the following is/are true?

P: Both M and N are multiple of 12.5.

Q: Value of 4(M – N) will be divisible by18.

R: Value of (M + N) is divisible by 12.5.

a. Only P and Q

b. Only Q and R

c. None is true

d. Only P

e. Only Q

3. Questions

If the ratio of monthly expenditure to monthly saving in 2017 is 19: 21, the difference between total expenditure and total saving in 2017 is Rs. 12000 and the monthly expenditure in 2019 is Rs. 3100 more than that in 2017, then the monthly saving in 2019 will be what per cent of the monthly salary in 2019?

a. 55%

b. 44%

c. 77%

d. 72%

e. 45%

4. Questions

A series is given below, whose;

1st term = increment in monthly salary from 2017 to 2018

2nd term = 70% of increment in monthly salary from 2018 to 2019

3rd term = 125% of increment in monthly salary from 2020 to 2021

4th term = increment in monthly salary from 2019 to 2020

5th term = double of increment in monthly salary from 2020 to 2021

6th term = 140% of increment in monthly salary from 2016 to 2017

Find the wrong term in the given series?

a. 2nd term

b. 5th term

c. 4th term

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d. 3rd term

e. 6th term

5. Questions

If the monthly salary in 2016 is Rs. 28500 and the increment in monthly salary from 2021 to 2022 is 5%, then what will be the per cent increment in monthly salary from 2015 to 2022?

a. 129%

b. 136.25%

c. 132.75%

d. 121%

e. 138. 5%

6. Questions

Study the following data carefully and answer the questions:

Three persons A, B and C started a business by investing in such a way that the ratio of A's investment to B's investment is 3: 5 and the difference between A's investment and B's investment is equal to C's investment. After 5 months, D also joined the business. After 1 more month, B left the business and E joined the business by investing Rs 10000 more than D's investment. After 15 months from the starting of the business, the ratio of C's profit share to D's profit share will be 4: 5 and the ratio of A's profit share to E's profit share will also be 4: 5.

If A had withdrawn 50% of his investment after 1 year from the starting of the business, B had withdrawn 50% of his investment after 6 months from the starting of the business and C had invested 87.5% more of his initial investment after 9 months from the starting of the business, then what would have been the ratio between the profit shares of A, B and C after 15 months?

a. 3: 5: 3

b. 18: 25: 18

c. 9: 14: 9

d. 6: 11: 6

e. 27: 35: 27

7. Questions

If B had withdrawn his whole investment after 6 months from the starting of the business and the difference between the profit shares of B and D, after 15 months from the starting of the business, were ₹1250, then what would have been the difference between the profit shares of D and E after 15 months from the starting of the business?

a. ₹ 3025

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b. ₹ 3125

c. ₹ 2525

d. ₹ 3225

e. ₹ 2825

8. Questions

Which of the following statements is/are not true?

P: C's investment is Rs. 4000 less than A's investment.

Q: Average of the investments of C, D and E is Rs. 15000.

R: Ratio of the total investments of A and B together to that of D and E together is 4: 5.

a. Only P and Q

b. All are true

c. Only Q

d. Only R

e. Only P and R

9. Questions

If the ratio of B's profit share after 8 months to C's profit share after 15 months is M: N, then find the ratio of A's profit share after 10 months to D's profit share after 6 months?

a. M: N

b. (N – 1): (M + 1)

c. (M + 1): (N – 1)

d. (M + 1): N

e. M: (N – 2)

10. Questions

If a new person F joined the business after 12 months from the starting of the business by investing 20% of the total investment of A, B, C, D and E together, then what will be the ratio of C's profit share to F's profit share after 20 months from the starting of the business?

a. 8: 7

b. 16: 15

c. 4: 3

d. 5: 4

e. 10: 9

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11. Questions

Study the following data carefully and answer the questions:

Data given below is related to the number of wooden doors, the number of iron doors and the number of plastic doors sold by six shops A, B, C, D, E and F. Cost of iron doors, wooden doors and plastic doors in each shop is different.

Table given below shows the following data:

Shop	Ratio of the no. of wooden doors to the no. of iron doors sold	Average of the no. of iron doors and plastic doors sold
A	3: 2	16
B	3: 5	17
C	5: 4	13
D	3: 5	17
E	2: 1	15
F	2: 3	18

Note: All the questions are related to each other in such a way that, the data, once calculated in a question, will be same for all the questions.

The costs of wooden doors and iron doors in shop F are Rs. 4500 per piece and Rs. 3000 per piece respectively and total amount received by shop F by selling wooden doors and iron doors is Rs. 90000. If the ratio of the number of plastic doors sold by shop E to that sold by shop F is 6: 7, then find the average of the number of wooden doors, iron doors and plastic doors sold by shop E?

a. 16

b. 20

c. 21

d. 18

e. 15

12. Questions

The number of wooden doors sold by shops D and E are respectively 60% and 120% of that sold by shop C and the cost of plastic doors in shop C is Rs. 300 per piece more than that in shop D. Find the difference between the total amount received by shops C and D by selling plastic doors?

a. Rs. 1400

b. Rs. 7000

c. Rs. 3500

d. Rs. 2800

e. Can't be determined

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13. Questions

The average number of wooden doors sold by shops B, D and E is 17 and the costs of wooden doors, iron doors and plastic doors sold by shop B are Rs. 4800 per piece, Rs. 3200 per piece and Rs. 3000 per piece, then find the total amount received by shop B by selling all the three types of doors?

a. Rs. 219000

b. Rs. 179000

c. Rs. 159000

d. Rs. 209000

e. Rs. 189000

14. Questions

Table given below shows the following data:

Shop	Cost of wooden doors (in Rs per piece)	Ratio of the cost of wooden doors to that of iron doors	Average of the cost of iron doors and that of plastic doors (in Rs per piece)
C	4500	9: 8	3400
D	5000	10: 7	3000
E	4500	5: 4	2800

Find the total amount received by shops C, D and E together by selling plastic doors only?

a. Rs. 99000

b. Rs. 89000

c. Rs. 90000

d. Rs. 79000

e. Rs. 109000

15. Questions

The average number of wooden doors sold by shops A and C is 19, the average number of iron doors sold by shops A and C is 14 and the average number of plastic doors sold by shops A and C is 15. Find the ratio between the number of wooden doors, iron doors and plastic doors respectively sold by shop A?

a. 9: 6: 20

b. 6: 5: 10

c. 3: 2: 5

d. 9: 6: 10

e. 6: 4: 5

16. Questions

Two series I and II are given below and both the series follow the same pattern:

I). 9, 19, 36, 75, 146, 297

II). 12, P, 48, 99, (8P – Q)

Find that Q is what per cent of P?

a. 36%

b. 12%

c. 30%

d. 24%

e. 18%

17. Questions

Two series I and II are given below and both the series follow the same pattern:

I). 455, 239, A, 50, B, 15

II). C, 233, 108, D, 17, 9

Find that which of the following statements is/are true?

P: A is 1 less than 5 times of B.

Q: C is 9 more than 10 times of D.

R: Value of (C – A)/(B + D) is 5.

a. All are true.

b. Only P and Q

c. Only Q and R

d. Only P and R

e. Only P

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18. Questions

Two series I and II are given below with missing terms A and B in series I and II respectively.

I: 441, 320, 239, 190, 165, A

II: 96, 73, 54, 37, 24, B

Which of the following statements is/are not true?

P: A is completely divisible by B.

Q: Average of next terms of both the series is 80.

R: Difference of A and B is divisible by 11.

a. Only P and Q

b. Only R

c. Only Q

d. None is true

e. Only Q and R

19. Questions

Three equations I, II and III are given below with missing term P in series I and missing term Q in series II:

I: 41, 42, 46, 55, 71, ?

II: 6, 13, 28, 59, 122, ?

III: Q, (2P + 1), (Q – P – 8), (P + 9), (Q – 30)/3

Find the average of P and Q?

a. 142.5

b. 192.5

c. 172.5

d. 182.5

e. 152.5

20. Questions

Find that the 6th term of series III is divisible by which of the following?

a. 7

b. 3

c. 11

d. 5

e. 8

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Explanations:

1. Questions

From the pie chart:

54 + (X + Y) + 36 + 60 + X + 48 = 360

2X + Y = 162 ----------(1)

Since, the sum of the increment in monthly salary from 2016 to 2017 and that from 2019 to 2020 is Rs. 13500.

So, the sum of the pie chart:

13500 × 360/(2X+Y)=13500 × 360/162= Rs. 30000

And the increment in monthly salary from 2015 to 2016 = 30000 * (54/360) = Rs. 4500

Since, the increment in monthly salary from 2016 to 2017 is Rs. 3000 more than that from 2015 to 2016.

So, the increment in monthly salary from 2016 to 2017 = 4500 + 3000 = Rs. 7500

And,

X+Y=54 × 7500/4500= 90

Y = 90 – X ----------(2)

From equations (1) and (2):

2X + 90 – X = 162

X = 72 and Y = 18

The degree distribution of the increment in monthly salary from 2016 to 2017 = 72° + 18° = 90°

The degree distribution of the increment in monthly salary from 2019 to 2020 = 72°

The increment in monthly salary from 2015 to 2016 = Rs. 4500

The increment in monthly salary from 2016 to 2017 = 4500 * (90/54) = Rs. 7500

The increment in monthly salary from 2017 to 2018 = 4500 * (36/54) = Rs. 3000

The increment in monthly salary from 2018 to 2019 = 4500 * (60/54) = Rs. 5000

The increment in monthly salary from 2019 to 2020 = 4500 * (72/54) = Rs. 6000

The increment in monthly salary from 2020 to 2021 = 4500 * (48/54) = Rs. 4000

Answer: D

Since, the ratio of monthly expenditure to monthly saving in 2017 = 9: 11

And the difference between monthly expenditure and monthly saving in 2017 = Rs. 3000

So, the monthly salary in 2017 = 3000 × (9+11)/(11-9) = Rs. 30000

The monthly salary in 2016 = 30000 – 7500 = Rs. 22500

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And the monthly salary in 2018 = 30000 + 3000 = Rs. 33000

Required ratio = 22500: 33000 = 15: 22

2. Questions

From the pie chart:

54 + (X + Y) + 36 + 60 + X + 48 = 360

2X + Y = 162 ----------(1)

Since, the sum of the increment in monthly salary from 2016 to 2017 and that from 2019 to 2020 is Rs. 13500.

So, the sum of the pie chart:

$13500 \times 360/(2X+Y) = 13500 \times 360/162 =$ Rs. 30000

And the increment in monthly salary from 2015 to 2016 = 30000 * (54/360) = Rs. 4500

Since, the increment in monthly salary from 2016 to 2017 is Rs. 3000 more than that from 2015 to 2016.

So, the increment in monthly salary from 2016 to 2017 = 4500 + 3000 = Rs. 7500

And,

$X+Y = 54 \times 7500/4500 = 90$

Y = 90 – X ----------(2)

From equations (1) and (2):

2X + 90 – X = 162

X = 72 and Y = 18

The degree distribution of the increment in monthly salary from 2016 to 2017 = 72° + 18° = 90°

The degree distribution of the increment in monthly salary from 2019 to 2020 = 72°

The increment in monthly salary from 2015 to 2016 = Rs. 4500

The increment in monthly salary from 2016 to 2017 = 4500 * (90/54) = Rs. 7500

The increment in monthly salary from 2017 to 2018 = 4500 * (36/54) = Rs. 3000

The increment in monthly salary from 2018 to 2019 = 4500 * (60/54) = Rs. 5000

The increment in monthly salary from 2019 to 2020 = 4500 * (72/54) = Rs. 6000

The increment in monthly salary from 2020 to 2021 = 4500 * (48/54) = Rs. 4000

Answer: E

If the monthly salary in 2015 is Rs. 25000:

So, the monthly salary in 2018 = 25000 + (4500 + 7500 + 3000) = Rs. 40000

And the monthly salary in 2021 = 40000 + (5000 + 6000 + 4000) = Rs. 55000

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Now, the per cent increment in monthly salary from 2015 to 2018:

M= (40000-25000)/25000 ×100=60%

And the per cent increment in monthly salary from 2018 to 2021:

N= (55000-40000)/40000 ×100=37.5%

From P:

Since, 37.5 is a multiple of 12.5, but 60 is not a multiple of 12.5.

So, P is not true.

From Q:

Value of 4(M – N) = 4(60 – 37.5) = 90

Since, 90 is divisible by 18.

So, Q is true.

From R:

Value of (M + N) = 60 + 37.5 = 97.5

Since, 97.5 is not divisible by 12.5.

So, R is not true,

Hence, only Q is true.

3. Questions

From the pie chart:

54 + (X + Y) + 36 + 60 + X + 48 = 360

2X + Y = 162 ----------(1)

Since, the sum of the increment in monthly salary from 2016 to 2017 and that from 2019 to 2020 is Rs. 13500.

So, the sum of the pie chart:

13500 × 360/(2X+Y)=13500 × 360/162= Rs. 30000

And the increment in monthly salary from 2015 to 2016 = 30000 * (54/360) = Rs. 4500

Since, the increment in monthly salary from 2016 to 2017 is Rs. 3000 more than that from 2015 to 2016.

So, the increment in monthly salary from 2016 to 2017 = 4500 + 3000 = Rs. 7500

And,

X+Y=54 × 7500/4500= 90

Y = 90 – X ----------(2)

From equations (1) and (2):

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$2X + 90 – X = 162$

$X = 72$ and $Y = 18$

The degree distribution of the increment in monthly salary from 2016 to 2017 = 72° + 18° = 90°

The degree distribution of the increment in monthly salary from 2019 to 2020 = 72°

The increment in monthly salary from 2015 to 2016 = Rs. 4500

The increment in monthly salary from 2016 to 2017 = 4500 * (90/54) = Rs. 7500

The increment in monthly salary from 2017 to 2018 = 4500 * (36/54) = Rs. 3000

The increment in monthly salary from 2018 to 2019 = 4500 * (60/54) = Rs. 5000

The increment in monthly salary from 2019 to 2020 = 4500 * (72/54) = Rs. 6000

The increment in monthly salary from 2020 to 2021 = 4500 * (48/54) = Rs. 4000

Answer: A

Since, the difference between total expenditure and total saving in 2017 = Rs. 12000

So, the difference between monthly expenditure and monthly saving in 2017 = 12000/12 = Rs. 1000

Since, the ratio of monthly expenditure to monthly saving in 2017 = 19: 21

So, the monthly salary in 2017 = $1000 \times (19+21)/(21-19)$ = Rs. 20000

And the monthly salary in 2019 = 20000 + (3000 + 5000) = Rs. 28000

Since, the monthly expenditure in 2017 = $20000 \times 19/(19+21)$ = Rs. 9500

So, the monthly expenditure in 2019 = 9500 + 3100 = Rs. 12600

And the monthly saving in 2019 = 28000 – 12600 = Rs. 15400

Required percentage = $15400/28000 \times 100 = 55\%$

4. Questions

From the pie chart:

$54 + (X + Y) + 36 + 60 + X + 48 = 360$

$2X + Y = 162$ ----------(1)

Since, the sum of the increment in monthly salary from 2016 to 2017 and that from 2019 to 2020 is Rs. 13500.

So, the sum of the pie chart:

$13500 \times 360/(2X+Y)=13500 \times 360/162$= Rs. 30000

And the increment in monthly salary from 2015 to 2016 = 30000 * (54/360) = Rs. 4500

Since, the increment in monthly salary from 2016 to 2017 is Rs. 3000 more than that from 2015 to 2016.

So, the increment in monthly salary from 2016 to 2017 = 4500 + 3000 = Rs. 7500

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And,

X+Y=54 × 7500/4500= 90

Y = 90 – X ----------(2)

From equations (1) and (2):

2X + 90 – X = 162

X = 72 and Y = 18

The degree distribution of the increment in monthly salary from 2016 to 2017 = 72° + 18° = 90°

The degree distribution of the increment in monthly salary from 2019 to 2020 = 72°

The increment in monthly salary from 2015 to 2016 = Rs. 4500

The increment in monthly salary from 2016 to 2017 = 4500 * (90/54) = Rs. 7500

The increment in monthly salary from 2017 to 2018 = 4500 * (36/54) = Rs. 3000

The increment in monthly salary from 2018 to 2019 = 4500 * (60/54) = Rs. 5000

The increment in monthly salary from 2019 to 2020 = 4500 * (72/54) = Rs. 6000

The increment in monthly salary from 2020 to 2021 = 4500 * (48/54) = Rs. 4000

Answer: D

1st term of the series = 3000

2nd term of the series = 70% of 5000 = 3500

3rd term of the series = 125% of 4000 = 5000

4th term of the series = 6000

5th term of the series = 2 * 4000 = 8000

6th term of the series = 140% of 7500 = 10500

The series: 3000, 3500, 5000, 6000, 8000, 10500

Logic in the series:

3000 + 500 = 3500

3500 + 1000 = 4500(5000)

4500 + 1500 = 6000

6000 + 2000 = 8000

8000 + 2500 = 10500

So, the wrong term in the series = 5000 = 3rd term

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5. Questions

From the pie chart:

54 + (X + Y) + 36 + 60 + X + 48 = 360

2X + Y = 162 ----------(1)

Since, the sum of the increment in monthly salary from 2016 to 2017 and that from 2019 to 2020 is Rs. 13500.

So, the sum of the pie chart:

$13500 \times 360/(2X+Y)=13500 \times 360/162=$ Rs. 30000

And the increment in monthly salary from 2015 to 2016 = 30000 * (54/360) = Rs. 4500

Since, the increment in monthly salary from 2016 to 2017 is Rs. 3000 more than that from 2015 to 2016.

So, the increment in monthly salary from 2016 to 2017 = 4500 + 3000 = Rs. 7500

And,

$X+Y=54 \times 7500/4500= 90$

Y = 90 – X ----------(2)

From equations (1) and (2):

2X + 90 – X = 162

X = 72 and Y = 18

The degree distribution of the increment in monthly salary from 2016 to 2017 = 72° + 18° = 90°

The degree distribution of the increment in monthly salary from 2019 to 2020 = 72°

The increment in monthly salary from 2015 to 2016 = Rs. 4500

The increment in monthly salary from 2016 to 2017 = 4500 * (90/54) = Rs. 7500

The increment in monthly salary from 2017 to 2018 = 4500 * (36/54) = Rs. 3000

The increment in monthly salary from 2018 to 2019 = 4500 * (60/54) = Rs. 5000

The increment in monthly salary from 2019 to 2020 = 4500 * (72/54) = Rs. 6000

The increment in monthly salary from 2020 to 2021 = 4500 * (48/54) = Rs. 4000

Answer: B

Since, the monthly salary in 2016 = Rs. 28500

So, the monthly salary in 2015 = 28500 – 4500 = Rs. 24000

And the monthly salary in 2021:

24000 + (4500 + 7500 + 3000 + 5000 + 6000 + 4000) = Rs. 54000

Since, the per cent increment in monthly salary from 2021 to 2022 is 5%.

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So, the monthly salary in 2022 = 105% of 54000 = Rs. 56700

And the required percentage = (56700-24000)/24000 ×100 = 136.25%

6. Questions

Let the investments of A and B be Rs. 3x and Rs 5xrespectively.

So, the investment of C = 5x – 3x = Rs. 2x

Also let the investment of D be Rs. y.

So, the investment of E = Rs (y + 10000)

Since, the ratio of C's profit share to D's profit share after 15 months will be 4: 5.

So,

(2x ×15)/(y ×(15-5))= 4/5

15x = 4y ---------(1)

Since, the ratio of A's profit share to E's profit share after 15 months will be 4: 5.

So,

$$\frac{3x \times 15}{(y+10000)\times(15-5-1)} = \frac{4}{5}$$

$$\frac{3x \times 15}{(y+10000)\times 9} = \frac{4}{5}$$

75x = 12y + 120000

25x = 4y + 40000 ---------(2)

From equations (1) and (2):

25x = 15x + 40000

x = 4000

From equation (1):

60000 = 4y

y = 15000

Now, A's investment = 3 * 4000 = Rs. 12000

B's investment = 5 * 4000 = Rs. 20000

C's investment = 2 * 4000 = Rs. 8000

D's investment = Rs. 15000

And E's investment = 15000 + 10000 = Rs. 25000

Answer: E

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A's initial investment = Rs. 12000

B's initial investment = Rs. 20000

C's initial investment = Rs. 8000

If A had withdrawn 50% after 1 year, B had withdrawn 50% after 6 months and C had invested 87.5% more after 9 months:

The ratio between profit shares of A, B and C after 15 months would have been:

[(12000 * 12) + (6000 * 3)]: [(20000 * 6) + (10000 * 9)]: [(8000 * 9) + (15000 * 6)]

[144000 + 18000]: [120000 + 90000]: [72000 + 90000]

162: 210: 162 = 27: 35: 27

7. Questions

Let the investments of A and B be Rs. 3x and Rs 5xrespectively.

So, the investment of C = 5x – 3x = Rs. 2x

Also let the investment of D be Rs. y.

So, the investment of E = Rs (y + 10000)

Since, the ratio of C's profit share to D's profit share after 15 months will be 4: 5.

So,

(2x ×15)/(y ×(15-5))= 4/5

15x = 4y ---------(1)

Since, the ratio of A's profit share to E's profit share after 15 months will be 4: 5.

So,

$$\frac{3x \times 15}{(y+10000)\times(15-5-1)} = \frac{4}{5}$$

$$\frac{3x \times 15}{(y+10000)\times 9} = \frac{4}{5}$$

75x = 12y + 120000

25x = 4y + 40000 ---------(2)

From equations (1) and (2):

25x = 15x + 40000

x = 4000

From equation (1):

60000 = 4y

y = 15000

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Now, A's investment = 3 * 4000 = Rs. 12000

B's investment = 5 * 4000 = Rs. 20000

C's investment = 2 * 4000 = Rs. 8000

D's investment = Rs. 15000

And E's investment = 15000 + 10000 = Rs. 25000

Answer: B

If B had withdrawn his whole investment after 6 months from the starting of the business:

The ratio between profit shares of B, D and E respectively after 15 months would have been:

[20000 * 6]: [15000 * (15 – 5)]: [25000 * (15 – 6)]

120000: 150000: 22500

8: 10: 15

Since, the difference between the profit shares of B and D after 15 months = ₹1250

So, the difference between the profit shares of D and E after 15 months:

$$1250 \times \frac{15-10}{10-8} = ₹3125$$

8. Questions

Let the investments of A and B be Rs. 3x and Rs 5xrespectively.

So, the investment of C = 5x – 3x = Rs. 2x

Also let the investment of D be Rs. y.

So, the investment of E = Rs (y + 10000)

Since, the ratio of C's profit share to D's profit share after 15 months will be 4: 5.

So,

(2x ×15)/(y ×(15-5))= 4/5

15x = 4y ---------(1)

Since, the ratio of A's profit share to E's profit share after 15 months will be 4: 5.

So,

$$\frac{3x \times 15}{(y+10000) \times (15-5-1)} = \frac{4}{5}$$

$$\frac{3x \times 15}{(y+10000) \times 9} = \frac{4}{5}$$

75x = 12y + 120000

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25x = 4y + 40000 ---------(2)

From equations (1) and (2):

25x = 15x + 40000

x = 4000

From equation (1):

60000 = 4y

y = 15000

Now, A's investment = 3 * 4000 = Rs. 12000

B's investment = 5 * 4000 = Rs. 20000

C's investment = 2 * 4000 = Rs. 8000

D's investment = Rs. 15000

And E's investment = 15000 + 10000 = Rs. 25000

Answer: C

From P:

Difference between C's investment and A's investment = 12000 – 8000 = Rs. 4000

So, P is true.

From Q:

Average of the investments of C, D and E:

(8000+15000+25000)/3= Rs.16000

So, Q is not true.

From R:

Ratio of the total investment of A and B together to that of D and E together:

(12000 + 20000): (15000 + 25000) = 32000: 40000 = 4: 5

So, R is true.

Hence, only Q is not true.

9. Questions

Let the investments of A and B be Rs. 3x and Rs 5xrespectively.

So, the investment of C = 5x – 3x = Rs. 2x

Also let the investment of D be Rs. y.

So, the investment of E = Rs (y + 10000)

Since, the ratio of C's profit share to D's profit share after 15 months will be 4: 5.

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So,

(2x ×15)/(y ×(15-5))= 4/5

15x = 4y ---------(1)

Since, the ratio of A's profit share to E's profit share after 15 months will be 4: 5.

So,

$$\frac{3x \times 15}{(y + 10000) \times (15 - 5 - 1)} = \frac{4}{5}$$

$$\frac{3x \times 15}{(y + 10000) \times 9} = \frac{4}{5}$$

75x = 12y + 120000

25x = 4y + 40000 ---------(2)

From equations (1) and (2):

25x = 15x + 40000

x = 4000

From equation (1):

60000 = 4y

y = 15000

Now, A's investment = 3 * 4000 = Rs. 12000

B's investment = 5 * 4000 = Rs. 20000

C's investment = 2 * 4000 = Rs. 8000

D's investment = Rs. 15000

And E's investment = 15000 + 10000 = Rs. 25000

Answer: A

Since, the ratio of B's profit share after 8 months to C's profit share after 15 months:

M: N = (20000 * 8): (8000 * 15)

M: N = 160000: 120000

M: N = 4: 3

So, the ratio of A's profit share after 10 months to D's profit share after 6 months:

(12000 * 10): (15000 * 6)

120000: 90000 = 4: 3 = M: N

10. Questions

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Let the investments of A and B be Rs. 3x and Rs 5xrespectively.

So, the investment of C = 5x – 3x = Rs. 2x

Also let the investment of D be Rs. y.

So, the investment of E = Rs (y + 10000)

Since, the ratio of C's profit share to D's profit share after 15 months will be 4: 5.

So,

(2x ×15)/(y ×(15-5))= 4/5

15x = 4y ---------(1)

Since, the ratio of A's profit share to E's profit share after 15 months will be 4: 5.

So,

$$\frac{3x \times 15}{(y + 10000) \times (15 - 5 - 1)} = \frac{4}{5}$$

$$\frac{3x \times 15}{(y + 10000) \times 9} = \frac{4}{5}$$

75x = 12y + 120000

25x = 4y + 40000 ---------(2)

From equations (1) and (2):

25x = 15x + 40000

x = 4000

From equation (1):

60000 = 4y

y = 15000

Now, A's investment = 3 * 4000 = Rs. 12000

B's investment = 5 * 4000 = Rs. 20000

C's investment = 2 * 4000 = Rs. 8000

D's investment = Rs. 15000

And E's investment = 15000 + 10000 = Rs. 25000

Answer: D

Since, the total investment of A, B, C, D and E together:

12000 + 20000 + 8000 + 15000 + 25000 = Rs 80000

So, the investment of F = 20% of 80000 = Rs 16000

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And the ratio of C's profit share to F's profit share after 20 months from the starting of the business:

[8000 * 20]: [16000 * (20 – 12)]

160000: 128000 = 5: 4

11. Questions

Answer: D

Let the number of wooden doors and iron doors sold by shop F be '2f' and '3f' respectively.

So,

(2f * 4500) + (3f * 3000) = 90000

f = 5

Since, the number of wooden doors sold by shop F = 2 * 5 = 10

And the number of iron doors sold by shop F = 3 * 5 = 15

So, the number of plastic doors sold by shop F = (2 * 18) – 15 = 21

The number of plastic doors sold by shop E = 21 * (6/7) = 18

The number of iron doors sold by shop E = (2 * 15) – 18 = 12

And the number of wooden doors sold by shop E = 12 * (2/1) = 24

Required average = (24 + 12 + 18)/3 = 18

12. Questions

Answer: E

From the above question:

Since, the number of wooden doors sold by shop E = 24

So, the number of wooden doors sold by shop C = 24 * (100/120) = 20

And the number of wooden doors sold by shop D = 60% of 20 = 12

The number of iron doors sold by shop C = 20 * (4/5) = 16

The number of plastic doors sold by shop C = (2 * 13) – 16 = 10

The number of iron doors sold by shop D = 12 * (5/3) = 20

The number of plastic doors sold by shop D = (2 * 17) – 20 = 14

Let the cost of plastic doors in shop D be 'Rs. d' per piece.

So, the cost of plastic doors in shop C = Rs. (d + 300)

Since, we can't find the value of 'x', so the required difference can't be determined.

13. Questions

Answer: B

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Since, the average number of wooden doors sold by shops B, D and E is 17.

So, the number of wooden doors sold by shop B = (3 * 17) – (12 + 24) = 15

The number of iron doors sold by shop B = 15 * (5/3) = 25

The number of plastic doors sold by shop B = (2 * 17) – 25 = 9

And the total amount received by shop B by selling all the three types of doors:

(15 * 4800) + (25 * 3200) + (9 * 3000) = 72000 + 80000 + 27000 = Rs. 179000

14. Questions

Answer: A

Shop	Cost of wooden doors (in Rs per piece)	Cost of iron doors (in Rs per piece)	Cost of plastic doors (in Rs per piece)
C	4500	4500 * (8/9) = 4000	2 * 3400 – 4000 = 2800
D	5000	5000 * (7/10) = 3500	2 * 3000 – 3500 = 2500
E	4500	4500 * (4/5) = 3600	2 * 2800 – 3600 = 2000

Since, the number of plastic doors sold by shops C, D and E are 10, 14 and 21 respectively.

So, the total amount received by shops C, D and E together by selling plastic doors = (10 * 2800) + (14 * 2500) + (18 * 2000) = Rs 99000

15. Questions

Answer: D

From the 2nd question:

The number of wooden doors sold by shop C = 20

The number of iron doors sold by shop C = 16

The number of plastic doors sold by shop C = 10

Since, the average number of wooden doors sold by shops A and C = 19

So, the number wooden doors sold by shop A = (2 * 19) – 20 = 18

Since, the average number of iron doors sold by shops A and C = 14

So, the number iron doors sold by shop A = (2 * 14) – 16 = 12

Since, the average number of plastic doors sold by shops A and C = 15

So, the number plastic doors sold by shop A = (2 * 15) – 10 = 20

Required ratio = 18: 12: 20 = 9: 6: 10

16. Questions

Answer: D

Since, the logic in series I:

9 * 2 + 1 = 19

19 * 2 – 2 = 36

36 * 2 + 3 = 75

75 * 2 – 4 = 146

146 * 2 + 5 = 297

So, series II:

12 * 2 + 1 = 25

25 * 2 – 2 = 48

48 * 2 + 3 = 99

99 * 2 – 4 = 194

Since, P = 25

And, (8P – Q) = 194

So, Q = 8 * 25 – 194

Q = 6

Required percentage = (6/25) * 100 = 24%

17. Questions

Answer: A

From series I:

455 - 6^3 = 239

239 - 5^3 = **114(A)**

114 - 4^3 = 50

50 - 3^3 = **23(B)**

23 - 2^3 = 15

From series II:

449(C) - 6^3 = 233

233 - 5^3 = 108

108 - 4^3 = **44(D)**

44 - 3^3 = 17

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$17 - 2^3 = 9$

A = 114, B = 23, C = 449, D = 44

From P:

Since, A is 1 less than 5 times of B.

So, A = 5B – 1

114 = 5 * 23 – 1

114 = 114

So, P is true.

From Q:

Since, C is 9 more than 10 times of D.

So, C = 10D + 9

449 = 10 * 44 + 9

449 = 449

So, Q is true.

From R:

Value of (C – A)/(B + D) = (449 – 114)/(23 + 44) = 5

So, R is true.

Hence, all are true

18. Questions

Answer: C

Logic in series I:

$441 - 11^2 = 320$

$320 - 9^2 = 239$

$239 - 7^2 = 190$

$190 - 5^2 = 165$

$165 - 3^2 = 156$

So, A = 156

And next term = $156 - 1^2 = 155$

Logic in series II:

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96 – 23 = 73

73 – 19 = 54

54 – 17 = 37

37 – 13 = 24

24 – 11 = 13

So, B = 13

And next term = 13 – 7 = 6

From P:

A/B = 156/13 = 12

Since, A is completely divisible by B.

So, P is true.

From Q:

Average of next terms of both the series = (155 + 6)/2 = 80.5

So, Q is not true.

From R:

The difference between A and B = 156 – 13 = 143

Since, 143 is divisible by 11.

So, R is true.

Hence, only Q is not true.

19. Questions

Logic in series I:

$41 + 1^2 = 42$

$42 + 2^2 = 46$

$46 + 3^2 = 55$

$55 + 4^2 = 71$

$71 + 5^2 = \mathbf{96}$

So, P = 96

Logic in series II:

6 + 7 = 13

13 + 15 = 28

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28 + 31 = 59

59 + 63 = 122

122 + 127 = **249**

So, Q = 249

Now, the 1st term of series III = Q = 249

The 2nd term of series III = (2P + 1) = (2 * 96 + 1) = 193

The 3rdterm of series III = (Q – P – 8) = (249 – 96 – 8) = 145

The 4th term of series III = (P + 9) = (96 + 9) = 105

The 5th term of series III = (Q – 30)/3 = (249 – 30)/3 = 73

Answer: C

The average of P and Q = (96 + 249)/2 = 172.5

20. Questions

Logic in series I:

$41 + 1^2 = 42$

$42 + 2^2 = 46$

$46 + 3^2 = 55$

$55 + 4^2 = 71$

$71 + 5^2 = \mathbf{96}$

So, P = 96

Logic in series II:

6 + 7 = 13

13 + 15 = 28

28 + 31 = 59

59 + 63 = 122

122 + 127 = **249**

So, Q = 249

Now, the 1st term of series III = Q = 249

The 2nd term of series III = (2P + 1) = (2 * 96 + 1) = 193

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The 3rd term of series III = (Q – P – 8) = (249 – 96 – 8) = 145

The 4th term of series III = (P + 9) = (96 + 9) = 105

The 5th term of series III = (Q – 30)/3 = (249 – 30)/3 = 73

Answer: A

Logic in series III:

249 – 56 = 193

193 – 48 = 145

145 – 40 = 105

105 – 32 = 73

So, the 6th term of series III = 73 – 24 = 49

And 49 is divisible by 7 only.

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1. Questions

Study the following data carefully and answer the questions:

Five workers A, B, C, D and E are hired to complete a work. They completed the work in such a way that, A and B started the work together. After some time, A left and C joined with B, after some time, B left and D joined with C and after some time, C left and E joined with D and the remaining work is completed by D and E together.

The table given below shows the part of work (in %) completed by the given pairs of workers and the time taken to complete the given part of work.

Pairs of workers	Part of the work completed	Time taken to complete the given part of work
A and B together	21%	2.7 hours
B and C together	24%	1.8 hours
C and D together	20%	1.35 hours
D and E together	35%	5.4 hours

Note: Ratio of A's efficiency to E's efficiency is 2: 1.

If they are paid in the ratio of the part of work completed by them and the maximum amount paid to a worker is Rs. 10500, then find the total amount paid to all the five workers together?

a. Rs. 36000

b. Rs. 32000

c. Rs. 38000

d. Rs. 34000

e. Rs. 30000

2. Questions

If A and C together can complete the work in 'm' hours and B and D together can complete the work in 'n' hours, then find the ratio between 'm' and 'n'?

a. 4: 15

b. 16: 45

c. 8: 25

d. 32: 75

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e. 2: 5

3. Questions

If they were paid in the ratio of their efficiencies and the difference between the amounts paid to C and E were Rs.11250, then what would have been the total amount paid to all the five workers together?

a. Rs.34250

b. Rs.30250

c. Rs.36250

d. Rs.32250

e. Rs.38250

4. Questions

If A had worked with 60% of its original efficiency, B had worked with its original efficiency and C had worked with 50% of its original efficiency, then what would have been the time taken by A, B and C together to finish the work?

a. 18 hours

b. 6 hours

c. 21 hours

d. 15 hours

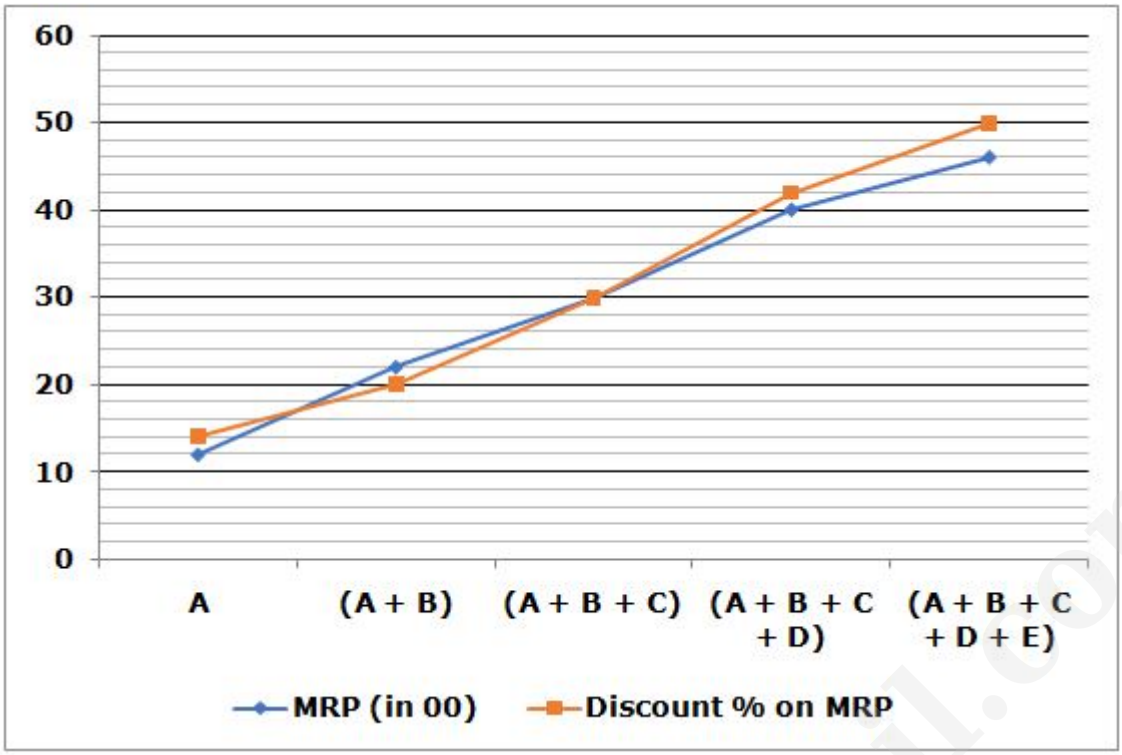
e. 9 hours

5. Questions

Study the following data carefully and answer the questions:

The line graph given below shows the cumulative data about the MRPs (Maximum retail price in Rs.) in hundreds, and discount per cent on the MRPs of five articles A, B, C, D and E.

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If the MRP of article B is 33(1/3)% above its cost price and the ratio of the cost price of article B to that of article E is 5: 3, then find the total profit per cent on the cost prices of articles B and E together?

a. 24.33%

b. 16.33%

c. 28.33%

d. 20.33%

e. 32.33%

6. Questions

If the average of discounts given on the MRPs of articles A, B and D is Rs. (100 + M + N) and M is 4 more than N, then find that the discount given on the MRP of article E is what per cent of discount given on the MRP of article C? (in terms of M and N)

a. (MN)%

b. (7M – 2N)%

c. 4(M +N)%

d. Can't be determined

e. (MN)/2%

7. Questions

The MRP of article A is 100% above its cost price. If an additional discount of 25% were given on article A, then what would have been the profit/loss per cent on the cost price of article A?

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a. 21%

b. 43%

c. 37%

d. 29%

e. 41%

8. Questions

If the average of the MRPs of articles A, B, C, D, E and F is Rs.1000 and the total discount given on articles A, B, C, D, E and F together is Rs.560, then find the discount per cent on the MRP of article F?

a. 6.5%

b. 8%

c. 6%

d. 7.5%

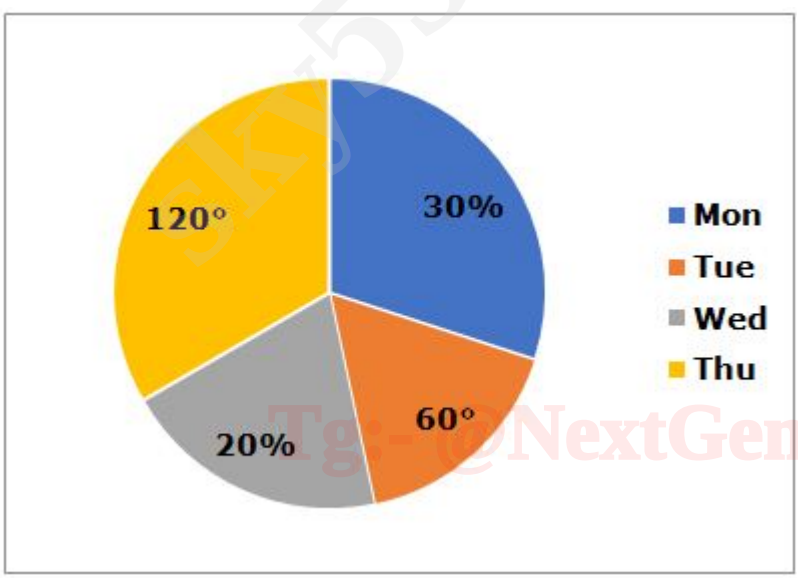
e. 10%

9. Questions

Study the following data carefully and answer the questions:

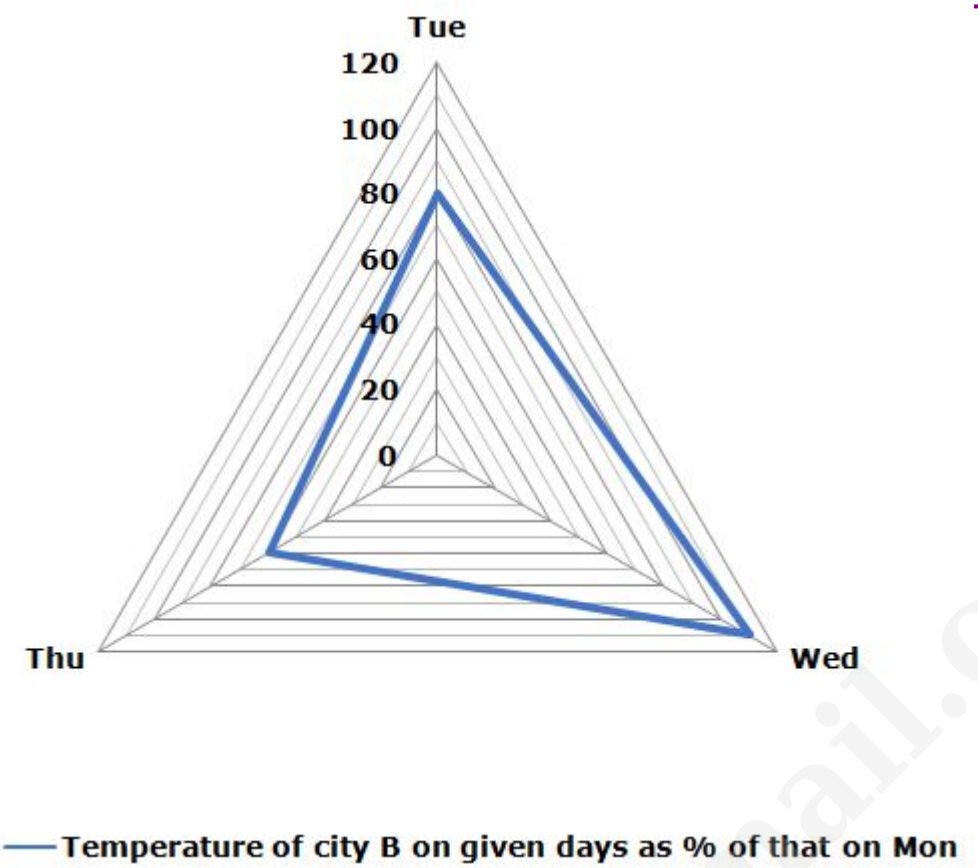
The data given below is related to the temperature (in °C) of two cities A and B recorded on four days: Mon, Tue, Wed and Thu.

The pie chart given below shows the percentage and degree distribution of temperature of city A on each of the given days:



The radar graph given below shows the temperature of city B on Tue, Wed and Thu as per cent of that on Mon.

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Note:

The difference between the temperature of city A on Mon and Tue is 20°C and the ratio of the temperature of city A on Mon to that of city B on Mon is 9: 8.

If the temperature of city B on Tue is M% more/less than that of city A on Tue and the temperature of city B on Thu is N% more/less than that of city A on Thu, then find the ratio of M to N?

a. 7: 13

b. 14: 39

c. 2: 3

d. 1: 2

e. 4: 13

10. Questions

Which of the following statements is/are not true?

P: The average temperature of city A on Mon, Tue and Thu is 40°C.

Q: The average temperature of city B on each of the given days is 35°C.

R: The ratio of the temperature of city A on Wed to that of city B on Wed is 6: 11.

a. All are true

b. Only P and R

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c. Only Q and R

d. Only Q

e. Only R

11. Questions

If the temperature of city A is decreased by 28% from Thu to Fri and the ratio of the temperature of city A on Fri to that of city B on Fri is 6: 5, then find that by what per cent the temperature of city B is increased/decreased from Thu to Fri?

a. 32.5%

b. 20%

c. 25%

d. 16.67%

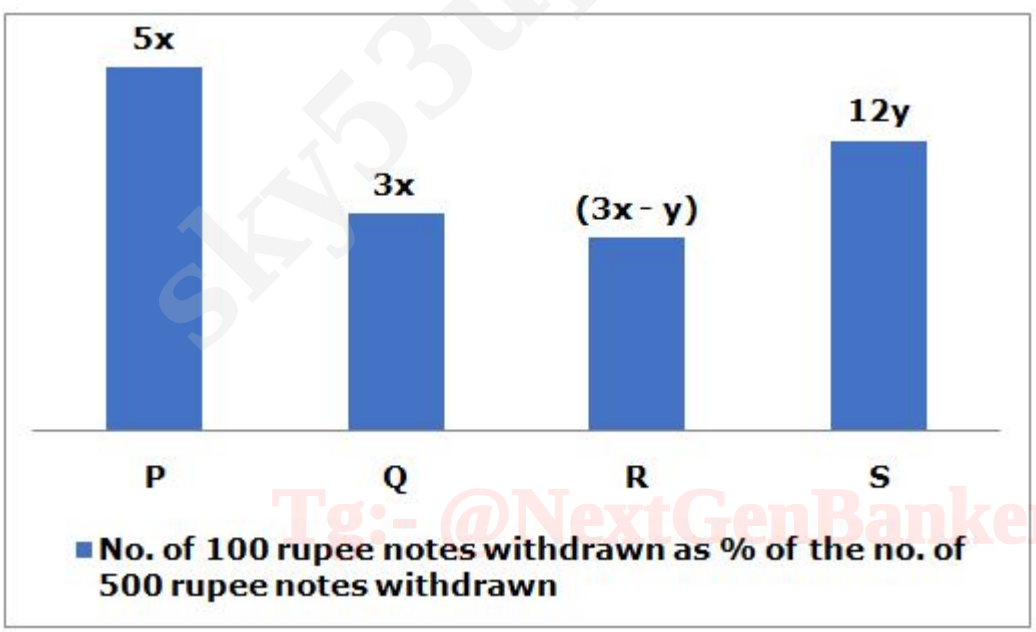
e. 30%

12. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of 100 rupee notes and the number of 500 rupee notes withdrawn by a person from four ATMs P, Q, R and S on a particular day. (Only 100 rupee notes and 500 rupee notes can be withdrawn from each ATM)

The bar graph given below shows the number of 100 rupee notes withdrawn as per cent of the number of 500 rupee notes withdrawn by the person from each ATM.



Notes:

1). The number of 100 rupee notes withdrawn from ATMs P and Q are 18 and 9 respectively.

2). The number of 500 rupee notes withdrawn from ATM P is 4 more than that withdrawn from ATM Q.

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3). Ratio of 'x' to 'y' is 3: 1.

If the number of 500 rupee notes withdrawn from ATM S is 'y' more than that withdrawn from ATM Q, then find the difference between the total amount withdrawn from ATM Q and that withdrawn from ATM S?

a. Rs. 2100
b. Rs. 2900
c. Rs. 3100
d. Rs. 2500
e. Rs. 3400

13. Questions

If the total amount withdrawn from ATM R is Rs.29700, then find the average of the number of 500 rupee notes withdrawn from ATMs P, Q and R?

a. 41
b. 29
c. 39
d. 43
e. 33

14. Questions

The total amount withdrawn from ATM P is approximately what per cent more/less than that of the total amount withdrawn from ATM Q?

a. 30.2%
b. 26.6%
c. 20%
d. 35.4%
e. 15%

15. Questions

Study the following data carefully and answer the questions:

The data given below is related to the number of fours and the number of sixes hit by two teams A and B in a cricket match. Some players hit only fours, some players hit only sixes and the remaining players hit both fours and sixes. (There are 11 players in each team)

50% of the total number of players in both the teams together hit both fours and sixes. The ratio of the total number of players, who hit fours, to the total number of players, who hit sixes, is 6: 5.

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If the number of players of team B, who hit only fours, is 40% of those of team A, who hit only fours and the number of players of team A, who hit only sixes, is 33(1/3)% of those of team B, who hit only sixes, then find that the number of players of team B, who hit both fours and sixes, is what per cent of the number of players of team A, who hit both fours and sixes?

a. Can't be determined

b. 83.33%

c. 120%

d. 57.14%

e. 175%

16. Questions

If the average number of fours hit by those players, who hit only fours, is 5 and the average number of fours hit by those players, who hit both fours and sixes, is 3, then find the number of runs scored by hitting fours only in that match?

a. 272

b. 204

c. 221

d. 238

e. 255

17. Questions

Approximately what per cent of the total number of players of both the teams together hit sixes?

a. 62%

b. 82.5%

c. 73.6%

d. 68.2%

e. 55%

18. Questions

In each of the following questions, read the given statement and compare Quantity I and Quantity II on its basis. (only quantity is to be compared)

Quantity I:

If $x^2 - 43x + 450 = 0$ and $y^2 - 14y + 45 = 0$, then find the possible difference between 'x' and 'y'?

Quantity II:

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The time taken by a boat to go 280m upstream is 12 seconds more than the time taken to come back. If the boat can cover 2.55 km in still water in 5 minutes, then find the difference between the speed of boat in still water and the speed of stream (in km/h)?

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II or relation can't be established

19. Questions

Quantity I:

A bag contains some black balls and white balls and the number of white balls is 2 more than the number of black balls. If the probability of taking out 1 white ball from the bag is 25% more than the probability of taking out 1 black ball from the bag, then find the probability (in %) of taking out 2 black balls from the bag?

Quantity II:

The average age of a group of three friends A, B and C is 20 years. When a friend D joined the group, the average age of the group remained the same as initially. If A is 2 years younger than D, then find that B's age is what per cent of C's age?

a. Quantity I > Quantity II
b. Quantity I < Quantity II
c. Quantity I ≥ Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I = Quantity II or relation can't be established

20. Questions

Bike A is running at 'x' m/s and crosses a 360 m long tunnel in (t + 5) seconds. Bike B is running at (x + 6) m/s in the same direction as bike A is running and crosses the same tunnel in 't' seconds.

Quantity I: Find the value of 'x'?

Quantity II: Find the value of 't'?

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a. Quantity I = Quantity II or relation can't be established.
b. Quantity I < Quantity II
c. Quantity I > Quantity II
d. Quantity I ≤ Quantity II
e. Quantity I ≥ Quantity II

Explanations:

1. Questions

Since, A and B together completed 21% part of work in 2.7 hours.

So, the time, in which A and B together can finish the whole work = 2.7 * (100/21) = (90/7) hours

And, 1/A+ 1/B= 7/90--------(1)

Since, B and C together completed 24% part of work in 1.8 hours.

So, the time, in which B and C together can finish the whole work = 1.8 * (100/24) = (15/2) hours

And, 1/B+ 1/C= 2/15--------(2)

Since, C and D together completed 20% part of work in 1.35 hours.

So, the time, in which C and D together can finish the whole work = 1.35 * (100/20) = (27/4) hours

And, 1/C+ 1/D= 4/27--------(3)

Since, D and E together completed 35% part of work in 5.4 hours.

So, the time, in which D and E together can finish the whole work = 5.4 * (100/35) = (108/7) hours

And, 1/D+ 1/E= 7/108--------(4)

By equation (3) – (4):

1/C- 1/E= 4/27- 7/108= (16-7)/108= 1/12--------(5)

By equation (2) – (5):

1/B+ 1/E= 2/15- 1/12= (8-5)/60= 1/20--------(6)

By equation (1) – (6):

1/A- 1/E= 7/90- 1/20= (14-9)/180= 1/36--------(7)

Since, the ratio of A's efficiency to E's efficiency is 2: 1.

So, 2A = E

From equation (7):

1/A- 1/2A=1/36

A = 18

The time, in which A alone can complete the whole work = 18 hours

From equation (1):

1/18+ 1/B= 7/90

B = 45

The time, in which B alone can complete the whole work = 45 hours

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From equation (2):

1/45+ 1/C= 2/15

C = 9

The time, in which C alone can complete the whole work = 9 hours

From equation (3):

1/9+ 1/D= 4/27

D = 27

The time, in which D alone can complete the whole work = 27 hours

From equation (4):

1/27+ 1/E= 7/108

E = 36

The time, in which E alone can complete the whole work = 36 hours

Answer: E

Since, the total work completed by A and B together = 21%

And the ratio of A's efficiency to B's efficiency = 45: 18 = 5: 2

So, the work completed by A = 21 * (5/7) = 15%

And the work completed by B, while working with A = 21 – 15 = 6%

Since, the total work completed by B and C together = 24%

And the ratio of B's efficiency to C's efficiency = 9: 45 = 1: 5

So, the work completed by B, while working with C = 24 * (1/6) = 4%

And the work complete by C, while working with B = 24 – 4 = 20%

Since, the total work completed by C and D together = 20%

And the ratio of C's efficiency to D's efficiency = 27: 9 = 3: 1

So, the work completed by C, while working with D = 20 * (3/4) = 15%

And the work completed by D, while working with C = 20 – 15 = 5%

Since, the total work completed by D and E together = 35%

And the ratio of D's efficiency to E's efficiency = 36: 27 = 4: 3

So, the work completed by D, while working with E = 35 * (4/7) = 20%

And the work completed by E, while working with D = 35 – 20 = 15%

The ratio between the parts of work completed by A, B, C, D and E respectively:

15%: (6 + 4)%: (20 + 15)%: (5 + 20)%: 15%

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15: 10: 35: 25: 15 = 3: 2: 7: 5: 3

Since, C completed the maximum part of work and the amount paid to C is Rs.10500.

So, the total amount paid to all the five workers together:

10500 × (3+2+7+5+3)/7 = Rs. 30000

2. Questions

Since, A and B together completed 21% part of work in 2.7 hours.

So, the time, in which A and B together can finish the whole work = 2.7 * (100/21) = (90/7) hours

And, 1/A+ 1/B= 7/90--------(1)

Since, B and C together completed 24% part of work in 1.8 hours.

So, the time, in which B and C together can finish the whole work = 1.8 * (100/24) = (15/2) hours

And, 1/B+ 1/C= 2/15--------(2)

Since, C and D together completed 20% part of work in 1.35 hours.

So, the time, in which C and D together can finish the whole work = 1.35 * (100/20) = (27/4) hours

And, 1/C+ 1/D= 4/27--------(3)

Since, D and E together completed 35% part of work in 5.4 hours.

So, the time, in which D and E together can finish the whole work = 5.4 * (100/35) = (108/7) hours

And, 1/D+ 1/E= 7/108--------(4)

By equation (3) – (4):

1/C- 1/E= 4/27- 7/108= (16-7)/108= 1/12--------(5)

By equation (2) – (5):

1/B+ 1/E= 2/15- 1/12= (8-5)/60= 1/20--------(6)

By equation (1) – (6):

1/A- 1/E= 7/90- 1/20= (14-9)/180= 1/36--------(7)

Since, the ratio of A's efficiency to E's efficiency is 2: 1.

So, 2A = E

From equation (7):

1/A- 1/2A=1/36

A = 18

The time, in which A alone can complete the whole work = 18 hours

From equation (1):

1/18+ 1/B= 7/90

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B = 45

The time, in which B alone can complete the whole work = 45 hours

From equation (2):

1/45+ 1/C= 2/15

C = 9

The time, in which C alone can complete the whole work = 9 hours

From equation (3):

1/9+ 1/D= 4/27

D = 27

The time, in which D alone can complete the whole work = 27 hours

From equation (4):

1/27+ 1/E= 7/108

E = 36

The time, in which E alone can complete the whole work = 36 hours

Answer: B

Since, the time, in which A alone can complete the work = 18 hours

And the time, in which C alone can complete the work = 9 hours

So,

1/m= 1/18+ 1/9= (1+2)/18= 1/6

m = 6 hours

Since, the time, in which B alone can complete the work = 45 hours

And the time, in which D alone can complete the work = 27 hours

So,

1/n= 1/45+ 1/27= (3+5)/135= 8/135

n = (135/8) hours

Required ratio = m: n = 6: (135/8) = 16: 45

3. Questions

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Since, A and B together completed 21% part of work in 2.7 hours.

So, the time, in which A and B together can finish the whole work = 2.7 * (100/21) = (90/7) hours

And, 1/A+ 1/B= 7/90--------(1)

Since, B and C together completed 24% part of work in 1.8 hours.

So, the time, in which B and C together can finish the whole work = 1.8 * (100/24) = (15/2) hours

And, 1/B+ 1/C= 2/15--------(2)

Since, C and D together completed 20% part of work in 1.35 hours.

So, the time, in which C and D together can finish the whole work = 1.35 * (100/20) = (27/4) hours

And, 1/C+ 1/D= 4/27--------(3)

Since, D and E together completed 35% part of work in 5.4 hours.

So, the time, in which D and E together can finish the whole work = 5.4 * (100/35) = (108/7) hours

And, 1/D+ 1/E= 7/108--------(4)

By equation (3) – (4):

1/C- 1/E= 4/27- 7/108= (16-7)/108= 1/12--------(5)

By equation (2) – (5):

1/B+ 1/E= 2/15- 1/12= (8-5)/60= 1/20--------(6)

By equation (1) – (6):

1/A- 1/E= 7/90- 1/20= (14-9)/180= 1/36--------(7)

Since, the ratio of A's efficiency to E's efficiency is 2: 1.

So, 2A = E

From equation (7):

1/A- 1/2A=1/36

A = 18

The time, in which A alone can complete the whole work = 18 hours

From equation (1):

1/18+ 1/B= 7/90

B = 45

The time, in which B alone can complete the whole work = 45 hours

From equation (2):

1/45+ 1/C= 2/15

C = 9

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The time, in which C alone can complete the whole work = 9 hours

From equation (3):

1/9+ 1/D= 4/27

D = 27

The time, in which D alone can complete the whole work = 27 hours

From equation (4):

1/27+ 1/E= 7/108

E = 36

The time, in which E alone can complete the whole work = 36 hours

Answer: A

Since, the ratio between the time, in which A alone, B alone, C alone, D alone and E alone respectively can complete the work:

18: 45: 9: 27: 36 = 2: 5: 1: 3: 4

So, the ratio between the efficiencies of A, B, C, D and E respectively:

60/2: 60/5: 60/1: 60/3: 60/4=30:12:60:20:15

Since, the difference between the amounts paid to C and E = Rs.11250

So, the total amount paid to all the five workers together:

11250 × (30+12+60+20+15)/(60-15)= Rs.34250

4. Questions

Since, A and B together completed 21% part of work in 2.7 hours.

So, the time, in which A and B together can finish the whole work = 2.7 * (100/21) = (90/7) hours

And, 1/A+ 1/B= 7/90--------(1)

Since, B and C together completed 24% part of work in 1.8 hours.

So, the time, in which B and C together can finish the whole work = 1.8 * (100/24) = (15/2) hours

And, 1/B+ 1/C= 2/15--------(2)

Since, C and D together completed 20% part of work in 1.35 hours.

So, the time, in which C and D together can finish the whole work = 1.35 * (100/20) = (27/4) hours

And, 1/C+ 1/D= 4/27--------(3)

Since, D and E together completed 35% part of work in 5.4 hours.

So, the time, in which D and E together can finish the whole work = 5.4 * (100/35) = (108/7) hours

And, 1/D+ 1/E= 7/108--------(4)

By equation (3) – (4):

1/C- 1/E= 4/27- 7/108= (16-7)/108= 1/12--------(5)

By equation (2) – (5):

1/B+ 1/E= 2/15- 1/12= (8-5)/60= 1/20--------(6)

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By equation (1) – (6):

1/A- 1/E= 7/90- 1/20= (14-9)/180= 1/36--------(7)

Since, the ratio of A's efficiency to E's efficiency is 2: 1.

So, 2A = E

From equation (7):

1/A- 1/2A=1/36

A = 18

The time, in which A alone can complete the whole work = 18 hours

From equation (1):

1/18+ 1/B= 7/90

B = 45

The time, in which B alone can complete the whole work = 45 hours

From equation (2):

1/45+ 1/C= 2/15

C = 9

The time, in which C alone can complete the whole work = 9 hours

From equation (3):

1/9+ 1/D= 4/27

D = 27

The time, in which D alone can complete the whole work = 27 hours

From equation (4):

1/27+ 1/E= 7/108

E = 36

The time, in which E alone can complete the whole work = 36 hours

Answer: E

Since, the time, in which A alone can finish the work with its original efficiency = 18 hours

So, the time, in which A alone will complete the work with its new efficiency:

18 * (100/60) = 30 hours

Since, the time, in which C alone can finish the work with its original efficiency = 9 hours

So, the time, in which C alone will complete the work with its new efficiency:

9 * (100/50) = 18 hours

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The time, in which B alone can finish the work with its original efficiency = 45 hours

Part of work completed by A, B and C together with their new efficiencies in 1 hour:

1/30+ 1/45+ 1/18= (3+2+5)/90= 1/9

So, the time, in which A, B and C together would have completed the work with their new efficiencies = 9 hours

5. Questions

Articles	MRP (in Rs.)	Discount % on MRP
A	1200	14%
A + B	2200	20%
A + B + C	3000	30%
A + B + C + D	4000	42%
A + B + C + D + E	4600	50%

Article	MRP (in Rs.)	Discount % on MRP
A	1200	14%
B	2200 – 1200 = 1000	20 – 14 = 6%
C	3000 – 2200 = 800	30 – 20 = 10%
D	4000 – 3000 = 1000	42 – 30 = 12%
E	4600 – 4000 = 600	50 – 42 = 8%

Answer: A

Since, the MRP of article B = Rs.1000

So, the CP of article B = 1000 * (300/400) = Rs.750

The CP of article E = 750 * (3/5) = Rs.450

And the total CP of both the articles together = 750 + 450 = Rs.1200

Since, the SP of article B = (100 – 6)% of 1000 = Rs.940

The SP of article E = (100 – 8)% of 600 = Rs.552

And the total SP of both the articles together = 940 + 552 = Rs.1492

So, the profit per cent on articles B and E together:

(1492-1200)/1200 ×100=24.33%

6. Questions

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Articles	MRP (in Rs.)	Discount % on MRP
A	1200	14%
A + B	2200	20%
A + B + C	3000	30%
A + B + C + D	4000	42%
A + B + C + D + E	4600	50%

Article	MRP (in Rs.)	Discount % on MRP
A	1200	14%
B	2200 – 1200 = 1000	20 – 14 = 6%
C	3000 – 2200 = 800	30 – 20 = 10%
D	4000 – 3000 = 1000	42 – 30 = 12%
E	4600 – 4000 = 600	50 – 42 = 8%

Answer: A

The discount given on the MRP of article A = 14% of 1200 = Rs. 168

The discount given on the MRP of article B = 6% of 1000 = Rs. 60

The discount given on the MRP of article D = 12% of 1000 = Rs. 120

So,

100+M+N= (168+60+120)/3

M + N = 16 --------(1)

Since, M is 4 more than N.

So, M – N = 4 --------(2)

From equations (1) and (2):

M = 10 and N = 6

Since, the discount given on the MRP of article C = 10% of 800 = Rs. 80

And the discount given on the MRP of article E = 8% of 600 = Rs. 48

So, the required percentage = 48/80 × 100=60%=(MN)%

7. Questions

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Articles	MRP (in Rs.)	Discount % on MRP
A	1200	14%
A + B	2200	20%
A + B + C	3000	30%
A + B + C + D	4000	42%
A + B + C + D + E	4600	50%

Article	MRP (in Rs.)	Discount % on MRP
A	1200	14%
B	2200 – 1200 = 1000	20 – 14 = 6%
C	3000 – 2200 = 800	30 – 20 = 10%
D	4000 – 3000 = 1000	42 – 30 = 12%
E	4600 – 4000 = 600	50 – 42 = 8%

Answer: D

Since, the MRP of article A = Rs.1200

So, the CP of article A = 1200 * (100/200) = Rs.600

Since, the discount given on article A is 14%.

So, the SP of article A = 86% of 1200 = Rs.1032

If an additional discount of 25% were given on article A:

The SP of article A would be = 75% of 1032 = Rs.774

And the profit per cent on article A would be = (774-600)/600 × 100 = 29%

8. Questions

Articles	MRP (in Rs.)	Discount % on MRP
A	1200	14%
A + B	2200	20%
A + B + C	3000	30%
A + B + C + D	4000	42%
A + B + C + D + E	4600	50%

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Article	MRP (in Rs.)	Discount % on MRP
A	1200	14%
B	2200 – 1200 = 1000	20 – 14 = 6%
C	3000 – 2200 = 800	30 – 20 = 10%
D	4000 – 3000 = 1000	42 – 30 = 12%
E	4600 – 4000 = 600	50 – 42 = 8%

Answer: C

Since, the total MRP of articles A, B, C, D and E together:

1200 + 1000 + 800 + 1000 + 600 = Rs.4600

So, the MRP of article F = 6 * 1000 – 4600 = Rs.1400

Since, the discount given on article A = 14% of 1200 = Rs.168

The discount given on article B = 6% of 1000 = Rs.60

The discount given on article C = 10% of 800 = Rs.80

The discount given on article D = 12% of 1000 = Rs.120

And the discount given on article E = 8% of 600 = Rs.48

So, the discount given on article F = 560 – (168 + 60 + 80 + 120 + 48) = Rs.84

And the discount given on article F = (84/1400) * 100 = 6%

9. Questions

The percentage distribution of the temperature of city A on Tue = 100 * (60/360) = (50/3)%

Ratio of the temperature of city A on Mon to that on Tue = 30: (50/3) = 9: 5

Since, the difference between the temperature of city A on Mon and Tue is 20°C.

So, the temperature of city A on Mon = 20 × 9/(9-5) = 45°C

The temperature of city A on Tue = 20 × 5/(9-5) = 25°C

The temperature of city A on Wed = 45 * (20/30) = 30°C

And the temperature of city A on Thu = 25 * (120/60) = 50°C

Since, the ratio of the temperature of city A on Mon to that of city B on Mon is 9: 8.

So, the temperature of city B on Mon = 45 * (8/9) = 40°C

The temperature of city B on Tue = 80% of 40 = 32°C

The temperature of city B on Wed = 110% of 40 = 44°C

And the temperature of city B on Thu = 60% of 40 = 24°C

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Days	Temperature of city A (in °C)	Temperature of city B (in °C)
Mon	45	40
Tue	25	32
Wed	30	44
Thu	50	24

Answer: A

The temperature of city A on Tue = 25°C

The temperature of city B on Tue = 32°C

So, M = (32-25)/25 ×100 = 28%

The temperature of city A on Thu = 50°C

The temperature of city B on Thu = 24°C

So, N = (50-24)/50 ×100 = 52%

Required ratio = 28: 52 = 7: 13

10. Questions

The percentage distribution of the temperature of city A on Tue = 100 * (60/360) = (50/3)%

Ratio of the temperature of city A on Mon to that on Tue = 30: (50/3) = 9: 5

Since, the difference between the temperature of city A on Mon and Tue is 20°C.

So, the temperature of city A on Mon = 20 × 9/(9-5) = 45°C

The temperature of city A on Tue = 20 × 5/(9-5) = 25°C

The temperature of city A on Wed = 45 * (20/30) = 30°C

And the temperature of city A on Thu = 25 * (120/60) = 50°C

Since, the ratio of the temperature of city A on Mon to that of city B on Mon is 9: 8.

So, the temperature of city B on Mon = 45 * (8/9) = 40°C

The temperature of city B on Tue = 80% of 40 = 32°C

The temperature of city B on Wed = 110% of 40 = 44°C

And the temperature of city B on Thu = 60% of 40 = 24°C

Days	Temperature of city A (in °C)	Temperature of city B (in °C)
Mon	45	40
Tue	25	32
Wed	30	44
Thu	50	24

Answer: E

From P:

Average temperature of city A on Mon, Tue and Thu is 40°C:

(45+25+50)/3=40℃

So, P is true.

From Q:

Average temperature of city B on each of the given days is 35°C:

(40+32+44+24)/4=35℃

So, Q is true.

From R:

Ratio of the temperature of city A on Wed to that of city B on Wed:

30: 44 = 15: 22

So, R is not true.

Hence, only R is not true.

11. Questions

The percentage distribution of the temperature of city A on Tue = 100 * (60/360) = (50/3)%

Ratio of the temperature of city A on Mon to that on Tue = 30: (50/3) = 9: 5

Since, the difference between the temperature of city A on Mon and Tue is 20°C.

So, the temperature of city A on Mon = 20 × 9/(9-5) = 45°C

The temperature of city A on Tue = 20 × 5/(9-5) = 25°C

The temperature of city A on Wed = 45 * (20/30) = 30°C

And the temperature of city A on Thu = 25 * (120/60) = 50°C

Since, the ratio of the temperature of city A on Mon to that of city B on Mon is 9: 8.

So, the temperature of city B on Mon = 45 * (8/9) = 40°C

The temperature of city B on Tue = 80% of 40 = 32°C

The temperature of city B on Wed = 110% of 40 = 44°C

And the temperature of city B on Thu = 60% of 40 = 24°C

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Days	Temperature of city A (in °C)	Temperature of city B (in °C)
Mon	45	40
Tue	25	32
Wed	30	44
Thu	50	24

Answer: C

Since, the temperature of city A on Thu = 50°C

So, the temperature of city A on Fri = (100 – 28)% of 50 = 36°C

And the temperature of city B on Fri = 36 * (5/6) = 30°C

Since, the temperature of city B on Thu = 24°C

So, the required percentage = (30-24)/24 ×100 = 25%

12. Questions

Since, the number of 100 rupee notes withdrawn from ATMs P and Q are 18 and 9 respectively.

And the number of 500 rupee notes withdrawn from ATM P is 4 more than that withdrawn from ATM Q.

So,

$18 \times (100/5x) - 9 \times (100/3x) = 4$

360 – 300 = 4x

x = 15

Since, the ratio of 'x' to 'y' is 3: 1.

So, the value of 'y' = 15 * (1/3) = 5

The number of 100 rupee notes withdrawn from ATM P = 18

The number of 500 rupee notes withdrawn from ATM P:

18 * (100/5x) = 18 * (100/75) = 24

The number of 100 rupee notes withdrawn from ATM Q = 9

The number of 500 rupee notes withdrawn from ATM Q:

9 * (100/3x) = 9 * (100/45) = 20

Answer: C

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Since, the number of 500 rupee notes withdrawn from ATM Q = 20

So, the number of 500 rupee notes withdrawn from ATM S = 20 + y = 20 + 5 = 25

And the number of 100 rupee notes withdrawn from ATM S = 12y% of 25 = 60% of 25 = 15

Since, the total amount withdrawn from ATM Q = (9 * 100) + (20 * 500) = Rs.10900

And the total amount withdrawn from ATM S = (15 * 100) + (25 * 500) = Rs.14000

So, the required difference = 14000 – 10900 = Rs.3100

13. Questions

Since, the number of 100 rupee notes withdrawn from ATMs P and Q are 18 and 9 respectively.

And the number of 500 rupee notes withdrawn from ATM P is 4 more than that withdrawn from ATM Q.

So,

$18 \times (100/5x)-9 \times (100/3x)=4$

360 – 300 = 4x

x = 15

Since, the ratio of 'x' to 'y' is 3: 1.

So, the value of 'y' = 15 * (1/3) = 5

The number of 100 rupee notes withdrawn from ATM P = 18

The number of 500 rupee notes withdrawn from ATM P:

18 * (100/5x) = 18 * (100/75) = 24

The number of 100 rupee notes withdrawn from ATM Q = 9

The number of 500 rupee notes withdrawn from ATM Q:

9 * (100/3x) = 9 * (100/45) = 20

Answer: E

Let the number of 500 rupee notes withdrawn from ATM R be '5r'.

So, the number of 100 rupee notes withdrawn from ATM R:

(3x – y)% of 5r = (45 – 5)% of 5r = 2r

Since, the total amount withdrawn from ATM R = Rs. 29700

So,

(2r * 100) + (5r * 500) = 29700

200r + 2500r = 29700

r = 11

The number of 500 rupee notes withdrawn from ATM R = 5 * 11 = 55

Since, the number of 500 rupee notes withdrawn from ATMs P and Q are 24 and 20 respectively.

So, the required average = (24+20+55)/3 = 33

14. Questions

Since, the number of 100 rupee notes withdrawn from ATMs P and Q are 18 and 9 respectively.

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And the number of 500 rupee notes withdrawn from ATM P is 4 more than that withdrawn from ATM Q.

So,

$18 \times (100/5x) - 9 \times (100/3x) = 4$

$360 - 300 = 4x$

$x = 15$

Since, the ratio of 'x' to 'y' is 3: 1.

So, the value of 'y' = 15 * (1/3) = 5

The number of 100 rupee notes withdrawn from ATM P = 18

The number of 500 rupee notes withdrawn from ATM P:

18 * (100/5x) = 18 * (100/75) = 24

The number of 100 rupee notes withdrawn from ATM Q = 9

The number of 500 rupee notes withdrawn from ATM Q:

9 * (100/3x) = 9 * (100/45) = 20

Answer: B

The total amount withdrawn from ATM P = (18 * 100) + (24 * 500) = Rs. 13800

The total amount withdrawn from ATM Q = (9 * 100) + (20 * 500) = Rs. 10900

Required percentage = $(13800-10900)/10900 \times 100 = 26.6\%$ (approx.)

15. Questions

Since, the total number of players in both the teams together = 11 + 11 = 22

So, the number of players, who hit both fours and sixes = 50% of 22 = 11

Let the number of players, who hit only fours and those, who hit only sixes, be 'x' and 'y' respectively.

So,

$x + y = 22 - 11$

$x + y = 11$ ---------(1)

Since, the ratio of the total number of players, who hit fours, to the total number of players, who hit sixes, is 6: 5.

So,

$(x+11)/(y+11) = 6/5$

$5x + 55 = 6y + 66$

$5x - 6y = 11$ ----------(2)

From equations (1) and (2):

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$5(11 - y) - 6y = 11$

$55 - 5y - 6y = 11$

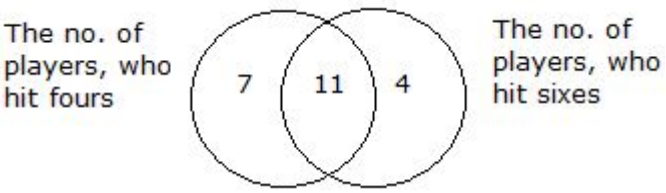
$y = 4$

From equation (1):

$x = 7$

The number of players, who hit only fours = 7

The number of players, who hit only sixes = 4



Answer: C

Ratio of the number of players of team A to those of team B, who hit only fours = 100: 40 = 5: 2

So, the number of players of team A, who hit only fours = 7 * (5/7) = 5

And the number of players of team B, who hit only fours = 7 – 5 = 2

Ratio of the number of players of team A to those of team B, who hit only sixes = (100/3): 100 = 1: 3

So, the number of players of team A, who hit only sixes = 4 * (1/4) = 1

And the number of players of team B, who hit only sixes = 4 – 1 = 3

Since, the number of players of team A, who hit both fours and sixes = 11 – 5 – 1 = 5

And the number of players of team B, who hit both fours and sixes = 11 – 2 – 3 = 6

So, the required percentage = (6/5) * 100 = 120%

16. Questions

Since, the total number of players in both the teams together = 11 + 11 = 22

So, the number of players, who hit both fours and sixes = 50% of 22 = 11

Let the number of players, who hit only fours and those, who hit only sixes, be ‘x’ and ‘y’ respectively.

So,

$x + y = 22 - 11$

$x + y = 11$ ---------(1)

Since, the ratio of the total number of players, who hit fours, to the total number of players, who hit sixes, is 6: 5.

So,

$(x+11)/(y+11) = 6/5$

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$5x + 55 = 6y + 66$

$5x - 6y = 11$ ----------(2)

From equations (1) and (2):

$5(11 - y) - 6y = 11$

$55 - 5y - 6y = 11$

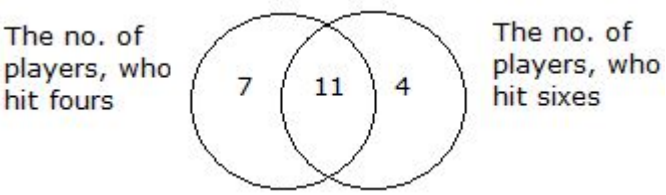
$y = 4$

From equation (1):

$x = 7$

The number of players, who hit only fours = 7

The number of players, who hit only sixes = 4



Answer: A

Since, the average number of fours hit by those players, who hit only fours, is 5.

And the average number of fours hit by those players, who hit both fours and sixes, is 3.

So, the total number of fours hit in that match = (7 * 5) + (11 * 3) = 68

And the number of runs scored by hitting fours only in that match = 68 * 4 = 272

17. Questions

Since, the total number of players in both the teams together = 11 + 11 = 22

So, the number of players, who hit both fours and sixes = 50% of 22 = 11

Let the number of players, who hit only fours and those, who hit only sixes, be 'x' and 'y' respectively.

So,

$x + y = 22 - 11$

$x + y = 11$ ---------(1)

Since, the ratio of the total number of players, who hit fours, to the total number of players, who hit sixes, is 6: 5.

So,

$(x+11)/(y+11) = 6/5$

$5x + 55 = 6y + 66$

$5x - 6y = 11$ ----------(2)

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From equations (1) and (2):

$5(11 - y) - 6y = 11$

$55 - 5y - 6y = 11$

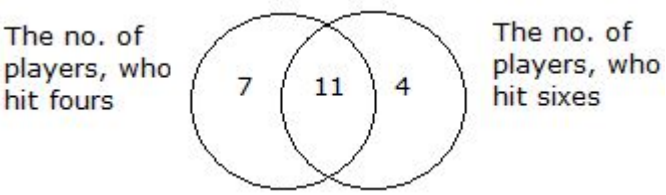
$y = 4$

From equation (1):

$x = 7$

The number of players, who hit only fours = 7

The number of players, who hit only sixes = 4



Answer: D

The total number of players, who hit sixes = 4 + 11 = 15

The total number of players in both the teams together = 11 + 11 = 22

Required percentage = (15/22) * 100 = 68.2% (approx.)

18. Questions

Answer: B

Quantity I:

$x^2 - 43x + 450 = 0$

$x^2 - 25x - 18x + 450 = 0$

$x(x - 25) - 18(x - 25) = 0$

$x = 18, 25$

$y^2 - 14y + 45 = 0$

$y^2 - 5y - 9y + 45 = 0$

$y(y - 5) - 9(y - 5) = 0$

$y = 5, 9$

Now, the possible differences between 'x' and 'y':

$18 - 5 = 13$

$18 - 9 = 9$

$25 - 5 = 20$

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25 – 9 = 16

Quantity II:

Since, the boat can cover 2.55 km in still water in 5 minutes.

So, the speed of the boat in still water = (2.55 ×60)/5 = 30.6 km/h = 8.5 m/s

Let the speed of the stream be 'x' m/s.

So,

$$\frac{280}{8.5-x}-\frac{280}{8.5+x}=12$$

$$\frac{70}{8.5-x}-\frac{70}{8.5+x}=3$$

$$\frac{595+70x-595+70x}{(8.5-x)(8.5+x)}=3$$

$140x = 3(72.25 – x^2)$

$140x = 216.75 – 3x^2$

$3x^2 + 140x – 216.75 = 0$

$3x^2 – 4.5x + 144.5x – 216.75 = 0$

$3x(x – 1.5) + 144.5(x – 1.5) = 0$

$x = 1.5$

The speed of the stream = 1.5 m/s

So, the required difference = 8.5 – 1.5 = 7 m/s = 25.2 km/h

Hence, Quantity I < Quantity II

19. Questions

Answer: E

Quantity I:

Let the number of black balls and the number of white balls in the bag be 'x' and (x + 2) respectively.

The probability of taking out 1 black ball from the bag:

$x/(x+x+2) \times 100 = 50x/(x+1)\%$

The probability of taking out 1 white ball from the bag:

$(x+2)/(x+x+2) \times 100 = (50x+100)/(x+1)\%$

So,

$(50x+100)/(x+1) - 50x/(x+1) = 25$

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$50x + 100 - 50x = 25(x + 1)$

$4 = x + 1$

$x = 3$

Since, the number of black balls in the bag = 3

And the number of white balls in the bag = 3 + 2 = 5

So, the probability of taking out 2 black balls from the bag:

$$\frac{3C2}{8C2} \times 100 = \frac{\frac{3!}{2! \times 1!}}{\frac{8!}{2! \times 6!}} \times 100 = \frac{3 \times 2}{8 \times 7} \times 100 = 10.7\% \text{ (approx.)}$$

Quantity II:

Since, average age of A, B and C is 20 years.

So, A + B + C = 3 * 20

A + B + C = 60 ----------(1)

Since, average age of A, B, C and D is 20 years.

So, A + B + C + D = 4 * 20

A + B + C + D = 80 ----------(2)

By equation (2) – (1):

D = 20 years

D's age = 20 years

Since, A is 2 years younger than D.

So, A's age = 20 – 2 = 18 years

From equation (1):

18 + B + C = 60

B + C = 42

Since, the equation can't be solved, so the required percentage can't be determined.

Hence, relation can't be established.

20. Questions

Answer: C

Since, bike A is running at 'x' m/s and crosses a 360 m long tunnel in (t + 5) seconds.

So,

x= 360/(t+5)------(1)

Since, bike B is running at (x + 6) m/s and crosses the same tunnel in 't' seconds.

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So,

x+6= 360/t------(2)

From equations (1) and (2):

$$\frac{360}{t+5}+6=\frac{360}{t}$$

$$\frac{360}{t}-\frac{360}{t+5}=6$$

$$\frac{60(t+5)-60t}{t(t+5)}=1$$

$60t + 300 – 60t = t^2 + 5t$

$t^2 + 5t – 300 = 0$

$t^2 + 20t – 15t – 300 = 0$

$t(t + 20) – 15(t + 20) = 0$

$t = 15$

From equation (1):

x= 360/(15+5)

x = 18

Quantity I:

The value of 'x' = 18

Quantity II:

The value of 't' = 15

Hence, Quantity I > Quantity II

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1. Questions

Study the following information carefully and answer the questions given below.

The given table shows the number of four items (headphones, cameras, chargers and Laptops) sold in five different shops: A, B, C, D and E. The total number of items sold in each shop and the total number of each item sold in all shops together is also given.

Shop	Number of Headphones sold	Number of Cameras sold	Number of Chargers sold	Number of Laptops sold	Total number of items sold
A	5X	4Y	_____	3M	416
B	4Z	3Y	4X	____	339
C	Z+M	_____	88	5Y	___
D	_____	____	3Y	85	269
E	_____	X+M	_____	101	343
Total	439	_____	452	____	1776

Notes:

I). 2X+3Y+4Z= 231,

II). 3X+5Y+4Z=296

III). 2X+Y+4Z=185

IV). M-Z=4

Find which of the following statements is true.

I). The average number of chargers sold in shops B and C together is 72.

II). The sum of headphones sold in shops A, C and E together is 234.

III). The difference between the number of cameras sold in shops D and E is 0.

a. Only I true

b. Only II false

c. Only III true

d. Only I false

e. Only III false

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2. Questions

The ratio of the number of watches to laptops sold in shop D is 7:5, and the ratio of the number of watches to laptops sold in shop C is 8:5. Find the possible difference between the number of watches sold in shops C and D.

I). 3X+8

II). 3Y-4

III). 2Z+3

a. Only I false

b. Only I true

c. All false

d. All true

e. Only II false

3. Questions

The average number of headphones sold in shops E and F is 6X, and the average number of cameras sold in shops B and F is 3Z. The average total number of items sold in shops E and F is 16Z. Find the total number of laptops and chargers sold in shop F.

a. 4M+6Y+3Z+26

b. 5M+6Y+7Z

c. 11X+12Y+3Z+22

d. 7X+8Y+9Z+23

e. More than one option is true

4. Questions

The difference between the number of headphones and cameras sold in shop C is 'N', and the difference between the number of chargers and laptops sold in shop A is 'P'. Find which of the following statements is true.

I). 2N>5P

II). N+P>X+Y

III). N-P<M-X

a. only I and II true

b. only II false

c. only III false

d. only I true

e. None of these

5. Questions

The ratio of the number of wired and wireless headphones sold in shop E is 2:1 and the ratio of the

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number of wired and wireless headphones sold in shop C is 5:1. Find the difference between the number of wired headphones sold in shops C and E.

a. 7

b. 6

c. 5

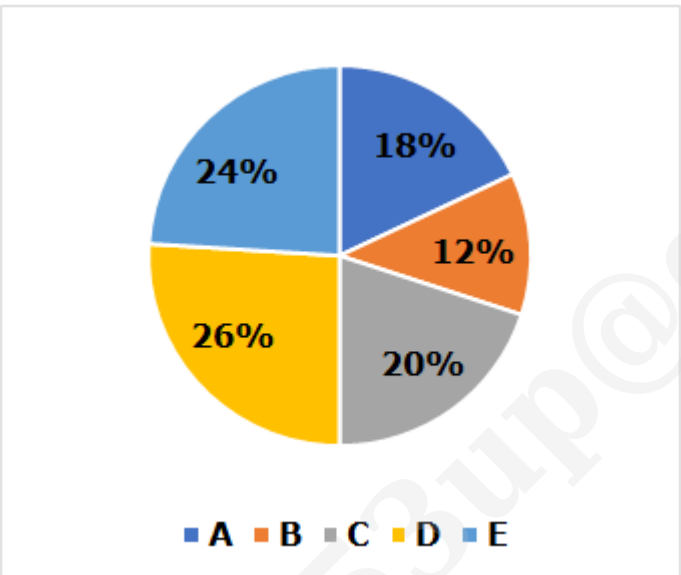
d. 4

e. None of these

6. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage distribution of the sum of the cost price and selling price of five items: A, B,C, D and E.



Note:

The profit percentage of items A, B, C, D and E is 20%, 200%, 20%, 20% and 66.66% respectively. The cost price of item E is Rs.99.

The ratio of the cost price of items A and F is _____ respectively. If item F is sold at a _____% profit, then the difference between the selling price of item F and C is Rs._____.

Find which of the following statement is true.

I). 9:11, 20, 142

II). 5:7, 40, 166.4

III). 6:11, 20, 188

a. All true

b. Only I true

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c. All false

d. Only I false

e. Only II true

7. Questions

Item C is marked up by 60% and sold at approximately x% discount and item D is marked up by 50% and sold at approximately y% discount. Find the sum of the values of x and y.

a. 55

b. 35

c. 40

d. 45

e. 50

8. Questions

The difference between the selling price of items A and C is m, and the difference between the cost price of items D and E is n. Find which of the following statements is true.

I). The selling price of item E is 12m+21.

II). The selling price of item C is 4n-4.

III). The selling price of item A is 9m.

a. Only I and III false

b. All true

c. Only II true

d. Only I false

e. All false

9. Questions

If the profit percentage of item D is (x+5)%, then the selling price of item D is Rs.32.5 more than the original selling price. Find the selling price of item B when item B is sold at x% profit.

a. 46.2

b. 32.5

c. 48.6

d. 56.5

e. None of these

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10. Questions

The difference between the cost price and the selling price of item A is x, and the difference between the cost price and the selling price of item B is y. The cost price of an item G is Rs. (x+y). Find the difference between the selling price of item C and G when item G sold at 10% profit.

a. 27.6

b. 24.5

c. 26.4

d. 29.3

e. None of these

11. Questions

Study the following information carefully and answer the questions given below.

There are five classes: A, B, C, D and E. Each class has different numbers of boys and girls. The ratio of boys to girls in class D is 11:7 respectively. The number of girls in class A is double the number of boys in class B. The ratio of boys to girls in class C is 15:8 respectively. The number of boys in class A is 21 less than the number of boys in class D. The total number of students in class A is 65, which is one less than the number of boys in class B and C together. The ratio of the number of girls in class B and C is 13:8 respectively. The number of boys and girls in class E is equal. The ratio of the number of boys in class C and E is 15:11 respectively. The total number of girls in all classes together is 166.

The ratio of boys and girls in class F is 4:3, and the ratio of boys and girls in class G is 3:2. The total number of students in class F is three times the number of boys in class B, and the total number of students in class G is the same as the total number of students in class B. Find the difference between the total number of boys and girls in both classes G and F.

a. 42

b. 32

c. 21

d. 40

e. 45

12. Questions

The average age of boys in class A is 24 years, and the average age of girls in class A is 22 years. Find the average age of all the students in class A.

a. 23.22 years

b. 22.22 years

c. 23.70 years

d. 22.70 years

e. 23.33 years

13. Questions

Out of the total boys of class C, 20% qualify for an exam, and out of the total girls of class C, 25% qualify for an exam. The total of 21 boys from C and D together qualify for the exam, and 50% of students from class D qualify for the exam. Find the number of girls who did not qualify for the exam from classes C and D.

a. 25

b. 22

c. 32

d. 24

e. None of these

14. Questions

The difference between the number of boys and girls in class D is x and the difference between the number of boys and girls in class C is y. Find which of the following equation is true.

a. 9x+11y=408

b. 11x+7y=258

c. 6x+9y=658

d. 7x+9y=328

e. none is true

15. Questions

Quantity I: The average weight of boys and girls of class B is 68 kg and 52 kg respectively. Find the total weight of class B.

Quantity II: The average weight for boys and girls of class A is 72 kg and 64 kg respectively. Find the total weight of class A.

a. Quantity: I < Quantity: II

b. Quantity: I ≥ Quantity: II

c. Quantity: II ≥ Quantity: I

d. Quantity: I > Quantity: II

e. Quantity I = Quantity II or relation can't be established

16. Questions

Point Q is a point between points P and R. Boat A starts from P towards Q with the stream, and Boat B starts from R towards Q against the stream at 8 a.m. The ratio of the speed of boats A and

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B in still water is 6:5, and the speed of the stream is 4 km/hr. Boat A return to P at 9.12 a.m., which is 22 minutes more than the time taken by boat B to return to R. Find the difference between the speed of the boat A and B, if the distance of PQ is 14 km and the distance of QR is 8 km.

a. 5

b. 6

c. 4

d. 4.5

e. 6.5

17. Questions

The curved surface area of a cylinder is 2376 m^2. The radius of the cylinder is equal to the side of the square, whose perimeter is 84 m. The radius of a cone is equal to the radius of cylinder, and the height of the cone is 2/3rd of the height of the cylinder. Find the difference between the volume of the cone and the volume of a cylinder.

a. 15842 m^3

b. 15640 m^3

c. 18404 m^3

d. 19404 m^3

e. 16404 m^3

18. Questions

The ratio of the amount invested in schemes A and B is 5:3 respectively. Scheme A provides simple interest at the rate of 25% per annum, while scheme B provides interest at the rate of 20% compounded annually. The difference between the interest amounts after 2 years is Rs. 1416. If the total amount is invested in scheme C, which provides interest at the rate of 30% compounded annually, then find the interest amount received from scheme C after 2 years.

a. Rs.6542

b. Rs.6624

c. Rs.8464

d. Rs.8462

e. None of these

19. Questions

In a container, there is 140 liters of milk. 'Y' litres of milk is taken out and the same amount of

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water is added then the ratio of milk and water in the mixture becomes 5:2 respectively. Again Y litres of mixture is taken out and same amount of water is added, now the ratio of milk and water in the mixture becomes m:n respectively. Find the value 7Y+8m+9n.

a. 696

b. 672

c. 854

d. 884

e. None of these

20. Questions

A and C can complete a work in 40/3 days, A and B can complete the work in 12 days, and B and C can complete the work in 120/7 days. X's efficiency is twice as that of A, and Y's efficiency is twice as that of C. X, Y, and Z can complete the work in 5 days. Find the number of days taken by Z alone to complete the work.

a. 20

b. 22

c. 30

d. 18

e. None of these

Explanations:

1. Questions

I). 2X+3Y+4Z= 231, **II).** 3X+5Y+4Z=296 **III).** 2X+Y+4Z=185

From I and II, 4Z=231-2X-3Y and 4Z=296-3X-5Y

So, 231-2X-3Y=296-3X-5Y

Or, X+2Y=65 ---(a)

From III, 2X+Y=185-231+2X+3Y

Or, 2Y=46, Y=23

X=65-46=19

4Z=296-3*19-5*23=124

Or, Z=124/4=31

So, M=31+4=35

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Shop	Number of Headphones sold	Number of Camera sold	Number of Chargers sold	Number of Laptops sold	Total number of items sold
A	5X=95	4Y=92	_____	3M=105	416
B	4Z=124	3Y=69	4X=76	____	339
C	Z+M=66	_____	88	5Y=115	___
D	_____	____	3Y=69	85	269
E	_____	X+M=54	_____	101	343
Total	439	_____	452	____	1776

The number of chargers sold in A = 416-95-92-105=124.

The number of laptops sold in B = 339-124-69-76=70.

The number of total items sold in C = 1776-343-269-339-416=409.

The number of cameras sold in C = 409-66-88-115=140.

The number of chargers sold in E = 452-69-88-124-76=95.

The number of headphones sold in E = 343-54-101-95=93.

The number of headphones sold in D = 439-95-124-66-93=61.

The number of cameras sold in D = 269-61-69-85=54.

The number of laptops sold in all shops together is

= 105+70+115+85+101=476

The number of cameras sold in all shops together is

= 1776-439-452-476=409.

The number of cameras sold in shop C is 409-54-54-69-92=140.

Shop	Number of Headphones sold	Number of Camera sold	Number of Chargers sold	Number of Laptops sold	Total number of items sold
A	5X=95	4Y=92	124	3M=105	416
B	4Z=124	3Y=69	4X=76	70	339
C	Z+M=66	140	88	5Y=115	409
D	61	54	3Y=69	85	269
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Answer: C

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I). The average number of chargers sold in shops B and C together is

=[76+88]=82.

II). The sum of headphones sold in shops A, C and E together is

=95+66+93=254.

III). The difference between the number of cameras sold in shops D and E is 54-54=0.

2. Questions

I). 2X+3Y+4Z= 231, **II).** 3X+5Y+4Z=296 **III).** 2X+Y+4Z=185

From I and II, 4Z=231-2X-3Y and 4Z=296-3X-5Y

So, 231-2X-3Y=296-3X-5Y

Or, X+2Y=65 ---(a)

From III, 2X+Y=185-231+2X+3Y

Or, 2Y=46, Y=23

X=65-46=19

4Z=296-3*19-5*23=124

Or, Z=124/4=31

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The number of headphones sold in D = 439-95-124-66-93=61.

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The number of cameras sold in D = 269-61-69-85=54.

The number of laptops sold in all shops together is

= 105+70+115+85+101=476

The number of cameras sold in all shops together is

= 1776-439-452-476=409.

The number of cameras sold in shop C is 409-54-54-69-92=140.

Shop	Number of Headphones sold	Number of Camera sold	Number of Chargers sold	Number of Laptops sold	Total number of items sold
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Answer: D

The number of watches sold in D is 85*7/5=119

The number of watches sold in C is 115*8/5=184

So, the difference is = 184-119=65

3X + 8= 3 * 19 + 8 = 65

3Y - 4= 3 * 23 – 4 = 65

2Z + 3 = 2 * 31 + 3 = 65

3. Questions

I). 2X+3Y+4Z= 231, **II).** 3X+5Y+4Z=296 **III).** 2X+Y+4Z=185

From I and II, 4Z=231-2X-3Y and 4Z=296-3X-5Y

So, 231-2X-3Y=296-3X-5Y

Or, X+2Y=65 ---(a)

From III, 2X+Y=185-231+2X+3Y

Or, 2Y=46, Y=23

X=65-46=19

4Z=296-3*19-5*23=124

Or, Z=124/4=31

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So, M=31+4=35

Shop	Number of Headphones sold	Number of Camera sold	Number of Chargers sold	Number of Laptops sold	Total number of items sold
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Answer: A

The number of headphones sold in F is = 6*19*2-93=135

The number of cameras sold in F is = 3*31*2-69=117

So, the total number of laptops and chargers sold in shop F is

= [16*31*2-343]-117-135=397=4M+6Y+3Z+26

4. Questions

I). 2X+3Y+4Z= 231, **II).** 3X+5Y+4Z=296 **III).** 2X+Y+4Z=185

From I and II, 4Z=231-2X-3Y and 4Z=296-3X-5Y

So, 231-2X-3Y=296-3X-5Y

Or, X+2Y=65 ---(a)

From III, 2X+Y=185-231+2X+3Y

Or, 2Y=46, Y=23

X=65-46=19

4Z=296-3*19-5*23=124

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So, M=31+4=35

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Answer: A

So, N=140-66=74

And P=124-105=19

So, only I and II are true

5. Questions

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From I and II, 4Z=231-2X-3Y and 4Z=296-3X-5Y

So, 231-2X-3Y=296-3X-5Y

Or, X+2Y=65 ---(a)

From III, 2X+Y=185-231+2X+3Y

Or, 2Y=46, Y=23

X=65-46=19

4Z=296-3*19-5*23=124

Or, Z=124/4=31

So, M=31+4=35

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E	93	X+M=54	95	101	343
Total	439	409	452	476	1776

Answer: A

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Required difference = 93*2/3 – 66*5/6=7

6. Questions

The selling price of item E is = 99*166.66/100=165

So, the total cost and selling price of item E is = 99+165=264

So, the total cost and selling price of all items is = [264/24]*100=1100

So, the total cost and selling price of item A is = 1100*18/100=198

The profit percentage of item A is 20%

So, the ratio of cost and selling price is 5:6.

So, the cost price of item A is 198*5/11=90

The selling price of item A is 198*6/11=108

Similarly, we can calculate the others' values also.

Item	Cost price	Selling price
A	90	108
B	33	99
C	100	120
D	130	156
E	99	165

Answer: C

For the value I,

The cost price of F = 90*11/9=110

So, selling price is = 110*120/100=132

So, it's false

Similarly, we can calculate others' values also.

7. Questions

The selling price of item E is = 99*166.66/100=165

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Item	Cost price	Selling price
A	90	108
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C	100	120
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E	99	165

Answer: D

So, the marked price of C is = 100*160/100=160

So, x=[160-120]*100/160=25%

Marked price of D is = 130*150/100=195

So, y=[195-156]*100/195=20%

So, the sum of x and y is 45

8. Questions

The selling price of item E is = 99*166.66/100=165

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So, the total cost and selling price of all items is = [264/24]*100=1100

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Similarly, we can calculate the others' values also.

Item	Cost price	Selling price
A	90	108
B	33	99
C	100	120
D	130	156
E	99	165

Answer: B

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So, m=120-108=12

And n=130-99=31

I). The Selling price of item E is 12m+21.

II). The selling price of item C is 4n-4.

III). The selling price of item A is 9m.

9. Questions

The selling price of item E is = 99*166.66/100=165

So, the total cost and selling price of item E is = 99+165=264

So, the total cost and selling price of all items is = [264/24]*100=1100

So, the total cost and selling price of item A is = 1100*18/100=198

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So, the cost price of item A is 198*5/11=90

The selling price of item A is 198*6/11=108

Similarly, we can calculate the others' values also.

Item	Cost price	Selling price
A	90	108
B	33	99
C	100	120
D	130	156
E	99	165

Answer: A

New selling price = 156+32.5=188.5

So, new profit percentage = [188.5-130]*100/130=45%

So, x=45-5=40%

So, the selling price of B = 33*140/100=46.2

10. Questions

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The selling price of item E is = 99*166.66/100=165

So, the total cost and selling price of item E is = 99+165=264

So, the total cost and selling price of all items is = [264/24]*100=1100

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Similarly, we can calculate the others' values also.

Item	Cost price	Selling price
A	90	108
B	33	99
C	100	120
D	130	156
E	99	165

Answer: A

So, x=108-90=18

And y=99-33=66

So, the cost price of G is = 18+66=84

So, selling price is = 84*110/100=92.4

Required answer = 120-92.4 = 27.6

11. Questions

Let, the number of boys and girls in class D be 11x and 7x.

The number of boys and girls in class C is 15y and 8y.

The number of girls in class B is 13y.

The number of boys in class E is 11y=the number of girls in class E.

The number of boys in class A is 11x-21.

The number of girls in class A is [65-11x+21] = 86-11x.

The number of boys in class B is [86-11x]/2.

So, 86-11x+13y+8y+7x+11y=166

Or, 32y-4x=80 (I)

[86-11x]/2 + 15y = 66

Or, 86-11x+30y=132

Or, 30y-11x=46 (II)

By solving (I) and (II), we get x = 4 and y = 3.

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The number of boys and girls in class D is 11*4=44 and 7*4=28, respectively.

The number of boys and girls in class C is 15*3=45 and 8*3=24, respectively.

The number of girls in class B is 13* 3 = 39.

The number of boys in class E is 11 * 3 = 33. Number of girls in class E.

The number of boys in class A is 11x-21=23.

The number of girls in class A is =86-11x=86-44=42.

The number of boys in class B is [86–11x]/2=42/2=21

Answer: C

The total number of students in class G is 60

The total number of students in class F is 21*3=63

So, the required difference

63*4/7+ 60*3/5 – 63*3/7-60*2/5 =[36+36]-[27+24]=21

12. Questions

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The number of boys in class A is 11x-21=23.

The number of girls in class A is =86-11x=86-44=42.

The number of boys in class B is [86–11x]/2=42/2=21

Answer: D

The average age of all the students of class A is

=[24*23+42*22]/65=22.70

13. Questions

Let, the number of boys and girls in class D be 11x and 7x.

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The number of boys in class A is 11x-21=23.

The number of girls in class A is =86-11x=86-44=42.

The number of boys in class B is [86–11x]/2=42/2=21

Answer: B

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The number of boys who qualify exam from D is = 21-45*20/100=21-9=12

So, the number of girls who did not qualify for the exam from C is 24/4*3 = 18

So, the number of girls who qualify exam from D is = 72*50/100-12=24

So, the number of girls who did not qualify for the exam from D is 28-24=4

The number of girls who did not qualify for the exams from C and D = 18+4 = 22

14. Questions

Let, the number of boys and girls in class D be 11x and 7x.

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The number of girls in class B is 13y.

The number of boys in class E is 11y=the number of girls in class E.

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[86-11x]/2 + 15y = 66

Or, 86-11x+30y=132

Or, 30y-11x=46 (II)

By solving (I) and (II), we get x = 4 and y = 3.

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The number of boys in class E is 11 * 3 = 33. Number of girls in class E.

The number of boys in class A is 11x-21=23.

The number of girls in class A is =86-11x=86-44=42.

The number of boys in class B is [86–11x]/2=42/2=21

Answer: E

So, x=44-28=16 and y=45-24=21

15. Questions

Let, the number of boys and girls in class D be 11x and 7x.

The number of boys and girls in class C is 15y and 8y.

The number of girls in class B is 13y.

The number of boys in class E is 11y=the number of girls in class E.

The number of boys in class A is 11x-21.

The number of girls in class A is [65-11x+21] = 86-11x.

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The number of boys in class B is [86-11x]/2.

So, 86-11x+13y+8y+7x+11y=166

Or, 32y-4x=80 (I)

[86-11x]/2 + 15y = 66

Or, 86-11x+30y=132

Or, 30y-11x=46 (II)

By solving (I) and (II), we get x = 4 and y = 3.

The number of boys and girls in class D is 11*4=44 and 7*4=28, respectively.

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The number of boys in class E is 11 * 3 = 33. Number of girls in class E.

The number of boys in class A is 11x-21=23.

The number of girls in class A is =86-11x=86-44=42.

The number of boys in class B is [86–11x]/2=42/2=21

Answer: A

Quantity I

Required weight =68*21+39*52=3456

Quantity II

Required weight = 23*72+42*64=4344

Quantity I < Quantity II

16. Questions

Answer: C

Total time taken by A is 9.12-8=1.12 hours = 72 minutes

The total time taken by B is 72-22=50 minutes.

Speed of boats A and B are 6x and 5x respectively.

So, [8/(5x-4)]+[8/(5x+4)]=50/60

Or, $480x=5*25x^2-80$

Or, $125x^2-480x-80=0$

By solving the above equation, we x=4

So, the difference between the speed of boats A and B is 6x-5x=x=4

17. Questions

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Answer: D

The curved surface area of a cylinder is 2*22/7*r*h

So, r= radius and h= height.

Side of square 84/4=21

So.2*22/7*r*h=2376

So, h=2376*7/[2*22*21]=18

So, the height of the cone is 18*2/3=12

Volume of cylinder = 22/7*21*21*18=24948 m^3

Volume of cone = 1/3*22/7*21*21*12=5544 m^3

So, difference = 24948-5544=19404 m^3

18. Questions

Answer: B

Let amount invested in A is 5x and amount invested in B is 3x.

So, 5x*25*2/100 – 3x*44/100=1416

[20% CI for 2 years equal to 44%].

So, 118x=141600

Or, x=141600/118=1200

So, the total amount invested is 8*1200=9600

So, required interest amount is = $9600*[1+30/100]^2-9600$

= 1.69*9600-9600=16224-9600=6624

19. Questions

Answer: A

So, We can say,

(140-Y)/Y=5/2

Or, 280-2Y=5Y

Or, 7Y=280

Or, Y=280/7=40

So, the final ratio of milk and water is [100-40*5/7]:[40-40*2/7+40]

=500:480=25:24=m:n

So, m=25 and n=24.

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7*40+8*25+9*24=696

20. Questions

Answer: A

One day's work for A and C is 3/40.

One day's work for A and B is 1/12.

One day's work for B and C is 7/120.

One day's work of A, B, and C together is

1/2[3/40+1/12+7/120]=[(9+10+7)]/120=13/120

So, one day's work for B is 13/120-3/40=4/120=1/30.

So, B can complete the work in 30 days.

Similarly, A and C can complete the work alone in 20 and 40 days.

Total work = LCM of 30, 20, and 40 = 120

The efficiency of A, B, and C is 6, 4, and 3, respectively.

The efficiency of X is 6*2=12.

The efficiency of Y is 3*2=6.

The total efficiency of X, Y, and Z is 120/5=24.

So, the efficiency of Z is 24-12-6=6.

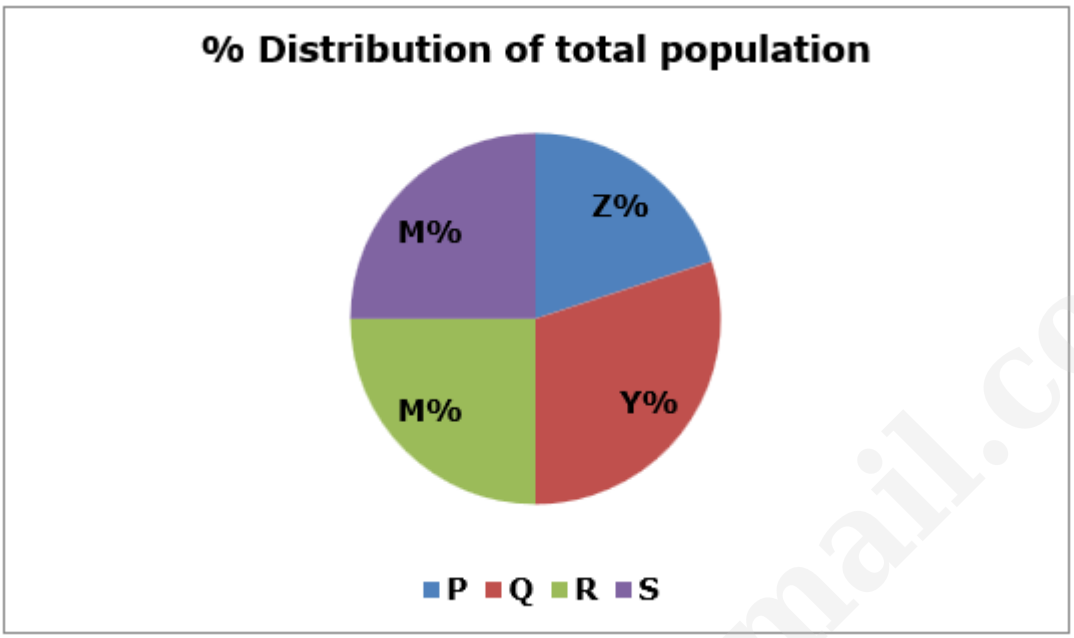
So, Z completed the work in 120/6=20 days.

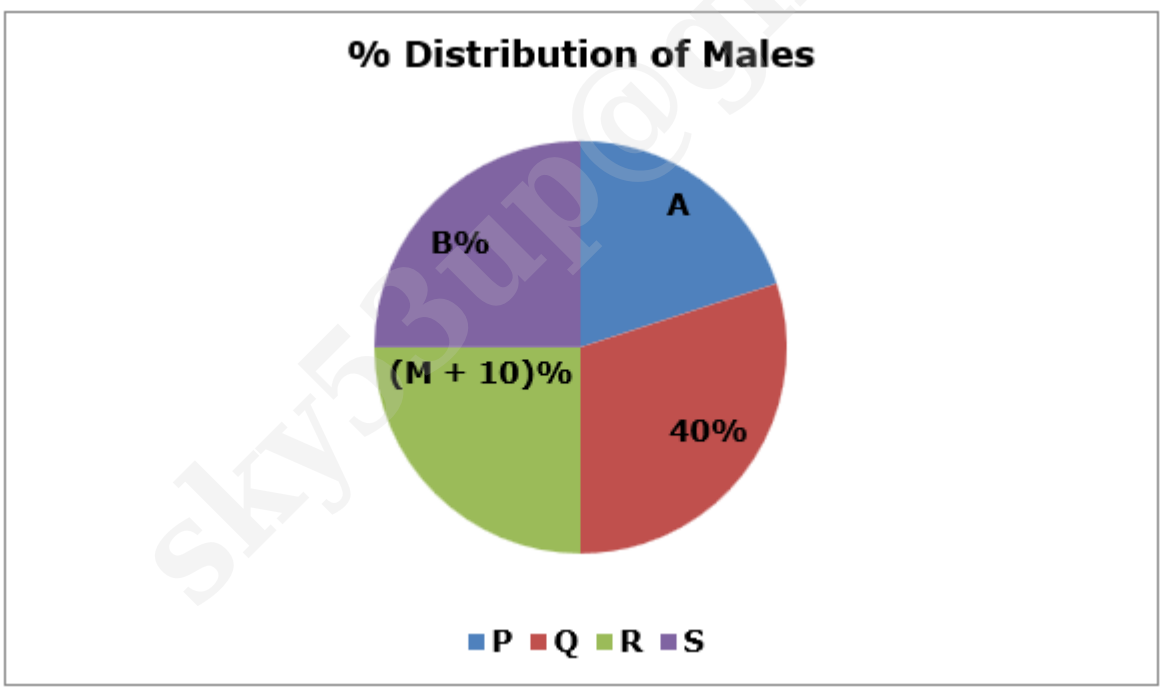
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1. Questions

Read the following information carefully and answer the questions based on it.

The charts given below show the % distribution of the total number of people in four different villages – P, Q, R, and S, and the % distribution of the number of males in these villages.





Notes:

a). The number of males in all cities together is 50% more than that of females.

b). The total population of Village P and Q together is the same as that of R and S together, and this statement is true for the male population as well.

c). If Z is expressed in degrees, it is equal to A, and the female population of P and Q together is 240.

The ratio of literate to illiterate males in village P is 3:1, and average of literate male in P and

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literate females in Q is 47. The number of illiterate females in village R is (5B – 5)% less than literate females in village Q. The number of illiterate males in village R is 148 more than that of literate males. Find literate population of village R.

a. 98

b. 84

c. 64

d. 88

e. None of these

2. Questions

The respective ratio of females in village R and males in village S is C^2: D. Find the value of (C^4/D^2).

a. 16/81

b. 16/25

c. 4/81

d. None of these

3. Questions

The average number of males in village R and T together is the same as the number of males in village Q, while the number of females in village T is M% more than that of village S. Find the total population of village P and T together.

a. 768

b. 824

c. 792

d. 804

e. None of these

4. Questions

The difference between males and females in village S is C, and the number of females in village Q is (D/2) % of that of males. Find the difference between the difference of C and D, and the value of 2B?

a. 14

b. 4

c. 16

d. 6

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e. None of these

5. Questions

Read the following information carefully and answer the questions based on it.

There are three schools – P, Q, and R. The data given below gives information about the number of students and the number of boys and girls in each school.

- The number of boys in school P is 3Z, which is 40% less than that in school Q.
- The total number of students in school Q is 200, and the difference between the number of students in school Q and R is M, which is twice the difference between the number of students in P and Q.
- The number of girls in school R is 4Y, which is 20% less than that in school Q, and the number of boys in school Q is the same as the total number of students in school P.
- Both Z and Y are exactly divisible by 10, such that Z > Y.

In another school S, the number of boys is 66.66% more than that in school R, and the number of girls in school S is 40% less than that of M. The number of girls in school S is how much % of the total number of students in school S?

a. 37.5%

b. 32.5%

c. 40%

d. 25%

e. Cannot be determined

6. Questions

The difference between the boys and girls in school P is 3A/2, while the difference between the boys and girls in school R is B, then A ____ B.

a. A > B

b. A < B

c. A = B or NO relation

d. A ≤ B

e. A ≥ B

7. Questions

Out of the total no. of girls in all given schools together, each girl likes at least one of the subjects among – A, B, and C. The No. of girls who like only A and only C is the same, which is thrice the no. of girls those who like only B. The no. of girls who like exactly two subjects is 65, and the difference between those who like exactly two subjects and those who like exactly one subject is 5% of M. The no. of girls who like both B and C, but not A (Same as those who like only C) is how much % more/less than those who like all three subjects?

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a. 100%

b. 60%

c. 80%

d. 66.66%

e. None of these

8. Questions

The number of students in school Q > The number of students in school R. Which of the following is/are true.

I). The difference between the boys in school P and Q is the same as the number of boys in school R.

II). M is a perfect square.

III). If the total number of students is represented in a pie chart, then the angle in degrees corresponding to the number of students in school R is (M – 20).

a. I and II only

b. I, II, and III

c. II only

d. I and III only

e. None of these

9. Questions

Read the following information carefully and answer the questions based on it.

There are five companies – P, Q, R, S, and T. Each company has three departments – A, B, and C.

Name of Company	Employees in A: Employees in B	% of employees in department C
P	-	M%
Q	7: Z	-
R	Y^2:5	20%
S	-	N%
T	3: Y^2	30%

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Note:

a). Employees in department B of company P are 25Z, which is 25% more than the employees in the same department of company R.

b). The difference between the total number of employees in company R and T is half the total number of employees in company S (480).

c). M – N = 5, and Y is a prime number. The number of employees in department B of company R is more than that in department A.

d). Employees in department B of company T are 50% more than those in the same department of company R.

For company Q, the total number of employees is K, and the number of employees in department B is [2(Z + Y)] % of the total number of employees in the company. Find the number of employees in department C of company Q.

a. 67.5%

b. 60%

c. 62.5%

d. 64%

e. Cannot be determined

10. Questions

The ratio of the number of employees in department A to B of company P is 7:2, and the number of employees in department A of the same company is 400 more than that of department C. Find the number of employees in department C of company S.

a. 120

b. 96

c. 72

d. 144

e. None of these

11. Questions

Twice the number of employees in department C of company R is L, while the number of employees in department B of company T is H. L is how much % more or less than that of H.

a. 40%

b. 30%

c. 25%

d. 37.5%

e. None of these

12. Questions

For company S, the number of employees in department B is 120, which is 9Z less than the number of employees in department A. Find the value of M?

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a. 45

b. 40

c. 35

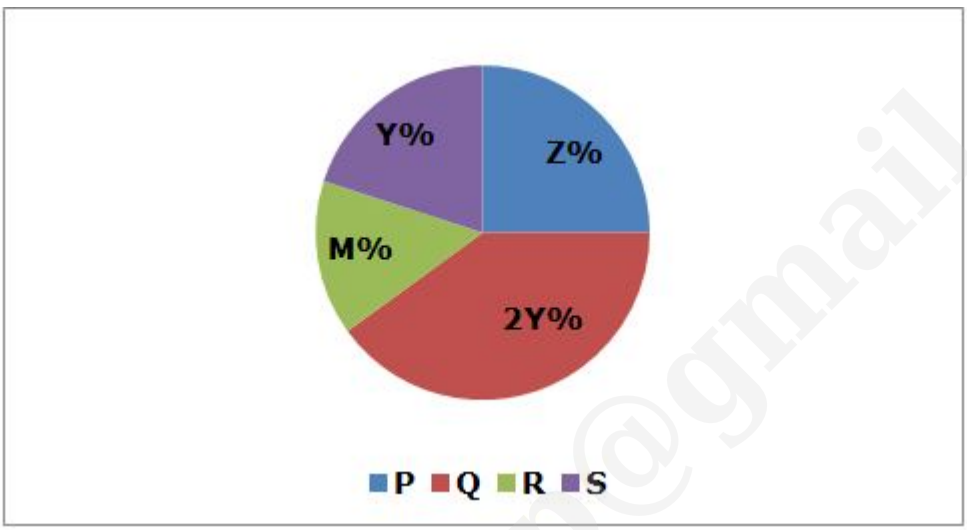
d. 30

e. None of these

13. Questions

Read the following information carefully and answer the questions based on it.

The pie chart given below shows the % or degree distribution of money invested by four people – P, Q, R, and S in scheme A, which offers simple interest.



Name	Rate of interest (%)	Time period of investment (in years)	Interest: Amount
P	20%	-	3:8
Q	-	5	9:19
R	25%	4	-
S	-	8	-

Note: Difference between money invested by Q and S is 16000.

After a given time, "R" withdraws the whole amount from scheme A and invests 2/3rd in scheme B, earning compound interest at 40% per annum, while the rest of the money is kept with him. The total amount with R after seven years (4years in Scheme A + 3 years in scheme B) is 51904. Find the value of 2Z if the money invested by R in scheme B is Rs. 4000 less than the money invested by P in scheme A.

a. 50

b. 30

c. 40

d. 60

e. None of these

14. Questions

The time period of investment of P is N years, and the money invested by T in scheme A is at (N + 2) % per annum. Find the interest earned by T if the value of Y is 2^3, and the ratio of money invested by P and R in scheme A is in the ratio 25:13 respectively. The time period of the investment of T is the same as that of S, and the investment of R is the same as that of T.

a. Rs. 22400

b. Rs. 19200

c. Rs. 20800

d. Rs. 24000

e. None of these

15. Questions

The interest earned by S from scheme A is 2/3 times the interest earned by Q from the same scheme. Find the compound interest earned by S after three years at the same rate of interest and on the same investment (as that of scheme A).

a. Rs. 8634

b. Rs. 11848

c. Rs. 11648

d. Rs. 8334

e. None of these

16. Questions

The investment of Q is 40% of the investment of all given people together in scheme A, and the difference between the investment of P and R is the same as the initial investment of S.

Quantity I: Find N times the initial investment of S, if N is the time of investment of P.

Quantity II: Find the amount received by R from scheme A.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II or no relation

d. Quantity I ≥ Quantity II

e. Quantity I ≤ Quantity II

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17. Questions

The investment of Q is the same as the amount (principal + interest) received by R from scheme A, and the investment of P is 50% of that of R. The investment of R in Scheme A is 12000. Find the value of N, if Z% = N^0 (degrees).

a. 40

b. 80

c. 60

d. 45

e. None of these

18. Questions

Quantity I: P, Q, and R entered into a partnership with the initial investments of Rs. 32000, Rs. 24000 and Rs. 40000 respectively. Q received 25% of the total profit at the end of the year for managing the business, and the rest of the profit was divided in the ratio of their capital. If the difference between the total profit of Q and R is Rs. 13500, then find the profit share of P.

Quantity II: P, Q, and R entered into a partnership. At the end of the year, the profit share of Q is twice that of R, while the profit share of P is the same as the total profit of Q and R together. If the total profit increased by 20%, then the profit share of P is Rs. 16200. Find the initial total profit of the business.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II

d. Quantity I ≤ Quantity II

e. Quantity I ≥ Quantity II

19. Questions

P and Q together can do a job in 40 days. If P worked at 40% less efficiency and Q worked at 50% more efficiency, then the work would finish in the same time as when they worked at their original efficiencies.

Quantity I: Find the time taken by the slower person (among P and Q) to complete 80% of the job.

Quantity II: If 6 people started the work (each is as efficient as P), and after one day one person left. Find the % of work left when all persons leave.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II

d. Quantity I ≤ Quantity II

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e. Quantity I ≥ Quantity II

20. Questions

A vessel contains M ml mixture of milk and water in the ratio of (A + 2): (B – 1). 25% of the mixture is removed and then 0.15M ml of water is added. Now, the amount of milk in the mixture becomes twice that of water.

Quantity I: Value of (4B – A) is Y/2, find the value of Y

Quantity II: % of water in initial mixture is Z%, find the value of (Z/2).

a. Quantity I < Quantity II

b. Quantity I = Quantity II

c. Quantity I ≤ Quantity II

d. Quantity I ≥ Quantity II

Explanations:

1. Questions

Total population of villages P and Q together = total population of villages R and S together

Total male population of villages P and Q together = Total male population of villages R and S together

So, it is certain that the total female population of villages (P + Q) = Total female population of villages (R + S)

So, total female population = 240 x 2 = 480

So, total male population = 480 x 3/2 = 720

Total population of all villages together = 480 + 720 = 1200

Male population of village Q = 40% x 720 = 288

So, male population of village P = 10% x 720 = 72

So, value of Z% in degrees = 72^0

So, Z in % = 20%

Value of M = 50/2 = 25%

Value of Y = 100 – 50 – 20 = 30%

Value of B = 50 – 25 – 10 = 15%

Now, we can find all the related data.

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Name of village	Total population	Number of males	Number of females
P	240	72	168
Q	360	288	72
R	300	252	48
S	300	108	192
Total	1200	720	480

Answer: D

According to the question,

Number of literate males in village P = 3/4 x 72 = 54

So, number of literate females in village Q = 47 x 2 – 54 = 40

So, illiterate females in R = 30% x 40 = 12

Literate females in R = 48 – 12 = 36

Number of literate males in R = (252 – 148)/2 = 52

So, literate population of village R = 52 + 36 = 88

Hence answer is option D

2. Questions

Total population of villages P and Q together = total population of villages R and S together

Total male population of villages P and Q together = Total male population of villages R and S together

So, it is certain that the total female population of villages (P + Q) = Total female population of villages (R + S)

So, total female population = 240 x 2 = 480

So, total male population = 480 x 3/2 = 720

Total population of all villages together = 480 + 720 = 1200

Male population of village Q = 40% x 720 = 288

So, male population of village P = 10% x 720 = 72

So, value of Z% in degrees = 72^0

So, Z in % = 20%

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Value of M = 50/2 = 25%

Value of Y = 100 – 50 – 20 = 30%

Value of B = 50 – 25 – 10 = 15%

Now, we can find all the related data.

Name of village	Total population	Number of males	Number of females
P	240	72	168
Q	360	288	72
R	300	252	48
S	300	108	192
Total	1200	720	480

Answer: A

Number of females in village R = 48

Number of males in village S = 108

Required ratio = 48/108 = 4/9

$C^2/D = 4/9$

So, $C^4/D = 16/81$

Hence answer is option A

3. Questions

Total population of villages P and Q together = total population of villages R and S together

Total male population of villages P and Q together = Total male population of villages R and S together

So, it is certain that the total female population of villages (P + Q) = Total female population of villages (R + S)

So, total female population = 240 x 2 = 480

So, total male population = 480 x 3/2 = 720

Total population of all villages together = 480 + 720 = 1200

Male population of village Q = 40% x 720 = 288

So, male population of village P = 10% x 720 = 72

So, value of Z% in degrees = 72^0

So, Z in % = 20%

Value of M = 50/2 = 25%

Value of Y = 100 – 50 – 20 = 30%

Value of B = 50 – 25 – 10 = 15%

Now, we can find all the related data.

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Name of village	Total population	Number of males	Number of females
P	240	72	168
Q	360	288	72
R	300	252	48
S	300	108	192
Total	1200	720	480

Answer: D

Number of males in village T = 288 x 2 – 252 = 324

Number of females in village T = 192 x 125% = 240

Total population of village T = 324 + 240 = 564

Total population of village P = 240

Required sum = 240 + 564 = 804

Hence answer is option D

4. Questions

Total population of villages P and Q together = total population of villages R and S together

Total male population of villages P and Q together = Total male population of villages R and S together

So, it is certain that the total female population of villages (P + Q) = Total female population of villages (R + S)

So, total female population = 240 x 2 = 480

So, total male population = 480 x 3/2 = 720

Total population of all villages together = 480 + 720 = 1200

Male population of village Q = 40% x 720 = 288

So, male population of village P = 10% x 720 = 72

So, value of Z% in degrees = 72^0

So, Z in % = 20%

Value of M = 50/2 = 25%

Value of Y = 100 – 50 – 20 = 30%

Value of B = 50 – 25 – 10 = 15%

Now, we can find all the related data.

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Name of village	Total population	Number of males	Number of females
P	240	72	168
Q	360	288	72
R	300	252	48
S	300	108	192
Total	1200	720	480

Answer: B

Value of C = 192 – 108 = 84

(D/2) % = 72/288 x 100 = 25%

So, value of D = 25 x 2 = 50

Value of 2B = 2 x 15 = 30

Required difference = (84 – 50) – 30 = 4

Hence answer is option B

5. Questions

Number of boys in school P = 3Z

So, number of boys in school Q = 3Z/60 x 100 = 5Z

Number of girls in school R = 4Y

So, number of girls in school Q = 4Y/80 x 100 = 5Y

Total number of students in school Q = 200

Now,

5Z + 5Y = 200

Z + Y = 40 …….... (1)

Both Z and Y are multiples of 10, such that Z > Y, so the only possible value of Y = 10

So, value of Z = 30

So, number of boys in school Q = 30 x 5 = 150

Number of girls in school Q = 5 x 10 = 50

Total number of students in school P = 150

Number of boys in school P = 3 x 30 = 90

So, Number of girls in school P = 150 – 90 = 60

Number of girls in school R = 4 x 10 = 40

Difference between the total number of students in school P and Q = 200 – 150 = 50

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So, difference between the number of students in school Q and R = M = 2 x 50 = 100

So, number of students in school R = 200 + 100 OR 200 – 100 = 300 or 100

So, number of boys in school R = 260 or 60

Now, we can find all the related data

Name of school	Total students	Number of boys	Number of girls
P	150	90	60
Q	200	150	50
R	300/100	260/60	40

Answer: A

According to question,

For school S

Number of boys = 5/3 x boys in school R

So, number of boys in school R must be multiple of 3, = 60

So, number of boys in school S = 5/3 x 60 = 100

Number of girls in school S = 60% x 100 = 60

So, required % = 60/160 x 100 = 37.5%

Hence answer is option A

6. Questions

Number of boys in school P = 3Z

So, number of boys in school Q = 3Z/60 x 100 = 5Z

Number of girls in school R = 4Y

So, number of girls in school Q = 4Y/80 x 100 = 5Y

Total number of students in school Q = 200

Now,

5Z + 5Y = 200

Z + Y = 40 …….... (1)

Both Z and Y are multiples of 10, such that Z > Y, so the only possible value of Y = 10

So, value of Z = 30

So, number of boys in school Q = 30 x 5 = 150

Number of girls in school Q = 5 x 10 = 50

Total number of students in school P = 150

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Number of boys in school P = 3 x 30 = 90

So, Number of girls in school P = 150 – 90 = 60

Number of girls in school R = 4 x 10 = 40

Difference between the total number of students in school P and Q = 200 – 150 = 50

So, difference between the number of students in school Q and R = M = 2 x 50 = 100

So, number of students in school R = 200 + 100 OR 200 – 100 = 300 or 100

So, number of boys in school R = 260 or 60

Now, we can find all the related data

Name of school	Total students	Number of boys	Number of girls
P	150	90	60
Q	200	150	50
R	300/100	260/60	40

Answer: D

Difference between the boys and girls in school P = 90 – 60 = 30

So, 3A/2 = 30

Value of A = 20

Value of B = (260 – 40) or (60 – 40) = 220 or 20

So, A ≤ B

Hence answer is option D

7. Questions

Number of boys in school P = 3Z

So, number of boys in school Q = 3Z/60 x 100 = 5Z

Number of girls in school R = 4Y

So, number of girls in school Q = 4Y/80 x 100 = 5Y

Total number of students in school Q = 200

Now,

5Z + 5Y = 200

Z + Y = 40 ……... (1)

Both Z and Y are multiples of 10, such that Z > Y, so the only possible value of Y = 10

So, value of Z = 30

So, number of boys in school Q = 30 x 5 = 150

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Number of girls in school Q = 5 x 10 = 50

Total number of students in school P = 150

Number of boys in school P = 3 x 30 = 90

So, Number of girls in school P = 150 – 90 = 60

Number of girls in school R = 4 x 10 = 40

Difference between the total number of students in school P and Q = 200 – 150 = 50

So, difference between the number of students in school Q and R = M = 2 x 50 = 100

So, number of students in school R = 200 + 100 OR 200 – 100 = 300 or 100

So, number of boys in school R = 260 or 60

Now, we can find all the related data

Name of school	Total students	Number of boys	Number of girls
P	150	90	60
Q	200	150	50
R	300/100	260/60	40

Answer: A

Let the number of girls who like Only A, Only B, and only C = 3a, a, 3a

Number of girls those who like exactly two = 65

So, number of girls who like exactly one subject is either 65 + 5 or 65 – 5 = 70 or 60

If (3a + a + 3a) = 60,

Then a = 60/7 (not possible)

So, 7a = 70,

and the value of a = 10

Thus, the number of girls who like only C = 3 x 10 = 30

The number of girls who like all three subjects = 150 – 70 – 65 = 15

Required % = (30 – 15)/15 x 100 = 100%

Hence answer is option A

8. Questions

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Number of boys in school P = 3Z

So, number of boys in school Q = 3Z/60 x 100 = 5Z

Number of girls in school R = 4Y

So, number of girls in school Q = 4Y/80 x 100 = 5Y

Total number of students in school Q = 200

Now,

5Z + 5Y = 200

Z + Y = 40 …….... (1)

Both Z and Y are multiples of 10, such that Z > Y, so the only possible value of Y = 10

So, value of Z = 30

So, number of boys in school Q = 30 x 5 = 150

Number of girls in school Q = 5 x 10 = 50

Total number of students in school P = 150

Number of boys in school P = 3 x 30 = 90

So, Number of girls in school P = 150 – 90 = 60

Number of girls in school R = 4 x 10 = 40

Difference between the total number of students in school P and Q = 200 – 150 = 50

So, difference between the number of students in school Q and R = M = 2 x 50 = 100

So, number of students in school R = 200 + 100 OR 200 – 100 = 300 or 100

So, number of boys in school R = 260 or 60

Now, we can find all the related data

Name of school	Total students	Number of boys	Number of girls
P	150	90	60
Q	200	150	50
R	300/100	260/60	40

Answer: B

I). Difference between the boys in school P and Q is the same as the number of boys in school R.

Difference between the boys in school P and R = 150 – 90 = 60

Number of boys in school R = 60

This statement is true.

II). M is a perfect square.

Value of M = 100 (10^2)

This statement is true.

III). If the total number of students is represented in a pie chart, then the angle in degrees corresponding to the number of students in school R is (M – 20).

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Required angle = 100/450 x 360 = 80 = (M – 20)

This statement is true.

Hence answer is option B

9. Questions

Ratio of the employees in department A to B of company R = Y^2:5

Y is a prime number, and $Y^2 < 5$. So only possible value of Y = 2.

So, ratio of the employees in department A to B of company R = 4:5

Number of employees in department B of company R = 5/9 x 80% x (total employees in company R) = 4/9 x (total employees in company R)

Number of employees in department B of company T = 4/7 x 70% x (total employees in company T) = 40% x total employees in T

Now,

40% x total employees in company T = 3/2 x 4/9 x (total employees in company R)

Total employees in company T / Total employees in company R = 5/3

Difference between the employees in company R and T = 240

So, total employees in company R = [240 / (5 – 3)] x 3 = 360

So, total employees in company T = 360 x 5/3 = 600

Employees in department B of company R = 5/9 x 80% x 360 = 160

So, 160 = 25Z x 4/5

So, value of Z = 8

Answer: C

According to question,

For company Q,

Number of employees in department B = 2(Z + Y) = 20% of total

Ratio of the employees in department A to B = 7:8 = 7:8

So, employees in department (A + B) = 15/8 x 20 = 37.5% of total

So, % of employees in department C = 100 – 37.5 = 62.5%

Hence answer is option C

10. Questions

Ratio of the employees in department A to B of company R = Y^2:5

Y is a prime number, and $Y^2 < 5$. So only possible value of Y = 2.

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So, ratio of the employees in department A to B of company R = 4:5

Number of employees in department B of company R = 5/9 x 80% x (total employees in company R) = 4/9 x (total employees in company R)

Number of employees in department B of company T = 4/7 x 70% x (total employees in company T) = 40% x total employees in T

Now,

40% x total employees in company T = 3/2 x 4/9 x (total employees in company R)

Total employees in company T / Total employees in company R = 5/3

Difference between the employees in company R and T = 240

So, total employees in company R = [240 / (5 – 3)] x 3 = 360

So, total employees in company T = 360 x 5/3 = 600

Employees in department B of company R = 5/9 x 80% x 360 = 160

So, 160 = 25Z x 4/5

So, value of Z = 8

Answer: B

For company P

Employees in department B = 25Z = 25 x 8 = 200

So, employees in department A = 7/2 x 200 = 700

So, employees in department C = 700- 400 = 300

So, M% = 300/1200 x 100 = 25%

So, value of N% = 25 – 5 = 20%

Total number of employees in company S = 480

Employees in department C of company S = 480 x 20% = 96

Hence answer is option B

11. Questions

Ratio of the employees in department A to B of company R = Y^2:5

Y is a prime number, and $Y^2 < 5$. So only possible value of Y = 2.

So, ratio of the employees in department A to B of company R = 4:5

Number of employees in department B of company R = 5/9 x 80% x (total employees in company R) = 4/9 x (total employees in company R)

Number of employees in department B of company T=4/7 x 70% x (total employees in company T)= 40% x total employees in T

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Now,

40% x total employees in company T = 3/2 x 4/9 x (total employees in company R)

Total employees in company T / Total employees in company R = 5/3

Difference between the employees in company R and T = 240

So, total employees in company R = [240 / (5 – 3)] x 3 = 360

So, total employees in company T = 360 x 5/3 = 600

Employees in department B of company R = 5/9 x 80% x 360 = 160

So, 160 = 25Z x 4/5

So, value of Z = 8

Answer: A

For value of L

Employees in department B of company R = 20 x 8 = 160

So, employees in department A of company R = 4/5 x 160 = 128

So, employees in department C of company R = (160 + 128)/80 x 20 = 72

So, value of L = 72 x 2 = 144

For value of H

Employees in department B of company T = 30Z = 30 x 8 = 240

Required % change = (240 – 144)/240 x 100 = 40%

Hence answer is option A

12. Questions

Ratio of the employees in department A to B of company R = Y^2:5

Y is a prime number, and $Y^2 < 5$. So only possible value of Y = 2.

So, ratio of the employees in department A to B of company R = 4:5

Number of employees in department B of company R = 5/9 x 80% x (total employees in company R) = 4/9 x (total employees in company R)

Number of employees in department B of company T=4/7 x 70% x (total employees in company T)= 40% x total employees in T

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Now,

40% x total employees in company T = 3/2 x 4/9 x (total employees in company R)

Total employees in company T / Total employees in company R = 5/3

Difference between the employees in company R and T = 240

So, total employees in company R = [240 / (5 – 3)] x 3 = 360

So, total employees in company T = 360 x 5/3 = 600

Employees in department B of company R = 5/9 x 80% x 360 = 160

So, 160 = 25Z x 4/5

So, value of Z = 8

Answer: B

For company S,

Total employees = 480

Employees in department B = 120

So, employees in department A = 120 + 9 x 8 = 192

So, employees in department C = 480 – 120 – 192 = 168

N% = 168/480 x 100 = 35%

So, value of M% = 35 + 5 = 40%

Hence answer is option B

13. Questions

For R,

Let Money invested in scheme A = 6a

Amount received from scheme A after four years = 6a x 25% x 4 = Rs. 12a

Money invested in scheme B = 2/3 x 12a = 8a

Now,

$1/3 \times 12a + 8a \times (1.4)^3 = 51904$

25.952a = 51904

So, value of a = 2000

So, money invested by R in scheme B = 8*2000= 16000

So, Money invested by P in scheme A = 16000+ 4000 = 20000

2Y% - Y% = 16000

Y% = 16000

Answer: A

For R,

Let Money invested in scheme A = 6a

Amount received from scheme A after four years = 6a x 25% x 4 = Rs. 12a

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Money invested in scheme B = 2/3 x 12a = 8a

Now,

$1/3 \times 12a + 8a \times (1.4)^3 = 51904$

25.952a = 51904

So, value of a = 2000

So, money invested by R in scheme B = 8*2000= 16000

So, Money invested by P in scheme A = 16000+ 4000 = 20000

2Y% - Y% = 16000

Y% = 16000

So, money invested by S = 16000

So, money invested by Q = 2 x 16000 = 32000

Money invested by R in scheme A = 12000

So, value of Z% = [20000 / (12000 + 20000 + 32000 + 16000)] x 100 = 25%

Required value = 2 x 25 = 50

Hence answer is option A

14. Questions

For R,

Let Money invested in scheme A = 6a

Amount received from scheme A after four years = 6a x 25% x 4 = Rs. 12a

Money invested in scheme B = 2/3 x 12a = 8a

Now,

$1/3 \times 12a + 8a \times (1.4)^3 = 51904$

25.952a = 51904

So, value of a = 2000

So, money invested by R in scheme B = 8*2000= 16000

So, Money invested by P in scheme A = 16000+ 4000 = 20000

2Y% - Y% = 16000

Y% = 16000

Answer: C

2Y% - Y% = 16000

Y% = 16000

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Money invested by S = Rs. 16000

So, total money invested = 16000/8 x 100 = Rs. 200000

Money invested by R = 12/37 x (100 – 8 – 16) % x 200000 = Rs. 52000

Money invested by T = Rs. 52000

Time period of investment of T = S = 8 years

Value of N = ([3 / (8 – 3)] x 100) / 20% = 3

So, rate of interest for T = (3 + 2) % = 5%

Required interest = 52000 x 5% x 8 = Rs. 20800

Hence answer is option C

15. Questions

For R,

Let Money invested in scheme A = 6a

Amount received from scheme A after four years = 6a x 25% x 4 = Rs. 12a

Money invested in scheme B = 2/3 x 12a = 8a

Now,

$1/3 \times 12a + 8a \times (1.4)^3 = 51904$

25.952a = 51904

So, value of a = 2000

So, money invested by R in scheme B = 8*2000= 16000

So, Money invested by P in scheme A = 16000+ 4000 = 20000

2Y% - Y% = 16000

Y% = 16000

Answer: D

Interest earned by Q from scheme A = [9/ (19 – 9)] x 100 = 90%

Interest earned by S = 2/3 x 90% x 32000 = Rs. 19200

Rate of interest for S = (19200/16000 x 8) x 100 = 15%

Required compound interest = $16000 \times [(23/20)^3 - 1]$ = Rs. 8334

Hence answer is option D

16. Questions

For R,

Let Money invested in scheme A = 6a

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Amount received from scheme A after four years = 6a x 25% x 4 = Rs. 12a

Money invested in scheme B = 2/3 x 12a = 8a

Now,

$1/3 \times 12a + 8a \times (1.4)^3 = 51904$

25.952a = 51904

So, value of a = 2000

So, money invested by R in scheme B = 8*2000= 16000

So, Money invested by P in scheme A = 16000+ 4000 = 20000

2Y% - Y% = 16000

Y% = 16000

Answer: D

Quantity I.

16000 x 3 = Rs. 48000

Quantity II

Total investment = 32000/40 x 100 = 80000

Investment of (P + R) = 80000 – 48000 = 32000

Difference between investment of P and R = 16000

So, investment of R = (32000 – 16000)/2 OR (32000 + 16000)/2 = 8000 or 24000.

Amount received by R = investment + 25% x 4 x investment = 2 x investment = 16000 or 48000

So, Quantity I ≥ Quantity II

Hence answer is option D

17. Questions

For R,

Let Money invested in scheme A = 6a

Amount received from scheme A after four years = 6a x 25% x 4 = Rs. 12a

Money invested in scheme B = 2/3 x 12a = 8a

Now,

$1/3 \times 12a + 8a \times (1.4)^3 = 51904$

25.952a = 51904

So, value of a = 2000

So, money invested by R in scheme B = 8*2000= 16000

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So, Money invested by P in scheme A = 16000+ 4000 = 20000

2Y% - Y% = 16000

Y% = 16000

Answer: A

Ratio of the investment of P and R = 1:2

So, investment of Q = Amount of R from scheme A = Rs. 24000

So, initial investment of P = 12000/2 = 6000

Total investment = 6000 + 24000 + 12000+12000 = 54000

Required values in degrees = 6000/54000 x 360 = 40^0

Hence answer is option A

18. Questions

Answer: C

Quantity I.

Ratio of investment = 32000:24000:40000 = 4:3:5

Time period of investment is same, so the ratio of profit is the same as the ratio of initial investment.

Profit share of Q = 25% + 3/12 x 75% = 43.75%

Profit share of R = 5/12 x 75% = 31.25%

Now,

(43.75% - 31.25%) of total profit = 13500

So, 12.5% of total profit = 13500

So, total profit = 13500/12.5 x 100 = Rs. 108000

Profit share of P = 4/12 x 75% x 108000 = Rs. 27000

Quantity II.

Ratio of profit share of Q and R = 2:1

Ratio of profit share of P, Q, and R = 3:2:1

Now,

Total profit after increased = 120/100 * 6a = 7.2a

3/6 x 7.2a = 16200

a = 4500

So, total initial profit = 4500 * 6 = Rs. 27000

So, Quantity I = Quantity II

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Hence answer is option C

19. Questions

Answer: A

According to the question,

(P + Q) x 40 = (60% x P + 150% x Q) x 40

P/Q = 5/4

Total work units = 9 x 40 = 360 units

Quantity I.

Required time = 80% x (360/4) = 72 days

Quantity II.

Amount of work done = 30 + 25 + 20 + 15 + 10 + 5 = 105 units

Required % = (360 – 105)/360 x 100 = 70.83%

So, quantity I > Quantity II

Hence answer is option A

20. Questions

Answer: A

Let initial quantity of milk and water in mixture = a and b

Now,

75% x a = 2 x [75% x b + 15% x (a + b)]

45% x a = 180% x b

So, (a/b) = 4/1

So, initial ratio of milk and water = 4:1

Now,

(A + 2) / (B – 1) = 4/1

A + 2 = 4B – 4

4B – A = 6……. (1)

Quantity I

Value of Y/2 = 6

Value of Y = 12

Quantity II

% of water initially = 1/5 x 100 = 20%

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So, required value = 20/2 = 10%

Value of Z = 10

So, Quantity I > Quantity II

Hence answer is option A

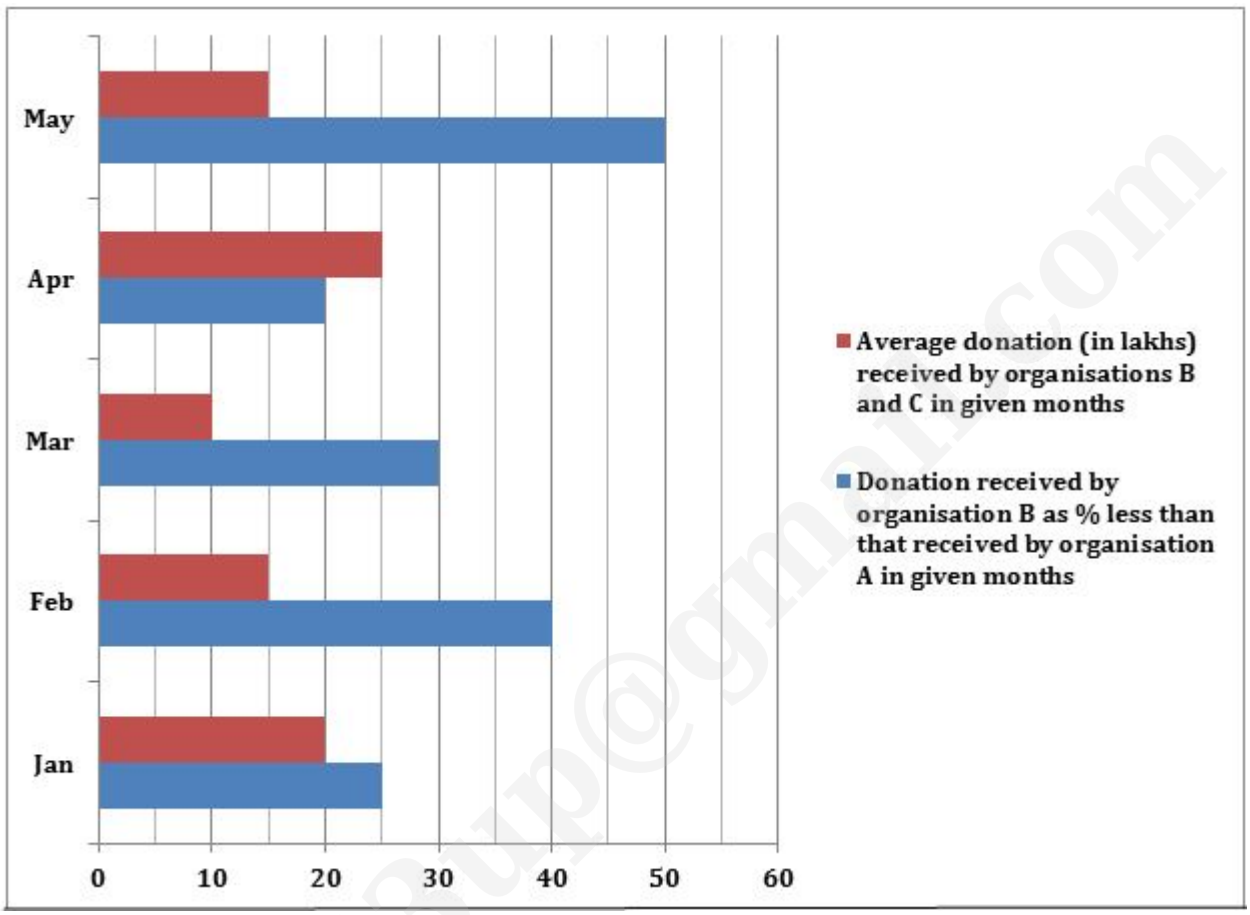
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1. Questions

Study the given data carefully and answer the following questions:

The data given below is related to the donations received by three different organisations A, B, and C in five months January, February, March, April, and May.

The bar graph given below shows the following data:



Notes:

1). Donations received by organisation A in the given months are ₹10 lakhs, ₹16 lakhs, ₹24 lakhs, ₹25 lakhs and ₹40 lakhs, but not necessarily in the same order.

2). Organisation A received the maximum donations in April, and the donations received in May are a multiple of 4 but not less than ₹20 lakhs.

3). The difference between the donations received by organisation A in February and May is ₹1 lakh.

4). The difference between the donations received by organisation A in March and May is a multiple of 2 but not less than 10.

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If the ratio of the total donations received by organisations A, B, and C together in March to that received in April is M: N, then find the ratio of the donations received by organisation C in February to that received in May?

a. 5M: 3N

b. (2N – M): 2N

c. (M + 1): N

d. 2M: N

e. None of these

2. Questions

If 33(1/3)% of the donations received by organisation B in January are received from person P and the remaining from person Q, 66(2/3)% of the donations received by organisation B in February are received from person P and remaining by person Q, and 57(1/7)% of the donations received by organisation B in March, are received from person P and the remaining from person Q, then find the ratio of the total donations received by organisation B from person P in January, February, and March together to that received from person Q in January, February and March together?

a. 2: 1

b. 9: 8

c. 3: 2

d. 6: 5

e. 18: 17

3. Questions

Which of the following statements is/are not true?

X: The average of donations received by organisation B in March, April and May is Rs.19 lakhs.

Y: The ratio of the donations received by organisation C in January to that received in February and March together is 14: 15.

Z: In February, the total donations received by organisations B and C together are 25% more than that received by organisation A.

a. Only X and Y

b. Only X and Z

c. None is true

d. Only X

e. Only Y and Z

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4. Questions

If the average of donations received by organisation A in April, May and June is Rs. 29 lakhs and the average of donations received by organisation C in March, April, and June is Rs. 18 lakhs, then find the ratio of the total donations received by organisation A in all the 6 months together to that received by organisation C in all the 6 months together?

a. 23: 22

b. 46: 43

c. 3: 1

d. 69: 65

e. 6: 5

5. Questions

If the donations received by organisation C in January is R% of the total donations received by all three organisations together in January and the donations received by organisation C in May is S% of the total donations received by all three organisations together in May, then find the value of [(R – S)/R]?

a. 1/3

b. 3.33

c. 6/7

d. 50/3

e. 3/50

6. Questions

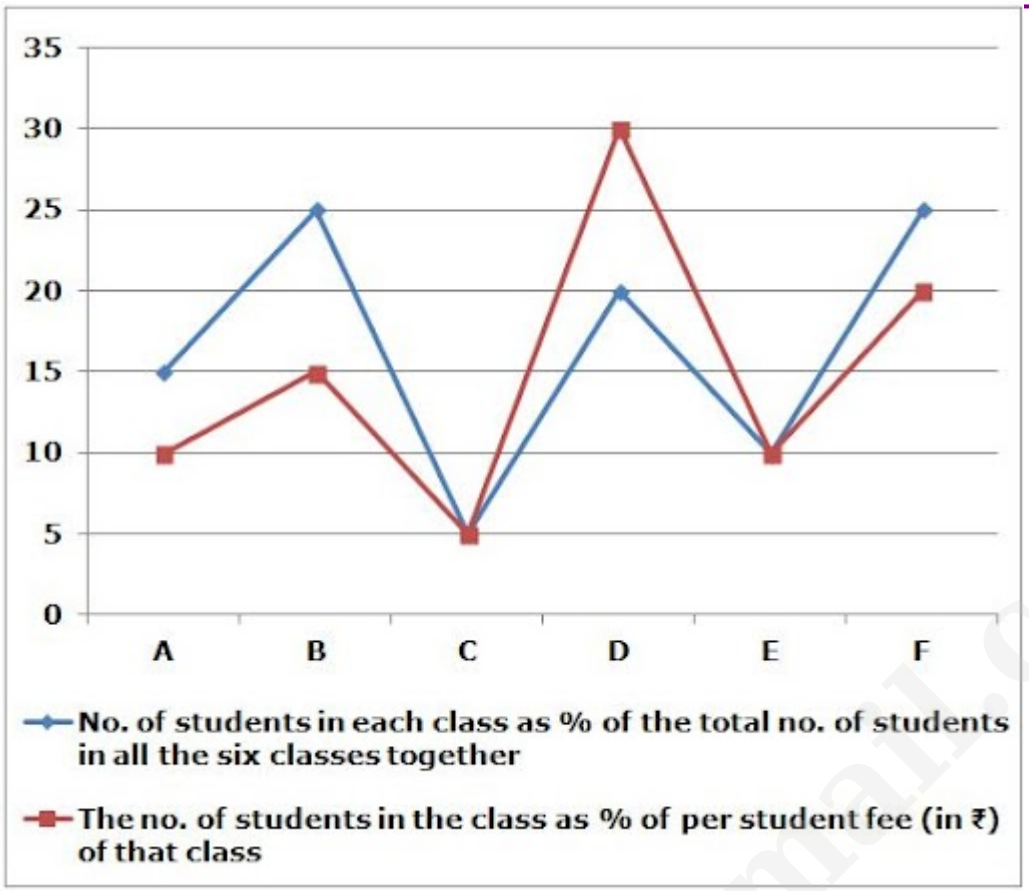
Study the given data carefully and answer the following questions:

The data given below is related to the number of students in six classes A, B, C, D, E, and F, and the per-student fee (in ₹) of each class.

The line graph given below shows the number of students in each class as a percentage of the total number of students out of all six classes together and the per-student fee as a percentage of the total number of students in the class.

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Note:

The total fee received from class A is ₹11040 less than that received from class B.

If the ratio of boys to girls in class B is 3: 2 and the ratio of boys to girls in class D is 5: 7, then find the difference between the total fee received from the boys of classes B and D together and that received from the girls of classes B and D together?

a. Rs. 3520

b. Rs. 3620

c. Rs. 3220

d. Rs. 4420

e. Rs. 4020

7. Questions

If the average of the total fee received from class A and that received from class C is Rs. M, then what will be the total fee (in terms of M) received from class D?

a. Rs. (M + 120)

b. Rs. (M – 240)

c. Rs. (M – 150)

d. Rs. (M + 160)

e. Rs. (M – 180)

8. Questions

If the ratio of boys to girls in class A is 5: 4 and the ratio of boys to girls in class C is 1: 2, then find the average of the fee received from all the boys of class A and that received from all the boys of class C?

a. Rs. 4180

b. Rs. 4680

c. Rs. 4280

d. Rs. 4480

e. Rs. 4080

9. Questions

Which of the following statements is/are true?

P: The average number of students in classes D, E, and F is 46.

Q: The difference between the total fee received from classes C and D is Rs. 4800.

R: The ratio of the per-student fee of class B to that of class E is 5: 4.

a. Only P and Q

b. Only Q and R

c. None is true

d. Only Q

e. Only R

10. Questions

If the average number of students in classes E, F, and G is 40 and the average per-student fee of classes E, F, and G is Rs. 230, then find the average of the total fees received from classes E, F, and G?

a. Rs. 9640

b. Rs. 9720

c. Rs. 9480

d. Rs. 9200

e. Rs. 9550

11. Questions

Study the given data carefully and answer the following questions:

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Four boats A, B, C, and D are sailing in a river, and the speed (in m/s) of each boat in still water is different.

The ratio of the speed of boat A in still water to that of boat B in still water is 5: 3, respectively, and the speed of boat C in still water is 5 m/s less than that of boat D in still water. The time taken by boat B to go 135 m in still water will be 4 seconds more than that taken by boat C to go 121 m in still water. The speed of the stream is 3 m/s, and the downstream speed of boat A is 5 m/s more than the upstream speed of boat D.

A person goes from point P to Q by boat B downstream and comes back by boat C. If he took a total time of 100 seconds, then find the distance (in meters) between points P and Q?

a. 360 m
b. 480 m
c. 320 m
d. 400 m
e. 440 m

12. Questions

A person has to cover a certain downstream distance in 2 minutes 10 seconds by boat D. After travelling for 65 seconds in the downstream direction by boat D, the stream started flowing against the boat. By what time will the person be late to his destination?

a. 30 seconds
b. 25 seconds
c. 10 seconds
d. 15 seconds
e. 20 seconds

13. Questions

Boats A and B are going downstream. After travelling for 15 seconds, the speed of the stream is decreased by 1 m/s, and both boats travelled for 25 seconds more. Find the difference between the total distances covered by boats A and B in 40 seconds?

a. 210 m
b. 300 m
c. 240 m
d. 270 m
e. 180 m

14. Questions

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If the average of the upstream distance covered by boat A in 15 seconds and that covered by boat B in 25 seconds is 'd' meters, then find the average of the upstream distance covered by boat C in 15 seconds and that covered by boat D in 20 seconds?

a. (d + 35) m

b. (d + 20) m

c. (d + 15) m

d. (d + 30) m

e. (d + 25) m

15. Questions

If the average of the speeds of boats C, D, and E in still water is 13 m/s, then find the average of the downstream distances covered by boats C, D, and E in 75 seconds?

a. 1400 m

b. 1200 m

c. 1600 m

d. 1100 m

e. 1300 m

16. Questions

In each of the following questions, read the given statement and compare Quantity I, Quantity II, and Quantity III based on it. (only the quantity is to be compared)

Quantity I: A's age is 40% more than B's age and also 25% more than C's age. If C's age is a whole number and a multiple of 7 between 20 and 30, then find B's age?

Quantity II: A boat is going upstream at 10 m/s. After travelling for 24 seconds, it came back to the starting point in 16 seconds. Find the speed of stream is what percentage of the speed of boat in still water?

Quantity III: A shopkeeper sold an item at Rs.30 discount on its marked price and received a 20% profit on its cost price. By what percentage was the item marked up?

a. Quantity I > Quantity II > Quantity III

b. Quantity I ≥ Quantity II > Quantity III

c. Quantity I > Quantity II ≥ Quantity III

d. Quantity I ≤ Quantity II < Quantity III

e. Quantity I = Quantity II = Quantity III or relation can't be established.

17. Questions

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A truck is following a car and a bike in such a way that the car is 240 m ahead of the truck and the bike is 200 m ahead of the car. The truck crosses the car and the bike in 96 seconds and 88 seconds, respectively, and the ratio of the speed of the truck to that of the bike is 4: 3, respectively.

Quantity I: In what time (in seconds), will the car cross the bike?

Quantity II: What is the speed of the truck (in km/h)?

Quantity III: If a bus is running 435 m behind the truck at 90 km/h, then in what time (in seconds), will the bus cross the car?

a. Quantity I > Quantity II > Quantity III
b. Quantity I < Quantity II < Quantity III
c. Quantity III > Quantity I > Quantity II
d. Quantity II > Quantity I > Quantity III
e. Quantity I > Quantity III > Quantity II

18. Questions

Two series A and B are given below, with one wrong term in series A and one missing term in series B, and both series follow the same pattern:

A: 2, 8, 19, 59, 122, 365

B: 4, 14, 31, ?, 193, 581

Quantity I: Find the wrong term in series A?

Quantity II: Find 125% of the missing term of series B?

Quantity III: Find 25% of the difference between the next terms of both series?

a. Quantity I > Quantity II > Quantity III
b. Quantity I < Quantity II < Quantity III
c. Quantity III > Quantity I > Quantity II
d. Quantity II > Quantity I > Quantity III
e. Quantity I > Quantity III > Quantity II

19. Questions

Three equations A, B and C are given below:

A: $x^2 - 13x - 90 = 0$

B: $y^2 - 55y + 700 = 0$

C: $z^2 - 14z - 72 = 0$

Quantity I: What is the minimum possible value of (x + y)?

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Quantity II: What is the 2nd maximum possible value of (x * z)?

Quantity III: What is the 2nd minimum possible value of (y – z)?

a. Quantity I > Quantity II > Quantity III
b. Quantity I < Quantity II < Quantity III
c. Quantity II > Quantity III > Quantity I
d. Quantity I > Quantity III > Quantity II

20. Questions

A and B entered into a partnership by investing in the ratio of 6: 7, respectively. After 'x' months, C joined them by investing 38(8/9)% more than A's investment. After 'y' more months, B left, and the ratio between profit shares of A, B, and C after 15 months is 9: 7: 10.

Quantity I: Find the value of (x + y)?

Quantity II: Find the value of $(x^3 – y^3)/(x + y)$?

Quantity III: If the ratio of the difference between compound interest and simple interest on Rs. 5000 at a 3R% rate after 2 years to that on Rs. 7500 at a 2R% rate after 2 years is 3: 2, then find the value of R?

a. Quantity I = Quantity II = Quantity III or relation can't be established.
b. Quantity I < Quantity II < Quantity III
c. Quantity II > Quantity I > Quantity III
d. Quantity III > Quantity I > Quantity II
e. Quantity I > Quantity III > Quantity II

Explanations:

1. Questions

Since, organisation A received the maximum donations in April,

the donations received by organisation A in April = Rs. 40 lakhs

Since, the donations received in May are a multiple of 4 but not less than Rs. 20 lakhs.

the only possible value of the donations received by organisation A in May = Rs. 24 lakhs

Since, the difference between the donations received by organisation A in February and May is Rs. 1 lakh,

the only possible value of the donations received by organisation A in February = 24 + 1 = Rs. 25 lakhs

Since, the difference between the donations received by organisation A in March and May is a multiple of 2 but not less than 10,

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We consider two cases,

Case I: If the donations received by organisation A in March are Rs. 10 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 10 = Rs.14 lakhs

Case II: If the donations received by organisation A in March are Rs. 16 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 16 = Rs. 8 lakhs

Since, 8 is a multiple of 2 but not more than 10,

the donations received by organisation A in March must be Rs. 10 lakhs.

Therefore, the donations received by organisation A in January = Rs. 16 lakhs

Month	Donations received by organisation A (in lakhs rupee)	Donations received by organisation B (in lakhs rupee)	Donations received by organisation C (in lakhs rupee)
Jan	16	75% of 16 = 12	2 * 20 – 12 = 28
Feb	25	60% of 25 = 15	2 * 15 – 15 = 15
Mar	10	70% of 10 = 7	2 * 10 – 7 = 13
Apr	40	80% of 40 = 32	2 * 25 – 32 = 18
May	24	50% of 24 = 12	2 * 15 – 12 = 18

Answer: B

Since the total donations received by organisations A, B, and C together in March = 10 + 7 + 13 = Rs. 30 lakhs,

and the total donations received by organisations A, B and C together in April = 40 + 32 + 18 = Rs. 90 lakhs

So the ratio M: N = 30: 90 = 1: 3

Since the donations received by organisation C in February = Rs. 15 lakhs,

and the donations received by organisation C in May = Rs. 18 lakhs

So, the required ratio = 15: 18 = 5: 6 = (2N – M): 2N

2. Questions

Since, organisation A received the maximum donations in April,

the donations received by organisation A in April = Rs. 40 lakhs

Since, the donations received in May are a multiple of 4 but not less than Rs. 20 lakhs.

the only possible value of the donations received by organisation A in May = Rs. 24 lakhs

Since, the difference between the donations received by organisation A in February and May is Rs. 1 lakh,

the only possible value of the donations received by organisation A in February = 24 + 1 = Rs. 25 lakhs

Since, the difference between the donations received by organisation A in March and May is a multiple of

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2 but not less than 10,

We consider two cases,

Case I: If the donations received by organisation A in March are Rs. 10 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 10 = Rs.14 lakhs

Case II: If the donations received by organisation A in March are Rs. 16 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 16 = Rs. 8 lakhs

Since, 8 is a multiple of 2 but not more than 10,

the donations received by organisation A in March must be Rs. 10 lakhs.

Therefore, the donations received by organisation A in January = Rs. 16 lakhs

Month	Donations received by organisation A (in lakhs rupee)	Donations received by organisation B (in lakhs rupee)	Donations received by organisation C (in lakhs rupee)
Jan	16	75% of 16 = 12	2 * 20 – 12 = 28
Feb	25	60% of 25 = 15	2 * 15 – 15 = 15
Mar	10	70% of 10 = 7	2 * 10 – 7 = 13
Apr	40	80% of 40 = 32	2 * 25 – 32 = 18
May	24	50% of 24 = 12	2 * 15 – 12 = 18

Answer: B

Since the total donations received by organisation B in January = Rs. 12 lakhs,

the donations received by organisation B from P in January = 33(1/3)% of 12 = Rs. 4 lakhs,

and the donations received by organisation B from Q in January = 12 – 4 = Rs. 8 lakhs

Since the total donations received by organisation B in February = Rs. 15 lakhs,

the donations received by organisation B from P in February = 66(2/3)% of 15 = Rs. 10 lakhs,

and the donations received by organisation B from Q in February = 15 – 10 = Rs. 5 lakhs

Since the total donations received by organisation B in March = Rs. 7 lakhs,

the donations received by organisation B from P in March = 57(1/7)% of 7 = Rs. 4 lakhs,

and the donations received by organisation B from Q in March = 7 – 4 = Rs. 3 lakhs

Therefore, the total donations received by organisation B from P in January, February and March together = 4 + 10 + 4 = Rs. 18 lakhs,

and the total donations received by organisation B from Q in January, February and March together = 8 + 5 + 3 = Rs. 16 lakhs

So, the required ratio = 18: 16 = 9: 8

3. Questions

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Since, organisation A received the maximum donations in April,

the donations received by organisation A in April = Rs. 40 lakhs

Since, the donations received in May are a multiple of 4 but not less than Rs. 20 lakhs.

the only possible value of the donations received by organisation A in May = Rs. 24 lakhs

Since, the difference between the donations received by organisation A in February and May is Rs. 1 lakh,

the only possible value of the donations received by organisation A in February = 24 + 1 = Rs. 25 lakhs

Since, the difference between the donations received by organisation A in March and May is a multiple of 2 but not less than 10,

We consider two cases,

Case I: If the donations received by organisation A in March are Rs. 10 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 10 = Rs.14 lakhs

Case II: If the donations received by organisation A in March are Rs. 16 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 16 = Rs. 8 lakhs

Since, 8 is a multiple of 2 but not more than 10,

the donations received by organisation A in March must be Rs. 10 lakhs.

Therefore, the donations received by organisation A in January = Rs. 16 lakhs

Month	Donations received by organisation A (in lakhs rupee)	Donations received by organisation B (in lakhs rupee)	Donations received by organisation C (in lakhs rupee)
Jan	16	75% of 16 = 12	2 * 20 – 12 = 28
Feb	25	60% of 25 = 15	2 * 15 – 15 = 15
Mar	10	70% of 10 = 7	2 * 10 – 7 = 13
Apr	40	80% of 40 = 32	2 * 25 – 32 = 18
May	24	50% of 24 = 12	2 * 15 – 12 = 18

Answer: C

From X:

Average of donations received by organisation B in Mar, Apr and May:

(7+32+12)/3= Rs.17 lakhs

So, X is not true.

From Y:

Ratio of the donations received by organisation C in January to that received on February and March together = 28: (15 + 13) = 1: 1

So, Y is not true.

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From Z:

The donations received by organisation A in February = Rs. 25 lakhs

The donations received by organisations B and C together in February = 15 + 15 = Rs. 30 lakhs

Required percentage = (30-25)/25 × 100 = 20%

So, Z is not true.

Hence, none is true.

4. Questions

Since, organisation A received the maximum donations in April,

the donations received by organisation A in April = Rs. 40 lakhs

Since, the donations received in May are a multiple of 4 but not less than Rs. 20 lakhs.

the only possible value of the donations received by organisation A in May = Rs. 24 lakhs

Since, the difference between the donations received by organisation A in February and May is Rs. 1 lakh,

the only possible value of the donations received by organisation A in February = 24 + 1 = Rs. 25 lakhs

Since, the difference between the donations received by organisation A in March and May is a multiple of 2 but not less than 10,

We consider two cases,

Case I: If the donations received by organisation A in March are Rs. 10 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 10 = Rs.14 lakhs

Case II: If the donations received by organisation A in March are Rs. 16 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 16 = Rs. 8 lakhs

Since, 8 is a multiple of 2 but not more than 10,

the donations received by organisation A in March must be Rs. 10 lakhs.

Therefore, the donations received by organisation A in January = Rs. 16 lakhs

Month	Donations received by organisation A (in lakhs rupee)	Donations received by organisation B (in lakhs rupee)	Donations received by organisation C (in lakhs rupee)
Jan	16	75% of 16 = 12	2 * 20 – 12 = 28
Feb	25	60% of 25 = 15	2 * 15 – 15 = 15
Mar	10	70% of 10 = 7	2 * 10 – 7 = 13
Apr	40	80% of 40 = 32	2 * 25 – 32 = 18
May	24	50% of 24 = 12	2 * 15 – 12 = 18

Answer: E

Since, the average of donations received by organisation A in April, May, and June is Rs. 29 lakhs,

the donations received by organisation A in June = (3 * 29) – (40 + 24) = Rs. 23 lakhs

Since the average of donations received by organisation C in March, April, and June is Rs. 18 lakhs.

the donations received by organisation C in June = (3 * 18) – (13 + 18) = Rs. 23 lakhs

Therefore, the total donations received by organisation A in all the six months together = 16 + 25 + 10 + 40 + 24 +23= Rs. 138 lakhs

And the total donations received by organisation C in all the six months together = 28 + 15 + 13 + 18 + 18 + 23 = Rs. 115 lakhs

So, the required ratio = 138: 115 = 6: 5

5. Questions

Since, organisation A received the maximum donations in April,

the donations received by organisation A in April = Rs. 40 lakhs

Since, the donations received in May are a multiple of 4 but not less than Rs. 20 lakhs.

the only possible value of the donations received by organisation A in May = Rs. 24 lakhs

Since, the difference between the donations received by organisation A in February and May is Rs. 1 lakh,

the only possible value of the donations received by organisation A in February = 24 + 1 = Rs. 25 lakhs

Since, the difference between the donations received by organisation A in March and May is a multiple of 2 but not less than 10,

We consider two cases,

Case I: If the donations received by organisation A in March are Rs. 10 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 10 = Rs.14 lakhs

Case II: If the donations received by organisation A in March are Rs. 16 lakhs, then the difference between the donations received by organisation A in March and May = 24 – 16 = Rs. 8 lakhs

Since, 8 is a multiple of 2 but not more than 10,

the donations received by organisation A in March must be Rs. 10 lakhs.

Therefore, the donations received by organisation A in January = Rs. 16 lakhs

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Month	Donations received by organisation A (in lakhs rupee)	Donations received by organisation B (in lakhs rupee)	Donations received by organisation C (in lakhs rupee)
Jan	16	75% of 16 = 12	2 * 20 – 12 = 28
Feb	25	60% of 25 = 15	2 * 15 – 15 = 15
Mar	10	70% of 10 = 7	2 * 10 – 7 = 13
Apr	40	80% of 40 = 32	2 * 25 – 32 = 18
May	24	50% of 24 = 12	2 * 15 – 12 = 18

Answer: A

Since, the donations received by organisation C in Jan = Rs. 28 lakhs,

and the donations received by organisations A, B, and C together in January = 16 + 12 + 28 = Rs. 56 lakhs

So, R = 28/56 ×100 = 50%

Since, the donations received by organisation C in May = Rs. 18 lakhs,

and the donations received by organisations A, B, and C together in May = 24 + 12 + 18 = Rs. 54 lakhs

So, S = 18/54 ×100 = (100/3)%

So, the value of (R-S)/R = $\frac{50-\frac{100}{3}}{50} = \frac{\frac{50}{3}}{50} = \frac{1}{3}$

6. Questions

Let the total number of students in all 6 classes together be 100x.

So, the number of students in classes A, B, C, D, E, and F are 15x, 25x, 5x, 20x, 10x, and 25x respectively.

Now, the per student fee of class A = 15x * (100/10) = Rs. 150x

The per student fee of class B = 25x * (100/15) = Rs. (500x/3)

The per student fee of class C = 5x * (100/5) = Rs. 100x

The per student fee of class D = 20x * (100/30) = Rs. (200x/3)

The per student fee of class E = 10x * (100/10) = Rs. 100x

And the per student fee of class F = 25x * (100/20) = Rs. 125x

Since the total fee received from class A is Rs. 11040 less than that received from class B.

$$25x \times \frac{500x}{3} - 15x \times 150x = 11040$$

$12500x^2 - 6750x^2 = 33120$

$5750x^2 = 33120$

x = 2.4

Class	No. of students in the class	Per student fee (in ₹)
A	15 * 2.4 = 36	150 * 2.4 = 360
B	25 * 2.4 = 60	(500 * 2.4)/3 = 400
C	5 * 2.4 = 12	100 * 2.4 = 240
D	20 * 2.4 = 48	(200 * 2.4)/3 = 160
E	10 * 2.4 = 24	100 * 2.4 = 240
F	25 * 2.4 = 60	125 * 2.4 = 300

Answer: A

Since the total number of students in class B is 60,

the number of boys in class B = 60 * (3/5) = 36

and the number of girls in class B = 60 – 36 = 24

Since the total number of students in class D is 48

the number of boys in class D = 48 * (5/12) = 20

and the number of girls in class D = 48 – 20 = 28

Therefore, the total fee received from the boys of classes B and D together:

(36 * 400) + (20 * 160) = 14400 + 3200 = Rs. 17600

And the total fee received from girls of classes B and D together:

(24 * 400) + (28 * 160) = 9600 + 4480 = Rs. 14080

So, the required difference = 17600 – 14080 = Rs. 3520

7. Questions

Let the total number of students in all 6 classes together be 100x.

So, the number of students in classes A, B, C, D, E, and F are 15x, 25x, 5x, 20x, 10x, and 25x respectively.

Now, the per student fee of class A = 15x * (100/10) = Rs. 150x

The per student fee of class B = 25x * (100/15) = Rs. (500x/3)

The per student fee of class C = 5x * (100/5) = Rs. 100x

The per student fee of class D = 20x * (100/30) = Rs. (200x/3)

The per student fee of class E = 10x * (100/10) = Rs. 100x

And the per student fee of class F = 25x * (100/20) = Rs. 125x

Since the total fee received from class A is Rs. 11040 less than that received from class B.

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$$25x \times \frac{500x}{3} - 15x \times 150x = 11040$$

$12500x^2 - 6750x^2 = 33120$

$5750x^2 = 33120$

$x = 2.4$

Class	No. of students in the class	Per student fee (in ₹)
A	15 * 2.4 = 36	150 * 2.4 = 360
B	25 * 2.4 = 60	(500 * 2.4)/3 = 400
C	5 * 2.4 = 12	100 * 2.4 = 240
D	20 * 2.4 = 48	(200 * 2.4)/3 = 160
E	10 * 2.4 = 24	100 * 2.4 = 240
F	25 * 2.4 = 60	125 * 2.4 = 300

Answer: B

Since the total fee received from class A = 36 * 360 = Rs. 12960,

and the total fee received from class C = 12 * 240 = Rs. 2880

So, M = (12960 + 2880)/2 = 7920.

Since the total fee received from class D = 48 * 160 = Rs. 7680,

So, the total fee received from class D (in terms of M) = Rs. (M – 240)

8. Questions

Let the total number of students in all 6 classes together be 100x.

So, the number of students in classes A, B, C, D, E, and F are 15x, 25x, 5x, 20x, 10x, and 25x respectively.

Now, the per student fee of class A = 15x * (100/10) = Rs. 150x

The per student fee of class B = 25x * (100/15) = Rs. (500x/3)

The per student fee of class C = 5x * (100/5) = Rs. 100x

The per student fee of class D = 20x * (100/30) = Rs. (200x/3)

The per student fee of class E = 10x * (100/10) = Rs. 100x

And the per student fee of class F = 25x * (100/20) = Rs. 125x

Since the total fee received from class A is Rs. 11040 less than that received from class B.

$$25x \times \frac{500x}{3} - 15x \times 150x = 11040$$

$12500x^2 - 6750x^2 = 33120$

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$5750x^2 = 33120$

$x = 2.4$

Class	No. of students in the class	Per student fee (in ₹)
A	15 * 2.4 = 36	150 * 2.4 = 360
B	25 * 2.4 = 60	(500 * 2.4)/3 = 400
C	5 * 2.4 = 12	100 * 2.4 = 240
D	20 * 2.4 = 48	(200 * 2.4)/3 = 160
E	10 * 2.4 = 24	100 * 2.4 = 240
F	25 * 2.4 = 60	125 * 2.4 = 300

Answer: E

Since, the total number of students in class A is 36,

the number of boys in class A = 36 * (5/9) = 20

and the fee received from all the boys of class A = 20 * 360 = Rs. 7200

Since the total number of students in class C is 12,

the number of boys in class C = 12 * (1/3) = 4, and the fee received from all the boys of class C = 4 * 240 = Rs. 960

Required average = (7200 + 960)/2 = Rs. 4080

9. Questions

Let the total number of students in all 6 classes together be 100x.

So, the number of students in classes A, B, C, D, E, and F are 15x, 25x, 5x, 20x, 10x, and 25x respectively.

Now, the per student fee of class A = 15x * (100/10) = Rs. 150x

The per student fee of class B = 25x * (100/15) = Rs. (500x/3)

The per student fee of class C = 5x * (100/5) = Rs. 100x

The per student fee of class D = 20x * (100/30) = Rs. (200x/3)

The per student fee of class E = 10x * (100/10) = Rs. 100x

And the per student fee of class F = 25x * (100/20) = Rs. 125x

Since the total fee received from class A is Rs. 11040 less than that received from class B.

$$25x \times \frac{500x}{3} - 15x \times 150x = 11040$$

$12500x^2 - 6750x^2 = 33120$

$5750x^2 = 33120$

x = 2.4

Class	No. of students in the class	Per student fee (in ₹)
A	15 * 2.4 = 36	150 * 2.4 = 360
B	25 * 2.4 = 60	(500 * 2.4)/3 = 400
C	5 * 2.4 = 12	100 * 2.4 = 240
D	20 * 2.4 = 48	(200 * 2.4)/3 = 160
E	10 * 2.4 = 24	100 * 2.4 = 240
F	25 * 2.4 = 60	125 * 2.4 = 300

Answer: D

From P:

Average number of students in classes D, E, and F:

(48+24+60)/3=44

So, P is not true.

From Q:

The total fee received from class C = 12 * 240 = Rs. 2880

The total fee received from class D = 48 * 160 = Rs. 7680

Required difference = 7680 – 2880 = Rs. 4800

So, Q is true.

From R:

Ratio of the per-student fee of class B to that of class E = 400: 240 = 5: 3

So, R is not true.

Hence, only Q is true.

10. Questions

Let the total number of students in all 6 classes together be 100x.

So, the number of students in classes A, B, C, D, E, and F are 15x, 25x, 5x, 20x, 10x, and 25x respectively.

Now, the per student fee of class A = 15x * (100/10) = Rs. 150x

The per student fee of class B = 25x * (100/15) = Rs. (500x/3)

The per student fee of class C = 5x * (100/5) = Rs. 100x

The per student fee of class D = 20x * (100/30) = Rs. (200x/3)

The per student fee of class E = 10x * (100/10) = Rs. 100x

And the per student fee of class F = 25x * (100/20) = Rs. 125x

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Since the total fee received from class A is Rs. 11040 less than that received from class B.

$$25x \times \frac{500x}{3} - 15x \times 150x = 11040$$

$12500x^2 - 6750x^2 = 33120$

$5750x^2 = 33120$

$x = 2.4$

Class	No. of students in the class	Per student fee (in ₹)
A	15 * 2.4 = 36	150 * 2.4 = 360
B	25 * 2.4 = 60	(500 * 2.4)/3 = 400
C	5 * 2.4 = 12	100 * 2.4 = 240
D	20 * 2.4 = 48	(200 * 2.4)/3 = 160
E	10 * 2.4 = 24	100 * 2.4 = 240
F	25 * 2.4 = 60	125 * 2.4 = 300

Answer: B

Since the average number of students in classes E, F, and G = 40

So the number of students in class G = (3 * 40) – (24 + 60) = 36

Since the average per-student fee of classes E, F, and G = Rs. 230

So the per-student fee of class G = (3 * 230) – (240 + 300) = Rs. 150

And the total fee received from class G = 36 * 150 = Rs. 5400

Since the total fee received from class E = 24 * 240 = Rs. 5760

and the total fee received from class F = 60 * 300 = Rs. 18000

So, the required average = (5760 + 18000 + 5400)/3 = Rs. 9720

11. Questions

Let the speeds of boats A and B in still water be '5x' m/s and '3x' m/s, respectively.

Also let the speed of boat C in still water be 'y' m/s.

So, the speed of boat D in still water = (y + 5) m/s

Since the time taken by boat B to go 135 m in still water will be 4 seconds more than that taken by boat C to go 121 m in still water.

So,

135/3x - 121/y=4

135y - 363x = 12xy ----------(1)

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Since the downstream speed of boat A is 5 m/s more than the upstream speed of boat D.

And the speed of stream = 3 m/s

So,

$(5x + 3) – (y + 5 – 3) = 5$

$5x + 3 – y – 2 = 5$

$5x – y = 4$ ----------(2)

From equations (1) and (2):

$135(5x – 4) – 363x = 12x(5x – 4)$

$675x – 540 – 363x = 60x^2 – 48x$

$60x^2 – 360x + 540 = 0$

$x^2 – 6x + 9 = 0$

$x^2 – 3x – 3x + 9 = 0$

$x(x – 3) – 3(x – 3) = 0$

$x = 3$

From equation (2):

$15 – y = 4$

$y = 11$

The speed of boat A in still water = 5 * 3 = 15 m/s

The speed of boat B in still water = 3 * 3 = 9 m/s

The speed of boat C in still water = 11 m/s

The speed of boat D in still water = 11 + 5 = 16 m/s

The speed of stream = 3 m/s

Answer: B

The downstream speed of boat B = 9 + 3 = 12 m/s

The upstream speed of boat C = 11 – 3 = 8 m/s

Since the ratio of the downstream speed of boat B to the upstream speed of boat C = 12: 8 = 3: 2

So the ratio of the time in which the person would reach point Q from point P to the time in which he would come back = 2: 3

So, the time taken by the person to reach point Q from point P by boat B:

100 * (2/5) = 40 seconds

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And the distance between points P and Q = 12 * 40 = 480 m

12. Questions

Let the speeds of boats A and B in still water be '5x' m/s and '3x' m/s, respectively.

Also let the speed of boat C in still water be 'y' m/s.

So, the speed of boat D in still water = (y + 5) m/s

Since the time taken by boat B to go 135 m in still water will be 4 seconds more than that taken by boat C to go 121 m in still water.

So,

135/3x- 121/y=4

135y – 363x = 12xy ----------(1)

Since the downstream speed of boat A is 5 m/s more than the upstream speed of boat D.

And the speed of stream = 3 m/s

So,

(5x + 3) – (y + 5 – 3) = 5

5x + 3 – y – 2 = 5

5x – y = 4 ----------(2)

From equations (1) and (2):

135(5x – 4) – 363x = 12x(5x – 4)

$675x – 540 – 363x = 60x^2 – 48x$

$60x^2 – 360x + 540 = 0$

$x^2 – 6x + 9 = 0$

$x^2 – 3x – 3x + 9 = 0$

x(x – 3) – 3(x – 3) = 0

x = 3

From equation (2):

15 – y = 4

y = 11

The speed of boat A in still water = 5 * 3 = 15 m/s

The speed of boat B in still water = 3 * 3 = 9 m/s

The speed of boat C in still water = 11 m/s

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The speed of boat D in still water = 11 + 5 = 16 m/s

The speed of stream = 3 m/s

Answer: A

Since the downstream speed of boat D = 16 + 3 = 19 m/s

So, the distance which the person has to cover in the downstream direction by boat D = 19 * (2 * 60 + 10) = 2470 m

The downstream distance covered by the person by boat D in 65 seconds = 19 * 65 = 1235 m

After travelling for 65 seconds, the stream started flowing against the boat and now boat D will be going upstream.

So, the time taken by boat D to cover the remaining distance in the upstream direction = (2470 – 1235)/(16 – 3) = 95 seconds

And the time by which the person will be late to his destination:

(65 + 95) – 130 = 30 seconds

13. Questions

Let the speeds of boats A and B in still water be '5x' m/s and '3x' m/s, respectively.

Also let the speed of boat C in still water be 'y' m/s.

So, the speed of boat D in still water = (y + 5) m/s

Since the time taken by boat B to go 135 m in still water will be 4 seconds more than that taken by boat C to go 121 m in still water.

So,

$135/3x - 121/y = 4$

$135y – 363x = 12xy$ ----------(1)

Since the downstream speed of boat A is 5 m/s more than the upstream speed of boat D.

And the speed of stream = 3 m/s

So,

$(5x + 3) – (y + 5 – 3) = 5$

$5x + 3 – y – 2 = 5$

$5x – y = 4$ ----------(2)

From equations (1) and (2):

$135(5x – 4) – 363x = 12x(5x – 4)$

$675x – 540 – 363x = 60x^2 – 48x$

$60x^2 – 360x + 540 = 0$

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$x^2 - 6x + 9 = 0$

$x^2 - 3x - 3x + 9 = 0$

$x(x - 3) - 3(x - 3) = 0$

$x = 3$

From equation (2):

$15 - y = 4$

$y = 11$

The speed of boat A in still water = 5 * 3 = 15 m/s

The speed of boat B in still water = 3 * 3 = 9 m/s

The speed of boat C in still water = 11 m/s

The speed of boat D in still water = 11 + 5 = 16 m/s

The speed of stream = 3 m/s

Answer: C

The speed of boat A in still water = 15 m/s

The speed of boat B in still water = 9 m/s

The speed of stream initially = 3 m/s

The downstream distance covered by boat A in 1st 15 seconds = (15 + 3) * 15 = 270 m

The downstream distance covered by boat B in 1st 15 seconds = (9 + 3) * 15 = 180 m

When the speed of the stream is decreased by 1 m/s:

The downstream distance covered by boat A in next 25 seconds = (15 + 3 – 1) * 25 = 425 m

The downstream distance covered by boat B in next 25 seconds = (9 + 3 – 1) * 25 = 275 m

Required difference = (270 + 425) – (180 + 275) = 240 m

14. Questions

Let the speeds of boats A and B in still water be '5x' m/s and '3x' m/s, respectively.

Also let the speed of boat C in still water be 'y' m/s.

So, the speed of boat D in still water = (y + 5) m/s

Since the time taken by boat B to go 135 m in still water will be 4 seconds more than that taken by boat C to go 121 m in still water.

So,

135/3x- 121/y=4

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$135y - 363x = 12xy$ ----------(1)

Since the downstream speed of boat A is 5 m/s more than the upstream speed of boat D.

And the speed of stream = 3 m/s

So,

$(5x + 3) - (y + 5 - 3) = 5$

$5x + 3 - y - 2 = 5$

$5x - y = 4$ ----------(2)

From equations (1) and (2):

$135(5x - 4) - 363x = 12x(5x - 4)$

$675x - 540 - 363x = 60x^2 - 48x$

$60x^2 - 360x + 540 = 0$

$x^2 - 6x + 9 = 0$

$x^2 - 3x - 3x + 9 = 0$

$x(x - 3) - 3(x - 3) = 0$

$x = 3$

From equation (2):

$15 - y = 4$

$y = 11$

The speed of boat A in still water = 5 * 3 = 15 m/s

The speed of boat B in still water = 3 * 3 = 9 m/s

The speed of boat C in still water = 11 m/s

The speed of boat D in still water = 11 + 5 = 16 m/s

The speed of stream = 3 m/s

Answer: E

Since the upstream distance covered by boat A in 15 seconds = (15 – 3) * 15 = 180 m

And the upstream distance covered by boat B in 25 seconds = (9 – 3) * 25 = 150 m

So, d = (180 + 150)/2 = 165 m

Since the upstream distance covered by boat C in 15 seconds = (11 – 3) * 15 = 120 m

And the upstream distance covered by boat D in 20 seconds = (16 – 3) * 20 = 260 m

So, the required average = (120 + 260)/2 = 190 m = (d + 25) m

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15. Questions

Let the speeds of boats A and B in still water be '5x' m/s and '3x' m/s, respectively.

Also let the speed of boat C in still water be 'y' m/s.

So, the speed of boat D in still water = (y + 5) m/s

Since the time taken by boat B to go 135 m in still water will be 4 seconds more than that taken by boat C to go 121 m in still water.

So,

135/3x- 121/y=4

135y – 363x = 12xy ----------(1)

Since the downstream speed of boat A is 5 m/s more than the upstream speed of boat D.

And the speed of stream = 3 m/s

So,

(5x + 3) – (y + 5 – 3) = 5

5x + 3 – y – 2 = 5

5x – y = 4 ----------(2)

From equations (1) and (2):

135(5x – 4) – 363x = 12x(5x – 4)

$675x – 540 – 363x = 60x^2 – 48x$

$60x^2 – 360x + 540 = 0$

$x^2 – 6x + 9 = 0$

$x^2 – 3x – 3x + 9 = 0$

x(x – 3) – 3(x – 3) = 0

x = 3

From equation (2):

15 – y = 4

y = 11

The speed of boat A in still water = 5 * 3 = 15 m/s

The speed of boat B in still water = 3 * 3 = 9 m/s

The speed of boat C in still water = 11 m/s

The speed of boat D in still water = 11 + 5 = 16 m/s

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The speed of stream = 3 m/s

Answer: B

Since the average of the speeds of boats C, D, and E in still water = 13 m/s

So the speed of boat E in still water = (3 * 13) – (11 + 16) = 12 m/s

Since the downstream distance covered by boat C in 75 seconds = (11 + 3) * 75 = 1050 m

The downstream distance covered by boat D in 75 seconds = (16 + 3) * 75 = 1425 m

And the downstream distance covered by boat E in 75 seconds = (12 + 3) * 75 = 1125 m

So, the required average = (1050 + 1425 + 1125)/3 = 1200 m

16. Questions

Answer: E

Quantity I:

Let B's age be '5x' years.

So A's age = 140% of 5x = '7x' years

And C's age = 7x * (100/125) = '5.6x' years

Since C's age is a whole number and a multiple of 7 between 20 and 30.

Therefore, If C's age is 21 years:

Then, 5.6x = 21

x = 3.75

And B's age = 5 * 3.75 = 18.75 years

If C's age is 28 years:

Then, 5.6x = 28

x = 5

And B's age = 5 * 5 = 25 years

So, B's age will be either 18.75 years or 25 years.

Quantity II:

The upstream speed of the boat = 10 m/s

Upstream distance covered by the boat in 24 seconds = 10 * 24 = 240 m

Since the time taken by the boat to come back = 1m 6 seconds

So the downstream speed of the boat = 240/16 = 15 seconds

Since the speed of boat in still water = (15 + 10)/2 = 12.5 m/s

And the speed of stream = (15 – 10)/2 = 2.5 m/s

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So, the required percentage = (2.5/12.5) * 100 = 20%

Quantity III:

Let the CP of the article be Rs.100x.

Since the article is sold at 20% profit.

So the SP of the article = 120% of 100x = Rs.120x

Since the discount given on its MP is Rs.30.

So the MP of the article = Rs. (120x + 30)

Since we can't find the value of 'x', so we can't determine the marked up percentage of the item.

Hence, the relation can't be established.

17. Questions

Answer: C

Let the speeds of the truck and the bike be '4x' m/s and '3x' m/s, respectively.

Also let the speed of the car be 'y' m/s.

Since the truck crosses the car in 96 seconds.

So,

(4x – y) = 240/96

4x – y = 2.5 ----------(1)

Since the truck crosses the bike in 88 seconds.

So,

(4x – 3x) = (240 + 200)/88

x = 5

From equation (1):

20 – y = 2.5

y = 17.5

The speed of the truck = 4 * 5 = 20 m/s

The speed of the car = 17.5 m/s

The speed of the bike = 3 * 5 = 15 m/s

Quantity I:

The time in which the car will cross the bike = 200/(17.5 – 15) = 80 seconds

Quantity II:

The speed of the truck = 20 m/s = 20 * (18/5) = 72 km/h

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Quantity III:

Since the speed of the bus = 90 km/h = 25 m/s

Since the bus is running 435 m behind the truck and the truck is running 240 m behind the car.

So the distance between the bus and the car = 435 + 240 = 675 m

Since the speed of the car = 17.5 m/s

So the time in which the bus will cross the car = 675/(25 – 17.5) = 90 seconds

Hence, Quantity III > Quantity I > Quantity II

18. Questions

Answer: A

Logic in series A:

2 * 3 + 2 = 8

8 * 2 + 3 = 19

19 * 3 + 2 = 59

59 * 2 + 3 = **121(122)**

121 * 3 + 2 = 365

So, the wrong term in series A = 122

And the next term of series A = 365 * 2 + 3 = 733

Logic in series B:

4 * 3 + 2 = 14

14 * 2 + 3 = 31

31 * 3 + 2 = **95**

95 * 2 + 3 = 193

193 * 3 + 2 = 581

So, the missing term in series B = 95

And the next term of series B = 581 * 2 + 3 = 1165

Quantity I:

The wrong term in series A = 122

Quantity II:

The 125% of the missing term of series B = 125% of 95 = 118.75

Quantity III:

The 25% of the difference between the next terms of both series:

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25% of (1165 – 733) = 108

Hence, Quantity I > Quantity II > Quantity III

19. Questions

Answer: D

From equation A:

$x^2 – 13x – 90 = 0$

$x^2 – 18x + 5x – 90 = 0$

$x(x – 18) + 5(x – 18) = 0$

$x = 18, -5$

From equation B:

$y^2 – 55y + 700 = 0$

$y^2 – 20y – 35y + 700 = 0$

$y(y – 20) – 35(y – 20) = 0$

$y = 20, 35$

From equation C:

$z^2 – 14z – 72 = 0$

$z^2 – 18z + 4z – 72 = 0$

$z(z – 18) + 4(z – 18) = 0$

$z = 18, -4$

Quantity I:

All the possible values of (x + y):

-5 + 20 = 15

-5 + 35 = 30

18 + 20 = 38

18 + 35 = 53

So, the minimum possible value of (x + y) = 15

Quantity II:

All the possible values of (x * z):

(-5) * (-4) = 20

(-5) * 18 = -90

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18 * (-4) = -72

18 * 18 = 324

So, the 2nd maximum possible value of (x * z) = 20

Quantity III:

All the possible values of (y – z):

20 – (-4) = 24

20 – 18 = 2

35 – (-4) = 39

35 – 18 = 17

So, the 2nd minimum possible value of (y – z) = 17

Hence, Quantity II > Quantity III > Quantity I

20. Questions

Answer: A

Quantity I:

Let the investments of A and B be '18a' and '21a', respectively.

So, the investment of C = 138(8/9)% of 18a = 25a

Since the ratio of A's investment to C's investment after 15 months is 9: 10.

So,

(18a×15)/(25a ×(15-x))= 9/10

12 = 15 – x

x = 3

Since the ratio of A's investment to B's investment after 15 months is 9: 7.

So,

(18a×15)/(21a ×(15-x-y))= 9/7

(18a×15)/(21a ×(15-3-y))= 9/7

10 = 12 – y

y = 2

So, the value of (x + y) = 3 + 2 = 5

Quantity II:

The value of $(x^3 - y^3)/(x+y)$ = (27-8)/5 = 3.8

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Quantity III:

Since the ratio of the difference between CI and SI on Rs.5000 at a 3R% rate after 2 years to that on Rs.7500 at a 2R% rate after 2 years is 3: 2.

So, $$\frac{5000 \times \left(\frac{3R}{100}\right)^2}{7500 \times \left(\frac{2R}{100}\right)^2} = \frac{3}{2}$$

Since the equation can't be solved.

So the value of R can't be determined.

Hence, relation can't established.

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1. Questions

Read the following information carefully and answer the questions based on it.

There are five different companies – P, Q, R, S and T- that manufactured three types of products – A, B, and C. The table given below shows the number of units of each type of article sold by each company.

Name of a company	Average of type A and type B articles sold	Number of type C articles sold	% of Type A articles sold out of the total articles sold by a company
P	Z	L	K%
Q	80	2L	(M/3) %
R	45	M	40%
S	Y	L	(2L + 20) %
T	3Z	M	25%

Note:

a). The number of type B and type C articles sold by company P is the same, while the difference between the number of type A and type B articles sold by company P is the same as that of type C articles sold by the company.

b). The total number of articles sold by company Q is 4K, and the number of type A articles sold by companies R and T is the same.

For company S, out of the total type B articles manufactured, 2/3rd remain unsold, and the number of type A articles manufactured is four times the number of type B articles that remain unsold. Find the difference between the number of type A articles that remain unsold and the number of type C articles sold if the total number of articles sold by the company is 2M.

a. 115

b. 95

c. 132

d. 125

e. None of these

2. Questions

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For company Q, out of the total articles A manufactured by the company, 40% are sold, and the number of type A articles remaining unsold is twice that of type B articles remaining unsold. Find the total number of articles manufactured by the company if all type C manufactured articles are sold.

a. 290

b. 320

c. 240

d. 360

e. None of these

3. Questions

Quantity I:

The average number of type A articles sold by companies P and R together.

Quantity II:

50% of the total number of articles sold by company R.

Find the difference between Quantity I and Quantity II as a percentage of M.

a. 33.33%

b. 37.5%

c. 43.75%

d. 41.67%

e. None of these

4. Questions

The average number of type A articles sold by companies R, S, and T together is 140. Find the revenue generated by company T from the sale of articles if the average selling price per unit of each type of article is Rs. (Y + 2Z).

a. Rs. 72000

b. Rs. 57600

c. Rs. 67200

d. Rs. 76800

e. None of these

5. Questions

Find the value of (2Z + 3L + 4M)?

a. 480

b. 300

c. 320

d. 540

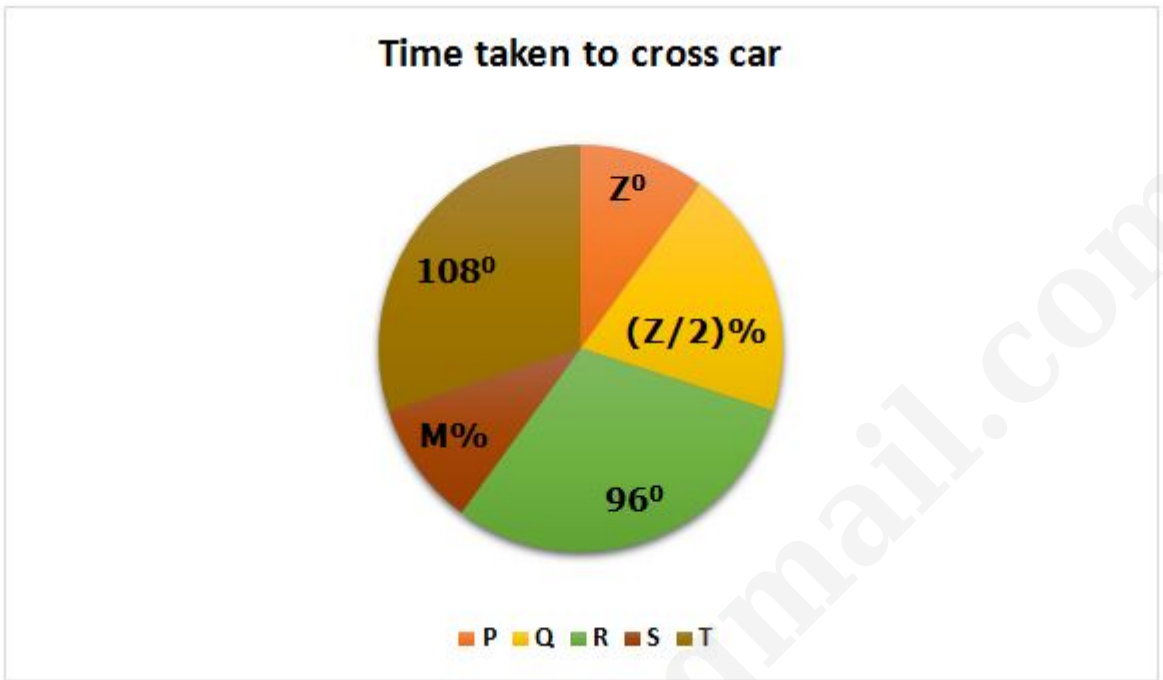
e. None of these

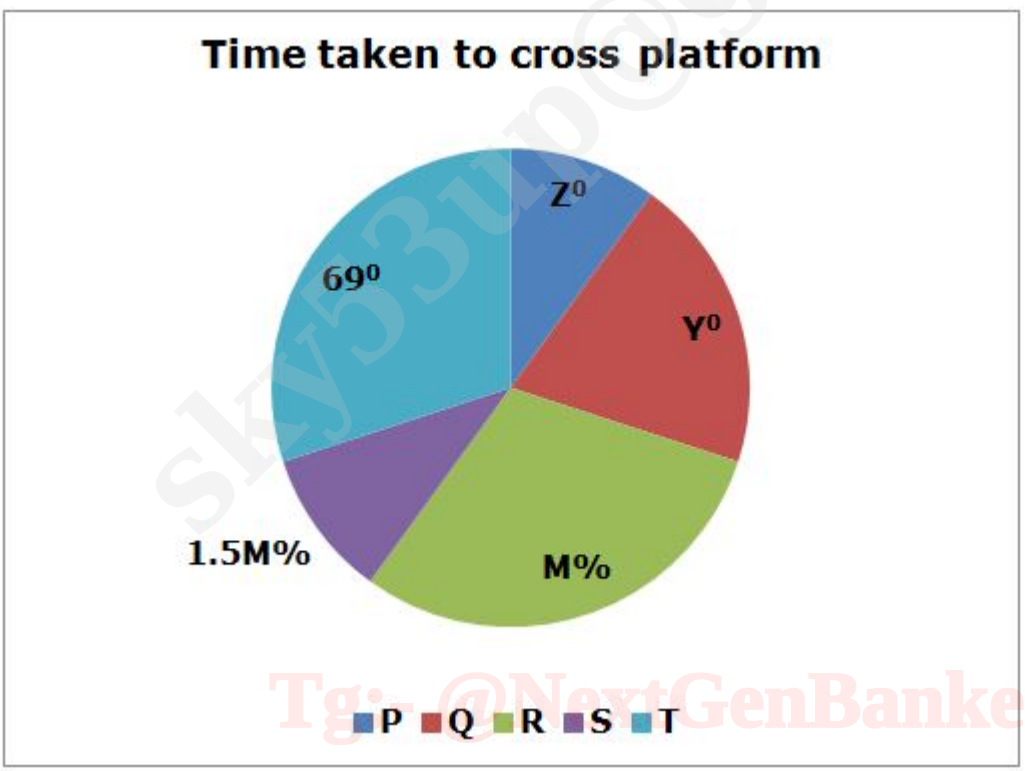
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6. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the percentage or degree distribution of the time taken by five trains – P, Q, R, S, and T to cross a car (standing car) and the distribution of time taken to cross a platform, which is the same as that of the length of train P.





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Note:

a). The length of train S is 200 meters, and the difference between the time taken by train T to cross a car and to cross a platform is 10 seconds.

b). The time taken by train Q to cross a platform is (Y – 27) seconds.

Train A and Train Q cross each other (in opposite directions) in 22.5 seconds. Find the speed of train A in km/h, if the length of train A is the same as that of train R?

a. 100 km/h
b. 120 km/h
c. 100/3 km/h
d. 80 km/h
e. None of these

7. Questions

Find the time taken by Train R and Train S to cross each other in the same direction (in seconds).

a. 60
b. 75
c. 48
d. 72
e. None of these

8. Questions

A = length of platform

B/2 = Length of train Q

a. A > B
b. A < B
c. A = B
d. (A + B) < 800
e. None of these

9. Questions

Find the difference between the speed of train Q and train R (in km/h).

a. 25
b. 45
c. 40
d. 50
e. None of these

10. Questions

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Which of the following is/are true.

I). Length of train T is 1440 meters

II). Speed of train P and train T are the same

III). Value of Y is 170% more than that of Z.

a. I and II only

b. I, II, and III

c. II and III only

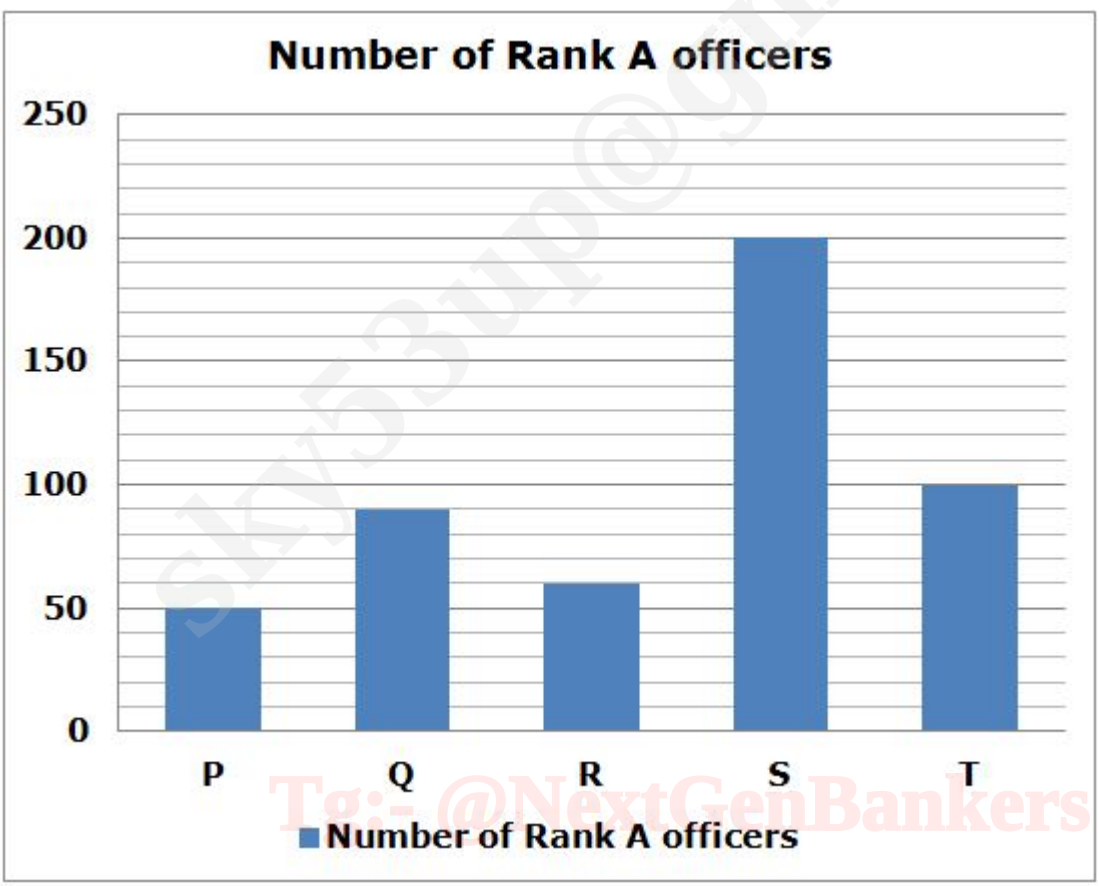
d. I and III only

e. None of these

11. Questions

Read the following information carefully and answer the questions based on it.

There are five police stations – P, Q, R, S, and T in Delhi. The graph given below shows the number of rank A police officers out of the total number of officers (Rank A + Rank B) in these five stations, and the table given below shows the number of total male and female officers in these five stations.



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Name of police stations	Male officers: Female officers
P	7: Z
Q	2: Y
R	5: (2Y – 1)
S	7: Z
T	3:1

Note:

a). In each police station, rank A officers constitute 25% of the total number of officers in the police station.

b). In each police station, there are more male officers than female officers.

c). The total number of officers (Rank A + Rank B) in all given police stations together is 2000, out of which male officers are 1440.

Out of the total rank A officers in police station T, 10(Z + Y) % are female officers, find the difference between Rank A male officers and Rank B female officers?

a. 40

b. 20

c. 0

d. 180

e. None of these

12. Questions

Out of the total number of Rank B officers in station R, 20% are promoted to Rank A officers. Now, the number of female officers in station P is approximately what percentage of the sum of the Rank A officers in station R and female officers in station P?

a. 40%

b. 35%

c. 42.5%

d. 37.5%

e. 45%

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13. Questions

For another station U, the number of Rank A officers is half that in station Q, and the number of Rank B officers is 20% more than that in station R. Find the total number of officers (Rank A + Rank B) in station U.

a. 251

b. 261

c. 241

d. 271

e. None of these

14. Questions

Quantity I:

The total female officers in stations P and R together

Quantity II:

30% of the difference between male and female officers in station S.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II or no relation

d. Quantity I ≥ Quantity II

e. Quantity I ≤ Quantity II

15. Questions

Which of the following is/are true.

I). The total number of officers in station R are the same as the female officers in station S.

II). The respective ratio of the difference between male and female officer in station P and male officers in station Q is 1:3

III). (Z + Y) is a perfect square.

a. I and II only

b. II and III only

c. I and III only

d. I, II, and III

e. None of these

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16. Questions

Boat P and Boat Q are running towards each other, starting from point A and point B, respectively, at the same time. The speed of boat P and Q in still water is Z km/h and Y km/h, and the speed of the stream is M km/h. The flow of the river is from A to B, and boat P and Q meet for the first time after T hours. Find the distance travelled by boat R, running in still water, in 4T hours. The speed of boat R in still water is the same as the average of the speeds of boat P and boat Q, if the distance between A and B is 70 km.

a. 210 km

b. 140 km

c. 280 km

d. 245 km

e. None of these

17. Questions

On selling article P at 80 % of its actual selling price , there is a loss of 4% , and when it is sold at the actual selling price, then the profit is Z%. The cost price of article Q is Rs. 1200, and it is marked up by 100% above its cost price and sold at Rs. 1344 after two successive discounts of (Z + 10) % and Y%, respectively.

Find which of the following is/are true.

I). $Z = Y$

II). $(Z + Y) > 40$

III). unit digit of $(Z + Y/4)^Z$ is 5.

a. I only

b. II and III only

c. I, II, and III

d. I and III only

e. None of these

18. Questions

The cost of fencing a square field of side (Z – 2) cm at Rs. Y/cm is 75% less than the cost of fencing a circular field of radius Z cm at Rs. 2Y/cm. Find the cost of fencing a rectangular field at Rs. (Y + 4)/cm, which has a length of (Z + 4) cm, which is twice its breadth. The cost of fencing the circular field is Rs. 480. Use $\pi = 3$.

a. Rs. 324

b. Rs. 288

c. Rs. 432

d. Rs. 384

e. None of these

19. Questions

The bag contains three red balls, some green balls, and some yellow balls. The number of green balls is _____, which is 50% more than the number of yellow balls. The probability of picking one

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red ball and one yellow ball from the bag is _____.

Find which of the following satisfies the blanks in the same order.

I). 18, 3/44

II). 3, 3/28

III). 9, 2/17

a. I and III only

b. II only

c. I, II, and III

d. II and III only

e. None of these

20. Questions

There are three persons in a family: P, Q, and R. The sum of the age of R and the average of P and Q together is 45 years, while the sum of the present age of Q and the average age of P and R together after 4 years is 55.5 years. Find the minimum possible sum of the present age of Q and R together. The age of each person is an integer value.

a. 13 years

b. 14 years

c. 15 years

d. 12 years

e. None of these

Explanations:

1. Questions

Number of type B articles sold by company P = type C articles sold by company P = L

Also, (Type A articles sold by company P – Type B articles sold by company P) = type C articles sold by company P

So, type A articles sold by company P = L + L = 2L

Also,

2 x Z = 2L + L

Z/L = 3/2 …….... (1)

Value of K% = 2L/4L x 100 = 50%

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Number of articles sold by company Q = 50 x 4 = 200

So, number of type C articles sold by company Q = 200 – 80 x 2 = 40 = 2L

So, L = 20

So, value of Z = 3/2 x 20 = 30

Number of type A articles sold by company R = 40% x (2 x 45 + M) = 40% x (90+M)

Number of type A articles sold by company T = 25% x (2 x 3 x 30 + M) = 25% x (180 + M)

Now,

40% x (90 + M) = 25% x (180 + M)

720 + 8M = 900 + 5M

3M = 180

Value of M = 60

Answer: C

According to the question,

For company S,

Total number of articles sold = 2M = 2 x 60 = 120

Number of type C articles sold = L = 20

So, number of type (A + B) articles sold = 120 – 20 = 100

So, type A articles sold = 120 x (2L + 20) % = 72

So, type B articles sold = 100 – 72 = 28

1/3 x (type B articles manufactured) = 28

So, type B articles remains unsold = 84-28 = 56

So, number of type A articles manufactured = 56 x 4 = 224

So, number of type A articles unsold = 224 – 72= 152

So, required difference = 152 – 20 = 132

Hence answer is option C

2. Questions

Number of type B articles sold by company P = type C articles sold by company P = L

Also, (Type A articles sold by company P – Type B articles sold by company P) = type C articles sold by company P

So, type A articles sold by company P = L + L = 2L

Also,

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2 x Z = 2L + L

Z/L = 3/2 ……... (1)

Value of K% = 2L/4L x 100 = 50%

Number of articles sold by company Q = 50 x 4 = 200

So, number of type C articles sold by company Q = 200 – 80 x 2 = 40 = 2L

So, L = 20

So, value of Z = 3/2 x 20 = 30

Number of type A articles sold by company R = 40% x (2 x 45 + M) = 40% x (90+M)

Number of type A articles sold by company T = 25% x (2 x 3 x 30 + M) = 25% x (180 + M)

Now,

40% x (90 + M) = 25% x (180 + M)

720 + 8M = 900 + 5M

3M = 180

Value of M = 60

Answer: A

For company Q

Total articles sold = 2 x 80 + 2 x 20 = 200

Number of type A articles sold = 200 x (60/3) % = 40

Number of type B articles sold = 160 – 40 = 120

Number of type A articles manufactured = 40/40 x 100 = 100

So, number of type B articles manufactured = 120 + 60/2 = 150

So, total number of articles manufactured = 100 + 150 + 40 = 290

Hence answer is option A

3. Questions

Number of type B articles sold by company P = type C articles sold by company P = L

Also, (Type A articles sold by company P – Type B articles sold by company P) = type C articles sold by company P

So, type A articles sold by company P = L + L = 2L

Also,

2 x Z = 2L + L

Z/L = 3/2 ……... (1)

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Value of K% = 2L/4L x 100 = 50%

Number of articles sold by company Q = 50 x 4 = 200

So, number of type C articles sold by company Q = 200 – 80 x 2 = 40 = 2L

So, L = 20

So, value of Z = 3/2 x 20 = 30

Number of type A articles sold by company R = 40% x (2 x 45 + M) = 40% x (90+M)

Number of type A articles sold by company T = 25% x (2 x 3 x 30 + M) = 25% x (180 + M)

Now,

40% x (90 + M) = 25% x (180 + M)

720 + 8M = 900 + 5M

3M = 180

Value of M = 60

Answer: D

Number of type A articles sold by company P = 80 x 50% = 40

Number of type A articles sold by company R = (45 x 2 + 60) x 40% = 60

Total number of articles sold by company R = 150

Quantity I = (40 + 60) = 100/2=50

Quantity II = 50% x 150 = 75

Required % = (75-50)/60 x 100 = 41.67%

Hence answer is option D

4. Questions

Number of type B articles sold by company P = type C articles sold by company P = L

Also, (Type A articles sold by company P – Type B articles sold by company P) = type C articles sold by company P

So, type A articles sold by company P = L + L = 2L

Also,

2 x Z = 2L + L

Z/L = 3/2 …….... (1)

Value of K% = 2L/4L x 100 = 50%

Number of articles sold by company Q = 50 x 4 = 200

So, number of type C articles sold by company Q = 200 – 80 x 2 = 40 = 2L

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So, L = 20

So, value of Z = 3/2 x 20 = 30

Number of type A articles sold by company R = 40% x (2 x 45 + M) = 40% x (90+M)

Number of type A articles sold by company T = 25% x (2 x 3 x 30 + M) = 25% x (180 + M)

Now,

40% x (90 + M) = 25% x (180 + M)

720 + 8M = 900 + 5M

3M = 180

Value of M = 60

Answer: A

Number of type A articles sold by company S = 140 x 3 – 60 – 60 = 300

Now,

(2Y + 20) x 60% = 300

Value of Y = 240

Average selling price per unit of each article sold by company T = 240 + 2 x 30 = Rs. 300

Total articles sold by company T = (6Z + M) = 6 x 30 + 60 = 240

Required revenue = 240 x 300 = Rs. 72000

Hence answer is option A

5. Questions

Number of type B articles sold by company P = type C articles sold by company P = L

Also, (Type A articles sold by company P – Type B articles sold by company P) = type C articles sold by company P

So, type A articles sold by company P = L + L = 2L

Also,

2 x Z = 2L + L

Z/L = 3/2 …….... (1)

Value of K% = 2L/4L x 100 = 50%

Number of articles sold by company Q = 50 x 4 = 200

So, number of type C articles sold by company Q = 200 – 80 x 2 = 40 = 2L

So, L = 20

So, value of Z = 3/2 x 20 = 30

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Number of type A articles sold by company R = 40% x (2 x 45 + M) = 40% x (90+M)

Number of type A articles sold by company T = 25% x (2 x 3 x 30 + M) = 25% x (180 + M)

Now,

40% x (90 + M) = 25% x (180 + M)

720 + 8M = 900 + 5M

3M = 180

Value of M = 60

Answer: E

Required value = 2 x 30 + 3 x 20 + 4 x 60 = 360

Hence answer is option E

6. Questions

Length of train P = length of platform

So, the time taken by train P to cross a platform = 2 x time taken by train P to cross a car

Z/360 x (Total time taken by all trains to cross a platform = 2 x Z/360 x (total time taken by all trains together to cross a car

Total time taken to cross the platform (2a) = 2 x (total time taken to cross the car)

Length of train S = speed of train S x M% x a…… (1)

Length of train S + length of platform = speed of train S x 1.5M% x 2a…….... (2)

From (1) and (2), we get

Length of train S/ (length of train S + Length of platform) = 1/3

Length of train S = 200 meters

So, length of platform = (3 – 1) x 200 = 400 meters

Since , the time taken to cover (length of train T + length of platform) – Length of train T = 10 seconds

Therefore, the time taken by cover 400 meters with the speed of train T = 4 seconds.

So, speed of train T = 400/10 = 40 m/s

Also,

69/360 x 2a – 108/360 x a = 10

So, a/12 = 10

So, a = 120

Time taken by train Q to cross the platform = (Y/360) x 2 x 120 = 2Y/3

So, (2Y/3) = Y – 27

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Y = 81

Now,

Z + [(Z/2) /100] x 360 + M/100 x 360 = 360 – 108 – 96 = 156

7Z + 9M = 390……. (3)

Also,

Z + (2.5M/100) x 360 = 360 – 81 – 69 = 210

Z + 9M = 210……. (4)

From (3) and (4), we get

6Z = 180

So, value of Z = 30

Value of M = 20,

So, the time taken by train P to cross the car = 30/360 x 120 = 10 seconds

So, time taken by train Q to cross the car = 15% x 120 = 18 seconds

Time taken by train S to cross the car = 20% x 120 = 24

Now, we can find all the related data

Name of the train	Length of train (meters)	Speed of train (m/s)
P	400	40
Q	200	100/9
R	800	25
S	200	25/3
T	1440	40

Answer: B

According to question,

(200 + 800) = 45/2 x (100/9 + speed of train A)

So, speed of train A = 100/3 x 18/5 = 120 km/h

Hence answer is option B

7. Questions

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Length of train P = length of platform

So, the time taken by train P to cross a platform = 2 x time taken by train P to cross a car

Z/360 x (Total time taken by all trains to cross a platform = 2 x Z/360 x (total time taken by all trains together to cross a car

Total time taken to cross the platform (2a) = 2 x (total time taken to cross the car)

Length of train S = speed of train S x M% x a…… (1)

Length of train S + length of platform = speed of train S x 1.5M% x 2a…….... (2)

From (1) and (2), we get

Length of train S/ (length of train S + Length of platform) = 1/3

Length of train S = 200 meters

So, length of platform = (3 – 1) x 200 = 400 meters

Since , the time taken to cover (length of train T + length of platform) – Length of train T = 10 seconds

Therefore, the time taken by cover 400 meters with the speed of train T = 4 seconds.

So, speed of train T = 400/10 = 40 m/s

Also,

69/360 x 2a – 108/360 x a = 10

So, a/12 = 10

So, a = 120

Time taken by train Q to cross the platform = (Y/360) x 2 x 120 = 2Y/3

So, (2Y/3) = Y – 27

Y = 81

Now,

Z + [(Z/2) /100] x 360 + M/100 x 360 = 360 – 108 – 96 = 156

7Z + 9M = 390……. (3)

Also,

Z + (2.5M/100) x 360 = 360 – 81 – 69 = 210

Z + 9M = 210……. (4)

From (3) and (4), we get

6Z = 180

So, value of Z = 30

Value of M = 20,

So, the time taken by train P to cross the car = 30/360 x 120 = 10 seconds

So, time taken by train Q to cross the car = 15% x 120 = 18 seconds

Time taken by train S to cross the car = 20% x 120 = 24

Now, we can find all the related data

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Name of the train	Length of train (meters)	Speed of train (m/s)
P	400	40
Q	200	100/9
R	800	25
S	200	25/3
T	1440	40

Answer: A

Required time = (800 + 200) / (25 – 25/3) = 60 seconds

Hence answer is option A

8. Questions

Length of train P = length of platform

So, the time taken by train P to cross a platform = 2 x time taken by train P to cross a car

Z/360 x (Total time taken by all trains to cross a platform = 2 x Z/360 x (total time taken by all trains together to cross a car

Total time taken to cross the platform (2a) = 2 x (total time taken to cross the car)

Length of train S = speed of train S x M% x a…… (1)

Length of train S + length of platform = speed of train S x 1.5M% x 2a…….... (2)

From (1) and (2), we get

Length of train S/ (length of train S + Length of platform) = 1/3

Length of train S = 200 meters

So, length of platform = (3 – 1) x 200 = 400 meters

Since , the time taken to cover (length of train T + length of platform) – Length of train T = 10 seconds

Therefore, the time taken by cover 400 meters with the speed of train T = 4 seconds.

So, speed of train T = 400/10 = 40 m/s

Also,

69/360 x 2a – 108/360 x a = 10

So, a/12 = 10

So, a = 120

Time taken by train Q to cross the platform = (Y/360) x 2 x 120 = 2Y/3

So, (2Y/3) = Y – 27

Y = 81

Now,

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Z + [(Z/2) /100] x 360 + M/100 x 360 = 360 – 108 – 96 = 156

7Z + 9M = 390……. (3)

Also,

Z + (2.5M/100) x 360 = 360 – 81 – 69 = 210

Z + 9M = 210……. (4)

From (3) and (4), we get

6Z = 180

So, value of Z = 30

Value of M = 20,

So, the time taken by train P to cross the car = 30/360 x 120 = 10 seconds

So, time taken by train Q to cross the car = 15% x 120 = 18 seconds

Time taken by train S to cross the car = 20% x 120 = 24

Now, we can find all the related data

Name of the train	Length of train (meters)	Speed of train (m/s)
P	400	40
Q	200	100/9
R	800	25
S	200	25/3
T	1440	40

Answer: C

Value of A = 400

Value of B = 200 x 2 = 400

9. Questions

Length of train P = length of platform

So, the time taken by train P to cross a platform = 2 x time taken by train P to cross a car

Z/360 x (Total time taken by all trains to cross a platform = 2 x Z/360 x (total time taken by all trains together to cross a car

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Total time taken to cross the platform (2a) = 2 x (total time taken to cross the car)

Length of train S = speed of train S x M% x a…… (1)

Length of train S + length of platform = speed of train S x 1.5M% x 2a……... (2)

From (1) and (2), we get

Length of train S/ (length of train S + Length of platform) = 1/3

Length of train S = 200 meters

So, length of platform = (3 – 1) x 200 = 400 meters

Since , the time taken to cover (length of train T + length of platform) – Length of train T = 10 seconds

Therefore, the time taken by cover 400 meters with the speed of train T = 4 seconds.

So, speed of train T = 400/10 = 40 m/s

Also,

69/360 x 2a – 108/360 x a = 10

So, a/12 = 10

So, a = 120

Time taken by train Q to cross the platform = (Y/360) x 2 x 120 = 2Y/3

So, (2Y/3) = Y – 27

Y = 81

Now,

Z + [(Z/2) /100] x 360 + M/100 x 360 = 360 – 108 – 96 = 156

7Z + 9M = 390……. (3)

Also,

Z + (2.5M/100) x 360 = 360 – 81 – 69 = 210

Z + 9M = 210……. (4)

From (3) and (4), we get

6Z = 180

So, value of Z = 30

Value of M = 20,

So, the time taken by train P to cross the car = 30/360 x 120 = 10 seconds

So, time taken by train Q to cross the car = 15% x 120 = 18 seconds

Time taken by train S to cross the car = 20% x 120 = 24

Now, we can find all the related data

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Name of the train	Length of train (meters)	Speed of train (m/s)
P	400	40
Q	200	100/9
R	800	25
S	200	25/3
T	1440	40

Answer: D

Required difference = 25 – 100/9 = 125/9 x 18/5 = 50 km/h

Hence answer is option D

10. Questions

Length of train P = length of platform

So, the time taken by train P to cross a platform = 2 x time taken by train P to cross a car

Z/360 x (Total time taken by all trains to cross a platform = 2 x Z/360 x (total time taken by all trains together to cross a car

Total time taken to cross the platform (2a) = 2 x (total time taken to cross the car)

Length of train S = speed of train S x M% x a…… (1)

Length of train S + length of platform = speed of train S x 1.5M% x 2a…….... (2)

From (1) and (2), we get

Length of train S/ (length of train S + Length of platform) = 1/3

Length of train S = 200 meters

So, length of platform = (3 – 1) x 200 = 400 meters

Since , the time taken to cover (length of train T + length of platform) – Length of train T = 10 seconds

Therefore, the time taken by cover 400 meters with the speed of train T = 4 seconds.

So, speed of train T = 400/10 = 40 m/s

Also,

69/360 x 2a – 108/360 x a = 10

So, a/12 = 10

So, a = 120

Time taken by train Q to cross the platform = (Y/360) x 2 x 120 = 2Y/3

So, (2Y/3) = Y – 27

Y = 81

Now,

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Z + [(Z/2) /100] x 360 + M/100 x 360 = 360 – 108 – 96 = 156

7Z + 9M = 390……. (3)

Also,

Z + (2.5M/100) x 360 = 360 – 81 – 69 = 210

Z + 9M = 210……. (4)

From (3) and (4), we get

6Z = 180

So, value of Z = 30

Value of M = 20,

So, the time taken by train P to cross the car = 30/360 x 120 = 10 seconds

So, time taken by train Q to cross the car = 15% x 120 = 18 seconds

Time taken by train S to cross the car = 20% x 120 = 24

Now, we can find all the related data

Name of the train	Length of train (meters)	Speed of train (m/s)
P	400	40
Q	200	100/9
R	800	25
S	200	25/3
T	1440	40

Answer: B

I). length of train T is 1440 meters (true)

II). speed of train P and train T are the same

Speed of train P = speed of train T = 40 m/s (true)

III). Value of Y is 170% more than that of Z.

Required % change = (81 – 30)/30 x 100 = 170% (true)

Hence answer is option B

11. Questions

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Total number of officers in station P = 50 x 4 = 200

Total number of officers in station Q = 90 x 4 = 360

Total number of officers in station R = 60 x 4 = 240

Total number of officers in station S = 200 x 4 = 800

Total number of officers in station T = 100 x 4 = 400

Total number of male officers = 14400

Number of male officers in each station > Number of female officers in each station

Ratio of male to female officer in station Q = 2: Y

So, only possible value of Y = 1

So, male officers in station Q = 2/3 x 360 = 240

Male officers in station R = 5/6 x 240 = 200

Male officers in station T = 3/4 x 400 = 300

So, male officers in station(P + S) = 1440 – 240 – 300 – 200 = 700

7/ (7 + Z) x [200 + 800] = 700

10 = 7 + Z

So, value of Z = 3

So, male officers in station P = 7/10 x 200 = 140

So, male officers in station Q = 7/10 x 800 = 560

Name of police station	Total officers	Number of male officers	Number of female officers
P	200	140	60
Q	360	240	120
R	240	200	40
S	800	560	240
T	400	300	100
Total	2000	1440	560

Answer: C

According to the question,

For station T,

Number of Rank A female officers = 40% x 100 = 40

Number of Rank A male officers = 100 – 40 = 60

So, number of Rank B female officers = 100 – 40 = 60

So, required difference = 60 – 60 = 0

Hence answer is option C

12. Questions

Total number of officers in station P = 50 x 4 = 200

Total number of officers in station Q = 90 x 4 = 360

Total number of officers in station R = 60 x 4 = 240

Total number of officers in station S = 200 x 4 = 800

Total number of officers in station T = 100 x 4 = 400

Total number of male officers = 14400

Number of male officers in each station > Number of female officers in each station

Ratio of male to female officer in station Q = 2: Y

So, only possible value of Y = 1

So, male officers in station Q = 2/3 x 360 = 240

Male officers in station R = 5/6 x 240 = 200

Male officers in station T = 3/4 x 400 = 300

So, male officers in station(P + S) = 1440 – 240 – 300 – 200 = 700

7/ (7 + Z) x [200 + 800] = 700

10 = 7 + Z

So, value of Z = 3

So, male officers in station P = 7/10 x 200 = 140

So, male officers in station Q = 7/10 x 800 = 560

Name of police station	Total officers	Number of male officers	Number of female officers
P	200	140	60
Q	360	240	120
R	240	200	40
S	800	560	240
T	400	300	100
Total	2000	1440	560

Answer: D

After the promotion of Rank B officers into Rank A officers in station R, the number of Rank A officers = 60 + 20% x 180 = 96

Number of female officers in station P = 60

Required % = [60 / (60 + 96)] x 100 = 38.4% = 37.5% (approx.., nearest).

Hence answer is option D

13. Questions

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Total number of officers in station P = 50 x 4 = 200

Total number of officers in station Q = 90 x 4 = 360

Total number of officers in station R = 60 x 4 = 240

Total number of officers in station S = 200 x 4 = 800

Total number of officers in station T = 100 x 4 = 400

Total number of male officers = 14400

Number of male officers in each station > Number of female officers in each station

Ratio of male to female officer in station Q = 2: Y

So, only possible value of Y = 1

So, male officers in station Q = 2/3 x 360 = 240

Male officers in station R = 5/6 x 240 = 200

Male officers in station T = 3/4 x 400 = 300

So, male officers in station(P + S) = 1440 – 240 – 300 – 200 = 700

7/ (7 + Z) x [200 + 800] = 700

10 = 7 + Z

So, value of Z = 3

So, male officers in station P = 7/10 x 200 = 140

So, male officers in station Q = 7/10 x 800 = 560

Name of police station	Total officers	Number of male officers	Number of female officers
P	200	140	60
Q	360	240	120
R	240	200	40
S	800	560	240
T	400	300	100
Total	2000	1440	560

Answer: B

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Number of Rank A officers in station U = 90/2 = 45

Number of Rank B officers in station U= 180 x 120% = 216

Required number of the total officers = 216 + 45 = 261

Hence answer is option B

14. Questions

Total number of officers in station P = 50 x 4 = 200

Total number of officers in station Q = 90 x 4 = 360

Total number of officers in station R = 60 x 4 = 240

Total number of officers in station S = 200 x 4 = 800

Total number of officers in station T = 100 x 4 = 400

Total number of male officers = 14400

Number of male officers in each station > Number of female officers in each station

Ratio of male to female officer in station Q = 2: Y

So, only possible value of Y = 1

So, male officers in station Q = 2/3 x 360 = 240

Male officers in station R = 5/6 x 240 = 200

Male officers in station T = 3/4 x 400 = 300

So, male officers in station(P + S) = 1440 – 240 – 300 – 200 = 700

7/ (7 + Z) x [200 + 800] = 700

10 = 7 + Z

So, value of Z = 3

So, male officers in station P = 7/10 x 200 = 140

So, male officers in station Q = 7/10 x 800 = 560

Name of police station	Total officers	Number of male officers	Number of female officers
P	200	140	60
Q	360	240	120
R	240	200	40
S	800	560	240
T	400	300	100
Total	2000	1440	560

Answer: A

Quantity I = 60 + 40 = 100

Quantity II = 30% x (560 – 240) = 96

So, Quantity I > Quantity II

Hence answer is option A

15. Questions

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Total number of officers in station P = 50 x 4 = 200

Total number of officers in station Q = 90 x 4 = 360

Total number of officers in station R = 60 x 4 = 240

Total number of officers in station S = 200 x 4 = 800

Total number of officers in station T = 100 x 4 = 400

Total number of male officers = 14400

Number of male officers in each station > Number of female officers in each station

Ratio of male to female officer in station Q = 2: Y

So, only possible value of Y = 1

So, male officers in station Q = 2/3 x 360 = 240

Male officers in station R = 5/6 x 240 = 200

Male officers in station T = 3/4 x 400 = 300

So, male officers in station(P + S) = 1440 – 240 – 300 – 200 = 700

7/ (7 + Z) x [200 + 800] = 700

10 = 7 + Z

So, value of Z = 3

So, male officers in station P = 7/10 x 200 = 140

So, male officers in station Q = 7/10 x 800 = 560

Name of police station	Total officers	Number of male officers	Number of female officers
P	200	140	60
Q	360	240	120
R	240	200	40
S	800	560	240
T	400	300	100
Total	2000	1440	560

Answer: D

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I). The total number of officers in station R is the same as the female officers in station S

Total number of officers in station R = Female officers in station S = 240

This statement is true.

II). The respective ratio of the difference between male and female officer in station P and male officers in station Q = 1:3

Required ratio = (140 – 60): 240 = 1:3

This statement is true.

III). (Z + Y) is a perfect square.

Value of (Z + Y) = (3 + 1) = 4 (perfect square)

Hence answer is option D

16. Questions

Answer: B

Time taken by boat P and Q when they meet for the first time = 70 / (Z + M + Y – M) = 70 / (Z + Y)

Speed of boat R in still water = (Z + Y)/2

Required distance = (Z + Y)/2 x 4 x 70 / (Z + Y) = 140 km

Hence answer is option B

17. Questions

Answer: D

80% x selling price of P = 96% x Cost price of P

CP/SP = 5/6

So, Z% = 1/5 x 100 = 20%

Cost price of article Q = Rs. 1200

Marked price of article Q = 2 x 1200 = Rs. 2400

Now,

2400 x (100 – 20 – 10) % x (100 – Y) % = 1344

(100 – Y) % = 80%

So, value of Y = 20%

I). Z = Y

Value of Z = Y = 20 (True)

II). (Z + Y) > 40

(Z + Y) = 40 (False)

III). unit digit of $(Z + Y/4)^{Z}$ is 5.

Required unit digit = $(20 + 20/4)^{20}$ = 5 (true)

So, only I and III are true

Hence answer is option D

18. Questions

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Answer: A

According to the question,

4 x (Z – 2) x Y/ 2 x 3 x Z x 2Y = 1/4

4Z – 8 = 3Z

Value of Z = 8 cm

So, 2 x 3 x 8 x 2Y = 480

Value of Y = 5

Length of rectangular field = 8 + 4 = 12 cm

Breadth of rectangular field = 12/2 = 6 cm

Required cost = 2 x (12 + 6) x (5 + 4) = Rs. 324

Hence answer is option A

19. Questions

Answer: A

Red balls = 3

Ratio of green ball to yellow ball = 3:2

I). 18, 3/44

Number of green balls = 18

Number of yellow balls = 18 x 2/3 = 12

Required probability = ${}^{3}C_1$ x ${}^{12}c_1$ / ${}^{33}C_2$ = 3/44

II). 3, 3/28

Number of green balls = 3

Number of yellow balls = 3 x 2/3 = 2

Required probability = ${}^{3}C_1$ x ${}^{2}c_1$ / ${}^{8}C_2$ = 3/14

III). 9, 3/20

Number of green balls = 9

Number of yellow balls = 9 x 2/3 = 6

Required probability = ${}^{3}C_1$ x ${}^{6}c_1$ / ${}^{18}C_2$ = 2/17

So, only I and III is true.

Hence answer is option A

20. Questions

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Answer: C

R + (P + Q)/2 = 45

2R + P + Q = 90…….... (1)

Q + (P + 4 + R + 4)/2 = 55.5

2Q + P + R = 103…….... (2)

Q – R = 13

Q = R + 13……. (3)

From (1) and (3), we get

3R + 13 + P = 90

3R + P = 77

For the minimum possible sum of the ages of Q and R, Age of R should also be minimum. And minimum possible value of R is 1

So, R = 1, P = 74

So, Q = 13 + 1 = 14

Required sum = 14 + 1 = 15

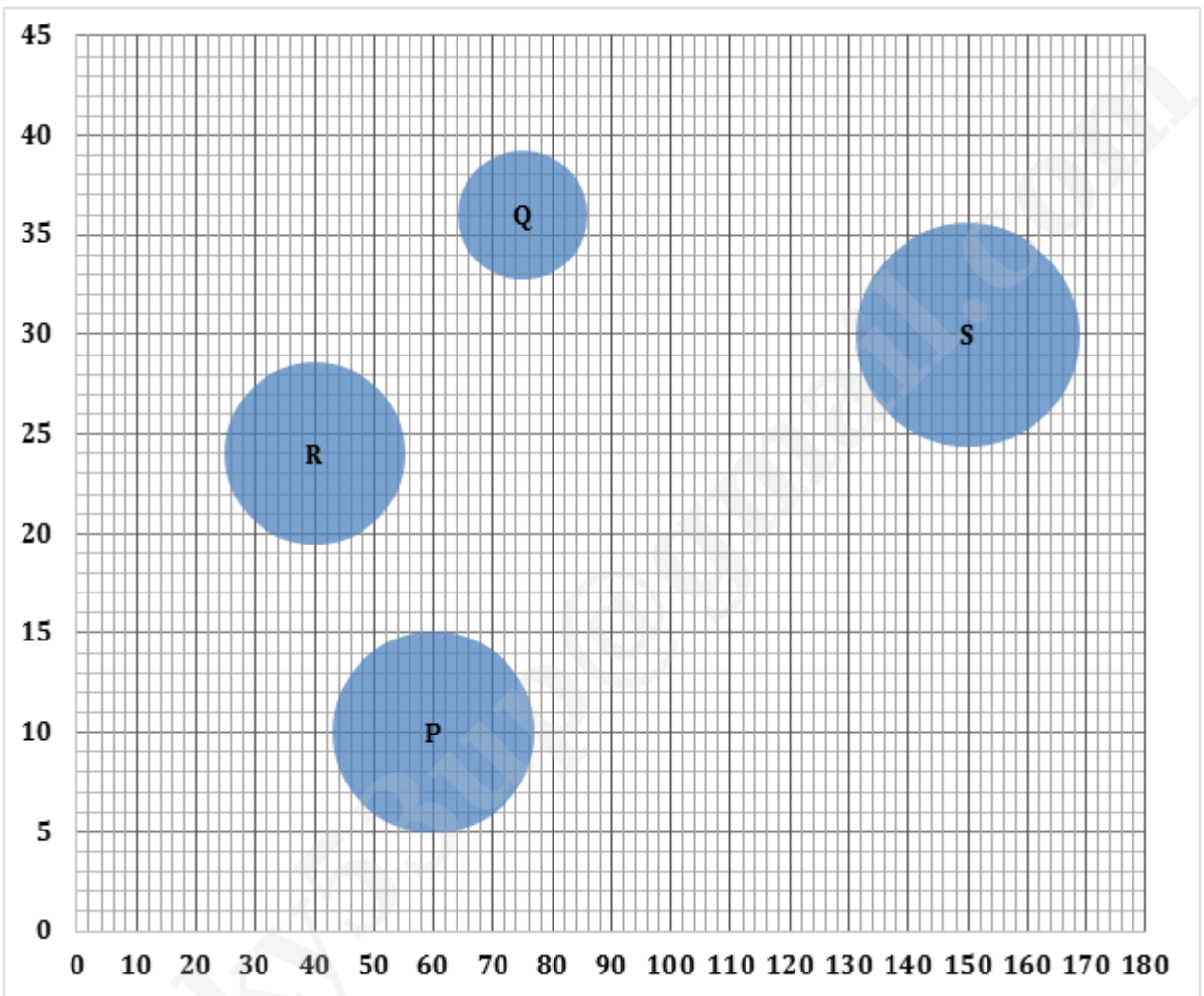
Hence answer is option C

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1. Questions

Read the following information and answer the questions based on it.

The chart given below shows partial information about the number of employees in two departments – A and B of four different companies – P, Q, R, and S. The point on the X axis shows the number of employees in department A of a company as a % of employees in department B of the same company, while the point on the Y axis shows the % of female employees in department A of a company out of the total number of female employees in department A of all the companies together.



Name of the company	Difference between males and Females in Department A	Males in Department B
P	20	80
Q	60	100
R	M	480
S	450	400

Note:

a). The number of females in department A of company S is 150, while the number of male and female employees in department A of company R is the same.

b). The number of males in department A of company Q is less than that of females in the same department, while for the other companies this statement is not true.

For company Q, in the next year, the number of male employees in department A increased by 50%, then the number of male employees in department A of company Q becomes Z.

Which of the following can be the possible value(s) of Z.

I). The number of female employees in department A of company Q in the given year.

II). 40% of the female employees in department B of company Q in the given year.

III). 80% more than the number of female employees in department B of company S in the given year.

a. II and III only
b. I and III only
c. I and II only
d. I, II, and III
e. None of these

2. Questions

The average number of female employees in department B of all the companies together is Z, while 80% of the total number of female employees in department A of all the companies together is Y. Find (Z + 40) is how much % less than that of (Y – 80)?

a. 37.5%
b. 40%
c. 60%
d. 62.5%
e. None of these

3. Questions

For company T, the number of female employees in department A is the same as the difference between female employees in both departments of company S. The number of male employees in department A is twice that of female employees in department B. Find the minimum possible number of male employees in department A, if the number of male employees in department B is not less than 250 and the total number of employees in department A is 50% less than that of B.

a. 100
b. 80
c. 75
d. 50
e. None of these

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4. Questions

Quantity I:

The total number of female employees in all the companies together

Quantity II:

The total number of employees in company S.

a. Quantity I > Quantity II

b. Quantity I < Quantity II

c. Quantity I = Quantity II

d. Quantity I ≥ Quantity II

e. None of these

5. Questions

For company S, if one third of the number of female employees in department A shifted to department B, and out of the total number of male employees in department B, 90% are non–managers. Find the number of female non – managers in department B of the company, if the total number of managers in department B of the company is the same as the males in department A of company P?

a. 70

b. 80

c. 150

d. 90

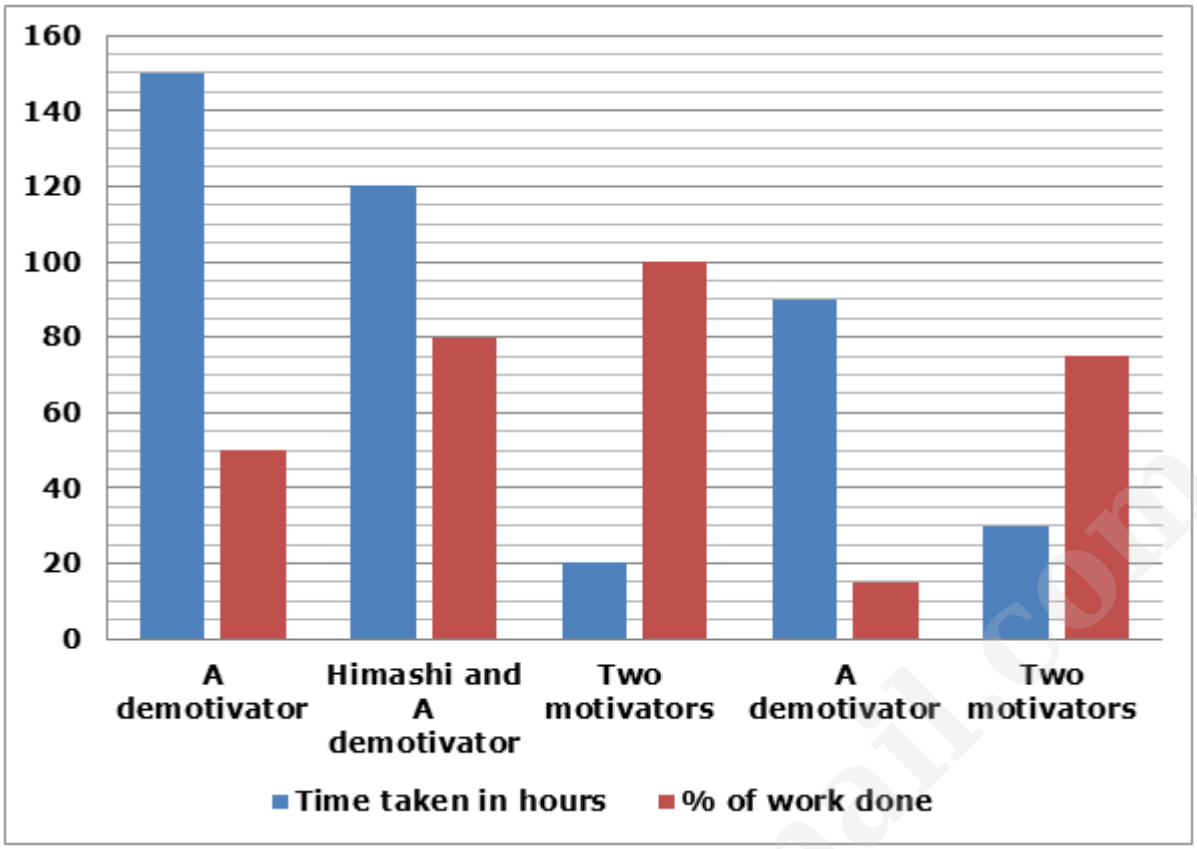
e. None of these

6. Questions

Read the following information carefully and answer the questions based on it.

There are two teams – Motivators (Arun, Minakshi, and Himanshi) and Demotivators – (P and Q). The chart given below shows the time taken in hours by some pairs to complete certain % of work. All members have different capabilities.

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Note:

The rate of destroying the work of Q is more than that of P, while the time taken by Arun to complete the work alone is less than any other member to complete or demolish the work alone. Arun worked on only one of the above works given in the bar.

A motivator is a positive worker, while a demotivator is a negative worker. The time taken by Himanshi to complete the whole work is > 100 hours

Quantity I:

If Q works at five times its efficiency, find the time taken to complete the whole work by Q and Arun together.

Quantity II:

If P works at five times its efficiency, find the time taken by P alone to demolish the work, which is 50% of the original work.

a. Quantity I < Quantity II

b. Quantity I > Quantity II

c. Quantity I = Quantity II

d. Quantity I ≥ Quantity II

e. Quantity I ≤ Quantity II

7. Questions

Find minimum possible difference between the time taken by Minakshi and the time taken by Arun

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to complete the task alone.

a. 30 hours

b. 24 hours

c. 36 hours

d. 20 hours

e. None of these

8. Questions

All members of the motivator team started the work, and after four hours, one of the members is replaced by one of the members of the demotivator team. After 10 hours from the start of the work, another member of the motivator team is replaced by a member of the demotivator team, and the rest of the work was completed by one member of the motivator team and two members of the demotivator team. Find the minimum possible time to complete the work. (Q worked at thrice its original efficiency).

a. 22.5 hours

b. 23.5 hours

c. 22.2727 hours

d. 23.11 hours

e. None of these

9. Questions

The efficiency of Minakshi is Z% more or less than that of Himanshi. The time taken by Kabir to complete the same job alone is Z hours. If Kabir, P and Q started the work in alternate manner such that for one hour Kabir worked and the next hour both P and Q worked together, find how many hours it will take to complete the work.

a. 397 hours

b. 200 hours

c. 199 hours

d. 399 hours

e. None of these

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10. Questions

When all members of both teams are working together, find the time taken (approx.) to complete the job if the time taken by Arun alone to complete the task > 25 hours?

a. 16 hours

b. 20 hours

c. 18 hours

d. 17 hours

e. 19 hours

11. Questions

Read the following information carefully and answer the questions based on it.

There are five boats – A, B, C, D, and E cover some downstream and upstream distance.

Boat A: The time taken by the boat to travel a certain upstream distance is twice as much as the time taken by the boat to travel the same downstream distance. Half of the sum of twice the speed of the boat in still water and the speed of the stream is 7 km/h.

Boat B: The speed of the boat in still water is P km/h, and the speed of the stream is Q km/h. The time taken by the boat to travel 60Q km downstream distance is 50% of the time taken by the boat to travel 16P km upstream. Also, the value of (2P + 3Q) = 13, where P and Q are integers.

Boat C: The time taken by the boat to travel 90 km downstream (6 hours) is 60% less than the time taken by the boat to travel M km upstream distance. If the speed of the boat in still water becomes three times and the speed of the stream becomes four times, now the time taken by the boat to travel 100 km downstream is 2 hours.

Boat D: The time taken by the boat to travel (4R + 8) km upstream distance is three times that of the boat to travel 2.5R km downstream distance. Also, the time taken by the boat to travel (4R + 8) km upstream distance is 20% of the time taken by boat C to travel 450 km downstream distance. The speed of the boat in still water is R km/h, and the speed of the stream is S km/h.

Boat E: The ratio of the speed of the boat in still water to the speed of the stream is 3: 1, while the difference between the downstream speed and upstream speed is 10 km/h.

Find the time taken by boat E to travel (M – 10) km upstream distance?

a. 14 hours

b. 5 hours

c. 6.5 hours

d. 10.5 hours

e. None of these

12. Questions

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The time taken by boat B to travel the given downstream distance is T hours, while the time taken by boat A to travel the given upstream distance is 4T hours. Find the total given distance covered by boat A(both Upstream and Downstream)?

a. 80 km

b. 120 km

c. 200 km

d. 320 km

e. None of these

13. Questions

Which of the following statements is/are must be true.

I). The downstream speed of boats D and E is the same.

II). The total given distance covered by boat C, with the original speed of the boat in still water and the stream, is 80 km less than that of boat A.

III). The upstream speed of boat A is S km/h

a. I and III only

b. I and II only

c. II and III only

d. I, II, and III

e. None of these

14. Questions

Quantity I:

Find the time taken by the boat B to travel 45 km upstream distance when the speed of the boat in still water is reduced by 20% and the speed of the stream becomes twice.

Quantity II:

Find the time taken by boat E to travel (4R + 11Q) km upstream distance with the original speed of the boat in still water and stream.

Find the difference between Quantity I and Quantity II.

a. 15

b. 12

c. 18.75

d. 17.5

e. None of these

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15. Questions

Find the sum of the original speed in still water of all given boats together.

a. 56 km/h

b. 62 km/h

c. 48 km/h

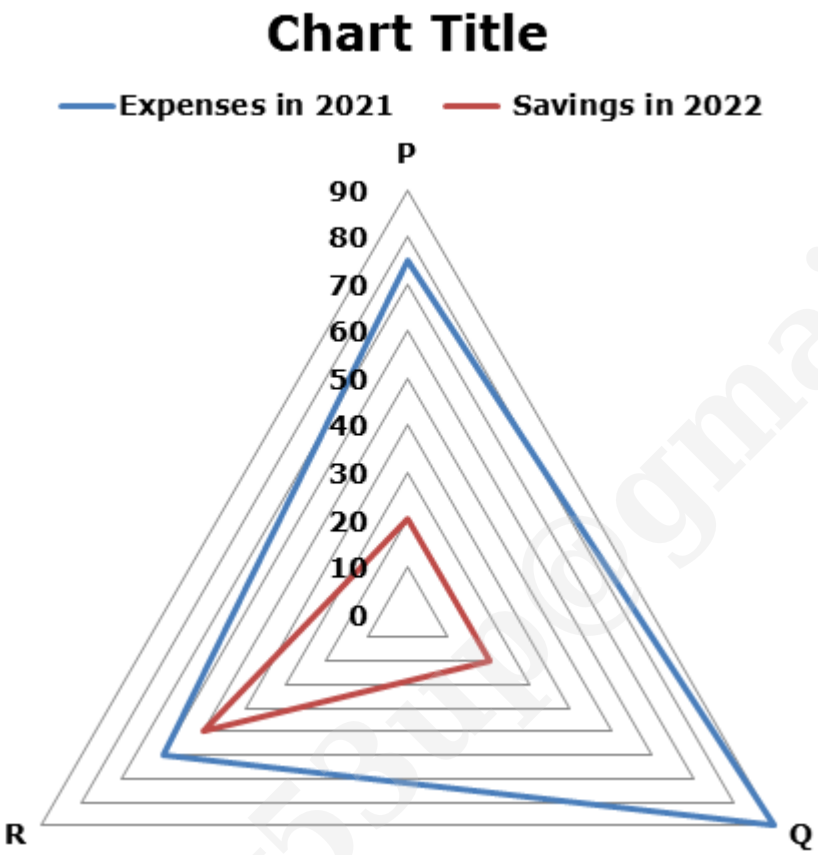
d. 54 km/h

e. None of these

16. Questions

Read the following information carefully and answer the questions based on it.

The chart given below shows the percentage of expenses and savings of three companies P, Q, and R in two different years – 2021 and 2022.



Savings of R in 2022 are 50% more than that in 2021, while the income of Q in 2022 is M% more than that in 2021. The average expense of both companies together in 2022 is Rs. 24Z more than that in 2021. Find the difference between the income Q in 2021 and the income Q in 2022 if the income of R in 2022 is M% more than that in 2021.

a. Rs. 144Z

b. Rs. 180Z

c. Rs. 160Z

d. Rs. 192Z

e. Cannot be determined

17. Questions

Expenses of P in 2022 are twice as much as those of Q in the same year. The difference between the

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expenses of P in 2021 and Q in 2022 is A, while the difference between the savings of P in 2021 and the expenses of Q in the same year is B. Find A is how much % more or less than that of B, if for each company income in both years is the same.

a. 40%

b. 75%

c. 62.5%

d. 32.5%

e. None of these

18. Questions

Expenses of R in 2021 are 60% less than those in 2022, while the expenses of P in 2022 are the average of the expenses of R in both years. The income of P in 2022 is 62.5% less than that in 2021. Find the income of R in 2021 is how much % more or less than one third of the expenses of P in the same year.

a. 12.5%

b. 32.5%

c. 42.85%

d. 14.28%

e. None of these

19. Questions

The average expenses of P in both years are Rs. 6200, and the income of R in 2021 is 50% more than that of P in 2022. Find the expenses of R in 2022 if the income of P in both years is the same, while the income of R in 2022 is 20% less than that in 2021?

a. Rs. 4800

b. Rs. 10800

c. Rs. 86400

d. Rs. 9720

e. None of these

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20. Questions

Expenses of Q in 2021 are Rs. 9000 more than the savings of R in 2022, while the savings of Q in 2022 are Rs. 3200 less than the expenses of R in 2021. The income of Q in both years is the same, while the income of R in 2022 is 50% more than that in 2021. Find the difference between the average income of Q in both years and the average income of R in both years.

a. Rs. 10000

b. Rs. 6000

c. Rs. 4800

d. Rs. 4000

e. None of these

Explanations:

1. Questions

Number of female employees in company S = 150 = (100 – 10 – 24 – 36) % x total number of female employees in department A of all the companies together

The total number of females in department A = 150/30 x 100 = 500

For company P

Ratio of the number of employees in department A to B = 3:5

Number of female employees in department A = 10% x 500 = 50

So, the number of male employees in department A = 50 + 20 = 70 or 50 – 20 = 30

So, the total number of employees in department A = 70 + 50 = 120 or 50 + 30 = 80 (80 is not possible)

So, the total number of employees in department B = 120 x 5/3 = 200

Number of male employees in department B = 80

So, the number of female employees in department B = 200 – 80 = 120

For company Q

Ratio of the number of employees in department A to B = 3:4

Number of female employees in department A = 36% x 500 = 180

So, the number of male employees in department A = (180 – 60) = 120

So, the total number of employees in department A = 120 + 180 = 300

So, the total number of employees in department B = 300 x 4/3 = 400

Number of male employees in department B = 100

So, the number of female employees in department B = 400 – 100 = 300

Similarly, we can find the data for the rest of the companies

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Name of the company	Department A (Male, Female)	Department B (Male, Female)	Total
P	120 [70, 50]	200 (80, 120)	320
Q	300 [120, 180]	400 (100, 300)	700
R	240 [120, 120]	600 (480, 120)	840
S	750 [600, 150]	500 (400, 100)	1250
Total	1410 [910, 500]	1700 [1060, 640]	3110

Answer: B

According to question,

Number of male employees in company Q in the next year = 120 x 150% = 180

I). Number of female employees in department A of company Q in the given year = 180

II). 40% of the number of female employees in department B of company Q in the given year = 120

III). 80% more than the number of female employees in department B of company S in the given year. [100 x 180% = 180]

Only I and III is true.

Hence answer is option B

2. Questions

Number of female employees in company S = 150 = (100 – 10 – 24 – 36) % x total number of female employees in department A of all the companies together

The total number of females in department A = 150/30 x 100 = 500

For company P

Ratio of the number of employees in department A to B = 3:5

Number of female employees in department A = 10% x 500 = 50

So, the number of male employees in department A = 50 + 20 = 70 or 50 – 20 = 30

So, the total number of employees in department A = 70 + 50 = 120 or 50 + 30 = 80 (80 is not possible)

So, the total number of employees in department B = 120 x 5/3 = 200

Number of male employees in department B = 80

So, the number of female employees in department B = 200 – 80 = 120

For company Q

Ratio of the number of employees in department A to B = 3:4

Number of female employees in department A = 36% x 500 = 180

So, the number of male employees in department A = (180 – 60) = 120

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So, the total number of employees in department A = 120 + 180 = 300

So, the total number of employees in department B = 300 x 4/3 = 400

Number of male employees in department B = 100

So, the number of female employees in department B = 400 – 100 = 300

Similarly, we can find the data for the rest of the companies

Name of the company	Department A (Male, Female)	Department B (Male, Female)	Total
P	120 [70, 50]	200 (80, 120)	320
Q	300 [120, 180]	400 (100, 300)	700
R	240 [120, 120]	600 (480, 120)	840
S	750 [600, 150]	500 (400, 100)	1250
Total	1410 [910, 500]	1700 [1060, 640]	3110

Answer: A

Value of Z =640/4 = 160

(Z + 40) = 200

Value of Y = 500*80/100

Y – 80 = 320

Required % change = (320 – 200)/320 x 100 = 37.5%

Hence answer is option A

3. Questions

Number of female employees in company S = 150 = (100 – 10 – 24 – 36) % x total number of female employees in department A of all the companies together

The total number of females in department A = 150/30 x 100 = 500

For company P

Ratio of the number of employees in department A to B = 3:5

Number of female employees in department A = 10% x 500 = 50

So, the number of male employees in department A = 50 + 20 = 70 or 50 – 20 = 30

So, the total number of employees in department A = 70 + 50 = 120 or 50 + 30 = 80 (80 is not possible)

So, the total number of employees in department B = 120 x 5/3 = 200

Number of male employees in department B = 80

So, the number of female employees in department B = 200 – 80 = 120

For company Q

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Ratio of the number of employees in department A to B = 3:4

Number of female employees in department A = 36% x 500 = 180

So, the number of male employees in department A = (180 – 60) = 120

So, the total number of employees in department A = 120 + 180 = 300

So, the total number of employees in department B = 300 x 4/3 = 400

Number of male employees in department B = 100

So, the number of female employees in department B = 400 – 100 = 300

Similarly, we can find the data for the rest of the companies

Name of the company	Department A (Male, Female)	Department B (Male, Female)	Total
P	120 [70, 50]	200 (80, 120)	320
Q	300 [120, 180]	400 (100, 300)	700
R	240 [120, 120]	600 (480, 120)	840
S	750 [600, 150]	500 (400, 100)	1250
Total	1410 [910, 500]	1700 [1060, 640]	3110

Answer: A

For company T,

Number of female employees in department A = 50

Minimum possible number of male employees in department B = 250

Let, the number of female employees in department B = a

So, the number of male employees in department A = 2 x a = 2a

Now,

(2a + 50) / (a + 250) = 1/2

4a + 100 = a + 250

Value of a = 50

So, the number of male employees in department A = 100

Hence answer is option A

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4. Questions

Number of female employees in company S = 150 = (100 – 10 – 24 – 36) % x total number of female employees in department A of all the companies together

The total number of females in department A = 150/30 x 100 = 500

For company P

Ratio of the number of employees in department A to B = 3:5

Number of female employees in department A = 10% x 500 = 50

So, the number of male employees in department A = 50 + 20 = 70 or 50 – 20 = 30

So, the total number of employees in department A = 70 + 50 = 120 or 50 + 30 = 80 (80 is not possible)

So, the total number of employees in department B = 120 x 5/3 = 200

Number of male employees in department B = 80

So, the number of female employees in department B = 200 – 80 = 120

For company Q

Ratio of the number of employees in department A to B = 3:4

Number of female employees in department A = 36% x 500 = 180

So, the number of male employees in department A = (180 – 60) = 120

So, the total number of employees in department A = 120 + 180 = 300

So, the total number of employees in department B = 300 x 4/3 = 400

Number of male employees in department B = 100

So, the number of female employees in department B = 400 – 100 = 300

Similarly, we can find the data for the rest of the companies

Name of the company	Department A (Male, Female)	Department B (Male, Female)	Total
P	120 [70, 50]	200 (80, 120)	320
Q	300 [120, 180]	400 (100, 300)	700
R	240 [120, 120]	600 (480, 120)	840
S	750 [600, 150]	500 (400, 100)	1250
Total	1410 [910, 500]	1700 [1060, 640]	3110

Answer: B

Quantity I = (640 + 500) = 1140

Quantity II = 750 + 500 = 1250

So, Quantity I < Quantity II

Hence answer is option B

5. Questions

Number of female employees in company S = 150 = (100 – 10 – 24 – 36) % x total number of female employees in department A of all the companies together

The total number of females in department A = 150/30 x 100 = 500

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For company P

Ratio of the number of employees in department A to B = 3:5

Number of female employees in department A = 10% x 500 = 50

So, the number of male employees in department A = 50 + 20 = 70 or 50 – 20 = 30

So, the total number of employees in department A = 70 + 50 = 120 or 50 + 30 = 80 (80 is not possible)

So, the total number of employees in department B = 120 x 5/3 = 200

Number of male employees in department B = 80

So, the number of female employees in department B = 200 – 80 = 120

For company Q

Ratio of the number of employees in department A to B = 3:4

Number of female employees in department A = 36% x 500 = 180

So, the number of male employees in department A = (180 – 60) = 120

So, the total number of employees in department A = 120 + 180 = 300

So, the total number of employees in department B = 300 x 4/3 = 400

Number of male employees in department B = 100

So, the number of female employees in department B = 400 – 100 = 300

Similarly, we can find the data for the rest of the companies

Name of the company	Department A (Male, Female)	Department B (Male, Female)	Total
P	120 [70, 50]	200 (80, 120)	320
Q	300 [120, 180]	400 (100, 300)	700
R	240 [120, 120]	600 (480, 120)	840
S	750 [600, 150]	500 (400, 100)	1250
Total	1410 [910, 500]	1700 [1060, 640]	3110

Answer: E

For company S, department B

Number of male employees = 400

Number of female employees = 100 + 1/3 x 150 = 150

Total number of managers = 70

Male managers = 10% x 400 = 40

So, female managers = 70 – 40 = 30

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So, female non–managers = 150 – 30 = 120

Hence answer is option E

6. Questions

One member of the demotivator team alone demolishes the work in = 150 x 2 = 300 hours

Another member of the demotivator team alone demolishes the work in = 90/15 x 100 = 600 hours

Q destroyed more work in 1 hour than P

So, time taken by Q alone = 300 hours

Time taken by P alone = 600 hours

Let, the total work = 600 units

So, rate of destroying of Q and P = 2 units and 1 unit per hours

Time taken by Himanshi and one demotivator together to complete the whole work = 120/80 x 100 = 150 hours

Efficiency of (Himanshi + One demotivator) = 600/150 = 4 units/ h

(Himanshi + Q or P) = 4 units/h

So, efficiency of Himanshi = 4 +1 or (4 + 2) = 5 units/h or 6 units/h

So, time taken by Himanshi alone to complete the work = 600/5 or 600/6 = 120 hours or 100 hours

So, time taken by Himanshi to complete the whole work alone = 120 hours

Two motivators, can complete the whole work in = 20 hours [efficiency = 30 units/hour]

Two motivators, can complete the whole work in = 30/75 x 100 = 40 hours [efficiency = 15 units/hour]

Arun is the most efficient compared to the other members. So, Arun is one of the two members who complete the work together in = 20 hours

So, (Himanshi + Minakshi) together must complete the work in = 40 hours

So, efficiency of Minakshi = 15 – 5 = 10 units/hr [Time taken = 600/10 = 60 hours

So, efficiency of Arun = (30 – 5) Or (30 – 10) = 25 or 20

So, time taken by Arun to complete the whole work = 600/25 OR 600/20 = 24 hours or 30 hours

Answer: E

According to the question,

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Quantity I:

Efficiency of Q = 2 x 5 = 10 units/hour

Efficiency of Arun = 25 or 20

So, Required time = 600 / (20 – 10) = 60 hours OR 600/ (25 – 10) = 40 hours

Quantity II:

Time taken by P = 50% x (600/5) = 60 hours

Quantity I ≤ Quantity II

Hence answer is option E

7. Questions

One member of the demotivator team alone demolishes the work in = 150 x 2 = 300 hours

Another member of the demotivator team alone demolishes the work in = 90/15 x 100 = 600 hours

Q destroyed more work in 1 hour than P

So, time taken by Q alone = 300 hours

Time taken by P alone = 600 hours

Let, the total work = 600 units

So, rate of destroying of Q and P = 2 units and 1 unit per hours

Time taken by Himanshi and one demotivator together to complete the whole work = 120/80 x 100 = 150 hours

Efficiency of (Himanshi + One demotivator) = 600/150 = 4 units/ h

(Himanshi + Q or P) = 4 units/h

So, efficiency of Himanshi = 4 +1 or (4 + 2) = 5 units/h or 6 units/h

So, time taken by Himanshi alone to complete the work = 600/5 or 600/6 = 120 hours or 100 hours

So, time taken by Himanshi to complete the whole work alone = 120 hours

Two motivators, can complete the whole work in = 20 hours [efficiency = 30 units/hour]

Two motivators, can complete the whole work in = 30/75 x 100 = 40 hours [efficiency = 15 units/hour]

Arun is the most efficient compared to the other members. So, Arun is one of the two members who complete the work together in = 20 hours

So, (Himanshi + Minakshi) together must complete the work in = 40 hours

So, efficiency of Minakshi = 15 – 5 = 10 units/hr [Time taken = 600/10 = 60 hours

So, efficiency of Arun = (30 – 5) Or (30 – 10) = 25 or 20

So, time taken by Arun to complete the whole work = 600/25 OR 600/20 = 24 hours or 30 hours

Answer: A

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Required minimum possible time = 60 – 30 = 30 hours.

Hence answer is option A

8. Questions

One member of the demotivator team alone demolishes the work in = 150 x 2 = 300 hours

Another member of the demotivator team alone demolishes the work in = 90/15 x 100 = 600 hours

Q destroyed more work in 1 hour than P

So, time taken by Q alone = 300 hours

Time taken by P alone = 600 hours

Let, the total work = 600 units

So, rate of destroying of Q and P = 2 units and 1 unit per hours

Time taken by Himanshi and one demotivator together to complete the whole work = 120/80 x 100 = 150 hours

Efficiency of (Himanshi + One demotivator) = 600/150 = 4 units/ h

(Himanshi + Q or P) = 4 units/h

So, efficiency of Himanshi = 4 +1 or (4 + 2) = 5 units/h or 6 units/h

So, time taken by Himanshi alone to complete the work = 600/5 or 600/6 = 120 hours or 100 hours

So, time taken by Himanshi to complete the whole work alone = 120 hours

Two motivators, can complete the whole work in = 20 hours [efficiency = 30 units/hour]

Two motivators, can complete the whole work in = 30/75 x 100 = 40 hours [efficiency = 15 units/hour]

Arun is the most efficient compared to the other members. So, Arun is one of the two members who complete the work together in = 20 hours

So, (Himanshi + Minakshi) together must complete the work in = 40 hours

So, efficiency of Minakshi = 15 – 5 = 10 units/hr [Time taken = 600/10 = 60 hours

So, efficiency of Arun = (30 – 5) Or (30 – 10) = 25 or 20

So, time taken by Arun to complete the whole work = 600/25 OR 600/20 = 24 hours or 30 hours

Answer: D

We need to find the minimum possible time, so Arun worked for the maximum time with more efficiency and Q worked for less hours.

Amount of work done in 10 hours = (5 + 10 + 25) x 4 + (10 + 25 – 1) x 6 = 364

Required time = 10 + (600 – 364)/ (25 – 1 – 2 x 3) = 23.11 hours

Hence answer is option D

9. Questions

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One member of the demotivator team alone demolishes the work in = 150 x 2 = 300 hours

Another member of the demotivator team alone demolishes the work in = 90/15 x 100 = 600 hours

Q destroyed more work in 1 hour than P

So, time taken by Q alone = 300 hours

Time taken by P alone = 600 hours

Let, the total work = 600 units

So, rate of destroying of Q and P = 2 units and 1 unit per hours

Time taken by Himanshi and one demotivator together to complete the whole work = 120/80 x 100 = 150 hours

Efficiency of (Himanshi + One demotivator) = 600/150 = 4 units/ h

(Himanshi + Q or P) = 4 units/h

So, efficiency of Himanshi = 4 +1 or (4 + 2) = 5 units/h or 6 units/h

So, time taken by Himanshi alone to complete the work = 600/5 or 600/6 = 120 hours or 100 hours

So, time taken by Himanshi to complete the whole work alone = 120 hours

Two motivators, can complete the whole work in = 20 hours [efficiency = 30 units/hour]

Two motivators, can complete the whole work in = 30/75 x 100 = 40 hours [efficiency = 15 units/hour]

Arun is the most efficient compared to the other members. So, Arun is one of the two members who complete the work together in = 20 hours

So, (Himanshi + Minakshi) together must complete the work in = 40 hours

So, efficiency of Minakshi = 15 – 5 = 10 units/hr [Time taken = 600/10 = 60 hours

So, efficiency of Arun = (30 – 5) Or (30 – 10) = 25 or 20

So, time taken by Arun to complete the whole work = 600/25 OR 600/20 = 24 hours or 30 hours

Answer: A

Value of Z = (10 – 5)/5 x 100 = 100%

So, time taken by Kabir to complete the work = 100 hours

Efficiency of Kabir = 600/100 = 6 units/hour

Amount of work done in first two hours = (6 – 3) = 3 units

Amount of work done in = 2 x 198 hours = 198 x 3 units

So, in 396 hours = 594 units done

So, amount of work done in = 396 + (600 – 594)/6 = 397 hours

Hence answer is option A

10. Questions

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One member of the demotivator team alone demolishes the work in = 150 x 2 = 300 hours

Another member of the demotivator team alone demolishes the work in = 90/15 x 100 = 600 hours

Q destroyed more work in 1 hour than P

So, time taken by Q alone = 300 hours

Time taken by P alone = 600 hours

Let, the total work = 600 units

So, rate of destroying of Q and P = 2 units and 1 unit per hours

Time taken by Himanshi and one demotivator together to complete the whole work = 120/80 x 100 = 150 hours

Efficiency of (Himanshi + One demotivator) = 600/150 = 4 units/ h

(Himanshi + Q or P) = 4 units/h

So, efficiency of Himanshi = 4 +1 or (4 + 2) = 5 units/h or 6 units/h

So, time taken by Himanshi alone to complete the work = 600/5 or 600/6 = 120 hours or 100 hours

So, time taken by Himanshi to complete the whole work alone = 120 hours

Two motivators, can complete the whole work in = 20 hours [efficiency = 30 units/hour]

Two motivators, can complete the whole work in = 30/75 x 100 = 40 hours [efficiency = 15 units/hour]

Arun is the most efficient compared to the other members. So, Arun is one of the two members who complete the work together in = 20 hours

So, (Himanshi + Minakshi) together must complete the work in = 40 hours

So, efficiency of Minakshi = 15 – 5 = 10 units/hr [Time taken = 600/10 = 60 hours

So, efficiency of Arun = (30 – 5) Or (30 – 10) = 25 or 20

So, time taken by Arun to complete the whole work = 600/25 OR 600/20 = 24 hours or 30 hours

Answer: E

Required time = 600 / (5 + 10 + 20 – 3) = 600/32 = 19 hours approx.

Hence answer is option E

11. Questions

For boat A

Ratio of upstream distance to downstream distance = 1:1

Ratio of upstream time to downstream time = 2:1

So, ratio of upstream speed to downstream speed = (1/2:1/1) = 1:2

So, ratio of the speed of the boat in still water to the speed of the stream = (1 +2)/2: (2 – 1)/2 = 3:1 = 3a, a

Now,

(2 x 3a + a)/2 = 7

7a = 14

Value of a = 2

So, speed of boat in still water and speed of stream = 6 km/hr and 2 km/hr

Downstream speed = 8 km/h

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Upstream speed = 4 km/h

Boat B

Time taken by the boat to travel 60Q downstream distance = 60Q/ (P +Q)

Now,

2 x 60Q / (P + Q) = 16P / (P – Q)

$2P^2 – 13PQ + 15Q^2 = 0$

(2P – 3Q) (P – 5Q) = 0

So, P = 5Q, and P = 3Q/2

Also,

2P + 3Q = 13

So, (2 x 5Q + 3Q) = 13

Value of Q = 1

Value of P = 5

If we put P = 3Q/2, we do not get an integral value.

For boat C

Time taken by the boat to travel 90 km downstream = 6 hours

So, downstream speed of boat = 90/6 = 15

(a + b) = 15…….... (1)

Also,

100/2 = (3 x a + 4 x b)

3a + 4b = 50……. (2)

On, solving both equations, we get

Value of b = 5, value of a = 10

so, upstream speed of boat = 10 – 5 = 5 km/h

upstream time taken = 6/40 x 100 = 15 hours

value of M = 5 x 15 =75 km

Boat D

Time taken by the boat to travel (4R + 8) km upstream = 20% x 450/15 = 6 hours

(4R + 8) = (R - S) x 6

2R + 4 = 3R - 3S

R – 3S = 4……. (1)

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Also,

2.5R = (R + S) x 6/3

R = 4S…….. (2)

On, solving (1) and (2), we get

4S – 3S = 4

Value of S = 4

Value of R = 16

Downstream distance = 2.5 x 16 = 40 km

Upstream distance = 4 x 16 + 8 = 72 km

Boat E

Ratio of the speed of the boat in still water to the speed of the stream = 3:1

So, ratio of upstream speed to downstream speed = 2:1

So, upstream speed = 1 x 10 = 10 km/h

Downstream speed = 2 x 10 = 20 km/h

So, speed of boat in still water = 15 km/h

Speed of stream = 5 km/h

Answer: C

According to the question,

For boat E,

Upstream speed of boat = 10 km/h

Required time = (75– 10)/10= 6.5 hours

Hence answer is option C

12. Questions

For boat A

Ratio of upstream distance to downstream distance = 1:1

Ratio of upstream time to downstream time = 2:1

So, ratio of upstream speed to downstream speed = (1/2:1/1) = 1:2

So, ratio of the speed of the boat in still water to the speed of the stream = (1 +2)/2: (2 – 1)/2 = 3:1 = 3a, a

Now,

(2 x 3a + a)/2 = 7

7a = 14

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Value of a = 2

So, speed of boat in still water and speed of stream = 6 km/hr and 2 km/hr

Downstream speed = 8 km/h

Upstream speed = 4 km/h

Boat B

Time taken by the boat to travel 60Q downstream distance = 60Q/ (P +Q)

Now,

2 x 60Q / (P + Q) = 16P / (P – Q)

$2P^2 – 13PQ + 15Q^2 = 0$

(2P – 3Q) (P – 5Q) = 0

So, P = 5Q, and P = 3Q/2

Also,

2P + 3Q = 13

So, (2 x 5Q + 3Q) = 13

Value of Q = 1

Value of P = 5

If we put P = 3Q/2, we do not get an integral value.

For boat C

Time taken by the boat to travel 90 km downstream = 6 hours

So, downstream speed of boat = 90/6 = 15

(a + b) = 15…….... (1)

Also,

100/2 = (3 x a + 4 x b)

3a + 4b = 50……. (2)

On, solving both equations, we get

Value of b = 5, value of a = 10

so, upstream speed of boat = 10 – 5 = 5 km/h

upstream time taken = 6/40 x 100 = 15 hours

value of M = 5 x 15 =75 km

Boat D

Time taken by the boat to travel (4R + 8) km upstream = 20% x 450/15 = 6 hours

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(4R + 8) = (R - S) x 6

2R + 4 = 3R - 3S

R – 3S = 4……. (1)

Also,

2.5R = (R + S) x 6/3

R = 4S…….. (2)

On, solving (1) and (2), we get

4S – 3S = 4

Value of S = 4

Value of R = 16

Downstream distance = 2.5 x 16 = 40 km

Upstream distance = 4 x 16 + 8 = 72 km

Boat E

Ratio of the speed of the boat in still water to the speed of the stream = 3:1

So, ratio of upstream speed to downstream speed = 2:1

So, upstream speed = 1 x 10 = 10 km/h

Downstream speed = 2 x 10 = 20 km/h

So, speed of boat in still water = 15 km/h

Speed of stream = 5 km/h

Answer: D

Value of T = 60 x 1 / (5 + 1) = 10

Time taken by boat A to travel the given upstream distance = (6 – 2) x 4 x 10 = 160 km

Required total distance covered by boat A = 160 + 160 = 320 km

Hence answer is option D

13. Questions

For boat A

Ratio of upstream distance to downstream distance = 1:1

Ratio of upstream time to downstream time = 2:1

So, ratio of upstream speed to downstream speed = (1/2:1/1) = 1:2

So, ratio of the speed of the boat in still water to the speed of the stream = (1 +2)/2: (2 – 1)/2 = 3:1 = 3a, a

Now,

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$(2 \times 3a + a)/2 = 7$

$7a = 14$

Value of $a = 2$

So, speed of boat in still water and speed of stream = 6 km/hr and 2 km/hr

Downstream speed = 8 km/h

Upstream speed = 4 km/h

Boat B

Time taken by the boat to travel 60Q downstream distance = 60Q/ (P +Q)

Now,

$2 \times 60Q / (P + Q) = 16P / (P - Q)$

$2P^2 - 13PQ + 15Q^2 = 0$

$(2P - 3Q) (P - 5Q) = 0$

So, $P = 5Q$, and $P = 3Q/2$

Also,

$2P + 3Q = 13$

So, $(2 \times 5Q + 3Q) = 13$

Value of $Q = 1$

Value of $P = 5$

If we put $P = 3Q/2$, we do not get an integral value.

For boat C

Time taken by the boat to travel 90 km downstream = 6 hours

So, downstream speed of boat = 90/6 = 15

$(a + b) = 15$…….. (1)

Also,

$100/2 = (3 \times a + 4 \times b)$

$3a + 4b = 50$……. (2)

On, solving both equations, we get

Value of $b = 5$, value of $a = 10$

so, upstream speed of boat = 10 – 5 = 5 km/h

upstream time taken = 6/40 x 100 = 15 hours

value of M = 5 x 15 =75 km

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Boat D

Time taken by the boat to travel (4R + 8) km upstream = 20% x 450/15 = 6 hours

(4R + 8) = (R - S) x 6

2R + 4 = 3R - 3S

R – 3S = 4……. (1)

Also,

2.5R = (R + S) x 6/3

R = 4S…….. (2)

On, solving (1) and (2), we get

4S – 3S = 4

Value of S = 4

Value of R = 16

Downstream distance = 2.5 x 16 = 40 km

Upstream distance = 4 x 16 + 8 = 72 km

Boat E

Ratio of the speed of the boat in still water to the speed of the stream = 3:1

So, ratio of upstream speed to downstream speed = 2:1

So, upstream speed = 1 x 10 = 10 km/h

Downstream speed = 2 x 10 = 20 km/h

So, speed of boat in still water = 15 km/h

Speed of stream = 5 km/h

Answer: A

I). The downstream speed of boats D and E is the same.

Downstream speed of boat D = downstream speed of boat E = 20 km/h

This statement is true.

II). The total given distance covered by boat C, with the original speed of the boat in still water and the stream, is 80 km less than that of boat A.

We cannot find the distance travelled by boat A

This statement does not have to be true.

III). The upstream speed of boat A is S km/h

Value of S = 4

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Upstream speed of boat A = 6 – 2 = 4 km/h

So, only I and III must be true

Hence answer is option A

14. Questions

For boat A

Ratio of upstream distance to downstream distance = 1:1

Ratio of upstream time to downstream time = 2:1

So, ratio of upstream speed to downstream speed = (1/2:1/1) = 1:2

So, ratio of the speed of the boat in still water to the speed of the stream = (1 +2)/2: (2 – 1)/2 = 3:1 = 3a, a

Now,

(2 x 3a + a)/2 = 7

7a = 14

Value of a = 2

So, speed of boat in still water and speed of stream = 6 km/hr and 2 km/hr

Downstream speed = 8 km/h

Upstream speed = 4 km/h

Boat B

Time taken by the boat to travel 60Q downstream distance = 60Q/ (P +Q)

Now,

2 x 60Q / (P + Q) = 16P / (P – Q)

$2P^2 – 13PQ + 15Q^2 = 0$

(2P – 3Q) (P – 5Q) = 0

So, P = 5Q, and P = 3Q/2

Also,

2P + 3Q = 13

So, (2 x 5Q + 3Q) = 13

Value of Q = 1

Value of P = 5

If we put P = 3Q/2, we do not get an integral value.

For boat C

Time taken by the boat to travel 90 km downstream = 6 hours

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So, downstream speed of boat = 90/6 = 15

(a + b) = 15…….... (1)

Also,

100/2 = (3 x a + 4 x b)

3a + 4b = 50……. (2)

On, solving both equations, we get

Value of b = 5, value of a = 10

so, upstream speed of boat = 10 – 5 = 5 km/h

upstream time taken = 6/40 x 100 = 15 hours

value of M = 5 x 15 =75 km

Boat D

Time taken by the boat to travel (4R + 8) km upstream = 20% x 450/15 = 6 hours

(4R + 8) = (R - S) x 6

2R + 4 = 3R - 3S

R – 3S = 4……. (1)

Also,

2.5R = (R + S) x 6/3

R = 4S…….... (2)

On, solving (1) and (2), we get

4S – 3S = 4

Value of S = 4

Value of R = 16

Downstream distance = 2.5 x 16 = 40 km

Upstream distance = 4 x 16 + 8 = 72 km

Boat E

Ratio of the speed of the boat in still water to the speed of the stream = 3:1

So, ratio of upstream speed to downstream speed = 2:1

So, upstream speed = 1 x 10 = 10 km/h

Downstream speed = 2 x 10 = 20 km/h

So, speed of boat in still water = 15 km/h

Speed of stream = 5 km/h

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Answer: A

Quantity I.

Required time = 45 / (80% x 5 – 1 x 2) = 22.5hours

Quantity II

Required time = (4 x 16 + 11 x 1)/ (15 – 5) = 7.5 hours

Required difference = (22.5 – 7.5) = 15 hours

Hence answer is option A

15. Questions

For boat A

Ratio of upstream distance to downstream distance = 1:1

Ratio of upstream time to downstream time = 2:1

So, ratio of upstream speed to downstream speed = (1/2:1/1) = 1:2

So, ratio of the speed of the boat in still water to the speed of the stream = (1 +2)/2: (2 – 1)/2 = 3:1 = 3a, a

Now,

(2 x 3a + a)/2 = 7

7a = 14

Value of a = 2

So, speed of boat in still water and speed of stream = 6 km/hr and 2 km/hr

Downstream speed = 8 km/h

Upstream speed = 4 km/h

Boat B

Time taken by the boat to travel 60Q downstream distance = 60Q/ (P +Q)

Now,

2 x 60Q / (P + Q) = 16P / (P – Q)

$2P^2 – 13PQ + 15Q^2 = 0$

(2P – 3Q) (P – 5Q) = 0

So, P = 5Q, and P = 3Q/2

Also,

2P + 3Q = 13

So, (2 x 5Q + 3Q) = 13

Value of Q = 1

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Value of P = 5

If we put P = 3Q/2, we do not get an integral value.

For boat C

Time taken by the boat to travel 90 km downstream = 6 hours

So, downstream speed of boat = 90/6 = 15

(a + b) = 15…….... (1)

Also,

100/2 = (3 x a + 4 x b)

3a + 4b = 50……. (2)

On, solving both equations, we get

Value of b = 5, value of a = 10

so, upstream speed of boat = 10 – 5 = 5 km/h

upstream time taken = 6/40 x 100 = 15 hours

value of M = 5 x 15 =75 km

Boat D

Time taken by the boat to travel (4R + 8) km upstream = 20% x 450/15 = 6 hours

(4R + 8) = (R - S) x 6

2R + 4 = 3R - 3S

R – 3S = 4……. (1)

Also,

2.5R = (R + S) x 6/3

R = 4S…….... (2)

On, solving (1) and (2), we get

4S – 3S = 4

Value of S = 4

Value of R = 16

Downstream distance = 2.5 x 16 = 40 km

Upstream distance = 4 x 16 + 8 = 72 km

Boat E

Ratio of the speed of the boat in still water to the speed of the stream = 3:1

So, ratio of upstream speed to downstream speed = 2:1

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So, upstream speed = 1 x 10 = 10 km/h

Downstream speed = 2 x 10 = 20 km/h

So, speed of boat in still water = 15 km/h

Speed of stream = 5 km/h

Answer: E

Required sum = 6 + 10 + 5 + 16 + 15 = 52 km/h

Hence answer is option E

16. Questions

Answer: C

Let income of R in 2021 = 100a

Savings of R in 2021 = 40% x 100a = 40a

So, savings of R in 2022 = 1.5 x 40a = 60a

So, expenses of R in 2022 = 60a

Income of R in 2022 = 120a

So, value of M = (120a – 100a)/100a x 100 = 20%

So, income of Q in 2022 is M% (20%) more than that in 2021

(Expenses of Q in 2022 + Expenses of R in 2022)/2 = 24Z + (expenses of Q in 2021 + expenses of R in 2021)/2

Expenses of R in 2021 = expenses of R in 2022

Expenses of Q in 2022 = 48Z + expenses of Q in 2021

Let income of Q in 2021 = 100b

So, income of Q in 2022 = 120b

120b x 80% = 48Z + 90% x 100b

6b = 48Z

Value of b = 8Z

Required difference = (120b – 100b) = 20b = 20 x 8Z = Rs. 160Z

Hence answer is option C

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17. Questions

Answer: B

Income of P in 2021 = Income of P in 2022 = 100a

Income of Q in 2021 = Income of Q in 2022 = 100b

80% x 100a = 2 x 80% x 100b

So, a = 2b

Value of A = (75% x 100 x 2b – 80% x 100b) = 70b

Value of B = (90% x b – 25% x 2b) = 40b

Required % change = (70b – 40b)/40b x 100 = 75%

Hence answer is option B

18. Questions

Answer: D

Let income of R in 2021 = 100a

So, expenses of R in 2021 = 60% x 100a = 60a

So, expenses of R in 2022 = 60a/40 x 100 = 150a

So, expenses of P in 2022 = (150a + 60a)/2 = 210a/2

Income of P in 2022 = 210a/2 x 5/4

Ratio of income of P in 2021 to 2022 = 8:3

So, expenses of P in 2021 = 75% x (8/3 x 210a/2 x 5/4) = (210a x 5/4)

Ratio of the income of R in 2021 to 1/3rd of the expenses of P in 2021 =

100a: 1/3 x 210a x 5/4 = 8:7

Required % change = (8 – 7)/7 x 100 = 14.28%

Hence answer is option D

19. Questions

Answer: A

Income of P in both years is the same. So,

(75% + 80%)/2 x income of P = 6200

So, income of P in 2021 = Income of P in 2022 = 8000

So, income of R in 2021 = 150% x 8000 = 12000

So, income of R in 2022 = 80% x 12000 = 9600

So, expenses of R in 2022 = 50% x 9600 = 4800

Hence answer is option A

20. Questions

Answer: E

Let, income of Q in each year = 100a

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Let, income of R in 2021 = 200b

So, income of R in 2022 = 200b x 150% = 300b

Now,

90% x 100a – 50% x 300b = 9000

3a – 5b = 300…… (1)

Also,

60% x 200b – 20% x 100a = 3200

6b – a = 160…… (2)

On solving both equations, we get

Value of a = 200, and b = 60

So, average income of Q = 200 x 100 = Rs. 20000

Average income of R = 250 x 60 = Rs. 15000

Required difference = 20000 – 15000 = Rs. 5000

Hence answer is option E.

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